

Appendix G

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Application Number	61/798,956
Filing Date	03-15-2013
First Named Inventor	Scott H. Milton
Title	HYBRID MEMORY SYSTEM WITH CONFIGURABLE ERROR THRESHOLD PARAMETERS
Art Unit	Not assigned
Examiner Name	Not assigned
Attorney Docket Number	0016.0020000

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	
Name	Khaled Shami	Registration Number	38,745
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Applicant Name (if Applicant is a juristic entity)			

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FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 17/138,766, filed December 30, 2020, titled "Flash-Dram Hybrid Memory", now U.S. Patent No. 11,016,918, which is a continuation of U.S. Patent Application No. 15/934,416, filed March 23, 2018, titled "Flash-Dram Hybrid Memory Module," which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled "Non-Volatile Memory Module," now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may be considered related to U.S. Patent Application No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed February 5, 2014, which is a continuation of U.S. Patent Application 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Patent 8,677,060, issued

March 18, 2014, which is a continuation of U.S. Patent Application 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed June 28, 2012, now U.S. Patent 8,516,187, issued August 20, 2013, which is a divisional of U.S. Patent Application 12/240,916, titled "Non-Volatile Memory Module," filed September 29, 2008, now U.S. Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007, the contents of which are incorporated by reference herein in their entirety.

[0003] This application may also be considered related to U.S. Patent Application No. 15/000,834, filed January 19, 2016 (abandoned), which is a continuation of U.S. Patent Application No. 14/489,332, filed September 17, 2014, now U.S. Patent No. 9,269,437, which is a continuation of U.S. Patent Application No. 14/173,219, filed February 5, 2014, now U.S. Patent No. 8,904,099, which is a continuation of U.S. Patent Application No. 13/905,048, filed May 29, 2013, now U.S. Patent No. 6,671,243, which is a continuation U.S. Patent Application 13/536,173 above.

[0004] This application may also be considered related to U.S. Patent Application No. 15/924,866, (abandoned), which is a continuation of U.S. Patent Application No. 14/489,281, filed September 17, 2014, now U.S. Patent No. 9,921,762, which is a continuation of U.S. Patent Application No. 13/625,563, filed September 24, 2012, now U.S. Patent No. 8,904,098, which claims the benefit of U.S. Provisional Application No. 61/583,775, filed September 23, 2011.

TECHNICAL FIELD

[0005] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0006] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0007] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high

speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0008] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0009] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a

data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MB/s for the EcoRAM vs. 104MB/s for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™, which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0010] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0011] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash

DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0012] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

[0013] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0014] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0015] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

[0016] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0017] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0018] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0019] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0020] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0021] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0022] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0024] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0025] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0026] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0027] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0029] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0030] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0031] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0032] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0033] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0034] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0035] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0036] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0037] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0038] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0039] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0040] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0041] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0042] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0043] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0044] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0045] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0047] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0048] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0049] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0050] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0052] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0053] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0054] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0055] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0056] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0057] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0060] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0061] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0062] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0063] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0064] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0065] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0066] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0067] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0068] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0069] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

[0070] Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to

allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

[0071] Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

[0072] Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0074] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0075] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to

implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0076] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0077] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only

Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0078] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0079] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0080] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0081] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0082] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or

commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0083] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0084] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0085] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then

the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0086] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0087] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0088] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory

module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0089] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In

the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0090] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0091] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH

510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0092] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0093] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the

DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0094] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0095] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate

and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0096] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0097] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or

provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0098] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0099] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[00100] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[00101] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[00102] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from

the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[00103] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[00104] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[00105] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a

first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[00106] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[00107] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00108] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502

monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00109] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061

field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A_0A_1 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00110] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00111] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash

memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00112] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'}))*(\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00113] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00114] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00115] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up

the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[00116] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[00117] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor

array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[00118] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[00119] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[00120] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[00121] FIG. 12 is a block diagram of an example memory system 1010 compatible with certain embodiments described herein. The memory system 1010 can be coupled to a host computer system and can include a volatile memory subsystem 1030, a non-volatile memory

subsystem 1040, and a controller 1062 operatively coupled to the non-volatile memory subsystem 1040. In certain embodiments, the memory system 1010 includes at least one circuit 1052 configured to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030.

[00122] In certain embodiments, the memory system 1010 comprises a memory module. The memory system 1010 may comprise a printed-circuit board (PCB) 1020. In certain embodiments, the memory system 1010 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 1010 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for

example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[00123] In certain preferred embodiments, the memory system 1010 is in electrical communication with the host system. In other embodiments, the memory system 1010 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 1010 can be in communication with a disk controller of a computer system, for example. The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.

[00124] The volatile memory subsystem 1030 comprises a plurality of volatile memory elements 1032 and the non-volatile memory subsystem 1040 comprises a plurality of non-volatile memory elements 1042. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem 1040 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 1030. In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and

synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00125] In certain embodiments, the second plurality of non-volatile memory elements 1042 comprises one or more flash memory elements. Types of flash memory elements 1042 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank 1040 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 1042. In addition, nonvolatile memory elements 1042 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 1042 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00126] FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements

1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

[00127] Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one

embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

[00128] In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

[00129] In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one

or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 1010 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example,

1010 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[00131] As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 1070 is also programmed to perform address/address translation between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 1042 because of their superior read speed and compact structure.

[00132] The memory system 1010 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 1062 and the non-volatile memory subsystem 1040 are operatively decoupled (e.g., isolated) from the

volatile memory subsystem 1030 by the at least one circuit 1052 and a second state in which the volatile memory subsystem 1030 is operatively coupled to the controller 1062 to allow data to be communicated between the volatile memory subsystem 1030 and the nonvolatile memory subsystem 1040 via the controller 1062. The memory system 1010 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 1010 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[00133] The memory system 1010 may further comprise a voltage monitor 1050. The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 1050 may transmit a signal to the controller 1062 indicative of the detected condition. The controller 1062 of certain embodiments responds to the signal from the voltage monitor circuit 1050 by transmitting a signal to the at least one circuit 1052 to operatively couple the controller to the volatile memory system 1030, such that the memory system 1010 enters the second state. For example, the voltage monitor 1050 may send a signal to the MCU 1060 which responds by accessing the data on the volatile memory system 1030 and by executing a write cycle on the nonvolatile memory subsystem 1040. During this write cycle, data is read from the volatile memory subsystem 1030 and is transferred to the non-volatile memory subsystem 1040 via the MCU 1060. In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.

[00134] The isolation or operational decoupling of the volatile memory subsystem 1030 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 1010 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 1030. For example, in one embodiment during such periods of operation, the controller 1062 and the nonvolatile memory subsystem 1040 do not add a significant capacitive load to the volatile memory system 1030 when the memory system 1010 is in the first state. In certain such embodiments, the capacitive load of the controller 1062 and the non-volatile memory subsystem 1040 do not significantly affect the signals propagating between the volatile memory subsystem 1030 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 1052 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 1052 comprises other types of isolation devices.

[00135] Power may be supplied to the volatile memory subsystem 1030 from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. In certain embodiments, the memory system 1010 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 1010 enters the second state in response to a trigger condition. In certain embodiments, the memory system 1010 has a third state in which the controller 1062 is operatively decoupled from the volatile memory subsystem 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory

subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.

[00136] In certain embodiments, the second power supply 1080 does not comprise a battery. Because a battery is not used, the second power supply 1080 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. 12-14, the second power supply 1080 comprises a step-up transformer 1082, a step-down transformer 1084, and a capacitor bank 1086 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 1080 is configured to power the volatile memory subsystem 1030 for less than thirty minutes. In certain embodiments, the second power supply 1080 may comprise a battery. For example, in certain embodiments, the second power supply 1080 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 1030 for no more than thirty minutes.

[00137] In certain embodiments, the capacitor bank 1086 of the second power supply 1080 is charged by the first power supply while the memory system 1010 is in the first state. As a result, the second power supply 1080 is fully charged when the memory system 1010 enters the second state. The memory system 1010 and the second power supply 1080 may be located on the same printed circuit board 1020. In other embodiments, the second power supply 1080 may not be on

the same printed circuit board 1020 and may be tethered to the printed circuit board 1020, for example.

[00138] When operating in the first state, in certain embodiments, the step-up transformer 1082 keeps the capacitor bank 1086 charged at a peak value. In certain embodiments, the step-down transformer 1084 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 1032 and 3.0V to the non-volatile flash memory elements 1042) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. 12-14, the memory module 1010 further comprises a switch 1090 (e.g., FET switch) that switches power provided to the controller 1062, the volatile memory subsystem 1030, and the non-volatile memory subsystem 1040, between the power from the second power supply 1080 and the power from the first power supply (e.g., system power) received via the interface 1022. For example, the switch 1090 may switch from the first power supply to the second power supply 1080 when the voltage monitor 1050 detects a low voltage condition. The switch 1090 of certain embodiments advantageously ensures that the volatile memory elements 1032 and non-volatile memory elements 1042 are powered long enough for the data to be transferred from the volatile memory elements 1032 and stored in the non-volatile memory elements 1042. In certain embodiments, after the data transfer is complete, the switch 1090 then switches back to the first power supply and the controller 1062 transmits a signal to the at least one circuit 1052 to operatively decouple the controller 1062 from the volatile memory subsystem 1030, such that the memory system 1010 reenters the first state.

[00139] When the memory system 1010 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 via the

controller 1062. The host system can then resume accessing the volatile memory subsystem 1030 of the memory module 1010. In certain embodiments, after the memory system 1010 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 1030 rather than the non-volatile memory subsystem 1040 because the volatile memory elements 1032 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 1030 to the nonvolatile memory bank 1040, or from the non-volatile memory bank 1040 to the volatile memory bank 1030, takes less than one minute per GB.

[00140] In certain embodiments, the memory system 1010 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 1010 copies the entire contents of the volatile memory subsystem 1030 into the nonvolatile memory subsystem 1040 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are copied back into the volatile memory subsystem 1030 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 1040 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 1010. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 1010 and can improve the performance of the memory system 1010. In certain other embodiments, the entire contents of the volatile memory subsystem 1030 are not copied into the non-volatile memory subsystem 1040 on each backup operation, but only a partial copy is performed. In certain embodiments, other

management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 1040 are performed in the controller 1062.

[00141] The memory system 1010 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 1030 which then writes the data to non-volatile storage which is not part of the memory system 1010, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 1010 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 1010 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 1010 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 1010 may be operated as a write-through cache or as some other type of cache.

[00142] FIG. 16 schematically illustrates an example power module 1100 of the memory system 1010 in accordance with certain embodiments described herein. The power module 1100 provides power to the various components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. In certain embodiments, the power module 1100 comprises one or more of the components described above with respect to FIG. 12. For example, in certain embodiments, the power module 1100 includes the second power supply 1080 and the switch 1090.

[00143] The power module 1100 provides a plurality of voltages to the memory system 1010 comprising non-volatile and volatile memory subsystems 1030, 1040. The plurality of voltages comprises at least a first voltage 1102 and a second voltage 1104. The power module 1100 comprises an input 1106 providing a third voltage 1108 to the power module 1100 and a voltage conversion element 1120 configured to provide the second voltage 1104 to the memory system 1010. The power module 1100 further comprises a first power element 1130 configured to selectively provide a fourth voltage 1110 to the conversion element 1120. In certain embodiments, the first power element 1130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 1130 is configured to receive a 1.8V input system voltage as the third voltage 1108 and to output a modulated 5V output as the fourth voltage 1110.

[00144] The power module 1100 further comprises a second power element 1140 can be configured to selectively provide a fifth voltage 1112 to the conversion element 1120. The power module 1100 can be configured to selectively provide the first voltage 1102 to the memory system 1010 either from the conversion element 1120 or from the input 1106.

[00145] The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element

1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.

[00146] In certain embodiments, the power module 1100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 1100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 1102 in the second state from the first power element 1130 rather than from the input 1106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 1102 from the first power element 1130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 1120 and the first and second power elements 1130, 1140 (e.g., the sources of the pre-regulated fourth voltage 1110 in the second state and the pre-regulated fifth voltage 1112 in the third state) can smooth out potential voltage spikes.

[00147] In certain embodiments, the second power element 1140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. 16, the second power element 1140 comprises a capacitor array 1142, a buck-boost converter 1144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 1146 which limits the charge current to the capacitor array 1142 and stops charging the capacitor array 1142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 1142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For

example, in one example embodiment, the buck-boost converter 1144 receives a 1.8V system voltage (first voltage 1108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 1146. The voltage/current limiter 1146 limits the current going to the capacitor array 1142 to 1A and stops charging the array 1142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 1140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 1142.

[00148] The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate

amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.

[00149] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 1120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.

[00150] FIG. 17 is a flowchart of an example method 1200 of providing a first voltage 1102 and a second voltage 1104 to a memory system 1010 including volatile and nonvolatile memory subsystems 1030, 1040. While the method 1200 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-15, other memory systems are also compatible with embodiments of the method 1200. During a first condition, the method 1200

comprises providing the first voltage 1102 to the memory system 1010 from an input power supply 1106 and providing the second voltage 1104 to the memory system 1010 from a first power subsystem in operational block 1210. For example, in one embodiment, the first power subsystem comprises the first power element 1130 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, other first power subsystems are used.

[00151] The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.

[00152] The method 1200 further comprises charging a second power subsystem in operational block 1240. In certain embodiments, the second power subsystem comprises the second power element 1140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 1140 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, some other second power subsystem is used.

[00153] The method 1200 further comprises detecting a third condition in an operational block 1250 and during the third condition, providing the first voltage 1102 and the second

voltage 1104 to the memory system 1010 from the second power subsystem 1140 in an operational block 1260. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 1200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 1140 is charged before detecting the second condition.

[00154] In certain embodiments, the memory system 1010 comprises a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 1010 also comprises a controller 1062 operatively coupled to the volatile memory subsystem 1030 and operatively coupled to the non-volatile memory subsystem 1040. The controller 1062 can be configured to allow data to be communicated between the volatile memory subsystem 1030 and the host system when the memory system 1010 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040 when the memory system 1010 is operating in a second state.

[00155] Although the memory system 1010 having extra storage capacity of the non-volatile memory subsystem 1040 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In various embodiments, there may be at least 200, 300, or 400 percent more

storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In other embodiments, the non-volatile memory subsystem 1040 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 1030. In some embodiments, the non-volatile memory subsystem 1040 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 1030. In one embodiment, the non-volatile memory subsystem 1040 includes less than 100 percent more storage capacity than does the volatile memory subsystem 1030.

[00156] The extra storage capacity of the non-volatile memory subsystem 1040 can be used to improve the backup capability of the memory system 1010. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 1040 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem 1040 allows the volatile memory subsystem 1030 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to backup the volatile memory subsystem 1030 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 1030 is copied to a first, previously erased portion of the nonvolatile memory subsystem 1040 via the controller 1062. Since the non-volatile memory subsystem 1040 has more storage capacity than does the volatile memory subsystem 1030, there is a second portion of the non-volatile memory subsystem 1040 which does not have data from the volatile memory subsystem 1030 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 1062 of the memory system 1010 restores the data to the volatile memory subsystem 1030 by copying the backed-up data from the non-volatile memory

subsystem 40 back to the volatile memory subsystem 1030. After the data is restored, the memory system 1010 erases the non-volatile memory subsystem 1040. While the first portion of the non-volatile memory subsystem 1040 is being erased, it may be temporarily inaccessible.

[00157] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 1040 is completely erased, the volatile memory subsystem 1030 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 1040 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 1040, the memory system 1010 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 1040.

[00158] FIG. 18 is a flowchart of an example method 1300 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. In certain embodiments, the non-volatile memory subsystem 1040 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 1300 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-14, the method 1300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 1310, the method 1300 comprises communicating data between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation. The method 1300 further comprises storing a first copy of data from the

volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a first time when the memory system 1010 is in a second mode of operation in an operational block 1320.

[00159] In an operational block 1330, the method 1300 comprises restoring the first copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030. The method 1300 further comprises erasing the first copy of data from the non-volatile memory subsystem 1040 in an operational block 1340. The method further comprises storing a second copy of data from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a second time when the memory system 1010 is in the second mode of operation in an operational block 1350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 1040.

[00160] In some embodiments, the memory system 1010 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 1040. The method 1300 can also include restoring the second copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 in an operational block 1360. The operational blocks of method 1300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem 1030 at operational block 1360 before the first copy of data is completely erased in the operational block 1340.

[00161] FIG. 19 schematically illustrates an example clock distribution topology 1400 of a memory system 1010 in accordance with certain embodiments described herein. The clock

distribution topology 1400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 1010. A clock source 1402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 1402 may feed a clock generator 1404 which provides a clock signal 1406 to the controller 1062, which may be an FPGA. In one embodiment, the clock generator 1404 generates a 125 MHz clock signal 1406. The controller 1062 receives the clock signal 1406 and uses it to clock the controller 1062 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 1062.

[00162] The clock signal 1406 can also be input into a clock divider 1410 which produces a frequency-divided version of the clock signal 1406. In an example embodiment, the clock divider 1410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 1406. A non-volatile memory phase-locked loop (PLL) block 1412 can be included (e.g., in the controller 1062) which distributes a series of clock signals to the non-volatile memory subsystem 1040 and to associated control logic. For example, a series of clock signals 1414 can be sent from the controller 1062 to the non-volatile memory subsystem 1040. Another clock signal 1416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 1040. For example, the clock signal 1416 may clock the portion of the controller 1062 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 1040. A feedback clock signal 1418 is fed back into the non-volatile memory PLL block 1412. In one embodiment, the PLL block 1412 compares the feedback clock 1418 to the reference clock 1411 and varies the phase and frequency of its output until the reference 1411 and feedback 1418 clocks are phase and frequency matched.

[00163] A version of the clock signal 1406 such as the backup clock signal 1408 may be sent from the controller to the volatile memory subsystem 1030. The clock signal 1408 may be, for example, a differential version of the clock signal 1406. As described herein, the backup clock signal 1408 may be used to clock the volatile memory subsystem 1030 when the memory system 1010 is backing up the data from the volatile memory subsystem 1030 into the non-volatile memory subsystem 1040. In certain embodiments, the backup clock signal 1408 may also be used to clock the volatile memory subsystem 1030 when the memory system 1010 is copying the backed-up data back into the volatile memory subsystem 1030 from the nonvolatile memory subsystem 1040 (also referred to as restoring the volatile memory subsystem 1030). The volatile memory subsystem 1030 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the nonvolatile memory subsystem 1040 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 1030 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 1040) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 1030 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 1010.

[00164] In one embodiment, the backup clock 1408 and the volatile memory system clock signal 1420 are received by a multiplexer 1422, as schematically illustrated by FIG. 19. The multiplexer 1422 can output either the volatile memory system clock signal 1420 or the backup clock signal 1408 depending on the backup state of the memory system 1010. For example, when the memory system 1010 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 1420 may be provided by the multiplexer 422 to the volatile memory PLL block 1424. When the memory system 1010 is performing a backup (or restore) operation, the backup clock signal 1408 may be provided.

[00165] The volatile memory PLL block 1424 receives the volatile memory reference clock signal 1423 from the multiplexer 1422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 1030 and associated control logic. For example, in one embodiment, the PLL block 1424 generates a series of clock signals 1426 which clock the volatile memory elements 1032. A clock signal 1428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 1430 may be sent to the controller 1062. A feedback clock signal 1432 is fed back into the volatile memory PLL block 1424. In one embodiment, the PLL block 1424 compares the feedback clock signal 1432 to the reference clock signal 1423 and varies the phase and frequency of its output until the reference clock signal 1423 and the feedback clock signal 1432 clocks are phase and frequency matched.

[00166] The clock signal 1430 may be used by the controller 1062 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 1030. For example, control logic in the controller 1062 may be used to control the volatile memory subsystem 1030 during a backup or restore operation. The clock signal 1430 may be used as a reference clock signal for the PLL block 1434 which can generate one or more clocks 1438 used by logic in the controller 1062. For example, the PLL block 1434 may generate one or more clock signals 1438 used to drive logic circuitry associated with

controlling the volatile memory subsystem 1030. In certain embodiments, the PLL block 1434 includes a feedback clock signal 1436 and operates in a similar manner to other PLL blocks described herein.

[00167] The clock signal 1430 may be used as a reference clock signal for the PLL block 1440 which may generate one or more clock signals used by a sub-block 1442 to generate one or more other clock signals 1444. In one embodiment, for example, the volatile memory subsystem 1030 comprises DDR2 SDRAM elements and the sub-block 1442 generates one or more DDR2 compatible clock signals 1444. A feedback clock signal 1446 is fed back into the PLL block 1440. In certain embodiments, the PLL block 1440 operates in a similar manner to other PLL blocks described herein.

[00168] While described with respect to the example embodiment of FIG. 19, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 1030 operates on the volatile memory clock signal 1420 and there is no backup clock signal 1408. In some embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a restore operation and not during a backup operation.

[00169] FIG. 20 is a flowchart of an example method 1500 of controlling a memory system 1010 operatively coupled to a host system. Although described with respect to the memory

system 1010 described herein, the method 1500 is compatible with other memory systems. The memory system 1010 may include a clock distribution topology 1400 similar to the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 1040.

[00170] In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[00171] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

[00172] FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKE signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice

from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[00173] In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[00174] In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630

into self-refresh mode by sending a signal via the CKe line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKe line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

[00175] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00176] FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation in an operational block 1610. For example, the memory

system 1010 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[00177] In an operational block 1620, the method further comprises transferring data words from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 when the memory system 1010 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.

[00178] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The

40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[00179] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[00180] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1040 write each backed-up data word to the controller 1062 which writes a first portion of the data word to the volatile memory subsystem 1030 and then a second portion of the data word to the volatile memory 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00181] The method 1600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile memory subsystem 1030 comprises DRAM elements, only the slice of the DRAM which is currently being accessed

(e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 1600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[00182] In addition, the connection between the DRAM elements and the controller 1062 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 1062 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 1062 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 1030. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[00183] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module comprising:
 - a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
 - a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;
 - a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;
 - a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;
 - a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and
 - a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:
 - a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and
 - at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes.

ABSTRACT

In certain embodiments, a memory module includes a printed circuit board (PCB) having an interface that couples it to a host system for provision of power, data, address and control signals. First, second, and third buck converters receive a pre-regulated input voltage and produce first, second and third regulated voltages. A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage. Synchronous dynamic random access memory (SDRAM) devices are coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, and a voltage monitor circuit monitors an input voltage and produces a signal in response to the input voltage having a voltage amplitude that is greater than a threshold voltage.

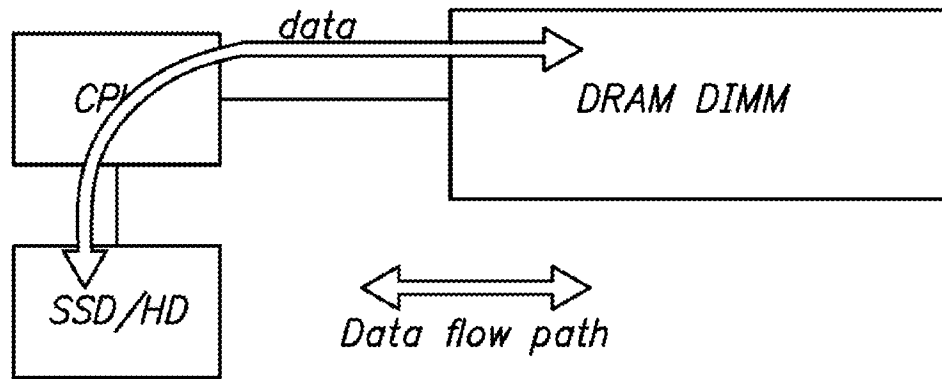


FIG. 1
(PRIOR ART)

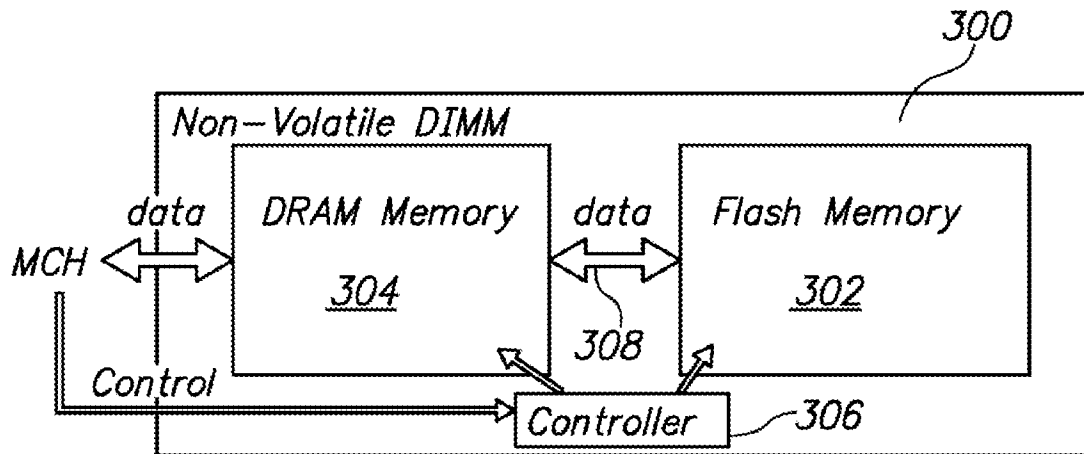
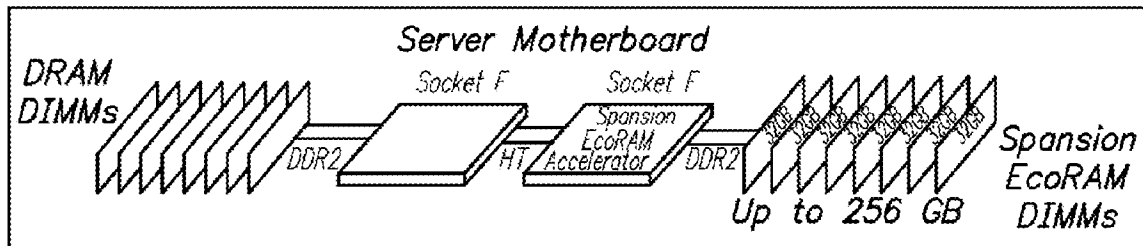


FIG. 3A

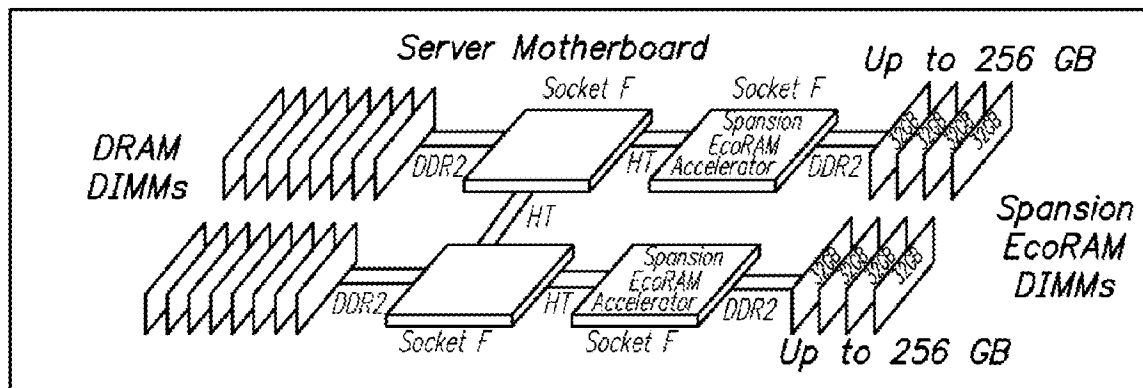
Spancion EcoRAM Configurations

256GB Spancion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

256GB Spancion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

FIG. 2
(PRIOR ART)

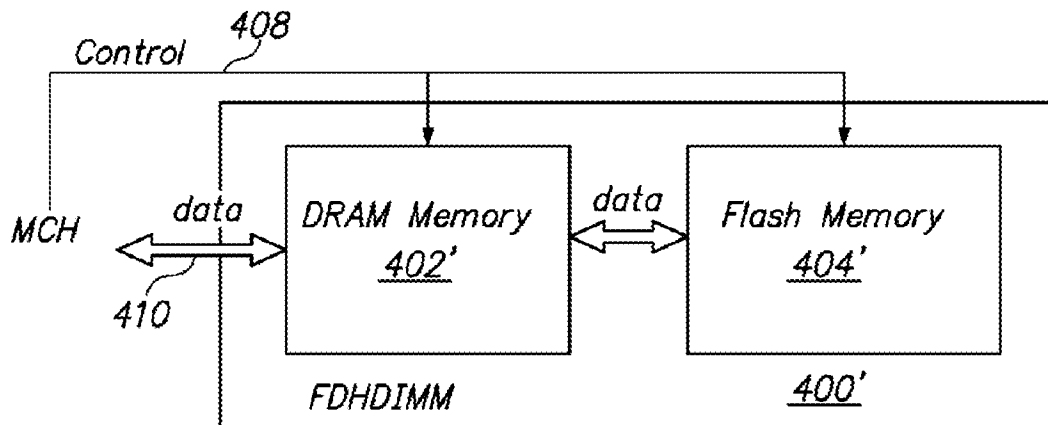


FIG. 4B

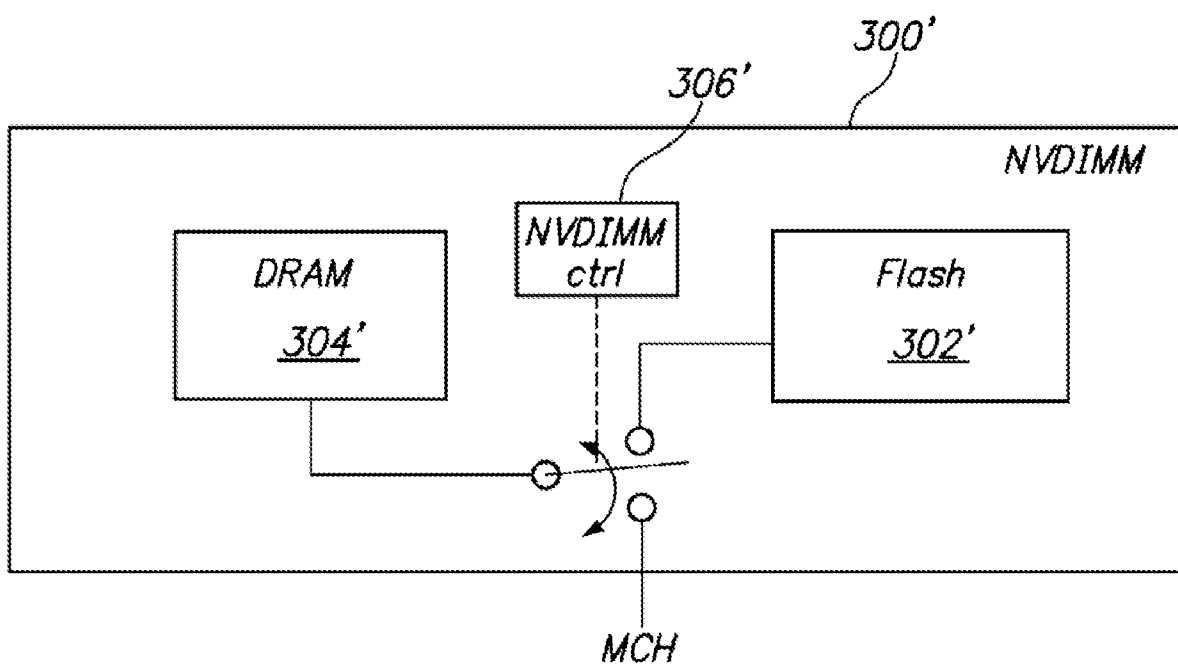


FIG. 3B

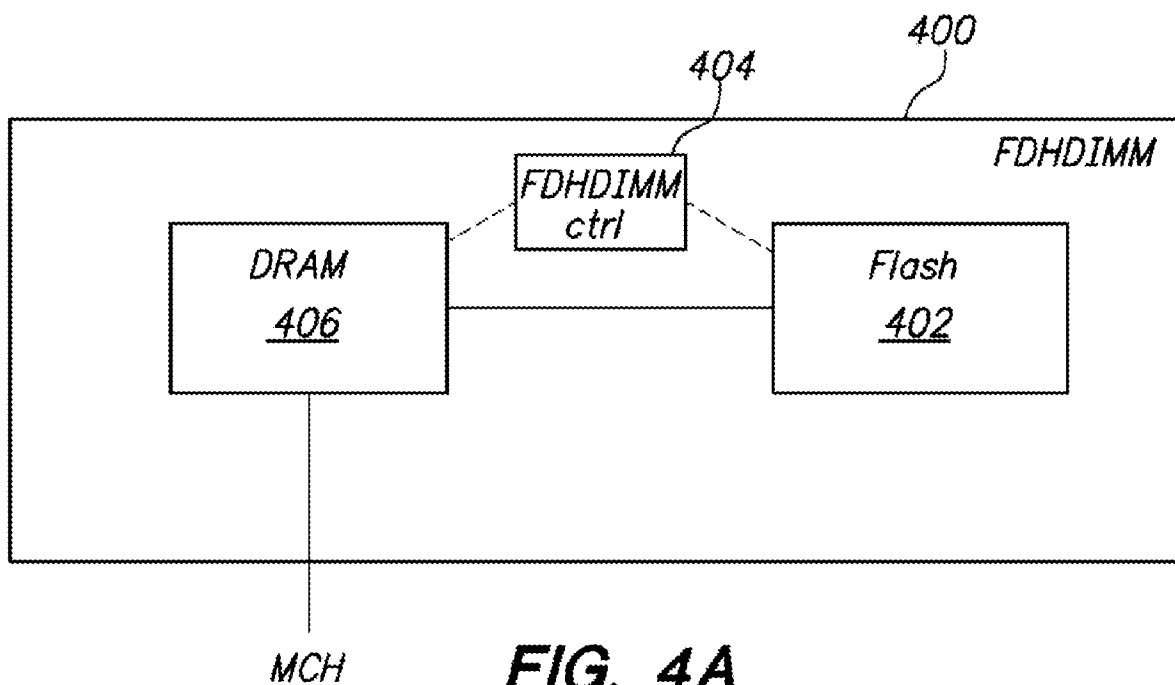


FIG. 4A

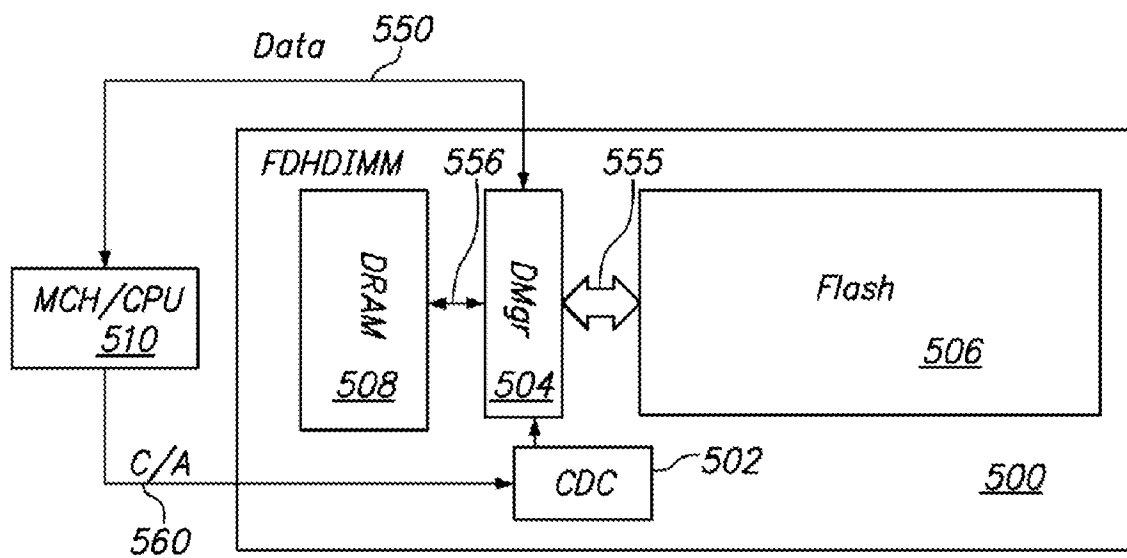


FIG. 5A

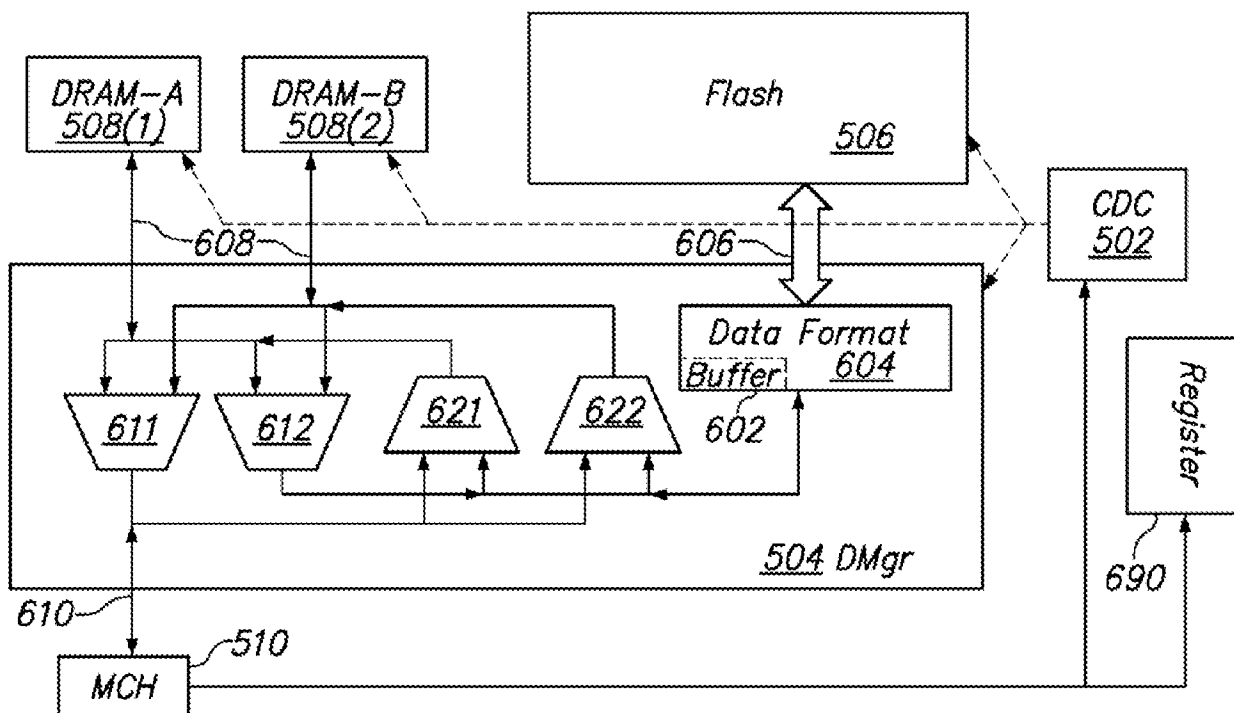


FIG. 6

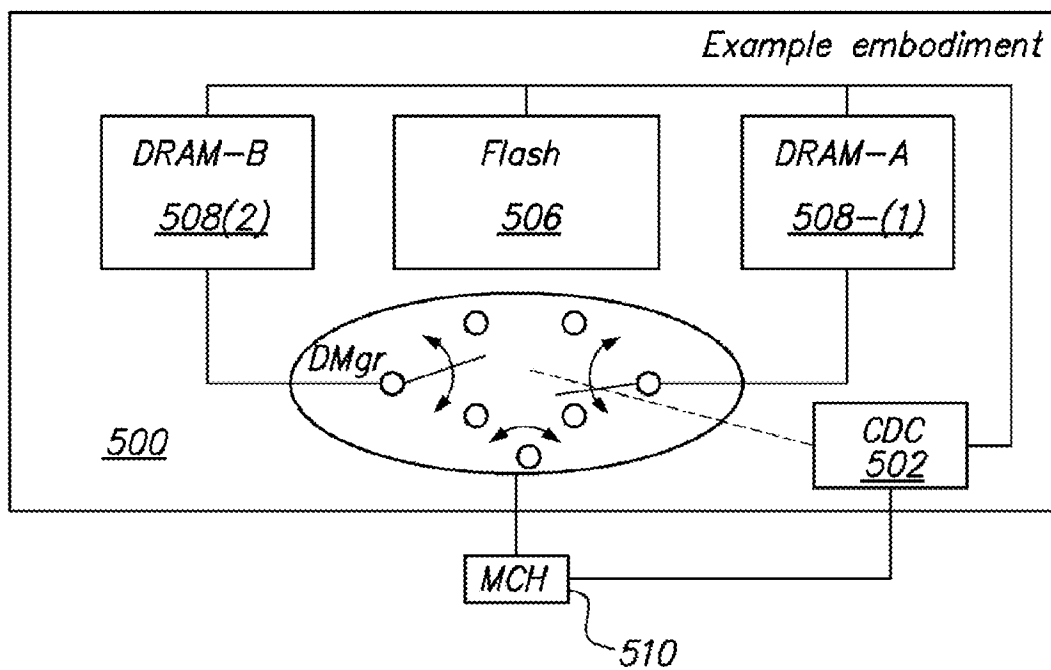


FIG. 5B

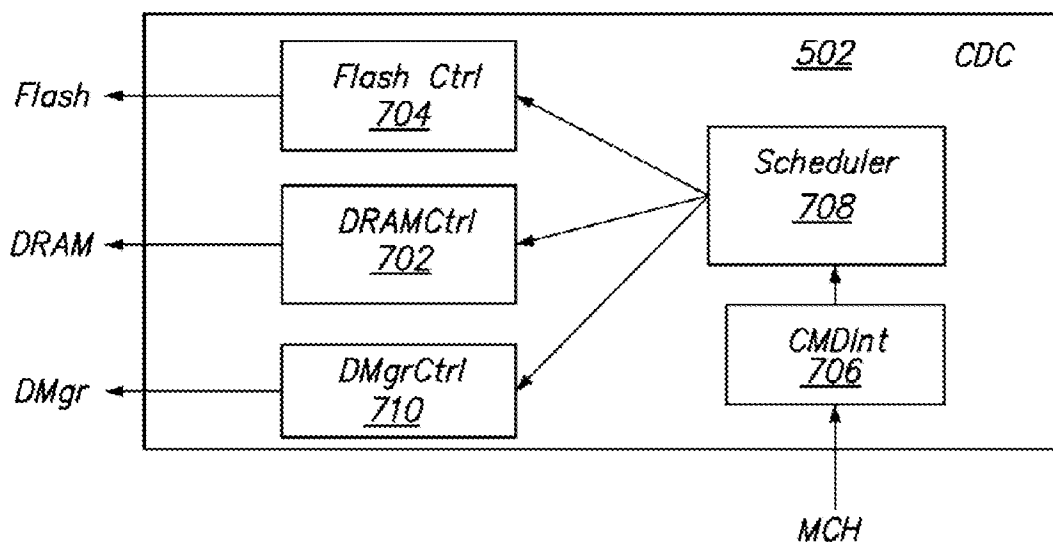


FIG. 7

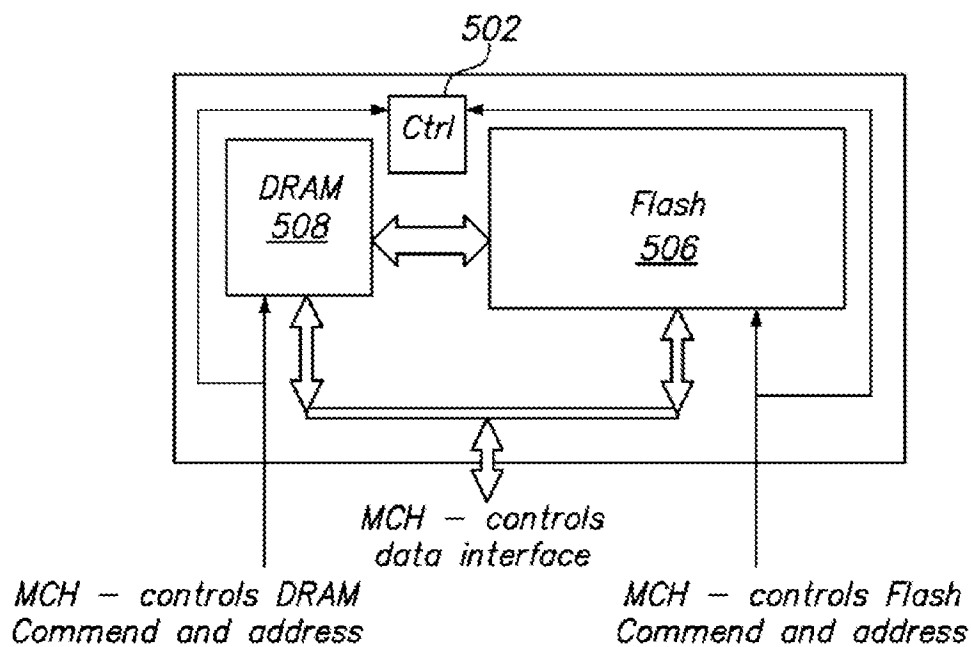


FIG. 8A

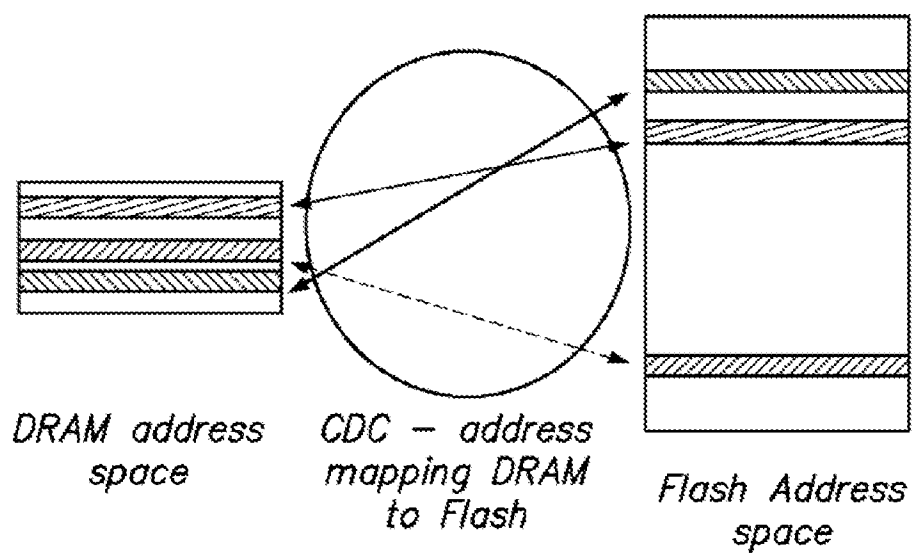
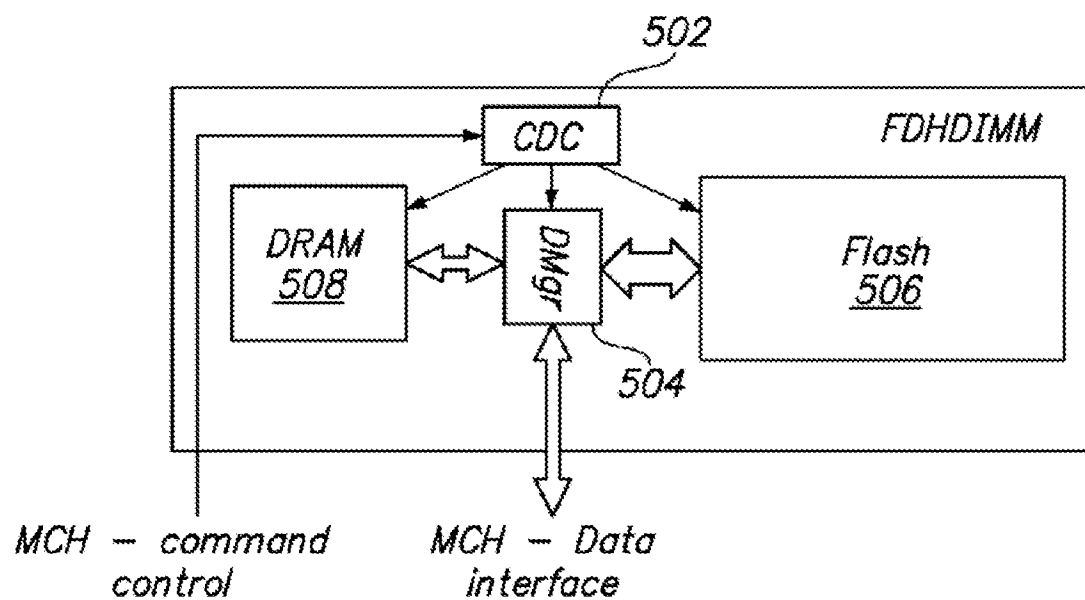


FIG. 8B

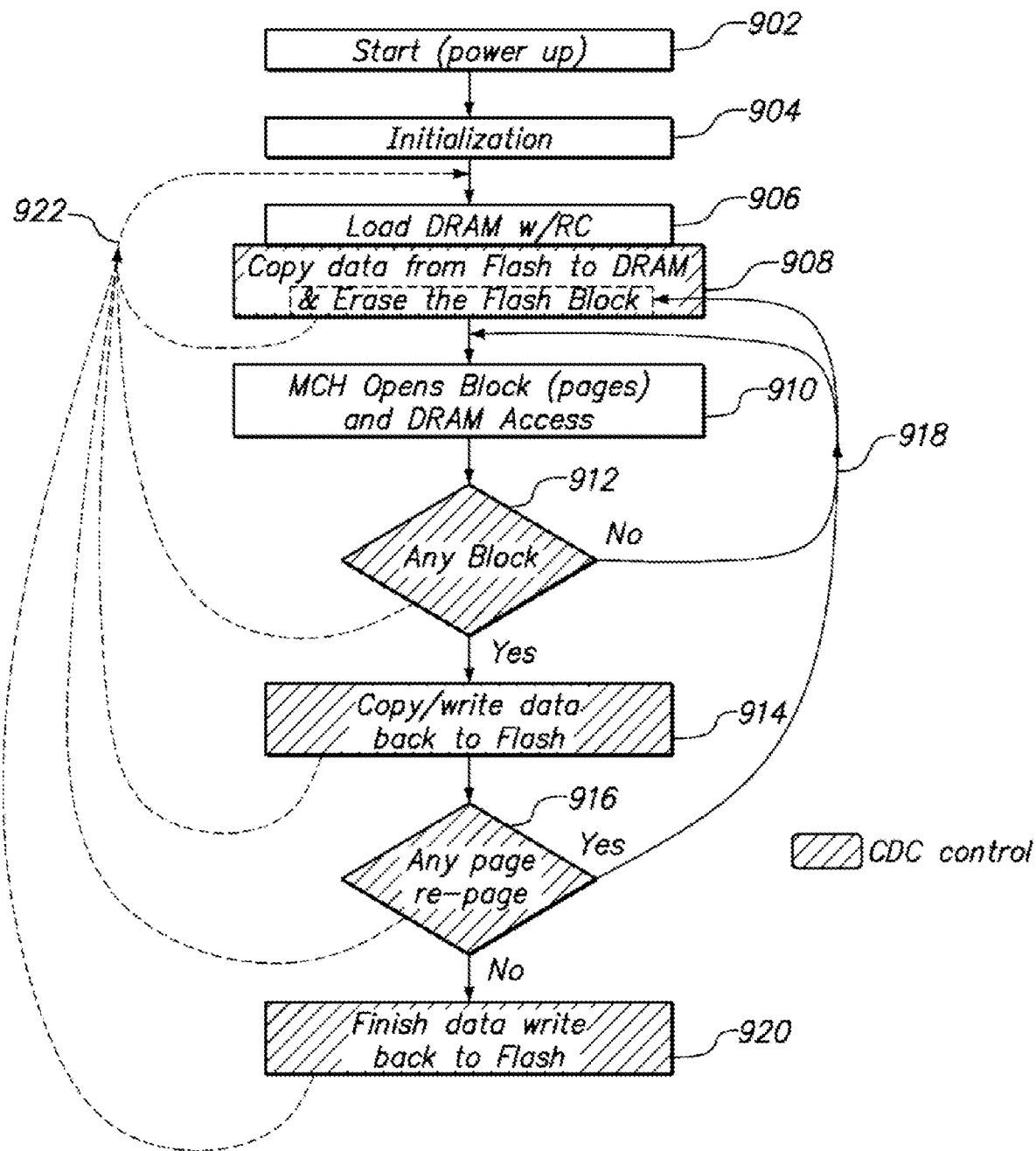


FIG. 9

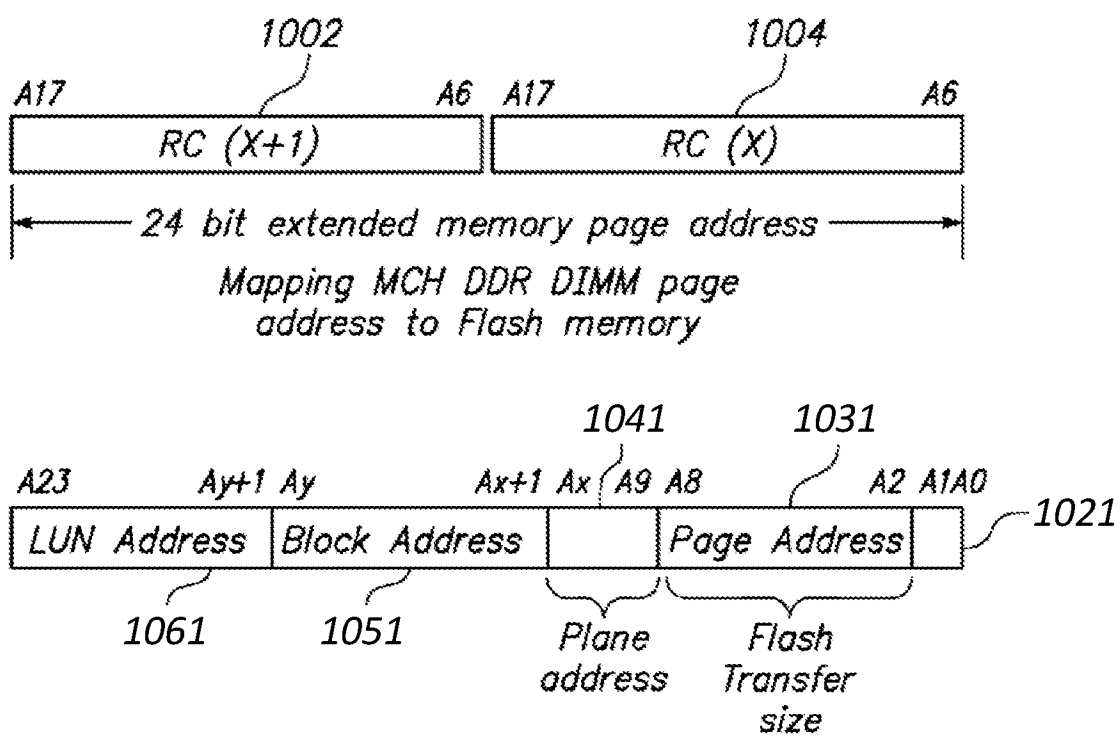
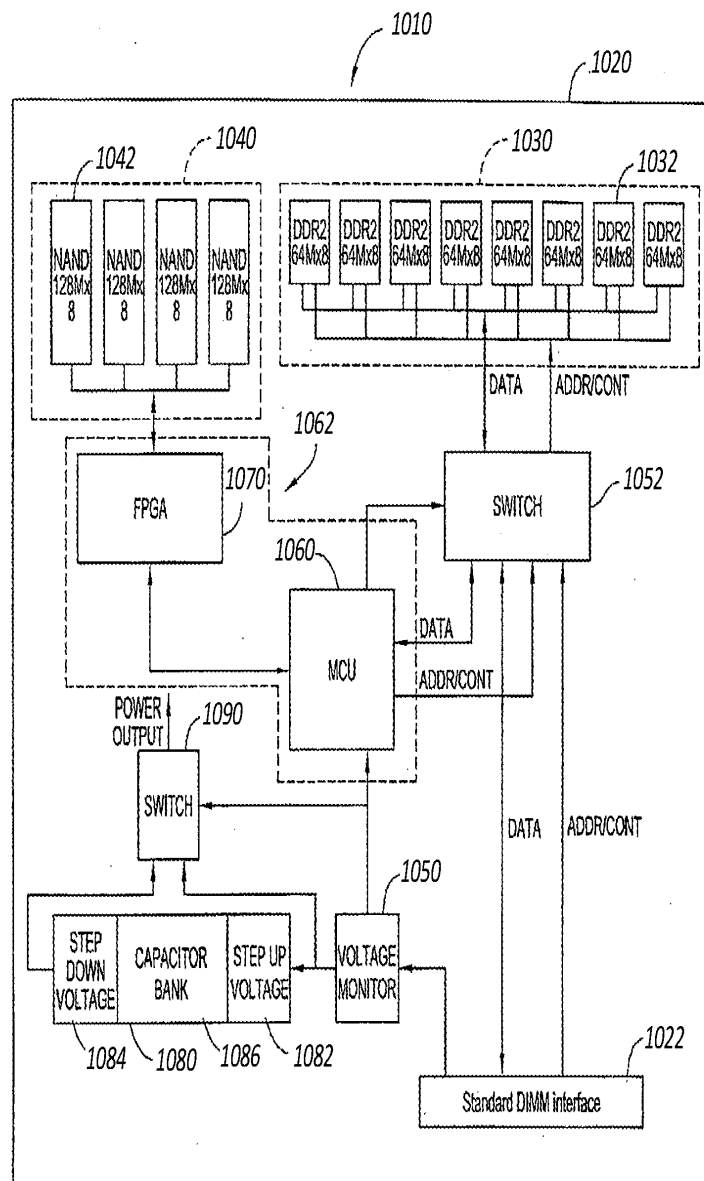


FIG. 10

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

FIG. 12



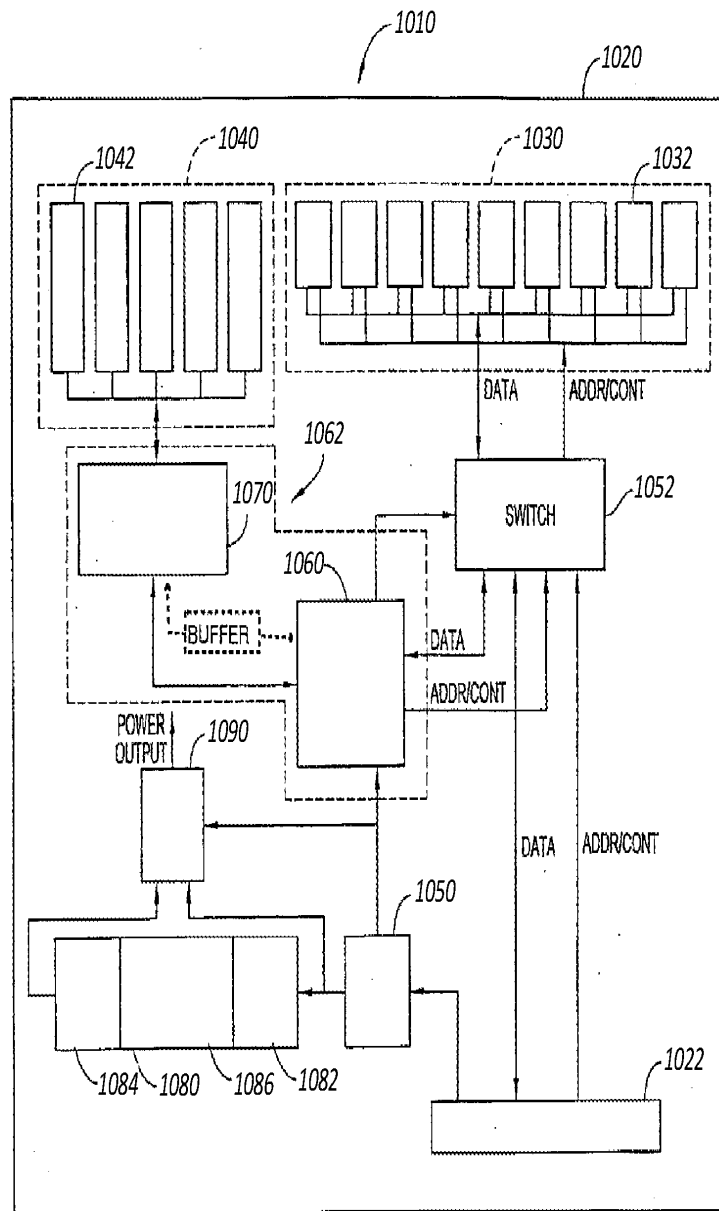


FIG. 13

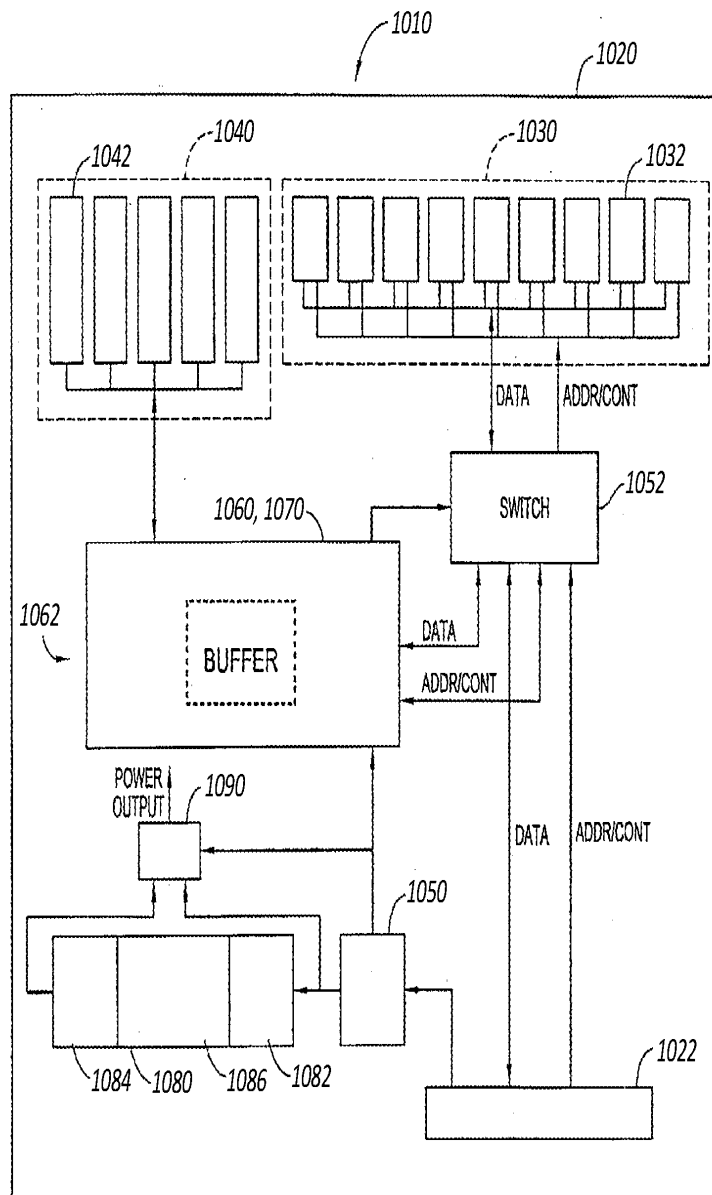


FIG. 14

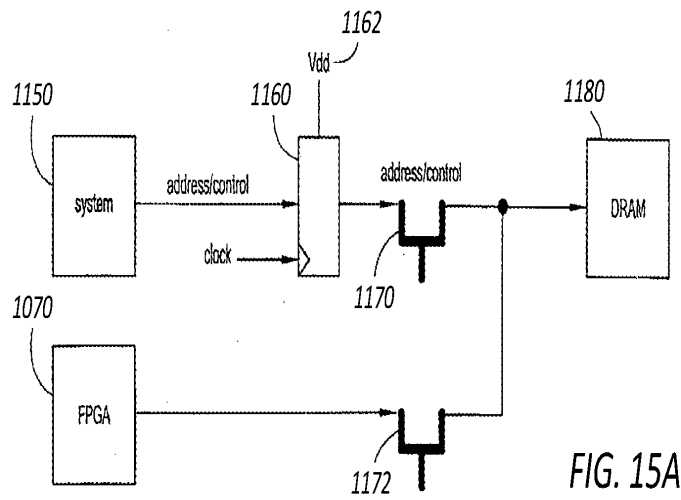


FIG. 15A

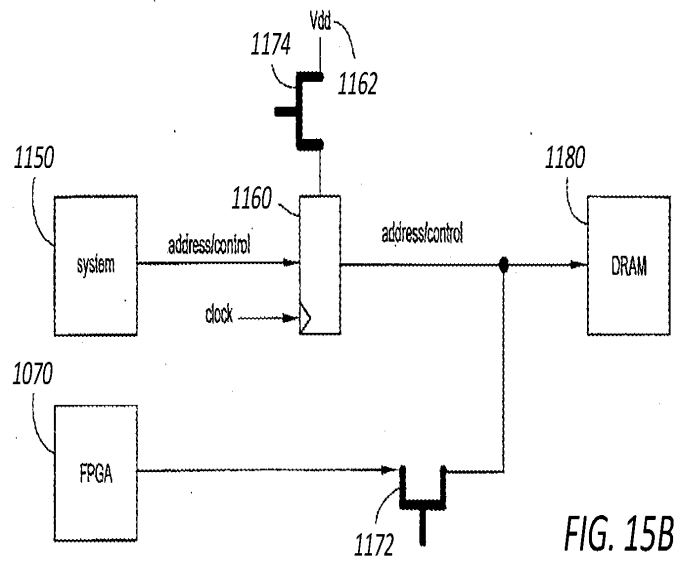


FIG. 15B

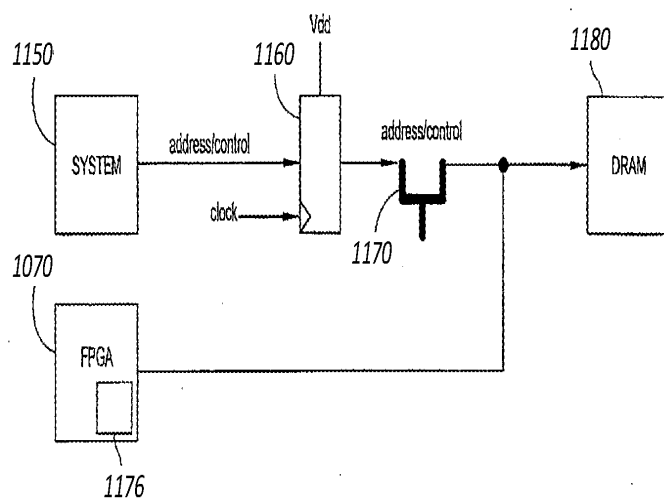


FIG. 15C

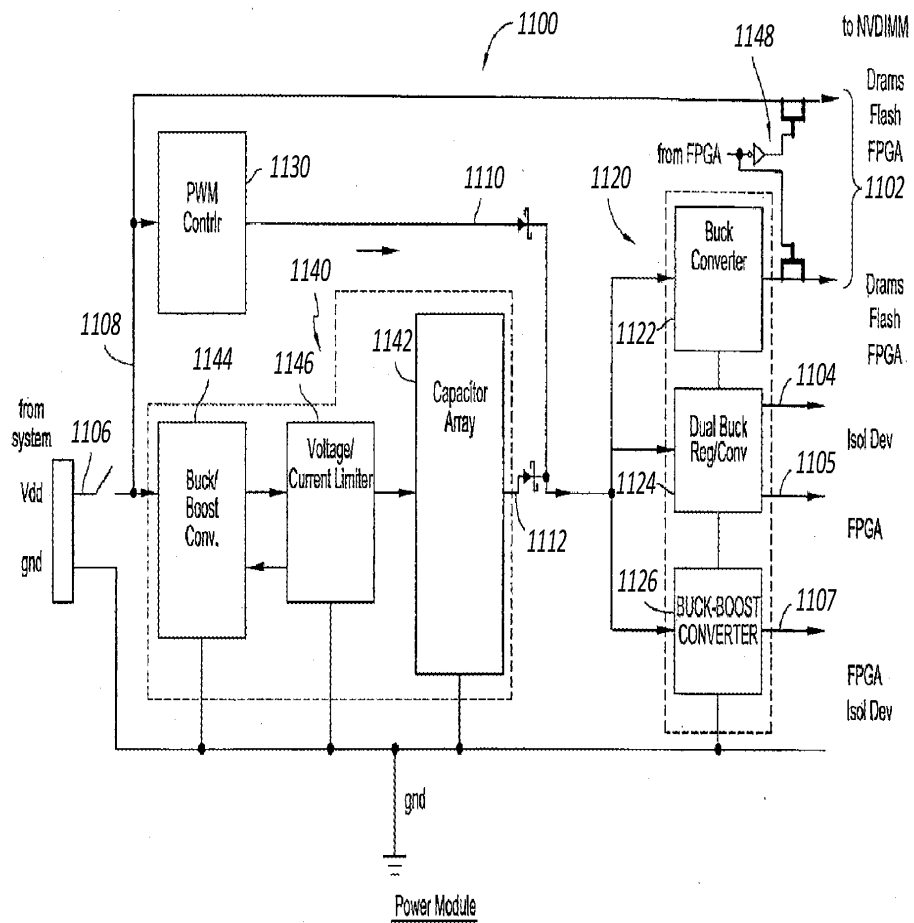


FIG. 16

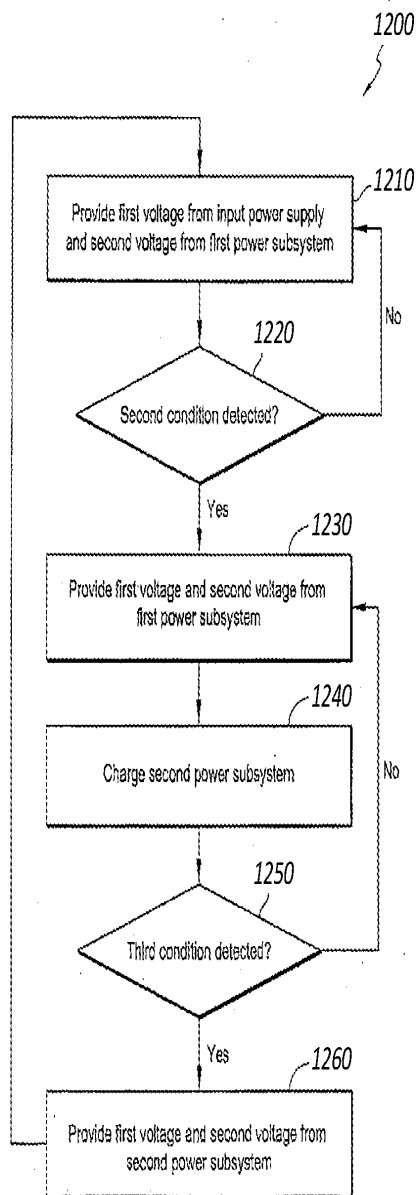


FIG. 17

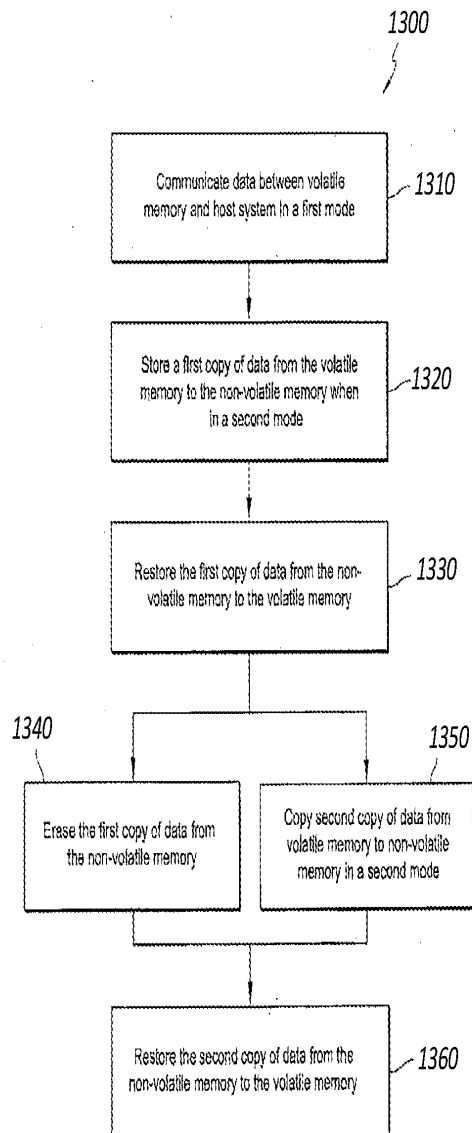


FIG. 18

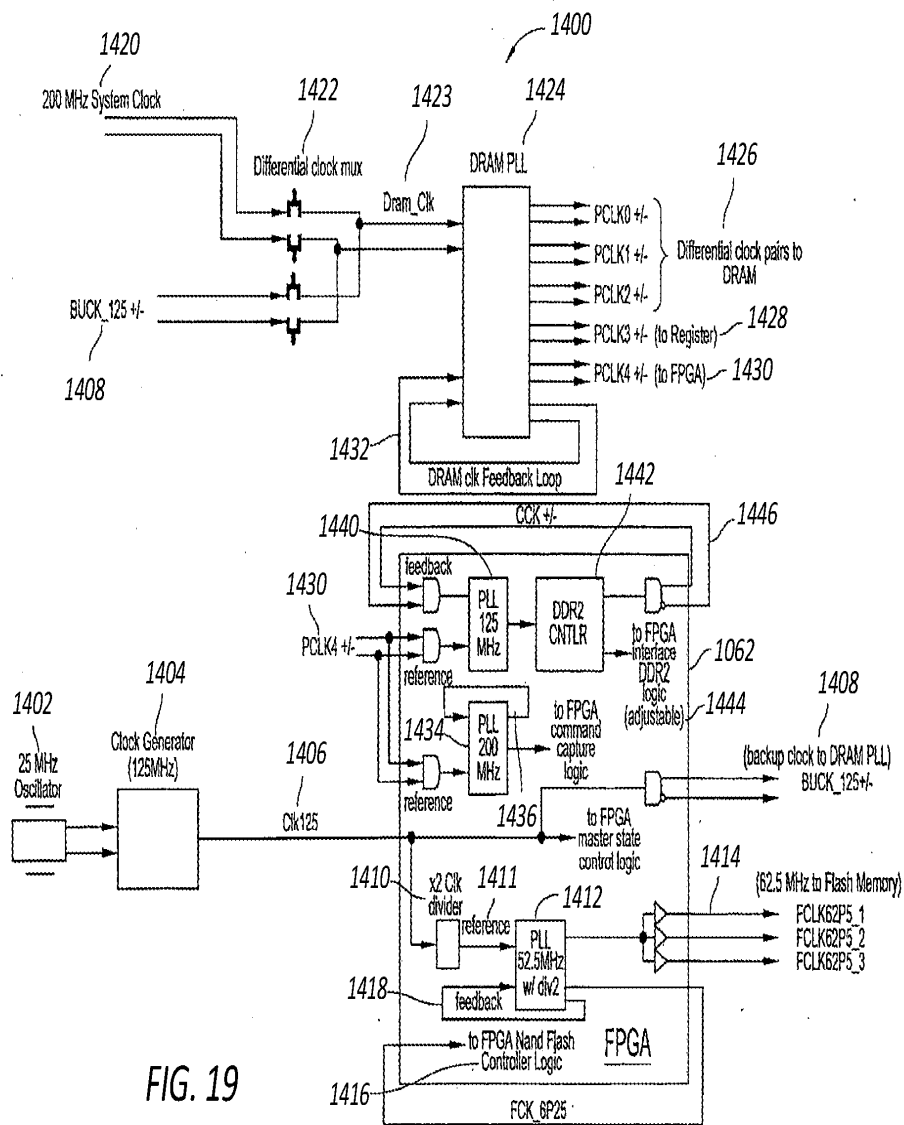


FIG. 19

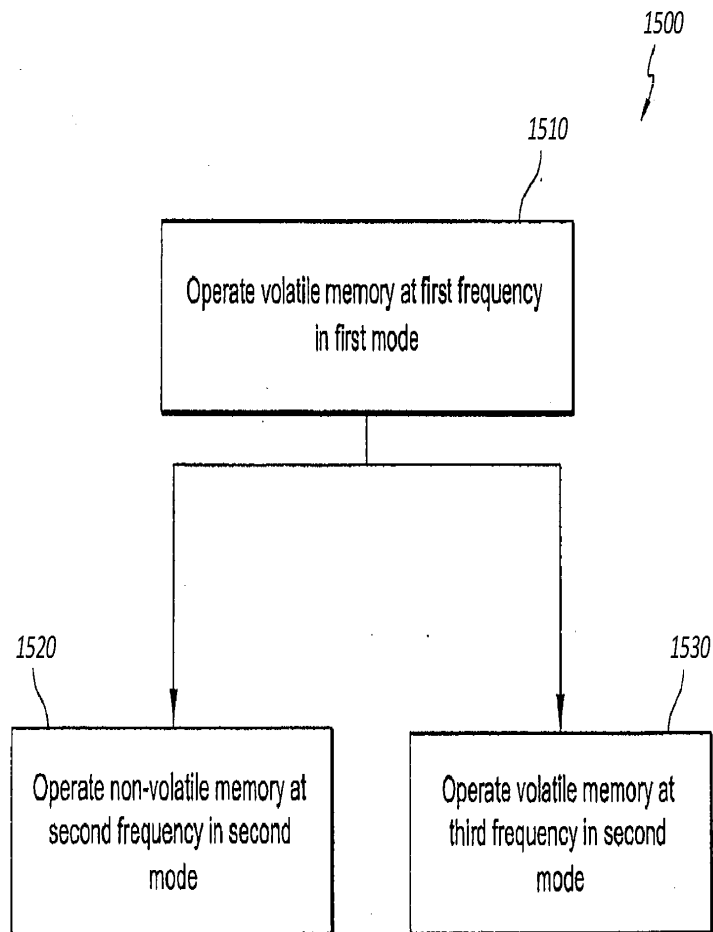


FIG. 20

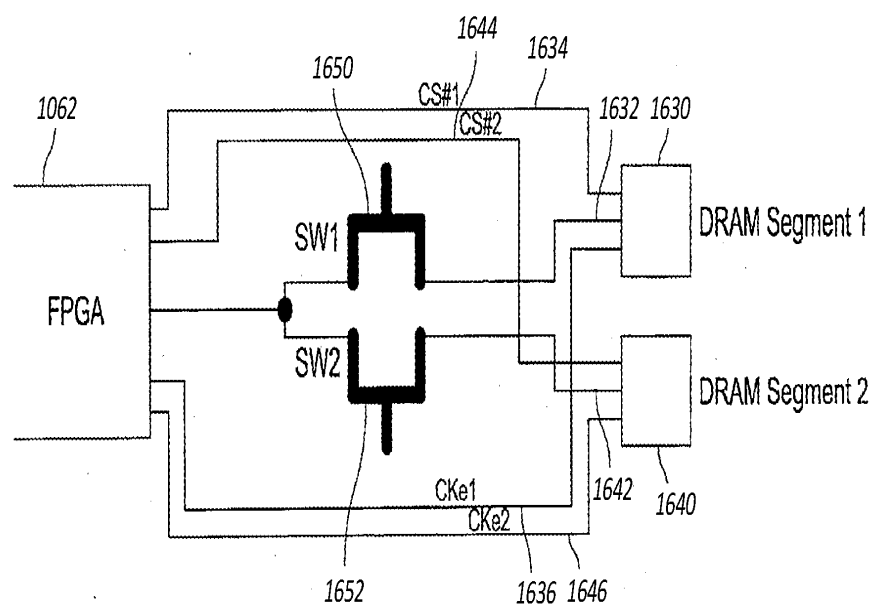


FIG. 21

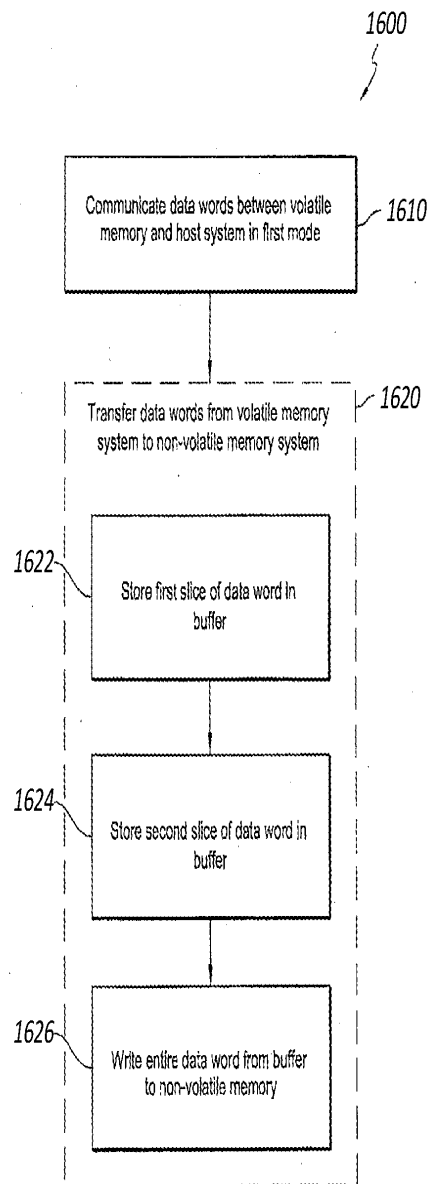


FIG. 22

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

**Title of
Invention**

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

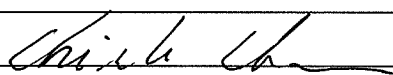
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LEGAL NAME OF INVENTOR

Inventor: Chi-She Chen

Date (Optional): Oct-17-2014

Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

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The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: Jeffrey C. Solomon

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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This declaration
is directed to:



The attached application, or



United States application or PCT international application number 14/489269

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LEGAL NAME OF INVENTOR

Inventor: Scott H. Milton

Date (Optional): 10/17/2014

Signature: Scott H. Milton

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: Jayesh Bhakta

Date (Optional): 9-17-2014

Signature: Jayesh Bhakta

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	TBA
Filing Date	TBA
First Named Inventor	Chi-She Chen
Title	FLASH-DRAM HYBRID MEMORY MODULE
Art Unit	TBA
Examiner Name	TBA
Attorney Docket Number	0016.001000I

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	
Name	Khaled Shami	Registration Number	38,745
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

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Electronic Patent Application Fee Transmittal

Application Number:				
Filing Date:				
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Susanh Perez			
Attorney Docket Number:	0016.001000I			
Filed as Large Entity				
Filing Fees for Track I Prioritized Examination - Nonprovisional Application under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	320	320
UTILITY SEARCH FEE	1111	1	700	700
UTILITY EXAMINATION FEE	1311	1	800	800
REQUEST FOR PRIORITIZED EXAMINATION	1817	1	4200	4200
Pages:				
Claims:				
Miscellaneous-Filing:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
PUBL. FEE- EARLY, VOLUNTARY, OR NORMAL	1504	1	0	0
PROCESSING FEE, EXCEPT PROV. APPLS.	1830	1	140	140
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				6160

Electronic Acknowledgement Receipt

EFS ID:	42797786
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Susanh Perez
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.0010001
Receipt Date:	24-MAY-2021
Filing Date:	
Time Stamp:	12:48:16
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$6160
RAM confirmation Number	E20215NC51133944
Deposit Account	602034
Authorized User	Susanh Perez

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37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	WebADS.pdf	173072	no	9
			024d0527756718ace8c8a6cf42912cad2b8d5f29		
Warnings:					
Information:					
2	TrackOne Request	0016001000I_TrackOneRequestForm_aia0424.pdf	130205	no	2
			f2be939f4d6bd7e584b64396a50033dece7c2cf5		
Warnings:					
Information:					
3	Power of Attorney	Signed_POA.pdf	880737	no	2
			6f585b39d647761dfeedc57b98cfd2a5aa8c777d		
Warnings:					
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Information:					
4		Application__0016001000I.pdf	476233	yes	77
			cbe99cd93ea85fac43989e7379f23180614092d9		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	75	
	Claims		76	76	
	Abstract		77	77	
Warnings:					
Information:					

5	Drawings-only black and white line drawings	Drwgs_0016001000l.pdf	4488651 4c32d20e5adfe5f5d06f6f4363693e2715b8f18c	no	22
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Information:					
6	Oath or Declaration filed	Inventors_Oath.pdf	372159 f091bc0e76caeabd5677e65b65d09f9c9b91d709	no	4
Warnings:					
Information:					
7	Transmittal Letter	POA_Transmittal_0016001000l.pdf	187769 eac6b27617adc227e82c9cf0957012218f329cf4	no	1
Warnings:					
Information:					
8	Fee Worksheet (SB06)	fee-info.pdf	39956 a9257abd068e08cce3929f6c0af7380317cb7eb5	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			6748782		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Chi-She		Chen		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Walnut	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1		944 Crystal Water Lane			
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country ⁱ	US		
Inventor 2					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1		6 Silver Fir			
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country ⁱ	US		
Inventor 3					
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Scott	H.	Milton		

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	49 Statehouse Place				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92602	Country ⁱ	US		

Inventor 4

Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:

Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country ⁱ	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	151145		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	0016.001000I	Small Entity Status Claimed	☐
Application Type			
Subject Matter			
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:						☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number		151145						
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove			
Registration Number								
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove			
Registration Number								
Additional Representative Information blocks may be generated within this form by selecting the Add button.								

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
	Continuation of		17138766	2020-12-30	

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
17138766	Continuation of		15934416	2018-03-23	

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
15934416	Continuation of	14840865	2015-08-31	9928186	2018-03-27

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14840865	Continuation of	14489269	2014-09-17	9158684	2015-10-13

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14489269	Continuation of	13559476	2012-07-26	8874831	2014-10-28

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
13559476	Claims benefit of provisional		61512871	2011-07-28	

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Prior Application Status	Abandoned	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
12240916	Continuation of	12131873	2008-06-02

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
12131873	Claims benefit of provisional	60941586	2007-06-01

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	0016.001000I
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

☐ Assignee ☐ Legal Representative under 35 U.S.C. 117 ☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign. ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information For Applicant:

Address 1	175 Technology Drive		
Address 2	Suite 150		
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1						
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.						
If the Assignee or Non-Applicant Assignee is an Organization check here.			☒			
Organization Name	Netlist, Inc.					
Mailing Address Information For Assignee including Non-Applicant Assignee:						
Address 1		175 Technology Drive				
Address 2		Suite 150				
City	Irvine	State/Province	CA			
Country i	US	Postal Code	92618			
Phone Number		Fax Number				
Email Address						
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000I
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Khaled Shami/			Date (YYYY-MM-DD)	
First Name	Khaled	Last Name	Shami	Registration Number	38745
Additional Signature may be generated within this form by selecting the Add button.					

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	Chi-She Chen	Nonprovisional Application Number (if known):	
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

- i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via EFS-Web.
---OR---
(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.
- ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- i. A request for continued examination has been filed with, or prior to, this form.
- ii. If the application is a utility application, this certification and request is being filed via EFS-Web.
- iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature /Khaled Shami/	Date May 24 2021
Name (Print/Typed) Khaled Shami	Practitioner Registration Number 38,745

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☐ *Total of _____ forms are submitted.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/328,019	05/24/2021		1820	0016.001000I	1	1

CONFIRMATION NO. 6552

FILING RECEIPT



0000000125973892

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Date Mailed: 06/02/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 151145

Domestic Priority data as claimed by applicant

This application is a CON of 17/138,766 12/30/2020 PAT 11016918
which is a CON of 15/934,416 03/23/2018
which is a CON of 14/840,865 08/31/2015 PAT 9928186
which is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/28/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/328,019**

Projected Publication Date: 09/09/2021

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign

patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/328,019

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	1 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/328,019	Filing Date 05/24/2021	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	06/04/2021	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 30	Minus	** 20	= 10	x \$100 =	1000
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$480 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	1000
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LDRC	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/EVA V GILLIS/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 6552

APPLICATION NO: 17/328,019

ART UNIT: Unassigned

FILING DATE: 05/024/2021

EXAMINER: Unassigned

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Commissioner:

Applicants respectfully request amendment of this application as directed herein.

Amendments to the Specification begin on page 2.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 3.

Remarks begin on page 9.

It is not believed that extensions of time are required beyond those that may otherwise be provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this Application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this Application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

In the Specification

Please amend paragraph [00130] as follows:

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is **[[1010]] 10** percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example, **[[1010]] 10** percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)

2. (New) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages;

a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and a first circuit that is coupled to the plurality of synchronous dynamic random access memory (SDRAM) devices and to a first set of edge connections of the plurality of edge connections, wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.

3. (New) The memory module of claim 2, wherein the first regulated voltage has a first voltage amplitude, and the second regulated voltage has a second voltage amplitude; and wherein a first one of the first and second voltage amplitudes is less than a second one of the first and second voltage amplitudes.

4. (New) The memory module of claim 2, wherein a third regulated voltage of the at least three regulated voltages has a voltage amplitude of 1.8 volts.

5. (New) The memory module of claim 2, further comprising:

a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude below a predetermined threshold voltage, wherein the predetermined threshold voltage is ten percent below a specified operating voltage.

6. (New) The memory module of claim 2, further comprising:

a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude above a predetermined threshold voltage, wherein the predetermined threshold voltage is ten percent above a specified operating voltage.

7. (New) The memory module of claim 2, further comprising:

a controller coupled to the PCB, the controller including a voltage monitor circuit configured to monitor an input voltage received from a second set of edge connections of the plurality of edge connections, wherein, in response to the voltage monitor circuit detecting a power threshold condition, the voltage monitor circuit transmits a signal to one or more portions of the controller.

8. (New) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting an amplitude of the input voltage being above a predetermined threshold voltage, and wherein the predetermined threshold voltage is above a specified operating voltage.

9. (New) The memory module of claim 8, wherein the predetermined threshold voltage is ten percent above the specified operating voltage.

10. (New) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting an amplitude of the input voltage being below a predetermined threshold voltage, and wherein the

predetermined threshold voltage is below a specified operating voltage.

11. (New) The memory module of claim 10, wherein the predetermined threshold voltage is ten percent below the specified operating voltage.

12. (New) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a power reduction condition or a low voltage condition of the input voltage.

13. (New) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a request by the host system.

14. (New) The memory module of claim 2, wherein the voltage conversion circuit includes at least three buck converters.

15. (New) The memory module of claim 14, wherein two of the at least three buck converters are configured to operate as a dual-buck converter.

16. (New) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system;

a voltage conversion circuit coupled to the PCB and configured to provide a plurality of regulated voltages;

a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the plurality of regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, the plurality of SDRAM devices coupled to a first regulated voltage of the plurality of regulated voltages; and

a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface, the voltage monitor circuit configured to detect a change in the input voltage, wherein, in response to the voltage

monitor detecting a change in the input voltage, the memory module transitions from a first operable state to a second operable state.

17. (New) The memory module of claim 16, wherein the voltage monitor circuit is configured to detect the input voltage being above a first predetermined threshold voltage and to detect the input voltage being below a second predetermined threshold voltage, and wherein the first predetermined threshold voltage is ten percent above a specified operating voltage, and the second predetermined threshold voltage is ten percent below the specified operating voltage.

18. (New) The memory module of claim 16, wherein, in the first operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

19. (New) The memory module of claim 18, wherein, in the first operable state, the first pre-regulated voltage is supplied to the voltage conversion circuit via a circuit, and wherein, in the second operable state, the second pre-regulated voltage is supplied to the voltage conversion circuit via the circuit.

20. (New) The memory module of claim 19, wherein the circuit includes a first diode having a first input and a first output, the first input is coupled to the first pre-regulated voltage and the first output is coupled to the voltage conversion circuit, and wherein the circuit includes a second diode having a second input and a second output, the second input is coupled to the second pre-regulated voltage and the second output is coupled to the first output and to the voltage conversion circuit.

21. (New) The memory module of claim 16, further comprising:

a first circuit having a first input, a second input and an output, the first input coupled to a first pre-regulated voltage, the second input coupled to a second pre-regulated voltage, the output coupled to the voltage conversion circuit, wherein the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a supply voltage, wherein, in the first operable state, the first circuit provides the supply voltage to the voltage conversion

circuit via the output using the first pre-regulated voltage, and wherein, in the second state, the first circuit provides the supply voltage to the voltage conversion circuit via the output using the second pre-regulated voltage.

22. (New) The memory module of claim 21, wherein the first circuit includes a first diode coupled between the first input and the output and a second diode coupled between the second input and the output.

23. (New) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system;

a voltage conversion circuit configured to provide a plurality of regulated voltages;

a plurality of components each coupled to at least one regulated voltage of the plurality of regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, wherein the plurality of SDRAM devices are coupled to a first regulated voltage of the plurality of regulated voltages; and

a controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface of the PCB, the voltage monitor circuit configured to monitor the input voltage, wherein the controller is configured to perform one or more operations in response to the voltage monitor circuit detecting a change in the input voltage.

24. (New) The memory module of claim 23, wherein, in response to the voltage monitor circuit detecting a change in the input voltage, the memory module transitions from a first state of operation to a second state of operation.

25. (New) The memory module of claim 24, wherein, in the first state of operation, the voltage conversion circuit provides the first regulated voltage to the SDRAM devices using a first pre-regulated voltage, and wherein, in the second state of operation, the voltage conversion circuit provides the first regulated voltage to the SDRAM devices using a second pre-regulated voltage.

26. (New) The memory module of claim 23, wherein the voltage monitor circuit is

configured to detect the input voltage being above a first predetermined threshold voltage.

27. (New) The memory module of claim 26, wherein the first predetermined threshold voltage is ten percent above a specified operating voltage.

28. (New) The memory module of claim 23, wherein the voltage monitor circuit is configured to detect the input voltage being below a second predetermined threshold voltage.

29. (New) The memory module of claim 28, wherein the second predetermined threshold voltage is ten percent below a specified operating voltage.

30. (New) The memory module of claim 25, wherein the first pre-regulated voltage is coupled to the voltage conversion circuit via a first diode.

31. (New) The memory module of claim 30, wherein the second pre-regulated voltage is coupled to the voltage conversion circuit via a second diode.

REMARKS

Entry of the foregoing amendment prior to substantive examination is respectfully requested. The pending claims subject to examination are 2 – 31, of which 2, 16, and 23 are independent.

Specification

The specification has been amended in order to correct a minor typographical error. Specifically, the numeral “1010” appearing in paragraph [00130]. Support for this change can be found in ancestor application 12/240,916, from which priority is claimed and whose contents are incorporated in the instant application by reference in their entirety. No new matter has been introduced.

Canceled Claims

Claim 1 has been canceled without prejudice or disclaimer of the subject matter contained therein.

Newly-Added Claims

Claims 2 – 31 have been added to further particularly point out and distinctly claim the subject matter of the claimed arrangement.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 60-2034.

Respectfully submitted,

Dated: June 4, 2021

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

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Washington, D.C. 20007
202-516-6900

Electronic Acknowledgement Receipt

EFS ID:	42907572
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Susanh Perez
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000I
Receipt Date:	04-JUN-2021
Filing Date:	24-MAY-2021
Time Stamp:	20:17:42
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Specification	Substitute_Spec_Annotated_016001000I.pdf	963765	no	75
			ef6ffe8adf9daddfedd9133c59e6bfa04dcb017		

Warnings:

Information:					
2	Specification	Substitute_Spec_Clean_0016001000l.pdf	1024741 138d5fdf9ccfb9975df8227aaf1f4b2ac13925e7	no	75
Warnings:					
Information:					
3		Preliminary_Amendment_FV_0016001000l.pdf	155598 be56600b31b7caef3acaa1e762c6a76e69db9cbb	yes	10
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Specification		2	2	
	Claims		3	8	
	Applicant Arguments/Remarks Made in an Amendment		9	10	
Warnings:					
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Total Files Size (in bytes):			2144104		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

SUBSTITUTE SPECIFICATION (ANNOTATED)

FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 17/138,766, filed December 30, 2020, titled “Flash-Dram Hybrid Memory”, now U.S. Patent No. 11,016,918, which is a continuation of U.S. Patent Application No 15/934,416, filed March 23, 2018, titled “Flash-Dram Hybrid Memory Module,” which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled “Non-Volatile Memory Module,” now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may be considered related to U.S. Patent Application No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed February 5, 2014, which is a continuation of U.S. Patent Application 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Patent 8,677,060, issued

March 18, 2014, which is a continuation of U.S. Patent Application 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed June 28, 2012, now U.S. Patent 8,516,187, issued August 20, 2013, which is a divisional of U.S. Patent Application 12/240,916, titled "Non-Volatile Memory Module," filed September 29, 2008, now U.S. Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007, the contents of which are incorporated by reference herein in their entirety.

[0003] This application may also be considered related to U.S. Patent Application No. 15/000,834, filed January 19, 2016 (abandoned), which is a continuation of U.S. Patent Application No. 14/489,332, filed September 17, 2014, now U.S. Patent No. 9,269,437, which is a continuation of U.S. Patent Application No. 14/173,219, filed February 5, 2014, now U.S. Patent No. 8,904,099, which is a continuation of U.S. Patent Application No. 13/905,048, filed May 29, 2013, now U.S. Patent No. 6,671,243, which is a continuation U.S. Patent Application 13/536,173 above.

[0004] This application may also be considered related to U.S. Patent Application No. 15/924,866, (abandoned), which is a continuation of U.S. Patent Application No. 14/489,281, filed September 17, 2014, now U.S. Patent No. 9,921,762, which is a continuation of U.S. Patent Application No. 13/625,563, filed September 24, 2012, now U.S. Patent No. 8,904,098, which claims the benefit of U.S. Provisional Application No. 61/583,775, filed September 23, 2011.

TECHNICAL FIELD

[0005] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0006] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0007] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high

speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0008] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0009] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a

data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MBs for the EcoRAM vs. 104MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™, which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0010] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0011] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash

DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0012] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

[0013] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0014] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0015] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

[0016] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0017] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0018] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0019] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0020] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0021] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0022] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0024] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0025] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0026] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0027] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0029] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0030] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0031] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0032] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0033] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0034] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0035] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0036] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0037] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0038] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0039] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0040] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0041] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0042] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0043] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0044] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0045] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0047] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0048] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0049] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0050] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0052] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0053] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0054] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0055] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0056] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0057] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0060] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0061] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0062] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0063] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0064] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0065] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0066] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0067] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0068] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0069] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

[0070] Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to

allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

[0071] Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

[0072] Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0074] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0075] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to

implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0076] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0077] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only

Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0078] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0079] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0080] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0081] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0082] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or

commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0083] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0084] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0085] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then

the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0086] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0087] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0088] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory

module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0089] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In

the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0090] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0091] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH

510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0092] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0093] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the

DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0094] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0095] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate

and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0096] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0097] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or

provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0098] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0099] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[00100] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[00101] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[00102] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from

the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[00103] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[00104] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[00105] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a

first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[00106] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[00107] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00108] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502

monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00109] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061

field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A_0A_1 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00110] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00111] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash

memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00112] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00113] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00114] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00115] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up

the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[00116] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[00117] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor

array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[00118] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[00119] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[00120] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[00121] FIG. 12 is a block diagram of an example memory system 1010 compatible with certain embodiments described herein. The memory system 1010 can be coupled to a host computer system and can include a volatile memory subsystem 1030, a non-volatile memory

subsystem 1040, and a controller 1062 operatively coupled to the non-volatile memory subsystem 1040. In certain embodiments, the memory system 1010 includes at least one circuit 1052 configured to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030.

[00122] In certain embodiments, the memory system 1010 comprises a memory module. The memory system 1010 may comprise a printed-circuit board (PCB) 1020. In certain embodiments, the memory system 1010 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 1010 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for

example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[00123] In certain preferred embodiments, the memory system 1010 is in electrical communication with the host system. In other embodiments, the memory system 1010 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 1010 can be in communication with a disk controller of a computer system, for example. The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.

[00124] The volatile memory subsystem 1030 comprises a plurality of volatile memory elements 1032 and the non-volatile memory subsystem 1040 comprises a plurality of non-volatile memory elements 1042. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem 1040 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 1030. In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and

synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00125] In certain embodiments, the second plurality of non-volatile memory elements 1042 comprises one or more flash memory elements. Types of flash memory elements 1042 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank 1040 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 1042. In addition, nonvolatile memory elements 1042 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 1042 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00126] FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements

1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

[00127] Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one

embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

[00128] In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

[00129] In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one

or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is ~~10~~ 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example,

~~1010~~ **10** percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[00131] As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 1070 is also programmed to perform address/address translation between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 1042 because of their superior read speed and compact structure.

[00132] The memory system 1010 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 1062 and the non-volatile memory subsystem 1040 are operatively decoupled (e.g., isolated) from the

volatile memory subsystem 1030 by the at least one circuit 1052 and a second state in which the volatile memory subsystem 1030 is operatively coupled to the controller 1062 to allow data to be communicated between the volatile memory subsystem 1030 and the nonvolatile memory subsystem 1040 via the controller 1062. The memory system 1010 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 1010 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[00133] The memory system 1010 may further comprise a voltage monitor 1050. The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 1050 may transmit a signal to the controller 1062 indicative of the detected condition. The controller 1062 of certain embodiments responds to the signal from the voltage monitor circuit 1050 by transmitting a signal to the at least one circuit 1052 to operatively couple the controller to the volatile memory system 1030, such that the memory system 1010 enters the second state. For example, the voltage monitor 1050 may send a signal to the MCU 1060 which responds by accessing the data on the volatile memory system 1030 and by executing a write cycle on the nonvolatile memory subsystem 1040. During this write cycle, data is read from the volatile memory subsystem 1030 and is transferred to the non-volatile memory subsystem 1040 via the MCU 1060. In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.

[00134] The isolation or operational decoupling of the volatile memory subsystem 1030 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 1010 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 1030. For example, in one embodiment during such periods of operation, the controller 1062 and the nonvolatile memory subsystem 1040 do not add a significant capacitive load to the volatile memory system 1030 when the memory system 1010 is in the first state. In certain such embodiments, the capacitive load of the controller 1062 and the non-volatile memory subsystem 1040 do not significantly affect the signals propagating between the volatile memory subsystem 1030 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 1052 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 1052 comprises other types of isolation devices.

[00135] Power may be supplied to the volatile memory subsystem 1030 from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. In certain embodiments, the memory system 1010 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 1010 enters the second state in response to a trigger condition. In certain embodiments, the memory system 1010 has a third state in which the controller 1062 is operatively decoupled from the volatile memory subsystem 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory

subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.

[00136] In certain embodiments, the second power supply 1080 does not comprise a battery. Because a battery is not used, the second power supply 1080 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. 12-14, the second power supply 1080 comprises a step-up transformer 1082, a step-down transformer 1084, and a capacitor bank 1086 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 1080 is configured to power the volatile memory subsystem 1030 for less than thirty minutes. In certain embodiments, the second power supply 1080 may comprise a battery. For example, in certain embodiments, the second power supply 1080 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 1030 for no more than thirty minutes.

[00137] In certain embodiments, the capacitor bank 1086 of the second power supply 1080 is charged by the first power supply while the memory system 1010 is in the first state. As a result, the second power supply 1080 is fully charged when the memory system 1010 enters the second state. The memory system 1010 and the second power supply 1080 may be located on the same printed circuit board 1020. In other embodiments, the second power supply 1080 may not be on

the same printed circuit board 1020 and may be tethered to the printed circuit board 1020, for example.

[00138] When operating in the first state, in certain embodiments, the step-up transformer 1082 keeps the capacitor bank 1086 charged at a peak value. In certain embodiments, the step-down transformer 1084 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 1032 and 3.0V to the non-volatile flash memory elements 1042) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. 12-14, the memory module 1010 further comprises a switch 1090 (e.g., FET switch) that switches power provided to the controller 1062, the volatile memory subsystem 1030, and the non-volatile memory subsystem 1040, between the power from the second power supply 1080 and the power from the first power supply (e.g., system power) received via the interface 1022. For example, the switch 1090 may switch from the first power supply to the second power supply 1080 when the voltage monitor 1050 detects a low voltage condition. The switch 1090 of certain embodiments advantageously ensures that the volatile memory elements 1032 and non-volatile memory elements 1042 are powered long enough for the data to be transferred from the volatile memory elements 1032 and stored in the non-volatile memory elements 1042. In certain embodiments, after the data transfer is complete, the switch 1090 then switches back to the first power supply and the controller 1062 transmits a signal to the at least one circuit 1052 to operatively decouple the controller 1062 from the volatile memory subsystem 1030, such that the memory system 1010 reenters the first state.

[00139] When the memory system 1010 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 via the

controller 1062. The host system can then resume accessing the volatile memory subsystem 1030 of the memory module 1010. In certain embodiments, after the memory system 1010 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 1030 rather than the non-volatile memory subsystem 1040 because the volatile memory elements 1032 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 1030 to the nonvolatile memory bank 1040, or from the non-volatile memory bank 1040 to the volatile memory bank 1030, takes less than one minute per GB.

[00140] In certain embodiments, the memory system 1010 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 1010 copies the entire contents of the volatile memory subsystem 1030 into the nonvolatile memory subsystem 1040 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are copied back into the volatile memory subsystem 1030 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 1040 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 1010. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 1010 and can improve the performance of the memory system 1010. In certain other embodiments, the entire contents of the volatile memory subsystem 1030 are not copied into the non-volatile memory subsystem 1040 on each backup operation, but only a partial copy is performed. In certain embodiments, other

management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 1040 are performed in the controller 1062.

[00141] The memory system 1010 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 1030 which then writes the data to non-volatile storage which is not part of the memory system 1010, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 1010 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 1010 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 1010 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 1010 may be operated as a write-through cache or as some other type of cache.

[00142] FIG. 16 schematically illustrates an example power module 1100 of the memory system 1010 in accordance with certain embodiments described herein. The power module 1100 provides power to the various components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. In certain embodiments, the power module 1100 comprises one or more of the components described above with respect to FIG. 12. For example, in certain embodiments, the power module 1100 includes the second power supply 1080 and the switch 1090.

[00143] The power module 1100 provides a plurality of voltages to the memory system 1010 comprising non-volatile and volatile memory subsystems 1030, 1040. The plurality of voltages comprises at least a first voltage 1102 and a second voltage 1104. The power module 1100 comprises an input 1106 providing a third voltage 1108 to the power module 1100 and a voltage conversion element 1120 configured to provide the second voltage 1104 to the memory system 1010. The power module 1100 further comprises a first power element 1130 configured to selectively provide a fourth voltage 1110 to the conversion element 1120. In certain embodiments, the first power element 1130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 1130 is configured to receive a 1.8V input system voltage as the third voltage 1108 and to output a modulated 5V output as the fourth voltage 1110.

[00144] The power module 1100 further comprises a second power element 1140 can be configured to selectively provide a fifth voltage 1112 to the conversion element 1120. The power module 1100 can be configured to selectively provide the first voltage 1102 to the memory system 1010 either from the conversion element 1120 or from the input 1106.

[00145] The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element

1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.

[00146] In certain embodiments, the power module 1100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 1100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 1102 in the second state from the first power element 1130 rather than from the input 1106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 1102 from the first power element 1130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 1120 and the first and second power elements 1130, 1140 (e.g., the sources of the pre-regulated fourth voltage 1110 in the second state and the pre-regulated fifth voltage 1112 in the third state) can smooth out potential voltage spikes.

[00147] In certain embodiments, the second power element 1140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. 16, the second power element 1140 comprises a capacitor array 1142, a buck-boost converter 1144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 1146 which limits the charge current to the capacitor array 1142 and stops charging the capacitor array 1142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 1142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For

example, in one example embodiment, the buck-boost converter 1144 receives a 1.8V system voltage (first voltage 1108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 1146. The voltage/current limiter 1146 limits the current going to the capacitor array 1142 to 1A and stops charging the array 1142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 1140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 1142.

[00148] The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate

amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.

[00149] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 1120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.

[00150] FIG. 17 is a flowchart of an example method 1200 of providing a first voltage 1102 and a second voltage 1104 to a memory system 1010 including volatile and nonvolatile memory subsystems 1030, 1040. While the method 1200 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-15, other memory systems are also compatible with embodiments of the method 1200. During a first condition, the method 1200

comprises providing the first voltage 1102 to the memory system 1010 from an input power supply 1106 and providing the second voltage 1104 to the memory system 1010 from a first power subsystem in operational block 1210. For example, in one embodiment, the first power subsystem comprises the first power element 1130 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, other first power subsystems are used.

[00151] The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.

[00152] The method 1200 further comprises charging a second power subsystem in operational block 1240. In certain embodiments, the second power subsystem comprises the second power element 1140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 1140 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, some other second power subsystem is used.

[00153] The method 1200 further comprises detecting a third condition in an operational block 1250 and during the third condition, providing the first voltage 1102 and the second

voltage 1104 to the memory system 1010 from the second power subsystem 1140 in an operational block 1260. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 1200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 1140 is charged before detecting the second condition.

[00154] In certain embodiments, the memory system 1010 comprises a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 1010 also comprises a controller 1062 operatively coupled to the volatile memory subsystem 1030 and operatively coupled to the non-volatile memory subsystem 1040. The controller 1062 can be configured to allow data to be communicated between the volatile memory subsystem 1030 and the host system when the memory system 1010 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040 when the memory system 1010 is operating in a second state.

[00155] Although the memory system 1010 having extra storage capacity of the non-volatile memory subsystem 1040 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In various embodiments, there may be at least 200, 300, or 400 percent more

storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In other embodiments, the non-volatile memory subsystem 1040 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 1030. In some embodiments, the non-volatile memory subsystem 1040 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 1030. In one embodiment, the non-volatile memory subsystem 1040 includes less than 100 percent more storage capacity than does the volatile memory subsystem 1030.

[00156] The extra storage capacity of the non-volatile memory subsystem 1040 can be used to improve the backup capability of the memory system 1010. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 1040 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem 1040 allows the volatile memory subsystem 1030 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to backup the volatile memory subsystem 1030 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 1030 is copied to a first, previously erased portion of the nonvolatile memory subsystem 1040 via the controller 1062. Since the non-volatile memory subsystem 1040 has more storage capacity than does the volatile memory subsystem 1030, there is a second portion of the non-volatile memory subsystem 1040 which does not have data from the volatile memory subsystem 1030 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 1062 of the memory system 1010 restores the data to the volatile memory subsystem 1030 by copying the backed-up data from the non-volatile memory

subsystem 40 back to the volatile memory subsystem 1030. After the data is restored, the memory system 1010 erases the non-volatile memory subsystem 1040. While the first portion of the non-volatile memory subsystem 1040 is being erased, it may be temporarily inaccessible.

[00157] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 1040 is completely erased, the volatile memory subsystem 1030 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 1040 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 1040, the memory system 1010 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 1040.

[00158] FIG. 18 is a flowchart of an example method 1300 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. In certain embodiments, the non-volatile memory subsystem 1040 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 1300 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-14, the method 1300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 1310, the method 1300 comprises communicating data between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation. The method 1300 further comprises storing a first copy of data from the

volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a first time when the memory system 1010 is in a second mode of operation in an operational block 1320.

[00159] In an operational block 1330, the method 1300 comprises restoring the first copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030. The method 1300 further comprises erasing the first copy of data from the non-volatile memory subsystem 1040 in an operational block 1340. The method further comprises storing a second copy of data from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a second time when the memory system 1010 is in the second mode of operation in an operational block 1350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 1040.

[00160] In some embodiments, the memory system 1010 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 1040. The method 1300 can also include restoring the second copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 in an operational block 1360. The operational blocks of method 1300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem 1030 at operational block 1360 before the first copy of data is completely erased in the operational block 1340.

[00161] FIG. 19 schematically illustrates an example clock distribution topology 1400 of a memory system 1010 in accordance with certain embodiments described herein. The clock

distribution topology 1400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 1010. A clock source 1402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 1402 may feed a clock generator 1404 which provides a clock signal 1406 to the controller 1062, which may be an FPGA. In one embodiment, the clock generator 1404 generates a 125 MHz clock signal 1406. The controller 1062 receives the clock signal 1406 and uses it to clock the controller 1062 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 1062.

[00162] The clock signal 1406 can also be input into a clock divider 1410 which produces a frequency-divided version of the clock signal 1406. In an example embodiment, the clock divider 1410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 1406. A non-volatile memory phase-locked loop (PLL) block 1412 can be included (e.g., in the controller 1062) which distributes a series of clock signals to the non-volatile memory subsystem 1040 and to associated control logic. For example, a series of clock signals 1414 can be sent from the controller 1062 to the non-volatile memory subsystem 1040. Another clock signal 1416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 1040. For example, the clock signal 1416 may clock the portion of the controller 1062 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 1040. A feedback clock signal 1418 is fed back into the non-volatile memory PLL block 1412. In one embodiment, the PLL block 1412 compares the feedback clock 1418 to the reference clock 1411 and varies the phase and frequency of its output until the reference 1411 and feedback 1418 clocks are phase and frequency matched.

[00163] A version of the clock signal 1406 such as the backup clock signal 1408 may be sent from the controller to the volatile memory subsystem 1030. The clock signal 1408 may be, for example, a differential version of the clock signal 1406. As described herein, the backup clock signal 1408 may be used to clock the volatile memory subsystem 1030 when the memory system 1010 is backing up the data from the volatile memory subsystem 1030 into the non-volatile memory subsystem 1040. In certain embodiments, the backup clock signal 1408 may also be used to clock the volatile memory subsystem 1030 when the memory system 1010 is copying the backed-up data back into the volatile memory subsystem 1030 from the nonvolatile memory subsystem 1040 (also referred to as restoring the volatile memory subsystem 1030). The volatile memory subsystem 1030 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the nonvolatile memory subsystem 1040 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 1030 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 1040) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 1030 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 1010.

[00164] In one embodiment, the backup clock 1408 and the volatile memory system clock signal 1420 are received by a multiplexer 1422, as schematically illustrated by FIG. 19. The multiplexer 1422 can output either the volatile memory system clock signal 1420 or the backup clock signal 1408 depending on the backup state of the memory system 1010. For example, when the memory system 1010 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 1420 may be provided by the multiplexer 422 to the volatile memory PLL block 1424. When the memory system 1010 is performing a backup (or restore) operation, the backup clock signal 1408 may be provided.

[00165] The volatile memory PLL block 1424 receives the volatile memory reference clock signal 1423 from the multiplexer 1422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 1030 and associated control logic. For example, in one embodiment, the PLL block 1424 generates a series of clock signals 1426 which clock the volatile memory elements 1032. A clock signal 1428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 1430 may be sent to the controller 1062. A feedback clock signal 1432 is fed back into the volatile memory PLL block 1424. In one embodiment, the PLL block 1424 compares the feedback clock signal 1432 to the reference clock signal 1423 and varies the phase and frequency of its output until the reference clock signal 1423 and the feedback clock signal 1432 clocks are phase and frequency matched.

[00166] The clock signal 1430 may be used by the controller 1062 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 1030. For example, control logic in the controller 1062 may be used to control the volatile memory subsystem 1030 during a backup or restore operation. The clock signal 1430 may be used as a reference clock signal for the PLL block 1434 which can generate one or more clocks 1438 used by logic in the controller 1062. For example, the PLL block 1434 may generate one or more clock signals 1438 used to drive logic circuitry associated with

controlling the volatile memory subsystem 1030. In certain embodiments, the PLL block 1434 includes a feedback clock signal 1436 and operates in a similar manner to other PLL blocks described herein.

[00167] The clock signal 1430 may be used as a reference clock signal for the PLL block 1440 which may generate one or more clock signals used by a sub-block 1442 to generate one or more other clock signals 1444. In one embodiment, for example, the volatile memory subsystem 1030 comprises DDR2 SDRAM elements and the sub-block 1442 generates one or more DDR2 compatible clock signals 1444. A feedback clock signal 1446 is fed back into the PLL block 1440. In certain embodiments, the PLL block 1440 operates in a similar manner to other PLL blocks described herein.

[00168] While described with respect to the example embodiment of FIG. 19, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 1030 operates on the volatile memory clock signal 1420 and there is no backup clock signal 1408. In some embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a restore operation and not during a backup operation.

[00169] FIG. 20 is a flowchart of an example method 1500 of controlling a memory system 1010 operatively coupled to a host system. Although described with respect to the memory

system 1010 described herein, the method 1500 is compatible with other memory systems. The memory system 1010 may include a clock distribution topology 1400 similar to the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 1040.

[00170] In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[00171] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

[00172] FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKE signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice

from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[00173] In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[00174] In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630

into self-refresh mode by sending a signal via the CKe line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKe line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

[00175] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00176] FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation in an operational block 1610. For example, the memory

system 1010 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[00177] In an operational block 1620, the method further comprises transferring data words from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 when the memory system 1010 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.

[00178] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The

40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[00179] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[00180] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1040 write each backed-up data word to the controller 1062 which writes a first portion of the data word to the volatile memory subsystem 1030 and then a second portion of the data word to the volatile memory 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00181] The method 1600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile memory subsystem 1030 comprises DRAM elements, only the slice of the DRAM which is currently being accessed

(e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 1600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[00182] In addition, the connection between the DRAM elements and the controller 1062 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 1062 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 1062 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 1030. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[00183] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

SUBSTITUTE SPECIFICATION (CLEAN)

FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 17/138,766, filed December 30, 2020, titled “Flash-Dram Hybrid Memory”, now U.S. Patent No. 11,016,918, which is a continuation of U.S. Patent Application No. 15/934,416, filed March 23, 2018, titled “Flash-Dram Hybrid Memory Module,” which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled “Flash-Dram Hybrid Memory Module,” now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled “Non-Volatile Memory Module,” now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may be considered related to U.S. Patent Application No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed February 5, 2014, which is a continuation of U.S. Patent Application 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Patent 8,677,060, issued

March 18, 2014, which is a continuation of U.S. Patent Application 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed June 28, 2012, now U.S. Patent 8,516,187, issued August 20, 2013, which is a divisional of U.S. Patent Application 12/240,916, titled "Non-Volatile Memory Module," filed September 29, 2008, now U.S. Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007, the contents of which are incorporated by reference herein in their entirety.

[0003] This application may also be considered related to U.S. Patent Application No. 15/000,834, filed January 19, 2016 (abandoned), which is a continuation of U.S. Patent Application No. 14/489,332, filed September 17, 2014, now U.S. Patent No. 9,269,437, which is a continuation of U.S. Patent Application No. 14/173,219, filed February 5, 2014, now U.S. Patent No. 8,904,099, which is a continuation of U.S. Patent Application No. 13/905,048, filed May 29, 2013, now U.S. Patent No. 6,671,243, which is a continuation U.S. Patent Application 13/536,173 above.

[0004] This application may also be considered related to U.S. Patent Application No. 15/924,866, (abandoned), which is a continuation of U.S. Patent Application No. 14/489,281, filed September 17, 2014, now U.S. Patent No. 9,921,762, which is a continuation of U.S. Patent Application No. 13/625,563, filed September 24, 2012, now U.S. Patent No. 8,904,098, which claims the benefit of U.S. Provisional Application No. 61/583,775, filed September 23, 2011.

TECHNICAL FIELD

[0005] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0006] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0007] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high

speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0008] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0009] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a

data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MB/s for the EcoRAM vs. 104MB/s for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™, which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0010] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0011] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash

DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0012] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

[0013] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0014] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0015] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

[0016] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0017] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0018] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0019] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0020] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0021] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0022] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0024] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0025] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0026] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0027] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0029] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0030] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0031] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0032] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0033] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0034] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0035] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0036] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0037] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0038] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0039] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0040] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0041] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0042] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0043] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0044] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0045] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0047] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0048] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0049] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0050] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0052] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0053] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0054] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0055] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0056] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0057] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0060] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0061] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0062] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0063] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0064] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0065] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0066] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0067] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0068] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0069] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

[0070] Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to

allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

[0071] Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

[0072] Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0074] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0075] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to

implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0076] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0077] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only

Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0078] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0079] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0080] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0081] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0082] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or

commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0083] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0084] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0085] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then

the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0086] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0087] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0088] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory

module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0089] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In

the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0090] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0091] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH

510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0092] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0093] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the

DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0094] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0095] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate

and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0096] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0097] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or

provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0098] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0099] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[00100] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[00101] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[00102] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from

the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[00103] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[00104] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[00105] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a

first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[00106] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[00107] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00108] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502

monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00109] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061

field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A_0A_1 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00110] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00111] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash

memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00112] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00113] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00114] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00115] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up

the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[00116] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[00117] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor

array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[00118] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[00119] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[00120] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[00121] FIG. 12 is a block diagram of an example memory system 1010 compatible with certain embodiments described herein. The memory system 1010 can be coupled to a host computer system and can include a volatile memory subsystem 1030, a non-volatile memory

subsystem 1040, and a controller 1062 operatively coupled to the non-volatile memory subsystem 1040. In certain embodiments, the memory system 1010 includes at least one circuit 1052 configured to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030.

[00122] In certain embodiments, the memory system 1010 comprises a memory module. The memory system 1010 may comprise a printed-circuit board (PCB) 1020. In certain embodiments, the memory system 1010 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 1010 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for

example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[00123] In certain preferred embodiments, the memory system 1010 is in electrical communication with the host system. In other embodiments, the memory system 1010 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 1010 can be in communication with a disk controller of a computer system, for example. The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.

[00124] The volatile memory subsystem 1030 comprises a plurality of volatile memory elements 1032 and the non-volatile memory subsystem 1040 comprises a plurality of non-volatile memory elements 1042. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem 1040 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 1030. In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and

synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00125] In certain embodiments, the second plurality of non-volatile memory elements 1042 comprises one or more flash memory elements. Types of flash memory elements 1042 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank 1040 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 1042. In addition, nonvolatile memory elements 1042 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 1042 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00126] FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements

1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

[00127] Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one

embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

[00128] In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

[00129] In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one

or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 10 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example,

10 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[00131] As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 1070 is also programmed to perform address/address translation between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 1042 because of their superior read speed and compact structure.

[00132] The memory system 1010 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 1062 and the non-volatile memory subsystem 1040 are operatively decoupled (e.g., isolated) from the

volatile memory subsystem 1030 by the at least one circuit 1052 and a second state in which the volatile memory subsystem 1030 is operatively coupled to the controller 1062 to allow data to be communicated between the volatile memory subsystem 1030 and the nonvolatile memory subsystem 1040 via the controller 1062. The memory system 1010 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 1010 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[00133] The memory system 1010 may further comprise a voltage monitor 1050. The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 1050 may transmit a signal to the controller 1062 indicative of the detected condition. The controller 1062 of certain embodiments responds to the signal from the voltage monitor circuit 1050 by transmitting a signal to the at least one circuit 1052 to operatively couple the controller to the volatile memory system 1030, such that the memory system 1010 enters the second state. For example, the voltage monitor 1050 may send a signal to the MCU 1060 which responds by accessing the data on the volatile memory system 1030 and by executing a write cycle on the nonvolatile memory subsystem 1040. During this write cycle, data is read from the volatile memory subsystem 1030 and is transferred to the non-volatile memory subsystem 1040 via the MCU 1060. In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.

[00134] The isolation or operational decoupling of the volatile memory subsystem 1030 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 1010 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 1030. For example, in one embodiment during such periods of operation, the controller 1062 and the nonvolatile memory subsystem 1040 do not add a significant capacitive load to the volatile memory system 1030 when the memory system 1010 is in the first state. In certain such embodiments, the capacitive load of the controller 1062 and the non-volatile memory subsystem 1040 do not significantly affect the signals propagating between the volatile memory subsystem 1030 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 1052 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 1052 comprises other types of isolation devices.

[00135] Power may be supplied to the volatile memory subsystem 1030 from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. In certain embodiments, the memory system 1010 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 1010 enters the second state in response to a trigger condition. In certain embodiments, the memory system 1010 has a third state in which the controller 1062 is operatively decoupled from the volatile memory subsystem 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory

subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.

[00136] In certain embodiments, the second power supply 1080 does not comprise a battery. Because a battery is not used, the second power supply 1080 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. 12-14, the second power supply 1080 comprises a step-up transformer 1082, a step-down transformer 1084, and a capacitor bank 1086 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 1080 is configured to power the volatile memory subsystem 1030 for less than thirty minutes. In certain embodiments, the second power supply 1080 may comprise a battery. For example, in certain embodiments, the second power supply 1080 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 1030 for no more than thirty minutes.

[00137] In certain embodiments, the capacitor bank 1086 of the second power supply 1080 is charged by the first power supply while the memory system 1010 is in the first state. As a result, the second power supply 1080 is fully charged when the memory system 1010 enters the second state. The memory system 1010 and the second power supply 1080 may be located on the same printed circuit board 1020. In other embodiments, the second power supply 1080 may not be on

the same printed circuit board 1020 and may be tethered to the printed circuit board 1020, for example.

[00138] When operating in the first state, in certain embodiments, the step-up transformer 1082 keeps the capacitor bank 1086 charged at a peak value. In certain embodiments, the step-down transformer 1084 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 1032 and 3.0V to the non-volatile flash memory elements 1042) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. 12-14, the memory module 1010 further comprises a switch 1090 (e.g., FET switch) that switches power provided to the controller 1062, the volatile memory subsystem 1030, and the non-volatile memory subsystem 1040, between the power from the second power supply 1080 and the power from the first power supply (e.g., system power) received via the interface 1022. For example, the switch 1090 may switch from the first power supply to the second power supply 1080 when the voltage monitor 1050 detects a low voltage condition. The switch 1090 of certain embodiments advantageously ensures that the volatile memory elements 1032 and non-volatile memory elements 1042 are powered long enough for the data to be transferred from the volatile memory elements 1032 and stored in the non-volatile memory elements 1042. In certain embodiments, after the data transfer is complete, the switch 1090 then switches back to the first power supply and the controller 1062 transmits a signal to the at least one circuit 1052 to operatively decouple the controller 1062 from the volatile memory subsystem 1030, such that the memory system 1010 reenters the first state.

[00139] When the memory system 1010 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 via the

controller 1062. The host system can then resume accessing the volatile memory subsystem 1030 of the memory module 1010. In certain embodiments, after the memory system 1010 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 1030 rather than the non-volatile memory subsystem 1040 because the volatile memory elements 1032 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 1030 to the nonvolatile memory bank 1040, or from the non-volatile memory bank 1040 to the volatile memory bank 1030, takes less than one minute per GB.

[00140] In certain embodiments, the memory system 1010 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 1010 copies the entire contents of the volatile memory subsystem 1030 into the nonvolatile memory subsystem 1040 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are copied back into the volatile memory subsystem 1030 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 1040 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 1010. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 1010 and can improve the performance of the memory system 1010. In certain other embodiments, the entire contents of the volatile memory subsystem 1030 are not copied into the non-volatile memory subsystem 1040 on each backup operation, but only a partial copy is performed. In certain embodiments, other

management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 1040 are performed in the controller 1062.

[00141] The memory system 1010 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 1030 which then writes the data to non-volatile storage which is not part of the memory system 1010, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 1010 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 1010 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 1010 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 1010 may be operated as a write-through cache or as some other type of cache.

[00142] FIG. 16 schematically illustrates an example power module 1100 of the memory system 1010 in accordance with certain embodiments described herein. The power module 1100 provides power to the various components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. In certain embodiments, the power module 1100 comprises one or more of the components described above with respect to FIG. 12. For example, in certain embodiments, the power module 1100 includes the second power supply 1080 and the switch 1090.

[00143] The power module 1100 provides a plurality of voltages to the memory system 1010 comprising non-volatile and volatile memory subsystems 1030, 1040. The plurality of voltages comprises at least a first voltage 1102 and a second voltage 1104. The power module 1100 comprises an input 1106 providing a third voltage 1108 to the power module 1100 and a voltage conversion element 1120 configured to provide the second voltage 1104 to the memory system 1010. The power module 1100 further comprises a first power element 1130 configured to selectively provide a fourth voltage 1110 to the conversion element 1120. In certain embodiments, the first power element 1130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 1130 is configured to receive a 1.8V input system voltage as the third voltage 1108 and to output a modulated 5V output as the fourth voltage 1110.

[00144] The power module 1100 further comprises a second power element 1140 can be configured to selectively provide a fifth voltage 1112 to the conversion element 1120. The power module 1100 can be configured to selectively provide the first voltage 1102 to the memory system 1010 either from the conversion element 1120 or from the input 1106.

[00145] The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element

1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.

[00146] In certain embodiments, the power module 1100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 1100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 1102 in the second state from the first power element 1130 rather than from the input 1106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 1102 from the first power element 1130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 1120 and the first and second power elements 1130, 1140 (e.g., the sources of the pre-regulated fourth voltage 1110 in the second state and the pre-regulated fifth voltage 1112 in the third state) can smooth out potential voltage spikes.

[00147] In certain embodiments, the second power element 1140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. 16, the second power element 1140 comprises a capacitor array 1142, a buck-boost converter 1144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 1146 which limits the charge current to the capacitor array 1142 and stops charging the capacitor array 1142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 1142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For

example, in one example embodiment, the buck-boost converter 1144 receives a 1.8V system voltage (first voltage 1108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 1146. The voltage/current limiter 1146 limits the current going to the capacitor array 1142 to 1A and stops charging the array 1142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 1140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 1142.

[00148] The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate

amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.

[00149] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 1120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.

[00150] FIG. 17 is a flowchart of an example method 1200 of providing a first voltage 1102 and a second voltage 1104 to a memory system 1010 including volatile and nonvolatile memory subsystems 1030, 1040. While the method 1200 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-15, other memory systems are also compatible with embodiments of the method 1200. During a first condition, the method 1200

comprises providing the first voltage 1102 to the memory system 1010 from an input power supply 1106 and providing the second voltage 1104 to the memory system 1010 from a first power subsystem in operational block 1210. For example, in one embodiment, the first power subsystem comprises the first power element 1130 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, other first power subsystems are used.

[00151] The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.

[00152] The method 1200 further comprises charging a second power subsystem in operational block 1240. In certain embodiments, the second power subsystem comprises the second power element 1140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 1140 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, some other second power subsystem is used.

[00153] The method 1200 further comprises detecting a third condition in an operational block 1250 and during the third condition, providing the first voltage 1102 and the second

voltage 1104 to the memory system 1010 from the second power subsystem 1140 in an operational block 1260. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 1200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 1140 is charged before detecting the second condition.

[00154] In certain embodiments, the memory system 1010 comprises a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 1010 also comprises a controller 1062 operatively coupled to the volatile memory subsystem 1030 and operatively coupled to the non-volatile memory subsystem 1040. The controller 1062 can be configured to allow data to be communicated between the volatile memory subsystem 1030 and the host system when the memory system 1010 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040 when the memory system 1010 is operating in a second state.

[00155] Although the memory system 1010 having extra storage capacity of the non-volatile memory subsystem 1040 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In various embodiments, there may be at least 200, 300, or 400 percent more

storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In other embodiments, the non-volatile memory subsystem 1040 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 1030. In some embodiments, the non-volatile memory subsystem 1040 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 1030. In one embodiment, the non-volatile memory subsystem 1040 includes less than 100 percent more storage capacity than does the volatile memory subsystem 1030.

[00156] The extra storage capacity of the non-volatile memory subsystem 1040 can be used to improve the backup capability of the memory system 1010. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 1040 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem 1040 allows the volatile memory subsystem 1030 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to backup the volatile memory subsystem 1030 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 1030 is copied to a first, previously erased portion of the nonvolatile memory subsystem 1040 via the controller 1062. Since the non-volatile memory subsystem 1040 has more storage capacity than does the volatile memory subsystem 1030, there is a second portion of the non-volatile memory subsystem 1040 which does not have data from the volatile memory subsystem 1030 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 1062 of the memory system 1010 restores the data to the volatile memory subsystem 1030 by copying the backed-up data from the non-volatile memory

subsystem 40 back to the volatile memory subsystem 1030. After the data is restored, the memory system 1010 erases the non-volatile memory subsystem 1040. While the first portion of the non-volatile memory subsystem 1040 is being erased, it may be temporarily inaccessible.

[00157] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 1040 is completely erased, the volatile memory subsystem 1030 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 1040 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 1040, the memory system 1010 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 1040.

[00158] FIG. 18 is a flowchart of an example method 1300 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. In certain embodiments, the non-volatile memory subsystem 1040 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 1300 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-14, the method 1300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 1310, the method 1300 comprises communicating data between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation. The method 1300 further comprises storing a first copy of data from the

volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a first time when the memory system 1010 is in a second mode of operation in an operational block 1320.

[00159] In an operational block 1330, the method 1300 comprises restoring the first copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030. The method 1300 further comprises erasing the first copy of data from the non-volatile memory subsystem 1040 in an operational block 1340. The method further comprises storing a second copy of data from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a second time when the memory system 1010 is in the second mode of operation in an operational block 1350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 1040.

[00160] In some embodiments, the memory system 1010 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 1040. The method 1300 can also include restoring the second copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 in an operational block 1360. The operational blocks of method 1300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem 1030 at operational block 1360 before the first copy of data is completely erased in the operational block 1340.

[00161] FIG. 19 schematically illustrates an example clock distribution topology 1400 of a memory system 1010 in accordance with certain embodiments described herein. The clock

distribution topology 1400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 1010. A clock source 1402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 1402 may feed a clock generator 1404 which provides a clock signal 1406 to the controller 1062, which may be an FPGA. In one embodiment, the clock generator 1404 generates a 125 MHz clock signal 1406. The controller 1062 receives the clock signal 1406 and uses it to clock the controller 1062 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 1062.

[00162] The clock signal 1406 can also be input into a clock divider 1410 which produces a frequency-divided version of the clock signal 1406. In an example embodiment, the clock divider 1410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 1406. A non-volatile memory phase-locked loop (PLL) block 1412 can be included (e.g., in the controller 1062) which distributes a series of clock signals to the non-volatile memory subsystem 1040 and to associated control logic. For example, a series of clock signals 1414 can be sent from the controller 1062 to the non-volatile memory subsystem 1040. Another clock signal 1416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 1040. For example, the clock signal 1416 may clock the portion of the controller 1062 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 1040. A feedback clock signal 1418 is fed back into the non-volatile memory PLL block 1412. In one embodiment, the PLL block 1412 compares the feedback clock 1418 to the reference clock 1411 and varies the phase and frequency of its output until the reference 1411 and feedback 1418 clocks are phase and frequency matched.

[00163] A version of the clock signal 1406 such as the backup clock signal 1408 may be sent from the controller to the volatile memory subsystem 1030. The clock signal 1408 may be, for example, a differential version of the clock signal 1406. As described herein, the backup clock signal 1408 may be used to clock the volatile memory subsystem 1030 when the memory system 1010 is backing up the data from the volatile memory subsystem 1030 into the non-volatile memory subsystem 1040. In certain embodiments, the backup clock signal 1408 may also be used to clock the volatile memory subsystem 1030 when the memory system 1010 is copying the backed-up data back into the volatile memory subsystem 1030 from the nonvolatile memory subsystem 1040 (also referred to as restoring the volatile memory subsystem 1030). The volatile memory subsystem 1030 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the nonvolatile memory subsystem 1040 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 1030 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 1040) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 1030 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 1010.

[00164] In one embodiment, the backup clock 1408 and the volatile memory system clock signal 1420 are received by a multiplexer 1422, as schematically illustrated by FIG. 19. The multiplexer 1422 can output either the volatile memory system clock signal 1420 or the backup clock signal 1408 depending on the backup state of the memory system 1010. For example, when the memory system 1010 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 1420 may be provided by the multiplexer 422 to the volatile memory PLL block 1424. When the memory system 1010 is performing a backup (or restore) operation, the backup clock signal 1408 may be provided.

[00165] The volatile memory PLL block 1424 receives the volatile memory reference clock signal 1423 from the multiplexer 1422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 1030 and associated control logic. For example, in one embodiment, the PLL block 1424 generates a series of clock signals 1426 which clock the volatile memory elements 1032. A clock signal 1428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 1430 may be sent to the controller 1062. A feedback clock signal 1432 is fed back into the volatile memory PLL block 1424. In one embodiment, the PLL block 1424 compares the feedback clock signal 1432 to the reference clock signal 1423 and varies the phase and frequency of its output until the reference clock signal 1423 and the feedback clock signal 1432 clocks are phase and frequency matched.

[00166] The clock signal 1430 may be used by the controller 1062 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 1030. For example, control logic in the controller 1062 may be used to control the volatile memory subsystem 1030 during a backup or restore operation. The clock signal 1430 may be used as a reference clock signal for the PLL block 1434 which can generate one or more clocks 1438 used by logic in the controller 1062. For example, the PLL block 1434 may generate one or more clock signals 1438 used to drive logic circuitry associated with

controlling the volatile memory subsystem 1030. In certain embodiments, the PLL block 1434 includes a feedback clock signal 1436 and operates in a similar manner to other PLL blocks described herein.

[00167] The clock signal 1430 may be used as a reference clock signal for the PLL block 1440 which may generate one or more clock signals used by a sub-block 1442 to generate one or more other clock signals 1444. In one embodiment, for example, the volatile memory subsystem 1030 comprises DDR2 SDRAM elements and the sub-block 1442 generates one or more DDR2 compatible clock signals 1444. A feedback clock signal 1446 is fed back into the PLL block 1440. In certain embodiments, the PLL block 1440 operates in a similar manner to other PLL blocks described herein.

[00168] While described with respect to the example embodiment of FIG. 19, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 1030 operates on the volatile memory clock signal 1420 and there is no backup clock signal 1408. In some embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a restore operation and not during a backup operation.

[00169] FIG. 20 is a flowchart of an example method 1500 of controlling a memory system 1010 operatively coupled to a host system. Although described with respect to the memory

system 1010 described herein, the method 1500 is compatible with other memory systems. The memory system 1010 may include a clock distribution topology 1400 similar to the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 30 and a non-volatile memory subsystem 1040.

[00170] In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[00171] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

[00172] FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice

from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[00173] In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[00174] In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630

into self-refresh mode by sending a signal via the CKe line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKe line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

[00175] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00176] FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation in an operational block 1610. For example, the memory

system 1010 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[00177] In an operational block 1620, the method further comprises transferring data words from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 when the memory system 1010 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.

[00178] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The

40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[00179] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[00180] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1040 write each backed-up data word to the controller 1062 which writes a first portion of the data word to the volatile memory subsystem 1030 and then a second portion of the data word to the volatile memory 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00181] The method 1600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile memory subsystem 1030 comprises DRAM elements, only the slice of the DRAM which is currently being accessed

(e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 1600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[00182] In addition, the connection between the DRAM elements and the controller 1062 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 1062 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 1062 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 1030. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[00183] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17328019
	Filing Date	2021-05-24
	First Named Inventor	Chi-She Chen
	Art Unit	
	Examiner Name	
	Attorney Docket Number	0016.0010001

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

1	Wong, A., The BIOS Optimization Guide, Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: 6552

SERIAL NO.: 17/328,019

ART UNIT: 2135

FILING DATE: 05-24-2021

EXAMINER: TBA

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Commissioner:

Notice of Related Litigation and/or Proceedings

Pursuant to MPEP 2001.06(c), Applicant notifies the Patent and Trademark Office of litigation and proceedings involving U.S. Patents related to the current patent application, the subject matter of which may be related to the present patent application.

Appendix A lists results of a Cases Search. The Cases Search was performed on June 16, 2021, on the Docket Navigator™ search tool for patent cases involving commonly-owned U.S. Patents No. 8,301,833, 8,516,187, 8,671,243, and 8,874,831. The Cases Search covered forums including U.S. federal courts, ITC, and PTAB. The Cases Search results presented includes information output by the search tool for the following fields: Case, Case Filing Date, Case Status Case Terminated, and Case Name.

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Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: June 22, 2021

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

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1000 Wisconsin Ave N.W., Suite 200
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Cases

12 Results

Patents: 8301833, 8516187, 8671243, 8874831

	Case	Case Filing Date	Case Status	Case Termination Date	Case Name
	Netlist, Inc. v. Micron Technology, Inc. et al 6-21-cv-00430 (WDTX)	Apr. 28, 2021	Active		Netlist, Inc. v. Micron Technology, Inc. et al
	<u>Number</u> 8301833	<u>Inventor</u> Chen	<u>Name</u> Non-volatile memory module		
	SK hynix Inc. et al v. Netlist, Inc. IPR2017-00692 (PTAB)	Jan. 17, 2017	Terminated	Jul. 05, 2018	SK hynix Inc. et al v. Netlist, Inc.
	<u>Number</u> 8874831	<u>Inventor</u> Lee	<u>Name</u> Flash-DRAM hybrid memory module		
	SK hynix Inc. v. Netlist, Inc. IPR2017-00649 (PTAB)	Jan. 13, 2017	Terminated	May. 29, 2018	SK hynix Inc. v. Netlist, Inc.
	<u>Number</u> 8301833	<u>Inventor</u> Chen	<u>Name</u> Non-volatile memory module		
	SK hynix Inc. et al v. Netlist, Inc. IPR2017-00587 (PTAB)	Jan. 06, 2017	Terminated	Jun. 20, 2018	SK hynix Inc. et al v. Netlist, Inc.
	<u>Number</u> 8671243	<u>Inventor</u> Chen	<u>Name</u> Isolation switching for backup memory		
	SMART Modular Technologies Inc. v. Netlist, Inc. IPR2014-01370 (PTAB)	Aug. 23, 2014	Terminated	Apr. 30, 2015	SMART Modular Technologies Inc. v. Netlist, Inc.
	<u>Number</u> 8301833	<u>Inventor</u> Chen	<u>Name</u> Non-volatile memory module		
	SMART Modular Technologies Inc. v. Netlist, Inc. IPR2014-01371 (PTAB)	Aug. 23, 2014	Terminated	Apr. 30, 2015	SMART Modular Technologies Inc. v. Netlist, Inc.
	<u>Number</u> 8516187	<u>Inventor</u> Chen	<u>Name</u> Data transfer scheme for non-volatile memory module		
	SanDisk Corporation v. Netlist, Inc. IPR2014-00994 (PTAB)	Jun. 20, 2014	Terminated	Jun. 02, 2015	SanDisk Corporation v. Netlist, Inc.
	<u>Number</u> 8301833	<u>Inventor</u> Chen	<u>Name</u> Non-volatile memory module		
	SanDisk Corporation v. Netlist, Inc. IPR2014-00982 (PTAB)	Jun. 19, 2014	Terminated	Jun. 03, 2015	SanDisk Corporation v. Netlist, Inc.
	<u>Number</u> 8516187	<u>Inventor</u> Chen	<u>Name</u> Data transfer scheme for non-volatile memory module		
	Netlist, Inc. v. Smart Modular Technologies, Inc. et al 4-13-cv-05889 (NDCA)	Dec. 19, 2013	Terminated	May. 19, 2020	Netlist, Inc. v. Smart Modular Technologies, Inc. et al
	<u>Number</u> 7381150	<u>Inventor</u> Solomon	<u>Name</u> Circuit providing load isolation and memory domain translation for memory module		
	8001434	Lee	Memory board with self-testing capability		
	8081536	Solomon	Circuit for memory module		
	8301833	Chen	Non-volatile memory module		
	8359501	Lee	Memory board with self-testing capability		
	8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module		
	8516187	Chen	Data transfer scheme for non-volatile memory module		

	Diablo Technologies, Inc. v. Netlist, Inc. 4-13-cv-03901 (NDCa)	Aug. 23, 2013	Terminated	Apr. 08, 2014	Diablo Technologies, Inc. v. Netlist, Inc.
	<u>Number</u>	<u>Inventor</u>	<u>Name</u>		
	7881150	Solomon	Circuit providing load isolation and memory domain translation for memory module		
	8001434	Lee	Memory board with self-testing capability		
	8081536	Solomon	Circuit for memory module		
	8301833	Chen	Non-volatile memory module		
	8359501	Lee	Memory board with self-testing capability		
	8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module		
	8516187	Chen	Data transfer scheme for non-volatile memory module		
	Smart Modular Technologies, Inc. v. Netlist, Inc. 4-13-cv-03916 (NDCa)	Aug. 23, 2013	Terminated	Feb. 12, 2014	Smart Modular Technologies, Inc. v. Netlist, Inc.
	<u>Number</u>	<u>Inventor</u>	<u>Name</u>		
	8001434	Lee	Memory board with self-testing capability		
	8301833	Chen	Non-volatile memory module		
	8359501	Lee	Memory board with self-testing capability		
	8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module		
	8516187	Chen	Data transfer scheme for non-volatile memory module		
	Netlist, Inc. v. Smart Modular Technologies, Inc. 8-13-cv-00996 (CDCA)	Jul. 01, 2013	Terminated	Dec. 23, 2013	Netlist, Inc. v. Smart Modular Technologies, Inc.
	<u>Number</u>	<u>Inventor</u>	<u>Name</u>		
	8001434	Lee	Memory board with self-testing capability		
	8301833	Chen	Non-volatile memory module		
	8359501	Lee	Memory board with self-testing capability		
	8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module		
	8516187	Chen	Data transfer scheme for non-volatile memory module		

Electronic Acknowledgement Receipt

EFS ID:	43017467
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Susanh Perez
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.0010001
Receipt Date:	22-JUN-2021
Filing Date:	24-MAY-2021
Time Stamp:	13:55:05
Application Type:	Utility under 35 USC 111(a)

Payment information:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	NPL2_1_OF_5_240pin_DDR2_ MetaSDRAM_Registered_DIMM .pdf	1937315 ce8d661768da4826c666025788c3c0f254d bdd54	no	32

Warnings:

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2	Non Patent Literature	NPL4_1_OF_5_Amazoncom_search_for_memory_module_downloaded_oct_12_2017.pdf	886882	no	6
			4ce418ee7dfc288864a1c0c0e70cfc25d3f73f20		
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3	Non Patent Literature	NPL8_1_OF_5_American_National_Standard_Dictionary_of_Electrical_Terms.pdf	1669432	no	6
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Warnings:					
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4	Non Patent Literature	NPL9_1_OF_5_Annotated_added_to_Russel_Jacob_Baker_Deposition_Exhibit_1026.pdf	174992	no	1
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5	Non Patent Literature	NPL10_1_OF_5_Annotated_added_to_Russel_Jacob_Baker_Deposition_Exhibit_1027.pdf	173996	no	1
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6	Non Patent Literature	NPL11_1_OF_5_Annotated_added_to_Russel_Jacob_Baker_Deposition_Exhibit_1028.pdf	1310232	no	1
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7	Non Patent Literature	NPL13_1_OF_5_Bonella_Provisional_Application_for_Advance .pdf	2290471	no	53
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8	Non Patent Literature	NPL14_1_OF_5_Bruce_Synchronous_DRAM_Architectures_Organizations.pdf	2371060	no	22
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9	Non Patent Literature	NPL15_1_OF_5_Catsoulis_Designing_Embedded_Hardware.pdf	6720819	no	67
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10	Non Patent Literature	NPL20_1_OF_5_Chiappetta_Kingston_Launches_667_MHz_DDR2SO_DIMM_Memory.pdf	471952	no	3
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11	Non Patent Literature	NPL21_1_OF_5_Copy_of_Notice_of_Final_Rejection_KR.pdf	80109	no	4
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12	Non Patent Literature	NPL22_1_OF_5_Copy_of_Notice_of_Submission_of_Opinion_KR.pdf	251796	no	6
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13	Non Patent Literature	NPL27_1_OF_5_David_H_Fully_Buffered_DIMM_FBDIMM_Design_Considerations.pdf	21690036	no	36
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14	Non Patent Literature	NPL28_1_OF_5_Decision_Denying_Institution_of_Inter_Parties_Review_PTAB_IPR2017_00649_7_july242017.pdf	247510	no	17
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15	Non Patent Literature	NPL34_1_OF_5_Decision_Instituting_Inter_Parties_Review_PTAB_IPR2017-00692_7.pdf	763751	no	29
			8388eddb68ce58cbb56e925c409f71e9e8f8428e		
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16	Non Patent Literature	NPL46_1_OF_5_Netlist_v_Smart_Storage_System_v_DiabloTechnologies_invalidity_CaseNo_413cv05889.pdf	10600473	no	109
			7ff547c9fc85d5fd7bec808fed93062f58202ecb		
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17	Non Patent Literature	NPL48_1_OF_5_ElmhurstD_A18_V_128MB_125MHZ_Multilevel_Cell.pdf	740560	no	5
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18	Non Patent Literature	NPL49_1_OF_5_Ex_1004_IPR2017_00587_Ron_Maltiel_CV.pdf	300416	no	7
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19	Non Patent Literature	NPL6_2_OF_5_Exhibit_12_Decision_Denying_Institution_of_Inter_Parties_Review.pdf	362253	no	17
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20	Non Patent Literature	NPL7_2_OF_5_Exhibit_9_Email_from_Defendants_Counsel.pdf	156463	no	6
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21	Non Patent Literature	NPL50_2_OF_5_JEDEC_Standard_FBDIMM_Specification_DDR2.pdf	2344051	no	129
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Warnings:					
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22	Non Patent Literature	NPL4_3_OF_5_MetaRAM_Develops_New_Technology.pdf	468176	no	3
			4328a50092b9eaa86184718f0f199ed050648ee8		
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23	Non Patent Literature	NPL9_3_OF_5_MutnuaryB_Analysis_of_Fully_Buffered_DIMM.pdf	1450795	no	7
			90ed23396568e94f53b629d3a89e55bc8a75f15a		
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24	Non Patent Literature	NPL19_3_OF_5_NOA_Appl_17138766_3_24_2021_H.pdf	556204	no	10
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25	Non Patent Literature	NPL34_3_OF_5_OA_Appl_15934416_4_2_2021_G.pdf	969167	no	21
			f1a17ef0d2bcbdf6f4baa6a4033161f7c90c5012		
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26	Non Patent Literature	NPL35_3_OF_5_OA_Appl_15934416_01_1_2021_G.pdf	717993	no	17
			4e2d25b2a6b9fd247caac69eba5bba16c959214		
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27	Non Patent Literature	NPL36_3_OF_5_OA_Appl_15934416_3_21_2019_G.pdf	285453	no	8
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30	Non Patent Literature	NPL44_3_OF_5_Order_Grants_Joint_Stipulation_of_Dismissal_NDCA_413cv05889_382.pdf	148422	no	2
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32	Non Patent Literature	NPL7_4_OF_5_Patterson_Computer_Organization_Design.pdf	8238976	no	71
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33	Non Patent Literature	NPL39_4_OF_5_SUPPLEMENTAL_DECLARATION_OF_DANIEL_E_ALBERTI.pdf	109313	no	3
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34	Non Patent Literature	NPL41_4_OF_5_SUPPLEMENTAL_DECLARATION_OF_STEVEN_J_CORRINSUPPORTOFSANDISK_case413cv058889.pdf	158968	no	3
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35	Non Patent Literature	NPL42_4_OF_5_Switches_DT_Online_PTAB_IPR2017_00587_Ex2022.pdf	281887	no	3
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36	Non Patent Literature	NPL48_4_OF_5_Using_Two_Chip_Selects_to_Enable_Quad_Rank.pdf	888238	no	2
			1b9858d81175a80c455ee67862df1d1017caff88		
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37	Non Patent Literature	NPL50_4_OF_5_Websters_IL_New_College_Dictionary.pdf	1024387	no	5
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39	Non Patent Literature	NPL26_1_OF_5_Data_Sheet_74F257A_Quad_2_line.pdf	619266	no	10
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40	Non Patent Literature	NPL50_1_OF_5_Exhibit_1_Claim_Chart_Comparing_Netlists_USPatent8001434.pdf	1401054	no	21
			bb30e92bbe043471a3c3b7a3da6850d524e4c154		
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42	Non Patent Literature	NPL33_3_OF_5_Intel_18Volt_Intel_StrataFlash.pdf	2219680	no	100
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43	Non Patent Literature	NPL32_2_OF_5_Innis_MPC8560_PowerQUICC.pdf	505730	no	25
			f3b5a1e20dd4b4057b7e2a680ad1708ddd72a9ee		
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44	Non Patent Literature	NPL31_2_OF_5_Horowitz_The_Art_of_Electronics.pdf	691449 9f78ad8989060f44b8d5e02cf2fa9e2fbc34362	no	4
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46	Information Disclosure Statement (IDS) Form (SB08)	IDS_20210616_0016001000I_2_OF_5.pdf	1055801 bd07e5548e5c87270596960461c6fc15c5460ae6	no	8
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	Filing Date		2021-05-24
	First Named Inventor	Chi-She Chen	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		0016.0010001

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Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

1	Joint Status Report Regarding Inter Partes Review, Netlist v. Smart Storage Systems, Inc. et. al., Case No. 4:13-CV-05889-YGR (NDCA), filed August 17, 2018, 4 pages.
2	Material Science & Engineering, Department of Material Science and Engineering, Stanford University, Exhibit 2011, Case No. IPR2017-00692, 2 pages.
3	Merriam-Webster's Collegiate Dictionary, Eleventh Ed., Merriam Webster Corporation, Springfield, MA, 2003, 7 pages.
4	MetaRAM Develops New Technology That Quadruples Memory Capacity of Servers and Workstations; Reduces Price by Up to 90 Percent, Press Release provided by MetaRAM and published on MarketWired.com, Feb. 25, 2008, 3 pages.
5	Microsoft Computer Dictionary Fifth Edition, 2002, 3 pages.
6	Microsoft Computer Dictionary Fifth Edition, 2002, 9 pages.
7	Microsoft Press, Computer Dictionary, Second Edition, 1994, 4 pages.
8	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
9	Mutnuary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conferencepp. 203-208.
10	Notice of Allowance in U.S. Appl. No. 12/240,916 dated Sep. 17, 2012, pp. 1-7.
11	Notice of Allowance in U.S. Appl. No. 12/240,916, mailed on Sep. 17, 2012.

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12	Notice of Allowance in U.S. Appl. No. 13/536,173, dated Jul. 2, 2013.
13	Notice of Allowance in U.S. Appl. No. 13/559,476, dated May 6, 2014.
14	Notice of Allowance in U.S. Appl. No. 13/559,476, dated Sep. 29, 2014.
15	Notice of Allowance in U.S. Appl. No. 13/905,048, dated Dec. 19, 2013, 8 pages.
16	Notice of Allowance in U.S. Appl. No. 13/905,053, dated Dec. 11, 2013.
17	Notice of Allowance in U.S. Appl. No. 14/173,219 dated Jul. 7, 2014.
18	Notice of Allowance in U.S. Appl. No. 14/489,269, dated Oct. 8, 2015.
19	Notice of Allowance in U.S. Appl. No. 17/138,766, dated Mar. 24, 2021, 10 pages.
20	Notice of Allowance, SanDisk Corporation v. Netlist, Application No. 13/536,173, IPR2014-00982 (PTAB), dated July 2, 2013, 8 pages.
21	Office Action dated Aug. 19, 2016 of the Chinese Patent Application No. 201280047758.X, 9 pages.
22	Office Action in U.S. Appl. No. 12/240,916, mailed Apr. 3, 2012, pp. 1-12.

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First Named Inventor	Chi-She Chen	
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23	Office Action in U.S. Appl. No. 12/240,916, mailed Feb. 1, 2012.
24	Office Action in U.S. Appl. No. 12/240,916, mailed on Apr. 3, 2012.
25	Office Action in U.S. Appl. No. 12/240,916, mailed on Jul. 29, 2011.
26	Office Action in U.S. Appl. No. 13/536,176, dated on Apr. 15, 2013.
27	Office Action in U.S. Appl. No. 13/625,563, dated Aug. 5, 2013.
28	Office Action in U.S. Appl. No. 13/625,563, dated May 9, 2014.
29	Office Action in U.S. Appl. No. 13/905,048, dated Aug. 1, 2013.
30	Office Action in U.S. Appl. No. 13/905,053, dated Aug. 1, 2013.
31	Office Action in U.S. Appl. No. 14/173,219, dated Mar. 13, 2014.
32	Office Action in U.S. Appl. No. 14/173,242, dated Mar. 14, 2014.
33	Office Action in U.S. Appl. No. 14/302,292, dated Dec. 21, 2015.

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34	Office Action in U.S. Appl. No. 15/934,416, dated on Apr. 2, 2020, 21 pages.
35	Office Action in U.S. Appl. No. 15/934,416, dated on Jan. 1, 2021, 17 pages.
36	Office Action in U.S. Appl. No. 15/934,416, dated on Mar. 21, 2019, 8 pages.
37	Office Action in U.S. Appl. No. 15/934,416, dated on Oct. 15, 2019, 13 pages.
38	Office Action in U.S. Appl. No. 17/138,766, dated on Mar. 2, 2021, 7 pages.
39	Office Action in U.S. Application No. 12/240,916, dated Feb. 1, 2012, 14 pages.
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41	Office Action, Application 12/240,916, IPR2014-0099-1005 (PTAB), dated July 29, 2011, 8 pages.
42	Office Action, Application 12/240,916, IPR2014-0099-1010 (PTAB), dated April 3, 2012, 11 pages.
43	Office Action, Application No. 13/536,173, IPR2014-00982 (PTAB), dated April 15, 2013, 9 pages.
44	Order Granting Joint Stipulation of Dismissal, Netlist v. Smart Storage Systems, Inc. et. al., Case No. 4:13-CV-05889-YGR (NDCA), filed May 15, 2020, 2 pages.

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Attorney Docket Number	0016.001000I	

45	Out of Stealth Mode, Start-Up MetaRAM Unveils New Technology That Quadruples DRAM Capacity, Press Release edited by Storage Newsletter on Feb. 28, 2008 at StorageNewsLetter.com, 8 pages.
46	Patent Owner's Demonstratives, Patent Number: 8,671,243, Case No. IPR 2017-00587-2023, (PTAB), 57 pages.
47	Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, Patent No. 8,671,243, Case No. IPR2017-00587 (PTAB), filed Jan. 29, 2018, 6 pages.
48	Patent Owner's Opposition to Petitioners' Motion to Exclude, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed April 2, 2018, 2017, 11 pages.
49	Patent Owner's Preliminary Response, Patent Number: 8,516,187, Case IPR2014-01371 (PTAB), filed Dec. 16, 2014, 66 pages.
50	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated October 2, 2014, 60 pages.

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	First Named Inventor	Chi-She Chen	
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

1	Exhibit, Application As Filed, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), 52 pages
2	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-3, filed March 17, 2015, 23 pages.
3	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-4, filed March 17, 2015, 20 pages.
4	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-5, filed March 17, 2015, 28 pages.
5	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-9, filed March 17, 2015, 17 pages.
6	Exhibit 12, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-7, filed March 17, 2015, 17 pages.
7	Exhibit 9, Email from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-17, filed March 10, 2015, 6 pages.
8	Exhibit, Institution of Inter Partes Review, Sandisk Corporation v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 316-9, filed March 24, 2015, 29 pages.
9	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
10	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
11	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-2, filed March 17, 2015, 23 pages.

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12	Exhibit, Letter from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-16, filed March 10, 2015, 3 pages.
13	Exhibit, Letter sent via email on December 6, 2013, Case 4:13-cv-03901-YGR Document 53-2, filed January 6, 2014
14	Exhibit, Order Denying Defendant's Motion to Stay Pending Inter Partes Review (Doc.59), The Procter and Gamble Company v. Team Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 316-6, filed March 24, 2015, 10 pages.
15	Exhibit, Patent Public Advisory Committee Quarterly Meeting, Appeals Statistics USPTO, Case No. 4:13-cv-05889-YGR Document 309-8, filed March 17, 2015, 23 pages.
16	Exhibit, Reporter's Transcript of Proceedings, Netlist, Inc v. Smart Modular Technologies, Inc., et al., Case 4:13-cv-05889-YGR Document 316-3, filed March 24, 2015, 15 pages.
17	Exhibit, Transcript of Official Electronic Sound Recording Proceeding, Netlist v. Smart Modular Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 305-7, filed March 10, 2015, 10 pages.
18	Extended European Search Report for European Application No. 12 817 751.6, dated Jun. 9, 2015.
19	File History for U.S. Appl. No. 12/240,916, filed Sep. 29, 2008, 320 pages.
20	File History for U.S. Appl. No. 13/905,048, filed May 29, 2013, 181 pages.
21	File History for U.S. Appl. No. 60/941,586, filed Jun. 1, 2007, 23 pages.
22	File History U.S. Pat. No. 8,671,243.

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23	Final Office Action, dated Jun. 15, 2016, dated in U.S. Appl. No. 14/489,281, 10 pages.
24	Final Office Action, dated Jun. 15, 2016, issued in U.S. Appl. No. 14/489,281, 10 pages.
25	Final Office Action, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-0099-1007 (PTAB), dated February 1, 2012, 13 pages
26	Final Written Decision, Patent 8,671,243, IPR2017-00587-34, Paper No. 34, entered June 20, 2018, 53 pages.
27	Final Written Decision, Patent 8,874,831, IPR2017-00692, Paper No. 25, entered July 5, 2018, 42 pages.
28	Final Written Decision, US Patent No. 7,881,150, Case No. Case IPR2014-00882, (PTAB), Paper 33, filed December 14, 2015, 51 pages.
29	First Amended Complaint for Patent Infringement, Netlist Inc. v Smart Modular Tech, Inc., et al., Case 8:13-cv-00996-DOC-JPR, Document 5, filed Aug. 23, 2013, 99 pages.
30	Hasan, J. et al. Efficient Use of Memory Bandwidth to Improve Network Processor Throughput, Proceedings of the 30th Annual International Symposium on Computer Architecture (ISCA'03), IEEE, 2003, 12 pages.
31	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.
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34	Inter Partes Review No. IPR2017-00692 (PTAB), Patent No. 8,874,831, filed July 26, 2012, 78 pages.
35	Inter Partes Review of U.S. Patent No. 8,874,831, Case IPR2017-00692 (PTAB), filed July 26, 2012, Paper No. 1, 78 pages.
36	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014.
37	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014, pp. 1-8.
38	International Search Report and Written Opinion in PCT/US12/48750, dated Oct. 10, 2012, pp. 1-10.
39	ISSCC 2006 / SESSION 7 / NON-VOLATILE MEMORY / 7.7, IEEE International Solid-State Circuits Conference, 2006, 10 pages.
40	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322, 58 pages.
41	Jandhyala, S. et al., "Design-For-Test Analysis of a Buffered SDRAM DIMM", Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, Aug. 13014, 1996, 15 pages.
42	JEDEC Definition of DIMM, Exhibit 1029, IPR No. 2017-00587, December 18, 2017, 2 pages.
43	JEDEC Global Standard for the Microelectronics Industry, Why JEDEC Standards Matter, 2014, 1 page.
44	JEDEC Standard 21-C, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.

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45	JEDEC Standard No. 21-C (Release 17), Annex J: Serial Presence Detects for DDR2 SDRAM (Revision 1.3), 60 pages.
46	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, Aug. 1999, 114 pages.
47	JEDEC Standard, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A) Jan. 2005, 113 pages.
48	JEDEC Standard, Double Data Rate (DDR) SDRAM Specification, JESD79, Jun. 2000, 77 pages.
49	JEDEC Standard, Double Data Rate (DDR): SDRAM Specification: JESD79C (Revision JESD79B), Mar. 2003, pp. 1-75.
50	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC Solid State Tech. Assoc., Mar. 2007, 129 pages.

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First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

CERTIFICATION STATEMENT

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That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

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SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

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	Filing Date		2021-05-24
	First Named Inventor	Chi-She Chen	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		0016.0010001

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Application Number	17328019
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First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

1	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Patent Number: 8,516,187, IPR2014-00982 (PTAB), dated September 26, 2014, 57 pages.
2	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,301,833, Case No. IPR2017-00649 (PTAB), filed May 1, 2017, 67 pages.
3	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed May 1, 2017, 48 pages.
4	Patent Owner's Preliminary Response, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, Case IPR2014-01370 (PTAB), filed Dec. 16, 2014, 66 pages.
5	Patent Owner's Response, Patent No. 8,671,243, Case No. IPR2017-00587-12 (PTAB), filed Oct. 13, 2017, 80 pages.
6	Patent Owner's Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692-12 (PTAB), filed Nov. 10, 2017, 77 pages.
7	Patterson et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998), 71 pages.
8	Petition for Inter Partes Review of Claims 1-30 of US.PATENT No. 8,301,833, IPR2014-01370 (PTAB), filed August 22, 2014, 68 pages.
9	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
10	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
11	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833, filed Sep. 29, 2008.

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12	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SanDisk, Corp.), filed Jun. 19, 2014.
13	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
14	Petition for Inter Partes Review of U.S. Pat. No. 8,671,243, filed May 29, 2013.
15	Petition for Inter Partes Review of U.S. Pat. No. 8,874,831, filed Jul. 26, 2012.
16	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
17	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
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19	Petition for Inter Partes Review of US Patent 8,516,187, IPR2014-00982 (PTAB), dated June 19, 2014, 67 pages.
20	Petition for Inter Partes Review of US Patent No. 8,671,243, Case No. IPR2017-00587-1(PTAB), May 29, 2013, 82 pages.
21	Petition for Inter Partes Review, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), filed June 20, 2014, 69 pages.
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24	Petitioners Demonstratives, SK hynix Inc., et al., v. Netlist, Inc., U.S. Patent No. 8,671,243, IPR2017-00587-1037, 89 pages.
25	Petitioners' Reply in Support of Its Motion to Exclude, U.S. Patent No. 8,874,831, Case No. IPR2017-00692, filed April 9, 2018, 8 pages.
26	Petitioners' Reply to Patent Owner's Response, Inter Partes Review No. IPR2017-00587-16, Patent No. 8,671,243, filed Jan. 12, 2018, 38 pages.
27	Petitioners' Reply, U.S. Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), Paper No. 15, filed March 2, 2018, 35 pages.
28	Petitioners' Request for Hearing, U.S. Patent No. 8,301,833, Case No. Case No. IPR2017-00649-2020 (PTAB), Paper No. 8, entered Aug, 23, 2017, 18 pages.
29	Prosecution History, Application No. 12,240,916, US Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date April 13, 2009, 320 pages.
30	Prosecution History, Application No. 13,559,476, US Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date Sep. 5, 2014, 312 pages.
31	Prosecution History, Provisional Application No. 60/941,586, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1005 (PTAB), received date Oct. 15, 2012, 23 pages.
32	Provisional Application for Advance Dynamic Disk Memory Module, Specification, SK hynix Inc., V. Netlist, Inc., Case No. IPR2017-00649-1006 (PTAB), dated Dec. 8, 2005, 53 pages.
33	Provisional Application No. 60/912,321 filed Apr. 17, 2007.

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34	Provisional Application No. 60/941,586 filed Jun. 1, 2007.
35	Requirement for Restriction Election, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994-1003 (PTAB), dated March 31, 2019, 7 pages.
36	Restriction Requirement in U.S. Appl. No. 12/240,916, dated Mar. 31, 2011.
37	Second Amended Answer and Counterclaims to Plaintiff's Complaint for Declaratory Judgment, Diablo Technologies, Inc. v. Netlist, Inc., Case No. 4:13-CV-03901 (NDCA), YGR, filed February 17, 2014, 21 pages.
38	Smart Storage Systems, Inc's Invalidity Contentions, Netlist, Inc. v Smart Storage Systems, et al., Case No. 4:13-cv-05889-YGR, dated Jun. 6, 2014, 538 pages.
39	Supplemental Declaration of Daniel E. Alberti in Response to Court Order [Docket No. 361] in Support of Motion to Withdraw as Counsel of Record, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 362, filed September 5, 2018, 3 pages.
40	Supplemental Declaration of Ronald H. Spuhler, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-1, filed March 10, 2015, 2 pages.
41	Supplemental Declaration of Steven J. Corr, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 309, filed March 17, 2015, 3 pages.
42	Switches - D&T Online, Switches, http://wiki.dtonline.org , Exhibit 1022, Case No. IPR2017-00587, accessed October 13, 2017, 3 pages.
43	U.S. Appl. No. 60/912,321, filed Apr. 17, 2007.
44	U.S. Office Action in U.S. Appl. No. 13/536,173, mailed on Apr. 15, 2013, pp. 1-10.

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45	Updated Joint Case Management Statement, Case No. 4:13-cv-03916-YGR (NDCA), Document 53, filed Jan. 6, 2014.
46	US Patent Application No. 12/240,916, Case No. IPR2017-00692-2018 (PTAB), dated Sep. 29, 2008, 52 pages.
47	US Patent Application No. 60/912,321, Case No. No. IPR2017-00692-1007 (PTAB), filed April 17, 2017, 42 pages.
48	Using Two Chip Selects to Enable Quad Rank, an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: Dec. 17, 2005, 2 pages.
49	Videotaped Deposition of Russel Jacob Baker, Patent No. 8,671,243, Case Nos. IPR2017-00587-1030 (PTAB), filed December 18, 2017, 268 pages.
50	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 257, 259; 1114,1115, Exhibit 1017, IPR2014-01370 (PTAB), Smart Modular Tech v Netlist Inc, Patent No. 8,301,833, Sep. 22, 2014, 5 pages.

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CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

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	First Named Inventor	Chi-She Chen	
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	Examiner Name		
	Attorney Docket Number		0016.0010001

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	1	Third Amended Complaint for Patent Infringement, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-CV-05889-YGR (NDCA), filed October 7, 2014, 21 pages.	
	2	240pin DDR2 MetaSDRAM Registered DIMM based on 1 GB version C, Hynix Semiconductor, Product Description Rev. 0.2, Sep. 2008, 32 pages.	
	3	Advisory Action in U.S. Appl. No. 12/240,916, dated Mar. 13, 2012.	
	4	Amazon.com, Search for "memory module", downloaded Oct. 12, 2017, 6 pages.	
	5	Amendment and Reply to Office Action, SanDisk Corporation v. Netlist, Application No. 13/536,173, IPR2014-00982 (PTAB), dated May 21, 2013, 24 pages.	
	6	Amendment and Response to Election Restriction, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994-1004 (PTAB), dated May 20, 2011, 9 pages.	
	7	American Heritage Dictionary of the English Language, Third Ed., Houghton Mifflin Company, Boston, MA, 1996, 7 pages.	
	8	American National Standard Dictionary of Electrical and Electrical Terms, IEEE, Fourth Edition, Revised, ANS/IEEE Std 100-1988, Institute of Electrical Engineers, Nov. 3, 1988, 6 pages.	
	9	Annotated added to Russel Jacob Baker Deposition, Exhibit 1026, p.78, Patent No. 8, 671,243, Case No. IPR-2017-00587 (PTAB), filed December 18, 2017, 1 page.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

10	Annotated added to Russel Jacob Baker Deposition, Exhibit 1027, p.79, Patent No. 8, 671,243, Case No. IPR-2017-00587 (PTAB), filed December 18, 2017, 1 page.
11	Annotated added to Russel Jacob Baker Deposition, Exhibit 1028, p.79, Patent No. 8, 671,243, Case No. IPR-2017-00587 (PTAB), filed December 18, 2017, 1 page.
12	Appeals from the USPTO, PTAB in Nos. IPR2014-00882, IPR2014-00883, IPR2014-01011, US Court of Appeals for the Federal Circuit, decided July 25 2017, 8 pages.
13	Bonella, R., Provisional Application for Advanced Dynamic Disk Memory Module, 53 pages.
14	Bruce, J., Synchronous DRAM Architectures, Organizations, and Alternate Technologies, Electrical and Computer Engineering Dept., Univ. of Maryland, Dec. 10, 2002, 22 pages.
15	Catsoulis, J., Designing Embedded Hardware: Create New Computers and Devices, O'Reilly Media, Inc. (2005), 67 pages.
16	Charles C., Multiplexers and Demultiplexers, Ex. 1024, IPR No. 2017-00587-1024(PTAB), 2003, 8 pages.
17	Charles C., Multiplexers and Demultiplexers, Ex. 1033, IPR No. 2017-00587-1033 (PTAB), 2003, 8 pages.
18	Charles C., Multiplexers and Demultiplexers, Ex. 1034, IPR No. 2017-00587-1033 (PTAB), 2003, 8 pages.
19	Charles C., Multiplexers and Demultiplexers, Ex. 1035, IPR No. 2017-00587-1035 (PTAB), 2003, 8 pages.
20	Chiappetta, M., Kingston Launches 667-MHz DDR2 SO-DIMM Memory, Oct. 11, 2005, 3 pages.

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Attorney Docket Number	0016.001000I

21	Copy of Notice of Final Rejection, South Korea Patent Office with English Translation, 4 pages.
22	Copy of Notice of Submission of Opinion, South Korea Patent Office with English Translation, 6 pages.
23	Corrected Petition for Inter Partes Review of Claims 1-30 of US Patent No. 8,301,833, Declaration of Michael F. Heafey, IPR2014-01370 (PTAB), filed September 22, 2014, 4 pages.
24	Corrected Petition for Inter Partes Review of Claims 1-32 of U.S. Patent No. 8,516,187, IPR2014-01371 (PTAB), September 22, 2014, 67 pages.
25	Corrected Petition for Inter Partes Review of Claims 1-32 of US Patent No. 8,516,187, Declaration of Dr. Nader Bagherzadeh, IPR2014-01371 (PTAB), filed September 22, 2014, 306 pages.
26	Data Sheet, 74F257A Quad 2-line to 1-line selector/multiplexer, non-inverting (3-State), Product specification, IC15 Data Handbook Mar. 31, 1995, 10 pages.
27	David, H. et al., Fully Buffered DIMM (FB-DIMM) Design Considerations, Intel Developer Forum, Intel Corp., Feb. 18, 2004, 36 pages.
28	Decision Denying Institution of Inter Partes Review, Patent 8,301,833, Case IPR2017-00649-7 (PTAB), Paper 7, entered July 24, 2017, 17 pages.
29	Decision Denying Institution of Inter Partes Review, Patent No. 8,516,187, IPR2014-01371 (PTAB), Paper 12, entered March 13, 2015, 22 pages.
30	Decision Denying Institution of Inter Partes Review, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated December 16, 2014, 16 pages.
31	Decision Denying Institution of Inter Partes Review, SanDisk Corporation v. Netlist, Patent 8,516,187, IPR2014-00982 (PTAB), Paper 9, dated December 22, 2014, 16 pages.

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Examiner Name	
Attorney Docket Number	0016.001000I

32	Decision Denying Institution of Inter Partes Review, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, IPR2014-01370 (PTAB), Paper 13, entered March 13, 2015, 19 pages.
33	Decision Instituting Inter Partes Review, Patent 8,671,243, Case No. IPR2017-00587-7(PTAB), Paper No. 7, entered June 22, 2017, 40 pages.
34	Decision Instituting Inter Partes Review, Patent 8,874,831, Case No. IPR2017-00692-7(PTAB), Paper No. 7, entered July 21, 2017, 29 pages.
35	Declaration of R. Jacob Baker, Patent No. 8,874,831, Case IPR2017-00692-2016 (PTAB), filed Nov. 10, 2017, 72 pages.
36	Declaration of R. Jacob Baker, Patent Number: 8,671,243, Case IPR2017-00587-2016 (PTAB), filed October 13, 2019, 109 pages.
37	Declaration of Dr. Nader Bagherzadeh, Patent No. 8,516,187, IPR2014-01371 (PTAB), filed September 22, 2014, 306 pages.
38	Declaration of Jeff McMullen, Netlist, Inc. v. Diablo Technologies, Inc., Case No. 4: I 3-CV-05889-YGR (NDCA), Document 362-1, filed September 5, 2018, 3 pages.
39	Declaration of Paul Min, In Inter Partes Review of U.S. Patent No. 8,301,833, IPR2014-00994-1020 (PTAB), filed September 29, 2008, 215 pages.
40	Declaration of Paul Min, In Inter Partes Review of U.S. Patent No. 8,516,187, IPR2014-00982-1013 (PTAB), filed June 28, 2012, pages 240.
41	Declaration of Ron Maltiel, Patent No. 8,671,243, No. IPR2017-00587-1003 (PTAB), filed May 29, 2013, 131 pages.
42	Declaration of Ron Maltiel, Patent No. 8,874,831, No. IPR2017-00692-1003 (PTAB), filed July 26, 2012 29, 172 pages.

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Application Number		17328019
Filing Date		2021-05-24
First Named Inventor	Chi-She Chen	
Art Unit		
Examiner Name		
Attorney Docket Number	0016.001000I	

43	Declaration of Steven J. Corr, Netlist, Inc. v. SMART Storage Systems, Inc., et al., Case 4:13-cv-05889-YGR (NDCA), Document 305-8, filed March 10, 2015, 3 pages.
44	Deposition of Baker, Exhibit 1030, p.78, Patent No. 8, 671,243, SK hynix Inc., et al. v. Netlist, Inc., Case No. IPR-2017-00587 (PTAB), filed December 18, 2017, 268 pages.
45	Deposition of Ron Maltiel, Patent No. 8874831, Case No. IPR2017-00692-2010 (PTAB), dated Sep. 27, 2017, 155 pages.
46	Netlist v Smart Storage System and Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889 YGR, dated Jun. 6, 2014, 109 pages.
47	Drawing by R. Jacob Bake regarding Address of Dram and Flash, Patent Number: 8,671,243, Case IPR2017-00587-1023 (PTAB), filed December 18, 2017, 1 page.
48	Elmhurst, D. et al., A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write, IEEE Journal of Solid-State Circuits, Vol. 38, No. 11, Nov. 2003, 5 pages.
49	Ex. 1004—IPR2017-00587 Ron Maltiel CV, 7 pages.
50	Exhibit 1: Claim Chart Comparing Netlist's US Patent No. 8,001,434 to Smart Storage Ulltradimm, Case4:13-cv-05889-YGR Document 193-1, filed Apr. 10, 2014, 21 pages.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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(Not for submission under 37 CFR 1.99)

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Filing Date	2021-05-24
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Art Unit	
Examiner Name	
Attorney Docket Number	0016.001000I

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	05/24/2021	Chi-She Chen	0016.001000I	6552
151145	7590	07/21/2021	EXAMINER	
Shami Messinger PLLC			CENTRAL, DOCKET	
1000 Wisconsin Ave. NW			ART UNIT	
Suite 200			PAPER NUMBER	
Washington, DC 20007			OPAP	
			NOTIFICATION DATE	DELIVERY MODE
			07/21/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

khaled@shamimessinger.com
mike@shamimessinger.com
stephanie@shamimessinger.com

<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 17/328,019	Applicant(s) Chen et al.	
	Examiner TERRI S JOHNSON	Art Unit OPET	AIA (FITF) Status No
1. THE REQUEST FILED <u>24 May 2021</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☒ for an original nonprovisional application (Track I). B. ☐ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to TERRI JOHNSON at (571)272-2991. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
/TERRI S JOHNSON/ Paralegal Specialist, OPET			



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/328,019	05/24/2021	Chi-She Chen	0016.001000I

CONFIRMATION NO. 6552

PUBLICATION NOTICE



OC000000128292336

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2021-0279194-A1

Publication Date:09/09/2021

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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17/328,019	05/24/2021	Chi-She Chen	0016.001000I	6552
151145	7590	09/27/2021	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW			ART UNIT	
9th Floor			PAPER NUMBER	
Washington, DC 20006-1152			2135	
			NOTIFICATION DATE	DELIVERY MODE
			09/27/2021	ELECTRONIC

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kshami@vorys.com
mvmessinger@vorys.com
vorys_docketing@cardinal-ip.com

Office Action Summary

Application No.

17/328,019

Applicant(s)

Chen et al.

Examiner

HASHEM FARROKH

Art Unit

2135

AIA (FITF) Status

No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 06/04/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 2-31 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☒ Claim(s) 16-19 and 21 is/are allowed.

7) ☒ Claim(s) 2-15,20,22-23 and 26-31 is/are rejected.

8) ☒ Claim(s) 24-25 is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 05/24/2021 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date 06/22/2021.

Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

The instant application having application No. 17/328,019 has a total of 30 claims pending in the application; there are 3 independent claim and 27 dependent claims, all of which are ready for examination by the examiner.

ACKNOWLEDGEMENT OF REFERENCES CITED BY APPLICANT

Information Disclosure Statement

*As required by **M.P.E.P. 2001.06(b)** and **37 C.F.R. 1.98(d)**, since the instant application has been identified as a continuation application of an earlier filed application and is relied upon for an earlier filing date under **35 U.S.C. 120**, the examiner has reviewed the prior art cited in the earlier related application as required by **M.P.E.P. 707.05** and **904** and as stated in **M.P.E.P. 2001.06(b)**, no separate citation of the same prior art need be made by the applicants in the instant application.*

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 06/22/2021 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the Examiner. However, the documents marked by lined-through have not been considered by because either copy of the documents

were not provided or the documents weren't readily identifiable from the IDS description of the corresponding documents.

INFORMATION CONCERNING CLAIMS:

Claim Interpretation

Claim limitations in this application do not use the word "means" (or "step"), thus, they are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

1. *Claims 20, 22, and 30-31 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement.*

The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or

a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

2. *Claim 20 recites the limitation(s):*

“wherein the circuit includes a first diode having a first input and a first output, the first input is coupled to the first pre-regulated voltage and the first output is coupled to the voltage conversion circuit, and wherein the circuit includes a second diode having a second input and a second output, the second input is coupled to the second pre-regulated voltage and the second output is coupled to the first output and to the voltage conversion circuit.” (*Emphasis added*).

3. *Claim 22 recites the limitation(s):*

“wherein the first circuit includes a first diode coupled between the first input and the output and a second diode coupled between the second input and the output.”

4. *Claim 30 recites the limitation(s):*

“wherein the first pre-regulated voltage is coupled to the voltage conversion circuit via a first diode.” (*Emphasis added*).

5. *Claim 31 recites the limitation(s):*

“wherein the second pre-regulated voltage is coupled to the voltage conversion circuit via a second diode.” (*Emphasis added*).

The claimed specification does not appear to describe or support the above limitations as claimed.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at

least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 et seq. for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For

more information about eTerminal Disclaimers, refer to

www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

6. Claims 2-15, 23, and 26 -29 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-30 of U.S. Patent No. 11,016,918 B2.

Although the claims at issue are not identical, they are not patentably distinct from each other because although, the language and order of limitations in claims are slightly different (e.g., see table below), they are not patentably distinct from each other.

7. **Claim 2 of instant application (Application No. 17/328,019) is compared to claim 1 of Patent No. 11,016,918 B2 in the following table:**

Patent No. 11,016,918 B2	Application No. 17/328,019
Claim 1: A memory module comprising: a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;	Claim 2: A memory module comprising: a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising: a plurality of synchronous dynamic random access memory (SDRAM)	a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages; a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, the plurality of components including
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devices coupled to the first regulated voltage, and at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.	a plurality of synchronous dynamic random access memory (SDRAM) devices and a first circuit that is coupled to the plurality of synchronous dynamic random access memory (SDRAM) devices and to a first set of edge connections of the plurality of edge connections, wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.
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Allowable Subject Matter

Claims 16-19 and 21 are allowed.

Claims 24-25 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

8. *The prior art of record does not disclose or suggest the distinct features “the voltage monitor circuit configured to detect a change in the input voltage, wherein, in response to the voltage monitor detecting a change in the input voltage, the memory module transitions from a first operable state to a second operable state”, as recited in the independent claim 16. Claims 17-19 and 21 are dependent from the independent claim 16.*

9. *The prior art of record does not disclose or suggest the distinct features “wherein, in response to the voltage monitor circuit detecting a change in the input voltage, the memory module transitions from a first state of operation to a second state of operation”, as recited in the independent claim 24. Claim 25 is dependent from claim 24.*

Conclusion

The prior art made of record and not relied upon are as follows:

1. *Chen et al. (US 20120271990 A1).*
2. *Bates (US 20070136523 A1).*
3. *Prete et al. (US 20080126624 A1).*

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2135*

September 22, 2021

<i>Notice of References Cited</i>	Application/Control No. 17/328,019		Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH		Art Unit 2135	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	10-2012	Chen; Chi-She	G06F12/00	711/103
*	B	US-20070136523-A1	06-2007	Bonella; Randy M.	G06F9/4401	711/113
*	C	US-20080126624-A1	05-2008	Prete; Edoardo	G11C7/106	710/53
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
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FOREIGN PATENT DOCUMENTS

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	R					
	S					
	T					

NON-PATENT DOCUMENTS

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	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM			DATE							
Final	Original	09/22/2021								
	1	-								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
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	22	✓								
	23	✓								
	24	O								
	25	O								
	26	✓								
	27	✓								
	28	✓								
	29	✓								
	30	✓								
	31	✓								

<i>Search Notes</i> 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G11C7/1072; G11C14/0018; G06F12/0246; G06F13/4243; G06F13/1694; G06F3/0659; G06F13/4027; G06F12/0638; G06F13/28; G06F1/185; G06F2212/205; G06F2212/7208; G06F3/0685; Y02D10/00	09/22/2021	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	09/22/2021	HF

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Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	09/22/2021	HF
EAST Text Search	09/22/2021	HF
IP Search	09/22/2021	HF
Google Search	09/22/2021	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17328019
	Filing Date		2021-05-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2135	
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number	0016.0010001	

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**INFORMATION DISCLOSURE
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Application Number		17328019
Filing Date		2021-05-24
First Named Inventor	Chi-She Chen	
Art Unit		
Examiner Name		
Attorney Docket Number	0016.0010001	

1	Joint Status Report Regarding Inter Partes Review, Nollist v. Smart Storage Systems, Inc. et al., Case No. 4:13 CV-05889-YGR (NDCA), filed August 17, 2018, 4 pages.
2	Material Science & Engineering, Department of Material Science and Engineering, Stanford University, Exhibit 2011, Case No. IPR2017-00692, 2 pages.
3	Merriam-Webster's Collegiate Dictionary, Eleventh Ed., Merriam-Webster Corporation, Springfield, MA, 2003, 7 pages.
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5	Microsoft Computer Dictionary Fifth Edition, 2002, 3 pages.
6	Microsoft Computer Dictionary Fifth Edition, 2002, 9 pages.
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9	Mutnuary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conferencepp. 203-208.
10	Notice of Allowance in U.S. Appl. No. 12/240,916 dated Sep. 17, 2012, pp. 1-7.
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Application Number	17328019
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First Named Inventor	Chi-She Chen
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Attorney Docket Number	0016.001000I

12	Notice of Allowance in U.S. Appl. No. 13/536,173, dated Jul. 2, 2013.
13	Notice of Allowance in U.S. Appl. No. 13/559,476, dated May 6, 2014.
14	Notice of Allowance in U.S. Appl. No. 13/559,476, dated Sep. 29, 2014.
15	Notice of Allowance in U.S. Appl. No. 13/905,048, dated Dec. 19, 2013, 8 pages.
16	Notice of Allowance in U.S. Appl. No. 13/905,053, dated Dec. 11, 2013.
17	Notice of Allowance in U.S. Appl. No. 14/173,219 dated Jul. 7, 2014.
18	Notice of Allowance in U.S. Appl. No. 14/489,269, dated Oct. 8, 2015.
19	Notice of Allowance in U.S. Appl. No. 17/138,766, dated Mar. 24, 2021, 10 pages.
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Attorney Docket Number	0016.001000I	

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29	Office Action in U.S. Appl. No. 13/905,048, dated Aug. 1, 2013.
30	Office Action in U.S. Appl. No. 13/905,053, dated Aug. 1, 2013.
31	Office Action in U.S. Appl. No. 14/173,219, dated Mar. 13, 2014.
32	Office Action in U.S. Appl. No. 14/173,242, dated Mar. 14, 2014.
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37	Office Action in U.S. Appl. No. 15/934,416, dated on Oct. 15, 2019, 13 pages.
38	Office Action in U.S. Appl. No. 17/138,766, dated on Mar. 2, 2021, 7 pages.
39	Office Action in U.S. Application No. 12/240,916, dated Feb. 1, 2012, 14 pages.
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41	Office Action, Application 12/240,916, IPR2014-0099-1005 (PTAB), dated July 29, 2011, 8 pages.
42	Office Action, Application 12/240,916, IPR2014-0099-1010 (PTAB), dated April 3, 2012, 11 pages.
43	Office Action, Application No. 13/536,173, IPR2014-00982 (PTAB), dated April 15, 2013, 9 pages.
44	Order Granting Joint Stipulation of Dismissal, Nollist v. Smart Storage Systems, Inc. et al., Case No. 4:13-cv-05889-YGR (NDCA), filed May 15, 2020, 2 pages.

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Examiner Name	
Attorney Docket Number	0016.0010001

45	Out of Stealth Mode, Start-Up MetaRAM Unveils New Technology That Quadruples DRAM Capacity, Press Release edited by Storage Newsletter on Feb. 28, 2008 at StorageNewsLetter.com, 8 pages.
46	Patent Owner's Demonstratives, Patent Number: 8,671,243, Case No. IPR 2017-00587-2023, (PTAB), 57 pages.
47	Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, Patent No. 8,671,243, Case No. IPR2017-00587 (PTAB), filed Jan. 29, 2018, 6 pages.
48	Patent Owner's Opposition to Petitioners' Motion to Exclude, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed April 2, 2018, 2017, 11 pages.
49	Patent Owner's Preliminary Response, Patent Number: 8,516,187, Case IPR2014-01371 (PTAB), filed Dec. 16, 2014, 66 pages.
50	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated October 2, 2014, 60 pages.

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EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	09/21/2021
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.0010001

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S413	4	"11016918".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/16 11:29
S414	1	"17328019"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/16 11:41
S415	3	"17138766"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/16 15:36
S416	380	((printed adj circuit adj board) or PCB) with (plurality near2 edge near2 connections)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 10:01
S417	10,693	((synchronous adj dynamic adj random adj access adj memory) or SDRAM) near2 (devices or chips or dies)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 10:05
S418	7,653	(regulated near2 voltages)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 10:06

S419	6	S417 same S418	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 10:06
S420	16	"8301833".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 10:08
S421	31	"12131873"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 10:12
S422	3	S419 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 10:24
S423	3	S416 same S417	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 10:47
S424	33	S416 and S417	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/09/20 11:03
S425	4	S424 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:03

S426	4	S425 and voltage	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:06
S427	0	S425 and (regulat\$3 near2 voltage)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:08
S428	3	((printed adj circuit adj board) or PCB) with (plurality near2 edge near2 connections) same SDRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:09
S429	7,625	(board same SDRAM)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:10
S430	43	S418 and S429	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:10
S431	19	S430 and connector	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:10
S432	19	S431 and (regulat\$3 near2 voltage)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:11

S433	2	S432 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/20 11:11
S434	220	("2043099" "3562555" "3916390" "4234920" "4607332" "4658204" "4884242" "4965828" "4420821" "4449205" "4882709" "5430742" "5444664" "5630096" "5757712" "58749955490155" "5519663" "5519831" "55638395917213" "5619644" "5675725" "5799200" "5813029" "5870350" "5890192" "5953215" "5991885" "6721860" "6026465" "6065092" "6216247" "6421279" "6614685" "6721212" "6799241" "6816982" "69106356948029" "6023421" "6112310" "6145068" "6158015" "6199142" "6269382" "6336174" "6336176" "6363450" "6459647" "6487102" "6487623").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 12:28
S435	281	("6571244" "6658507" "6691209" "6693840" "6769081" "6799244" "6810513" "6839774" "6944042" "6952368" "7062618" "7107480" "7136978" "7421552" "7467251" "7600142" "7881150" "7952179" "7053470" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7353325" "7409491" "7409590" "7411859" "7519754" "7613877" "7716411" "7818488" "7873750" "8001434" "8081536" "8301833" "8874831" "8086955" "8102614" "8214616" "8233303" "8407395" "8412879" "8516187" "8671243" "8677060" "8880791" "8904098" "8904099" "9043677" "9158684" "9361250" "9436600" "9921762" "20020053944" "20020083368" "20020199061" "20030028733" "20030076726" "20030158995" "20030204776" "20030210601").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:14
S436	101	("20030147297" "20030206478" "20030137881" "20040163027" "20040088508" "20040190210"	US-PGPUB; USPAT; USOCR;	OR	ON	2021/09/21 14:34

		"20050044302" "20050060488" "20050132250" "20050141273" "20050144418" "20050185472" "20050204091" "20050249011" "20050273548" "20060080515" "20060126369" "20060294295" "20060039197" "20060069896" "20060212651" "20070276995" "20070070669" "20070136523" "20070147115" "20070192627" "20070255898" "20070288683" "20080126690" "20080104344" "20080126624" "20080147968" "20080189479" "20080195806" "20080235443" "20080291727" "20090235038" "20090031099" "20100122200" "20100110748" "20100274953" "20100322020" "20110078496" "20110161569").pn.	FPRS; EPO; JPO; DERWENT; IBM_TDB			
S437	34	("20110320804" "20120110417" "20120271990" "20120117402" "20120204079" "20120265952" "20120317433" "20130019076" "20130086309" "20130254456" "20140059170" "20140156919" "20140156920" "20150058701").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:39
S438	72	((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:42
S439	220	("2043099" "3562555" "3916390" "4234920" "4607332" "4658204" "4884242" "4965828" "4420821" "4449205" "4882709" "5430742" "5444664" "5630096" "5757712" "58749955490155" "5519663" "5519831" "55638395917213" "5619644" "5675725" "5799200" "5813029" "5870350" "5890192" "5953215" "5991885" "6721860" "6026465" "6065092" "6216247" "6421279" "6614685" "6721212" "6799241" "6816982" "69106356948029" "6023421" "6112310" "6145068" "6158015" "6199142" "6269382" "6336174" "6336176" "6363450" "6459647" "6487102" "6487623").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:43

S440	13	S438 and (S439 or S435 or S436 or S437)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:43
S441	0	S440 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/21 14:47
S442	1,356	Lee-Hyun.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S443	79	Chen-Chi-She.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S444	75	Solomon-Jeffrey-C.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S445	36	Milton-Scott-H.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S446	39	Bhakta-Jayesh.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16

S447	37	((transfer\$3 or cop\$4 or transmi\$5 or retriev\$3 or read\$3) near5 ?DRAM near5 (flash or non\$1volatile)).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S448	0	S447 and (S442 or S443 or S444 or S445 or S446)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:16
S449	12,048	(transfer\$4 or cop\$4 or transmi\$5 or mov\$3) with data with (non\$1volatile or flash or SSD or NVM) with (volatile or ?RAM or ?DRAM memory).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:18
S450	226	NETLIST.as.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:18
S451	21	S449 and S450	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:18
S452	4	((((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:21
S453	72	((((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:22

S454	3	S452 and (S442 or S443 or S444 or S445 or S446)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:23
S455	59	S450 and S453	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:23
S456	3	S455 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:24
S457	15	S455 and (regulat\$3 near2 voltage)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:25
S458	0	S457 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:26
S459	48	S453 AND ((G11C7/1072 OR G11C14/0018 OR G06F12/0246 OR G06F13/4243 OR G06F13/1694 OR G06F3/0659 OR G06F13/4027 OR G06F12/0638 OR G06F13/28 OR G06F1/185 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0685 OR Y02D10/00).CPC. OR (711/110).CCLS. OR (G06F13/16 OR G06F3/06 OR G06F12/02 OR G06F13/42 OR G06F12/06 OR G06F13/40 OR G06F13/28 OR G06F1/18 OR G11C7/10 OR G11C14/00).IPCR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 10:47
S460	2	S459 and (@ad<"20070601")	US-PGPUB; USPAT;	OR	ON	2021/09/22 10:47

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S461	1	(respons\$3 or based or according or "when") with (voltage adj monitor) with (detect\$3 or sens\$3 or recogniz\$3) with (chang\$3 or transit\$3) with (input adj voltage) with (memory adj module) with (transition\$3 or chang\$3) with (first adj (state or status or condition) adj operation) with (second adj (state or status or condition) adj operation)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 12:02
S462	1	(respons\$3 or based or according or "when") same (voltage adj monitor) same (detect\$3 or sens\$3 or recogniz\$3) same (chang\$3 or transit\$3) same (input adj voltage) same (memory adj module) same (transition\$3 or chang\$3) same (first adj (state or status or condition) adj operation) same (second adj (state or status or condition) adj operation)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 12:04
S463	1	(respons\$3 or based or according or "when") same (voltage adj monitor) same (detect\$3 or sens\$3 or recogniz\$3) same (chang\$3 or transit\$3) same (input adj voltage) same (memory or storage) same (transition\$3 or chang\$3) same (first adj (state or status or condition) adj operation) same (second adj (state or status or condition) adj operation)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 12:06
S464	1	(respons\$3 or based or according or "when") same (voltage adj monitor) same (detect\$3 or sens\$3 or recogniz\$3) same (chang\$3 or transit\$3) same (input adj voltage) same (memory or storage) same ((transition\$3 or chang\$3) near5 ((state or status or condition) adj operation))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/09/22 12:08

9/22/2021 12:32:08 PM

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17328019
	Filing Date		2021-05-24
	First Named Inventor		Chi-She Chen
	Art Unit		2135
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number		0016.0010001

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17328019
Filing Date		2021-05-24
First Named Inventor	Chi-She Chen	
Art Unit		
Examiner Name		
Attorney Docket Number	0016.001000I	

1	Wong, A., The BIOS Optimization Guide, Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.
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EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	09/21/2021
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	
Examiner Name	
Attorney Docket Number	0016.0010001

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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19,912,151 results

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1. Mother board and computer system capable of flexibly using synchronous dynamic random access memory and double data rate dynamic random access memory

A mother board and a computer system capable of flexibly using the SDRAM and the DDRAM. The mother board has several memory module slots, a voltage comparator, a clock generator and a chip set. Each of the memory module slots comprises a reference voltage pin, and the reference voltage pins of the...

CURRENT ASSIGNEES: VIA TECH INC
US6813157 | US PATENTS | 02-NOV-2004

2. Mother board and computer system capable of flexibly using synchronous dynamic random access memory and double data rate dynamic random access memory

A mother board and a computer system capable of flexibly using the SDRAM and the DDRAM. The mother board has several memory module slots, a voltage comparator, a clock generator and a chip set. Each of the memory module slots comprises a reference voltage pin, and the reference voltage pins of the...

CURRENT ASSIGNEES: CHANG NAI-SHUNG
US20020196650 | US APPLICATIONS | 26-DEC-2002

3. Mother board and computer system capable of flexibly using synchronous dynamic random access memory and double data rate dynamic random access memory

A mother board and a computer system capable of flexibly using the SDRAM and the DDRAM. The mother board has several memory module slots, a voltage comparator, a clock generator and a chip set. Each of the memory module slots comprises a reference voltage pin, and the reference voltage pins of the...

CURRENT ASSIGNEES: VIA TECH INC
US6424555 | US PATENTS | 23-JUL-2002

4. Mainboard and computer system using elastic memory and double rate memory

The present invention relates to mainboard and computer system with elastic synchronous DRAM and double data rate DRAM. By means of setting voltage regulator and several terminal resistors in terminal circuit module, the present invention has decreased printed circuit board area and production cost...

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CN1365036A | CHINA APPLICATIONS | 21-AUG-2002

5. Terminating circuit module used in a computer system

Relevance

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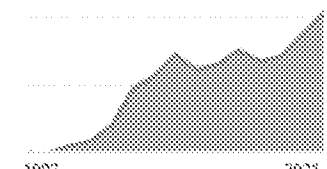
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
Art Unit	2135
Examiner Name	Hashem Farrokh
Attorney Docket Number	0016.001000I

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28	Decision Denying Institution of Inter Partes Review, Patent 8,301,833, Case IPR2017-00649-7 (PTAB), Paper 7, entered July 24, 2017, 17 pages.
29	Decision Denying Institution of Inter Partes Review, Patent No. 8,516,187, IPR2014-01371 (PTAB), Paper 12, entered March 13, 2015, 22 pages.
30	Decision Denying Institution of Inter Partes Review, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated December 16, 2014, 16 pages.
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**INFORMATION DISCLOSURE
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Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen
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32	Decision Denying Institution of Inter Partes Review, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, IPR2014-01370 (PTAB), Paper 13, entered March 13, 2015, 19 pages.
33	Decision Instituting Inter Partes Review, Patent 8,671,243, Case No. IPR2017-00587-7(PTAB), Paper No. 7, entered June 22, 2017, 40 pages.
34	Decision Instituting Inter Partes Review, Patent 8,874,831, Case No. IPR2017-00692-7(PTAB), Paper No. 7, entered July 21, 2017, 29 pages.
35	Declaration of R. Jacob Baker, Patent No. 8,874,831, Case IPR2017-00692-2016 (PTAB), filed Nov. 10, 2017, 72 pages.
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43	Declaration of Steven J. Corr, Netlist, Inc. v. SMART Storage Systems, Inc., et al., Case 4:13-cv-05889-YGR (NDCA), Document 305-8, filed March 10, 2015, 3 pages.
44	Deposition of Baker, Exhibit 1030, p.78, Patent No. 8, 671,243, SK hynix Inc., et al. v. Netlist, Inc., Case No. IPR-2017-00587 (PTAB), filed December 18, 2017, 268 pages.
45	Deposition of Ron Maltiel, Patent No. 8874831, Case No. IPR2017-00692-2010 (PTAB), dated Sep. 27, 2017, 155 pages.
46	Netlist v Smart Storage System and Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889 YGR, dated Jun. 6, 2014, 109 pages.
47	Drawing by R. Jacob Bake regarding Address of Dram and Flash, Patent Number: 8,671,243, Case IPR2017-00587-1023 (PTAB), filed December 18, 2017, 1 page.
48	Elmhurst, D. et al., A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write, IEEE Journal of Solid-State Circuits, Vol. 38, No. 11, Nov. 2003, 5 pages.
49	Ex. 1004—IPR2017-00587 Ron Maltiel CV, 7 pages.
50	Exhibit 1: Claim Chart Comparing Netlist's US Patent No. 8,001,434 to Smart Storage Ulltradimm, Case4:13-cv-05889-YGR Document 193-1, filed Apr. 10, 2014, 21 pages.

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Memory module having voltage regulator module - Google.az

A memory module includes a circuit board having socket mating contacts at a socket interface and VRM contacts at a VRM interface. Memory devices are coupled ...

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US8151009B2 - Serial connection external interface from printed ...

A translator of an apparatus in an example through a serial connection external interface of a printed circuit board (PCB) communicates between a serial ...

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US20050093120A1 - Detachable on package voltage regulation ...

An IC die is flip-bounded to a substrate having a plurality of connections to couple to a socket or to be mounted directly to a circuit board. An integrated ...

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Printed circuit board - Wikipedia

A printed circuit board (PCB) is a laminated sandwich structure of conductive and insulating layers. PCBs have two complementary functions.

Missing: plurality | Must include: plurality

<https://www.nxp.com/docs/application-note/AN2826.pdf>

AN2826: DDR-SDRAM Layout Considerations for MCF547x ...

This application note discusses the critical parts of a DDR SDRAM memory sub-system design, particularly those related to printed circuit board (PCB) layout. 1.

20 pages

<https://www.gerberlabs.com/what-is-a-printed-circuit-board/>

What is a Printed Circuit Board (PCB) | Gerber Labs

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Missing: SDRAM regulators

<https://www.electromaker.io/blog/article/what-is-a-pcb/>

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Missing: SDRAM | Must include: SDRAM

<https://www.xilinx.com/support/documentation/ug583.pdf>

UltraScale Architecture PCB Design User Guide (UG583) - Xilinx

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	Filing Date		2021-05-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2135	
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number	0016.0010001	

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First Named Inventor	Chi-She Chen	
Art Unit		
Examiner Name		
Attorney Docket Number		0016.0010001

1	Exhibit, Application As Filed, Sandisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00004 (PTAB), 52 pages.
2	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-3, filed March 17, 2015, 23 pages.
3	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-4, filed March 17, 2015, 20 pages.
4	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-5, filed March 17, 2015, 28 pages.
5	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-9, filed March 17, 2015, 17 pages.
6	Exhibit 12, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-7, filed March 17, 2015, 17 pages.
7	Exhibit 9, Email from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-17, filed March 10, 2015, 6 pages.
8	Exhibit, Institution of Inter Partes Review, Sandisk Corporation v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 316-9, filed March 24, 2015, 29 pages.
9	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
10	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
11	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-2, filed March 17, 2015, 23 pages.

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12	Exhibit, Letter from Defendant's Counsel, <i>Netlist v. Smart Storage Systems, Inc. et. al.</i> , Case 4:13-cv-05889-YGR Document 305-16, filed March 10, 2015, 3 pages.
13	Exhibit, Letter sent via email on December 6, 2013, Case 4:13-cv-03901-YGR Document 53-2, filed January 6, 2014
14	Exhibit, Order Denying Defendant's Motion to Stay Pending Inter Partes Review (Doc.59), <i>The Procter and Gamble Company v. Team Technologies, Inc, et al.</i> , Case 4:13-cv-05889-YGR Document 316-6, filed March 24, 2015, 10 pages.
15	Exhibit, Patent Public Advisory Committee Quarterly Meeting, Appeals Statistics USPTO, Case No. 4:13-cv-05889-YGR Document 309-8, filed March 17, 2015, 23 pages.
16	Exhibit, Reporter's Transcript of Proceedings, <i>Netlist, Inc v. Smart Modular Technologies, Inc., et al.</i> , Case 4:13-cv-05889-YGR Document 316-3, filed March 24, 2015, 15 pages.
17	Exhibit, Transcript of Official Electronic Sound Recording Proceeding, <i>Netlist v. Smart Modular Technologies, Inc, et al.</i> , Case 4:13-cv-05889-YGR Document 305-7, filed March 10, 2015, 10 pages.
18	Extended European Search Report for European Application No. 12 817 751.6, dated Jun. 9, 2015.
19	File History for U.S. Appl. No. 12/240,046, filed Sep. 20, 2008, 320 pages.
20	File History for U.S. Appl. No. 13/005,048, filed May 20, 2013, 181 pages.
21	File History for U.S. Appl. No. 60/941,586, filed Jun. 1, 2007, 23 pages.
22	File History U.S. Pat. No. 8,674,243.

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23	Final Office Action, dated Jun. 15, 2016, dated in U.S. Appl. No. 14/489,281, 16 pages.
24	Final Office Action, dated Jun. 15, 2016, issued in U.S. Appl. No. 14/489,281, 16 pages.
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26	Final Written Decision, Patent 8,674,243, IPR2017-00587-34, Paper No. 34, entered June 20, 2018, 53 pages.
27	Final Written Decision, Patent 8,874,831, IPR2017-00692, Paper No. 25, entered July 5, 2018, 42 pages.
28	Final Written Decision, US Patent No. 7,884,150, Case No. Case IPR2014-06882, (PTAB), Paper 33, filed December 14, 2015, 51 pages.
29	First Amended Complaint for Patent Infringement, Netlist Inc. v Smart Modular Tech, Inc., et al., Case 8:13-cv-00996-DOC-JPR, Document 5, filed Aug. 23, 2013, 99 pages.
30	Hasan, J. et al. Efficient Use of Memory Bandwidth to Improve Network Processor Throughput, Proceedings of the 30th Annual International Symposium on Computer Architecture (ISCA'03), IEEE, 2003, 12 pages.
31	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.
32	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.
33	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18), 2003, 100 pages.

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34	Inter Partes Review No. IPR2017-00692 (PTAB), Patent No. 8,874,831, filed July 26, 2012, 78 pages.
35	Inter Partes Review of U.S. Patent No. 8,874,831, Case IPR2017-00692 (PTAB), filed July 26, 2012, Paper No. 1, 78 pages.
36	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014.
37	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014, pp. 1-8.
38	International Search Report and Written Opinion in PCT/US12/48750, dated Oct. 10, 2012, pp. 1-10.
39	ISSCC 2006 / SESSION 7 / NON-VOLATILE MEMORY / 7.7, IEEE International Solid State Circuits Conference, 2006, 10 pages.
40	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322, 58 pages.
41	Jandhvala, S. et al. "Design-For-Test Analysis of a Buffered SDRAM DIMM". Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, Aug. 13014, 1996, 15 pages.
42	JEDEC Definition of DIMM, Exhibit 1029, IPR No. 2017-00587, December 18, 2017, 2 pages.
43	JEDEC Global Standard for the Microelectronics Industry, Why JEDEC Standards Matter, 2014, 1 page.
44	JEDEC Standard 24-G, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.

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45	JEDEC Standard No. 21-C (Release 17), Annex J: Serial Presence Detects for DDR2 SDRAM (Revision 1.3), 60 pages.
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47	JEDEC Standard, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A) Jan. 2006, 143 pages.
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50	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC Solid State Tech. Assoc., Mar. 2007, 129 pages.

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Attorney Docket Number	0016.0010001

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17328019
	Filing Date		2021-05-24
	First Named Inventor	Chi-She Chen	
	Art Unit	2135	
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number	0016.0010001	

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Application Number		17328019
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Attorney Docket Number		0016.0010001

1	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Patent Number: 8,516,187, IPR2014-00982 (PTAB), dated September 26, 2014, 57 pages.
2	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,301,833, Case No. IPR2017-00649 (PTAB), filed May 1, 2017, 67 pages.
3	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed May 1, 2017, 48 pages.
4	Patent Owner's Preliminary Response, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, Case IPR2014-01370 (PTAB), filed Dec. 16, 2014, 66 pages.
5	Patent Owner's Response, Patent No. 8,671,243, Case No. IPR2017-00587-12 (PTAB), filed Oct. 13, 2017, 80 pages.
6	Patent Owner's Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692-12 (PTAB), filed Nov. 10, 2017, 77 pages.
7	Patterson et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998), 71 pages.
8	Petition for Inter Partes Review of Claims 1-30 of U.S. PATENT No. 8,301,833, IPR2014-01370 (PTAB), filed August 22, 2014, 68 pages.
9	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
10	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
11	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833, filed Sep. 29, 2008.

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	12	Petition for Inter Partes Review of U.S. Pat. No. 8,510,187 (on behalf of SanDisk, Corp.), filed Jun. 19, 2014.
	13	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
	14	Petition for Inter Partes Review of U.S. Pat. No. 8,671,243, filed May 29, 2013.
	15	Petition for Inter Partes Review of U.S. Pat. No. 8,874,834, filed Jul. 26, 2012.
	16	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
	17	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
	18	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed Jun. 10, 2014.
	19	Petition for Inter Partes Review of U.S. Patent 8,516,187, IPR2014-00982 (PTAB), dated June 10, 2014, 67 pages.
	20	Petition for Inter Partes Review of US Patent No. 8,671,243, Case No. IPR2017-00587-1 (PTAB), May 29, 2013, 82 pages.
	21	Petition for Inter Partes Review, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), filed June 20, 2014, 69 pages.
	22	Petition for Inter Parties Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.

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23	Petition for Inter Parties Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed Jun. 19, 2014.
24	Petitioners Demonstratives, SK hynix Inc., et al., v. Netlist, Inc., U.S. Patent No. 8,671,243, IPR2017-00587-1037, 89 pages.
25	Petitioners' Reply in Support of Its Motion to Exclude, U.S. Patent No. 8,874,831, Case No. IPR2017-00692, filed April 9, 2018, 8 pages.
26	Petitioners' Reply to Patent Owner's Response, Inter Parties Review No. IPR2017-00587-16, Patent No. 8,671,243, filed Jan. 12, 2018, 38 pages.
27	Petitioners' Reply, U.S. Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), Paper No. 15, filed March 2, 2018, 35 pages.
28	Petitioners' Request for Hearing, U.S. Patent No. 8,301,833, Case No. Case No. IPR2017-00649-2020 (PTAB), Paper No. 8, entered Aug. 23, 2017, 18 pages.
29	Prosecution History, Application No. 12,240,916, U.S. Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date April 13, 2009, 320 pages.
30	Prosecution History, Application No. 13,559,470, US Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date Sep. 5, 2014, 312 pages.
31	Prosecution History, Provisional Application No. 60/941,586, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1005 (PTAB), received date Oct. 15, 2012, 23 pages.
32	Provisional Application for Advance Dynamic Disk Memory Module, Specification, SK hynix Inc., V. Netlist, Inc., Case No. IPR2017-00649-1006 (PTAB), dated Dec. 8, 2005, 53 pages.
33	Provisional Application No. 60/912,321 filed Apr. 17, 2007.

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34	Provisional Application No. 60/941,586 filed Jun. 1, 2007.
35	Requirement for Restriction Election, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994, 1003 (PTAB), dated March 31, 2019, 7 pages.
36	Restriction Requirement in U.S. Appl. No. 12/240,916, dated Mar. 31, 2011.
37	Second Amended Answer and Counterclaims to Plaintiffs Complaint for Declaratory Judgment, Diablo Technologies, Inc. v. Netlist, Inc., Case No. 4:13-CV-03901 (NDCA), YGR, filed February 17, 2014, 21 pages.
38	Smart Storage Systems, Inc's Invalidity Contentions, Netlist, Inc. v Smart Storage Systems, et al., Case No. 4:13-cv-05889-YGR, dated Jun. 6, 2014, 538 pages.
39	Supplemental Declaration of Daniel E. Alberti in Response to Court Order [Docket No. 361] in Support of Motion to Withdraw as Counsel of Record, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 362, filed September 5, 2018, 3 pages.
40	Supplemental Declaration of Ronald H. Spuhler, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-1, filed March 10, 2015, 2 pages.
41	Supplemental Declaration of Steven J. Corr, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 309, filed March 17, 2015, 3 pages.
42	Switches - D&T Online, Switches, http://wiki.dtonline.org , Exhibit 1022, Case No. IPR2017-00587, accessed October 13, 2017, 3 pages.
43	U.S. Appl. No. 60/942,321, filed Apr. 17, 2007.
44	U.S. Office Action in U.S. Appl. No. 13/536,173, mailed on Apr. 15, 2013, pp. 1-10.

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Attorney Docket Number		0016.0010001

45	Updated Joint Case Management Statement, Case No. 4:13-cv-03916-YGP (NDCA), Document 53, filed Jan. 6, 2014.
46	US Patent Application No. 12/240,916, Case No. IPR2017-00692-2018 (PTAB), dated Sep. 29, 2008, 52 pages.
47	US Patent Application No. 60/912,321, Case No. No. IPR2017-00692-1007 (PTAB), filed April 17, 2017, 42 pages.
48	Using Two Chip Selects to Enable Quad Rank, an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: Dec. 17, 2005, 2 pages.
49	Videotaped Deposition of Russel Jacob Baker, Patent No. 8,671,243, Case Nos. IPR2017-00587-1030 (PTAB), filed December 18, 2017, 268 pages.
50	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 257, 259; 1114,1115, Exhibit 1017, IPR2014-01370 (PTAB), Smart Modular Tech v Netlist Inc, Patent No. 8,301,833, Sep. 22, 2014, 5 pages.

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EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	09/21/2021
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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-06-22
Name/Print	Khaled Shami	Registration Number	38745

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 6552

APPLICATION NO: 17/328,019

ART UNIT: 2135

FILING DATE: 05/24/2021

EXAMINER: HASHEM FARROKH

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

AMENDMENT AND RESPONSE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

This paper is submitted in response to the Non-Final Office Action dated September 27, 2021, making a response due on or before December 27, 2021. Accordingly, this paper is timely filed.

Amendments to the Claims begin on page 2 of this paper.

Remarks begin on page 9 of this paper.

A Conclusion appears at page 13 of this paper.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Canceled)
2. (Currently Amended) A memory module comprising:
a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages;
a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and a first circuit that is coupled to the plurality of ~~synchronous dynamic random access memory (SDRAM)~~ devices and to a first set of edge connections of the plurality of edge connections, wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.
3. (Previously presented) The memory module of claim 2, wherein the first regulated voltage has a first voltage amplitude, and the second regulated voltage has a second voltage amplitude; and wherein a first one of the first and second voltage amplitudes is less than a second one of the first and second voltage amplitudes.
4. (Previously presented) The memory module of claim 2, wherein a third regulated voltage of the at least three regulated voltages has a voltage amplitude of 1.8 volts.

5. (Currently Amended) The memory module of claim 2, further comprising:
a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude below a predetermined threshold voltage, wherein the memory module transitions from a first operable state to a second operable state in response to the trigger signal ~~the predetermined threshold voltage is ten percent below a specified operating voltage.~~

6. (Currently Amended) The memory module of claim 2, further comprising:
a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude above a predetermined threshold voltage, wherein the memory module transitions from a first operable state to a second operable state in response to the trigger signal ~~the predetermined threshold voltage is ten percent above a specified operating voltage.~~

7. (Previously presented) The memory module of claim 2, further comprising:
a controller coupled to the PCB, the controller including a voltage monitor circuit configured to monitor an input voltage received from a second set of edge connections of the plurality of edge connections, wherein, in response to the voltage monitor circuit detecting a power threshold condition, the voltage monitor circuit transmits a signal to one or more portions of the controller.

8. (Currently Amended) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting an amplitude of the input voltage being above a first predetermined threshold voltage, and wherein the first predetermined threshold voltage is above a specified operating voltage.

9. (Currently Amended) The memory module of claim 8, wherein the first predetermined threshold voltage is ten percent above the specified operating voltage.

10. (Currently Amended) The memory module of claim ~~[[7]]~~8, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting an amplitude of the input voltage being below a second predetermined threshold voltage, ~~and wherein the second predetermined threshold voltage is below a the specified operating voltage, and wherein the memory module transitions from a first operable state to a second operable state in response to the signal.~~

11. (Currently Amended) The memory module of claim 10, wherein the second predetermined threshold voltage is ten percent below the specified operating voltage.

12. (Previously presented) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a power reduction condition or a low voltage condition of the input voltage.

13. (Previously presented) The memory module of claim 7, wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a request by the host system.

14. (Canceled)

15. (Currently Amended) The memory module of claim ~~[[14]]~~2, wherein two of the at least three buck converters are configured to operate as a dual-buck converter.

16. (Currently Amended) A memory module comprising:
a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system;
a voltage conversion circuit coupled to the PCB and configured to provide a plurality of regulated voltages, wherein the voltage conversion circuit includes three buck converters each of which is configured to produce a regulated voltage of the plurality of regulated voltages;
a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the plurality of regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory

(SDRAM) devices, the plurality of SDRAM devices coupled to a first regulated voltage of the plurality of regulated voltages; and

a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface, the voltage monitor circuit configured to detect an amplitude change in the input voltage, wherein, in response to the voltage monitor detecting an amplitude change in the input voltage, the memory module transitions from a first operable state to a second operable state.

17. (Currently Amended) The memory module of claim 16, wherein the voltage monitor circuit is configured to detect the input voltage being above a first predetermined threshold voltage and to detect the input voltage being below a second predetermined threshold voltage, ~~and wherein the first predetermined threshold voltage is ten percent above a specified operating voltage, and the second predetermined threshold voltage is ten percent below the specified operating voltage.~~

18. (Previously presented) The memory module of claim 16, wherein, in the first operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

19. (Previously presented) The memory module of claim 18, wherein, in the first operable state, the first pre-regulated voltage is supplied to the voltage conversion circuit via a circuit, and wherein, in the second operable state, the second pre-regulated voltage is supplied to the voltage conversion circuit via the circuit.

20. (Previously presented) The memory module of claim 19, wherein the circuit includes a first diode having a first input and a first output, the first input is coupled to the first pre-regulated voltage and the first output is coupled to the voltage conversion circuit, and wherein the circuit includes a second diode having a second input and a second output, the second input is coupled to the second pre-regulated voltage and the second output is coupled to the first output and to the voltage conversion circuit.

21. (Previously presented) The memory module of claim 16, further comprising:

a first circuit having a first input, a second input and an output, the first input coupled to a first pre-regulated voltage, the second input coupled to a second pre-regulated voltage, the output coupled to the voltage conversion circuit, wherein the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a supply voltage, wherein, in the first operable state, the first circuit provides the supply voltage to the voltage conversion circuit via the output using the first pre-regulated voltage, and wherein, in the second state, the first circuit provides the supply voltage to the voltage conversion circuit via the output using the second pre-regulated voltage.

22. (Previously presented) The memory module of claim 21, wherein the first circuit includes a first diode coupled between the first input and the output and a second diode coupled between the second input and the output.

23. (Currently Amended) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system;

a voltage conversion circuit configured to provide a plurality of regulated voltages, wherein the voltage conversion circuit includes three buck converters each of which is configured to produce a regulated voltage of the plurality of regulated voltages;

a plurality of components each coupled to at least one regulated voltage of the plurality of regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, wherein the plurality of SDRAM devices are coupled to a first regulated voltage of the plurality of regulated voltages; and

a controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface of the PCB, the voltage monitor circuit configured to monitor the input voltage, wherein the controller is configured to perform one or more operations in response to the voltage monitor circuit detecting an amplitude change in the input voltage, and wherein the one or more operations include a write operation to transfer data into non-volatile memory.

24. (Currently Amended) The memory module of claim 23, wherein, in response to

the voltage monitor circuit detecting an amplitude change in the input voltage, the memory module transitions from a first operable state of operation of operation to a second operable state of operation of operation.

25. (Currently Amended) The memory module of claim 24, wherein, in the first operable state of operation of operation, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state of operation of operation, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

26. (Currently Amended) The memory module of claim 23, wherein the voltage monitor circuit is configured to detect the input voltage being above a first predetermined threshold voltage or below a second predetermined threshold voltage.

27. (Currently Amended) The memory module of claim 26, wherein the first predetermined threshold voltage is ~~ten percent ten percent~~ above a specified operating voltage, and the second predetermined threshold voltage is below the specified operating voltage.

28 - 29. (Canceled)

30. (Previously presented) The memory module of claim 25, wherein the first pre-regulated voltage is coupled to the voltage conversion circuit via a first diode.

31. (Previously presented) The memory module of claim 30, wherein the second pre-regulated voltage is coupled to the voltage conversion circuit via a second diode.

32. (New) The memory module of claim 5, further comprising:
a controller coupled to the voltage monitor circuit; wherein, in response to the trigger signal, the controller is configured to perform one or more operations including a write operation to transfer data to non-volatile memory.

33. (New) The memory module of claim 6, further comprising:
a controller coupled to the voltage monitor circuit; wherein, in response to the trigger signal, the controller is configured to perform one or more operations including a write operation to transfer data to non-volatile memory.
34. (New) The memory module of claim 16, wherein the voltage monitor circuit is configured to produce a trigger signal in response to detecting an amplitude change in the input voltage; and wherein, in response to the trigger signal, the controller is configured to perform one or more operations including a write operation to transfer data to non-volatile memory.

REMARKS

The Non-Final Office Action issued September 27, 2021, has been carefully considered.

Reconsideration in view of the following remarks is respectfully requested.

Claims 2, 5-6, 8-11, 15-17, 23-27 have been amended, and claims 32-34 have been added.

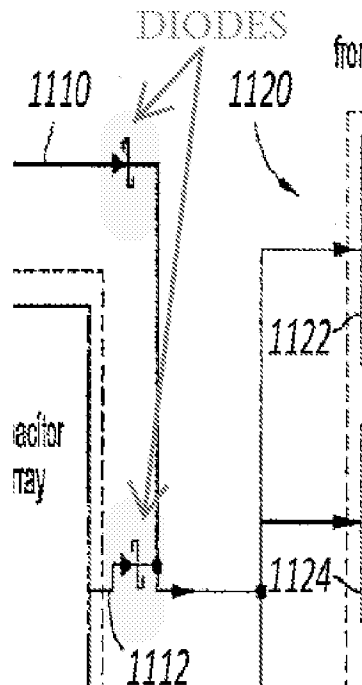
Claim Interpretation

Applicant acknowledges that the Examiner is not interpreting the claims under 35 USC 112(f), or under pre-AIA paragraph six.

Rejection under 35 USC 112

Claims 20, 22, and 30-31 stand rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-A/A), first paragraph, as allegedly failing to comply with the written description requirement. According to the Office Action, the “claimed specification does not appear to describe or support” the claimed first and second diodes.

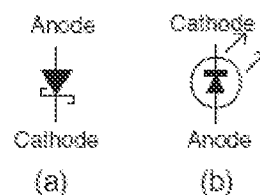
Applicant respectfully disagrees, and directs the Examiner’s attention to FIG. 16, and specifically to the symbols therein highlighted below, which represent diodes as is known in the electrical arts.



Referring for example to Chapter 5 of the Practical Electronics Handbook, Sixth Edition, 2007, it illustrates, in FIG. 5.4, a Schottky diode as follows:

Figure 5.4

Symbols: (a) Schottky diode, (b) LED.



It will be appreciated that the symbols highlighted in FIG. 16 are identical to those in the Practical Electronics Handbook that are used to represent a Schottky diode, and the drawings therefore illustrate the claimed diodes. Namely, in FIG. 16, and consistently with the claims, one diode is depicted having an input coupled to a pre-regulated fourth voltage 1110 from first power conversion element 1130 and an output coupled to voltage conversion element 1120; and another diode is depicted having an input coupled to a pre-regulated fifth voltage 1112 from second power element 1140 and an output coupled to voltage conversion element 1120.¹ The MPEP states that the fundamental inquiry behind compliance with the written description requirement (under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-A/A), first paragraph) “is whether the specification conveys with reasonable clarity to those skilled in the art that, as of the filing date sought, applicant was in possession of the invention as now claimed. See, e.g., *Vas-Cath, Inc. v. Mahurkar*, 935 F.2d 1555, 1563-64, 19 USPQ2d 1111, 1117 (Fed. Cir. 1991).”² The MPEP then explains that “An applicant shows possession of the claimed invention by describing the claimed invention with all of its limitations using such descriptive means as words, structures, **figures**, **diagrams**, and formulas that fully set forth the claimed invention. (Relying on *Lockwood v. Am. Airlines, Inc.*, 107 F.3d 1565, 1572, 41 USPQ2d 1961, 1966 (Fed. Cir. 1997), emphasis added).³

For these reasons at least, Applicant respectfully maintains that claims 20, 22, and 30-31 are in compliance with 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-A/A), first paragraph, and the rejection should be withdrawn.

Double Patenting

Timely submitted herewith is a Terminal Disclaimer over U.S. Patent No. 11,016,918 overcoming the alleged non-statutory double patenting rejection of claims 2-15, 23, and 26 -29.

Newly-Added Claims

Claims 32-34 have been added to further particularly point out and distinctly claim the subject matter of the claimed arrangement.

¹ See parenthetical clause at end of paragraph [00146] expressly characterizing the voltages 1110 and 1112 as “pre-regulated.”

² MPEP 2163.02 Standard for Determining Compliance With the Written Description Requirement

³ *Id.*

Comments on the Statement of Reasons for the Indication of Allowable Subject Matter

Applicant gratefully acknowledges the indication that claims 16-19 are 21 are allowed and that claim 24-25 contain allowable subject matter. Of these allowed and allowable claims, claims 16-17 and 24-25 have been amended, and Applicant respectfully maintains that they continue to be allowable at least for the reasons cited by the Examiner. Further, pursuant to 37 C.F.R. § 1.104, Applicant respectfully submits the instant statement commenting on the Examiner's reasons for allowance.

Applicant maintains that additional and/or alternative reasons for allowance may exist apart from those expressed by the Examiner and the Applicant, and these reasons may be independently sufficient to establish the patentability of each of the allowed dependent or independent claims.

Applicant respectfully reserves the right to introduce, articulate, or otherwise comment on any such additional reasons for allowance as may be appropriate in any future proceedings concerning the one or more claimed embodiments.

To the extent that the Examiner's reasons for allowance are inconsistent with applicable case law, statutes, and regulations, Applicant respectfully disagrees with them. Furthermore, Applicant does not admit to any characterization or limitation of the claims or to any characterization of a reference by the Examiner, particularly any that are inconsistent with the language of the claims considered in their entirety and including all of their constituent limitations.

Conclusion

In view of the foregoing, Applicant submits that the application is in condition for immediate allowance. Early notice to that effect is earnestly solicited. The Examiner is invited to contact the undersigned attorney if it is believed that such contact will expedite the prosecution of the application.

In the event this paper is not timely filed, Applicant petitions for an appropriate extension of time. Please charge any fee deficiency or credit any overpayment to Deposit Account No. 22-0585.

Respectfully submitted,

VORYS, SATER, SEYMOUR and PEASE LLP

/Khaled Shami/
Khaled Shami, Registration No. 38745
Customer No.: 151145

Date: November 19, 2021

VORYS, SATER, SEYMOUR and PEASE LLP

1909 K Street, NW, 9th Floor
Washington, DC 20006-1152
Tel: (202) 467-8800

Electronic Acknowledgement Receipt

EFS ID:	44332358
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000047
Receipt Date:	19-NOV-2021
Filing Date:	24-MAY-2021
Time Stamp:	17:26:21
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		022955-000047-RNFOA.pdf	593855	yes	13
			23ffb5a30a1b55b47a496627ee3c54c66c563c2f		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Claims	2	8
	Applicant Arguments/Remarks Made in an Amendment	9	13

Warnings:

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Doc Code: DIST.E.FILE Document Description: Electronic Terminal Disclaimer - Filed		PTO/SB/26 U.S. Patent and Trademark Office Department of Commerce	
Electronic Petition Request	TERMINAL DISCLAIMER TO OBIATE A DOUBLE PATENTING REJECTION OVER A "PRIOR" PATENT		
Application Number	17328019		
Filing Date	24-May-2021		
First Named Inventor	Chi-She Chen		
Attorney Docket Number	022955-000047		
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
☒ Filing of terminal disclaimer does not obviate requirement for response under 37 CFR 1.111 to outstanding Office Action ☒ This electronic Terminal Disclaimer is not being used for a Joint Research Agreement.			
Owner		Percent Interest	
Netlist, Inc.		100%	
The owner(s) with percent interest listed above in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of prior patent number(s) 11016918 as the term of said prior patent is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the prior patent are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns. In making the above disclaimer, the owner does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the prior patent, "as the term of said prior patent is presently shortened by any terminal disclaimer," in the event that said prior patent later: - expires for failure to pay a maintenance fee; - is held unenforceable; - is found invalid by a court of competent jurisdiction; - is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321; - has all claims canceled by a reexamination certificate; - is reissued; or - is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer. ☒ Terminal disclaimer fee under 37 CFR 1.20(d) is included with Electronic Terminal Disclaimer request.			

☐ I certify, in accordance with 37 CFR 1.4(d)(4), that the terminal disclaimer fee under 37 CFR 1.20(d) required for this terminal disclaimer has already been paid in the above-identified application.	
Applicant claims the following fee status: ☐ Small Entity ☐ Micro Entity ☒ Regular Undiscounted	
I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.	
THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES I certify, in accordance with 37 CFR 1.4(d)(4) that I am: ☒ An attorney or agent registered to practice before the Patent and Trademark Office who is of record in this application Registration Number <u>38745</u> ☐ A sole inventor ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application ☐ A joint inventor; all of whom are signing this request	
Signature	/Khaled Shami/
Name	Khaled Shami

*Statement under 37 CFR 3.73(b) is required if terminal disclaimer is signed by the assignee (owner).
 Form PTO/SB/96 may be used for making this certification. See MPEP § 324.

Electronic Patent Application Fee Transmittal

Application Number:	17328019			
Filing Date:	24-May-2021			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Yayoi Bourne			
Attorney Docket Number:	022955-000047			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
STATUTORY OR TERMINAL DISCLAIMER	1814	1	170	170
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				170

Doc Code: DISQ.E.FILE

Document Description: Electronic Terminal Disclaimer – Approved

Application No.: 17328019

Filing Date: 24-May-2021

Applicant/Patent under Reexamination: Chen

Electronic Terminal Disclaimer filed on November 19, 2021

☒ APPROVED

This patent is subject to a terminal disclaimer

☐ DISAPPROVED

Approved/Disapproved by: Electronic Terminal Disclaimer automatically approved by EFS-Web

U.S. Patent and Trademark Office

Electronic Acknowledgement Receipt

EFS ID:	44332141
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000047
Receipt Date:	19-NOV-2021
Filing Date:	24-MAY-2021
Time Stamp:	17:23:37
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 170
RAM confirmation Number	E2021AIH23341883
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Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	37630	no	2
			7a27c9b856f3944653656aab4ad49be4c9cd55ac		
Warnings:					
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Total Files Size (in bytes):			78792		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/328,019		Filing Date 05/24/2021		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	11/19/2021	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 30	Minus	** 30	= 0			x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0			x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/VENESSA JONES/			
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This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17328019
	Filing Date	2021-05-24
	First Named Inventor	Chi-She Chen, et al.
	Art Unit	2135
	Examiner Name	HASHEM FARROKH
	Attorney Docket Number	022955-000047

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen, et al.
Art Unit	2135
Examiner Name	HASHEM FARROKH
Attorney Docket Number	022955-000047

1	an R. Sinclair and John Dunton, Practical Electronics Handbook, Sixth Edition 2007, ELSEVIER, <http://al-marefah.blogspot.com>, retrieved from the Internet November 18, 2021
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Examiner Signature		Date Considered	
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen, et al.
Art Unit	2135
Examiner Name	HASHEM FARROKH
Attorney Docket Number	022955-000047

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-11-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	17328019			
Filing Date:	24-May-2021			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Yayoi Bourne			
Attorney Docket Number:	022955-000047			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260

Electronic Acknowledgement Receipt

EFS ID:	44342728
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000047
Receipt Date:	22-NOV-2021
Filing Date:	24-MAY-2021
Time Stamp:	15:41:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 260
RAM confirmation Number	E2021ALF42182208
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	022955-000047-IDS6-SB08.pdf	1041828	no	4
			fo71bac4e9b043ff842a49f647066ff677bb10ccf		
Warnings:					
Information:					
A U.S. Patent Number Citation or a U.S. Publication Number Citation is required in the Information Disclosure Statement (IDS) form for autoloading of data into USPTO systems. You may remove the form to add the required data in order to correct the Informational Message if you are citing U.S. References. If you chose not to include U.S. References, the image of the form will be processed and be made available within the Image File Wrapper (IFW) system. However, no data will be extracted from this form. Any additional data such as Foreign Patent Documents or Non Patent Literature will be manually reviewed and keyed into USPTO systems.					
2	Non Patent Literature	Chapter5_practical_electronics_handbook_sixth_edition.pdf	807816	no	84
			50c00ca37f885e0f9a6bd5155f774d5543f73b6b		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	37865	no	2
			3370a6da9c7d4f0a0de929fabe365dd7c27c585a		
Warnings:					
Information:					
Total Files Size (in bytes):			1887509		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

151145 7590 12/10/2021
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

EXAMINER	
FARROKH, HASHEM	
ART UNIT	PAPER NUMBER
2135	

DATE MAILED: 12/10/2021

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	05/24/2021	Chi-She Chen	022955-000047	6552

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/10/2022

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 12/10/2021
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	05/24/2021	Chi-She Chen	022955-000047	6552

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/10/2022

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2135	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	05/24/2021	Chi-She Chen	022955-000047	6552
151145	7590	12/10/2021	EXAMINER	
Vorys, Sater, Seymour and Pease LLP			FARROKH, HASHEM	
1909 K Street, NW				
9th Floor			ART UNIT	
Washington, DC 20006-1152			PAPER NUMBER	
			2135	

DATE MAILED: 12/10/2021

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/328,019	Applicant(s) Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2135	AIA (FITF) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 11/199/2021.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-13,15-27 and 30-34. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>11/22/2021</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
---	--

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
---	--

Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

Terminal Disclaimer

1. *The terminal disclaimer filed on 11/19/2021 disclaiming the terminal portion of any patent granted on this application which would extend beyond the expiration date of Application 17/138,766 now US patent 11,016,918 has been reviewed and is accepted. The terminal disclaimer has been recorded.*

INFORMATION CONCERNING IDS:

2. *The information disclosure statement (IDS) submitted on 11/22/2021 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement has been considered by the examiner.*

Allowable Subject Matter

3. *Claims 2-13, 15-27, and 30-34 allowed.*

The following is an examiner's statement of reasons for allowance:

4. *The prior art of record does not teach or suggest the distinct features "a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three*

regulated voltages”, recited in the independent claims 2 and 23. The dependent claims 3-13, 15, and 32-33 are allowed by virtue of their dependency from the independent claim 2. The dependent claims 24-27 and 30-31 are allowed by virtue of their dependency from the independent claim 23.

5. The prior art of record does not teach or suggest the distinct features “a controller coupled to the PCB, the controller including a voltage monitor circuit coupled to an input voltage received from the host system via the interface, the voltage monitor circuit configured to detect an amplitude change in the input voltage, wherein, in response to the voltage monitor detecting an amplitude change in the input voltage, the memory module transitions from a first operable state to a second operable state”, recited in the independent claim 16. The dependent claims 17-22 and 34 are allowed by virtue of their dependency from the independent claim 16.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”

Conclusion

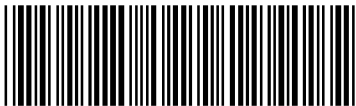
Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached on 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2135*

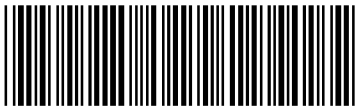
November 30, 2021

Issue Classification 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2135	

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135 (Primary Examiner) _____ (Date) _____		30 November 2021 O.G. Print Claim(s) 1	O.G. Print Figure 6

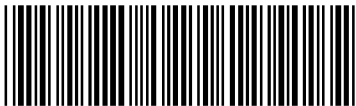
<i>Issue Classification</i> 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	13	/ 28
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	110

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	30 November 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	6

Issue Classification 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☒ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1	11	10	19	19		28								
1	2	12	11	20	20		29								
2	3	13	12	21	21	27	30								
3	4	14	13	22	22	28	31								
4	5		14	24	23	5	32								
6	6	15	15	25	24	7	33								
8	7	16	16	26	25	23	34								
9	8	17	17	29	26										
10	9	18	18	30	27										

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135 (Primary Examiner) _____ (Date) 30 November 2021		O.G. Print Claim(s) 1	O.G. Print Figure 6

<i>Search Notes</i> 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G11C7/1072; G11C14/0018; G06F12/0246; G06F13/4243; G06F13/1694; G06F3/0659; G06F13/4027; G06F12/0638; G06F13/28; G06F1/185; G06F2212/205; G06F2212/7208; G06F3/0685; Y02D10/00	09/22/2021	HF
G11C7/1072; G11C14/0018; G06F12/0246; G06F13/4243; G06F13/1694; G06F3/0659; G06F13/4027; G06F12/0638; G06F13/28; G06F2212/205; G06F2212/7208; G06F3/0613; G06F3/0685; Y02D10/00	11/30/2021	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	09/22/2021	HF
711	110	11/30/2021	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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<i>Search Notes</i> 	Application/Control No. 17/328,019	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

Search Notes		
Search Notes	Date	Examiner
Inventor/Assignee Search	09/22/2021	HF
EAST Text Search	09/22/2021	HF
IP Search	09/22/2021	HF
Google Search	09/22/2021	HF
Inventor/Assignee Search Updated	11/30/2021	HF
EAST Text Search Updated	11/30/2021	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G11C7/1072	G11C7/00	11/30/2021	HF
G06F13/1694	G06F13/00	11/30/2021	HF
G06F2212/205	G06F2212/00	11/30/2021	HF

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	72	((((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 14:02
L2	48	L1 AND ((G11C7/1072 OR G11C14/0018 OR G06F12/0246 OR G06F13/4243 OR G06F13/1694 OR G06F3/0659 OR G06F13/4027 OR G06F12/0638 OR G06F13/28 OR G06F1/185 OR G06F2212/205 OR G06F2212/7208 OR G06F3/0685 OR Y02D10/00).CPC. OR (711/110).CCLS. OR (G06F13/16 OR G06F3/06 OR G06F12/02 OR G06F13/42 OR G06F12/06 OR G06F13/40 OR G06F13/28 OR G06F1/18 OR G11C7/10 OR G11C14/00).IPCR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 14:02

11/30/2021 2:04:19 PM

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CONFIRMATION NO. 6552

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
17/328,019	05/24/2021	711	2135	022955-000047	
RULE					
APPLICANTS Netlist, Inc., Irvine, CA INVENTORS Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;					
** CONTINUING DATA ***** This application is a CON of 17/138,766 12/30/2020 PAT 11016918 which is a CON of 15/934,416 03/23/2018 ABN which is a CON of 14/840,865 08/31/2015 PAT 9928186 which is a CON of 14/489,269 09/17/2014 PAT 9158684 which is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 05/28/2021					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/HASHEM FARROKH/</u> Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 22	TOTAL CLAIMS 30	INDEPENDENT CLAIMS 3
ADDRESS Vorys, Sater, Seymour and Pease LLP 1909 K Street, NW 9th Floor Washington, DC 20006-1152 UNITED STATES					
TITLE FLASH-DRAM HYBRID MEMORY MODULE					
FILING FEE RECEIVED 1820	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____		

	☐ Credit
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S465	1	(respons\$3 or based or according or "when") with (voltage adj monitor) with (detect\$3 or sens\$3 or recogniz\$3) with (chang\$3 or transit\$3) with (input adj voltage) with (memory adj module) with (transition\$3 or chang\$3) with (first adj (state or status or condition) adj operation)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 09:36
S466	11	((printed adj circuit adj board) or PCB) with (regulat\$3 near2 voltages) with (buck near2 converters)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 10:20
S467	2	S466 and (@ad<"20070601")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 10:23
S468	0	((printed adj circuit adj board) or PCB) with (regulat\$3 near2 voltages) with (buck near2 converters)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/11/30 10:24
S469	3	((printed adj circuit adj board) or PCB) same (regulat\$3 near2 voltages) same (buck near2 converters)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/11/30 10:27
S470	1,361	Lee-Hyun.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S471	79	Chen-Chi-She.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2021/11/30 11:46

			DERWENT; IBM_TDB			
S472	75	Solomon-Jeffrey-C.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S473	36	Milton-Scott-H.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S474	41	Bhakta-Jayesh.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S475	4	((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S476	3	S475 and (S470 or S471 or S472 or S473 or S474)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:46
S477	12,186	(transfer\$4 or cop\$4 or transmi\$5 or mov\$3) with data with (non\$1volatile or flash or SSD or NVM) with (volatile or ?RAM or ?DRAM memory).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:48
S478	229	NETLIST.as.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:48
S479	22	S477 and S478	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2021/11/30 11:48

			DERWENT; IBM_TDB			
S480:	72	((((printed adj circuit adj board) or PCB) and (plurality near2 edge near2 connections) and SDRAM)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/11/30 11:52

11/30/2021 1:56:11 PM

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	17328019
	Filing Date	2021-05-24
	First Named Inventor	Chi-She Chen, et al.
	Art Unit	2135
	Examiner Name	HASHEM FARROKH
	Attorney Docket Number	022955-000047

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen, et al.
Art Unit	2135
Examiner Name	HASHEM FARROKH
Attorney Docket Number	022955-000047

1	an R. Sinclair and John Dunton, Practical Electronics Handbook, Sixth Edition 2007, ELSEVIER, <http://al-marefah.blogspot.com>, retrieved from the Internet November 18, 2021
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EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	11/30/2021
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17328019
Filing Date	2021-05-24
First Named Inventor	Chi-She Chen, et al.
Art Unit	2135
Examiner Name	HASHEM FARROKH
Attorney Docket Number	022955-000047

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-11-22
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

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Commissioner for Patents
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By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

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Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
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Washington, DC 20006-1152

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I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	05/24/2021	Chi-She Chen	022955-000047	6552

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/10/2022

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2135	711-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Vorys, Sater, Seymour and Pease LLP

2. _____

3. _____

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

NETLIST, INC.

IRVINE, CALIFORNIA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

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☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 220585

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

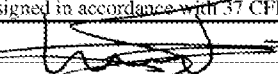
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature: 

Date: December 11, 2021

Typed or printed name: Khaled Shami

Registration No. 38745

Electronic Patent Application Fee Transmittal

Application Number:	17328019			
Filing Date:	24-May-2021			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Yayoi Bourne			
Attorney Docket Number:	022955-000047			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	44492396
Application Number:	17328019
International Application Number:	
Confirmation Number:	6552
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Yayoi Bourne
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	022955-000047
Receipt Date:	11-DEC-2021
Filing Date:	24-MAY-2021
Time Stamp:	09:58:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E2021BA058351021
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	IFTransmittal.pdf	1521478	no	1
			f6bb439a0cb580c525070081789fbc9c642b0f4c		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	37643	no	2
			a7aa19502086d62deb034f9db9337ce137e14146		

Warnings:

Information:

Total Files Size (in bytes):			1559121
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152



**Courtesy Reminder for
Application Serial No: 17/328,019**

Attorney Docket No: 022955-000047

Customer Number: 151145

Date of Electronic Notification: 12/10/2021

This is a courtesy reminder that new correspondence is available for this application. If you have not done so already, please review the correspondence. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:

kshami@vorys.com

mvmessinger@vorys.com

vorys_docketing@cardinal-ip.com

To view your correspondence online or update your email addresses, please visit us anytime at **<https://ppair-my.uspto.gov/pair/PrivatePair>**.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/328,019	01/25/2022	11232054	022955-000047	6552

151145 7590 01/05/2022
Vorys, Sater, Seymour and Pease LLP
1909 K Street, NW
9th Floor
Washington, DC 20006-1152

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

INVENTOR(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional inventors):

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Netlist, Inc., Irvine, CA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

The United States Patent and Trademark Office
PATENT TRIAL AND APPEAL BOARD



A petition has been filed in Patent Number 11/232,054, Application Number 17/328,019
on 5/17/22 (Date).

The Case Number is IPR2022-00999.
(IPR, CBM, PGR, DER #)

To view the documents filed in this petition, go to <https://ptab.uspto.gov>.

Use the Search PTAB tab and enter the Patent Number or the Trial or Case Number and select the Search button.

Questions regarding this notice should be directed to the Patent Trial and Appeal Board at 571-272-7822.

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Eastern District of Texas - Marshall Division on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO.	DATE FILED 6/10/2022	U.S. DISTRICT COURT for the Eastern District of Texas - Marshall Division
PLAINTIFF Netlist, Inc.		DEFENDANT Micron Technology, Inc., et al
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,860,506	12/8/2020	Netlist, Inc.
2 10,949,339	3/16/2021	Netlist, Inc.
3 11,016,918	5/25/2021	Netlist, Inc.
4 11,232,054	1/25/2022	Netlist, Inc.
5 8,787,060	7/22/2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED 6/10/2022	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☒ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 9,318,160	4/19/2016	Netlist, Inc.
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00999
Patent 11,232,054 B2

Before JON M. JURGOVAN, DANIEL J. GALLIGAN, and
NABEEL U. KHAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314

I. INTRODUCTION

Samsung Electronics Co., Ltd. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) to institute an *inter partes* review of claims 1–30 of U.S. Patent 11,232,054 B2, issued on January 25, 2022 (Ex. 1001, “the ’054 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition.

Institution of an *inter partes* review is authorized when “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a).

Based on the current record, and for the reasons explained below, we determine that Petitioner has established a reasonable likelihood that it would prevail with respect to at least one challenged claim, and we institute an *inter partes* review as to all the challenged claims and grounds raised in the Petition.

II. BACKGROUND

A. *Real Parties in Interest*

Petitioner identifies Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. as the real parties in interest. Pet. 1. Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

B. *Related Matters*

The parties advise that the ’054 patent is related to *Samsung Electronics Co., Ltd., et al. v. Netlist, Inc.*, IPR2022-00996; *Netlist, Inc. v. Samsung Electronics Co., Ltd., et al.*, Case No. 2-21-cv-00463 (E.D. Tex.); *Samsung Electronics Co., Ltd. v. Netlist, Inc.*, Case No. 1:21-cv-01453 (D. Del.); and U.S. Appl. No. 17/582,797. Pet. 1; Paper 4, 1.

The parties advise that the ’054 patent is related to the following legal proceeding, which is no longer pending: *SK hynix Inc. v. Netlist, Inc.*, IPR2017-00692. Pet. 1; Paper 4, 1.

C. *The ’054 Patent (Ex. 1001)*

The ’054 patent is titled “Flash-DRAM Hybrid Memory Module.” Ex. 1001, code (54). Figure 12 of the ’054 patent is reproduced below.

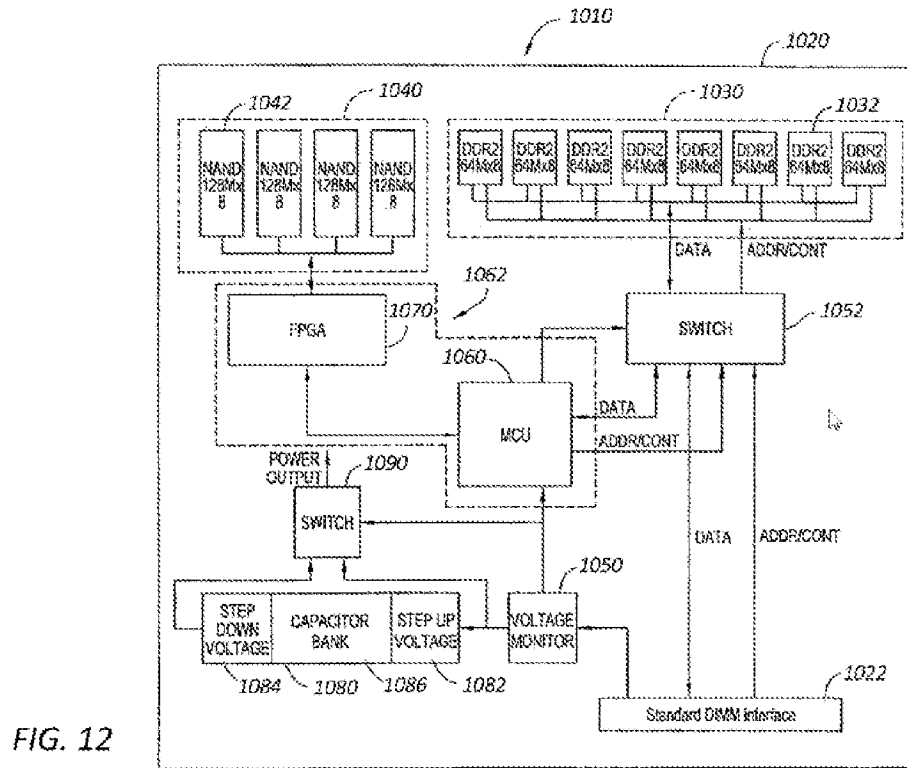


FIG. 12

Figure 12 shows an example memory system 1010 of the '054 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. *Id.* at 21:16–20. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022

for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. *Id.* at 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '054 patent refers to this as the “first state”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '054 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g. a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. *Id.* at 25:54–58. Second power supply 1080 may comprise step-up transformer

1082, step-down transformer 1084, and capacitor bank 1086 with one or more capacitors. *Id.* at 26:3–13.

The memory system 1010 further has a third state in which controller 1062 is decoupled from the volatile memory system 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). *Id.* at 25:62–66. The third power supply may provide power to volatile memory subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred. *Id.* at 25:66–26:3.

Figure 16 of the '054 patent is reproduced below.

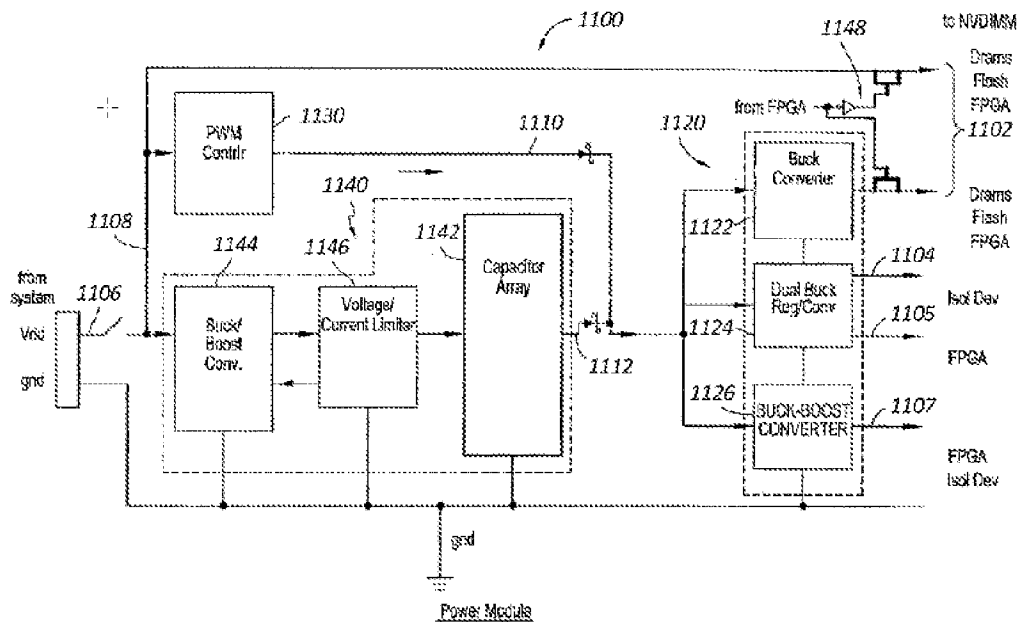


FIG. 16

Figure 16 shows power module 1100 of memory system 1010. *Id.* at 27:59–61. Power module 1100 comprises conversion element 1120, first power element 1130, and second power element 1140. *Id.* at 28:7–15, 28:20–22. Conversion element 1120 comprises buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 generating respective

voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 28:62–67. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. Power module 1100 provides power to components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. *Id.* at 27:61–65.

Specifically, in a first state, first voltage 1102 is provided to memory system 1010 from input 1106 and fourth voltage 1102 is provided to conversion element 1120 from the first power element 1130. *Id.* at 28:27–31. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. *Id.* at 28:31–34. In the third state, the fifth voltage is provided to conversion element 1120 from second power element 1140 and the first voltage 1104 is provided to memory system 1010 from conversion element 1120. *Id.* at 28:34–38. Transition from the first state to the second state may occur when power module 1100 detects a power failure is about to occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

D. Illustrative Claim

Of the challenged claims, claims 1, 16, and 24 are independent. Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages;

[1.d] [1.d.1] a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, [1.d.2] the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and [1.d.3] a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections, [1.d.4] wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and [1.d.5] wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.

Ex. 1001, 38:19–44.

E. Evidence

Petitioner relies on the following references (*see* Pet. 3, 9–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published August 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published Apr. 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued February 15, 2005

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 4):

Claims Challenged	35 U.S.C. §	References
1–3, 15	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

III. ANALYSIS OF CHALLENGED GROUNDS

We turn now to Petitioner’s asserted grounds of unpatentability and Patent Owner’s arguments in its Preliminary Response to determine whether Petitioner has met the threshold standard of 35 U.S.C. § 314(a).

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see also KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic*, 815 F.3d at 1363 (citing 35 U.S.C. § 312(a)(3)); *see also Intelligent Bio-Sys.*, 821 F.3d at 1369 (“It is of the utmost importance that petitioners in the IPR proceedings adhere to the requirement that the initial petition identify ‘with particularity’ the ‘evidence that supports the grounds for the challenge to each claim.’” (quoting 35 U.S.C. § 312(a)(3))). Therefore, to prevail in an *inter partes* review, Petitioner must explain how the proposed combinations of prior art would have rendered the challenged claims unpatentable. At this preliminary stage, we determine whether the information presented in the Petition shows there is a reasonable likelihood that Petitioner would prevail in establishing that at least one of the challenged claims would have been obvious over the proposed combinations of prior art.

A. *Level of Ordinary Skill in the Art*

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 7–8 (citing Ex. 1003 ¶ 61). Petitioner contends that additional training can substitute for educational or research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a

computer system, including standard communication busses and protocols, such as PCI and SMBus busses and protocols. *Id.* Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

In the Preliminary Response, Patent Owner applies the level of ordinary skill in the art proposed by Petitioner. Prelim. Resp. 7.

Among the factors that may be considered in determining the level of ordinary skill in the art are the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995) (citing *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962–63 (Fed. Cir. 1986)). The evidence presented mostly applies to the educational level of workers in the field, but also touches upon other factors as well. We find Petitioner’s proposal is consistent with the level of ordinary skill in the art reflected by the ’054 patent and the prior art of record, and, therefore, we adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least,” for purposes of this Decision.

B. Claim Construction

We construe each claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent,” the same standard used to construe the claim in a civil action. 37 C.F.R. § 42.100(b).

Petitioner contends that “no express constructions are needed for this proceeding” but notes that Patent Owner has broadly interpreted some claim terms for purposes of infringement even though a narrower interpretation may be more reasonable. Pet. 8–9 (citing Ex. 1071, 39–46; Ex. 1073, 55–62). Petitioner further contends it is not necessary to determine whether claim terms are governed by § 112, 6th paragraph because at least one of the prior art combinations matches the disclosure of the ’054 patent. *Id.* at 9.

At this stage in the proceeding, we need only construe the claims to the extent necessary to determine whether to institute *inter partes* review. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))). We determine that, at this stage of this proceeding, there is no need to expressly construe these claim terms in order to determine whether or not to institute review because the parties do not dispute the claimed terms’ meanings, and their ordinary and customary meanings suffice for our analysis. However, we do invite the parties to construe the term “pre-regulated voltage” at trial for reasons explained in Section III.E.11, *infra*.

C. Patent Owner's Preliminary Arguments

Patent Owner argues that Petitioner makes conclusory statements and then cites to pages in Dr. Wolfe's declaration (Ex. 1003) without explanation, improperly incorporating several hundred pages of additional briefing. Prelim. Resp. 14 (citing Pet. 57–58, 68–70, 110–112, 123–125; *Cisco Systems, Inc. v. C-Cation Techs., LLC*, IPR2014-00454, Paper 12 (PTAB Aug. 29, 2014) (informative); 37 C.F.R. § 42.6(a)(3)). Petitioner argues that we should reject these alleged improper attempts to incorporate by reference Dr. Wolfe's testimony. *Id.*

In our analysis below, we will determine for ourselves whether the Petition adequately supports the ground and the weight to be accorded the declarant's testimony.

Patent Owner also takes issue with Petitioner's assertion that elements 7b–7c of claim 7 are similar to elements 5b–5c of claim 5. Prelim. Resp. 14–15. Specifically, Patent Owner contends claim 5 pertains to performing a write operation in response to an undervoltage condition whereas claim 7 pertains to an overvoltage condition and Petitioner's cross-reference allegedly provides no analysis why a write operation would be conducted in response to an overvoltage condition. *Id.* at 15. We address Patent Owner's argument in Section III.E.6, *infra*.

D. Ground 1: Obviousness Over Harris and FBDIMM Standards

Petitioner contends claims 1–3 and 15 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–40. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's arguments and,

therefore, establishes a reasonable likelihood of prevailing with respect to this ground at this stage of the proceeding.

1. *Harris (Ex. 1023)*

Harris is titled “Voltage Distribution System and Method for a Memory Assembly.” Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 9.

Harris’s Figure 1A is reproduced below.

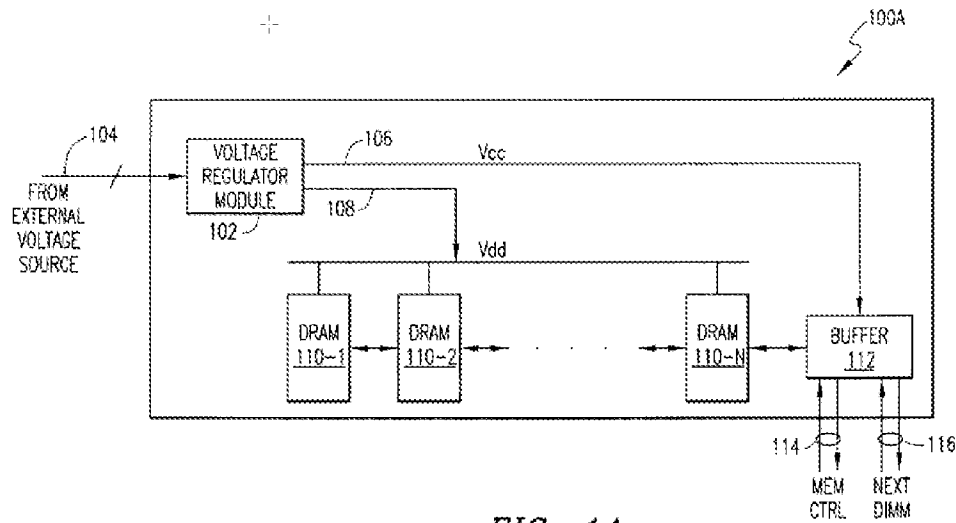


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface 116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Modules (DIMM) wherein each of the memory devices 100-1 to 100-N comprises a Double

Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. FBDIMM Standards (Exs. 1027, 1028)

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029 ¶¶ 134–137. Petitioner refers to these standards collectively as the “FBDIMM Standards.” Pet. 10. Petitioner contends the FBDIMM Standards are prior art under § 102(a).

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DS}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	0.453* V_{DD}	0.5* V_{DD}	0.547* V_{DD}	(DRAM Interface V_{TT}) This supply should track as 0.5 * 1.8 volt supply
	3.0	3.3	3.6	(V_{DD3FBD})
Buffer interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM interface	SSTL_18			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.

Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)

An Unloaded system should supply V_{TT} of 0.48* V_{DD} /0.52* V_{DD} to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
$\overline{\text{RESET}}$ [1]		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM.
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	I	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground.
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by “x” are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies VCC, VCCFBD, VDD, VSS, and VDDSPD.

3. Motivation to Combine

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16. Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and

thus would naturally look to them for more details about the “fully buffered DIMM” that Harris describes as compatible with his on-board voltage regulator module (VRM). *Id.*

Patent Owner does not dispute Petitioner’s motivation to combine Harris and the FBDIMM Standards. *See* Prelim. Resp.

Petitioner has sufficiently shown that a person of ordinary skill in the art would have combined Harris and the FBDIMM standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶ 9. We agree that one of ordinary skill in the art would have recognized this as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9.

4. *Analysis of Independent Claim 1*

a) *Limitation 1.a: “A memory module comprising:”*

Petitioner asserts that Harris’s memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed “memory module.” Pet. 19–20 (citing Ex. 1023 ¶¶ 9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 217–223; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1 in Figure 3.

Patent Owner does not specifically respond to Petitioner’s contention for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Harris teaches the preamble for purposes of institution.¹

b) Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. For example, Petitioner notes that Harris discloses that its double-rank DIMMs may be accommodated on both sides of a printed circuit board (PCB). Pet. 20 (citing Ex. 1023 ¶ 13). Petitioner further notes that a PCB may be referred to as a ‘memory board’ or ‘raw card.’ *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 224–226).

As for the PCB “having an interface configured to fit into a corresponding slot connector of a host system,” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 227–228).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “configured to coupled

¹ Neither party argues whether the preamble limits claim 1. Although we find that the evidence supports that the prior art teaches the preamble, we make no determination at this stage of the proceeding that the preamble of claim 1 is limiting.

power, data, address and control signals between the memory module and the host system.” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A (Ex. 1023 ¶ 10, 12, 19). Petitioner contends that the Harris’s buffer 112 in Figure 1A is called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 22. Petitioner indicates that buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Harris’s Figure 1A. Pet. 22–25 (citing Ex. 1023 ¶ 9 (“buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data space at least for a portion of memory devices 110-1 through 110-N”)). In addition, Petitioner argues that the FBDIMM Standards indicate buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

Patent Owner argues that Petitioner has not made a *prima facie* case that Harris discloses a memory module having a PCB interface that receives power from the host system. Prelim. Resp. 15–21. Harris states, however, that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Harris also discloses that “external voltage sources may comprise any combination of *known* or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14 (emphasis added). Patent Owner does not argue that voltage supplied by a host system is not a ‘known’ voltage supply as referenced by Harris. Furthermore, Petitioner indicates that the FBDIMM Standards show that the buffer AMB may be connected to a host, suggesting that the FBDIMM may derive its power from the host. Pet. 24 (citing

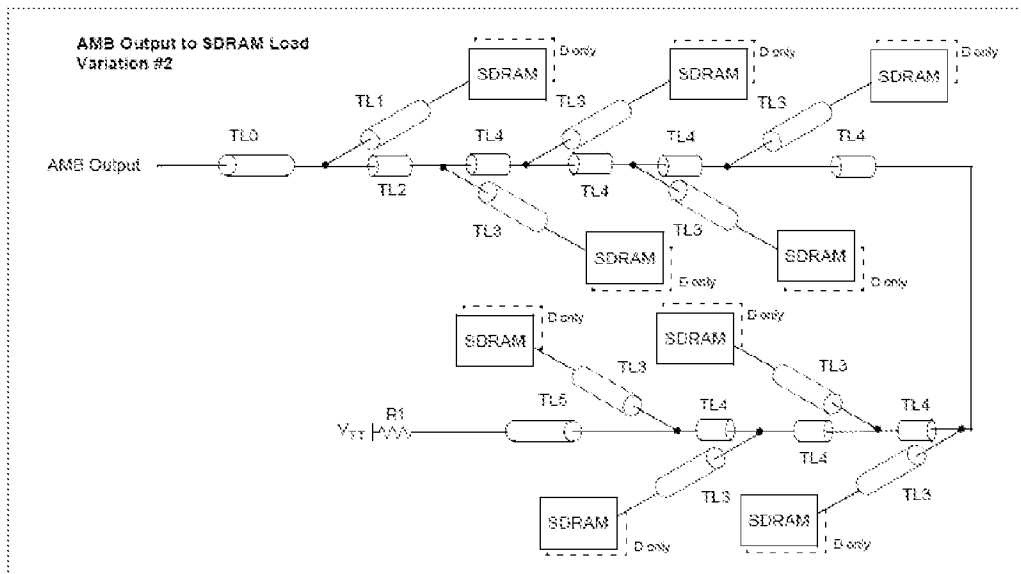
Ex. 1027, 4). These facts point to the conclusion that Harris's external voltage source may be the host system notwithstanding Patent Owner's arguments to the contrary.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

c) Limitation 1.c: "a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages"

Petitioner contends that Harris and the FBDIMM Standards disclose limitation 1.c of claim 1 of the '054 patent. Pet. 26–30. Specifically, Petitioner contends that Harris's Voltage Regulator Module 102 corresponds to the claimed "voltage conversion circuit." Pet. 26 (citing Ex. 1023 ¶ 10, Fig. 1A). Petitioner contends that the claimed "at least three regulated voltages" correspond to voltage 106 (V_{cc}) and voltage 108 (V_{dd}) as shown in Harris's Figure 1A, and a third voltage (V_{TT}) which is not actually shown in Harris's Figure but is shown in the FBDIMM Figure below.

DDR2 SDRAM Fully Buffered DIMM Design Specification DDR2 Fully Buffered DIMM Wiring Details
Net Structure Routing for Address/Command to SDRAM (Raw Cards B, C)



Ex. 1028, 68. The figure above shows voltage V_{TT} terminating an array of SDRAM devices via resistor R1.

As to the voltages being “regulated,” Petitioner argues Harris discloses that “at least one on-board voltage regulator” is “capable of generating tightly-controlled voltage levels.” Pet. 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 236).

As to the “at least three buck converters,” Petitioner contends that Harris teaches using buck converters to provide the three regulated voltages above. Pet. 28 (citing Ex. 1003 ¶¶ 238–241). Petitioner contends Harris teaches using a “high-frequency switching voltage converter capable of generating tightly-controlled voltage levels” to provide each needed on-board regulated voltage. *Id.* (citing Ex. 1023 ¶¶ 10, 12). Petitioner contends it would have been obvious to use a “buck converter” to convert higher input voltage (e.g., 12V) to the lower output voltage (3.5V or less), and there would have been a reasonable expectation of success because buck

converters were well-known “switching” devices commonly used to step down the voltage between its input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 147–150, 239–241; Ex. 1058, 3, 5, 12–16 (1995 Lenk textbook showing buck, boost, and buck-boost circuits); Ex. 1032, 161 (1995 Mohan textbook showing various step-down (buck) converters); Ex. 1030, 2:32–43, 5:39–44; Ex. 1024, Fig. 6 (Amidi showing DC/DC buck converter; Ex. 1050, 1:21 (identifying buck converters as one of “the most basic building blocks in power electronics”). Petitioner contends buck converters were well-known for their efficiency in generating step down voltages without generating excess heat or requiring large cooling devices. *Id.* at 29 (citing Ex. 1003 ¶ 240; Ex. 1059, 5:23–30; Ex. 1058, 5; Ex. 1040, 1, 23–24; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1062, 11; Ex. 1064 ¶ 101). Petitioner contends that it would have been obvious to use at least three converters in Harris given the need for at least three different voltages in the FBDIMM Standards. Pet. 29–30 (citing Ex. 1028, 17–20; Ex. 1026, 2–3, 9; Ex. 1003 ¶¶ 248–249, 255; Ex. 1062, 13; Ex. 1028, 30–32).

Patent Owner argues that Harris requires at most two buck converters to provide the two voltages needed and thus does not disclose “at least three buck converters” as claimed. Prelim. Resp. 24–26. However, as explained, Petitioner relies on Harris for disclosing two voltages V_{cc} and V_{dd} , and the FBDIMM Standards to provide the third voltage V_{TT} such that three buck converters would be needed. *See* Pet. 26–30.

Patent Owner further contends that Petitioner has not made out a case to use two or more buck converters to provide voltages having the same level. Prelim. Resp. 27–30. Petitioner explained, however, that the

FBDIMM standards identify the voltages V_{DD} , V_{DDQ} and V_{DDL} as well as V_{CC} and V_{CCFBD} as separate voltages with separate pins that can be turned on and off separately. Pet. 30.

Patent Owner further contends that Petitioner did not make the case that a person of ordinary skill in the art would have used a third buck converter to provide termination voltage V_{TT} . Prelim. Resp. 30–35. Patent Owner contends that Petitioner does not explain why a person of ordinary skill in the art would have generated the voltage V_{TT} on the module as opposed to obtaining V_{TT} from interface pins. *Id.* at 31. Harris’s Figure 1A shows that the voltages V_{cc} and V_{dd} are generated on the module. We agree with Petitioner that it would logically follow to generate V_{TT} on the module using the same voltage regulator module 102 as used to generate voltages V_{cc} and V_{dd} . *See* Pet. 26. In addition, “[when] there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp.” *KSR*, 550 U.S. at 421. Here, there are only two options—generate the voltage V_{TT} on the module, as Petitioner indicates, or obtain the voltage V_{TT} from interface pins. Petitioner’s choice of the former of two options does not negate its showing of obviousness.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches limitation 1.c of claim 1 for purposes of institution.

d) *Limitation 1.d.1: “a plurality of components coupled to the PCB, each component of the plurality of*

components coupled to at least one regulated voltage of the at least three regulated voltages”

Petitioner asserts that Harris discloses a “a plurality of components coupled to the PCB” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 261–266). Petitioner further contends that these components are each coupled to at least one regulated voltage of the at least three regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches this limitation for purposes of institution.

e) Limitation 1.d.2: “the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 31–32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 267–271). Petitioner contends that a person of ordinary skill in the art would have known that according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* at 31–32 (citing Ex. 1028, 9; Ex. 1045, cover; Ex. 1026, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches this limitation for purposes of institution.

f) Limitation 1.d.3: “a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections”

Petitioner contends that the “first circuit” corresponds to buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–33 (citing Pet. 20–25; Ex. 1003 ¶¶ 272–277). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8.

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

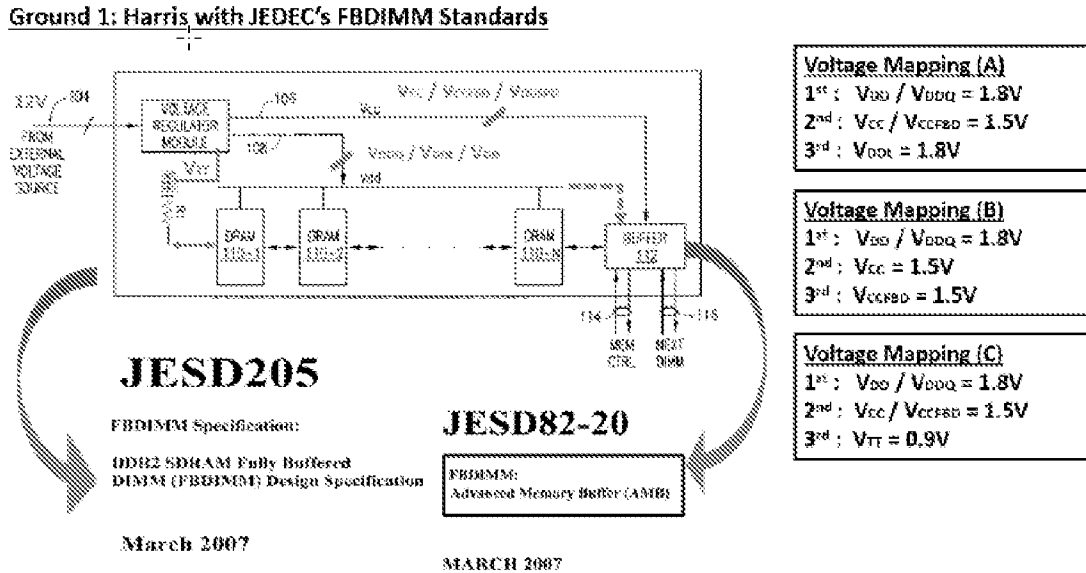
Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

g) Limitation 1.d.4: “wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and”

Petitioner contends that Harris’s buffer corresponds to the first circuit and is coupled to “first” (e.g., V_{DD} or $V_{DDQ} = 1.8V$) and “second” (e.g., V_{CC}

or $V_{CCFBD} = 1.5V$) “regulated voltages of the at least three regulated voltages.” Pet. 33–34 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 278–284).

Petitioner provides an annotated version of Harris’s Figure 1A shown below.



As shown above, Petitioner fills in Harris’s Figure 1A with additional information from the FBDIMM Standards to show that Harris’s buffer 112 would be understood by a person of ordinary skill considering the FBDIMM standards to couple to the voltages V_{DD} or V_{DDQ} and V_{CC} or V_{CCFBD} .

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Prelim. Resp.*

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards teaches this limitation for purposes of institution.

h) Limitation 1.d.5: “wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 285–287). Specifically, Petitioner contends that the SDRAM devices from Harris’s Figure 1A are coupled to the first regulated voltage (V_{DD} or $V_{DDQ} = 1.8V$) of the “at least three regulated voltages.” *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards teaches this limitation for purposes of institution.

i) Determination for Claim 1

Petitioner further shows sufficiently that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown sufficiently that the combination of Harris and the FBDIMM Standards teaches each limitation of claim 1. Accordingly, Petitioner has established a reasonable likelihood to prevail in showing that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. Analysis of Dependent Claims

Claim 2 depends from claim 1 and recites “wherein the first regulated voltage has a first voltage amplitude, and the second regulated voltage has a second voltage amplitude, and wherein a first one of the first and second

voltage amplitudes is less than a second one of the first and second voltage amplitudes.” Ex. 1001, 38:45–50. Petitioner contends that, as taught by the FBDIMM Standards, the first voltage amplitude may be 1.8V and the second voltage amplitude may be 1.5V which is less than the first voltage amplitude of 1.8V. Pet. 34 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 288–296).

Claim 3 depends from claim 1 and recites “wherein a third regulated voltage of the at least three regulated voltages has a voltage amplitude of 1.8 volts.” Petitioner contends the third regulated voltage is V_{DDL} with a value of 1.8V as shown in the FBDIMM Standards (see previous figure—“voltage mapping (A)”). Pet. 35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 297–301).

Claim 15 depends from claim 1 and recites “wherein two of the at least three buck converters are configured to operate as a dual-buck converter.” Ex. 1001, 39:59–61. Petitioner contends at the time there were many commercially available products that could output two (or more) regulated voltages using buck converters, and thus “it would be obvious to implement any two of the regulated voltages as a ‘*dual buck converter*’ to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design.” Pet. 36 (citing Ex. 1003 ¶ 439; *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 797–99 (Fed. Cir. 2021)). Petitioner contends that the Murata MPD4S014S dual buck converter, Texas Instruments TPS51020 dual buck converter, and Fairchild FAN5026 dual-output PWM controller were commercially available and would have been suitable to implement for use in Harris in view of the FBDIMM Standards. Pet. 37–40.

Patent Owner does not specifically respond to Petitioner’s contentions as to claims 2, 3 and 15. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Harris in combination with the FBDIMM Standards teaches the limitations of claims 2, 3, and 15. Based on the current record, Petitioner has established a reasonable likelihood to prevail in showing unpatentability of claims 2, 3, and 15.

6. *Determination for Challenge Ground 1*

Petitioner has shown a reasonable likelihood to prevail in showing unpatentability of claims 1–3 and 15 as obvious over the combination of Harris and the FBDIMM Standards for the reasons explained.

E. *Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi*

Petitioner contends claims 1–30 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 41–70. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's arguments and, therefore, establishes a reasonable likelihood of prevailing with respect to this ground at this stage of the proceeding.

1. *Amidi (Ex. 1024)*

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled "Clock and Power Fault Detection for Memory Modules." Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 11.

Amidi's Figure 5 is reproduced below.

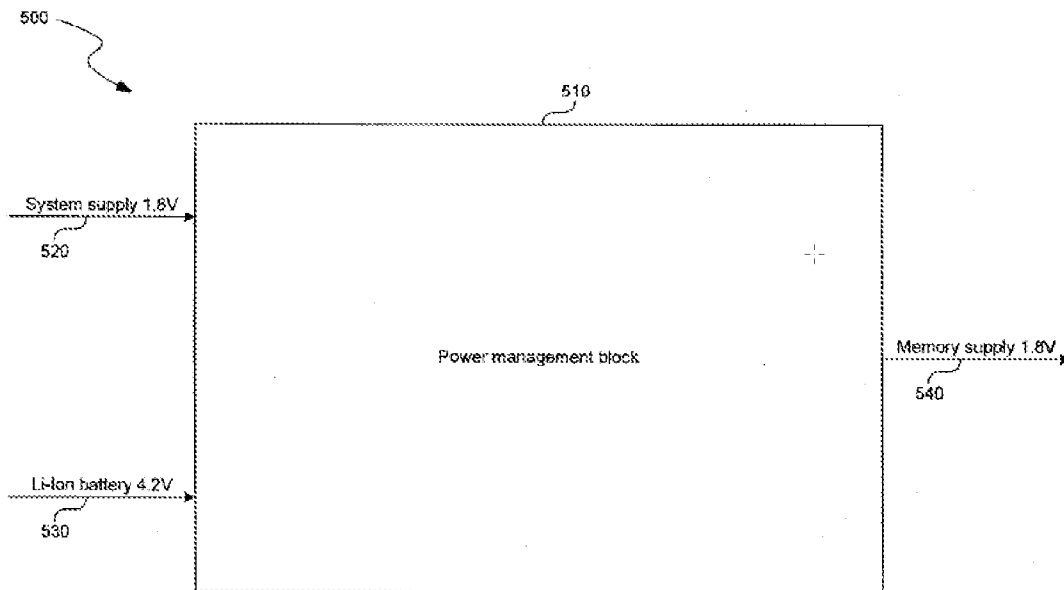


FIG. 5

Amidi's Figure 5 above illustrates a power management block 510 that receives an incoming system supply 520, and incoming battery supply 530, and generates an outgoing memory power supply 540 which is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Fig. 5, Fig. 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 41. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 41–42 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 171–177). Petitioner further notes that

Amidi's power management block could be modified easily to work with Harris's FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 42 (citing Ex. 1024, Fig. 5; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 173–174). Petitioner contends that to a person of ordinary skill in the art it would have been obvious to use the 12V external supply stepped-down with a buck converter to a 5V supply for charging Amidi's battery, and that Amidi's battery voltage would be stepped-up with a boost converter to the 12V level used by Harris's memory module. *Id.* at 42–43 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 173–174). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed, and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 43 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi's battery backup mode is similar to the S3 power-saving mode of Harris's FBDIMM memory module and the FBDIMM standards, such that a person of ordinary skill in the art would have been motivated to combine their teachings. *Id.* at 43–45.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:

Ground 2: Ground 1 and Battery Backup of Amidi

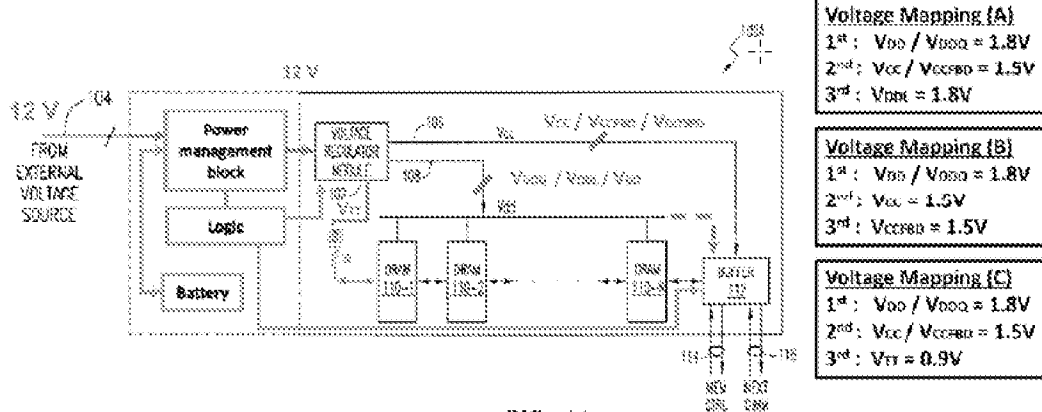


FIG. 1A

Pet. 45. Above, Petitioner has annotated Harris’s Figure 1A to show features added from the teachings of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue).

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Prelim. Resp. 35–39. Harris does not appear to describe, however, switching to an alternate voltage source in response to power loss, nor does it explicitly mention a battery as an alternate voltage source, whereas Amidi does. Ex. 1024, code (57). Thus, Patent Owner’s argument does not undermine Petitioner’s motivation to combine the references.

Patent Owner also argues that “Petitioner has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the combination would even fit onto Harris’s circuit board, particularly where Petitioner has already proposed to add additional converters.” Prelim. Resp. 38–39. Patent Owner’s declarant, Dr. Sunil Khatri states that space on the board to fit all of the components that Petitioner proposes would have been a concern. Ex. 2001 ¶¶ 78, n.3, 79. As explained above however, Petitioner

argues that one of ordinary skill would have had reasonable expectation of success in making the combination and supports its argument with credible testimony from Dr. Wolfe who explains how one of ordinary skill would have been able to incorporate Amidi's functionality in the FBDIMM memory module of Harris and that such modification would have been well within the level of skill at the time. Ex. 1003 ¶¶ 171–177.

Accordingly, we determine Petitioner has adequately shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3 and 15*

We agree with Petitioner that the addition of Amidi does not negate Petitioner's showing with respect to the previous ground. Pet. 45–46 (citing Ex. 1003 ¶¶ 216–301, 431–448). Accordingly, Petitioner has shown a reasonable likelihood to prevail in demonstrating claims 1–3 and 15 unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claim 4*

Claim 4 depends from claim 1 and recites that the memory module further comprises:

a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude below a predetermined threshold voltage, wherein the memory module transitions from a first operable state to a second operable state in response to the trigger signal.

Ex. 1001, 38:54–64.

Petitioner points to Amidi's power supervisory module/block 480/665/800 as claim 4's "voltage monitor circuit" coupled to Harris's edge connections for power. Pet. 46. According to Petitioner, Amidi's power supervisory module/block monitors the external system power supply 605/825 in Amidi. *Id.* Petitioner contends that Amidi's power supervisory module/block generates a "trigger signal" corresponding to Amidi's signal 670/858/868. *Id.* Petitioner contends the "predetermined threshold voltage" corresponds to Amidi's reference voltage 675/820 set to 5% or 10% below the nominal voltage of 12V. *Id.* at 46–49 (citing Ex. 1023 ¶¶ 12, 13; Ex. 1003 ¶¶ 303–314; Ex. 1024, 4:8–11, 4:44–52, 5:25–43, 5:31–62, 8:30–62, 8:23–29, 9:8–12, Figs. 4–6, 14, 15). Petitioner further states that the trigger signal causes the memory module to transition from normal operations using an external voltage in the normal range ("the first operable state") to self-refresh operations using a backup battery when the external voltage is outside the normal range ("the second operable state"). *Id.* at 49–50 (citing Ex. 1024, 4:44–52, 8:30–62, 9:8–22, Figs. 6-8, 14, 15; Ex. 1003 ¶¶ 315–321).

Patent Owner does not present any argument specific to claim 4 at this time. *See* Prelim. Resp.

Petitioner has shown sufficiently for institution that claim 4 is obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

5. *Claim 5*

Claim 5 depends from claim 4 and recites

a controller coupled to the voltage monitor circuit;
wherein, in response to the trigger signal, the controller is configured to perform one or more operations including a write operation to transfer data to non-volatile memory.

Ex. 1001, 38:65–39:3.

Petitioner contends that the logic for controlling the S3 sleep mode (Pet. 43–44) corresponds to the claimed “controller” that is coupled to the voltage monitor circuit. Pet. 50–51. Petitioner contends that S3 configuration information is stored in non-volatile memory before entering the S3 sleep mode in response to the trigger signal, satisfying the “wherein” clause of claim 4. *Id.* at 51 (citing Ex. 1003 ¶¶ 322–340).

Patent Owner does not provide any argument specific to claim 5. *See* Prelim. Resp.

Petitioner shows sufficiently that claim 5 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

6. *Claims 6, 7, 9–12, and 17*

Claims 6, 7, 9–12, and 17 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being above a predetermined threshold voltage. Ex. 1001, 39:4–20, 39:30–49, 40:21–25. Patent Owner contends that the Petition provides no analysis to explain why one would have configured a memory module to perform a write operation in response to an overvoltage condition. Prelim. Resp. 14–15. Of these claims, only claim 7 requires a write operation to be performed in response to the trigger signal. And Amidi seeks to prevent data loss by storing data in response to detection of an undervoltage situation. On this record, Petitioner’s showing for these claims is sufficient for institution. We invite further development of the record at trial before addressing patentability of these claims in a final written decision.

7. *Claim 8*

Claim 8 depends from claim 1 and recites

a controller coupled to the PCB, the controller including a voltage monitor circuit configured to monitor an input voltage received from a second set of edge connections of the plurality of edge connections, wherein, in response to the voltage monitor circuit detecting a power threshold condition, the voltage monitor circuit transmits a signal to one or more portions of the controller.

Ex. 1001, 39:21–29.

Petitioner contends that the “controller” corresponds to logic for controlling the S3 sleep mode described in the FBDIMM Standards. Pet. 55 (citing Pet. 43–44). Petitioner contends the controller is coupled to the PCB and includes a “voltage monitor circuit” corresponding to Amidi’s power supervisory module/block 480/665/800. *Id.* Petitioner contends Amidi’s power supervisory module/block monitors the external system power supply 605/835 (“input voltage”) received from Harris’s edge connections for power (“a second set of edge connections of the plurality of edge connections”). Pet. 55–56 (citing Pet. 20–25; Ex. 1023, Fig. 1A; Ex. 1024, Fig. 4; Ex. 1027, Ex. 1003 ¶¶ 373–379). Petitioner contends when the external system power supply voltage is too low, Amidi’s power supervisory module/block generates signal 670/858/868 (“a signal to one or more portions of the controller”) to indicate a power disruption to switch to battery backup and initiate the S3 sleep mode to preserve data in the SDRAMs while conserving power (the “wherein” clause in claim 4). Pet. 56–57 (citing Pet. 46–54; Ex. 1003 ¶¶ 380–285).

Patent Owner does not present any arguments specific to claim 8.

We determine Petitioner shows sufficiently for institution that claim 8 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

8. *Claim 13*

Petitioner contends that claim 13 recites a limitation substantially similar to one addressed with respect to claim 4. Pet. 58 (citing Ex. 1003 ¶¶ 419–423). Patent Owner does not provide specific argument for claim 13. We determine that Petitioner has shown sufficiently that claim 13 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

9. *Claim 14*

Claim 14 recites

wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a request by the host system.

Ex. 1001, 39:55–58.

Petitioner contends that a power disruption detected by Amidi's voltage supervisory block 800 causes the system to switch to battery backup. Pet. 61 (citing Pet. 55–57). Petitioner contends that Amidi's voltage supervisory block 800 detects a request by the host system to return control to the host. *Id.* at 61–62 (citing Ex. 1003 ¶¶ 424–430). Petitioner asserts that Amidi “discloses that, if the power is restored after a failure, the voltage supervisory block detects whether the host has cleared the backup-mode bit (*‘detecting a request by the host system’*) and subsequently, if so, returns control of the module to the host, including handling of the power signal.” *Id.* at 62 (citing Ex. 1003 ¶¶ 311, 429; Ex. 1024, 5:31–62, 8:30–62, Fig. 14

9:11–39, Figs. 8, 14 (1460–1490), 15 (steps 1530, 1540, 1570, 1580; Ex. 1003 ¶ 312).

Patent Owner does not separately argue claim 14.

Petitioner shows sufficiently for institution that claim 14 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

10. Claim 16

Petitioner contends that claim 16 is disclosed by the combination of Harris, the FBDIMM Standards, and Amidi for the reasons previously stated for claims 1, 8, and 9, noting that Amidi’s voltage supervisory block (e.g., 665) (“the voltage monitor circuit”) is “configured to detect an amplitude change in the input voltage” when the external system supply changes by crossing the reference voltage. Pet. 58 n.4 (emphasis omitted).

Patent Owner presents no separate arguments for claim 16 apart from those previously discussed.

Petitioner has shown sufficiently for institution that claim 16 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

11. Claims 18–22 and 26–28

Claim 18 depends from claim 16 and recites

wherein, in the first operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

Ex. 1001, 40:26–32.

Petitioner contends that this limitation is taught by the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 63–65. Petitioner contends that “pre-regulated voltage” means either that the voltage is within the 12V +/- 15% limits described in Harris, or must be pre-regulated on the memory board itself. Pet. 63; Ex. 1023 ¶ 13. For purposes of this Decision, we interpret “pre-regulated voltage” to mean that the voltage is regulated before conversion to a stepped up or down level. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). Petitioner contends that Harris discloses external voltage sources that are regulated. Pet. 63 (citing Ex. 1023 ¶¶ 13–14); *see* Ex. 1023 ¶ 14 (“It should be readily recognized that the external voltage sources may comprise any combination of known or heretofore unknown voltage supplies, either *regulated* or unregulated, and even including variable voltages.” (emphasis added)). Harris’s disclosure of regulated voltage sources appears to teach a “pre-regulated voltage,” as we understand that term. Petitioner also offers an alternative argument based on Amidi, but we need not address it at this point because Petitioner’s first argument is sufficiently persuasive. Pet. 63–64. We welcome any further development of the record on this issue, including the proper interpretation of “pre-regulated voltage.”

12. *Claim 23*

Claim 23 depends from claim 16 and recites

wherein the voltage monitor circuit is configured to produce a trigger signal in response to detecting an amplitude change in the input voltage; and wherein, in response to the trigger signal, the controller is configured to perform one or more operations including a write operation to transfer data to non-volatile memory.

Ex. 1001, 40:65–41:4.

Petitioner contends that claim 23 is substantially similar to limitations included in claims 4 to 6. We agree that Petitioner's showing for claims 4 and 5 is sufficient for claim 23 because determining an undervoltage condition per claim 4 is detecting a voltage amplitude change per claim 23.

Patent Owner does not submit any argument specific to claim 23. *See* Prelim. Resp.

Petitioner shows sufficiently that claim 23 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

13. Claim 24

Petitioner contends that the limitations of claim 24 were all addressed with respect to claims 1, 16, and 23. Pet. 69–70.

Patent Owner's arguments with respect to claim 24 were previously addressed. *See* Sect. III.E.2, *supra*.

Petitioner has shown sufficiently for institution that claim 24 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

14. Claim 25

Petitioner contends that the limitations of claim 25 were addressed with respect to claims 4 and 16. Pet. 58, 70.

Patent Owner provides no argument specific to claim 25. *See* Prelim. Resp.

Petitioner has shown sufficiently for institution that claim 25 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

15. Claim 29

Claim 29 depends from claim 24 and recites that the voltage monitor circuit is configured to detect whether the input voltage is above or below respective thresholds. Ex. 1001, 42:19–23. Petitioner contends that the limitations of claim 29 were addressed with respect to claims 4, 6, and 17. Pet. 58, 70.

Patent Owner provides no argument specific to claim 29. *See Prelim. Resp.*

Petitioner’s showing with respect to the undervoltage condition in claim 4 is sufficient to institute for claim 29.

16. Claim 30

Claim 30 depends from claim 29 and recites that the predetermined threshold voltages are above and below a “specified operating voltage.” Ex. 1001, 42:24–27. Petitioner contends that the “specified operating voltage” corresponds to Harris’s 12V nominal external voltage. Pet. 57, n.2 (citing Pet. 41–45; Ex. 1023 ¶¶ 12, 13).

Patent Owner does not present any arguments specific to claim 30. *See Prelim. Resp.*

Petitioner has shown sufficiently that claim 30 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

17. Determination for Challenge Ground 2

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations of at least one claim in this challenge

ground are taught by the combination. Accordingly, under this ground, Petitioner has demonstrated a reasonable likelihood to prevail in showing unpatentability of at least one claim of the '054 patent as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

F. Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck

Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

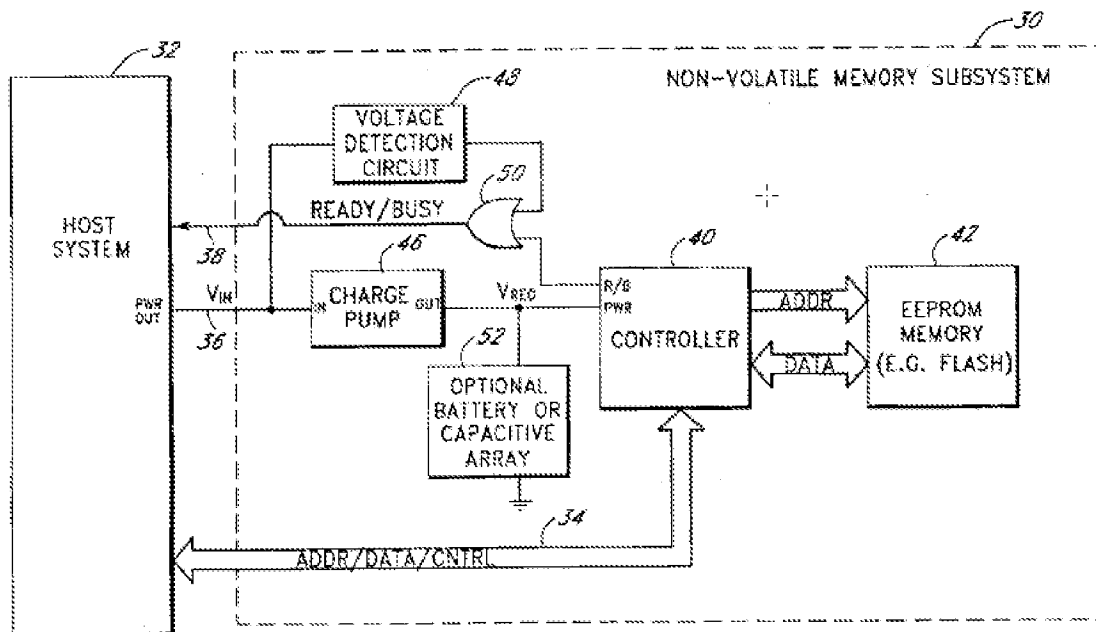
Pet. 70–72.

1. Hajeck (Ex. 1038)

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1001, code (54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck’s Figure 1 is reproduced below.



In Hajeck's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–33.

2. Motivation to Combine

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajeck with Harris, the FBDIMM Standards, and Amidi because one would have appreciated the desirability of switching to backup power in both undervoltage and overvoltage conditions.

Pet. 71–72. Hajeck teaches to use a battery backup in response to power loss from the host system to complete outstanding operations and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57). Hajeck also teaches a voltage detection circuit generate a “busy” signal in response to undervoltage or overvoltage conditions. *Id.* at 3:30–43. Accordingly, Petitioner’s motivation to combine is sufficient for purposes of institution. We invite further development of the record on the parties’ views of on the combination of Hajeck with Harris, the FBDIMM Standards, and Amidi.

3. *Determination for Ground 3*

Petitioner has shown a reasonable likelihood to prevail in demonstrating that claims 1–30 of the ’054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

G. *Ground 4: Obviousness over Spiers and Amidi*

Petitioner contends that claims 1–30 are obvious over the combination of Spiers and Amidi. Pet. 72–125.

1. *Spiers (Ex. 1025)*

Spiers is titled “Non-Volatile Memory Backup for Network Storage System.” Ex. 1025, code (54). Spiers was filed on October 12, 2004, and published on April 13, 2006 as U.S. Patent Pub. No. 2006/0080515 A1. *Id.* at codes (10), (43). Petitioner contends that Spiers is prior art under § 102(b). Pet. 13.

Spiers’s Figure 4 is reproduced below.

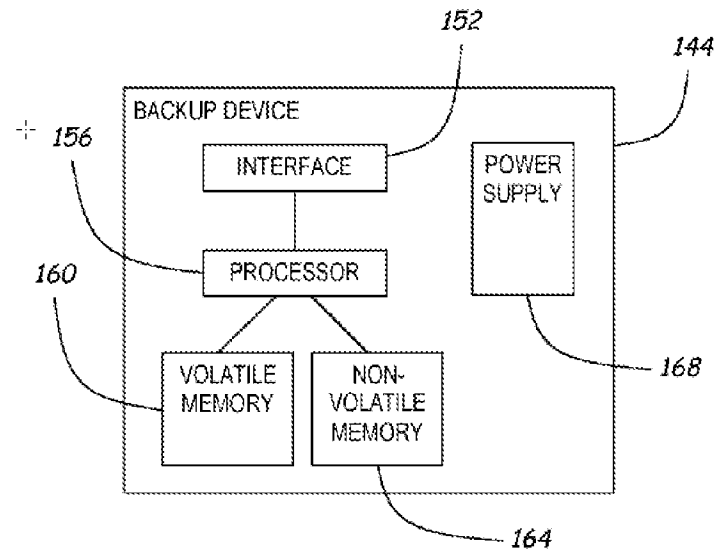


FIG.4

Spiers's Figure 4 above shows backup device 144. Ex. 1025 ¶ 36. Backup device 144 comprises an interface 152, backup device processor 156, volatile memory 160, non-volatile memory 164, and power supply 168. *Id.* Interface 152 communicates with storage controller 132 (not shown). *Id.* Interface 152 connects to processor 156 which controls operations within backup device 144. *Id.* Processor 156 connects to volatile memory 160 and non-volatile memory 164. *Id.* Volatile memory 160 may be an SDRAM used to store data from storage controller 132 during typical write operations. *Id.* Non-volatile memory 164 may be flash memory and is used in the event of a power failure detection. *Id.* Upon detecting a power failure, processor 156 switches backup device 144 to power supply 168 (capacitors or batteries) and moves data in volatile memory 160 to non-volatile memory 164. *Id.*

2. Motivation to Combine

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Spiers and Amidi. Pet. 73–77. Petitioner

contends one would have used Amidi to learn implementation details, such as specific type of SDRAM devices and voltage regulators, useful for Spiers. *Id.* at 73 (citing Ex. 1003 ¶¶ 197–201). Petitioner contends while Spiers teaches SDRAM devices, Amidi specifically discloses DDR SDRAM devices. *Id.* at 74 (citing Ex. 1024, claims 4–5). Petitioner contends a person of ordinary skill in the art would have been motivated to implement Spiers with commercially available DDR2 or DDR3 SDRAMs powered according to relevant JEDEC standards with a reasonable expectation of success when using such well-known, standardized technology. *Id.* (citing Ex. 1003 ¶¶ 199–201).

Patent Owner argues that Spiers only mentions two voltage regulators, and that Petitioner provides no reason why a person of ordinary skill in the art would have used DDR2 or DDR3 devices, which require more than two regulated voltages, instead of the SDR SDRAM devices specified by Spiers. Prelim. Resp. 39–40. Petitioner contends that Amidi provides a reason to use DDR2 or DDR3 SDRAMs powered according to relevant JEDEC standards. Pet. 74 (citing Ex. 1003 ¶¶ 199–201). Patent Owner contends that Spiers's writes to SDRAM occur with every system write, and that Petitioner makes no showing that DDR2/DDR3 devices would have been used for such a high write-to-read ratio application. Prelim. Resp. 40 (Ex. 2001 ¶ 90; Ex. 1025 ¶¶ 34, 36, 40, 41, Figs. 11, 12).

Patent Owner's evidence does not establish that DDR2 or DDR3 devices would not work in Petitioner's combination, only that they would be sub-optimal in that application. On the current record, we cannot determine that such a solution would be sufficiently sub-optimal that one of ordinary skill in the art not have pursued it, but instead would have looked to some

other unidentified solution. *See also Novartis Pharms. Corp. v. West-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (noting that an obviousness showing “does not require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation for the current invention.” (quoting *In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004))).

Patent Owner also argues that SDRAM devices could operate at or above the data rates of a PCI interface used by Spiers, so replacing Spiers’s SDRAM devices with faster DDR2 or DDR3 would not bring about a gain in data rate or justify the added expense of additional voltage regulators to accommodate DDR2/DDR3 devices. Prelim. Resp. 41 (citing Ex. 2001 ¶¶ 91–95).

Patent Owner’s argument assumes that Spiers is limited to a PCI interface, but a PCI backup device is described as one embodiment in Spiers. Ex. 1025 ¶ 19. Spiers also discloses an “interface 152” in more general terms, which does not appear limited to a PCI interface. *Id.* ¶ 36. Patent Owner has not shown that the interfaces that one of ordinary skill in the art would have considered to use in light of Spiers would not have been able to accommodate faster DDR2/DDR3 devices. Accordingly, this argument is not persuasive on the record as thus far developed.

Petitioner has shown sufficiently that one of ordinary skill in the art would have combined Spiers and Amidi.

3. *Claim 1*

a) *Limitation 1.a: “A memory module comprising:”*

Petitioner contends that Spiers discloses this limitation. Pet. 77–78. Petitioner contends that Spiers’s backup device 144 corresponds to the

claimed “memory module.” *Id.* at 77. Spiers’s backup module 144 has both volatile memory (SDRAMs 190) and non-volatile memory (NAND flash 194). *Id.* at 77–78.

Patent Owner does not specifically respond to these arguments. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Spiers teaches this limitation for purposes of institution.

b) Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”

Petitioner contends that Spiers discloses this limitation. Pet. 79–81. Petitioner asserts that Spiers’s PCI card corresponds to “a printed circuit board (PCB).” *Id.* at 79. Petitioner contends that Spiers’s PCI card has an interface configured to fit into the slot connector of a host system, and including a plurality of edge connections. *Id.* (citing Ex. 1025, Fig. 5). Petitioner contends these edge connections are configured to couple power, data, address and control signals between the memory module and host system. *Id.* (citing Ex. 1025 ¶¶ 37, 39, 54; Ex. 1031, 1, 7, 21–25, 146–150).

Patent Owner submits no argument specific to this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Spiers teaches this limitation for purposes of institution.

c) Limitation 1.c: “a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages”

Petitioner contends that Spiers’s power supply 168 discloses a voltage conversion circuit coupled to the PCB and configured to provide three regulated voltages when backup device 144 is implemented with DDR2 or DDR3 DRAMs according to Petitioner’s voltage mappings. Pet. 81–84. Petitioner relies on Amidi to disclose use of buck converters to supply the regulated voltages specified by the JEDEC standards for DDR2 or DDR3 DRAMs. *Id.* at 85–86. In addition, Petitioner identifies commercially available buck converters for use in implementing the proposed combination. *Id.* at 87–88.

Patent Owner contends that Petitioner has not shown why one would have required three buck converters for use in Spiers. Prelim. Resp. 52–70. As mentioned, Petitioner contends that to supply the regulated voltages specified in the JEDEC standards for DDR2 or DDR3 DRAMs or FBDIMMs, at least three buck converters would be required. Pet. 81–86.

Patent Owner contends that Petitioner did not provide any evidence why one would need to use V_{TT} in the DDR2 or DDR3 standards. Prelim. Resp. 53–56. However, even if Patent Owner is correct, Petitioner did not rely solely on V_{TT} as the third regulated voltage. *See* Pet. 82.

Patent Owner argues there is no evidence that one would have generated V_{TT} using a buck converter. Prelim. Resp. 56–59. Petitioner explained that the reason for using a buck converter for regulated voltages comes from Amidi. *See* Pet. 86 (citing Ex. 1024, Fig. 6 (640), 4:38–41).

Patent Owner further argues that Petitioner provided no reason for equipping Spiers with multiple 1.8V regulators. Prelim. Resp. 59–60. Assuming Patent Owner is correct, Petitioner also provided Voltage Mapping A which does not require the same voltage to be used for the regulated voltages. *See* Pet. 82. Petitioner also explained that the JEDEC standards provide for separate pins with the same voltage levels, and that certain voltages should be isolated and separately controlled to provide independence, stability, and flexibility for power management. Pet. 30.

Patent Owner further argues there are other ways to generate regulated voltages than providing a buck converter for each voltage level, and Petitioner did not show why one would not have pursued these other options. Prelim. Resp. 60–70. Petitioner did explain, however, that using a buck converter for each regulated voltage would “achieve high efficiency, reliability, and flexible power conversion.” Pet. 85 (citing Pet. 29–30).

Petitioner has shown sufficiently for institution that the combination of Spiers and Amidi teaches this claim limitation.

d) Limitation 1.d.1: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages”

Petitioner contends that Spiers’s DRAM, FPGA/processor, termination resistors, and non-volatile memory are components coupled to the PCI card, which are each coupled to at least one of three regulated voltages. Pet. 89.

Patent Owner provides no argument specific to this limitation. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

e) Limitation 1.d.2: “the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and”

Petitioner asserts that Spiers discloses SDRAMs 190 that may be implemented as DDR2 or DDR3 devices. Pet. 90.

Patent Owner provides no argument specific to this limitation. See Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

f) Limitation 1.d.3: “a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections”

Petitioner contends that the “first circuit” corresponds to Spiers FPGA processor 198 coupled to the SDRAM devices 190, 218 and to the PCI interface 172 for data, address and control signals. Pet. 90–91.

Patent Owner does not specifically respond to Petitioner’s contention. See Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

g) Limitation 1.d.4: “wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and”

Petitioner contends that Spiers’s FPGA processor 198 couples to voltage V_{DD}/V_{DDQ} (1.5V) and V_{CCFPGA} (1.8V). Pet. 91–92.

Patent Owner does not specifically address Petitioner’s contention for this limitation. See Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

h) Limitation 1.d.5: “wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages”

Petitioner contends that Spiers’s SDRAM devices are coupled to the regulated voltage V_{DD}/V_{DDQ} (1.5V or 1.8V). Pet. 92.

Patent Owner does not specifically respond to Petitioner’s contention. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

i) Determination for Claim 1

Petitioner shows sufficiently that one of ordinary skill in the art would have had reason to combine Spiers and Amidi with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown sufficiently that the combination of Spiers and Amidi teaches each limitation of claim 1. Accordingly, Petitioner has established a reasonable likelihood to prevail in showing that claim 1 is unpatentable as obvious over the combination of Spiers and Amidi.

4. Claims 2—30

We have reviewed Petitioner’s contentions for each of claims 2–30, and find them sufficient to disclose the limitations recited in at least claims 2, 5, 8, 13, 14, 16, 23, 25, 29 and 30. Pet. 92–125.

Other than the arguments previously addressed under claim 1, Patent Owner does not present further any argument specific to these claims. *See* Prelim. Resp.

Petitioner has shown sufficiently that the combination of Spiers and Amidi discloses at least one of these claims.

Accordingly, Petitioner has established a reasonable likelihood to prevail in showing at least one of these claims is unpatentable as obvious over the combination of Spiers and Amidi under this ground.

H. Ground 5: Obviousness over Spiers, Amidi, and Hajeck

Petitioner contends that a person of ordinary skill in the art would have combined Hajeck with Spiers and Amidi because Hajeck teaches the importance of detecting both undervoltage and overvoltage conditions to avoid data loss. Pet. 26. Hajeck teaches generating a busy signal for a host in response to undervoltage or overvoltage conditions. Ex. 1038, 3:44–61. Hajeck also teaches that its charge pump protects its controller from being damaged by spikes and surges in a power signal provided by a host. *Id.* at 3:13–16. Petitioner has shown sufficiently that one of ordinary skill in the art would have been motivated to combine Hajeck with Spiers and Amidi to detect undervoltage and overvoltage conditions. We invite elaboration of the parties views’ concerning the combination of Spiers, Amidi and Hajeck at trial.

On this record, Petitioner’s showing is sufficient to present a reasonable likelihood that claims 1–30 are unpatentable as obvious over the combination of Spiers, Amidi, and Hajeck. Pet. 125–127. And this ground subsumes the previous one, and there is no persuasive evidence that Hajeck negates Spiers and Amidi. Accordingly, Petitioner has presented a reasonable likelihood to prevail in showing at least one of claims 1–30 is unpatentable as obvious over the combination of Spiers, Amidi, and Hajeck under this ground.

IV. CONCLUSION

After considering the evidence and arguments presented in the Petition and Preliminary Response, we determine that Petitioner has established a reasonable likelihood of prevailing on its assertion that at least one claim of the '054 patent is unpatentable. Accordingly, we institute an *inter partes* review on all the challenged claims and all of the grounds presented in the Petition. At this stage of the proceeding, we have not made a final determination as to the patentability of these challenged claims.

V. ORDER

In consideration of the foregoing, it is hereby

ORDERED that pursuant to 35 U.S.C. § 314, *inter partes* review is instituted as to the challenged claims of the '054 patent with respect to all grounds of unpatentability presented in the Petition; and

FURTHER ORDERED that *inter partes* review is commenced on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial.

IPR2022-00999
Patent 11,232,054 B2

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

VOLKSWAGEN GROUP OF AMERICA, INC.,
Petitioner,

v.

YECHEZKAL EVAN SPERO,
Patent Owner.

IPR2023-00315
Patent 9,955,551 B2

Before JON M. JURGOVAN, JASON W. MELVIN, and
AARON W. MOORE, *Administrative Patent Judges*.

MELVIN, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314

I. INTRODUCTION

Volkswagen Group of America, Inc. (“Petitioner”) filed a Petition (Paper 2, “Pet.”) requesting *inter partes* review of claims 24–55 of U.S. Patent No. 9,955,551 B2 (Ex. 1001, “the ’551 patent”). Yechezkal Evan Spero (“Patent Owner”) filed a Preliminary Response. (Paper 6, “Prelim. Resp.”). As authorized, Petitioner filed a Preliminary Reply (Paper 7 (“Prelim. Reply”)), and Patent Owner filed a Preliminary Sur-Reply (Paper 8 (“Prelim. Sur-Reply”)). Pursuant to 35 U.S.C. § 314 and 37 C.F.R. § 42.4(a), we have authority to determine whether to institute review.

An *inter partes* review may not be instituted unless “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). For the reasons set forth below, we conclude that Petitioner has shown a reasonable likelihood it will prevail in establishing the unpatentability of at least one challenged claim, and we institute *inter partes* review.

A. REAL PARTIES IN INTEREST

Petitioner identifies itself as the real party in interest. Pet. 89. Patent Owner also identifies itself as the real party in interest, noting that Torchlight Technologies LLC is the exclusive licensee of the ’551 patent. Paper 4, 1 (Patent Owner’s Mandatory Notices).

B. RELATED MATTERS

Patent Owner identifies the following related federal district court litigations of the ’551 patent: *Torchlight Techs. LLC v. Daimler AG et al.*, Case No. 1:22-cv-

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00751 (D. Del.); *Torchlight Technologies LLC v. General Motors LLC et al.*, No. 1:22-cv-00752 (D. Del). Paper 4, 1.

Patent Owner identifies the following PTAB *inter parte* reviews of the '551 patent: *Volkswagen Group of America, Inc., v. Yechezkal Evan Spero*, IPR2023-00318 (pending); *Volkswagen Group of America, Inc., v. Yechezkal Evan Spero*, IPR2023-00336 (pending). Paper 4, 2.¹

C. THE '551 PATENT

The '551 patent is titled “Detector Controlled Illuminating System” and relates to an “illuminating device” whereby “illumination intensity and spectrum” are varied to illuminate “areas according to the principles of correct lighting practice,” e.g., “adapt to ambient lighting, movement, visual tasks being performed, and environmental and personal conditions.” Ex. 1001, codes (54, 57). The specification generally describes a lighting device that “incorporates one or more discrete light sources and their ancillary optical and electrical control equipment in an integrated illuminating element.” *Id.* at 13:34–36. The combined unit is referred to as a Digital Lighting Fixture (DLF). *Id.* at 18:18–22.

The specification further describes transportation vehicle applications. *Id.* at 50:38–57:37. One such application involves a DLF headlamp device that includes a cluster of LEDs for flexible illumination. *Id.* at 50:50–58. With LEDs having a variety of aims, the headlamp’s light distribution pattern may be controlled based on a number of factors, including detected oncoming vehicles. *See id.* at 51:43–47, 52:59–65.

¹ Patent Owner further lists matters involving two different patents related to the '029 patent, including several IPRs, a pending reissue, and a completed *ex parte* reexamination. Paper 4, 2 (identifying IPR2022-01500, IPR2022-01586, IPR2023-00197, IPR2023-00328, , IPR2023-00335, 16/858,342, 90/,014,815).

D. CHALLENGED CLAIMS

Petitioner challenges claims 24–55 of the '551 patent. Pet. 24. Claim 1 is independent and is reproduced, below, with limitation identifiers reproduced from the Petition's claim appendix (Pet., Appx B):

- [.p] 24. An illuminating device having automatic control of light provided to an illuminated area comprising:
 - [.1] a structure for incorporating illuminating device elements, wherein the structure comprises a motor vehicle;
 - [.2] a plurality of individually controllable LEDs incorporated in the structure via at least one headlamp mounted to the structure, each headlamp including at least one of the LEDs, the LEDs in communication with a power source;
 - [.3] electronic circuitry apparatus for the controlled powering of the LEDs and other of the illuminating device elements;
 - [.4] one or more processors;
 - [.5] one or more sensors, including one or more cameras, placed on the structure for detecting coordinate-specific information about the illuminated area, at least one of said cameras in communication with power, and in communication with at least one of the one or more processors;
 - [.6] a controller, including at least one of the one or more processors, integrated with the structure via mounting, in communication with the LED controlling power electronic circuitry apparatus, said controller arranged to process the information communicated from the cameras and to automatically control the light provided to the illuminated area; and
 - [.7] memory storing instructions that, when executed by one or more processors of the controller, enable the controller to:
 - [.8] determine coordinates corresponding to another vehicle, based on the coordinate-specific information;
 - [.9] select a cluster of the LEDs that emit light toward at least a portion of the other vehicle based on the LEDs having emission aimings associated with the first coordinates; and

[.10] change emission of the selected cluster of LEDs relative to emission of other LEDs not included in the cluster to create headlamp illumination that preserves a first level of illumination illuminating past the other vehicle while preventing glare to a driver of the other vehicle.

Ex. 1035 (Reexam. Cert.), 1:21–24; Ex. 1002, 1011 (Certificate of Correction). Claims 32 (*id.* at 2:42–3:19), 40 (*id.* at 4:20–5:5), and 48 (*id.* at 6:24–7:5) are the other independent claims and Petitioner asserts that each recites only limitations similar to the limitations of independent claim 24 (Pet. 22–33). Remaining claims 25–31 (*id.* at 1:64–2:41), 33–39 (*id.* at 3:20–4:19), 41–47 (*id.* at 5:6–6:3), and 49–55 (*id.* at 5:6–6:3) depend, directly or indirectly, respectively (as grouped) from independent claims 24, 32, 40, and 48.

E. PRIOR ART AND ASSERTED GROUNDS

Petitioner asserts the following unpatentability grounds:

	Claims Challenged	35 U.S.C. §	References/Basis
1	24, 25, 30–33, 38–41, 46–49, 54–55	103	Beam, ² Satonaka ³
2	26–29, 34–37, 42–45, 50–53	103	Beam, Satonaka, Kobayashi ⁴
3	24, 25, 30–33, 38–41, 46–49, 54–55	103	Karlsson, ⁵ Nakamura ⁶
4	26–29, 34–37, 42–45, 50–53	103	Karlsson, Nakamura, Gotou ⁷

Pet. 4–5. Petitioner also relies on the Declaration of Dr. Jianzhong Jiao. Ex. 1003.

Petitioner asserts that each reference—Beam, Satonaka, Kobayashi, Karlsson, Nakamura, and Gotou—qualifies as prior art under pre-AIA 35 U.S.C. § 102(b) (2006). Pet. 3–4; *see also* pre-AIA 35 U.S.C. § 102(b) (“patent[] or . . . printed publication . . . more than one year prior to the date of the application for patent in the United States”).⁸ Patent Owner does not contest those assertions at

² Beam, U.S. Patent No. 6,144,158, issued Nov. 7, 2000. Ex. 1005, code (45).

³ English translation of Satonaka et al., Japan Patent Publication No. H7-101291, published Apr. 18, 1995. Ex. 1014, 0001, code (43).

⁴ Kobayashi, U.S. Patent No. 6,049,749, issued Apr. 11, 2000. Ex. 1008, code (45).

⁵ Karlsson, WIPO Patent Publication No. 98/54030, published December 3, 1998. Ex. 1010, code (43).

⁶ English translation of Nakamura et al., Japan Patent Publication No. H7-65603, published Mar. 10, 1995. Ex. 1013, 0001, code (43).

⁷ Gotou, U.S. Patent No. 5,588,733, issued Dec. 31, 1996. Ex. 1012, code (45).

⁸ The ’551 patent claims priority as follows: “The present application claims the benefit under 35 U.S.C. [§] 119(e) of U.S. Provisional Application[s] No. 60/395,308 filed Jul. 12, 2002[and] . . . No. 61/535,981, filed Oct. 17, 2011 [T]his application is a continuation-in-part application of, and claims the

this stage. *See, e.g.*, Prelim. Resp. Based on the current record and for purposes of institution, we adopt Petitioner’s assertion of prior art.

II. ANALYSIS

A. LEVEL OF ORDINARY SKILL IN THE ART

Petitioner asserts that an ordinarily skilled artisan “would have had a bachelor’s degree (B.S.) in mechanical engineering, electrical engineering, optical engineering, applied physics, or an equivalent field, as well as at least 2 years of industry experience in the area of automotive lighting and lighting-control systems” and “may work as part of a team” listing an example. Pet. 7 (citing Ex. 1003 ¶¶ 44–46, 67). Patent Owner presents a different description, proposing that a person of the level of ordinary skill in the art:

would have had a Master’s of Science Degree (or a similar technical Master’s Degree, or higher degree) in an academic area emphasizing electrical engineering, computer engineering, or computer science with experience or education in optics and imaging systems or, alternatively, a Bachelor’s Degree (or higher degree) in an academic area emphasizing electrical, computer engineering or computer science and having two or more years of experience in the field of optical and imaging systems.

Prelim. Resp. 9. Neither party, however, explains the significance of its proposed level of ordinary skill. To the extent either party believes the level of ordinary skill affects this proceeding, that party should explain how in the instituted trial.

benefit under 35 U.S.C. [§]120 from, U.S. [App.] No. 10/604,360, . . . filed on Jul. 14, 2003.” Ex. 1001, 1:4–19.

B. CLAIM CONSTRUCTION

The parties agree that no express claim construction is required at this stage. Pet. 7–10; Prelim. Resp. 9–10. We agree and therefore do not construe the claims for this decision. *Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) (“The Board is required to construe ‘only those terms that . . . are in controversy, and only to the extent necessary to resolve the controversy.’”) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

To the extent the scope of a particular claim term impacts a party’s argument during trial, the party should propose an express construction and show how the record supports it.

C. GROUPINGS LIMITATIONS FOR INDEPENDENT CLAIMS 24, 32, 40, AND 48

Petitioner groups contentions for independent claims 24, 32, 40, and 48 together because:

Each independent claim recites “an illuminating device” with several identical or patentably indistinct structural elements, including “memory storing instructions.” EX1003, ¶111. The only differences between these claims are the stored instructions—and even the instructions recite similar or patentably indistinct elements. *Id.*

Pet. 21. Petitioner summarizes the similarities and supplements this analysis with a claim chart corresponding the claims’ limitations. *Id.* at 22, n.3 (explaining Appx’s A, B). Petitioner reads claims 24, 32, 40, and 48 as a group to apply the claim language on the prior art, with structural limitations identified and grouped in the Appendices. *E.g., compare* Pet. 24 (§ VI.B.2, “[24.1]/[32.1]/[40.1]/[48.1]”) with App’x A (same grouping in “Claim Comparison Tables”).

Although Patent Owner challenges Petitioner’s groupings on procedural grounds (Prelim. Resp. 29–30 (addressed *infra* at 18 (§ II.H))), Patent Owner does

not presently assert that a grouping is inaccurate, i.e., that differences among claims 24, 32, 40, and 48 prevents Petitioner's asserted, substantive application of prior art to the grouped limitations. *See generally* Prelim. Resp. Further, to the extent Patent Owner addresses particular limitations of the independent claims, Patent Owner addresses a set of limitations as identified and grouped by Petitioner. *Compare* Prelim. Resp. 61–63 (addressing “[24.5]/[32.5]/[40.5]/[48.5]”) with Pet. 62–64 (same grouping).

Based on the present record, we adopt Petitioner's groupings of limitations for purposes of institution.

D. GROUND 1: OBVIOUSNESS OVER BEAM AND SATONAKA

Petitioner asserts that claims 24, 25, 30–33, 38–41, 46–49, and 54–55 are unpatentable as obvious over Beam and Satonaka. Because we find that Petitioner shows a reasonable likelihood that it will prevail as to at least the independent claims, we institute review.

Beam is titled “Adaptive/Anti-Blinding Headlights” and is directed to a system that automatically adjusts a vehicle's headlights to avoid blinding oncoming drivers. Ex. 1005, code (54), 1:5–18. Beam describes how its adaptive/anti-blinding headlight (“AABH”) system contains an optical system 106, a sensor 107, a controller 108, and an illuminator 109. Oncoming light enters the system through the optical system. *Id.* at 4:9–25. The sensor detects the light and converts the light into data, which is sent to the controller. *Id.* The controller then uses the data to adjust the output of light from the illuminator so that “[t]he resulting beam is at full intensity except in areas that would dazzle a driver ahead.” *Id.* at 4:32–59.

Satonaka is titled “Vehicle Headlamp Device” and is directed to “optimal brightness for driver vision without inflicting glare on other vehicles.” Ex. 1014, code (57). Referencing a flow diagram of steps (*id.*, Fig. 6), Satonaka discloses that its device operates as follows:

When the headlamps are illuminated, the vehicular gap S_L up to the preceding vehicle and the vehicular gap S_R up to the oncoming vehicle as determined by the image processing apparatus are read to assess the combination of other vehicles (202–206). The lamp control voltage value V_H is set to the maximum voltage value V_{max} when there are no other vehicles (208), it is set to the voltage value V_L according to the vehicular gap S_R from the preceding vehicle anti-glare map when there is only a preceding vehicle (210), it is set to the voltage value V_R according to the vehicular gap S_R based on the oncoming vehicle anti-glare map when there is only an oncoming vehicle (212), and it is set to a low voltage value after the derivation of the voltage value V_L and voltage value V_R from each map when there are both preceding vehicles and oncoming vehicles (214). As a result, it will be possible to illuminate the front of the vehicle with light from the headlamp with brightness according to the existence of other vehicles and the vehicular gap.

Id., code (57); *see also id.* ¶¶ 40–51 (describing Fig. 6).

1. Independent Claims 24, 32, 40, and 48

Patent Owner does not presently contest Petitioner’s contentions that Beam and its combination with Satonaka teaches every limitation. *See generally* Prelim. Resp. Patent Owner contends that Beam teaches away from the asserted combination because it “explains that mounting the sensor within the headlamp assembly is critical to ensure accuracy.” Prelim. Resp. 42. That argument relies on Patent Owner’s assertion that the combination is “modifying Beam to include Satonaka’s camera 22 that is mounted to the top of the windshield.” *Id.* at 43; *accord id.* at 40 (“Petitioner relies on Satonaka’s hardware, *i.e.*, its remote camera

22 and electrical connections for satisfying the ‘*camera*’ limitations.”). But that assertion is inaccurate; Petitioner relies on Satonaka for “determining and storing coordinates.” Pet. 18; *accord id.* at 19 (“modify Beam’s algorithm to include the coordinates of the pixels as taught by Satonaka”), 29 (same). Petitioner relies on Beam for the claimed camera. *Id.* at 28. Petitioner further asserts that, to the extent Beam does not disclose a powered connection for its camera, skilled artisans “would have been motivated to connect Beam’s AABH to a power source, like the battery in Satonaka, to provide power to the electrical components in the AABH, including Beam’s camera.” *Id.* at 28. Neither assertion relies on Satonaka’s camera and Patent Owner’s argument is therefore inapposite.

Relatedly, Patent Owner argues that the proposed combination renders Beam inoperable by using Satonaka’s remote camera rather than Beam’s camera. Prelim. Resp. 47–48. For the same reason discussed above, the argument is not persuasive because Petitioner does not rely on a combination using Satonaka’s remote camera. Pet. 18, 20, 28.

Next, Patent Owner argues that Petitioner does not explain “how an image from Beam’s co-located and boresighted camera would be used by Satonaka’s vehicle recognition zone image processing strategy—a processing strategy that is based on a remote camera.” Prelim. Resp. 49–52. In particular, Patent Owner points out that the different viewpoint would shift the perspective. *Id.* at 50. Based on the present record, however, Petitioner has the better argument that because “Satonaka provides details of how to implement the coordinate mapping of its pixels,” skilled artisans would have understood the programming changes required to accommodate the combination. Pet. 20 (citing Ex. 1014 ¶¶ 22, 27; Ex. 1003 ¶ 110). Patent Owner does not contest that Satonaka discloses “how to implement the coordinate mapping of its pixels” but distinguishes that from “how to map

coordinates to light sources.” Prelim. Resp. 51. The claim language does not require a particular mapping between coordinate and light sources. Rather, it requires “detecting coordinate-specific information about the illuminated area” and, based on that information, “determin[ing] coordinates corresponding to an other vehicle.” Satonaka describes how oncoming vehicles are detected within the coordinate frame of its image, and we conclude based on those disclosures that Satonaka satisfies the claim language. Ex. 1014 ¶¶ 30–34.

Patent Owner argues also that the Petition lacks an adequate reason for the combination. Prelim. Resp. 57–58. According to Patent Owner, the Petition is deficient because it lacks “evidence that Beam does not already accurately ‘track the location of the upcoming vehicle.’” *Id.* at 57 (quoting Pet. 20). But Beam already tracking upcoming vehicles does not undermine a reason to also use Satonaka’s teachings. *See Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373, 1381 (Fed. Cir. 2023) (noting that a challenger need not show the combination “was an ‘improvement’ in a categorical sense” but only that it “was a ‘suitable option’”). For the same reason, we do not find persuasive Patent Owner’s arguments that Petitioner does not “explain any deficiency in Beam’s teaching of tracking the location of the oncoming vehicle” or “any new or different functionality provided by this combination.” *See* Prelim. Resp. 58.

We have reviewed Petitioner’s contentions and evidence, along with Patent Owner’s arguments against institution, and conclude that Petition has shown a reasonable likelihood of prevailing with respect to unpatentability of the independent claims over Beam and Satonaka.

2. Dependent Claims

We have reviewed Petitioner’s contentions for the challenges to dependent claims 25, 30, 31, 33, 38, 39, 41, 46, 47, 49, 54, and 55 under Ground 1. Patent Owner has not yet substantively addressed those claims. Because we find Petitioner meets its burden for institution as to the independent claims, we do not address the dependent claims at this time.

E. GROUND 2: OBVIOUSNESS OVER BEAM, SATONAKA, AND KOBAYASHI

Petitioner asserts that dependent claims 26–29, 34–37, 42–45, and 50–53 are unpatentable as obvious over Beam and Satonaka in view of Kobayashi. Pet. 48–53. We have reviewed Petitioner’s contentions under Ground 2. Patent Owner has not yet substantively addressed those contentions. Because we find Petitioner to meet its burden for institution as to the independent claims, we do not address the dependent claims at this time.

F. GROUND 3: OBVIOUSNESS OVER KARLSSON AND NAKAMURA

Petitioner asserts that claims 24, 25, 30–33, 38–41, 46–49, 54, and 55 are unpatentable as obvious over Karlsson and Nakamura. Pet. 54–79. Because we find that Petitioner shows a reasonable likelihood that it will prevail as to at least the independent claims, we institute review.

Karlsson is titled “A Lighting Device Having a Controllable Lighting Pattern” and is directed to a lighting device that automatically adjusts the lighting pattern and intensity of a vehicle’s headlights to reduce blinding oncoming traffic. Ex. 1010, codes (54), (57). Karlsson’s system uses “a plurality of controllable spotlight beams . . . such as LEDs, which can preferably be controlled either individually or in groups.” *Id.* at 13:24–14:4. “[T]he pattern of the emitted light beam is varied automatically in an efficient manner and with maximum sensitivity

in response to at least one control signal delivered by light-sensitive sensor.” *Id.* at 2:30–35. Karlsson sought to improve road safety, noting that the desire for increased illumination leads to “the problem of blinding oncoming traffic which, in turn, may lead to a hazardous situation.” *Id.* at 1:21–26. Karlsson describes that its controllable beams may be responsive to a sensor to provide a desired pattern. *Id.* at 4:25–31. That sensor may detect light from oncoming traffic, including “precisely the direction from which light is being detected, so that subsequently a precise adjustment of the pattern of the light beam can be carried out, for example so as not to emit light in the direction of the detected light.” *Id.* at 5:16–32; *accord id.* at 9:18–26 (“glaring or blinding of oncoming traffic is effectively prevented”). Karlsson discloses that it may use a variety of sensors to detect light from oncoming traffic, including “a matrix of photo diodes” and “a photo-sensitive plate as used in modern video cameras.” *Id.* at 12:19–24.

Nakamura is titled “Vehicle Headlamp Device” and is directed to “illuminat[ing] [in] front of a vehicle without any reduction of driver visibility even when the vehicle is traveling along a . . . winding road.” Ex. 1013, code (57). Referencing a flow diagram of steps (*id.*, Fig. 12), Nakamura discloses that its device operates as follows:

The positions of the preceding vehicle and oncoming vehicle, the deviation angle, and the output signals of the beam-switching switch are read (202, 204), and the low-beam lamps are controlled to the deflection angle according to the curved road when the high-beam lamps are illuminated (light distribution control B, 212). When the high-beam lamps are turned off and the low-beam lamps are not disconnected (208, 210), there will be control to the deflection angle according to the curved road (light distribution control A, 214). When the low-beam lamps are disconnected, the deflection angle will be set to the initial angle when proceeding straight forward (light distribution control C, 216). In this way, it will be possible to perform

deflection to the low-beam according to the road shape even when the high-beams are turned on, improving visibility near the vehicle.

Id., code (57); *see also id.* ¶¶ 54–60 (describing Fig. 12).

1. Independent Claims 24, 32, 40, and 48

Petitioner relies on Karlsson’s sensors as the claimed camera. Pet. 62–64. Patent Owner argues that Karlsson does not teach “one or more cameras” as claimed. Prelim. Resp. 61–63. We agree with Petitioner that the claim language reads on Karlsson’s sensors, which it describes as including “a photo sensitive plate as used in modern video cameras.” Ex. 1010, 12:22–24. Patent Owner contends that Karlsson “does not disclose that its sensors have the full capability of cameras, specifically with regards to capturing images” (Prelim. Resp. 62) but does not explain what the claims require beyond Karlsson’s devices. Based on the present record, Karlsson discloses the claimed camera. *See* Ex. 1010, 12:19–22 (“[T]he light-sensitive sensors 9 and 23 are preferably designed to be capable of precisely determining the direction and the intensity of the detected light 11.”).

Other than the claimed camera, Patent Owner challenges the combination of Karlsson and Nakamura for the same reasons discussed above regarding Beam and Satonaka. As an initial matter, it is not clear that Karlsson itself lacks the disclosures required to render the claims obvious. Thus, challenging the combination with Nakamura does not appear dispositive on the present record.

Somewhat confusingly, despite agreeing that Petitioner relies on Karlsson’s camera (Prelim. Resp. 61–62), Patent Owner submits that Petitioner relies on Nakamura’s remote camera 22 as the claimed camera (*id.* at 43). Although Patent Owner asserts that Karlsson teaches away from such a remote camera (*id.* at 43–46), Petitioner unwaveringly relies on Karlsson’s camera (Pet. 62–64) and Patent Owner’s argument is therefore inapposite. Similarly, Patent Owner’s argument that

Nakamura's remote camera would render Karlsson inoperable (Prelim. Resp. 48–49) is unpersuasive because Petitioner relies on Karlsson's camera (Pet. 62–64).

Patent Owner argues additionally that the Petition does not adequately explain how Nakamura's processing strategy, which is based on a remote camera, would use the information sensed by Karlsson's lamp-located sensor. Prelim. Resp. 52–56. Patent Owner points out that the different viewpoint would shift the perspective. *Id.* at 53. Based on the present record, however, Petitioner has the better argument that skilled artisans would have understood how to map Karlsson's image with coordinates as taught by Nakamura. Pet. 63 (citing Ex. 1013 ¶¶ 36–47; Ex. 1003 ¶¶ 258–260 (citing Ex. 1013 ¶¶ 23, 25, 36–47)). According to Patent Owner, Petitioner agrees that Karlsson does not teach how to determine which LEDs to dim. Prelim. Resp. 54–55. The record does not support that view. Petitioner asserts that Karlsson discloses processing an image “to determine which areas of the image contains a vehicle to control the headlights.” Pet. 55 (citing Ex. 1003 ¶ 238). Petitioner asserts that using Nakamura's method of associating an image's pixels with a coordinate system to plot the oncoming cars' locations would have “provided the algorithm details that are already implicit in Karlsson's system, therefore allowing the coordinates of the traffic ahead to be determined based on the video sensor data.” Pet. 56 (citing Ex. 1003 ¶¶ 235, 239). Those assertions and Petitioner's support for them are adequate at this stage to support institution because Petitioner relies on Nakamura as supplemental to Karlsson's existing method.

Finally, Patent Owner challenges Petitioner's reason for the combination, arguing that Petitioner has shown no improvement that Nakamura would bring to Karlsson. Prelim. Resp. 58–61. We find that Petitioner's reasons for the combination support institution. First, as discussed above, an obviousness

combination need not show improvement in a categorical sense. *See PACT XPP Schweiz*, 61 F.4th at 1381. Petitioner’s contentions justify that skilled artisans would have had reason to use Nakamura’s coordinate-based approach with Karlsson’s system. For example, Petitioner reasons that the combination would have used Nakamura’s known coordinate technique to improve Karlsson’s similar system. *Id.* at 57. And Petitioner asserts the combination would “further ensure Karlsson’s system accurately identifies traffic ahead and dims its headlights appropriately, improving illumination quality and safety for drivers.” *Id.* Thus, although Karlsson was already directed at dimming its headlights for oncoming traffic, Nakamura’s teachings would have readily applied to Karlsson’s system and would have worked towards Karlsson’s goals.

We have reviewed Petitioner’s contentions and evidence, along with Patent Owner’s arguments against institution, and conclude that Petitioner has shown a reasonable likelihood of prevailing with respect to unpatentability of the independent claims over Karlsson and Nakamura.

2. Dependent Claims

We have reviewed Petitioner’s contentions for the challenges to dependent claims 25, 30, 31, 33, 38, 39, 41, 46, 47, 49, 54, and 55 under Ground 3. *See* Pet. 69–79. Patent Owner has not yet substantively addressed those claims. Because we find Petitioner meets its burden for institution as to the independent claims, we do not address the dependent claims at this time.

G. GROUND 4: OBVIOUSNESS OVER KARLSSON, NAKAMURA, AND GOTOU

Petitioner asserts that dependent claims 26–29, 34–37, 42–45, and 50–53 are unpatentable as obvious over Karlsson and Nakamura in view of Gotou. Pet. 79–83. We have reviewed Petitioner’s for the challenges to those claims under

Ground 4. Patent Owner has not yet substantively addressed those claims. Because we find Petitioner meets its burden for institution as to the independent claims, we do not address the dependent claims at this time.

H. DISCRETIONARY DENIAL UNDER § 314(a)

Patent Owner argues we should exercise our discretion to decline institution under § 314(a) because the Petition lacks particularity. Prelim. Resp. 26–35. First Patent Owner asserts that Petitioner’s claim identifiers are unclear and that the Petition falls short of what is required by paraphrasing limitations. *Id.* at 26–30. While Petitioner’s use of claim appendixes presents its challenges in a cumbersome manner, the Petition is sufficiently particular.

Patent Owner argues also that because Petitioner challenges the independent claims based on a single reference and alternatively based on a combination, the Petition is unclear by asserting both anticipation and obviousness. *Id.* at 30–32. We do not agree. Petitioner presents only obviousness grounds, which may rely on a single reference. Petitioner’s alternative contentions are clearly stated for each relevant limitation. *See, e.g.*, Pet. 28–29 (relying on Satonaka’s power connection to a camera to the extent not disclosed by Beam), 29–30 (relying on Satonaka’s processing that maps pixels to coordinates), 33–34 (same). Thus, we do not agree that the challenge raised anticipation, or that it inappropriately proposes uncertain grounds. *See* Prelim. Resp. 32–35.

I. DISCRETIONARY DENIAL UNDER § 316(b)

Patent Owner argues we should exercise our discretion to decline institution under § 316(b) because Petitioner has not justified three petitions against the ’551 patent. Prelim. Resp. 35–37. We do not agree, because we find three petitions justified in view of the large number of claims and the differences in claim scope.

III. CONCLUSION

For the reasons discussed above, we conclude Petitioner has shown a reasonable likelihood of prevailing with respect to at least one claim. We have evaluated all of the parties' submissions and determine that the record supports institution.

Our determination at this stage of the proceeding is based on the evidentiary record currently before us. This decision to institute trial is not a final decision as to patentability of any claim for which *inter partes* review has been instituted. Our final decision will be based on the full record developed during trial.

IV. ORDER

Accordingly, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a), *inter partes* review of claims 24–55 of the '551 patent is instituted on the grounds set forth in the Petition; and

FURTHER ORDERED that, pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial commencing on the entry date of this decision.

IPR2023-00315
Patent 9,955,551 B2

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD., MICRON TECHNOLOGY, INC.,
MICRON SEMICONDUCTOR PRODUCTS, INC., and MICRON
TECHNOLOGY TEXAS LLC,¹
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00999
Patent 11,232,054 B2

Before PATRICK M. BOUCHER, JON M. JURGOVAN, and
DANIEL J. GALLIGAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Final Written Decision
Determining All Challenged Claims Unpatentable
Dismissing Petitioner's Motion to Exclude
35 U.S.C. § 318(a)

¹ Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC filed a motion for joinder and a petition in IPR2023-00405 and have been joined as petitioners in this proceeding. *See* Paper 27.

I. INTRODUCTION

A. *Background and Summary*

Samsung Electronics Co., Ltd. (“Samsung”) filed a Petition (Paper 1, “Pet.”) for *inter partes* review of claims 1–30 (“challenged claims”) of U.S. Patent 11,232,054 B2 (Ex. 1001, “the ’054 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition. We instituted *inter partes* review under 35 U.S.C. § 314(a). Paper 11 (“Inst. Dec.”).

During the trial, Patent Owner filed a Response (Paper 22, “Resp.”), Petitioner filed a Reply (Paper 26), and Patent Owner filed a Sur-Reply (Paper 33). We joined Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC as petitioners in this proceeding, and we refer to Samsung and these entities collectively as “Petitioner.” *See* Paper 27.

Petitioner and Patent Owner requested oral argument (Papers 29 and 30). A hearing was conducted on September 11, 2023. Paper 49 (“Tr.”).

Petitioner objected to evidence (Papers 23, 34) and filed a Motion to Exclude (Paper 35). Patent Owner filed an Opposition to Petitioner’s Motion to Exclude (Paper 36), and Petitioner filed a Reply (Paper 39) in support of its Motion to Exclude.

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is entered pursuant to 35 U.S.C. § 318(a). Having reviewed the complete trial record, we determine that Petitioner has shown, by a preponderance of the evidence, that the challenged claims are unpatentable.

B. Real Parties in Interest

Petitioner entities, Samsung Electronics Co., Ltd., Samsung Semiconductor, Inc., Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC identify themselves as real parties in interest. Pet. 1; IPR2023-00405, Paper 3, 1.

Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

C. Related Matters

The parties advise that the '054 patent is related to the following pending matters:

- *Samsung Electronics Co., Ltd. et al. v Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021)
- *Netlist, Inc. v. Samsung Electronics Co., Ltd. et al.*, No. 2:21-cv-00463 (E.D. Tex. filed Dec. 20, 2021)
- IPR2012-00996 (U.S. Patent No. 11,016,918)
- IPR2017-00692 (U.S. Patent No. 8,874,831)
- U.S. Patent Application No. 17/582,797

Pet. 1; Paper 4, 1.

D. Overview of the '054 Patent (Ex. 1001)

The '054 patent is titled “Flash-DRAM Hybrid Memory Module.” Ex. 1001, code (54). Figure 12 of the '054 patent is reproduced below.

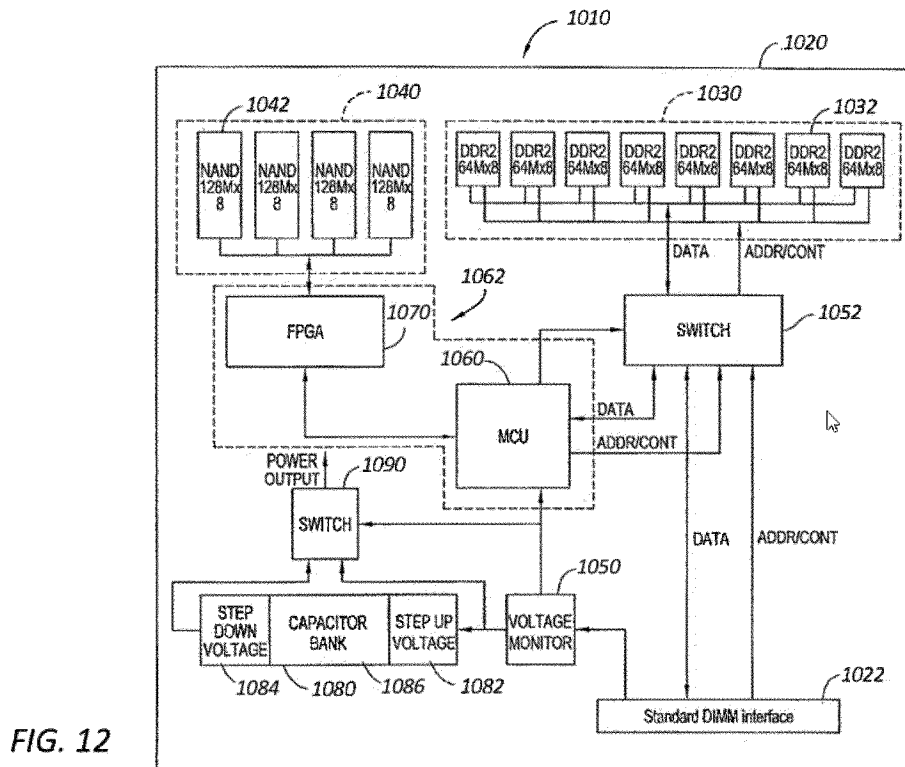


FIG. 12

Figure 12 shows an example memory system 1010 of the '054 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. *Id.* at 21:16–20. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022

for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. Ex. 1001, 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '054 patent refers to this as the “first state”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '054 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g. a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. *Id.* at 25:54–58. Second power supply 1080 may comprise step-up transformer

1082, step-down transformer 1084, and capacitor bank 1086 with one or more capacitors. *Id.* at 26:3–13.

The memory system 1010 further has a third state in which controller 1062 is decoupled from the volatile memory system 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). *Id.* at 25:62–66. The third power supply may provide power to volatile memory subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred. *Id.* at 25:66–26:3.

Figure 16 of the '054 patent is reproduced below.

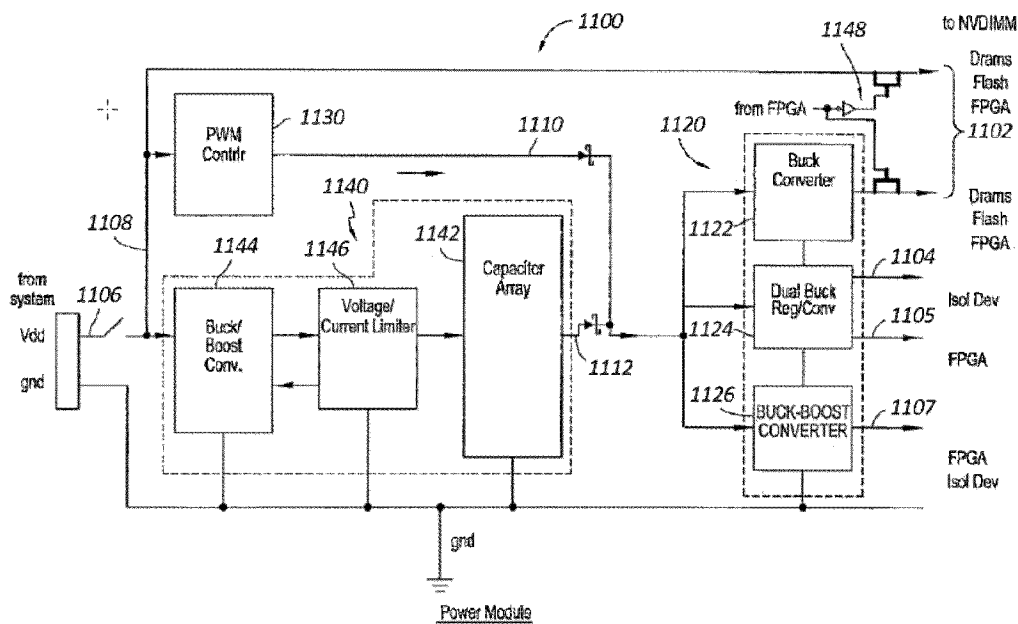


FIG. 16

Figure 16 shows power module 1100 of memory system 1010. Ex. 1001, 27:59–61. Power module 1100 comprises conversion element 1120, first power element 1130, and second power element 1140. *Id.* at 28:7–15, 28:20–22. Conversion element 1120 comprises buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 generating respective

voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 28:62–67. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. Power module 1100 provides power to components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. *Id.* at 27:61–65.

Specifically, in a first state, first voltage 1102 is provided to memory system 1010 from input 1106 and fourth voltage 1102 is provided to conversion element 1120 from the first power element 1130. Ex. 1001, 28:27–31. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. *Id.* at 28:31–34. In the third state, the fifth voltage is provided to conversion element 1120 from second power element 1140 and the first voltage 1104 is provided to memory system 1010 from conversion element 1120. *Id.* at 28:34–38. Transition from the first state to the second state may occur when power module 1100 detects a power failure is about to occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

E. Illustrative Claim

Of the challenged claims, claims 1, 16, and 24 are independent.

Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages;

[1.d] [1.d.1] a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, [1.d.2] the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and [1.d.3] a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections, [1.d.4] wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and [1.d.5] wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.

Ex. 1001, 38:19–44.

F. Evidence

Petitioner relies on the following references (*see* Pet. 3, 9–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published Aug. 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published Apr. 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued Feb. 15, 2005

G. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 3):

Claims Challenged	35 U.S.C. §	References
1–3, 15	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

II. ANALYSIS OF CHALLENGED GROUNDS

We now consider Petitioner’s asserted grounds of unpatentability and Patent Owner’s arguments to determine whether Petitioner has demonstrated by a preponderance of the evidence that the challenged claims would have been obvious under 35 U.S.C. § 103. *See* 35 U.S.C. § 316(e) (providing that

petitioner has the burden of proving unpatentability by a preponderance of the evidence).

A. Principles of Law

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination based on the teachings of multiple references requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see also KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re*

NuVasive, Inc., 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Factors pertinent to a determination of the level of ordinary skill in the art include “(1) the educational level of the inventor; (2) type of problems encountered in the art; (3) prior art solutions to those problems; (4) rapidity with which innovations are made; (5) sophistication of the technology; and (6) educational level of active workers in the field.” *Envtl. Designs, Ltd. v. Union Oil Co. of Cal.*, 713 F.2d 693, 696 (Fed. Cir. 1983) (citing *Orthopedic Equip. Co. v. All Orthopedic Appliances, Inc.*, 707 F.2d 1376, 1381–82 (Fed. Cir. 1983)). “Not all such factors may be present in every case, and one or more of these or other factors may predominate in a particular case.” *Id.* at 696–97. The prior art may reflect an appropriate level of skill. *Okajima v. Bourdeau*, 261 F.3d 1350, 1354–55 (Fed. Cir. 2001).

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 7–8 (citing Ex. 1003 ¶61). Petitioner contends that additional training can substitute for educational or research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a computer system, including standard communication busses and protocols,

such as PCI and SMBus busses and protocols. *Id.* Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

Patent Owner indicates “[f]or the purposes of this [Response], Patent Owner is applying the level of ordinary skill in the art proposed by Petitioner.” Resp. 2.

The evidence that Petitioner presents mostly applies to the educational level and experience of workers in the field, but also touches upon other factors as well, including problems and solutions in the prior art and complexity of the technology. *See* Pet. 7–8. We find Petitioner’s proposal is consistent with the level of skill in the art reflected by the ’054 patent and the prior art of record, and, therefore, we adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least,” which introduces ambiguity and may encompass skill levels beyond ordinary.

C. *Claim Construction*

We construe claim terms “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2021). There is a presumption that claim

terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art in the context of the specification. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Nonetheless, if the specification “reveal[s] a special definition given to a claim term by the patentee that differs from the meaning it would otherwise possess[,] . . . the inventor’s lexicography governs.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1316 (Fed. Cir. 2005) (en banc) (citing *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002)). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Only disputed claim terms must be construed, and then only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

1. “Memory Module”

Patent Owner contends as follows:

The term “memory module” appears in the preamble of the claims, and provides antecedent basis for the same term in the body of claims 1, 4, 6, 11, 16 and 25. The preamble is thus limiting as the District Court found after reviewing the specification in detail. Based on the intrinsic evidence, the Court also concluded that a memory module is a modular computer accessory that “includes the structure necessary to connect to a memory controller” of a host system.

Resp. 2–3 (citing Ex. 2032, 28). As support for its argument, Patent Owner quotes the District Court for the Eastern District of Texas as follows:

While the claims recite many of the structural requirements of a “memory module,” the claims arguably read on other modular computer devices, such as a video card or network controller, despite no evidence the inventors intended to encompass such devices by the claims. To the contrary, as the Overview section explains, the invention “is couplable to a memory controller of a host system,” ’918 Patent at 3:66–67 (emphasis added), not just the host system as recited in the claims. *See also id.* at 1:66–67 (“[t]he present disclosure relates generally to computer memory devices”). Thus, a skilled artisan would understand ***a “memory module” is distinct from, and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller.*** *See* Memory Systems: Cache, DRAM, Disk [EX2034] ... at 319 (depicting, in FIG. 7.6, a memory controller connected to two memory modules). Accordingly, the preambles are limiting.

Resp. 3 (citing Ex. 2032, 28) (emphasis in original). Patent Owner argues that Petitioner’s expert, Dr. Wolfe, testified that the term “memory modules” typically refers to “main memory modules,” which “are designed to connect to the primary memory controller for the purpose of holding general purpose code and data in a computer system.” Resp. 3–4 (citing Ex. 2060, 123:14–25; Ex. 2056, 100:15–101:8 (by 2004–2005, a memory module “was intended to go into a dedicated memory slot and not a general-purpose IO slot”)). Patent Owner contends that the ’054 patent’s usage of “memory module” is consistent with that understanding, and asserts that we should adopt the District Court’s claim construction, including that the claimed “memory module” includes structures necessary to connect to a memory controller. *Id.* at 4 (citing Ex. 2061 ¶¶ 51–53).

In Reply, Petitioner contends that we properly found that Harris, the FBDIMM Standards, Amidi, and Spiers all disclose a “memory module” in

our Institution Decision. Reply 1–2 (citing Inst. Dec. 14–15, 17–18, 29–33, 47–48; Pet. 19–20, 77–78; Reply 24–27). Petitioner contends that the District Court’s construction did not limit “memory module” to only “*main* memory modules . . . designed to connect to the *primary* memory controller.” *Id.* at 2 (emphasis in original) (citing Resp. 3–4). Petitioner contends that the District Court never used the words “main” or “primary,” and that District Court’s construction was simply that the “memory module” in the preamble was “limiting” without more. *Id.* (citing Ex. 2032, 26–28, 35). More specifically, Petitioner contends that the District Court did not further limit “memory module” to only “main memory modules . . . designed to connect to the primary memory controller.” *Id.* at 2 (citing Resp. 3–4). Petitioner further contends that Patent Owner misleadingly quotes Dr. Wolfe’s statements about “main memory modules,” but asserts that Dr. Wolfe explained in the context of the ’054 patent that a “memory module” is not limited to main memory or to any specific connection. *Id.* (citing Ex. 2060, 125:12–127:13 (“memory module” is a “circuit board that connects to a host computer that includes memory”); Ex. 2056, 100:15–101:19).

In Sur-Reply, Patent Owner disagrees with Petitioner’s contention that “memory module” is not limited to “*main memory* modules . . . designed to connect to the primary memory controller.” Sur-Reply 1 (emphasis in original) (citing Reply 2). Patent Owner argues that Petitioner’s contention “ignores the District Court’s finding based on intrinsic evidence—which [Petitioner] did not object to—that “the invention [i.e., the claimed memory module] “is couplable to a memory controller of a host system,” ’918 Patent at 3:66–67 . . . , not just the host system as recited in the claims.”” *Id.* (citing

Ex. 2032, 28; Ex. 1001, 3:66–67, 4:5–12, 4:14–24, 4:35–39, 4:45–51, 5:4–20, 4:36–50, 6:4–6, 6:25–29, 6:61–66, 7:1–4, 7:21–25, 7:29–34, 7:43–49, 7:54–57, 21:24–25, 22:53–58, 23:19–22, 26:43–51, 23:41–44).

The District Court’s Claim Construction Order (Ex. 2032) issued after our Institution Decision (Paper 11). Although Petitioner argued in the District Court litigation that the claim preambles were not limiting, Petitioner does not appear to maintain that argument here following the District Court’s determination that the preambles are limiting. Ex. 2032, 35. Since the term is recited in both the preambles and bodies of these claims, there is evidence that the inventors of the ’054 patent intended the term to limit the claim, as the District Court determined. *See Catalina Mktg. Int’l., Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell Commc’ns Rsch., Inc. v. Vitalink Communications Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995) (“dependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention.”)).

We agree with Petitioner’s contention that “memory module” is not limited to “**main** memory modules . . . designed to connect to the **primary** memory controller.” *See* Reply 2. The ’054 patent states

In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the **main memory**.

Ex. 1001, 11:3–6 (emphases added). This is the only mention of “main memory” in the ’054 patent. That the Flash memory serves as “main memory” in “certain embodiments” implies that there may be other embodiments in which this is not the case. Accordingly, we agree with

Petitioner that the term “memory module” is not limited to “*main* memory modules . . . designed to connect to the *primary* memory controller.”

We agree with Patent Owner that the “memory module” includes structure to connect with a “host system” including its memory controller. *See, e.g.*, Ex. 1001, code (57), 12:52–59. The claims do not mention a “memory controller,” but they do mention that the “memory module” has an “interface” to connect with a “host system.” Ex. 1001, 38:20–25 (claim 1), 39:63–65 (claim 16), 41:6–8 (claim 24). That the “memory module” has structure for connecting to the host system’s memory controller may thus be implied from the claims.

We agree with Petitioner that the District Court’s determination was that the preambles (presumably of claims 1, 4, 6, 11, 16 and 25) are limiting. Ex. 2032, 35. To the extent that the District Court’s determination construed “memory module” to include structure necessary to connect to a memory controller, our construction of “memory module” is consistent.

D. Ground 1: Obviousness Over Harris and FBDIMM Standards

Petitioner contends claims 1–3 and 15 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–40. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe’s testimony, supports Petitioner’s showing and establishes by a preponderance of the evidence that these claims are unpatentable.

1. *Harris (Ex. 1023²)*

Harris is titled “Voltage Distribution System and Method for a Memory Assembly.” Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 9.

Harris’s Figure 1A is reproduced below.

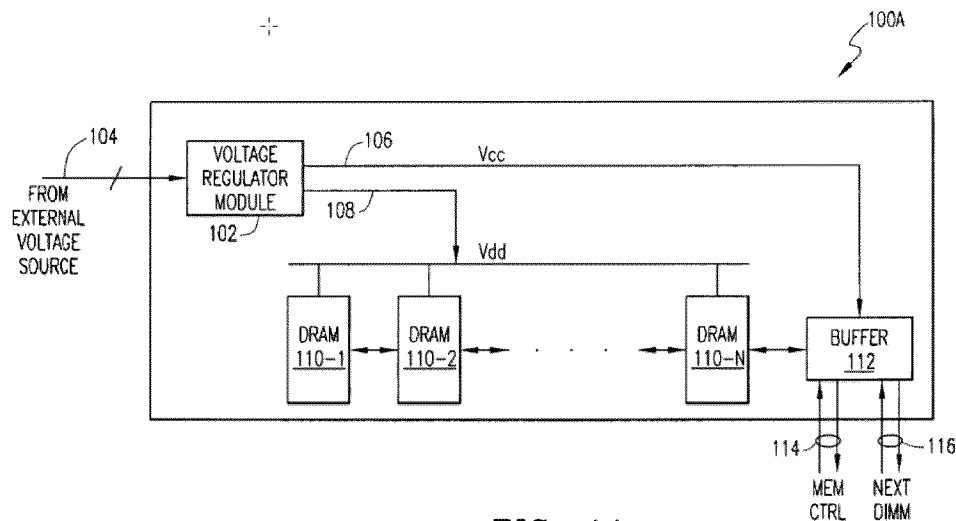


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface 116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Module (DIMM)

² Exhibit 1023 is incorrectly identified in the Board’s P-TACTS system as “Micron Tech., Inc. et al. v. Netlist, Inc., IPR2022-00418, Paper No. 2 (PTAB January 14, 2022) (831 Patent IPR Petition),” i.e., incorrectly repeating the identification provided for Exhibit 1022.

wherein each of the memory devices 100-1 to 100-N comprises a Double Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. FBDIMM Standards (Exs. 1027, 1028)

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029 ¶¶ 134–137. Petitioner refers to these standards collectively as the “FBDIMM Standards.” Pet. 10. Petitioner contends the FBDIMM Standards are prior art under § 102(a).

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max. thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max. thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DD}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	0.453* V_{DD}	0.5* V_{DD}	0.547* V_{DD}	(DRAM Interface V_{TT}) This supply should track as 0.5 * 1.8 volt supply
	3.0	3.3	3.6	(V_{DDSPD})
Buffer Interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM Interface	SSTL_18			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.

Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)

An Unloaded system should supply V_{TT} of 0.48* V_{DD} /0.52* V_{DD} to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
RESET $\bar{I}$		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	I	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by “a” are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies V_{CC} , V_{CCFBD} , V_{DD} , V_{SS} , and V_{DDSPD} .

3. Motivation to Combine

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–166). Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and thus would naturally look

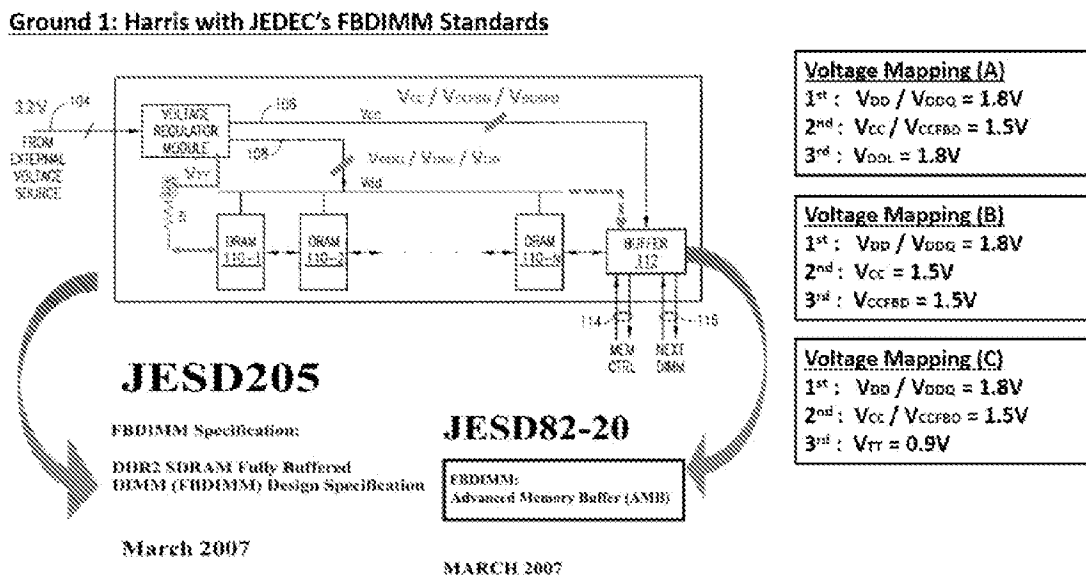
to them for more details about the “fully buffered DIMM” that Harris describes as compatible with the disclosed on-board voltage regulator module (VRM). *Id.*

To explain its combination, Petitioner provides the following table:

Voltage Mappings (Grounds 1-3)			
	<u>A</u>	<u>B</u>	<u>C</u>
“first”:	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} or $V_{DDQ} = 1.8V$
“second”:	V_{CC} or $V_{CCFB0} = 1.5V$	$V_{CC} = 1.5V$	V_{CC} or $V_{CCFB0} = 1.5V$
“third”:	$V_{DDQ} = 1.8V$	$V_{CCFB0} = 1.5V$	$V_{TT} = 0.9V$

The table above shows Petitioner’s Voltage Mappings A, B, and C from the FBDIMM Standards for Grounds 1–3, and how they disclose the “first,” “second,” and “third regulated voltages” in the claims. Pet. 27.

For Ground 1, Petitioner provides the illustration below:



This illustration shows how Petitioner is combining the teachings of Harris and the FBDIMM Standards to arrive at the claims. *Id.* at 15. Specifically,

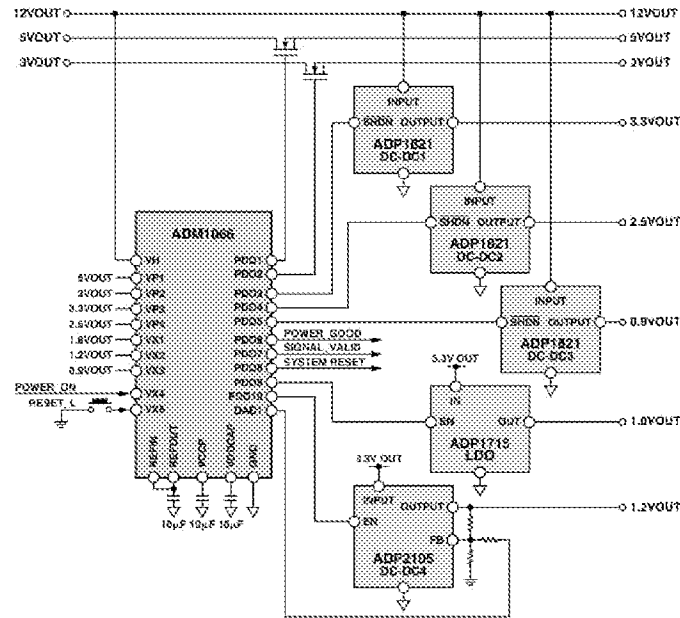
FBDIMM Standard JESD205 applies to Harris’s memory module and JESD82-20 applies to the Harris’s buffer. *Id.* at 26.

a) Modifying Harris to Have Three Converters

Patent Owner asserts that a person of ordinary skill in the art would not have modified Harris’s memory module to have three converters, as Petitioner asserts. Resp. 18–20. Patent Owner contends that Petitioner did not demonstrate “a reason to modify Harris with the FBDIMM Standards,” or “show a reason for Harris to use separate regulators for each of the alleged regulated voltages across all of Petitioner’s Voltage Mappings A–C.” *Id.* at 20. Patent Owner argues that it would not “have been obvious to use as many buck converters as possible, especially given that Harris expressly discloses a switching regulator outputting multiple voltage levels, consistent with the state of art.” Sur-Reply 10–11 (citing Reply 11; Ex. 1023 ¶ 10; Ex. 2020, 5, Table 1-1). Patent Owner asserts that Petitioner “has not articulated any advantage of using one converter per voltage if Harris already uses a single converter for at least two voltages.” *Id.* at 11 (citing Reply 11–14; Ex. 1023 ¶ 10).

As to why a person of ordinary skill in the art would use separate regulators for each of the voltages, Petitioner indicates that this was well-known in the art. Specifically, Petitioner contends that “buck converters were well-known switching devices commonly used to step down an input voltage to a lower output voltage.” Reply 11 (citing Pet. 28–29; Ex. 1075, 103:21–107:14, 112:12–113:19). Petitioner contends that it “would have been obvious to a [person of ordinary skill in the art]—and consistent with Harris—to use three buck converters to convert the 12V input to the three required voltages on the memory board.” *Id.* (citing Pet. 27–30; Ex. 2060,

53:16–54:14). Petitioner contends that “[t]his was common” as shown in the figure below:



Id. at 11–12 (citing Ex. 1062, 15, Fig. 4³). The above figure shows an ADM 1066 sequencer and three ADP182 DC-to-DC converters for generating 3.3V, 2.5V and 0.9 voltages outputs. Ex. 1062, 15. Petitioner also points to the data sheet for the ADP1821 titled “Step-Down DC-to-DC Controller,” which describes a “synchronous, pulse-width-modulated (PWM), voltage-mode, step-down controller.” Ex. 1078, 1; *see* Reply 11 (citing Ex. 1078, 1). Petitioner further relies on deposition testimony of

³ Exhibit 1062 is titled “Analog Dialogue,” Vol. 40, No. 2, published in 2006 by Analog Devices, Inc. “Analog Dialogue” contains several papers, one of which is titled “Power-Supply Management—Principles, Problems, and Parts” by Alan Moloney. The figure above was extracted from this paper. Although Exhibit 1062 was not identified in the Petition, it is part of the background knowledge that one of ordinary skill in the art would have had. *KSR*, 550 U.S. at 401 (obviousness analysis requires an assessment of the background knowledge possessed by person of ordinary skill in the art).

Patent Owner's expert, Dr. Mangione-Smith, who confirmed that the ADP1821 is a DC-to-DC converter and that it has structural elements and switching that are in common with buck converters to generate voltages. Reply 11 (citing Ex. 1075, 132:21–141:23). Patent Owner's argument that one of ordinary skill in the art would not have modified Harris to include three buck converters does not undermine Petitioner's showing of a motivation to combine Harris with the FBDIMM Standards when the background knowledge expressly teaches using three buck converters to generate voltages used in FBDIMMs. Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–166); Reply 11–12 (citing Ex. 1062, 15, Fig. 4; Ex. 1078, 1).

Patent Owner argues that Harris's mention of “a high-frequency switching voltage regulator” means only one, and that Petitioner's contention that “a” means “one or more” applies to the claims, not the specification. Sur-Reply 11 (citing *Baldwin Graphic Sys., Inc. v. Siebert, Inc.*, 512 F.3d 1338, 1342–43 (Fed. Cir. 2008); Ex. 1023 ¶ 14). Patent Owner further argues that “Harris's reference to ‘a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels’ (plural) means a single converter with multiple outputs, not multiple converters.” *Id.* at 15–16 (citing Ex. 2061 ¶¶ 79–80).

Patent Owner does not address Petitioner's argument that Harris refers to “at least one on-board voltage regulator module (VRM)” in the specification and “at least one voltage regulator module” in claims 1 and 30. Reply 12 (citing Ex. 1023 ¶ 10, Fig. 1A). Patent Owner's argument does not undermine Petitioner's showing.

Patent Owner further asserts that space issues would have influenced any design decisions. Sur-Reply 12 (citing *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1368 (Fed. Cir. 2016)). Patent Owner argues there are good reasons why a person of ordinary skill in the art would have minimized the number of buck converters, including space constraints, and that Petitioner has not shown how multiple voltage converters would fit in the one square-inch space on both sides of the circuit board described as available for a voltage regulator module (VRM) in Harris. *Id.* at 12–13 (citing Ex. 1023 ¶ 13; Ex. 1078, 20–23; Ex. 2101, 22, 25–26).

Harris states that “[i]t is contemplated that local supply voltage conversion for double-rank DIMMs can be accommodated with a form factor design of approximately about one square inch (both sides of the printed circuit board).” Ex. 1023 ¶ 13. Thus, Harris indicates that the VRM circuitry to generate two regulated voltages can fit within this space. Patent Owner argues that Petitioner’s evidence shows that each ADP1821 converter package would be about 0.193 by 0.236 inches. Sur-Reply 12 (citing Ex. 1078, 20–23). This means three ADP1821 converters would occupy about 40% of the available space. Patent Owner does not demonstrate that the remaining 60% of available space would have been insufficient to accommodate the additional components required for three voltage converters. *Id.* at 12–13. In any case, we do not view Harris’s disclosure of “approximately about one square inch” to limit the applicability of its teachings to a person of ordinary skill in the art. In the same paragraph, Harris states that the form factor “may be suitably modified.” Ex. 1023 ¶ 13.

Patent Owner further argues that “independent voltage sources for different voltages would require specific control circuitry to delay and more precisely control each voltage source’s ramping rate, adding complexity and cost without commensurate benefits.” Sur-Reply 14 (citing Ex. 2061 ¶ 90). However, Dr. Wolfe states that it is simpler to use multiple regulators for multiple voltages since a single regulator that generates multiple voltages is more complicated than multiple regulators. Ex. 2060, 53:16–54:14, 89:1–21, *cited in* Reply 14. Dr. Wolfe further indicates that buck converters can be in some applications “extremely small.” Ex. 2060, 89:11–15. Dr. Wolfe further testifies that multiple smaller converters may be less costly than a larger single converter because of component costs, separated spaces available for converters on circuit board, and other “ordinary engineering factors.” *Id.* at 45:21–46:10.

Although Patent Owner may be correct that individual control of multiple voltage regulators would have required additional control circuitry, this does not meaningfully undermine the benefits that Petitioner mentions including permitting power sequencing, independent control of voltages, improved efficiency without generating excess heat or requiring cooling devices, and saving power. Pet. 29–30 (citing *id.* at 14–19; Ex. 1028, 17–20; Ex. 1026, 2–3, 9; Ex. 1028, 30–32; Ex. 1040, 1, 23–24 (Figs. 22–25); Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1058, 5; Ex. 1059, 5:23–30; Ex. 1062, 11, 13; Ex. 1064 ¶ 101; Ex. 1003 ¶¶ 240, 255). Dr. Wolfe testified that, in at least some circumstances, multiple converters may be less complicated, less costly, and take up less space. Ex. 2060, 45:21–46:10, 89:11–15. Patent Owner’s argument that a single converter generating multiple voltages would be better in some circumstances does not negate that Petitioner’s

proposed use of multiple converters for multiple voltages would have been preferred in other circumstances that a person of ordinary skill in the art would have recognized. *See Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021) (“Our caselaw is clear. It’s not necessary to show that a combination is ‘the *best* option, only that it be a *suitable* option.’” quoting *PAR Pharm., Inc. v. TWI Pharms., Inc.*, 773 F.3d 1186, 1197–98 (Fed. Cir. 2014)). Thus, Patent Owner’s argument does not undermine Petitioner’s showing of a motivation to combine Harris and the FBDIMM Standards.

b) *Use of Separate Converters to Generate $V_{DD}/V_{DDQ}/V_{DDL}$ or V_{CC}/V_{CCFBD}*

Patent Owner argues against Petitioner’s showing of a motivation to combine by arguing that neither Harris nor the FBDIMM Standards suggest separate power rails for V_{DD}/V_{DDQ} and V_{DDL} or V_{CC} and V_{CCFBD} for Voltage Mappings A and B. Resp. 24–28. Patent Owner argues that the DDR2 Specification that Petitioner cited requires that the voltages V_{DD} , V_{DDQ} and V_{DDL} are driven from a single power converter output. *Id.* (citing Ex. 1026, 9; Ex. 2061 ¶ 87). The excerpt from the DDR2 Specification (JESD79-2B) in question states as follows:

2.3.1 Power-up and initialization sequence

The following sequence is required for POWER UP and Initialization.

- a) Apply power and attempt to maintain CKE below $0.2 \cdot V_{DDQ}$ and ODT^{*1} at a low state (all other inputs may be undefined.) The power voltage ramp time must be no greater than 20mS; and during the ramp, $V_{DD} > V_{DDL} > V_{DDQ}$ and $V_{DD} - V_{DDQ} < 0.3$ volts.

- V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output. AND
- VTT is limited to 0.95 V max. AND
- Vref tracks $V_{DDQ}/2$.

or

- Apply VDD without any slope reversal before or at the same time as VDDL.
- Apply VDDL without any slope reversal before or at the same time as VDDQ.
- Apply VDDQ without any slope reversal before or at the same time as VTT & Vref.

at least one of these two sets of conditions must be met.

Ex. 1026, 9 (DDR2); *see also* Ex. 1046, 15 (same for DDR3).⁴ The above excerpt from the DDR2 Specification states a “first option” (boxed in blue⁵) that requires V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output, and a “second option” (boxed in red) the defines timings for when the voltages V_{DD} , V_{DDL} and V_{DDQ} must be applied relative to V_{DDL} , V_{DDQ} , and V_{TT} & V_{ref} , respectively. Patent Owner’s argument focuses on the “first option” (boxed in blue above), which specifies that V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output. Resp. 24–28. However, Petitioner contends that the “second option” (boxed in red above) does not specify any converter. Petitioner contends that V_{DD} , V_{DDQ} and V_{DDL} would have been generated with separate converters because they are identified as separate voltages with separate pins (Ex. 1028, 17–20), JEDEC indicates that they can be turned on and off independently (Ex. 1026, 9), and JEDEC indicates that V_{DDL} should use an isolated voltage source (Ex. 1026, 2–3). Pet. 29–30 (citing Ex. 1003 ¶¶ 248–249; Ex. 1062, 13). Petitioner shows that one of ordinary skill in the art would have understood the second option

⁴ Although Petitioner does not expressly rely on JEDEC’s JESD79-2B (DDR2) (Ex. 1026) and JESD79-3A (DDR3) (Ex. 1046) in its combinations, the parties seem to be in agreement that these specifications are part of the background knowledge that a person of ordinary skill in the art would have had. In an obviousness analysis, the background knowledge is properly considered together with the demands known in the design community, and the inferences and creative steps that a person of ordinary skill in the art would employ. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013).

⁵ ⁵ In our opinion, Petitioner should have drawn the blue box to include the following two lines “ V_{TT} is limited to 0.95 V max, AND V_{ref} tracks $V_{DDQ}/2$ ” because of the way the standard presents the two sets of conditions with the word “or” between them and refers to them as “two sets of conditions.” We refer to the two sets of conditions as “options” herein.

(boxed in red above) and other parts of the JESD78-2B specification suggest the possibility of using separate voltage regulators for V_{DD} , V_{DDQ} and V_{DDL} .

Patent Owner argues that a single power converter was required so that the voltages track one another during power up. Resp. 25–26 (citing Ex. 2006, 4; Ex. 2061 ¶ 88; Ex. 1023 ¶ 12). But Dr. Wolfe indicates that the voltages of separate converters can be made to track one another by coupling their feedback, which was a known option. Ex. 2060, 54:25–55:13. Patent Owner does not refute Dr. Wolfe’s testimony.

Patent Owner asserts that Petitioner’s contention that an FBDIMM has separate pins for V_{DD} , V_{DDQ} and V_{DDL} does not support its contention of separate converters, “particularly where it was known that there are stability and timing benefits when using a single source.” Resp. 27 (citing Ex. 2006, 4). Dr. Wolfe explains that the FBDIMM module of Petitioner’s combination would not have separate pins to receive V_{DD} , V_{DDQ} and V_{DDL} but would derive them onboard the module from the 12-volt input pin. Ex. 2060, 35:25–36:15. As to stability, Dr. Wolfe testifies that a feedback loop maintains stability in light of current variations, temperature changes, and other factors. *Id.* at 64:25–65:10. As to timing, Dr. Wolfe indicates that separate converters can track one another by coupling their feedback, and that this was a known option. *Id.* at 54:25–55:13.

Patent Owner argues that it was common for persons of ordinary skill in the art to supply all three voltages using a single power converter. Resp. 28 (citing Ex. 2060, 133:5–134:6, 39:11–40:24). Although using a single power converter may have been the most common approach, this does not negate Petitioner’s showing that a person of ordinary skill in the art would have considered separate converters desirable under certain circumstances.

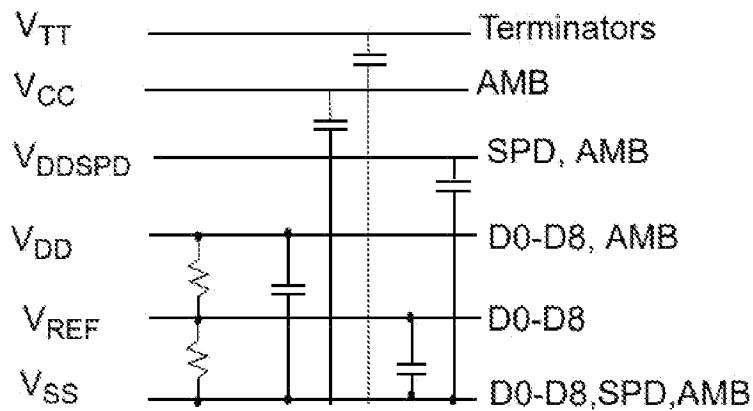
See Elekta Ltd. v. ZAP Surgical Systems, Inc., 81 F.4th 1368 (Fed. Cir. 2023) (quoting *Novartis Pharms. Corp. v. West-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (“Nor does an obviousness showing ‘require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation’”).

Patent Owner similarly argues that V_{CC} and V_{CCFBD} are separate voltages with separate pins, but asserts that Petitioner does not explain why a single on-board V_{CC} power source is insufficient for V_{CC} and V_{CCFBD} power rails when a single set of V_{CC} interface pins providing power from the host met the power needs for both V_{CC} and V_{CCFBD} . Resp. 29–30 (citing Pet. 30–31; Ex. 1028, 11–12; Ex. 2061 ¶ 92). For similar reasons as explained for V_{DD} , V_{DDQ} and V_{DDL} , we disagree. As Petitioner indicates, using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. Reply 15 (citing Ex. 2060, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15; Ex. 2012, 73; Ex. 1003 ¶ 255).

Patent Owner’s arguments do not undercut Petitioner’s showing of a motivation to combine.

c) Motivation for Third Buck Converter for V_{TT}

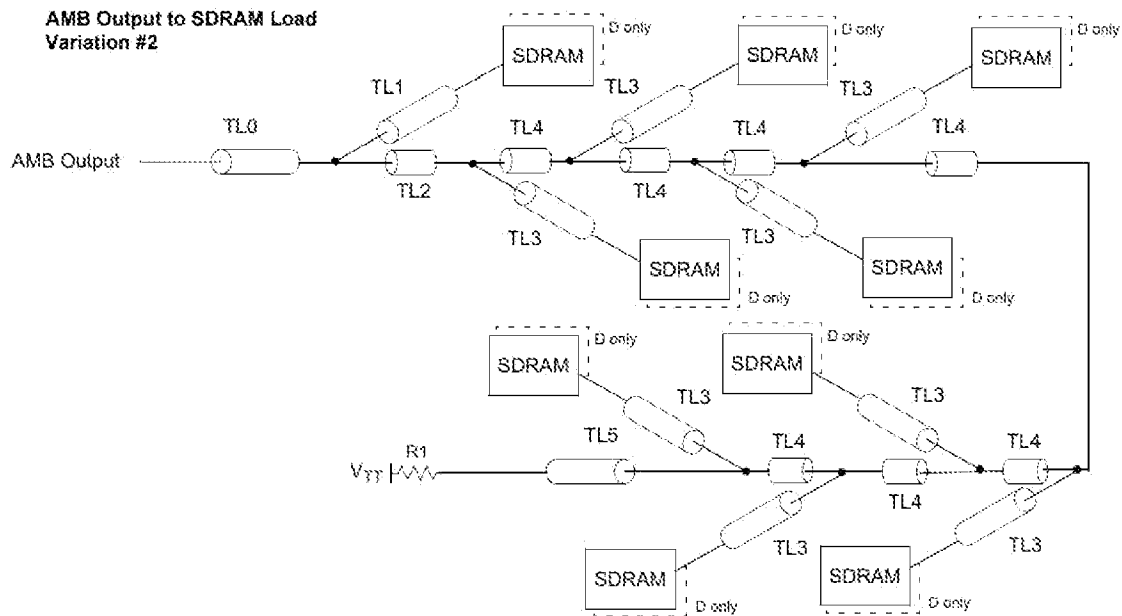
Patent Owner contends that Petitioner’s Voltage Mapping C relies on three different voltages, including a termination voltage, V_{TT} , connected to terminators. Resp. 30 (citing Pet. 26; Ex. 1028, 13, 15–16). For context, the relevant figure from JESD205 is reproduced below:



Ex. 1028, 13. The above figure shows various voltages V_{TT} , V_{CC} , V_{DDSPD} , V_{DD} , V_{REF} , V_{SS} and their relationship to one another according to the JEDEC Specification JESD205. *Id.*

Patent Owner argues that Harris does not disclose generating V_{TT} on its module and argues that it would be supplied from the motherboard instead because this would ensure that all DIMMs have the same termination voltages and eliminate ground loops that could degrade signal integrity. Resp. 31–32 (citing Ex. 2061 ¶ 96). Patent Owner further argues that design complexity favors having a single regulator for a group of DIMMs rather than one regulator per DIMM. *Id.* at 32 (citing Ex. 1027, 20; Ex. 2061 ¶ 97; Ex. 2012, 72). In addition, Patent Owner argues that Petitioner touted JEDEC’s move of voltage regulation from the motherboard to the DIMM as a “major design improvement” and Patent Owner asks, “If it had been so obvious, why did it take the industry so long to make that design improvement?” *Id.* at 32–33. Patent Owner further argues that generating V_{TT} is not one of two choices, and that V_{TT} is generally not provided to DDR2 modules; passive termination is used instead. *Id.* at 33 (citing Ex. 2061 ¶ 98; Ex. 2044, 6; Ex. 2045, 6; Ex. 2046, 4.20.11-6; Ex. 2006, 5).

In Reply, Petitioner argues that Harris teaches generating all of the voltages on the module, including all of the voltages for an FBDIMM, which would include V_{TT} . Reply 16 (citing Inst. Dec. 20–23; Ex. 1023 ¶ 12; Ex. 2060, 30:15–20, 72:22–73:7, 103:11–104:23, 109:5–111:10, 239:8–20). Petitioner contends that “Harris’s module provides the voltages for the buffer 112 to send address and control signals to the DDR memory devices, which the FBDIMM Standards make clear require not just V_{DD} , but also V_{TT} to terminate those signals.” *Id.* at 16–17 (underlining omitted) (citing Pet. 17–18; Ex. 1028, 9, 15, 68). Petitioner reproduces the figure below:



Pet. 17 (citing Ex. 1028, 68). The above figure is from the FBDIMM Standards and shows the net structure routing for addresses and commands from the advanced memory buffer (AMB) to the SDRAMs and how they are terminated by V_{TT} . *Id.* Petitioner contends that V_{TT} is required by the JEDEC Standards to track V_{DD} (Ex. 1028, 15) which is why dual buck converters for generating V_{DD} and V_{TT} were readily available. Reply 17–19

(citing Ex. 1028, 9; Ex. 2060, 72:22–73:7, 196:3–197:7; Ex. 1040, 1, 11; Ex. 1041, 1; Ex. 1048).

Petitioner further argues that Patent Owner’s “suggestion of producing V_{TT} on the motherboard—where V_{TT} could change with each new generation of memory devices—would defeat the benefit of Harris’s invention, e.g., ‘a technology-independent voltage distribution scheme’ to ‘provide upgrades to next generation DRAM technology in a cost-effective manner.’” *Id.* at 19 (citing Ex. 1023 ¶¶ 2, 19).

We agree with Petitioner that the combination of Harris and the FBDIMM Standards at least suggests that V_{TT} can be generated onboard the memory module. Reply 16–19 (citing Pet. 17–18; Inst. Dec. 20–23; Ex. 1023 ¶¶ 2, 12, 19; Ex. 1040, 1, 11; Ex. 1041, 1–2, 7–9; Ex. 1048; Ex. 1075, 112:12–114:8; Ex. 2060, 30:15–20, 57:17–20, 58:18–59:10, 72:22–73:7, 103:11–104:23, 109:5–111:10, 140:15–24, 239:8–20). Harris indicates that its on-board voltage regulator converts externally supplied voltage into appropriate local voltage levels for powering memory devices of the memory assembly module. Ex. 1023, code (57), ¶¶ 12, 16, Fig. 2, step 204. Petitioner shows that V_{TT} is needed onboard a memory module to terminate address and control signals. Reply 16–17 (citing Pet. 17–18; Ex. 1028, 9, 15, 68). In addition, Petitioner shows that V_{TT} is required by JEDEC to track V_{DD} , which is why dual buck converters for generating these voltages were commercially available. Reply 17–19 (citing Ex. 1028, 9; Ex. 2060, 72:22–73:7, 196:3–197:7; Ex. 1040; Ex. 1041; Ex. 1048). In light of these facts and the teachings of Harris and the FBDIMM Standards, Petitioner has shown that a person of ordinary skill in the art would have considered adding to the memory module a third buck converter to generate V_{TT} .

Patent Owner further argues that “even if V_{TT} is generated on module, there is no reason that it should be generated by a buck converter.”

Resp. 33. Patent Owner contends that, “[a]s Micron noted, an LDO would be sufficient to power V_{TT} .” *Id.* (citing Ex. 2006, 7; Ex. 2007–2010; Ex. 2050). Patent Owner contends that an LDO is much smaller than a buck converter, including a controller and discrete components such as an inductor, which makes the LDO preferable given the space constraint. *Id.* (citing Ex. 2061 ¶ 99). Patent Owner also argues that an LDO is preferred when the current load is below 1A. *Id.* (citing Ex. 2061 ¶ 99; Ex. 2047, 21; Ex. 2048, 23; Ex. 2049, 20; Ex. 1040, 23–24, Figs. 22–25).

In Reply, Petitioner contends that buck converters were commercially available for V_{TT} and provided high efficiency compared to an LDO. Reply 19 (citing Ex. 1075, 112:12–114:8 (LDO may be only 10% efficient); Ex. 2060, 57:17–20, 58:18–59:10 (buck converters up to 98% efficient), 140:15–24 (trend has been to use buck converters)). We agree with Petitioner that, under appropriate circumstances, one of ordinary skill in the art would have chosen a buck converter over an LDO. Indeed, Patent Owner’s arguments simply underscore that the person of ordinary skill in the art would have been aware of, and capable of using, various known options for voltage conversion.

d) Conclusion on Motivation to Combine Harris and the FBDIMM Standards

Petitioner has sufficiently shown that a person of ordinary skill in the art would have combined Harris and the FBDIMM standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶ 9. We agree that one of ordinary skill in the art would have

recognized this as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9. For these reasons and others set forth above, we determine that one of ordinary skill in the art would have combined Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at the claims.

4. *Analysis of Independent Claim 1*

a) *Limitation 1.a: “A memory module comprising:”*

Petitioner asserts that Harris’s memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed “memory module.” Pet. 19–20 (citing Ex. 1023 ¶¶ 9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 217–223; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1 in Figure 3.

Patent Owner does not specifically respond to Petitioner’s contention for the preamble. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that Harris teaches the preamble.

b) *Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”*

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. For example, Petitioner notes that Harris discloses that its double-rank DIMMs may be accommodated on both sides of a printed circuit board (PCB). Pet. 20 (citing Ex. 1023 ¶ 13). Petitioner

further notes that a PCB may be referred to as a “memory board” or “raw card.” *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 224–226).

As for the PCB “having an interface configured to fit into a corresponding slot connector of a host system,” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 227–228).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “configured to couple power, data, address and control signals between the memory module and the host system.” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A (Ex. 1023 ¶ 10, 12, 19). Petitioner contends that the Harris’s buffer 112 in Figure 1A is called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 22. Petitioner indicates that buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Harris’s Figure 1A. Pet. 22–25 (citing Ex. 1023 ¶ 9 (disclosing that “buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data space at least for a portion of memory devices 110-1 through 110-N”)). In addition, Petitioner argues that the FBDIMM Standards indicate buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

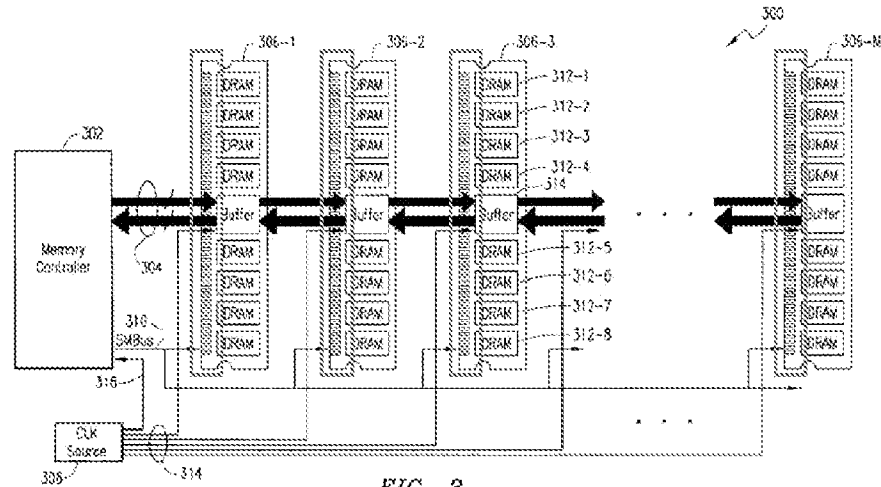
(1) *Power Via Edge Connection*

Claim 1 recites “the interface including a plurality of edge connections configured to couple power . . . between the memory module and the host system.” Ex. 1001, 38:22–25. Patent Owner argues that “all claims of the ’054 patent require that the memory module includes a module interface that fits into a memory slot of a host system and receives voltage/power from the host system via that interface.” Resp. 4–5.

According to Patent Owner, Harris does not disclose such a memory module. *Id.* at 5. Specifically, Patent Owner argues that “Harris supposedly provides a ‘technology-independent voltage distribution scheme for memory devices wherein system board power supply and associated voltage plane(s) are eliminated.’” *Id.* at 6 (citing Ex. 1023 ¶ 19) (emphasis omitted). Patent Owner argues that Harris’s memory module connects to the external voltage source from the side and not from its edge connections. *Id.* at 7. Patent Owner contends that supplying power to a DIMM from its side was known in the art. *Id.* (citing Ex. 2035; Ex. 2036, 41–42; Ex. 2061 ¶ 59).

Petitioner contends that Patent Owner does not dispute that the FBDIMM Standards teach supplying power via edge connections, which Patent Owner’s expert, Dr. Mangione-Smith, admits were “standard.” Reply 3 (citing Pet. 16–21; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9). Petitioner further contends that Patent Owner argues that Harris’s “externally supplied voltage” requires power external to the entire host system (*id.* (citing Resp. 4–8)), when “external” relates to the DIMM memory module and does not exclude power from the host system, including from edge connections. *Id.* (citing Ex. 2060, 66:7–19, 67:20–68:21, 91:22–92:7, 129:24–130:17; Pet. 21–22). Petitioner disagrees with

Patent Owner's assertion that Harris's memory module would never utilize edge connections for power. Reply 3-4. To explain its position, Petitioner provides an annotated version of Harris's Figure 3 below.



Petitioner contends that Harris's Figure 3 above shows memory boards 306 connected to the host system only through their edge connections (annotated by Petitioner in red), implying that power would also come from those edge connections. Reply 4 (citing Ex. 1023, Fig. 3). Petitioner contends that Harris's Figure 3 is nearly identical to an Intel drawing where memory modules admittedly receive power through the edge connections from the host system. *Id.* (citing Ex. 1075, 171:3-17; Ex. 2101, 4).

Petitioner also contends that Harris discloses that each memory board receives a supply voltage which may be sourced from either the host system or a separate voltage source. *Id.* at 4-5 (citing Ex. 1075, 167:23-168:1; Ex. 1023 ¶ 17). Petitioner asserts that there is no dispute that the supply voltage was commonly provided through edge connectors alongside memory controller signals. *Id.* at 5 (citing Ex. 2060, 131:3-5).

We agree with Petitioner that Harris's "external voltage source" covers sources originating from the host system (as well as sources external

to the host system). Harris states as background that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Harris also discloses that “external voltage sources may comprise any combination of *known* or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14 (emphasis added). The parties agree that supplying power from a host system to a memory module was ‘known’ in the art, as referenced by Harris. Ex. 2060, 131:3–5; Ex. 1075, 162:18–25. And, as Petitioner notes, Harris’s Figure 3 shows that the memory boards have only edge connections, suggesting they receive their power from them. Ex. 1023, Fig. 3; Ex. 1003 ¶ 229. These facts support the conclusion that Harris’s external voltage source may originate from the host system. Accordingly, we do not agree with Patent Owner’s arguments that Harris is limited to using only external voltage sources unrelated to the host system. Resp. 4–14; Sur-Reply 2–6.

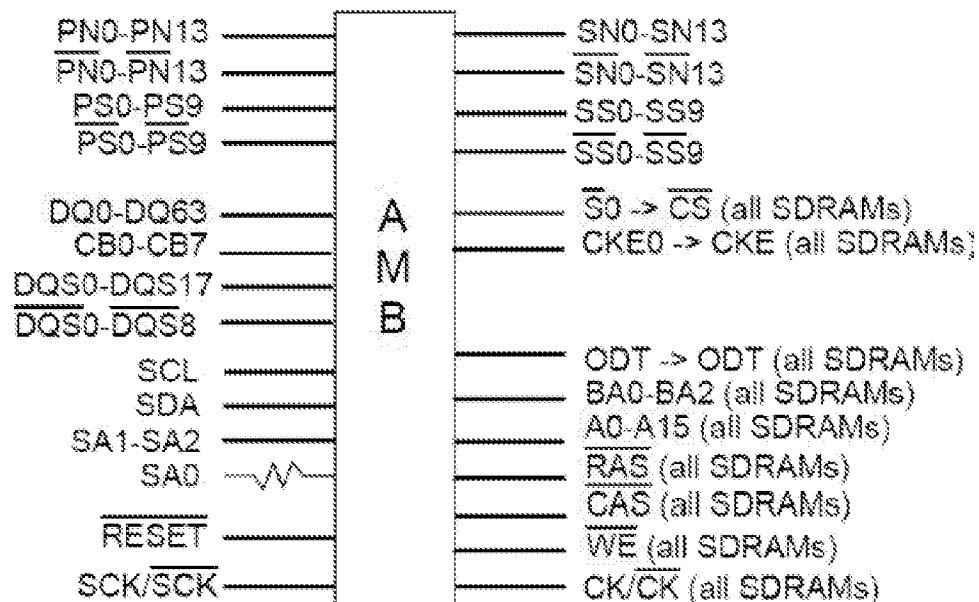
Furthermore, Petitioner demonstrates that the FBDIMM Standards show that memory modules receive power from their edge connections, and Petitioner relied on the FBDIMM Standards as disclosing this feature of the claims. *See* Pet. 16–21; Ex. 2060, 131:3–5; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9; Ex. 1028, 97 (GF-9), 103.

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards discloses this limitation.

*(2) Data, Address, and Control Signals Via
Edge Connection*

Claim 1 recites an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the

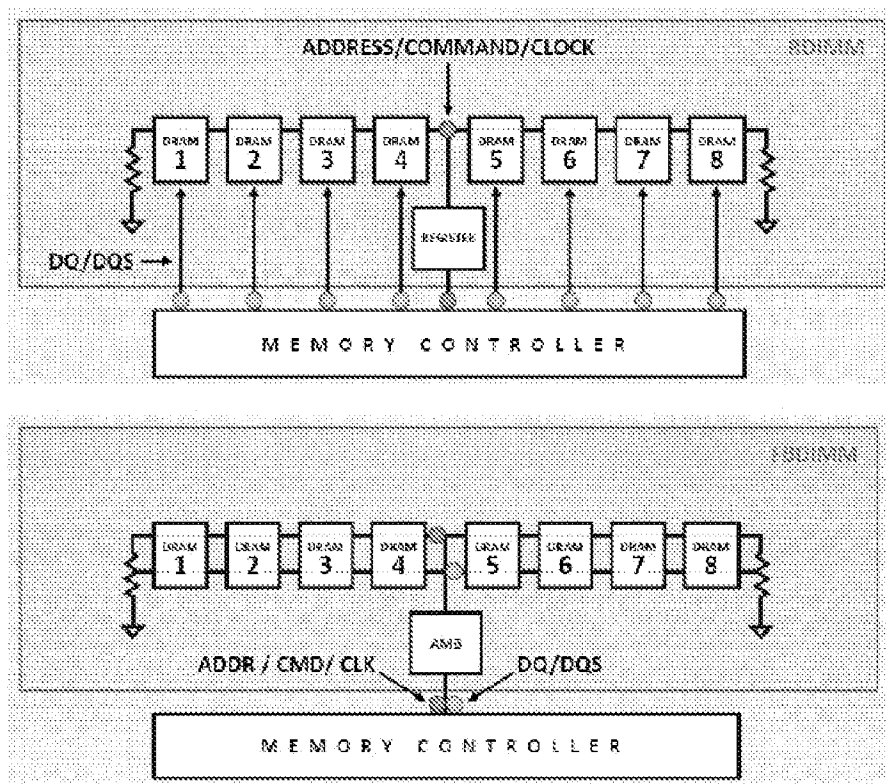
memory module and the host system.” Ex. 1001, 38:23–25. Patent Owner argues that the DIMM and AMB do not receive data signals (DQ0–DQ63) or address and control signals (A0–A15, RAS, CAS, WE) from the host. Resp. 14 (citing Pet. 22–25). According to Patent Owner, these signals are instead generated by the AMB based on decoded FBDIMM channel signals (PS[9:0] and PS[9:0]bar). *Id.* at 14–18 (citing Ex. 2061 ¶¶ 72; Ex. 1028, 29; Ex. 1027, 3–4; Ex. 2039, 2; Ex. 2101, 4; Ex. 2040, 1). Patent Owner argues that Petitioner’s evidence does not show that the data, address and control signals are received from the host. *Id.* at 18. Specifically, Patent Owner relies on the figure below from the FBDIMM Standards (*see* Sur-Reply 7 (citing Ex. 1028, 13)).



According to Patent Owner, the above figure shows that data, address and controls signals (highlighted in yellow) are generated and outputted by the buffer, and are not received from the DIMM interface. Sur-Reply 6–7 (citing Pet. 23–25; Resp. 14–18; Ex. 2061 ¶¶ 72–74). Patent Owner contends this does not comport with the claim language which requires an

“interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” *Id.* at 7 (citing Ex. 1001, 38:23–25). Patent Owner argues that Dr. Wolfe merely stated that the AMB outputs “information” to the DRAMs but the claims require specific “signals,” not just “information.” *Id.* at 8 (citing Ex. 1003 ¶ 230; Ex. 2060, 7:20–11:6; Ex. 1001, 38:23–25). Patent Owner argues that the outputs from the AMB are not transmitted over FBDIMM’s edge connections. *Id.*

Petitioner argues that Patent Owner’s technology tutorial indicates that an FBDIMM receives address and control signals similar to an RDIMM, as shown below:



Reply 9 (citing Ex. 1077, 8–9; Ex. 1075, 91:23–92:19, 95:14–96:13, 97:16–98:18). The figures above show that the memory controllers transmit

address, command and clock signals (shown in red), or information containing these signals, to the register (RDIMM) or AMB (FBDIMM), which transmits them to the DRAMs. *Id.*; Sur-Reply 10.

Petitioner contends that the host system provides address, command and clock signals to the AMB encoded as a packetized, serialized signal on fewer wires than would be required to receive them as separate, parallel signals. Reply 9–10 (citing Resp. 14; Ex. 2061 ¶ 31; Ex. 1075, 155:22–157:1, 212:3–8, 213:3–215:20, 219:13–220:9, 226:7–228:8; Ex. 2060, 8:3–11:6). Petitioner alleges that Patent Owner is in essence attempting to rewrite the claims to require dedicated pins for these signals, which the claims do not require. *Id.*

Patent Owner argues that it is not attempting to rewrite the claims to require “dedicated pins” as Petitioner alleges. Sur-Reply 9 (citing Reply 9–10). Patent Owner asserts that the pin names indicate what “signals” are exchanged at the interface between the host and the FBDIMM, as shown in the figure below.

Pin Name	Pin Description	Pin
FB-OBMM Chemical Signals		
SDN	System Clock (note: positive logic)	1
SDN	System Clock Input (negative logic)	2
PSD(1:32)	Primary Northbound Data (positive logic)	3-34
PSD(1:32)	Primary Northbound Data (negative logic)	35-66
PSD(33)	Primary Southbound Data (positive logic)	67
PSD(33)	Primary Southbound Data (negative logic)	68
SDN(1:32)	Secondary Northbound Data (positive logic)	69-100
SDN(1:32)	Secondary Northbound Data (negative logic)	101-132
SDS(33)	Secondary Southbound Data (positive logic)	133
SDS(33)	Secondary Southbound Data (negative logic)	134
PSD(33:2)	For an external precision calibration, spectral calibration up limit	135
OBMM Interface Signals		
SDS(33:2)	Data Receive (positive logic)	136
SDS(33:2)	Data Receive (negative logic)	137
PSD(1:32)/SDS(33)	Data Receive (pin 1:32 only), positive logic. These signals are shared due to no CSNMS on analog	138-149
SDS(1:32)	Data Receive (pin 33:65 only), negative logic	150-161
CSNMS(33)	None	162
CSNMS(33)	CSNMS	163
ATD(1:32), ATD(33)	Addressed, ATD is part of the are change command	164-165
BSN(1:32), BSN(33)	Block Addressing	166-167
SDN(1), SDN(3)	Start of command with SDN(1), SDN(3) and CSNMS	168
SDN(4), SDN(5)	Part of command with SDN(1), SDN(3) and CSNMS	169
SDN(6), SDN(7)	Part of command with SDN(1), SDN(3) and CSNMS	170
SDN(8), SDN(9)	End of Termination Block	171
CSNMS(1:32), CSNMS(33)	Clock Enable line per pin	172-183
CSNMS(1:32), CSNMS(33)	High Current (one pin per)	184-195
CLK(33)	CLK(33) used on 1 and 10 module OBMMs, CLK(33:2) used on 33 module OBMMs. CLK(33:2) enables the bus clock (33:33:2) when CLK(33) is low	196
CLK(33:2)	Negative lines for CLK(33:2)	197
SDN(1), CSNMS(1:32)	OBMM Temperature. Common return per for SDN(1), CSNMS(1:32)	198
SDN(1), CSNMS(1:32)	OBMM Temperature. Receiver connected to common return per CSNMS(1:32)	199
SDN(1), CSNMS(1:32)	OBMM Temperature. Receiver connected to common return per CSNMS(1:32)	200
SDN(1), CSNMS(1:32)	OBMM Temperature. Receiver connected to CSNMS(1:32)	201
SDN(1), CSNMS(1:32)	OBMM Temperature. Receiver connected to CSNMS(1:32)	202

Ex. 1028, 29. The figure above shows Patent Owner’s highlighting to show the “FB-DIMM Channel Signals” including PS[9:0] and PS[9:0]bar, and the “DDR2 Interface Signals” including DQ0–DQ63, A0–A15, RAS, CAS, and WE. Patent Owner states that all memory control for the DRAM resides in the host, and argues that the fact that all read, write, and configuration accesses are addressed to the DIMM does not undermine that the signals Petitioner relies on are generated and outputted by the buffer, and not exchanged at the interface between the FBDIMM and the host. Sur-Reply 9–10 (citing Ex. 2060, 7:20–11:6; Ex. 2060 ¶ 230).⁶

⁶ Patent Owner’s citation to “Ex. 2060 ¶ 230” appears to be an error and we cannot determine from the context what Patent Owner intended to cite.

Petitioner further argues that the FBDIMM is a preferred embodiment in the '054 patent, and excluding a preferred embodiment from the claims is “rarely, if ever correct.” Reply 10 (citing *Kaufman v. Microsoft Corp.*, 34 F.4th 1360, 1372 (Fed. Cir. 2022)). To the contrary, Patent Owner argues that the claims need not encompass FBDIMM embodiments, which are disclosed but unclaimed subject matter. Sur-Reply 10 (citing *Maxwell v. J. Baker, Inc.*, 86 F.3d 1098, 1107 (Fed. Cir. 1996); Ex. 1001, 21:38–55).

According to Petitioner, Patent Owner concedes that the signals (PS[9:0] and PS[9:0]bar) received by the AMB result in data, address, and control signals needed by DDR2 SDRAMS. Reply 10 (citing Ex. 2061 ¶ 31). Petitioner further notes that the FBDIMM Standards state that all memory control for the DRAM resides in the host, including memory request initiation, and that the AMB acts as a DRAM memory buffer for all read, write, and configuration accesses addressed to the DIMM. *Id.* (citing Ex. 1027, 1). As a buffer for all such commands to the FBDIMM, Petitioner asserts, the AMB must necessarily couple data, address, and control signals from the host system to the memory module as the claims require. *Id.*

Petitioner and Patent Owner agree that FBDIMM AMB receives data, address, and control signals encoded in packetized, serialized form at its edge connections via the signals PS[9:0] and PS[9:0]bar, and that the AMB uses these signals to generate data, address and command signals including DQ0–DQ63, A0–A15, RAS, CAS, and WE provided to the DIMMs onboard the memory module. Pet. 22–25; Sur-Reply 10. The claims merely require an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” Ex. 1001, 38:23–25. We determine that the data, address, and

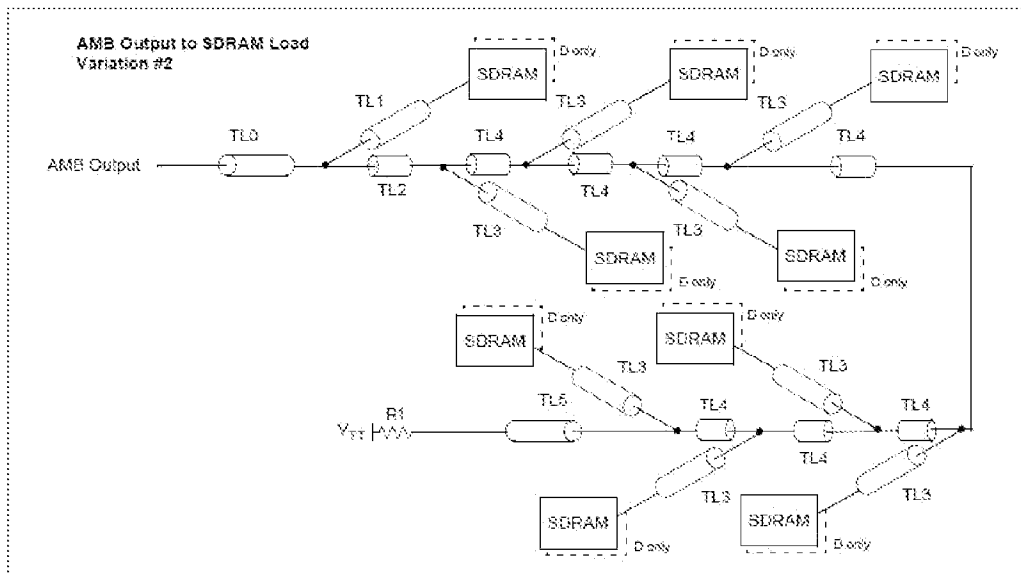
control signals of Harris and the FBDIMM Standards as received at edge connections coupling the memory module and host system satisfy this claim limitation, even though the signals are in encoded, packetized, or serialized form and thus may be received at the same pin or pins. That the signals are encoded, packetized, and serialized does not change the fact that they are data, address, and control signals. The claims do not require these signals to be received at the edge connections in any particular form or on any particular pins. *See In re Self*, 671 F.2d 1344, 1348 (CCPA 1982) (stating that limitations not appearing in the claims cannot be relied upon for patentability).

Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitation 1.b of claim 1, and Patent Owner's arguments do not undermine Petitioner's showing.

c) Limitation 1.c: "a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages"

Petitioner contends that Harris and the FBDIMM Standards disclose limitation 1.c of claim 1 of the '054 patent. Pet. 26–30. Specifically, Petitioner contends that Harris's Voltage Regulator Module 102 corresponds to the claimed "voltage conversion circuit." Pet. 26 (citing Ex. 1023 ¶ 10, Fig. 1A). Petitioner contends that the claimed "at least three regulated voltages" correspond to voltage 106 (V_{cc}) and voltage 108 (V_{dd}) as shown in Harris's Figure 1A, and a third voltage (V_{TT}) which is not actually shown in Harris's Figure but is shown in the FBDIMM Figure below.

DDR2 SDRAM Fully Buffered DIMM Design Specification DDR2 Fully Buffered DIMM Wiring Details
Net Structure Routing for Address/Command to SDRAM (Raw Cards B, C)



Ex. 1028, 68. The figure above shows voltage V_{TT} terminating an array of SDRAM devices via resistor R1.

As to the voltages being “regulated,” Petitioner argues Harris discloses that “at least one on-board voltage regulator” is “capable of generating tightly-controlled voltage levels.” Pet. 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 236).

As to the “at least three buck converters,” Petitioner contends that Harris teaches using buck converters to provide the three regulated voltages above. Pet. 28 (citing Ex. 1003 ¶¶ 238–241). Petitioner contends Harris teaches using a “high-frequency switching voltage converter capable of generating tightly-controlled voltage levels” to provide each needed on-board regulated voltage. *Id.* (citing Ex. 1023 ¶¶ 10, 12). Petitioner contends it would have been obvious to use a “buck converter” to convert higher input voltage (e.g., 12V) to the lower output voltage (3.5V or less), and there would have been a reasonable expectation of success because buck

converters were well-known “switching” devices commonly used to step down the voltage between its input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 147–150, 239–241; Ex. 1058, 3, 5, 12–16 (1995 Lenk textbook showing buck, boost, and buck-boost circuits); Ex. 1032, 161 (1995 Mohan textbook showing various step-down (buck) converters); Ex. 1030, 2:32–43, 5:39–44; Ex. 1024, Fig. 6 (Amidi showing DC/DC buck converter); Ex. 1050, 1:21 (identifying buck converters as one of “the most basic building blocks in power electronics”)). Petitioner contends buck converters were well-known for their efficiency in generating step down voltages without generating excess heat or requiring large cooling devices. *Id.* at 29 (citing Ex. 1003 ¶ 240; Ex. 1059, 5:23–30; Ex. 1058, 5; Ex. 1040, 1, 23–24; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1062, 11; Ex. 1064 ¶ 101). Petitioner contends that it would have been obvious to use at least three converters in Harris given the need for at least three different voltages in the FBDIMM Standards. Pet. 29–30 (citing Ex. 1028, 17–20; Ex. 1026, 2–3, 9; Ex. 1003 ¶¶ 248–249, 255; Ex. 1062, 13; Ex. 1028, 30–32).

For the reasons discussed above in Section II.D.3, we disagree with Patent Owner’s arguments regarding the use of three buck converters.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitation 1.c of claim 1.

d) Limitation 1.d.1: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages”

Petitioner asserts that Harris discloses a “a plurality of components coupled to the PCB” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 261–266). Petitioner further contends that these components are each coupled to at least one regulated voltage of the at least three regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches limitation 1.d.1 of claim 1.

e) Limitation 1.d.2: “the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 31–32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 267–271). Petitioner contends that a person of ordinary skill in the art would have known that according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* (citing Ex. 1028, 9; Ex. 1045, cover; Ex. 1026, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

f) Limitation 1.d.3: “a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections”

Petitioner contends that the “first circuit” corresponds to buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–33 (citing *id.* at 20–25; Ex. 1023, Figs. 1A, 3; Ex. 1003 ¶¶ 272–277). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8.

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

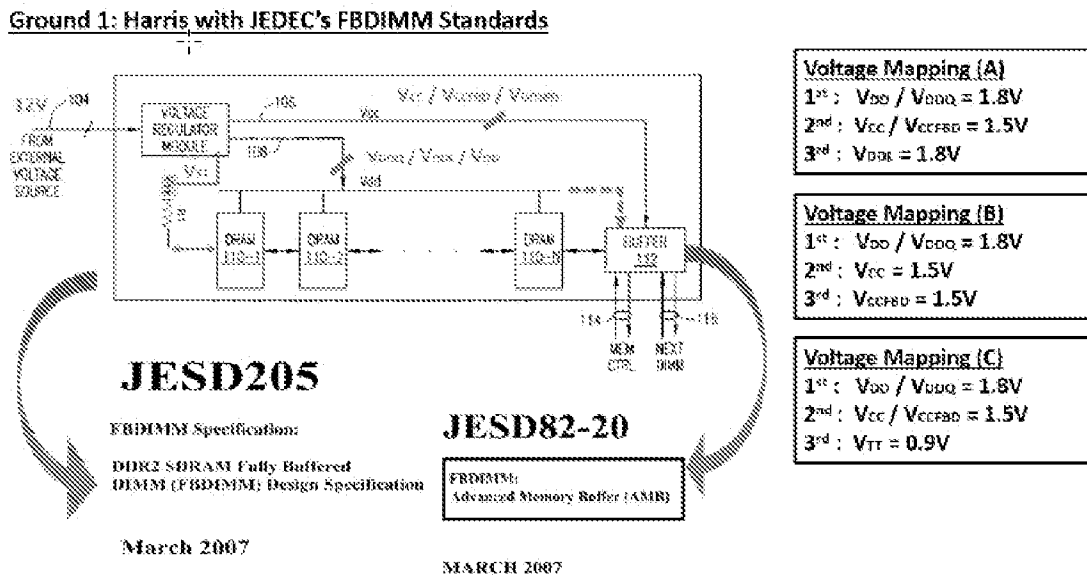
Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

g) Limitation 1.d.4: “wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and”

Petitioner contends that Harris’s buffer corresponds to the first circuit and is coupled to “first” (e.g., V_{DD} or $V_{DDQ} = 1.8V$) and “second” (e.g., V_{CC} or $V_{CCFBD} = 1.5V$) “regulated voltages of the at least three regulated

voltages.” Pet. 33–34 (citing *id.* 14–19, 27; Ex. 1003 ¶¶ 278–284).

Petitioner provides an annotated version of Harris’s Figure 1A shown below.



As shown above, Petitioner fills in Harris’s Figure 1A with additional information from the FBDIMM Standards to show that Harris’s buffer 112 would be understood by a person of ordinary skill considering the FBDIMM Standards to couple to the voltages V_{DD} or V_{DDQ} and V_{CC} or V_{CCFBD} .

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation 1.d.4 of claim 1.

h) Limitation 1.d.5: “wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34 (citing *id.* 14–19, 27; Ex. 1003 ¶¶ 285–287).

Specifically, Petitioner contends that the SDRAM devices from Harris’s

Figure 1A are coupled to the first regulated voltage (V_{DD} or $V_{DDQ} = 1.8V$) of the “at least three regulated voltages.” *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches limitation 1.d.5 of claim 1.

i) Conclusion for Claim 1

For the reasons explained in Section II.D.3, Petitioner shows that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches each limitation of claim 1. Accordingly, Petitioner has shown that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. Analysis of Dependent Claims

Claim 2 depends from claim 1 and recites “wherein the first regulated voltage has a first voltage amplitude, and the second regulated voltage has a second voltage amplitude, and wherein a first one of the first and second voltage amplitudes is less than a second one of the first and second voltage amplitudes.” Ex. 1001, 38:45–50. Petitioner contends that, as taught by the FBDIMM Standards, the first voltage amplitude may be 1.8V and the second voltage amplitude may be 1.5V which is less than the first voltage amplitude of 1.8V. Pet. 34–35 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 288–296). Patent Owner presents no arguments specific to claim 2.

Claim 3 depends from claim 1 and recites “wherein a third regulated voltage of the at least three regulated voltages has a voltage amplitude of 1.8 volts.” Petitioner contends that the third regulated voltage is V_{DDL} with a value of 1.8V as shown in the FBDIMM Standards (see previous figure—“voltage mapping(A)”). Pet. 35 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 297–301).

Patent Owner argues that because claim 3 relies solely on Voltage Mapping A, Petitioner fails to show obviousness over the combination of Harris and the FBDIMM Standards. Resp. 28. Patent Owner argues that “[a]s for Mapping A, Petitioner does not explain why a single on-board V_{CC} power source is insufficient for $[V_{CC}]$ and $[V_{CCFBD}]$ power rails when a single set of $[V_{CC}]$ interface pins providing power from the host met the power needs for both $[V_{CC}]$ and $[V_{CCFBD}]$.” *Id.* at 29. We addressed this argument in Section II.D.3.c explaining that Petitioner shows that using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. We find no flaw in Petitioner’s reliance on Voltage Mapping A for claim 3.

Claim 15 depends from claim 1 and recites “wherein two of the at least three buck converters are configured to operate as a dual-buck converter.” Ex. 1001, 39:59–61. Petitioner contends at the time there were many commercially available products that could output two (or more) regulated voltages using buck converters, and thus “it would be obvious to implement any two of the regulated voltages as a ‘*dual buck converter*’ to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design.” Pet. 35–36 (citing Ex. 1003

¶ 439; *Qualcomm*, 21 F.4th at 797–99 (Fed. Cir. 2021)). Petitioner contends that the Murata MPD4S014S dual buck converter, Texas Instruments TPS51020 dual buck converter, and Fairchild FAN5026 dual-output PWM controller were commercially available and would have been suitable to implement for use in Harris in view of the FBDIMM Standards. Pet. 37–40 (citing Ex. 1003 ¶¶ 439, 443–444, 446–447; Ex. 1023 ¶ 10; Ex. 1032, 161–64; Ex. 1040, 1, 10–11; Ex. 1041, 1–2, 7–9; Ex. 1042, 16; Ex. 1047, 3:49–50, 4:7–10, 4:37–56, 5:4–13, 5:49–59, Fig. 2; Ex. 1048, 1–2; Ex. 1050, 2:19–20, 3:36–37; Ex. 1058, 5; Ex. 1073, 49–50, 58).

Patent Owner argues that claim 15 of the '054 patent provides “two of the at least three buck converters are configured to operate as a dual-buck converter,” making clear that the “dual buck converter” includes two buck converters. Resp. 14 (citing Ex. 1001, 39:59–61). Patent Owner argues that Harris does not characterize its “high-frequency switching voltage converter” as a “dual” converter. *Id.* (citing Ex. 1023 ¶ 10, Fig. 1A). Patent Owner also contends Harris’s use of the indefinite article “a” before mentioning this converter means that only a single converter is used in Harris’s voltage regulator module. *Id.*

Petitioner correctly asserts that Harris states that “at least one on-board voltage regulator module (VRM)” may be used to generate the voltages V_{DD} and V_{CC} . Reply 12 (citing Ex. 1023 ¶ 10, Fig. 1A). Also, Harris’s claims 1 and 10 recite “at least one voltage regulator module” which covers the possibility of multiple regulators. Although Harris expresses a preference for implementing a voltage regulator module using “a high-frequency switching voltage converter” (emphasis added) which may be understood to mean a single converter, Harris does not preclude using

multiple regulators to generate the voltages V_{DD} and V_{CC} . Ex. 1023 ¶ 10. As Petitioner states, Harris’s voltage regulator module receives one input voltage and generates two output voltages. Reply 12 (citing Ex. 1023, Fig. 1A). One of ordinary skill in the art would have understood that there are two options: to use a single converter to generate both voltages, or to use multiple converters to generate the voltages. “When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill in the art has good reason to pursue the known options within his or her technical grasp.” *KSR*, 550 U.S. at 421. Petitioner shows that dual buck converters capable of generating the two voltages from the one input were known in the art and used for DIMMs, including the TPS51020 dual buck converter (Ex. 1040), the FAN5026 dual-output PWM controller (Ex. 1041), and the MPD4S014S dual buck converter (Ex. 1042). Armed with the foregoing knowledge, a person of ordinary skill in the art would have understood one option for implementing Harris’s voltage regulator module would be to use a dual buck converter notwithstanding Patent Owner’s arguments to the contrary.

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris in combination with the FBDIMM Standards teaches the limitations of claims 2, 3, and 15.

6. *Determination for Ground 1*

Petitioner has shown that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at the memory modules recited in claims 1–3 and 15. Accordingly, Petitioner has shown that claims 1–3

and 15 are unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

E. Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi

Petitioner contends claims 1–30 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 41–70. For the reasons that follow, Petitioner shows by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

1. Amidi (Ex. 1024)

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled “Clock and Power Fault Detection for Memory Modules.” Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 11.

Amidi’s Figure 5 is reproduced below.

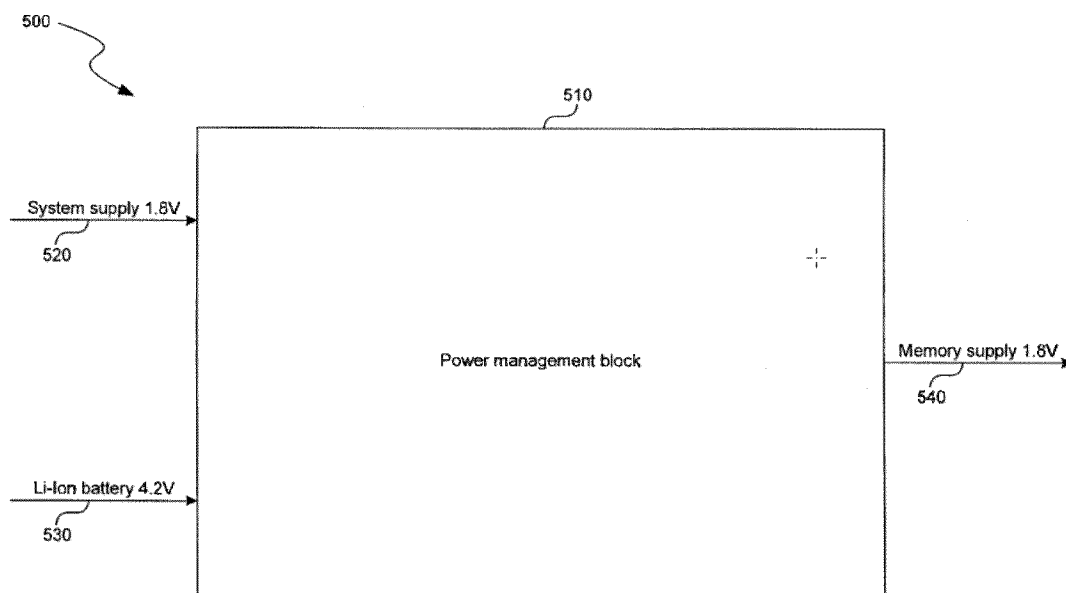


FIG. 5

Amidi's Figure 5 above illustrates power management block 510 that receives incoming system supply 520 and incoming battery supply 530, and generates outgoing memory power supply 540 which is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Fig. 5, Fig. 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 41. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 41–42 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 171–177). Petitioner further notes that Amidi's power management block could be modified easily to work with Harris's FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 42 (citing Ex. 1024, Fig. 5; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 173–174). According to Petitioner, a person of ordinary skill in the art would have considered it obvious that the 12V external supply would be stepped-down with a buck converter to a 5V supply for charging Amidi's battery and that Amidi's battery voltage would be stepped-up with a boost converter to the 12V level used by Harris's memory module. *Id.* at 42–43 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 173–174). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed,

and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 43 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi’s battery backup mode is similar to the S3 power-saving mode of Harris’s FBDIMM memory module and the FBDIMM standards, such that a person of ordinary skill in the art would have been motivated to combine their teachings. *Id.* at 43–45.

In sum, Petitioner contends that a person of ordinary skill in the art “would have been motivated to implement Harris’s FBDIMM memory module with the functionality of Amidi’s power management and logic blocks for detecting power disruptions and switching over to battery backup when needed.” Pet. 44 (citing Ex. 1003 ¶¶ 171–176). Petitioner contends that this “straightforward modification of Harris’s memory module in view of Amidi and the knowledge of a [person of ordinary skill in the art] simply uses a known technique (e.g., Amidi’s battery backup techniques) to improve a similar device (e.g., Harris’s memory module) in the same way (e.g., to provide a backup power supply using a battery).” *Id.* at 44 (citing Ex. 1003 ¶ 177); *see KSR*, 550 U.S. at 418. Petitioner further contends that “the modification merely applies a known technique (e.g., providing a backup power supply) to a known device (e.g., a memory module) that is ready for improvement to yield predictable results (e.g., redundancy when the system supply or clock fails).” Pet. 44 (citing Ex. 1003 ¶ 177); *see KSR*, 550 U.S. at 417.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:

Ground 2: Ground 1 and Battery Backup of Amidi

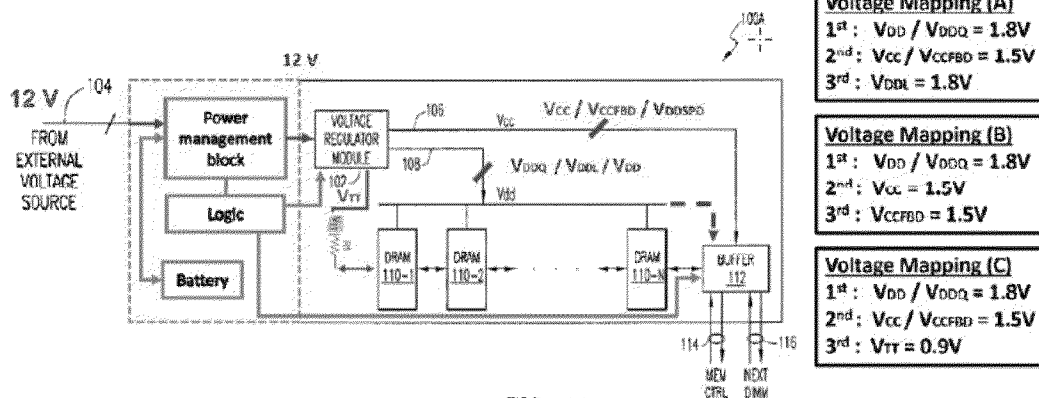


FIG. 1A

Pet. 45. Above, Petitioner has annotated Harris’s Figure 1A to show features added from the teachings of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue).

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Resp. 34–40. Specifically, Patent Owner argues that Harris has “logic 124” that selects a different voltage source if power becomes unavailable on one of the supply voltage paths 120-1 through 120-K. *Id.* at 34–35 (citing Ex. 1023, Fig. 1B; Ex. 2061 ¶ 106). Patent Owner further argues that substituting Amidi’s battery backup solution for Harris’s redundant power sources would require substantial space to accommodate the lithium battery and associated battery charging and monitoring circuits and power management block. *Id.* at 37–38 (citing Ex. 1024, Figs. 4–6; Ex. 1023 ¶¶ 2, 13; Ex. 1061 ¶ 106, n.5; Ex. 2061 ¶ 107). Patent Owner argues that Petitioner “has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the combination would even fit onto Harris’ circuit board” and that “Petitioner has also presented no evidence that a

[person of ordinary skill in the art] would expect Harris’ memory module to be able to accommodate three buck converters, a lithium battery and its associated charging and monitoring circuits, and Amidi’s power management module, at least not without undergoing substantial redesign.” *Id.* at 38–40 (citing Ex. 2101, 20; Ex. 1055, Fig. 4; Ex. 2061 ¶ 110).

We agree with Petitioner that Amidi provides a redundant power supply on the memory module that is independent from the host, and that a person of ordinary skill in the art would have seen the value of adding a redundant power supply onboard Harris’s memory module at least for some practical applications. Reply 19–20 (citing Pet. 41–42; Ex. 1003 ¶ 172; Ex. 1024, 1:28–35, 2:6–26). Although Harris mentions an “external voltage source,” it does not mention the term “battery.” Hence, Patent Owner’s arguments that Amidi provides a solution to a problem that Harris already solves is not correct. *See* Ex. 1003 ¶ 172 (“Amidi recognizes that it is useful to keep data on the memory module with a backup power supply on the module itself, such as a battery supply, when the entire system surrounding the memory module loses power”) (citing Ex. 1024, 1:28–35; 2:6–26). Furthermore, Petitioner shows that memory modules with battery backup were known in the art and readily available. Reply 20 (citing Ex. 2035, 9; Ex. 1075, 165:10–167:7). As to the space issue, Petitioner notes that even if the battery backup occupies one side of an FBDIMM board, the other side still can include one rank or two ranks using stacked memory devices. *Id.* at 20–21 (Ex. 1028, 36; Ex. 1075, 74:22–75:25, 77:10–17).

Accordingly, we determine that Petitioner has shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3 and 15*

We agree with Petitioner that the addition of Amidi does not negate Petitioner’s showing with respect to the previous ground. Pet. 45–46 (citing Ex. 1003 ¶¶ 216–301, 431–448). Accordingly, Petitioner has shown that claims 1–3 and 15 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claim 4*

Claim 4 depends from claim 1 and recites that the memory module further comprises:

a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude below a predetermined threshold voltage, wherein the memory module transitions from a first operable state to a second operable state in response to the trigger signal.

Ex. 1001, 38:54–64.

Petitioner points to Amidi’s power supervisory module/block 480/665/800 as claim 4’s “voltage monitor circuit” coupled to Harris’s edge connections for power. Pet. 46. According to Petitioner, Amidi’s power supervisory module/block monitors the external system power supply 605/825 in Amidi. *Id.* Petitioner contends that Amidi’s power supervisory module/block generates a “trigger signal” corresponding to Amidi’s signal

670/858/868. *Id.* Petitioner contends the “predetermined threshold voltage” corresponds to Amidi’s reference voltage 675/820 set to 5% or 10% below the nominal voltage of 12V. *Id.* at 46–49 (citing Ex. 1023 ¶¶ 12, 13; Ex. 1003 ¶¶ 303–314; Ex. 1024, 4:8–11, 4:44–52, 5:25–43, 5:31–62, 8:30–62, 8:23–29, 9:8–12, Figs. 4–6, 14, 15). Petitioner further states that the trigger signal causes the memory module to transition from normal operations using an external voltage in the normal range (“the first operable state”) to self-refresh operations using a backup battery when the external voltage is outside the normal range (“the second operable state”). *Id.* at 49–50 (citing Ex. 1024, 4:44–52, 8:30–62, 9:8–22, Figs. 6–8, 14, 15; Ex. 1003 ¶¶ 315–321).

Patent Owner does not present any argument specific to claim 4. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 4 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

5. *Claims 5, 7, 8, 23 and 24*

Claim 5 depends from claim 4 and recites

a controller coupled to the voltage monitor circuit;
wherein, in response to the trigger signal, the controller is
configured to perform one or more operations including a write
operation to transfer data to non-volatile memory.

Ex. 1001, 38:65–39:3. Claims 7, 23 and 24 recite the same or similar limitations. Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi discloses these claims. Pet. 50–52 (claim 5); Pet. 55–57 (claim 8); Pet. 57–59 (claim 7); Pet. 69 (claims 23 and

24). We consider Petitioner's contentions and Patent Owner's arguments for claim 5 to be representative of the issues presented for all of these claims.

Petitioner contends that the logic for controlling the S3 sleep mode (Pet. 43–44) corresponds to the claimed “controller” that is coupled to the voltage monitor circuit (Amidi's voltage supervisory block). Pet. 50–51. Petitioner contends that “Amidi's battery backup mode is similar to the S3 power-saving mode of Harris's FBDIMM memory module, because both modes put the SDRAMs in a self-refresh state to preserve data while conserving power.” Pet. 51 (citing *id.* at 43–44). According to Petitioner, “in the event of a power disruption causing Amidi's ‘voltage monitor circuit’ to produce a ‘trigger signal’ . . . , a [person of ordinary skill in the art] would have been motivated to use the S3 sleep mode discussed above . . . to conserve power when using Amidi's battery backup.” *Id.* (citing Ex. 1003 ¶¶ 322–333). Petitioner contends that S3 configuration information is stored in non-volatile memory before entering the S3 sleep mode in response to the trigger signal, satisfying the “wherein” clause of claim 4 from which claim 5 depends. *Id.* at 51–52 (citing Ex. 1003 ¶¶ 322–340; Ex. 1027, 25, 117; Ex. 1028, 13; Ex. 1066, 26:64–27:4; Ex. 1067, 1–2; Ex. 1023 ¶ 19).

Patent Owner argues that each of these claims recites that the “controller” is part of the memory module. Resp. 46. However, according to Patent Owner, Dr. Wolfe stated that the S3 sleep mode is “always determined by something other than the memory module.” *Id.* (citing Ex. 2060, 282:22–284:25; Ex. 2006, 3 (S3 controlled by main controller in CPU); Ex. 2061 ¶ 119. Patent Owner asserts that the memory controller outside of the memory module controls the S3 sleep mode, not a controller

onboard the memory module. *Id.* at 47 (citing Ex. 2061 ¶ 120); *see also* Sur-Reply 23–24.

Petitioner replies that Patent Owner’s argument ignores the combination of Harris’s FBDIMM S3 sleep mode with Amidi’s teaching of backup power management and logic functionality on the memory module to “maintain memory (through refresh, for example).” Reply 23 (citing Pet. 43–44, 50–52; Inst. Dec. 35–36; Ex. 1024, 2:16–19, Fig. 11; Ex. 1003 ¶¶ 175–177). Petitioner contends that Amidi’s logic disconnects the system memory controller from the module, so a person of ordinary skill in the art “would have been motivated to use the S3 mode of the buffer 112 [on Harris’s module] when implementing the backup power supply and logic functionality as disclosed in Amidi, because S3 mode performs the function of refreshing the memory (e.g., self refresh), just like in Amidi.” Ex. 1003 ¶¶ 329–330.

From the foregoing, it is clear that Petitioner is relying on Amidi’s logic detecting a low voltage condition to initiate the S3 sleep mode of Harris’s FBDIMM module, rather than a signal from the system memory controller. Petitioner otherwise shows the limitations of these claims are disclosed by the combination of Harris, the FBDIMM Standards, and Amidi.

Petitioner has shown by a preponderance of the evidence that claims 5, 7, 8, 23 and 24 are obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

6. *Claims 6, 7, 9–12, and 17*

Claims 6, 7, 9–12, and 17 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being above a predetermined threshold voltage. Ex. 1001,

39:4–20, 39:30–49, 40:21–25. We address Petitioner’s contentions and Patent Owner’s arguments for claim 6, which is representative of these claims.

Petitioner contends that claim 6 is similar to claim 4 except it concerns *over*voltage detection rather than *under*voltage detection as in claim 4. Pet. 52. Petitioner contends that Harris teaches to detect both undervoltage and overvoltage conditions. *Id.* at 53 (citing Ex. 1023 ¶ 13). In this regard, Harris states that the tolerance of the +12V power supply is “around +/- 15%.” Ex. 1023 ¶ 13. Petitioner asserts that, in the combination, Amidi’s power supervisory module/block 480/665/800 is coupled to Harris’s edge connections for power. Pet. 53 (citing *id.* at 20–25). Petitioner further asserts that Amidi’s signal 670/858/868 is the claimed “trigger signal” generated in response to the input voltage having an amplitude above the predetermined threshold voltage. Pet. 53 (citing *id.* at 46–49). Petitioner contends the predetermined threshold voltage corresponds to Harris’s teaching of 15% above the nominal value of +12V, which is 13.8V. *Id.* (citing Ex. 1023 ¶ 13; Ex. 1003 ¶¶ 341–355).

Petitioner contends that Harris teaches both overvoltage and undervoltage detection of “+/- 15%.” *Id.* (citing Ex. 1023 ¶ 13). Petitioner contends that a person of ordinary skill in the art would have been motivated by Harris to include overvoltage detection in Amidi’s “voltage monitor circuit” with a reasonable expectation of success. *Id.* (citing Ex. 1003 ¶¶ 347–348). Petitioner contends that circuitry for both undervoltage and overvoltage detection was well known and commercially available. *Id.* at 54 (citing Ex. 1061, 15; Ex. 1062, 15; Ex. 1063, 1–2; Ex. 1065, code (57), ¶¶ 14, 18–19, Figs. 1, 5; Ex. 1003 ¶¶ 349–352).

As for the transition between the “first” and “second operable states,” Petitioner contends that the “first operable state” in its combination is “e.g., normal operations using an external voltage when its amplitude is within a normal range,” and the “second operable state” is “e.g., self-refresh operations using battery backup when the external voltage amplitude is outside the normal range.” *Id.* at 54–55 (citing Ex. 1003 ¶¶ 356–359).

Patent Owner argues that neither Harris nor Amidi teaches detecting an overvoltage. Resp. 40. However, Harris discloses that its +12V power supply has a tolerance of “+/- 15%” which indicates it would be a concern if the power supply was greater than 15%. Ex. 1023 ¶ 13. Amidi indicates concern with “power faults.” Ex. 1024, codes (54), (57). Although Patent Owner points out that Amidi discloses detection of undervoltage (Ex. 1024, Fig. 14), “power fault” is broad enough to encompass overvoltage as well. We agree with Petitioner that a person of ordinary skill in the art considering these teachings would have considered it obvious to detect overvoltage and generate a trigger signal to transition between operable states.

Patent Owner argues that the voltage regulators on which Petitioner relies do not detect input overvoltage because they can accommodate a wide range of input voltages, and there is a low probability that they would exceed their operational range. Resp. 42 (citing Ex. 2060, 58:24–61:12; Ex. 1040; Ex. 1041; Ex. 2061 ¶ 114). Petitioner replies that power surges were a known problem that could cause data loss or corruption or damage circuitry in memory boards. Reply 22 (citing Pet. 70–71; Ex. 1003 ¶¶ 138, 185). Petitioner contends that Harris discloses that its module can accommodate only “+/- 15%” of the nominal supply voltage, which is consistent with maximum ratings for commercial converters. *Id.* at 23

(citing Ex. 1041, 4). We agree with Petitioner that power surges would have been a problem in at least some circumstances for which one of ordinary skill in the art would have seen the value of voltage detection in order to take preventive action to avoid loss of data and damage to the memory module. Dr. Wolfe credibly explains that overvoltage was a danger known to persons of ordinary skill in the art, citing industry datasheets that specify overvoltage parameters. Ex. 1003 ¶¶ 351–352 (citing Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15). Indeed, the cited evidence uses the phrase “an overvoltage fault” (Ex. 1061, 15), which shows that power faults are not limited to undervoltage situations.

On this record, we determine that Petitioner has shown that claims 6, 7, 9–12, and 17 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi notwithstanding Patent Owner’s arguments.

7. *Claim 13*

Claim 13 depends from claims 1 and 8, and recites “wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a power reduction condition or a low voltage condition of the input voltage.” Ex. 1001, 39:50–54. Petitioner contends that claim 13 recites a limitation substantially similar to one addressed with respect to claim 4. Pet. 58 (citing Ex. 1003 ¶¶ 419–423). Patent Owner does not provide specific argument for claim 13. Petitioner has shown by a preponderance of the evidence that claim 13 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

8. *Claim 14*

Claim 14 recites

wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a request by the host system.

Ex. 1001, 39:55–58.

Petitioner contends that a power disruption detected by Amidi's voltage supervisory block 800 causes the system to switch to battery backup. Pet. 61 (citing Pet. 55–57). Petitioner contends that Amidi's voltage supervisory block 800 detects a request by the host system to return control to the host. *Id.* at 61–62 (citing Ex. 1003 ¶¶ 424–430). Petitioner asserts that Amidi “discloses that, if the power is restored after a failure, the voltage supervisory block detects whether the host has cleared the backup-mode bit (*‘detecting a request by the host system’*) and subsequently, if so, returns control of the module to the host, including handling of the power signal.” *Id.* at 62 (citing Ex. 1003 ¶¶ 311, 429; Ex. 1024, 5:31–62, 8:30–62, Fig. 14 9:11–39, Figs. 8, 14 (1460–1490), 15 (steps 1530, 1540, 1570, 1580; Ex. 1003 ¶ 312).

Patent Owner does not separately argue claim 14.

Petitioner shows by a preponderance of the evidence that claim 14 is unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

9. *Claim 16*

Petitioner contends that claim 16 is disclosed by the combination of Harris, the FBDIMM Standards, and Amidi for the reasons previously stated for claims 1, 8, and 9, noting that Amidi's voltage supervisory block (e.g., 665) (“the voltage monitor circuit”) is “configured to detect an amplitude

change in the input voltage” when the external system supply changes by crossing the reference voltage. Pet. 58 n.4 (emphasis omitted)(citing Ex. 1003 ¶ 471).

Patent Owner presents no separate arguments for claim 16 apart from those previously discussed.

Petitioner has shown by a preponderance of the evidence that claim 16 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

10. Claims 18–22 and 26–28

Claim 18 depends from claim 16 and recites

wherein, in the first operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

Ex. 1001, 40:26–32. Claim 19 requires that the pre-regulated voltages are provided to the voltage conversion circuit “via a circuit.” *Id.* at 40:33–37. Claim 20 recites that the “circuit” includes respective “first” and “second diodes” coupling respective “first” and “second pre-regulated voltages” to the “voltage conversion circuit.” *Id.* at 40:38–46. Claims 21, 22, 27, and 28 recite limitations from claims that have been addressed or will be addressed in this section.

Petitioner contends that the limitation of claim 18 is taught by the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 63–65. Petitioner contends that “pre-regulated voltage” means either that the voltage is within the 12V +/- 15% limits described in Harris, or must be pre-regulated on the memory board itself. Pet. 63; Ex. 1023 ¶ 13. We interpret

“pre-regulated voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage conversion unit. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). Petitioner contends that Harris discloses external voltage sources that are regulated. Pet. 63 (citing Ex. 1023 ¶¶ 13–14); *see* Ex. 1023 ¶ 14 (“It should be readily recognized that the external voltage sources may comprise any combination of known or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” (emphasis added)); Ex. 1003 ¶ 484. Thus, Harris’s regulated voltage sources generate “pre-regulated voltages” as recited in these claims.

Petitioner contends that the combination discloses that in the first operable state, which is, for example, normal operation when the external voltage amplitude is within normal range, Amidi’s voltage conversion circuit provides the first regulated voltage ($V_{DD}/V_{DDQ}=1.8V$) to the SDRAM devices. Pet. 63 (citing *id.* at 14–19, 27, 31–32, 45, 49–50; Ex. 1003 ¶¶ 480–487).

Petitioner further contends that the combination discloses that in the second operable state, which is, for example, self-refresh operations using battery backup when the external voltage amplitude is outside the normal range, Amidi’s voltage conversion circuit provides the first regulated voltage to the SDRAM devices using a second pre-regulated voltage from a boost converter in the power management block that converts and pre-regulates the 4.2V battery voltage to the regulated output voltage of 12V. Pet. 65 (citing *id.* at 41–43, 49–50; Ex. 1003 ¶¶ 488–492).

For claim 19, Petitioner contends that the claimed “circuit” corresponds to the power switch multiplexer circuit shown in Amidi’s

Figure 7 which supplies pre-regulated “System_VDD” in the first operable state (normal operation), and supplies pre-regulated and boosted “DC_18V” using the battery supply. Pet. 65–67 (citing *id.* at 26; Ex. 1023, Fig. 1A; Ex. 1024, 4:23–29, 4:56–5:24, Figs. 5–7; Ex. 1003 ¶¶ 493–500).

For claim 20, Petitioner contends that the first and second diodes as claimed are disclosed in Harris. Pet. 67–68 (citing Ex. 1024, 5:3 (first diode), 5:17 (second diode); Ex. 1024, 4:64–5:24, Fig. 7; Ex. 1003 ¶¶ 501–511).

For claims 21, 22, and 26–28, Petitioner relies on its contentions presented for claim limitations previously discussed. Pet. 68–70 (citing ¶¶ 450–452, 480–492, 494–500, 502–511, 513–526 (claim 21); ¶¶ 502–511; 527–530 (claim 22); ¶¶ 480–492, 571–576 (claim 26); ¶¶ 502–506, 577–580 (claim 27); ¶¶ 507–511, 581–584 (claim 28)).

Patent Owner does not present any arguments specific to these claims.

After reviewing the record, we determine that Petitioner has shown by a preponderance of the evidence that claims 18–22 and 26–28 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

11. Claim 25

Petitioner contends that the limitations of claim 25 were addressed with respect to claims 4 and 16. Pet. 58, 70.

Patent Owner provides no argument specific to claim 25. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 25 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

12. Claim 29

Claim 29 depends from claim 24 and recites that the voltage monitor circuit is configured to detect whether the input voltage is above or below respective thresholds. Ex. 1001, 42:19–23. Petitioner contends that the limitations of claim 29 were addressed with respect to claims 4, 6, and 17. Pet. 58, 70.

Patent Owner provides no argument specific to claim 29. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 29 is obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

13. Claim 30

Claim 30 depends from claim 29 and recites that the predetermined threshold voltages are above and below a “specified operating voltage.” Ex. 1001, 42:24–27. Petitioner contends that the “specified operating voltage” corresponds to Harris’s +12V nominal external voltage. Pet. 57 n.2 (citing Pet. 41–45; Ex. 1023 ¶¶ 12, 13).

Patent Owner does not present any arguments specific to claim 30. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 30 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

14. Determination for Ground 2

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations are taught or at least suggested by the

combination. Accordingly, under this ground, Petitioner has demonstrated by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

F. Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck

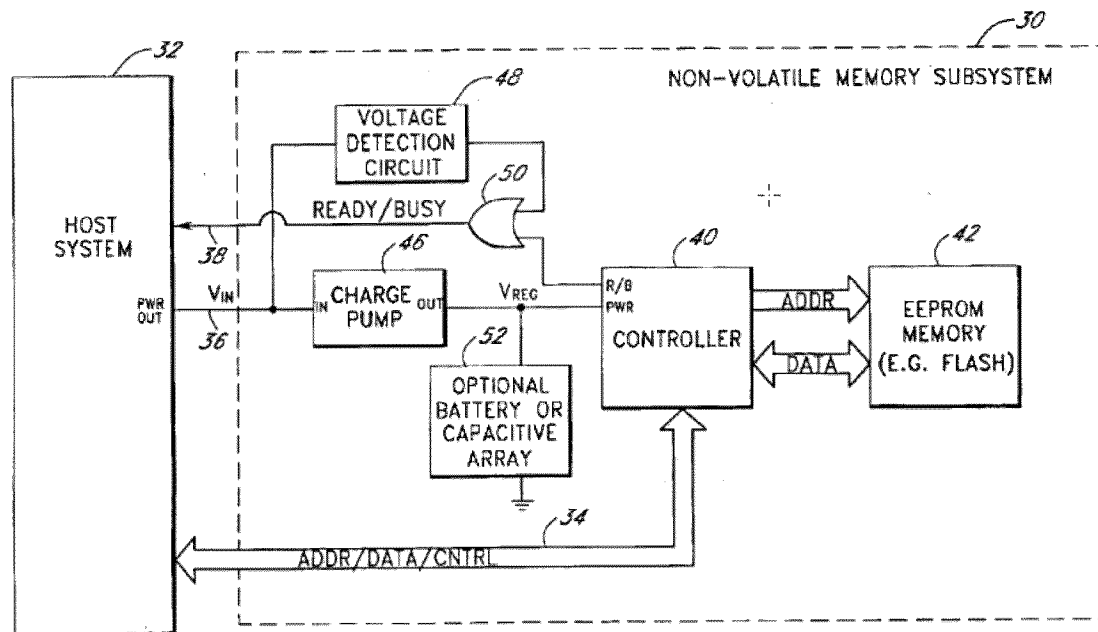
Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck. Pet. 70–72.

1. Hajeck (Ex. 1038)

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1001, code (54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck’s Figure 1 is reproduced below.



In Hajec's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–33.

2. Motivation to Combine

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajec with Harris, the FBDIMM Standards, and Amidi because one would have appreciated the desirability of switching to backup power in both undervoltage and overvoltage conditions.

Pet. 71–72. Hajeck teaches using a battery backup in response to power loss from the host system to complete outstanding operations and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57). Hajeck also teaches a voltage detection circuit generate a “busy” signal in response to undervoltage or overvoltage conditions. *Id.* at 3:30–43.

3. *Claims 7, 9–12, 17, 29, and 30*

Petitioner asserts that claims 1–30 would have been obvious under this ground for the same reasons provided for Ground 2. Pet. 71 (citing *id.* at 45–70). Petitioner contends that to the extent one would argue that Ground 2 fails to teach *over*voltage detection, Hajeck discloses this feature. Pet. 71 (citing *id.* at 52–54). Specifically, Petitioner contends that Hajeck discloses overvoltage detection as required by claims 7, 9–12, 17, 29, and 30. Pet. 71 (citing *id.* at 52–54; Ex. 1003 ¶¶ 354–355). Petitioner contends a person of ordinary skill in the art would have looked to Hajeck’s teachings about voltage anomalies when implementing Amidi’s voltage supervisory block to detect both undervoltage and overvoltage anomalies. *Id.* at 71–72 (citing Ex. 1003 ¶¶ 186–188). In the combination, “Amidi’s voltage supervisory block would be modified to detect voltage anomalies and switch to the backup power not only ‘[i]f system supply 605 has a magnitude lower than reference voltage 675’ as disclosed by Amidi (Ex. 1024, 4:44–52), but also ‘when the voltage exceeds a certain level’ as taught by Hajeck (Ex. 1038, 3:30–43; *see also id.* at code (57), 1:10–18, 1:28–31, 1:62–2:7, 3:30–4:9, 4:62–65, Fig. 1).” *Id.* at 72. Petitioner contends that this ground teaches both overvoltage and undervoltage detection and protection, further

rendering obvious claims 6, 9, 17, and 29 and their dependent claims 7, 10–12, and 30. *Id.* (citing Ex. 1003 ¶¶ 354–355, 389, 478, 588).

Patent Owner argues that “Harris does not teach the need to detect undervoltage and overvoltage anomalies” and that Hajeck “does not teach or suggest a voltage detection circuit that produces a trigger signal that causes transitions to different operating states in over-voltage conditions or otherwise even relates to switching power sources” or “sending the trigger signal to the memory system.” Resp. 42–43 (citing Pet. 48, 51). Patent Owner argues that “the only signal generated by Hajeck’s voltage detection circuit 48 is a ‘busy signal’ 38 sent back to host 32 to prevent it from performing write operations to the memory subsystem 30 when voltage anomalies are detected.” *Id.* at 43 (citing Ex. 1038, 3:30–57, Fig. 2). Patent Owner argues that Hajeck does not switch to a different operating state or a different power supply, but instead its charge pump 46 continues to provide a near-constant voltage V_{REG} to controller 40. *Id.* (citing Ex. 1038, 3:6–10, 3:66–4:3; Ex. 2061 ¶ 116). Patent Owner argues that Hajeck’s charge pump protects controller 40 and other circuits from damage caused by voltage surges and spikes. *Id.* at 44 (citing Ex. 1038, 4:7–9).

Petitioner replies that Patent Owner’s arguments ignore the Petition’s extensive explanation and evidence about a person of ordinary skill in the art’s understanding of how to handle power anomalies, including both undervoltage and overvoltage conditions, with a reasonable expectation of success. Reply 21 (citing Pet. 12–13, 41–45, 53–54, 70–72; Ex. 1003 ¶¶ 138, 345–355). Petitioner contends that the Petition also explained how the combination teaches a trigger signal to switch to back-up power and avoid data loss in case of a “power fault” including undervoltage and

overvoltage conditions. *Id.* (citing Pet. 49–50; Ex. 1003 ¶¶ 315–321; Ex. 2060, 226:16–22, 230:17–232:6, 251:10–254:2).

We agree with Petitioner that Patent Owner’s arguments attack the references individually and do not properly consider what the combination of references would have signified to a person of ordinary skill in the art. *In re Merck & Co.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981)). Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition to the extent that the combination of Harris, the FBDIMM Standards, and Amidi were deemed not to already disclose that feature. Pet. 72 (citing Ex. 1038, 3:37–40). Thus, the features that Patent Owner argues are missing from Hajeck are not ones for which Petitioner relied on Hajeck.

For similar reasons, we also do not agree with Patent Owner’s argument that Hajeck does not suggest that data loss was a concern because its charge pump could provide the desired V_{REF} indefinitely. Resp. 44–45. Petitioner relies on Amidi for switching operational states and the voltage supply to battery backup in the event of an overvoltage. Reply 21 (citing Pet. 49–50). This argument also considers Hajeck individually, and not in combination with Harris, the FBDIMM Standards, and Amidi, as set forth in the Petition.

4. *Determination for Ground 3*

Petitioner has shown that a person of ordinary skill in the art would have combined Harris, the FBDIMM Standards, Amidi and Hajeck with a reasonable expectation of success. Petitioner has also shown that all of the limitations of claims 1–30 are taught or at least suggested by the combination. On this record, Petitioner has shown by a preponderance of

the evidence that claims 1–30 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

G. Grounds 4 and 5: Obviousness over Spiers and Amidi

As we have determined that claims 1–30 are unpatentable as obvious under Grounds 1 to 3, we do not reach Grounds 4 and 5.

H. Motion to Exclude

Petitioner seeks to exclude materials referenced in three URLs submitted with Patent Owner's Sur-Reply on August 4, 2023. Paper 35 (citing Paper 33, 1 n.2, 24, 26). Because we do not rely on any of these URLs in a manner adverse to Petitioner, we dismiss the Motion to Exclude as moot.

III. CONCLUSION

For the foregoing reasons, we determine that Petitioner establishes by a preponderance of the evidence that claims 1–30 of the '054 patent are unpatentable.

IV. ORDER

Accordingly, it is:

ORDERED that claims 1–30 of the '054 patent have been shown to be unpatentable;

FURTHER ORDERED that Petitioner's Motion to Exclude is dismissed as moot;

FURTHER ORDERED that Petitioner will contact the Board within seven (7) business days to correctly identify Exhibit 1023 as Harris in the Board's P-TACTS system.

FURTHER ORDERED that any party seeking judicial review must comply with the notice and service requirements of 37 C.F.R. § 90.2.⁷

⁷ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner's attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–3, 15	103	Harris, FBDIMM Standards	1–3, 15	
1–30	103	Harris, FBDIMM Standards, Amidi	1–30	
1–30	103	Harris, FBDIMM Standards, Amidi, Hajeck	1–30	
1–30	103	Spiers, Amidi ⁸		
1–30	103	Spiers, Amidi, Hajeck		
Overall Outcome			1–30	

⁸ Because we determine that claims 1–30 are unpatentable on other grounds, we do not reach the remaining grounds.

IPR2022-00999
Patent 11,232,054 B2

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD., MICRON TECHNOLOGY, INC.,
MICRON SEMICONDUCTOR PRODUCTS, INC., and MICRON
TECHNOLOGY TEXAS LLC,¹
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00999
Patent 11,232,054 B2

Before PATRICK M. BOUCHER, JON M. JURGOVAN, and
DANIEL J. GALLIGAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Final Written Decision
Determining All Challenged Claims Unpatentable
Dismissing Petitioner's Motion to Exclude
35 U.S.C. § 318(a)

¹ Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC filed a motion for joinder and a petition in IPR2023-00405 and have been joined as petitioners in this proceeding. *See* Paper 27.

I. INTRODUCTION

A. *Background and Summary*

Samsung Electronics Co., Ltd. (“Samsung”) filed a Petition (Paper 1, “Pet.”) for *inter partes* review of claims 1–30 (“challenged claims”) of U.S. Patent 11,232,054 B2 (Ex. 1001, “the ’054 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition. We instituted *inter partes* review under 35 U.S.C. § 314(a). Paper 11 (“Inst. Dec.”).

During the trial, Patent Owner filed a Response (Paper 22, “Resp.”), Petitioner filed a Reply (Paper 26), and Patent Owner filed a Sur-Reply (Paper 33). We joined Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC as petitioners in this proceeding, and we refer to Samsung and these entities collectively as “Petitioner.” *See* Paper 27.

Petitioner and Patent Owner requested oral argument (Papers 29 and 30). A hearing was conducted on September 11, 2023. Paper 49 (“Tr.”).

Petitioner objected to evidence (Papers 23, 34) and filed a Motion to Exclude (Paper 35). Patent Owner filed an Opposition to Petitioner’s Motion to Exclude (Paper 36), and Petitioner filed a Reply (Paper 39) in support of its Motion to Exclude.

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is entered pursuant to 35 U.S.C. § 318(a). Having reviewed the complete trial record, we determine that Petitioner has shown, by a preponderance of the evidence, that the challenged claims are unpatentable.

B. Real Parties in Interest

Petitioner entities, Samsung Electronics Co., Ltd., Samsung Semiconductor, Inc., Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC identify themselves as real parties in interest. Pet. 1; IPR2023-00405, Paper 3, 1.

Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

C. Related Matters

The parties advise that the '054 patent is related to the following pending matters:

- *Samsung Electronics Co., Ltd. et al. v Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021)
- *Netlist, Inc. v. Samsung Electronics Co., Ltd. et al.*, No. 2:21-cv-00463 (E.D. Tex. filed Dec. 20, 2021)
- IPR2012-00996 (U.S. Patent No. 11,016,918)
- IPR2017-00692 (U.S. Patent No. 8,874,831)
- U.S. Patent Application No. 17/582,797

Pet. 1; Paper 4, 1.

D. Overview of the '054 Patent (Ex. 1001)

The '054 patent is titled “Flash-DRAM Hybrid Memory Module.” Ex. 1001, code (54). Figure 12 of the '054 patent is reproduced below.

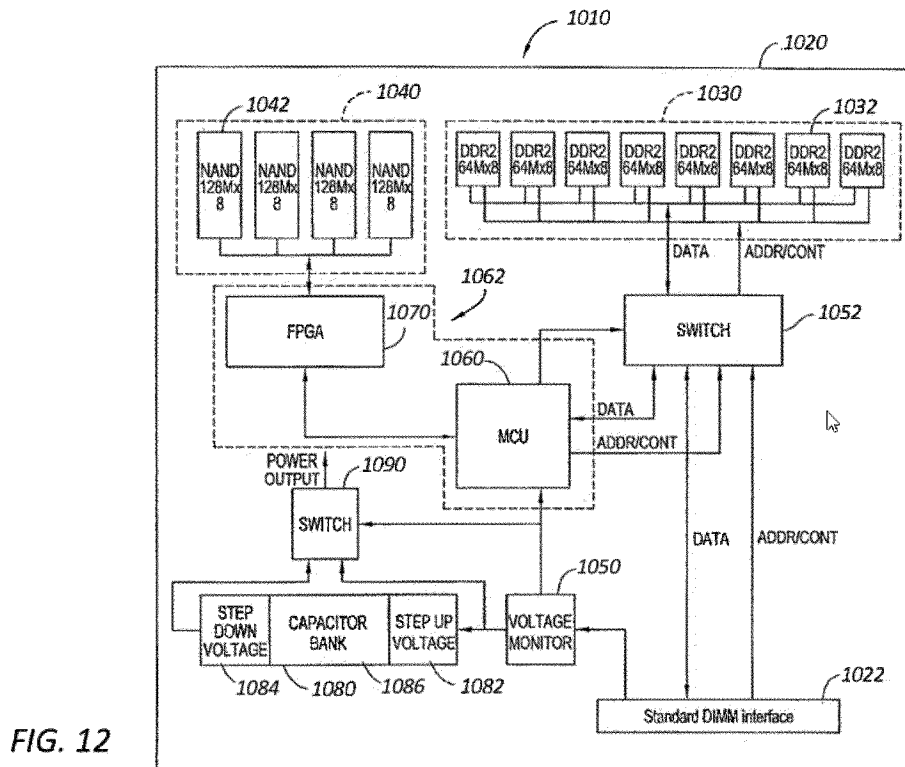


FIG. 12

Figure 12 shows an example memory system 1010 of the '054 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. *Id.* at 21:16–20. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022

for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. Ex. 1001, 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '054 patent refers to this as the “first state”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '054 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g. a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. *Id.* at 25:54–58. Second power supply 1080 may comprise step-up transformer

1082, step-down transformer 1084, and capacitor bank 1086 with one or more capacitors. *Id.* at 26:3–13.

The memory system 1010 further has a third state in which controller 1062 is decoupled from the volatile memory system 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). *Id.* at 25:62–66. The third power supply may provide power to volatile memory subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred. *Id.* at 25:66–26:3.

Figure 16 of the '054 patent is reproduced below.

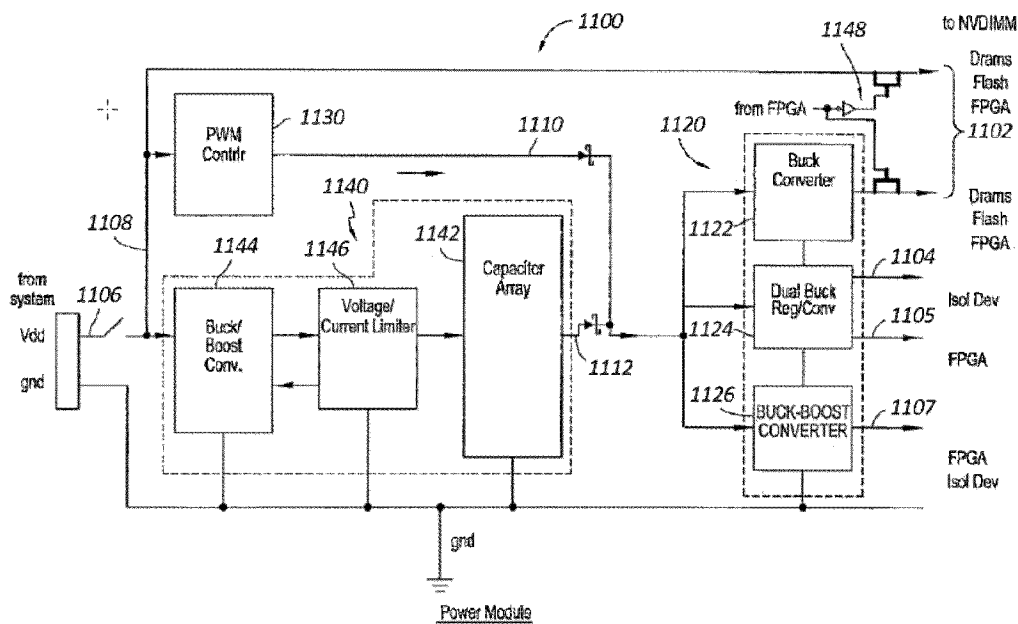


FIG. 16

Figure 16 shows power module 1100 of memory system 1010. Ex. 1001, 27:59–61. Power module 1100 comprises conversion element 1120, first power element 1130, and second power element 1140. *Id.* at 28:7–15, 28:20–22. Conversion element 1120 comprises buck converter 1122, dual buck converter 1124, and buck-boost converter 1126 generating respective

voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 28:62–67. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. Power module 1100 provides power to components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. *Id.* at 27:61–65.

Specifically, in a first state, first voltage 1102 is provided to memory system 1010 from input 1106 and fourth voltage 1102 is provided to conversion element 1120 from the first power element 1130. Ex. 1001, 28:27–31. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. *Id.* at 28:31–34. In the third state, the fifth voltage is provided to conversion element 1120 from second power element 1140 and the first voltage 1104 is provided to memory system 1010 from conversion element 1120. *Id.* at 28:34–38. Transition from the first state to the second state may occur when power module 1100 detects a power failure is about to occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

E. Illustrative Claim

Of the challenged claims, claims 1, 16, and 24 are independent.

Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages;

[1.d] [1.d.1] a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages, [1.d.2] the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and [1.d.3] a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections, [1.d.4] wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and [1.d.5] wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages.

Ex. 1001, 38:19–44.

F. Evidence

Petitioner relies on the following references (*see* Pet. 3, 9–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published Aug. 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published Apr. 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued Feb. 15, 2005

G. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 3):

Claims Challenged	35 U.S.C. §	References
1–3, 15	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

II. ANALYSIS OF CHALLENGED GROUNDS

We now consider Petitioner’s asserted grounds of unpatentability and Patent Owner’s arguments to determine whether Petitioner has demonstrated by a preponderance of the evidence that the challenged claims would have been obvious under 35 U.S.C. § 103. *See* 35 U.S.C. § 316(e) (providing that

petitioner has the burden of proving unpatentability by a preponderance of the evidence).

A. Principles of Law

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination based on the teachings of multiple references requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see also KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re*

NuVasive, Inc., 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Factors pertinent to a determination of the level of ordinary skill in the art include “(1) the educational level of the inventor; (2) type of problems encountered in the art; (3) prior art solutions to those problems; (4) rapidity with which innovations are made; (5) sophistication of the technology; and (6) educational level of active workers in the field.” *Envtl. Designs, Ltd. v. Union Oil Co. of Cal.*, 713 F.2d 693, 696 (Fed. Cir. 1983) (citing *Orthopedic Equip. Co. v. All Orthopedic Appliances, Inc.*, 707 F.2d 1376, 1381–82 (Fed. Cir. 1983)). “Not all such factors may be present in every case, and one or more of these or other factors may predominate in a particular case.” *Id.* at 696–97. The prior art may reflect an appropriate level of skill. *Okajima v. Bourdeau*, 261 F.3d 1350, 1354–55 (Fed. Cir. 2001).

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 7–8 (citing Ex. 1003 ¶ 61). Petitioner contends that additional training can substitute for educational or research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a computer system, including standard communication busses and protocols,

such as PCI and SMBus busses and protocols. *Id.* Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

Patent Owner indicates “[f]or the purposes of this [Response], Patent Owner is applying the level of ordinary skill in the art proposed by Petitioner.” Resp. 2.

The evidence that Petitioner presents mostly applies to the educational level and experience of workers in the field, but also touches upon other factors as well, including problems and solutions in the prior art and complexity of the technology. *See* Pet. 7–8. We find Petitioner’s proposal is consistent with the level of skill in the art reflected by the ’054 patent and the prior art of record, and, therefore, we adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least,” which introduces ambiguity and may encompass skill levels beyond ordinary.

C. *Claim Construction*

We construe claim terms “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2021). There is a presumption that claim

terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art in the context of the specification. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Nonetheless, if the specification “reveal[s] a special definition given to a claim term by the patentee that differs from the meaning it would otherwise possess[,] . . . the inventor’s lexicography governs.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1316 (Fed. Cir. 2005) (en banc) (citing *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002)). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Only disputed claim terms must be construed, and then only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

1. “Memory Module”

Patent Owner contends as follows:

The term “memory module” appears in the preamble of the claims, and provides antecedent basis for the same term in the body of claims 1, 4, 6, 11, 16 and 25. The preamble is thus limiting as the District Court found after reviewing the specification in detail. Based on the intrinsic evidence, the Court also concluded that a memory module is a modular computer accessory that “includes the structure necessary to connect to a memory controller” of a host system.

Resp. 2–3 (citing Ex. 2032, 28). As support for its argument, Patent Owner quotes the District Court for the Eastern District of Texas as follows:

While the claims recite many of the structural requirements of a “memory module,” the claims arguably read on other modular computer devices, such as a video card or network controller, despite no evidence the inventors intended to encompass such devices by the claims. To the contrary, as the Overview section explains, the invention “is couplable to a memory controller of a host system,” ’918 Patent at 3:66–67 (emphasis added), not just the host system as recited in the claims. *See also id.* at 1:66–67 (“[t]he present disclosure relates generally to computer memory devices”). Thus, a skilled artisan would understand *a “memory module” is distinct from, and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller.* *See* Memory Systems: Cache, DRAM, Disk [EX2034] ... at 319 (depicting, in FIG. 7.6, a memory controller connected to two memory modules). Accordingly, the preambles are limiting.

Resp. 3 (citing Ex. 2032, 28) (emphasis in original). Patent Owner argues that Petitioner’s expert, Dr. Wolfe, testified that the term “memory modules” typically refers to “main memory modules,” which “are designed to connect to the primary memory controller for the purpose of holding general purpose code and data in a computer system.” Resp. 3–4 (citing Ex. 2060, 123:14–25; Ex. 2056, 100:15–101:8 (by 2004–2005, a memory module “was intended to go into a dedicated memory slot and not a general-purpose IO slot”)). Patent Owner contends that the ’054 patent’s usage of “memory module” is consistent with that understanding, and asserts that we should adopt the District Court’s claim construction, including that the claimed “memory module” includes structures necessary to connect to a memory controller. *Id.* at 4 (citing Ex. 2061 ¶¶ 51–53).

In Reply, Petitioner contends that we properly found that Harris, the FBDIMM Standards, Amidi, and Spiers all disclose a “memory module” in

our Institution Decision. Reply 1–2 (citing Inst. Dec. 14–15, 17–18, 29–33, 47–48; Pet. 19–20, 77–78; Reply 24–27). Petitioner contends that the District Court’s construction did not limit “memory module” to only “*main* memory modules . . . designed to connect to the *primary* memory controller.” *Id.* at 2 (emphasis in original) (citing Resp. 3–4). Petitioner contends that the District Court never used the words “main” or “primary,” and that District Court’s construction was simply that the “memory module” in the preamble was “limiting” without more. *Id.* (citing Ex. 2032, 26–28, 35). More specifically, Petitioner contends that the District Court did not further limit “memory module” to only “main memory modules . . . designed to connect to the primary memory controller.” *Id.* at 2 (citing Resp. 3–4). Petitioner further contends that Patent Owner misleadingly quotes Dr. Wolfe’s statements about “main memory modules,” but asserts that Dr. Wolfe explained in the context of the ’054 patent that a “memory module” is not limited to main memory or to any specific connection. *Id.* (citing Ex. 2060, 125:12–127:13 (“memory module” is a “circuit board that connects to a host computer that includes memory”); Ex. 2056, 100:15–101:19).

In Sur-Reply, Patent Owner disagrees with Petitioner’s contention that “memory module” is not limited to “*main memory* modules . . . designed to connect to the primary memory controller.” Sur-Reply 1 (emphasis in original) (citing Reply 2). Patent Owner argues that Petitioner’s contention “ignores the District Court’s finding based on intrinsic evidence—which [Petitioner] did not object to—that “the invention [i.e., the claimed memory module] “is couplable to a memory controller of a host system,” ’918 Patent at 3:66–67 . . . , not just the host system as recited in the claims.”” *Id.* (citing

Ex. 2032, 28; Ex. 1001, 3:66–67, 4:5–12, 4:14–24, 4:35–39, 4:45–51, 5:4–20, 4:36–50, 6:4–6, 6:25–29, 6:61–66, 7:1–4, 7:21–25, 7:29–34, 7:43–49, 7:54–57, 21:24–25, 22:53–58, 23:19–22, 26:43–51, 23:41–44).

The District Court’s Claim Construction Order (Ex. 2032) issued after our Institution Decision (Paper 11). Although Petitioner argued in the District Court litigation that the claim preambles were not limiting, Petitioner does not appear to maintain that argument here following the District Court’s determination that the preambles are limiting. Ex. 2032, 35. Since the term is recited in both the preambles and bodies of these claims, there is evidence that the inventors of the ’054 patent intended the term to limit the claim, as the District Court determined. *See Catalina Mktg. Int’l., Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell Commc’ns Rsch., Inc. v. Vitalink Communications Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995) (“dependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention.”)).

We agree with Petitioner’s contention that “memory module” is not limited to “**main** memory modules . . . designed to connect to the **primary** memory controller.” *See* Reply 2. The ’054 patent states

In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the **main memory**.

Ex. 1001, 11:3–6 (emphases added). This is the only mention of “main memory” in the ’054 patent. That the Flash memory serves as “main memory” in “certain embodiments” implies that there may be other embodiments in which this is not the case. Accordingly, we agree with

Petitioner that the term “memory module” is not limited to “*main* memory modules . . . designed to connect to the *primary* memory controller.”

We agree with Patent Owner that the “memory module” includes structure to connect with a “host system” including its memory controller. *See, e.g.*, Ex. 1001, code (57), 12:52–59. The claims do not mention a “memory controller,” but they do mention that the “memory module” has an “interface” to connect with a “host system.” Ex. 1001, 38:20–25 (claim 1), 39:63–65 (claim 16), 41:6–8 (claim 24). That the “memory module” has structure for connecting to the host system’s memory controller may thus be implied from the claims.

We agree with Petitioner that the District Court’s determination was that the preambles (presumably of claims 1, 4, 6, 11, 16 and 25) are limiting. Ex. 2032, 35. To the extent that the District Court’s determination construed “memory module” to include structure necessary to connect to a memory controller, our construction of “memory module” is consistent.

D. Ground 1: Obviousness Over Harris and FBDIMM Standards

Petitioner contends claims 1–3 and 15 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–40. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe’s testimony, supports Petitioner’s showing and establishes by a preponderance of the evidence that these claims are unpatentable.

1. *Harris (Ex. 1023²)*

Harris is titled “Voltage Distribution System and Method for a Memory Assembly.” Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 9.

Harris’s Figure 1A is reproduced below.

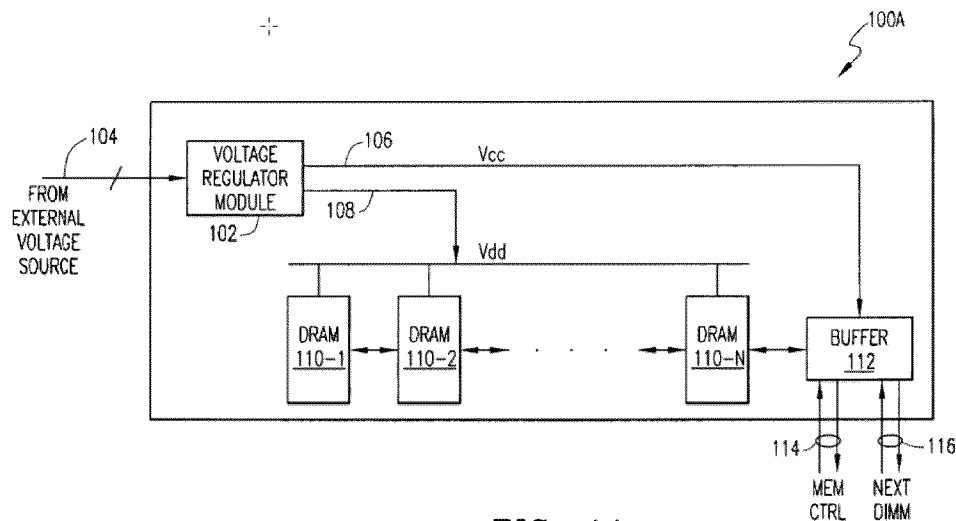


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface 116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Module (DIMM)

² Exhibit 1023 is incorrectly identified in the Board’s P-TACTS system as “Micron Tech., Inc. et al. v. Netlist, Inc., IPR2022-00418, Paper No. 2 (PTAB January 14, 2022) (831 Patent IPR Petition),” i.e., incorrectly repeating the identification provided for Exhibit 1022.

wherein each of the memory devices 100-1 to 100-N comprises a Double Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. FBDIMM Standards (Exs. 1027, 1028)

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029 ¶¶ 134–137. Petitioner refers to these standards collectively as the “FBDIMM Standards.” Pet. 10. Petitioner contends the FBDIMM Standards are prior art under § 102(a).

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max. thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max. thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DD}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	0.453* V_{DD}	0.5* V_{DD}	0.547* V_{DD}	(DRAM Interface V_{TT}) This supply should track as 0.5 * 1.8 volt supply
	3.0	3.3	3.6	(V_{DDSPD})
Buffer Interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM Interface	SSTL_18			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.

Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)

An Unloaded system should supply V_{TT} of 0.48* V_{DD} /0.52* V_{DD} to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
RESET $\bar{I}$		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	I	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by “a” are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies V_{CC} , V_{CCFBD} , V_{DD} , V_{SS} , and V_{DDSPD} .

3. Motivation to Combine

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–166). Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and thus would naturally look

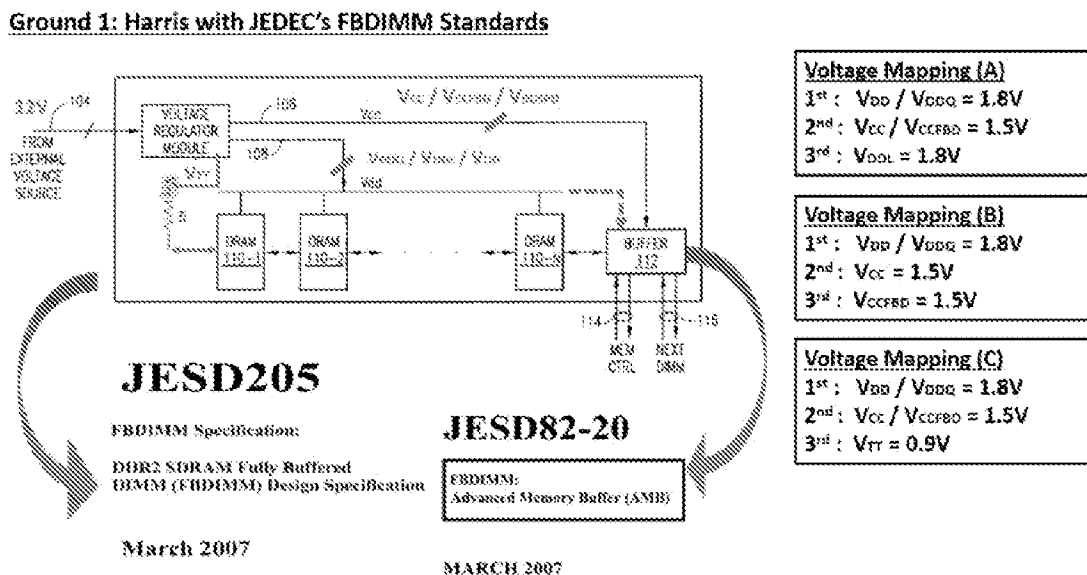
to them for more details about the “fully buffered DIMM” that Harris describes as compatible with the disclosed on-board voltage regulator module (VRM). *Id.*

To explain its combination, Petitioner provides the following table:

Voltage Mappings (Grounds 1-3)			
	<u>A</u>	<u>B</u>	<u>C</u>
“first”:	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} or $V_{DDQ} = 1.8V$
“second”:	V_{CC} or $V_{CCFB0} = 1.5V$	$V_{CC} = 1.5V$	V_{CC} or $V_{CCFB0} = 1.5V$
“third”:	$V_{DDT} = 1.8V$	$V_{CCFB0} = 1.5V$	$V_{TT} = 0.9V$

The table above shows Petitioner’s Voltage Mappings A, B, and C from the FBDIMM Standards for Grounds 1–3, and how they disclose the “first,” “second,” and “third regulated voltages” in the claims. Pet. 27.

For Ground 1, Petitioner provides the illustration below:



This illustration shows how Petitioner is combining the teachings of Harris and the FBDIMM Standards to arrive at the claims. *Id.* at 15. Specifically,

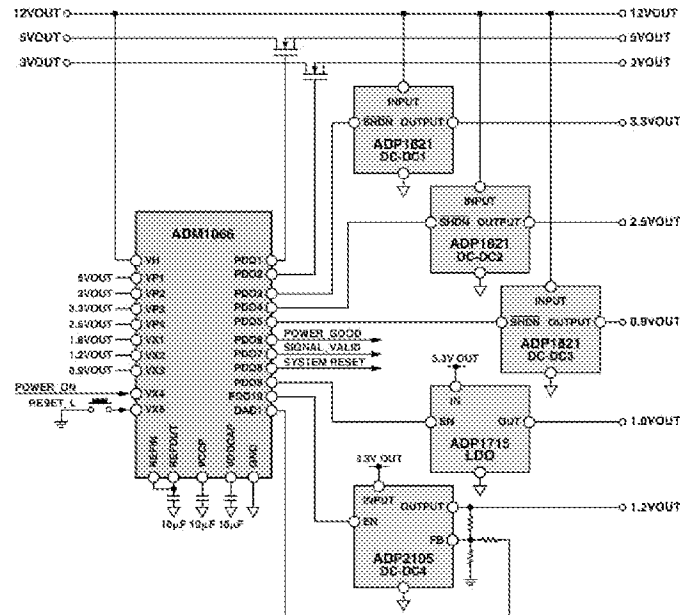
FBDIMM Standard JESD205 applies to Harris's memory module and JESD82-20 applies to the Harris's buffer. *Id.* at 26.

a) Modifying Harris to Have Three Converters

Patent Owner asserts that a person of ordinary skill in the art would not have modified Harris's memory module to have three converters, as Petitioner asserts. Resp. 18–20. Patent Owner contends that Petitioner did not demonstrate “a reason to modify Harris with the FBDIMM Standards,” or “show a reason for Harris to use separate regulators for each of the alleged regulated voltages across all of Petitioner's Voltage Mappings A–C.” *Id.* at 20. Patent Owner argues that it would not “have been obvious to use as many buck converters as possible, especially given that Harris expressly discloses a switching regulator outputting multiple voltage levels, consistent with the state of art.” Sur-Reply 10–11 (citing Reply 11; Ex. 1023 ¶ 10; Ex. 2020, 5, Table 1-1). Patent Owner asserts that Petitioner “has not articulated any advantage of using one converter per voltage if Harris already uses a single converter for at least two voltages.” *Id.* at 11 (citing Reply 11–14; Ex. 1023 ¶ 10).

As to why a person of ordinary skill in the art would use separate regulators for each of the voltages, Petitioner indicates that this was well-known in the art. Specifically, Petitioner contends that “buck converters were well-known switching devices commonly used to step down an input voltage to a lower output voltage.” Reply 11 (citing Pet. 28–29; Ex. 1075, 103:21–107:14, 112:12–113:19). Petitioner contends that it “would have been obvious to a [person of ordinary skill in the art]—and consistent with Harris—to use three buck converters to convert the 12V input to the three required voltages on the memory board.” *Id.* (citing Pet. 27–30; Ex. 2060,

53:16–54:14). Petitioner contends that “[t]his was common” as shown in the figure below:



Id. at 11–12 (citing Ex. 1062, 15, Fig. 4³). The above figure shows an ADM 1066 sequencer and three ADP182 DC-to-DC converters for generating 3.3V, 2.5V and 0.9 voltages outputs. Ex. 1062, 15. Petitioner also points to the data sheet for the ADP1821 titled “Step-Down DC-to-DC Controller,” which describes a “synchronous, pulse-width-modulated (PWM), voltage-mode, step-down controller.” Ex. 1078, 1; *see* Reply 11 (citing Ex. 1078, 1). Petitioner further relies on deposition testimony of

³ Exhibit 1062 is titled “Analog Dialogue,” Vol. 40, No. 2, published in 2006 by Analog Devices, Inc. “Analog Dialogue” contains several papers, one of which is titled “Power-Supply Management—Principles, Problems, and Parts” by Alan Moloney. The figure above was extracted from this paper. Although Exhibit 1062 was not identified in the Petition, it is part of the background knowledge that one of ordinary skill in the art would have had. *KSR*, 550 U.S. at 401 (obviousness analysis requires an assessment of the background knowledge possessed by person of ordinary skill in the art).

Patent Owner's expert, Dr. Mangione-Smith, who confirmed that the ADP1821 is a DC-to-DC converter and that it has structural elements and switching that are in common with buck converters to generate voltages. Reply 11 (citing Ex. 1075, 132:21–141:23). Patent Owner's argument that one of ordinary skill in the art would not have modified Harris to include three buck converters does not undermine Petitioner's showing of a motivation to combine Harris with the FBDIMM Standards when the background knowledge expressly teaches using three buck converters to generate voltages used in FBDIMMs. Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–166); Reply 11–12 (citing Ex. 1062, 15, Fig. 4; Ex. 1078, 1).

Patent Owner argues that Harris's mention of “a high-frequency switching voltage regulator” means only one, and that Petitioner's contention that “a” means “one or more” applies to the claims, not the specification. Sur-Reply 11 (citing *Baldwin Graphic Sys., Inc. v. Siebert, Inc.*, 512 F.3d 1338, 1342–43 (Fed. Cir. 2008); Ex. 1023 ¶ 14). Patent Owner further argues that “Harris's reference to ‘a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels’ (plural) means a single converter with multiple outputs, not multiple converters.” *Id.* at 15–16 (citing Ex. 2061 ¶¶ 79–80).

Patent Owner does not address Petitioner's argument that Harris refers to “at least one on-board voltage regulator module (VRM)” in the specification and “at least one voltage regulator module” in claims 1 and 30. Reply 12 (citing Ex. 1023 ¶ 10, Fig. 1A). Patent Owner's argument does not undermine Petitioner's showing.

Patent Owner further asserts that space issues would have influenced any design decisions. Sur-Reply 12 (citing *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1368 (Fed. Cir. 2016)). Patent Owner argues there are good reasons why a person of ordinary skill in the art would have minimized the number of buck converters, including space constraints, and that Petitioner has not shown how multiple voltage converters would fit in the one square-inch space on both sides of the circuit board described as available for a voltage regulator module (VRM) in Harris. *Id.* at 12–13 (citing Ex. 1023 ¶ 13; Ex. 1078, 20–23; Ex. 2101, 22, 25–26).

Harris states that “[i]t is contemplated that local supply voltage conversion for double-rank DIMMs can be accommodated with a form factor design of approximately about one square inch (both sides of the printed circuit board).” Ex. 1023 ¶ 13. Thus, Harris indicates that the VRM circuitry to generate two regulated voltages can fit within this space. Patent Owner argues that Petitioner’s evidence shows that each ADP1821 converter package would be about 0.193 by 0.236 inches. Sur-Reply 12 (citing Ex. 1078, 20–23). This means three ADP1821 converters would occupy about 40% of the available space. Patent Owner does not demonstrate that the remaining 60% of available space would have been insufficient to accommodate the additional components required for three voltage converters. *Id.* at 12–13. In any case, we do not view Harris’s disclosure of “approximately about one square inch” to limit the applicability of its teachings to a person of ordinary skill in the art. In the same paragraph, Harris states that the form factor “may be suitably modified.” Ex. 1023 ¶ 13.

Patent Owner further argues that “independent voltage sources for different voltages would require specific control circuitry to delay and more precisely control each voltage source’s ramping rate, adding complexity and cost without commensurate benefits.” Sur-Reply 14 (citing Ex. 2061 ¶ 90). However, Dr. Wolfe states that it is simpler to use multiple regulators for multiple voltages since a single regulator that generates multiple voltages is more complicated than multiple regulators. Ex. 2060, 53:16–54:14, 89:1–21, *cited in* Reply 14. Dr. Wolfe further indicates that buck converters can be in some applications “extremely small.” Ex. 2060, 89:11–15. Dr. Wolfe further testifies that multiple smaller converters may be less costly than a larger single converter because of component costs, separated spaces available for converters on circuit board, and other “ordinary engineering factors.” *Id.* at 45:21–46:10.

Although Patent Owner may be correct that individual control of multiple voltage regulators would have required additional control circuitry, this does not meaningfully undermine the benefits that Petitioner mentions including permitting power sequencing, independent control of voltages, improved efficiency without generating excess heat or requiring cooling devices, and saving power. Pet. 29–30 (citing *id.* at 14–19; Ex. 1028, 17–20; Ex. 1026, 2–3, 9; Ex. 1028, 30–32; Ex. 1040, 1, 23–24 (Figs. 22–25); Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1058, 5; Ex. 1059, 5:23–30; Ex. 1062, 11, 13; Ex. 1064 ¶ 101; Ex. 1003 ¶¶ 240, 255). Dr. Wolfe testified that, in at least some circumstances, multiple converters may be less complicated, less costly, and take up less space. Ex. 2060, 45:21–46:10, 89:11–15. Patent Owner’s argument that a single converter generating multiple voltages would be better in some circumstances does not negate that Petitioner’s

proposed use of multiple converters for multiple voltages would have been preferred in other circumstances that a person of ordinary skill in the art would have recognized. *See Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021) (“Our caselaw is clear. It’s not necessary to show that a combination is ‘the *best* option, only that it be a *suitable* option.’” quoting *PAR Pharm., Inc. v. TWI Pharms., Inc.*, 773 F.3d 1186, 1197–98 (Fed. Cir. 2014)). Thus, Patent Owner’s argument does not undermine Petitioner’s showing of a motivation to combine Harris and the FBDIMM Standards.

b) *Use of Separate Converters to Generate $V_{DD}/V_{DDQ}/V_{DDL}$ or V_{CC}/V_{CCFBD}*

Patent Owner argues against Petitioner’s showing of a motivation to combine by arguing that neither Harris nor the FBDIMM Standards suggest separate power rails for V_{DD}/V_{DDQ} and V_{DDL} or V_{CC} and V_{CCFBD} for Voltage Mappings A and B. Resp. 24–28. Patent Owner argues that the DDR2 Specification that Petitioner cited requires that the voltages V_{DD} , V_{DDQ} and V_{DDL} are driven from a single power converter output. *Id.* (citing Ex. 1026, 9; Ex. 2061 ¶ 87). The excerpt from the DDR2 Specification (JESD79-2B) in question states as follows:

2.3.1 Power-up and initialization sequence

The following sequence is required for POWER UP and Initialization.

- a) Apply power and attempt to maintain CKE below $0.2 \cdot V_{DDQ}$ and ODT^{*1} at a low state (all other inputs may be undefined.) The power voltage ramp time must be no greater than 20mS; and during the ramp, $V_{DD} > V_{DDL} > V_{DDQ}$ and $V_{DD} - V_{DDQ} < 0.3$ volts.

- V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output. AND
- VTT is limited to 0.95 V max. AND
- Vref tracks $V_{DDQ}/2$.

or

- Apply VDD without any slope reversal before or at the same time as VDDL.
- Apply VDDL without any slope reversal before or at the same time as VDDQ.
- Apply VDDQ without any slope reversal before or at the same time as VTT & Vref.

at least one of these two sets of conditions must be met.

Ex. 1026, 9 (DDR2); *see also* Ex. 1046, 15 (same for DDR3).⁴ The above excerpt from the DDR2 Specification states a “first option” (boxed in blue⁵) that requires V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output, and a “second option” (boxed in red) the defines timings for when the voltages V_{DD} , V_{DDL} and V_{DDQ} must be applied relative to V_{DDL} , V_{DDQ} , and V_{TT} & V_{ref} , respectively. Patent Owner’s argument focuses on the “first option” (boxed in blue above), which specifies that V_{DD} , V_{DDL} and V_{DDQ} are driven from a single power converter output. Resp. 24–28. However, Petitioner contends that the “second option” (boxed in red above) does not specify any converter. Petitioner contends that V_{DD} , V_{DDQ} and V_{DDL} would have been generated with separate converters because they are identified as separate voltages with separate pins (Ex. 1028, 17–20), JEDEC indicates that they can be turned on and off independently (Ex. 1026, 9), and JEDEC indicates that V_{DDL} should use an isolated voltage source (Ex. 1026, 2–3). Pet. 29–30 (citing Ex. 1003 ¶¶ 248–249; Ex. 1062, 13). Petitioner shows that one of ordinary skill in the art would have understood the second option

⁴ Although Petitioner does not expressly rely on JEDEC’s JESD79-2B (DDR2) (Ex. 1026) and JESD79-3A (DDR3) (Ex. 1046) in its combinations, the parties seem to be in agreement that these specifications are part of the background knowledge that a person of ordinary skill in the art would have had. In an obviousness analysis, the background knowledge is properly considered together with the demands known in the design community, and the inferences and creative steps that a person of ordinary skill in the art would employ. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013).

⁵ ⁵ In our opinion, Petitioner should have drawn the blue box to include the following two lines “ V_{TT} is limited to 0.95 V max, AND V_{ref} tracks $V_{DDQ}/2$ ” because of the way the standard presents the two sets of conditions with the word “or” between them and refers to them as “two sets of conditions.” We refer to the two sets of conditions as “options” herein.

(boxed in red above) and other parts of the JESD78-2B specification suggest the possibility of using separate voltage regulators for V_{DD} , V_{DDQ} and V_{DDL} .

Patent Owner argues that a single power converter was required so that the voltages track one another during power up. Resp. 25–26 (citing Ex. 2006, 4; Ex. 2061 ¶ 88; Ex. 1023 ¶ 12). But Dr. Wolfe indicates that the voltages of separate converters can be made to track one another by coupling their feedback, which was a known option. Ex. 2060, 54:25–55:13. Patent Owner does not refute Dr. Wolfe’s testimony.

Patent Owner asserts that Petitioner’s contention that an FBDIMM has separate pins for V_{DD} , V_{DDQ} and V_{DDL} does not support its contention of separate converters, “particularly where it was known that there are stability and timing benefits when using a single source.” Resp. 27 (citing Ex. 2006, 4). Dr. Wolfe explains that the FBDIMM module of Petitioner’s combination would not have separate pins to receive V_{DD} , V_{DDQ} and V_{DDL} but would derive them onboard the module from the 12-volt input pin. Ex. 2060, 35:25–36:15. As to stability, Dr. Wolfe testifies that a feedback loop maintains stability in light of current variations, temperature changes, and other factors. *Id.* at 64:25–65:10. As to timing, Dr. Wolfe indicates that separate converters can track one another by coupling their feedback, and that this was a known option. *Id.* at 54:25–55:13.

Patent Owner argues that it was common for persons of ordinary skill in the art to supply all three voltages using a single power converter. Resp. 28 (citing Ex. 2060, 133:5–134:6, 39:11–40:24). Although using a single power converter may have been the most common approach, this does not negate Petitioner’s showing that a person of ordinary skill in the art would have considered separate converters desirable under certain circumstances.

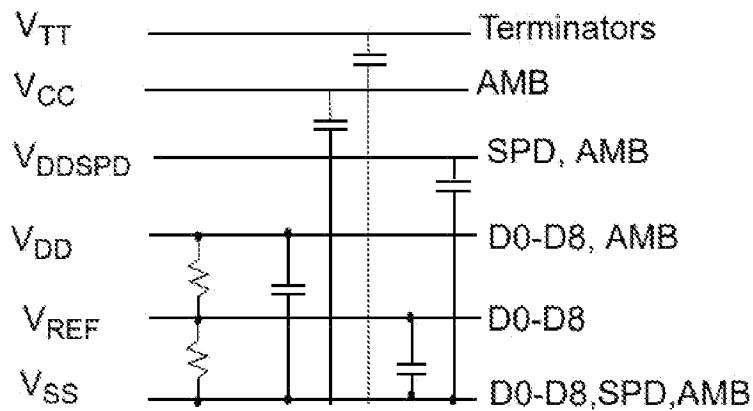
See Elekta Ltd. v. ZAP Surgical Systems, Inc., 81 F.4th 1368 (Fed. Cir. 2023) (quoting *Novartis Pharms. Corp. v. West-Ward Pharms. Int'l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (“Nor does an obviousness showing ‘require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation’”).

Patent Owner similarly argues that V_{CC} and V_{CCFBD} are separate voltages with separate pins, but asserts that Petitioner does not explain why a single on-board V_{CC} power source is insufficient for V_{CC} and V_{CCFBD} power rails when a single set of V_{CC} interface pins providing power from the host met the power needs for both V_{CC} and V_{CCFBD} . Resp. 29–30 (citing Pet. 30–31; Ex. 1028, 11–12; Ex. 2061 ¶ 92). For similar reasons as explained for V_{DD} , V_{DDQ} and V_{DDL} , we disagree. As Petitioner indicates, using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. Reply 15 (citing Ex. 2060, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15; Ex. 2012, 73; Ex. 1003 ¶ 255).

Patent Owner’s arguments do not undercut Petitioner’s showing of a motivation to combine.

c) Motivation for Third Buck Converter for V_{TT}

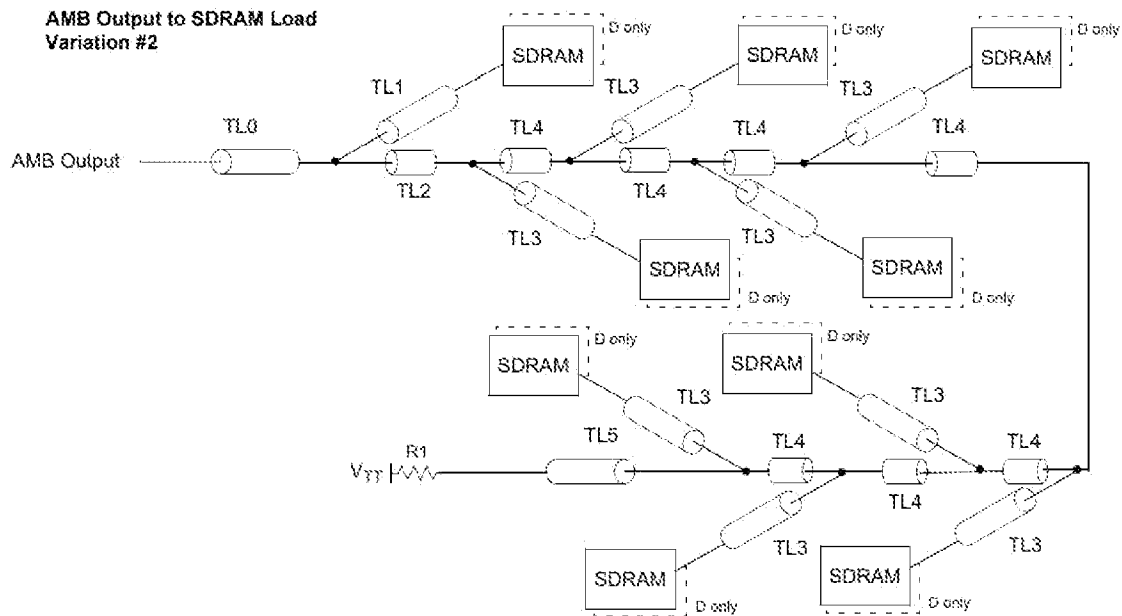
Patent Owner contends that Petitioner’s Voltage Mapping C relies on three different voltages, including a termination voltage, V_{TT} , connected to terminators. Resp. 30 (citing Pet. 26; Ex. 1028, 13, 15–16). For context, the relevant figure from JESD205 is reproduced below:



Ex. 1028, 13. The above figure shows various voltages V_{TT} , V_{CC} , V_{DDSPD} , V_{DD} , V_{REF} , V_{SS} and their relationship to one another according to the JEDEC Specification JESD205. *Id.*

Patent Owner argues that Harris does not disclose generating V_{TT} on its module and argues that it would be supplied from the motherboard instead because this would ensure that all DIMMs have the same termination voltages and eliminate ground loops that could degrade signal integrity. Resp. 31–32 (citing Ex. 2061 ¶ 96). Patent Owner further argues that design complexity favors having a single regulator for a group of DIMMs rather than one regulator per DIMM. *Id.* at 32 (citing Ex. 1027, 20; Ex. 2061 ¶ 97; Ex. 2012, 72). In addition, Patent Owner argues that Petitioner touted JEDEC’s move of voltage regulation from the motherboard to the DIMM as a “major design improvement” and Patent Owner asks, “If it had been so obvious, why did it take the industry so long to make that design improvement?” *Id.* at 32–33. Patent Owner further argues that generating V_{TT} is not one of two choices, and that V_{TT} is generally not provided to DDR2 modules; passive termination is used instead. *Id.* at 33 (citing Ex. 2061 ¶ 98; Ex. 2044, 6; Ex. 2045, 6; Ex. 2046, 4.20.11-6; Ex. 2006, 5).

In Reply, Petitioner argues that Harris teaches generating all of the voltages on the module, including all of the voltages for an FBDIMM, which would include V_{TT} . Reply 16 (citing Inst. Dec. 20–23; Ex. 1023 ¶ 12; Ex. 2060, 30:15–20, 72:22–73:7, 103:11–104:23, 109:5–111:10, 239:8–20). Petitioner contends that “Harris’s module provides the voltages for the buffer 112 to send address and control signals to the DDR memory devices, which the FBDIMM Standards make clear require not just V_{DD} , but also V_{TT} to terminate those signals.” *Id.* at 16–17 (underlining omitted) (citing Pet. 17–18; Ex. 1028, 9, 15, 68). Petitioner reproduces the figure below:



Pet. 17 (citing Ex. 1028, 68). The above figure is from the FBDIMM Standards and shows the net structure routing for addresses and commands from the advanced memory buffer (AMB) to the SDRAMs and how they are terminated by V_{TT} . *Id.* Petitioner contends that V_{TT} is required by the JEDEC Standards to track V_{DD} (Ex. 1028, 15) which is why dual buck converters for generating V_{DD} and V_{TT} were readily available. Reply 17–19

(citing Ex. 1028, 9; Ex. 2060, 72:22–73:7, 196:3–197:7; Ex. 1040, 1, 11; Ex. 1041, 1; Ex. 1048).

Petitioner further argues that Patent Owner’s “suggestion of producing V_{TT} on the motherboard—where V_{TT} could change with each new generation of memory devices—would defeat the benefit of Harris’s invention, e.g., ‘a technology-independent voltage distribution scheme’ to ‘provide upgrades to next generation DRAM technology in a cost-effective manner.’” *Id.* at 19 (citing Ex. 1023 ¶¶ 2, 19).

We agree with Petitioner that the combination of Harris and the FBDIMM Standards at least suggests that V_{TT} can be generated onboard the memory module. Reply 16–19 (citing Pet. 17–18; Inst. Dec. 20–23; Ex. 1023 ¶¶ 2, 12, 19; Ex. 1040, 1, 11; Ex. 1041, 1–2, 7–9; Ex. 1048; Ex. 1075, 112:12–114:8; Ex. 2060, 30:15–20, 57:17–20, 58:18–59:10, 72:22–73:7, 103:11–104:23, 109:5–111:10, 140:15–24, 239:8–20). Harris indicates that its on-board voltage regulator converts externally supplied voltage into appropriate local voltage levels for powering memory devices of the memory assembly module. Ex. 1023, code (57), ¶¶ 12, 16, Fig. 2, step 204. Petitioner shows that V_{TT} is needed onboard a memory module to terminate address and control signals. Reply 16–17 (citing Pet. 17–18; Ex. 1028, 9, 15, 68). In addition, Petitioner shows that V_{TT} is required by JEDEC to track V_{DD} , which is why dual buck converters for generating these voltages were commercially available. Reply 17–19 (citing Ex. 1028, 9; Ex. 2060, 72:22–73:7, 196:3–197:7; Ex. 1040; Ex. 1041; Ex. 1048). In light of these facts and the teachings of Harris and the FBDIMM Standards, Petitioner has shown that a person of ordinary skill in the art would have considered adding to the memory module a third buck converter to generate V_{TT} .

Patent Owner further argues that “even if V_{TT} is generated on module, there is no reason that it should be generated by a buck converter.”

Resp. 33. Patent Owner contends that, “[a]s Micron noted, an LDO would be sufficient to power V_{TT} .” *Id.* (citing Ex. 2006, 7; Ex. 2007–2010; Ex. 2050). Patent Owner contends that an LDO is much smaller than a buck converter, including a controller and discrete components such as an inductor, which makes the LDO preferable given the space constraint. *Id.* (citing Ex. 2061 ¶ 99). Patent Owner also argues that an LDO is preferred when the current load is below 1A. *Id.* (citing Ex. 2061 ¶ 99; Ex. 2047, 21; Ex. 2048, 23; Ex. 2049, 20; Ex. 1040, 23–24, Figs. 22–25).

In Reply, Petitioner contends that buck converters were commercially available for V_{TT} and provided high efficiency compared to an LDO. Reply 19 (citing Ex. 1075, 112:12–114:8 (LDO may be only 10% efficient); Ex. 2060, 57:17–20, 58:18–59:10 (buck converters up to 98% efficient), 140:15–24 (trend has been to use buck converters)). We agree with Petitioner that, under appropriate circumstances, one of ordinary skill in the art would have chosen a buck converter over an LDO. Indeed, Patent Owner’s arguments simply underscore that the person of ordinary skill in the art would have been aware of, and capable of using, various known options for voltage conversion.

d) Conclusion on Motivation to Combine Harris and the FBDIMM Standards

Petitioner has sufficiently shown that a person of ordinary skill in the art would have combined Harris and the FBDIMM standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶ 9. We agree that one of ordinary skill in the art would have

recognized this as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9. For these reasons and others set forth above, we determine that one of ordinary skill in the art would have combined Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at the claims.

4. *Analysis of Independent Claim 1*

a) *Limitation 1.a: “A memory module comprising:”*

Petitioner asserts that Harris’s memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed “memory module.” Pet. 19–20 (citing Ex. 1023 ¶¶ 9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 217–223; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1 in Figure 3.

Patent Owner does not specifically respond to Petitioner’s contention for the preamble. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that Harris teaches the preamble.

b) *Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”*

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. For example, Petitioner notes that Harris discloses that its double-rank DIMMs may be accommodated on both sides of a printed circuit board (PCB). Pet. 20 (citing Ex. 1023 ¶ 13). Petitioner

further notes that a PCB may be referred to as a “memory board” or “raw card.” *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 224–226).

As for the PCB “having an interface configured to fit into a corresponding slot connector of a host system,” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 227–228).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “configured to couple power, data, address and control signals between the memory module and the host system.” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A (Ex. 1023 ¶ 10, 12, 19). Petitioner contends that the Harris’s buffer 112 in Figure 1A is called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 22. Petitioner indicates that buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Harris’s Figure 1A. Pet. 22–25 (citing Ex. 1023 ¶ 9 (disclosing that “buffer/logic component 112 is provided for buffering command/address(C/A) space as well as data space at least for a portion of memory devices 110-1 through 110-N”)). In addition, Petitioner argues that the FBDIMM Standards indicate buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

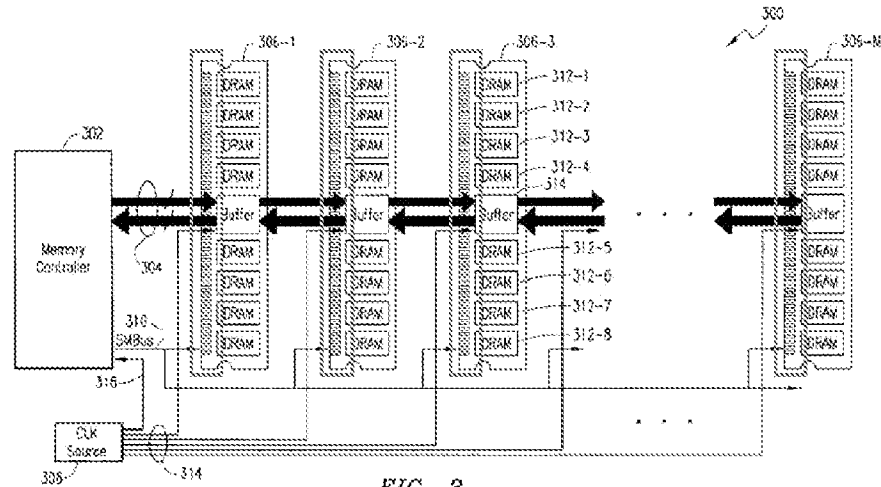
(1) *Power Via Edge Connection*

Claim 1 recites “the interface including a plurality of edge connections configured to couple power . . . between the memory module and the host system.” Ex. 1001, 38:22–25. Patent Owner argues that “all claims of the ’054 patent require that the memory module includes a module interface that fits into a memory slot of a host system and receives voltage/power from the host system via that interface.” Resp. 4–5.

According to Patent Owner, Harris does not disclose such a memory module. *Id.* at 5. Specifically, Patent Owner argues that “Harris supposedly provides a ‘technology-independent voltage distribution scheme for memory devices wherein system board power supply and associated voltage plane(s) are eliminated.’” *Id.* at 6 (citing Ex. 1023 ¶ 19) (emphasis omitted). Patent Owner argues that Harris’s memory module connects to the external voltage source from the side and not from its edge connections. *Id.* at 7. Patent Owner contends that supplying power to a DIMM from its side was known in the art. *Id.* (citing Ex. 2035; Ex. 2036, 41–42; Ex. 2061 ¶ 59).

Petitioner contends that Patent Owner does not dispute that the FBDIMM Standards teach supplying power via edge connections, which Patent Owner’s expert, Dr. Mangione-Smith, admits were “standard.” Reply 3 (citing Pet. 16–21; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9). Petitioner further contends that Patent Owner argues that Harris’s “externally supplied voltage” requires power external to the entire host system (*id.* (citing Resp. 4–8)), when “external” relates to the DIMM memory module and does not exclude power from the host system, including from edge connections. *Id.* (citing Ex. 2060, 66:7–19, 67:20–68:21, 91:22–92:7, 129:24–130:17; Pet. 21–22). Petitioner disagrees with

Patent Owner's assertion that Harris's memory module would never utilize edge connections for power. Reply 3-4. To explain its position, Petitioner provides an annotated version of Harris's Figure 3 below.



Petitioner contends that Harris's Figure 3 above shows memory boards 306 connected to the host system only through their edge connections (annotated by Petitioner in red), implying that power would also come from those edge connections. Reply 4 (citing Ex. 1023, Fig. 3). Petitioner contends that Harris's Figure 3 is nearly identical to an Intel drawing where memory modules admittedly receive power through the edge connections from the host system. *Id.* (citing Ex. 1075, 171:3-17; Ex. 2101, 4).

Petitioner also contends that Harris discloses that each memory board receives a supply voltage which may be sourced from either the host system or a separate voltage source. *Id.* at 4-5 (citing Ex. 1075, 167:23-168:1; Ex. 1023 ¶ 17). Petitioner asserts that there is no dispute that the supply voltage was commonly provided through edge connectors alongside memory controller signals. *Id.* at 5 (citing Ex. 2060, 131:3-5).

We agree with Petitioner that Harris's "external voltage source" covers sources originating from the host system (as well as sources external

to the host system). Harris states as background that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Harris also discloses that “external voltage sources may comprise any combination of *known* or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14 (emphasis added). The parties agree that supplying power from a host system to a memory module was ‘known’ in the art, as referenced by Harris. Ex. 2060, 131:3–5; Ex. 1075, 162:18–25. And, as Petitioner notes, Harris’s Figure 3 shows that the memory boards have only edge connections, suggesting they receive their power from them. Ex. 1023, Fig. 3; Ex. 1003 ¶ 229. These facts support the conclusion that Harris’s external voltage source may originate from the host system. Accordingly, we do not agree with Patent Owner’s arguments that Harris is limited to using only external voltage sources unrelated to the host system. Resp. 4–14; Sur-Reply 2–6.

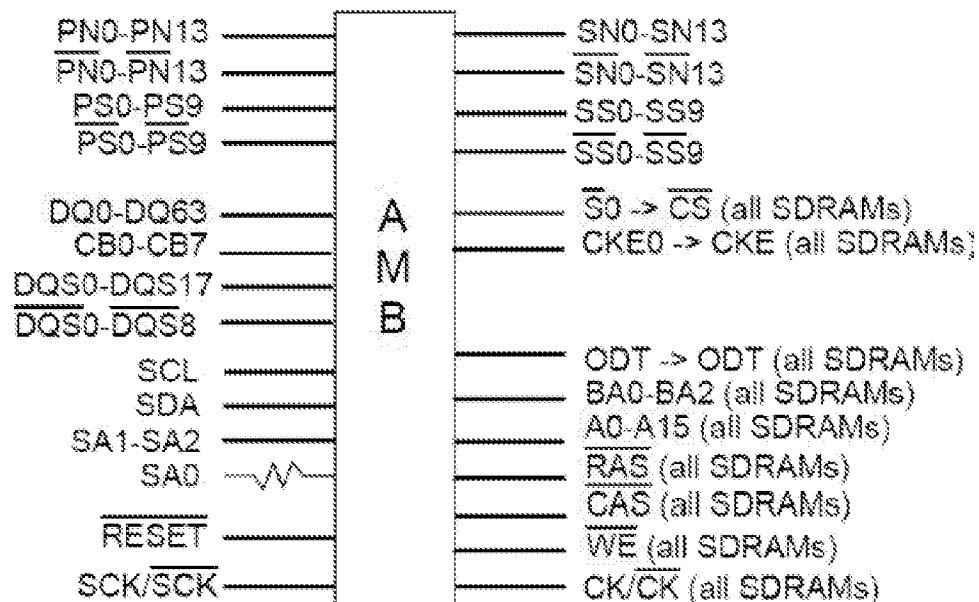
Furthermore, Petitioner demonstrates that the FBDIMM Standards show that memory modules receive power from their edge connections, and Petitioner relied on the FBDIMM Standards as disclosing this feature of the claims. *See* Pet. 16–21; Ex. 2060, 131:3–5; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9; Ex. 1028, 97 (GF-9), 103.

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards discloses this limitation.

*(2) Data, Address, and Control Signals Via
Edge Connection*

Claim 1 recites an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the

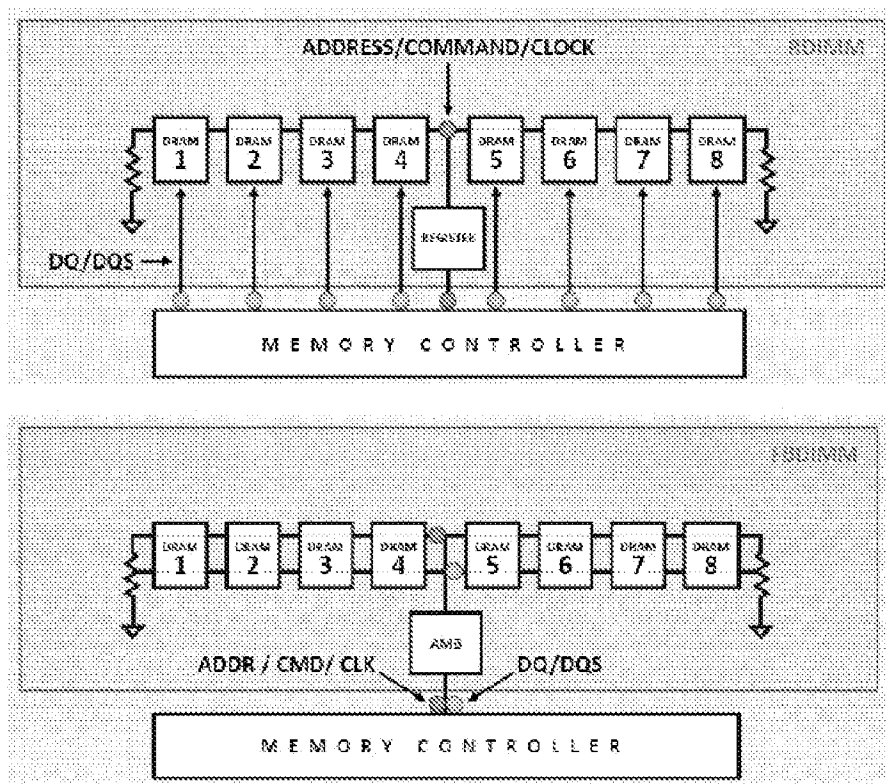
memory module and the host system.” Ex. 1001, 38:23–25. Patent Owner argues that the DIMM and AMB do not receive data signals (DQ0–DQ63) or address and control signals (A0–A15, RAS, CAS, WE) from the host. Resp. 14 (citing Pet. 22–25). According to Patent Owner, these signals are instead generated by the AMB based on decoded FBDIMM channel signals (PS[9:0] and PS[9:0]bar). *Id.* at 14–18 (citing Ex. 2061 ¶¶ 72; Ex. 1028, 29; Ex. 1027, 3–4; Ex. 2039, 2; Ex. 2101, 4; Ex. 2040, 1). Patent Owner argues that Petitioner’s evidence does not show that the data, address and control signals are received from the host. *Id.* at 18. Specifically, Patent Owner relies on the figure below from the FBDIMM Standards (*see* Sur-Reply 7 (citing Ex. 1028, 13)).



According to Patent Owner, the above figure shows that data, address and controls signals (highlighted in yellow) are generated and outputted by the buffer, and are not received from the DIMM interface. Sur-Reply 6–7 (citing Pet. 23–25; Resp. 14–18; Ex. 2061 ¶¶ 72–74). Patent Owner contends this does not comport with the claim language which requires an

“interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” *Id.* at 7 (citing Ex. 1001, 38:23–25). Patent Owner argues that Dr. Wolfe merely stated that the AMB outputs “information” to the DRAMs but the claims require specific “signals,” not just “information.” *Id.* at 8 (citing Ex. 1003 ¶ 230; Ex. 2060, 7:20–11:6; Ex. 1001, 38:23–25). Patent Owner argues that the outputs from the AMB are not transmitted over FBDIMM’s edge connections. *Id.*

Petitioner argues that Patent Owner’s technology tutorial indicates that an FBDIMM receives address and control signals similar to an RDIMM, as shown below:



Reply 9 (citing Ex. 1077, 8–9; Ex. 1075, 91:23–92:19, 95:14–96:13, 97:16–98:18). The figures above show that the memory controllers transmit

address, command and clock signals (shown in red), or information containing these signals, to the register (RDIMM) or AMB (FBDIMM), which transmits them to the DRAMs. *Id.*; Sur-Reply 10.

Petitioner contends that the host system provides address, command and clock signals to the AMB encoded as a packetized, serialized signal on fewer wires than would be required to receive them as separate, parallel signals. Reply 9–10 (citing Resp. 14; Ex. 2061 ¶ 31; Ex. 1075, 155:22–157:1, 212:3–8, 213:3–215:20, 219:13–220:9, 226:7–228:8; Ex. 2060, 8:3–11:6). Petitioner alleges that Patent Owner is in essence attempting to rewrite the claims to require dedicated pins for these signals, which the claims do not require. *Id.*

Patent Owner argues that it is not attempting to rewrite the claims to require “dedicated pins” as Petitioner alleges. Sur-Reply 9 (citing Reply 9–10). Patent Owner asserts that the pin names indicate what “signals” are exchanged at the interface between the host and the FBDIMM, as shown in the figure below.

Pin Name	Pin Description	Pin
FB-OBMM Chemical Signals		
SDN	System Clock (note: positive logic)	1
SDN	System Clock Input (negative logic)	2
PSD(1:32)	Primary Northbound Data (positive logic)	3-34
PSD(1:32)	Primary Northbound Data (negative logic)	35-66
PSD(33)	Primary Southbound Data (positive logic)	67
PSD(33)	Primary Southbound Data (negative logic)	68
SDN(1:32)	Secondary Northbound Data (positive logic)	69-100
SDN(1:32)	Secondary Northbound Data (negative logic)	101-132
SDS(33)	Secondary Southbound Data (positive logic)	133
SDS(33)	Secondary Southbound Data (negative logic)	134
PSD(33:2)	For an external precision calibration, spectral calibration up limit	135
OBMM Interface Signals		
SDS(33:2)	Data Receive (positive logic)	3
SDS(33:2)	Data Receive (negative logic)	4
SDS(33:2) (OBMM only)	Data Receive pin (OBMM only) (positive logic) (these signals are always low to all OBMMs on a ring)	5
SDS(33:2)	Data Receive pin (OBMM only) (negative logic)	6
SDS(33:2)	None	136
SDS(33)	Low-Z Drive	137
ATD(1:32), ATD(33)	Addressed ATD is part of the one change word and	138
SDS(1:32), SDS(33)	Reset Addressing	139
SDS(1), SDS(3)	Start of command with SDS(1), SDS(2) and SDS(3)	140
SDS(4), SDS(5)	Start of command with SDS(4), SDS(5) and SDS(6)	141
SDS(6), SDS(7)	Start of command with SDS(6), SDS(7) and SDS(8)	142
SDS(8), SDS(9)	Start of command with SDS(8), SDS(9) and SDS(10)	143
SDS(10), SDS(11)	Start of command with SDS(10), SDS(11) and SDS(12)	144
SDS(12), SDS(13)	Start of command with SDS(12), SDS(13) and SDS(14)	145
SDS(14), SDS(15)	Start of command with SDS(14), SDS(15) and SDS(16)	146
SDS(16), SDS(17)	Start of command with SDS(16), SDS(17) and SDS(18)	147
SDS(18), SDS(19)	Start of command with SDS(18), SDS(19) and SDS(20)	148
SDS(20), SDS(21)	Start of command with SDS(20), SDS(21) and SDS(22)	149
SDS(22), SDS(23)	Start of command with SDS(22), SDS(23) and SDS(24)	150
SDS(24), SDS(25)	Start of command with SDS(24), SDS(25) and SDS(26)	151
SDS(26), SDS(27)	Start of command with SDS(26), SDS(27) and SDS(28)	152
SDS(28), SDS(29)	Start of command with SDS(28), SDS(29) and SDS(30)	153
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SDS(82), SDS(83)	Start of command with SDS(82), SDS(83) and SDS(84)	180
SDS(84), SDS(85)	Start of command with SDS(84), SDS(85) and SDS(86)	181
SDS(86), SDS(87)	Start of command with SDS(86), SDS(87) and SDS(88)	182
SDS(88), SDS(89)	Start of command with SDS(88), SDS(89) and SDS(90)	183
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SDS(100), SDS(101)	Start of command with SDS(100), SDS(101) and SDS(102)	189
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SDS(118), SDS(119)	Start of command with SDS(118), SDS(119) and SDS(120)	198
SDS(120), SDS(121)	Start of command with SDS(120), SDS(121) and SDS(122)	199
SDS(122), SDS(123)	Start of command with SDS(122), SDS(123) and SDS(124)	200

Ex. 1028, 29. The figure above shows Patent Owner’s highlighting to show the “FB-DIMM Channel Signals” including PS[9:0] and PS[9:0]bar, and the “DDR2 Interface Signals” including DQ0–DQ63, A0–A15, RAS, CAS, and WE. Patent Owner states that all memory control for the DRAM resides in the host, and argues that the fact that all read, write, and configuration accesses are addressed to the DIMM does not undermine that the signals Petitioner relies on are generated and outputted by the buffer, and not exchanged at the interface between the FBDIMM and the host. Sur-Reply 9–10 (citing Ex. 2060, 7:20–11:6; Ex. 2060 ¶ 230).⁶

⁶ Patent Owner’s citation to “Ex. 2060 ¶ 230” appears to be an error and we cannot determine from the context what Patent Owner intended to cite.

Petitioner further argues that the FBDIMM is a preferred embodiment in the '054 patent, and excluding a preferred embodiment from the claims is “rarely, if ever correct.” Reply 10 (citing *Kaufman v. Microsoft Corp.*, 34 F.4th 1360, 1372 (Fed. Cir. 2022)). To the contrary, Patent Owner argues that the claims need not encompass FBDIMM embodiments, which are disclosed but unclaimed subject matter. Sur-Reply 10 (citing *Maxwell v. J. Baker, Inc.*, 86 F.3d 1098, 1107 (Fed. Cir. 1996); Ex. 1001, 21:38–55).

According to Petitioner, Patent Owner concedes that the signals (PS[9:0] and PS[9:0]bar) received by the AMB result in data, address, and control signals needed by DDR2 SDRAMS. Reply 10 (citing Ex. 2061 ¶ 31). Petitioner further notes that the FBDIMM Standards state that all memory control for the DRAM resides in the host, including memory request initiation, and that the AMB acts as a DRAM memory buffer for all read, write, and configuration accesses addressed to the DIMM. *Id.* (citing Ex. 1027, 1). As a buffer for all such commands to the FBDIMM, Petitioner asserts, the AMB must necessarily couple data, address, and control signals from the host system to the memory module as the claims require. *Id.*

Petitioner and Patent Owner agree that FBDIMM AMB receives data, address, and control signals encoded in packetized, serialized form at its edge connections via the signals PS[9:0] and PS[9:0]bar, and that the AMB uses these signals to generate data, address and command signals including DQ0–DQ63, A0–A15, RAS, CAS, and WE provided to the DIMMs onboard the memory module. Pet. 22–25; Sur-Reply 10. The claims merely require an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” Ex. 1001, 38:23–25. We determine that the data, address, and

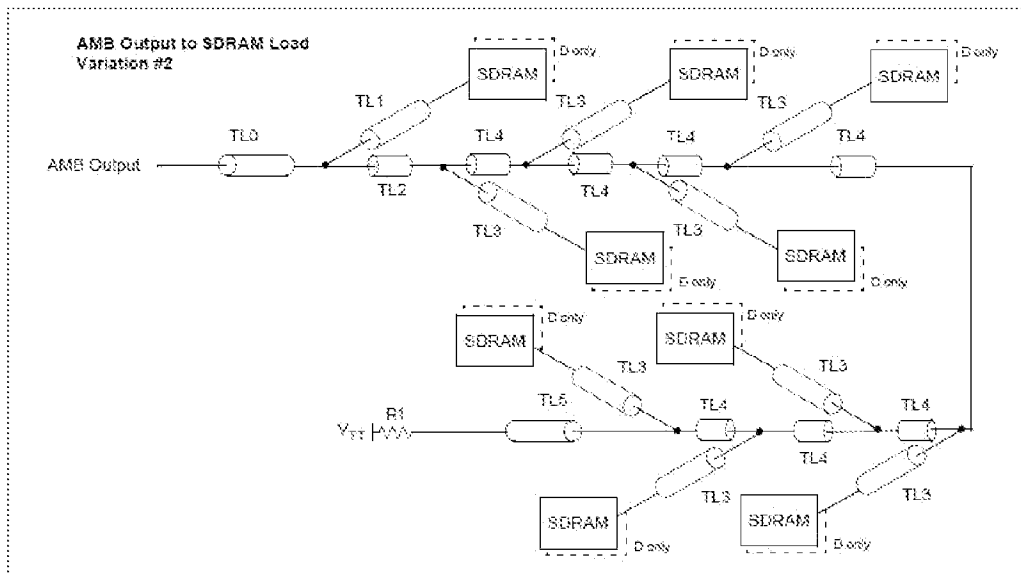
control signals of Harris and the FBDIMM Standards as received at edge connections coupling the memory module and host system satisfy this claim limitation, even though the signals are in encoded, packetized, or serialized form and thus may be received at the same pin or pins. That the signals are encoded, packetized, and serialized does not change the fact that they are data, address, and control signals. The claims do not require these signals to be received at the edge connections in any particular form or on any particular pins. *See In re Self*, 671 F.2d 1344, 1348 (CCPA 1982) (stating that limitations not appearing in the claims cannot be relied upon for patentability).

Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitation 1.b of claim 1, and Patent Owner's arguments do not undermine Petitioner's showing.

c) Limitation 1.c: "a voltage conversion circuit coupled to the PCB and configured to provide at least three regulated voltages, wherein the voltage conversion circuit includes at least three buck converters each of which is configured to produce a regulated voltage of the at least three regulated voltages"

Petitioner contends that Harris and the FBDIMM Standards disclose limitation 1.c of claim 1 of the '054 patent. Pet. 26–30. Specifically, Petitioner contends that Harris's Voltage Regulator Module 102 corresponds to the claimed "voltage conversion circuit." Pet. 26 (citing Ex. 1023 ¶ 10, Fig. 1A). Petitioner contends that the claimed "at least three regulated voltages" correspond to voltage 106 (V_{cc}) and voltage 108 (V_{dd}) as shown in Harris's Figure 1A, and a third voltage (V_{TT}) which is not actually shown in Harris's Figure but is shown in the FBDIMM Figure below.

DDR2 SDRAM Fully Buffered DIMM Design Specification DDR2 Fully Buffered DIMM Wiring Details
Net Structure Routing for Address/Command to SDRAM (Raw Cards B, C)



Ex. 1028, 68. The figure above shows voltage V_{TT} terminating an array of SDRAM devices via resistor R1.

As to the voltages being “regulated,” Petitioner argues Harris discloses that “at least one on-board voltage regulator” is “capable of generating tightly-controlled voltage levels.” Pet. 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 236).

As to the “at least three buck converters,” Petitioner contends that Harris teaches using buck converters to provide the three regulated voltages above. Pet. 28 (citing Ex. 1003 ¶¶ 238–241). Petitioner contends Harris teaches using a “high-frequency switching voltage converter capable of generating tightly-controlled voltage levels” to provide each needed on-board regulated voltage. *Id.* (citing Ex. 1023 ¶¶ 10, 12). Petitioner contends it would have been obvious to use a “buck converter” to convert higher input voltage (e.g., 12V) to the lower output voltage (3.5V or less), and there would have been a reasonable expectation of success because buck

converters were well-known “switching” devices commonly used to step down the voltage between its input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 147–150, 239–241; Ex. 1058, 3, 5, 12–16 (1995 Lenk textbook showing buck, boost, and buck-boost circuits); Ex. 1032, 161 (1995 Mohan textbook showing various step-down (buck) converters); Ex. 1030, 2:32–43, 5:39–44; Ex. 1024, Fig. 6 (Amidi showing DC/DC buck converter); Ex. 1050, 1:21 (identifying buck converters as one of “the most basic building blocks in power electronics”)). Petitioner contends buck converters were well-known for their efficiency in generating step down voltages without generating excess heat or requiring large cooling devices. *Id.* at 29 (citing Ex. 1003 ¶ 240; Ex. 1059, 5:23–30; Ex. 1058, 5; Ex. 1040, 1, 23–24; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1062, 11; Ex. 1064 ¶ 101). Petitioner contends that it would have been obvious to use at least three converters in Harris given the need for at least three different voltages in the FBDIMM Standards. Pet. 29–30 (citing Ex. 1028, 17–20; Ex. 1026, 2–3, 9; Ex. 1003 ¶¶ 248–249, 255; Ex. 1062, 13; Ex. 1028, 30–32).

For the reasons discussed above in Section II.D.3, we disagree with Patent Owner’s arguments regarding the use of three buck converters.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitation 1.c of claim 1.

d) Limitation 1.d.1: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to at least one regulated voltage of the at least three regulated voltages”

Petitioner asserts that Harris discloses a “a plurality of components coupled to the PCB” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 261–266). Petitioner further contends that these components are each coupled to at least one regulated voltage of the at least three regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches limitation 1.d.1 of claim 1.

e) Limitation 1.d.2: “the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 31–32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 267–271). Petitioner contends that a person of ordinary skill in the art would have known that according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* (citing Ex. 1028, 9; Ex. 1045, cover; Ex. 1026, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

f) Limitation 1.d.3: “a first circuit that is coupled to the plurality of SDRAM devices and to a first set of edge connections of the plurality of edge connections”

Petitioner contends that the “first circuit” corresponds to buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–33 (citing *id.* at 20–25; Ex. 1023, Figs. 1A, 3; Ex. 1003 ¶¶ 272–277). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8.

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

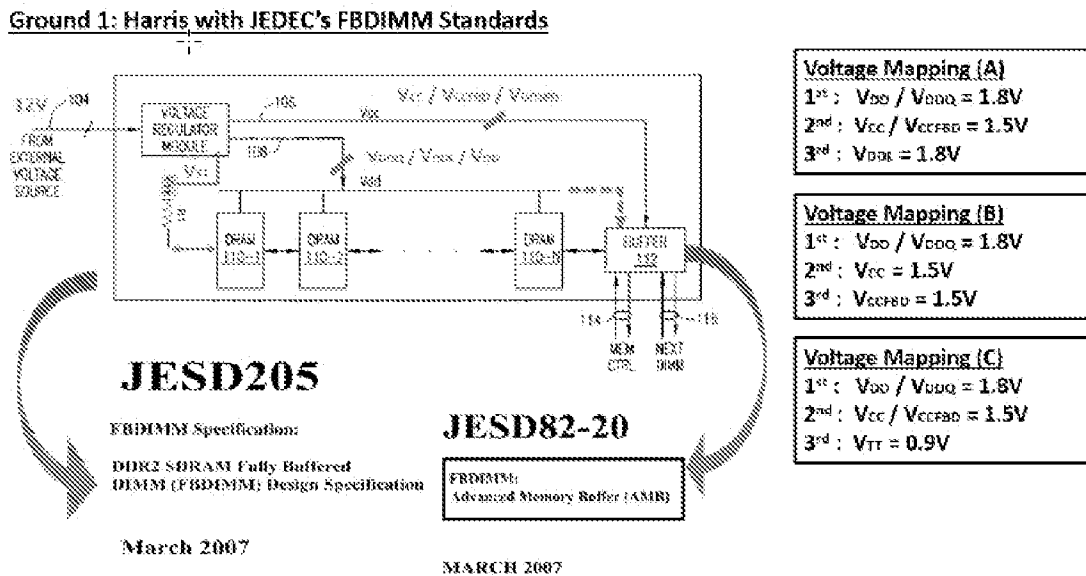
Based on our review and consideration of the record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

g) Limitation 1.d.4: “wherein the first circuit is coupled to first and second regulated voltages of the at least three regulated voltages, and”

Petitioner contends that Harris’s buffer corresponds to the first circuit and is coupled to “first” (e.g., V_{DD} or $V_{DDQ} = 1.8V$) and “second” (e.g., V_{CC} or $V_{CCFBD} = 1.5V$) “regulated voltages of the at least three regulated

voltages.” Pet. 33–34 (citing *id.* 14–19, 27; Ex. 1003 ¶¶ 278–284).

Petitioner provides an annotated version of Harris’s Figure 1A shown below.



As shown above, Petitioner fills in Harris’s Figure 1A with additional information from the FBDIMM Standards to show that Harris’s buffer 112 would be understood by a person of ordinary skill considering the FBDIMM Standards to couple to the voltages V_{DD} or V_{DDQ} and V_{CC} or V_{CCFBD} .

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation 1.d.4 of claim 1.

h) Limitation 1.d.5: “wherein the plurality of SDRAM devices are coupled to the first regulated voltage of the at least three regulated voltages”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34 (citing *id.* 14–19, 27; Ex. 1003 ¶¶ 285–287).

Specifically, Petitioner contends that the SDRAM devices from Harris’s

Figure 1A are coupled to the first regulated voltage (V_{DD} or $V_{DDQ} = 1.8V$) of the “at least three regulated voltages.” *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches limitation 1.d.5 of claim 1.

i) Conclusion for Claim 1

For the reasons explained in Section II.D.3, Petitioner shows that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches each limitation of claim 1. Accordingly, Petitioner has shown that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. Analysis of Dependent Claims

Claim 2 depends from claim 1 and recites “wherein the first regulated voltage has a first voltage amplitude, and the second regulated voltage has a second voltage amplitude, and wherein a first one of the first and second voltage amplitudes is less than a second one of the first and second voltage amplitudes.” Ex. 1001, 38:45–50. Petitioner contends that, as taught by the FBDIMM Standards, the first voltage amplitude may be 1.8V and the second voltage amplitude may be 1.5V which is less than the first voltage amplitude of 1.8V. Pet. 34–35 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 288–296). Patent Owner presents no arguments specific to claim 2.

Claim 3 depends from claim 1 and recites “wherein a third regulated voltage of the at least three regulated voltages has a voltage amplitude of 1.8 volts.” Petitioner contends that the third regulated voltage is V_{DDL} with a value of 1.8V as shown in the FBDIMM Standards (see previous figure—“voltage mapping(A)”). Pet. 35 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 297–301).

Patent Owner argues that because claim 3 relies solely on Voltage Mapping A, Petitioner fails to show obviousness over the combination of Harris and the FBDIMM Standards. Resp. 28. Patent Owner argues that “[a]s for Mapping A, Petitioner does not explain why a single on-board V_{CC} power source is insufficient for $[V_{CC}]$ and $[V_{CCFBD}]$ power rails when a single set of $[V_{CC}]$ interface pins providing power from the host met the power needs for both $[V_{CC}]$ and $[V_{CCFBD}]$.” *Id.* at 29. We addressed this argument in Section II.D.3.c explaining that Petitioner shows that using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. We find no flaw in Petitioner’s reliance on Voltage Mapping A for claim 3.

Claim 15 depends from claim 1 and recites “wherein two of the at least three buck converters are configured to operate as a dual-buck converter.” Ex. 1001, 39:59–61. Petitioner contends at the time there were many commercially available products that could output two (or more) regulated voltages using buck converters, and thus “it would be obvious to implement any two of the regulated voltages as a ‘*dual buck converter*’ to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design.” Pet. 35–36 (citing Ex. 1003

¶ 439; *Qualcomm*, 21 F.4th at 797–99 (Fed. Cir. 2021)). Petitioner contends that the Murata MPD4S014S dual buck converter, Texas Instruments TPS51020 dual buck converter, and Fairchild FAN5026 dual-output PWM controller were commercially available and would have been suitable to implement for use in Harris in view of the FBDIMM Standards. Pet. 37–40 (citing Ex. 1003 ¶¶ 439, 443–444, 446–447; Ex. 1023 ¶ 10; Ex. 1032, 161–64; Ex. 1040, 1, 10–11; Ex. 1041, 1–2, 7–9; Ex. 1042, 16; Ex. 1047, 3:49–50, 4:7–10, 4:37–56, 5:4–13, 5:49–59, Fig. 2; Ex. 1048, 1–2; Ex. 1050, 2:19–20, 3:36–37; Ex. 1058, 5; Ex. 1073, 49–50, 58).

Patent Owner argues that claim 15 of the '054 patent provides “two of the at least three buck converters are configured to operate as a dual-buck converter,” making clear that the “dual buck converter” includes two buck converters. Resp. 14 (citing Ex. 1001, 39:59–61). Patent Owner argues that Harris does not characterize its “high-frequency switching voltage converter” as a “dual” converter. *Id.* (citing Ex. 1023 ¶ 10, Fig. 1A). Patent Owner also contends Harris’s use of the indefinite article “a” before mentioning this converter means that only a single converter is used in Harris’s voltage regulator module. *Id.*

Petitioner correctly asserts that Harris states that “at least one on-board voltage regulator module (VRM)” may be used to generate the voltages V_{DD} and V_{CC} . Reply 12 (citing Ex. 1023 ¶ 10, Fig. 1A). Also, Harris’s claims 1 and 10 recite “at least one voltage regulator module” which covers the possibility of multiple regulators. Although Harris expresses a preference for implementing a voltage regulator module using “a high-frequency switching voltage converter” (emphasis added) which may be understood to mean a single converter, Harris does not preclude using

multiple regulators to generate the voltages V_{DD} and V_{CC} . Ex. 1023 ¶ 10. As Petitioner states, Harris's voltage regulator module receives one input voltage and generates two output voltages. Reply 12 (citing Ex. 1023, Fig. 1A). One of ordinary skill in the art would have understood that there are two options: to use a single converter to generate both voltages, or to use multiple converters to generate the voltages. "When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill in the art has good reason to pursue the known options within his or her technical grasp." *KSR*, 550 U.S. at 421. Petitioner shows that dual buck converters capable of generating the two voltages from the one input were known in the art and used for DIMMs, including the TPS51020 dual buck converter (Ex. 1040), the FAN5026 dual-output PWM controller (Ex. 1041), and the MPD4S014S dual buck converter (Ex. 1042). Armed with the foregoing knowledge, a person of ordinary skill in the art would have understood one option for implementing Harris's voltage regulator module would be to use a dual buck converter notwithstanding Patent Owner's arguments to the contrary.

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris in combination with the FBDIMM Standards teaches the limitations of claims 2, 3, and 15.

6. *Determination for Ground 1*

Petitioner has shown that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at the memory modules recited in claims 1–3 and 15. Accordingly, Petitioner has shown that claims 1–3

and 15 are unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

E. Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi

Petitioner contends claims 1–30 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 41–70. For the reasons that follow, Petitioner shows by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

1. Amidi (Ex. 1024)

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled “Clock and Power Fault Detection for Memory Modules.” Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 11.

Amidi’s Figure 5 is reproduced below.

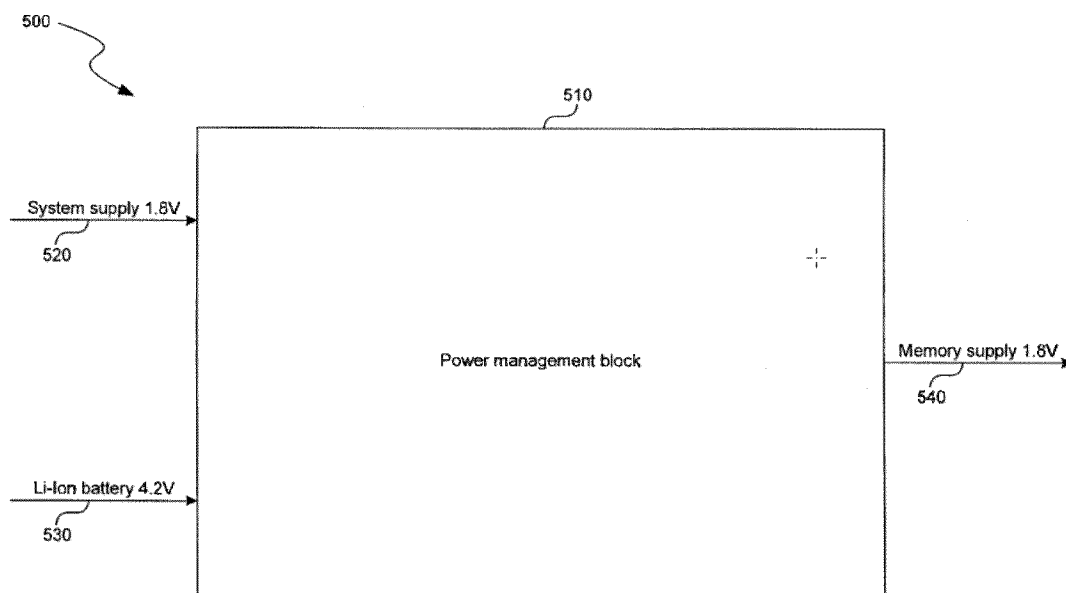


FIG. 5

Amidi's Figure 5 above illustrates power management block 510 that receives incoming system supply 520 and incoming battery supply 530, and generates outgoing memory power supply 540 which is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Fig. 5, Fig. 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 41. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 41–42 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 171–177). Petitioner further notes that Amidi's power management block could be modified easily to work with Harris's FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 42 (citing Ex. 1024, Fig. 5; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 173–174). According to Petitioner, a person of ordinary skill in the art would have considered it obvious that the 12V external supply would be stepped-down with a buck converter to a 5V supply for charging Amidi's battery and that Amidi's battery voltage would be stepped-up with a boost converter to the 12V level used by Harris's memory module. *Id.* at 42–43 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 173–174). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed,

and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 43 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi’s battery backup mode is similar to the S3 power-saving mode of Harris’s FBDIMM memory module and the FBDIMM standards, such that a person of ordinary skill in the art would have been motivated to combine their teachings. *Id.* at 43–45.

In sum, Petitioner contends that a person of ordinary skill in the art “would have been motivated to implement Harris’s FBDIMM memory module with the functionality of Amidi’s power management and logic blocks for detecting power disruptions and switching over to battery backup when needed.” Pet. 44 (citing Ex. 1003 ¶¶ 171–176). Petitioner contends that this “straightforward modification of Harris’s memory module in view of Amidi and the knowledge of a [person of ordinary skill in the art] simply uses a known technique (e.g., Amidi’s battery backup techniques) to improve a similar device (e.g., Harris’s memory module) in the same way (e.g., to provide a backup power supply using a battery).” *Id.* at 44 (citing Ex. 1003 ¶ 177); *see KSR*, 550 U.S. at 418. Petitioner further contends that “the modification merely applies a known technique (e.g., providing a backup power supply) to a known device (e.g., a memory module) that is ready for improvement to yield predictable results (e.g., redundancy when the system supply or clock fails).” Pet. 44 (citing Ex. 1003 ¶ 177); *see KSR*, 550 U.S. at 417.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:

Ground 2: Ground 1 and Battery Backup of Amidi

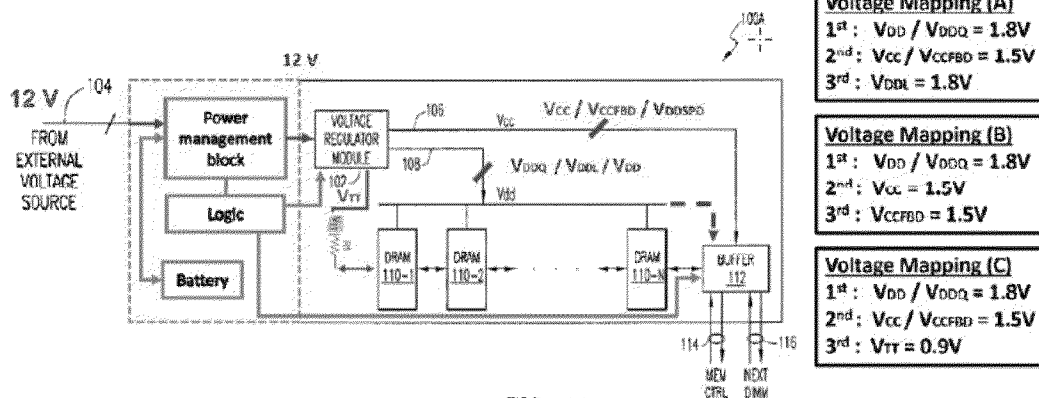


FIG. 1A

Pet. 45. Above, Petitioner has annotated Harris’s Figure 1A to show features added from the teachings of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue).

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Resp. 34–40. Specifically, Patent Owner argues that Harris has “logic 124” that selects a different voltage source if power becomes unavailable on one of the supply voltage paths 120-1 through 120-K. *Id.* at 34–35 (citing Ex. 1023, Fig. 1B; Ex. 2061 ¶ 106). Patent Owner further argues that substituting Amidi’s battery backup solution for Harris’s redundant power sources would require substantial space to accommodate the lithium battery and associated battery charging and monitoring circuits and power management block. *Id.* at 37–38 (citing Ex. 1024, Figs. 4–6; Ex. 1023 ¶¶ 2, 13; Ex. 1061 ¶ 106, n.5; Ex. 2061 ¶ 107). Patent Owner argues that Petitioner “has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the combination would even fit onto Harris’ circuit board” and that “Petitioner has also presented no evidence that a

[person of ordinary skill in the art] would expect Harris’ memory module to be able to accommodate three buck converters, a lithium battery and its associated charging and monitoring circuits, and Amidi’s power management module, at least not without undergoing substantial redesign.” *Id.* at 38–40 (citing Ex. 2101, 20; Ex. 1055, Fig. 4; Ex. 2061 ¶ 110).

We agree with Petitioner that Amidi provides a redundant power supply on the memory module that is independent from the host, and that a person of ordinary skill in the art would have seen the value of adding a redundant power supply onboard Harris’s memory module at least for some practical applications. Reply 19–20 (citing Pet. 41–42; Ex. 1003 ¶ 172; Ex. 1024, 1:28–35, 2:6–26). Although Harris mentions an “external voltage source,” it does not mention the term “battery.” Hence, Patent Owner’s arguments that Amidi provides a solution to a problem that Harris already solves is not correct. *See* Ex. 1003 ¶ 172 (“Amidi recognizes that it is useful to keep data on the memory module with a backup power supply on the module itself, such as a battery supply, when the entire system surrounding the memory module loses power”) (citing Ex. 1024, 1:28–35; 2:6–26). Furthermore, Petitioner shows that memory modules with battery backup were known in the art and readily available. Reply 20 (citing Ex. 2035, 9; Ex. 1075, 165:10–167:7). As to the space issue, Petitioner notes that even if the battery backup occupies one side of an FBDIMM board, the other side still can include one rank or two ranks using stacked memory devices. *Id.* at 20–21 (Ex. 1028, 36; Ex. 1075, 74:22–75:25, 77:10–17).

Accordingly, we determine that Petitioner has shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3 and 15*

We agree with Petitioner that the addition of Amidi does not negate Petitioner’s showing with respect to the previous ground. Pet. 45–46 (citing Ex. 1003 ¶¶ 216–301, 431–448). Accordingly, Petitioner has shown that claims 1–3 and 15 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claim 4*

Claim 4 depends from claim 1 and recites that the memory module further comprises:

a voltage monitor circuit coupled to the PCB and to a second set of edge connections of the plurality of edge connections, the voltage monitor circuit configured to monitor an input voltage received from the second set of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the input voltage having a voltage amplitude below a predetermined threshold voltage, wherein the memory module transitions from a first operable state to a second operable state in response to the trigger signal.

Ex. 1001, 38:54–64.

Petitioner points to Amidi’s power supervisory module/block 480/665/800 as claim 4’s “voltage monitor circuit” coupled to Harris’s edge connections for power. Pet. 46. According to Petitioner, Amidi’s power supervisory module/block monitors the external system power supply 605/825 in Amidi. *Id.* Petitioner contends that Amidi’s power supervisory module/block generates a “trigger signal” corresponding to Amidi’s signal

670/858/868. *Id.* Petitioner contends the “predetermined threshold voltage” corresponds to Amidi’s reference voltage 675/820 set to 5% or 10% below the nominal voltage of 12V. *Id.* at 46–49 (citing Ex. 1023 ¶¶ 12, 13; Ex. 1003 ¶¶ 303–314; Ex. 1024, 4:8–11, 4:44–52, 5:25–43, 5:31–62, 8:30–62, 8:23–29, 9:8–12, Figs. 4–6, 14, 15). Petitioner further states that the trigger signal causes the memory module to transition from normal operations using an external voltage in the normal range (“the first operable state”) to self-refresh operations using a backup battery when the external voltage is outside the normal range (“the second operable state”). *Id.* at 49–50 (citing Ex. 1024, 4:44–52, 8:30–62, 9:8–22, Figs. 6–8, 14, 15; Ex. 1003 ¶¶ 315–321).

Patent Owner does not present any argument specific to claim 4. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 4 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

5. *Claims 5, 7, 8, 23 and 24*

Claim 5 depends from claim 4 and recites

a controller coupled to the voltage monitor circuit;
wherein, in response to the trigger signal, the controller is
configured to perform one or more operations including a write
operation to transfer data to non-volatile memory.

Ex. 1001, 38:65–39:3. Claims 7, 23 and 24 recite the same or similar limitations. Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi discloses these claims. Pet. 50–52 (claim 5); Pet. 55–57 (claim 8); Pet. 57–59 (claim 7); Pet. 69 (claims 23 and

24). We consider Petitioner's contentions and Patent Owner's arguments for claim 5 to be representative of the issues presented for all of these claims.

Petitioner contends that the logic for controlling the S3 sleep mode (Pet. 43–44) corresponds to the claimed “controller” that is coupled to the voltage monitor circuit (Amidi's voltage supervisory block). Pet. 50–51. Petitioner contends that “Amidi's battery backup mode is similar to the S3 power-saving mode of Harris's FBDIMM memory module, because both modes put the SDRAMs in a self-refresh state to preserve data while conserving power.” Pet. 51 (citing *id.* at 43–44). According to Petitioner, “in the event of a power disruption causing Amidi's ‘voltage monitor circuit’ to produce a ‘trigger signal’ . . . , a [person of ordinary skill in the art] would have been motivated to use the S3 sleep mode discussed above . . . to conserve power when using Amidi's battery backup.” *Id.* (citing Ex. 1003 ¶¶ 322–333). Petitioner contends that S3 configuration information is stored in non-volatile memory before entering the S3 sleep mode in response to the trigger signal, satisfying the “wherein” clause of claim 4 from which claim 5 depends. *Id.* at 51–52 (citing Ex. 1003 ¶¶ 322–340; Ex. 1027, 25, 117; Ex. 1028, 13; Ex. 1066, 26:64–27:4; Ex. 1067, 1–2; Ex. 1023 ¶ 19).

Patent Owner argues that each of these claims recites that the “controller” is part of the memory module. Resp. 46. However, according to Patent Owner, Dr. Wolfe stated that the S3 sleep mode is “always determined by something other than the memory module.” *Id.* (citing Ex. 2060, 282:22–284:25; Ex. 2006, 3 (S3 controlled by main controller in CPU); Ex. 2061 ¶ 119. Patent Owner asserts that the memory controller outside of the memory module controls the S3 sleep mode, not a controller

onboard the memory module. *Id.* at 47 (citing Ex. 2061 ¶ 120); *see also* Sur-Reply 23–24.

Petitioner replies that Patent Owner’s argument ignores the combination of Harris’s FBDIMM S3 sleep mode with Amidi’s teaching of backup power management and logic functionality on the memory module to “maintain memory (through refresh, for example).” Reply 23 (citing Pet. 43–44, 50–52; Inst. Dec. 35–36; Ex. 1024, 2:16–19, Fig. 11; Ex. 1003 ¶¶ 175–177). Petitioner contends that Amidi’s logic disconnects the system memory controller from the module, so a person of ordinary skill in the art “would have been motivated to use the S3 mode of the buffer 112 [on Harris’s module] when implementing the backup power supply and logic functionality as disclosed in Amidi, because S3 mode performs the function of refreshing the memory (e.g., self refresh), just like in Amidi.” Ex. 1003 ¶¶ 329–330.

From the foregoing, it is clear that Petitioner is relying on Amidi’s logic detecting a low voltage condition to initiate the S3 sleep mode of Harris’s FBDIMM module, rather than a signal from the system memory controller. Petitioner otherwise shows the limitations of these claims are disclosed by the combination of Harris, the FBDIMM Standards, and Amidi.

Petitioner has shown by a preponderance of the evidence that claims 5, 7, 8, 23 and 24 are obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

6. *Claims 6, 7, 9–12, and 17*

Claims 6, 7, 9–12, and 17 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being above a predetermined threshold voltage. Ex. 1001,

39:4–20, 39:30–49, 40:21–25. We address Petitioner’s contentions and Patent Owner’s arguments for claim 6, which is representative of these claims.

Petitioner contends that claim 6 is similar to claim 4 except it concerns *over*voltage detection rather than *under*voltage detection as in claim 4. Pet. 52. Petitioner contends that Harris teaches to detect both undervoltage and overvoltage conditions. *Id.* at 53 (citing Ex. 1023 ¶ 13). In this regard, Harris states that the tolerance of the +12V power supply is “around +/- 15%.” Ex. 1023 ¶ 13. Petitioner asserts that, in the combination, Amidi’s power supervisory module/block 480/665/800 is coupled to Harris’s edge connections for power. Pet. 53 (citing *id.* at 20–25). Petitioner further asserts that Amidi’s signal 670/858/868 is the claimed “trigger signal” generated in response to the input voltage having an amplitude above the predetermined threshold voltage. Pet. 53 (citing *id.* at 46–49). Petitioner contends the predetermined threshold voltage corresponds to Harris’s teaching of 15% above the nominal value of +12V, which is 13.8V. *Id.* (citing Ex. 1023 ¶ 13; Ex. 1003 ¶¶ 341–355).

Petitioner contends that Harris teaches both overvoltage and undervoltage detection of “+/- 15%.” *Id.* (citing Ex. 1023 ¶ 13). Petitioner contends that a person of ordinary skill in the art would have been motivated by Harris to include overvoltage detection in Amidi’s “voltage monitor circuit” with a reasonable expectation of success. *Id.* (citing Ex. 1003 ¶¶ 347–348). Petitioner contends that circuitry for both undervoltage and overvoltage detection was well known and commercially available. *Id.* at 54 (citing Ex. 1061, 15; Ex. 1062, 15; Ex. 1063, 1–2; Ex. 1065, code (57), ¶¶ 14, 18–19, Figs. 1, 5; Ex. 1003 ¶¶ 349–352).

As for the transition between the “first” and “second operable states,” Petitioner contends that the “first operable state” in its combination is “e.g., normal operations using an external voltage when its amplitude is within a normal range,” and the “second operable state” is “e.g., self-refresh operations using battery backup when the external voltage amplitude is outside the normal range.” *Id.* at 54–55 (citing Ex. 1003 ¶¶ 356–359).

Patent Owner argues that neither Harris nor Amidi teaches detecting an overvoltage. Resp. 40. However, Harris discloses that its +12V power supply has a tolerance of “+/- 15%” which indicates it would be a concern if the power supply was greater than 15%. Ex. 1023 ¶ 13. Amidi indicates concern with “power faults.” Ex. 1024, codes (54), (57). Although Patent Owner points out that Amidi discloses detection of undervoltage (Ex. 1024, Fig. 14), “power fault” is broad enough to encompass overvoltage as well. We agree with Petitioner that a person of ordinary skill in the art considering these teachings would have considered it obvious to detect overvoltage and generate a trigger signal to transition between operable states.

Patent Owner argues that the voltage regulators on which Petitioner relies do not detect input overvoltage because they can accommodate a wide range of input voltages, and there is a low probability that they would exceed their operational range. Resp. 42 (citing Ex. 2060, 58:24–61:12; Ex. 1040; Ex. 1041; Ex. 2061 ¶ 114). Petitioner replies that power surges were a known problem that could cause data loss or corruption or damage circuitry in memory boards. Reply 22 (citing Pet. 70–71; Ex. 1003 ¶¶ 138, 185). Petitioner contends that Harris discloses that its module can accommodate only “+/- 15%” of the nominal supply voltage, which is consistent with maximum ratings for commercial converters. *Id.* at 23

(citing Ex. 1041, 4). We agree with Petitioner that power surges would have been a problem in at least some circumstances for which one of ordinary skill in the art would have seen the value of voltage detection in order to take preventive action to avoid loss of data and damage to the memory module. Dr. Wolfe credibly explains that overvoltage was a danger known to persons of ordinary skill in the art, citing industry datasheets that specify overvoltage parameters. Ex. 1003 ¶¶ 351–352 (citing Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15). Indeed, the cited evidence uses the phrase “an overvoltage fault” (Ex. 1061, 15), which shows that power faults are not limited to undervoltage situations.

On this record, we determine that Petitioner has shown that claims 6, 7, 9–12, and 17 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi notwithstanding Patent Owner’s arguments.

7. *Claim 13*

Claim 13 depends from claims 1 and 8, and recites “wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a power reduction condition or a low voltage condition of the input voltage.” Ex. 1001, 39:50–54. Petitioner contends that claim 13 recites a limitation substantially similar to one addressed with respect to claim 4. Pet. 58 (citing Ex. 1003 ¶¶ 419–423). Patent Owner does not provide specific argument for claim 13. Petitioner has shown by a preponderance of the evidence that claim 13 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

8. *Claim 14*

Claim 14 recites

wherein the voltage monitor circuit detecting a power threshold condition includes the voltage monitor circuit detecting a request by the host system.

Ex. 1001, 39:55–58.

Petitioner contends that a power disruption detected by Amidi's voltage supervisory block 800 causes the system to switch to battery backup. Pet. 61 (citing Pet. 55–57). Petitioner contends that Amidi's voltage supervisory block 800 detects a request by the host system to return control to the host. *Id.* at 61–62 (citing Ex. 1003 ¶¶ 424–430). Petitioner asserts that Amidi “discloses that, if the power is restored after a failure, the voltage supervisory block detects whether the host has cleared the backup-mode bit (*‘detecting a request by the host system’*) and subsequently, if so, returns control of the module to the host, including handling of the power signal.” *Id.* at 62 (citing Ex. 1003 ¶¶ 311, 429; Ex. 1024, 5:31–62, 8:30–62, Fig. 14 9:11–39, Figs. 8, 14 (1460–1490), 15 (steps 1530, 1540, 1570, 1580; Ex. 1003 ¶ 312).

Patent Owner does not separately argue claim 14.

Petitioner shows by a preponderance of the evidence that claim 14 is unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

9. *Claim 16*

Petitioner contends that claim 16 is disclosed by the combination of Harris, the FBDIMM Standards, and Amidi for the reasons previously stated for claims 1, 8, and 9, noting that Amidi's voltage supervisory block (e.g., 665) (“the voltage monitor circuit”) is “configured to detect an amplitude

change in the input voltage” when the external system supply changes by crossing the reference voltage. Pet. 58 n.4 (emphasis omitted)(citing Ex. 1003 ¶ 471).

Patent Owner presents no separate arguments for claim 16 apart from those previously discussed.

Petitioner has shown by a preponderance of the evidence that claim 16 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

10. Claims 18–22 and 26–28

Claim 18 depends from claim 16 and recites

wherein, in the first operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a first pre-regulated voltage, and wherein, in the second operable state, the voltage conversion circuit provides the first regulated voltage to the plurality of SDRAM devices using a second pre-regulated voltage.

Ex. 1001, 40:26–32. Claim 19 requires that the pre-regulated voltages are provided to the voltage conversion circuit “via a circuit.” *Id.* at 40:33–37. Claim 20 recites that the “circuit” includes respective “first” and “second diodes” coupling respective “first” and “second pre-regulated voltages” to the “voltage conversion circuit.” *Id.* at 40:38–46. Claims 21, 22, 27, and 28 recite limitations from claims that have been addressed or will be addressed in this section.

Petitioner contends that the limitation of claim 18 is taught by the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 63–65. Petitioner contends that “pre-regulated voltage” means either that the voltage is within the 12V +/- 15% limits described in Harris, or must be pre-regulated on the memory board itself. Pet. 63; Ex. 1023 ¶ 13. We interpret

“pre-regulated voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage conversion unit. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). Petitioner contends that Harris discloses external voltage sources that are regulated. Pet. 63 (citing Ex. 1023 ¶¶ 13–14); *see* Ex. 1023 ¶ 14 (“It should be readily recognized that the external voltage sources may comprise any combination of known or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” (emphasis added)); Ex. 1003 ¶ 484. Thus, Harris’s regulated voltage sources generate “pre-regulated voltages” as recited in these claims.

Petitioner contends that the combination discloses that in the first operable state, which is, for example, normal operation when the external voltage amplitude is within normal range, Amidi’s voltage conversion circuit provides the first regulated voltage ($V_{DD}/V_{DDQ}=1.8V$) to the SDRAM devices. Pet. 63 (citing *id.* at 14–19, 27, 31–32, 45, 49–50; Ex. 1003 ¶¶ 480–487).

Petitioner further contends that the combination discloses that in the second operable state, which is, for example, self-refresh operations using battery backup when the external voltage amplitude is outside the normal range, Amidi’s voltage conversion circuit provides the first regulated voltage to the SDRAM devices using a second pre-regulated voltage from a boost converter in the power management block that converts and pre-regulates the 4.2V battery voltage to the regulated output voltage of 12V. Pet. 65 (citing *id.* at 41–43, 49–50; Ex. 1003 ¶¶ 488–492).

For claim 19, Petitioner contends that the claimed “circuit” corresponds to the power switch multiplexer circuit shown in Amidi’s

Figure 7 which supplies pre-regulated “System_VDD” in the first operable state (normal operation), and supplies pre-regulated and boosted “DC_18V” using the battery supply. Pet. 65–67 (citing *id.* at 26; Ex. 1023, Fig. 1A; Ex. 1024, 4:23–29, 4:56–5:24, Figs. 5–7; Ex. 1003 ¶¶ 493–500).

For claim 20, Petitioner contends that the first and second diodes as claimed are disclosed in Harris. Pet. 67–68 (citing Ex. 1024, 5:3 (first diode), 5:17 (second diode); Ex. 1024, 4:64–5:24, Fig. 7; Ex. 1003 ¶¶ 501–511).

For claims 21, 22, and 26–28, Petitioner relies on its contentions presented for claim limitations previously discussed. Pet. 68–70 (citing ¶¶ 450–452, 480–492, 494–500, 502–511, 513–526 (claim 21); ¶¶ 502–511; 527–530 (claim 22); ¶¶ 480–492, 571–576 (claim 26); ¶¶ 502–506, 577–580 (claim 27); ¶¶ 507–511, 581–584 (claim 28)).

Patent Owner does not present any arguments specific to these claims.

After reviewing the record, we determine that Petitioner has shown by a preponderance of the evidence that claims 18–22 and 26–28 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

11. Claim 25

Petitioner contends that the limitations of claim 25 were addressed with respect to claims 4 and 16. Pet. 58, 70.

Patent Owner provides no argument specific to claim 25. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 25 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

12. Claim 29

Claim 29 depends from claim 24 and recites that the voltage monitor circuit is configured to detect whether the input voltage is above or below respective thresholds. Ex. 1001, 42:19–23. Petitioner contends that the limitations of claim 29 were addressed with respect to claims 4, 6, and 17. Pet. 58, 70.

Patent Owner provides no argument specific to claim 29. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 29 is obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

13. Claim 30

Claim 30 depends from claim 29 and recites that the predetermined threshold voltages are above and below a “specified operating voltage.” Ex. 1001, 42:24–27. Petitioner contends that the “specified operating voltage” corresponds to Harris’s +12V nominal external voltage. Pet. 57 n.2 (citing Pet. 41–45; Ex. 1023 ¶¶ 12, 13).

Patent Owner does not present any arguments specific to claim 30. *See* Resp.

Petitioner has shown by a preponderance of the evidence that claim 30 would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

14. Determination for Ground 2

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations are taught or at least suggested by the

combination. Accordingly, under this ground, Petitioner has demonstrated by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

F. Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck

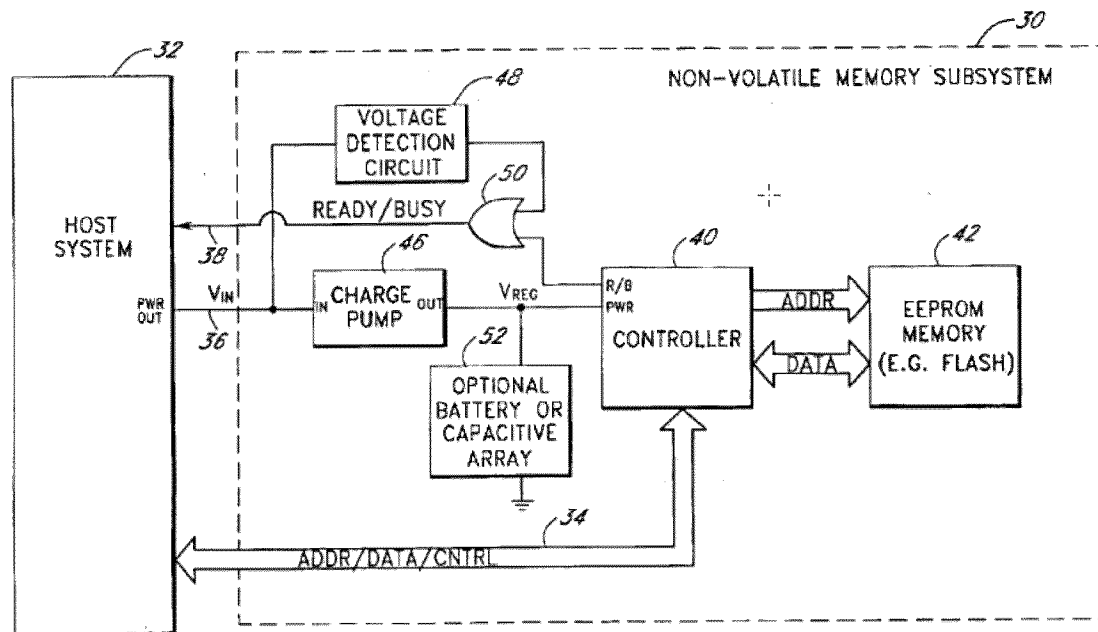
Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck. Pet. 70–72.

1. Hajeck (Ex. 1038)

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1001, code (54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck’s Figure 1 is reproduced below.



In Hajec's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–33.

2. Motivation to Combine

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajec with Harris, the FBDIMM Standards, and Amidi because one would have appreciated the desirability of switching to backup power in both undervoltage and overvoltage conditions.

Pet. 71–72. Hajeck teaches using a battery backup in response to power loss from the host system to complete outstanding operations and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57). Hajeck also teaches a voltage detection circuit generate a “busy” signal in response to undervoltage or overvoltage conditions. *Id.* at 3:30–43.

3. *Claims 7, 9–12, 17, 29, and 30*

Petitioner asserts that claims 1–30 would have been obvious under this ground for the same reasons provided for Ground 2. Pet. 71 (citing *id.* at 45–70). Petitioner contends that to the extent one would argue that Ground 2 fails to teach *over*voltage detection, Hajeck discloses this feature. Pet. 71 (citing *id.* at 52–54). Specifically, Petitioner contends that Hajeck discloses overvoltage detection as required by claims 7, 9–12, 17, 29, and 30. Pet. 71 (citing *id.* at 52–54; Ex. 1003 ¶¶ 354–355). Petitioner contends a person of ordinary skill in the art would have looked to Hajeck’s teachings about voltage anomalies when implementing Amidi’s voltage supervisory block to detect both undervoltage and overvoltage anomalies. *Id.* at 71–72 (citing Ex. 1003 ¶¶ 186–188). In the combination, “Amidi’s voltage supervisory block would be modified to detect voltage anomalies and switch to the backup power not only ‘[i]f system supply 605 has a magnitude lower than reference voltage 675’ as disclosed by Amidi (Ex. 1024, 4:44–52), but also ‘when the voltage exceeds a certain level’ as taught by Hajeck (Ex. 1038, 3:30–43; *see also id.* at code (57), 1:10–18, 1:28–31, 1:62–2:7, 3:30–4:9, 4:62–65, Fig. 1).” *Id.* at 72. Petitioner contends that this ground teaches both overvoltage and undervoltage detection and protection, further

rendering obvious claims 6, 9, 17, and 29 and their dependent claims 7, 10–12, and 30. *Id.* (citing Ex. 1003 ¶¶ 354–355, 389, 478, 588).

Patent Owner argues that “Harris does not teach the need to detect undervoltage and overvoltage anomalies” and that Hajeck “does not teach or suggest a voltage detection circuit that produces a trigger signal that causes transitions to different operating states in over-voltage conditions or otherwise even relates to switching power sources” or “sending the trigger signal to the memory system.” Resp. 42–43 (citing Pet. 48, 51). Patent Owner argues that “the only signal generated by Hajeck’s voltage detection circuit 48 is a ‘busy signal’ 38 sent back to host 32 to prevent it from performing write operations to the memory subsystem 30 when voltage anomalies are detected.” *Id.* at 43 (citing Ex. 1038, 3:30–57, Fig. 2). Patent Owner argues that Hajeck does not switch to a different operating state or a different power supply, but instead its charge pump 46 continues to provide a near-constant voltage V_{REG} to controller 40. *Id.* (citing Ex. 1038, 3:6–10, 3:66–4:3; Ex. 2061 ¶ 116). Patent Owner argues that Hajeck’s charge pump protects controller 40 and other circuits from damage caused by voltage surges and spikes. *Id.* at 44 (citing Ex. 1038, 4:7–9).

Petitioner replies that Patent Owner’s arguments ignore the Petition’s extensive explanation and evidence about a person of ordinary skill in the art’s understanding of how to handle power anomalies, including both undervoltage and overvoltage conditions, with a reasonable expectation of success. Reply 21 (citing Pet. 12–13, 41–45, 53–54, 70–72; Ex. 1003 ¶¶ 138, 345–355). Petitioner contends that the Petition also explained how the combination teaches a trigger signal to switch to back-up power and avoid data loss in case of a “power fault” including undervoltage and

overvoltage conditions. *Id.* (citing Pet. 49–50; Ex. 1003 ¶¶ 315–321; Ex. 2060, 226:16–22, 230:17–232:6, 251:10–254:2).

We agree with Petitioner that Patent Owner’s arguments attack the references individually and do not properly consider what the combination of references would have signified to a person of ordinary skill in the art. *In re Merck & Co.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981)). Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition to the extent that the combination of Harris, the FBDIMM Standards, and Amidi were deemed not to already disclose that feature. Pet. 72 (citing Ex. 1038, 3:37–40). Thus, the features that Patent Owner argues are missing from Hajeck are not ones for which Petitioner relied on Hajeck.

For similar reasons, we also do not agree with Patent Owner’s argument that Hajeck does not suggest that data loss was a concern because its charge pump could provide the desired V_{REF} indefinitely. Resp. 44–45. Petitioner relies on Amidi for switching operational states and the voltage supply to battery backup in the event of an overvoltage. Reply 21 (citing Pet. 49–50). This argument also considers Hajeck individually, and not in combination with Harris, the FBDIMM Standards, and Amidi, as set forth in the Petition.

4. *Determination for Ground 3*

Petitioner has shown that a person of ordinary skill in the art would have combined Harris, the FBDIMM Standards, Amidi and Hajeck with a reasonable expectation of success. Petitioner has also shown that all of the limitations of claims 1–30 are taught or at least suggested by the combination. On this record, Petitioner has shown by a preponderance of

the evidence that claims 1–30 of the '054 patent would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

G. Grounds 4 and 5: Obviousness over Spiers and Amidi

As we have determined that claims 1–30 are unpatentable as obvious under Grounds 1 to 3, we do not reach Grounds 4 and 5.

H. Motion to Exclude

Petitioner seeks to exclude materials referenced in three URLs submitted with Patent Owner's Sur-Reply on August 4, 2023. Paper 35 (citing Paper 33, 1 n.2, 24, 26). Because we do not rely on any of these URLs in a manner adverse to Petitioner, we dismiss the Motion to Exclude as moot.

III. CONCLUSION

For the foregoing reasons, we determine that Petitioner establishes by a preponderance of the evidence that claims 1–30 of the '054 patent are unpatentable.

IV. ORDER

Accordingly, it is:

ORDERED that claims 1–30 of the '054 patent have been shown to be unpatentable;

FURTHER ORDERED that Petitioner's Motion to Exclude is dismissed as moot;

FURTHER ORDERED that Petitioner will contact the Board within seven (7) business days to correctly identify Exhibit 1023 as Harris in the Board's P-TACTS system.

FURTHER ORDERED that any party seeking judicial review must comply with the notice and service requirements of 37 C.F.R. § 90.2.⁷

⁷ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner's attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–3, 15	103	Harris, FBDIMM Standards	1–3, 15	
1–30	103	Harris, FBDIMM Standards, Amidi	1–30	
1–30	103	Harris, FBDIMM Standards, Amidi, Hajeck	1–30	
1–30	103	Spiers, Amidi ⁸		
1–30	103	Spiers, Amidi, Hajeck		
Overall Outcome			1–30	

⁸ Because we determine that claims 1–30 are unpatentable on other grounds, we do not reach the remaining grounds.

IPR2022-00999
Patent 11,232,054 B2

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**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	Hyun Lee	Nonprovisional Application Number (if known):	
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

- i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via EFS-Web.
---OR---
(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.
- ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- i. A request for continued examination has been filed with, or prior to, this form.
- ii. If the application is a utility application, this certification and request is being filed via EFS-Web.
- iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature /Khaled Shami/	Date Dec. 30, 2020
Name (Print/Typed) Khaled Shami	Practitioner Registration Number 38,745

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☐ *Total of _____ forms are submitted.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Postal Code	92694	Country ⁱ	US	
Inventor 2				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Walnut	State/Province	CA	Country of Residence ⁱ US
Mailing Address of Inventor:				
Address 1		944 Crystal Water Lane		
Address 2				
City	Walnut	State/Province	CA	
Postal Code	91789	Country ⁱ	US	
Inventor 3				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jeffrey	C.	Solomon	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		6 Silver Fir			
Address 2					
City	Irvine	State/Province		CA	
Postal Code	92604	Country ⁱ	US		

Inventor 4				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Scott	H.	Milton	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		49 Statehouse Place			
Address 2					
City	Irvine	State/Province		CA	
Postal Code	92602	Country ⁱ	US		

Inventor 5				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		12220 Rose Street			
Address 2					
City	Cerritos	State/Province		CA	
Postal Code	90703	Country ⁱ	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

☐ An Address is being provided for the correspondence information of this application.

Customer Number	151145		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	0016.001000H	Small Entity Status Claimed	☐
Application Type			
Subject Matter			
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	151145		

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Prefix	Given Name	Middle Name	Family Name	Suffix	Remove
Registration Number					
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove
Registration Number					
Additional Representative Information blocks may be generated within this form by selecting the Add button.					

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
	Continuation of	15934416	2018-03-23		
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
15934416	Continuation of	14840865	2015-08-31	9928186	2018-03-27
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14840865	Continuation of	14489269	2014-09-17	9158684	2015-10-13
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14489269	Continuation of	13559476	2012-07-26	8874831	2014-10-28
Prior Application Status		Expired		Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
13559476	Claims benefit of provisional	61512871	2011-07-28		

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Prior Application Status		Patented	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30

Prior Application Status		Abandoned	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		

Prior Application Status		Expired	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12131873	Claims benefit of provisional	60941586	2007-06-01		

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

☐ NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	0016.001000H
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

- ☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.
- ☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

[Clear](#)

☒ Assignee ☐ Legal Representative under 35 U.S.C. 117 ☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign. ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information For Applicant:

Address 1	175 Technology Drive		
Address 2	Suite 150		
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1						
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.						
If the Assignee or Non-Applicant Assignee is an Organization check here.			☒			
Organization Name	Netlist, Inc.					
Mailing Address Information For Assignee including Non-Applicant Assignee:						
Address 1		175 Technology Drive				
Address 2		Suite 150				
City	Irvine	State/Province	CA			
Country i	US	Postal Code	92618			
Phone Number		Fax Number				
Email Address						
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Khaled Shami/			Date (YYYY-MM-DD)	
First Name	Khaled	Last Name	Shami	Registration Number	38745
Additional Signature may be generated within this form by selecting the Add button.					

Electronic Patent Application Fee Transmittal

Application Number:				
Filing Date:				
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Stephanie Semler			
Attorney Docket Number:	0016.001000H			
Filed as Large Entity				
Filing Fees for Track I Prioritized Examination - Nonprovisional Application under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	320	320
UTILITY SEARCH FEE	1111	1	700	700
UTILITY EXAMINATION FEE	1311	1	800	800
REQUEST FOR PRIORITIZED EXAMINATION	1817	1	4200	4200
Pages:				
Claims:				
Miscellaneous-Filing:				
PROCESSING FEE, EXCEPT PROV. APPLS.	1830	1	140	140

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				6160

Electronic Acknowledgement Receipt

EFS ID:	41526926
Application Number:	17138766
International Application Number:	
Confirmation Number:	8804
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	151145
Filer:	Khaled Shami/Stephanie Semler
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000H
Receipt Date:	30-DEC-2020
Filing Date:	
Time Stamp:	21:04:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$6160
RAM confirmation Number	E2020BTL11438211
Deposit Account	602034
Authorized User	Stephanie Semler

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	NetlistPostAIPOA.pdf	711728	no	1
			e45cea84246703abf611c09bfc51150cafd7cb72		

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

2	Oath or Declaration filed	0016001000H_InvDec.pdf	448715	no	5
			e3ef024451ed3cd6e1c63ed595c4a0ad7a22b58		

Warnings:

Information:

3	Drawings-only black and white line drawings	0016001000H_Drwgs.pdf	4555960	no	22
			10944aad8e0cf3b0b62a67fd9fd555d91238f91a		

Warnings:

Information:

4		0016001000H_Application.pdf	579398	yes	83
			655826dcaacfa39c576f09b511b5596571f5251b		

Multipart Description/PDF files in .zip description

	Document Description	Start	End
	Specification	1	75
	Claims	76	82
	Abstract	83	83

Warnings:

Information:

5	Power of Attorney	0016001000H_POA_Transmittal.pdf	195754 dc1b846e0e8d107ff5950cc5e22af92bbdb8de18	no	1
Warnings:					
Information:					
6	TrackOne Request	0016001000H_TrackOneRequestForm_aia0424.pdf	129042 5c434a352f92938e97546479f254e9cca34651d5	no	2
Warnings:					
Information:					
7	Application Data Sheet	WebADS.pdf	179961 64330f64bb5b9af022c5c24dd0ba25bc33aeb7eb	no	9
Warnings:					
Information:					
8	Fee Worksheet (SB06)	fee-info.pdf	38288 fa419f8e8f3b24170d0188bb4e59632311ef315b	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			6838846		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA/82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	To Be Assigned
Filing Date	To Be Assigned
First Named Inventor	Hyun Lee
Title	FLASH-DRAM HYBRID MEMORY MODULE
Art Unit	To Be Assigned
Examiner Name	To Be Assigned
Attorney Docket Number	0016.001000H

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	
Name	Khaled Shami	Registration Number	38,745
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number
Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

151145

OR

I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

The address associated with the above-mentioned Customer Number

OR

The address associated with Customer Number:

OR

Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Netlist, Inc.

- Inventor or Joint Inventor (title not required below)
Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

12/5/17

Name

Noel B. Whitley

Title

VP, IP & Litigation

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

Total of forms are submitted.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

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LEGAL NAME OF INVENTOR

Inventor: Jeffrey C. Solomon

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: Scott H. Milton

Date (Optional): 10/17/2014

Signature: Scott H. Milton

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Hyun Lee

Date (Optional) : _____

Signature: _____

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LEGAL NAME OF INVENTOR

Inventor: Jayesh Bhakta

Date (Optional): 9-17-2014

Signature: Jayesh Bhakta

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LEGAL NAME OF INVENTOR

Inventor: Chi-She Chen

Date (Optional): Oct-17-2014

Signature: Chi-She Chen

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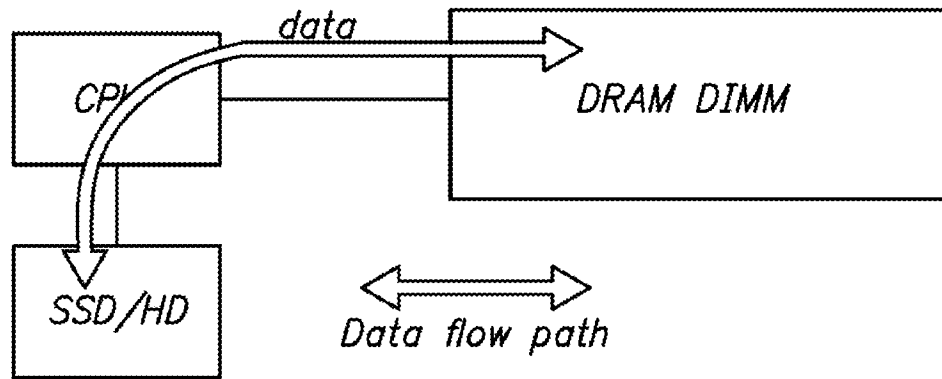


FIG. 1
(PRIOR ART)

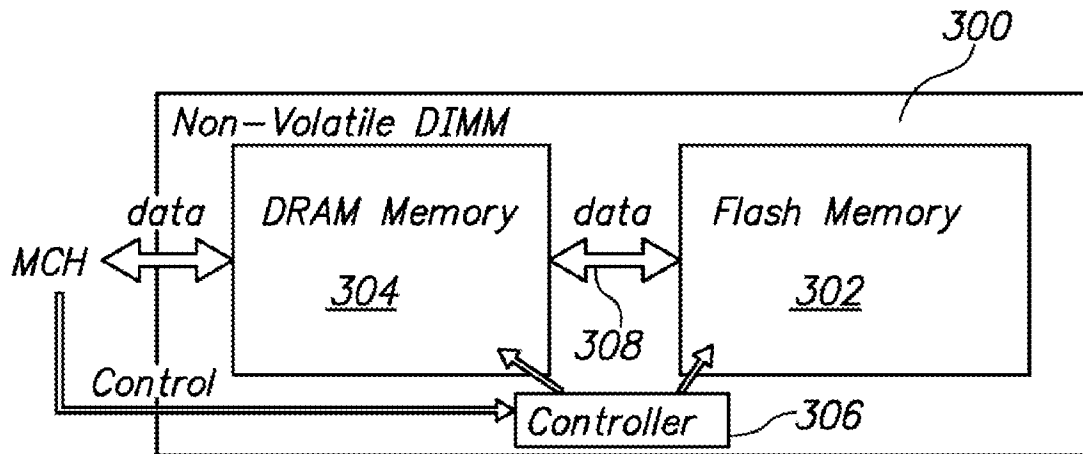
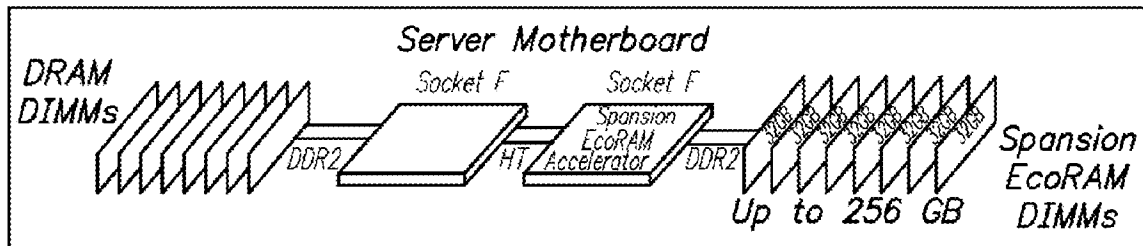


FIG. 3A

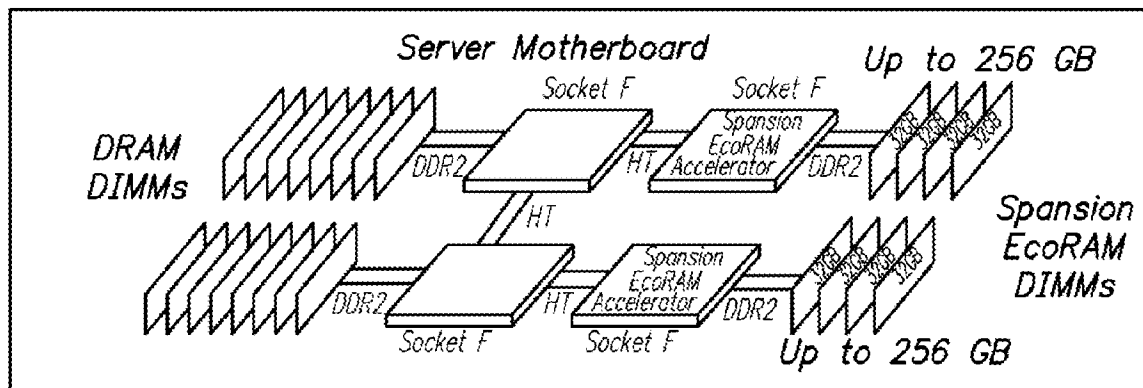
Spancion EcoRAM Configurations

256GB Spancion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

256GB Spancion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

FIG. 2
(PRIOR ART)

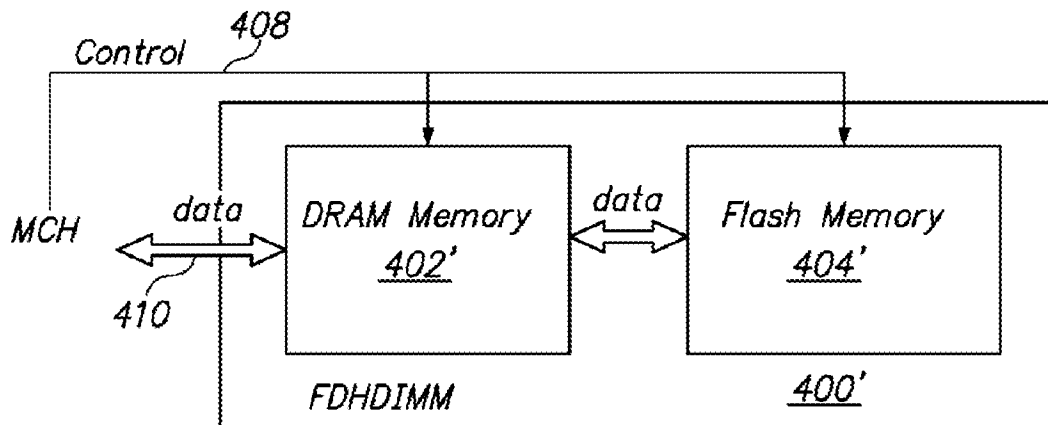


FIG. 4B

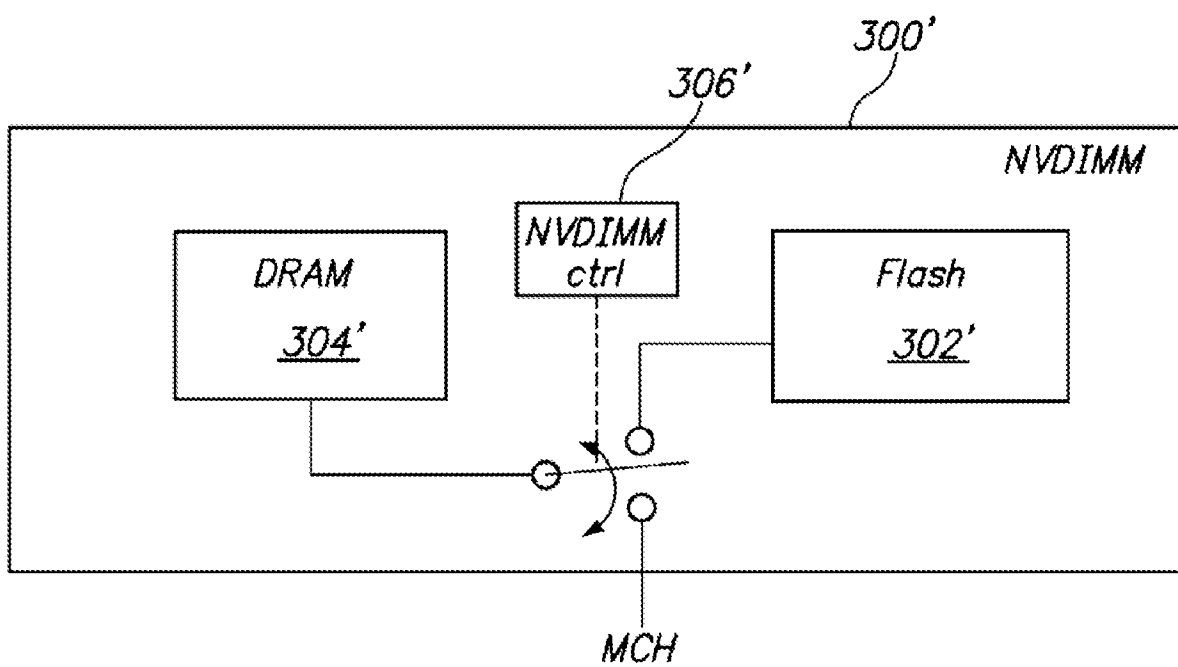


FIG. 3B

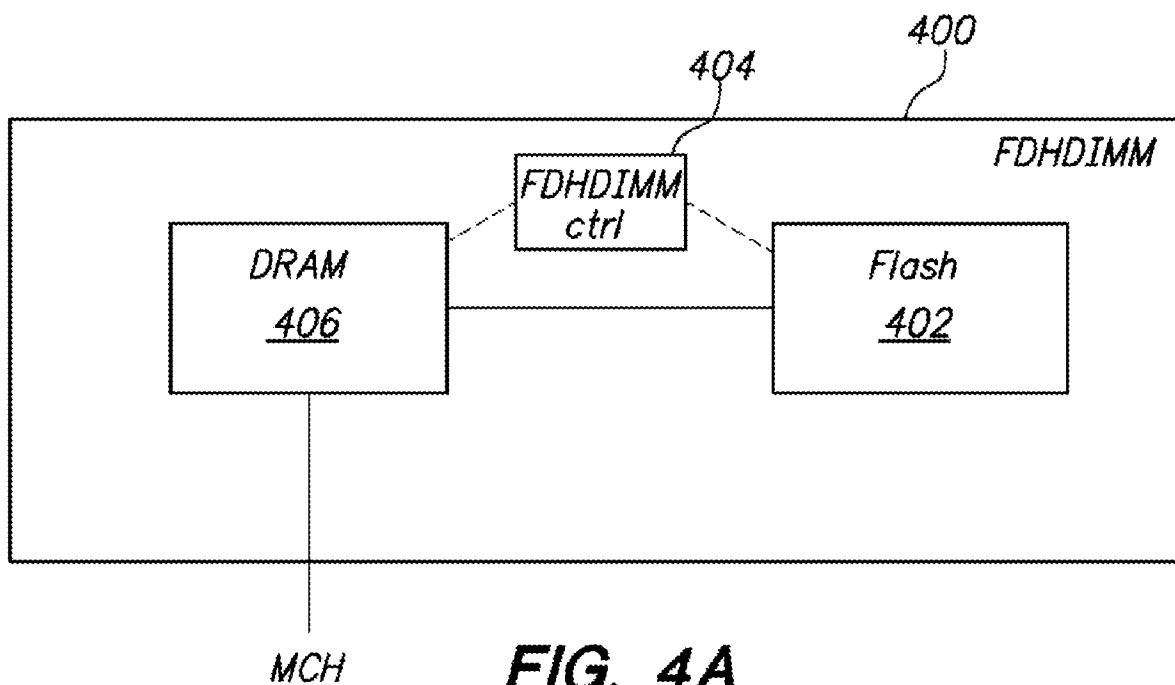


FIG. 4A

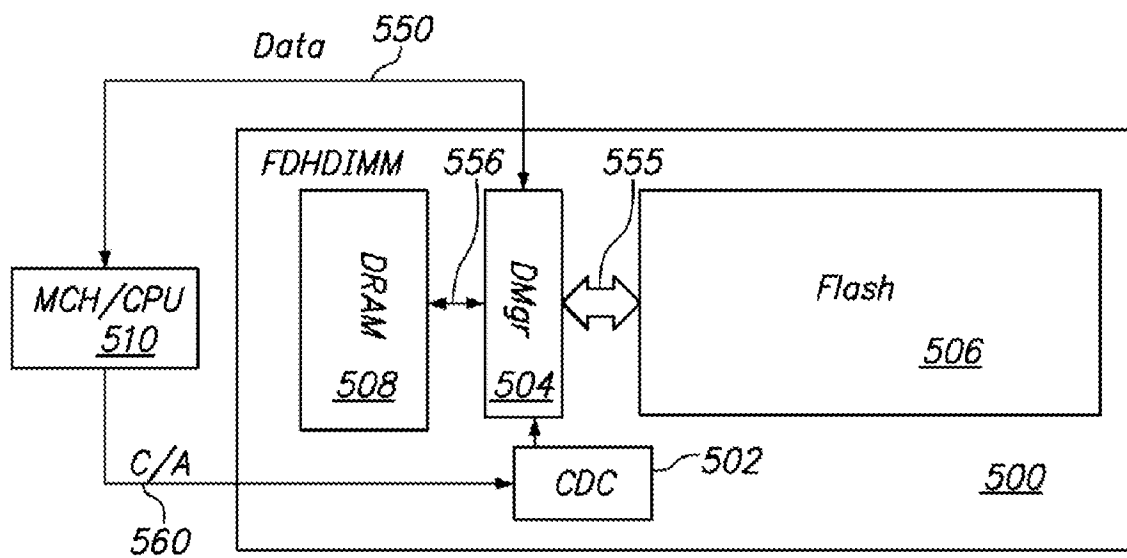


FIG. 5A

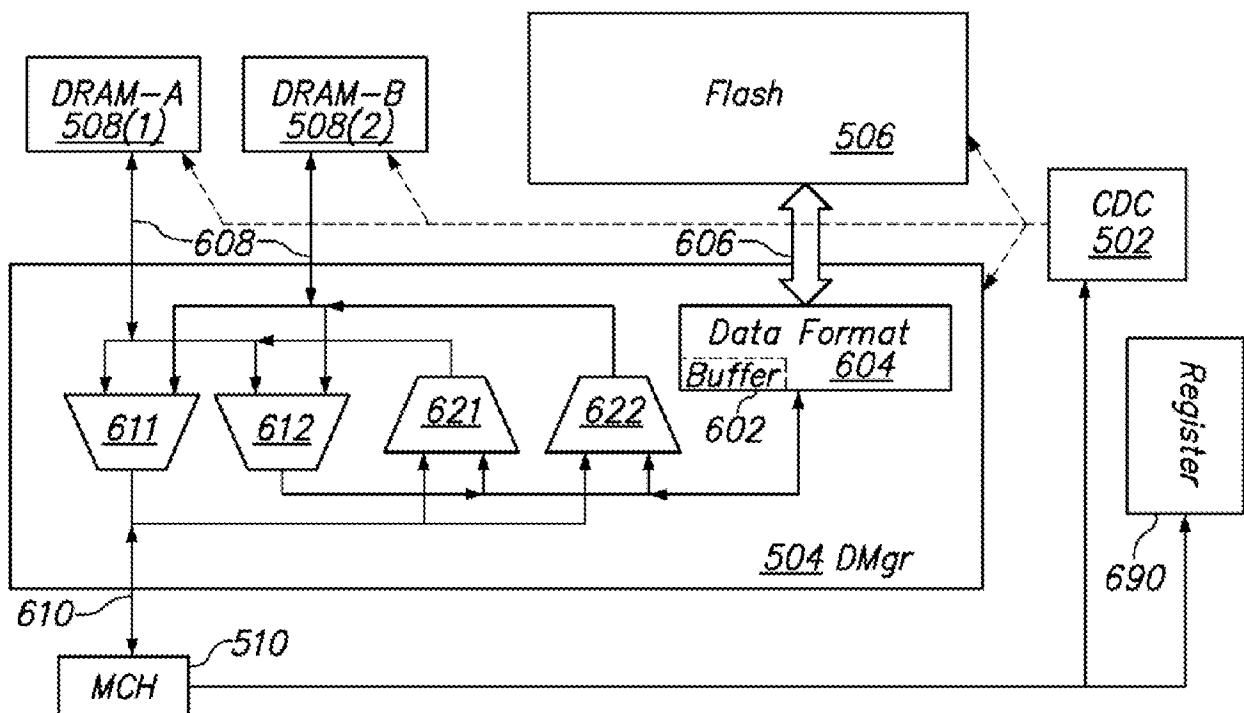


FIG. 6

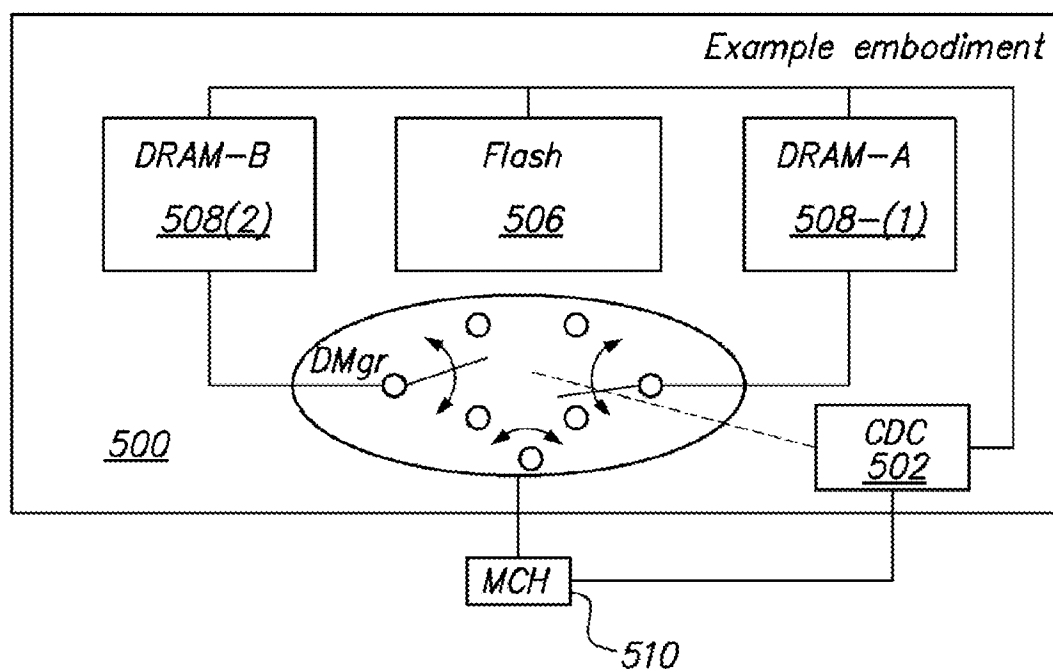


FIG. 5B

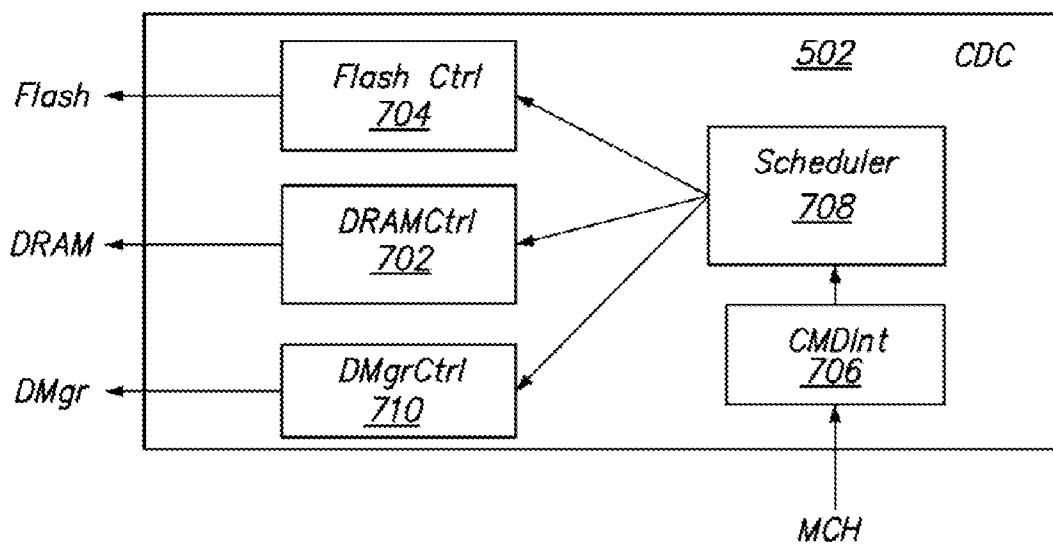


FIG. 7

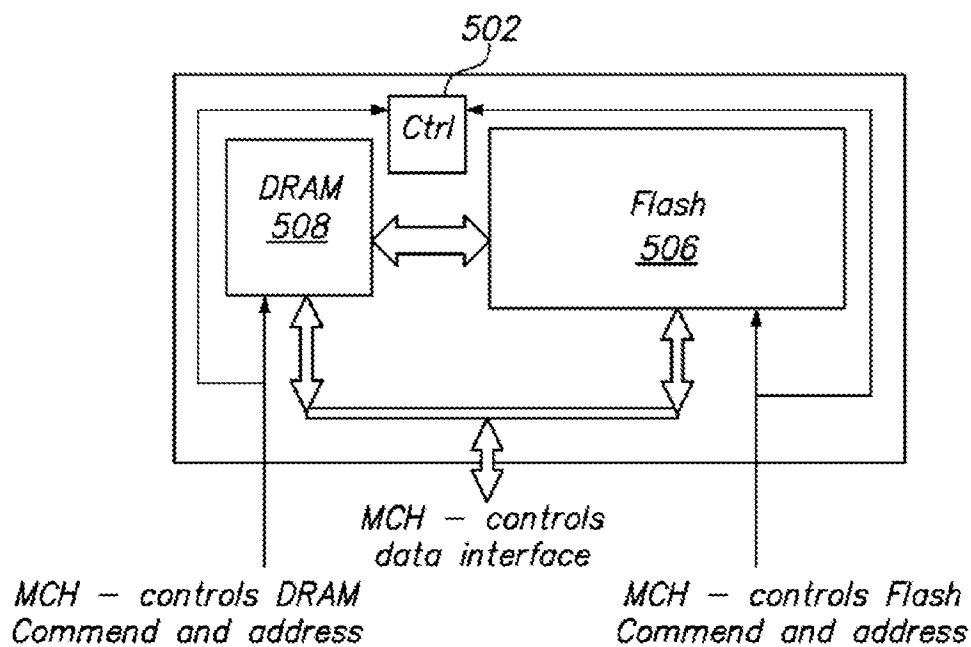


FIG. 8A

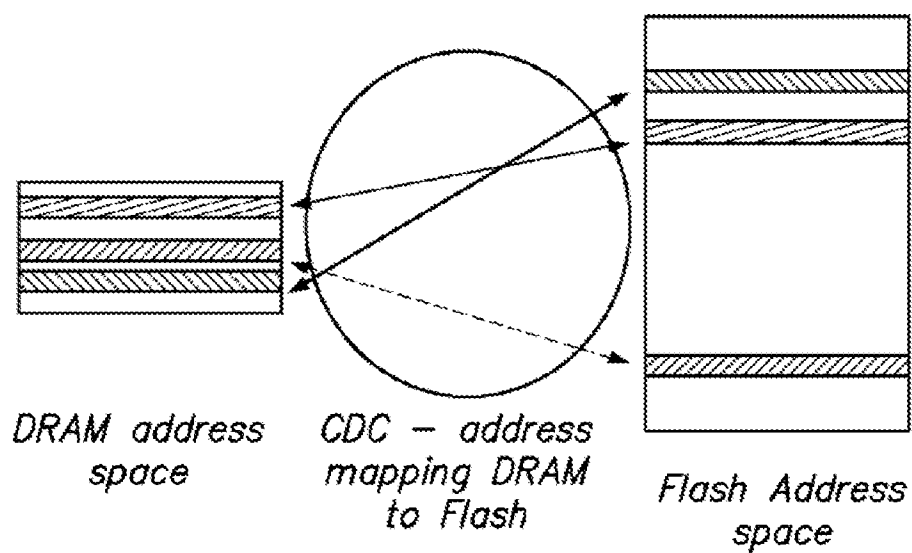
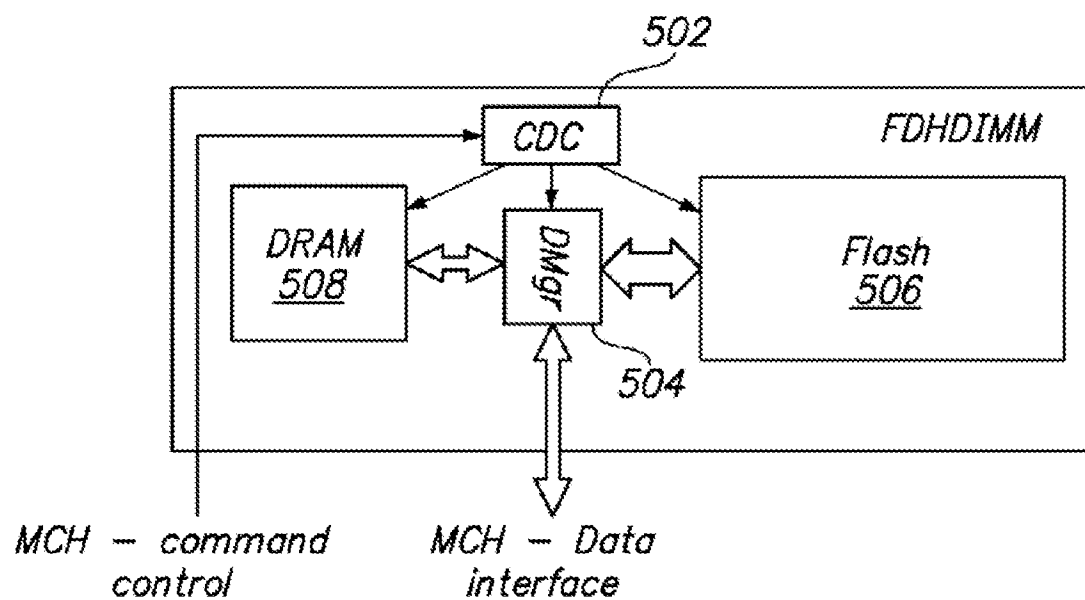


FIG. 8B

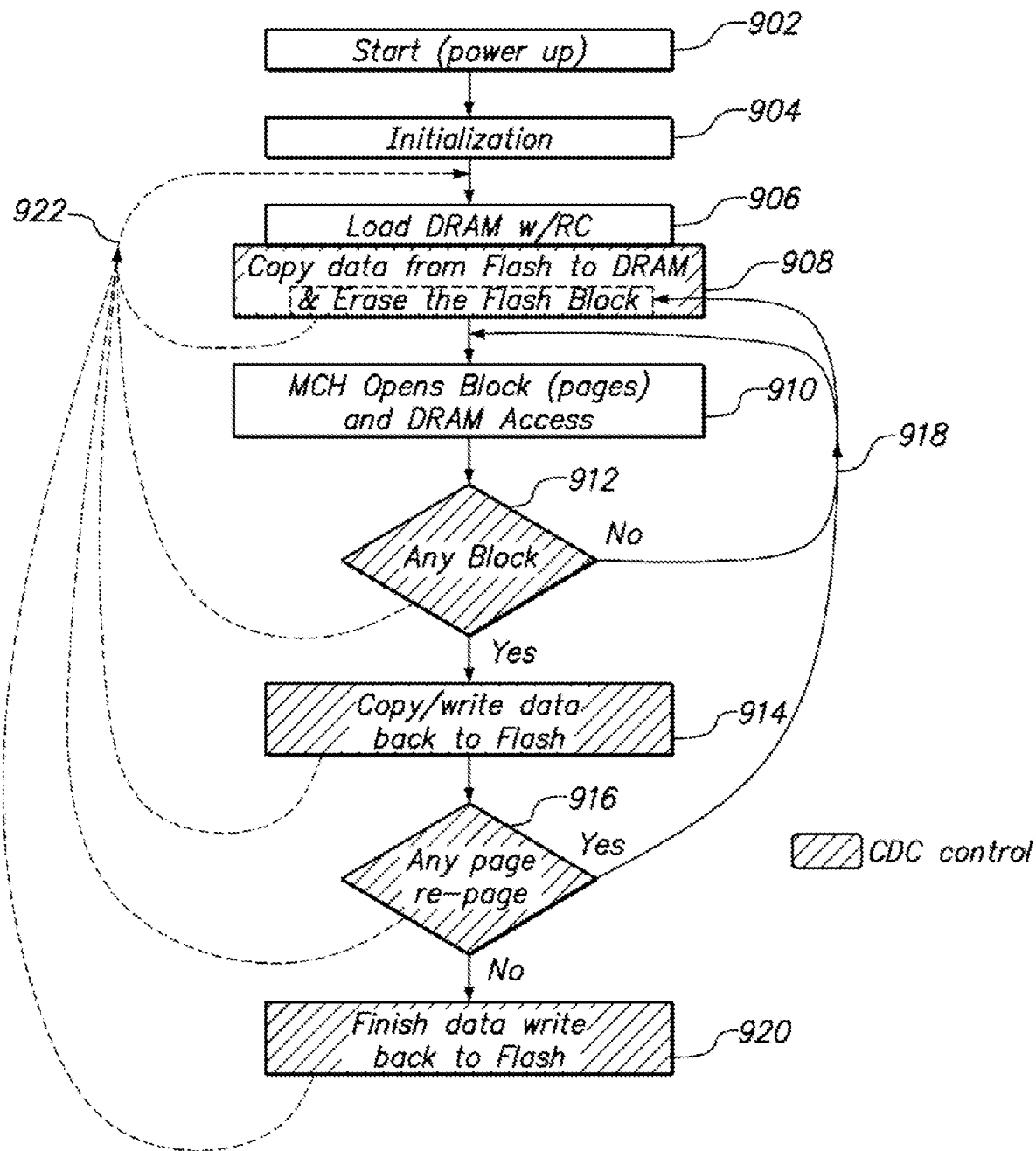


FIG. 9

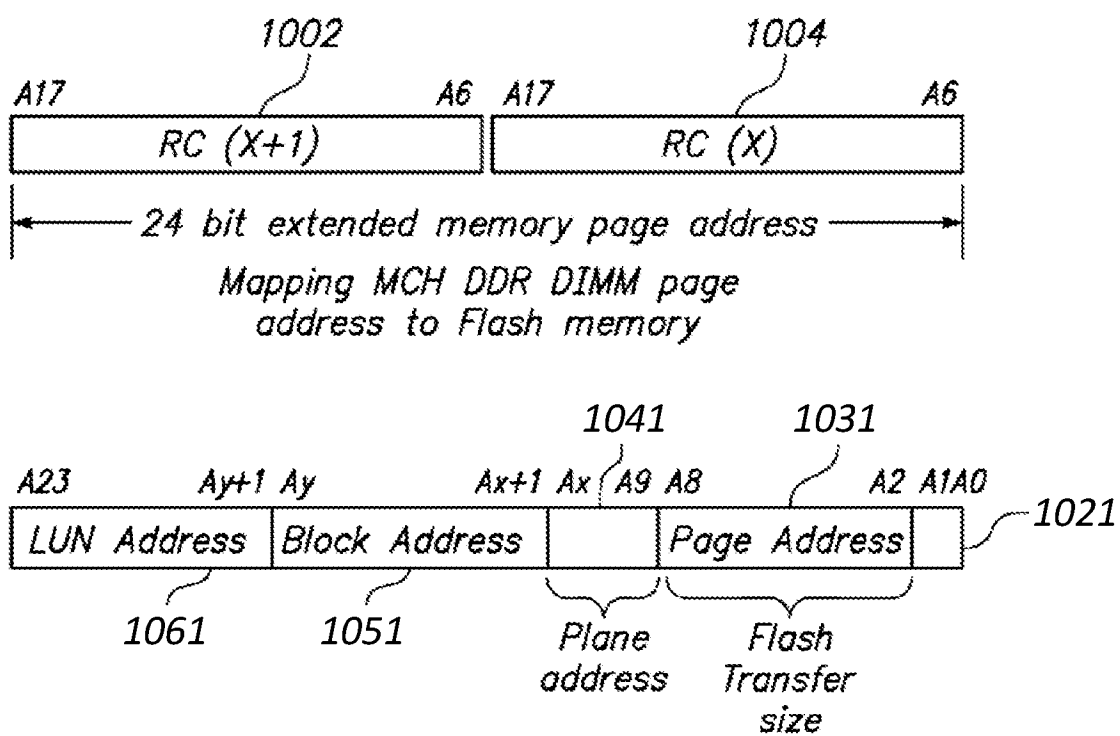


FIG. 10

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

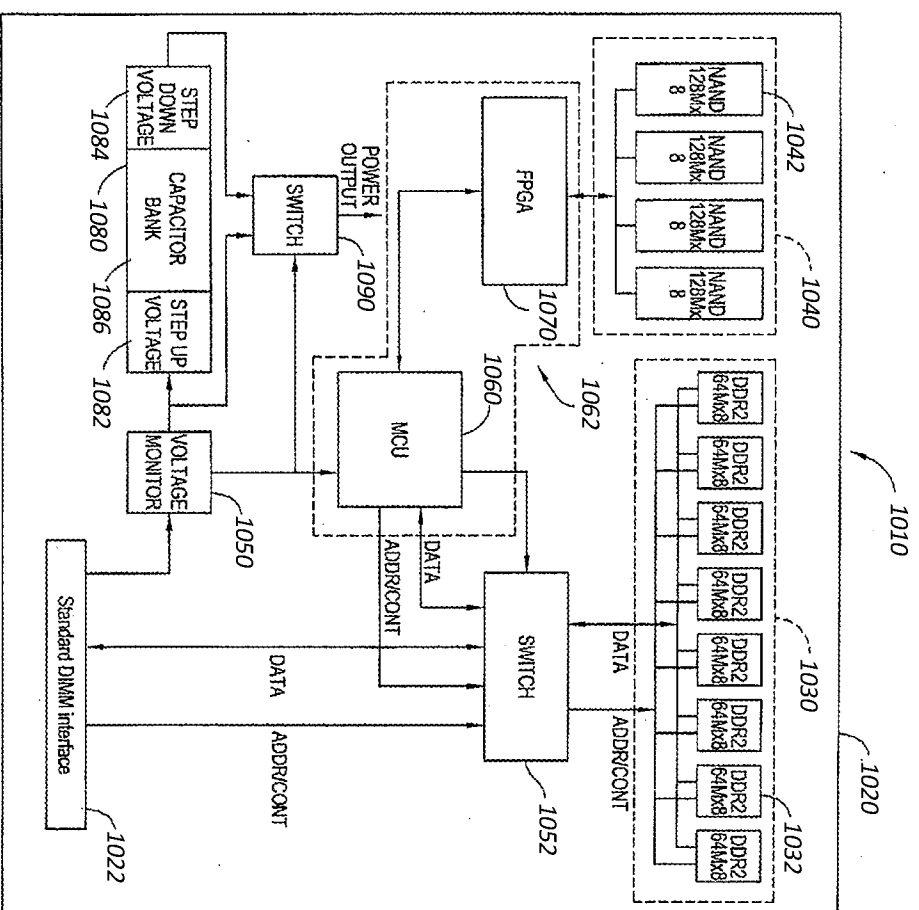


FIG. 12

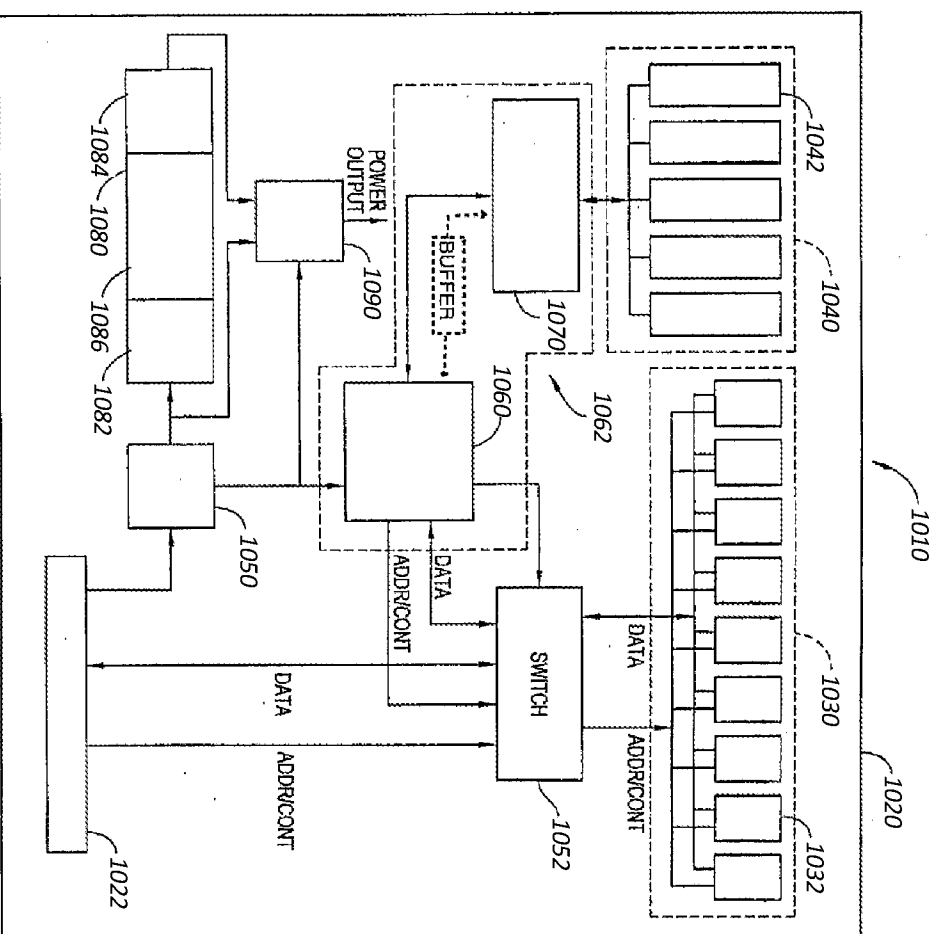


FIG. 13

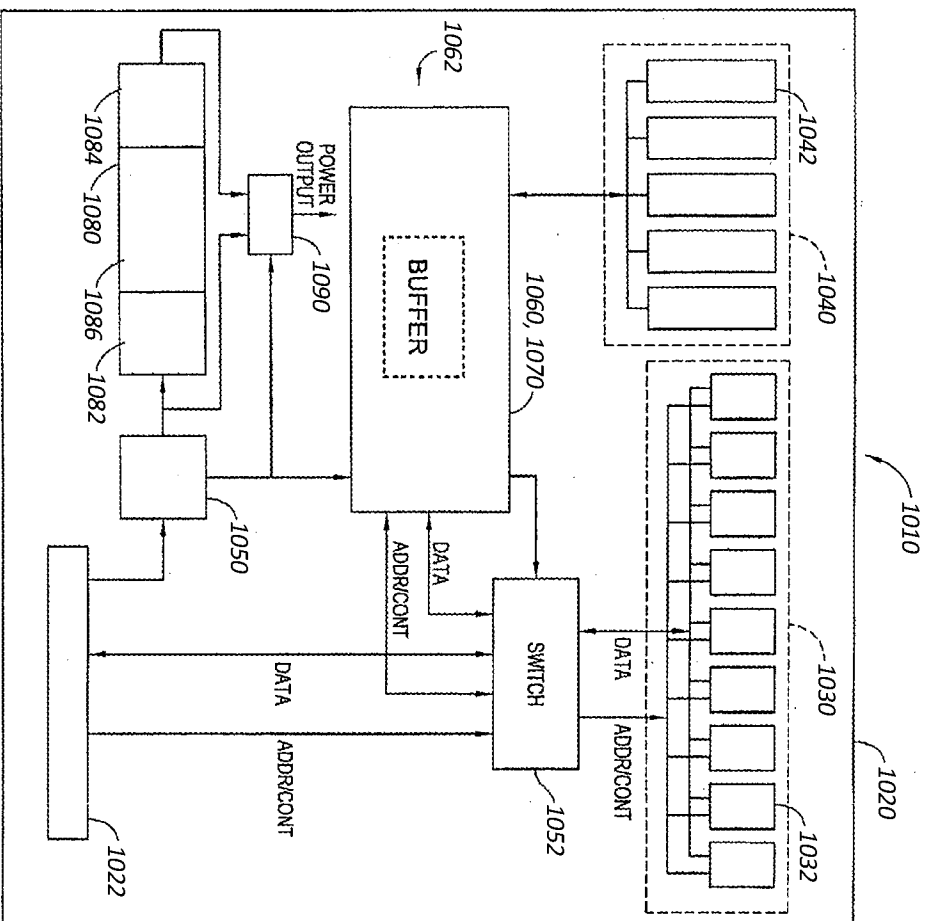


FIG. 14

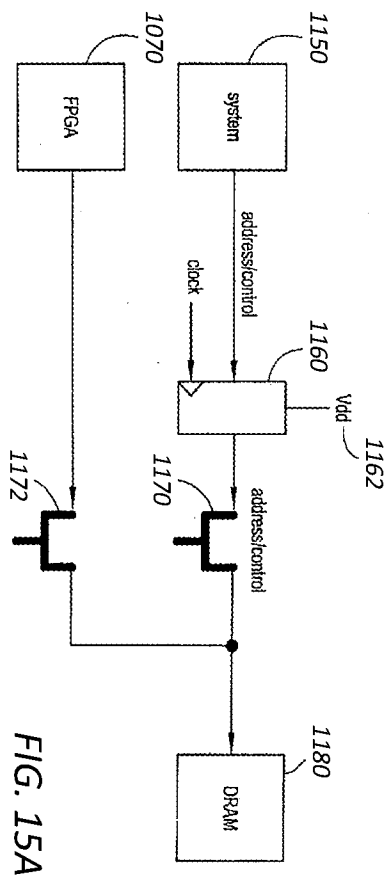


FIG. 15A

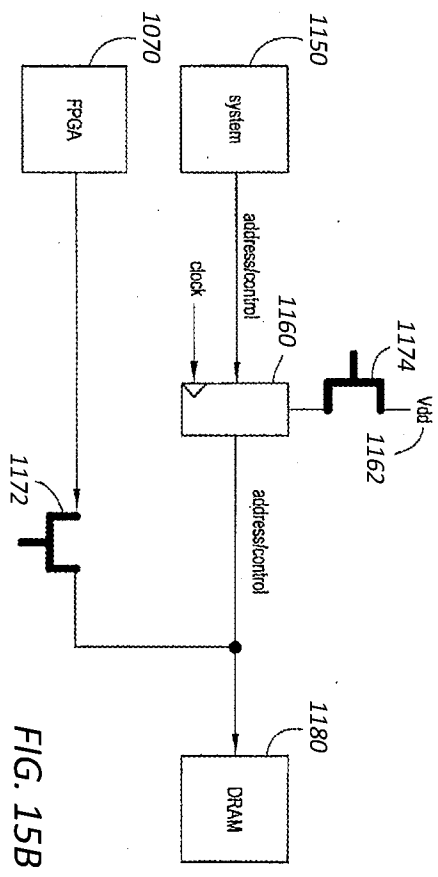


FIG. 15B

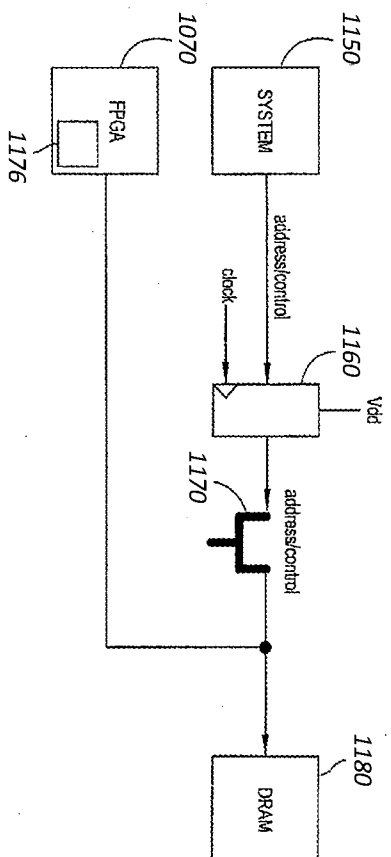
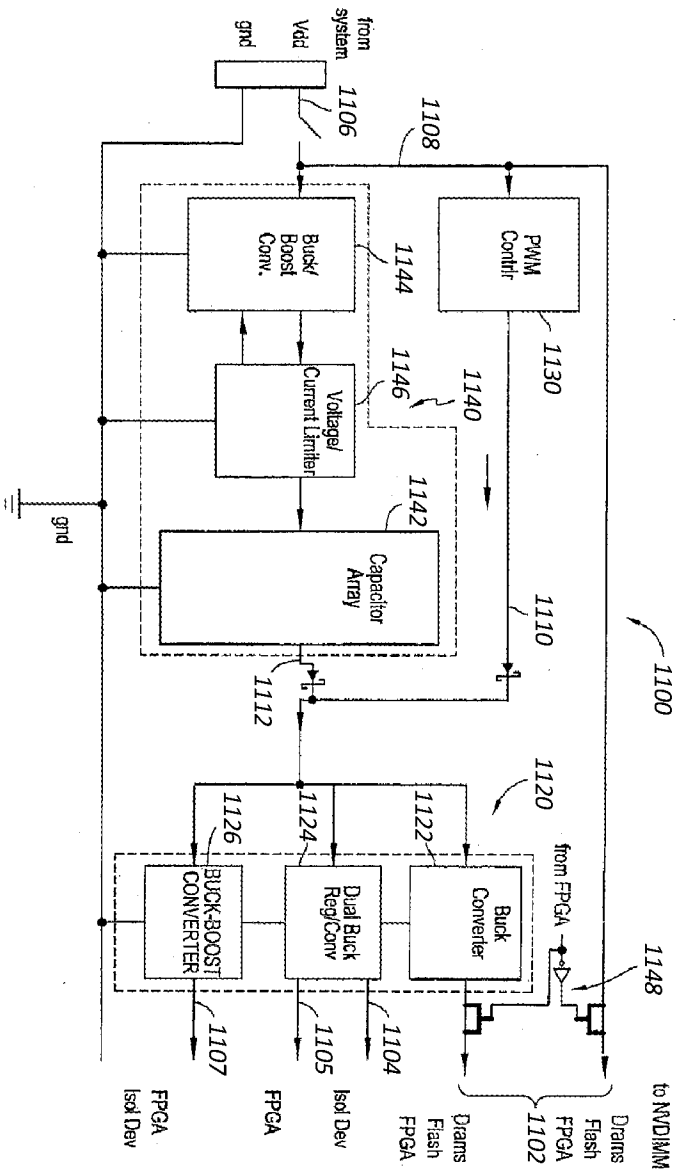


FIG. 15C



Power Module
FIG. 16

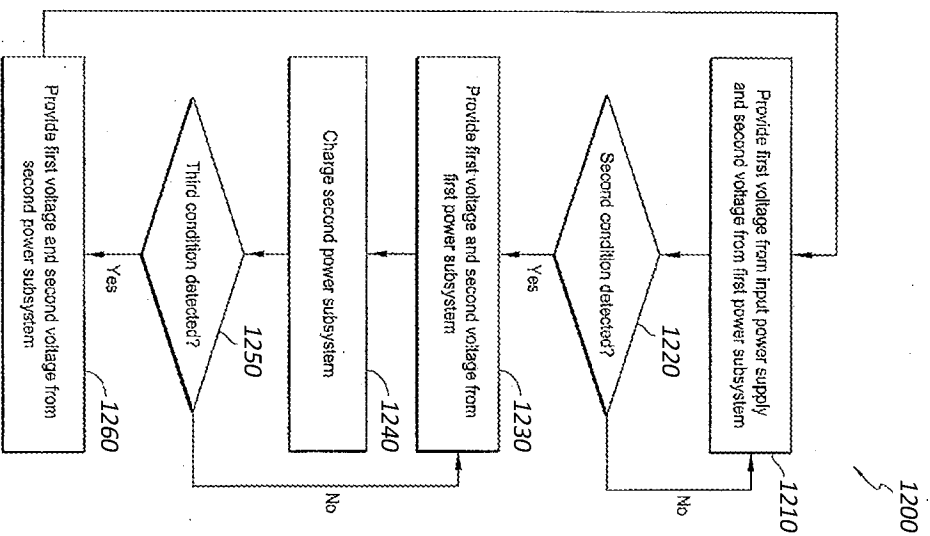


FIG. 17

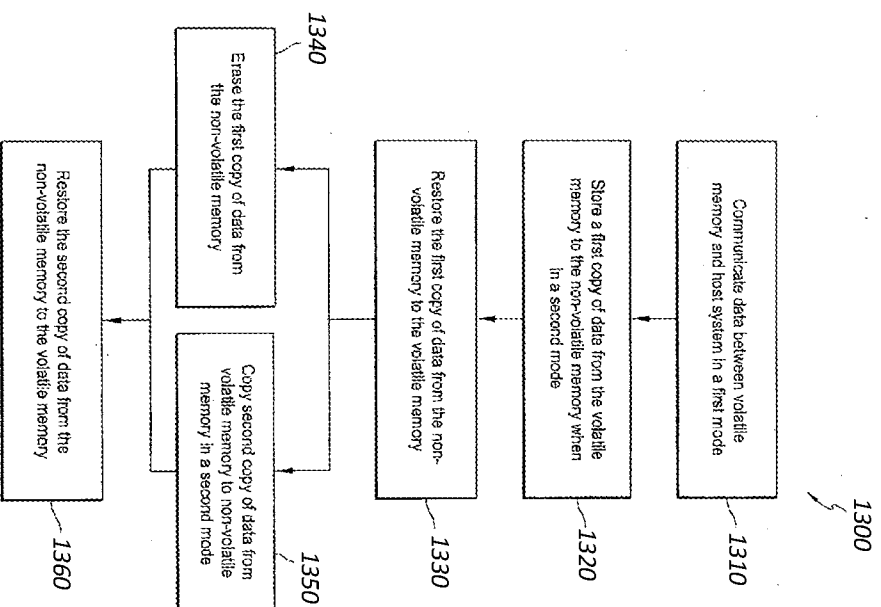


FIG. 18

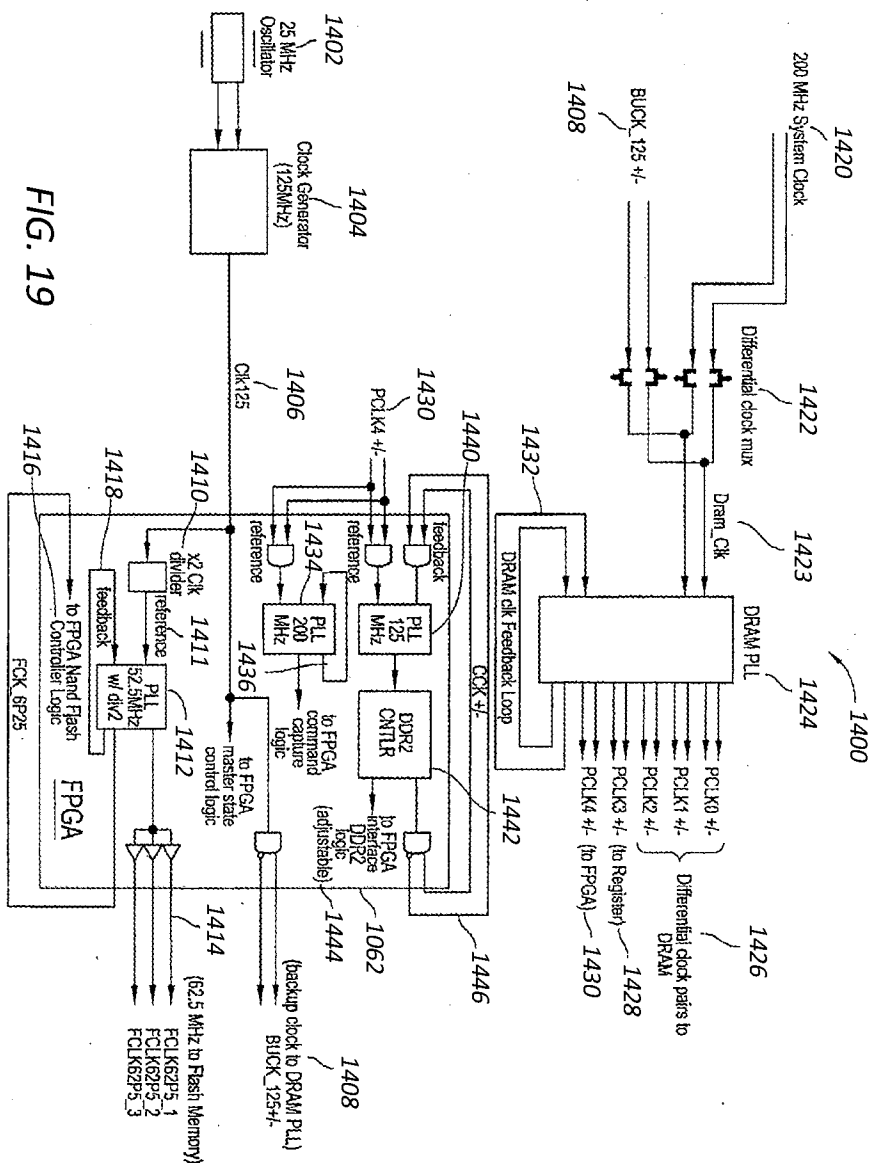


FIG. 19

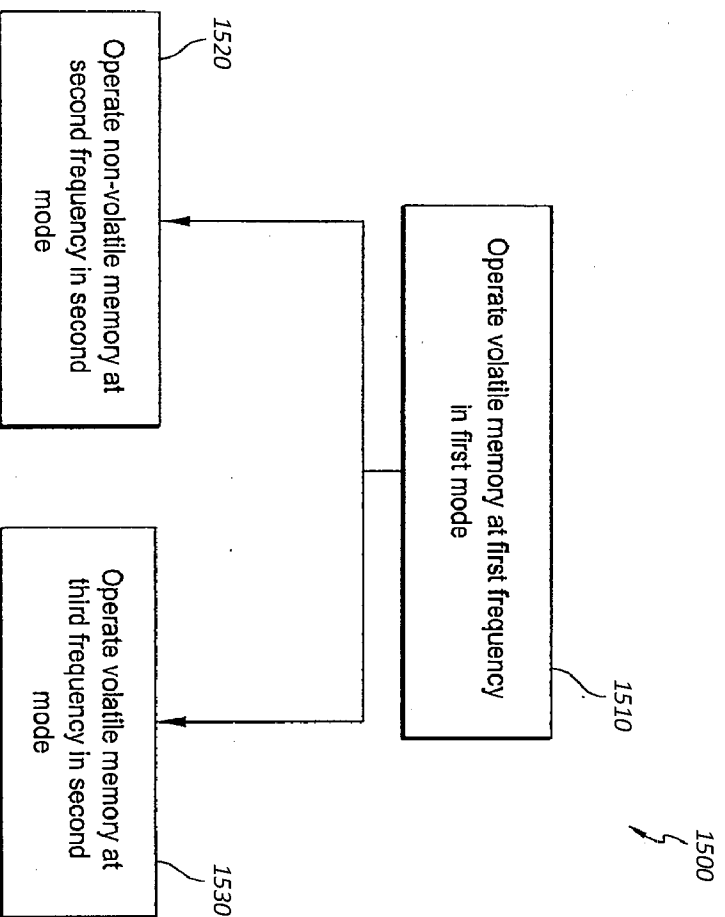


FIG. 20

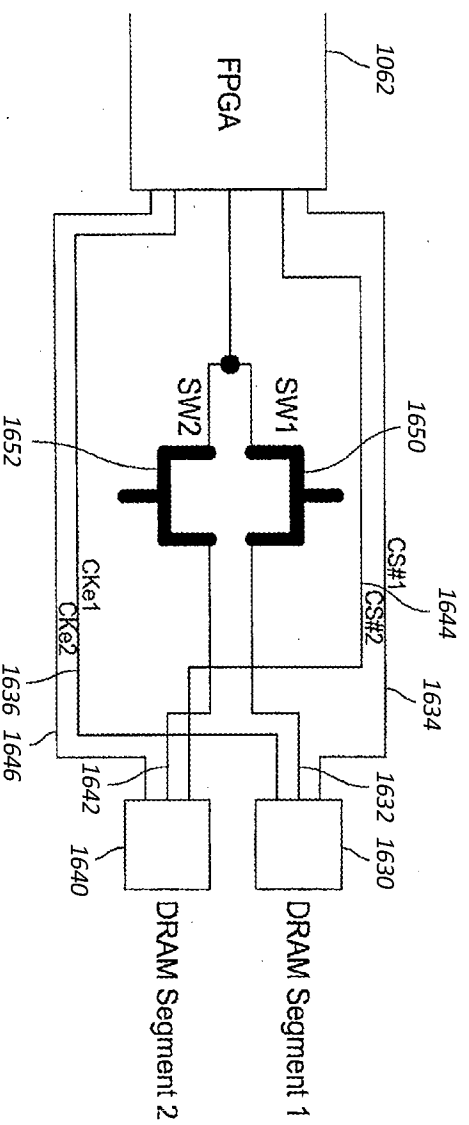


FIG. 21

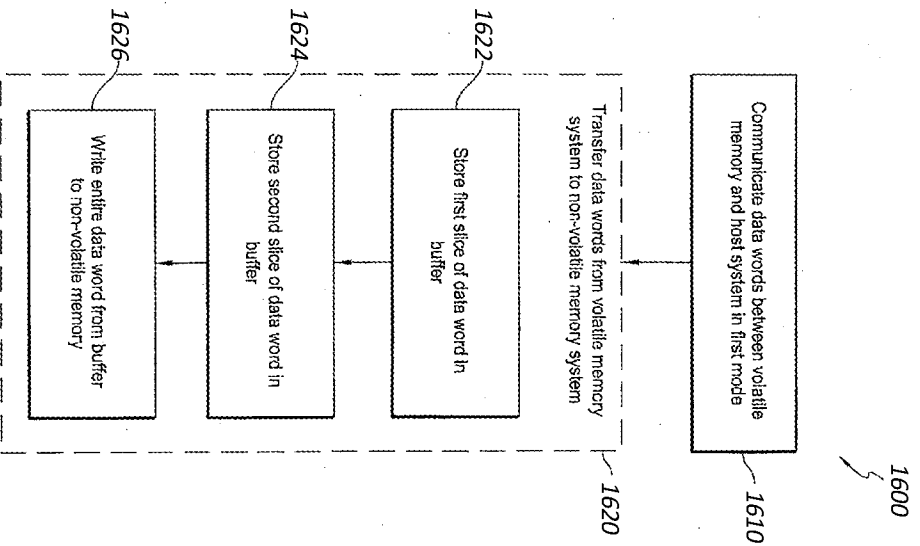


FIG. 22

FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 15/934,416, filed March 23, 2018, titled "Flash-Dram Hybrid Memory Module," which is a continuation of U.S. Patent Application No. 14/840,865, filed August 31, 2015, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,928,186, which is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 9,158,684, which is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled "Flash-Dram Hybrid Memory Module," now U.S. Patent No. 8,874,831, which claims the benefit of U. S. Provisional Patent Application No. 61/512,871, filed July 28, 2011, and is a continuation-in-part of U.S. Patent Application No. 12/240,916, filed September 29, 2008, titled "Non-Volatile Memory Module," now U.S. Patent No. 8,301,833, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, which claims the benefit of U. S. Provisional Patent Application No. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may be considered related to U.S. Patent Application No. 14/173,242, titled "Isolation Switching For Backup Of Registered Memory," filed February 5, 2014, which is a continuation of U.S. Patent Application 13/905,053, titled "Isolation Switching For Backup Of Registered Memory," filed May 29, 2013, now U.S. Patent 8,677,060, issued March 18, 2014, which is a continuation of U.S. Patent Application 13/536,173, titled "Data Transfer Scheme For Non-Volatile Memory Module," filed June 28, 2012, now U.S. Patent

8,516,187, issued August 20, 2013, which is a divisional of U.S. Patent Application 12/240,916, titled "Non-Volatile Memory Module," filed September 29, 2008, now U.S. Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. Patent Application No. 12/131,873, filed June 2, 2008, now abandoned, which claims the benefit of U.S. Provisional Application No. 60/941,586, filed June 1, 2007, the contents of which are incorporated by reference herein in their entirety.

[0003] This application may also be considered related to U.S. Patent Application No. 15/000,834, filed January 19, 2016 (abandoned), which is a continuation of U.S. Patent Application No. 14/489,332, filed September 17, 2014, now U.S. Patent No. 9,269,437, which is a continuation of U.S. Patent Application No. 14/173,219, filed February 5, 2014, now U.S. Patent No. 8,904,099, which is a continuation of U.S. Patent Application No. 13/905,048, filed May 29, 2013, now U.S. Patent No. 6,671,243, which is a continuation U.S. Patent Application 13/536,173 above.

[0004] This application may also be considered related to U.S. Patent Application No. 15/924,866, (abandoned), which is a continuation of U.S. Patent Application No. 14/489,281, filed September 17, 2014, now U.S. Patent No. 9,921,762, which is a continuation of U.S. Patent Application No. 13/625,563, filed September 24, 2012, now U.S. Patent No. 8,904,098, which claims the benefit of U.S. Provisional Application No. 61/583,775, filed September 23, 2011.

TECHNICAL FIELD

[0005] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0006] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0007] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high

speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0008] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0009] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a

data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MB/s for the EcoRAM vs. 104MB/s for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™), which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0010] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0011] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash

DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0012] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

[0013] Certain types of memory modules comprise a plurality of dynamic random-access memory (DRAM) devices mounted on a printed circuit board (PCB). These memory modules are typically mounted in a memory slot or socket of a computer system (e.g., a server system or a personal computer) and are accessed by the computer system to provide volatile memory to the computer system.

[0014] Volatile memory generally maintains stored information only when it is powered. Batteries have been used to provide power to volatile memory during power failures or interruptions. However, batteries may require maintenance, may need to be replaced, are not environmentally friendly, and the status of batteries can be difficult to monitor.

[0015] Non-volatile memory can generally maintain stored information while power is not applied to the non-volatile memory. In certain circumstances, it can therefore be useful to backup volatile memory using non-volatile memory.

OVERVIEW

[0016] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0017] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0018] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0019] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0020] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0021] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0022] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0024] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0025] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0026] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0027] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0029] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0030] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0031] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0032] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0033] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0034] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0035] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0036] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0037] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0038] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0039] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0040] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0041] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0042] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0043] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0044] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0045] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0047] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0048] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0049] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0050] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0052] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0053] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0054] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0055] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0056] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0057] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0060] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0061] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0062] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0063] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0064] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0065] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0066] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0067] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0068] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0069] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

[0070] Also described herein is a memory system having a volatile memory subsystem, a non-volatile memory subsystem, a controller coupled to the non-volatile memory subsystem, and a circuit coupled to the volatile memory subsystem, to the controller, and to a host system. In a first mode of operation, the circuit is operable to selectively isolate the controller from the volatile memory subsystem, and to selectively couple the volatile memory subsystem to the host system to allow data to be communicated between the volatile memory subsystem and the host system. In a second mode of operation, the circuit is operable to selectively couple the controller to the volatile memory subsystem to

allow data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem using the controller, and the circuit is operable to selectively isolate the volatile memory subsystem from the host system.

[0071] Also described herein is a method for operating a memory system. The method includes coupling a circuit to a host system, a volatile memory subsystem, and a controller, wherein the controller is coupled to a non-volatile memory subsystem. In a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, the circuit is used to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, the circuit is used to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

[0072] Also described herein is a nontransitory computer readable storage medium storing one or more programs configured to be executed by one or more computing devices. The programs, when executing on the one or more computing devices, cause a circuit that is coupled to a host system, to a volatile memory subsystem, and to a controller that is coupled to a nonvolatile memory subsystem, to perform a method in which, in a first mode of operation that allows data to be communicated between the volatile memory subsystem and the host system, operating the circuit to (i) selectively isolate the controller from the volatile memory subsystem, and (ii) selectively couple the volatile memory subsystem to the host system. In a second mode of operation that allows data to be communicated between the volatile memory subsystem and the nonvolatile memory subsystem via the controller, operating the circuit to (i) selectively couple the controller to the volatile memory subsystem, and (ii) selectively isolate the volatile memory subsystem from the host system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0073] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0074] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

FIG. 12 is a block diagram of an example memory system compatible with certain embodiments described herein.

FIG. 13 is a block diagram of an example memory module with ECC (error-correcting code) having a volatile memory subsystem with nine volatile memory elements and a non-volatile memory subsystem with five non-volatile memory elements in accordance with certain embodiments described herein.

FIG. 14 is a block diagram of an example memory module having a microcontroller unit and logic element integrated into a single device in accordance with certain embodiments described herein.

FIGS. 15A-15C schematically illustrate example embodiments of memory systems having volatile memory subsystems comprising registered dual in-line memory modules in accordance with certain embodiments described herein.

FIG. 16 schematically illustrates an example power module of a memory system in accordance with certain embodiments described herein.

FIG. 17 is a flowchart of an example method of providing a first voltage and a second voltage to a memory system including volatile and non-volatile memory subsystems.

FIG. 18 is a flowchart of an example method of controlling a memory system operatively coupled to a host system and which includes at least 100 percent more storage capacity in non-volatile memory than in volatile memory.

FIG. 19 schematically illustrates an example clock distribution topology of a memory system in accordance with certain embodiments described herein.

FIG. 20 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including operating a volatile memory subsystem at a reduced rate in a back-up mode.

FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments of a volatile memory subsystem of a memory system to a controller of the memory system.

FIG. 22 is a flowchart of an example method of controlling a memory system operatively coupled to a host system, the method including backing up and/or restoring a volatile memory subsystem in slices.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0075] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to

implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0076] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0077] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only

Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0078] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0079] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0080] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0081] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0082] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or

commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0083] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0084] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0085] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then

the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0086] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0087] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0088] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory

module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0089] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In

the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0090] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0091] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH

510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0092] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0093] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the

DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0094] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0095] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate

and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0096] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0097] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or

provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0098] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0099] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[00100] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[00101] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash-DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[00102] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM). As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 506 and the other for the DRAM memory DRAM 508. The MCH treats the DRAM 508 and Flash 506 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path between Flash 506 and DRAM 508 allows for direct data transfer to occur between the Flash 506 and the DRAM 508 in response to control information from Ctrl 502. In this embodiment, this data transfer mechanism provides direct support for executing commands from

the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 506 to the DRAM 508. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[00103] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[00104] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[00105] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a

first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[00106] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[00107] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00108] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502

monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00109] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1061

field, a Block address 1051 field, a Plane address 1041, a Page address 1031, and a group of least significant address bits A_0A_1 1021. The Plane address 1041 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00110] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00111] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash

memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00112] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00113] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00114] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00115] Certain embodiments described herein include a memory system which can communicate with a host system such as a disk controller of a computer system. The memory system can include volatile and non-volatile memory, and a controller. The controller backs up

the volatile memory using the non-volatile memory in the event of a trigger condition. Trigger conditions can include, for example, a power failure, power reduction, request by the host system, etc. In order to power the system in the event of a power failure or reduction, the memory system can include a secondary power source which does not comprise a battery and may include, for example, a capacitor or capacitor array.

[00116] In certain embodiments, the memory system can be configured such that the operation of the volatile memory is not adversely affected by the non-volatile memory or by the controller when the volatile memory is interacting with the host system. For example, one or more isolation devices may isolate the non-volatile memory and the controller from the volatile memory when the volatile memory is interacting with the host system and may allow communication between the volatile memory and the non-volatile memory when the data of the volatile memory is being restored or backed-up. This configuration generally protects the operation of the volatile memory when isolated while providing backup and restore capability in the event of a trigger condition, such as a power failure.

[00117] In certain embodiments described herein, the memory system includes a power module which provides power to the various components of the memory system from different sources based on a state of the memory system in relation to a trigger condition (e.g., a power failure). The power module may switch the source of the power to the various components in order to efficiently provide power in the event of the power failure. For example, when no power failure is detected, the power module may provide power to certain components, such as the volatile memory, from system power while charging a secondary power source (e.g., a capacitor

array). In the event of a power failure or other trigger condition, the power module may power the volatile memory elements using the previously charged secondary power source.

[00118] In certain embodiments, the power module transitions relatively smoothly from powering the volatile memory with system power to powering it with the secondary power source. For example, the power system may power volatile memory with a third power source from the time the memory system detects that power failure is likely to occur until the time the memory system detects that the power failure has actually occurred.

[00119] In certain embodiments, the volatile memory system can be operated at a reduced frequency during backup and/or restore operations which can improve the efficiency of the system and save power. In some embodiments, during backup and/or restore operations, the volatile memory communicates with the non-volatile memory by writing and/or reading data words in bit-wise slices instead of by writing entire words at once. In certain embodiments, when each slice is being written to or read from the volatile memory the unused slice(s) of volatile memory is not active, which can reduce the power consumption of the system.

[00120] In yet other embodiments, the non-volatile memory can include at least 100 percent more storage capacity than the volatile memory. This configuration can allow the memory system to efficiently handle subsequent trigger conditions.

[00121] FIG. 12 is a block diagram of an example memory system 1010 compatible with certain embodiments described herein. The memory system 1010 can be coupled to a host computer system and can include a volatile memory subsystem 1030, a non-volatile memory

subsystem 1040, and a controller 1062 operatively coupled to the non-volatile memory subsystem 1040. In certain embodiments, the memory system 1010 includes at least one circuit 1052 configured to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030.

[00122] In certain embodiments, the memory system 1010 comprises a memory module. The memory system 1010 may comprise a printed-circuit board (PCB) 1020. In certain embodiments, the memory system 1010 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other volatile memory capacities are also compatible with certain embodiments described herein. In certain embodiments, the memory system 1010 has a non-volatile memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, 16-GB, or 32-GB. Other non-volatile memory capacities are also compatible with certain embodiments described herein. In addition, memory systems 1010 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 1020 has an industry-standard form factor. For example, the PCB 1020 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 1020 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 1020 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), miniDIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for

example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[00123] In certain preferred embodiments, the memory system 1010 is in electrical communication with the host system. In other embodiments, the memory system 1010 may communicate with a host system using some other type of communication, such as, for example, optical communication. Examples of host systems include, but are not limited to, blade servers, 1U servers, personal computers (PCs), and other applications in which space is constrained or limited. The memory system 1010 can be in communication with a disk controller of a computer system, for example. The PCB 1020 can comprise an interface 1022 that is configured to be in electrical communication with the host system (not shown). For example, the interface 1022 can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system. The interface 1022 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory system 1010 and the host system. For example, the interface 1022 can comprise a standard 240-pin DDR2 edge connector.

[00124] The volatile memory subsystem 1030 comprises a plurality of volatile memory elements 1032 and the non-volatile memory subsystem 1040 comprises a plurality of non-volatile memory elements 1042. Certain embodiments described herein advantageously provide nonvolatile storage via the non-volatile memory subsystem 1040 in addition to high-performance (e.g., high speed) storage via the volatile memory subsystem 1030. In certain embodiments, the first plurality of volatile memory elements 1032 comprises two or more dynamic random-access memory (DRAM) elements. Types of DRAM elements 1032 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and

synchronous DRAM (SDRAM). For example, in the block diagram of FIG. 12, the first memory bank 1030 comprises eight 64Mx8 DDR2 SDRAM elements 1032. The volatile memory elements 1032 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory elements 1032 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Volatile memory elements 1032 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBOA), micro-BOA (1.1,BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00125] In certain embodiments, the second plurality of non-volatile memory elements 1042 comprises one or more flash memory elements. Types of flash memory elements 1042 compatible with certain embodiments described herein include, but are not limited to, NOR flash, NAND flash, ONE-NAND flash, and multi-level cell (MLC). For example, in the block diagram of FIG. 12, the second memory bank 1040 comprises 512 MB of flash memory organized as four 128Mbx8 NAND flash memory elements 1042. In addition, nonvolatile memory elements 1042 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Non-volatile memory elements 1042 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BOA), fine-pitch BOA (FBGA), micro-BOA (POA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[00126] FIG. 13 is a block diagram of an example memory module 10 with ECC (error-correcting code) having a volatile memory subsystem 1030 with nine volatile memory elements

1032 and a non-volatile memory subsystem 1040 with five non-volatile memory elements 1042 in accordance with certain embodiments described herein. The additional memory element 1032 of the first memory bank 1030 and the additional memory element 1042 of the second memory bank 1040 provide the ECC capability. In certain other embodiments, the volatile memory subsystem 1030 comprises other numbers of volatile memory elements 1032 (e.g., 2, 3, 4, 5, 6, 7, more than 9). In certain embodiments, the non-volatile memory subsystem 1040 comprises other numbers of nonvolatile memory elements 1042 (e.g., 2, 3, more than 5).

[00127] Referring to FIG. 12, in certain embodiments, the logic element 1070 comprises a field-programmable gate array (FPGA). In certain embodiments, the logic element 1070 comprises an FPGA available from Lattice Semiconductor Corporation which includes an internal flash. In certain other embodiments, the logic element 1070 comprises an FPOA available from another vendor. The internal flash can improve the speed of the memory system 1010 and save physical space. Other types of logic elements 1070 compatible with certain embodiments described herein include, but are not limited to, a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a custom-designed semiconductor device, a complex programmable logic device (CPLD). In certain embodiments, the logic element 1070 is a custom device. In certain embodiments, the logic element 1070 comprises various discrete electrical elements, while in certain other embodiments, the logic element 1070 comprises one or more integrated circuits. FIG. 14 is a block diagram of an example memory module 1010 having a microcontroller unit 1060 and logic element 1070 integrated into a single controller 1062 in accordance with certain embodiments described herein. In certain embodiments, the controller 1062 includes one or more other components. For example, in one

embodiment, an FPGA without an internal flash is used and the controller 1062 includes a separate flash memory component which stores configuration information to program the FPGA.

[00128] In certain embodiments, the at least one circuit 1052 comprises one or more switches coupled to the volatile memory subsystem 1030, to the controller 1062, and to the host computer (e.g., via the interface 1022, as schematically illustrated by FIGS. 12-14). The one or more switches are responsive to signals (e.g., from the controller 1062) to selectively operatively decouple the controller 1062 from the volatile memory subsystem 1030 and to selectively operatively couple the controller 1062 to the volatile memory subsystem 1030. In addition, in certain embodiments, the at least one circuit 1052 selectively operatively couples and decouples the volatile memory subsystem 1030 and the host system.

[00129] In certain embodiments, the volatile memory subsystem 1030 can comprise a registered DIMM subsystem comprising one or more registers 1160 and a plurality of DRAM elements 1180, as schematically illustrated by FIG. 15A. In certain such embodiments, the at least one circuit 1052 can comprise one or more switches 1172 coupled to the controller 1062 (e.g., logic element 1070) and to the volatile memory subsystem 1030 which can be actuated to couple and decouple the controller 1062 to and from the volatile memory subsystem 1030, respectively. The memory system 1010 further comprises one or more switches 1170 coupled to the one or more registers 1160 and to the plurality of DRAM elements 1180 as schematically illustrated by FIG. 15A. The one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. In certain other embodiments, as schematically illustrated by FIG. 15B, the one or more switches 1174 are also coupled to the one or more registers 1160 and to a power source 1162 for the one

or more registers 1160. The one or more switches 1174 can be selectively switched to turn power on or off to the one or more registers 1160, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150. As schematically illustrated by FIG. 15C, in certain embodiments the at least one circuit 1052 comprises a dynamic on-die termination (ODT) 1176 circuit of the logic element 1070. For example, the logic element 1070 can comprise a dynamic ODT circuit 1176 which selectively operatively couples and decouples the logic element 1070 to and from the volatile memory subsystem 1030, respectively. In addition, and similar to the example embodiment of FIG. 15A described above, the one or more switches 1170 can be selectively switched, thereby selectively operatively coupling the volatile memory subsystem 1030 to the host system 1150.

[00130] Certain embodiments described herein utilize the non-volatile memory subsystem 1040 as a flash “mirror” to provide backup of the volatile memory subsystem 1030 in the event of certain system conditions. For example, the non-volatile memory subsystem 1040 may backup the volatile memory subsystem 1030 in the event of a trigger condition, such as, for example, a power failure or power reduction or a request from the host system. In one embodiment, the nonvolatile memory subsystem 1040 holds intermediate data results in a noisy system environment when the host computer system is engaged in a long computation. In certain embodiments, a backup may be performed on a regular basis. For example, in one embodiment, the backup may occur every millisecond in response to a trigger condition. In certain embodiments, the trigger condition occurs when the memory system 1010 detects that the system voltage is below a certain threshold voltage. For example, in one embodiment, the threshold voltage is 1010 percent below a specified operating voltage. In certain embodiments, a trigger condition occurs when the voltage goes above a certain threshold value, such as, for example,

1010 percent above a specified operating voltage. In some embodiments, a trigger condition occurs when the voltage goes below a threshold or above another threshold. In various embodiments, a backup and/or restore operation may occur in reboot and/or non-reboot trigger conditions.

[00131] As schematically illustrated by FIGS. 12 and 13, in certain embodiments, the controller 1062 may comprise a microcontroller unit (MCU) 1060 and a logic element 1070. In certain embodiments, the MCU 1060 provides memory management for the non-volatile memory subsystem 1040 and controls data transfer between the volatile memory subsystem 30 and the nonvolatile memory subsystem 1040. The MCU 1060 of certain embodiments comprises a 16-bit microcontroller, although other types of microcontrollers are also compatible with certain embodiments described herein. As schematically illustrated by FIGS. 12 and 13, the logic element 1070 of certain embodiments is in electrical communication with the non-volatile memory subsystem 1040 and the MCU 1060. The logic element 1070 can provide signal level translation between the volatile memory elements 1032 (e.g., 1.8V SSTL-2 for DDR2 SDRAM elements) and the non-volatile memory elements 1042 (e.g., 3V TTL for NAND flash memory elements). In certain embodiments, the logic element 1070 is also programmed to perform address/address translation between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. In certain preferred embodiments, 1-NAND type flash are used for the non-volatile memory elements 1042 because of their superior read speed and compact structure.

[00132] The memory system 1010 of certain embodiments is configured to be operated in at least two states. The at least two states can comprise a first state in which the controller 1062 and the non-volatile memory subsystem 1040 are operatively decoupled (e.g., isolated) from the

volatile memory subsystem 1030 by the at least one circuit 1052 and a second state in which the volatile memory subsystem 1030 is operatively coupled to the controller 1062 to allow data to be communicated between the volatile memory subsystem 1030 and the nonvolatile memory subsystem 1040 via the controller 1062. The memory system 1010 may transition from the first state to the second state in response to a trigger condition, such as when the memory system 1010 detects that there is a power interruption (e.g., power failure or reduction) or a system hang-up.

[00133] The memory system 1010 may further comprise a voltage monitor 1050. The voltage monitor circuit 1050 monitors the voltage supplied by the host system via the interface 1022. Upon detecting a low voltage condition (e.g., due to a power interruption to the host system), the voltage monitor circuit 1050 may transmit a signal to the controller 1062 indicative of the detected condition. The controller 1062 of certain embodiments responds to the signal from the voltage monitor circuit 1050 by transmitting a signal to the at least one circuit 1052 to operatively couple the controller to the volatile memory system 1030, such that the memory system 1010 enters the second state. For example, the voltage monitor 1050 may send a signal to the MCU 1060 which responds by accessing the data on the volatile memory system 1030 and by executing a write cycle on the nonvolatile memory subsystem 1040. During this write cycle, data is read from the volatile memory subsystem 1030 and is transferred to the non-volatile memory subsystem 1040 via the MCU 1060. In certain embodiments, the voltage monitor circuit 1050 is part of the controller 1062 (e.g., part of the MCU 1060) and the voltage monitor circuit 1050 transmits a signal to the other portions of the controller 1062 upon detecting a power threshold condition.

[00134] The isolation or operational decoupling of the volatile memory subsystem 1030 from the non-volatile memory subsystem in the first state can preserve the integrity of the operation of the memory system 1010 during periods of operation in which signals (e.g., data) are transmitted between the host system and the volatile memory subsystem 1030. For example, in one embodiment during such periods of operation, the controller 1062 and the nonvolatile memory subsystem 1040 do not add a significant capacitive load to the volatile memory system 1030 when the memory system 1010 is in the first state. In certain such embodiments, the capacitive load of the controller 1062 and the non-volatile memory subsystem 1040 do not significantly affect the signals propagating between the volatile memory subsystem 1030 and the host system. This can be particularly advantageous in relatively high-speed memory systems where loading effects can be significant. In one preferred embodiment, the at least one circuit 1052 comprises an FSA1208 Low-Power, Eight-Port, Hi-Speed Isolation Switch from Fairchild Semiconductor. In other embodiments, the at least one circuit 1052 comprises other types of isolation devices.

[00135] Power may be supplied to the volatile memory subsystem 1030 from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a second power supply 1080 when the memory system 1010 is in the second state. In certain embodiments, the memory system 1010 is in the first state when no trigger condition (e.g., a power failure) is present and the memory system 1010 enters the second state in response to a trigger condition. In certain embodiments, the memory system 1010 has a third state in which the controller 1062 is operatively decoupled from the volatile memory subsystem 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). For example, in one embodiment the third power supply may provide power to the volatile memory

subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.

[00136] In certain embodiments, the second power supply 1080 does not comprise a battery. Because a battery is not used, the second power supply 1080 of certain embodiments may be relatively easy to maintain, does not generally need to be replaced, and is relatively environmentally friendly. In certain embodiments, as schematically illustrated by FIGS. 12-14, the second power supply 1080 comprises a step-up transformer 1082, a step-down transformer 1084, and a capacitor bank 1086 comprising one or more capacitors (e.g., double-layer capacitors). In one example embodiment, capacitors may take about three to four minutes to charge and about two minutes to discharge. In other embodiments, the one or more capacitors may take a longer time or a shorter time to charge and/or discharge. For example, in certain embodiments, the second power supply 1080 is configured to power the volatile memory subsystem 1030 for less than thirty minutes. In certain embodiments, the second power supply 1080 may comprise a battery. For example, in certain embodiments, the second power supply 1080 comprises a battery and one or more capacitors and is configured to power the volatile memory subsystem 1030 for no more than thirty minutes.

[00137] In certain embodiments, the capacitor bank 1086 of the second power supply 1080 is charged by the first power supply while the memory system 1010 is in the first state. As a result, the second power supply 1080 is fully charged when the memory system 1010 enters the second state. The memory system 1010 and the second power supply 1080 may be located on the same printed circuit board 1020. In other embodiments, the second power supply 1080 may not be on

the same printed circuit board 1020 and may be tethered to the printed circuit board 1020, for example.

[00138] When operating in the first state, in certain embodiments, the step-up transformer 1082 keeps the capacitor bank 1086 charged at a peak value. In certain embodiments, the step-down transformer 1084 acts as a voltage regulator to ensure that regulated voltages are supplied to the memory elements (e.g., 1.8V to the volatile DRAM elements 1032 and 3.0V to the non-volatile flash memory elements 1042) when operating in the second state (e.g., during power down). In certain embodiments, as schematically illustrated by FIGS. 12-14, the memory module 1010 further comprises a switch 1090 (e.g., FET switch) that switches power provided to the controller 1062, the volatile memory subsystem 1030, and the non-volatile memory subsystem 1040, between the power from the second power supply 1080 and the power from the first power supply (e.g., system power) received via the interface 1022. For example, the switch 1090 may switch from the first power supply to the second power supply 1080 when the voltage monitor 1050 detects a low voltage condition. The switch 1090 of certain embodiments advantageously ensures that the volatile memory elements 1032 and non-volatile memory elements 1042 are powered long enough for the data to be transferred from the volatile memory elements 1032 and stored in the non-volatile memory elements 1042. In certain embodiments, after the data transfer is complete, the switch 1090 then switches back to the first power supply and the controller 1062 transmits a signal to the at least one circuit 1052 to operatively decouple the controller 1062 from the volatile memory subsystem 1030, such that the memory system 1010 reenters the first state.

[00139] When the memory system 1010 re-enters the first state, data may be transferred back from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 via the

controller 1062. The host system can then resume accessing the volatile memory subsystem 1030 of the memory module 1010. In certain embodiments, after the memory system 1010 enters or re-enters the first state (e.g., after power is restored), the host system accesses the volatile memory subsystem 1030 rather than the non-volatile memory subsystem 1040 because the volatile memory elements 1032 have superior read/write characteristics. In certain embodiments, the transfer of data from the volatile memory bank 1030 to the nonvolatile memory bank 1040, or from the non-volatile memory bank 1040 to the volatile memory bank 1030, takes less than one minute per GB.

[00140] In certain embodiments, the memory system 1010 protects the operation of the volatile memory when communicating with the host-system and provides backup and restore capability in the event of a trigger condition such as a power failure. In certain embodiments, the memory system 1010 copies the entire contents of the volatile memory subsystem 1030 into the nonvolatile memory subsystem 1040 on each backup operation. Moreover, in certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are copied back into the volatile memory subsystem 1030 on each restore operation. In certain embodiments, the entire contents of the non-volatile memory subsystem 1040 are accessed for each backup and/or restore operation, such that the non-volatile memory subsystem 1040 (e.g., flash memory subsystem) is used generally uniformly across its memory space and wear-leveling is not performed by the memory system 1010. In certain embodiments, avoiding wear-leveling can decrease cost and complexity of the memory system 1010 and can improve the performance of the memory system 1010. In certain other embodiments, the entire contents of the volatile memory subsystem 1030 are not copied into the non-volatile memory subsystem 1040 on each backup operation, but only a partial copy is performed. In certain embodiments, other

management capabilities such as bad-block management and error management for the flash memory elements of the non-volatile memory subsystem 1040 are performed in the controller 1062.

[00141] The memory system 1010 generally operates as a write-back cache in certain embodiments. For example, in one embodiment, the host system (e.g., a disk controller) writes data to the volatile memory subsystem 1030 which then writes the data to non-volatile storage which is not part of the memory system 1010, such as, for example, a hard disk. The disk controller may wait for an acknowledgment signal from the memory system 1010 indicating that the data has been written to the hard disk or is otherwise secure. The memory system 1010 of certain embodiments can decrease delays in the system operation by indicating that the data has been written to the hard disk before it has actually done so. In certain embodiments, the memory system 1010 will still be able to recover the data efficiently in the event of a power outage because of the backup and restore capabilities described herein. In certain other embodiments, the memory system 1010 may be operated as a write-through cache or as some other type of cache.

[00142] FIG. 16 schematically illustrates an example power module 1100 of the memory system 1010 in accordance with certain embodiments described herein. The power module 1100 provides power to the various components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition. In certain embodiments, the power module 1100 comprises one or more of the components described above with respect to FIG. 12. For example, in certain embodiments, the power module 1100 includes the second power supply 1080 and the switch 1090.

[00143] The power module 1100 provides a plurality of voltages to the memory system 1010 comprising non-volatile and volatile memory subsystems 1030, 1040. The plurality of voltages comprises at least a first voltage 1102 and a second voltage 1104. The power module 1100 comprises an input 1106 providing a third voltage 1108 to the power module 1100 and a voltage conversion element 1120 configured to provide the second voltage 1104 to the memory system 1010. The power module 1100 further comprises a first power element 1130 configured to selectively provide a fourth voltage 1110 to the conversion element 1120. In certain embodiments, the first power element 1130 comprises a pulse-width modulation power controller. For example, in one example embodiment, the first power element 1130 is configured to receive a 1.8V input system voltage as the third voltage 1108 and to output a modulated 5V output as the fourth voltage 1110.

[00144] The power module 1100 further comprises a second power element 1140 can be configured to selectively provide a fifth voltage 1112 to the conversion element 1120. The power module 1100 can be configured to selectively provide the first voltage 1102 to the memory system 1010 either from the conversion element 1120 or from the input 1106.

[00145] The power module 1100 can be configured to be operated in at least three states in certain embodiments. In a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130. In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120. In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element

1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.

[00146] In certain embodiments, the power module 1100 transitions from the first state to the second state upon detecting that a trigger condition is likely to occur and transitions from the second state to the third state upon detecting that the trigger condition has occurred. For example, the power module 1100 may transition to the second state when it detects that a power failure is about to occur and transitions to the third state when it detects that the power failure has occurred. In certain embodiments, providing the first voltage 1102 in the second state from the first power element 1130 rather than from the input 1106 allows a smoother transition from the first state to the third state. For example, in certain embodiments, providing the first voltage 1102 from the first power element 1130 has capacitive and other smoothing effects. In addition, switching the point of power transition to be between the conversion element 1120 and the first and second power elements 1130, 1140 (e.g., the sources of the pre-regulated fourth voltage 1110 in the second state and the pre-regulated fifth voltage 1112 in the third state) can smooth out potential voltage spikes.

[00147] In certain embodiments, the second power element 1140 does not comprise a battery and may comprise one or more capacitors. For example, as schematically illustrated in FIG. 16, the second power element 1140 comprises a capacitor array 1142, a buck-boost converter 1144 which adjusts the voltage for charging the capacitor array and a voltage/current limiter 1146 which limits the charge current to the capacitor array 1142 and stops charging the capacitor array 1142 when it has reached a certain charge voltage. In one example embodiment, the capacitor array 1142 comprises two 50 farad capacitors capable of holding a total charge of 4.6V. For

example, in one example embodiment, the buck-boost converter 1144 receives a 1.8V system voltage (first voltage 1108) and boosts the voltage to 4.3V which is outputted to the voltage current limiter 1146. The voltage/current limiter 1146 limits the current going to the capacitor array 1142 to 1A and stops charging the array 1142 when it is charged to 4.3V. Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the second power element 1140 may include alternative embodiments. For example, different components and/or different value components may be used. For example, in other embodiments, a pure boost converter may be used instead of a buck-boost converter. In another embodiment, only one capacitor may be used instead of a capacitor array 1142.

[00148] The conversion element 1120 can comprise one or more buck converters and/or one or more buck-boost converters. The conversion element 1120 may comprise a plurality of sub-blocks 1122, 1124, 1126 as schematically illustrated by FIG. 16, which can provide more voltages in addition to the second voltage 1104 to the memory system 1010. The sub-blocks may comprise various converter circuits such as buck-converters, boost converters, and buck-boost converter circuits for providing various voltage values to the memory system 1010. For example, in one embodiment, sub-block 1122 comprises a buck converter, sub-block 1124 comprises a dual buck converter, and sub-block 1126 comprises a buck-boost converter as schematically illustrated by FIG. 16. Various other components for the sub-blocks 1122, 1124, 1126 of the conversion element 1120 are also compatible with certain embodiments described herein. In certain embodiments, the conversion element 1120 receives as input either the fourth voltage 1110 from the first power element 1130 or the fifth voltage 1112 from the second power element 1140, depending on the state of the power module 1100, and reduces the input to an appropriate

amount for powering various components of the memory system. For example, the buck-converter of sub-block 1122 can provide 1.8V at 2A for about 60 seconds to the volatile memory elements 1032 (e.g., DRAM), the non-volatile memory elements 1042 (e.g., flash), and the controller 1062 (e.g., an FPGA) in one embodiment. The sub-block 1124 can provide the second voltage 1104 as well as another reduced voltage 1105 to the memory system 1010. In one example embodiment, the second voltage 1104 is 2.5V and is used to power the at least one circuit 1052 (e.g., isolation device) and the other reduced voltage 1105 is 1.2V and is used to power the controller 1062 (e.g., FPGA). The subblock 1126 can provide yet another voltage 1107 to the memory system 1010. For example, the voltage 1107 may be 3.3V and may be used to power both the controller 1062 and the at least one circuit 1052.

[00149] Although described with respect to certain example embodiments, one of ordinary skill will recognize from the disclosure herein that the conversion element 1120 may include alternative embodiments. For example, there may be more or less sub-blocks which may comprise other types of converters (e.g., pure boost converters) or which may produce different voltage values. In one embodiment, the volatile memory elements 1032 and nonvolatile memory elements 1042 are powered using independent voltages and are not both powered using the first voltage 1102.

[00150] FIG. 17 is a flowchart of an example method 1200 of providing a first voltage 1102 and a second voltage 1104 to a memory system 1010 including volatile and nonvolatile memory subsystems 1030, 1040. While the method 1200 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-15, other memory systems are also compatible with embodiments of the method 1200. During a first condition, the method 1200

comprises providing the first voltage 1102 to the memory system 1010 from an input power supply 1106 and providing the second voltage 1104 to the memory system 1010 from a first power subsystem in operational block 1210. For example, in one embodiment, the first power subsystem comprises the first power element 1130 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, other first power subsystems are used.

[00151] The method 1200 further comprises detecting a second condition in operational block 1220. In certain embodiments, detecting the second condition comprises detecting that a trigger condition is likely to occur. During the second condition, the method 1200 comprises providing the first voltage 1102 and the second voltage 1104 to the memory system 1010 from the first power subsystem in an operational block 1230. For example, referring to FIG. 16, a switch 1148 can be toggled to provide the first voltage 1102 from the conversion element 1120 rather than from the input power supply.

[00152] The method 1200 further comprises charging a second power subsystem in operational block 1240. In certain embodiments, the second power subsystem comprises the second power element 1140 or another power supply that does not comprise a battery. For example, in one embodiment, the second power subsystem comprises the second power element 1140 and the voltage conversion element 1120 described above with respect to FIG. 16. In other embodiments, some other second power subsystem is used.

[00153] The method 1200 further comprises detecting a third condition in an operational block 1250 and during the third condition, providing the first voltage 1102 and the second

voltage 1104 to the memory system 1010 from the second power subsystem 1140 in an operational block 1260. In certain embodiments, detecting the third condition comprises detecting that the trigger condition has occurred. The trigger condition may comprise various conditions described herein. In various embodiments, for example, the trigger condition comprises a power reduction, power failure, or system hang-up. The operational blocks of the method 1200 may be performed in different orders in various embodiments. For example, in certain embodiments, the second power subsystem 1140 is charged before detecting the second condition.

[00154] In certain embodiments, the memory system 1010 comprises a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040 comprising at least 100 percent more storage capacity than does the volatile memory subsystem. The memory system 1010 also comprises a controller 1062 operatively coupled to the volatile memory subsystem 1030 and operatively coupled to the non-volatile memory subsystem 1040. The controller 1062 can be configured to allow data to be communicated between the volatile memory subsystem 1030 and the host system when the memory system 1010 is operating in a first state and to allow data to be communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040 when the memory system 1010 is operating in a second state.

[00155] Although the memory system 1010 having extra storage capacity of the non-volatile memory subsystem 1040 has been described with respect to certain embodiments, alternative configurations exist. For example, in certain embodiments, there may be more than 100 percent more storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In various embodiments, there may be at least 200, 300, or 400 percent more

storage capacity in the non-volatile memory subsystem 1040 than in the volatile memory subsystem 1030. In other embodiments, the non-volatile memory subsystem 1040 includes at least some other integer multiples of the storage capacity of the volatile memory subsystem 1030. In some embodiments, the non-volatile memory subsystem 1040 includes a non-integer multiple of the storage capacity of the volatile memory subsystem 1030. In one embodiment, the non-volatile memory subsystem 1040 includes less than 100 percent more storage capacity than does the volatile memory subsystem 1030.

[00156] The extra storage capacity of the non-volatile memory subsystem 1040 can be used to improve the backup capability of the memory system 1010. In certain embodiments in which data can only be written to portions of the non-volatile memory subsystem 1040 which do not contain data (e.g., portions which have been erased), the extra storage capacity of the nonvolatile memory subsystem 1040 allows the volatile memory subsystem 1030 to be backed up in the event of a subsequent power failure or other trigger event. For example, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to backup the volatile memory subsystem 1030 efficiently in the event of multiple trigger conditions (e.g., power failures). In the event of a first power failure, for example, the data in the volatile memory system 1030 is copied to a first, previously erased portion of the nonvolatile memory subsystem 1040 via the controller 1062. Since the non-volatile memory subsystem 1040 has more storage capacity than does the volatile memory subsystem 1030, there is a second portion of the non-volatile memory subsystem 1040 which does not have data from the volatile memory subsystem 1030 copied to it and which remains free of data (e.g., erased). Once system power is restored, the controller 1062 of the memory system 1010 restores the data to the volatile memory subsystem 1030 by copying the backed-up data from the non-volatile memory

subsystem 1040 back to the volatile memory subsystem 1030. After the data is restored, the memory system 1010 erases the non-volatile memory subsystem 1040. While the first portion of the non-volatile memory subsystem 1040 is being erased, it may be temporarily inaccessible.

[00157] If a subsequent power failure occurs before the first portion of the non-volatile memory subsystem 1040 is completely erased, the volatile memory subsystem 1030 can be backed-up or stored again in the second portion of the non-volatile memory subsystem 1040 as described herein. In certain embodiments, the extra storage capacity of the non-volatile memory subsystem 1040 may allow the memory system 1010 to operate more efficiently. For example, because of the extra storage capacity of the non-volatile memory subsystem 1040, the memory system 1010 can handle a higher frequency of trigger events that is not limited by the erase time of the non-volatile memory subsystem 1040.

[00158] FIG. 18 is a flowchart of an example method 1300 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. In certain embodiments, the non-volatile memory subsystem 1040 comprises at least 100 percent more storage capacity than does the volatile memory subsystem 30 as described herein. While the method 1300 is described herein by reference to the memory system 1010 schematically illustrated by FIGS. 12-14, the method 1300 can be practiced using other memory systems in accordance with certain embodiments described herein. In an operational block 1310, the method 1300 comprises communicating data between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation. The method 1300 further comprises storing a first copy of data from the

volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a first time when the memory system 1010 is in a second mode of operation in an operational block 1320.

[00159] In an operational block 1330, the method 1300 comprises restoring the first copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030. The method 1300 further comprises erasing the first copy of data from the non-volatile memory subsystem 1040 in an operational block 1340. The method further comprises storing a second copy of data from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 at a second time when the memory system 1010 is in the second mode of operation in an operational block 1350. Storing the second copy begins before the first copy is completely erased from the non-volatile memory subsystem 1040.

[00160] In some embodiments, the memory system 1010 enters the second mode of operation in response to a trigger condition, such as a power failure. In certain embodiments, the first copy of data and the second copy of data are stored in separate portions of the nonvolatile memory subsystem 1040. The method 1300 can also include restoring the second copy of data from the non-volatile memory subsystem 1040 to the volatile memory subsystem 1030 in an operational block 1360. The operational blocks of method 1300 referred to herein may be performed in different orders in various embodiments. For example, in some embodiments, the second copy of data is restored to the volatile memory subsystem 1030 at operational block 1360 before the first copy of data is completely erased in the operational block 1340.

[00161] FIG. 19 schematically illustrates an example clock distribution topology 1400 of a memory system 1010 in accordance with certain embodiments described herein. The clock

distribution topology 1400 generally illustrates the creation and routing of the clock signals provided to the various components of the memory system 1010. A clock source 1402 such as, for example, a 25 MHz oscillator, generates a clock signal. The clock source 1402 may feed a clock generator 1404 which provides a clock signal 1406 to the controller 1062, which may be an FPGA. In one embodiment, the clock generator 1404 generates a 125 MHz clock signal 1406. The controller 1062 receives the clock signal 1406 and uses it to clock the controller 1062 master state control logic. For example, the master state control logic may control the general operation of an FPGA controller 1062.

[00162] The clock signal 1406 can also be input into a clock divider 1410 which produces a frequency-divided version of the clock signal 1406. In an example embodiment, the clock divider 1410 is a divide by two clock divider and produces a 62.5 MHz clock signal in response to the 125 MHz clock signal 1406. A non-volatile memory phase-locked loop (PLL) block 1412 can be included (e.g., in the controller 1062) which distributes a series of clock signals to the non-volatile memory subsystem 1040 and to associated control logic. For example, a series of clock signals 1414 can be sent from the controller 1062 to the non-volatile memory subsystem 1040. Another clock signal 1416 can be used by the controller logic which is dedicated to controlling the non-volatile memory subsystem 1040. For example, the clock signal 1416 may clock the portion of the controller 1062 which is dedicated to generating address and/or control lines for the non-volatile memory subsystem 1040. A feedback clock signal 1418 is fed back into the non-volatile memory PLL block 1412. In one embodiment, the PLL block 1412 compares the feedback clock 1418 to the reference clock 1411 and varies the phase and frequency of its output until the reference 1411 and feedback 1418 clocks are phase and frequency matched.

[00163] A version of the clock signal 1406 such as the backup clock signal 1408 may be sent from the controller to the volatile memory subsystem 1030. The clock signal 1408 may be, for example, a differential version of the clock signal 1406. As described herein, the backup clock signal 1408 may be used to clock the volatile memory subsystem 1030 when the memory system 1010 is backing up the data from the volatile memory subsystem 1030 into the non-volatile memory subsystem 1040. In certain embodiments, the backup clock signal 1408 may also be used to clock the volatile memory subsystem 1030 when the memory system 1010 is copying the backed-up data back into the volatile memory subsystem 1030 from the nonvolatile memory subsystem 1040 (also referred to as restoring the volatile memory subsystem 1030). The volatile memory subsystem 1030 may normally be run at a higher frequency (e.g., DRAM running at 400MHz) than the nonvolatile memory subsystem 1040 (e.g., flash memory running at 62.5MHz) when communicating with the host system (e.g., when no trigger condition is present). However, in certain embodiments the volatile memory subsystem 1030 may be operated at a reduced frequency (e.g., at twice the frequency of the non-volatile memory subsystem 1040) without introducing significant delay into the system during backup operation and/or restore operations. Running the volatile memory subsystem 1030 at the reduced frequency during a backup and/or restore operation may advantageously reduce overall power consumption of the memory system 1010.

[00164] In one embodiment, the backup clock 1408 and the volatile memory system clock signal 1420 are received by a multiplexer 1422, as schematically illustrated by FIG. 19. The multiplexer 1422 can output either the volatile memory system clock signal 1420 or the backup clock signal 1408 depending on the backup state of the memory system 1010. For example, when the memory system 1010 is not performing a backup or restore operation and is

communicating with the host system (e.g., normal operation), the volatile memory system clock signal 1420 may be provided by the multiplexer 422 to the volatile memory PLL block 1424. When the memory system 1010 is performing a backup (or restore) operation, the backup clock signal 1408 may be provided.

[00165] The volatile memory PLL block 1424 receives the volatile memory reference clock signal 1423 from the multiplexer 1422 and can generate a series of clock signals which are distributed to the volatile memory subsystem 1030 and associated control logic. For example, in one embodiment, the PLL block 1424 generates a series of clock signals 1426 which clock the volatile memory elements 1032. A clock signal 1428 may be used to clock control logic associated with the volatile memory elements, such as one or more registers (e.g., the one or more registers of a registered DIMM). Another clock signal 1430 may be sent to the controller 1062. A feedback clock signal 1432 is fed back into the volatile memory PLL block 1424. In one embodiment, the PLL block 1424 compares the feedback clock signal 1432 to the reference clock signal 1423 and varies the phase and frequency of its output until the reference clock signal 1423 and the feedback clock signal 1432 clocks are phase and frequency matched.

[00166] The clock signal 1430 may be used by the controller 1062 to generate and distribute clock signals which will be used by controller logic which is configured to control the volatile memory subsystem 1030. For example, control logic in the controller 1062 may be used to control the volatile memory subsystem 1030 during a backup or restore operation. The clock signal 1430 may be used as a reference clock signal for the PLL block 1434 which can generate one or more clocks 1438 used by logic in the controller 1062. For example, the PLL block 1434 may generate one or more clock signals 1438 used to drive logic circuitry associated with

controlling the volatile memory subsystem 1030. In certain embodiments, the PLL block 1434 includes a feedback clock signal 1436 and operates in a similar manner to other PLL blocks described herein.

[00167] The clock signal 1430 may be used as a reference clock signal for the PLL block 1440 which may generate one or more clock signals used by a sub-block 1442 to generate one or more other clock signals 1444. In one embodiment, for example, the volatile memory subsystem 1030 comprises DDR2 SDRAM elements and the sub-block 1442 generates one or more DDR2 compatible clock signals 1444. A feedback clock signal 1446 is fed back into the PLL block 1440. In certain embodiments, the PLL block 1440 operates in a similar manner to other PLL blocks described herein.

[00168] While described with respect to the example embodiment of FIG. 19, various alternative clock distribution topologies are possible. For example, one or more of the clock signals have a different frequency in various other embodiments. In some embodiments, one or more of the clocks shown as differential signals are single ended signals. In one embodiment, the volatile memory subsystem 1030 operates on the volatile memory clock signal 1420 and there is no backup clock signal 1408. In some embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a backup operation and not during a restore operation. In other embodiments, the volatile memory subsystem 1030 is operated at a reduced frequency during a restore operation and not during a backup operation.

[00169] FIG. 20 is a flowchart of an example method 1500 of controlling a memory system 1010 operatively coupled to a host system. Although described with respect to the memory

system 1010 described herein, the method 1500 is compatible with other memory systems. The memory system 1010 may include a clock distribution topology 1400 similar to the one described above with respect to FIG. 19 or another clock distribution topology. The memory system 1010 can include a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040.

[00170] In an operational block 1510, the method 1500 comprises operating the volatile memory subsystem 1030 at a first frequency when the memory system 1010 is in a first mode of operation in which data is communicated between the volatile memory subsystem 1030 and the host system. In an operational block 1520, the method 1500 comprises operating the non-volatile memory subsystem 1040 at a second frequency when the memory system 1010 is in a second mode of operation in which data is communicated between the volatile memory subsystem 1030 and the non-volatile memory subsystem 1040. The method 1500 further comprises operating the volatile memory subsystem 1030 at a third frequency in an operational block 1530 when the memory system 1010 is in the second mode of operation. In certain embodiments, the memory system 1010 is not powered by a battery when it is in the second mode of operation. The memory system 1010 may switch from the first mode of operation to the second mode of operation in response to a trigger condition. The trigger condition may be any trigger condition described herein such as, for example, a power failure condition. In certain embodiments, the second mode of operation includes both backup and restore operations as described herein. In other embodiments, the second mode of operation includes backup operations but not restore operations. In yet other embodiments, the second mode of operation includes restore operations but not backup operations.

[00171] The third frequency can be less than the first frequency. For example, the third frequency can be approximately equal to the second frequency. In certain embodiments, the reduced frequency operation is an optional mode. In yet other embodiments, the first, second and/or third frequencies are configurable by a user or by the memory system 1010.

[00172] FIG. 21 schematically illustrates an example topology of a connection to transfer data slices from two DRAM segments 1630, 1640 of a volatile memory subsystem 1030 of a memory system 1010 to a controller 1062 of the memory system 1010. While the example of FIG. 21 shows a topology including two DRAM segments 1630, 1640 for the purposes of illustration, each address location of the volatile memory subsystem 1030 comprises more than the two segments in certain embodiments. The data lines 1632, 1642 from the first DRAM segment 1630 and the second DRAM segment 1640 of the volatile memory subsystem 1030 are coupled to switches 1650, 1652 which are coupled to the controller 1062 (e.g., logic element 1070) of the memory system 1010. The chip select lines 1634, 1644 and the self-refresh lines 1636, 1646 (e.g., CKe signals) of the first and second DRAM segments 1630, 1640, respectively, are coupled to the controller 1062. In certain embodiments, the controller 1062 comprises a buffer (not shown) which is configured to store data from the volatile memory subsystem 1030. In certain embodiments, the buffer is a first-in, first out buffer (FIFO). In certain embodiments, data slices from each DRAM segment 1630, 1640 comprise a portion of the volatile memory subsystem data bus. In one embodiment, for example, the volatile memory subsystem 1030 comprises a 72-bit data bus (e.g., each data word at each addressable location is 72 bits wide and includes, for example, 64 bits of accessible SDRAM and 8 bits of ECC), the first data slice from the first DRAM segment 1630 may comprise 40 bits of the data word, and the second data slice

from the second DRAM segment 1640 may comprise the remaining 32 bits of the data word. Certain other embodiments comprise data buses and/or data slices of different sizes.

[00173] In certain embodiments, the switches 1650, 1652 can each be selectively switched to selectively operatively couple the data lines 1632, 1642, respectively from the first and second DRAM segments 1630, 1640 to the controller 1062. The chip select lines 1634, 1644 enable the first and second DRAM segments 1630, 1640, respectively, of the volatile memory subsystem 1030, and the self-refresh lines 1636, 1646 toggle the first and second DRAM segments 1630, 1640, respectively, from self-refresh mode to active mode. In certain embodiments, the first and second DRAM segments 1630, 1640 maintain stored information but are not accessible when they are in self-refresh mode, and maintain stored information and are accessible when they are in active mode.

[00174] In certain embodiments, when the memory system 1010 is backing up the volatile memory system 1030, data slices from only one of the two DRAM segments 1630, 1640 at a time are sent to the controller 1062. For example, when the first slice is being written to the controller 1062 during a back-up, the controller 1062 sends a signal via the CKe line 1636 to the first DRAM segment 1630 to put the first DRAM segment 1630 in active mode. In certain embodiments, the data slice from the first DRAM segment 1630 for multiple words (e.g., a block of words) is written to the controller 1062 before writing the second data slice from the second DRAM segment 1640 to the controller 1062. While the first data slice is being written to the controller 1062, the controller 1062 also sends a signal via the CKe line 1646 to put the second DRAM segment 1640 in self-refresh mode. Once the first data slice for one word or for a block of words is written to the controller 1062, the controller 1062 puts the first DRAM segment 1630

into self-refresh mode by sending a signal via the CKE line 1636 to the first DRAM segment 1640. The controller 1062 also puts the second DRAM segment 1640 into active mode by sending a signal via the CKE line 1646 to the DRAM segment 1640. The second slice for a word or for a block of words is written to the controller 1062. In certain embodiments, when the first and second data slices are written to the buffer in the controller 1062, the controller 1062 combines the first and second data slices 1630, 1640 into complete words or blocks of words and then writes each complete word or block of words to the non-volatile memory subsystem 1040. In certain embodiments, this process is called "slicing" the volatile memory subsystem 1030.

[00175] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1042 write each backed-up data word to the controller 1062 which writes a first slice of the data word to the volatile memory subsystem 1030 and then a second slice of the data word to the volatile memory subsystem 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00176] FIG. 22 is a flowchart of an example method 1600 of controlling a memory system 1010 operatively coupled to a host system and which includes a volatile memory subsystem 1030 and a non-volatile memory subsystem 1040. Although described with respect to the memory system 1010 described herein with respect to FIGS. 12-14 and 21, the method 1600 is compatible with other memory systems. The method 1600 comprises communicating data words between the volatile memory subsystem 1030 and the host system when the memory system 1010 is in a first mode of operation in an operational block 1610. For example, the memory

system 1010 may be in the first mode of operation when no trigger condition has occurred and the memory system is not performing a backup and/or restore operation or is not being powered by a secondary power supply.

[00177] In an operational block 1620, the method further comprises transferring data words from the volatile memory subsystem 1030 to the non-volatile memory subsystem 1040 when the memory system 1010 is in a second mode of operation. In certain embodiments, each data word comprises the data stored in a particular address of the memory system 1010. The memory system 1010 may enter the second mode of operation, for example, when a trigger condition (e.g., a power failure) occurs. In certain embodiments, transferring each data word comprises storing a first portion (also referred to as a slice) of the data word in a buffer in an operational block 1622, storing a second portion of the data word in the buffer in an operational block 1624, and writing the entire data word from the buffer to the non-volatile memory subsystem 1040 in an operational block 1626.

[00178] In one example embodiment, the data word may be a 72 bit data word (e.g., 64 bits of accessible SDRAM and 8 bits of ECC), the first portion (or "slice") may comprise 40 bits of the data word, and the second portion (or "slice") may comprise the remaining 32 bits of the data word. In certain embodiments, the buffer is included in the controller 1062. For example, in one embodiment, the buffer is a first-in, first-out buffer implemented in the controller 1062 which comprises an FPGA. The method 1600 may generally be referred to as "slicing" the volatile memory during a backup operation. In the example embodiment, the process of "slicing" the volatile memory during a backup includes bringing the 32-bit slice out of self-refresh, reading a 32-bit block from the slice into the buffer, and putting the 32-bit slice back into self-refresh. The

40-bit slice is then brought out of self-refresh and a 40-bit block from the slice is read into a buffer. Each block may comprise a portion of multiple words. For example, each 32-bit block may comprise 32-bit portions of multiple 72-bit words. In other embodiments, each block comprises a portion of a single word. The 40-bit slice is then put back into self-refresh in the example embodiment. The 32-bit and 40-bit slices are then combined into a 72-bit block by the controller 1062 and ECC detection/correction is performed on each 72-bit word as it is read from the buffer and written into the non-volatile memory subsystem (e.g., flash).

[00179] In some embodiments, the entire data word may comprise more than two portions. For example, the entire data word may comprise three portions instead of two and transferring each data word further comprises storing a third portion of each data word in the buffer. In certain other embodiments, the data word may comprise more than three portions.

[00180] In certain embodiments, the data may be sliced in a restore operation as well as, or instead of, during a backup operation. For example, in one embodiment, the nonvolatile memory elements 1040 write each backed-up data word to the controller 1062 which writes a first portion of the data word to the volatile memory subsystem 1030 and then a second portion of the data word to the volatile memory 1030. In certain embodiments, slicing the volatile memory subsystem 1030 during a restore operation may be performed in a manner generally inverse to slicing the volatile memory subsystem 1030 during a backup operation.

[00181] The method 1600 can advantageously provide significant power savings and can lead to other advantages. For example, in one embodiment where the volatile memory subsystem 1030 comprises DRAM elements, only the slice of the DRAM which is currently being accessed

(e.g., written to the buffer) during a backup is configured in full-operational mode. The slice or slices that are not being accessed may be put in self-refresh mode. Because DRAM in self-refresh mode uses significantly less power than DRAM in full-operational mode, the method 1600 can allow significant power savings. In certain embodiments, each slice of the DRAM includes a separate self-refresh enable (e.g., CKe) signal which allows each slice to be accessed independently.

[00182] In addition, the connection between the DRAM elements and the controller 1062 may be as large as the largest slice instead of as large as the data bus. In the example embodiment, the connection between the controller 1062 and the DRAM may be 40 bits instead of 72 bits. As a result, pins on the controller 1062 may be used for other purposes or a smaller controller may be used due to the relatively low number of pin-outs used to connect to the volatile memory subsystem 1030. In certain other embodiments, the full width of the data bus is connected between the volatile memory subsystem 1030 and the controller 1062 but only a portion of it is used during slicing operations. For example, in some embodiments, memory slicing is an optional mode.

[00183] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module comprising:
 - a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
 - a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;
 - a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;
 - a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;
 - a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and
 - a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:
 - a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and
 - at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes.
2. The memory module of claim 1, wherein the first and third buck converters are further configured to operate as a dual buck converter.

3. The memory module of claim 1, wherein the first voltage amplitude is 1.8 volts.
4. The memory module of claim 1, wherein the second, third, and fourth voltage amplitudes are 2.5 volts, 1.2 volts, and 3.3 volts, respectively.
5. The memory module of claim 1, further comprising:
a voltage monitor circuit configured to monitor a power input voltage received via a second portion of the plurality of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage.
6. The memory module of claim 1, the plurality of components further comprising:
one or more registers coupled to one of the first, second, third and fourth regulated voltages, the one or more registers configured to register, in response to a clock, the first plurality of address and control signals, wherein the one of the first, second, third and fourth regulated voltages is selectively switched off to turn power off to the one or more registers while one or more components of the plurality of components are powered on.
7. The memory module of claim 5, wherein the voltage monitor circuit is further configured to produce the trigger signal in response to the power input voltage having a voltage amplitude that is less than a second threshold voltage.
8. The memory module of claim 6, wherein, in response to selectively switching on the one of the first, second, third and fourth regulated voltages to the one or more registers, the one or more registers is configured to output the registered first plurality of address and control signals to the plurality of SDRAM devices.
9. The memory module of claim 5, wherein the power input voltage is coupled to the first, second, and third buck converters and the converter circuit.

10. The memory module of claim 5, wherein the first threshold voltage corresponds to a voltage level that is ten percent greater than a specified operating voltage.
11. The memory module of claim 7, wherein the second threshold voltage corresponds to a voltage level that is ten percent less than a specified operating voltage.
12. The memory module of claim 5, the plurality of components further comprising:
a logic element including a non-volatile memory, the non-volatile memory is configured to store configuration information.
13. The memory module of claim 12, wherein, in response to the trigger signal, the logic element writes information into the non-volatile memory.
14. The memory module of claim 1, the plurality of components further comprising:
a logic element including one or more integrated circuits and discrete electrical elements, the one or more integrated circuit including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information.
15. The memory module of claim 5, the plurality of components further comprising:
a non-volatile memory; and
a controller configured to receive the trigger signal, wherein, in response to the trigger signal, the controller performs a write operation to the non-volatile memory.
16. A memory module comprising:
a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively;

a converter circuit configured to reduce the pre-regulated input voltage to provide a fourth regulated voltage, wherein the first, second, third and fourth regulated voltages have first, second, third, and fourth voltage amplitudes, respectively;

a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages; and

a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage.

17. The memory module of claim 16, wherein the second and third buck converters are configured to operate as a dual buck converter.

18. The memory module of claim 16, the plurality of components further including:
a controller coupled to the voltage monitor circuit and configured to receive the signal, wherein the controller executes a write operation in response to the signal.

19. The memory module of claim 18, wherein the write operation includes writing data information into non-volatile memory.

20. The memory module of claim 16, wherein the plurality of SDRAM devices are configured to receive at least one of the first, second, third and fourth regulated voltages having a voltage amplitude of 1.8 volts.

21. The memory module of claim 16, the plurality of components further including:
at least one circuit coupled between the interface and the plurality of SDRAM devices, the at least one circuit operable to receive a first plurality of address and control signals via a second portion of the plurality of edge connections and to output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both

the second regulated voltage and the fourth regulated voltage, wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes.

22. The memory module of claim 16, the plurality of components further including:
a logic element including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information, wherein the configuration information is used to program the logic element.

23. A memory module comprising:
a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;
a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of first, second, third and fourth regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and one or more registers, the plurality of SDRAM devices coupled to the first regulated voltage, the one or more registers coupled to (i) the second regulated voltage, (ii) a portion of the plurality of edge connections, and (iii) the plurality of SDRAM devices, wherein a plurality of address and control signals are coupled to the one or more registers via the portion of the plurality of edge connections;
first, second, and third buck converters configured to provide the first, second and third regulated voltages, respectively; and
a converter circuit configured to provide the fourth regulated voltage,
wherein the second regulated voltage is configured to be selectively switched on or off to the one or more registers while at least the plurality of SDRAM devices are powered on,
wherein if the second regulated voltage is switched on while at least the plurality of SDRAM devices are powered on, the one or more registers are configured to couple the first plurality of address and control signals to the plurality of SDRAM devices, and

wherein if the second regulated voltage is switched off while the plurality of SDRAM devices are powered on, the one or more registers are configured to decouple the plurality of SDRAM devices from the first plurality of address and control signals.

24. The memory module of claim 23, further comprising:
a voltage monitor circuit configured to monitor an input voltage received from the host system via the interface, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage.
25. The memory module of claim 24, wherein the voltage monitor circuit is further configured to produce the signal in response to the input voltage having a voltage amplitude that is less than a second threshold voltage.
26. The memory module of claim 23, wherein the second and third buck converters are configured to operate as a dual buck converter.
27. The memory module of claim 24, the plurality of components further including:
a controller coupled to the voltage monitor circuit and configured to receive the signal, wherein, in response to the signal, the controller executes a write operation.
28. The memory module of claim 27, wherein the write operation includes writing data information to non-volatile memory.
29. The memory module of claim 23, wherein the plurality of SDRAM devices are configured to receive at least one of the first, second, third and fourth regulated voltages having a voltage amplitude of 1.8 volts.
30. The memory module of claim 23, wherein the first, second, and third buck converters are configured to receive a pre-regulated input voltage and to provide the first, second and third

regulated voltages, respectively, and wherein the converter circuit is configured to reduce the pre-regulated voltage input to provide the fourth regulated voltage.

ABSTRACT

In certain embodiments, a memory module includes a printed circuit board (PCB) having an interface that couples it to a host system for provision of power, data, address and control signals. First, second, and third buck converters receive a pre-regulated input voltage and produce first, second and third regulated voltages. A converter circuit reduces the pre-regulated input voltage to provide a fourth regulated voltage. Synchronous dynamic random access memory (SDRAM) devices are coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, and a voltage monitor circuit monitors an input voltage and produces a signal in response to the input voltage having a voltage amplitude that is greater than a threshold voltage.



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Application or Docket Number
17/138,766

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	30 minus 20 =	* 10
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	320
N/A	700
N/A	800
x 100 =	1000
x 480 =	0.00
	0.00
	0.00
TOTAL	2820

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*		Minus	**	=
Independent (37 CFR 1.16(h))	*		Minus	***	=
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/138,766	12/30/2020	1731	2820	0016.001000H	30	3

CONFIRMATION NO. 8804

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

FILING RECEIPT



0000000122702518

Date Mailed: 01/19/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 151145

Domestic Priority data as claimed by applicant

This application is a CON of 15/934,416 03/23/2018
which is a CON of 14/840,865 08/31/2015 PAT 9928186
which is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/14/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/138,766**

Projected Publication Date: 04/29/2021

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

501

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804
151145	7590	01/26/2021	EXAMINER	
Shami Messinger PLLC 1000 Wisconsin Ave. NW Suite 200 Washington, DC 20007			ART UNIT	PAPER NUMBER
			2135	
			NOTIFICATION DATE	DELIVERY MODE
			01/26/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

khaled@shamimessinger.com
mike@shamimessinger.com
stephanie@shamimessinger.com

<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 17/138,766	Applicant(s) Lee et al.	
	Examiner CHERYL P GIBSON BAYLOR	Art Unit OPET	AIA (FITF) Status No
1. THE REQUEST FILED <u>30 December 2020</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☒ for an original nonprovisional application (Track I). B. ☐ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to CHERYL GIBSON BAYLOR at (571)272-3213. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
/CHERYL GIBSON BAYLOR/ Paralegal Specialist, OPET			



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804
151145	7590	03/02/2021	EXAMINER	
Shami Messinger PLLC			FARROKH, HASHEM	
1000 Wisconsin Ave. NW			ART UNIT	
Suite 200			PAPER NUMBER	
Washington, DC 20007			2135	
			NOTIFICATION DATE	
			DELIVERY MODE	
			03/02/2021	
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The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

khaled@shamimessinger.com
mike@shamimessinger.com
stephanie@shamimessinger.com

Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

The instant application having application No. 17/138,766 has a total of 30 claims pending in the application; there are 3 independent claim and 27 dependent claims, all of which are ready for examination by the examiner.

ACKNOWLEDGEMENT OF REFERENCES CITED BY APPLICANT

Information Disclosure Statement

*As required by **M.P.E.P. 2001.06(b)** and **37 C.F.R. 1.98(d)**, since the instant application has been identified as a continuation application of an earlier filed application and is relied upon for an earlier filing date under **35 U.S.C. 120**, the examiner has reviewed the prior art cited in the earlier related application as required by **M.P.E.P. 707.05** and **904** and as stated in **M.P.E.P. 2001.06(b)**, no separate citation of the same prior art need be made by the applicants in the instant application.*

INFORMATION CONCERNING CLAIMS:

Claim Interpretation

Claim limitations in this application do not use the word "means" (or "step"), thus, they are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 112

1. *Claims 1-15 and 21 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.*

2. *The independent claim 1 in lines 24-25 recite limitation:*

wherein one of the second and fourth voltage amplitudes is less than the other one of the second and fourth voltage amplitudes."

It is not clear what is meant by "the other one of the second and fourth voltage amplitudes". Dependent claims 2-15 are dependent from the independent claim 1 and are rejected by virtue of their dependency from the independent claim 1. Claim 21 recites a similar limitation and rejected based on the same ground of rejection.

3. *Claims 1-15 and 21 are rejected because of lack sufficient antecedent basis.*

4. *Claim 1 recites the limitation "the other one of the second and fourth voltage amplitudes" in 25. There is insufficient antecedent basis for this limitation in the claim. Dependent claims 2-15 are dependent from the independent claim 1 and are rejected by virtue of their dependency from the independent claim 1. Claim 21 recites a similar limitation and rejected based on the same ground of rejection.*

Allowable Subject Matter

5. *Claim 16-20 and 22-30 are allowed.*

The following is an examiner's statement of reasons for allowance:

6. *The primary reasons for allowance of claims 16-20 in the instant application is the combination with the inclusion of the following limitations: "first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively; a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages; and a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage."*

7. *The primary reasons for allowance of claims 23-30 in the instant application is the combination with the inclusion of the following limitations: "a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of first, second, third and fourth regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and one or*

more registers, the plurality of SDRAM devices coupled to the first regulated voltage, the one or more registers coupled to (i) the second regulated voltage, (ii) a portion of the plurality of edge connections, and (iii) the plurality of SDRAM devices, wherein a plurality of address and control signals are coupled to the one or more registers via the portion of the plurality of edge connections; first, second, and third buck converters configured to provide the first, second and third regulated voltages, respectively”

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”

Conclusion

The prior art made of record and not relied upon are as follows:

1. *Chen et al. (US 20120271990 A1).*
2. *Bates (US 20070136523 A1).*
3. *Prete et al. (US 20080126624 A1).*

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached Monday through Friday from 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2135*

February 25, 2021

<i>Notice of References Cited</i>	Application/Control No. 17/138,766		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner HASHEM FARROKH		Art Unit 2135	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120271990-A1	10-2012	Chen; Chi-She	G06F12/00	711/103
*	B	US-20070136523-A1	06-2007	Bonella; Randy M.	G06F9/4401	711/113
*	C	US-20080126624-A1	05-2008	Prete; Edoardo	G11C7/106	710/53
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	E					
	F					
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	I					
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FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G06F13/28; G06F13/4027; G06F13/1694; G06F13/4223; G06F12/0638; G06F12/0246; G06F1/185; G06F3/0613; G06F3/0685 OR G06F3/0659; G06F2212/7208; G06F2212/205; G11C7/1072; G11C14/0018	02/24/2021	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	103	02/25/2021	HF
711	104	02/25/2021	HF
711	105	02/25/2021	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Assignee/Inventor Search	02/25/2021	HF
EAST Text Search	02/24/2021	HF
IP Search	02/25/2021	HF
Google Search	02/25/2021	HF

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<i>Search Notes</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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<i>Index of Claims</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	02/25/2021								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
	7	✓								
	8	✓								
	9	✓								
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	24	=								
	25	=								
	26	=								
	27	=								
	28	=								
	29	=								
	30	=								

Discover

PM

SDRAM and buck converters

+ 0 mods

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STIC

Visuals

Query

Results

0 Selected

Relevance

None

None

1 - 50

77,040 results

Top 1500 results

1. Voltage regulator with multi-level, multi-phase buck architecture

A voltage regulator having a multi-level, multi-phase architecture is disclosed. The circuit includes a two-level buck converter and an N-level buck converter each coupled to an output node, wherein N is an integer value of three or more. During operation, the two-level buck converter provides one...

CURRENT ASSIGNEES: APPLE INC
US10866851 | US PATENTS | 05-JAN-2021

2. Systems, devices, and methods for providing backup power to a load

Systems, devices, and methods for providing backup power to a load are disclosed. A power converter may comprise a capacitor array comprising a plurality of capacitors and configured to store a charge from an input during a charge mode of operation and provide a charge to an output during a...

CURRENT ASSIGNEES: RENESAS ELTNC AMERICA INC
WO201103354A2 | WIPO APPLICATIONS | 25-AUG-2011

3. Systems, devices, and methods for providing backup power to a load

Systems, devices, and methods for providing backup power to a load are disclosed. A power converter may comprise a capacitor array comprising a plurality of capacitors and configured to store a charge from an input during a charge mode of operation and provide a charge to an output during a...

CURRENT ASSIGNEES: RENESAS ELTNC AMERICA INC
US2010198831 | US APPLICATIONS | 18-AUG-2011

4. Systems, devices, and methods for providing backup power to a load

Systems, devices, and methods for providing backup power to a load are disclosed. A power converter may comprise a capacitor array comprising a plurality of capacitors and configured to store a charge from an input during a charge mode of operation and provide a charge to an output during a...

CURRENT ASSIGNEES: RENESAS ELTNC AMERICA INC
US8638010 | US PATENTS | 28-JAN-2014

5. DC-DC converter with dynamically adaptive load line

The invention provides a DC-DC converter with a dynamic adaptive load line. Systems, devices, and methods are described that efficiently generate a stable output for a transient load of one or more components. In

Relevance

VIEW

0 Results

531 Results

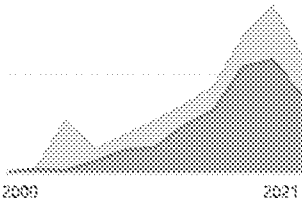
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Publication Date

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20092021

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APPLE INC (84)

RENESAS ELTNC AMERICA I... (8)

MICRON TECH INC (54)

HTC CORP (7)

TELEFON AB LM ERICSSON... (8)

QUALCOMM INC (27)

More Visuals

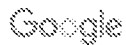
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1/1



pcb and sdram and buck converter



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Webinar: EMC Optimized Buck Converter Layout

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Guide to Choosing the Best DC-to-DC Converter for Your ...

Mar 14, 2019 — If both cases are required, a BUCKBOOST converter is used ... under a no current load condition (e.g. CMOS RAM keepalive applications). ... Also, check the package type and size and PCB design constraints provided by the ...

[ieeexplore.ieee.org](#) › [document](#)

Addressing PCB effects in the design of a buck converter ...

Addressing PCB effects in the design of a buck converter. Abstract: ... Ihsan Erdin; Ram Achar ... such as open-loop, input-to-output and control-to-output transfer functions including distributed circuit parameters that account for PCB parasitics.

[www.eetimes.com](#) › [proper-board-layout-makes-switch...](#)

Proper board layout makes switchers work right the first time ...

These are the critical traces for any switcher PCB layout ... For example, a buck-boost IC construction will usually be different from that of a buck IC, even though ...

[en.wikipedia.org](#) › [wiki](#) › [Buck_converter](#)

Buck converter - Wikipedia

A buck converter (step-down converter) is a DC-to-DC power converter which steps down ... (bulk) supply voltage (often 12 V) down to lower voltages needed by USB, DRAM and the CPU (1.8 V or less). ... (conduction) losses in the wires or PCB traces, as well as in the switches and inductor, as in any electrical circuit.

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Switching Regulator Series. PCB Layout Techniques of Buck Converter. PCB layout design for switching power supply IC is as important as the circuit design.

Missing: sdram | Must include: sdram

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Design Considerations for EMI Reduction in Automotive ...

May 26, 2017 — The Buck topology DC/DC converter constitutes a circuit with the input ... the components selection, and the PCB layout as the method that does not ... thereof, WICED, PSoC, CapSense, EZ-USB, F-RAM, and Traveo are.

onlinelibrary.wiley.com/doi/full/10.1002/etn.2018-0551

Efficiency improvement of a DC/DC converter using LTCC ...

Jun 21, 2019 — Efficiency improvement of a DC/DC converter using LTCC substrate ... The power conversion efficiencies of the LTCC- and PCB-based synchronous ... bioMEMS/biochips, surface micromachining, DRAM metallization, and ...

www.analog.com/technical-documentation/data-sheets/PDF/

Wide VIN, Multi-Output DC/DC Converter and PowerPath ...

efficiency switching DC/DC converters which seamlessly transition from battery to ... 125°C, $\theta_{JA} = 37^{\circ}\text{C/W}$. EXPOSED PAD (PIN 25) IS GND, MUST BE SOLDERED TO PCB ... such as a 1.8V rail for SDRAM and a 1.2V rail to supply the system ...

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Design Procedure for AC/DC Circuits Switching Regulator Basics and More. Switching Regulator Basics by ROHM, AC/DC Converter Designs and more. Highlights: Engineering Tips Available, Design Support Tools Available.

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S347	132	(buck\$1converter or (buck near5 convert\$3)) with ((regulated adj voltage) near2 (amplitude or value or potential))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/23 15:55
S348	0	S347 and SDRAM	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/23 16:00
S349	0	((first near5 (buck adj converter)) with (second near5 (buck adj converter)) with (thirsd near5 (buck adj converter))) or ((plurality or multiple) near5 (buck adj converters))) with (supp\$3 or produc\$3 or generat\$3 or creat\$3) with ((first near5 second near5 third) near5 (regulated adj voltages))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 14:27
S350	2	(buck adj converters) with (regulated adj voltages)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 14:32
S351	0	((printed\$1circuit\$1board or (printed adj circuit adj board)) or PCB) with ((plurality or multiple) near5 ((synchronous adj dynamic adj random adj access adj memor\$3) or SDRAM)) with (regulated adj voltages)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 14:55
S352	0	(monitor\$3 or detect\$3) with (input adj voltage) with receiv\$3 with (edge adj connections)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:00

S353	0	(monitor\$3 or detect\$3) with (input adj voltage) with receiv\$3 with (edge adj connections)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 15:00
S354	33	(monitor\$3 or detect\$3) with (input adj voltage) with receiv\$3 with connections	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 15:03
S355	0	((first near5 (buck adj converter)) same (second near5 (buck adj converter)) same (thirsd near5 (buck adj converter))) or ((plurality or multiple) near5 (buck adj converters))) same (supp\$3 or produc\$3 or generat\$3 or creat\$3) same ((first near5 second near5 third) near5 (regulated adj voltages))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:06
S356	0	((first near5 (buck adj converter)) same (second near5 (buck adj converter)) same (thirsd near5 (buck adj converter))) or ((plurality or multiple) near5 (buck adj converters))) same (supp\$3 or produc\$3 or generat\$3 or creat\$3) same ((first near5 second near5 third) near5 (regulated adj voltages))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 15:07
S357	0	(monitor\$3 or detect\$3) with (input adj voltage) with receiv\$3 with (edge adj connect\$4)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:08
S358	0	((printed\$1circuit\$1board or (printed adj circuit adj board)) or PCB) with ((plurality or muliple) near5 ((synchronous adj dynamic adj random adj access adj memor\$3) or SDRAM)) with voltages	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 15:11
S359	0	((printed\$1circuit\$1board or (printed adj circuit adj board)) or PCB) same ((plurality or muliple) near5 ((synchronous adj dynamic adj	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	OFF	2021/02/24 15:12

		random adj access adj memor\$3) or SDRAM)) same voltages	JPO; DERWENT; IBM_TDB			
S360	32	((printed\$1circuit\$1board or (printed adj circuit adj board)) or PCB) same ((plurality or multiple) near5 ((synchronous adj dynamic adj random adj access adj memor\$3) or SDRAM))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2021/02/24 15:13
S361	128,557	(G06F13/28 OR G06F13/4027 OR G06F13/1694 OR G06F13/4223 OR G06F12/0638 OR G06F12/0246 OR G06F1/185 OR G06F3/0613 OR G06F3/0685 OR G06F3/0659 OR G06F2212/7208 OR G06F2212/205 OR G11C7/1072 OR G11C14/0018).cpc. OR 711/100.CCLS OR G06F3/06.icpr	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:39
S362	14	S361 and (SDRAM and (buck near5 converter))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:41
S363	0	(SDRAM and (buck near5 converter)).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/24 15:51
S364	151	((transfer\$3 or cop\$4 or transmi\$5 or retriev\$3 or read\$3) near5 DRAM near5 (flash or non\$1volatile)).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S365	1,328	Lee-Hyun.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S366	76	Chen-Chi-She.in.	US-PGPUB; USPAT; USOCR;	OR	ON	2021/02/25 11:58

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S367	72	Solomon-Jeffrey-C.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S368	33	Milton-Scott-H.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S369	36	Bhakta-Jayesh.in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S370	0	S364 and (S365 or S366 or S367 or S368 or S369)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 11:58
S371	11,623	(transfer\$4 or cop\$4 or transmi\$5 or mov\$3) with data with (non\$1volatile or flash or SSD or NVM) with (volatile or ?RAM or ?DRAM memory).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:01
S372	215	NETLIST.as.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:01
S373	21	S371 and S372	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	2021/02/25 12:01

			JPO; DERWENT; IBM_TDB			
S374	128,598	(G06F13/28 OR G06F13/4027 OR G06F13/1694 OR G06F13/4223 OR G06F12/0638 OR G06F12/0246 OR G06F1/185 OR G06F3/0613 OR G06F3/0685 OR G06F3/0659 OR G06F2212/7208 OR G06F2212/205 OR G11C7/1072 OR G11C14/0018).cpc. OR 711/100.CCLS OR G06F3/06.icpr	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:04
S375	14	S374 and (SDRAM and (buck near5 converter))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:04
S376	0	((synchronous adj dynamic adj random adj access adj memor\$3) or SDRAM) and (buck near5 converter)).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:09
S377	702	((synchronous adj dynamic adj random adj access adj memor\$3) or SDRAM) and (buck near5 converter))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:10
S378	14	S374 and S377	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:12
S379	12	S372 and S378	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/02/25 12:12

2/25/2021 1:49:43 PM

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Electronic Acknowledgement Receipt

EFS ID:	42152924
Application Number:	17138766
International Application Number:	
Confirmation Number:	8804
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	151145
Filer:	Khaled Shami/Stephanie Semler
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000H
Receipt Date:	11-MAR-2021
Filing Date:	30-DEC-2020
Time Stamp:	19:29:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	IDS3_NPL3_Ex_Stanford's_Materials_Science_department_website_PTAB_IPR2017_00692_2011.pdf	1098382 dd0a1972017d81f2b4ff992fb9f259aa0eda73e1e	no	2

Warnings:

Information:					
2	Non Patent Literature	IDS3_NPL6_Ex_MS_Computer_Dictionary.pdf	1551595	no	9
			a3e376f0fe699011d8aa8ffbae8c8533fc2800ae		
Warnings:					
Information:					
3	Non Patent Literature	IDS3_NPL33_ORDER_Grantlyng_Joint_Stipulation_NDCA_413cv05889_382.pdf	148436	no	2
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4	Non Patent Literature	IDS3_NPL34_Patent_Owners_Demonstratives_PTAB_IPR2017_00587_2023.pdf	5671795	no	57
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5	Non Patent Literature	IDS3_NPL35_Patent_Owners_Listing_ofNew_Argumentsand_PTAB_IPR2017_00587_23.pdf	111913	no	6
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6	Non Patent Literature	IDS3_NPL36_PATENT_OWENERS_OPOSITIONTOPETITIONER_PTAB_IPR2017_00692_20.pdf	67333	no	11
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7	Non Patent Literature	IDS3_NPL37_Patent_Owners_Preliminary_Response_PTAB_IPR2014_01371_10.pdf	1468755	no	66
			212cb60a34ee7a642f08a25c7284cefef72e85cd		
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8	Non Patent Literature	IDS3_NPL38_Patent_Owners_Preliminary_Response_PTAB_IPR2014_00994_7.pdf	1220159	no	60
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9	Non Patent Literature	IDS3_NPL39_Patent_Owners_Preliminary_Response_PTAB_IPR2014_00982_7.pdf	1257355	no	57
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11	Non Patent Literature	IDS3_NPL41_PATENT_OWENERS_PRELIMINARY_RESPONSE_PTAB_IPR2017_00692_6.pdf	715159	no	48
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12	Non Patent Literature	IDS3_NPL42_Patent_Owners_Preliminary_Response_PTAB_IPR2014_01370_11.pdf	1091571	no	66
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13	Non Patent Literature	IDS3_NPL43_Patent_Owners_Response_PTAB_IPR2017_00587_12.pdf	1625403	no	80
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14	Non Patent Literature	IDS3_NPL44_PATENT_OWENERS_RESPONSE_PTAB_IPR2017_0092_12.pdf	1227588	no	77
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15	Non Patent Literature	IDS3_NPL46_PETITION_FOR_INTER_PARTES_REVIEWPAT8301833_PTAB_IPR2014_01370_1.pdf	682144	no	68
			04a89f746f6a10c8f84876f858271e45036bb9ca		
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16	Non Patent Literature	IDS4_NPL7_Petition_for_Inter-Partes_Review_PTAB_IPR2014_00982_1.pdf	675276	no	67
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17	Non Patent Literature	IDS4_NPL8_Petition_for_Inter-Partes_Review_ofPat8671243_PTAB_IPR2017_00587_1.pdf	1245700	no	82
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18	Non Patent Literature	IDS4_NPL9_PETITION_FOR_INTER_PARTES_REVIEW_PAT8301833_PTAB_IPR2014_00994.pdf	749563	no	69
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19	Non Patent Literature	IDS4_NPL12_Petitioners_Demonstratives_PTAB_IPR2017_00587_1037.pdf	15554459	no	89
			72a5a4810671fc80c9e07759f81e6ac10ec818be		
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20	Non Patent Literature	IDS4_NPL14_Petitioners_Reply_to_Patent_Owners_Response_PTAB_IPR2017_00587_16.pdf	551197	no	38
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21	Non Patent Literature	IDS4_NPL15_PETITIONERS_REPLY_PTAB_IPR2017_00692_15.pdf	430614	no	35
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22	Non Patent Literature	IDS4_NPL20_Ex_US20070136523A1_ProvisionalSpecification_60749267.pdf	8203054	no	53
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23	Non Patent Literature	IDS4_NPL23_Ex_1003_OA_Requirement_for_Restriction.pdf	524947	no	7
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24	Non Patent Literature	IDS4_NPL26_Search_Report_Updated_Nov242020_G123P120S.pdf	3229408	no	179
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25	Non Patent Literature	IDS4_NPL27_Search_Report_Nov202020_G123P120S.pdf	663014	no	38
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26	Non Patent Literature	IDS4_NPL28_SECOND_AMENDED_ANSWER_AND_COUNTERCLAIMS_NDCA413cv03901_73.pdf	542838	no	21
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27	Non Patent Literature	IDS4_NPL30_SUPPLEMENTAL_DECLARATION_OF DANIEL_E_ALBERTI.pdf	117001	no	3
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28	Non Patent Literature	IDS4_NPL31_SUPPLEMENTAL_DECLARATION_OF RONALD_H_SPUHLER_Case413cv05889YGRDocument305-1.pdf	43325	no	2
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29	Non Patent Literature	IDS4_NPL32_SUPPLEMENTAL_DECLARATION_OF STEVEN_J_CORRINSUPPORTOFSANDISK_Case413cv058889.pdf	122342	no	3
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30	Non Patent Literature	IDS4_NPL17_Ex_Prosecution_History_8301833App12240916_1.pdf	4146193	no	320
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31	Non Patent Literature	IDS2_NPL44_JEDEC_Definition_of_DIMM_Exhibit_1029ofBakerDepo12182017_PTAB_IPR2017_00587_1029.pdf	1301305	no	2
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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17138766
	Filing Date		2020-12-30
	First Named Inventor	Hyun Lee	
	Art Unit	1731	
	Examiner Name		
	Attorney Docket Number	0016.001000H	

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

1	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
2	Petition for Inter Partes Review of U.S. Pat. No. 8,671,243, filed May 29, 2013.
3	Petition for Inter Partes Review of U.S. Pat. No. 8,874,831, filed Jul. 26, 2012.
4	Wong, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.
5	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
6	US Patent Application No. 60/912,321, Case No. No. IPR2017-00692-1007 (PTAB), filed April 17, 2017, 42 pages.
7	Petition for Inter Partes Review of US Patent 8,516,187, IPR2014-00982 (PTAB), dated June 19, 2014, 67 pages.
8	Petition for Inter Partes Review of US Patent No. 8,671,243, Case No. IPR2017-00587-1(PTAB), May 29, 2013, 82 pages.
9	Petition for Inter Partes Review, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), filed June 20, 2014, 69 pages.
10	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1115.
11	Videotaped Deposition of Russel Jacob Baker, Patent No. 8,671,243, Case Nos. IPR2017-00587-1030 (PTAB), filed December 18, 2017, 268 pages.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

12	Petitioners Demonstratives, SK hynix Inc., et al., v. Netlist, Inc., U.S. Patent No. 8,671,243, IPR2017-00587-1037, 89 pages.
13	Petitioners' Reply in Support of Its Motion to Exclude, U.S. Patent No. 8,874,831, Case No. IPR2017-00692, filed April 9, 2018, 8 pages.
14	Petitioners' Reply to Patent Owner's Response, Inter Partes Review No. IPR2017-00587-16, Patent No. 8,671,243, filed Jan. 12, 2018, 38 pages.
15	Petitioners' Reply, U.S. Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), Paper No. 15, filed March 2, 2018, 35 pages.
16	Petitioners' Request for Hearing, U.S. Patent No. 8,301,833, Case No. Case No. IPR2017-00649-2020 (PTAB), Paper No. 8, entered Aug, 23, 2017, 18 pages.
17	Prosecution History, Application No. 12,240,916, US Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date April 13, 2009, 320 pages.
18	Prosecution History, Application No. 13,559,476, US Patent No. 8,301,833, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1002 (PTAB), received date Sep. 5, 2014, 312 pages.
19	Prosecution History, Provisional Application No. 60/941,586, SK hynix Inc v Netlist Inc, Case No. IPR2017-00649-1005 (PTAB), received date Oct. 15, 2012, 23 pages.
20	Provisional Application for Advance Dynamic Disk Memory Module, Specification, SK hynix Inc., V. Netlist, Inc., Case No. IPR2017-00649-1006 (PTAB), dated Dec. 8, 2005, 53 pages.
21	Provisional Application No. 60/912,321 filed Apr. 17, 2007.
22	Provisional Application No. 60/941,586 filed Jun. 1, 2007.

**INFORMATION DISCLOSURE
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Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

23	Requirement for Restriction Election, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994-1003 (PTAB), dated March 31, 2019, 7 pages.
24	Restriction Requirement in U.S. Appl. No. 12/240,916, dated Mar. 31, 2011.
25	Search on Amazon Website for "memory module", downloaded Oct. 12, 2017, 6 pages.
26	Search Report (Updated) Prior Art Search for U.S. Patent No. 8,301,833, Global Patent Solutions, Nov. 24, 2020, 179 pages.
27	Search Report Prior Art Search for U.S. Patent No. 8,301,833, Global Patent Solutions, Nov. 2, 2020, 38 pages.
28	Second Amended Answer and Counterclaims to Plaintiff's Complaint for Declaratory Judgment, Diablo Technologies, Inc. v. Netlist, Inc., Case No. 4:13-CV-03901 (NDCA), YGR, filed February 17, 2014, 21 pages.
29	Smart Storage Systems, Inc's Invalidity Contentions, Case No. 4:13-cv-05889-YGR, dated Jun. 6, 2014.
30	Supplemental Declaration of Daniel E. Alberti in Response to Court Order [Docket No. 361] in Support of Motion to Withdraw as Counsel of Record, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 362, filed September 5, 2018, 13 pages.
31	Supplemental Declaration of Ronald H. Spuhler, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-1, filed March 10, 2015, 2 pages.
32	Supplemental Declaration of Steven J. Corr, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 309, filed March 17, 2015, 3 pages.
33	Switches - DT Online, Switches, Exhibit 1022, Case No. IPR2017-00587, accessed October 13, 2017, 3 pages.

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34	J.S. Appl. No. 60/912,321, filed Apr. 17, 2007.
35	J.S. Office Action in U.S. Appl. No. 13/536,173, mailed on Apr. 15, 2013, pp. 1-10.
36	UPDATED JOINT CASE MANAGEMENT STATEMENT, U.S. DISTRICT COURT NORTHERN DISTRICT OF CALIFORNIA, Case No. 4:13-cv-03916-YGR, Document 53, filed January 6, 2014.
37	US Patent Application No. 12/240,916, Case No. IPR2017-00692-2018 (PTAB), dated Sep. 29, 2008, 52 pages.

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Attorney Docket Number	0016.001000H

1	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC Solid State Tech. Assoc., Mar. 2007, 9 pages.
2	Joint Status Report Regarding Inter Partes Review, Netlist v. Smart Storage Systems, Inc. et. al., Case No. 4:13-CV-05889-YGR (NDCA), filed August 17, 2018, 4 pages.
3	Material Science & Engineering, Department of Material Science and Engineering, Stanford University, Exhibit 2011, Case No. IPR2017-00692, 2 pages.
4	Merriam-Webster's Collegiate Dictionary, Eleventh Ed., Merriam Webster Corporation, Springfield, MA, 2003, 7 pages.
5	Microsoft Computer Dictionary Fifth Edition, 2002, 3 pages.
6	Microsoft Computer Dictionary Fifth Edition, 2002, 9 pages.
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8	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
9	Mutnuary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.
10	Notice of Allowance in U.S. Appl. No. 12/240,916, dated Sep. 17, 2012.
11	Notice of Allowance in U.S. Appl. No. 13/536,173, dated Jul. 2, 2013.

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12	Notice of Allowance in U.S. Appl. No. 13/559,476, dated May 6, 2014.
13	Notice of Allowance in U.S. Appl. No. 13/559,476, dated Sep. 29, 2014.
14	Notice of Allowance in U.S. Appl. No. 13/905,048, dated Dec. 19, 2013, 8 pages.
15	Notice of Allowance in U.S. Appl. No. 13/905,053, dated Dec. 11, 2013.
16	Notice of Allowance in U.S. Appl. No. 14/173,219 dated Jul. 7, 2014.
17	Notice of Allowance in U.S. Appl. No. 14/489,269, dated Oct. 8, 2015.
18	Notice of Allowance, SanDisk Corporation v. Netlist, Application No. 13/536,173, IPR2014-00982 (PTAB), dated July 2, 2013, 8 pages.
19	Office Action dated Aug. 19, 2016 of the Chinese Patent Application No. 201280047758.X, 9 pages.
20	Office Action in U.S. Appl. No. 12/240,916, mailed on Apr. 3, 2012.
21	Office Action in U.S. Appl. No. 13/536,176, dated on Apr. 15, 2013.
22	Office Action in U.S. Appl. No. 13/625,563, dated Aug. 5, 2013.

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23	Office Action in U.S. Appl. No. 13/625,563, dated May 9, 2014.
24	Office Action in U.S. Appl. No. 13/905,048, dated Aug. 1, 2013.
25	Office Action in U.S. Appl. No. 13/905,053, dated Aug. 1, 2013.
26	Office Action in U.S. Appl. No. 14/173,219, dated Mar. 13, 2014.
27	Office Action in U.S. Appl. No. 14/302,292, dated Dec. 21, 2015.
28	Office Action in U.S. Application No. 12/240,916, dated Feb. 1, 2012, 14 pages.
29	Office Action in U.S. Application No. 14/173,242, dated March 14, 2014, 7 pages.
30	Office Action, Application 12/240,916, IPR2014-0099-1005 (PTAB), dated July 29, 2011, 8 pages.
31	Office Action, Application 12/240,916, IPR2014-0099-1010 (PTAB), dated April 3, 2012, 11 pages.
32	Office Action, Application No. 13/536,173, IPR2014-00982 (PTAB), dated April 15, 2013, 9 pages.
33	Order Granting Joint Stipulation of Dismissal, Netlist v. Smart Storage Systems, Inc. et. al., Case No. 4:13-CV-05889-YGR (NDCA), filed May 15, 2020, 2 pages.

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34	Patent Owner's Demonstratives, Patent Number: 8,671,243, Case No. IPR 2017-00587-2023, (PTAB), 57 pages.
35	Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, Patent No. 8,671,243, Case No. IPR2017-00587 (PTAB), filed Jan. 29, 2018, 6 pages.
36	Patent Owner's Opposition to Petitioners' Motion to Exclude, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed April 2, 2018, 2017, 11 pages.
37	Patent Owner's Preliminary Response, Patent Number: 8,516,187, Case IPR2014-01371 (PTAB), filed Dec. 16, 2014, 66 pages.
38	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated October 2, 2014, 60 pages.
39	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Patent Number: 8,516,187, IPR2014-00982 (PTAB), dated September 26, 2014, 57 pages.
40	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,301,833, Case No. IPR2017-00649 (PTAB), filed May 1, 2017, 67 pages.
41	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed May 1, 2017, 48 pages.
42	Patent Owner's Preliminary Response, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, Case IPR2014-01370 (PTAB), filed Dec. 16, 2014, 66 pages.
43	Patent Owner's Response, Patent No. 8,671,243, Case No. IPR2017-00587-12 (PTAB), filed Oct. 13, 2017, 80 pages.
44	Patent Owner's Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692-12 (PTAB), filed Nov. 10, 2017, 77 pages.

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45	Patterson et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998), 71 pages.
46	Petition for Inter Partes Review of Claims 1-30 of US.PATENT No. 8,301,833, IPR2014-01370 (PTAB), filed August 22, 2014, 68 pages.
47	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
48	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
49	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833, filed Sep. 29, 2008.
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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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Attorney Docket Number	0016.001000H

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

- ☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 8804

APPLICATION NO: 17/138,766

ART UNIT: 2135

FILING DATE: 12/30/20

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Commissioner:

In response to the non-final Office Action issued March 2, 2021, Applicant submits the following Amendment and Remarks.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 10.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

LISTING OF CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the application.

1. (Currently amended) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;

a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;

a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;

a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and

a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:

a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and

at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth

regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than the ~~other~~ a second one of the second and fourth voltage amplitudes.

2. (Original) The memory module of claim 1, wherein the first and third buck converters are further configured to operate as a dual buck converter.

3. (Original) The memory module of claim 1, wherein the first voltage amplitude is 1.8 volts.

4. (Original) The memory module of claim 1, wherein the second, third, and fourth voltage amplitudes are 2.5 volts, 1.2 volts, and 3.3 volts, respectively.

5. (Original) The memory module of claim 1, further comprising:

a voltage monitor circuit configured to monitor a power input voltage received via a second portion of the plurality of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage.

6. (Original) The memory module of claim 1, the plurality of components further comprising:

one or more registers coupled to one of the first, second, third and fourth regulated voltages, the one or more registers configured to register, in response to a clock, the first plurality of address and control signals, wherein the one of the first, second, third and fourth regulated voltages is selectively switched off to turn power off to the one or more registers while one or more components of the plurality of components are powered on.

7. (Original) The memory module of claim 5, wherein the voltage monitor circuit is further configured to produce the trigger signal in response to the power input voltage having a voltage amplitude that is less than a second threshold voltage.

8. (Original) The memory module of claim 6, wherein, in response to selectively switching on the one of the first, second, third and fourth regulated voltages to the one or more registers, the one or more registers is configured to output the registered first plurality of address and control signals to the plurality of SDRAM devices.

9. (Original) The memory module of claim 5, wherein the power input voltage is coupled to the first, second, and third buck converters and the converter circuit.

10. (Original) The memory module of claim 5, wherein the first threshold voltage corresponds to a voltage level that is ten percent greater than a specified operating voltage.

11. (Original) The memory module of claim 7, wherein the second threshold voltage corresponds to a voltage level that is ten percent less than a specified operating voltage.

12. (Original) The memory module of claim 5, the plurality of components further comprising:

a logic element including a non-volatile memory, the non-volatile memory is configured to store configuration information.

13. (Original) The memory module of claim 12, wherein, in response to the trigger signal, the logic element writes information into the non-volatile memory.

14. (Original) The memory module of claim 1, the plurality of components further comprising:

a logic element including one or more integrated circuits and discrete electrical elements, the one or more integrated circuit including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information.

15. (Original) The memory module of claim 5, the plurality of components further comprising:

a non-volatile memory; and

a controller configured to receive the trigger signal, wherein, in response to the trigger signal, the controller performs a write operation to the non-volatile memory.

16. (Original) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively;

a converter circuit configured to reduce the pre-regulated input voltage to provide a fourth regulated voltage, wherein the first, second, third and fourth regulated voltages have first, second, third, and fourth voltage amplitudes, respectively;

a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages; and

a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage.

17. (Original) The memory module of claim 16, wherein the second and third buck converters are configured to operate as a dual buck converter.

18. (Original) The memory module of claim 16, the plurality of components further including:

a controller coupled to the voltage monitor circuit and configured to receive the signal, wherein the controller executes a write operation in response to the signal.

19. (Original) The memory module of claim 18, wherein the write operation includes writing data information into non-volatile memory.

20. (Original) The memory module of claim 16, wherein the plurality of SDRAM devices are configured to receive at least one of the first, second, third and fourth regulated voltages having a voltage amplitude of 1.8 volts.

21. (Currently amended) The memory module of claim 16, the plurality of

components further including:

at least one circuit coupled between the interface and the plurality of SDRAM devices, the at least one circuit operable to receive a first plurality of address and control signals via a second portion of the plurality of edge connections and to output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than ~~the other~~ a second one of the second and fourth voltage amplitudes.

22. (Original) The memory module of claim 16, the plurality of components further including:

a logic element including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information, wherein the configuration information is used to program the logic element.

23. (Original) A memory module comprising:

a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of first, second, third and fourth regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and one or more registers, the plurality of SDRAM devices coupled to the first regulated voltage, the one or more registers coupled to (i) the second regulated voltage, (ii) a portion of the plurality of edge connections, and (iii) the plurality of SDRAM devices, wherein a plurality of address and control signals are coupled to the one or

more registers via the portion of the plurality of edge connections;

first, second, and third buck converters configured to provide the first, second and third regulated voltages, respectively; and

a converter circuit configured to provide the fourth regulated voltage,

wherein the second regulated voltage is configured to be selectively switched on or off to the one or more registers while at least the plurality of SDRAM devices are powered on,

wherein if the second regulated voltage is switched on while at least the plurality of SDRAM devices are powered on, the one or more registers are configured to couple the first plurality of address and control signals to the plurality of SDRAM devices, and

wherein if the second regulated voltage is switched off while the plurality of SDRAM devices are powered on, the one or more registers are configured to decouple the plurality of SDRAM devices from the first plurality of address and control signals.

24. (Original) The memory module of claim 23, further comprising:

a voltage monitor circuit configured to monitor an input voltage received from the host system via the interface, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage.

25. (Original) The memory module of claim 24, wherein the voltage monitor circuit is further configured to produce the signal in response to the input voltage having a voltage amplitude that is less than a second threshold voltage.

26. (Original) The memory module of claim 23, wherein the second and third buck converters are configured to operate as a dual buck converter.

27. (Original) The memory module of claim 24, the plurality of components further including:

a controller coupled to the voltage monitor circuit and configured to receive the signal, wherein, in response to the signal, the controller executes a write operation.

28. (Original) The memory module of claim 27, wherein the write operation includes writing data information to non-volatile memory.

29. (Original) The memory module of claim 23, wherein the plurality of SDRAM devices are configured to receive at least one of the first, second, third and fourth regulated voltages having a voltage amplitude of 1.8 volts.

30. (Original) The memory module of claim 23, wherein the first, second, and third buck converters are configured to receive a pre-regulated input voltage and to provide the first, second and third regulated voltages, respectively, and wherein the converter circuit is configured to reduce the pre-regulated voltage input to provide the fourth regulated voltage.

REMARKS

The non-final Office Action issued March 2, 2021, has been carefully considered.

Reconsideration in view of the following remarks is respectfully requested.

Upon entry of the foregoing amendment, claims 1-30 are pending in this application, with claims 1, 16, and 23 being the independent claims.

Subject Matter Indicated Allowed or Allowable

Applicant gratefully acknowledges the indication of allowance of claims 16-20 and 22-30.

Rejection(s) Under 35 U.S.C. § 112(b) (or Pre-AIA 35 U.S.C. § 112, Second Paragraph)

Claims 1-15 and 21 stand rejected under 35 U.S.C. § 112(b) (or pre-AIA 35 U.S.C. § 112, second paragraph), as allegedly being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or, for pre-AIA, the Applicant regards as the invention.

Claims 1 and 21 have been amended to recite “a first one of the second and fourth voltage amplitudes is less than ~~the other~~ a second one of the second and fourth voltage amplitudes.” In this manner the term “other,” which seems to be the source of the alleged indefiniteness, is eliminated. While applicant believes that the usage of the term “other” as it appeared in the claim is grammatically correct, applicant has elected to remove it in the interest of advancing prosecution and gaining allowance.

It is respectfully submitted that claims 1 and 21 are now in compliance with 35 U.S.C. § 112(b) (or pre-AIA 35 U.S.C. § 112, second paragraph), and the rejection claims 1-15 and 21 should be withdrawn.

Amendment to Claims

Claims 1 and 21 have been amended for improved clarity and grammatical accuracy. The amendment does not raise any issues beyond those already considered by the Examiner. No amendment made is related to the statutory requirements of patentability unless expressly stated herein. No amendment is made for the purpose of narrowing the scope of any claim, unless Applicant argues herein that such amendment is made to distinguish over a particular identified reference or combination of references. Any remarks made herein with respect to a given claim or amendment is intended only in the context of that specific claim or amendment, and should not be applied to other claims, amendments or aspects of the claimed arrangements.

Conclusion

In view of the preceding discussion, Applicant respectfully urges that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution of the present application, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 60-2034.

Respectfully submitted,

Dated: March 11, 2021

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

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202-516-6900

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: 8804

SERIAL NO.: 17/138,766

ART UNIT: 2135

FILING DATE: 12-30-2020

EXAMINER: FARROKH, HASHEM

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Commissioner:

Notice of Related Litigation and/or Proceedings

Pursuant to MPEP 2001.06(c), Applicant notifies the Patent and Trademark Office of litigation and proceedings involving U.S. Patents related to the current patent application, the subject matter of which may be related to the present patent application.

Appendix A lists results of a Cases Search. The Cases Search was performed on January 29, 2021, on the Docket Navigator™ search tool for patent cases involving commonly-owned U.S. Patents No. 8,301,833, 8,516,187, 8,671,243, and 8,874,831. The Cases Search covered forums including U.S. federal courts, ITC, and PTAB. The Cases Search results presented includes information output by the search tool for the following fields: Case, Case Filing Date, Case Terminated, and Case Status.

Notification of the cases and proceedings identified in the attached Notice in no way operates as an admission that documents cited or filed in those cases and proceedings are material to the

patentability of the present application or that the subject matter of this application is related to the specific patents subjected to suits and/or proceedings listed in the Notice.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 60-2034.

Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: March 11, 2021

/Khaled Shami/
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Appendix A

Cases

11 Results

Patents: 8301833, 8516187, 8671243, 8874831

Case

Case F...

▼

Case Ter...

Case ...

Court

Case Name

Case Nu...

SK hynix Inc. et al v. Netlist, Inc.
IPR2017-00692 (PTAB)

Jan. 17,
2017

Jul. 05,
2018

Termin
ated

PTAB

SK hynix Inc. et al v. Netlist, Inc.

IPR2017-
00692

PARTIES

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Party Role

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SK Hynix Memory Solutions Inc.

Party Role

PETITIONER

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Joseph A Micallef, Samuel A Dillon

Number

8874831

Inventor

Lee

Name

Flash-DRAM hybrid memory module

SK hynix Inc. v. Netlist, Inc.
IPR2017-00649 (PTAB)

Jan. 13,
2017

May. 29,
2018

Termin
ated

PTAB

SK hynix Inc. v. Netlist, Inc.

IPR2017-
00649

PARTIES

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Party Role

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Number

8301833

Inventor

Chen

Name

Non-volatile memory module

SK hynix Inc. et al v. Netlist, Inc.
IPR2017-00587 (PTAB)

Jan. 06,
2017

Jun. 20,
2018

Termin
ated

PTAB

SK hynix Inc. et al v. Netlist, Inc.

IPR2017-
00587

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Number

8671243

Inventor

Chen

Name

Isolation switching for backup memory

Appendix A

	SMART Modular Technologies Inc. v. Netlist, Inc. IPR2014-01370 (PTAB)	Aug. 23, 2014	Apr. 30, 2015	Termin ated	PTAB	SMART Modular Technologies Inc. v. Netlist, Inc.	IPR2014- 01370
PARTIES							
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Netlist, Inc.	PATENT OWNER	McAndrews Held & Malloy	Gregory C Schodde, Ronald H Spuhler, Scott P McBride, Thomas J Wimbiscus, Wayne H Bradley				
Smart Modular Technologies, Inc.	PETITIONER	King & Spalding	Michael F Heafey, Sanjiva K Reddy				
PATENTS							
<u>Number</u>	<u>Inventor</u>	<u>Name</u>					
8301833	Chen	Non-volatile memory module					

	SMART Modular Technologies Inc. v. Netlist, Inc. IPR2014-01371 (PTAB)	Aug. 23, 2014	Apr. 30, 2015	Termin ated	PTAB	SMART Modular Technologies Inc. v. Netlist, Inc.	IPR2014- 01371
PARTIES							
<u>Party</u>	<u>Party Role</u>	<u>Firm</u>	<u>Attorneys</u>				
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Smart Modular Technologies, Inc.	PETITIONER	King & Spalding	Michael F Heafey, Sanjiva K Reddy				
PATENTS							
<u>Number</u>	<u>Inventor</u>	<u>Name</u>					
8516187	Chen	Data transfer scheme for non-volatile memory module					

	SanDisk Corporation v. Netlist, Inc. IPR2014-00994 (PTAB)	Jun. 20, 2014	Jun. 02, 2015	Termin ated	PTAB	SanDisk Corporation v. Netlist, Inc.	IPR2014- 00994
PARTIES							
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PATENTS							
<u>Number</u>	<u>Inventor</u>	<u>Name</u>					
8301833	Chen	Non-volatile memory module					

	SanDisk Corporation v. Netlist, Inc. IPR2014-00982 (PTAB)	Jun. 19, 2014	Jun. 03, 2015	Termin ated	PTAB	SanDisk Corporation v. Netlist, Inc.	IPR2014- 00982
PARTIES							
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SanDisk LLC f/k/a SanDisk Corp.	PETITIONER	Jones Day	David B Cochran, Joseph M Sauer, Joshua R Nightingale				
PATENTS							
<u>Number</u>	<u>Inventor</u>	<u>Name</u>					
8516187	Chen	Data transfer scheme for non-volatile memory module					

 Netlist, Inc. v. Smart Modular Technologies, Inc. et al 4-13-cv-05889 (NDCa)	Dec. 19, 2013	May. 19, 2020	Termin ated	NDCa	Netlist, Inc. v. Smart Modular Technologies, Inc. et al	4-13-cv- 05889
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PATENTS

<u>Number</u>	<u>Inventor</u>	<u>Name</u>
7881150	Solomon	Circuit providing load isolation and memory domain translation for memory module
8001434	Lee	Memory board with self-testing capability
8081536	Solomon	Circuit for memory module
8301833	Chen	Non-volatile memory module
8359501	Lee	Memory board with self-testing capability
8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module
8516187	Chen	Data transfer scheme for non-volatile memory module

 Diablo Technologies, Inc. v. Netlist, Inc. 4-13-cv-03901 (NDCA)	Aug. 23, 2013	Apr. 08, 2014	Termin ated	NDCA	Diablo Technologies, Inc. v. Netlist, Inc.	4-13-cv- 03901
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PARTIES

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Netlist, Inc.	DEFENDANT, COUNTER CLAIMANT		Gregory C Schodde
Smart Modular Technologies, Inc.	THIRD-PARTY PLAINTIFF	King & Spalding	Michael F Heafey

PATENTS

<u>Number</u>	<u>Inventor</u>	<u>Name</u>
7881150	Solomon	Circuit providing load isolation and memory domain translation for memory module
8001434	Lee	Memory board with self-testing capability
8081536	Solomon	Circuit for memory module
8301833	Chen	Non-volatile memory module
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8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module
8516187	Chen	Data transfer scheme for non-volatile memory module

 Smart Modular Technologies, Inc. v. Netlist, Inc. 4-13-cv-03916 (NDCA)	Aug. 23, 2013	Feb. 12, 2014	Termin ated	NDCA	Smart Modular Technologies, Inc. v. Netlist, Inc.	4-13-cv- 03916
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PARTIES

<u>Party</u>	<u>Party Role</u>	<u>Firm</u>	<u>Attorneys</u>
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Netlist, Inc.	DEFENDANT, COUNTER CLAIMANT	DLA Piper	Erin P Gibson, Rajiv S Dharnidharka, Ryan W Cobb, Sean C Cunningham, Stanley J Panikowski, III
Netlist, Inc.	DEFENDANT, COUNTER CLAIMANT	McAndrews Held & Malloy	Gregory C Schodde, Thomas J Wimbiscus, Wayne H Bradley

PATENTS

<u>Number</u>	<u>Inventor</u>	<u>Name</u>
8001434	Lee	Memory board with self-testing capability
8301833	Chen	Non-volatile memory module
8359501	Lee	Memory board with self-testing capability
8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module
8516187	Chen	Data transfer scheme for non-volatile memory module

 Netlist, Inc. v. Smart Modular Technologies, Inc. 8-13-cv-00996 (CDCA)	Jul. 01, 2013	Dec. 23, 2013	Terminated	CDCA	Netlist, Inc. v. Smart Modular Technologies, Inc.	8-13-cv-00996
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PARTIES

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Netlist, Inc.	PLAINTIFF	McAndrews Held & Malloy	Gregory C Schodde, Thomas J Wimbiscus, Wayne H Bradley
Diablo Technologies, Inc.	DEFENDANT	King & Spalding	Jason Blake Cunningham, Sanjeet K Dutta
Diablo Technologies, Inc.	DEFENDANT	Orrick Herrington & Sutcliffe	Michael F Heafey
Diablo Technologies, Inc.	DEFENDANT	Stephoe & Johnson	William F Abrams
Smart Modular Technologies, Inc.	DEFENDANT	Orrick Herrington & Sutcliffe	Howard M Ullman, Matthew H Poppe, Michael F Heafey, Monte M F Cooper, Robert A Rosenfeld
Smart Modular Technologies, Inc.	DEFENDANT	Stephoe & Johnson	William F Abrams
Smart Storage Systems, Inc.	DEFENDANT	Jones Day	Gregory L Lippetz, Kathleen D Lynott, Steven J Corr
Smart Worldwide Holdings Inc.	DEFENDANT	Orrick Herrington & Sutcliffe	Howard M Ullman, Matthew H Poppe, Monte M F Cooper, Robert A Rosenfeld

PATENTS

<u>Number</u>	<u>Inventor</u>	<u>Name</u>
8001434	Lee	Memory board with self-testing capability
8301833	Chen	Non-volatile memory module
8359501	Lee	Memory board with self-testing capability
8516185	Lee	System and method utilizing distributed byte-wise buffers on a memory module
8516187	Chen	Data transfer scheme for non-volatile memory module

Electronic Patent Application Fee Transmittal

Application Number:	17138766			
Filing Date:	30-Dec-2020			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Stephanie Semler			
Attorney Docket Number:	0016.001000H			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260

Electronic Acknowledgement Receipt

EFS ID:	42149047
Application Number:	17138766
International Application Number:	
Confirmation Number:	8804
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	151145
Filer:	Khaled Shami/Stephanie Semler
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000H
Receipt Date:	11-MAR-2021
Filing Date:	30-DEC-2020
Time Stamp:	19:24:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 260
RAM confirmation Number	E20213AJ27051960
Deposit Account	602034
Authorized User	Stephanie Semler

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	20210309_IDS_0016001000H_1_OF_4.pdf	1061508	no	25
			d2087e571cff87c57ecbbcc51581f4f17f87fded		
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	20210309_IDS_0016001000H_2_OF_4.pdf	1056127	no	8
			09e3bd6e50f7527c0c545b481049e3abbe6873a7		
Warnings:					
Information:					
3	Information Disclosure Statement (IDS) Form (SB08)	20210309_IDS_0016001000H_3_OF_4.pdf	1055481	no	8
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Warnings:					
Information:					
4	Information Disclosure Statement (IDS) Form (SB08)	20210309_IDS_0016001000H_4_OF_4.pdf	1055382	no	7
			9bc18ca9eb17b63d6c4cdce0858d48e87cd6db09		
Warnings:					
Information:					
5	Foreign Reference	IDS1_FP2_KR0130873Y1_and_Translation.pdf	402370	no	8
			dbc2f7b0071a4338cd2e0c7607a40023f94526c0		
Warnings:					
Information:					
6	Foreign Reference	IDS1_FP3_KR100606242B1_and_Translation.pdf	965386	no	32
			8cfce5bf3cd020ec029d60b7f5b58d314e7165d9		
Warnings:					
Information:					

7	Non Patent Literature	IDS1_NPL2_DrawingbyDr_Baker_AddressofDRAM_and_FLASH_PTAB_IPR2017_00587_1023.pdf	133927	no	1
			479777e8b46a22b09082e7bf9e99d524a9e662a0		
Warnings:					
Information:					
8	Non Patent Literature	IDS1_NPL1_THIRD_AMENDED_COMPLAINT_FOR_PATENTINFRINGEMENT_NDCA413cv05889_267.pdf	203729	no	21
			f95c9bc00e62e08ead951ceda614fd985c81fe99		
Warnings:					
Information:					
9	Non Patent Literature	IDS1_NPL8_Amendment_and_Reply_to_Office_Action_IPR2014_00982.pdf	557562	no	24
			5016da648e134cf23081d7d93b9beba47f1016ae		
Warnings:					
Information:					
10	Non Patent Literature	IDS1_NPL9_Ex_1004_Amendment_and_ReplyElection_Rest.pdf	913188	no	9
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Warnings:					
Information:					
11	Non Patent Literature	IDS1_NPL10_Excerpts_from_the_American_Heritage_Dictionary1996_PTAB_IPR2017_00587_2014.pdf	1788746	no	7
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Information:					
12	Non Patent Literature	IDS1_NPL10_Excerpts_from_the_AmericanHeritageDictionary1996_PTAB_IPR2017_00587_2014.pdf	1358996	no	7
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Information:					
13	Non Patent Literature	IDS1_NPL12_Ex2016_134_Annotated_Exhibit1026ofBakerDepos12182017_PTAB_IPR2017_00587_1026.pdf	174992	no	1
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Warnings:					
Information:					

14	Non Patent Literature	IDS1_NPL13_Ex_2016_135_Annotated_Exhibit_1027ofBakerDeposits12182017_PTAB_IPR2017_00587_1027.pdf	173996	no	1
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16	Non Patent Literature	IDS1_NPL22_Charles_Multiplexers_and_Demultiplexers_Ex_1035_PTAB_IPR2017_00587_1035.pdf	1506358	no	8
			9881437f56136a93a324aae7fde4b7be0ed05490		
Warnings:					
Information:					
17	Non Patent Literature	IDS1_NPL26_DECLARATION_OF_MICHAEL_FHEAFEY_PTABIPR2014013701011.pdf	569636	no	4
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18	Non Patent Literature	IDS1_NPL27_d_Petition_For_Inter_Parties_Review_PTAB_IPR2014_01371_7.pdf	1049330	no	67
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Warnings:					
Information:					
19	Non Patent Literature	IDS1_NPL28_Ex_Corrected_Petition_DECLARATIONOFDRNADE RBAGHERZADEH_PTAB_IPR2014_01371_012.pdf	5546098	no	306
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20	Non Patent Literature	IDS1_NPL31_Decision_Denying_Institution_of_Inter_Parties_Review_PTAB_IPR2014_01371_12.pdf	393606	no	22
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21	Non Patent Literature	IDS1_NPL33_Decision_Denyin g_Institution_of_Inter_PartесR evieu_PTAB_IPR2014_00994_8 .pdf	392031	no	16
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Information:					
24	Non Patent Literature	IDS1_NPL36_Decision_Institute d_Document_PTAB_IPR2017_0 0587_7.pdf	444263	no	40
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25	Non Patent Literature	IDS1_NPL38_Ex_Declaration_o f_R_Jacob_BakerPhDPE_PTAB_ IPR2017_00692_2016.pdf	2006241	no	72
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27	Non Patent Literature	IDS1_NPL40_Ex_Declaration_o f_Dr_Nader_PTAB_IPR2014-01 30_1010.pdf	5984516	no	333
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Warnings:					
Information:					

28	Non Patent Literature	IDS1_NPL41_DECLARATION_OF_JEFF_MCMILLEN.pdf	100607	no	3
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Information:					
29	Non Patent Literature	IDS1_NPL42_Ex_Declaration_of_Paul_Min_PTAB_IPR201400994_1020.pdf	4188828	no	215
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30	Non Patent Literature	IDS1_NPL43_Declaration_of_Paul_Min_PTAB_IPR2014_00982_1013.pdf	3562443	no	240
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31	Non Patent Literature	IDS1_NPL44_Declaration_of_Ron_Maltiel_RegardingUSPatentNo_8671243_PTAB_IPR2017_00587_1003.pdf	2554536	no	131
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32	Non Patent Literature	IDS1_NPL45_Declaration_of_RonMaltielre831Patent_PTAB_IPR2017_00692_1003.pdf	2188026	no	172
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34	Non Patent Literature	IDS1_NPL47_Baker_Deposition_Transcript_December182017_PTAB_IPR2017_00587_1030.pdf	1144729	no	268
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Information:					

35	Non Patent Literature	IDS1_NPL48_Ex_Deposition_transcript_of_Ron_Maltiel_PTAB_IPR2017_00692_2010.pdf	6994059	no	155
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36	Non Patent Literature	IDS1_NPL19_Charles_Multiplexers_andDemultiplexersPTAB_IPR2017_00587_024.pdf	959653	no	10
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37	Non Patent Literature	IDS1_NPL20_Charles_Multiplexers_andDemultiplexers_Ex_1033_PTAB_IPR201700587_1033.pdf	1695832	no	8
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38	Non Patent Literature	IDS1_NPL21_Charles_Multiplexers_and_Demultiplexers_Ex1034_PTAB_IPR2017-00587_1034.pdf	2320813	no	8
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39	Non Patent Literature	IDS1_NPL15_Federal_Circuit_Appeal_2016_1742_PTAB_IPR2017_00587_2015.pdf	174201	no	8
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40	Non Patent Literature	IDS2_NPL5_Ex_1002_Application_As_Filed_PTAB_IPR2014_00994_1002.pdf	2780890	no	52
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41	Non Patent Literature	IDS2_NPL6_Ex_8_Denying_Institution_of_Inter_Parties_Review.pdf	381534	no	23
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42	Non Patent Literature	IDS2_NPL7_Decision_Case413cv05889_YGR_Document309_4_Filed031715.pdf	406089	no	20
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46	Non Patent Literature	IDS2_NPL13_Ex_6_Institution_of_Inter_Parties_Review.pdf	470815	no	22
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47	Non Patent Literature	IDS2_NPL16_Ex_Letter_Case413cv05889YGRDocument305_16_Filed031015.pdf	106828	no	3
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49	Non Patent Literature	IDS2_NPL18_ORDER_DENYING_DEFENDANTS_MOTION_TO_STAY_PENDING_INTER_PARTES_REVIEW.pdf	167274	no	10
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52	Non Patent Literature	IDS2_NPL22_Ex_TRANSCRIPT_OF_OFFICIAL_ELECTRONIC_SOUND_RECORDING_OF_PROCEEDINGS.pdf	121364	no	10
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53	Non Patent Literature	IDS2_NPL30_FINAL_WRITTEN_DECISION_PTAB_IPR2017_00692_25.pdf	871272	no	42
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54	Non Patent Literature	IDS2_NPL33_Ex_Hasan_etal_PTAB_IPR2017_00692_1026.pdf	258448	no	12
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55	Non Patent Literature	IDS2_NPL37_PETITION_FOR_INTERPARTESREVIEW_PTAB_IPR2017_00692_1.pdf	764405	no	78
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56		RespNFOA_0016001000H.pdf	176445 9243ae4ea7a31323272bc94a40f0317b9e192440	yes	12
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	9	
	Applicant Arguments/Remarks Made in an Amendment		10	12	
Warnings:					
Information:					
57	Notice of concurrent proceedings / decisions	0016001000H_Notice_Related_Litigation_and_Appendix_A2.pdf	292145 966de13502c43242708d63e6245e0dd30aa56ab7	no	7
Warnings:					
Information:					
58	Fee Worksheet (SB06)	fee-info.pdf	30208 10e88fe64c995c30fe30caa498080e1c179dd5a8	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			69946053		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/138,766		Filing Date 12/30/2020		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	03/11/2021	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 30	Minus	** 30	= 0			x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0			x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/CRYSTAL QUEEN/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

151145 7590 03/24/2021
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

EXAMINER

FARROKH, HASHEM

ART UNIT

PAPER NUMBER

2135

DATE MAILED: 03/24/2021

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	06/24/2021

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

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I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 03/24/2021
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	06/24/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2135	710-308000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804
151145	7590	03/24/2021	EXAMINER	
Shami Messinger PLLC 1000 Wisconsin Ave. NW Suite 200 Washington, DC 20007			FARROKH, HASHEM	
			ART UNIT	PAPER NUMBER
			2135	
DATE MAILED: 03/24/2021				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/138,766	Applicant(s) Lee et al.	
	Examiner HASHEM FARROKH	Art Unit 2135	AIA (FITF) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 03/11/2021.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-30. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>03/11/2021</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
---	--

/HASHEM FARROKH/
Primary Examiner, Art Unit 2135

Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 03/11/2021 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the examiner. However, a copy foreign priority patents listed as KR100606242 and WO2013016723 have not been provided. Therefore, these foreign priority patent documents have not been considered. Furthermore, NPL documents shown by lined-through have not been considered by the Examiner because a copy of the line-through documents have not been provide or they are not readily identifiable from title of documents listed.

PART I THE EXAMINER'S EVALUATION OF THE APPLICATION

SECTION A RELEVANT ART CITED BY THE EXAMINER

1. *Chen et al. (US 20120271990 A1).*
2. *Bates (US 20070136523 A1).*
3. *Prete et al. (US 20080126624 A1).*

SECTION B DISTINGUISHING FEATURES RECITED IN THE CLAIMS

The following is an Examiner's Statement of Reasons for Allowance See MPEP 1302.14

4. *The primary reasons for allowance of claims 1-15 in the instant application is the combination with the inclusion of the following limitations: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than the a second one of the second and fourth voltage amplitudes.”*

5. *The primary reasons for allowance of claims 16-20 in the instant application is the combination with the inclusion of the following limitations: “first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively; a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices,*

each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages; and a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage”

6. *The primary reasons for allowance of claims 23-30 in the instant application is the combination with the inclusion of the following limitations: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of first, second, third and fourth regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and one or more registers, the plurality of SDRAM devices coupled to the first regulated voltage, the one or more registers coupled to (i) the second regulated voltage, (is) a portion of the plurality of edge connections, and (iii) the plurality of SDRAM devices, wherein a plurality of address and control signals are coupled to the one or more registers via the portion of the plurality of edge connections; first, second, and third buck converters configured to provide the first, second and third regulated voltages, respectively”*

The prior art of record including the disclosures of Chen et al. (US 20120271990 A1), Bates (US 20070136523 A1), and Prete et al. (US 20080126624 A1) neither anticipates nor renders obvious the above-recited combination.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays should be clearly labeled "Comments on Statement of Reasons for Allowance"

PART II THE ATTENTION OF FUTURE CORRESPONDENT

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached on 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO

Application/Control Number: 17/138,766
Art Unit: 2135

Page 6

Customer Service Representative or access to the automated information system, call
800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HASHEM FARROKH/
Primary Examiner, Art Unit 2135

March 17, 2021

<i>Search Notes</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G06F13/28; G06F13/4027; G06F13/1694; G06F13/4223; G06F12/0638; G06F12/0246; G06F1/185; G06F3/0613; G06F3/0685 OR G06F3/0659; G06F2212/7208; G06F2212/205; G11C7/1072; G11C14/0018	02/24/2021	HF
G06F3/0659; G06F12/0246; G06F2212/7208; G06F13/4027; G06F13/28; G06F3/0613; G06F2212/205; G06F3/0685; G11C7/1072	03/17/2021	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	103	02/25/2021	HF
711	104	02/25/2021	HF
711	105	02/25/2021	HF
710	138	03/17/2021	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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<i>Search Notes</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

Search Notes		
Search Notes	Date	Examiner
Assignee/Inventor Search	02/25/2021	HF
EAST Text Search	02/24/2021	HF
IP Search	02/25/2021	HF
Google Search	02/25/2021	HF
Assignee/Inventor Search Updated	03/17/2021	03/17/202
Assignee/Inventor Search Updated	03/17/2021	03/17/202

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	03/17/202	03/17/202
G06F12/0246	G06F12/00	03/17/202	03/17/202
G06F2212/205	G06F2212/00	03/17/202	03/17/202

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner HASHEM FARROKH	Art Unit 2135	

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	18 March 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	12

<i>Issue Classification</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	13	/ 36
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
710	308

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	18 March 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	12

Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	9	10	19	19	26	28								
2	2	7	11	20	20	29	29								
3	3	10	12	21	21	30	30								
4	4	11	13	22	22										
5	5	15	14	23	23										
8	6	12	15	24	24										
6	7	16	16	27	25										
14	8	17	17	28	26										
8	9	18	18	25	27										

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	18 March 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	12

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17138766
	Filing Date		2020-12-30
	First Named Inventor		Hyun Lee
	Art Unit		1731 2135
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number		0016.001000H

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	9928186	B2	2018-03-27	Lee et al.	Entire Document
If you wish to add additional U.S. Patent citation information please click the Add button.						Add

U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1					
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	1							
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NON-PATENT LITERATURE DOCUMENTS			Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17138766
Filing Date		2020-12-30
First Named Inventor	Hyun Lee	
Art Unit		1731
Examiner Name		
Attorney Docket Number		0016.001000H

1	Elmhurst et al. "A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write", IEEE Journal of Solid-State Circuits 38(11):1929-1933 (2003).
2	Elmhurst, D. et al., A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write, IEEE Journal of Solid-State Circuits, Vol. 38, No. 11, Nov. 2003, 5 pages.
3	Ex. 1004—IPR2017-00587 Ron Maltel CV, 7 pages.
4	Exhibit 1: Claim Chart Comparing Netlist's US Patent No. 8,001,434 to Smart Storage Ultradimm, Case 4:13-cv-05889-YGR Document 193-1, filed Apr. 10, 2014, 21 pages.
5	Exhibit, Application As Filed, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), 52 pages.
6	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-3, filed March 17, 2015, 23 pages.
7	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-4, filed March 17, 2015, 20 pages.
8	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-5, filed March 17, 2015, 28 pages.
9	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-9, filed March 17, 2015, 17 pages.
10	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-7, filed March 17, 2015, 17 pages.
11	Exhibit, Email from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-17, filed March 10, 2015, 6 pages.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		17138766
Filing Date		2020-12-30
First Named Inventor	Hyun Lee	
Art Unit		1731
Examiner Name		
Attorney Docket Number		0016.001000H

12	Exhibit, Institution of Inter Partes Review, Sandisk Corporation v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 316-9, filed March 24, 2015, 29 pages.
13	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
14	JEDEC Standard, Double Data Rate (DDR) SDRAM Specification, JESD79, Jun. 2000, 77 pages.
15	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-2, filed March 17, 2015, 23 pages.
16	Exhibit, Letter from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et al., Case 4:13-cv-05889-YGR Document 305-16, filed March 10, 2015, 3 pages.
17	Exhibit, Letter sent via email on December 6, 2013, Case 4:13-cv-03901-YGR Document 53-2, filed January 6, 2014
18	Exhibit, Order Denying Defendant's Motion to Stay Pending Inter Partes Review (Doc.59), The Procter and Gamble Company v. Team Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 316-6, filed March 24, 2015, 10 pages.
19	Exhibit, Patent Public Advisory Committee Quarterly Meeting, Appeals Statistics USPTO, Case No. 4:13-cv-05889-YGR Document 309-8, filed March 17, 2015, 23 pages.
20	Exhibit, Reporter's Transcript of Proceedings, Netlist, Inc v. Smart Modular Technologies, Inc., et al., Case 4:13-cv-05889-YGR Document 316-3, filed March 24, 2015, 15 pages.
21	Exhibit, Transcript of Official Electronic Sound Recording Proceeding, Netlist v. Smart Modular Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 305-7, filed March 10, 2015, 10 pages.
22	Extended European Search Report for European Application No. 12 817 751.6, dated Jun. 9, 2015.

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23	File History for U.S. Appl. No. 12/240,916, filed Sep. 29, 2008, 320 pages.
24	File History for U.S. Appl. No. 13/905,048, filed May 29, 2013, 181 pages.
25	File History for U.S. Appl. No. 60/941,586, filed Jun. 1, 2007, 23 pages.
26	File History U.S. Pat. No. 8,671,243.
27	Final Office Action, dated Jun. 15, 2016, issued in U.S. Appl. No. 14/489,281, 10 pages.
28	Final Office Action, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-0099-1007 (PTAB), dated February 1, 2012, 13 pages
29	Final Written Decision, Patent 8,671,243, IPR2017-00587-34, Paper No. 34, entered June 20, 2018, 53 pages.
30	Final Written Decision, Patent 8,874,831, IPR2017-00692, Paper No. 25, entered July 5, 2018, 42 pages.
31	Final Written Decision, US Patent No. 7,881,150, Case No. Case IPR2014-00882, (PTAB), Paper 33, filed December 14, 2015, 51 pages.
32	First Amended Complaint for Patent Infringement, Netlist Inc. v Smart Modular Tech. Inc. et al. Case 8:13-cv-00996-DOC-JPR, Document 5, filed Aug. 23, 2013, 99 pages.
33	Hasan, J. et al. Efficient Use of Memory Bandwidth to Improve Network Processor Throughput, Proceedings of the 30th Annual International Symposium on Computer Architecture (ISCA'03), IEEE, 2003, 12 pages.

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34	Horowitz, P. et al. "The Art of Electronics". Cambridge University Press 2nd Ed. 1989. pp. 471495-496.
35	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.
36	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18), 2003, 100 pages.
37	Inter Partes Review No. IPR2017-00692 (PTAB), Patent No. 8,874,831, filed July 26, 2012, 78 pages.
38	Inter Partes Review of U.S. Patent No. 8,874,831, Case IPR2017-00692 (PTAB), filed July 26, 2012, Paper No. 1, 78 pages.
39	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014pp. 1-8.
40	International Search Report and Written Opinion in PCT/US12/48750, dated Oct. 10, 2012pp. 1-10.
41	ISSCC 2006 / SESSION 7 / NON-VOLATILE MEMORY / 7.7, IEEE International Solid-State Circuits Conference, 2006, 10 pages.
42	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322, 58 pages.
43	Jandhyala, S. et al., "Design-For-Test Analysis of a Buffered SDRAM DIMM", Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, August 13014, 1996, 15 pages.
44	JEDEC Definition of DIMM, Exhibit 1029, IPR No. 2017-00587, December 18, 2017, 2 pages.

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45	JEDEC Global Standard for the Microelectronics Industry, Why JEDEC Standards Matter, 2014, 1 page.
46	JEDEC Standard 21-C "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.
47	JEDEC Standard No. 21-C (Release 17), Annex J: Serial Presence Detects for DDR2 SDRAM (Revision 1.3), 60 pages.
48	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, Aug. 1999, 144 pages.
49	JEDEC Standard, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A), Jan. 2005, 113 pages.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	03/15/2021
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Art Unit	1731
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CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
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S410	426	S404 AND ((G06F3/0659 OR G06F12/0246 OR G06F2212/7208 OR G06F13/4027 OR G06F13/28 OR G06F3/0613 OR G06F2212/205 OR G06F3/0685 OR G11C7/1072).CPC. OR (710/308).CCLS. OR (G06F3/06).IPCR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2021/03/18 12:15
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S412	12	S411 and (S407 or S408 or S409)	US-PGPUB; USPAT;	OR	ON	2021/03/18 12:15

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17138766
	Filing Date		2020-12-30
	First Named Inventor		Hyun Lee
	Art Unit		1731 2135
	Examiner Name		Hashem Farrokh
	Attorney Docket Number		0016.001000H

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Application Number	17138766
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First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

1	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
2	Petition for Inter Partes Review of U.S. Pat. No. 8,671,243, filed May 29, 2013.
3	Petition for Inter Partes Review of U.S. Pat. No. 8,874,831, filed Jul. 26, 2012.
4	Wong, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.
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6	US Patent Application No. 60/912,321, Case No. No. IPR2017-00692-1007 (PTAB), filed April 17, 2017, 42 pages.
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Attorney Docket Number		0016.001000H

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Examiner Signature	/HASHEM FARROKH/	Date Considered	03/15/2021
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First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

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Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	7731 2135
Examiner Name	Hashem Farrokh
Attorney Docket Number	0016.001000H

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First Named Inventor	Hyun Lee	
Art Unit		1731
Examiner Name		
Attorney Docket Number		0016.001000H

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



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UNITED STATES DEPARTMENT OF COMMERCE
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 www.uspto.gov

BIB DATA SHEET

CONFIRMATION NO. 8804

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
17/138,766	12/30/2020	710	2135	0016.001000H
RULE				
APPLICANTS Netlist, Inc., Irvine, CA;				
INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;				
** CONTINUING DATA ***** This application is a CON of 15/934,416 03/23/2018 which is a CON of 14/840,865 08/31/2015 PAT 9928186 which is a CON of 14/489,269 09/17/2014 PAT 9158684 which is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007				
** FOREIGN APPLICATIONS *****				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 01/14/2021				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/HASHEM FARROKH/</u> Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 22	TOTAL CLAIMS 30
INDEPENDENT CLAIMS 3				
ADDRESS Shami Messinger PLLC 1000 Wisconsin Ave. NW Suite 200 Washington, DC 20007 UNITED STATES				
TITLE FLASH-DRAM HYBRID MEMORY MODULE				
FILING FEE RECEIVED 2820	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____	

	☐ Credit
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17138766
	Filing Date		2020-12-30
	First Named Inventor		Hyun Lee
	Art Unit		1731 2135
	Examiner Name	Hashem Farrokh	
	Attorney Docket Number		0016.001000H

U.S.PATENTS						Remove
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

1	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC Solid State Tech. Assoc., Mar. 2007, 9 pages.
2	Joint Status Report Regarding Inter Partes Review, Netlist v. Smart Storage Systems, Inc. et. al., Case No. 4:13-CV-05889-YGR (NDCA), filed August 17, 2018, 4 pages.
3	Material Science & Engineering, Department of Material Science and Engineering, Stanford University, Exhibit 2011, Case No. IPR2017-00692, 2 pages.
4	Merriam-Webster's Collegiate Dictionary, Eleventh Ed., Merriam Webster Corporation, Springfield, MA, 2003, 7 pages.
5	Microsoft Computer Dictionary Fifth Edition, 2002, 3 pages.
6	Microsoft Computer Dictionary Fifth Edition, 2002, 9 pages.
7	Microsoft Press, Computer Dictionary, Second Edition, 1994, 4 pages.
8	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
9	Mutnuary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp. xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.
10	Notice of Allowance in U.S. Appl. No. 12/240,916, dated Sep. 17, 2012.
11	Notice of Allowance in U.S. Appl. No. 13/536,173, dated Jul. 2, 2013.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

12	Notice of Allowance in U.S. Appl. No. 13/559,476, dated May 6, 2014.
13	Notice of Allowance in U.S. Appl. No. 13/559,476, dated Sep. 29, 2014.
14	Notice of Allowance in U.S. Appl. No. 13/905,048, dated Dec. 19, 2013, 8 pages.
15	Notice of Allowance in U.S. Appl. No. 13/905,053, dated Dec. 11, 2013.
16	Notice of Allowance in U.S. Appl. No. 14/173,219 dated Jul. 7, 2014.
17	Notice of Allowance in U.S. Appl. No. 14/489,269, dated Oct. 8, 2015.
18	Notice of Allowance, SanDisk Corporation v. Netlist, Application No. 13/536,173, IPR2014-00982 (PTAB), dated July 2, 2013, 8 pages.
19	Office Action dated Aug. 19, 2016 of the Chinese Patent Application No. 201280047758.X, 9 pages.
20	Office Action in U.S. Appl. No. 12/240,916, mailed on Apr. 3, 2012.
21	Office Action in U.S. Appl. No. 13/536,176, dated on Apr. 15, 2013.
22	Office Action in U.S. Appl. No. 13/625,563, dated Aug. 5, 2013.

**INFORMATION DISCLOSURE
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Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

23	Office Action in U.S. Appl. No. 13/625,563, dated May 9, 2014.
24	Office Action in U.S. Appl. No. 13/905,048, dated Aug. 1, 2013.
25	Office Action in U.S. Appl. No. 13/905,053, dated Aug. 1, 2013.
26	Office Action in U.S. Appl. No. 14/173,219, dated Mar. 13, 2014.
27	Office Action in U.S. Appl. No. 14/302,292, dated Dec. 21, 2015.
28	Office Action in U.S. Application No. 12/240,916, dated Feb. 1, 2012, 14 pages.
29	Office Action in U.S. Application No. 14/173,242, dated March 14, 2014, 7 pages.
30	Office Action, Application 12/240,916, IPR2014-0099-1005 (PTAB), dated July 29, 2011, 8 pages.
31	Office Action, Application 12/240,916, IPR2014-0099-1010 (PTAB), dated April 3, 2012, 11 pages.
32	Office Action, Application No. 13/536,173, IPR2014-00982 (PTAB), dated April 15, 2013, 9 pages.
33	Order Granting Joint Stipulation of Dismissal, Netlist v. Smart Storage Systems, Inc. et al., Case No. 4:13-cv-05889-YGR (NDCA), filed May 15, 2020, 2 pages.

**INFORMATION DISCLOSURE
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Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

34	Patent Owner's Demonstratives, Patent Number: 8,671,243, Case No. IPR 2017-00587-2023, (PTAB), 57 pages.
35	Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, Patent No. 8,671,243, Case No. IPR2017-00587 (PTAB), filed Jan. 29, 2018, 6 pages.
36	Patent Owner's Opposition to Petitioners' Motion to Exclude, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed April 2, 2018, 2017, 11 pages.
37	Patent Owner's Preliminary Response, Patent Number: 8,516,187, Case IPR2014-01371 (PTAB), filed Dec. 16, 2014, 66 pages.
38	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated October 2, 2014, 60 pages.
39	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Patent Number: 8,516,187, IPR2014-00982 (PTAB), dated September 26, 2014, 57 pages.
40	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,301,833, Case No. IPR2017-00649 (PTAB), filed May 1, 2017, 67 pages.
41	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed May 1, 2017, 48 pages.
42	Patent Owner's Preliminary Response, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, Case IPR2014-01370 (PTAB), filed Dec. 16, 2014, 66 pages.
43	Patent Owner's Response, Patent No. 8,671,243, Case No. IPR2017-00587-12 (PTAB), filed Oct. 13, 2017, 80 pages.
44	Patent Owner's Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692-12 (PTAB), filed Nov. 10, 2017, 77 pages.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
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Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

45	Patterson et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998), 71 pages.
46	Petition for Inter Partes Review of Claims 1-30 of US.PATENT No. 8,301,833, IPR2014-01370 (PTAB), filed August 22, 2014, 68 pages.
47	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
48	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
49	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833, filed Sep. 29, 2008.
50	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SanDisk, Corp.), filed Jun. 19, 2014.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/HASHEM FARROKH/	Date Considered	03/15/2021
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 8804

APPLICATION NO: 17/138,766

ART UNIT: 2135

FILING DATE: 12/30/20

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

**Mail Stop Issue Fee
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

**REQUEST FOR CONSIDERATION OF TIMELY CITED REFERENCES AND
COMMENTS ON STATEMENT OF REASONS FOR ALLOWANCE**

Commissioner:

Applicant gratefully acknowledges the indication of allowance of Claims 1-30 in the Notice of Allowance mailed March 24, 2021.

The Notice of Allowance indicates that foreign references KR100606242 and WO2013016723 were not considered because copies thereof were not provided. Applicant respectfully disagrees. A copy of KR100606242 was submitted in the Information Disclosure Statement of March 11, 2021. In addition, a copy of WO2013016723 was cited in the information disclosure statement filed March 23, 2018 in parent application 15/934,416. Applicant respectfully requests consideration of these timely filed references and acknowledgement of same in a suitably marked Form PTO-892.

In addition, page 5 of the Notice of Allowance states:

The prior art of record including the disclosures of Chen et al. (US 20120271990 A1), Bates (US 20070136523A1), and Prete et al. (US 2008012662A1) neither anticipates nor renders obvious the above-recited combination.

Applicant respectfully agrees that US 2008012662A1 and US 20070136523A1 do not anticipate nor render obvious the presently claimed invention. Applicant respectfully points out that the first named inventor of US 20070136523A1 is Bonnella, not Bates.

In addition, Applicant respectfully points out that Chen et al. (US 20120271990 A1) is not prior art at least because it is an issued patent from a commonly-owned ancestor application of the instant application and shares a common priority date with the instant application.

Pursuant to 37 C.F.R. § 1.104, Applicant respectfully submits the instant statement commenting on the Examiner's reasons for allowance.

Applicant maintains that additional and/or alternative reasons for allowance may exist apart from those expressed by the Examiner and the Applicant, and these reasons may be independently sufficient to establish the patentability of each of the allowed dependent or independent claims.

Applicant respectfully reserves the right to introduce, articulate, or otherwise comment on any such additional reasons for allowance as may be appropriate in any future proceedings concerning the one or more claimed embodiments.

To the extent that the Examiner's reasons for allowance are inconsistent with applicable case law, statutes, and regulations, Applicant respectfully disagrees with them. Furthermore, Applicant does not admit to any characterization or limitation of the claims or to any characterization of a reference by the Examiner, particularly any that are inconsistent with the language of the claims considered in their entirety and including all of their constituent limitations.

Respectfully submitted,

Dated: April 8, 2021

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

Shami Messinger PLLC
1000 Wisconsin Ave N.W., Suite 200
Washington, D.C. 20007
202-516-6900

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONF. NO: 8804

APPLICATION NO: 17/138,766

ART UNIT: 2135

FILING DATE: 12/30/20

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Correction of Inventorship
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

CORRECTION OF INVENTORSHIP UNDER 37 C.F.R. § 1.48
PURSUANT TO 35 U.S.C. 116

Commissioner:

Applicants respectfully requests correction of inventorship under 37 C.F.R. § 1.48 pursuant to 35 U.S.C. 116 in the above-captioned application. The actual inventors are:

- 1. CHEN, Chi-She**
- 2. SOLOMON, Jeffery C.**
- 3. MILTON, Scott H.**
- 4. BHAKTA, Jayesh**

Submitted herewith in support of the instant request are:

- 1. A corrected Application Data Sheet (ADS) in accordance with 37 C.F.R. § 1.76 that identifies each inventor by his or her legal name**
- 2. The processing fee set forth in 37 C.F.R. § 1.17(i)**
- 3. The fee set forth in 37 C.F.R. § 1.17(d)**

It is not believed that additional fees are required in support of this request. However, authorization is hereby granted to charge additional fees to our Deposit Account No. 60-2034 should such additional fees be required.

Respectfully submitted,

Dated: April 8, 2021

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

SHAMI MESSINGER PLLC
1000 Wisconsin Ave N.W., Suite 200
Washington, D.C. 20007
202-516-6900

**REQUEST FOR CORRECTION IN A
PATENT APPLICATION RELATING TO
INVENTORSHIP OR AN INVENTOR
NAME, OR ORDER OF NAMES, OTHER
THAN IN A REISSUE APPLICATION (37
CFR 1.48)**

Application Number	17/138,766
Filing Date	Dec. 30, 2020
First Named Inventor	CHEN, Chi-She
Art Unit	2135
Examiner Name	FARROKH, Hashem
Practitioner Docket Number	0016.001000H

**To: Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Applicant hereby **requests that the inventorship be corrected or changed, or that the name of the inventor or a joint inventor, or the order of the names of joint inventors, be changed**, in the above-identified application. **Note:** 37 CFR 1.48 applies to any request to correct inventorship filed on or after September 16, 2012, regardless of the application filing date. **Do not submit this form after payment of the issue fee or if the application has been patented.** See 37 CFR 1.324 for correction of inventorship in a patent.

Please check the applicable box(es) below.

For a nonprovisional application:

☒ 1. This request is to correct or change the inventorship in a **nonprovisional** application (**under 37 CFR 1.48(a)**) and includes:

☒ An application data sheet (ADS) in accordance with 37 CFR 1.76(c) with the corrected or updated information shown with markings (e.g., underlining for insertions, strikethrough for deletions). See the Manual of Patent Examining Procedure (MPEP) section 601.05(a) for information about filing an ADS in an application filed on/after September 16, 2012. For information about filing a Supplemental ADS in an application filed before September 16, 2012, see MPEP 601.05(b).

☒ The processing fee set forth in 37 CFR 1.17(i).

\$ **140**

☐ An inventor is being added. An inventor's oath or declaration by any actual inventor who has not yet executed an oath or declaration is required (see 37 CFR 1.48(b)). See MPEP 602.01(a) for information about an inventor's oath or declaration for an application filed on/after September 16, 2012 (e.g., form **PTO/AIA/01**). For information about an inventor's oath or declaration for an application filed before September 16, 2012 (e.g., form **PTO/SB/01**), see MPEP 602.01(b).

☒ This request is being filed after the first Office action on the merits has been given or mailed (see 37 CFR 1.48(c) and 1.17(d)). Check one of the following:

☐ This request to correct or change the inventorship is due solely to the cancellation of claims in the application.

OR

☒ The fee set forth in 37 CFR 1.17(d) is due (in addition to the fee set forth in 37 CFR 1.17(i)).

\$ **640**

[Page 1 of 2]

This collection of information is required by 37 CFR 1.48. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

REQUEST FOR CORRECTION IN A PATENT APPLICATION RELATING TO INVENTORSHIP OR AN INVENTOR NAME, OR ORDER OF NAMES, OTHER THAN IN A REISSUE APPLICATION (37 CFR 1.48)

- ☐ 2. This request is to correct or update the name of the inventor or a joint inventor, or the order of names of joint inventors, in a **nonprovisional** application (under 37 CFR 1.48(f)) and includes:

- ☐ An application data sheet in accordance with 37 CFR 1.76(c) identifying the complete inventive entity, including the corrected or updated name of the inventor, or the new order of names shown with markings (e.g., underlining for insertions, strikethrough for deletions). See the MPEP 601.05(a) for information about filing an ADS in an application filed on/after September 16, 2012. For information about filing a Supplemental ADS in an application filed before September 16, 2012, see MPEP 601.05(b).

- ☐ The processing fee set forth in 37 CFR 1.17(i). \$ _____

For a provisional application:

- ☐ This request is to change or correct the inventorship, or correct or update the name of the inventor or a joint inventor, in a **provisional** application (under 37 CFR 1.48(d)) and includes:

- ☐ **Attached hereto** is a document that is signed by a party set forth in 37 CFR 1.33(b) and identifies each inventor by his or her legal name, in the preferred order. Note: the document may be an application data sheet in accordance with 37 CFR 1.76(c) that identifies the changes with markings (underlining for insertions, strikethrough for deletions).

- ☐ The processing fee set forth in 37 CFR 1.17(q). \$ _____

Fee Payment Information:

- ☐ Applicant asserts small entity status. See 37 CFR 1.27.
- ☐ Applicant certifies micro entity status. See 37 CFR 1.29.
Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously
- ☐ A check in the amount of the fee is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment to Deposit Account No. 60-2034.
- ☒ Payment made via EFS-Web.

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

I am the

- ☐ Applicant* ☒ attorney or agent of record ☐ attorney or agent acting under 37 CFR 1.34
- Registration number 38,745 Registration number _____

Signature /Khaled Shami/

Typed or printed name Khaled Shami

Date April 8, 2021

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. *Juristic entities must be represented by a patent practitioner (See 37 CFR 1.31, applicable to any paper filed on or after September 16, 2012 that is presented on behalf of a juristic entity, regardless of application filing date). Submit multiple forms if more than one signature is required, see below**.

- ☐ ** Total of _____ forms are submitted.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Hyun		Lee	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Ladera Ranch	State/Province	CA	Country of Residence ⁱ US
Mailing Address of Inventor:				
Address 1		21 Thalia Street		
Address 2				
City	Ladera Ranch	State/Province	CA	
Postal Code	92694	Country ⁱ	US	
Inventor 2				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Walnut	State/Province	CA	Country of Residence ⁱ US
Mailing Address of Inventor:				
Address 1		944 Crystal Water Lane		
Address 2				
City	Walnut	State/Province	CA	
Postal Code	91789	Country ⁱ	US	
Inventor 3				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jeffrey	C.	Solomon	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		6 Silver Fir			
Address 2					
City	Irvine	State/Province		CA	
Postal Code	92604	Country ⁱ	US		

Inventor 4				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Scott	H.	Milton	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Irvine	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		49 Statehouse Place			
Address 2					
City	Irvine	State/Province		CA	
Postal Code	92602	Country ⁱ	US		

Inventor 5				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One)		☒ US Residency	☐ Non US Residency	☐ Active US Military Service	
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US

Mailing Address of Inventor:					
Address 1		12220 Rose Street			
Address 2					
City	Cerritos	State/Province		CA	
Postal Code	90703	Country ⁱ	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

☐ An Address is being provided for the correspondence information of this application.

Customer Number	151145		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	0016.001000H	Small Entity Status Claimed	☐
Application Type			
Subject Matter			
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	151145		

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H	
		Application Number		
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE			

Prefix	Given Name	Middle Name	Family Name	Suffix	Remove
Registration Number					
Prefix	Given Name	Middle Name	Family Name	Suffix	Remove
Registration Number					

Additional Representative Information blocks may be generated within this form by selecting the **Add** button.

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
	Continuation of		15934416	2018-03-23	

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
15934416	Continuation of	14840865	2015-08-31	9928186	2018-03-27

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14840865	Continuation of	14489269	2014-09-17	9158684	2015-10-13

Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14489269	Continuation of	13559476	2012-07-26	8874831	2014-10-28

Prior Application Status		Expired		Remove	
Application Number	Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)	
13559476	Claims benefit of provisional		61512871	2011-07-28	

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Prior Application Status		Patented	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30

Prior Application Status		Abandoned	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		

Prior Application Status		Expired	Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
12131873	Claims benefit of provisional	60941586	2007-06-01		

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

☐ NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	0016.001000H
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	0016.001000H
	Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE	

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

Clear

- ☒ Assignee ☐ Legal Representative under 35 U.S.C. 117 ☐ Joint Inventor
- ☐ Person to whom the inventor is obligated to assign. ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information For Applicant:

Address 1	175 Technology Drive		
Address 2	Suite 150		
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1					
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.					
If the Assignee or Non-Applicant Assignee is an Organization check here.			☒		
Organization Name	Netlist, Inc.				
Mailing Address Information For Assignee including Non-Applicant Assignee:					
Address 1		175 Technology Drive			
Address 2		Suite 150			
City		Irvine	State/Province CA		
Country i	US	Postal Code	92618		
Phone Number		Fax Number			
Email Address					
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	0016.001000H
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Khaled Shami/			Date (YYYY-MM-DD)	<u>April 09, 2021</u>
First Name	Khaled	Last Name	Shami	Registration Number	38745
Additional Signature may be generated within this form by selecting the Add button.					

Electronic Patent Application Fee Transmittal

Application Number:	17138766			
Filing Date:	30-Dec-2020			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Stephanie Semler			
Attorney Docket Number:	0016.001000H			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
PROCESSING FEE, EXCEPT PROV. APPLS.	1830	1	140	140
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
CORRECTION OF INVENTORSHIP ON MERITS	1819	1	640	640
Total in USD (\$)				780

Electronic Acknowledgement Receipt

EFS ID:	42400224
Application Number:	17138766
International Application Number:	
Confirmation Number:	8804
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	151145
Filer:	Khaled Shami/Stephanie Semler
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000H
Receipt Date:	09-APR-2021
Filing Date:	30-DEC-2020
Time Stamp:	22:03:46
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 780
RAM confirmation Number	E202149M04584434
Deposit Account	602034
Authorized User	Stephanie Semler

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	Comments_on_Rsns_08_04_21_for_Allowance_03_24_21_0016001000H.pdf	116857	no	2
			b53482352ee793d3d55f8527db473bf927920812		
Warnings:					
Information:					
2	Request under Rule 48 correcting inventorship	00160001000H_Correction_of_Inventorship.pdf	114198	no	2
			a697d0ddc462a6397b09830c9c0d472f2205ee22		
Warnings:					
Information:					
3	Request under Rule 48 correcting inventorship	0016001000H_aia0040_04_08_21.pdf	199981	no	3
			150981ef7e9a033a16a0c9981227754c7a023e2d		
Warnings:					
Information:					
4	Application Data Sheet	Corrected_ADS_0016001000H.pdf	510157	no	9
			918aa5a64fb14e5289831fd7fb1b91e500f3315b		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
5	Fee Worksheet (SB06)	fee-info.pdf	32354	no	2
			8d509ec842da660c049f1f26da9f06bfe9d730eb		
Warnings:					
Information:					
Total Files Size (in bytes):			973547		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/138,766	12/30/2020	Chi-She Chen	0016.001000H

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

CONFIRMATION NO. 8804
37 CFR 1.48 ACKNOWLEDGEMENT
LETTER



OC000000124856466

Date Mailed: 04/14/2021

NOTICE OF ACCEPTANCE OF REQUEST UNDER 37 CFR 1.48(a)

This is in response to the applicant's request under 37 CFR 1.48(a) submitted on 04/09/2021.

The request under 37 CFR 1.48(a) to correct the inventorship, to correct or update the name of an inventor, or to correct the order of names of joint inventors is accepted.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ttran/



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/138,766	12/30/2020	2135	2820	0016.001000H	30	3

CONFIRMATION NO. 8804

UPDATED FILING RECEIPT

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007



Date Mailed: 04/14/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 151145

Domestic Priority data as claimed by applicant

This application is a CON of 15/934,416 03/23/2018
which is a CON of 14/840,865 08/31/2015 PAT 9928186
which is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

Projected Publication Date: 04/29/2021

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

710

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor

community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Chi-She Chen	0016.001000H	8804
151145	7590	04/22/2021	EXAMINER	
Shami Messinger PLLC			FARROKH, HASHEM	
1000 Wisconsin Ave. NW			ART UNIT	
Suite 200			PAPER NUMBER	
Washington, DC 20007			2135	
			NOTIFICATION DATE	
			DELIVERY MODE	
			04/22/2021	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

khaled@shamimessinger.com
mike@shamimessinger.com
stephanie@shamimessinger.com

**Complimentary
Notice of Allowability**

Application No.
17/138,766

Applicant(s)
Chen et al.

Examiner
HASHEM FARROKH

Art Unit
2135

AIA (FITF) Status
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 03/11/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-30. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.

☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)

2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date 03/11/2021.

3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material _____.

4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date. _____.

5. ☐ Examiner's Amendment/Comment

6. ☒ Examiner's Statement of Reasons for Allowance

7. ☐ Other _____.

/HASHEM FARROKH/
Primary Examiner, Art Unit 2135

**UNITED STATES DEPARTMENT OF COMMERCE****U.S. Patent and Trademark Office**

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P.O. Box 1450

Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR/ PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
17/138,766	12/30/2020	Chen et al.	0016.001000H

Shami Messinger PLLC 1000 Wisconsin Ave. NW Suite 200 Washington, DC 20007		EXAMINER	
		HASHEM FARROKH	
		ART UNIT	PAPER
		2135	20210419

DATE MAILED: _____

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

This communication is in response to the Applicant Letter received on 04/09/2021. The letter indicates that KR100606242 was submitted in the Information disclosure Statement (IDS) of March 11, 2021; and a copy WO2013016723 was cited in the IDS filed March 23, 2018. Examiner agrees, a signed copy of IDS' are enclosed. This NOA states that the prior art does not appear to teach or suggest the combination of features or limitations included in the claims, instead of the prior art of record as was stated in the NOA issued on 03/24/2021. The Letter also states that "additional and/or alternative reasons may exist apart from those expressed by the Examiner and the Applicant". The Examiner would agree. However, the Examiner has chosen to include the combination of features or limitations that appear not being disclosed or render obvious by the prior art searched. using combination features or limitations in the claims reason for allowance is consistent with MPEP (e.g., see MPEP 1302.14).

/HASHEM FARROKH/
Primary Examiner, Art Unit 2135

Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

INFORMATION CONCERNING IDS:

The information disclosure statements (IDS') submitted on 03/11/2021 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements have been considered by the examiner. However, a copy NPL documents shown by lined-through have not been considered by the Examiner because a copy of the lined-through documents have not been provide or they are not readily identifiable from title of documents listed.

Reason for Allowance

1. *The primary reasons for allowance of claims 1-15 in the instant application is the combination with the inclusion of the following limitations: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, the at*

Art Unit: 2135

least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, wherein a first one of the second and fourth voltage amplitudes is less than the a second one of the second and fourth voltage amplitudes.”

2. *The primary reasons for allowance of claims 16-20 in the instant application is the combination with the inclusion of the following limitations: “first, second, and third buck converters configured to receive a pre-regulated input voltage and to produce first, second and third regulated voltages, respectively; a plurality of components coupled to the PCB, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages; and a voltage monitor circuit configured to monitor an input voltage received via a first portion of the plurality of edge connections, the voltage monitor circuit configured to produce a signal in response to the input voltage having a voltage amplitude that is greater than a first threshold voltage”*

3. *The primary reasons for allowance of claims 23-30 in the instant application is the combination with the inclusion of the following limitations: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to*

Art Unit: 2135

one or more regulated voltages of first, second, third and fourth regulated voltages, the plurality of components including a plurality of synchronous dynamic random access memory (SDRAM) devices and one or more registers, the plurality of SDRAM devices coupled to the first regulated voltage, the one or more registers coupled to (i) the second regulated voltage, (is) a portion of the plurality of edge connections, and (iii) the plurality of SDRAM devices, wherein a plurality of address and control signals are coupled to the one or more registers via the portion of the plurality of edge connections; first, second, and third buck converters configured to provide the first, second and third regulated voltages, respectively”

The prior art does not appear to teach or suggest the above-recited combinations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays should be clearly labeled “Comments on Statement of Reasons for Allowance”

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays should be clearly labeled “Comments on Statement of Reasons for Allowance”

Conclusion

Art Unit: 2135

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HASHEM FARROKH whose telephone number is (571)272-4193. The examiner can normally be reached on 8:30 am - 5:00 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mr. Sanjiv Shah can be reached on (571)272-4098. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. For questions regarding access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

*/HASHEM FARROKH/
Primary Examiner, Art Unit 2135*

April 19, 2021

Search Notes 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G06F13/28; G06F13/4027; G06F13/1694; G06F13/4223; G06F12/0638; G06F12/0246; G06F1/185; G06F3/0613; G06F3/0685 OR G06F3/0659; G06F2212/7208; G06F2212/205; G11C7/1072; G11C14/0018	02/24/2021	HF
G06F3/0659; G06F12/0246; G06F2212/7208; G06F13/4027; G06F13/28; G06F3/0613; G06F2212/205; G06F3/0685; G11C7/1072	03/17/2021	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	103	02/25/2021	HF
711	104	02/25/2021	HF
711	105	02/25/2021	HF
710	138	03/17/2021	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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<i>Search Notes</i> 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

Search Notes		
Search Notes	Date	Examiner
Assignee/Inventor Search	02/25/2021	HF
EAST Text Search	02/24/2021	HF
IP Search	02/25/2021	HF
Google Search	02/25/2021	HF
Assignee/Inventor Search Updated	03/17/2021	03/17/202
Assignee/Inventor Search Updated	03/17/2021	03/17/202

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06F3/0685	G06F3/00	03/17/202	03/17/202
G06F12/0246	G06F12/00	03/17/202	03/17/202
G06F2212/205	G06F2212/00	03/17/202	03/17/202

/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	
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Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Chen et al.	
	Examiner HASHEM FARROKH	Art Unit 2135	

CPC						
Symbol					Type	Version
G06F	/	13	/	28	F	2013-01-01
G06F	/	12	/	0246	I	2013-01-01
G06F	/	13	/	1694	I	2013-01-01
G06F	/	1	/	185	I	2013-01-01
G06F	/	12	/	0638	I	2013-01-01
G06F	/	13	/	4243	I	2013-01-01
G11C	/	7	/	1072	I	2013-01-01
G11C	/	14	/	0018	I	2013-01-01
G06F	/	3	/	0613	I	2013-01-01
G06F	/	3	/	0659	I	2013-01-01
G06F	/	3	/	0685	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01
G06F	/	2212	/	205	A	2013-01-01
G06F	/	2212	/	7208	A	2013-01-01

CPC Combination Sets								
Symbol					Type	Set	Ranking	Version
	/		/					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	19 April 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	12

Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

INTERNATIONAL CLASSIFICATION					
CLAIMED					
G06F	/	13	/	36	
NON-CLAIMED					
	/		/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
710	308

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135	19 April 2021	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	12

Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Chen et al.
	Examiner HASHEM FARROKH	Art Unit 2135

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	9	10	19	19	26	28								
2	2	7	11	20	20	29	29								
3	3	10	12	21	21	30	30								
4	4	11	13	22	22										
5	5	15	14	23	23										
8	6	12	15	24	24										
6	7	16	16	27	25										
14	8	17	17	28	26										
8	9	18	18	25	27										

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135 (Primary Examiner) _____ (Date) 19 April 2021		O.G. Print Claim(s) 1	O.G. Print Figure 12

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731 2135
Examiner Name	Hashem Farrokh
Attorney Docket Number	0016.001000H

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731
Examiner Name	
Attorney Docket Number	0016.001000H

1	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
2	Petition for Inter Partes Review of U.S. Pat. No. 8,671,243, filed May 29, 2013.
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Examiner Signature	/HASHEM FARROKH/	Date Considered	03/15/2021
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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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See attached certification statement.

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A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

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STATEMENT BY APPLICANT**
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Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	7731 2135
Examiner Name	Hashem Farrokh
Attorney Docket Number	0016.001000H

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
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Art Unit		1731
Examiner Name		
Attorney Docket Number		0016.001000H

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Name/Print	Khaled Shami	Registration Number	38745

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	3	Ex. 1004—IPR2017-00587 Ron Maltel CV, 7 pages.	
	4	Exhibit 1: Claim Chart Comparing Netlist's US Patent No. 8,001,434 to Smart Storage Ultradimm, Case 4:13-cv-05889-YGR Document 193-1, filed Apr. 10, 2014, 21 pages.	
	5	Exhibit, Application As Filed, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), 52 pages.	
	6	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-3, filed March 17, 2015, 23 pages.	
	7	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-4, filed March 17, 2015, 20 pages.	
	8	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-5, filed March 17, 2015, 28 pages.	
	9	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-9, filed March 17, 2015, 17 pages.	
	10	Exhibit, Decision Denying Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-7, filed March 17, 2015, 17 pages.	
	11	Exhibit, Email from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et. al., Case 4:13-cv-05889-YGR Document 305-17, filed March 10, 2015, 6 pages.	

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Application Number		17138766
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First Named Inventor	Hyun Lee	
Art Unit		1731
Examiner Name		
Attorney Docket Number		0016.001000H

12	Exhibit, Institution of Inter Partes Review, Sandisk Corporation v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 316-9, filed March 24, 2015, 29 pages.
13	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case 4:13-cv-05889-YGR Document 309-1, filed March 17, 2015, 22 pages.
14	JEDEC Standard, Double Data Rate (DDR) SDRAM Specification, JESD79, Jun. 2000, 77 pages.
15	Exhibit, Institution of Inter Partes Review, Smart Modular Technologies, Inc. v. Netlist, Inc., Case No. 4:13-cv-05889-YGR Document 309-2, filed March 17, 2015, 23 pages.
16	Exhibit, Letter from Defendant's Counsel, Netlist v. Smart Storage Systems, Inc. et al., Case 4:13-cv-05889-YGR Document 305-16, filed March 10, 2015, 3 pages.
17	Exhibit, Letter sent via email on December 6, 2013, Case 4:13-cv-03901-YGR Document 53-2, filed January 6, 2014
18	Exhibit, Order Denying Defendant's Motion to Stay Pending Inter Partes Review (Doc.59), The Procter and Gamble Company v. Team Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 316-6, filed March 24, 2015, 10 pages.
19	Exhibit, Patent Public Advisory Committee Quarterly Meeting, Appeals Statistics USPTO, Case No. 4:13-cv-05889-YGR Document 309-8, filed March 17, 2015, 23 pages.
20	Exhibit, Reporter's Transcript of Proceedings, Netlist, Inc v. Smart Modular Technologies, Inc., et al., Case 4:13-cv-05889-YGR Document 316-3, filed March 24, 2015, 15 pages.
21	Exhibit, Transcript of Official Electronic Sound Recording Proceeding, Netlist v. Smart Modular Technologies, Inc, et al., Case 4:13-cv-05889-YGR Document 305-7, filed March 10, 2015, 10 pages.
22	Extended European Search Report for European Application No. 12 817 751.6, dated Jun. 9, 2015.

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23	File History for U.S. Appl. No. 12/240,916, filed Sep. 29, 2008, 320 pages.
24	File History for U.S. Appl. No. 13/905,048, filed May 29, 2013, 181 pages.
25	File History for U.S. Appl. No. 60/941,586, filed Jun. 1, 2007, 23 pages.
26	File History U.S. Pat. No. 8,671,243.
27	Final Office Action, dated Jun. 15, 2016, issued in U.S. Appl. No. 14/489,281, 10 pages.
28	Final Office Action, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-0099-1007 (PTAB), dated February 1, 2012, 13 pages
29	Final Written Decision, Patent 8,671,243, IPR2017-00587-34, Paper No. 34, entered June 20, 2018, 53 pages.
30	Final Written Decision, Patent 8,874,831, IPR2017-00692, Paper No. 25, entered July 5, 2018, 42 pages.
31	Final Written Decision, US Patent No. 7,881,150, Case No. Case IPR2014-00882, (PTAB), Paper 33, filed December 14, 2015, 51 pages.
32	First Amended Complaint for Patent Infringement, Netlist Inc. v Smart Modular Tech. Inc. et al. Case 8:13-cv-00996-DOC-JPR, Document 5, filed Aug. 23, 2013, 99 pages.
33	Hasan, J. et al. Efficient Use of Memory Bandwidth to Improve Network Processor Throughput, Proceedings of the 30th Annual International Symposium on Computer Architecture (ISCA'03), IEEE, 2003, 12 pages.

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34	Horowitz, P. et al. "The Art of Electronics". Cambridge University Press 2nd Ed. 1989, pp. 471-495-496.
35	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.
36	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18), 2003, 100 pages.
37	Inter Partes Review No. IPR2017-00692 (PTAB), Patent No. 8,874,831, filed July 26, 2012, 78 pages.
38	Inter Partes Review of U.S. Patent No. 8,874,831, Case IPR2017-00692 (PTAB), filed July 26, 2012, Paper No. 1, 78 pages.
39	International Preliminary Report on Patentability in PCT/US12/48750, mailed Apr. 3, 2014pp. 1-8.
40	International Search Report and Written Opinion in PCT/US12/48750, dated Oct. 10, 2012pp. 1-10.
41	ISSCC 2006 / SESSION 7 / NON-VOLATILE MEMORY / 7.7, IEEE International Solid-State Circuits Conference, 2006, 10 pages.
42	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322, 58 pages.
43	Jandhyala, S. et al., "Design-For-Test Analysis of a Buffered SDRAM DIMM", Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, August 13-14, 1996, 15 pages.
44	JEDEC Definition of DIMM, Exhibit 1029, IPR No. 2017-00587, December 18, 2017, 2 pages.

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45	JEDEC Global Standard for the Microelectronics Industry, Why JEDEC Standards Matter, 2014, 1 page.
46	JEDEC Standard 21-C "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.
47	JEDEC Standard No. 21-C (Release 17), Annex J: Serial Presence Detects for DDR2 SDRAM (Revision 1.3), 60 pages.
48	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, Aug. 1999, 114 pages.
49	JEDEC Standard, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A), Jan. 2005, 113 pages.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

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Application Number	17138766
Filing Date	2020-12-30
First Named Inventor	Hyun Lee
Art Unit	1731 2135
Examiner Name	Hashem Farrokh
Attorney Docket Number	0016.001000H

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Examiner Name	
Attorney Docket Number	0016.001000H

1	JEDEC Standard, EBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (EBDIMM) Design Specification: JESD205, JEDEC Solid State Tech. Assoc., Mar. 2007, 9 pages.
2	Joint Status Report Regarding Inter Partes Review, <i>Netlist v. Smart Storage Systems, Inc. et. al.</i> , Case No. 4:13-CV-05889-YGR (NDCA), filed August 17, 2018, 4 pages.
3	Material Science & Engineering, Department of Material Science and Engineering, Stanford University, Exhibit 2011, Case No. IPR2017-00692, 2 pages.
4	Merriam-Webster's Collegiate Dictionary, Eleventh Ed., Merriam Webster Corporation, Springfield, MA, 2003, 7 pages.
5	Microsoft Computer Dictionary Fifth Edition, 2002, 3 pages.
6	Microsoft Computer Dictionary Fifth Edition, 2002, 9 pages.
7	Microsoft Press, Computer Dictionary, Second Edition, 1994, 4 pages.
8	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
9	Mutnary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications". IBM Corp. xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.
10	Notice of Allowance in U.S. Appl. No. 12/240,916, dated Sep. 17, 2012.
11	Notice of Allowance in U.S. Appl. No. 13/536,173, dated Jul. 2, 2013.

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12	Notice of Allowance in U.S. Appl. No. 13/559,476, dated May 6, 2014.
13	Notice of Allowance in U.S. Appl. No. 13/559,476, dated Sep. 29, 2014.
14	Notice of Allowance in U.S. Appl. No. 13/905,048, dated Dec. 19, 2013, 8 pages.
15	Notice of Allowance in U.S. Appl. No. 13/905,053, dated Dec. 11, 2013.
16	Notice of Allowance in U.S. Appl. No. 14/173,219 dated Jul. 7, 2014.
17	Notice of Allowance in U.S. Appl. No. 14/489,269, dated Oct. 8, 2015.
18	Notice of Allowance, SanDisk Corporation v. Netlist, Application No. 13/536,173, IPR2014-00982 (PTAB), dated July 2, 2013, 8 pages.
19	Office Action dated Aug. 19, 2016 of the Chinese Patent Application No. 201280047758.X, 9 pages.
20	Office Action in U.S. Appl. No. 12/240,916, mailed on Apr. 3, 2012.
21	Office Action in U.S. Appl. No. 13/536,176, dated on Apr. 15, 2013.
22	Office Action in U.S. Appl. No. 13/625,563, dated Aug. 5, 2013.

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23	Office Action in U.S. Appl. No. 13/625,563, dated May 9, 2014.
24	Office Action in U.S. Appl. No. 13/905,048, dated Aug. 1, 2013.
25	Office Action in U.S. Appl. No. 13/905,053, dated Aug. 1, 2013.
26	Office Action in U.S. Appl. No. 14/173,219, dated Mar. 13, 2014.
27	Office Action in U.S. Appl. No. 14/302,292, dated Dec. 21, 2015.
28	Office Action in U.S. Application No. 12/240,916, dated Feb. 1, 2012, 14 pages.
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30	Office Action, Application 12/240,916, IPR2014-0099-1005 (PTAB), dated July 29, 2011, 8 pages.
31	Office Action, Application 12/240,916, IPR2014-0099-1010 (PTAB), dated April 3, 2012, 11 pages.
32	Office Action, Application No. 13/536,173, IPR2014-00982 (PTAB), dated April 15, 2013, 9 pages.
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35	Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, Patent No. 8,671,243, Case No. IPR2017-00587 (PTAB), filed Jan. 29, 2018, 6 pages.
36	Patent Owner's Opposition to Petitioners' Motion to Exclude, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed April 2, 2018, 2017, 11 pages.
37	Patent Owner's Preliminary Response, Patent Number: 8,516,187, Case IPR2014-01371 (PTAB), filed Dec. 16, 2014, 66 pages.
38	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Inc., Patent 8,301,833, IPR2014-00994 (PTAB), Paper 8, dated October 2, 2014, 60 pages.
39	Patent Owner's Preliminary Response, SanDisk Corporation v. Netlist, Patent Number: 8,516,187, IPR2014-00982 (PTAB), dated September 26, 2014, 57 pages.
40	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,301,833, Case No. IPR2017-00649 (PTAB), filed May 1, 2017, 67 pages.
41	Patent Owner's Preliminary Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692 (PTAB), filed May 1, 2017, 48 pages.
42	Patent Owner's Preliminary Response, Smart Modular Tech v Netlist Inc., Patent No. 8,301,833, Case IPR2014-01370 (PTAB), filed Dec. 16, 2014, 66 pages.
43	Patent Owner's Response, Patent No. 8,671,243, Case No. IPR2017-00587-12 (PTAB), filed Oct. 13, 2017, 80 pages.
44	Patent Owner's Response, SK hynix Inc., et al., v. Netlist, Patent No. 8,874,831, Case No. IPR2017-00692-12 (PTAB), filed Nov. 10, 2017, 77 pages.

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45	Patterson et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998), 71 pages.
46	Petition for Inter Partes Review of Claims 1-30 of US.PATENT No. 8,301,833, IPR2014-01370 (PTAB), filed August 22, 2014, 68 pages.
47	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SanDisk, Corp.), filed Jun. 20, 2014.
48	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed Aug. 22, 2014.
49	Petition for Inter Partes Review of U.S. Pat. No. 8,301,833, filed Sep. 29, 2008.
50	Petition for Inter Partes Review of U.S. Pat. No. 8,516,187 (on behalf of SanDisk, Corp.), filed Jun. 19, 2014.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2021-03-11
Name/Print	Khaled Shami	Registration Number	38745

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

151145 7590 03/24/2021
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

	(Typed or printed name)
	(Signature)
	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	12/30/2020	Hyun Lee	0016.001000H	8804

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	06/24/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
FARROKH, HASHEM	2135	710-308000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 SHAMI MESSINGER PLLC

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 60-2034

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

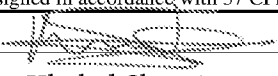
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature 

Date April 23, 2021

Typed or printed name Khaled Shami

Registration No. 38,745

Electronic Patent Application Fee Transmittal

Application Number:	17138766			
Filing Date:	30-Dec-2020			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Chi-She Chen			
Filer:	Khaled Shami/Susanh Perez			
Attorney Docket Number:	0016.001000H			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	42534404
Application Number:	17138766
International Application Number:	
Confirmation Number:	8804
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Chi-She Chen
Customer Number:	151145
Filer:	Khaled Shami/Susanh Perez
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000H
Receipt Date:	23-APR-2021
Filing Date:	30-DEC-2020
Time Stamp:	11:48:37
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E20214MB51461131
Deposit Account	602034
Authorized User	Susanh Perez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	IF_0016001000H.pdf	175227	no	1
			f536f334c83cf7490843936f582f0dabe2a7c eaa		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30113	no	2
			e7f39ebeb81aaf3e89d982b4e61313a05be 5de71		

Warnings:

Information:

Total Files Size (in bytes):	205340
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/138,766	12/30/2020	Chi-She Chen	0016.001000H

CONFIRMATION NO. 8804

PUBLICATION NOTICE



OC000000125250853

151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2021-0124701-A1

Publication Date:04/29/2021

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Issue Classification 	Application/Control No. 17/138,766	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	9	10	19	19	26	28								
2	2	7	11	20	20	29	29								
3	3	10	12	21	21	30	30								
4	4	11	13	22	22										
5	5	15	14	23	23										
8	6	12	15	24	24										
6	7	16	16	27	25										
14	8	17	17	28	26										
13	9	18	18	25	27										

Change(s) applied
 to document,
 /N.B.H./
 4/5/2021

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/HASHEM FARROKH/ Primary Examiner, Art Unit 2135 (Primary Examiner) _____ (Date) 18 March 2021		O.G. Print Claim(s) 1	O.G. Print Figure 12



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/138,766	05/25/2021	11016918	0016.001000H	8804

151145 7590 05/05/2021
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Chi-She Chen, Walnut, CA;
Netlist, Inc., Irvine, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

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AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court EASTERN DISTRICT OF TEXAS MARSHALL DIVISION on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 2:21-cv-463	DATE FILED 12/20/2021	U.S. DISTRICT COURT EASTERN DISTRICT OF TEXAS MARSHALL DIVISION
PLAINTIFF NETLIST, INC.		DEFENDANT Samsung Electronics Co., Ltd., Samsung Electronics America, Inc. Samsung Semiconductor, Inc.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,860,506	12/8/2020	Netlist, Inc.
2 10,949,339	3/16/2021	Netlist, Inc.
3 11,016,918	5/25/2021	Netlist Inc.
4		
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
-------	-------------------	------

Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court **EASTERN DISTRICT OF TEXAS MARSHALL DIVISION** on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 2:21-cv-463	DATE FILED 12/20/2021	U.S. DISTRICT COURT EASTERN DISTRICT OF TEXAS MARSHALL DIVISION
PLAINTIFF NETLIST, INC.		DEFENDANT Samsung Electronics Co., Ltd., Samsung Electronics America, Inc. Samsung Semiconductor, Inc.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,860,506	12/8/2020	Netlist, Inc.
2 10,949,339	3/16/2021	Netlist, Inc.
3 11,016,918	5/25/2021	Netlist Inc.
4		
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been
 filed in the U.S. District Court for the Eastern District of Texas - Marshall Division on the following
☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO.	DATE FILED 6/10/2022	U.S. DISTRICT COURT for the Eastern District of Texas - Marshall Division
PLAINTIFF Netlist, Inc.		DEFENDANT Micron Technology, Inc., et al
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,860,506	12/8/2020	Netlist, Inc.
2 10,949,339	3/16/2021	Netlist, Inc.
3 11,016,918	5/25/2021	Netlist, Inc.
4 11,232,054	1/25/2022	Netlist, Inc.
5 8,787,060	7/22/2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED 6/10/2022	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☒ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 9,318,160	4/19/2016	Netlist, Inc.
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
-------	-------------------	------

Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00996
Patent 11,016,918 B2

Before JON M. JURGOVAN, DANIEL J. GALLIGAN, and
NABEEL U. KHAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314

I. INTRODUCTION

Samsung Electronics Co., Ltd. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) to institute an *inter partes* review of claims 1–30 of U.S. Patent 11,016,918 B2, issued on May 25, 2021 (Ex. 1001, “the ’918 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition.

Institution of an *inter partes* review is authorized when “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a).

Based on the current record, and for the reasons explained below, we determine that Petitioner has established a reasonable likelihood that it would prevail with respect to at least one challenged claim, and we institute an *inter partes* review as to all the challenged claims and grounds raised in the Petition.

II. BACKGROUND

A. *Real Parties in Interest*

Petitioner identifies Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. as the real parties in interest. Pet. 1. Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

B. *Related Matters*

The parties advise that the ’918 patent is related to *Samsung Electronics Co., Ltd., et al. v. Netlist, Inc.*, IPR2022-00999 (U.S. Patent No. 11,232,054 B2; *Netlist, Inc. v. Samsung Electronics Co., Ltd., et al.*, Case No. 2-21-cv-00463 (E.D. Tex.); *Samsung Electronics Co., Ltd. v. Netlist, Inc.*, Case No. 1:21-cv-01453 (D. Del.); and U.S. Appl. No. 17/582,797. Pet. 1; Paper 4, 1.

The parties advise that the ’918 patent is related to the following legal proceeding, which is no longer pending: *SK hynix Inc. v. Netlist, Inc.*, IPR2017-00692 (U.S. Patent No. 8,874,831 B2). Pet. 1; Paper 4, 1.

C. The '918 Patent (Ex. 1001)

The '918 patent is titled "Flash-DRAM Hybrid Memory Module." Ex. 1001, code (54). Figure 12 of the '918 patent is reproduced below.

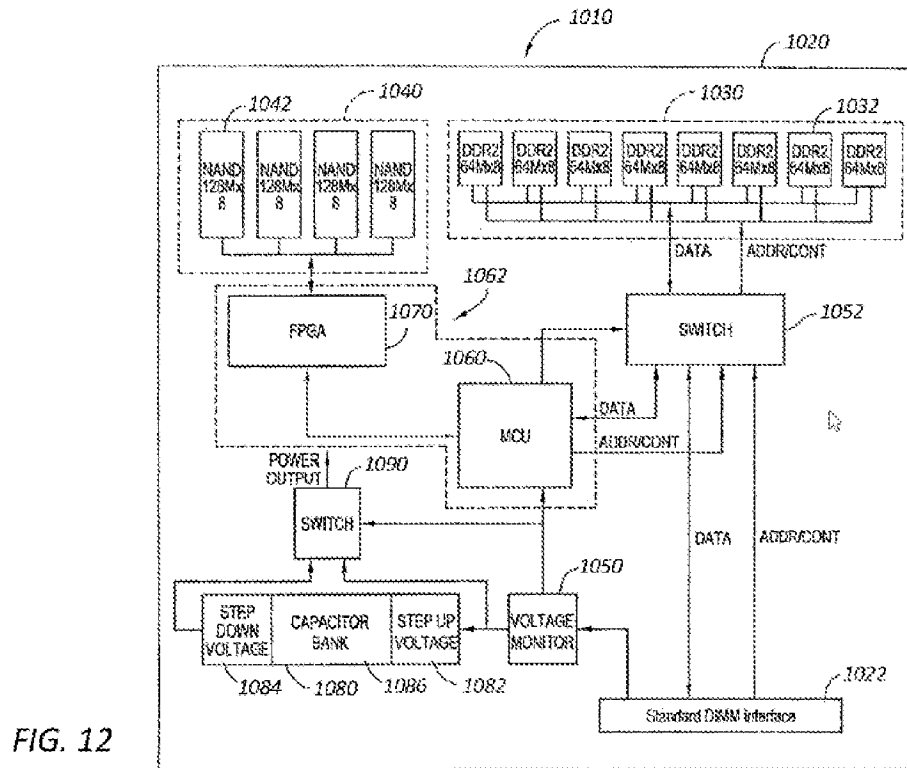


FIG. 12

Figure 12 shows an example memory system 1010 of the '918 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and selectively coupled to the non-volatile memory subsystem 1040. *Id.* at 21:16–22. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level

cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board (PCB) 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022 for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. *Id.* at 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '918 patent refers to this as the “first state.”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '918 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g. a system power supply) when the memory system 1010 is in the first state and from a second

power supply 1080 when the memory system 1010 is in the second state. *Id.* at 25:54–58. Second power supply 1080 may comprise step-up transformer 1082, step-down transformer 1084, and capacitor bank 1086 with one or more capacitors. *Id.* at 26:3–13.

The memory system 1010 further has a third state in which controller 1062 is decoupled from the volatile memory system 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). *Id.* at 25:62–69. “[T]he third power supply may provide power to the volatile memory subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.” *Id.* at 25:66–26:3.

Figure 16 of the '918 patent is reproduced below.

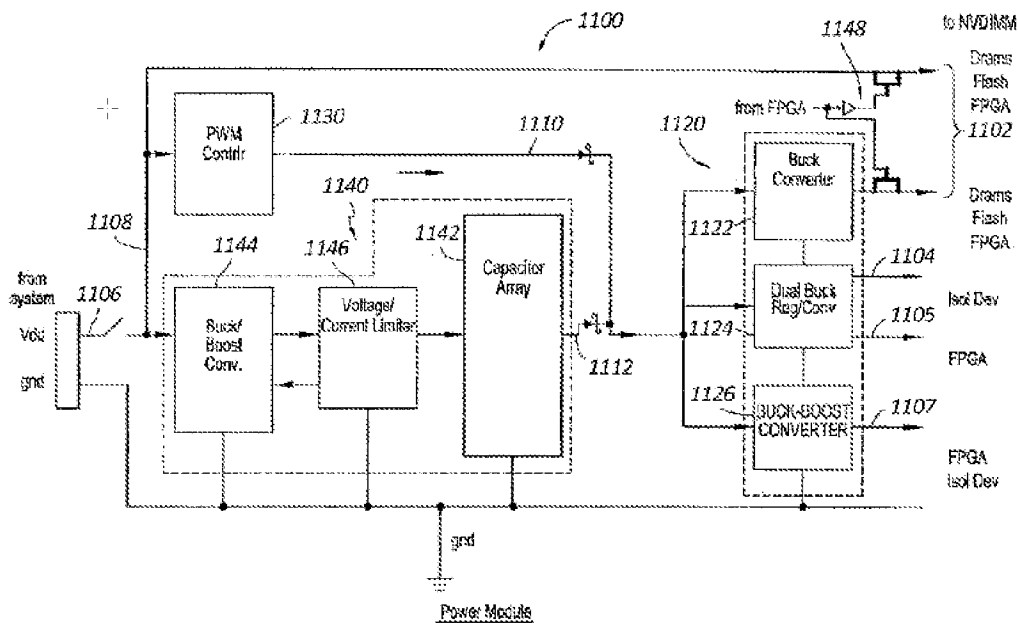


FIG. 16

Figure 16 shows power module 1100 of memory system 1010. *Id.* at 27:59–61. Power module 1100 comprises conversion element 1120, first power element 1130, and second power element 1140. *Id.* at 28:7–15,

28:20–22. Conversion element 1120 comprises sub-blocks 1122, 1124, 1126 comprised of various converter circuits such as buck converters, boost converters, and buck-boost converters for providing various voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 29:18–54. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. “[P]ower module 1100 provides power to [] components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition.” *Id.* at 27:61–65.

Specifically, “[i]n a first state, first voltage 1102 is provided to memory system 1010 from input 1106 and fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130.” *Id.* at 28:27–31. “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” *Id.* at 28:32–37. “In the third state, the fifth voltage 1112 is provided to conversion element 1120 from second power element 1140 and the first voltage 1104 is provided to memory system 1010 from conversion element 1120.” *Id.* at 28:34–38. Transition from the first state to the second state may occur when power module 1100 detects a power failure is about to

occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

D. Illustrative Claim

Of the challenged claims, claims 1, 16, and 23 are independent. Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;

[1.d] a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;

[1.e] a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;

[1.f] a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and

[1.g] a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:

[1.h] a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and

[1.i] [1.i.1] at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, [1.i.2] the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, [1.i.3] the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage, [1.i.4] wherein a first one of the second and fourth voltage

amplitudes is less than a second one of the second and fourth voltage amplitudes.

Ex. 1001, 38:19–44.

E. Evidence

Petitioner relies on the following references (*see* Pet. 3, 10–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published August 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published Apr. 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued Feb. 15, 2005

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 4):

Claims Challenged	35 U.S.C. §	References
1–3, 8, 14, 15, 23	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

III. ANALYSIS OF CHALLENGED GROUNDS

We turn now to Petitioner's asserted grounds of unpatentability and Patent Owner's arguments in its Preliminary Response to determine whether Petitioner has met the threshold standard of 35 U.S.C. § 314(a).

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations.¹ *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see also KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere

¹ Neither party has identified any objective evidence of nonobviousness on the record to be considered at this time.

conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic*, 815 F.3d at 1363 (citing 35 U.S.C. § 312(a)(3)); *see also Intelligent Bio-Sys.*, 821 F.3d at 1369 (“It is of the utmost importance that petitioners in the IPR proceedings adhere to the requirement that the initial petition identify ‘with particularity’ the ‘evidence that supports the grounds for the challenge to each claim.’” (quoting 35 U.S.C. § 312(a)(3))). Therefore, to prevail in an *inter partes* review, Petitioner must explain how the proposed combinations of prior art would have rendered the challenged claims unpatentable. At this preliminary stage, we determine whether the information presented in the Petition shows there is a reasonable likelihood that Petitioner would prevail in establishing that at least one of the challenged claims would have been obvious over the proposed combinations of prior art.

A. *Level of Ordinary Skill in the Art*

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 8–9 (citing Ex. 1003 ¶ 67). Petitioner contends that additional training can substitute for educational or

research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a computer system, including standard communication busses and protocols, such as PCI and SMBus busses and protocols. *Id.* at 8–9. Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* at 9. Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

In the Preliminary Response, Patent Owner applies the level of ordinary skill in the art proposed by Petitioner. Prelim. Resp. 6.

Among the factors that may be considered in determining the level of ordinary skill in the art are “the type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995) (citing *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962–63 (Fed. Cir. 1986)). The evidence presented mostly applies to the educational level of workers in the field, but also touches upon other factors

as well. We find Petitioner’s proposal is consistent with the level of ordinary skill in the art reflected by the ’918 patent and the prior art of record, and we, therefore, adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least,” for purposes of this Decision.

B. Claim Construction

We construe each claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent,” the same standard used to construe the claim in a civil action. 37 C.F.R. § 42.100(b).

Petitioner contends that “no express constructions are needed for this proceeding” but notes that Patent Owner has broadly interpreted some claim terms for purposes of infringement even though a narrower interpretation may be more reasonable. Pet. 9 (citing Ex. 1071, 39–46; Ex. 1073, 46–63; Ex. 1003 ¶ 128). Petitioner further contends it is not necessary to determine whether claim terms are governed by § 112, 6th paragraph because at least one of the prior art combinations matches the disclosure of the ’918 patent. *Id.*

At this stage in the proceeding, we need only construe the claims to the extent necessary to determine whether to institute *inter partes* review. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) (“The Board is required to construe ‘only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))). We determine that, at this stage of this proceeding, there is no need to expressly construe any claim terms in order to determine

whether or not to institute review because the parties do not dispute the claimed terms' meanings, and their ordinary and customary meanings suffice for our analysis. However, we do invite the parties to construe the term “pre-regulated voltage” at trial for reasons explained in Section III.D.4, *infra*.

C. Ground 1: Obviousness Over Harris and FBDIMM Standards

Petitioner contends claims 1–3, 8, 14, 15, and 23 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–51. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's arguments and, therefore, establishes a reasonable likelihood of prevailing with respect to this ground at this stage of the proceeding.

1. Harris (Ex. 1023)

Harris is titled “Voltage Distribution System and Method for a Memory Assembly.” Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 10.

Harris's Figure 1A is reproduced below.

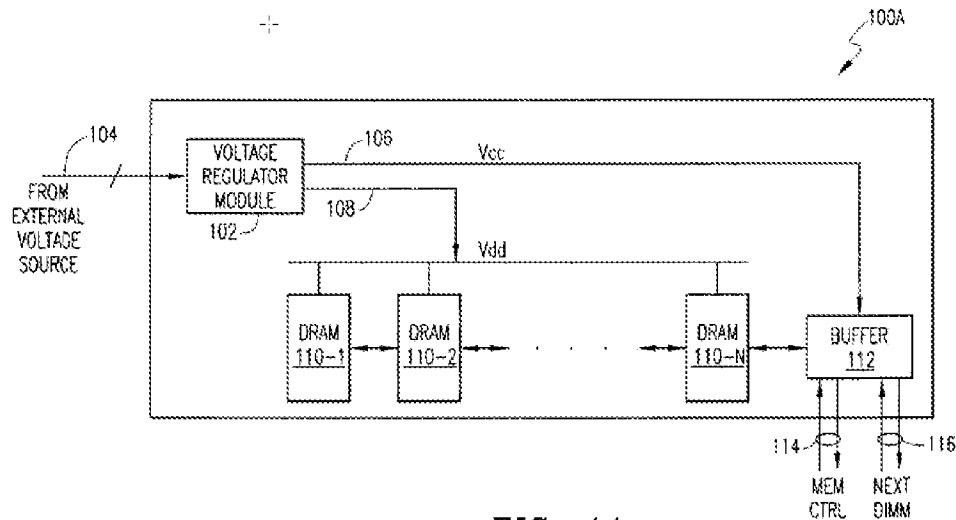


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface 116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Modules (DIMM) wherein each of the memory devices 100-1 to 100-N comprises a Double Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. FBDIMM Standards (Exs. 1027, 1028)

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029 ¶¶ 134–137. Petitioner refers to these standards

collectively as the “FBDIMM Standards.” Pet. 11. Petitioner contends the FBDIMM Standards are prior art under § 102(a). *Id.*

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DD}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	0.453* V_{DD}	0.5* V_{DD}	0.547* V_{DD}	(DRAM Interface V_{TT}) This supply should track as 0.5 * 1.8 volt supply
	3.0	3.3	3.6	(V_{DDSPD})
Buffer interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM interface	SSTL_18			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.
Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)
An Unloaded system should supply V_{TT} of 0.48*V_{DD}/0.52*V_{DD} to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
RESET [1]		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM.
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	I	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground.
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by "a" are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies V_{CC} , V_{CCFBD} , V_{DD} , V_{SS} , and V_{DDSPD} .

3. Motivation to Combine

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16. Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and thus, would naturally look to them for more details about the “fully buffered DIMM” that Harris describes as compatible with his on-board voltage regulator module (VRM). *Id.*

Patent Owner does not dispute Petitioner's motivation to combine Harris and the FBDIMM Standards. *See* Prelim. Resp.

Petitioner has sufficiently shown that a person of ordinary skill in the art would have combined Harris and the FBDIMM Standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶ 9. We agree that one of ordinary skill in the art would have recognized this as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9. Petitioner has shown sufficiently that one of ordinary skill in the art would have combined Harris and the FBDIMM Standards with a reasonable expectation of success.

4. *Analysis of Independent Claim 1*

a) *Limitation 1.a: "A memory module comprising:"*

Petitioner asserts that Harris's memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed "memory module." Pet. 19–20 (citing Ex. 1023 ¶¶ 9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 217–223; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1, for example, in Figure 3.

Patent Owner does not specifically respond to Petitioner's contention for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Harris teaches the preamble for purposes of institution.²

b) Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. Petitioner contends that Harris and the FBDIMM standard disclose printed circuit boards (PCBs). Pet. 20 (citing Ex. 1023 ¶ 13, Figs. 1A, 3). Petitioner also contends that a PCB may be referred to as a ‘memory board’ or ‘raw card.’ *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 223–225).

As for the PCB “having an interface configured to fit into a corresponding slot connector of a host system,” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 227–228).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “configured to couple power, data, address and control signals between the memory module and

² Neither party argues whether the preamble limits claim 1. Although we find that the evidence supports that the prior art teaches the preamble, we make no determination at this stage of the proceeding that the preamble of claim 1 is limiting.

the host system.” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A. *Id.* at 21 (citing Ex. 1023 ¶¶ 10, 12, 19). Petitioner contends that Harris’s buffer 112 in Figure 1A is called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 23–24. Petitioner indicates that Harris’s buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Figure 1A. Pet. 22–25 (citing Ex. 1023 ¶ 9 (“buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data space at least for a portion of memory devices 110-1 through 110-N”). In addition, Petitioner argues that the FBDIMM Standards indicate that buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

Patent Owner argues that Petitioner has not made a *prima facie* case that Harris discloses a memory module having a PCB interface that receives power from the host system. Prelim. Resp. 14–20. Harris states, however, that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Harris also discloses that “external voltage sources may comprise any combination of *known* or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14 (emphasis added). Patent Owner does not argue that voltage supplied by a host system is not a “known” voltage supply as referenced by Harris. Furthermore, Petitioner indicates that the FBDIMM Standards show that the buffer AMB may be connected to a host, suggesting that the FBDIMM may derive its power from the host. Pet. 24 (showing figure at Ex. 1027, 4). These facts point to the conclusion that Harris’s

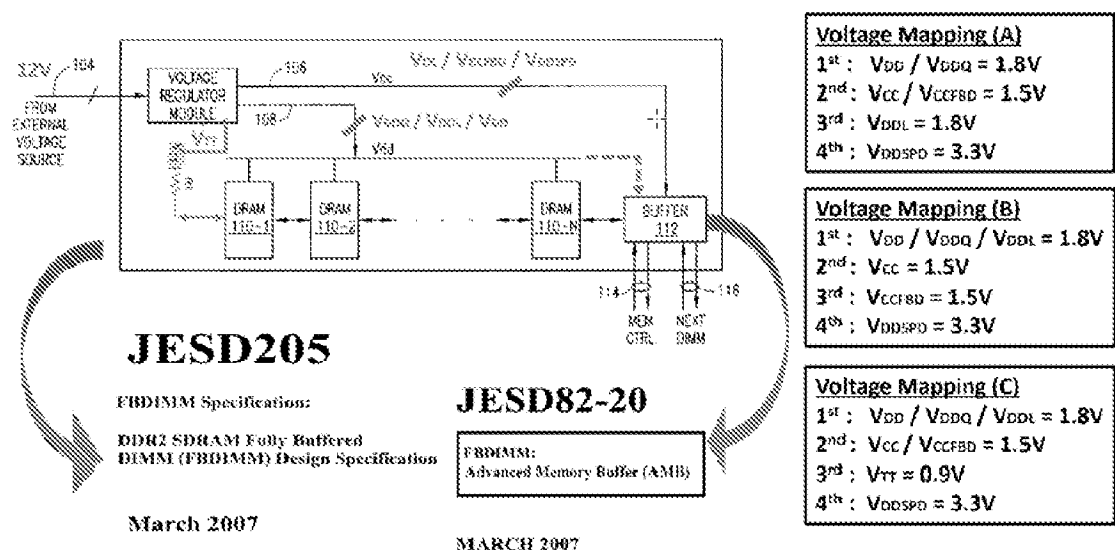
external voltage source may be the host system notwithstanding Patent Owner's arguments to the contrary.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

c) Limitations 1.c to 1.f: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude;”

Petitioner contends that Harris and the FBDIMM Standards disclose limitations 1.c to 1.f. Pet. 26–31. Petitioner relies on an annotated version of Harris's Figure 1A, shown below.

Ground 1: Harris with JEDEC's FBDIMM Standards



Above is Harris's Figure 1A annotated with information one would have obtained from the FBDIMM Standards according to Petitioner. Pet. 26 (citing Pet. 14–19; Ex. 1003 ¶¶ 232–243, 250–257, 262–276, 281–287). Specifically, the above figure shows voltage mappings A, B, and C of four voltages specified by Harris and the FBDIMM Standards. Petitioner contends the voltage range described in Harris (0.5V–3.5V) is the same range as the voltages described in the FBDIMM Standards. *Id.* at 27 (citing Ex. 1023 ¶ 9). Petitioner contends the voltages would be “regulated” because Harris indicates “tightly regulated power” is a problem to be solved, and proposes “at least one on-board voltage regulator” that is “capable of generating tightly-controlled voltage levels” as the solution. *Id.* at 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 234).

Petitioner further contends that Harris teaches using “buck converters” to provide the four regulated voltages. *Id.* (citing Ex. 1003 ¶¶ 236, 237, 252, 264, 286). Specifically, Petitioner contends that Harris discloses “replacing [the prior art] power supply interface pins with as few as six +12V pins (from an external voltage source)” and then using “a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels” to provide each needed on-board regulated voltage. *Id.* (alteration by Petitioner; emphasis omitted; citing Ex. 1023 ¶¶ 10, 12). Petitioner contends that a “buck converter” was a conventional device for implementing such a “voltage-reducing switching converter.” *Id.* (emphasis omitted; citing Ex. 1030, 2:41–43). Although the claim states that a converter circuit generates the fourth voltage, Petitioner notes that a buck converter would be an obvious way to implement a converter circuit. *Id.* at 28, n.2. Petitioner further contends a buck converter would have been an obvious way to step

down a higher input voltage of 12V to a lower output voltage of 3.5V or less, and that there would have been a reasonable expectation of success because buck converters were well-known switching devices commonly used to step down the voltage between input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 146–149, 236–237; Ex. 1058, 3, 5, 12–16; Ex. 1032, 161, 164; Ex. 1024, Fig. 6; Ex. 1050, 1:21).

Petitioner further notes that “buck converters” were well-known as a highly-efficient way to step down voltages without generating excess heat or requiring large cooling devices, providing further motivation to use buck converters. *Id.* at 29–30 (citing Ex. 1003 ¶ 237; Ex. 1040, 1, 23, 24, Figs. 22–25; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1058, 5; Ex. 1059, 5:23–30; Ex. 1062, 11; Ex. 1064 ¶ 101).

Petitioner further contends that it would have been obvious to use at least four converters in Harris given the need for four different voltages (e.g., 0.9V, 1.5V, 1.8V, 3.3V) in the FBDIMM Standards. *Id.* at 30 (citing Pet. 14–19). Petitioner asserts that it would have been obvious to use four converters because the voltages at the same level in Petitioner’s voltage mappings A and B are described as separate voltages with separate pins that are separately controllable in the FBDIMM Standards. *Id.* at 30–31 (citing Ex. 1028, 17–20, 30–32; Ex. 1026, 9; Ex. 1003 ¶¶ 242, 256; Ex. 1062, 13).

Patent Owner argues that Harris requires at most two or three buck converters to provide the voltages needed and thus, does not disclose the four claimed converters. Prelim. Resp. 20–25. Specifically, Patent Owner contends that Harris discloses a single converter generating two regulated voltages, so Harris does not disclose four converters as claimed. Prelim. Resp. 21–22. Petitioner showed sufficiently that the FBDIMM Standards

mentioned in Harris call for at least four voltages, and that given Harris's teaching of a converter, it would have been obvious to one of ordinary skill in the art to use multiple converters, including well-known buck converters, to generate the four voltages needed. Pet. 26–31. Patent Owner's argument thus, appears to be an attack on Harris alone without considering what the combination of Harris and the FBDIMM Standards would have suggested to a person of ordinary skill in the art. "Non-obviousness cannot be established by attacking references individually where the rejection is based upon the teachings of a combination of references." *In re Merck & Co.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981)).

Patent Owner further contends that Petitioner has not made out a case to use two or more buck converters to provide voltages having the same level. Prelim. Resp. 25–29. Petitioner explained sufficiently, however, that the FBDIMM Standards identify V_{DD} , V_{DDQ} and V_{DDL} as well as V_{CC} and V_{CCFBD} as separate voltages of the same level with separate pins that can be turned on and off independently of one another. Pet. 30–31. Petitioner contends this provides independence for the power supplies with improved stability and flexibility for power management. *Id.* at 31. Petitioner also relies on voltage mapping C (Pet. 27) which has different voltage levels for the four voltages used, so Patent Owner's argument, even if correct, would not negate Petitioner's showing with respect to this voltage mapping.

Patent Owner contends that Petitioner did not make the case that a person of ordinary skill in the art would have used a third buck converter, as opposed to a linear regulator, to provide termination voltage V_{TT} . Prelim. Resp. 29–33. Petitioner explained sufficiently that "buck converters" were

well-known as a highly-efficient way to step down voltages without generating excess heat or requiring large cooling devices, providing further motivation to use buck converters. Pet. 29–30.

Patent Owner contends that Petitioner does not explain why a person of ordinary skill in the art would have generated the voltage V_{TT} on the module as opposed to obtaining V_{TT} from interface pins. Prelim. Resp. 29–30. Patent Owner similarly argues that Petitioner does not explain why one of ordinary skill in the art would have generated V_{DDSPD} on the module instead of continuing to use the regulated voltage supplied via the V_{DDSPD} pin on a standard DIMM. Prelim. Resp. 33–34.

Harris’s Figure 1A shows that the voltages V_{cc} and V_{dd} are generated on the module. On this record, we agree with Petitioner that it would logically follow to generate V_{TT} on the module using the same voltage regulator module 102 as used to generate voltages V_{cc} and V_{dd} . *See* Pet. 15–19. In addition, “when there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp.” *KSR*, 550 U.S. at 421. Here, there are only two options—generate the voltage V_{TT} on the module, as Petitioner indicates, or obtain the voltage V_{TT} from interface pins. Petitioner’s choice of the former of the two options does not negate its showing of obviousness.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches limitations 1.c to 1.f of claim 1 for purposes of institution.

d) Limitation 1.g: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:”

Petitioner asserts that Harris discloses a “a plurality of components coupled to the PCB” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 288–90, 293–297). Petitioner further contends that these components are each coupled to at least one or more of the second, third and fourth regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches this limitation for purposes of institution.

e) Limitation 1.h: “a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 298–303). Petitioner contends that a person of ordinary skill in the art would have known that according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* (citing Ex. 1028, 9; Ex. 1045, cover; Ex. 1026, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMMs Standards teaches this limitation for purposes of institution.

f) Limitation 1.i.1: “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices”

Petitioner contends that the “first circuit” corresponds to Harris’s buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–34 (citing Pet. 20–25; Ex. 1003 ¶¶ 304–317). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8. *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

g) Limitation 1.i.2: “the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge

connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices”

Petitioner relies on the same showing for limitation 1.i.2 as for limitation 1.i.1. Pet. 32–34.

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris and the FBDIMM Standards teaches this limitation for purposes of institution.

h) Limitation 1.i.3: “the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34–35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 318–323). Specifically, Petitioner contends that Harris’s buffer 112 in Figure 1A is coupled to “the second regulated voltage” (e.g., V_{CC} or $V_{CCFBD}=1.5V$) and “the fourth regulated voltage” (e.g., $V_{DDSPD}=3.3V$). *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards teaches this limitation for purposes of institution.

i) Limitation 1.i.4: “wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.”

Petitioner contends that “one of the second and fourth voltages” (e.g., 1.5V) “is less than a second one of the second and fourth voltages” (e.g. 3.3V). Pet. 35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 324–327).

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards teaches this limitation for purposes of institution.

j) Determination for Claim 1

Petitioner shows sufficiently that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown sufficiently that the combination of Harris and the FBDIMM Standards teaches each limitation of claim 1. Accordingly, Petitioner has established a reasonable likelihood to prevail in showing that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. Claims 2, 3, 8, 14, and 15

Claim 2 depends from claim 1 and recites “wherein the first and third buck converters are further configured to operate as a dual buck converter.” EX. 1001, 38:53–55. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 35–40. Specifically, Petitioner contends that at the time there were many commercially available

products that could output two regulated voltages using buck converters, and thus, it would have been obvious to implement two or more of the regulated voltages such as the “first” and “third” voltages as a “dual buck converter” to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design. Pet. 36 (Ex. 1003 ¶ 338).

Petitioner contends that commercially available devices included the Murata MPD4S014S dual buck converter (EX. 1042, 6; Ex. 1048, 1–2; Ex. 1058, 5), the Texas Instruments TPS51020 Dual Step-Down Controller (Ex. 1040, 1, 11) and the Fairchild Semiconductor FAN5026 Dual-Output PWM Controller” (Ex. 1041, 1, 2, 9). *See* Pet. 36–40.

Patent Owner presents no argument specific to claim 2. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards renders claim 2 obvious for purposes of institution.

Claims 3 depends from claim 1 and recites “wherein the first voltage amplitude is 1.8 volts.” Ex. 1001, 38:56–57. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 41 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 348–353).

Patent Owner presents no argument specific to claim 3. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards renders claim 3 obvious for purposes of institution.

Claim 8 depends from claim 1 and recites

the plurality of components further comprising:
one or more registers coupled to one of the first, second, third and fourth regulated voltages, the one or more registers configured to register, in response to a clock, the first plurality of address and control signals, wherein the one of the first, second, third and fourth regulated voltages is selectively switched off to turn power off to the one or more registers while one or more components of the plurality of components are powered on.

Ex. 1001, 39:9–19.

Petitioner contends the combination of Harris and the FBDIMM Standards discloses claim 8. Pet. 41–46. Specifically, Petitioner contends that Harris’s buffer 112 is a component that has registers to register incoming signals and output them according to a “clock” signal, as shown in Harris’s Figures 1A and 3. *Id.* at 41. Petitioner contends Harris’s buffer 112 is coupled to the voltage V_{CC} or V_{CCFBD} as the “second regulated voltage.” *Id.* (citing Pet. 14–19, 27). Petitioner contends Harris discloses “buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data” which a person of ordinary skill in the art would understand involves registers for registering the address, control, and data signals in response to a “clock” signal. *Id.* at 42 (citing Pet. 22–25; Ex. 1023 ¶¶ 9, 17, Fig. 3; Ex. 1003 ¶¶ 400–402; Ex. 1027, 19) (emphasis omitted).

Petitioner further contends that the combination of Harris and the FBDIMM Standards discloses an S3 sleep mode that selectively switches off the second voltage to turn off the power to the one or more registers while the DRAMs remain powered on to refresh data in the DRAMs. Pet. 45 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 410–411, 414–420; Ex. 1012 ¶ 9; Ex. 1027, 21, 39).

Patent Owner presents no arguments specific to claim 8. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards renders obvious claim 8 for purposes of institution.

Claim 14 depends from claim 8 and recites

wherein, in response to selectively switching on the one of the first, second, third and fourth regulated voltages to the one or more registers, the one or more registers is configured to output the registered first plurality of address and control signals to the plurality of SDRAM devices.

Ex. 1001, 39:40–45.

Petitioner contends that when the S3 sleep mode ends, normal DRAM transactions begin again and the second voltage which powers the input/output registers switches on and the registers output the registered address and control signals to the SDRAM devices. Pet. 46 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 462–466).

Patent Owner presents no argument specific to claim 14. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards renders claim 14 obvious for purposes of institution.

Claim 15 depends from claim 1 and recites

the plurality of components further comprising:
a logic element including one or more integrated circuits
and discrete electrical elements, the one or more integrated

circuit including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information.

Ex. 1001, 39:46–52.

Petitioner contends that the combination of Harris and the FBDIMM Standards discloses claim 15. Pet. 47–50. Specifically, Petitioner contends logic in Harris’s buffer includes an integrated circuit, as does its serial presence detect (SPD), including non-volatile memory. Pet. 47 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 435–437, 468–476). Petitioner further contends the logic element includes discrete elements such as resistors and capacitors to terminate voltages for Harris’s buffer. *Id.* Petitioner asserts that the S3 sleep mode of the FBDIMM Standards requires S3 Recovery Configuration Registers. *Id.* According to Petitioner, an FBDIMM like Harris’s memory module will store configuration information in non-volatile memory before entering into S3 sleep mode. *Id.* (citing Ex. 1027, 25, 95, 96, 141).

Petitioner further contends that a person of ordinary skill in the art would have understood that the “non-volatile memory” can be implemented in the SPD device separate from the integrated circuit implements the AMB (Advanced Memory Buffer) where the SPD is used to store configuration information in the non-volatile memory. *Id.* at 48–49 (citing Ex. 1023 ¶ 19; Ex. 1027, 117; Ex. 1028, 13; Ex. 1066, 26:64–27:4; Ex. 1067, p.1-1; Ex. 1003 ¶ 436). Petitioner further contends a person of ordinary skill in the art would have understood that the logic element includes discrete electrical elements such as resistors and capacitors, as taught by the FBDIMM Standards. *Id.* at 49–50 (citing Ex. 1003 ¶ 475; Ex. 1023 ¶ 9, Fig. 1A; Ex. 1028, 13, 42–45).

Patent Owner presents no argument specific to claim 15. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that the combination of Harris with the FBDIMM Standards renders claim 15 obvious for purposes of institution.

6. *Claim 23*

Claim 23 is an independent claim. Ex. 1001, 40:50–41:21. Petitioner contends that the limitations of claim 23 are substantially identical to earlier limitations, and are thus obvious for the reasons stated above. Pet. 50–51.

Patent Owner presents no additional argument for claim 23 other than arguments previously discussed. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Harris in combination with the FBDIMM Standards renders obvious claim 23.

7. *Determination for Ground 1*

Petitioner has shown a reasonable likelihood to prevail in showing unpatentability of claims 1–3, 8, 14, 15, and 23 as obvious over the combination of Harris and the FBDIMM Standards for the reasons explained.

D. *Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi*

Petitioner contends that claims 1–30 of the '918 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 41–70. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's

arguments and, therefore, establishes a reasonable likelihood of prevailing with respect to this ground at this stage of the proceeding.

1. *Amidi (Ex. 1024)*

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled “Clock and Power Fault Detection for Memory Modules.” Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 12.

Amidi’s Figure 5 is reproduced below.

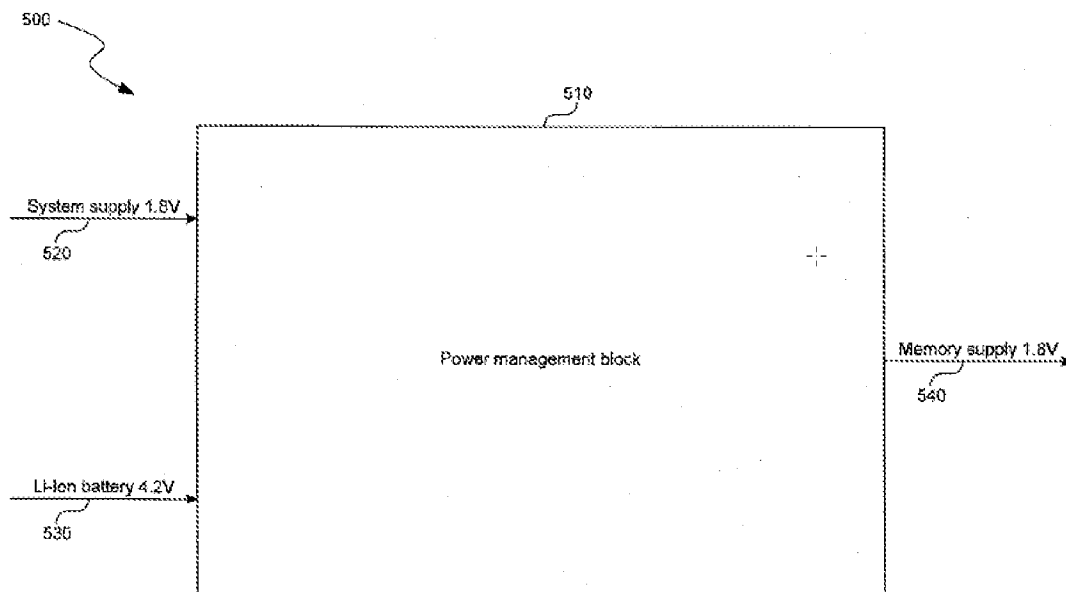


FIG. 5

Amidi’s Figure 5 above illustrates a power management block 510 that receives an incoming system supply 520, and incoming battery supply 530, and generates an outgoing memory power supply 540 which is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Figs. 5, 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 52. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 52–53 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 171–177). Petitioner further notes that Amidi’s power management block could be modified easily to work with Harris’s FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 53 (citing Ex. 1024, Figs. 5, 6; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 173–174). Petitioner contends that to a person of ordinary skill in the art it would have been obvious to use the 12V external supply stepped-down with a buck converter to a 5V supply for charging Amidi’s battery, and that Amidi’s battery voltage would be stepped-up with a boost converter to the 12V level used by Harris’s memory module. *Id.* at 53–54 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 171–172). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed, and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 54 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi’s battery backup mode is similar to the S3 power-saving mode of Harris’s FBDIMM memory

module and the FBDIMM standards, such that a person of ordinary skill in the art would have been motivated to combine their teachings. *Id.* at 54–56.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:

Ground 2: Ground 1 and Battery Backup of Amidi

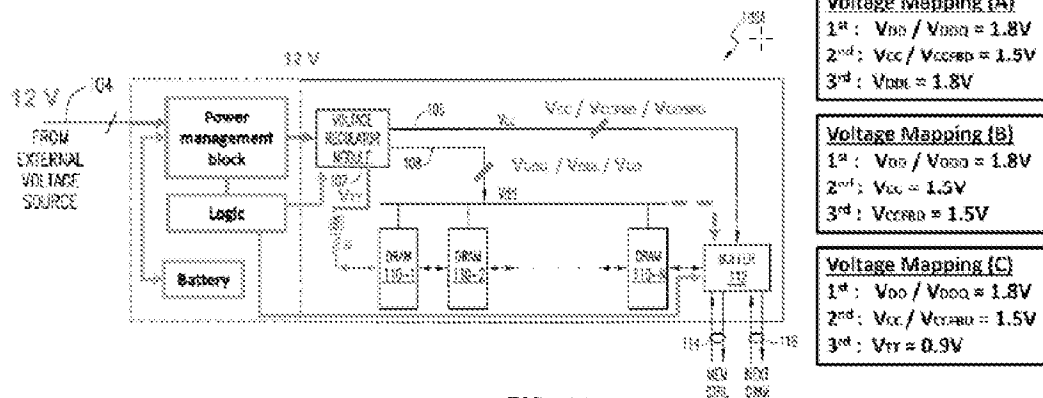


FIG. 14

Pet. 56. Above, Petitioner has annotated Harris's Figure 1A to show features added from the teachings of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue).

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Prelim. Resp. 35–39. Harris does not appear to describe, however, switching to an alternate voltage source in response to power loss, nor does it explicitly mention a battery as an alternate voltage source, whereas Amidi does. Ex. 1024, code (57). Thus, Patent Owner’s argument does not undermine Petitioner’s motivation to combine the references.

Patent Owner also argues that “Petitioner has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the

combination would even fit onto Harris’s circuit board, particularly where Petitioner has already proposed to add additional converters.” Prelim. Resp. 38–39. Patent Owner’s declarant, Dr. Sunil Khatri, states that space on the board to fit all of the components that Patent Owner proposes would have been a concern. Ex. 2001 ¶¶ 78, n.3, 79. As explained above, however, Petitioner argues that one of ordinary skill would have had reasonable expectation of success in making the combination and supports its argument with credible testimony from Dr. Wolfe who explains how one of ordinary skill would have been able to incorporate Amidi’s functionality in the FBDIMM memory module of Harris and that such modification would have been well within the level of skill at the time. Ex. 1003 ¶¶ 170–179.

Accordingly, we determine Petitioner has adequately shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3, 8, 14, 15, and 23*

We agree with Petitioner that the addition of Amidi does not negate Petitioner’s showing with respect to the previous ground. Pet. 56, 66–67 (citing Ex. 1003 ¶¶ 215–353, 412–413, 467–476, 557–597). Accordingly, Petitioner has shown a reasonable likelihood to prevail in demonstrating claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claims 16–22 and 30*

Claims 16–22 recite a “pre-regulated input voltage.” Ex. 1001, 39:61, 42:23. Petitioner identifies at least two meanings for this claim term, one meaning within pre-determined limits and the other meaning the voltage

must be pre-regulated on the memory board itself. Pet. 73. For purposes of this decision, we interpret “pre-regulated voltage” to mean that the voltage is regulated before conversion to a stepped up or down level. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). Patent Owner further argues that Amidi does not disclose a “pre-regulated voltage” but offers no interpretation for the term. Prelim. Resp. 39–40. We invite development of the record at trial to explain the meaning of “pre-regulated input voltage.” In any case, we find Petitioner’s showing of obviousness with respect to these claims sufficient for institution.

5. *Claims 5–7, 9–13, 16–22, and 24–27*

Claims 5–7, 9–13, 16–22 and 24–27 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being greater than a predetermined threshold voltage. Ex. 1001, 38:61–67, 40:7–13, 41:22–27, 42:21–26. Claims 5–7, 9–13, 16, 17, 20–22, and 24 require generation of a trigger signal, and we determine that Petitioner sufficiently shows for institution that these claims would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Claims 18, 19, 25, and 26 require that the trigger signal causes a controller to execute a write operation. We determine that Petitioner’s showing of obviousness is sufficient for these claims for purposes of institution, although we invite further explanation of the parties’ views of the combination of Harris, the FBDIMM Standards, and Amidi at trial.

6. *Determination for Ground 2*

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and

Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations of at least one claim in this ground are taught or suggested by the combination. Accordingly, under this ground, Petitioner has demonstrated a reasonable likelihood to prevail in showing unpatentability of at least one claim of the '918 patent as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

E. Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck

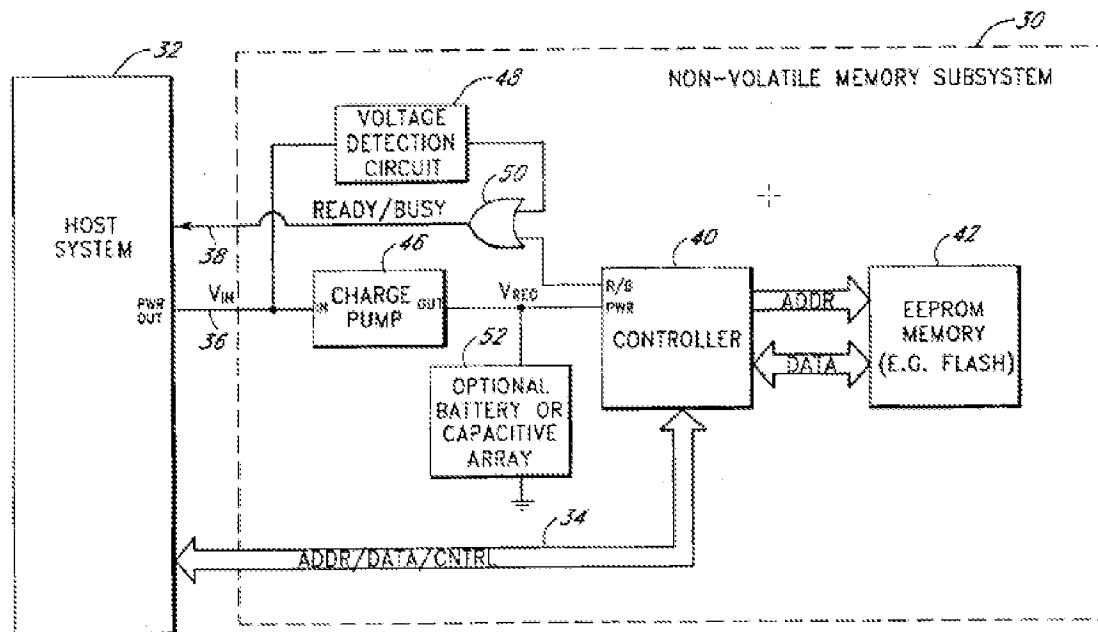
Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck. Pet. 75–77.

1. Hajeck (Ex. 1038)

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1001, code (54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck’s Figure 1 is reproduced below.



In Hajeck's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage, including undervoltage and overvoltage conditions, and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–43.

2. Motivation to Combine

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajeck with Harris, the FBDIMM Standards, and Amidi because one would have appreciated the desirability of switching to backup power in both undervoltage and overvoltage conditions. Pet.

75–76. Hajeck teaches to use a backup power supply, such as a charge pump, battery, or capacitive array, in response to power loss from the host system to complete outstanding operations, and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57), 1:53–61. To the extent that Hajeck and the other prior art would have been used to back up data to prevent its loss during a power interruption, one would have been able to make the combination to arrive at the claims requiring a write operation to non-volatile memory in response to a signal. *See, e.g.*, claims 11 and 12. We invite further development of the record on the reasons to combine Hajeck with Harris, the FBDIMM Standards, and Amidi. For purposes of institution, Petitioner shows sufficiently that a person of ordinary skill in the art would have combined Harris, the FMDIMM Standards, Amidi, and Hajeck with a reasonable expectation of success.

3. *Claims 1–3, 8, 14, 15, and 23*

We are not aware of any reason why the addition of Hajeck would negate Petitioner’s showing with respect to the previous grounds which incorporate Harris, the FBDIMM Standards, and Amidi. Accordingly, Petitioner has shown a reasonable likelihood to prevail in demonstrating claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck for the reasons stated.

4. *Determination for Ground 3*

Petitioner demonstrates a reasonable likelihood to prevail in showing that at least one claim of the ’918 patent is unpatentable as obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

F. Ground 4: Obviousness over Spiers and Amidi

Petitioner contends that claims 1–30 are obvious over the combination of Spiers and Amidi. Pet. 77–127.

1. Spiers (Ex. 1025)

Spiers is titled “Non-Volatile Memory Backup for Network Storage System.” Ex. 1025, code (54). Spiers was filed on October 12, 2004, and published on April 13, 2006 as U.S. Patent Pub. No. 2006/0080515 A1. *Id.* at codes (10), (43). Petitioner contends that Spiers is prior art under § 102(b). Pet. 13.

Spiers’s Figure 4 is reproduced below.

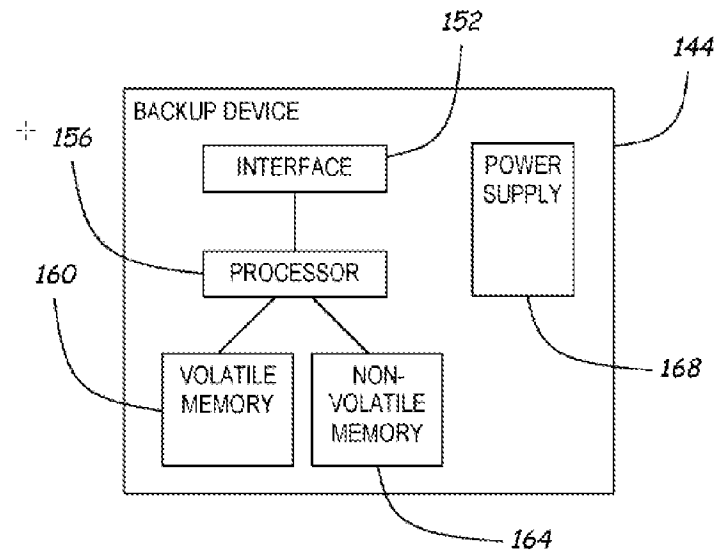


FIG.4

Spiers’s Figure 4 above shows backup device 144. Ex. 1025 ¶ 36. Backup device 144 comprises an interface 152, backup device processor 156, volatile memory 160, non-volatile memory 164, and power supply 168. *Id.* Interface 152 communicates with storage controller 132 (not shown). *Id.* Interface 152 connects to processor 156 which controls operations within backup device 144. *Id.* Processor 156 connects to volatile memory 160 and

non-volatile memory 164. *Id.* Volatile memory 160 may be an SDRAM used to store data from storage controller 132 during typical write operations. *Id.* Non-volatile memory 164 may be flash memory and is used in the event of a power failure detection. *Id.* Upon detecting a power failure, processor 156 switches backup device 144 to power supply 168 (capacitors or batteries) and moves data in volatile memory 160 to non-volatile memory 164. *Id.*

2. *Motivation to Combine*

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Spiers and Amidi. Pet. 78–81. Petitioner contends one would have used Amidi to learn implementation details, such as specific type of SDRAM devices and voltage regulators, useful for Spiers. *Id.* at 78–79 (citing Ex. 1003 ¶¶ 196–200). Petitioner contends while Spiers teaches SDRAM devices, Amidi specifically discloses DDR SDRAM devices. *Id.* at 79 (citing Ex. 1024, claims 4–5). Petitioner contends a person of ordinary skill in the art would have been motivated to implement Spiers with commercially available DDR2 or DDR3 SDRAMs powered according to relevant JEDEC standards with a reasonable expectation of success when using such well-known, standardized technology. *Id.* (citing Ex. 1003 ¶¶ 198–200).

Patent Owner argues that Spiers only mentions two voltage regulators, and that Petitioner provides no reason why a person of ordinary skill in the art would have used DDR2 or DDR3 devices, which require more than two regulated voltages, instead of the SDR SDRAM devices specified by Spiers. Prelim. Resp. 42–43. Petitioner contends that Amidi provides a reason to use DDR2 or DDR3 SDRAMs powered according to relevant JEDEC

standards. Pet. 79 (citing Ex. 1003 ¶¶ 198–200). Patent Owner contends that Spiers’s writes to SDRAM occur with every system write, and that Petitioner makes no showing that DDR2/DDR3 devices would have been used for such a high write-to-read ratio application. Prelim. Resp. 41 (Ex. 2001 ¶ 90; Ex. 1025 ¶¶ 34, 36, 40, 41, Figs. 11, 12).

Patent Owner’s evidence does not establish that DDR2 or DDR3 devices would not work in Petitioner’s combination, only that they would be sub-optimal in that application. On the current record, we cannot determine that such a solution would be sufficiently sub-optimal that one of ordinary skill in the art not have pursued it, but instead would have looked to some other unidentified solution. *See also Novartis Pharms. Corp. v. West-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (noting that an obviousness showing “does not require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation for the current invention.” (quoting *In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004))).

Patent Owner also argues that SDRAM devices could operate at or above the data rates of a PCI interface used by Spiers, so replacing Spiers’s SDRAM devices with faster DDR2 or DDR3 would not bring about a gain in data rate or justify the added expense of additional voltage regulators to accommodate DDR2/DDR3 devices. Prelim. Resp. 41–42 (citing Ex. 2001 ¶¶ 90–92).

Patent Owner’s argument assumes that Spiers is limited to a PCI interface, but a PCI backup device is described as one embodiment in Spiers. Ex. 1025 ¶ 19. Spiers also discloses an “interface 152” in more general terms, which does not appear limited to a PCI interface. *Id.* ¶ 36. Patent

Owner has not shown that the interfaces that one of ordinary skill in the art would have considered to use in light of Spiers would not have been able to accommodate faster DDR2/DDR3 devices. Accordingly, we disagree with this argument on the record as thus far developed.

Petitioner has shown sufficiently that one of ordinary skill in the art would have combined Spiers and Amidi with reasonable expectation of success.

3. *Claim 1*

a) *Limitation 1.a: “A memory module comprising:”*

Petitioner contends that Spiers discloses the preamble of claim 1. Pet. 82. Petitioner contends that Spiers’s backup device 144 corresponds to the claimed “memory module.” *Id.* at 77. Spiers’s backup module 144 has both volatile memory (SDRAMs 190) and non-volatile memory (NAND flash 194). *Id.* (citing Ex. 1025 ¶¶ 34, 37, Figs. 3, 5; Ex. 1003 ¶¶ 643–646).

Patent Owner does not specifically respond to these arguments. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Spiers teaches the preamble for purposes of institution.

b) *Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”*

Petitioner contends that Spiers discloses this limitation. Pet. 83–85. Petitioner asserts that Spiers’s PCI card of backup device 144 corresponds to “a printed circuit board (PCB).” *Id.* at 83. Petitioner contends that Spiers’s

PCI card has an interface configured to fit into the slot connector of a host system, and including a plurality of edge connections. *Id.* (citing Ex. 1025, Fig. 5). Petitioner contends these edge connections are configured to couple power, data, address, and control signals between the memory module and host system. *Id.* at 84 (citing Ex. 1025 ¶¶ 37, 39, 54; Ex. 1031, 1, 7, 21–25, 146–150; Ex. 1003 ¶¶ 652–654).

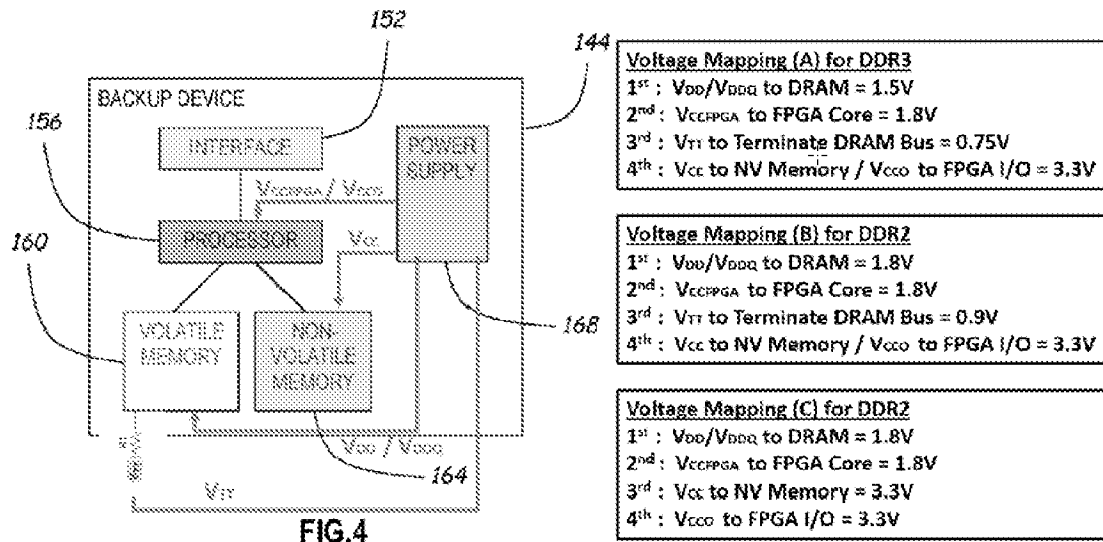
Patent Owner submits no argument specific to this limitation. *See* Prelim. Resp.

Based on our review and consideration of the current record, we determine that Petitioner has adequately shown that Spiers teaches this limitation for purposes of institution.

c) Limitations 1.c to 1.f: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude;”

Petitioner contends that the combination of Spiers and Amidi discloses these limitations. Pet. 86–93. Petitioner provides the following figure to explain how the combination teaches these limitations:

Ground 4: Spiers in view of Amidi, with DDR2 or DDR3 DRAMs



Pet. 86. The above figure shows Petitioner’s view of what a person of ordinary skill in the art would have understood from the combined teachings of Spiers and Amidi in light of the JEDEC standards that would have been known to those of ordinary skill in the art. Pet. 86 (citing Ex. 1003 ¶¶ 655–700). Specifically, Petitioner indicates correspondences between the first, second, third and fourth regulated voltages and those used in Spiers’s backup device 144 shown as voltage mappings A, B, and C. *Id.*

Petitioner contends that Spiers’s PCI card uses a +5V power supply of the PCI bus to power components of the card, including the SDRAM devices (190, 218) required power supply voltages such as V_{DD}/V_{DDQ} (“first regulated voltage”) with amplitudes depending upon the specific DRAM type (DDR2, DDR3, etc.). *Id.* at 87. Petitioner contends that a person of ordinary skill in the art would have recognized that terminal voltage V_{TT} (“third regulated voltage” in voltage mappings A and B) is required for communicating with DRAM devices. *Id.* at 87–88 (citing Ex. 1003 ¶¶ 680–684; Ex. 1025 ¶¶ 37, 46, Fig. 9; Ex. 1028, 68). Petitioner also

contends that Spiers's voltage regulator 206 provides $V_{CCFPGA}=1.8V$ regulated power ("second regulated voltage") to the FPGA core. Pet. 88 (citing Ex. 1025 ¶ 37, Fig. 5; Ex. 1003 ¶¶ 670–673). Petitioner further contends that a person of ordinary skill in the art would have understood that different FPGAs have different voltage requirements for the FPGA core. Pet. 88 (citing Ex. 1042, 2).

Petitioner also contends that Spiers discloses a voltage regulator 184 configured to step down a 5V backup voltage to a 3.3V output voltage used to transfer data from the volatile to non-volatile memories 194, and that a person of ordinary skill in the art would have understood that the 3.3V can be used as V_{CC} power to the non-volatile memories ("third regulated voltage" in mapping C and "fourth regulated voltage" in mappings A and B). *Id.* at 88–89 (citing Ex. 1025, Fig. 5; Ex. 1049, 38; ; Ex. 1003 ¶¶ 685–687, 691–698).

Petitioner further contends a person of ordinary skill in the art would have understood that a converter circuit such as voltage regulator 184 or similar, is configured to provide regulated voltage V_{CCO} of 3.3V ("fourth regulated voltage" in mapping C, "third regulated voltage" in mappings A and B) to an interface of processor 198 to transfer data to and from the non-volatile memories 194. Pet. 89 (citing Ex. 1042, 2, 16; Ex. 1003 ¶¶ 691–700).

Petitioner further contends it would have been obvious to use buck converters to generate these regulated voltages to achieve high efficiency, reliability, and flexible power conversion. Pet. 90 (citing Pet. 29–31; Ex. 1025, Fig. 14 (step 752); Ex. 1062, 11; Ex. 1003 ¶¶ 662–666). Petitioner contends that buck converters powering SDRAM and FPGA

devices from higher voltage sources were well known and commercially available at the time, as disclosed by Amidi. *Id.* at 90–91 (citing Ex. 1024, 4:38–41, Fig. 6 (640)). Petitioner further contends that there were commercially available buck converters to implement the proposed combination. *Id.* at 91–93 (citing Ex. 1047, 1:28–32, 2:47–55, 5:56–59, 7:6–14; Ex. 1042, 1, 2, 9; Ex. 1042, 16, Ex. 1048, 1, 2; Ex. 1058, 5).

Patent Owner argues that Spiers describes its memory devices as SDR SDRAMs, not DDR2 or DDR3 devices. Prelim. Resp. 42–46. Patent Owner further contends that Petitioner has not shown why one would have replaced Spiers’s SDRAM with DDR2 or DDR3 devices. *Id.* at 47–53. Petitioner contends, however, that Amidi specifically discloses DDR SDRAM devices. Pet. 79 (citing Ex. 1024, claims 4–5). Petitioner further contends the person of ordinary skill in the art would have been familiar with the JEDEC standards for DDR2 and DDR3 devices, which specify the four or more voltages required for those memory devices. *Id.* Pet. 79 (citing Ex. 1026; Ex. 1046).

Patent Owner contends that Petitioner did not provide any evidence why one would need to use V_{TT} in the DDR2 or DDR3 standards. Prelim. Resp. 53–56. Even if Patent Owner is correct, Petitioner did not rely solely on V_{TT} as the third regulated voltage. *See* Pet. 86.

Patent Owner argues there is no evidence that one would have generated V_{TT} using a buck converter. Prelim. Resp. 53–56. Petitioner explained that the reason for using a buck converter for regulated voltages comes from Amidi. *See* Pet. 91 (citing Ex. 1024, 4:38–41, Fig. 6 (640)).

Patent Owner further argues that Petitioner provided no reason for equipping Spiers with multiple 1.8V regulators. Prelim. Resp. 59–60.

Petitioner explained that the JEDEC standards provide for separate pins with the same voltage levels, and that certain voltages should be isolated and separately controlled to provide independence, stability, and flexibility for power management. Pet. 30–31.

Patent Owner further argues there are other ways to generate regulated voltages than providing a buck converter for each voltage level, and Petitioner did not show why one would not have pursued these other options. Prelim. Resp. 60–69. Petitioner did explain, however, that using a buck converter for each regulated voltage would “achieve high efficiency, reliability, and flexible power conversion.” Pet. 90 (citing Pet. 29–31; Ex. 1025, Fig. 14; Ex. 1062, 11; Ex. 1003 ¶¶ 662–666).

Petitioner has shown sufficiently for institution that the combination of Spiers and Amidi teaches this claim limitation.

d) Limitation 1.g: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:”

Petitioner contends that Spiers’s DRAM, FPGA/processor, resistors, and non-volatile memory are components coupled to the PCI card, which are each coupled to at least one of three regulated voltages. Pet. 94 (citing Pet. 78–81; Ex. 1003 ¶¶ 701–713).

Patent Owner provides no argument specific to this limitation. See Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

e) Limitation h: “a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and”

Petitioner asserts that Spiers discloses SDRAMs 190 that may be implemented as DDR2 or DDR3 devices. Pet. 94 (citing Pet. 32, 78–81; Ex. 1003 ¶¶ 714–718).

Patent Owner provides no argument specific to this limitation. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

f) Limitation 1.i.1: “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices,”

Petitioner contends that the “at least one circuit” corresponds to Spiers FPGA processor 198 coupled to the SDRAM devices 190, 218 and to the PCI interface 172 for data, address and control signals. Pet. 95.

Patent Owner does not specifically respond to Petitioner’s contention. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

g) Limitation 1.i.2: “the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices,”

Petitioner contends that Spiers’s FPGA processor 198 receives read and write commands from the PCI interface 172 and outputs address and control signals on DDR SDRAM interface between 198 and 190 to the

SDRAM devices. Pet. 95–98 (citing Ex. 1025 ¶¶ 37, 39, 44–46, Figs. 5, 9; Ex. 1031, 1, 7, 21, 146–149, Ex. 1003 ¶¶ 724–729).

Patent Owner does not specifically address Petitioner’s contention for this limitation. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

h) Limitation 1.i.3: “the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage,”

Petitioner contends that Spiers’s FPGA processor is coupled to both the second regulated voltage ($V_{CCFPGA}=1.8V$) and the fourth regulated voltage ($V_{CCO}=3.3V$). Pet. 98–99 (citing Pet. 78–81; Ex. 1003 ¶¶ 730–734).

Patent Owner does not specifically respond to Petitioner’s contention. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

i) Limitation 1.i.4: “wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes”

Petitioner contends that Spiers discloses that one of the second and fourth amplitudes (e.g., 1.8V) is less than a second one of the second and fourth amplitudes (e.g., 3.3V). Pet. 99 (citing Pet. 78–81, 87; Ex. 1003 ¶¶ 735–738).

Patent Owner does not specifically respond to Petitioner’s contention. *See* Prelim. Resp.

Petitioner has adequately shown that the combination of Spiers and Amidi teaches this limitation for purposes of institution.

j) Determination for Claim 1

Petitioner shows sufficiently that one of ordinary skill in the art would have had reason to combine Spiers and Amidi with a reasonable expectation of success in arriving at claim 1. Furthermore, Petitioner has shown sufficiently that the combination of Spiers and Amidi teaches each limitation of claim 1. Accordingly, Petitioner has established a reasonable likelihood to prevail in showing that claim 1 is unpatentable as obvious over the combination of Spiers and Amidi.

4. Claims 2–30

We have reviewed Petitioner's contentions for each of claims 2–30. Pet. 99–130. We find them sufficient to disclose the limitations recited in at least claims 2–4, 8, 14, 15, 23, and 28–30.

Patent Owner contends that claim 23 requires a second voltage to be selectively switched on or off, and that Petitioner maps the second voltage as the voltage supply to the processor core. Prelim. Resp. 71 (citing Pet. 99). Patent Owner contends that Petitioner's mapping relates to turning off the supply voltage to the system bus, not the processor core. *Id.* (citing Pet. 114–15). Patent Owner does not explain why turning off the voltage supply on the system bus would not also turn off the power to the processor core. Although Patent Owner further argues that the processor must remain on while backing up data in the SDRAM, Patent Owner does not indicate what language in claim 23 requires the processor core to be switched on during data backup.

Other than these arguments previously addressed, Patent Owner does not present any further argument specific to these claims. *See* Prelim. Resp.

Petitioner has shown sufficiently that the combination of Spiers and Amidi discloses at least one of these claims.

Accordingly, Petitioner has established a reasonable likelihood to prevail in showing at least one of these claims is unpatentable as obvious over the combination of Spiers and Amidi under this ground.

G. Ground 5: Obviousness over Spiers, Amidi, and Hajeck

Petitioner contends that a person of ordinary skill in the art would have combined Hajeck with Spiers and Amidi because Hajeck teaches the importance of detecting both undervoltage and overvoltage conditions to avoid data loss. Pet. 128–29. Hajeck teaches generating a busy signal for a host in response to undervoltage or overvoltage conditions, and also teaches that its charge pump protects its controller from being damaged by spikes and surges in a power signal provided by a host. Ex. 1038, 1:64–67, 3:13–16, 3:30–43.

On this record, Petitioner’s showing is sufficient to present a reasonable likelihood that at least one of claims 1–30 is unpatentable as obvious over the combination of Spiers, Amidi, and Hajeck. Pet. 128–30. Also, this ground subsumes the previous one, and there is no persuasive evidence that Hajeck negates Spiers and Amidi, so Petitioner has presented a reasonable likelihood to prevail in showing at least one of claims 1–30 is unpatentable as obvious over the combination of Spiers, Amidi, and Hajeck under this ground.

IV. CONCLUSION

After considering the evidence and arguments presented in the Petition and Preliminary Response, we determine that Petitioner has

established a reasonable likelihood of prevailing on its assertion that at least one claim of the '918 patent is unpatentable. We institute an *inter partes* review on all the challenged claims and all of the grounds presented in the Petition. At this stage of the proceeding, we have not made a final determination as to the patentability of these challenged claims.

V. ORDER

In consideration of the foregoing, it is hereby

ORDERED that pursuant to 35 U.S.C. § 314, *inter partes* review is instituted as to the challenged claims of the '918 patent with respect to all grounds of unpatentability presented in the Petition; and

FURTHER ORDERED that *inter partes* review is commenced on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial.

IPR2022-00996
Patent 11,016,918 B2

FOR PETITIONER:

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MICRON TECHNOLOGY, INC., MICRON SEMICONDUCTOR
PRODUCTS, INC., and MICRON TECHNOLOGY TEXAS LLC,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2023-00406
Patent 11,016,918 B2

Before PATRICK M. BOUCHER, JON M. JURGOVAN, and
DANIEL J. GALLIGAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314
Granting Motion for Joinder
35 U.S.C. § 315(c); 37 C.F.R. § 42.122

I. DISCUSSION

Inter partes review of U.S. Patent 11,016,918 B2 (“’918 patent”) was instituted in IPR2022-00996 (“’996 IPR”) based on a petition filed by Samsung Electronics Co., Ltd. (“Samsung”). ’996 IPR, Papers 1, 10. In this proceeding, Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC (collectively “Petitioner”) concurrently filed a Petition (Paper 1) and a Motion for Joinder (Paper 3) seeking to be joined as a petitioner in the ’996 IPR. Petitioner represents that Samsung, the petitioner in the ’996 IPR, does not oppose joinder. Motion 2. Patent Owner did not file an opposition to the Motion.

The statute governing *inter partes* review joinder states the following:

JOINDER.—If the Director institutes an *inter partes* review, the Director, in his or her discretion, may join as a party to that *inter partes* review any person who properly files a petition under section 311 that the Director, after receiving a preliminary response under section 313 or the expiration of the time for filing such a response, determines warrants the institution of an *inter partes* review under section 314.

35 U.S.C. § 315(c).

The time for filing a preliminary response to the Petition has expired, and Patent Owner did not file one. *See* 37 C.F.R. 42.107(b) (“The preliminary response must be filed no later than three months after the date of a notice indicating that the request to institute an *inter partes* review has been granted a filing date.”); Paper 4 (filing date notice entered on January 23, 2023).

The standard for instituting an *inter partes* review is set forth in 35 U.S.C. § 314(a), which provides that an *inter partes* review may not be instituted unless the information presented in the Petition and any preliminary response shows “there is a reasonable likelihood that the

petitioner would prevail with respect to at least 1 of the claims challenged in the petition.”

The patentability challenges in the 996 IPR met the “reasonable likelihood” standard of § 314(a). 996 IPR, Paper 10 at 55. Petitioner represents that the Petition “is substantively identical to” the petition in the 996 IPR. Motion 1. Petitioner presents the following grounds in this Petition, which are identical to the instituted grounds in the 996 IPR:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–3, 8, 14, 15, 23	103(a) ¹	Harris ² , FBDIMM Standards ³
1–30	103(a)	Harris, FBDIMM Standards, Amidi ⁴
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck ⁵
1–30	103(a)	Spiers ⁶ , Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

Pet. 3; *see* 996 IPR, Paper 10 at 8, 55 (identifying same grounds and instituting *inter partes* review).

¹ The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, 125 Stat. 284, 287–88 (2011), amended 35 U.S.C. § 103 and became effective March 16, 2013. For this proceeding, Petitioner assumes that the ’918 patent has an effective priority date before this date (Pet. 3–4) and applies the pre-AIA version of § 103.

² US 2006/0174140 A1, published Aug. 3, 2006 (Ex. 1023).

³ JESD82-20 and JESD205 standards published March 2007 (Ex. 1027, Ex. 1028).

⁴ US Patent No. 7,724,604 B2, issued May 25, 2010 (Ex. 1024).

⁵ US Patent No. 6,856,556 B1, published Apr. 13, 2006 (Ex. 1038).

⁶ US 2006/0080515 A1, published Apr. 13, 2006 (Ex. 1025).

We determine that Petitioner has demonstrated a reasonable likelihood of prevailing with respect to at least one challenged claim of the '918 patent for the reasons given in the institution decision in the 996 IPR and that the Petition warrants institution. *See* 996 IPR, Paper 10 at 8–55.

As discussed above, Petitioner's Motion is unopposed by Samsung or Patent Owner. We have reviewed the Motion, and we determine that it is appropriate under these circumstances to join Petitioner as a party to the 996 IPR.

II. ORDER

Accordingly, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a) and 37 C.F.R. § 42.4, an *inter partes* review is hereby instituted on the challenge raised in the Petition;

FURTHER ORDERED that Petitioner's Motion for Joinder with IPR2022-00996 is *granted*, and Petitioner is hereby joined as a petitioner in IPR2022-00996;

FURTHER ORDERED that the grounds on which trial in IPR2022-00996 was instituted are unchanged, and no other grounds are added in IPR2022-00996;

FURTHER ORDERED that the Scheduling Order entered in IPR2022-00996 (Paper 11) and the Parties' stipulation to modify Dates 1, 2, and 3 (Paper 15) shall govern the trial schedule in IPR2022-00996;

FURTHER ORDERED that Petitioner's role in IPR2022-00996 shall be limited as stated by Petitioner in the Motion for Joinder (Paper 3 at 5–9) unless and until Samsung is terminated from that proceeding;

IPR2023-00406
Patent 11,016,918 B2

FURTHER ORDERED that the case caption in IPR2022-00996 shall be changed to reflect joinder of Petitioner in accordance with the attached example;

FURTHER ORDERED that a copy of this Decision be entered into the record of IPR2022-00996; and

FURTHER ORDERED that all further filings shall be made in IPR2022-00996.

IPR2023-00406
Patent 11,016,918 B2

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD, MICRON TECHNOLOGY, INC.,
MICRON SEMICONDUCTOR PRODUCTS, INC., and
MICRON TECHNOLOGY TEXAS LLC,⁷
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NETLIST, INC.,
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IPR2022-00996
Patent 11,016,918 B2

⁷ Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC filed a motion for joinder and a petition in IPR2023-00406 and have been joined as petitioners in this proceeding.

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD., MICRON TECHNOLOGY, INC.,
MICRON SEMICONDUCTOR PRODUCTS, INC., and MICRON
TECHNOLOGY TEXAS LLC,¹
Petitioner,

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NETLIST, INC.,
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IPR2022-00996
Patent 11,016,918 B2

Before PATRICK M. BOUCHER, JON M. JURGOVAN, and
DANIEL J. GALLIGAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Final Written Decision
Determining All Challenged Claims Unpatentable
Dismissing Petitioner's Motion to Exclude
35 U.S.C. § 318(a)

¹ Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC filed a motion for joinder and a petition in IPR2023-00406 and have been joined as petitioners in this proceeding. *See* Paper 26.

I. INTRODUCTION

Samsung Electronics Co., Ltd. (“Samsung”) filed a Petition (Paper 1, “Pet.”) for *inter partes* review of claims 1–30 (“challenged claims”) of U.S. Patent 11,016,918 B2 (Ex. 1001, “the ’918 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition. We instituted *inter partes* review under 35 U.S.C. § 314(a). Paper 10 (“Inst. Dec.”).

During the trial, Patent Owner filed a Response (Paper 21, “Resp.”), Petitioner filed a Reply (Paper 25), and Patent Owner filed a Sur-Reply (Paper 31). We joined Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC as petitioners in this proceeding, and we refer to Samsung and these entities collectively as “Petitioner.” *See* Paper 26.

Petitioner and Patent Owner requested oral argument (Papers 28 and 29). A hearing was conducted on September 11, 2023. Paper 47 (“Tr.”).

Petitioner objected to evidence (Paper 32) and filed a Motion to Exclude (Paper 33). Patent Owner filed an Opposition to Petitioner’s Motion to Exclude (Paper 34), and Petitioner filed a Reply (Paper 37) in support of its Motion to Exclude.

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is entered pursuant to 35 U.S.C. § 318(a). Having reviewed the complete trial record, we determine that Petitioner has shown, by a preponderance of the evidence, that the challenged claims are unpatentable.

II. BACKGROUND

A. Real Parties in Interest

The identified real parties in interest on the Petitioner side are the following entities: Samsung Electronics Co., Ltd., Samsung Semiconductor, Inc., Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC. Pet. 1; IPR2023-00406, Paper 3, 1.

Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

B. Related Matters

The parties advise that the '918 patent is related to the following pending matters:

- *Samsung Electronics Co., Ltd. et al. v Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021)
- *Netlist, Inc. v. Samsung Electronics Co., Ltd. et al.*, No. 2:21-cv-00463 (E.D. Tex. filed Dec. 20, 2021)
- *Netlist, Inc. v. Micron Technology, Inc.*, No. 2:22-cv-00203 (E.D. Tex. filed June 13, 2022)
- IPR2012-00999 (U.S. Patent No. 11,232,054)
- U.S. Patent Application No. 17/582,797

Pet. 1; Paper 4, 1; Paper 9, 2; Paper 12, 1.

The parties advise that the following related proceeding is no longer pending:

- IPR2017-00692 (U.S. Patent No. 8,874,831)

Pet. 1; Paper 4, 1.

C. Overview of the '918 Patent (Ex. 1001)

The '918 patent is titled “Flash-DRAM Hybrid Memory Module.”

Ex. 1001, code (54). Figure 12 of the '918 patent is reproduced below.

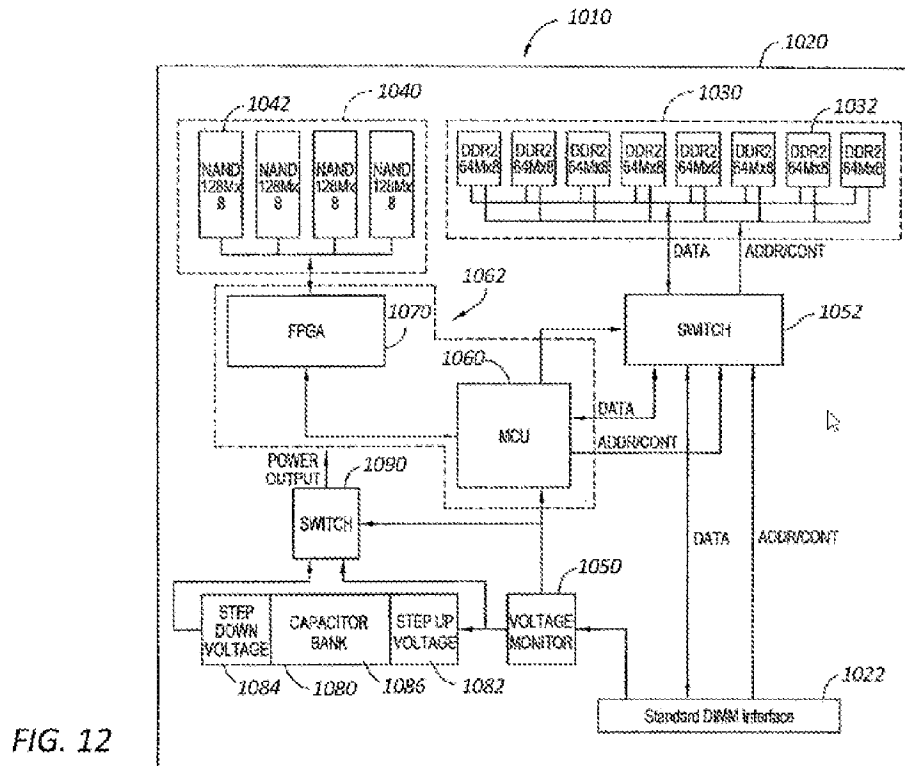


FIG. 12

Figure 12 shows an example memory system 1010 of the '918 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and selectively coupled to the non-volatile memory subsystem 1040. *Id.* at 21:16–22. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level

cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board (PCB) 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022 for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. *Id.* at 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '918 patent refers to this as the “first state.”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '918 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a

28:20–22. Conversion element 1120 comprises sub-blocks 1122, 1124, 1126 comprised of various converter circuits such as buck converters, boost converters, and buck-boost converters for providing various voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 29:18–54. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. “[P]ower module 1100 provides power to [] components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition.” *Id.* at 27:61–65.

Specifically, “[i]n a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130.” *Id.* at 28:27–31. “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” *Id.* at 28:32–37. “In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element 1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.” *Id.* at 28:34–38.

Transition from the first state to the second state may occur when power

module 1100 detects a power failure is about to occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

D. Illustrative Claim

Of the challenged claims, claims 1, 16, and 23 are independent. Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;

[1.d] a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;

[1.e] a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;

[1.f] a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and

[1.g] a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:

[1.h] a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and

[1.i] [1.i.1] at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, [1.i.2] the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, [1.i.3] the at least one circuit coupled to both

the second regulated voltage and the fourth regulated voltage, [1.i.4] wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.

Ex. 1001, 38:19–44.

E. Evidence

Petitioner relies on the following references (*see* Pet. 3, 10–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published August 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published April 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued February 15, 2005

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 3):

Claims Challenged	35 U.S.C. §	References
1–3, 8, 14, 15, 23	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

III. ANALYSIS OF CHALLENGED GROUNDS

We now consider Petitioner's asserted grounds of unpatentability and Patent Owner's arguments to determine whether Petitioner has demonstrated by a preponderance of the evidence that the challenged claims would have been obvious under 35 U.S.C. § 103. *See* 35 U.S.C. § 316(e) (petitioner has the burden of proving unpatentability by a preponderance of the evidence).

A. Principles of Law

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination based on the teachings of multiple references requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation

omitted); *see also* *KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Factors pertinent to a determination of the level of ordinary skill in the art include “(1) the educational level of the inventor; (2) type of problems encountered in the art; (3) prior art solutions to those problems; (4) rapidity with which innovations are made; (5) sophistication of the technology; and (6) educational level of active workers in the field.” *Envtl. Designs, Ltd. v. Union Oil Co. of Cal.*, 713 F.2d 693, 696 (Fed. Cir. 1983) (citing *Orthopedic Equip. Co. v. All Orthopedic Appliances, Inc.*, 707 F.2d 1376, 1381–82 (Fed. Cir. 1983)). “Not all such factors may be present in every case, and one or more of these or other factors may predominate in a particular case.” *Id.* at 696–697. The prior art may reflect an appropriate level of skill. *Okajima v. Bourdeau*, 261 F.3d 1350, 1354–55 (Fed. Cir. 2001).

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 8–9 (citing Ex. 1003 ¶67). Petitioner contends that additional training can substitute for educational or research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry

standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a computer system, including standard communication busses and protocols, such as PCI and SMBus busses and protocols. *Id.* at 8–9. Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* at 9. Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

Patent Owner indicates “[f]or the purposes of this [Response], Patent Owner is applying the level of ordinary skill in the art proposed by Petitioner” for this proceeding. Resp. 1.

The evidence that Petitioner presents mostly applies to the educational level and experience of workers in the field, but also touches upon other factors as well, including problems and solutions in the prior art and complexity of the technology. *See* Pet. 7–8. We find Petitioner’s proposal is consistent with the level of ordinary skill in the art reflected by the ’918 patent and the prior art of record, and, therefore, we adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least” which introduces ambiguity and may encompass skill levels beyond ordinary. *See Okajima v. Bourdeau*, 261 F.3d 1350,

1355 (Fed. Cir. 2001) (the prior art may reflect an appropriate level of skill in the art).

C. Claim Construction

We construe claim terms “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2021). There is a presumption that claim terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art in the context of the specification. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Nonetheless, if the specification “reveal[s] a special definition given to a claim term by the patentee that differs from the meaning it would otherwise possess[,] . . . the inventor’s lexicography governs.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1316 (Fed. Cir. 2005) (en banc) (citing *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002)). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Only disputed claim terms must be construed, and then only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

1. “Memory Module”

Patent Owner contends as follows:

The term “memory module” appears in the preamble of the independent claims, which provides antecedent basis for the same term in the body of the text. The preamble is thus limiting

as the District Court found after reviewing the specification in detail. The Court further explained that “a skilled artisan would understand a ‘memory module’ is distinct from, and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller.

Resp. 1–2 (citing Ex. 2032, 28; Ex. 2034, 319). Patent Owner argues that the “court’s ruling is consistent with the specification which states in the Overview section that the invention is particularly ‘couplable to a *memory controller* of a host system.’” *Id.* at 2 (citing Ex. 1001, 1:66–67, 3:66–67). Patent Owner states that Petitioner “objected to the court’s finding that the memory module is limiting, but does not dispute the Court’[s] finding on what a ‘memory module’ means to a [person of ordinary skill in the art].” *Id.* (citing Ex. 2033, 4).

As support for its argument, Patent Owner relies on a claim construction order issued by the District Court for the Eastern District of Texas, which states as follows:

Having reviewed the entire patent “to gain an understanding of what the inventors actually invented and intended to encompass by the claim[s],” *Catalina Mktg. Int’l*, 289 F.3d at 808–09, the Court finds the preamble limiting. While the claims recite many of the structural requirements of a “memory module,” the claims arguably read on other modular computer devices, such as a video card or network controller, despite no evidence the inventors intended to encompass such devices by the claims. To the contrary, as the Overview section [of the Court’s claim construction order] explains, the invention “is couplable to a *memory controller* of a host system,” [Ex. 1001] at 3:66–67 (emphasis added), not just the host system as recited in the claims. *See also id.* at 1:66–67 (“[t]he present disclosure relates generally to computer memory devices”). Thus, a skilled artisan would understand a “memory module” is distinct from,

and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller. *See Memory Systems: Cache, DRAM, Disk*, Dkt. No. 76-17 at 319 (depicting, in FIG. 7.6, a memory controller connected to two memory modules). Accordingly, the preambles are limiting.

Ex. 2032, 28) (alterations in original), *cited in* Resp. 2.

Patent Owner argues that Petitioner's expert, Dr. Wolfe, testified that the term "memory modules" typically refers to "main memory modules," which "are *designed to connect to the primary memory controller* for the purpose of holding general purpose code and data in a computer system." *Id.* at 2 (citing Ex. 2030, 123:14–25; Ex. 2056, 100:15–101:8 (by 2004–2005, a memory module "was intended to go into a dedicated memory slot and not a general-purpose IO slot")). Patent Owner contends that the '918 patent's usage of "memory module" is consistent with that understanding, and asserts that we should adopt the District Court's claim construction, including that the claimed "memory module" includes structures necessary to connect to a memory controller. *Id.* at 2–3 (citing Ex. 2031 ¶¶ 51–53); *accord* Sur-Reply 1.

In Reply, Petitioner contends that we properly found that Harris, the FBDIMM Standards, Amidi, and Spiers all disclose a "memory module" in our Institution Decision. Reply 1–2 (citing Inst. Dec. 14–15, 17–18, 34–37, 45; Pet. 19–20, 82–83; Reply 23–26). Petitioner contends that the District Court's construction did not limit "memory module" to only "*main* memory modules . . . designed to connect to the *primary* memory controller." *Id.* at 1–2 (emphasis in original) (citing Resp. 2; Ex. 2032, 26–28, 35; Ex. 2030, 125:12–127:13; Ex. 2056, 100:15–101:19).

In Sur-Reply, Patent Owner disagrees with Petitioner’s contention that “memory module” is not limited to “main memory modules . . . designed to connect to the primary memory controller.” Sur-Reply 1 (emphasis in original) (citing Reply 1). Patent Owner argues that Petitioner’s contention “ignores the District Court’s finding based on intrinsic evidence—which [Petitioner] did not object to—that the ‘invention [i.e., the claimed memory module] ‘is couplable to a memory controller of a host system,’ [Ex. 1001] at 3:66–67 . . . , not just the host system as recited in the claims.’” *Id.* (citing Ex. 2032, 28; Ex. 1001, 4:5–12, 4:14–24, 4:35–39, 4:45–51, 5:4–20, 4:36–50, 6:4–6, 6:25–29, 6:61–66, 7:1–4, 7:21–25, 7:29–34, 7:43–49, 7:54–57, 21:24–25, 22:53–58, 23:19–22, 26:43–51, 23:41–44).

The District Court’s Claim Construction Order (Ex. 2032) issued after our Institution Decision (Paper 10). Although Petitioner argued in the District Court litigation that the claim preambles were not limiting, Petitioner does not appear to maintain that argument here following the District Court’s determination that the preambles were limiting. Ex. 2032, 28. Since the term is recited in both the preambles and bodies of these claims, there is evidence that the inventors of the ’918 patent intended the term to limit the claim, as the District Court determined. *See Catalina Mktg. Int’l., Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell Commc’ns Rsch., Inc. v. Vitalink Communications Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995) (“dependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention.”)).

We agree with Petitioner's contention that "memory module" is not limited to "*main* memory modules . . . designed to connect to the *primary* memory controller." Reply 1. The '918 patent states

In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the **main memory**.

Ex. 1001, 11:3–6 (emphases added). This is the only mention of "main memory" in the '918 patent. That the Flash memory serves as "main memory" in "certain embodiments" implies that there may be other embodiments in which this is not the case. Accordingly, we agree with Petitioner that the term "memory module" is not limited to "*main* memory modules . . . designed to connect to the *primary* memory controller." See Reply 1.

We agree with Patent Owner that the "memory module" includes structure to connect with a "host system" including its memory controller. See, e.g., Ex. 1001, code (57), 12:52–59. The claims do not mention a "memory controller" but they do mention that the "memory module" has an "interface" to connect with a "host system." Ex. 1001, 38:19–24 (claim 1), 39:54–59 (claim 16), 40:51–56 (claim 23). That the "memory module" has structure for connecting to the host system's memory controller may thus be implied from the claims.

We agree with Petitioner that the District Court's determination was that the preambles of all claims of the '918 patent are limiting. Ex. 2032, 35. To the extent that the District Court's determination construed "memory module" to include structure necessary to connect to a memory controller, our construction of "memory module" is consistent.

2. *“Pre-Regulated Input Voltage”*

Claims 16–22 recite a “pre-regulated input voltage.” Ex. 1001, 39:61, 42:23. Petitioner identifies at least two meanings for this claim term, one meaning within pre-determined limits and the other meaning the voltage must be pre-regulated on the memory board itself. Pet. 73. Petitioner contends that the District Court construed “pre-regulated input voltage” to have its plain and ordinary meaning, and rejected Samsung’s argument that the voltage must be pre-regulated on the memory module. Reply 20, n.3 (citing Ex. 2032, 21–22, 34; Inst. Dec. 37–38; Ex. 2030, 268:18–270:5). Petitioner contends that Patent Owner’s expert does not dispute the District Court’s construction. *Id.* (citing Ex. 1075, 142:10–144:18). According to Petitioner, both experts agree that the V12V input to Harris can be pre-regulated under that construction. Reply 20 (citing *id.* at 5, n.2).

Patent Owner argues that a “pre-regulated input voltage is just a regulated voltage provided to these voltage conversion circuits.” Resp. 38.

Under its ordinary and plain meaning in the context of the ’918 patent, we understand “pre-regulated input voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage converters. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). This is consistent with Patent Owner’s proposed construction.

D. *Ground 1: Obviousness Over Harris and FBDIMM Standards*

Petitioner contends claims 1–3, 8, 14, 15, and 23 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–51. For the reasons that follow, we are persuaded that the evidence, including

Dr. Wolfe's testimony, supports Petitioner's showing and establishes by a preponderance of the evidence that these claims are unpatentable.

1. *Harris (Ex. 1023)*

Harris is titled "Voltage Distribution System and Method for a Memory Assembly." Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 10.

Harris's Figure 1A is reproduced below.

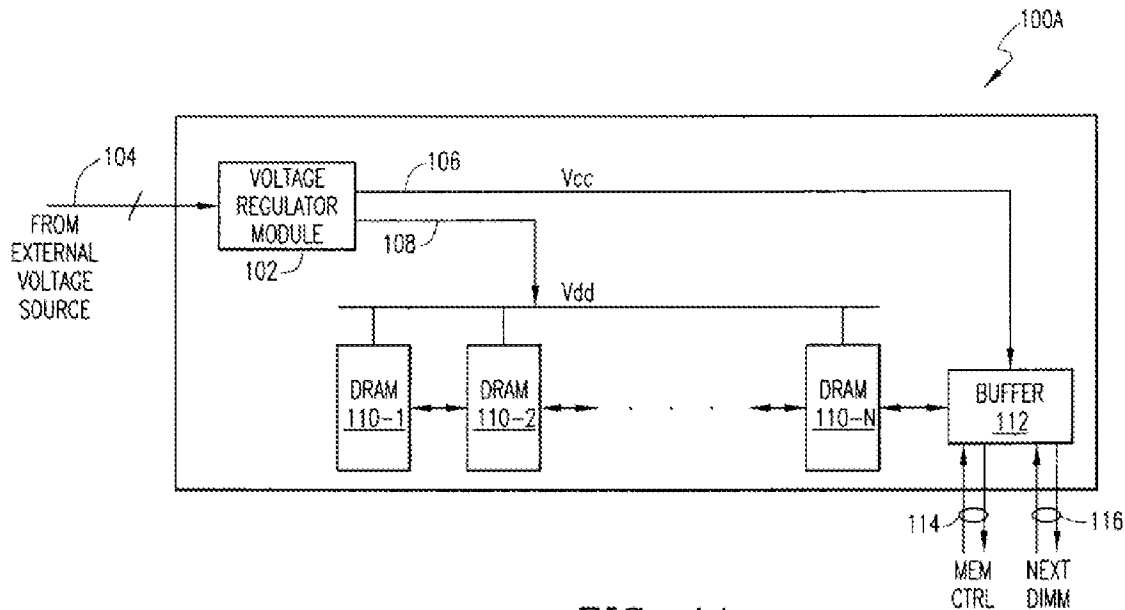


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface

116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Modules (DIMM) wherein each of the memory devices 100-1 to 100-N comprises a Double Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. *FBDIMM Standards (Exs. 1027, 1028)*

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029; Ex. 1003 ¶¶ 134–136. Petitioner refers to these standards collectively as the “FBDIMM Standards.” Pet. 11. Petitioner contends the FBDIMM Standards are prior art under § 102(a). *Id.*

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max. thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max. thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DD}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	$0.453 \cdot V_{DD}$	$0.5 \cdot V_{DD}$	$0.547 \cdot V_{DD}$	(DRAM interface V_{TT}) This supply should track as $0.5 \cdot 1.8$ volt supply
	3.0	3.3	3.6	(V_{DDSPD})
Buffer interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM interface	SSTL ₁₈			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.

Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)

An Unloaded system should supply V_{TT} of $0.48 \cdot V_{DD}/0.52 \cdot V_{DD}$ to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
RESET		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	1	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by “a” are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies V_{CC} , V_{CCFBD} , V_{DD} , V_{SS} , and V_{DDSPD} .

3. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–165). Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and thus would naturally look to them for more details about the “fully buffered DIMM” that Harris

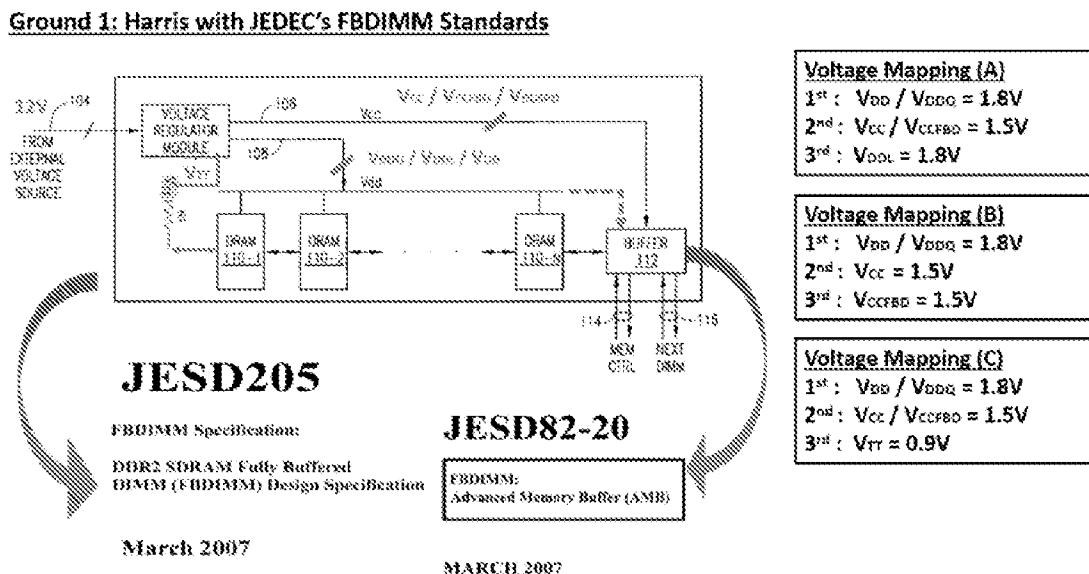
describes as compatible with his on-board voltage regulator module (VRM).
Id.

To explain its combination, Petitioner provides the following table:

Voltage Mappings (Grounds 1-3)			
	<u>A</u>	<u>B</u>	<u>C</u>
“first”:	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} , V_{DDQ} , or $V_{DDL} = 1.8V$	V_{DD} , V_{DDQ} , or $V_{DDL} = 1.8V$
“second”:	V_{CC} or $V_{CCFBD} = 1.5V$	$V_{CC} = 1.5V$	V_{CC} or $V_{CCFBD} = 1.5V$
“third”:	$V_{DDL} = 1.8V$	$V_{CCFBD} = 1.5V$	$V_{TT} = 0.9V$
“fourth”:	$V_{DDSPD} = 3.3V$	$V_{DDSPD} = 3.3V$	$V_{DDSPD} = 3.3V$

Pet. 27. The table above shows Petitioner’s Voltage Mappings A, B, and C from the FBDIMM Standards for Grounds 1–3, and how they disclose the “first,” “second,” “third,” and “fourth regulated voltages” in the claims. *Id.*

For Ground 1, Petitioner provides the illustration below:



Pet. 15. This illustration shows how Petitioner is combining the teachings of Harris and the FBDIMM Standards to arrive at the claims. Specifically,

JEDEC FBDIMM Standard JESD205 applies to Harris's memory module and JESD82-20 applies to the Harris's buffer.

a) Modifying Harris to Have Four Converters

Patent Owner asserts that a person of ordinary skill in the art would not have modified Harris's memory module to have four converters, as Petitioner asserts. Resp. 17–21. Patent Owner argues that Harris discloses a single converter providing at least two voltages in each of Petitioner's voltage mappings. *Id.* at 17 (citing Ex. 2031 ¶¶ 75–79). Patent Owner asserts that Harris discloses “a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels.” (Emphases original) *Id.* (citing Ex. 1023 ¶ 10, Fig. 1A); *see also* Sur-Reply 15–17.

Petitioner replies that the Petition explained the motivation for multiple converters, including that “the JEDEC standards treat those voltages separately, thus permitting power-on sequencing; providing independent control; improving efficiency; and saving power.” Reply 14 (citing Pet. 30–31; Ex. 2030, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15). In addition, Dr. Wolfe explains that smaller, lower-power regulators may be more desirable than larger, higher-power regulators because they can be less costly. Ex. 2030, 45:21–25. He also stated that space availability on the memory board can require separate regulators. *Id.* at 46:1–4. In sum, Dr. Wolfe stated that “a number of ordinary engineering factors” would be considered when deciding to use one supply for both voltages or two separate supplies. *Id.* at 46:5–10.

Petitioner provides the following excerpt from the JEDEC standards to explain its position:

2.3.1 Power-up and initialization sequence

The following sequence is required for POWER UP and Initialization.

- a) Apply power and attempt to maintain CKE below $0.2 \cdot VDDQ$ and ODT^{n1} at a low state (all other inputs may be undefined.) The power voltage ramp time must be no greater than 20ns, and during the ramp, $VDD > VDDL > VDDQ$ and $VDD - VDDQ < 0.3$ volts.

- VDD, VDDL and VDDQ are driven from a single power converter output, AND
- VTT is limited to 0.95 V max, AND
- Vref tracks $VDDQ/2$.

OR

- Apply VDD without any slope reversal before or at the same time as VDDL.
- Apply VDDL without any slope reversal before or at the same time as VDDQ.
- Apply VDDQ without any slope reversal before or at the same time as VTT & Vref.

at least one of these two sets of conditions must be met.

Reply 14 (citing Resp. 25; Ex. 1026, 9 (DDR2); Ex. 1046, 15 (same for DDR3); Pet. 30–31. The excerpt from JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3)² above shows two options for compliance with these standards, one shown partially with a blue box,³ the other with a red box. Although the first option (blue box) indicates that a single converter generates the voltages, Petitioner correctly contends that the second option (red box) does not list that any converter provides these voltages, but leaves open how they are generated. *Id.* at 13–14. A person of ordinary skill in the

² Although Petitioner does not expressly rely on JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3) in its combinations, the parties seem to be in agreement that these specifications are part of the background knowledge that a person of ordinary skill in the art would have had. In an obviousness analysis, the background knowledge is properly considered together with the demands known in the design community, and the inferences and creative steps that a person of ordinary skill in the art would employ. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013).

³ In our opinion, Petitioner should have drawn the blue box to include the following two lines “VTT is limited to 0.95 V max, AND Vref tracks $VDDQ/2$ ” because of the way the standard presents the two sets of conditions with the word “or” between them and refers to them as “two sets of conditions.” We refer to the two sets of conditions as “options” herein.

art would have understood that the listed voltages may be generated with respective converters. *See* Ex. 1026, 9; Ex. 1046, 15; Ex. 1003 ¶ 242 (Dr. Wolfe testifying that JESD79-2B’s disclosure that the voltage supplies can be turned on and off separately suggests to a person of ordinary skill in the art that the voltages are provided by separate converters); *see also KSR*, 550 U.S. at 418 (“a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.”).

Moreover, Harris discloses “at least one on-board voltage regulator module (VRM) is provided as part of the memory board assembly module 100A for converting externally supplied voltage level available on external source path 104 into appropriate local voltage levels that power the first and second voltage paths, i.e., the V_{dd} and V_{cc} .” Ex. 1023 ¶ 10 (emphases added). Also, Harris’s claim 1 recites “at least one voltage regulator module for converting an externally supplied voltage level into appropriate voltage levels that power said first and second voltage paths, respectively.” Ex. 1023, claim 1. Thus, Harris discloses that multiple converters can be used to generate respective multiple voltage levels.

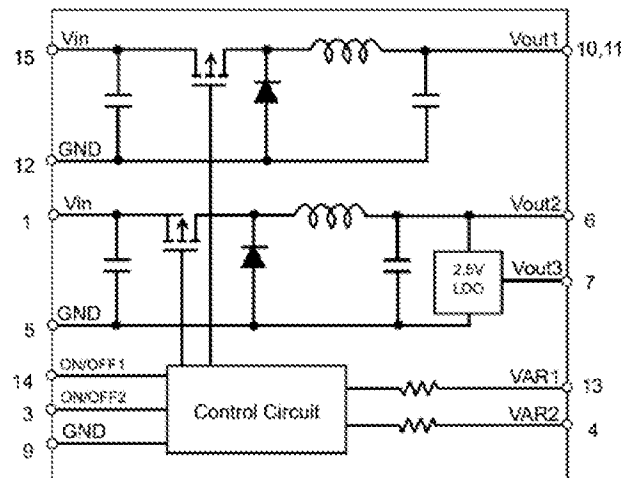
Furthermore, Petitioner contends that Patent Owner’s examples relate to a SODIMM, not an FBDIMM like Harris. Reply 14 (citing Resp. 23; Ex. 2006, 4 (“SODIMM”); Ex. 1075, 102:17–103:9 (SODIMM is unbuffered as shown in Ex. 2045 and Ex. 2046, 4.20.11-23)). Although Harris discloses unbuffered DIMMs may be an option to use, Petitioner relies upon Harris’s FBDIMM teachings in its Petition. *See, e.g.*, Pet. 16, Ex. 1023 ¶ 9, claim 24.

Patent Owner further argues that single buck converters with multiple voltage outputs were known in the art. Resp. 18 (citing Ex. 2031 ¶ 80;

Ex. 2003; Ex. 2004). As Petitioner shows, however, single buck converters for generating single voltages were also known in the art. Pet. 91 (citing Ex. 1023, Fig. 6 (640)); Ex. 1047, 1:28–32, 2:47–55, 5:56–59, 7:6–14; Ex. 1041, 1–2, 9; Ex. 1042, 16; Ex. 1048, 1–2; Ex. 1058, 5). Although Patent Owner is correct that Petitioner must show a reason for using separate buck converters, as already explained, Petitioner did so. Reply 15 (sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints); Ex. 2030, 45:21–46:10 (smaller regulators may be less costly than larger ones, and may be desirable when separate spaces are available on the memory board). Thus, we do not agree with Patent Owner’s argument that Petitioner “has not articulated any advantage of using one converter per voltage if Harris already uses a single converter for at least two voltages.” Sur-Reply 11 (citing Reply 11–14; Ex. 1023 ¶ 10).

We also find Patent Owner’s distinction between a single converter generating two voltages and two separate converters is not meaningful because of how the single converter may be internally constructed. For example, Petitioner presents the following illustration:

DC-DC Converter Specification(DRAFT) MPD4S014S



Reply 13 (citing Ex. 1075, 127:21–129:24, 228:25–232:19; Ex. 1042, 16 (corresponding to EX1048, 2)). The figure above shows two buck converters receiving two V_{in} inputs and generating respective outputs V_{out1} and V_{out2} (the LDO V_{out3} can be disregarded in this discussion). Externally, this chip resembles a single converter, and has effectively only one input if the pins 1 and 15 are tied together, but internally it has two buck converters generating respective voltages. Thus, the presence of multiple outputs suggests, to a person of ordinary skill in the art, that multiple buck converters are being used or, at least, that such a configuration is a suitable option.

Patent Owner next argues that a tightly spaced FBDIMM does not allow for the large inductors that multiple buck converters would require, and a person of ordinary skill in the art would not have modified Harris to add more components. Resp. 18–21 (citing Ex. 2031 ¶¶ 83–84; Ex. 2101, 20 (illustration of FBDIMM prototype); Ex. 2042, 7 (illustration of FBDIMM)).

Patent Owner also argues that one could use a single-inductor four-output buck converter instead of separate ones. *Id.* at 20–21.

Petitioner counters that defining the number of converters based on the number of inductors lacks any support in the '918 patent. Reply 12 (citing Ex. 1001; Ex. 2032). Petitioner argues that “it was well known that a single chip can include multiple buck converters (and inductors).” *Id.* (citing Ex. 1048, 1–2; Ex. 1075, 106:24–109:8 (discussing use of inductor in buck converter); Ex. 2030, 98:8–18 (same); Ex. 1058, 5, 14–15 (similar)). Petitioner also contends that space for multiple buck converters is not mentioned in the '918 patent or its claims, since solving any such problem was within the level of ordinary skill in the art. Reply 13 (citing *id.* at 19–20; Ex. 2030, 53:16–54:14, 89:1–21).

Petitioner has shown that one of ordinary skill in the art would have implemented a memory module with four buck converters to generate respective voltages in at least some situations. Patent Owner may be right that four buck converters using a single inductor would have been the more typical or preferred approach, as Dr. Mangione-Smith implied. *See, e.g.*, Ex. 2031 ¶¶ 76, 81–84. However, “it is not necessary to show that a combination of prior art references is the best option, only that it be a suitable option.” *Intel Corp. v. PACTXPP Schweiz AG*, 61 F.4th 1373, 1380 (Fed. Cir. 2023); *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021); *PAR Pharm., Inc. v TWI Pharms., Inc.*, 773 F.3d 1186, 1197–98 (Fed. Cir. 2014).

*b) Use of Separate Converters to Generate
 $V_{DD}/V_{DDQ}/V_{DDL}$ or V_{CC}/V_{CCFBD}*

Patent Owner argues against Petitioner's showing of a motivation to combine by arguing that neither Harris nor the FBDIMM Standards suggest separate power rails for V_{DD}/V_{DDQ} and V_{DDL} or V_{CC} and V_{CCFBD} for Voltage Mappings A and B. Resp. 21–30. Patent Owner argues that the DDR2 Specification that Petitioner cited requires that the voltages V_{DD} , V_{DDQ} and V_{DDL} are driven from a single power converter output. *Id.* at 22 (citing Ex. 1026, 9; Ex. 2061 ¶ 87). As discussed previously, Patent Owner argues that the DDR2 Specification requires a single converter (see text in blue box of Exhibit 1026, 9 above) whereas Petitioner relies on the second option (see text in red box) which does not specify how the voltages are to be generated. Although Patent Owner's argument is correct for the first option, it is not for the second option, which is the one on which Petitioner relies.

Patent Owner argues that Dr. Wolfe concedes that in all standard FBDIMMs that he was aware of, V_{DDL} , V_{DD} and V_{DDQ} for the DRAMs are connected to a single power converter, and are not individually powered on or off. Resp. 22–23 (citing Ex. 2030, 133:5–134:6, 39:11–40:24). Dr. Wolfe stated “I'm not aware of any [FBDIMMs with three different power rails] on the market. Obviously, the teaching is right here in the JEDEC specification.” Ex. 2030, 133:10–17. We understand Dr. Wolfe to be referring to the second option boxed in red in the above excerpt from JEDEC's JESD79-2B (DDR2) (Ex. 1026) and JESD79-3A (DDR3) (Ex. 1046), on which Petitioner relies. Patent Owner argues that Petitioner is suggesting a design that persons of ordinary skill in the art were aware of, but rejected, and that this is evidence of non-obviousness. Resp. 22–23.

We do not agree that a person of ordinary skill in the art would have rejected the design, however, when the JEDEC specification allows for it and Harris discloses that it is an option for generating V_{CC} and V_{DD} . Ex. 1026, 9; Ex. 1023 ¶ 9. Extending Harris’s teaching to generate four voltages with respective voltage regulators is within the level of ordinary skill in the art. *See* Ex. 2030, 43:10–14 (Dr. Wolfe stating four voltages specified by JEDEC would require use of four power supplies in most situations); 160:2–3 (“it would be obvious to use four power supplies”); Ex. 1003 ¶¶ 146–149 (credible testimony showing that buck converters were well known to persons of ordinary skill in the art). “A person of ordinary skill is also a person of ordinary creativity, not an automaton.” *KSR*, 550 U.S. at 421.

Patent Owner argues that a single power converter was required so that the voltages track one another during power up. Resp. 23–24 (citing Ex. 2006, 4; Ex. 2061 ¶ 88; Ex. 1023 ¶ 12). But Dr. Wolfe indicates that the voltages of separate converters can be made to track one another by coupling their feedback, which was a known option. Ex. 2030, 54:25–55:13. Patent Owner does not refute Dr. Wolfe’s testimony.

Patent Owner argues that JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3) do not address power generation on the memory module, only the inputs and outputs of the memory device. Resp. 24–25. Petitioner replies that Harris teaches a module that supplies all the voltages needed on the module, which would include all those needed by the memory devices. Reply 14–15 (citing Ex. 2030, 37:13–38:5); *see also* Ex. 1023 ¶ 9. Patent Owner’s argument views these JEDEC specifications, which would have been general knowledge of one of ordinary skill in the art, in isolation

without considering them together in the combination of Harris and the FBDIMM Standards. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013) (“[T]he Court required an analysis that reads the prior art in context, taking account of ‘demands known to the design community,’ ‘*the background knowledge possessed by a person having ordinary skill in the art*,’ and ‘the inferences and creative steps that a person of ordinary skill in the art would employ.’” (emphasis added)); *In re Merck & Co.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981) (“Non-obviousness cannot be established by attacking references individually where the rejection is based upon the teachings of a combination of references.”)).

Patent Owner asserts that Petitioner’s contention that an FBDIMM has separate pins for V_{DD} , V_{DDQ} and V_{DDL} does not support its contention of separate converters, “particularly where it was known that there are stability and timing benefits when using a single source.” Resp. 24 (citing Ex. 2006, 4). Dr. Wolfe explains that the FBDIMM module of Petitioner’s combination would not have separate pins to receive V_{DD} , V_{DDQ} and V_{DDL} but would derive them onboard the module from the 12-volt input pin. Ex. 2030, 35:25–36:15. As to stability, Dr. Wolfe testifies that a feedback loop maintains stability in light of current variations, temperature changes, and other factors. *Id.* at 64:25–65:10. As to timing, Dr. Wolfe indicated that separate converters can track one another by coupling their feedback, and that this was a known option. *Id.* at 54:25–55:13.

Patent Owner further argues that sequencing the different voltages does not require separate converters. Resp. 24–27. Petitioner contends that this was the obvious way to implement the JEDEC specifications (boxed in

red above) with several benefits. Reply 15 (citing Ex. 1003 ¶ 256). We agree with Petitioner that separate converters were an option that one of ordinary skill in the art would have used in some practical circumstances for reasons of cost, space availability, etc.

Patent Owner argues that it was actual practice for persons of ordinary skill in the art to use a single power converter to generate the necessary voltages. Resp. 25 (citing Ex. 2030, 133:5–134:6, 39:11–40:24). Although using a single power converter may have been the most common approach, this does not negate Petitioner’s showing that a person of ordinary skill in the art would have considered separate converters desirable under certain circumstances. *See Elekta Ltd. v. ZAP Surgical Systems, Inc.*, 81 F.4th 1368 (Fed. Cir. 2023) (quoting *Novartis Pharms. Corp. v. West-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (“Nor does an obviousness showing ‘require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation’”).

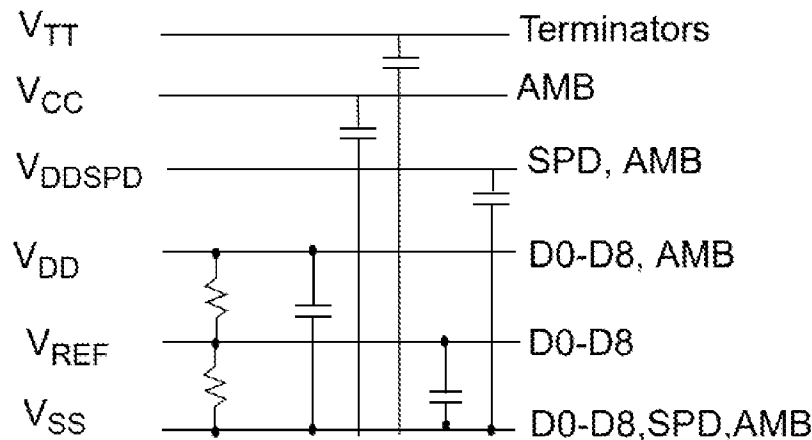
Patent Owner similarly argues that V_{CC} and V_{CCFBD} are separate voltages with separate pins, but asserts that Petitioner does not explain why a single on-board V_{CC} power source is insufficient for V_{CC} and V_{CCFBD} power rails when a single set of V_{CC} interface pins providing power from the host met the power needs for both V_{CC} and V_{CCFBD} . Resp. 25–27 (citing Pet. 30–31; Ex. 1028, 11–12; Ex. 2031 ¶ 92). For similar reasons as explained for V_{DD} , V_{DDQ} and V_{DDL} , we disagree. As Petitioner indicates, using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. Reply 15

(citing Ex. 2060, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15; Ex. 2012, 73; Ex. 1003 ¶ 255).

Patent Owner's arguments do not undercut Petitioner's showing of a motivation to combine.

c) Use of Third Buck Converter for V_{TT}

Patent Owner contends that Petitioner's Voltage Mapping C relies on three different voltages, including a termination voltage, V_{TT} , connected to terminators. Resp. 27 (citing Pet. 26, 30; Ex. 1028, 13, 15–16). For context, the relevant figure from the JEDEC JESD205 specification is reproduced below:

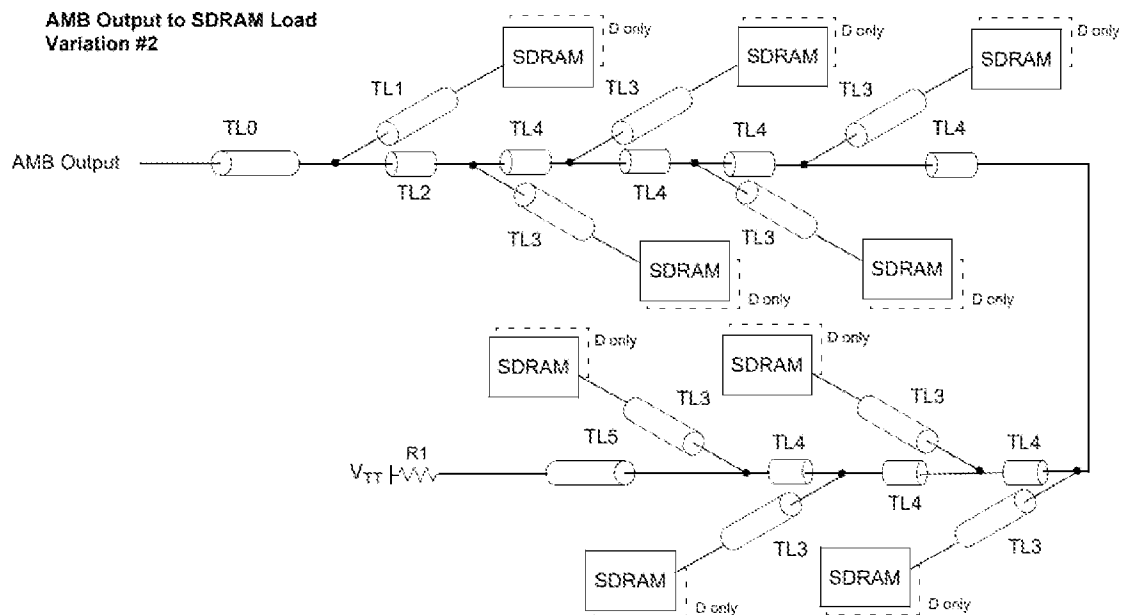


Ex. 1028, 13. The above figure shows various voltages V_{TT} , V_{CC} , V_{DDSPD} , V_{DD} , V_{REF} , V_{SS} and their relationship to one another according to the JEDEC JESD205 specification. *Id.*

Patent Owner argues that Harris does not disclose generating V_{TT} on its module and argues that it would be supplied from the motherboard instead because this would ensure that all DIMMs have the same termination voltages and eliminate ground loops that could degrade signal integrity. Resp. 27–28 (citing Ex. 2061 ¶ 96); Sur-Reply 17–18. Patent Owner further argues that design complexity favors having a single regulator for a group of

DIMMs rather than one regulator per DIMM. Resp. 28–29 (citing Ex. 1027, 20; Ex. 2031 ¶ 97; Ex. 2012, 72). In addition, Patent Owner argues that Petitioner touted JEDEC’s move of voltage regulation from the motherboard to the DIMM as a “major design improvement” and Patent Owner asks, “If it had been so obvious, why did it take the industry so long to make that design improvement?” *Id.* at 29. Patent Owner further argues that generating V_{TT} is not one of two choices, and that V_{TT} is generally not provided to DDR2 modules; passive termination is used instead. *Id.* at 33 (citing Ex. 2061 ¶ 98; Ex. 2044, 6; Ex. 2045, 6; Ex. 2046, 4.20.11-6; Ex. 2006, 5).

In Reply, Petitioner contends that Harris teaches generating all of the voltages on the module, including all of the voltages for an FBDIMM, which would include V_{TT} . Reply 15 (citing Inst. Dec. 20–23; Ex. 1023 ¶ 12; Ex. 2030, 30:15–20, 72:22–73:7, 103:11–104:23, 109:5–111:10, 239:8–20). Petitioner contends that “Harris’s module provides the voltages for the buffer 112 to send address and control signals to the DDR memory devices, which the FBDIMM Standards make clear require not just V_{DD} , but also V_{TT} to terminate those signals.” *Id.* (underlining omitted) (citing Pet. 17–18; Ex. 1028, 9, 15, 68). Petitioner reproduces the figure below:



Reply 16 (reproducing the figure at Ex. 1028, 68). The above figure is from the FBDIMM Standards and shows the net structure routing for addresses and commands from the advanced memory buffer (AMB) to the SDRAMs and how they are terminated by V_{TT} . *Id.* Petitioner contends that V_{TT} is required by the JEDEC Standards to track V_{DD} (Ex. 1028, 15) which is why dual buck converters for generating V_{DD} and V_{TT} were readily available. Reply 16–17 (citing Ex. 1028, 9; Ex. 2030, 72:22–73:7, 196:3–197:7; Ex. 1040, 1, 11; Ex. 1041, 1; Ex. 1048).

We agree with Petitioner that the combination of Harris and the FBDIMM Standards at least suggests that V_{TT} can be generated onboard the memory module. Reply 15–17 (citing Pet. 17–18; Inst. Dec. 20–24; Ex. 1023 ¶ 12; Ex. 1028, 9, 15, 68; Ex. 1040, 1, 11; Ex. 1041, 1–2, 7–9; Ex. 1048; Ex. 1075, 112:12–114:8; Ex. 2030, 30:15–20, 57:17–20, 58:18–59:10, 72:22–73:7, 103:11–104:23, 109:5–111:10, 140:15–24, 196:3–197:7, 239:8–20). Harris indicates that its on-board voltage regulator converts externally supplied voltage into appropriate local voltage levels for powering

memory devices of the memory assembly module. Ex. 1023, code (57), ¶¶ 12, 16, Fig. 2, step 204. Petitioner shows that V_{TT} is needed onboard a memory module to terminate address and control signals. Reply 15 (citing Pet. 17–18; Ex. 1028, 9, 15, 68). In addition, Petitioner shows that V_{TT} is required by JEDEC to track V_{DD} , which is why dual buck converters for generating these voltages were commercially available. Reply 16–17 (citing Ex. 1028, 9; Ex. 2030, 72:22–73:7, 196:3–197:7; Ex. 1040; Ex. 1041, Ex. 1048). In light of these facts and the teachings of Harris and the FBDIMM Standards, Petitioner has shown that a person of ordinary skill in the art would have added to the memory module a third buck converter to generate V_{TT} .

Patent Owner further argues that “even if V_{TT} is generated on module, there is no reason that it should be generated by a buck converter.”

Resp. 29. Patent Owner contends that, “[a]s Micron noted, . . . an LDO would be sufficient to power V_{TT} .” *Id.* (citing Ex. 2006, 7; Ex. 2007–2010; Ex. 2050). Patent Owner contends that an LDO is much smaller than a buck converter, which requires a controller and discrete components such as an inductor, which makes the LDO preferable given the space constraint. *Id.* (citing Ex. 2061 ¶ 99). Patent Owner also argues that an LDO is preferred when the current load is below 1A. *Id.* at 29–30 (citing Ex. 2031 ¶ 99; Ex. 2047, 21; Ex. 2048, 23; Ex. 2049, 20; Ex. 1040, 23–24, Figs. 22–25).

In Reply, Petitioner contends that buck converters were commercially available for V_{TT} and provided high efficiency compared to an LDO. Reply 17 (citing Ex. 1075, 112:12–114:8 (LDO may be only 10% efficient); Ex. 2030, 57:17–20, 58:18–59:10 (buck converters up to 98% efficient), 140:15–24 (trend has been to use buck converters). We agree with

Petitioner that, under some circumstances, one of ordinary skill in the art would have chosen a buck converter over an LDO for the reasons that Petitioner mentions.

d) Use of Buck Converter to Provide V_{DDSPD}

Patent Owner argues that Petitioner's voltage mappings A–C all relate the recited “fourth regulated voltage” to the 3.3V V_{DDSPD} . Resp. 30 (citing Pet. 26). Patent Owner contends that Harris's reasons for on-module regulation do not apply to V_{DDSPD} . *Id.* Specifically, Patent Owner argues that the “3.3V-supply voltage is sufficiently high such that a regulator on the motherboard could readily provide the needed 300mV tolerance; and the same SPD [serial presence detect] can be used for different generations of DRAMs.” *Id.* (citing Ex. 2031 ¶ 102); *see also* Sur-Reply 18–19.

Petitioner replies that Harris teaches generating all of the voltage onboard the module, including all of the voltages for an FBDIMM, which would include V_{DDSPD} . Reply 18 (citing Ex. 1023 ¶ 12; Pet. 16–18). Petitioner further contends there are only two options—generating the voltages on the motherboard or on the module—rendering either option obvious. *Id.* (citing Inst. Dec. 24).

We agree with Petitioner that Harris teaches generating voltages onboard the memory module, which suggests this would include V_{DDSPD} as it is one of the voltages required in the FBDIMM standards. Ex. 1023 ¶¶ 9–12; Ex. 1028, 15. As Petitioner contends, Harris teaches generating the voltages required in the FBDIMM standards on the memory module, and the FBDIMM standards set out what those voltages are. Ex. 1023 ¶¶ 9–12; Ex. 1028, 15.

We also agree with Petitioner that there are only two options: either V_{DDSPD} is generated on the motherboard or on the module, rendering either option obvious. Reply 18. “[W]hen there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp.” *KSR*, 550 U.S. at 421.

Patent Owner argues, if V_{DDSPD} ’s are supplied both on the motherboard and individual DIMMs, they cannot track one another, which would lead to communication problems when they are at different voltage potentials. Resp. 30–31 (citing Ex. 2031 ¶ 102). Patent Owner also argues that “3.3V is a common power rail on the motherboard such that it can be supplied to FBDIMMs without the increased cost and problems associated with providing a regulator for each of Harris’ modified FBDIMM memory boards.” *Id.* at 31 (citing Ex. 2031 ¶ 102; Ex. 2038, 13 (3.3V PSU output)).

As previously discussed, Dr. Wolfe indicated that separate converters can track one another by coupling their feedback. Ex. 2030, 54:25–55:13. We recognize, however, that this may require connection and coordination across the motherboard-DIMM interface.

Harris mentions in its background a voltage supply level of 3.3V, which is the nominal value for V_{DDSPD} in the JEDEC standards. Ex. 1028, 9. Harris also states that it is one of the values that “keeps getting lower” with each generation of DIMM/DRAM technology, implying that it should be generated not on the motherboard, but on the FBDIMM. Ex. 1023 ¶ 2. One of ordinary skill in the art thus would have recognized that V_{DDSPD} may be generated onboard the memory module when considering Harris’s teachings. Ex. 1023 ¶¶ 9–12.

We also note that the JEDEC FBDIMM standards provide that V_{DDSPD} can vary within the range of 3.3V +/- 0.3 V. Ex. 1029, 9. As long as the voltage is maintained within this range on both the motherboard and the FBDIMM, the implementation is compliant with the JEDEC standards.

Patent Owner further argues that, for the low currents required for components using V_{DDSPD} , buck converters would have been highly inefficient. Resp. 31 (citing Ex. 2051, D-1 (up to 5mA active power supply current); Ex. 2052, 6, Table III (I2C power consumption less than 15mW on average; and @3.3V, current is less than 5mA); Ex. 1040, Figs. 22–25 (efficiency less than 20% at 10mA current); Ex. 2031 ¶ 103 (an LDO with a 12V-input-3.3V-output would be 26% efficient); Ex. 1062, 14 (using LDO to generate 3.3V from 12V)). Patent Owner asserts that a person of ordinary skill in the art would have used LDOs to generate V_{DDSPD} . *Id.*

Petitioner replies that Patent Owner concedes that the claimed “converter circuit” can be an LDO (Resp. 30, n.5). Reply 18. Petitioner also contends that “using a buck converter . . . was an obvious, highly efficient, and clearly ‘suitable’ option for converting 12V to 3.3V for V_{DDSPD} . . . regardless of whether [Patent Owner] considers it ‘inferior’ to an LDO.” Reply 18 (citing Ex. 1048, 2; Ex. 1003 ¶¶ 146–49, 286; Ex. 2030, 76:4–19, 78:18–79:7, 79:24–80:11, 82:7–85:10, 88:4–20, 89:1–21, 138:14–21, 140:15–24; *Dome Pat. L.P. v. Lee*, 799 F.3d 1372, 1381 (Fed. Cir. 2016).

From the foregoing discussion, it is apparent that there are advantages and disadvantages to generating V_{DDSPD} onboard the memory module. However, “simultaneous advantages and disadvantages . . . do[] not necessarily obviate motivation to combine.” *Qualcomm*, 21 F.4th 784 (Fed. Cir. 2021) (citing *Medichem, S.A. v. Rolabo, S.L.*, 437 F.3d 1157, 1165 (Fed.

Cir. 2006)). Petitioner has shown that the advantages outweigh the disadvantages in at least some scenarios that the person of ordinary skill in the art would have considered desirable to pursue. Petitioner has shown that a person of ordinary skill in the art would have had reasons to combine Harris and the FBDIMM Standards with a reasonable expectation of success.

e) Conclusion on Motivation to Combine Harris and the FBDIMM Standards

On this record, Petitioner has shown by a preponderance of the evidence that a person of ordinary skill in the art would have combined Harris and the FBDIMM standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶9. We agree that one of ordinary skill in the art would have recognized Harris's mention of FBDIMMs as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9. For these reasons and others set forth above, we determine that one of ordinary skill in the art would have combined Harris and the FBDIMM Standards with a reasonable expectation of success.

4. Analysis of Independent Claim 1

a) Limitation 1.a: "A memory module comprising:"

Petitioner asserts that Harris's memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed "memory module." Pet. 19–20 (citing Ex. 1023 ¶¶9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 216–222; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1, for example, in Figure 3.

Patent Owner does not dispute that the preamble is met by Harris.
See Resp.

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris teaches the preamble.

b) Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. Petitioner contends that Harris and the FBDIMM Standards disclose printed circuit boards (PCBs). Pet. 20–21 (citing Ex. 1023 ¶¶ 13, Figs. 1A, 3). Petitioner also contends that a PCB may be referred to as a “memory board” or “raw card.” *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 223–225).

As for the PCB “*having an interface configured to fit into a corresponding slot connector of a host system,*” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 226–227).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “*configured to couple power, data, address and control signals between the memory module and the host system.*” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A. *Id.* at 21 (citing Ex. 1023 ¶¶ 10, 12, 19). Petitioner contends that Harris’s buffer 112 in Figure 1A is

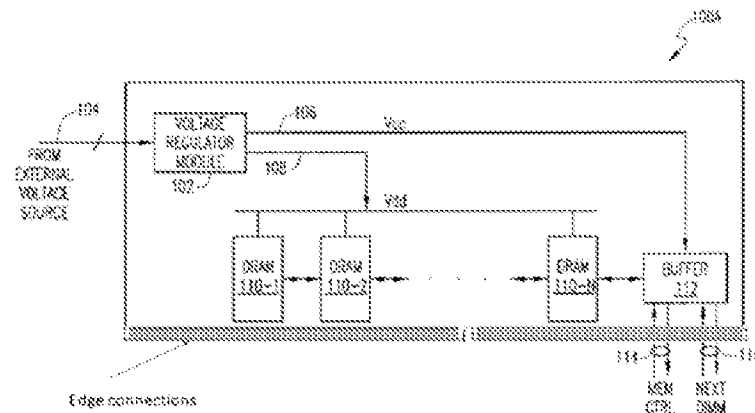
called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 22–23. Petitioner indicates that Harris’s buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Figure 1A. Pet. 22 (citing Ex. 1023 ¶ 9 (“buffer/logic component 112 is provided for buffering *command/address* (C/A) space as well as *data* space at least for a portion of memory devices 110-1 through 110-N”). In addition, Petitioner argues that the FBDIMM Standards indicate that buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

(1) Power from the Interface with the Host System

Patent Owner argues that all challenged claims require a memory module that includes a printed circuit board (PCB) having an interface with edge connections configured to couple power between the memory module and the host system. Resp. 3. According to Patent Owner, Harris does not disclose such a memory module. *Id.* Instead, Patent Owner asserts, Harris replaces the power supply interface pins with as few as six +12V non-interface pins from an external power source which is unregulated and has a wide tolerance, which does not comport with the regulated sources associated with a host system. *Id.* at 4, 6, 7 (citing Ex. 1023 ¶¶ 12, 16; Ex. 2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2031 ¶¶ 60–66; Ex. 2037, 22; Ex. 2038, 13, Table 2); *see also* Sur-Reply 2–6. Patent Owner further argues that Harris supposedly provides a “technology-independent voltage distribution scheme for memory devices wherein system board power supply and associated voltage plane(s) are eliminated.” Resp. 4 (citing Ex. 1023

¶ 19). Patent Owner argues that Harris effectively decouples the host system's voltage supply from the memory module. *Id.* at 5 (citing Ex. 2031 ¶ 38).

To illustrate its points, Patent Owner provides the following annotated figure:



Resp. 5. In the figure above, Patent Owner has modified Harris's Figure 1A to show edge connections along the bottom edge of memory board 100A, and highlights that the external voltage is connected to the side of the memory board. Patent Owner asserts that external sources connected to the side or top of a memory module were known. *Id.* at 5 (citing Ex. 2035; Ex. 2036, 41–42; Ex. 2031 ¶ 59).

In addition, Patent Owner argues that Petitioner's expert, Dr. Wolfe, testified that, in computer systems using FBDIMMs, the power supply units provide regulated voltages. *Id.* at 6 (citing Ex. 2030, EX2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2037, 22, Table 15; Ex. 2038, 13, Table 2). According to Patent Owner, it would be “inadvisable” to provide unregulated voltages via a memory board to a memory slot because the “unregulated voltages could cause random noise patterns and degrade signal integrity.” *Id.* at 7 (citing Ex. 2031 ¶¶ 65–66). Even with regulated voltage

sources, Patent Owner asserts, the voltage was not supplied via the DIMM interface. *Id.* (citing Ex. 2031 ¶ 66). Patent Owner argues that Petitioner’s modified FBDIMM is powered from an external power source “wherein system board power supply and associated voltage plane(s) are eliminated.” Resp. 8 (citing Ex. 1023 ¶¶ 10, 12, 19). Patent Owner further argues that neither Dr. Wolfe nor Dr. Mangione-Smith was aware of any memory controllers that supplied power supply voltages to DIMMs, and that Patent Owner did not provide evidence that a person of ordinary skill in the art would have known how to design a memory controller that supplied power to DIMMs. *Id.* at 8–9 (citing Ex. 1023 ¶ 17; Ex. 2030, 130:19–131:10; Ex. 2031 ¶ 67).

Although Patent Owner acknowledges that Harris discloses that its external voltage sources can be any “known or heretofore unknown voltage supplies, either regulated or unregulated,” Patent Owner argues that this would sweep in the type of voltage supply that Harris seeks to eliminate. *Id.* at 9 (citing Ex. 2031 ¶ 68). Patent Owner further asserts that “nothing suggests that the voltage paths are via the memory slot to the interface edge connections.” *Id.* at 9–10 (citing Ex. 2031 ¶ 69). Furthermore, Patent Owner asserts that different edge pins would be needed for different voltage sources independently distributed through the motherboard, increasing, rather than decreasing, the cost for power distribution in the motherboard and pin count on the DIMM connector to the motherboard. *Id.* at 10 (citing Ex. 1023 ¶¶ 13, 19–20; Ex. 2031 ¶ 70). Patent Owner summarizes its argument, stating that Harris teaches to eliminate power distribution to the DIMM slot, a feature which Petitioner does not propose to modify. *Id.* at 10–11; *see also* Sur-Reply 6. Patent Owner argues that Harris does not

couple power between the module and the host system, and therefore does not disclose elements 1.b, 16.b, 23.b and 16.f. *See* Pet. xi (1.b), xiii (16.b), xiv (16.d), and xvii (23.b). Patent Owner argues that the specific benefits Harris teaches can be achieved by decoupling power between the motherboard and memory module. Resp. 11 (citing Ex. 1023 ¶¶ 19, 20; Ex. 2031 ¶¶ 26, 58, 70). Patent Owner argues that the same arguments apply to claim 13. *Id.*

Petitioner replies that Patent Owner does not dispute that the FBDIMM Standards teach supplying power via edge connection, and that Patent Owner's expert, Dr. Mangione-Smith, admits that such edge connections were "standard." Reply 2 (citing Pet. 16–21; Resp. 3–11; Inst Dec. 18–20; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9). Petitioner also contends that Harris's reference to externally supplied voltage requires power external to the entire host system, but Harris merely requires power external to the DIMM memory module. *Id.* (citing Pet. 21–22; Resp. 3–11; Ex. 2030, 66:7–19, 67:20–68:21, 91:22–92:7, 129:24–130:17).

Petitioner further replies that Patent Owner wrongly modifies Harris's Figure 1A (above) to suggest that Harris would never use edge connections at the bottom for power. Reply 3. To illustrate its points, Petitioner reproduces the following annotated figure:

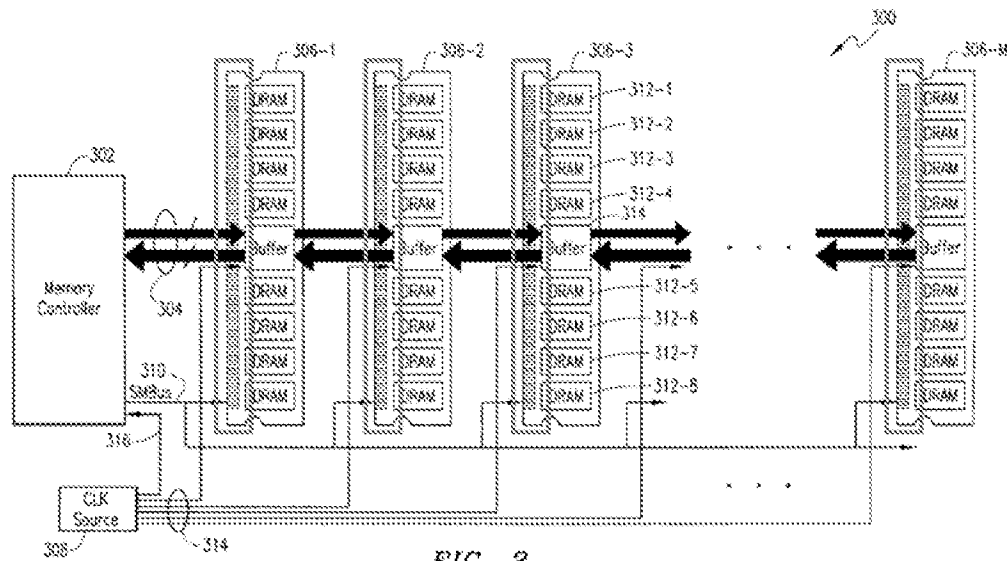


FIG. 3

Reply 4 (citing Ex. 1023, Fig. 3). In Harris’s Figure 3 as annotated by Petitioner, above, Petitioner shows that Harris’s memory modules 306-1 to 306-M are connected to the memory controller only through edge connections (boxed in red), meaning that power must come from those edge connections. *Id.* at 3–4. Petitioner further asserts that Harris’s Figure 3 is nearly identical to an Intel drawing where the memory modules admittedly receive power through the edge connections from the host system. *Id.* at 4 (citing Ex. 1075, 171:4–17; Ex. 2101, 4). According to Petitioner, Harris confirms that, “[a]lthough not explicitly shown in this FIGURE [3, above], each memory board also receives a supply voltage ... [which] may be sourced from the memory controller 302 [which is part of the host system, Ex. 1075, 167:23–168:1] or from a separate voltage source.” *Id.* (quoting Ex. 1023 ¶ 17) (alterations in original). Petitioner’s expert, Dr. Wolfe, explained that it was common for the host to supply power through edge connectors “alongside the memory controller signals.” *Id.* (quoting Ex. 2030, 131:3–5).

Petitioner contends that Patent Owner misinterprets Harris to conclude that the system power supply is eliminated entirely. *Id.* at 4–5 (citing Ex. 1023 ¶ 19; Resp. 4–5, 8, 10–11). To the contrary, Petitioner argues, Harris proposes avoiding the need for different system board voltages such as “3.3V, 2.5V, 1.8V, 1.5V and beyond,” by simply supplying a single voltage (i.e., “12V”) to the memory module so that an “on-board voltage regulator module[e.g., 102 above] [can] generate appropriate local voltage levels” on the memory module. *Id.* at 5 (citing Ex. 1023 ¶¶ 2, 3, 12–13, 19). Petitioner contends, as Dr. Wolfe explained, Harris’s “technology-independent voltage distribution scheme” eliminates the need for a “system-board-specific power supply,” not all power. *Id.* (citing Ex. 2030, 116:10–117:6).

Petitioner contends that using edge connectors like those shown in Harris’s Figure 3 is consistent with Harris’s use of the known FBDIMM design, except with “few[er]” power pins given the higher 12V input voltage. Reply 6 (citing Ex. 1023 ¶ 12; Ex. 2030, 100:12–101:19, 102:17–25, 103:11–104:23). Petitioner argues that to prevent “accidental damage” due to this change in the power pins of the edge connector, Harris teaches changing “the board’s connector keyway” so that it is “not interchangeable with the standard DIMM.” *Id.* (citing Ex. 1023 ¶ 13; Ex. 2030, 117:7–21). According to Petitioner, this was a standard technique for memory modules receiving power from the host via the edge connections. *Id.* (citing Ex. 2016, 6–7; Ex. 2101, 21–22; Ex. 1075, 171:21–175:20). Patent Owner disagrees, stating that the keyway connector does not mean the 12V pins are added to the edge connection. Sur-Reply 5.

Petitioner asserts that Patent Owner’s argument that “[s]upplying power... from [the] side... was known” “misses the point: regardless of

whether a side connection was possible, that would not negate the obviousness of using edge connections for power, as was highly common.” Reply 7 (citing *Dome Pat. L.P. v. Lee*, 799 F.3d 1372, 1381 (Fed. Cir. 2015) (existence of a better alternative “does not mean that an inferior combination is inapt for obviousness purposes”). Petitioner states that Patent Owner’s “expert admitted that using **both** was a known option, consistent with the combination for Grounds 2 to 3.” *Id.* (emphasis Petitioner’s). According to Petitioner, the side connection could be used for battery backup, as taught by Amidi, and the edge connections could be used for power from the host system during normal operation, as taught by Harris. *Id.* (citing Ex. 2035, 39; Ex. 1075, 165:10–166:12).

Considering the foregoing, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards discloses limitation 1.b (and claim 13). Harris states that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Although this is presented in the background of Harris, and Patent Owner argues that it is technology over which Harris seeks to improve, it is nonetheless information that a person of ordinary skill in the art would have known. Harris’s teaching of “replacing” power supply interface pins with as few as six +12V pins” means that those six pins take the place of the 72 pins in the interface, implying that power still comes from the interface with the host system. *Id.* ¶ 12. Harris reinforces this by disclosing that “the supply voltage may be sourced from the memory controller 302 or from a separate voltage source.” *Id.* ¶ 17. Although Patent Owner argues that Dr. Wolfe nor Dr. Mangione-Smith could recall any DIMM memory module powered by a memory controller, Harris expressly discloses this feature. *Id.*

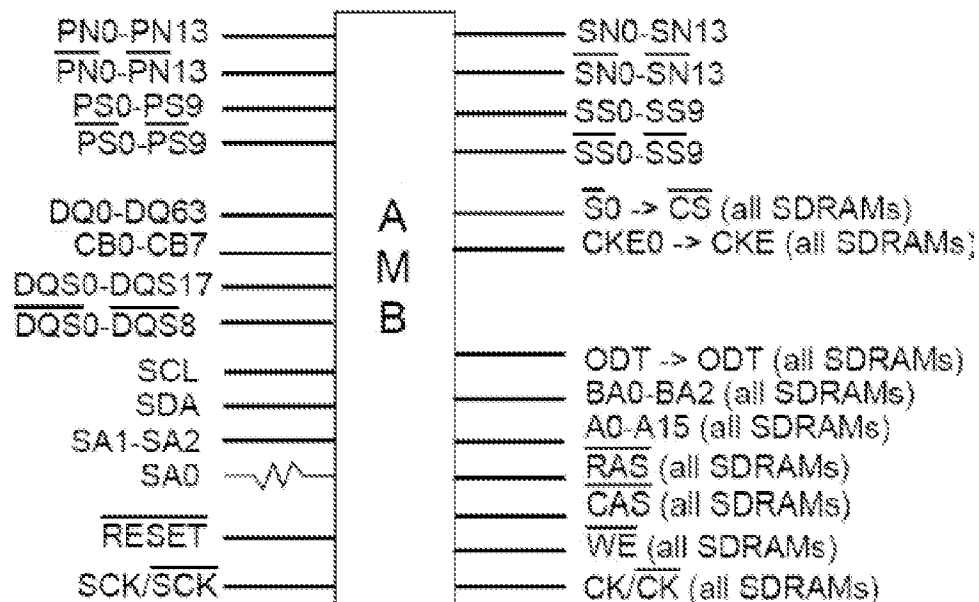
Harris further teaches that “external voltage sources may comprise any combination of known or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14. This means there may be more than one voltage supply, either regulated or unregulated, as Petitioner contends (which may be the host system and a backup source, for example). Dr. Wolfe explains that Harris’s statement that “system board power supply and associated voltage plane(s) are eliminated” means that “supplies such as the memory [V_{DD}] and the memory [V_{CC}] that are specific to a particular generation of memory chips . . . can be supplied on the DIMM rather than being supplied from the motherboard.” Ex. 2030, 116:25–117:6.

Furthermore, Petitioner indicates that the FBDIMM Standards show that the buffer AMB may be connected to a host, suggesting that the FBDIMM derives its power from the host. Pet. 24 (citing Ex. 1027, 4). The experts agree that it was common at the time for the host system to provide the regulated 12V power supply to the FBDIMM, and commercially available devices are in evidence. Ex. 2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2037, 22, Table 15, Ex. 2038, 13, Table 2; Ex. 2031 ¶ 65; Ex. 1075, 180:14–183:3; Ex. 2038, 8, 13. These facts point to the conclusion that the external voltage source may be the host system in the combination of Harris with the FBDIMM Standards notwithstanding Patent Owner’s arguments to the contrary.

*(2) Data, Address, and Control Signals Via
Edge Connection*

Claim 1 recites an “interface including a plurality of edge connections configured to couple . . . data, address and control signals between the

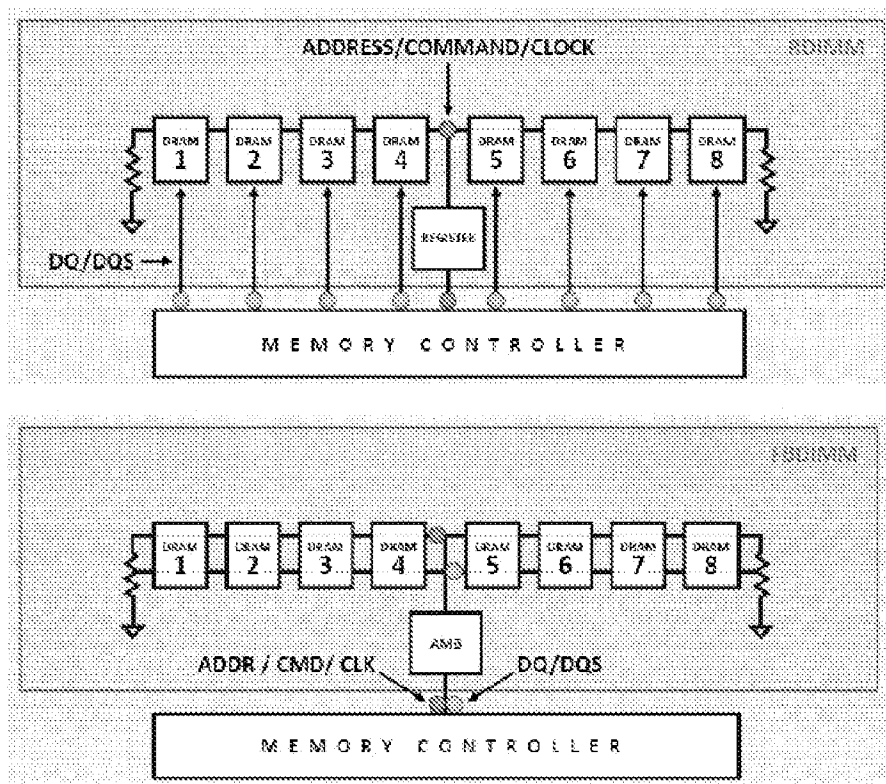
memory module and the host system.” Ex. 1001, 38:23–25. Patent Owner argues that the DIMM and AMB do not receive data signals (DQ0–DQ63) or address and control signals (A0–A15, RAS, CAS, WE) from the host. Resp. 11 (citing Pet. 22–25). According to Patent Owner, these signals are instead generated by the AMB based on decoded FBDIMM channel signals (PS[9:0] and PS[9:0]bar). *Id.* at 11–12 (citing Ex. 2031 ¶¶ 72; Ex. 1027, 3–4; Ex. 1028, 29; Ex. 2039, 2; Ex. 2101, 4; Ex. 2040, 1). Patent Owner argues that Petitioner’s evidence does not show that the data, address and control signals are received from the host. *Id.* at 15. Specifically, Patent Owner relies on the figure below from the FBDIMM Standards (*see* Sur-Reply 7 (citing Ex. 1028, 13)).



According to Patent Owner, the above figure shows that data, address and controls signals (highlighted in yellow) are generated and outputted by the buffer, and are not received from the DIMM interface. Sur-Reply 6–7 (citing Pet. 22–23; Resp. 11–15; Ex. 2031 ¶¶ 72–74). Patent Owner contends this does not comport with the claim language which requires an

“interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” *Id.* at 8 (citing Ex. 1001, 22:1–6). Patent Owner argues that Dr. Wolfe merely stated that the AMB outputs “information” to the DRAMs but the claims require specific “signals,” not just “information.” *Id.* at 8 (citing Ex. 1003 ¶ 230; Ex. 2030, 7:20–11:6). Patent Owner argues that the outputs from the AMB are not transmitted over FBDIMM’s edge connections. *Id.*

Petitioner argues that Patent Owner’s technology tutorial indicates that an FBDIMM receives address and control signals similar to an RDIMM, as shown below:



Reply 7–8 (citing Ex. 1077, 8–9; Ex. 1075, 91:23–92:19, 95:14–96:13, 97:16–98:18). The figures above show that the memory controllers transmit

address, command and clock signals (shown in red), or information containing these signals, to the register (RDIMM) or AMB (FBDIMM), which transmits them to the DRAMs. *Id.*; Sur-Reply 9.

Petitioner contends that the host system provides address, command and clock signals to the AMB encoded as a packetized, serialized signal on fewer wires than would be required to receive them as separate, parallel signals. Reply 8–9 (citing Resp. 11–12; Ex. 2031 ¶ 31; Ex. 1075, 155:22–157:1, 212:3–8, 213:3–215:20, 219:13–220:9, 226:7–228:8; Ex. 2030, 8:3–11:6). Petitioner alleges that Patent Owner is in essence attempting to rewrite the claims to require dedicated pins for these signals, which the claims do not require. *Id.* at 9.

Patent Owner argues that it is not attempting to rewrite the claims to require “dedicated pins” as Petitioner alleges. Sur-Reply 8–9 (citing Reply 8–9). Patent Owner asserts that the pin names indicate what “signals” are exchanged at the interface between the host and the FBDIMM, as shown in the figure below.

Pin Name	Pin Description	Pin
FB-OBMM Chemical Signals		
SDN	System Clock (note: positive logic)	1
SDN	System Clock Input (negative logic)	2
PSD(1:32)	Primary Northbound Data (positive logic)	3-34
PSD(1:32)	Primary Northbound Data (negative logic)	35-66
PSD(33)	Primary Southbound Data (positive logic)	67
PSD(33)	Primary Southbound Data (negative logic)	68
SDN(1:32)	Secondary Northbound Data (positive logic)	69-100
SDN(1:32)	Secondary Northbound Data (negative logic)	101-132
SDS(33)	Secondary Southbound Data (positive logic)	133
SDS(33)	Secondary Southbound Data (negative logic)	134
PSD(33:2)	For an external precision half-duplex receiver connected to Pin 135	135
DBF2 Interface Signals		
SDS(1:32)	Data Sources (positive logic)	136
SDS(1:32)	Data Sinks (negative logic)	137
PSD(1:32)/SDS(33)	Data Sources and Sinks only (positive logic). These signals are shared due to no DBF2 on analog	138-149
SDS(1:32)	Data Sources and Sinks only (negative logic)	150-161
SDS(33)	Data	162
SDS(33)	Control	163
ATD(1:32), ATD(33)	Addressed ATD is part of the air change command	164-165
SDS(1:32), SDS(33)	Event Addressing	166-167
SDS(1), SDS(3)	Start of command with SDS(1), SDS(3) and SDS(3)	168
SDS(4), SDS(6)	Part of command with SDS(1), SDS(3) and SDS(3)	169
SDS(8), SDS(6)	Part of command with SDS(1), SDS(3) and SDS(3)	170
SDS(7), SDS(3)	On-die Termination Switch	171
SDS(1:32), SDS(33)	Clock Enable line per pin	172-183
SDS(1:32), SDS(33)	High Current sense pin	184
CLK(33)	CLK(33) used on 1 and 10 module DBFMs; CLK(33:2) used on 33 module DBFMs. CLK(33:2) enables the bus clock (DBF2) when CLK(33) is 1.	185
CLK(33:2)	Negative lines for CLK(33:2)	186
SDS(1), SDS(3)	DBF2 Temperature. Common return per for SDS(1), SDS(3) and SDS(3)	187
SDS(1), SDS(3)	DBF2 Temperature. Receiver connected to common return per SDS(1), SDS(3)	188
SDS(4), SDS(6)	DBF2 Temperature. Receiver connected to common return per SDS(4), SDS(6)	189
SDS(8), SDS(6)	DBF2 Temperature. Receiver connected to V_{DD}	190
SDS(7), SDS(3)	DBF2 Temperature. Receiver connected to V_{DD}	191

Ex. 1028, 29. The figure above shows Patent Owner’s highlighting to show the “FB-DIMM Channel Signals” including PS[9:0] and PS[9:0]bar, and the “DDR2 Interface Signals” including DQ0–DQ63, A0–A15, RAS, CAS, and WE. Patent Owner states that all memory control for the DRAM resides in the host, and argues that the fact that all read, write, and configuration accesses are addressed to the DIMM does not undermine that the signals Petitioner relies on are generated and outputted by the buffer, and not exchanged at the interface between the FBDIMM and the host. Sur-Reply 9 (citing Ex. 2030, 7:20–11:6; Ex. 2030 ¶ 230).⁴

⁴ Patent Owner’s citation to “Ex. 2030 ¶ 230” appears to be an error and we cannot determine from the context what Patent Owner intended to cite.

Petitioner further argues that the FBDIMM is a preferred embodiment in the '918 patent, and excluding a preferred embodiment from the claims is “rarely, if ever correct.” Reply 9 (citing *Kaufman v. Microsoft Corp.*, 34 F.4th 1360, 1372 (Fed. Cir. 2022)). To the contrary, Patent Owner argues that the claims need not encompass FBDIMM embodiments, which are disclosed but unclaimed subject matter. Sur-Reply 10 (citing *Maxwell v. J. Baker, Inc.*, 86 F.3d 1098, 1107 (Fed. Cir. 1996); Ex. 1001, 21:38–55).

According to Petitioner, Patent Owner concedes that the signals (PS[9:0] and PS[9:0]bar) received by the AMB result in data, address, and control signals needed by DDR2 SDRAMS. Reply 9 (citing Ex. 2031 ¶ 31). Petitioner further notes that the FBDIMM Standards state that all memory control for the DRAM resides in the host, including memory request initiation, and that the AMB acts as a DRAM memory buffer for all read, write, and configuration accesses addressed to the DIMM. *Id.* (citing Ex. 1027, 1). As a buffer for all such commands to the FBDIMM, Petitioner asserts, the AMB must necessarily couple data, address, and control signals from the host system to the memory module as the claims require. *Id.*

Petitioner and Patent Owner agree that FBDIMM AMB receives information that contains data, address, and control signals encoded in packetized, serialized form at its edge connections via the signals PS[9:0] and PS[9:0]bar, and that the AMB uses these signals to generate data, address and command signals including DQ0–DQ63, A0–A15, RAS, CAS, and WE provided to the DIMMs onboard the memory module. Pet. 21–23; Sur-Reply 10. The claims merely require an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” Ex. 1001, 38:23–

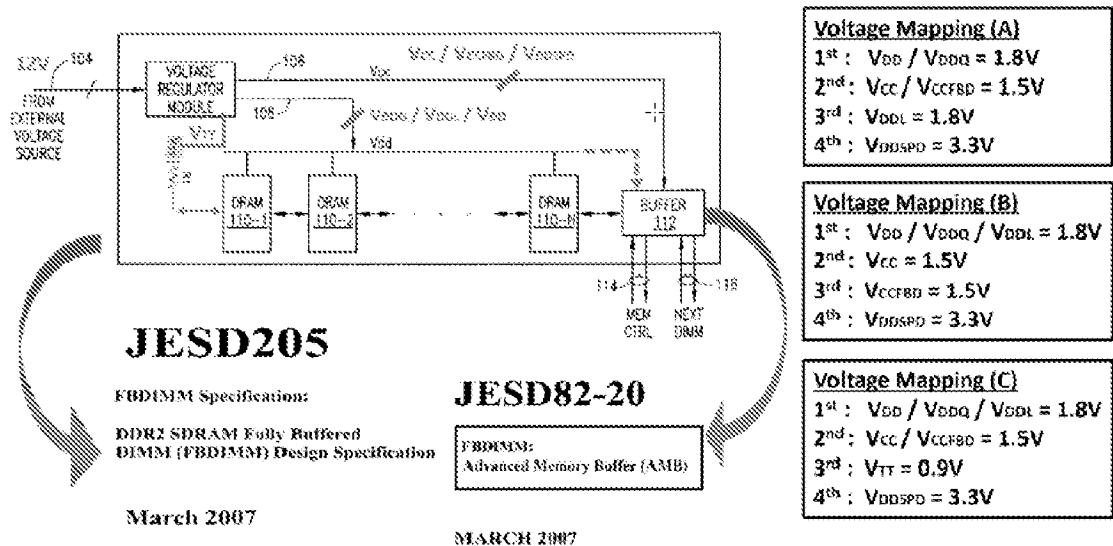
25. We determine that the data, address, and control signals of Harris and the FBDIMM Standards as received at edge connections coupling the memory module and host system satisfy this claim limitation, even though the signals are in encoded, packetized, or serialized form and thus may be received at the same pin or pins. That the signals are encoded, packetized, and serialized does not change the fact that they are data, address, and control signals. The claims do not require these signals to be received at the edge connections in any particular form or on any particular pins. *See In re Self*, 671 F.2d 1344, 1348 (CCPA 1982) (stating that limitations not appearing in the claims cannot be relied upon for patentability).

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation.

c) Limitations 1.c to 1.f: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude;”

Petitioner contends that Harris and the FBDIMM Standards disclose limitations 1.c to 1.f. Pet. 26–31. Petitioner relies on an annotated version of Harris’s Figure 1A, shown below.

Ground 1: Harris with JEDEC's FBDIMM Standards



Above is Harris’s Figure 1A annotated with information one would have obtained from the FBDIMM Standards according to Petitioner. Pet. 26 (citing Pet. 14–19; Ex. 1003 ¶¶ 232–243, 250–257, 262–276, 281–287). Specifically, the above figure shows voltage mappings A, B, and C of four voltages specified by Harris and the FBDIMM Standards. Petitioner contends the voltage range described in Harris (0.5V–3.5V) is the same range as the voltages described in the FBDIMM Standards. *Id.* at 27 (citing Ex. 1023 ¶ 9). Petitioner contends the voltages would be “regulated” because Harris indicates “tightly *regulated* power” is a problem to be solved, and proposes “at least one *on-board* voltage *regulator*” that is “capable of generating tightly-controlled voltage levels” as the solution. *Id.* at 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 234).

Petitioner further contends that Harris teaches using “buck converters” to provide the four regulated voltages. *Id.* (citing Ex. 1003 ¶¶ 236, 237, 252, 264, 286). Specifically, Petitioner contends that Harris discloses “replacing [the prior art] power supply interface pins with as few as six +12V pins

(from an external voltage source)” and then using “*a high-frequency switching voltage converter* capable of generating *tightly-controlled* voltage levels” to provide each needed on-board regulated voltage. *Id.* (alteration by Petitioner; emphasis omitted; citing Ex. 1023 ¶¶ 10, 12). Petitioner contends that a “buck converter” was a conventional device for implementing such a “voltage-reducing switching converter.” *Id.* (emphasis omitted; citing Ex. 1030, 2:41–43). Although the claim states that a converter circuit generates the fourth voltage, Dr. Wolfe testifies that a buck converter would be an obvious way to implement a converter circuit. Ex. 1003 ¶ 286 (cited by Pet. 28, n.2). Petitioner further contends a buck converter would have been an obvious way to step down a higher input voltage of 12V to a lower output voltage of 3.5V or less, and that there would have been a reasonable expectation of success because buck converters were well-known switching devices commonly used to step down the voltage between input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 146–149, 236–237; Ex. 1058, 3, 5, 12–16; Ex. 1032, 161, 164; Ex. 1024, Fig. 6; Ex. 1050, 1:21).

Petitioner further notes that “*buck converters*” were well-known as a highly-efficient way to step down voltages without generating excess heat or requiring large cooling devices, providing further motivation to use buck converters. *Id.* at 29–30 (citing Ex. 1003 ¶ 237; Ex. 1040, 1, 23–24, Figs. 22–25; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1058, 5; Ex. 1059, 5:23–30; Ex. 1062, 11; Ex. 1064 ¶ 101).

Additionally, Petitioner contends that it would have been obvious to use at least four converters in Harris given the need for four different voltages (e.g., 0.9V, 1.5V, 1.8V, 3.3V) in the FBDIMM Standards. *Id.* at 30

(citing Pet. 14–19). Petitioner asserts that it would have been obvious to use four converters because the voltages at the same level in Petitioner’s voltage mappings A and B are described as separate voltages with separate pins that are separately controllable in the FBDIMM Standards. *Id.* at 30–31 (citing Ex. 1028, 17–20, 30–32; Ex. 1026, 2–3, 9; Ex. 1003 ¶¶ 242, 256; Ex. 1062, 13).

For the reasons discussed above in Section III.D.3, we disagree with Patent Owner’s arguments regarding the use of buck converters.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitations 1.c to 1.f of claim 1.

d) Limitation 1.g: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:”

Petitioner asserts that Harris discloses a “*a plurality of components coupled to the PCB*” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 288–90, 293–297). Petitioner further contends that

these components are each coupled to at least one or more of the second, third, and fourth regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

e) Limitation 1.h: “a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 298–303). Petitioner contends that a person of ordinary skill in the art would have known that, according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* (citing Ex. 1026, cover; Ex. 1028, 9; Ex. 1045, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

f) Limitation 1.i.1: “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices”

Petitioner contends that the “first circuit” corresponds to Harris’s buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–34 (citing Pet. 20–25; Ex. 1023, Figs. 1A, 3; Ex. 1003 ¶¶ 304–317). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8. *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation.

g) Limitation 1.i.2: “the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices”

Petitioner relies on the same showing for limitation 1.i.2 as for limitation 1.i.1. Pet. 32–34.

For the reasons stated with respect to limitation 1.b, the combination of Harris and the FBDIMM Standards discloses this limitation notwithstanding Patent Owner’s arguments. *See Section III.D.4.b.2.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation for.

h) Limitation 1.i.3: “the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34–35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 318–323). Specifically, Petitioner contends that Harris’s buffer 112 in Figure 1A is coupled to “the second regulated voltage” (e.g., V_{CC} or $V_{CCFBD}=1.5V$) and “the fourth regulated voltage” (e.g., $V_{DDSPD}=3.3V$). *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation.

i) Limitation 1.i.4: “wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.”

Petitioner contends that “one of the second and fourth voltage[s]” (e.g., 1.5V) “is less than a second one of the second and fourth voltage[s]” (e.g., 3.3V). Pet. 35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 324–327).

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the current record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation.

j) Determination for Claim 1

Petitioner has shown that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable

expectation of success. Petitioner has further shown that the combination of Harris and the FBDIMM Standards meets each limitation of claim 1. Accordingly, Petitioner has established by a preponderance of the evidence that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. *Claims 2, 3, 8, 14, and 15*

Claim 2 depends from claim 1 and recites “wherein the first and third buck converters are further configured to operate as a dual buck converter.” EX. 1001, 38:53–55. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 35–40. Specifically, Petitioner contends that at the time there were many commercially available products that could output two regulated voltages using buck converters, and thus, it would have been obvious to implement two or more of the regulated voltages such as the “first” and “third” voltages as a “dual buck converter” to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design. Pet. 36 (Ex. 1003 ¶ 338). Petitioner contends that commercially available devices included the Murata MPD4S014S dual buck converter (EX. 1042, 6; Ex. 1048, 1–2; Ex. 1058, 5), the Texas Instruments TPS51020 Dual Step-Down Controller (Ex. 1040, 1, 11) and the Fairchild Semiconductor FAN5026 Dual-Output PWM Controller” (Ex. 1041, 1, 2, 9). *See* Pet. 36–40.

Patent Owner presents no argument specific to claim 2. *See* Resp.

Based on our review and consideration of the current record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 2 obvious.

Claims 3 depends from claim 1 and recites “wherein the first voltage amplitude is 1.8 volts.” Ex. 1001, 38:56–57. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 41 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 348–353).

Patent Owner presents no argument specific to claim 3. *See* Resp.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 3 obvious.

Claim 8 depends from claim 1 and recites

the plurality of components further comprising:
one or more registers coupled to one of the first, second, third and fourth regulated voltages, the one or more registers configured to register, in response to a clock, the first plurality of address and control signals, wherein the one of the first, second, third and fourth regulated voltages is selectively switched off to turn power off to the one or more registers while one or more components of the plurality of components are powered on.

Ex. 1001, 39:9–19.

Petitioner contends the combination of Harris and the FBDIMM Standards discloses claim 8. Pet. 41–46. Specifically, Petitioner contends that Harris’s buffer 112 is a component that has registers to register incoming signals and output them according to a “clock” signal, as shown in Harris’s Figures 1A and 3. *Id.* at 41. Petitioner contends Harris’s buffer 112 is coupled to the voltage V_{CC} or V_{CCFBD} as the “second regulated voltage.” *Id.* (citing Pet. 14–19, 27). Petitioner contends Harris discloses “buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data,” which a person of ordinary skill in the art would have understood involves registers for registering the address, control, and data

signals in response to a “clock” signal. *Id.* at 42 (citing Pet. 22–25; Ex. 1023 ¶¶ 9, 17, Fig. 3; Ex. 1003 ¶¶ 400–402; Ex. 1027, 19) (emphasis omitted).

Petitioner further contends that the combination of Harris and the FBDIMM Standards discloses an S3 sleep mode that selectively switches off the second voltage to turn off the power to the one or more registers while the DRAMs remain powered on to refresh data in the DRAMs. Pet. 45 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 410–411, 414–420; Ex. 1012 ¶ 9; Ex. 1027, 21, 39).

We have already addressed Patent Owner’s arguments for patentability of claim 8 with respect to limitation 1.b and determine they do not undermine Petitioner’s showing. *See* Section III.D.4.b.2.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders obvious claim 8.

Claim 14 depends from claim 8 and recites

wherein, in response to selectively switching on the one of the first, second, third and fourth regulated voltages to the one or more registers, the one or more registers is configured to output the registered first plurality of address and control signals to the plurality of SDRAM devices.

Ex. 1001, 39:40–45.

Petitioner contends that when the S3 sleep mode ends, normal DRAM transactions begin again and the second voltage which powers the input/output registers switches on and the registers output the registered address and control signals to the SDRAM devices. Pet. 46 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 462–466).

For the reasons stated for claim 8, we determine that Patent Owner's arguments do not undermine Petitioner's showing. Resp. 11–15.⁵

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 14 obvious.

Claim 15 depends from claim 1 and recites

the plurality of components further comprising:
a logic element including one or more integrated circuits and discrete electrical elements, the one or more integrated circuit including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information.

Ex. 1001, 39:46–52.

Petitioner contends that the combination of Harris and the FBDIMM Standards discloses claim 15. Pet. 47–50. Specifically, Petitioner contends logic in Harris's buffer includes an integrated circuit, as does its serial presence detect (SPD), including non-volatile memory. Pet. 47 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 435–437, 468–476). Petitioner further contends the logic element includes discrete elements such as resistors and capacitors to terminate voltages for Harris's buffer. *Id.* Petitioner asserts that the S3 sleep mode of the FBDIMM Standards requires S3 Recovery Configuration Registers. *Id.* According to Petitioner, an FBDIMM like Harris's memory module will store configuration information in non-volatile memory before entering into S3 sleep mode. *Id.* (citing Ex. 1027, 25, 95–96, 141).

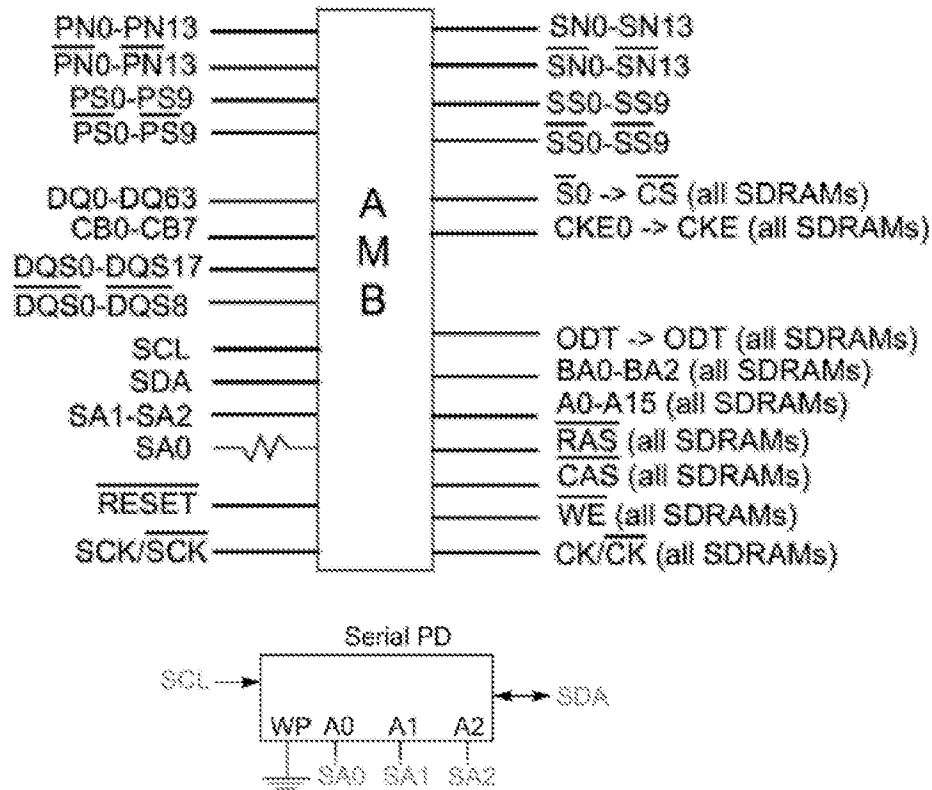
Petitioner further contends that a person of ordinary skill in the art would

⁵ Patent Owner makes this same argument for claim 21 challenged under Ground 2, which we find does not undermine Petitioner's showing for the same reasons as stated for claim 8. Resp. 11–15.

have understood that the “non-volatile memory” can be implemented in the SPD device separate from the integrated circuit implementing the AMB (Advanced Memory Buffer) where the SPD is used to store configuration information in the non-volatile memory. *Id.* at 48–49 (citing Ex. 1023 ¶ 19; Ex. 1027, 117; Ex. 1028, 13; Ex. 1066, 26:64–27:4; Ex. 1067, 1-1; Ex. 1003 ¶ 436). Petitioner further contends a person of ordinary skill in the art would have understood that the logic element includes discrete electrical elements such as resistors and capacitors, as taught by the FBDIMM Standards. *Id.* at 49–50 (citing Ex. 1003 ¶ 475; Ex. 1023 ¶ 9, Fig. 1A; Ex. 1028, 13, 42–45).

Patent Owner argues that Dr. Wolfe acknowledged that the AMB neither inherently nor expressly disclosed having a non-volatile memory. Resp. 43 (citing Ex. 2030, 285:23–286:17; *see also* Ex. 2031 ¶ 122); Sur-Reply 22. Patent Owner further argues that the statement in the FBDIMM Standards that the Control and Status Register (CSR) for the S3 sleep mode should be stored in non-volatile memory does not mean that the non-volatile memory is in the AMB. Resp. 43 (citing Ex. 2030, 285:23–286:17); Ex. 1003 ¶ 435 (CSR meaning). Patent Owner further notes that Dr. Wolfe stated that he had no knowledge that any AMB included a non-volatile memory. *Id.* (citing Ex. 2030, 292:10–20, 293:1–6). Patent Owner argues that the CSRs that the Petition references at page 47 are stored in volatile, not non-volatile, memory. *Id.* (citing Ex. 2030, 293:8–294:2). Patent Owner further argues that SPD and AMB are two different components. *Id.* (citing Ex. 2030, 293:1–6, 292:21–25). For these reasons, Patent Owner argues that Petitioner “has not cited any reference showing a logic element with a non-volatile memory.” *Id.*

To explain his views, Dr. Wolfe reproduces the figure below from the FBDIMM Standards:



Ex. 1003 ¶ 437 (citing Ex. 1028, 13) (highlighting Dr. Wolfe’s). Dr. Wolfe shows that the SPD is connected to the AMB to exchange the signals indicated in yellow highlighting. He further states that “the ‘logic element’ includes the logic in the buffer 112 and non-volatile memory (e.g., internal or external to the buffer) for storing configuration information as required by the S3 sleep mode.” *Id.*

There is no dispute on this record that the SPD is non-volatile memory—the FBDIMM Standards require it to be. Ex. 1027, 25. Dr. Wolfe expounds that the SPD can contain non-volatile memory, such as an EEPROM, for storing configuration information using a higher voltage supply at 3.3V. Ex. 1003 ¶ 436. Dr. Wolfe stated that “it would have been

obvious to a [person of ordinary skill in the art] to implement the buffer 112 with an internal nonvolatile memory because it would decrease the number of components, interfaces and interconnections on the memory module, resulting in a cheaper, faster, and more secure solution.” Ex. 1003 ¶ 436. Dr. Wolfe further states that the SPD and AMB are tied to the same voltage V_{DDSPD} . *Id.* In other words, the SPD and AMB are closely related.

Dr. Mangione-Smith states that the SPD and AMB are two different components. Ex. 2031 ¶ 123. The claims do not preclude, however, that the “logic element” may be parts of the SPD and AMB (Harris’s FBDIMM buffer) related to the SPD functionality. Ex. 1023 ¶ 9; Ex. 1028, 105; 1003 ¶ 474. In fact, as Petitioner notes, claim 15 recites that the logic element may include “*one or more* integrated circuits.” Reply 23 (citing Ex. 1001, 23:1–18; Pet. 47–49; Ex. 1003 ¶¶ 434–436). Accordingly, Petitioner has shown that the combination discloses a “logic element including a non-volatile memory” as claimed. Pet. 47–50, 67–68, 72; Reply 23.

There is no dispute that the CSRs, which are stored in the SPD before the memory module enters the S3 sleep mode, constitute “configuration information” as recited in claim 15. Pet. 47 (citing Ex. 1027, 25 (DRC, MTR, DREFTC, DAREFTC, S3RESTORE[15:0], SPDPAR[15:0] – SPDI personality bytes), 95–96, 141; Ex. 1003 ¶ 435)).

Further, there is no dispute that Harris’s buffer and the SPD are “integrated circuits,” and that the buffer is connected to “discrete electrical elements” (e.g., resistors and capacitors) as recited in claim 15. Ex. 1023, Figs. 1A, 3 (showing buffer is integrated circuit on DIMM); Ex. 1027, 25 (non-volatile memory is an integrated circuit); Ex. 1028, 43 (showing resistors and capacitors connected to AMB).

Thus, we agree with Petitioner that the features of these claims are taught or at least suggested by the combination of Harris and the FBDIMM Standards. Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 15 obvious.

6. *Claim 23*

Claim 23 is an independent claim. Ex. 1001, 40:50–41:21. Petitioner contends that the limitations of claim 23 are substantially identical to earlier limitations, and are thus obvious for the reasons stated above. Pet. 50–51.

Patent Owner presents no additional argument for claim 23 other than arguments previously discussed. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris in combination with the FBDIMM Standards renders obvious claim 23.

7. *Determination for Ground 1*

Petitioner has shown by a preponderance of the evidence that claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris and the FBDIMM Standards for the reasons explained.

E. Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi

Petitioner contends that claims 1–30 of the '918 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 51–75. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's arguments and, therefore, establishes by a preponderance of the evidence

that claims 1–30 of the '918 patent are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

1. *Amidi (Ex. 1024)*

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled “Clock and Power Fault Detection for Memory Modules.” Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 12.

Amidi’s Figure 5 is reproduced below.

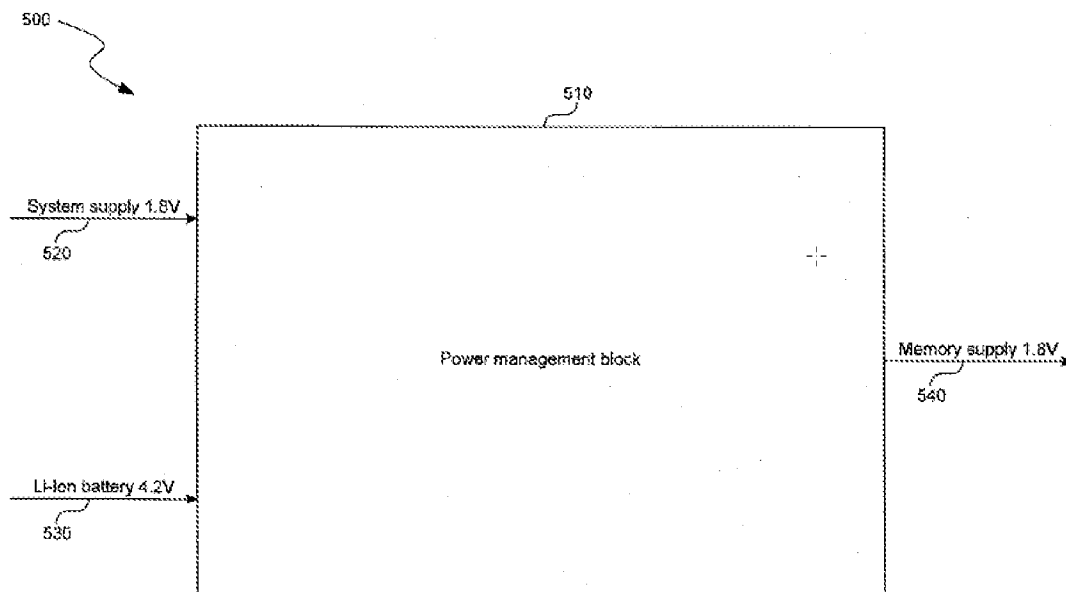


FIG. 5

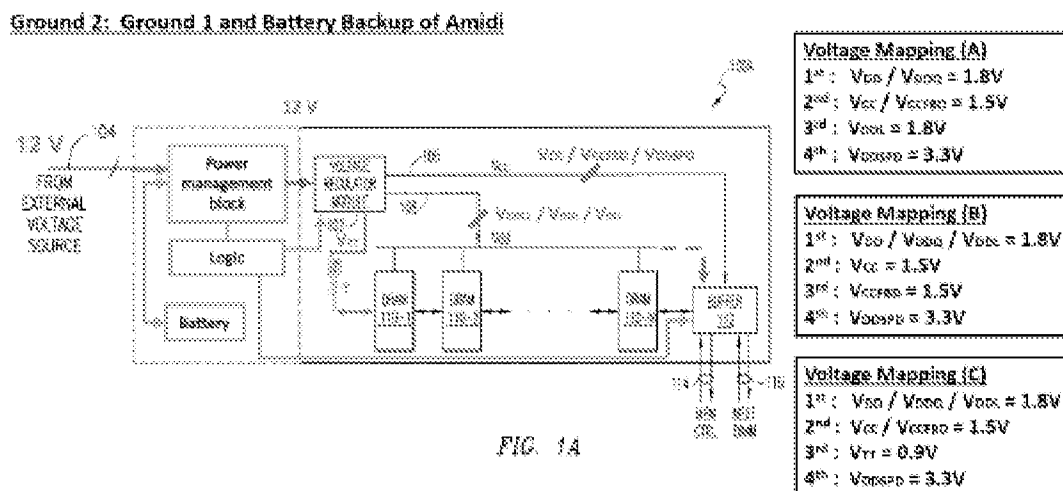
Amidi’s Figure 5 above illustrates a power management block 510 that receives an incoming system supply 520, and incoming battery supply 530, and generates an outgoing memory power supply 540 that is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Figs. 5, 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 52. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 52–53 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 170–175). Petitioner further notes that Amidi’s power management block could be modified easily to work with Harris’s FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 53 (citing Ex. 1024, Figs. 5, 6; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 171–172). Petitioner contends that, to a person of ordinary skill in the art, it would have been obvious to use the 12V external supply stepped-down with a buck converter to a 5V supply for charging Amidi’s battery, and that Amidi’s battery voltage would be stepped-up with a boost converter to the 12V level used by Harris’s memory module. *Id.* at 53–54 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 171–172). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed, and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 54 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi’s battery backup mode is similar to the S3 power-saving mode of Harris’s FBDIMM memory module and the FBDIMM Standards, such that a person of ordinary skill in

the art would have been motivated to combine their teachings. *Id.* at 54–56. Petitioner contends this is a straightforward modification of Harris’s memory module in view of Amidi and the knowledge of a person of ordinary skill in the art and simply uses a known technique (e.g., Amidi’s battery backup techniques) to improve a similar device (e.g., Harris’s memory module) in the same way (e.g., to provide a backup power supply using a battery). *Id.* at 55 (citing Ex. 1003 ¶ 175). In addition, Petitioner contends, the modification merely applies a known technique (e.g., providing a backup power supply) to a known device (e.g., a memory module) that is ready for the improvement to yield predictable results (e.g., redundancy when the system supply or clock fails). *Id.*

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:



Pet. 55–56 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 174–179). Above, Petitioner has annotated Harris’s Figure 1A to show features added from the teachings

of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue). *Id.*

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Resp. 31–38; Sur-Reply 19–20. Specifically, Patent Owner argues that Harris has “logic 124” that selects a different voltage source if power becomes unavailable on one of the supply voltage paths 120-1 through 120-K. Resp. at 32–33 (citing Ex. 1023, Fig. 1B; Ex. 2031 ¶ 106). Patent Owner further argues that substituting Amidi’s battery backup solution for Harris’s redundant power sources would require substantial space to accommodate the lithium battery and associated battery charging and monitoring circuits and power management block. *Id.* at 35–38 (citing Ex. 1024, Figs. 4–6; Ex. 1023 ¶¶ 2, 13; Ex. 2031 ¶¶ 108 n.5, 109). Patent Owner argues that Petitioner “has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the combination would even fit onto Harris’ circuit board” and that “Petitioner has presented no evidence that a [person of ordinary skill in the art] would expect Harris’ memory module to be able to accommodate three buck converters, a lithium battery and its associated charging and monitoring circuits, and Amidi’s power management module, at least not without undergoing a substantial redesign.” *Id.* at 35–38 (citing Ex. 1055, Fig. 4; Ex. 2101, 20; Ex. 2042, 7; Ex. 2031 ¶ 112).

We agree with Petitioner that Amidi provides a redundant power supply on the memory module that is independent from the host, and that a person of ordinary skill in the art would have seen the value of adding a

redundant power supply onboard Harris's memory module at least for some practical applications. Reply 19–20 (citing Pet. 52–53; Ex. 1003 ¶ 170; Ex. 1024, 1:28–35, 2:6–26). Although Harris mentions an “external voltage source,” it does not mention the term “battery.” Hence, Patent Owner's arguments that Amidi provides a solution to a problem that Harris already solves is not correct. *See* Ex. 1003 ¶ 170 (“Amidi recognizes that it is useful to keep data on the memory module with a backup power supply on the module itself, such as a battery supply, when the entire system surrounding the memory module loses power.”)(citing Ex. 1024, 1:28–35; 2:6–26). Furthermore, Petitioner shows that memory modules with battery backup were known in the art and readily available. Reply 19 (citing Ex. 2035, 39; Ex. 1075, 165:10–167:7). As to the space issue, Petitioner notes that even if the battery backup occupies one side of an FBDIMM board, the other side still can include one rank or two ranks using stacked memory devices. *Id.* at 19–20 (Ex. 1028, 36; Ex. 1075, 74:22–75:25, 77:10–17).

Accordingly, we determine that Petitioner has shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3, 8, 14, 15, and 23*

We agree with Petitioner that the addition of Amidi does not negate Petitioner's showing with respect to the previous ground. Pet. 56, 66–67 (citing Ex. 1003 ¶¶ 215–353, 412–413, 467–476, 557–597). Accordingly, Petitioner has shown by a preponderance of the evidence that claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claims 16–22 and 30 (“pre-regulated input voltage”)*

Claims 16–22 and 30 recite a “pre-regulated input voltage.” Ex. 1001, 39:61, 42:23. As explained in Section III.C.2, we understand “pre-regulated input voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage converters. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112).

Petitioner contends that, with the exception of the “pre-regulated input voltage” limitation, claims 16–22 and 30 are combinations of limitations previously discussed which are rendered obvious for the reasons for reasons earlier presented in the Petition. Pet. 70–75. Petitioner provides a table showing the correspondences of the limitations of these claims to others discussed earlier in the Petition. *Id.* at 71–73. We agree that each limitation of these claims is taught or suggested by the combination of Harris, the FBDIMM Standards, and Amidi for the reasons previously discussed in the Petition for those same limitations.

Turning to the “pre-regulated input voltage” limitation, Petitioner contends that Harris discloses an input that is “pre-regulated” because it is “within pre-determined limits” and because it can be “regulated.” Pet. 73 (citing Ex. 1023 ¶¶ 13–14; Ex. 1003 ¶¶ 487–488). Alternatively, Petitioner contends that Amidi teaches a boost converter in its power management block that converts (and thus pre-regulates) the battery voltage to an output of 12 Volts. Pet. 73–74 (citing *id.* at 52–56; Ex. 1003 ¶ 489; Ex. 1032, 161 (explaining that switch-mode converters, including boost converters, “convert the unregulated dc input into a controlled dc output at a desired voltage level”))).

Patent Owner states that Harris's 12V input with a "+/-15%" tolerance is an unregulated voltage source. Resp. 38 (citing Ex. 1023 ¶ 13). Patent Owner does not explain what distinguishes a regulated voltage from an unregulated one, but, in any event, Harris expressly discloses that the external voltage supply may be "regulated." Ex. 1023 ¶ 14. This means "pre-regulated" in the context of the claims. Pet. 73–74 (citing Ex. 1023 ¶¶ 13–14; Ex. 1003 ¶¶ 487–489). Because the voltage that is received at the voltage converters has already been regulated externally, this meets our adopted construction (which, as we note above, is consistent with Patent Owner's proposed construction). *See* Resp. 38 ("A pre-regulated input voltage is just a regulated voltage provided to these voltage conversion circuits.").

Patent Owner further argues that "when a power supply other than the 12V external power source is used, Harris teaches the use of corresponding VRMs." Resp. 38 (citing Ex. 1023 ¶ 14, Fig. 1B). Therefore, Patent Owner argues, there is no reason to generate a pre-regulated 12V voltage, as Petitioner contends. *Id.* But, as explained, Harris expressly discloses that a "regulated" voltage may be input to its memory module and, therefore, Harris teaches converters that are "configured to receive a pre-regulated input voltage," as recited in the claims. Ex. 1023 ¶ 14.

After reviewing the record, we determine that Petitioner has shown by a preponderance of the evidence that the combination of Harris, the FBDIMM Standards, and Amidi teaches the "pre-regulated input voltage" subject matter of claims 16–22.

5. *Claims 5–7, 9–13, 16–22, and 24–27*

Claims 5–7, 9–13, 16–22 and 24–27 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being **greater than** a predetermined threshold voltage. Ex. 1001, 38:61–67, 40:7–13, 41:22–27, 42:21–26. Claims 11, 12, 18, 19, 25, and 26 require certain actions to be taken in response to the “trigger signal.”

a) *Trigger Signal on Overvoltage—Claims 5, 16, and 24*

Claim 5 recites

a voltage monitor circuit configured to monitor a power input voltage received via a second portion of the plurality of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage.

Ex. 1001, 38:61–67. Claims 16 and 24 recite similar limitations. *Id.* at 41:22–27, 40:7–12.

We address claim 5 as representative of this group of claims. For claim 5, Petitioner contends that Harris teaches detecting both undervoltage and overvoltage conditions. Pet. 62 (citing Ex. 1023 ¶ 13). In this regard, Harris states that the tolerance of the +12V power supply is “around +/- 15%.” Ex. 1023 ¶ 13. Petitioner asserts that, in the combination, Amidi’s power supervisory module/block 665 (“voltage monitor circuit”) monitors system supply 605 (“power input voltage”) received from Harris’s edge connections. Pet. 60–61 (citing Ex. 1003 ¶¶ 359–376). Petitioner further asserts that Amidi’s signal 670 is the claimed “trigger signal” to switch to battery power. *Id.* at 61–62 (Ex. 1024, 4:44–52, 5:25–43, 8:23–29, 9:8–12, Figs. 5, 6, 14–15). Petitioner contends that such circuitry for either

overvoltage and undervoltage protection was well known and commercially available. *Id.* at 62–63 (citing Ex. 1063, 1–2; Ex. 1061, 15 (circuit for “undervoltage” and “overvoltage” detection); Ex. 1062, 15 (same); Ex. 1065, code (57), ¶¶ 14, 18–19, Figs. 1, 5 (similar)).

Patent Owner argues that neither Harris nor Amidi teaches detecting an overvoltage. Resp. 39; Sur-Reply 20–22. However, Harris discloses that its +12V power supply has a tolerance of “+/- 15%” which indicates it would be a concern if the power supply was greater than 15%. Ex. 1023 ¶ 13. Amidi indicates concern with “power fault[s].” Ex. 1024, codes (54), (57). Although Patent Owner points out that Amidi discloses detection of undervoltage (Ex. 1024, Fig. 14), “power fault” is broad enough to encompass overvoltage as well. We agree with Petitioner that a person of ordinary skill in the art considering these teachings would have considered it obvious to detect overvoltage and generate a trigger signal in response to the detection.

Patent Owner argues that the voltage regulators on which Petitioner relies do not detect input overvoltage because they can accommodate a wide range of input voltages, and there is a low probability that they would exceed their operational range. Resp. 40 (citing Ex. 2030, 58:24–61:12; Ex. 1040; Ex. 1041; Ex. 2031 ¶ 117). Petitioner replies that power surges were a known problem that could cause data loss or corruption or damage circuitry in memory boards. Reply 21–22 (citing Ex. 1003 ¶¶ 371–373; Ex. 2030, 254:24–256:11, 258:12–20; Ex. 1065, 18–19; Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15; Ex. 1075, 196:10–197:13). Petitioner contends that Harris discloses that its module can accommodate only “+/- 15%” of the

nominal supply voltage, which is consistent with maximum ratings for commercial converters. *Id.* at 22 (citing Ex. 1041, 4).

We agree with Petitioner that power surges would have been a problem in at least some circumstances for which one of ordinary skill in the art would have seen the value of voltage detection in order to take preventive action to avoid loss of data and damage to the memory module. Dr. Wolfe credibly explains that overvoltage was a danger known to persons of ordinary skill in the art, citing industry datasheets that specify overvoltage parameters. Ex. 1003 ¶¶ 371–373 (citing Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15). Indeed, the cited evidence uses the phrase “an overvoltage fault” (Ex. 1061, 15), which shows that power faults are not limited to undervoltage situations.

b) Logic Element with Non-Volatile Memory—Claims 10, 11, 15, and 22

Claims 10, 11, 15, and 22 recite a logic element that includes a non-volatile memory. Ex. 1001, 39:23–27, 39:28–30, 39:46–52, 40:43–49. Claim 10 further recites that the non-volatile memory is “configured to store configuration information.” Claim 11 depends from claim 10 and recites that the logic element writes information into the non-volatile memory in response to the trigger signal. Claim 15 further recites that logic element includes one or more integrated circuits including an internal non-volatile memory, and discrete electrical elements. Claim 22 omits mentions of integrated circuits or discrete elements, but recites that the configuration information is used to program the logic element.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi discloses these claims. Pet. 47–50, 67–68, 72; Reply

23. Specifically, Petitioner contends that the “non-volatile memory” for the “logic” can be either in Harris buffer (e.g., an AMB) or in an SPD connected to that buffer. Pet. 47–50, 67–68, 72.

Patent Owner argues that the FBDIMM AMB is not disclosed as having a non-volatile memory, and that, although the SPD contains a non-volatile memory according to the FBDIMM standards, the AMB and SPD are two different components. Resp. 43.

For the reasons explained for claim 15 under Ground 1, Patent Owner’s arguments do not undermine Petitioner’s showing. Thus, we agree with Petitioner that the features of these claims are taught or at least suggested by the combination of Harris, the FBDIMM Standards, and Amidi.

c) Write Operation on Overvoltage—Claims 11, 12, 18, 19, 25, and 26

Claim 11 recites “wherein, in response to the trigger signal, the logic element writes information into the non-volatile memory.” Ex. 1001, 39:28–30. Claim 12 depends from claim 11 and recites a non-volatile memory and a controller, and the controller performs a write operation to the non-volatile memory in response to the trigger signal. *Id.* at 39:31–39. Claim 18 recites a controller that executes a write operation in response to a (trigger) signal. *Id.* at 40:17–21. Claim 19 recites that the write operation includes writing data information into non-volatile memory. *Id.* at 40:22–24. Claim 25 recites a controller that executes a write operation in response to the trigger signal. *Id.* at 42:1–5. Claim 26 recites that the write operation includes writing data information to non-volatile memory. *Id.* at 42:6–8.

For claim 11, Petitioner contends that the trigger signal was explained with respect to claim 5 and may be a power disruption. Pet. 68. Petitioner further contends that the logic element writes S3 configuration information as discussed for claims 10 and 15, to allow sleep mode to conserve power during Amidi's battery backup mode. *Id.* (citing Ex. 1003 ¶¶ 439–443). Petitioner contends that Amidi's battery backup mode is similar to the S3 power-saving mode of Harris's FBDIMM memory module, because both modes put the SDRAMs in a self-refresh state to preserve data while conserving power. *Id.* (citing Pet. 45–46; Ex. 1024, 2:16–19, Fig. 11). In the event of a power disruption causing a trigger signal, Petitioner contends that a person of ordinary skill in the art would have been motivated to use the S3 sleep mode (Pet. 45–46) to conserve power during Amidi's battery backup mode, as discussed in claims 10 (Pet. 67) and 15 (Pet. 47–50), and that the S3 sleep mode requires writing S3 configuration information into the non-volatile memory in the SPD before going to sleep. *Id.* (citing Ex. 1003 ¶ 442). For claim 12, Petitioner contends that the “controller” corresponds to Harris's buffer, which stores S3 configuration information in non-volatile memory in the SPD before entering S3 sleep mode. Pet. 47–50, 67–68; Ex. 1071, 39–46; Ex. 1073, 46–63; Ex. 1003 ¶¶ 444–456. Petitioner relies on similar disclosures for claims 18, 19, 25, and 26. Pet. 71–73.

Patent Owner argues that S3 sleep mode is a low power state mode whereas overvoltage does not result in low or lost power. Resp. 44 (citing Ex. 2031 ¶ 124); Sur-Reply 22–23. Patent Owner also argues that known converters could operate even at high input overvoltage. Resp. 44 (citing Resp. Sect. IV.F.1).

Petitioner counters that overvoltage conditions can cause data loss, thus motivating a person of ordinary skill in the art to switch to backup power, where the S3 mode would be used. Reply 23 (citing Pet. 68–69).

Patent Owner argues that a memory module cannot decide to enter the S3 sleep mode upon detection of an overvoltage. Sur-Reply 22. Instead, according to Patent Owner, the CPU determines when the memory module should enter the S3 sleep mode. Sur-Reply 22–23 (citing Ex. 2030, 282:22–284:25; Ex. 2031 ¶¶ 124–125; Ex. 2006, 3). Patent Owner argues there is no evidence that such a modification could be predictably made or result in an operable implementation. *Id.* at 23 (citing Ex. 2031 ¶ 125).

As Petitioner notes, Amidi provides the teaching of generating the trigger signal *onboard the memory module* in response to detecting a “power fault” or “disruption.” Pet. 12, 52, 55, 61–62; Ex. 1003 ¶¶ 132, 171, 174, 368–374. Considering the prior art teachings together, we agree with Petitioner that one of ordinary skill in the art would have understood that overvoltage detection and generating the trigger signal onboard the memory module was an option to pursue. This option would have been both predictable and pursued with a reasonable expectation of success since the prior art combination, Amidi in particular, teaches how it could be implemented.

6. *Claims 4, 28, and 29*

Claim 4 recites that the second, third, and fourth voltage amplitudes are 2.5V, 1.2V, and 3.3V, respectively. Ex. 1001, 38:58–60. Petitioner contends that Harris discloses voltages within the range of 0.5V to 3.3V, and that Voltage Mapping D from the FBDIMM Standards discloses these voltages. Pet. 57–60.

Claim 28 recites that the second and third buck converters are configured to operate as a dual buck converter. Ex. 1001, 42:14–16. Petitioner contends this feature is disclosed by the combination. Pet. 73.

Claim 29 recites that the SDRAMs devices are configured to receive one of the first through fourth regulated voltages at 1.8V. Petitioner contends this feature is disclosed by the combination. *Id.*

Patent Owner presents no arguments specific to these claims. Petitioner has shown that these claims are disclosed by the combination.

7. *Determination for Ground 2*

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations of claims 1–30 are taught or suggested by the combination. Accordingly, Petitioner has demonstrated by a preponderance of the evidence that claims 1–30 of the ’918 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

F. *Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck*

Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck. Pet. 75–77.

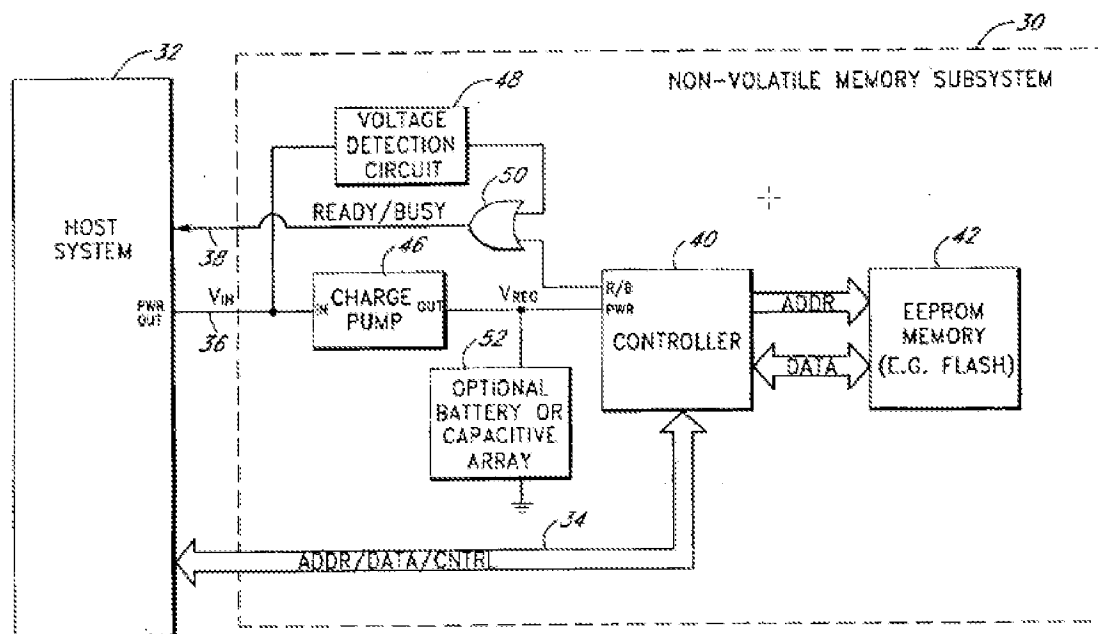
1. *Hajeck (Ex. 1038)*

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1038, code

(54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck's Figure 1 is reproduced below.



In Hajeck's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage, including

undervoltage and overvoltage conditions, and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–43.

2. *Motivation to Combine*

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajeck with Harris, the FBDIMM Standards, and Amidi because such person would have appreciated the desirability of switching to backup power in both undervoltage and **overvoltage** conditions. Pet. 75–76. Hajeck teaches to use a backup power supply, such as a charge pump, battery, or capacitive array, in response to power loss from the host system to complete outstanding operations, and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57), 1:53–61.

Petitioner also contends that overvoltage caused by anomalies such as power surges “can cause data loss, data corruption or circuitry damage in memory boards—as anyone with a surge protector at home knows.” Reply 22 (citing Pet. 75–76; Ex. 1003 ¶¶ 137, 184). Although Patent Owner argues that Hajeck’s charge pump protects against power surges (Resp. 41), Petitioner relies on Hajeck for its “teachings related to overvoltage anomalies when implementing Amidi’s voltage supervisory block in Ground 2 so that it detects both undervoltage and overvoltage anomalies.” Pet. 76 (emphasis omitted) (citing Ex. 1003 ¶¶ 185–187).

We agree with Petitioner that Patent Owner’s arguments attack the references individually and do not properly consider what the combination of references would have signified to a person of ordinary skill in the art. Reply 21 (citing Resp. 39–42); *In re Merck & Co.*, 800 F.2d 1091, 1097

(Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981)). Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition to the extent that the combination of Harris, the FBDIMM Standards, and Amidi were deemed not to already disclose that feature. Pet. 76–77. Thus, the features that Patent Owner argues are missing from Hajeck are not ones for which Petitioner relied on Hajeck.

For similar reasons, we also do not agree with Patent Owner’s argument that Hajeck does not suggest that data loss was a concern because its charge pump could provide the desired V_{REF} indefinitely. Resp. 42 (citing Ex. 2031 ¶¶ 120–121). Petitioner does not rely on Hajeck for its teaching of a charge pump, nor does Petitioner propose that one is included in the prior art combination. Pet. 75–76. In any case, Hajeck teaches that the charge pump provides power to its storage subsystem *temporarily* and that a battery or capacitive array may provide additional sources of backup power, so we do not agree that Hajeck teaches that the charge pump can supply power indefinitely. Ex. 1038, 1:56–61. Petitioner relies on Amidi for switching operational states and the voltage supply to battery backup in the event of an overvoltage. Reply 21 (citing Pet. 49–50). Thus, this argument too considers Hajeck individually, and not in combination with Harris, the FBDIMM Standards, and Amidi, as set forth in the Petition.

3. *Overvoltage—Claims 5, 16, and 24*

Claims 5, 16, and 24 recite detecting that the voltage monitor circuit is configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage. Ex. 1001, 38:61–67, 40:7–12, 41:22–28. Petitioner contends that these claims are disclosed by the combination of Harris, the FBDIMM Standards,

Amidi, and Hajeck. Pet. 76–77. Patent Owner argues that these claims are not disclosed. Resp. 40–42.

Patent Owner argues that Hajeck does not teach a voltage detection circuit that produces a trigger signal that causes transitions to a different operable state or otherwise relates to switching power sources. Resp. 40–42 (citing EX1038, 3:6–10, 3:30–57, 3:66–4:9, Ex. 2031 ¶¶ 119–121). This attacks Hajeck individually and is an improper approach to the obviousness analysis. *See Merck, Keller, supra*. Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition, in combination with the teachings of Harris, the FBDIMM Standards, and Amidi which teach the voltage detection circuit configured to produce the trigger signal. Pet. 12–13, 60–66, 75–77; Ex. 1003 ¶¶ 137, 180–87, 363–84; Ex. 2030, 226:16–22, 230:17–232:6, 251:10–254:2.

Accordingly, the combination of Harris, the FBDIMM Standards, Amidi and Hajeck teaches, or at least suggests, the limitations of claims 5, 16, and 24.

4. *Remaining Claims*

We find no reason suggesting that Hajeck would negate any of the teachings of Harris, the FBDIMM Standards, and Amidi as shown under Ground 2 (Sect. III.E). Thus, the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck teach or suggest the limitations of the remaining claims under this Ground 3 for the same reasons stated for Ground 2.

5. *Determination for Ground 3*

Petitioner has shown that a person of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, Amidi,

and Hajeck with a reasonable expectation of success. Petitioner also shows that each limitation of claims 1–30 is taught or at least suggested by the prior art combination. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

G. Grounds 4 and 5: Obviousness over Spiers and Amidi

As we have determined that claims 1–30 are unpatentable as obvious under Grounds 1 to 3, we do not reach Grounds 4 and 5.

H. Motion to Exclude

Petitioner seeks to exclude materials referenced in three URLs submitted with Patent Owner’s Sur-Reply on August 4, 2023. Paper 33 (citing Paper 31, 1 n.2, 24, 26). Because we do not rely on any of these URLs in a manner adverse to Petitioner, we dismiss the Motion to Exclude as moot.

IV. CONCLUSION

For the foregoing reasons, we determine that Petitioner establishes by a preponderance of the evidence that claims 1–30 of the ’918 patent are unpatentable.

V. ORDER

Accordingly, it is:

ORDERED that claims 1–30 of the ’918 patent have been shown to be unpatentable;

FURTHER ORDERED that Petitioner’s Motion to Exclude is dismissed as moot;

FURTHER ORDERED that any party seeking judicial review must

comply with the notice and service requirements of 37 C.F.R. § 90.2.⁶

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–3, 8, 14, 15, 23	103	Harris, FBDIMM Standards	1–3, 8, 14, 15, 23	
1–30	103	Harris, FBDIMM Standards, Amidi	1–30	
1–30	103	Harris, FBDIMM Standards, Amidi, Hajeck	1–30	
1–30	103	Spiers, Amidi		
1–30	103	Spiers, Amidi, Hajeck		
Overall Outcome			1–30	

⁶ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner’s attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

IPR2022-00996
Patent 11,016,918 B2

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD., MICRON TECHNOLOGY, INC.,
MICRON SEMICONDUCTOR PRODUCTS, INC., and MICRON
TECHNOLOGY TEXAS LLC,¹
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00996
Patent 11,016,918 B2

Before PATRICK M. BOUCHER, JON M. JURGOVAN, and
DANIEL J. GALLIGAN, *Administrative Patent Judges*.

JURGOVAN, *Administrative Patent Judge*.

DECISION
Final Written Decision
Determining All Challenged Claims Unpatentable
Dismissing Petitioner's Motion to Exclude
35 U.S.C. § 318(a)

¹ Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC filed a motion for joinder and a petition in IPR2023-00406 and have been joined as petitioners in this proceeding. *See* Paper 26.

I. INTRODUCTION

Samsung Electronics Co., Ltd. (“Samsung”) filed a Petition (Paper 1, “Pet.”) for *inter partes* review of claims 1–30 (“challenged claims”) of U.S. Patent 11,016,918 B2 (Ex. 1001, “the ’918 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 7, “Prelim. Resp.”) to the Petition. We instituted *inter partes* review under 35 U.S.C. § 314(a). Paper 10 (“Inst. Dec.”).

During the trial, Patent Owner filed a Response (Paper 21, “Resp.”), Petitioner filed a Reply (Paper 25), and Patent Owner filed a Sur-Reply (Paper 31). We joined Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC as petitioners in this proceeding, and we refer to Samsung and these entities collectively as “Petitioner.” *See* Paper 26.

Petitioner and Patent Owner requested oral argument (Papers 28 and 29). A hearing was conducted on September 11, 2023. Paper 47 (“Tr.”).

Petitioner objected to evidence (Paper 32) and filed a Motion to Exclude (Paper 33). Patent Owner filed an Opposition to Petitioner’s Motion to Exclude (Paper 34), and Petitioner filed a Reply (Paper 37) in support of its Motion to Exclude.

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is entered pursuant to 35 U.S.C. § 318(a). Having reviewed the complete trial record, we determine that Petitioner has shown, by a preponderance of the evidence, that the challenged claims are unpatentable.

II. BACKGROUND

A. Real Parties in Interest

The identified real parties in interest on the Petitioner side are the following entities: Samsung Electronics Co., Ltd., Samsung Semiconductor, Inc., Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC. Pet. 1; IPR2023-00406, Paper 3, 1.

Patent Owner identifies itself as the sole real party in interest. Paper 4, 1.

B. Related Matters

The parties advise that the '918 patent is related to the following pending matters:

- *Samsung Electronics Co., Ltd. et al. v Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021)
- *Netlist, Inc. v. Samsung Electronics Co., Ltd. et al.*, No. 2:21-cv-00463 (E.D. Tex. filed Dec. 20, 2021)
- *Netlist, Inc. v. Micron Technology, Inc.*, No. 2:22-cv-00203 (E.D. Tex. filed June 13, 2022)
- IPR2012-00999 (U.S. Patent No. 11,232,054)
- U.S. Patent Application No. 17/582,797

Pet. 1; Paper 4, 1; Paper 9, 2; Paper 12, 1.

The parties advise that the following related proceeding is no longer pending:

- IPR2017-00692 (U.S. Patent No. 8,874,831)

Pet. 1; Paper 4, 1.

C. Overview of the '918 Patent (Ex. 1001)

The '918 patent is titled “Flash-DRAM Hybrid Memory Module.”

Ex. 1001, code (54). Figure 12 of the '918 patent is reproduced below.

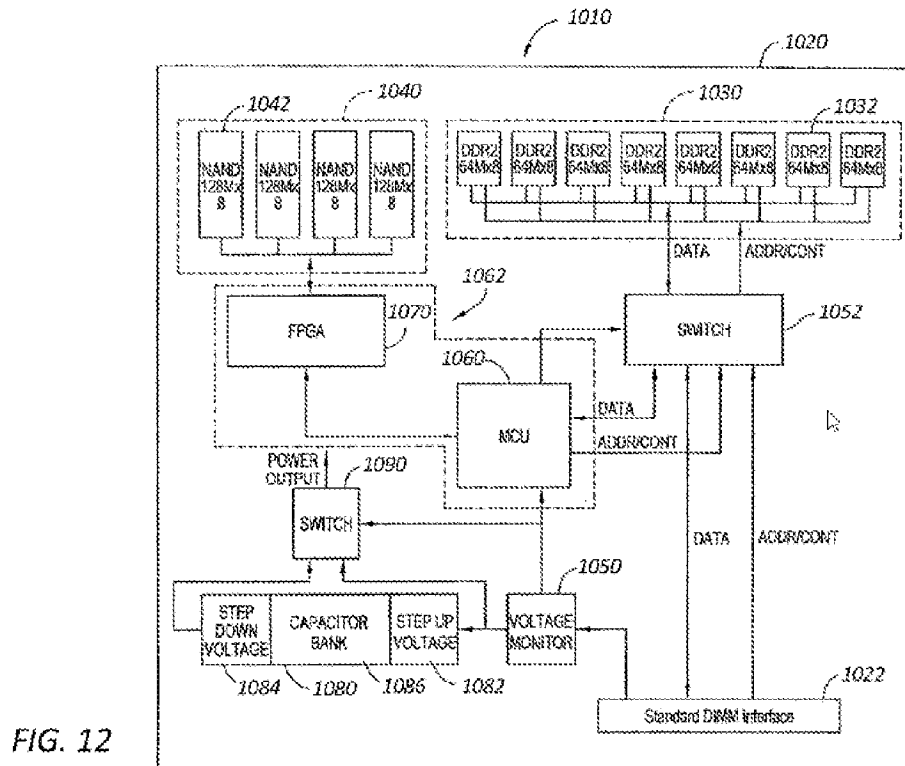


FIG. 12

Figure 12 shows an example memory system 1010 of the '918 patent. *Id.* at 21:14–16. The memory system 1010 includes a volatile memory subsystem 1030, a non-volatile memory subsystem 1040, and a controller 1062 operatively coupled to the volatile memory subsystem 1030 and selectively coupled to the non-volatile memory subsystem 1040. *Id.* at 21:16–22. Volatile memory 1030 may comprise elements 1032 of two or more dynamic random access memory (DRAM) elements such as double data rate (DDR), DDR2, DDR3, and synchronous DRAM (SDRAM). *Id.* at 22:16–19. Non-volatile memory 1040 may comprise elements 1042 of flash memory elements such as NOR, NAND, ONE-NAND flash and multi-level

cell (MLC). *Id.* at 22:35–40. Memory system 1010 may comprise a memory module or printed circuit board (PCB) 1020. *Id.* at 21:24–26. Memory system 1010 has an interface 1022 for power voltage, data, address, and control signal transfer between memory system 1010 and a host system. *Id.* at 22:3–6.

Controller 1062 may include microcontroller unit 1060 and FPGA logic 1070, either as separate devices or integrated together. *Id.* at 24:35–37, 23:19–22, Fig. 14. Microcontroller 1060 may transfer data between the volatile memory 1030 and non-volatile memory 1040. *Id.* at 24:35–41. Logic element 1070 provides signal level translation and address translation between the volatile memory and the non-volatile memory. *Id.* at 24:45–56.

When the system is operating normally, controller 1062 controls switch 1052 to decouple the volatile memory 1030 from controller 1062 and the non-volatile memory 1040 (the '918 patent refers to this as the “first state.”). *Id.* at 24:60–25:7. In response to a power interruption, for example, controller 1062 controls switch 1052 to couple the volatile memory 1030 to itself and non-volatile memory 1040, and transfers data from the volatile memory to the non-volatile memory to prevent its loss (the '918 patent refers to this as the “second state”). *Id.* at 25:9–20.

Memory system 1010 may comprise a voltage monitor 1050 to monitor voltage supplied from the host system via interface 1022. *Id.* at 25:8–10. When the voltage monitor 1050 detects a low voltage condition, the voltage monitor transmits a signal to the controller 1062 to indicate the detected condition. *Id.* at 25:11–15.

Power may be supplied from a first power supply (e.g., a system power supply) when the memory system 1010 is in the first state and from a

second power supply 1080 when the memory system 1010 is in the second state. *Id.* at 25:54–58. Second power supply 1080 may comprise step-up transformer 1082, step-down transformer 1084, and capacitor bank 1086 with one or more capacitors. *Id.* at 26:3–13.

The memory system 1010 further has a third state in which controller 1062 is decoupled from the volatile memory system 1030 and power is supplied to the volatile memory subsystem 1030 from a third power supply (not shown). *Id.* at 25:62–69. “[T]he third power supply may provide power to the volatile memory subsystem 1030 when the memory system 1010 detects that a trigger condition is likely to occur but has not yet occurred.” *Id.* at 25:66–26:3.

Figure 16 of the '918 patent is reproduced below.

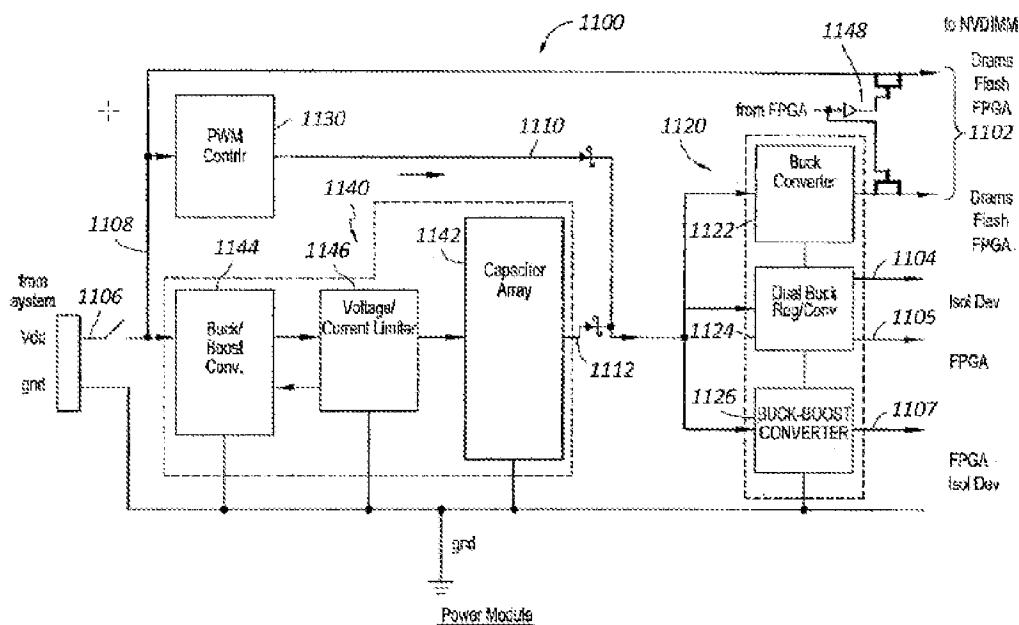


FIG. 16

Figure 16 shows power module 1100 of memory system 1010. *Id.* at 27:59–61. Power module 1100 comprises conversion element 1120, first power element 1130, and second power element 1140. *Id.* at 28:7–15,

28:20–22. Conversion element 1120 comprises sub-blocks 1122, 1124, 1126 comprised of various converter circuits such as buck converters, boost converters, and buck-boost converters for providing various voltages 1102, 1104, 1105, 1107 from the outputs of the first and second power elements 1130, 1140. *Id.* at 29:18–54. The first power element 1130 may comprise a pulse-width modulation power controller generating voltage 1110 from voltages 1106, 1108. *Id.* at 28:13–15. Second power element 1140 may comprise capacitor array 1142, buck-boost converter 1144 receiving voltages 1106, 1108 and adjusting the voltage for charging the capacitor array, and voltage/current limiter 1146, which limits charge current to the capacitor array and stops charging the capacitor array 1142 when it reaches a certain charge voltage. *Id.* at 28:62–67. “[P]ower module 1100 provides power to [] components of the memory system 1010 using different elements based on a state of the memory system 1010 in relation to a trigger condition.” *Id.* at 27:61–65.

Specifically, “[i]n a first state, the first voltage 1102 is provided to the memory system 1010 from the input 1106 and the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130.” *Id.* at 28:27–31. “In a second state, the fourth voltage 1110 is provided to the conversion element 1120 from the first power element 1130 and the first voltage 1102 is provided to the memory system 1010 from the conversion element 1120.” *Id.* at 28:32–37. “In the third state, the fifth voltage 1112 is provided to the conversion element 1120 from the second power element 1140 and the first voltage 1104 is provided to the memory system 1010 from the conversion element 1120.” *Id.* at 28:34–38.

Transition from the first state to the second state may occur when power

module 1100 detects a power failure is about to occur, and transition from the second state to the third state may occur when it detects a power failure has occurred. *Id.* at 28:39–47.

D. Illustrative Claim

Of the challenged claims, claims 1, 16, and 23 are independent. Independent claim 1, reproduced below with brackets noting Petitioner's identifiers, is illustrative of the claimed subject matter.

1. [1.a] A memory module comprising:

[1.b] a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system;

[1.c] a first buck converter configured to provide a first regulated voltage having a first voltage amplitude;

[1.d] a second buck converter configured to provide a second regulated voltage having a second voltage amplitude;

[1.e] a third buck converter configured to provide a third regulated voltage having a third voltage amplitude;

[1.f] a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude; and

[1.g] a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:

[1.h] a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and

[1.i] [1.i.1] at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices, [1.i.2] the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices, [1.i.3] the at least one circuit coupled to both

the second regulated voltage and the fourth regulated voltage, [1.i.4] wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.

Ex. 1001, 38:19–44.

E. Evidence

Petitioner relies on the following references (*see* Pet. 3, 10–14), as well as the Declaration of Dr. Andrew Wolfe (Ex. 1003).

Reference	Exhibit No.	Patent/Printed Publication
Harris	1023	U.S. Patent Pub. No. 2006/0174140 A1 to Harris, published August 3, 2006
Amidi	1024	U.S. Patent No. 7,724,604 B2, issued May 25, 2010
Spiers	1025	U.S. Patent Pub. No. 2006/0080515 A1, published April 13, 2006
FBDIMM Standards	1027, 1028	JESD82-20 and JESD205 standards published March 2007
Hajeck	1038	U.S. Patent No. 6,856,556 B1 to Hajeck, issued February 15, 2005

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–30 are unpatentable on the following Grounds (Pet. 3):

Claims Challenged	35 U.S.C. §	References
1–3, 8, 14, 15, 23	103(a)	Harris, FBDIMM Standards
1–30	103(a)	Harris, FBDIMM Standards, Amidi
1–30	103(a)	Harris, FBDIMM Standards, Amidi, Hajeck
1–30	103(a)	Spiers, Amidi
1–30	103(a)	Spiers, Amidi, Hajeck

III. ANALYSIS OF CHALLENGED GROUNDS

We now consider Petitioner's asserted grounds of unpatentability and Patent Owner's arguments to determine whether Petitioner has demonstrated by a preponderance of the evidence that the challenged claims would have been obvious under 35 U.S.C. § 103. *See* 35 U.S.C. § 316(e) (petitioner has the burden of proving unpatentability by a preponderance of the evidence).

A. *Principles of Law*

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination based on the teachings of multiple references requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation

omitted); *see also* *KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Factors pertinent to a determination of the level of ordinary skill in the art include “(1) the educational level of the inventor; (2) type of problems encountered in the art; (3) prior art solutions to those problems; (4) rapidity with which innovations are made; (5) sophistication of the technology; and (6) educational level of active workers in the field.” *Envtl. Designs, Ltd. v. Union Oil Co. of Cal.*, 713 F.2d 693, 696 (Fed. Cir. 1983) (citing *Orthopedic Equip. Co. v. All Orthopedic Appliances, Inc.*, 707 F.2d 1376, 1381–82 (Fed. Cir. 1983)). “Not all such factors may be present in every case, and one or more of these or other factors may predominate in a particular case.” *Id.* at 696–697. The prior art may reflect an appropriate level of skill. *Okajima v. Bourdeau*, 261 F.3d 1350, 1354–55 (Fed. Cir. 2001).

Petitioner asserts a person of ordinary skill in the art “would have had an advanced degree in electrical or computer engineering, or a related field, and two years working or studying in the field of design or development of memory systems, or a bachelor’s degree in such engineering disciplines and at least three years working in the field.” Pet. 8–9 (citing Ex. 1003 ¶67). Petitioner contends that additional training can substitute for educational or research experience, and vice versa. *Id.* at 8. Petitioner asserts that such a hypothetical person would have been familiar with the JEDEC industry

standards, and knowledgeable about the design and operation of standardized DRAM and SDRAM memory devices and memory modules and how they interacted with a memory controller and other parts of a computer system, including standard communication busses and protocols, such as PCI and SMBus busses and protocols. *Id.* at 8–9. Petitioner further contends that such “a hypothetical person would also have been familiar with the structure and operation of circuitry used to access and control computer memories, including sophisticated circuits such as ASICs, FPGAs, and CPLDs, and more low-level circuits such as tri-state buffers.” *Id.* at 9. Petitioner further asserts that such “a hypothetical person would further have been familiar with voltage supply requirements of such structures (e.g., memory modules, memory devices, memory controller, and associated access and control circuitry), including voltage conversion and voltage regulation circuitry.” *Id.*

Patent Owner indicates “[f]or the purposes of this [Response], Patent Owner is applying the level of ordinary skill in the art proposed by Petitioner” for this proceeding. Resp. 1.

The evidence that Petitioner presents mostly applies to the educational level and experience of workers in the field, but also touches upon other factors as well, including problems and solutions in the prior art and complexity of the technology. *See* Pet. 7–8. We find Petitioner’s proposal is consistent with the level of ordinary skill in the art reflected by the ’918 patent and the prior art of record, and, therefore, we adopt Petitioner’s proposed level of ordinary skill in the art, with the exception of the open-ended language “at least” which introduces ambiguity and may encompass skill levels beyond ordinary. *See Okajima v. Bourdeau*, 261 F.3d 1350,

1355 (Fed. Cir. 2001) (the prior art may reflect an appropriate level of skill in the art).

C. Claim Construction

We construe claim terms “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2021). There is a presumption that claim terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art in the context of the specification. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Nonetheless, if the specification “reveal[s] a special definition given to a claim term by the patentee that differs from the meaning it would otherwise possess[,] . . . the inventor’s lexicography governs.” *Phillips v. AWH Corp.*, 415 F.3d 1303, 1316 (Fed. Cir. 2005) (en banc) (citing *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002)). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Only disputed claim terms must be construed, and then only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

1. “Memory Module”

Patent Owner contends as follows:

The term “memory module” appears in the preamble of the independent claims, which provides antecedent basis for the same term in the body of the text. The preamble is thus limiting

as the District Court found after reviewing the specification in detail. The Court further explained that “a skilled artisan would understand a ‘memory module’ is distinct from, and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller.

Resp. 1–2 (citing Ex. 2032, 28; Ex. 2034, 319). Patent Owner argues that the “court’s ruling is consistent with the specification which states in the Overview section that the invention is particularly ‘couplable to a *memory controller* of a host system.’” *Id.* at 2 (citing Ex. 1001, 1:66–67, 3:66–67). Patent Owner states that Petitioner “objected to the court’s finding that the memory module is limiting, but does not dispute the Court’[s] finding on what a ‘memory module’ means to a [person of ordinary skill in the art].” *Id.* (citing Ex. 2033, 4).

As support for its argument, Patent Owner relies on a claim construction order issued by the District Court for the Eastern District of Texas, which states as follows:

Having reviewed the entire patent “to gain an understanding of what the inventors actually invented and intended to encompass by the claim[s],” *Catalina Mktg. Int’l*, 289 F.3d at 808–09, the Court finds the preamble limiting. While the claims recite many of the structural requirements of a “memory module,” the claims arguably read on other modular computer devices, such as a video card or network controller, despite no evidence the inventors intended to encompass such devices by the claims. To the contrary, as the Overview section [of the Court’s claim construction order] explains, the invention “is couplable to a *memory controller* of a host system,” [Ex. 1001] at 3:66–67 (emphasis added), not just the host system as recited in the claims. *See also id.* at 1:66–67 (“[t]he present disclosure relates generally to computer memory devices”). Thus, a skilled artisan would understand a “memory module” is distinct from,

and has essential structural requirements not necessarily found in, other modular computer accessories. That includes the structure necessary to connect to a memory controller. *See Memory Systems: Cache, DRAM, Disk*, Dkt. No. 76-17 at 319 (depicting, in FIG. 7.6, a memory controller connected to two memory modules). Accordingly, the preambles are limiting.

Ex. 2032, 28) (alterations in original), *cited in* Resp. 2.

Patent Owner argues that Petitioner's expert, Dr. Wolfe, testified that the term "memory modules" typically refers to "main memory modules," which "are *designed to connect to the primary memory controller* for the purpose of holding general purpose code and data in a computer system." *Id.* at 2 (citing Ex. 2030, 123:14–25; Ex. 2056, 100:15–101:8 (by 2004–2005, a memory module "was intended to go into a dedicated memory slot and not a general-purpose IO slot")). Patent Owner contends that the '918 patent's usage of "memory module" is consistent with that understanding, and asserts that we should adopt the District Court's claim construction, including that the claimed "memory module" includes structures necessary to connect to a memory controller. *Id.* at 2–3 (citing Ex. 2031 ¶¶ 51–53); *accord* Sur-Reply 1.

In Reply, Petitioner contends that we properly found that Harris, the FBDIMM Standards, Amidi, and Spiers all disclose a "memory module" in our Institution Decision. Reply 1–2 (citing Inst. Dec. 14–15, 17–18, 34–37, 45; Pet. 19–20, 82–83; Reply 23–26). Petitioner contends that the District Court's construction did not limit "memory module" to only "*main* memory modules . . . designed to connect to the *primary* memory controller." *Id.* at 1–2 (emphasis in original) (citing Resp. 2; Ex. 2032, 26–28, 35; Ex. 2030, 125:12–127:13; Ex. 2056, 100:15–101:19).

In Sur-Reply, Patent Owner disagrees with Petitioner’s contention that “memory module” is not limited to “main memory modules . . . designed to connect to the primary memory controller.” Sur-Reply 1 (emphasis in original) (citing Reply 1). Patent Owner argues that Petitioner’s contention “ignores the District Court’s finding based on intrinsic evidence—which [Petitioner] did not object to—that the ‘invention [i.e., the claimed memory module] ‘is couplable to a memory controller of a host system,’ [Ex. 1001] at 3:66–67 . . . , not just the host system as recited in the claims.’” *Id.* (citing Ex. 2032, 28; Ex. 1001, 4:5–12, 4:14–24, 4:35–39, 4:45–51, 5:4–20, 4:36–50, 6:4–6, 6:25–29, 6:61–66, 7:1–4, 7:21–25, 7:29–34, 7:43–49, 7:54–57, 21:24–25, 22:53–58, 23:19–22, 26:43–51, 23:41–44).

The District Court’s Claim Construction Order (Ex. 2032) issued after our Institution Decision (Paper 10). Although Petitioner argued in the District Court litigation that the claim preambles were not limiting, Petitioner does not appear to maintain that argument here following the District Court’s determination that the preambles were limiting. Ex. 2032, 28. Since the term is recited in both the preambles and bodies of these claims, there is evidence that the inventors of the ’918 patent intended the term to limit the claim, as the District Court determined. *See Catalina Mktg. Int’l., Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell Commc’ns Rsch., Inc. v. Vitalink Communications Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995) (“dependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention.”)).

We agree with Petitioner’s contention that “memory module” is not limited to “*main* memory modules . . . designed to connect to the *primary* memory controller.” Reply 1. The ’918 patent states

In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the **main memory**.

Ex. 1001, 11:3–6 (emphases added). This is the only mention of “main memory” in the ’918 patent. That the Flash memory serves as “main memory” in “certain embodiments” implies that there may be other embodiments in which this is not the case. Accordingly, we agree with Petitioner that the term “memory module” is not limited to “*main* memory modules . . . designed to connect to the *primary* memory controller.” See Reply 1.

We agree with Patent Owner that the “memory module” includes structure to connect with a “host system” including its memory controller. See, e.g., Ex. 1001, code (57), 12:52–59. The claims do not mention a “memory controller” but they do mention that the “memory module” has an “interface” to connect with a “host system.” Ex. 1001, 38:19–24 (claim 1), 39:54–59 (claim 16), 40:51–56 (claim 23). That the “memory module” has structure for connecting to the host system’s memory controller may thus be implied from the claims.

We agree with Petitioner that the District Court’s determination was that the preambles of all claims of the ’918 patent are limiting. Ex. 2032, 35. To the extent that the District Court’s determination construed “memory module” to include structure necessary to connect to a memory controller, our construction of “memory module” is consistent.

2. *“Pre-Regulated Input Voltage”*

Claims 16–22 recite a “pre-regulated input voltage.” Ex. 1001, 39:61, 42:23. Petitioner identifies at least two meanings for this claim term, one meaning within pre-determined limits and the other meaning the voltage must be pre-regulated on the memory board itself. Pet. 73. Petitioner contends that the District Court construed “pre-regulated input voltage” to have its plain and ordinary meaning, and rejected Samsung’s argument that the voltage must be pre-regulated on the memory module. Reply 20, n.3 (citing Ex. 2032, 21–22, 34; Inst. Dec. 37–38; Ex. 2030, 268:18–270:5). Petitioner contends that Patent Owner’s expert does not dispute the District Court’s construction. *Id.* (citing Ex. 1075, 142:10–144:18). According to Petitioner, both experts agree that the V12V input to Harris can be pre-regulated under that construction. Reply 20 (citing *id.* at 5, n.2).

Patent Owner argues that a “pre-regulated input voltage is just a regulated voltage provided to these voltage conversion circuits.” Resp. 38.

Under its ordinary and plain meaning in the context of the ’918 patent, we understand “pre-regulated input voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage converters. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112). This is consistent with Patent Owner’s proposed construction.

D. *Ground 1: Obviousness Over Harris and FBDIMM Standards*

Petitioner contends claims 1–3, 8, 14, 15, and 23 would have been obvious over the combination of Harris and FBDIMM Standards and relies on the Declaration of Dr. Andrew Wolfe (Ex. 1003) in support. Pet. 14–51. For the reasons that follow, we are persuaded that the evidence, including

Dr. Wolfe's testimony, supports Petitioner's showing and establishes by a preponderance of the evidence that these claims are unpatentable.

1. *Harris (Ex. 1023)*

Harris is titled "Voltage Distribution System and Method for a Memory Assembly." Ex. 1023, code (54). Harris was published as U.S. Patent Pub. No. 2006/0174140 A1 on August 3, 2006. Petitioner contends Harris is prior art under § 102(a). Pet. 10.

Harris's Figure 1A is reproduced below.

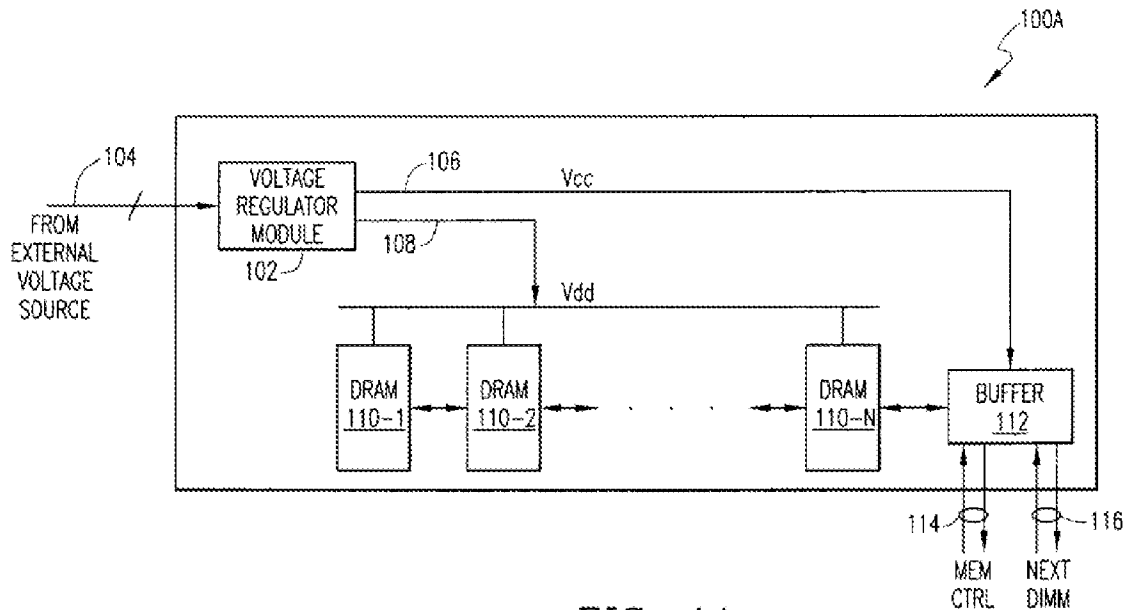


FIG. 1A

As shown in Figure 1A, Harris discloses a memory module 100A including on-board regulator 102 for converting an externally supplied voltage 104 to appropriate local voltage levels 106 (V_{cc}), 108 (V_{dd}), such as 0.5V to 3.5V. Ex. 1023, code (57), ¶¶ 9–10. Voltage 106 is supplied to buffer/logic component 112 which may be connected to a memory controller via interface 114 and daisy-chained with other memory assemblies via interface

116. *Id.* ¶ 9. Voltage 108 powers memory devices 110-1 to 110-N. *Id.* The memory module 100A may be a Dual In-Line Memory Modules (DIMM) wherein each of the memory devices 100-1 to 100-N comprises a Double Data Rate (DDR), DDR2, or DDR3 device. *Id.* The memory module 100A may be an unbuffered, registered or fully buffered DIMM. *Id.*

2. *FBDIMM Standards (Exs. 1027, 1028)*

In March 2007, the Joint Electron Device Engineering Council (JEDEC) published standards for Fully Buffered DIMM (FBDIMM) memory modules titled “JESD82-20” (Ex. 1027) and “JESD205” (Ex. 1028). Ex. 1029; Ex. 1003 ¶¶ 134–136. Petitioner refers to these standards collectively as the “FBDIMM Standards.” Pet. 11. Petitioner contends the FBDIMM Standards are prior art under § 102(a). *Id.*

The FBDIMM Standards specify voltages for components on the memory module as follows:

Product Family Attributes

DIMM organization	x72 ECC			
DIMM dimensions (nominal)	30.35mm (height) x 133.35mm (width) x 8.2 mm (max thickness) MO-256 variation AB 30.35mm (height) x 133.35mm (width) x 8.8 mm (max thickness) MO-256 variation BB			
Pin count	240			
SDRAMs supported	256Mb, 512Mb, 1Gb, 2Gb, 4Gb			
Capacity	256MB, 512MB, 1GB, 2GB, 4GB, 8GB, 16GB			
Serial PD	Consistent with JC 45			
Supply voltages (nominal)	min	typ	max	
	1.7	1.8	1.9	(DRAM V_{DD}/V_{DDQ} , AMB V_{DDQ})
	1.455 ¹	1.5	1.575 ¹	(AMB V_{CC}/V_{CCFBD})
	$0.453 \cdot V_{DD}$	$0.5 \cdot V_{DD}$	$0.547 \cdot V_{DD}$	(DRAM interface V_{TT}) This supply should track as $0.5 \cdot 1.8$ volt supply
	3.0	3.3	3.6	(V_{DDSPD})
Buffer interface	High-speed Differential Point-to-point Link at 1.5 volt			
DRAM interface	SSTL ₁₈			

Note 1: Approximate DC values, refer to AMB Component Specification for actual DC and AC values and conditions.

Note 2: V_{TT} range accommodates measurable offset due to complementary CA bus current paths. (See V_{TT} section)

An Unloaded system should supply V_{TT} of $0.48 \cdot V_{DD}/0.52 \cdot V_{DD}$ to Dimm socket

Ex. 1028, 9. The above table shows values for supply voltages including DRAM V_{DD} , AMB V_{CC} , DRAM interface V_{TT} , and V_{DDSPD} . The FBDIMMM Standards further specify the following voltages for various power supplies:

Table 9.2 — Pin Description (Sheet 3 of 3)

Signal	Type	Description
RESET		Power Good Reset
Miscellaneous Test		
TEST (4 pins)	NC	Pin for debug and test. Must be floated on DIMM.
TESTLO (5 pins)	A	Pin for debug and test. Must be tied to Ground on DIMM
TESTLO_AB20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
TESTLO_AC20	A	Pin for debug and test. Connected to two resistors. One resistor is connected to VCCFBD, the other resistor is connected to VSS.
Power Supplies		
VCC (24 pins)	A	1.5V nominal supply for core IO
VCCFBD (8 pins)	A	1.5V nominal supply for FBD high speed IO
VDD (24 pins)	A	1.8V nominal supply for DDR IO
VSS (156 pins)	A	Ground
VDDSPD	A	3.3V nominal supply for SMB receivers and ESD diodes
Other Pins		
BFUNC	1	Buffer Function Bit: When BFUNC = 0, AMB is used as a regular buffer on FB-DIMM. When BFUNC = 1, AMB is used as either a repeater or a buffer for LAI function. On FB-DIMM, BFUNC is tied to Ground
RFU (18 pins)	NC	Reserved for Future Use. Must be floated on DIMM. RFU pins denoted by “a” are reserved for forwarded clocks in future AMB implementations.
Other No Connect Pins		
NC (129 pins)	NC	No Connect pins

Ex. 1027, 83. The table above sets voltage levels for power supplies V_{CC}, V_{CCFBD}, V_{DD}, V_{SS}, and V_{DDSPD}.

3. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris with the FBDIMM Standards with a reasonable expectation of success because Harris states that its Figure 1A may be a “fully buffered DIMM” (FBDIMM or FBD). Pet. 16 (citing Ex. 1023 ¶¶ 9–13; Ex. 1003 ¶¶ 158–165). Petitioner contends that a person of ordinary skill in the art would have understood that this type of DIMM is standardized in JEDEC’s FBDIMM Standards and thus would naturally look to them for more details about the “fully buffered DIMM” that Harris

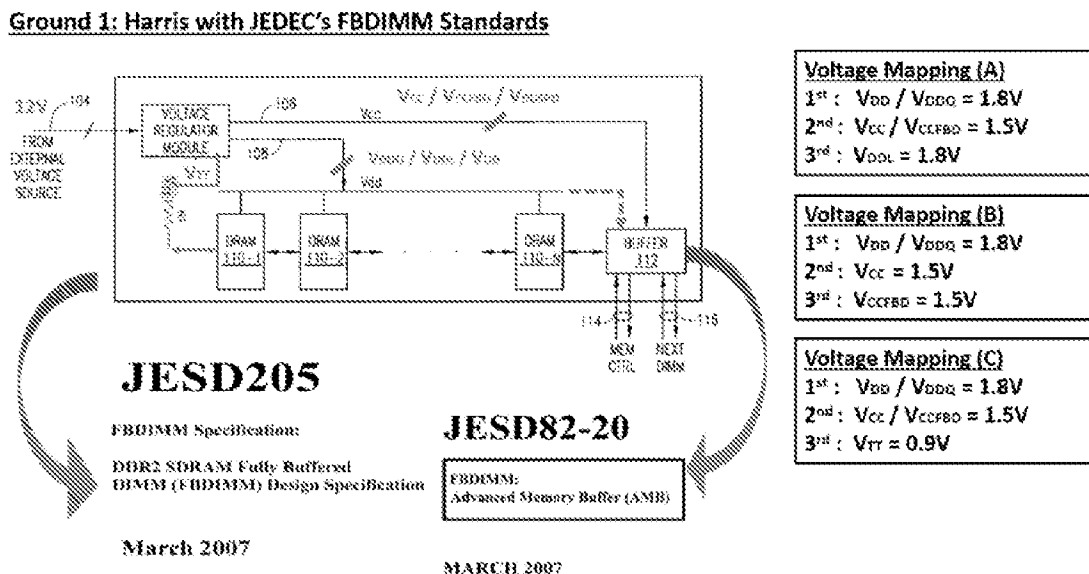
describes as compatible with his on-board voltage regulator module (VRM).
Id.

To explain its combination, Petitioner provides the following table:

Voltage Mappings (Grounds 1-3)			
	<u>A</u>	<u>B</u>	<u>C</u>
“first”:	V_{DD} or $V_{DDQ} = 1.8V$	V_{DD} , V_{DDQ} , or $V_{DDL} = 1.8V$	V_{DD} , V_{DDQ} , or $V_{DDL} = 1.8V$
“second”:	V_{CC} or $V_{CCFBD} = 1.5V$	$V_{CC} = 1.5V$	V_{CC} or $V_{CCFBD} = 1.5V$
“third”:	$V_{DDL} = 1.8V$	$V_{CCFBD} = 1.5V$	$V_{TT} = 0.9V$
“fourth”:	$V_{DDSPD} = 3.3V$	$V_{DDSPD} = 3.3V$	$V_{DDSPD} = 3.3V$

Pet. 27. The table above shows Petitioner’s Voltage Mappings A, B, and C from the FBDIMM Standards for Grounds 1–3, and how they disclose the “first,” “second,” “third,” and “fourth regulated voltages” in the claims. *Id.*

For Ground 1, Petitioner provides the illustration below:



Pet. 15. This illustration shows how Petitioner is combining the teachings of Harris and the FBDIMM Standards to arrive at the claims. Specifically,

JEDEC FBDIMM Standard JESD205 applies to Harris's memory module and JESD82-20 applies to the Harris's buffer.

a) Modifying Harris to Have Four Converters

Patent Owner asserts that a person of ordinary skill in the art would not have modified Harris's memory module to have four converters, as Petitioner asserts. Resp. 17–21. Patent Owner argues that Harris discloses a single converter providing at least two voltages in each of Petitioner's voltage mappings. *Id.* at 17 (citing Ex. 2031 ¶¶ 75–79). Patent Owner asserts that Harris discloses “a high-frequency switching voltage converter capable of generating tightly-controlled voltage levels.” (Emphases original) *Id.* (citing Ex. 1023 ¶ 10, Fig. 1A); *see also* Sur-Reply 15–17.

Petitioner replies that the Petition explained the motivation for multiple converters, including that “the JEDEC standards treat those voltages separately, thus permitting power-on sequencing; providing independent control; improving efficiency; and saving power.” Reply 14 (citing Pet. 30–31; Ex. 2030, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15). In addition, Dr. Wolfe explains that smaller, lower-power regulators may be more desirable than larger, higher-power regulators because they can be less costly. Ex. 2030, 45:21–25. He also stated that space availability on the memory board can require separate regulators. *Id.* at 46:1–4. In sum, Dr. Wolfe stated that “a number of ordinary engineering factors” would be considered when deciding to use one supply for both voltages or two separate supplies. *Id.* at 46:5–10.

Petitioner provides the following excerpt from the JEDEC standards to explain its position:

2.3.1 Power-up and initialization sequence

The following sequence is required for POWER UP and Initialization.

- a) Apply power and attempt to maintain CKE below $0.2 \cdot VDDQ$ and ODT^{n1} at a low state (all other inputs may be undefined.) The power voltage ramp time must be no greater than 20ns, and during the ramp, $VDD > VDDL > VDDQ$ and $VDD - VDDQ < 0.3$ volts.

- VDD, VDDL and VDDQ are driven from a single power converter output, AND
- VTT is limited to 0.95 V max, AND
- Vref tracks $VDDQ/2$.

OR

- Apply VDD without any slope reversal before or at the same time as VDDL.
- Apply VDDL without any slope reversal before or at the same time as VDDQ.
- Apply VDDQ without any slope reversal before or at the same time as VTT & Vref.

at least one of these two sets of conditions must be met.

Reply 14 (citing Resp. 25; Ex. 1026, 9 (DDR2); Ex. 1046, 15 (same for DDR3); Pet. 30–31. The excerpt from JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3)² above shows two options for compliance with these standards, one shown partially with a blue box,³ the other with a red box. Although the first option (blue box) indicates that a single converter generates the voltages, Petitioner correctly contends that the second option (red box) does not list that any converter provides these voltages, but leaves open how they are generated. *Id.* at 13–14. A person of ordinary skill in the

² Although Petitioner does not expressly rely on JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3) in its combinations, the parties seem to be in agreement that these specifications are part of the background knowledge that a person of ordinary skill in the art would have had. In an obviousness analysis, the background knowledge is properly considered together with the demands known in the design community, and the inferences and creative steps that a person of ordinary skill in the art would employ. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013).

³ In our opinion, Petitioner should have drawn the blue box to include the following two lines “VTT is limited to 0.95 V max, AND Vref tracks $VDDQ/2$ ” because of the way the standard presents the two sets of conditions with the word “or” between them and refers to them as “two sets of conditions.” We refer to the two sets of conditions as “options” herein.

art would have understood that the listed voltages may be generated with respective converters. *See* Ex. 1026, 9; Ex. 1046, 15; Ex. 1003 ¶ 242 (Dr. Wolfe testifying that JESD79-2B’s disclosure that the voltage supplies can be turned on and off separately suggests to a person of ordinary skill in the art that the voltages are provided by separate converters); *see also KSR*, 550 U.S. at 418 (“a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.”).

Moreover, Harris discloses “at least one on-board voltage regulator module (VRM) is provided as part of the memory board assembly module 100A for converting externally supplied voltage level available on external source path 104 into appropriate local voltage levels that power the first and second voltage paths, i.e., the V_{dd} and V_{cc} .” Ex. 1023 ¶ 10 (emphases added). Also, Harris’s claim 1 recites “at least one voltage regulator module for converting an externally supplied voltage level into appropriate voltage levels that power said first and second voltage paths, respectively.” Ex. 1023, claim 1. Thus, Harris discloses that multiple converters can be used to generate respective multiple voltage levels.

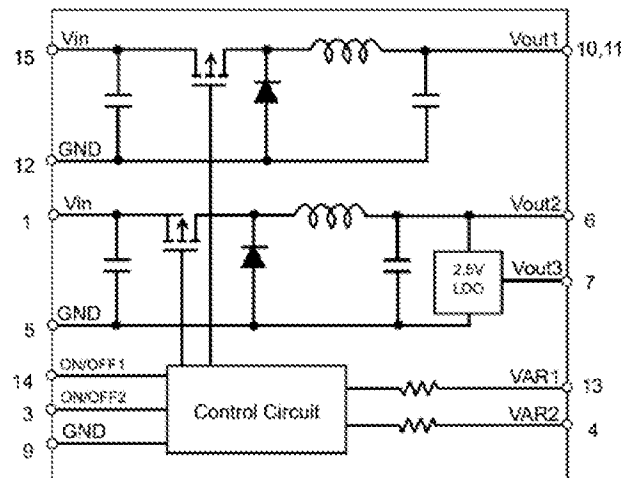
Furthermore, Petitioner contends that Patent Owner’s examples relate to a SODIMM, not an FBDIMM like Harris. Reply 14 (citing Resp. 23; Ex. 2006, 4 (“SODIMM”); Ex. 1075, 102:17–103:9 (SODIMM is unbuffered as shown in Ex. 2045 and Ex. 2046, 4.20.11-23)). Although Harris discloses unbuffered DIMMs may be an option to use, Petitioner relies upon Harris’s FBDIMM teachings in its Petition. *See, e.g.*, Pet. 16, Ex. 1023 ¶ 9, claim 24.

Patent Owner further argues that single buck converters with multiple voltage outputs were known in the art. Resp. 18 (citing Ex. 2031 ¶ 80;

Ex. 2003; Ex. 2004). As Petitioner shows, however, single buck converters for generating single voltages were also known in the art. Pet. 91 (citing Ex. 1023, Fig. 6 (640)); Ex. 1047, 1:28–32, 2:47–55, 5:56–59, 7:6–14; Ex. 1041, 1–2, 9; Ex. 1042, 16; Ex. 1048, 1–2; Ex. 1058, 5). Although Patent Owner is correct that Petitioner must show a reason for using separate buck converters, as already explained, Petitioner did so. Reply 15 (sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints); Ex. 2030, 45:21–46:10 (smaller regulators may be less costly than larger ones, and may be desirable when separate spaces are available on the memory board). Thus, we do not agree with Patent Owner’s argument that Petitioner “has not articulated any advantage of using one converter per voltage if Harris already uses a single converter for at least two voltages.” Sur-Reply 11 (citing Reply 11–14; Ex. 1023 ¶ 10).

We also find Patent Owner’s distinction between a single converter generating two voltages and two separate converters is not meaningful because of how the single converter may be internally constructed. For example, Petitioner presents the following illustration:

DC-DC Converter Specification(DRAFT) MPD4S014S



Reply 13 (citing Ex. 1075, 127:21–129:24, 228:25–232:19; Ex. 1042, 16 (corresponding to EX1048, 2)). The figure above shows two buck converters receiving two V_{in} inputs and generating respective outputs V_{out1} and V_{out2} (the LDO V_{out3} can be disregarded in this discussion). Externally, this chip resembles a single converter, and has effectively only one input if the pins 1 and 15 are tied together, but internally it has two buck converters generating respective voltages. Thus, the presence of multiple outputs suggests, to a person of ordinary skill in the art, that multiple buck converters are being used or, at least, that such a configuration is a suitable option.

Patent Owner next argues that a tightly spaced FBDIMM does not allow for the large inductors that multiple buck converters would require, and a person of ordinary skill in the art would not have modified Harris to add more components. Resp. 18–21 (citing Ex. 2031 ¶¶ 83–84; Ex. 2101, 20 (illustration of FBDIMM prototype); Ex. 2042, 7 (illustration of FBDIMM)).

Patent Owner also argues that one could use a single-inductor four-output buck converter instead of separate ones. *Id.* at 20–21.

Petitioner counters that defining the number of converters based on the number of inductors lacks any support in the '918 patent. Reply 12 (citing Ex. 1001; Ex. 2032). Petitioner argues that “it was well known that a single chip can include multiple buck converters (and inductors).” *Id.* (citing Ex. 1048, 1–2; Ex. 1075, 106:24–109:8 (discussing use of inductor in buck converter); Ex. 2030, 98:8–18 (same); Ex. 1058, 5, 14–15 (similar)). Petitioner also contends that space for multiple buck converters is not mentioned in the '918 patent or its claims, since solving any such problem was within the level of ordinary skill in the art. Reply 13 (citing *id.* at 19–20; Ex. 2030, 53:16–54:14, 89:1–21).

Petitioner has shown that one of ordinary skill in the art would have implemented a memory module with four buck converters to generate respective voltages in at least some situations. Patent Owner may be right that four buck converters using a single inductor would have been the more typical or preferred approach, as Dr. Mangione-Smith implied. *See, e.g.*, Ex. 2031 ¶¶ 76, 81–84. However, “it is not necessary to show that a combination of prior art references is the best option, only that it be a suitable option.” *Intel Corp. v. PACTXPP Schweiz AG*, 61 F.4th 1373, 1380 (Fed. Cir. 2023); *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021); *PAR Pharm., Inc. v TWI Pharms., Inc.*, 773 F.3d 1186, 1197–98 (Fed. Cir. 2014).

*b) Use of Separate Converters to Generate
 $V_{DD}/V_{DDQ}/V_{DDL}$ or V_{CC}/V_{CCFBD}*

Patent Owner argues against Petitioner's showing of a motivation to combine by arguing that neither Harris nor the FBDIMM Standards suggest separate power rails for V_{DD}/V_{DDQ} and V_{DDL} or V_{CC} and V_{CCFBD} for Voltage Mappings A and B. Resp. 21–30. Patent Owner argues that the DDR2 Specification that Petitioner cited requires that the voltages V_{DD} , V_{DDQ} and V_{DDL} are driven from a single power converter output. *Id.* at 22 (citing Ex. 1026, 9; Ex. 2061 ¶ 87). As discussed previously, Patent Owner argues that the DDR2 Specification requires a single converter (see text in blue box of Exhibit 1026, 9 above) whereas Petitioner relies on the second option (see text in red box) which does not specify how the voltages are to be generated. Although Patent Owner's argument is correct for the first option, it is not for the second option, which is the one on which Petitioner relies.

Patent Owner argues that Dr. Wolfe concedes that in all standard FBDIMMs that he was aware of, V_{DDL} , V_{DD} and V_{DDQ} for the DRAMs are connected to a single power converter, and are not individually powered on or off. Resp. 22–23 (citing Ex. 2030, 133:5–134:6, 39:11–40:24). Dr. Wolfe stated “I’m not aware of any [FBDIMMs with three different power rails] on the market. Obviously, the teaching is right here in the JEDEC specification.” Ex. 2030, 133:10–17. We understand Dr. Wolfe to be referring to the second option boxed in red in the above excerpt from JEDEC's JESD79-2B (DDR2) (Ex. 1026) and JESD79-3A (DDR3) (Ex. 1046), on which Petitioner relies. Patent Owner argues that Petitioner is suggesting a design that persons of ordinary skill in the art were aware of, but rejected, and that this is evidence of non-obviousness. Resp. 22–23.

We do not agree that a person of ordinary skill in the art would have rejected the design, however, when the JEDEC specification allows for it and Harris discloses that it is an option for generating V_{CC} and V_{DD} . Ex. 1026, 9; Ex. 1023 ¶ 9. Extending Harris’s teaching to generate four voltages with respective voltage regulators is within the level of ordinary skill in the art. *See* Ex. 2030, 43:10–14 (Dr. Wolfe stating four voltages specified by JEDEC would require use of four power supplies in most situations); 160:2–3 (“it would be obvious to use four power supplies”); Ex. 1003 ¶¶ 146–149 (credible testimony showing that buck converters were well known to persons of ordinary skill in the art). “A person of ordinary skill is also a person of ordinary creativity, not an automaton.” *KSR*, 550 U.S. at 421.

Patent Owner argues that a single power converter was required so that the voltages track one another during power up. Resp. 23–24 (citing Ex. 2006, 4; Ex. 2061 ¶ 88; Ex. 1023 ¶ 12). But Dr. Wolfe indicates that the voltages of separate converters can be made to track one another by coupling their feedback, which was a known option. Ex. 2030, 54:25–55:13. Patent Owner does not refute Dr. Wolfe’s testimony.

Patent Owner argues that JEDEC’s JESD79-2B (DDR2) and JESD79-3A (DDR3) do not address power generation on the memory module, only the inputs and outputs of the memory device. Resp. 24–25. Petitioner replies that Harris teaches a module that supplies all the voltages needed on the module, which would include all those needed by the memory devices. Reply 14–15 (citing Ex. 2030, 37:13–38:5); *see also* Ex. 1023 ¶ 9. Patent Owner’s argument views these JEDEC specifications, which would have been general knowledge of one of ordinary skill in the art, in isolation

without considering them together in the combination of Harris and the FBDIMM Standards. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362 (Fed. Cir. 2013) (“[T]he Court required an analysis that reads the prior art in context, taking account of ‘demands known to the design community,’ ‘*the background knowledge possessed by a person having ordinary skill in the art*,’ and ‘the inferences and creative steps that a person of ordinary skill in the art would employ.’” (emphasis added)); *In re Merck & Co.*, 800 F.2d 1091, 1097 (Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981) (“Non-obviousness cannot be established by attacking references individually where the rejection is based upon the teachings of a combination of references.”)).

Patent Owner asserts that Petitioner’s contention that an FBDIMM has separate pins for V_{DD} , V_{DDQ} and V_{DDL} does not support its contention of separate converters, “particularly where it was known that there are stability and timing benefits when using a single source.” Resp. 24 (citing Ex. 2006, 4). Dr. Wolfe explains that the FBDIMM module of Petitioner’s combination would not have separate pins to receive V_{DD} , V_{DDQ} and V_{DDL} but would derive them onboard the module from the 12-volt input pin. Ex. 2030, 35:25–36:15. As to stability, Dr. Wolfe testifies that a feedback loop maintains stability in light of current variations, temperature changes, and other factors. *Id.* at 64:25–65:10. As to timing, Dr. Wolfe indicated that separate converters can track one another by coupling their feedback, and that this was a known option. *Id.* at 54:25–55:13.

Patent Owner further argues that sequencing the different voltages does not require separate converters. Resp. 24–27. Petitioner contends that this was the obvious way to implement the JEDEC specifications (boxed in

red above) with several benefits. Reply 15 (citing Ex. 1003 ¶ 256). We agree with Petitioner that separate converters were an option that one of ordinary skill in the art would have used in some practical circumstances for reasons of cost, space availability, etc.

Patent Owner argues that it was actual practice for persons of ordinary skill in the art to use a single power converter to generate the necessary voltages. Resp. 25 (citing Ex. 2030, 133:5–134:6, 39:11–40:24). Although using a single power converter may have been the most common approach, this does not negate Petitioner’s showing that a person of ordinary skill in the art would have considered separate converters desirable under certain circumstances. *See Elekta Ltd. v. ZAP Surgical Systems, Inc.*, 81 F.4th 1368 (Fed. Cir. 2023) (quoting *Novartis Pharms. Corp. v. West-Ward Pharms. Int’l Ltd.*, 923 F.3d 1051, 1059 (Fed. Cir. 2019) (“Nor does an obviousness showing ‘require that a particular combination must be the preferred, or the most desirable, combination described in the prior art in order to provide motivation’”).

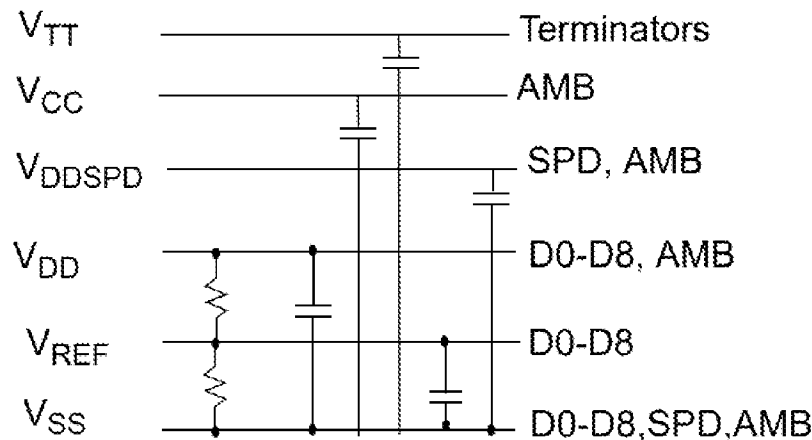
Patent Owner similarly argues that V_{CC} and V_{CCFBD} are separate voltages with separate pins, but asserts that Petitioner does not explain why a single on-board V_{CC} power source is insufficient for V_{CC} and V_{CCFBD} power rails when a single set of V_{CC} interface pins providing power from the host met the power needs for both V_{CC} and V_{CCFBD} . Resp. 25–27 (citing Pet. 30–31; Ex. 1028, 11–12; Ex. 2031 ¶ 92). For similar reasons as explained for V_{DD} , V_{DDQ} and V_{DDL} , we disagree. As Petitioner indicates, using separate converters for V_{CC} and V_{CCFBD} provides the benefits of sequencing the power, turning power on and off independently, saving cost, eliminating cross-coupling of noise, and solving space constraints. Reply 15

(citing Ex. 2060, 39:2–10, 44:25–46:10; Ex. 1075, 134:22–136:2, 194:23–195:7; Ex. 1062, 13–15; Ex. 2012, 73; Ex. 1003 ¶ 255).

Patent Owner's arguments do not undercut Petitioner's showing of a motivation to combine.

c) Use of Third Buck Converter for V_{TT}

Patent Owner contends that Petitioner's Voltage Mapping C relies on three different voltages, including a termination voltage, V_{TT} , connected to terminators. Resp. 27 (citing Pet. 26, 30; Ex. 1028, 13, 15–16). For context, the relevant figure from the JEDEC JESD205 specification is reproduced below:

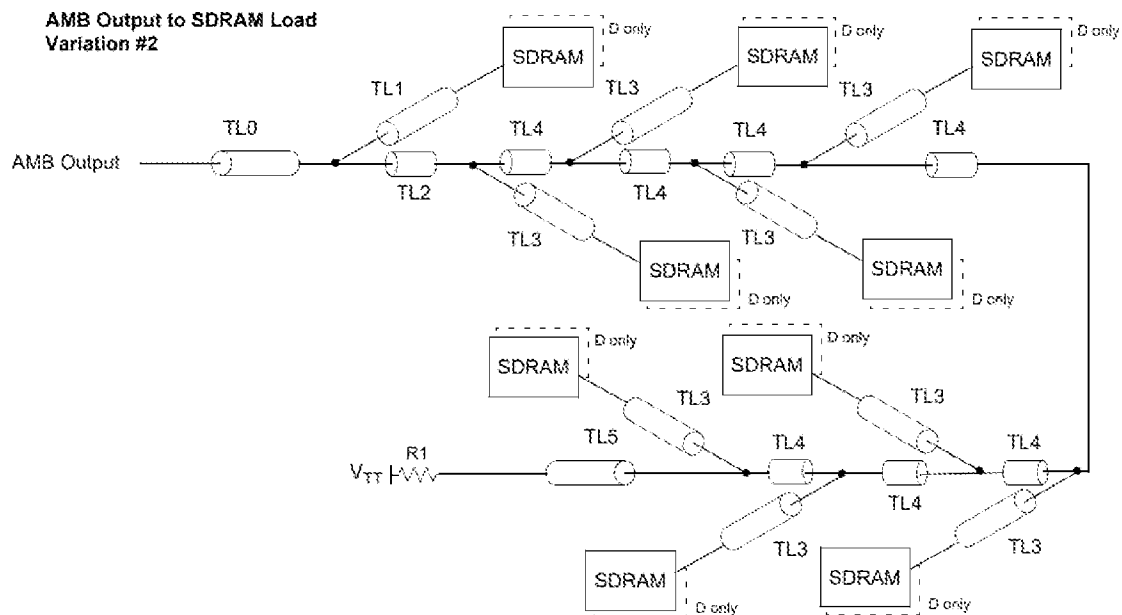


Ex. 1028, 13. The above figure shows various voltages V_{TT} , V_{CC} , V_{DDSPD} , V_{DD} , V_{REF} , V_{SS} and their relationship to one another according to the JEDEC JESD205 specification. *Id.*

Patent Owner argues that Harris does not disclose generating V_{TT} on its module and argues that it would be supplied from the motherboard instead because this would ensure that all DIMMs have the same termination voltages and eliminate ground loops that could degrade signal integrity. Resp. 27–28 (citing Ex. 2061 ¶ 96); Sur-Reply 17–18. Patent Owner further argues that design complexity favors having a single regulator for a group of

DIMMs rather than one regulator per DIMM. Resp. 28–29 (citing Ex. 1027, 20; Ex. 2031 ¶ 97; Ex. 2012, 72). In addition, Patent Owner argues that Petitioner touted JEDEC’s move of voltage regulation from the motherboard to the DIMM as a “major design improvement” and Patent Owner asks, “If it had been so obvious, why did it take the industry so long to make that design improvement?” *Id.* at 29. Patent Owner further argues that generating V_{TT} is not one of two choices, and that V_{TT} is generally not provided to DDR2 modules; passive termination is used instead. *Id.* at 33 (citing Ex. 2061 ¶ 98; Ex. 2044, 6; Ex. 2045, 6; Ex. 2046, 4.20.11-6; Ex. 2006, 5).

In Reply, Petitioner contends that Harris teaches generating all of the voltages on the module, including all of the voltages for an FBDIMM, which would include V_{TT} . Reply 15 (citing Inst. Dec. 20–23; Ex. 1023 ¶ 12; Ex. 2030, 30:15–20, 72:22–73:7, 103:11–104:23, 109:5–111:10, 239:8–20). Petitioner contends that “Harris’s module provides the voltages for the buffer 112 to send address and control signals to the DDR memory devices, which the FBDIMM Standards make clear require not just V_{DD} , but also V_{TT} to terminate those signals.” *Id.* (underlining omitted) (citing Pet. 17–18; Ex. 1028, 9, 15, 68). Petitioner reproduces the figure below:



Reply 16 (reproducing the figure at Ex. 1028, 68). The above figure is from the FBDIMM Standards and shows the net structure routing for addresses and commands from the advanced memory buffer (AMB) to the SDRAMs and how they are terminated by V_{TT} . *Id.* Petitioner contends that V_{TT} is required by the JEDEC Standards to track V_{DD} (Ex. 1028, 15) which is why dual buck converters for generating V_{DD} and V_{TT} were readily available. Reply 16–17 (citing Ex. 1028, 9; Ex. 2030, 72:22–73:7, 196:3–197:7; Ex. 1040, 1, 11; Ex. 1041, 1; Ex. 1048).

We agree with Petitioner that the combination of Harris and the FBDIMM Standards at least suggests that V_{TT} can be generated onboard the memory module. Reply 15–17 (citing Pet. 17–18; Inst. Dec. 20–24; Ex. 1023 ¶ 12; Ex. 1028, 9, 15, 68; Ex. 1040, 1, 11; Ex. 1041, 1–2, 7–9; Ex. 1048; Ex. 1075, 112:12–114:8; Ex. 2030, 30:15–20, 57:17–20, 58:18–59:10, 72:22–73:7, 103:11–104:23, 109:5–111:10, 140:15–24, 196:3–197:7, 239:8–20). Harris indicates that its on-board voltage regulator converts externally supplied voltage into appropriate local voltage levels for powering

memory devices of the memory assembly module. Ex. 1023, code (57), ¶¶ 12, 16, Fig. 2, step 204. Petitioner shows that V_{TT} is needed onboard a memory module to terminate address and control signals. Reply 15 (citing Pet. 17–18; Ex. 1028, 9, 15, 68). In addition, Petitioner shows that V_{TT} is required by JEDEC to track V_{DD} , which is why dual buck converters for generating these voltages were commercially available. Reply 16–17 (citing Ex. 1028, 9; Ex. 2030, 72:22–73:7, 196:3–197:7; Ex. 1040; Ex. 1041, Ex. 1048). In light of these facts and the teachings of Harris and the FBDIMM Standards, Petitioner has shown that a person of ordinary skill in the art would have added to the memory module a third buck converter to generate V_{TT} .

Patent Owner further argues that “even if V_{TT} is generated on module, there is no reason that it should be generated by a buck converter.”

Resp. 29. Patent Owner contends that, “[a]s Micron noted, . . . an LDO would be sufficient to power V_{TT} .” *Id.* (citing Ex. 2006, 7; Ex. 2007–2010; Ex. 2050). Patent Owner contends that an LDO is much smaller than a buck converter, which requires a controller and discrete components such as an inductor, which makes the LDO preferable given the space constraint. *Id.* (citing Ex. 2061 ¶ 99). Patent Owner also argues that an LDO is preferred when the current load is below 1A. *Id.* at 29–30 (citing Ex. 2031 ¶ 99; Ex. 2047, 21; Ex. 2048, 23; Ex. 2049, 20; Ex. 1040, 23–24, Figs. 22–25).

In Reply, Petitioner contends that buck converters were commercially available for V_{TT} and provided high efficiency compared to an LDO. Reply 17 (citing Ex. 1075, 112:12–114:8 (LDO may be only 10% efficient); Ex. 2030, 57:17–20, 58:18–59:10 (buck converters up to 98% efficient), 140:15–24 (trend has been to use buck converters). We agree with

Petitioner that, under some circumstances, one of ordinary skill in the art would have chosen a buck converter over an LDO for the reasons that Petitioner mentions.

d) Use of Buck Converter to Provide V_{DDSPD}

Patent Owner argues that Petitioner's voltage mappings A–C all relate the recited “fourth regulated voltage” to the 3.3V V_{DDSPD} . Resp. 30 (citing Pet. 26). Patent Owner contends that Harris's reasons for on-module regulation do not apply to V_{DDSPD} . *Id.* Specifically, Patent Owner argues that the “3.3V-supply voltage is sufficiently high such that a regulator on the motherboard could readily provide the needed 300mV tolerance; and the same SPD [serial presence detect] can be used for different generations of DRAMs.” *Id.* (citing Ex. 2031 ¶ 102); *see also* Sur-Reply 18–19.

Petitioner replies that Harris teaches generating all of the voltage onboard the module, including all of the voltages for an FBDIMM, which would include V_{DDSPD} . Reply 18 (citing Ex. 1023 ¶ 12; Pet. 16–18). Petitioner further contends there are only two options—generating the voltages on the motherboard or on the module—rendering either option obvious. *Id.* (citing Inst. Dec. 24).

We agree with Petitioner that Harris teaches generating voltages onboard the memory module, which suggests this would include V_{DDSPD} as it is one of the voltages required in the FBDIMM standards. Ex. 1023 ¶¶ 9–12; Ex. 1028, 15. As Petitioner contends, Harris teaches generating the voltages required in the FBDIMM standards on the memory module, and the FBDIMM standards set out what those voltages are. Ex. 1023 ¶¶ 9–12; Ex. 1028, 15.

We also agree with Petitioner that there are only two options: either V_{DDSPD} is generated on the motherboard or on the module, rendering either option obvious. Reply 18. “[W]hen there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp.” *KSR*, 550 U.S. at 421.

Patent Owner argues, if V_{DDSPD} ’s are supplied both on the motherboard and individual DIMMs, they cannot track one another, which would lead to communication problems when they are at different voltage potentials. Resp. 30–31 (citing Ex. 2031 ¶ 102). Patent Owner also argues that “3.3V is a common power rail on the motherboard such that it can be supplied to FBDIMMs without the increased cost and problems associated with providing a regulator for each of Harris’ modified FBDIMM memory boards.” *Id.* at 31 (citing Ex. 2031 ¶ 102; Ex. 2038, 13 (3.3V PSU output)).

As previously discussed, Dr. Wolfe indicated that separate converters can track one another by coupling their feedback. Ex. 2030, 54:25–55:13. We recognize, however, that this may require connection and coordination across the motherboard-DIMM interface.

Harris mentions in its background a voltage supply level of 3.3V, which is the nominal value for V_{DDSPD} in the JEDEC standards. Ex. 1028, 9. Harris also states that it is one of the values that “keeps getting lower” with each generation of DIMM/DRAM technology, implying that it should be generated not on the motherboard, but on the FBDIMM. Ex. 1023 ¶ 2. One of ordinary skill in the art thus would have recognized that V_{DDSPD} may be generated onboard the memory module when considering Harris’s teachings. Ex. 1023 ¶¶ 9–12.

We also note that the JEDEC FBDIMM standards provide that V_{DDSPD} can vary within the range of 3.3V +/- 0.3 V. Ex. 1029, 9. As long as the voltage is maintained within this range on both the motherboard and the FBDIMM, the implementation is compliant with the JEDEC standards.

Patent Owner further argues that, for the low currents required for components using V_{DDSPD} , buck converters would have been highly inefficient. Resp. 31 (citing Ex. 2051, D-1 (up to 5mA active power supply current); Ex. 2052, 6, Table III (I2C power consumption less than 15mW on average; and @3.3V, current is less than 5mA); Ex. 1040, Figs. 22–25 (efficiency less than 20% at 10mA current); Ex. 2031 ¶ 103 (an LDO with a 12V-input-3.3V-output would be 26% efficient); Ex. 1062, 14 (using LDO to generate 3.3V from 12V)). Patent Owner asserts that a person of ordinary skill in the art would have used LDOs to generate V_{DDSPD} . *Id.*

Petitioner replies that Patent Owner concedes that the claimed “converter circuit” can be an LDO (Resp. 30, n.5). Reply 18. Petitioner also contends that “using a buck converter . . . was an obvious, highly efficient, and clearly ‘suitable’ option for converting 12V to 3.3V for V_{DDSPD} . . . regardless of whether [Patent Owner] considers it ‘inferior’ to an LDO.” Reply 18 (citing Ex. 1048, 2; Ex. 1003 ¶¶ 146–49, 286; Ex. 2030, 76:4–19, 78:18–79:7, 79:24–80:11, 82:7–85:10, 88:4–20, 89:1–21, 138:14–21, 140:15–24; *Dome Pat. L.P. v. Lee*, 799 F.3d 1372, 1381 (Fed. Cir. 2016)).

From the foregoing discussion, it is apparent that there are advantages and disadvantages to generating V_{DDSPD} onboard the memory module. However, “simultaneous advantages and disadvantages . . . do[] not necessarily obviate motivation to combine.” *Qualcomm*, 21 F.4th 784 (Fed. Cir. 2021) (citing *Medichem, S.A. v. Rolabo, S.L.*, 437 F.3d 1157, 1165 (Fed.

Cir. 2006)). Petitioner has shown that the advantages outweigh the disadvantages in at least some scenarios that the person of ordinary skill in the art would have considered desirable to pursue. Petitioner has shown that a person of ordinary skill in the art would have had reasons to combine Harris and the FBDIMM Standards with a reasonable expectation of success.

e) Conclusion on Motivation to Combine Harris and the FBDIMM Standards

On this record, Petitioner has shown by a preponderance of the evidence that a person of ordinary skill in the art would have combined Harris and the FBDIMM standards since Harris states that its memory module may include fully buffered DIMMs (FBDs). Ex. 1023 ¶9. We agree that one of ordinary skill in the art would have recognized Harris's mention of FBDIMMs as a standard and looked to the FBDIMM Standards for information concerning the voltage values standardized for use in an FBDIMM. Ex. 1027, 83; Ex. 1028, 9. For these reasons and others set forth above, we determine that one of ordinary skill in the art would have combined Harris and the FBDIMM Standards with a reasonable expectation of success.

4. Analysis of Independent Claim 1

a) Limitation 1.a: "A memory module comprising:"

Petitioner asserts that Harris's memory module 100A in Figure 1A or memory module 306-1 in Figure 3 corresponds to the claimed "memory module." Pet. 19–20 (citing Ex. 1023 ¶¶9, 17, 20, Figs. 1A, 3; Ex., 1003 ¶¶ 216–222; Ex. 1028, 38). Harris does indeed disclose a memory module with multiple memory devices such as DRAMs 110-1 to 110-N in Figure 1A or 306-1, for example, in Figure 3.

Patent Owner does not dispute that the preamble is met by Harris.
See Resp.

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris teaches the preamble.

b) Limitation 1.b: “a printed circuit board (PCB) having an interface configured to fit into a corresponding slot connector of a host system, the interface including a plurality of edge connections configured to couple power, data, address and control signals between the memory module and the host system”

Petitioner asserts that Harris and the FBDIMM Standards disclose this limitation. Pet. 20–25. Petitioner contends that Harris and the FBDIMM Standards disclose printed circuit boards (PCBs). Pet. 20–21 (citing Ex. 1023 ¶¶ 13, Figs. 1A, 3). Petitioner also contends that a PCB may be referred to as a “memory board” or “raw card.” *Id.* at 20–21 (citing Ex. 1023 ¶ 9, Ex. 1028, 10, 38, 84; Ex. 1003 ¶¶ 223–225).

As for the PCB “*having an interface configured to fit into a corresponding slot connector of a host system,*” Petitioner notes that Harris’s Figure 3 shows that each memory module 306-1 to 306-M includes an edge connection for fitting into a corresponding slot of a host system. Pet. 21 (citing Ex. 1023 ¶¶ 2, 12, 13, 19, Figs. 3, 4; Ex. 1028, 38, 84; Ex. 1003 ¶¶ 226–227).

Petitioner further contends that Harris, consistent with the FBDIMM Standards, discloses that the edge connections are “*configured to couple power, data, address and control signals between the memory module and the host system.*” Pet. 21–25. Petitioner contends that the power signal corresponds to Harris’s voltage 104 in Figure 1A. *Id.* at 21 (citing Ex. 1023 ¶¶ 10, 12, 19). Petitioner contends that Harris’s buffer 112 in Figure 1A is

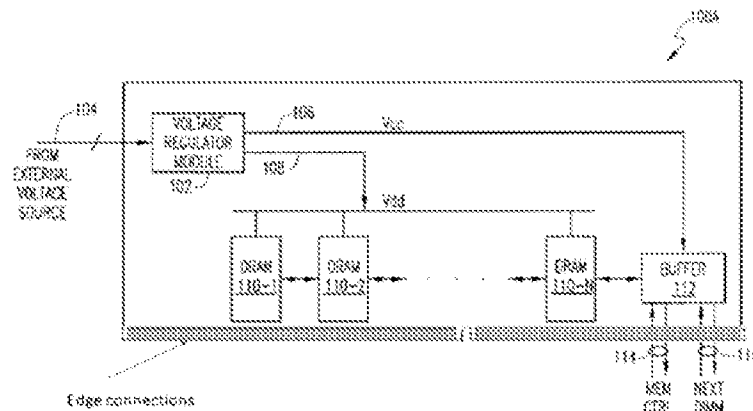
called “AMB” (Advanced Memory Buffer) in the FBDIMM Standards. Pet. 22–23. Petitioner indicates that Harris’s buffer 112 receives data, address, and control signals via memory controller interface 114 and transmits these signals to DRAMs 110-1 to 110-N in Figure 1A. Pet. 22 (citing Ex. 1023 ¶ 9 (“buffer/logic component 112 is provided for buffering *command/address* (C/A) space as well as *data* space at least for a portion of memory devices 110-1 through 110-N”). In addition, Petitioner argues that the FBDIMM Standards indicate that buffer AMB receives data signals DQ0–DQ63; address signals A0–A15; and control signals RAS, CAS, WE, CS, etc. Pet. 22–23 (citing Ex. 1028, 13).

(1) Power from the Interface with the Host System

Patent Owner argues that all challenged claims require a memory module that includes a printed circuit board (PCB) having an interface with edge connections configured to couple power between the memory module and the host system. Resp. 3. According to Patent Owner, Harris does not disclose such a memory module. *Id.* Instead, Patent Owner asserts, Harris replaces the power supply interface pins with as few as six +12V non-interface pins from an external power source which is unregulated and has a wide tolerance, which does not comport with the regulated sources associated with a host system. *Id.* at 4, 6, 7 (citing Ex. 1023 ¶¶ 12, 16; Ex. 2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2031 ¶¶ 60–66; Ex. 2037, 22; Ex. 2038, 13, Table 2); *see also* Sur-Reply 2–6. Patent Owner further argues that Harris supposedly provides a “technology-independent voltage distribution scheme for memory devices wherein system board power supply and associated voltage plane(s) are eliminated.” Resp. 4 (citing Ex. 1023

¶ 19). Patent Owner argues that Harris effectively decouples the host system's voltage supply from the memory module. *Id.* at 5 (citing Ex. 2031 ¶ 38).

To illustrate its points, Patent Owner provides the following annotated figure:



Resp. 5. In the figure above, Patent Owner has modified Harris's Figure 1A to show edge connections along the bottom edge of memory board 100A, and highlights that the external voltage is connected to the side of the memory board. Patent Owner asserts that external sources connected to the side or top of a memory module were known. *Id.* at 5 (citing Ex. 2035; Ex. 2036, 41–42; Ex. 2031 ¶ 59).

In addition, Patent Owner argues that Petitioner's expert, Dr. Wolfe, testified that, in computer systems using FBDIMMs, the power supply units provide regulated voltages. *Id.* at 6 (citing Ex. 2030, EX2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2037, 22, Table 15; Ex. 2038, 13, Table 2). According to Patent Owner, it would be “inadvisable” to provide unregulated voltages via a memory board to a memory slot because the “unregulated voltages could cause random noise patterns and degrade signal integrity.” *Id.* at 7 (citing Ex. 2031 ¶¶ 65–66). Even with regulated voltage

sources, Patent Owner asserts, the voltage was not supplied via the DIMM interface. *Id.* (citing Ex. 2031 ¶ 66). Patent Owner argues that Petitioner’s modified FBDIMM is powered from an external power source “wherein system board power supply and associated voltage plane(s) are eliminated.” Resp. 8 (citing Ex. 1023 ¶¶ 10, 12, 19). Patent Owner further argues that neither Dr. Wolfe nor Dr. Mangione-Smith was aware of any memory controllers that supplied power supply voltages to DIMMs, and that Patent Owner did not provide evidence that a person of ordinary skill in the art would have known how to design a memory controller that supplied power to DIMMs. *Id.* at 8–9 (citing Ex. 1023 ¶ 17; Ex. 2030, 130:19–131:10; Ex. 2031 ¶ 67).

Although Patent Owner acknowledges that Harris discloses that its external voltage sources can be any “known or heretofore unknown voltage supplies, either regulated or unregulated,” Patent Owner argues that this would sweep in the type of voltage supply that Harris seeks to eliminate. *Id.* at 9 (citing Ex. 2031 ¶ 68). Patent Owner further asserts that “nothing suggests that the voltage paths are via the memory slot to the interface edge connections.” *Id.* at 9–10 (citing Ex. 2031 ¶ 69). Furthermore, Patent Owner asserts that different edge pins would be needed for different voltage sources independently distributed through the motherboard, increasing, rather than decreasing, the cost for power distribution in the motherboard and pin count on the DIMM connector to the motherboard. *Id.* at 10 (citing Ex. 1023 ¶¶ 13, 19–20; Ex. 2031 ¶ 70). Patent Owner summarizes its argument, stating that Harris teaches to eliminate power distribution to the DIMM slot, a feature which Petitioner does not propose to modify. *Id.* at 10–11; *see also* Sur-Reply 6. Patent Owner argues that Harris does not

couple power between the module and the host system, and therefore does not disclose elements 1.b, 16.b, 23.b and 16.f. *See* Pet. xi (1.b), xiii (16.b), xiv (16.d), and xvii (23.b). Patent Owner argues that the specific benefits Harris teaches can be achieved by decoupling power between the motherboard and memory module. Resp. 11 (citing Ex. 1023 ¶¶ 19, 20; Ex. 2031 ¶¶ 26, 58, 70). Patent Owner argues that the same arguments apply to claim 13. *Id.*

Petitioner replies that Patent Owner does not dispute that the FBDIMM Standards teach supplying power via edge connection, and that Patent Owner's expert, Dr. Mangione-Smith, admits that such edge connections were "standard." Reply 2 (citing Pet. 16–21; Resp. 3–11; Inst Dec. 18–20; Ex. 1075, 97:16–98:18, 163:16–20; Ex. 1077, 9). Petitioner also contends that Harris's reference to externally supplied voltage requires power external to the entire host system, but Harris merely requires power external to the DIMM memory module. *Id.* (citing Pet. 21–22; Resp. 3–11; Ex. 2030, 66:7–19, 67:20–68:21, 91:22–92:7, 129:24–130:17).

Petitioner further replies that Patent Owner wrongly modifies Harris's Figure 1A (above) to suggest that Harris would never use edge connections at the bottom for power. Reply 3. To illustrate its points, Petitioner reproduces the following annotated figure:

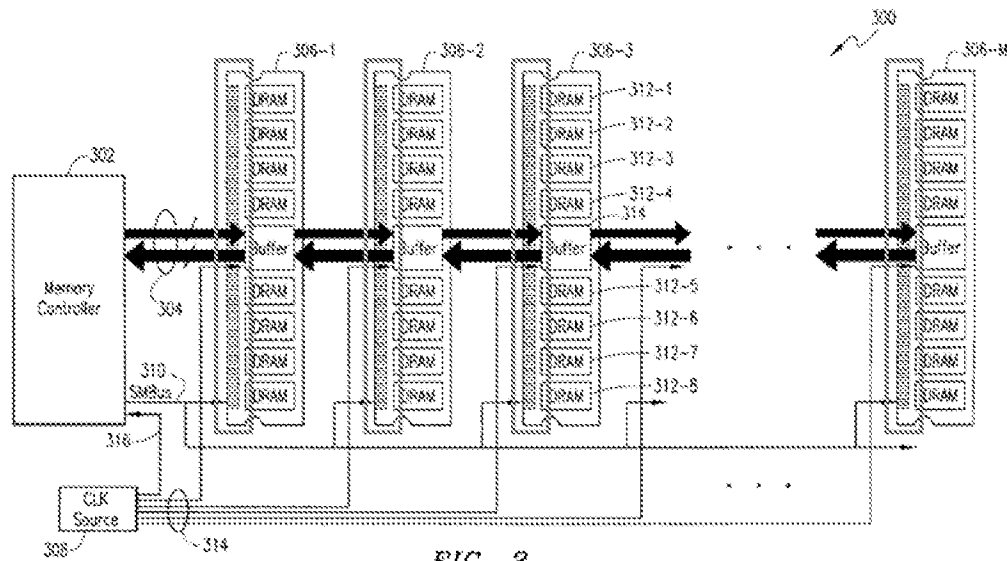


FIG. 3

Reply 4 (citing Ex. 1023, Fig. 3). In Harris’s Figure 3 as annotated by Petitioner, above, Petitioner shows that Harris’s memory modules 306-1 to 306-M are connected to the memory controller only through edge connections (boxed in red), meaning that power must come from those edge connections. *Id.* at 3–4. Petitioner further asserts that Harris’s Figure 3 is nearly identical to an Intel drawing where the memory modules admittedly receive power through the edge connections from the host system. *Id.* at 4 (citing Ex. 1075, 171:4–17; Ex. 2101, 4). According to Petitioner, Harris confirms that, “[a]lthough not explicitly shown in this FIGURE [3, above], each memory board also receives a supply voltage ... [which] may be sourced from the memory controller 302 [which is part of the host system, Ex. 1075, 167:23–168:1] or from a separate voltage source.” *Id.* (quoting Ex. 1023 ¶ 17) (alterations in original). Petitioner’s expert, Dr. Wolfe, explained that it was common for the host to supply power through edge connectors “alongside the memory controller signals.” *Id.* (quoting Ex. 2030, 131:3–5).

Petitioner contends that Patent Owner misinterprets Harris to conclude that the system power supply is eliminated entirely. *Id.* at 4–5 (citing Ex. 1023 ¶ 19; Resp. 4–5, 8, 10–11). To the contrary, Petitioner argues, Harris proposes avoiding the need for different system board voltages such as “3.3V, 2.5V, 1.8V, 1.5V and beyond,” by simply supplying a single voltage (i.e., “12V”) to the memory module so that an “on-board voltage regulator module[e.g., 102 above] [can] generate appropriate local voltage levels” on the memory module. *Id.* at 5 (citing Ex. 1023 ¶¶ 2, 3, 12–13, 19). Petitioner contends, as Dr. Wolfe explained, Harris’s “technology-independent voltage distribution scheme” eliminates the need for a “system-board-specific power supply,” not all power. *Id.* (citing Ex. 2030, 116:10–117:6).

Petitioner contends that using edge connectors like those shown in Harris’s Figure 3 is consistent with Harris’s use of the known FBDIMM design, except with “few[er]” power pins given the higher 12V input voltage. Reply 6 (citing Ex. 1023 ¶ 12; Ex. 2030, 100:12–101:19, 102:17–25, 103:11–104:23). Petitioner argues that to prevent “accidental damage” due to this change in the power pins of the edge connector, Harris teaches changing “the board’s connector keyway” so that it is “not interchangeable with the standard DIMM.” *Id.* (citing Ex. 1023 ¶ 13; Ex. 2030, 117:7–21). According to Petitioner, this was a standard technique for memory modules receiving power from the host via the edge connections. *Id.* (citing Ex. 2016, 6–7; Ex. 2101, 21–22; Ex. 1075, 171:21–175:20). Patent Owner disagrees, stating that the keyway connector does not mean the 12V pins are added to the edge connection. Sur-Reply 5.

Petitioner asserts that Patent Owner’s argument that “[s]upplying power... from [the] side... was known” “misses the point: regardless of

whether a side connection was possible, that would not negate the obviousness of using edge connections for power, as was highly common.” Reply 7 (citing *Dome Pat. L.P. v. Lee*, 799 F.3d 1372, 1381 (Fed. Cir. 2015) (existence of a better alternative “does not mean that an inferior combination is inapt for obviousness purposes”). Petitioner states that Patent Owner’s “expert admitted that using **both** was a known option, consistent with the combination for Grounds 2 to 3.” *Id.* (emphasis Petitioner’s). According to Petitioner, the side connection could be used for battery backup, as taught by Amidi, and the edge connections could be used for power from the host system during normal operation, as taught by Harris. *Id.* (citing Ex. 2035, 39; Ex. 1075, 165:10–166:12).

Considering the foregoing, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards discloses limitation 1.b (and claim 13). Harris states that DRAM devices may be “powered from system board or main board voltage sources.” Ex. 1023 ¶ 2. Although this is presented in the background of Harris, and Patent Owner argues that it is technology over which Harris seeks to improve, it is nonetheless information that a person of ordinary skill in the art would have known. Harris’s teaching of “replacing” power supply interface pins with as few as six +12V pins” means that those six pins take the place of the 72 pins in the interface, implying that power still comes from the interface with the host system. *Id.* ¶ 12. Harris reinforces this by disclosing that “the supply voltage may be sourced from the memory controller 302 or from a separate voltage source.” *Id.* ¶ 17. Although Patent Owner argues that Dr. Wolfe nor Dr. Mangione-Smith could recall any DIMM memory module powered by a memory controller, Harris expressly discloses this feature. *Id.*

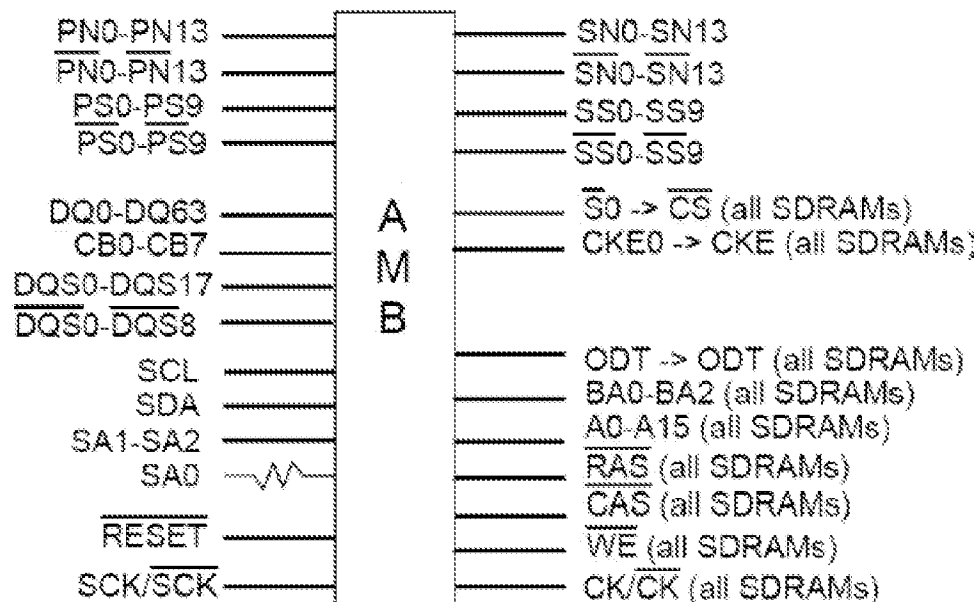
Harris further teaches that “external voltage sources may comprise any combination of known or heretofore unknown voltage supplies, either regulated or unregulated, and even including variable voltages.” Ex. 1023 ¶ 14. This means there may be more than one voltage supply, either regulated or unregulated, as Petitioner contends (which may be the host system and a backup source, for example). Dr. Wolfe explains that Harris’s statement that “system board power supply and associated voltage plane(s) are eliminated” means that “supplies such as the memory [V_{DD}] and the memory [V_{CC}] that are specific to a particular generation of memory chips . . . can be supplied on the DIMM rather than being supplied from the motherboard.” Ex. 2030, 116:25–117:6.

Furthermore, Petitioner indicates that the FBDIMM Standards show that the buffer AMB may be connected to a host, suggesting that the FBDIMM derives its power from the host. Pet. 24 (citing Ex. 1027, 4). The experts agree that it was common at the time for the host system to provide the regulated 12V power supply to the FBDIMM, and commercially available devices are in evidence. Ex. 2030, 62:21–25, 63:1–8, 92:9–95:8; Ex. 2037, 22, Table 15, Ex. 2038, 13, Table 2; Ex. 2031 ¶ 65; Ex. 1075, 180:14–183:3; Ex. 2038, 8, 13. These facts point to the conclusion that the external voltage source may be the host system in the combination of Harris with the FBDIMM Standards notwithstanding Patent Owner’s arguments to the contrary.

*(2) Data, Address, and Control Signals Via
Edge Connection*

Claim 1 recites an “interface including a plurality of edge connections configured to couple . . . data, address and control signals between the

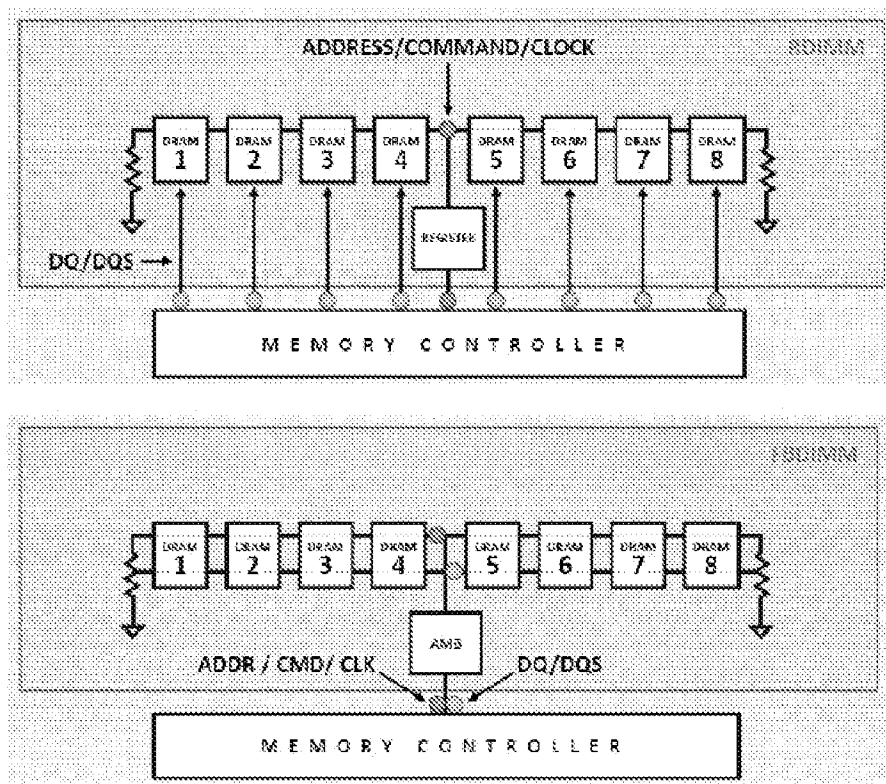
memory module and the host system.” Ex. 1001, 38:23–25. Patent Owner argues that the DIMM and AMB do not receive data signals (DQ0–DQ63) or address and control signals (A0–A15, RAS, CAS, WE) from the host. Resp. 11 (citing Pet. 22–25). According to Patent Owner, these signals are instead generated by the AMB based on decoded FBDIMM channel signals (PS[9:0] and PS[9:0]bar). *Id.* at 11–12 (citing Ex. 2031 ¶¶ 72; Ex. 1027, 3–4; Ex. 1028, 29; Ex. 2039, 2; Ex. 2101, 4; Ex. 2040, 1). Patent Owner argues that Petitioner’s evidence does not show that the data, address and control signals are received from the host. *Id.* at 15. Specifically, Patent Owner relies on the figure below from the FBDIMM Standards (*see* Sur-Reply 7 (citing Ex. 1028, 13)).



According to Patent Owner, the above figure shows that data, address and controls signals (highlighted in yellow) are generated and outputted by the buffer, and are not received from the DIMM interface. Sur-Reply 6–7 (citing Pet. 22–23; Resp. 11–15; Ex. 2031 ¶¶ 72–74). Patent Owner contends this does not comport with the claim language which requires an

“interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” *Id.* at 8 (citing Ex. 1001, 22:1–6). Patent Owner argues that Dr. Wolfe merely stated that the AMB outputs “information” to the DRAMs but the claims require specific “signals,” not just “information.” *Id.* at 8 (citing Ex. 1003 ¶ 230; Ex. 2030, 7:20–11:6). Patent Owner argues that the outputs from the AMB are not transmitted over FBDIMM’s edge connections. *Id.*

Petitioner argues that Patent Owner’s technology tutorial indicates that an FBDIMM receives address and control signals similar to an RDIMM, as shown below:



Reply 7–8 (citing Ex. 1077, 8–9; Ex. 1075, 91:23–92:19, 95:14–96:13, 97:16–98:18). The figures above show that the memory controllers transmit

address, command and clock signals (shown in red), or information containing these signals, to the register (RDIMM) or AMB (FBDIMM), which transmits them to the DRAMs. *Id.*; Sur-Reply 9.

Petitioner contends that the host system provides address, command and clock signals to the AMB encoded as a packetized, serialized signal on fewer wires than would be required to receive them as separate, parallel signals. Reply 8–9 (citing Resp. 11–12; Ex. 2031 ¶ 31; Ex. 1075, 155:22–157:1, 212:3–8, 213:3–215:20, 219:13–220:9, 226:7–228:8; Ex. 2030, 8:3–11:6). Petitioner alleges that Patent Owner is in essence attempting to rewrite the claims to require dedicated pins for these signals, which the claims do not require. *Id.* at 9.

Patent Owner argues that it is not attempting to rewrite the claims to require “dedicated pins” as Petitioner alleges. Sur-Reply 8–9 (citing Reply 8–9). Patent Owner asserts that the pin names indicate what “signals” are exchanged at the interface between the host and the FBDIMM, as shown in the figure below.

Pin Name	Pin Description	Pin
FB-ORIMM Channel Signals		
SDN	System Clock (note: positive logic)	1
SDN	System Clock Input (negative logic)	2
PS0(1:0)	Primary Northbound Data (positive logic)	3,4
PS0(1:0)	Primary Northbound Data (negative logic)	5,6
PS0(1:0)	Primary Southbound Data (positive logic)	7,8
PS0(1:0)	Primary Southbound Data (negative logic)	9,10
SD0(1:0)	Secondary Northbound Data (positive logic)	11,12
SD0(1:0)	Secondary Northbound Data (negative logic)	13,14
SD0(1:0)	Secondary Southbound Data (positive logic)	15,16
SD0(1:0)	Secondary Southbound Data (negative logic)	17,18
PS0(0:0)	For an external precision calibration, spectral calibration up limit	19
DBF2 Interface Signals		
SD0(0:0)	Data Receive (positive logic)	3
SD0(0:0)	Data Receive (negative logic)	4
PS0(0:0) (DBF0:0)	Data Receive (no DBMS only) (positive logic) (these signals are always low to all DBMS on any)	5
PS0(0:0)	Data Receive (no DBMS only) (negative logic)	6
SD0(0:0)	None	19
DBF0:0	Low-Z Drive	2
AT0:000, AT0:000	Addresser, AT0 is part of the one-chip receiver	30
DBF1:000, DBF1:000	Reset Addresser	31
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	32
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	33
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	34
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	35
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	36
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	37
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	38
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	39
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	40
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	41
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	42
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	43
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	44
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	45
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	46
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	47
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	48
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	49
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	50
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	51
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	52
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	53
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	54
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	55
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	56
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	57
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	58
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	59
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	60
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	61
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	62
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	63
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	64
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	65
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	66
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	67
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	68
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	69
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	70
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	71
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	72
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	73
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	74
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	75
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	76
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	77
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	78
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	79
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	80
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	81
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	82
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	83
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	84
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	85
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	86
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	87
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	88
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	89
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	90
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	91
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	92
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	93
DBF1:0, DBF1:0	Start of command with DBF0, DBF0, and DBF1:0	94

Ex. 1028, 29. The figure above shows Patent Owner’s highlighting to show the “FB-DIMM Channel Signals” including PS[9:0] and PS[9:0]bar, and the “DDR2 Interface Signals” including DQ0–DQ63, A0–A15, RAS, CAS, and WE. Patent Owner states that all memory control for the DRAM resides in the host, and argues that the fact that all read, write, and configuration accesses are addressed to the DIMM does not undermine that the signals Petitioner relies on are generated and outputted by the buffer, and not exchanged at the interface between the FBDIMM and the host. Sur-Reply 9 (citing Ex. 2030, 7:20–11:6; Ex. 2030 ¶ 230).⁴

⁴ Patent Owner’s citation to “Ex. 2030 ¶ 230” appears to be an error and we cannot determine from the context what Patent Owner intended to cite.

Petitioner further argues that the FBDIMM is a preferred embodiment in the '918 patent, and excluding a preferred embodiment from the claims is “rarely, if ever correct.” Reply 9 (citing *Kaufman v. Microsoft Corp.*, 34 F.4th 1360, 1372 (Fed. Cir. 2022)). To the contrary, Patent Owner argues that the claims need not encompass FBDIMM embodiments, which are disclosed but unclaimed subject matter. Sur-Reply 10 (citing *Maxwell v. J. Baker, Inc.*, 86 F.3d 1098, 1107 (Fed. Cir. 1996); Ex. 1001, 21:38–55).

According to Petitioner, Patent Owner concedes that the signals (PS[9:0] and PS[9:0]bar) received by the AMB result in data, address, and control signals needed by DDR2 SDRAMS. Reply 9 (citing Ex. 2031 ¶ 31). Petitioner further notes that the FBDIMM Standards state that all memory control for the DRAM resides in the host, including memory request initiation, and that the AMB acts as a DRAM memory buffer for all read, write, and configuration accesses addressed to the DIMM. *Id.* (citing Ex. 1027, 1). As a buffer for all such commands to the FBDIMM, Petitioner asserts, the AMB must necessarily couple data, address, and control signals from the host system to the memory module as the claims require. *Id.*

Petitioner and Patent Owner agree that FBDIMM AMB receives information that contains data, address, and control signals encoded in packetized, serialized form at its edge connections via the signals PS[9:0] and PS[9:0]bar, and that the AMB uses these signals to generate data, address and command signals including DQ0–DQ63, A0–A15, RAS, CAS, and WE provided to the DIMMs onboard the memory module. Pet. 21–23; Sur-Reply 10. The claims merely require an “interface including a plurality of edge connections configured to couple ... data, address and control signals between the memory module and the host system.” Ex. 1001, 38:23–

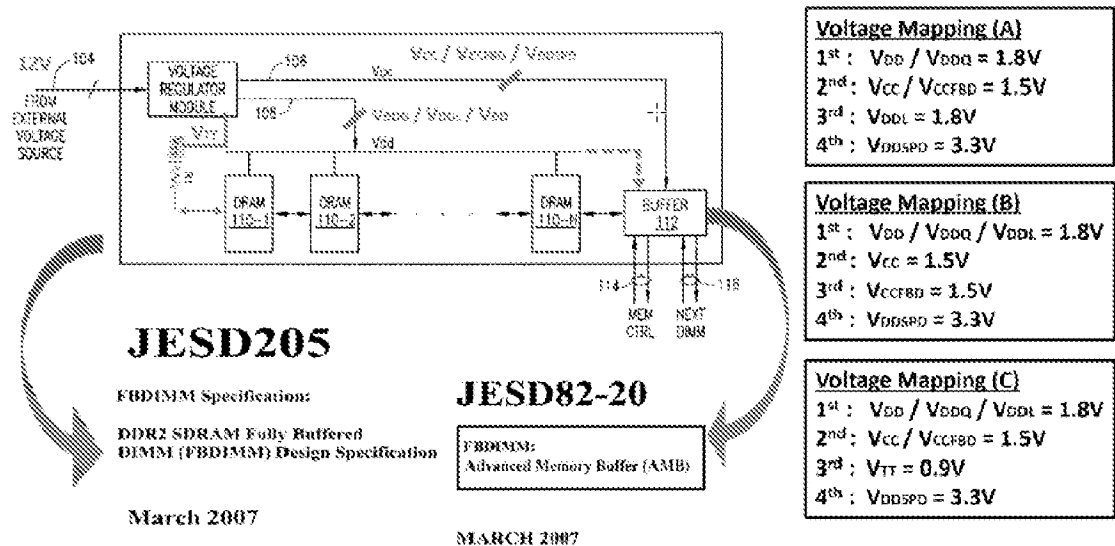
25. We determine that the data, address, and control signals of Harris and the FBDIMM Standards as received at edge connections coupling the memory module and host system satisfy this claim limitation, even though the signals are in encoded, packetized, or serialized form and thus may be received at the same pin or pins. That the signals are encoded, packetized, and serialized does not change the fact that they are data, address, and control signals. The claims do not require these signals to be received at the edge connections in any particular form or on any particular pins. *See In re Self*, 671 F.2d 1344, 1348 (CCPA 1982) (stating that limitations not appearing in the claims cannot be relied upon for patentability).

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation.

c) Limitations 1.c to 1.f: “a first buck converter configured to provide a first regulated voltage having a first voltage amplitude; a second buck converter configured to provide a second regulated voltage having a second voltage amplitude; a third buck converter configured to provide a third regulated voltage having a third voltage amplitude; a converter circuit configured to provide a fourth regulated voltage having a fourth voltage amplitude;”

Petitioner contends that Harris and the FBDIMM Standards disclose limitations 1.c to 1.f. Pet. 26–31. Petitioner relies on an annotated version of Harris’s Figure 1A, shown below.

Ground 1: Harris with JEDEC's FBDIMM Standards



Above is Harris’s Figure 1A annotated with information one would have obtained from the FBDIMM Standards according to Petitioner. Pet. 26 (citing Pet. 14–19; Ex. 1003 ¶¶ 232–243, 250–257, 262–276, 281–287). Specifically, the above figure shows voltage mappings A, B, and C of four voltages specified by Harris and the FBDIMM Standards. Petitioner contends the voltage range described in Harris (0.5V–3.5V) is the same range as the voltages described in the FBDIMM Standards. *Id.* at 27 (citing Ex. 1023 ¶ 9). Petitioner contends the voltages would be “regulated” because Harris indicates “tightly *regulated* power” is a problem to be solved, and proposes “at least one *on-board* voltage *regulator*” that is “capable of generating tightly-controlled voltage levels” as the solution. *Id.* at 28 (citing Ex. 1023 ¶¶ 2, 3, 9–11; Ex. 1003 ¶ 234).

Petitioner further contends that Harris teaches using “buck converters” to provide the four regulated voltages. *Id.* (citing Ex. 1003 ¶¶ 236, 237, 252, 264, 286). Specifically, Petitioner contends that Harris discloses “replacing [the prior art] power supply interface pins with as few as six +12V pins

(from an external voltage source)” and then using “*a high-frequency switching voltage converter* capable of generating *tightly-controlled* voltage levels” to provide each needed on-board regulated voltage. *Id.* (alteration by Petitioner; emphasis omitted; citing Ex. 1023 ¶¶ 10, 12). Petitioner contends that a “buck converter” was a conventional device for implementing such a “voltage-reducing switching converter.” *Id.* (emphasis omitted; citing Ex. 1030, 2:41–43). Although the claim states that a converter circuit generates the fourth voltage, Dr. Wolfe testifies that a buck converter would be an obvious way to implement a converter circuit. Ex. 1003 ¶ 286 (cited by Pet. 28, n.2). Petitioner further contends a buck converter would have been an obvious way to step down a higher input voltage of 12V to a lower output voltage of 3.5V or less, and that there would have been a reasonable expectation of success because buck converters were well-known switching devices commonly used to step down the voltage between input and output, as had long been taught in textbooks. *Id.* at 28–29 (citing Ex. 1003 ¶¶ 146–149, 236–237; Ex. 1058, 3, 5, 12–16; Ex. 1032, 161, 164; Ex. 1024, Fig. 6; Ex. 1050, 1:21).

Petitioner further notes that “*buck converters*” were well-known as a highly-efficient way to step down voltages without generating excess heat or requiring large cooling devices, providing further motivation to use buck converters. *Id.* at 29–30 (citing Ex. 1003 ¶ 237; Ex. 1040, 1, 23–24, Figs. 22–25; Ex. 1041, 1, 13; Ex. 1048, 3; Ex. 1058, 5; Ex. 1059, 5:23–30; Ex. 1062, 11; Ex. 1064 ¶ 101).

Additionally, Petitioner contends that it would have been obvious to use at least four converters in Harris given the need for four different voltages (e.g., 0.9V, 1.5V, 1.8V, 3.3V) in the FBDIMM Standards. *Id.* at 30

(citing Pet. 14–19). Petitioner asserts that it would have been obvious to use four converters because the voltages at the same level in Petitioner’s voltage mappings A and B are described as separate voltages with separate pins that are separately controllable in the FBDIMM Standards. *Id.* at 30–31 (citing Ex. 1028, 17–20, 30–32; Ex. 1026, 2–3, 9; Ex. 1003 ¶¶ 242, 256; Ex. 1062, 13).

For the reasons discussed above in Section III.D.3, we disagree with Patent Owner’s arguments regarding the use of buck converters.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches limitations 1.c to 1.f of claim 1.

d) Limitation 1.g: “a plurality of components coupled to the PCB, each component of the plurality of components coupled to one or more regulated voltages of the first, second, third and fourth regulated voltages, the plurality of components comprising:”

Petitioner asserts that Harris discloses a “*a plurality of components coupled to the PCB*” including a Buffer and DRAMs shown in Harris’s Figure 3, and a Serial Presence Detect (SPD) and resistors. Pet. 31 (citing Pet. 14–19; Ex. 1003 ¶¶ 288–90, 293–297). Petitioner further contends that

these components are each coupled to at least one or more of the second, third, and fourth regulated voltages. *Id.*

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

e) Limitation 1.h: “a plurality of synchronous dynamic random access memory (SDRAM) devices coupled to the first regulated voltage, and”

Petitioner asserts that Harris discloses that the plurality of components includes a plurality of DDR DRAM devices 110-1 to 110-N as shown in Harris’s Figure 1A and DRAM devices 312-1 to 312-8 for each of the memory modules shown in Harris’s Figure 3. Pet. 32 (citing Ex. 1023 ¶¶ 9, 11, Figs. 1A, 3; Ex. 1003 ¶¶ 298–303). Petitioner contends that a person of ordinary skill in the art would have known that, according to the JEDEC standards, DDR memory devices are “synchronous” DRAM devices. *Id.* (citing Ex. 1026, cover; Ex. 1028, 9; Ex. 1045, cover; Ex. 1046, cover).

Patent Owner does not dispute that the combination of Harris and the FBDIMM Standards discloses this feature. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMMs Standards teaches this limitation.

f) Limitation 1.i.1: “at least one circuit coupled between a first portion of the plurality of edge connections and the plurality of SDRAM devices”

Petitioner contends that the “first circuit” corresponds to Harris’s buffer 112 as shown in Harris’s Figure 1A, as well as the buffer of memory module 306-1 in Harris’s Figure 3. Pet. 32–34 (citing Pet. 20–25; Ex. 1023, Figs. 1A, 3; Ex. 1003 ¶¶ 304–317). Petitioner contends that Harris’s buffers are coupled to receive data, address, and control signals via memory controller interface 114 across edge connections, and transmits them to DRAMs 110-1 to 110-N or DRAMs 312-1 to 312-8. *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation.

g) Limitation 1.i.2: “the at least one circuit operable to (i) receive a first plurality of address and control signals via the first portion of the plurality of edge connections, and (ii) output a second plurality of address and control signals to the plurality of SDRAM devices”

Petitioner relies on the same showing for limitation 1.i.2 as for limitation 1.i.1. Pet. 32–34.

For the reasons stated with respect to limitation 1.b, the combination of Harris and the FBDIMM Standards discloses this limitation notwithstanding Patent Owner’s arguments. *See Section III.D.4.b.2.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris and the FBDIMM Standards teaches this limitation for.

h) Limitation 1.i.3: “the at least one circuit coupled to both the second regulated voltage and the fourth regulated voltage”

Petitioner contends that Harris with the FBDIMM standards discloses this limitation. Pet. 34–35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 318–323). Specifically, Petitioner contends that Harris’s buffer 112 in Figure 1A is coupled to “the second regulated voltage” (e.g., V_{CC} or $V_{CCFBD}=1.5V$) and “the fourth regulated voltage” (e.g., $V_{DDSPD}=3.3V$). *Id.*

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation.

i) Limitation 1.i.4: “wherein a first one of the second and fourth voltage amplitudes is less than a second one of the second and fourth voltage amplitudes.”

Petitioner contends that “one of the second and fourth voltage[s]” (e.g., 1.5V) “is less than a second one of the second and fourth voltage[s]” (e.g., 3.3V). Pet. 35 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 324–327).

Patent Owner does not specifically respond to Petitioner’s contentions for this limitation. *See Resp.*

Based on our review and consideration of the current record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards teaches this limitation.

j) Determination for Claim 1

Petitioner has shown that one of ordinary skill in the art would have had reason to combine Harris and the FBDIMM Standards with a reasonable

expectation of success. Petitioner has further shown that the combination of Harris and the FBDIMM Standards meets each limitation of claim 1. Accordingly, Petitioner has established by a preponderance of the evidence that claim 1 is unpatentable as obvious over the combination of Harris and the FBDIMM Standards.

5. *Claims 2, 3, 8, 14, and 15*

Claim 2 depends from claim 1 and recites “wherein the first and third buck converters are further configured to operate as a dual buck converter.” EX. 1001, 38:53–55. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 35–40. Specifically, Petitioner contends that at the time there were many commercially available products that could output two regulated voltages using buck converters, and thus, it would have been obvious to implement two or more of the regulated voltages such as the “first” and “third” voltages as a “dual buck converter” to reduce the number of integrated circuits, pins, and interconnections on the module, therefore simplifying the design. Pet. 36 (Ex. 1003 ¶ 338). Petitioner contends that commercially available devices included the Murata MPD4S014S dual buck converter (EX. 1042, 6; Ex. 1048, 1–2; Ex. 1058, 5), the Texas Instruments TPS51020 Dual Step-Down Controller (Ex. 1040, 1, 11) and the Fairchild Semiconductor FAN5026 Dual-Output PWM Controller” (Ex. 1041, 1, 2, 9). *See* Pet. 36–40.

Patent Owner presents no argument specific to claim 2. *See* Resp.

Based on our review and consideration of the current record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 2 obvious.

Claims 3 depends from claim 1 and recites “wherein the first voltage amplitude is 1.8 volts.” Ex. 1001, 38:56–57. Petitioner contends that the combination of Harris and the FBDIMM Standards discloses this limitation. Pet. 41 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 348–353).

Patent Owner presents no argument specific to claim 3. *See* Resp.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 3 obvious.

Claim 8 depends from claim 1 and recites

the plurality of components further comprising:
one or more registers coupled to one of the first, second, third and fourth regulated voltages, the one or more registers configured to register, in response to a clock, the first plurality of address and control signals, wherein the one of the first, second, third and fourth regulated voltages is selectively switched off to turn power off to the one or more registers while one or more components of the plurality of components are powered on.

Ex. 1001, 39:9–19.

Petitioner contends the combination of Harris and the FBDIMM Standards discloses claim 8. Pet. 41–46. Specifically, Petitioner contends that Harris’s buffer 112 is a component that has registers to register incoming signals and output them according to a “clock” signal, as shown in Harris’s Figures 1A and 3. *Id.* at 41. Petitioner contends Harris’s buffer 112 is coupled to the voltage V_{CC} or V_{CCFBD} as the “second regulated voltage.” *Id.* (citing Pet. 14–19, 27). Petitioner contends Harris discloses “buffer/logic component 112 is provided for buffering command/address (C/A) space as well as data,” which a person of ordinary skill in the art would have understood involves registers for registering the address, control, and data

signals in response to a “clock” signal. *Id.* at 42 (citing Pet. 22–25; Ex. 1023 ¶¶ 9, 17, Fig. 3; Ex. 1003 ¶¶ 400–402; Ex. 1027, 19) (emphasis omitted).

Petitioner further contends that the combination of Harris and the FBDIMM Standards discloses an S3 sleep mode that selectively switches off the second voltage to turn off the power to the one or more registers while the DRAMs remain powered on to refresh data in the DRAMs. Pet. 45 (citing Pet. 14–19, 27; Ex. 1003 ¶¶ 410–411, 414–420; Ex. 1012 ¶ 9; Ex. 1027, 21, 39).

We have already addressed Patent Owner’s arguments for patentability of claim 8 with respect to limitation 1.b and determine they do not undermine Petitioner’s showing. *See* Section III.D.4.b.2.

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders obvious claim 8.

Claim 14 depends from claim 8 and recites

wherein, in response to selectively switching on the one of the first, second, third and fourth regulated voltages to the one or more registers, the one or more registers is configured to output the registered first plurality of address and control signals to the plurality of SDRAM devices.

Ex. 1001, 39:40–45.

Petitioner contends that when the S3 sleep mode ends, normal DRAM transactions begin again and the second voltage which powers the input/output registers switches on and the registers output the registered address and control signals to the SDRAM devices. Pet. 46 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 462–466).

For the reasons stated for claim 8, we determine that Patent Owner's arguments do not undermine Petitioner's showing. Resp. 11–15.⁵

Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 14 obvious.

Claim 15 depends from claim 1 and recites

the plurality of components further comprising:
a logic element including one or more integrated circuits and discrete electrical elements, the one or more integrated circuit including an internal non-volatile memory, wherein the non-volatile memory is configured to store configuration information.

Ex. 1001, 39:46–52.

Petitioner contends that the combination of Harris and the FBDIMM Standards discloses claim 15. Pet. 47–50. Specifically, Petitioner contends logic in Harris's buffer includes an integrated circuit, as does its serial presence detect (SPD), including non-volatile memory. Pet. 47 (citing Ex. 1027, 25; Ex. 1003 ¶¶ 435–437, 468–476). Petitioner further contends the logic element includes discrete elements such as resistors and capacitors to terminate voltages for Harris's buffer. *Id.* Petitioner asserts that the S3 sleep mode of the FBDIMM Standards requires S3 Recovery Configuration Registers. *Id.* According to Petitioner, an FBDIMM like Harris's memory module will store configuration information in non-volatile memory before entering into S3 sleep mode. *Id.* (citing Ex. 1027, 25, 95–96, 141).

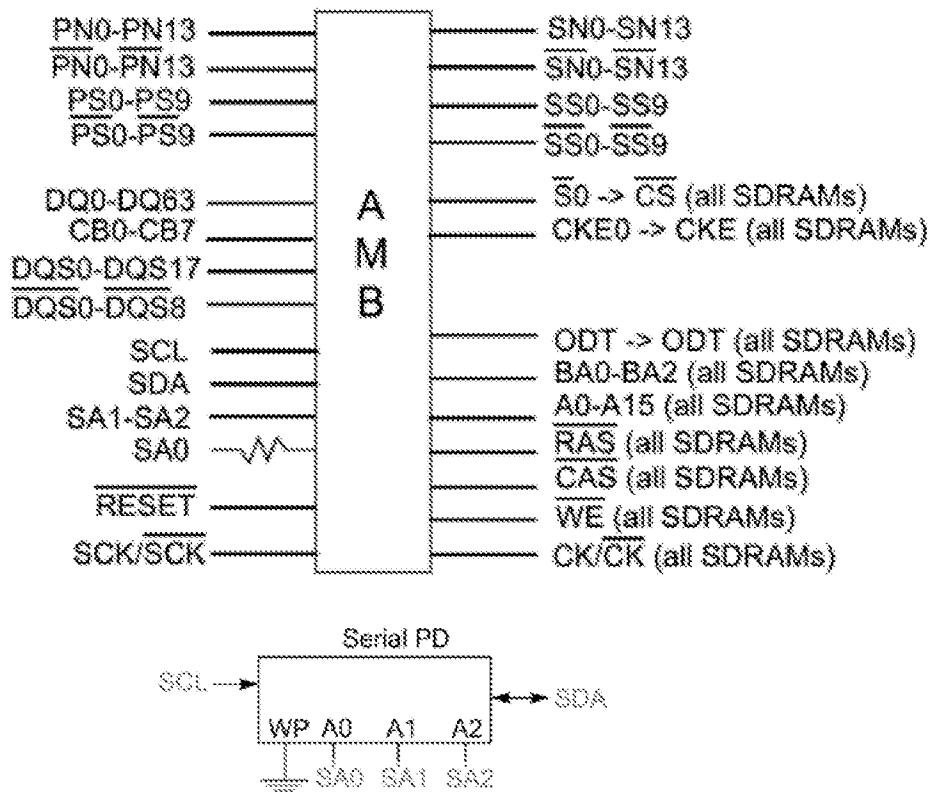
Petitioner further contends that a person of ordinary skill in the art would

⁵ Patent Owner makes this same argument for claim 21 challenged under Ground 2, which we find does not undermine Petitioner's showing for the same reasons as stated for claim 8. Resp. 11–15.

have understood that the “non-volatile memory” can be implemented in the SPD device separate from the integrated circuit implementing the AMB (Advanced Memory Buffer) where the SPD is used to store configuration information in the non-volatile memory. *Id.* at 48–49 (citing Ex. 1023 ¶ 19; Ex. 1027, 117; Ex. 1028, 13; Ex. 1066, 26:64–27:4; Ex. 1067, 1-1; Ex. 1003 ¶ 436). Petitioner further contends a person of ordinary skill in the art would have understood that the logic element includes discrete electrical elements such as resistors and capacitors, as taught by the FBDIMM Standards. *Id.* at 49–50 (citing Ex. 1003 ¶ 475; Ex. 1023 ¶ 9, Fig. 1A; Ex. 1028, 13, 42–45).

Patent Owner argues that Dr. Wolfe acknowledged that the AMB neither inherently nor expressly disclosed having a non-volatile memory. Resp. 43 (citing Ex. 2030, 285:23–286:17; *see also* Ex. 2031 ¶ 122); Sur-Reply 22. Patent Owner further argues that the statement in the FBDIMM Standards that the Control and Status Register (CSR) for the S3 sleep mode should be stored in non-volatile memory does not mean that the non-volatile memory is in the AMB. Resp. 43 (citing Ex. 2030, 285:23–286:17); Ex. 1003 ¶ 435 (CSR meaning). Patent Owner further notes that Dr. Wolfe stated that he had no knowledge that any AMB included a non-volatile memory. *Id.* (citing Ex. 2030, 292:10–20, 293:1–6). Patent Owner argues that the CSRs that the Petition references at page 47 are stored in volatile, not non-volatile, memory. *Id.* (citing Ex. 2030, 293:8–294:2). Patent Owner further argues that SPD and AMB are two different components. *Id.* (citing Ex. 2030, 293:1–6, 292:21–25). For these reasons, Patent Owner argues that Petitioner “has not cited any reference showing a logic element with a non-volatile memory.” *Id.*

To explain his views, Dr. Wolfe reproduces the figure below from the FBDIMM Standards:



Ex. 1003 ¶ 437 (citing Ex. 1028, 13) (highlighting Dr. Wolfe’s). Dr. Wolfe shows that the SPD is connected to the AMB to exchange the signals indicated in yellow highlighting. He further states that “the ‘logic element’ includes the logic in the buffer 112 and non-volatile memory (e.g., internal or external to the buffer) for storing configuration information as required by the S3 sleep mode.” *Id.*

There is no dispute on this record that the SPD is non-volatile memory—the FBDIMM Standards require it to be. Ex. 1027, 25. Dr. Wolfe expounds that the SPD can contain non-volatile memory, such as an EEPROM, for storing configuration information using a higher voltage supply at 3.3V. Ex. 1003 ¶ 436. Dr. Wolfe stated that “it would have been

obvious to a [person of ordinary skill in the art] to implement the buffer 112 with an internal nonvolatile memory because it would decrease the number of components, interfaces and interconnections on the memory module, resulting in a cheaper, faster, and more secure solution.” Ex. 1003 ¶ 436. Dr. Wolfe further states that the SPD and AMB are tied to the same voltage V_{DDSPD} . *Id.* In other words, the SPD and AMB are closely related.

Dr. Mangione-Smith states that the SPD and AMB are two different components. Ex. 2031 ¶ 123. The claims do not preclude, however, that the “logic element” may be parts of the SPD and AMB (Harris’s FBDIMM buffer) related to the SPD functionality. Ex. 1023 ¶ 9; Ex. 1028, 105; 1003 ¶ 474. In fact, as Petitioner notes, claim 15 recites that the logic element may include “*one or more* integrated circuits.” Reply 23 (citing Ex. 1001, 23:1–18; Pet. 47–49; Ex. 1003 ¶¶ 434–436). Accordingly, Petitioner has shown that the combination discloses a “logic element including a non-volatile memory” as claimed. Pet. 47–50, 67–68, 72; Reply 23.

There is no dispute that the CSRs, which are stored in the SPD before the memory module enters the S3 sleep mode, constitute “configuration information” as recited in claim 15. Pet. 47 (citing Ex. 1027, 25 (DRC, MTR, DREFTC, DAREFTC, S3RESTORE[15:0], SPDPAR[15:0] – SPDI personality bytes), 95–96, 141; Ex. 1003 ¶ 435)).

Further, there is no dispute that Harris’s buffer and the SPD are “integrated circuits,” and that the buffer is connected to “discrete electrical elements” (e.g., resistors and capacitors) as recited in claim 15. Ex. 1023, Figs. 1A, 3 (showing buffer is integrated circuit on DIMM); Ex. 1027, 25 (non-volatile memory is an integrated circuit); Ex. 1028, 43 (showing resistors and capacitors connected to AMB).

Thus, we agree with Petitioner that the features of these claims are taught or at least suggested by the combination of Harris and the FBDIMM Standards. Based on our review and consideration of the record, we determine that Petitioner has shown that the combination of Harris with the FBDIMM Standards renders claim 15 obvious.

6. *Claim 23*

Claim 23 is an independent claim. Ex. 1001, 40:50–41:21. Petitioner contends that the limitations of claim 23 are substantially identical to earlier limitations, and are thus obvious for the reasons stated above. Pet. 50–51.

Patent Owner presents no additional argument for claim 23 other than arguments previously discussed. *See* Resp.

Based on our review and consideration of the record, we determine that Petitioner has shown that Harris in combination with the FBDIMM Standards renders obvious claim 23.

7. *Determination for Ground 1*

Petitioner has shown by a preponderance of the evidence that claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris and the FBDIMM Standards for the reasons explained.

E. Ground 2: Obviousness Over Harris, the FBDIMM Standards, and Amidi

Petitioner contends that claims 1–30 of the '918 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi. Pet. 51–75. For the reasons that follow, we are persuaded that the evidence, including Dr. Wolfe's testimony, sufficiently supports Petitioner's arguments and, therefore, establishes by a preponderance of the evidence

that claims 1–30 of the '918 patent are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

1. *Amidi (Ex. 1024)*

Amidi was filed on October 25, 2006, issued on May 25, 2010, and is titled “Clock and Power Fault Detection for Memory Modules.” Ex. 1024, codes (22), (45), (54). Petitioner contends Amidi is prior art under § 102(e). Pet. 12.

Amidi’s Figure 5 is reproduced below.

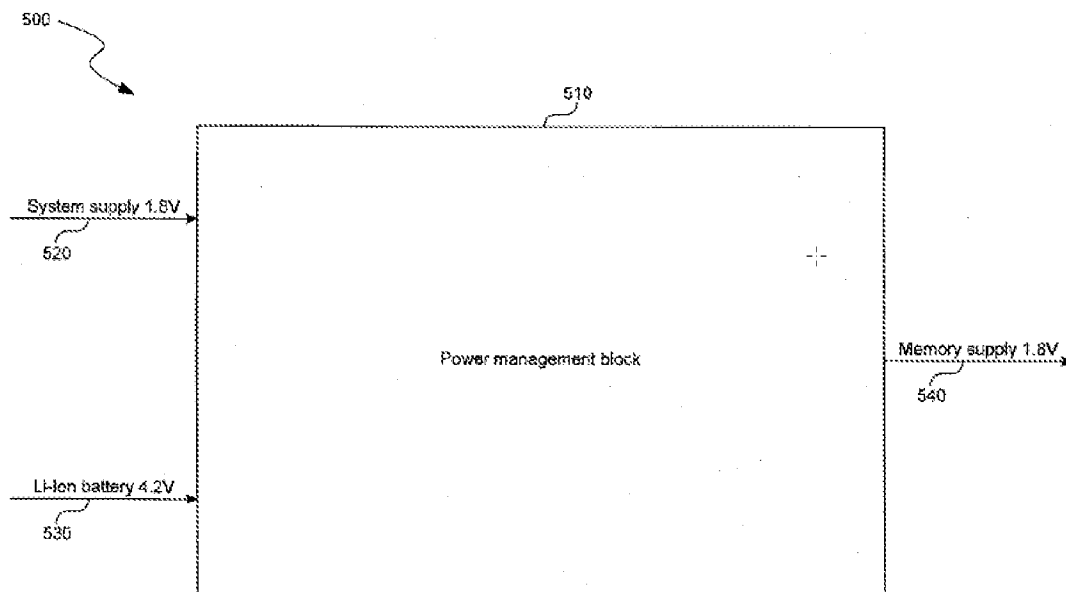


FIG. 5

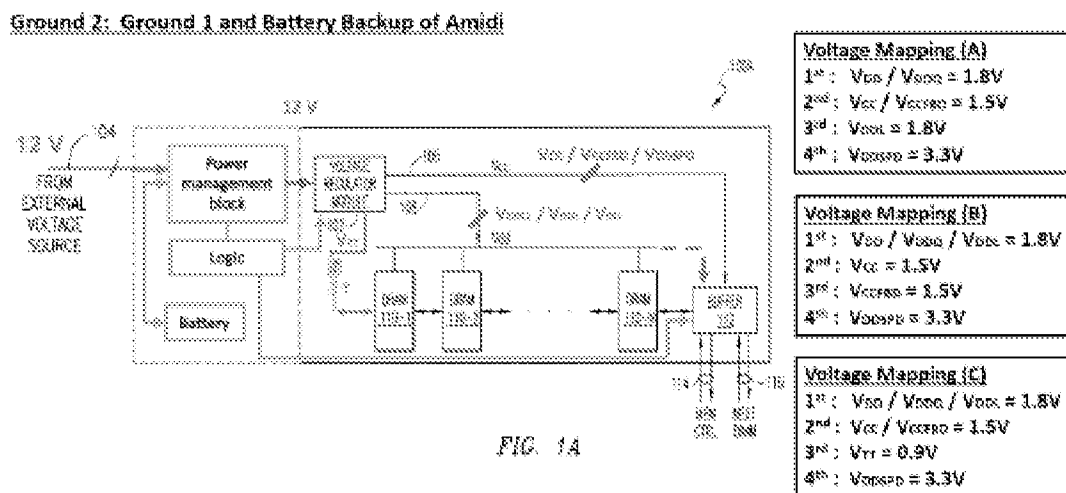
Amidi’s Figure 5 above illustrates a power management block 510 that receives an incoming system supply 520, and incoming battery supply 530, and generates an outgoing memory power supply 540 that is stabilized in the face of disruptions to the system supply 520 using the battery supply 530. Ex. 1024, 4:14–22, 8:23–36, Figs. 5, 14; Ex. 1003 ¶¶ 131–132.

2. *Motivation to Combine*

Petitioner contends that a person of ordinary skill in the art would have been motivated to combine Harris and the FBDIMM Standards with Amidi with a reasonable expectation of success. Pet. 52. Specifically, Petitioner contends that Harris recognizes concerns with power reliability and proposes the use of a redundant power source. *Id.* (citing Ex. 1023 ¶¶ 12–14, 16, Figs. 1B, 2). Petitioner notes that Amidi teaches a redundant power source (a battery on the memory module) for maintaining data during power disruption. *Id.* at 52–53 (citing Ex. 1024, code (57), 1:28–35, 2:6–26, 4:14–60, Figs. 5–6; Ex. 1003 ¶¶ 170–175). Petitioner further notes that Amidi’s power management block could be modified easily to work with Harris’s FBDIMM memory module by changing the system supply 520 and memory supply 540 to 12V as taught by Harris. *Id.* at 53 (citing Ex. 1024, Figs. 5, 6; Ex. 1023 ¶ 12; Ex. 1003 ¶¶ 171–172). Petitioner contends that, to a person of ordinary skill in the art, it would have been obvious to use the 12V external supply stepped-down with a buck converter to a 5V supply for charging Amidi’s battery, and that Amidi’s battery voltage would be stepped-up with a boost converter to the 12V level used by Harris’s memory module. *Id.* at 53–54 (citing Ex. 1024, Fig. 6 (620); Ex. 1003 ¶¶ 171–172). Petitioner contends that Amidi discloses that its power management block uses “buck” converters to step-down voltages as needed, and “boost” converters to step-up voltages as needed, as had long been taught in textbooks. *Id.* at 54 (citing Ex. 1024, 4:27–32, 4:38–40, Figs. 5, 6; Ex. 1058, 3; Ex. 1032, 161). Petitioner further contends that Amidi’s battery backup mode is similar to the S3 power-saving mode of Harris’s FBDIMM memory module and the FBDIMM Standards, such that a person of ordinary skill in

the art would have been motivated to combine their teachings. *Id.* at 54–56. Petitioner contends this is a straightforward modification of Harris’s memory module in view of Amidi and the knowledge of a person of ordinary skill in the art and simply uses a known technique (e.g., Amidi’s battery backup techniques) to improve a similar device (e.g., Harris’s memory module) in the same way (e.g., to provide a backup power supply using a battery). *Id.* at 55 (citing Ex. 1003 ¶ 175). In addition, Petitioner contends, the modification merely applies a known technique (e.g., providing a backup power supply) to a known device (e.g., a memory module) that is ready for the improvement to yield predictable results (e.g., redundancy when the system supply or clock fails). *Id.*

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi would result in the following configuration and voltage mappings:



Pet. 55–56 (citing *id.* at 14–19, 27; Ex. 1003 ¶¶ 174–179). Above, Petitioner has annotated Harris’s Figure 1A to show features added from the teachings

of the FBDIMM Standards (red) and features added from the teachings of Amidi (blue). *Id.*

Patent Owner argues that Harris provides alternate voltage sources to power a memory module in the event of a power interruption, so Harris already provides a solution for the alleged problem that Amidi addresses. Resp. 31–38; Sur-Reply 19–20. Specifically, Patent Owner argues that Harris has “logic 124” that selects a different voltage source if power becomes unavailable on one of the supply voltage paths 120-1 through 120-K. Resp. at 32–33 (citing Ex. 1023, Fig. 1B; Ex. 2031 ¶ 106). Patent Owner further argues that substituting Amidi’s battery backup solution for Harris’s redundant power sources would require substantial space to accommodate the lithium battery and associated battery charging and monitoring circuits and power management block. *Id.* at 35–38 (citing Ex. 1024, Figs. 4–6; Ex. 1023 ¶¶ 2, 13; Ex. 2031 ¶¶ 108 n.5, 109). Patent Owner argues that Petitioner “has entirely ignored the issue of whether the essential parts of Amidi it seeks to apply to the combination would even fit onto Harris’ circuit board” and that “Petitioner has presented no evidence that a [person of ordinary skill in the art] would expect Harris’ memory module to be able to accommodate three buck converters, a lithium battery and its associated charging and monitoring circuits, and Amidi’s power management module, at least not without undergoing a substantial redesign.” *Id.* at 35–38 (citing Ex. 1055, Fig. 4; Ex. 2101, 20; Ex. 2042, 7; Ex. 2031 ¶ 112).

We agree with Petitioner that Amidi provides a redundant power supply on the memory module that is independent from the host, and that a person of ordinary skill in the art would have seen the value of adding a

redundant power supply onboard Harris's memory module at least for some practical applications. Reply 19–20 (citing Pet. 52–53; Ex. 1003 ¶ 170; Ex. 1024, 1:28–35, 2:6–26). Although Harris mentions an “external voltage source,” it does not mention the term “battery.” Hence, Patent Owner's arguments that Amidi provides a solution to a problem that Harris already solves is not correct. *See* Ex. 1003 ¶ 170 (“Amidi recognizes that it is useful to keep data on the memory module with a backup power supply on the module itself, such as a battery supply, when the entire system surrounding the memory module loses power.”)(citing Ex. 1024, 1:28–35; 2:6–26). Furthermore, Petitioner shows that memory modules with battery backup were known in the art and readily available. Reply 19 (citing Ex. 2035, 39; Ex. 1075, 165:10–167:7). As to the space issue, Petitioner notes that even if the battery backup occupies one side of an FBDIMM board, the other side still can include one rank or two ranks using stacked memory devices. *Id.* at 19–20 (Ex. 1028, 36; Ex. 1075, 74:22–75:25, 77:10–17).

Accordingly, we determine that Petitioner has shown that one of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success.

3. *Claims 1–3, 8, 14, 15, and 23*

We agree with Petitioner that the addition of Amidi does not negate Petitioner's showing with respect to the previous ground. Pet. 56, 66–67 (citing Ex. 1003 ¶¶ 215–353, 412–413, 467–476, 557–597). Accordingly, Petitioner has shown by a preponderance of the evidence that claims 1–3, 8, 14, 15, and 23 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, and Amidi for the reasons stated in the previous ground.

4. *Claims 16–22 and 30 (“pre-regulated input voltage”)*

Claims 16–22 and 30 recite a “pre-regulated input voltage.” Ex. 1001, 39:61, 42:23. As explained in Section III.C.2, we understand “pre-regulated input voltage” to mean that the voltage is regulated before conversion to a stepped up or down level by the voltage converters. *See* Ex. 1001, code (57), 28:53–58, Fig. 16 (1110, 1112).

Petitioner contends that, with the exception of the “pre-regulated input voltage” limitation, claims 16–22 and 30 are combinations of limitations previously discussed which are rendered obvious for the reasons for reasons earlier presented in the Petition. Pet. 70–75. Petitioner provides a table showing the correspondences of the limitations of these claims to others discussed earlier in the Petition. *Id.* at 71–73. We agree that each limitation of these claims is taught or suggested by the combination of Harris, the FBDIMM Standards, and Amidi for the reasons previously discussed in the Petition for those same limitations.

Turning to the “pre-regulated input voltage” limitation, Petitioner contends that Harris discloses an input that is “pre-regulated” because it is “within pre-determined limits” and because it can be “regulated.” Pet. 73 (citing Ex. 1023 ¶¶ 13–14; Ex. 1003 ¶¶ 487–488). Alternatively, Petitioner contends that Amidi teaches a boost converter in its power management block that converts (and thus pre-regulates) the battery voltage to an output of 12 Volts. Pet. 73–74 (citing *id.* at 52–56; Ex. 1003 ¶ 489; Ex. 1032, 161 (explaining that switch-mode converters, including boost converters, “convert the unregulated dc input into a controlled dc output at a desired voltage level”))).

Patent Owner states that Harris's 12V input with a "+/-15%" tolerance is an unregulated voltage source. Resp. 38 (citing Ex. 1023 ¶ 13). Patent Owner does not explain what distinguishes a regulated voltage from an unregulated one, but, in any event, Harris expressly discloses that the external voltage supply may be "regulated." Ex. 1023 ¶ 14. This means "pre-regulated" in the context of the claims. Pet. 73–74 (citing Ex. 1023 ¶¶ 13–14; Ex. 1003 ¶¶ 487–489). Because the voltage that is received at the voltage converters has already been regulated externally, this meets our adopted construction (which, as we note above, is consistent with Patent Owner's proposed construction). *See* Resp. 38 ("A pre-regulated input voltage is just a regulated voltage provided to these voltage conversion circuits.").

Patent Owner further argues that "when a power supply other than the 12V external power source is used, Harris teaches the use of corresponding VRMs." Resp. 38 (citing Ex. 1023 ¶ 14, Fig. 1B). Therefore, Patent Owner argues, there is no reason to generate a pre-regulated 12V voltage, as Petitioner contends. *Id.* But, as explained, Harris expressly discloses that a "regulated" voltage may be input to its memory module and, therefore, Harris teaches converters that are "configured to receive a pre-regulated input voltage," as recited in the claims. Ex. 1023 ¶ 14.

After reviewing the record, we determine that Petitioner has shown by a preponderance of the evidence that the combination of Harris, the FBDIMM Standards, and Amidi teaches the "pre-regulated input voltage" subject matter of claims 16–22.

5. *Claims 5–7, 9–13, 16–22, and 24–27*

Claims 5–7, 9–13, 16–22 and 24–27 recite, or depend from a claim that recites, that the voltage monitor circuit generates the trigger signal in response to the input voltage being **greater than** a predetermined threshold voltage. Ex. 1001, 38:61–67, 40:7–13, 41:22–27, 42:21–26. Claims 11, 12, 18, 19, 25, and 26 require certain actions to be taken in response to the “trigger signal.”

a) *Trigger Signal on Overvoltage—Claims 5, 16, and 24*

Claim 5 recites

a voltage monitor circuit configured to monitor a power input voltage received via a second portion of the plurality of edge connections, the voltage monitor circuit configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage.

Ex. 1001, 38:61–67. Claims 16 and 24 recite similar limitations. *Id.* at 41:22–27, 40:7–12.

We address claim 5 as representative of this group of claims. For claim 5, Petitioner contends that Harris teaches detecting both undervoltage and overvoltage conditions. Pet. 62 (citing Ex. 1023 ¶ 13). In this regard, Harris states that the tolerance of the +12V power supply is “around +/- 15%.” Ex. 1023 ¶ 13. Petitioner asserts that, in the combination, Amidi’s power supervisory module/block 665 (“voltage monitor circuit”) monitors system supply 605 (“power input voltage”) received from Harris’s edge connections. Pet. 60–61 (citing Ex. 1003 ¶¶ 359–376). Petitioner further asserts that Amidi’s signal 670 is the claimed “trigger signal” to switch to battery power. *Id.* at 61–62 (Ex. 1024, 4:44–52, 5:25–43, 8:23–29, 9:8–12, Figs. 5, 6, 14–15). Petitioner contends that such circuitry for either

overvoltage and undervoltage protection was well known and commercially available. *Id.* at 62–63 (citing Ex. 1063, 1–2; Ex. 1061, 15 (circuit for “undervoltage” and “overvoltage” detection); Ex. 1062, 15 (same); Ex. 1065, code (57), ¶¶ 14, 18–19, Figs. 1, 5 (similar)).

Patent Owner argues that neither Harris nor Amidi teaches detecting an overvoltage. Resp. 39; Sur-Reply 20–22. However, Harris discloses that its +12V power supply has a tolerance of “+/- 15%” which indicates it would be a concern if the power supply was greater than 15%. Ex. 1023 ¶ 13. Amidi indicates concern with “power fault[s].” Ex. 1024, codes (54), (57). Although Patent Owner points out that Amidi discloses detection of undervoltage (Ex. 1024, Fig. 14), “power fault” is broad enough to encompass overvoltage as well. We agree with Petitioner that a person of ordinary skill in the art considering these teachings would have considered it obvious to detect overvoltage and generate a trigger signal in response to the detection.

Patent Owner argues that the voltage regulators on which Petitioner relies do not detect input overvoltage because they can accommodate a wide range of input voltages, and there is a low probability that they would exceed their operational range. Resp. 40 (citing Ex. 2030, 58:24–61:12; Ex. 1040; Ex. 1041; Ex. 2031 ¶ 117). Petitioner replies that power surges were a known problem that could cause data loss or corruption or damage circuitry in memory boards. Reply 21–22 (citing Ex. 1003 ¶¶ 371–373; Ex. 2030, 254:24–256:11, 258:12–20; Ex. 1065, 18–19; Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15; Ex. 1075, 196:10–197:13). Petitioner contends that Harris discloses that its module can accommodate only “+/- 15%” of the

nominal supply voltage, which is consistent with maximum ratings for commercial converters. *Id.* at 22 (citing Ex. 1041, 4).

We agree with Petitioner that power surges would have been a problem in at least some circumstances for which one of ordinary skill in the art would have seen the value of voltage detection in order to take preventive action to avoid loss of data and damage to the memory module. Dr. Wolfe credibly explains that overvoltage was a danger known to persons of ordinary skill in the art, citing industry datasheets that specify overvoltage parameters. Ex. 1003 ¶¶ 371–373 (citing Ex. 1063, 1–2; Ex. 1061, 15; Ex. 1062, 15). Indeed, the cited evidence uses the phrase “an overvoltage fault” (Ex. 1061, 15), which shows that power faults are not limited to undervoltage situations.

b) Logic Element with Non-Volatile Memory—Claims 10, 11, 15, and 22

Claims 10, 11, 15, and 22 recite a logic element that includes a non-volatile memory. Ex. 1001, 39:23–27, 39:28–30, 39:46–52, 40:43–49. Claim 10 further recites that the non-volatile memory is “configured to store configuration information.” Claim 11 depends from claim 10 and recites that the logic element writes information into the non-volatile memory in response to the trigger signal. Claim 15 further recites that logic element includes one or more integrated circuits including an internal non-volatile memory, and discrete electrical elements. Claim 22 omits mentions of integrated circuits or discrete elements, but recites that the configuration information is used to program the logic element.

Petitioner contends that the combination of Harris, the FBDIMM Standards, and Amidi discloses these claims. Pet. 47–50, 67–68, 72; Reply

23. Specifically, Petitioner contends that the “non-volatile memory” for the “logic” can be either in Harris buffer (e.g., an AMB) or in an SPD connected to that buffer. Pet. 47–50, 67–68, 72.

Patent Owner argues that the FBDIMM AMB is not disclosed as having a non-volatile memory, and that, although the SPD contains a non-volatile memory according to the FBDIMM standards, the AMB and SPD are two different components. Resp. 43.

For the reasons explained for claim 15 under Ground 1, Patent Owner’s arguments do not undermine Petitioner’s showing. Thus, we agree with Petitioner that the features of these claims are taught or at least suggested by the combination of Harris, the FBDIMM Standards, and Amidi.

c) Write Operation on Overvoltage—Claims 11, 12, 18, 19, 25, and 26

Claim 11 recites “wherein, in response to the trigger signal, the logic element writes information into the non-volatile memory.” Ex. 1001, 39:28–30. Claim 12 depends from claim 11 and recites a non-volatile memory and a controller, and the controller performs a write operation to the non-volatile memory in response to the trigger signal. *Id.* at 39:31–39. Claim 18 recites a controller that executes a write operation in response to a (trigger) signal. *Id.* at 40:17–21. Claim 19 recites that the write operation includes writing data information into non-volatile memory. *Id.* at 40:22–24. Claim 25 recites a controller that executes a write operation in response to the trigger signal. *Id.* at 42:1–5. Claim 26 recites that the write operation includes writing data information to non-volatile memory. *Id.* at 42:6–8.

For claim 11, Petitioner contends that the trigger signal was explained with respect to claim 5 and may be a power disruption. Pet. 68. Petitioner further contends that the logic element writes S3 configuration information as discussed for claims 10 and 15, to allow sleep mode to conserve power during Amidi's battery backup mode. *Id.* (citing Ex. 1003 ¶¶ 439–443). Petitioner contends that Amidi's battery backup mode is similar to the S3 power-saving mode of Harris's FBDIMM memory module, because both modes put the SDRAMs in a self-refresh state to preserve data while conserving power. *Id.* (citing Pet. 45–46; Ex. 1024, 2:16–19, Fig. 11). In the event of a power disruption causing a trigger signal, Petitioner contends that a person of ordinary skill in the art would have been motivated to use the S3 sleep mode (Pet. 45–46) to conserve power during Amidi's battery backup mode, as discussed in claims 10 (Pet. 67) and 15 (Pet. 47–50), and that the S3 sleep mode requires writing S3 configuration information into the non-volatile memory in the SPD before going to sleep. *Id.* (citing Ex. 1003 ¶ 442). For claim 12, Petitioner contends that the “controller” corresponds to Harris's buffer, which stores S3 configuration information in non-volatile memory in the SPD before entering S3 sleep mode. Pet. 47–50, 67–68; Ex. 1071, 39–46; Ex. 1073, 46–63; Ex. 1003 ¶¶ 444–456. Petitioner relies on similar disclosures for claims 18, 19, 25, and 26. Pet. 71–73.

Patent Owner argues that S3 sleep mode is a low power state mode whereas overvoltage does not result in low or lost power. Resp. 44 (citing Ex. 2031 ¶ 124); Sur-Reply 22–23. Patent Owner also argues that known converters could operate even at high input overvoltage. Resp. 44 (citing Resp. Sect. IV.F.1).

Petitioner counters that overvoltage conditions can cause data loss, thus motivating a person of ordinary skill in the art to switch to backup power, where the S3 mode would be used. Reply 23 (citing Pet. 68–69).

Patent Owner argues that a memory module cannot decide to enter the S3 sleep mode upon detection of an overvoltage. Sur-Reply 22. Instead, according to Patent Owner, the CPU determines when the memory module should enter the S3 sleep mode. Sur-Reply 22–23 (citing Ex. 2030, 282:22–284:25; Ex. 2031 ¶¶ 124–125; Ex. 2006, 3). Patent Owner argues there is no evidence that such a modification could be predictably made or result in an operable implementation. *Id.* at 23 (citing Ex. 2031 ¶ 125).

As Petitioner notes, Amidi provides the teaching of generating the trigger signal *onboard the memory module* in response to detecting a “power fault” or “disruption.” Pet. 12, 52, 55, 61–62; Ex. 1003 ¶¶ 132, 171, 174, 368–374. Considering the prior art teachings together, we agree with Petitioner that one of ordinary skill in the art would have understood that overvoltage detection and generating the trigger signal onboard the memory module was an option to pursue. This option would have been both predictable and pursued with a reasonable expectation of success since the prior art combination, Amidi in particular, teaches how it could be implemented.

6. *Claims 4, 28, and 29*

Claim 4 recites that the second, third, and fourth voltage amplitudes are 2.5V, 1.2V, and 3.3V, respectively. Ex. 1001, 38:58–60. Petitioner contends that Harris discloses voltages within the range of 0.5V to 3.3V, and that Voltage Mapping D from the FBDIMM Standards discloses these voltages. Pet. 57–60.

Claim 28 recites that the second and third buck converters are configured to operate as a dual buck converter. Ex. 1001, 42:14–16. Petitioner contends this feature is disclosed by the combination. Pet. 73.

Claim 29 recites that the SDRAMs devices are configured to receive one of the first through fourth regulated voltages at 1.8V. Petitioner contends this feature is disclosed by the combination. *Id.*

Patent Owner presents no arguments specific to these claims. Petitioner has shown that these claims are disclosed by the combination.

7. *Determination for Ground 2*

Petitioner has demonstrated that a person of ordinary skill in the art would have had reason to combine Harris, the FBDIMM Standards, and Amidi with a reasonable expectation of success. Petitioner has also demonstrated that all of the limitations of claims 1–30 are taught or suggested by the combination. Accordingly, Petitioner has demonstrated by a preponderance of the evidence that claims 1–30 of the '918 patent would have been obvious over the combination of Harris, the FBDIMM Standards, and Amidi.

F. *Ground 3: Obviousness over Harris, the FBDIMM Standards, Amidi, and Hajeck*

Petitioner contends claims 1–30 would have been obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck. Pet. 75–77.

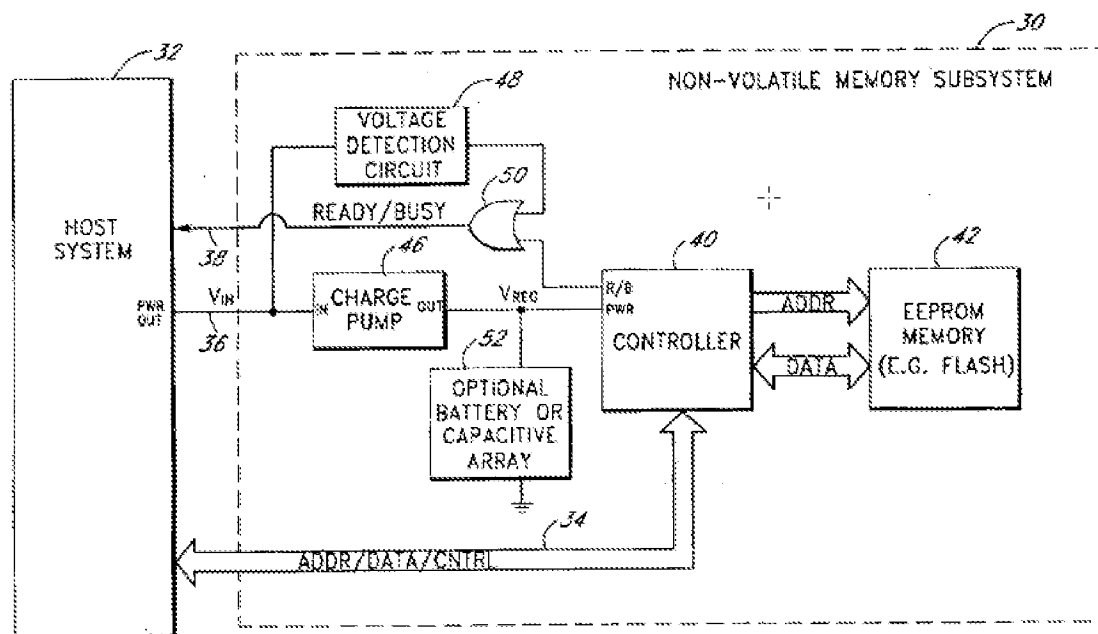
1. *Hajeck (Ex. 1038)*

Hajeck is titled “Storage Subsystem with Embedded Circuit for Protecting Against Anomalies in Power Signal from Host.” Ex. 1038, code

(54). Hajeck issued as U.S. Patent No. 6,856,556 B1 on February 15, 2005. Petitioner contends Hajeck is prior art under § 102(b). Pet. 12.

Hajeck seeks to protect storage subsystems from damage and data loss caused by irregularities in a power signal provided by a host. Ex. 1038, 1:10–13. Specifically, Hajeck seeks to prevent data loss due to loss of power from a host system, and to prevent power surges or spikes from damaging circuitry of the storage subsystem. *Id.* at 1:15–31.

Hajeck's Figure 1 is reproduced below.



In Hajeck's Figure 1, non-volatile memory subsystem 30 receives power from host system 32 at charge pump 46 which supplies regulated voltage to controller 40. *Id.* at 2:64–67. In the event of a power surge or spike, charge pump 46 protects controller 40 from damage. *Id.* at 3:12–16. In the event of a voltage drop, charge pump 46 with battery and capacitive array 52 provides sustained voltage to controller 40. *Id.* at 2:60–63, 3:10–13. Voltage detection circuit 48 detects anomalies in the input voltage, including

undervoltage and overvoltage conditions, and generates a “busy” signal provided to the host system 32 to block the host system from performing write operations to the storage subsystem. *Id.* at 1:64–67, 3:30–43.

2. *Motivation to Combine*

Petitioner contends that one of ordinary skill in the art would have been motivated to combine Hajeck with Harris, the FBDIMM Standards, and Amidi because such person would have appreciated the desirability of switching to backup power in both undervoltage and **overvoltage** conditions. Pet. 75–76. Hajeck teaches to use a backup power supply, such as a charge pump, battery, or capacitive array, in response to power loss from the host system to complete outstanding operations, and to use the charge pump to protect the controller from surges or spikes in the power supply. *See, e.g.*, Ex. 1038, code (57), 1:53–61.

Petitioner also contends that overvoltage caused by anomalies such as power surges “can cause data loss, data corruption or circuitry damage in memory boards—as anyone with a surge protector at home knows.” Reply 22 (citing Pet. 75–76; Ex. 1003 ¶¶ 137, 184). Although Patent Owner argues that Hajeck’s charge pump protects against power surges (Resp. 41), Petitioner relies on Hajeck for its “teachings related to overvoltage anomalies when implementing Amidi’s voltage supervisory block in Ground 2 so that it detects both undervoltage and overvoltage anomalies.” Pet. 76 (emphasis omitted) (citing Ex. 1003 ¶¶ 185–187).

We agree with Petitioner that Patent Owner’s arguments attack the references individually and do not properly consider what the combination of references would have signified to a person of ordinary skill in the art. Reply 21 (citing Resp. 39–42); *In re Merck & Co.*, 800 F.2d 1091, 1097

(Fed. Cir. 1986) (citing *In re Keller*, 642 F.2d 413, 425 (CCPA 1981)). Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition to the extent that the combination of Harris, the FBDIMM Standards, and Amidi were deemed not to already disclose that feature. Pet. 76–77. Thus, the features that Patent Owner argues are missing from Hajeck are not ones for which Petitioner relied on Hajeck.

For similar reasons, we also do not agree with Patent Owner’s argument that Hajeck does not suggest that data loss was a concern because its charge pump could provide the desired V_{REF} indefinitely. Resp. 42 (citing Ex. 2031 ¶¶ 120–121). Petitioner does not rely on Hajeck for its teaching of a charge pump, nor does Petitioner propose that one is included in the prior art combination. Pet. 75–76. In any case, Hajeck teaches that the charge pump provides power to its storage subsystem *temporarily* and that a battery or capacitive array may provide additional sources of backup power, so we do not agree that Hajeck teaches that the charge pump can supply power indefinitely. Ex. 1038, 1:56–61. Petitioner relies on Amidi for switching operational states and the voltage supply to battery backup in the event of an overvoltage. Reply 21 (citing Pet. 49–50). Thus, this argument too considers Hajeck individually, and not in combination with Harris, the FBDIMM Standards, and Amidi, as set forth in the Petition.

3. *Overvoltage—Claims 5, 16, and 24*

Claims 5, 16, and 24 recite detecting that the voltage monitor circuit is configured to produce a trigger signal in response to the power input voltage having a voltage amplitude that is greater than a first threshold voltage. Ex. 1001, 38:61–67, 40:7–12, 41:22–28. Petitioner contends that these claims are disclosed by the combination of Harris, the FBDIMM Standards,

Amidi, and Hajeck. Pet. 76–77. Patent Owner argues that these claims are not disclosed. Resp. 40–42.

Patent Owner argues that Hajeck does not teach a voltage detection circuit that produces a trigger signal that causes transitions to a different operable state or otherwise relates to switching power sources. Resp. 40–42 (citing EX1038, 3:6–10, 3:30–57, 3:66–4:9, Ex. 2031 ¶¶ 119–121). This attacks Hajeck individually and is an improper approach to the obviousness analysis. *See Merck, Keller, supra*. Petitioner relies on Hajeck for its teaching of detecting an overvoltage condition, in combination with the teachings of Harris, the FBDIMM Standards, and Amidi which teach the voltage detection circuit configured to produce the trigger signal. Pet. 12–13, 60–66, 75–77; Ex. 1003 ¶¶ 137, 180–87, 363–84; Ex. 2030, 226:16–22, 230:17–232:6, 251:10–254:2.

Accordingly, the combination of Harris, the FBDIMM Standards, Amidi and Hajeck teaches, or at least suggests, the limitations of claims 5, 16, and 24.

4. *Remaining Claims*

We find no reason suggesting that Hajeck would negate any of the teachings of Harris, the FBDIMM Standards, and Amidi as shown under Ground 2 (Sect. III.E). Thus, the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck teach or suggest the limitations of the remaining claims under this Ground 3 for the same reasons stated for Ground 2.

5. *Determination for Ground 3*

Petitioner has shown that a person of ordinary skill in the art would have been motivated to combine Harris, the FBDIMM Standards, Amidi,

and Hajeck with a reasonable expectation of success. Petitioner also shows that each limitation of claims 1–30 is taught or at least suggested by the prior art combination. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Harris, the FBDIMM Standards, Amidi, and Hajeck.

G. Grounds 4 and 5: Obviousness over Spiers and Amidi

As we have determined that claims 1–30 are unpatentable as obvious under Grounds 1 to 3, we do not reach Grounds 4 and 5.

H. Motion to Exclude

Petitioner seeks to exclude materials referenced in three URLs submitted with Patent Owner’s Sur-Reply on August 4, 2023. Paper 33 (citing Paper 31, 1 n.2, 24, 26). Because we do not rely on any of these URLs in a manner adverse to Petitioner, we dismiss the Motion to Exclude as moot.

IV. CONCLUSION

For the foregoing reasons, we determine that Petitioner establishes by a preponderance of the evidence that claims 1–30 of the ’918 patent are unpatentable.

V. ORDER

Accordingly, it is:

ORDERED that claims 1–30 of the ’918 patent have been shown to be unpatentable;

FURTHER ORDERED that Petitioner’s Motion to Exclude is dismissed as moot;

FURTHER ORDERED that any party seeking judicial review must

comply with the notice and service requirements of 37 C.F.R. § 90.2.⁶

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–3, 8, 14, 15, 23	103	Harris, FBDIMM Standards	1–3, 8, 14, 15, 23	
1–30	103	Harris, FBDIMM Standards, Amidi	1–30	
1–30	103	Harris, FBDIMM Standards, Amidi, Hajeck	1–30	
1–30	103	Spiers, Amidi		
1–30	103	Spiers, Amidi, Hajeck		
Overall Outcome			1–30	

⁶ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this Decision, we draw Patent Owner’s attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

IPR2022-00996
Patent 11,016,918 B2

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: To be assigned

SERIAL NO.: Filed herewith

ART UNIT: To be assigned

FILING DATE: 03-23-2018

EXAMINER: To be assigned

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Information Disclosure Statement

Commissioner:

Listed on accompanying IDS Forms PTO/SB/08a equivalent and PTO/SB/08b equivalent are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). This Information Disclosure Statement is being filed within three months of the date of filing of a national application other than a continued prosecution application (CPA),. No statement or fee is required.

Copies of documents **FP1-FP4** are submitted herewith. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications cited as documents **US1-US141** on the attached IDS Forms are submitted.

A concise explanation of the relevance of the non-English language document(s) appears below in accordance with 37 C.F.R. § 1.98(a)(3).

Document **US139** (9,361,250 B2) is submitted herewith as an English-language counterpart to document **FP2** (CN 102110057 A).

Document **US119** (7,519,754 B2) is submitted herewith as an English-language counterpart to document **FP3** (CN 101017460 A).

Copies of the **NPL1-NPL72** documents were cited by or submitted to the Office in an IDS that complies with 37 C.F.R. § 1.98(a)-(c) in Application No. 14/840,865, filed 08-31-2015 (now pending), which is relied upon for an earlier filing date under 35 U.S.C. § 120. Thus, copies of these documents are not attached. 37 C.F.R. § 1.98(d).

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 14/840,865, filed 08-31-215 (now pending); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next communication

from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

In accordance with the Federal Circuit decision in *Dayco Prods., Inc. v. Total Containment, Inc.* 329 F.3d 1358 (Fed. Cir. 2003), Applicants submit herewith Office Actions from the co-pending, commonly assigned U.S. Patent Applications as documents **NPL36-NPL43** and **NPL45-NPL56**.

The identification of these Office Actions is not to be construed as a waiver of secrecy as to those applications now or upon issuance of the present application as a patent. The Examiner is respectfully requested to consider the cited applications and the art cited therein during examination.

It is respectfully requested that the Examiner initial and return a copy of the enclosed IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: 03-23-2018

/ Khaled Shami /
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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	Filed herewith
				Filing Date	03-23-2018
				First Named Inventor	Hyun Lee
				Art Unit	To be assigned
				Examiner Name	To be assigned
Sheet	1	of	8	Attorney Docket Number	0016.001000G

U.S. PATENT DOCUMENTS					
Examiner initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
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Examiner initials*	Cite No. ¹	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T 6
		Country Code ³ -Number ⁴ - Kind Code ⁵ (if known)				
	FP1	WO 2013/016723 A3	01-31-2013	Netlist, Inc.		
	FP2	CN 102110057 A	06-29-2011	Lanqui Semiconductor Shanghai Co., Ltd.		X
	FP3	CN 101017460 A	08-15-2007	Silicon Storage Tech Inc.		X
	FP4	EP 2737383 A2	06-04-2014	Netlist, Inc.		

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patent Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

(Use as many sheets as necessary)

Sheet	2	of	8
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Attorney Docket Number	0016.001000G
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Examiner initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
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Sheet	3	of	8
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Application Number	Filed herewith
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Filing Date	03-23-2018
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First Named Inventor	Hyun Lee
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Art Unit	To be assigned
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Examiner Name	To be assigned
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Attorney Docket Number	0016.001000G
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Examiner initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
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		Number-Kind Code ² (if known)			
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		Number-Kind Code ² (if known)			
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	US137	9,043,677 B2	05-26-2015	Kong et al.	
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	US139	9,361,250 B2	06-07-2016	Shan et al.	
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Filing Date	03-23-2018
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First Named Inventor	Hyun Lee
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Art Unit	To be assigned
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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
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				Filing Date	03-23-2018
				First Named Inventor	Hyun Lee
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				Examiner Name	To be assigned
Sheet	1	of	8	Attorney Docket Number	0016.001000G

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published	T ²
	NPL1	240pin DDR2 MetaSDRAM Registered DIMM based on 1 GB version C, Hynix Semiconductor, Product Description Rev. 0.2, September 2008, 32 pages.	
	NPL2	Advisory Action in U.S. Patent Application No. 12/240,916, mailed March 13,2012.	
	NPL3	American National Standard Dictionary of Electrical and Electrical Terms, IEEE, Fourth Edition, Revised, ANS/IEEE Std 100-1988, Institute of Electrical Engineers, November 3, 1988, pp. 215,722, 964 and 1103.	
	NPL4	Bonella, Provisional Application for "ADVANCED DYNAMIC DISK MEMORY MODULE", 53 pages.	
	NPL5	Bruce, J., "Synchronous DRAM Architectures, Organizations, and Alternate Technologies", Electrical and Computer Engineering Dept., Univ. of Maryland, December 10, 2002, 22 pages.	
	NPL6	CATSOULIS, "Designing Embedded Hardware: Create New Computers and Devices", O'Reilly Media, Inc. (2005). (67 pages)	
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	NPL10	Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889 YGR, dated June 6, 2014.	

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	NPL12	Ex. 1004 - IPR2017-00587 Ron Maltiel CV, 7 pages.	
	NPL13	Extended European Search Report for European Application No. 12 817 751.6, mailed June 9, 2015	
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	NPL15	File History for Application No. 13/905,048 filed May 29, 2013, 181 pages.	
	NPL16	File History for Provisional Application 60/941,586, filed June 1, 2007, 23 pages.	
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	NPL19	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.	
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	NPL21	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18).	
	NPL22	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014, pp.1-8.	
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	NPL24	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322.	
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	NPL26	JEDEC Standard 21-C, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.	
	NPL27	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, August 1999, 114 pages.	
	NPL28	JEDEC STANDARD, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A) January 2005, 113 pages.	
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	NPL30	JEDEC Standard, Double Data Rate (DDR): SDRAM Specification: JESD79C (Revision JESD79B), March 2003, pp.1-75.	

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	NPL31	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC SOLID STATE TECH. ASSOC., March 2007, pp. 1-129.	
	NPL32	MetaRAM Develops New Technology That Quadruples Memory Capacity of Servers and Workstations; Reduces Price by Up to 90 Percent", Press Release provided by MetaRAM and published on MarketWired.com, February 25, 2008, 3 pages.	
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	NPL36	Notice of Allowance in U.S. Application No. 12/240,916, mailed on September 17, 2012.	
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	NPL38	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed May 6, 2014.	
	NPL39	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed September 29, 2014.	
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	NPL41	Notice of Allowance in U.S. Patent Application No. 13/905,053, mailed December 11, 2013.	
	NPL42	Notice of Allowance in U.S. Patent Application No. 14/173,219 dated July 7, 2014.	
	NPL43	Notice of Allowance in U.S. Patent Application No. 14/489,269, mailed October 8, 2015.	
	NPL44	Office Action dated August 19, 2016 of the Chinese Patent Application No. 201280047758.X	
	NPL45	Office Action in U.S. Application No. 13/536,176, mailed on April 15, 2013.	
	NPL46	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012 pp.1-12.	
	NPL47	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012 pp 1-14.	
	NPL48	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011 pp.1-8.	
	NPL49	Office Action in U.S. Patent Application No. 13/536,173, mailed April 15, 2013.	
	NPL50	Office Action in U.S. Patent Application No. 13/625,563, mailed August 5, 2013.	

Examiner Signature		Date Considered	
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¹ Applicant's unique citation designation number (optional). ² Applicant is to place a check mark here if English language Translation is attached.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	Filed herewith
				Filing Date	03-23-2018
				First Named Inventor	Hyun Lee
				Art Unit	To be assigned
				Examiner Name	To be assigned
Sheet	6	of	8	Attorney Docket Number	0016.001000G

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume number, publisher, city and/or country where published	T ²
	NPL51	Office Action in U.S. Patent Application No. 13/625,563, mailed May 9, 2014.	
	NPL52	Office Action in U.S. Patent Application No. 13/905,048, mailed August 1, 2013.	
	NPL53	Office Action in U.S. Patent Application No. 13/905,053, mailed August 1, 2013.	
	NPL54	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014, pp.1-7.	
	NPL55	Office Action in U.S. Patent Application No. 14/173,242, mailed March 14, 2014, pp.1-7	
	NPL56	Office Action in U.S. Patent Application No. 14/302,292, mailed December 21, 2015.	
	NPL57	Out of Stealth Mode, Start-Up MetaRAM Unveils New Technology That Quadruples DRAM Capacity, Press Release edited by Storage Newsletter on February 28, 2008 at StorageNewsLetter.com, 8 pages.	
	NPL58	PATTERSON et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998).	
	NPL59	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.	
	NPL60	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	

Examiner Signature		Date Considered	
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				First Named Inventor	Hyun Lee
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Sheet	7	of	8	Attorney Docket Number	0016.001000G

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	NPL61	Petition for Inter Partes Review of U.S. Patent No. 8,301,833, filed September 29, 2008.	
	NPL62	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed June 19, 2014.	
	NPL63	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	
	NPL64	Petition for Inter Partes Review of U.S. Patent No. 8,671,243, filed May 29, 2013.	
	NPL65	Petition for Inter Partes Review of U.S. Patent No. 8,874,831, filed July 26, 2012.	
	NPL66	Provisional Application No. 60/912,321 filed April 17, 2007	
	NPL67	Provisional Application No. 60/941,586, filed June 1, 2007	
	NPL68	Restriction Requirement in U.S. Patent Application No. 12/240,916, mailed March 31, 2011.	
	NPL69	Smart Storage Systems, Inc's Invalidity Contentions, Case No. 4:13-cv-05889-YGR, dated June 6, 2014.	
	NPL70	Using Two Chip Selects to Enable Quad Rank, an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: December 17, 2005, 2 pages.	

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Sheet	8	of	8	Attorney Docket Number	0016.001000G

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	NPL71	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1115.	
	NPL72	WONG, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.	

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UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/934,416	03/23/2018	Hyun Lee	0016.001000G

CONFIRMATION NO. 2728

FORMALITIES LETTER



151145
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

Date Mailed: 04/19/2018

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **300** basic filing fee.
- \$ **160** surcharge.
- \$ **660** search fee.
- \$ **760** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **1880** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at
(571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/jmilani/

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: 2728

SERIAL NO.: 15/934,416

ART UNIT: 2185

FILING DATE: 03-23-2018

EXAMINER: TBA

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Commissioner:

Notice of Prior and Concurrent Proceedings

Applicant hereby calls to the attention of the Patent and Trademark Office the following *inter partes* review proceedings involving patents that are commonly-assigned with the patent in the above-identified patent application:

Proceeding	Status
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00365 (P.T.A.B.), filed December 27, 2017	Institution Granted 8/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00364 (P.T.A.B.), filed December 27, 2018	Institution Granted 8/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00363 (P.T.A.B.), filed December 22, 2017	Institution Granted 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00362 (P.T.A.B.), filed December 22, 2017	Institution Granted 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed December 14, 2018	Institution Granted 5/24/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00730 (P.T.A.B.), filed January 20, 2017	Terminated 5/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00692 (P.T.A.B.), filed January 17, 2017	Terminated 7/5/2018

Proceeding	Status
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00668 (P.T.A.B.), filed January 13, 2017	Terminated 7/18/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00667 (P.T.A.B.), filed January 13, 2017	Terminated 7/5/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00649 (P.T.A.B.), filed January 13, 2017	Terminated 5/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00587 (P.T.A.B.), filed January 6, 2017	Terminated 7/20/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00577 (P.T.A.B.), filed January 5, 2017	Terminated 7/5/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00560 (P.T.A.B.), filed January 3, 2017	Terminated 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00562 (P.T.A.B.), filed January 17, 2017	Terminated 9/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00561 (P.T.A.B.), filed January 17, 2017	Terminated 9/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00548 (P.T.A.B.), filed January 17, 2017	Terminated 5/3/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00549 (P.T.A.B.), filed January 17, 2017	Terminated 5/3/2018

Information Disclosure Statement

Listed on accompanying IDS Forms PTO/SB/08a equivalent and/or PTO/SB/08b equivalent are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). This Information Disclosure Statement is being filed before the mailing date of a first Office Action on the merits. No statement or fee is required.

Copies of documents **NPL6 – NPL37** are submitted. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications are submitted.

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 14/840,865, filed 08-31-215 (now U.S. Pat. No. 9,928,186); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next communication from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

In accordance with the Federal Circuit decision in *Dayco Prods., Inc. v. Total Containment, Inc.* 329 F.3d 1358 (Fed. Cir. 2003), Applicants submit herewith Office Actions from co-pending U.S. Patent Application No(s). 14/214,652, filed March 24, 2014 as documents **NPL1 – NPL5**.

The identification of these Office Actions is not to be construed as a waiver of secrecy as to those applications now or upon issuance of the present application as a patent. The Examiner is respectfully requested to consider the cited applications and the art cited therein during examination.

It is respectfully requested that the Examiner initial and return a copy of the enclosed IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 60-2034.

Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: October 17, 2018

/Khaled Shami/

Khaled Shami

Reg. No. 38,745

SHAMI MESSINGER PLLC
1000 Wisconsin Ave N.W., Suite 200
Washington, D.C. 20007
Tel: 202-516-6900

[illegible][illegible]

Examiner Signature		Date Considered	
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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	15/934,416
				Filing Date	03-23-2018
				First Named Inventor	Hyun Lee
				Art Unit	2185
				Examiner Name	To be assigned
Sheet	1	of	4	Attorney Docket Number	0016.001000G

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published	T ²
	NPL1	Copy of Non-Final Office Action for U.S. Appl. No. 14/214,652, mailed March 8, 2018; 41 pages.	
	NPL2	Copy of Examiner Search Notes for U.S. Appl. No. 14/214,652, mailed March 8, 2018; 3 pages.	
	NPL3	Copy of Examiner Notice of References Cited for U.S. Appl. No. 14/214,652, mailed March 8, 2018; 1 page.	
	NPL4	Extended European search report issuing from European Patent Application 17191878.2, August 2, 2018, 9 pgs.	
	NPL5	Copy of Final Office Action for U.S. Appl. No. 15/000,834, mailed May 3, 2018; 16 pages.	
	NPL6	Patent Owner's Preliminary Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00365 (P.T.A.B.), filed May 7, 2018; 47 pages.	
	NPL7	Patent Owner's Preliminary Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00364 (P.T.A.B.), filed May 7, 2018; 47 pages.	
	NPL8	Patent Owner's Preliminary Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00363 (P.T.A.B.), filed May 7, 2018; 42 pages.	
	NPL9	Patent Owner's Preliminary Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00362 (P.T.A.B.), filed May 7, 2018; 41 pages.	
	NPL10	Patent Owner's Supplemental Mandatory Notices, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed September 17, 2018; 3 pages.	

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				Art Unit	2185	
				Examiner Name	To be assigned	
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	NPL11	Patent Owner's Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed August 10, 2018; 65 pages.	
	NPL12	Patent Owner's Preliminary Response, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed March 19, 2018; 59 pages.	
	NPL13	Netlist Inc.'s Notice of Appeal to the U.S. Court of Appeals for the Federal Circuit and Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00562 (P.T.A.B.), filed September 6, 2018; 56 pages.	
	NPL14	Netlist Inc.'s Notice of Appeal to the U.S. Court of Appeals for the Federal Circuit and Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00561 (P.T.A.B.), filed September 6, 2018; 67 pages.	
	NPL15	Netlist Inc.'s Notice of Appeal to the U.S. Court of Appeals for the Federal Circuit and Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00560 (P.T.A.B.), filed June 29, 2018; 22 pages.	
	NPL16	Patent Owner's Supplemental Mandatory Notices, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00549 (P.T.A.B.), filed April 19, 2018; 4 pages.	
	NPL17	Patent Owner's Supplemental Mandatory Notices, filed in <i>Sandisk Corp. v. Netlist, Inc.</i> , Case No. IPR2015-01020 (P.T.A.B.), filed April 19, 2018; 3 pages.	
	NPL18	Netlist Inc.'s Notice of Appeal and Decision on Remand, filed in <i>Diablo Technologies, Inc. v. Netlist, Inc.</i> , Case No. IPR2014-01011 (P.T.A.B.), filed May 30, 2018; 33 pages.	
	NPL19	Netlist Inc.'s Notice of Appeal and Decision on Remand, filed in <i>Diablo Technologies, Inc. v. Netlist, Inc.</i> , Case No. IPR2014-00883 (P.T.A.B.), filed May 30, 2018; 28 pages.	
	NPL20	Netlist Inc.'s Notice of Appeal and Decision on Remand, filed in <i>Diablo Technologies, Inc. v. Netlist, Inc.</i> , Case No. IPR2014-00882 (P.T.A.B.), filed May 30, 2018; 28 pages.	

Examiner Signature		Date Considered	
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				Art Unit	2185
				Examiner Name	To be assigned
Sheet	3	of	4	Attorney Docket Number	0016.001000G

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	NPL21	Supplemental Declaration of Robert J. Murphy, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed August 10, 2018; 76 pages.	
	NPL22	Deposition of Donald Alpert, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed July 20, 2018; 218 pages.	
	NPL23	Joint List of Claim Terms For Construction and Proposed Constructions, filed as Exhibit 2007 in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed August 10, 2018; 23 pages.	
	NPL24	Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00362 and IPR2018-00363 (P.T.A.B.), entered June 29, 2018; 33 pages.	
	NPL25	"The Authoritative Dictionary of IEEE Standards Terms," 7th Ed. 2000, pp. 1 - 10.	
	NPL26	"The Authoritative Dictionary of IEEE Standards Terms," 6th Ed. 2000, pp. 1 - 8.	
	NPL27	"Orbit Micro News and Notes," available at http://www.orbitmicro.com/company/blog/87 ; 10 pages.	
	NPL28	Jacob, B. et al, "Memory Systems: Cache, DRAM, Disk, Chapter 30" New York, NY: Morgan Kaufmann Press, 2008; 40 pages.	
	NPL29	Declaration of Robert J. Murphy, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed March 19, 2001; 71 pages.	
	NPL30	Decision Instituting <i>Inter Partes</i> Review of U.S. Patent No. 9,535,623, Entered May 24, 2018, 18 pages.	

Examiner Signature		Date Considered	
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NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume number, publisher, city and/or country where published	T ²
	NPL31	Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00587 (P.T.A.B.), entered June 20, 2018; 53 pages.	
	NPL32	Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00692 (P.T.A.B.), entered July 5, 2018; 44 pages.	
	NPL33	Patent Owner's Motion to Strike, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00562 (P.T.A.B.), entered March 28, 2018; 8 pages.	
	NPL34	Petitioner's Opposition to Patent Owner's Motion to Strike, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00562 (P.T.A.B.), entered April 4, 28, 2018; 8 pages.	
	NPL35	Record of Oral Hearing, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00562 (P.T.A.B.), held April 6, 28, 2018; 68 pages.	
	NPL36	Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00548 (P.T.A.B.), entered May 3, 2018; 18 pages.	
	NPL37	Final Written Decision, filed in <i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00549 (P.T.A.B.), entered May 3, 2018; 16 pages.	
	NPL38		
	NPL39		
	NPL40		

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ Applicant's unique citation designation number (optional). ² Applicant is to place a check mark here if English language Translation is attached.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	15/934,416
				Filing Date	03-23-2018
				First Named Inventor	Hyun Lee
				Art Unit	2185
				Examiner Name	To be assigned
Sheet	1	of	1	Attorney Docket Number	0016.001000G

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published	T ²
	NPL1	Copy of Notice of Submission of Opinion, South Korea Patent Office with English Translation, 6 pages	X
	NPL2	Cop of Notice of Final Rejection, South Korea Patent Office with English Translation, 4 pages.	X
	NPL3		
	NPL4		
	NPL5		
	NPL6		
	NPL7		
	NPL8		
	NPL9		
	NPL10		

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ Applicant's unique citation designation number (optional). ² Applicant is to place a check mark here if English language Translation is attached.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO.: 2728

SERIAL NO.: 15/934,416

ART UNIT: 2185

FILING DATE: 03-23-2018

EXAMINER: TBA

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Commissioner:

Notice of Prior and Concurrent Proceedings

Applicant hereby calls to the attention of the Patent and Trademark Office the following *inter partes* review proceedings involving patents that are commonly-assigned with the patent in the above-identified patent application:

Proceeding	Status
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00365 (P.T.A.B.), filed December 27, 2017	Institution Granted 8/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00364 (P.T.A.B.), filed December 27, 2018	Institution Granted 8/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00363 (P.T.A.B.), filed December 22, 2017	Institution Granted 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00362 (P.T.A.B.), filed December 22, 2017	Institution Granted 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2018-00303 (P.T.A.B.), filed December 14, 2018	Institution Granted 5/24/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00730 (P.T.A.B.), filed January 20, 2017	Terminated 5/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00692 (P.T.A.B.), filed January 17, 2017	Terminated 7/5/2018

Proceeding	Status
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00668 (P.T.A.B.), filed January 13, 2017	Terminated 7/18/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00667 (P.T.A.B.), filed January 13, 2017	Terminated 7/5/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00649 (P.T.A.B.), filed January 13, 2017	Terminated 5/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00587 (P.T.A.B.), filed January 6, 2017	Terminated 7/20/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00577 (P.T.A.B.), filed January 5, 2017	Terminated 7/5/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00560 (P.T.A.B.), filed January 3, 2017	Terminated 6/29/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00562 (P.T.A.B.), filed January 17, 2017	Terminated 9/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00561 (P.T.A.B.), filed January 17, 2017	Terminated 9/6/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00548 (P.T.A.B.), filed January 17, 2017	Terminated 5/3/2018
<i>SK HYNIX et al. v. Netlist, Inc.</i> , Case No. IPR2017-00549 (P.T.A.B.), filed January 17, 2017	Terminated 5/3/2018

Information Disclosure Statement

Listed on accompanying IDS Forms PTO/SB/08a equivalent and/or PTO/SB/08b equivalent are documents that may be considered material to the patentability of this application as defined in 37 C.F.R. §1.56, and in compliance with the duty of disclosure requirements of 37 C.F.R. §§ 1.97 and 1.98.

Applicant has listed publication dates on the attached IDS Forms based on information presently available to the undersigned. However, the listed publication dates should not be construed as an admission that the information was actually published on the date indicated.

Applicant reserves the right to establish the patentability of the claimed invention over any of the information provided herewith, and/or to prove that this information may not be prior art, and/or to prove that this information may not be enabling for the teachings purportedly offered.

This statement should not be construed as a representation that a search has been made, or that information more material to the examination of the present patent application does not exist. The Examiner is specifically requested not to rely solely on the material submitted herewith.

Filing under 37 C.F.R. § 1.97(b). This Information Disclosure Statement is being filed before the mailing date of a first Office Action on the merits. No statement or fee is required.

Copies of documents **NPL1** and **NPL2** are submitted. However, in accordance with 37 C.F.R. § 1.98(a)(2)(ii), no copies of U.S. patents and patent application publications are submitted.

It is expected that the examiner will review the prosecution and cited art in the parent application no(s). 14/840,865, filed 08-31-215 (now U.S. Pat. No. 9,928,186); 14/489,269, filed 09-17-2014 (now U.S. Pat. No. 9,158,685); 13/559,476, filed 07-26-2012 (now U.S. Pat. No. 8,874,831); 12/240,916, filed 09-29-2008 (now U.S. Pat. No. 8,301,833); and 12/131,873, filed 06-02-2008 (now abandoned) in accordance with MPEP 2001.06(b), and indicate in the next communication from the office that the art cited in the earlier prosecution history has been reviewed in connection with the present application.

It is respectfully requested that the Examiner initial and return a copy of the enclosed IDS Forms, and indicate in the official file wrapper of this patent application that the documents have been considered.

Appl. No. 15/934,416
Atty. Docket No. 0016.0010000G

The U.S. Patent and Trademark Office is hereby authorized to charge any fee deficiency, or credit any overpayment, to our Deposit Account No. 60-2034.

Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: December 17, 2018

/Khaled Shami/

Shami
Reg. No. 38,745

Khaled

SHAMI MESSINGER PLLC
1000 Wisconsin Ave N.W., Suite 200
Washington, D.C. 20007
Tel: 202-516-6900

<i>Search Notes</i> 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2131

CPC - Searched*		
Symbol	Date	Examiner
G06F13/1694; G06F12/0246; G06F13/28; G06F13/4243; G06F2212/7208; G06F3/0685; G06F12/0638; G06F13/4027; G06F1/185; G06F2212/205; G06F3/0613; G06F3/0659; G11C7/1072; G11C14/0018	03/17/2019	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	03/17/2019	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PALM and Inventor Search	03/16/2019	HF
EAST Text Search	03/16/2019	HF
Google Search	03/17/2019	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO: 2728

SERIAL NO: 15/934,416

ART UNIT: 2131

FILING DATE: 3/23/18

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Commissioner:

In response to the Office Action mailed March 21, 2019, Applicant submits the following Amendment and Remarks.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 6.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

Search Notes 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2131

CPC - Searched*		
Symbol	Date	Examiner
G06F13/1694; G06F12/0246; G06F13/28; G06F13/4243; G06F2212/7208; G06F3/0685; G06F12/0638; G06F13/4027; G06F1/185; G06F2212/205; G06F3/0613; G06F3/0659; G11C7/1072; G11C14/0018	03/17/2019	HF
G06F12/0246; G06F3/0685; G06F12/0638; G06F13/28; G06F2212/7208; G06F3/0613; G06F3/0659; G11C7/1072	10/09/2019	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	03/17/2019	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PALM and Inventor Search	03/16/2019	HF
EAST Text Search	03/16/2019	HF
Google Search	03/17/2019	HF
PALM and Inventor Search Updated	10/09/2019	HF
EAST Text Search Updated	10/09/2019	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	15934416	Filing Date	2018-03-23	Docket Number (if applicable)	0016.001000G	Art Unit	2131
First Named Inventor	Hyun Lee			Examiner Name	FARROKH, HASHEM		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 602034

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	Khaled Shami/	Date (YYYY-MM-DD)	2020-03-12
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO: 2728

SERIAL NO: 15/934,416

ART UNIT: 2131

FILING DATE: 3/23/18

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Commissioner:

In response to the Office Action mailed October 15, 2019, Applicant submits the following Amendment and Remarks.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 6.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

<i>Search Notes</i> 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2131

CPC - Searched*		
Symbol	Date	Examiner
G06F13/1694; G06F12/0246; G06F13/28; G06F13/4243; G06F2212/7208; G06F3/0685; G06F12/0638; G06F13/4027; G06F1/185; G06F2212/205; G06F3/0613; G06F3/0659; G11C7/1072; G11C14/0018	03/17/2019	HF
G06F12/0246; G06F3/0685; G06F12/0638; G06F13/28; G06F2212/7208; G06F3/0613; G06F3/0659; G11C7/1072	10/09/2019	HF
G06F12/0246; G06F3/0685; G06F13/1694; G06F2212/205; G06F3/0659; G06F13/28; G06F2212/7208; G06F3/0613; G06F12/0638; G06F1/185; G11C14/0018; G11C7/1072	03/28/2020	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	03/17/2019	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

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<i>Search Notes</i> 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2131

Search Notes		
Search Notes	Date	Examiner
PALM and Inventor Search	03/16/2019	HF
EAST Text Search	03/16/2019	HF
Google Search	03/17/2019	HF
PALM and Inventor Search Updated	10/09/2019	HF
EAST Text Search Updated	10/09/2019	HF
PALM and Inventor Search Updated	03/28/2020	HF
EAST Text Search Updated	03/27/2020	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711	110	03/28/2020	HF

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		15934416
	Filing Date		2018-03-23
	First Named Inventor	Hyun Lee	
	Art Unit	2131	
	Examiner Name	Hashem FARROKH	
	Attorney Docket Number	0016.001000G	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	2043099	A	1936-06-02	Hanna	
	2	3562555	A	1971-02-09	Ahrons	
	3	3916390	A	1975-10-28	Chang et al.	
	4	4234920	A	1980-11-18	Van Ness et al.	
	5	4420821	A	1983-12-13	Hoffman	
	6	4449205	A	1984-05-15	Hoffman	
	7	4882709	A	1989-11-21	Wyland	
	8	4965828	A	1990-10-23	Ergott Jr et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		15934416
Filing Date		2018-03-23
First Named Inventor	Hyun Lee	
Art Unit		2131
Examiner Name	Hashem FARROKH	
Attorney Docket Number		0016.001000G

9	5430742	A	1995-07-04	Jeddeloh et al.	
10	5490155	A	1996-02-06	Abdoo et al.	
11	5519663	A	1996-05-21	Harper Jr et al.	
12	5519831	A	1996-05-21	Holzhammer	
13	5563839	A	1996-10-08	Herd et al.	
14	5577213	A	1996-11-19	Avery et al.	
15	5619644	A	1997-04-08	Crockett et al.	
16	5675725	A	1997-10-07	Malcolm	
17	5799200	A	1998-08-25	Brant et al.	
18	5813029	A	1998-09-22	Klein	
19	5870350	A	1999-02-09	Bertin et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		15934416
Filing Date		2018-03-23
First Named Inventor	Hyun Lee	
Art Unit		2131
Examiner Name	Hashem FARROKH	
Attorney Docket Number		0016.001000G

20	5874995	A	1999-02-23	Naimpally et al.	
21	5890192	A	1999-03-30	Lee et al.	
22	5953215	A	1999-09-14	Karabatsos	
23	5991885	A	1999-11-23	Chang et al.	
24	6023421	A	2000-02-08	Madanza et al.	
25	6026465	A	2000-02-15	Mills et al.	
26	6065092	A	2000-05-16	Roy	
27	6112310	A	2000-08-29	Jun et al.	
28	6145068	A	2000-11-07	Lewis	
29	6158015	A	2000-12-05	Klein	
30	6199142	B1	2001-03-06	Saulsbury et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		15934416
Filing Date		2018-03-23
First Named Inventor	Hyun Lee	
Art Unit		2131
Examiner Name	Hashem FARROKH	
Attorney Docket Number		0016.001000G

31	6216247	B1	2001-04-10	Creta et al.	
32	6269382	B1	2001-07-31	Cabrera et al.	
33	6336174	B1	2002-01-01	Li et al.	
34	6336176	B1	2002-01-01	Leyda et al.	
35	6363450	B1	2002-03-26	Lash et al.	Entire Document
36	6421279	B1	2002-07-16	Tobita et al.	
37	6459647	B1	2002-10-01	Kengeri	
38	6487102	B1	2002-11-26	Halbert et al.	
39	6487623	B1	2002-11-26	Emerson et al.	
40	6571244	B1	2003-05-27	Larson	
41	6614685	B2	2003-09-02	Wong	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		15934416
Filing Date		2018-03-23
First Named Inventor	Hyun Lee	
Art Unit		2131
Examiner Name	Hashem FARROKH	
Attorney Docket Number		0016.001000G

42	6658507	B1	2003-12-02	Chan	
43	6691209	B1	2004-02-10	O'connell	
44	6693840	B2	2004-02-17	Shimada et al.	
45	6721860	B2	2004-04-13	Klein	
46	6769081	B1	2004-07-27	Parulkar	
47	6799241	B2	2004-09-28	Kahn et al.	
48	6799244	B2	2004-09-28	Tanaka et al.	
49	6810513	B1	2004-10-26	Vest	
50	6816982	B2	2004-11-09	Ravid	
51	6839774	B1	2005-01-04	Ahn et al.	Entire Document
52	6944042	B2	2005-09-13	Komatsuzaki	

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53	6948029	B2	2005-09-20	Yano	
54	6952368	B2	2005-10-04	Miura et al.	
55	7053470	B1	2006-05-30	Sellers et al.	
56	7062618	B2	2006-06-13	Tsunoda et al.	
57	7089412	B2	2006-08-08	Chen	
58	7102391	B1	2006-09-05	Sun et al.	
59	7111142	B2	2006-09-19	Spencer et al.	
60	7136978	B2	2006-11-14	Miura et al.	
61	7155627	B2	2006-12-26	Matsui	
62	7200021	B2	2007-04-03	Raghuram	
63	7234099	B2	2007-06-19	Gower et al.	

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64	7353325	B2	2008-04-01	Lofgren et al.	
65	7409491	B2	2008-08-05	Doblar et al.	
66	7409590	B2	2008-08-05	Moshayedi et al.	
67	7411859	B2	2008-08-12	Sohn et al.	
68	7421552	B2	2008-09-02	Long	
69	7467251	B2	2008-12-16	Park et al.	
70	7519754	B2	2009-04-14	Wang et al.	
71	7600142	B2	2009-10-06	Ichikawa	
72	7716411	B2	2010-05-11	Panabaker et al.	
73	7818488	B2	2010-10-19	Park et al.	
74	7873750	B2	2011-01-18	Yabuta et al.	

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75	8086955	B2	2011-12-27	Zhou et al.	
76	8102614	B2	2012-01-24	Song et al.	
77	8233303	B2	2012-07-31	Best et al.	
78	8301833	B1	2012-10-30	Chen et al.	
79	8407395	B2	2013-03-26	Kim et al.	
80	8412879	B2	2013-04-02	Chang et al.	
81	8516187	B2	2013-08-20	Chen et al.	
82	8671243	B2	2014-03-11	Chen et al.	
83	8677060	B2	2014-03-18	Chen et al.	
84	8874831	B2	2014-10-28	Lee et al.	
85	8880791	B2	2014-11-04	Chen et al.	

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86	8904098	B2	2014-12-02	Amidi et al.	
87	8904099	B2	2014-12-02	Chen et al.	
88	9043677	B2	2015-05-26	Kong et al.	
89	9158684	B2	2015-10-13	Lee et al.	
90	9361250	B2	2016-06-07	Shan et al.	
91	9436600	B2	2016-09-06	Lee	Entire Document
92	9921762	B2	2018-03-20	Amidi et al.	Entire Document
93	9928186	B2	2018-03-27	Lee et al.	Entire Document

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20130086309	A1	2013-04-04	Lee et al.	

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2	20130254456	A1	2013-09-26	Chen et al.
3	20130254497	A1	2013-09-26	Chen et al.
4	20140032820	A1	2014-01-30	Harasawa et al.
5	20140059170	A1	2014-02-27	Gasparakis et al.
6	20140156919	A1	2014-06-05	Chen et al.
7	20140156920	A1	2014-06-05	Chen et al.
8	20150058701	A1	2015-02-26	Xing et al.

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1115, 2 pages.	
	2	Wong, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.	

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Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Application Number	15934416
Filing Date	2018-03-23
First Named Inventor	Hyun Lee
Art Unit	2131
Examiner Name	Hashem FARROKH
Attorney Docket Number	0016.001000G

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2020-06-01
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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UNITED STATES PATENT AND TRADEMARK OFFICE

USPTO Automated Interview Request (AIR)

Jun 28 2020

This paper requesting to schedule and/or conduct an interview is appropriate because:

This submission is requested to be accepted as an authorization for this interview to communicate via the internet. Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned concerning scheduling of the interview via video conference, instant messaging, or electronic mail, and to conduct the interview in accordance with office practice including video conferencing.

Name(s) :
Khaled Shami

S-signature:
/Khaled Shami/

Registration Number:
38745

U.S. Application Number:
15934416

Confirmation Number:
2728

E-mail Address:
khaled@shamimessinger.com

Phone Number:
+1 6507455353

Proposed Time of Interview:
7-9-2020 10:00 AM ET

Alternative Proposed Time(s) of Interview:
7-10-2020 10:00 AM ET

Alternative Proposed Time(s) of Interview:
7-13-2020 10:00 AM ET

Preferred Interview Type:
Video Conference

I am the applicant or applicant's representative for this application.

Topic for Discussion:
Rejections under 35 USC 112 and 103



UNITED STATES
PATENT AND TRADEMARK OFFICE

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Netlist, Inc.

CONFIRMATION NO: 2728

SERIAL NO: 15/934,416

ART UNIT: 2131

FILING DATE: 3/23/18

EXAMINER: FARROKH, Hashem

TITLE: FLASH-DRAM HYBRID MEMORY MODULE

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Commissioner:

In response to the non-final Office Action mailed April 2, 2020, Applicant submits the following Amendment and Remarks.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 8.

Applicant does not believe that extensions of time are required beyond those provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

<i>Search Notes</i> 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

CPC - Searched*		
Symbol	Date	Examiner
G06F13/1694; G06F12/0246; G06F13/28; G06F13/4243; G06F2212/7208; G06F3/0685; G06F12/0638; G06F13/4027; G06F1/185; G06F2212/205; G06F3/0613; G06F3/0659; G11C7/1072; G11C14/0018	03/17/2019	HF
G06F12/0246; G06F3/0685; G06F12/0638; G06F13/28; G06F2212/7208; G06F3/0613; G06F3/0659; G11C7/1072	10/09/2019	HF
G06F12/0246; G06F3/0685; G06F13/1694; G06F2212/205; G06F3/0659; G06F13/28; G06F2212/7208; G06F3/0613; G06F12/0638; G06F1/185; G11C14/0018; G11C7/1072	03/28/2020	HF

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	110	03/17/2019	HF

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

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<i>Search Notes</i> 	Application/Control No. 15/934,416	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner HASHEM FARROKH	Art Unit 2135

Search Notes		
Search Notes	Date	Examiner
PALM and Inventor Search	03/16/2019	HF
EAST Text Search	03/16/2019	HF
Google Search	03/17/2019	HF
PALM and Inventor Search Updated	10/09/2019	HF
EAST Text Search Updated	10/09/2019	HF
PALM and Inventor Search Updated	03/28/2020	HF
EAST Text Search Updated	03/27/2020	HF
PALM and Inventor Search Updated	12/29/2020	HF
EAST Text Search Updated	12/29/2020	HF

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711	110	03/28/2020	HF

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		2015-08-31
	First Named Inventor	Hyun Lee	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	062453-036	

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	2043099		1936-02-02	Hanna	
	2	3562555		1971-02-09	Ahrons	
	3	3916390		1975-10-28	Chang et al.	
	4	4234920		1980-11-18	Van Ness et al.	
	5	4420821		1983-12-13	Hoffman	
	6	4449205		1984-05-15	Hoffman	
	7	4965828		1990-10-23	Ergott, Jr. et al.	
	8	5430742		1995-07-04	Jeddeloh et al.	

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Examiner Name		
Attorney Docket Number	062453-036	

	9	5519663		1996-05-21	Harper, Jr. et al.	
	10	5519831	A	1996-05-21	Holzhammer	
	11	5563839		1996-10-08	Herd et al.	
	12	5577213		1996-11-19	Avery et al.	
	13	5619644		1997-04-08	Crockett et al.	
	14	5675725		1997-10-07	Malcolm	
	15	5870350		1999-02-09	Bertin et al.	
	16	5874995		1999-02-23	Naimpally et al.	
	17	5890192		1999-03-30	Lee et al.	
	18	5953215		1999-09-14	Karabatsos	
	19	6023421		2000-02-08	Clinton et al.	

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	20	6112310		2000-08-29	Jun et al.	
	21	6145068		2000-11-07	Lewis	
	22	6158015		2000-12-05	Klein	
	23	6199142	B1	2001-03-06	Saulsbury et al.	
	24	6216247	B1	2001-04-10	Creta et al.	
	25	6269382		2001-07-31	Cabrera et al.	
	26	6336174		2002-01-01	Li et al.	
	27	6336176		2002-01-01	Leyda et al.	
	28	6421279	B1	2002-07-16	Tobita et al.	
	29	6459647	B1	2002-10-01	Kengeri	
	30	6487623		2002-11-26	Emerson et al.	

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	31	6658507		2003-12-02	Chan	
	32	6691209		2004-02-10	O'Connell	
	33	6721860		2004-02-10	Klein	
	34	6769081	B1	2004-07-27	Parulkar	
	35	6799241	B2	2004-09-28	Kahn et al.	
	36	6799244		2004-09-28	Tanaka et al.	
	37	6816982		2004-11-09	Ravid	
	38	6487102	B1	2002-11-26	Halbert et al.	
	39	6944042	B2	2005-09-13	Komatsuzaki	
	40	6948029	B2	2005-09-20	Yano	
	41	6952368		2005-10-04	Miura et al.	

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	42	7053470	B1	2006-05-30	Sellers et al.	
	43	7062618	B2	2006-06-13	Tsunoda et al.	
	44	7089412	B2	2006-08-08	Chen	
	45	7102391	B1	2006-09-05	Sun et al.	
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	47	7155627	B2	2006-12-26	Matsui	
	48	7200021	B2	2007-04-03	Raghuram	
	49	7234099	B2	2007-06-19	Gower et al.	
	50	7353325		2008-04-01	Lofgren et al.	
	51	7409491	B2	2008-12-05	Doblar et al.	
	52	7409590		2008-08-05	Moshayedi et al.	

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	53	7411859	B2	2008-08-12	Sohn et al.	
	54	7421552	B2	2008-09-02	Long	
	55	7467251	B2	2008-12-16	Park et al.	
	56	7600142		1990-10-17	Groos	
	57	7716411	B2	2010-05-11	Panabaker et al.	
	58	7818488	B2	2010-10-19	Park et al.	
	59	8086955		2011-12-27	Zhou et al.	
	60	8102614		2012-01-24	Song et al.	
	61	8233303	B2	2012-07-31	Best et al.	
	62	8301833		2012-10-30	Chen et al.	
	63	8407395		2013-03-26	Kim et al.	

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	64	8412879		2013-04-02	Chang et al.	
	65	8516187		2013-08-20	Chen et al.	
	66	8671243		2014-03-11	Chen et al.	
	67	8677060		2014-03-18	Chen et al.	
	68	8874831	B2	2014-10-28	Lee et al.	
	69	8880791	B2	2014-11-04	Chen et al.	
	70	8904098	B2	2014-12-02	Amidi et al.	
	71	8904099	B2	2014-12-02	Chen et al.	

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	1	20020083368		2002-06-27	Abe et al.	

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	2	20020199061		2002-12-26	Friedman et al.	
	3	20030158995	A1	2003-08-21	Lee et al.	
	4	20040088508		2004-05-06	Ballard et al.	
	5	20040163027	A1	2004-08-19	MacLaren et al.	
	6	20040190210		2004-09-30	Leete	
	7	20050044302	A1	2005-02-25	Pauley et al.	
	8	20050060488	A1	2005-03-17	Poechmueller	
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	12	20060069896	A1	2006-03-30	Sanders	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

	13	20060080515	A1	2006-04-13	Spiers et al.	
	14	20060294295	A1	2006-12-28	Fukuzo	
	15	20070136523		2007-06-14	Bonella et al.	
	16	20070192627		2007-08-16	Oshikiri	
	17	20080104344		2008-05-01	Shimozono et al.	
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	20	20100274953		2010-10-28	Lee et al.	
	21	20110320804		2011-12-29	Chan et al.	
	22	20120204079		2012-08-09	Takefman et al.	
	23	20120271990		2012-10-25	Chen et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
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Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

	24	20130019076		2013-01-17	Amidi et al.	
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	26	20130254456		2013-09-26	Chen et al.	
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	29	20140156919	A1	2014-06-05	Chen et al.	
	30	20140156920	A1	2014-06-05	Chen et al.	

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FOREIGN PATENT DOCUMENTS								Remove
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	2013016723	WO	A3	2013-01-31	NETLIST INC.		☐
	2	2737383	EP	A2	2014-06-04	NETLIST INC.		☐

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Application Number		
Filing Date		2015-08-31
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Examiner Name		
Attorney Docket Number	062453-036	

NON-PATENT LITERATURE DOCUMENTS			Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011.	☐
	2	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012.	☐
	3	Advisory Action in U.S. Patent Application No. 12/240,916, mailed March 13, 2012.	☐
	4	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012.	☐
	5	Office Action in U.S. Patent Application No. 13/536,173, mailed April 15, 2013.	☐
	6	Office Action in U.S. Patent Application No. 13/905,048, mailed August 1, 2013.	☐
	7	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed May 6, 2014.	☐
	8	International Search Report and Written Opinion in PCT/US12/48750, dated October 10, 2012.	☐
	9	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

10	Office Action in U.S. Patent Application No. 13/625,563, mailed August 5, 2013.	☐
11	Office Action in U.S. Patent Application No. 13/625,563, mailed May 9, 2014.	☐
12	Office Action in U.S. Patent Application No. 13/905,053, mailed August 1, 2013.	☐
13	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014.	☐
14	Office Action in U.S. Patent Application No. 14/173,242, mailed March 14, 2014.	☐
15	JEDEC Standard 21–C, “Configurations for Solid State Memories,” pp. 4.5.5–1 to 4.5.5–18.	☐
16	Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889 YGR, dated June 6, 2014.	☐
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19	WONG, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.	☐
20	American National Standard Dictionary of Electrical and Electrical Terms, IEEE, Fourth Edition, Revised, ANS/IEEE Std 100-1988, Institute of Electrical Engineers, November 3, 1988, pp. 215, 722, 964 and 1103.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

21	Webster's II New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1115.	☐
22	Bruce, J., "Synchronous DRAM Architectures, Organizations, and Alternate Technologies", Electrical and Computer Engineering Dept., Univ. of Maryland, December 10, 2002, 22 pages.	☐
23	David, H. et al., "Fully Buffered DIMM (FB-DIMM) Design Considerations", Intel Developer Forum, Intel Corp., February 18, 2004, 36 pages.	☐
24	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.	☐
25	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.	☐
26	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322.	☐
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30	Mutnary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.	☐
31	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed June 19, 2014.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

32	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.	☐
33	"Using Two Chip Selects to Enable Quad Rank", an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: December 17, 2005, 2 pages.	☐
34	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	☐
35	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	☐
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39	Notice of Allowance in U.S. Patent Application No. 14/489,269, mailed October 8, 2015.	☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	2015-08-31
First Named Inventor	Hyun Lee
Art Unit	
Examiner Name	
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-08-31
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Inventor Information:

Inventor 1					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Hyun		Lee			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Ladera Ranch	State/Province	CA	Country of Residence i	US	
Mailing Address of Inventor:						
Address 1		21 Thalia Street				
Address 2						
City	Ladera Ranch	State/Province	CA			
Postal Code	92694	Country i	US			
Inventor 2					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Chi-She		Chen			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Walnut	State/Province	CA	Country of Residence i	US	
Mailing Address of Inventor:						
Address 1		944 Crystal Water Lane				
Address 2						
City	Walnut	State/Province	CA			
Postal Code	91789	Country i	US			
Inventor 3					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Jeffrey	C.	Solomon			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

City	Irvine	State/Province	CA	Country of Residence i	US
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Mailing Address of Inventor:

Address 1	6 Silver Fir				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country i	US		

Inventor 4[Remove](#)**Legal Name**

Prefix	Given Name	Middle Name	Family Name	Suffix
	Scott	H.	Milton	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Irvine	State/Province	CA	Country of Residence i	US
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Mailing Address of Inventor:

Address 1	49 Statehouse Place				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92602	Country i	US		

Inventor 5[Remove](#)**Legal Name**

Prefix	Given Name	Middle Name	Family Name	Suffix
	Jayesh		Bhakta	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Cerritos	State/Province	CA	Country of Residence i	US
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Mailing Address of Inventor:

Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.[Add](#)**Correspondence Information:**Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Customer Number	22204		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	062453-036	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	10	Suggested Figure for Publication (if any)	

Filing By Reference :

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	22204		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the application number blank.

Prior Application Status	Pending	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
	Continuation of	14/489269	2014-09-17		
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14/489269	Continuation of	13/559476	2012-07-26	8874831	2014-10-28
Prior Application Status	Expired	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
13/559476	Claims benefit of provisional	61512871	2011-07-28		
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13/559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30
Prior Application Status	Abandoned	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		
Prior Application Status	Expired	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12131873	Claims benefit of provisional	60941586	2007-06-01		
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Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Remove			
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ^j (if applicable)

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

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Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Authorization to Permit Access:

☒ Authorization to Permit Access to the Instant Application by the Participating Offices

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Applicant 1				Remove	
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.					
Clear					
☒ Assignee		☐ Legal Representative under 35 U.S.C. 117		☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.			☐ Person who shows sufficient proprietary interest		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-036
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Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

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FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled, “FLASH-DRAM HYBRID MEMORY MODULE”, which is a continuation of U.S. Patent No. 8,874,831, issued, October 28, 2014, titled , “FLASH-DRAM HYBRID MEMORY MODULE“, which claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and is a continuation-in-part of US Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

TECHNICAL FIELD

[0003] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0004] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0005] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0006] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or

solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0007] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem

using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MBs for the EcoRAM vs. 104MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™), which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0008] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0009] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0010] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read

operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

OVERVIEW

[0011] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0012] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0013] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0014] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0015] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0016] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0017] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0018] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0019] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0020] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0021] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0022] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0024] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0025] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0026] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0027] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0029] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0030] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0031] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0032] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0033] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0034] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0035] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0036] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0037] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0038] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0039] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0040] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0041] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0042] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0043] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0044] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0045] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0047] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0048] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0049] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0050] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0052] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0053] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0054] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0055] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0056] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0057] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0058] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0060] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0061] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0062] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0063] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0064] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0065] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0066] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0067] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0068] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be

a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0069] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0070] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0071] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one

arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0072] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0073] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0074] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or

is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0075] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly,

when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0076] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0077] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0078] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0079] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term

“read/write access” means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0080] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0081] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0082] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly

(while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0083] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH 510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0084] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0085] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0086] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr

504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0087] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0088] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating

at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0089] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0090] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0091] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[0092] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[0093] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash–DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[0094] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM) 801. As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 803 and the other for the DRAM memory DRAM 805. The MCH 810 treats the DRAM 805 and Flash 803 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path 807 between Flash 803 and DRAM 805 allows for direct data transfer to occur between the Flash 803 and the DRAM 805 in response to control information from Ctrl 830. In this embodiment, this data transfer mechanism provides direct support for executing commands from the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 803 to the DRAM 805. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[0095] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate)

protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[0096] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[0097] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in

accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[0098] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one

embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[0099] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which

was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00100] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502 monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00101] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately

4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1060 field, a Block address 1050 field, a Plane address 1040, a Page address 1030, and a group of least significant address bits A_0A_1 1020. The Plane address 1040 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00102] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00103] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508

to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00104] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00105] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00106] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00107] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module comprising:
 - a first plurality of data signal lines forming a first data bus;
 - a second plurality of data signal lines forming a second data bus;
 - a third plurality of data signal lines forming a data bus;
 - a data manager coupled to the data bus, the first data bus, and the second data bus,wherein the memory module is couplable to a memory controller of a host system using the data bus, a control bus, and an address bus;
 - a non-volatile memory subsystem coupled to the data manager using the first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus;
 - a volatile memory subsystem coupled to the data manager using the second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus; and
 - a controller operable to receive one or more memory access commands from the memory controller of the host system by way of the control bus and the address bus, the controller operable to generate at least one of a first, second and third plurality of signals in response to the one or more memory access commands received from the memory controller of the host system, the controller operable to direct (i) operation of the non-volatile memory subsystem using the first plurality of signals, (ii) operation of the volatile memory subsystem using the second plurality of signals, and (iii) operation of the data manager using the third plurality of signals.

ABSTRACT

A memory module that is couplable to a memory controller hub (MCH) of a host system includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive read/write commands from the MCH and to direct transfer of data between any two or more of the MCH, the volatile memory subsystem, and the non-volatile memory subsystem based on the commands.

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Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/George Hinton
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Drawings-only black and white line drawings	20150831_Drawings_062453-036.pdf	932503 1cfa192aee65744bca4fc2e66b66b833bad b7474	no	10

Warnings:

Information:

2	Oath or Declaration filed	20150831_Declaration_062453-036.pdf	452725	no	5
			8d46a1fbbc611ec470054168069dd74b0f1d2d22		
Warnings:					
Information:					
3	Information Disclosure Statement (IDS) Form (SB08)	20150831_IDS_062453-036.pdf	625037	no	16
			70d0d218d78d655a8018ff572b857c37a769aa92		
Warnings:					
Information:					
4	Application Data Sheet	20150831_ADS_062453-036.pdf	1819865	no	8
			a103c4b4ce56a9e2ba68f3812f79c9d30c256b3d		
Warnings:					
Information:					
5		20150831_Specification_062453-036.pdf	164709	yes	38
			9b31d15d6bcb49c33d7a005af0f5c08245e00777		
	Multipart Description/PDF files in .zip description				
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

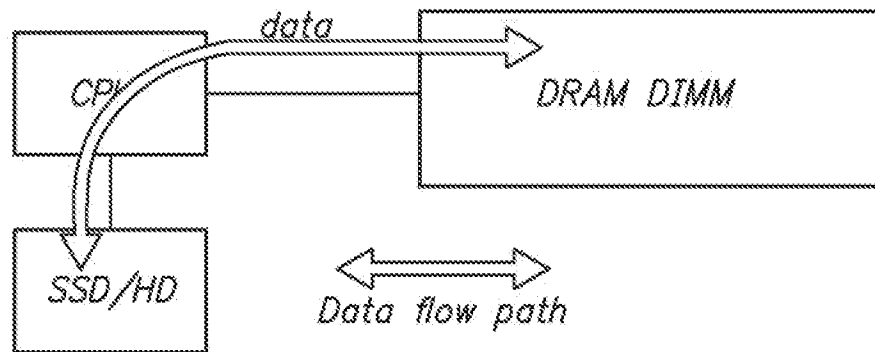


FIG. 1
(PRIOR ART)

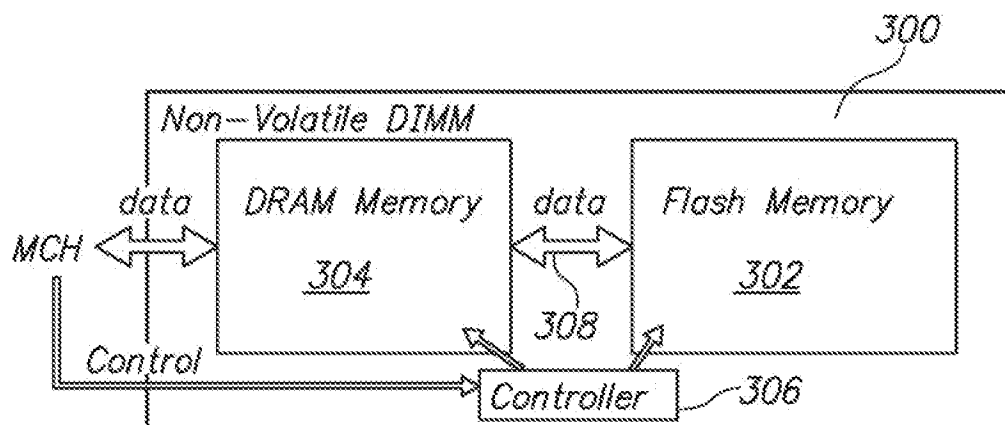
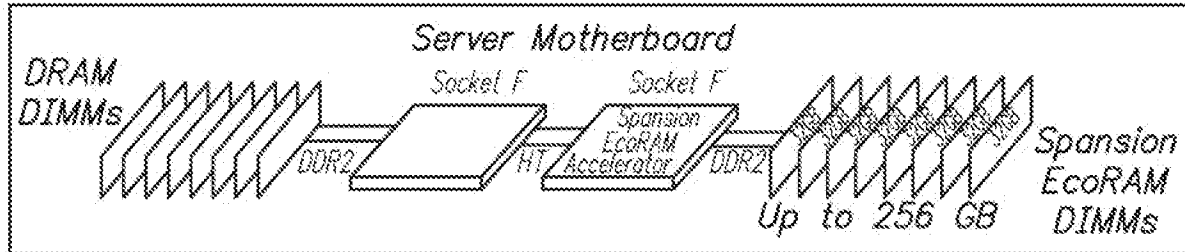


FIG. 3A

2/10

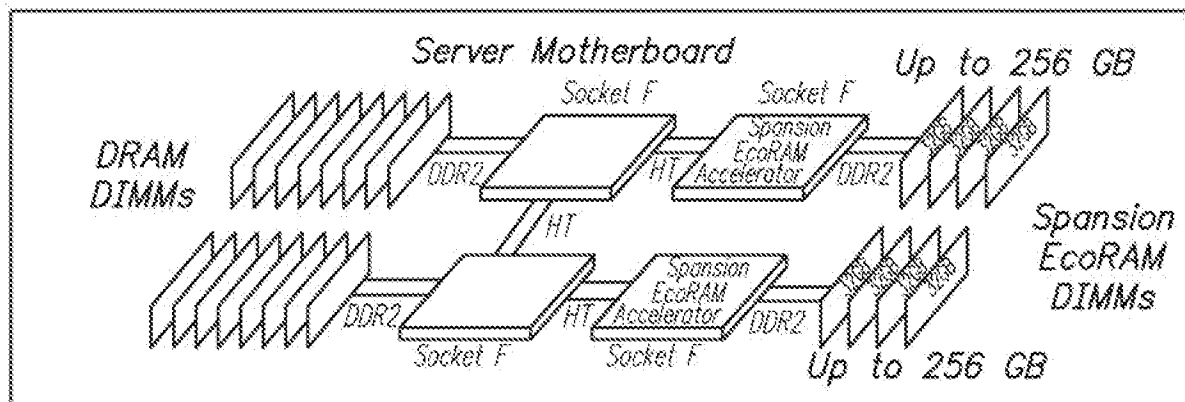
Spanion EcoRAM Configurations

256GB Spanion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spanion EcoRAM Solution

256GB Spanion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spanion EcoRAM Solution

FIG. 2
(PRIOR ART)

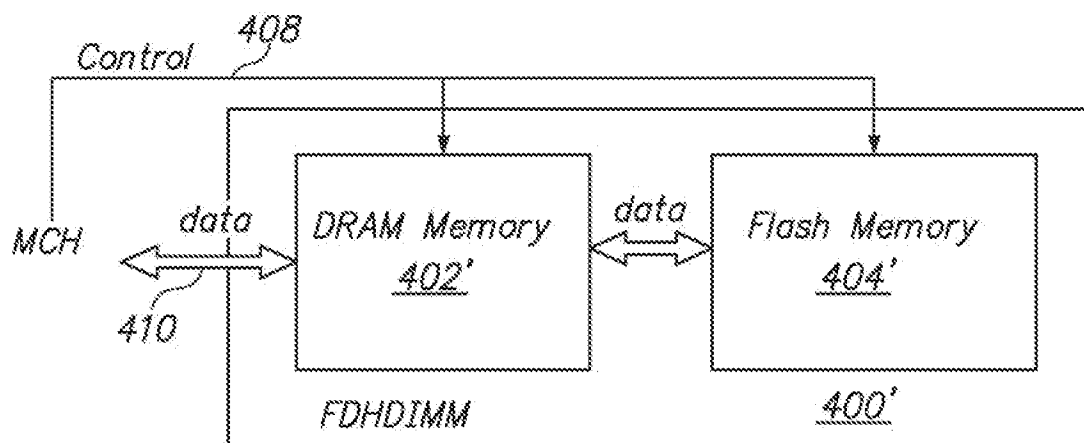


FIG. 4B

3/10

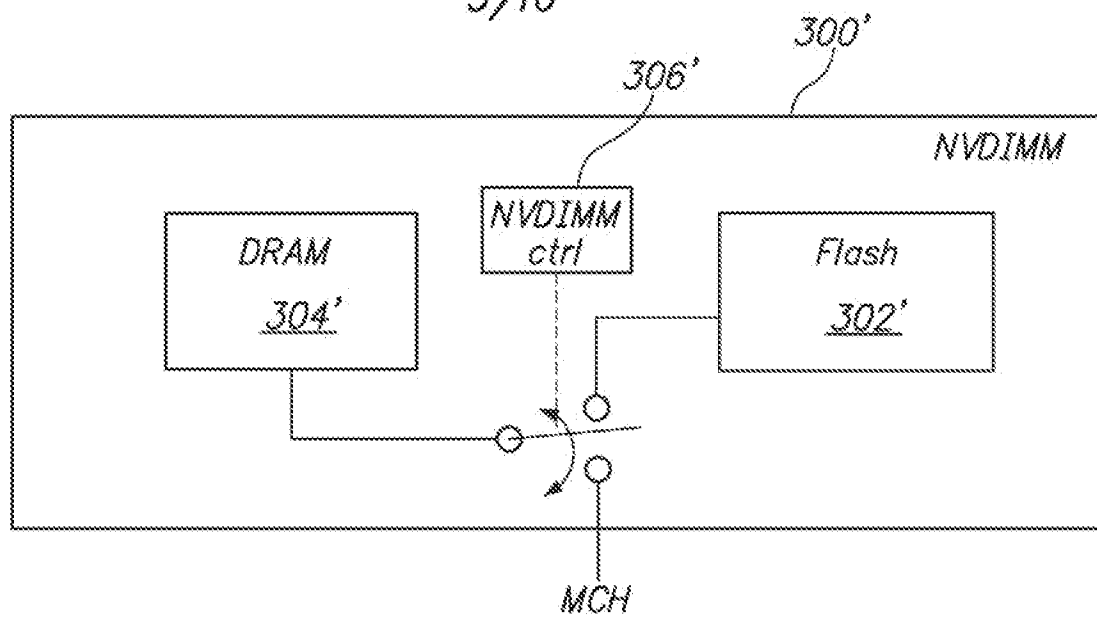


FIG. 3B

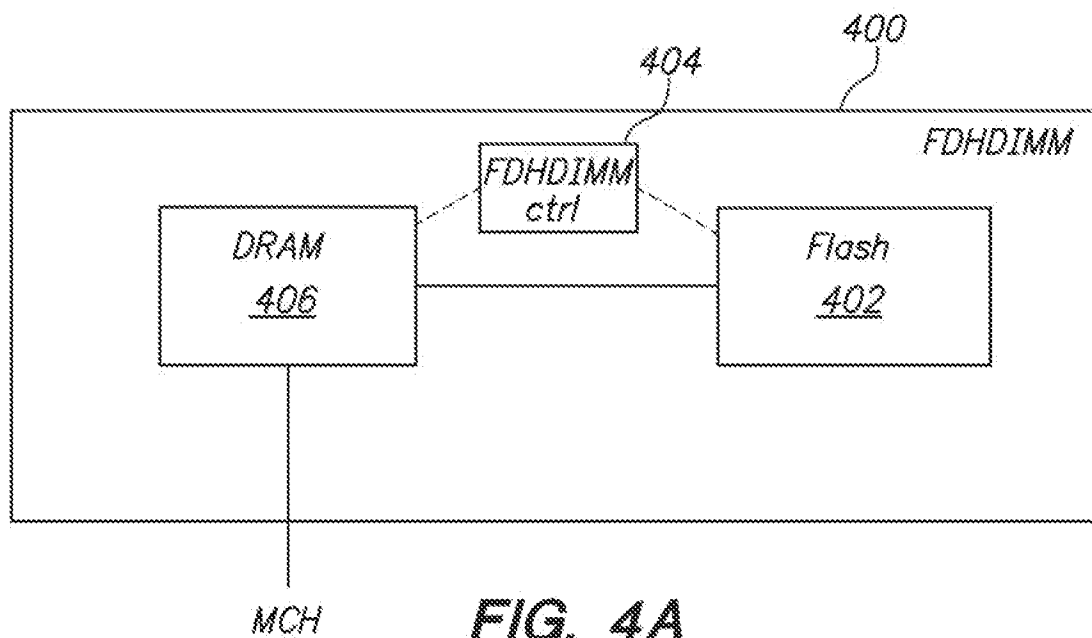


FIG. 4A

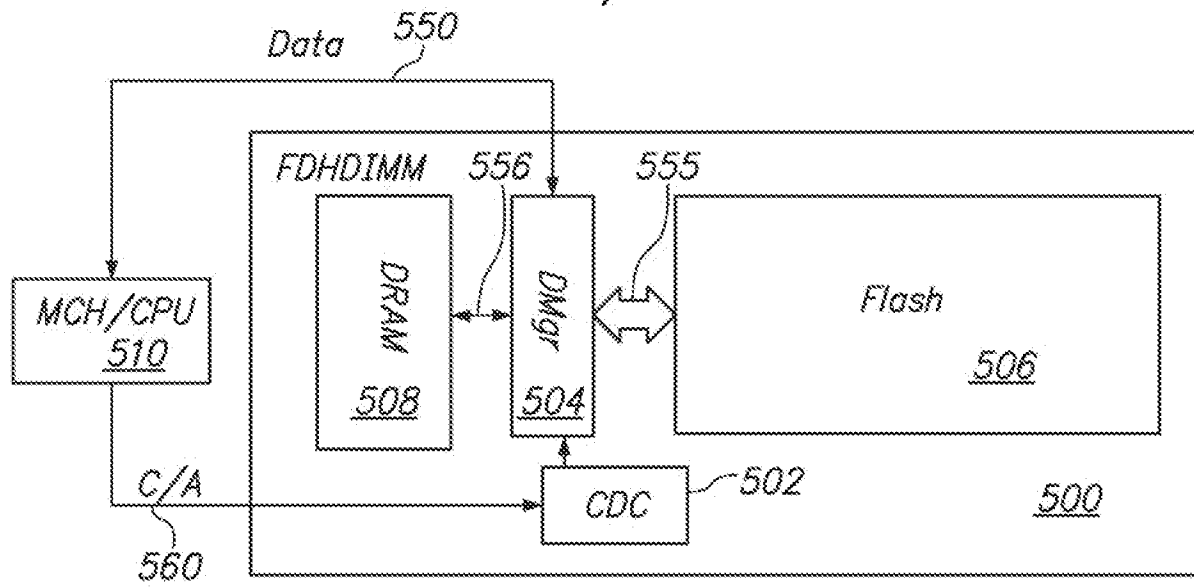


FIG. 5A

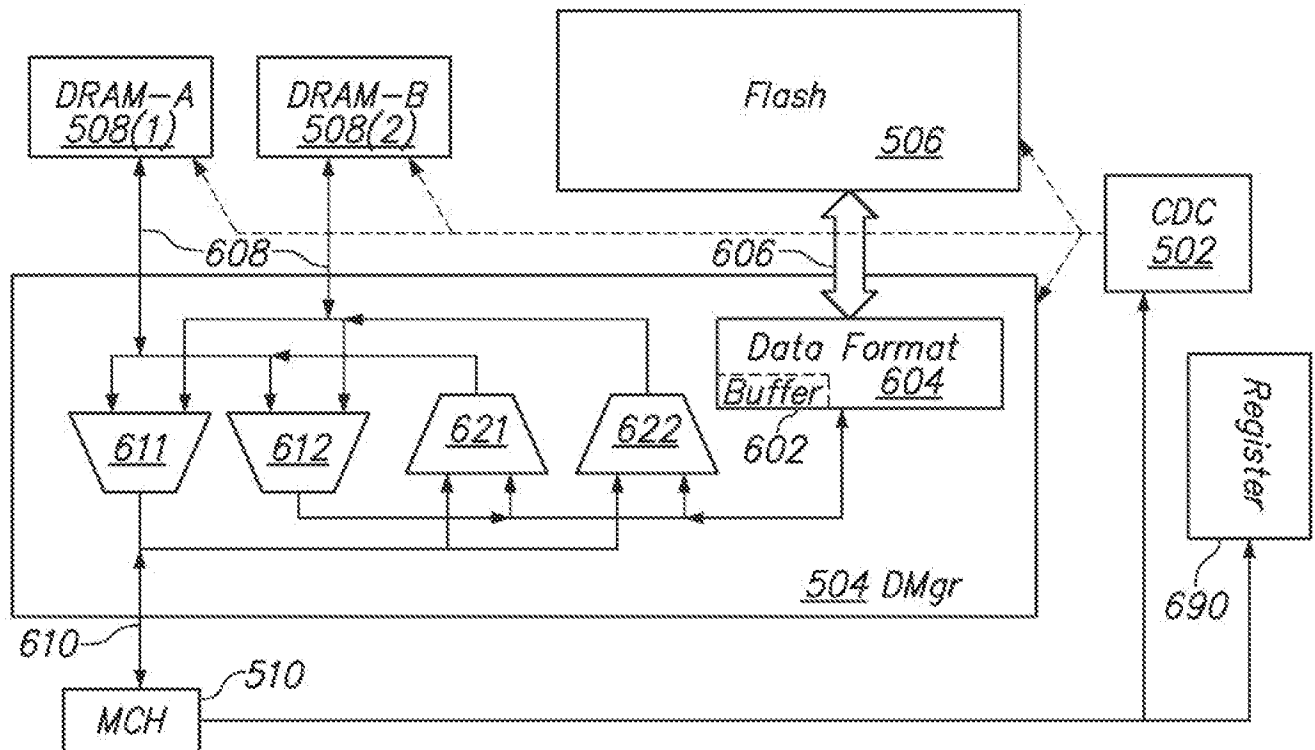
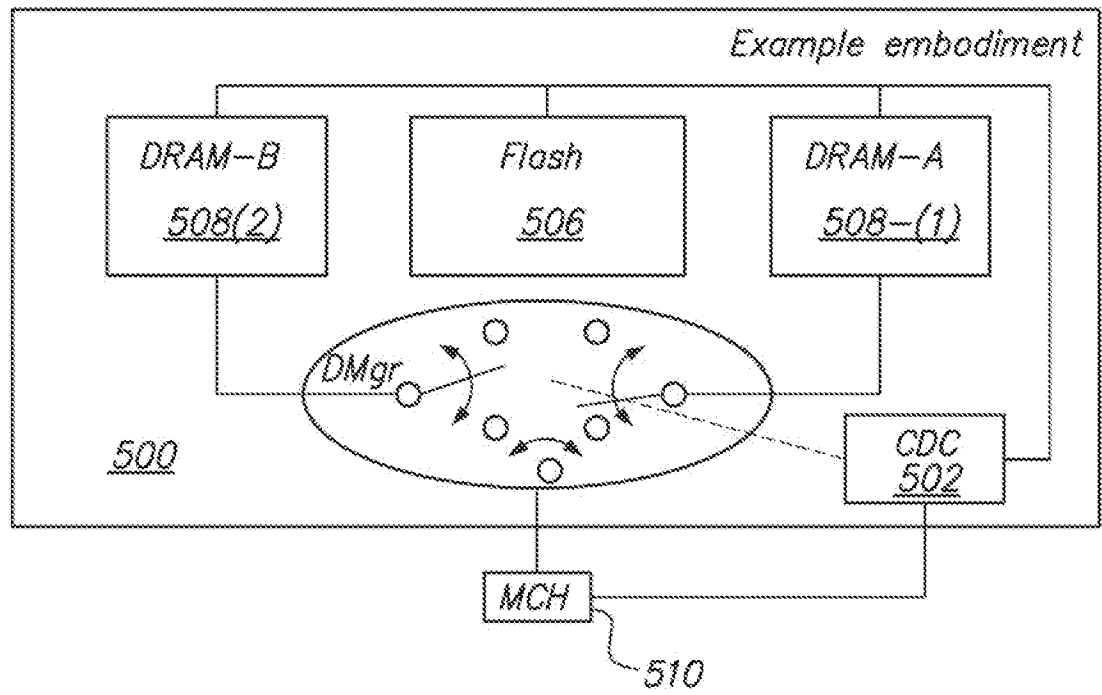


FIG. 6

**FIG. 5B**

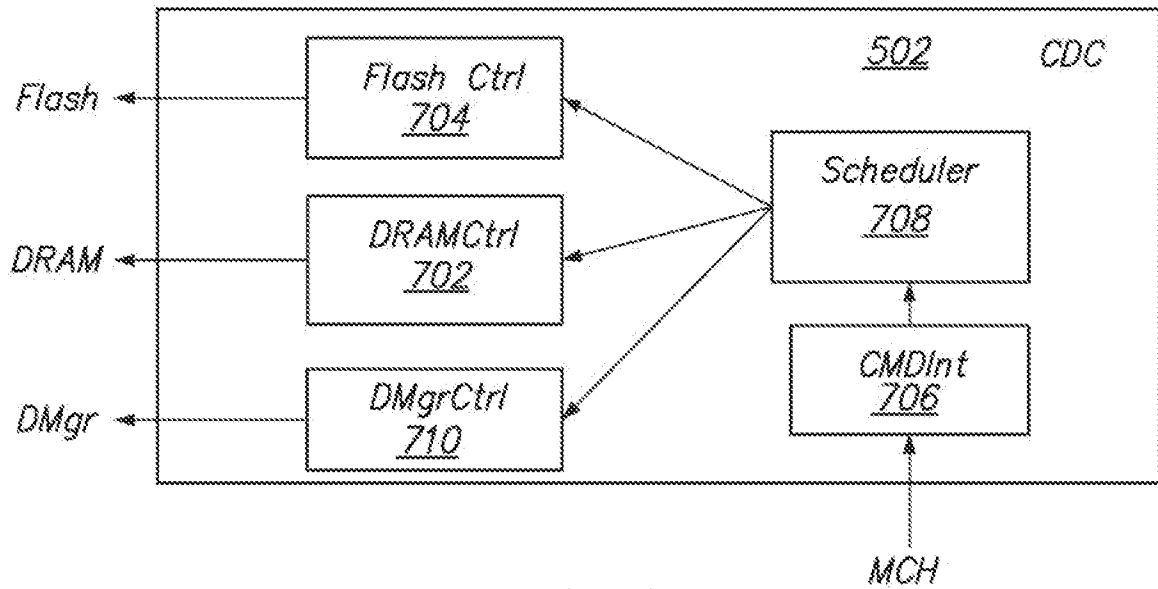


FIG. 7

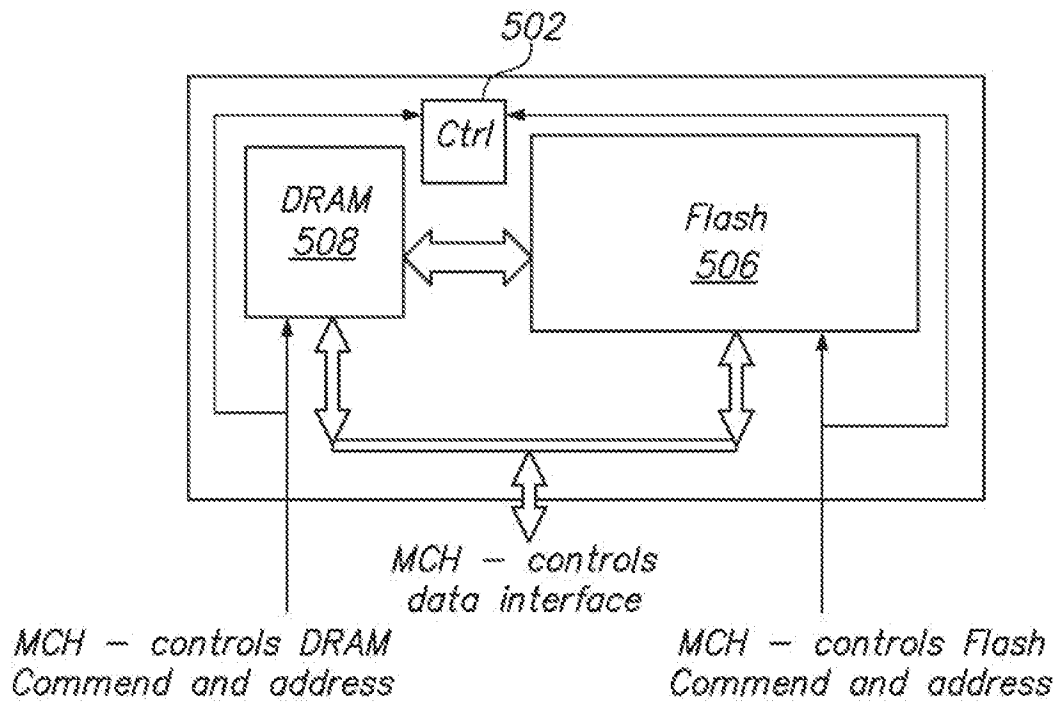
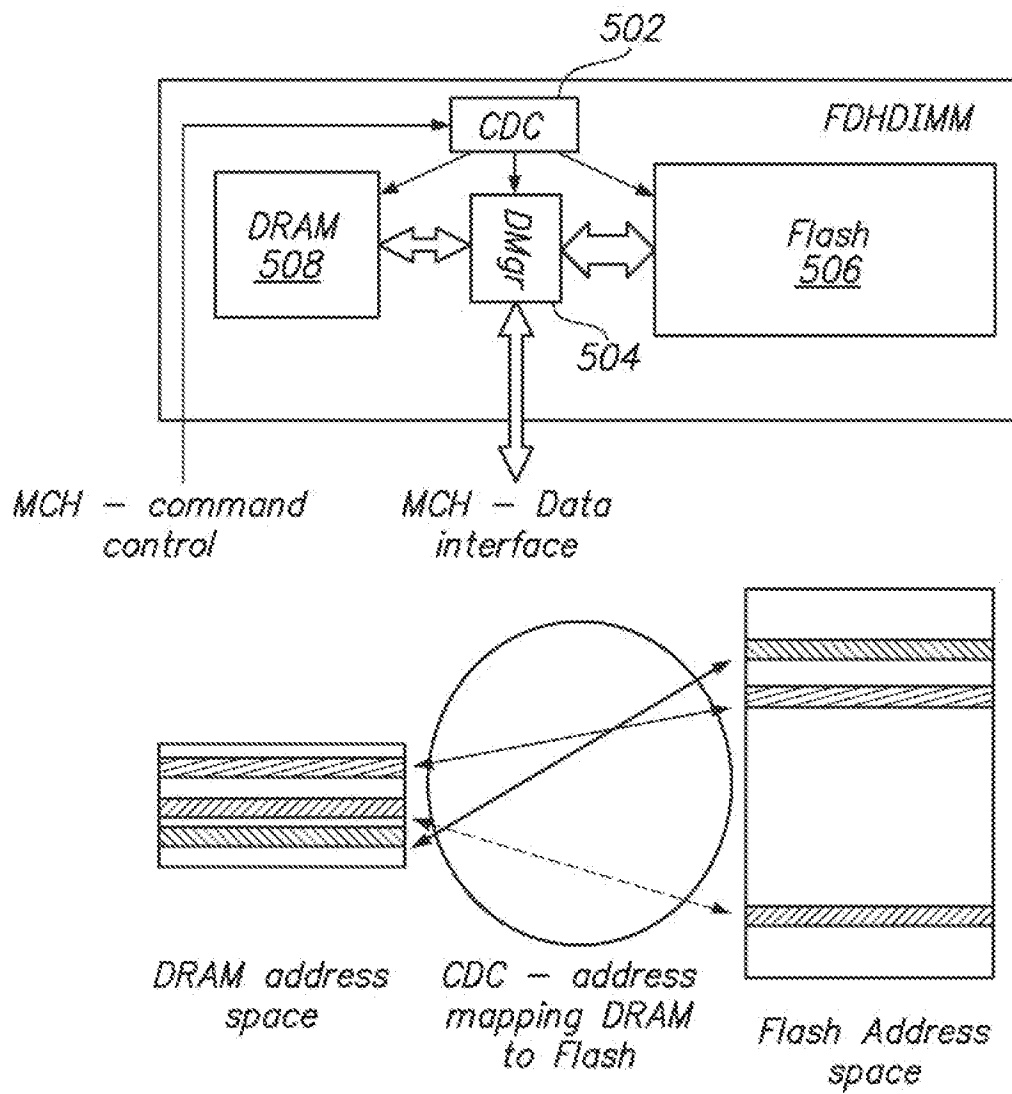
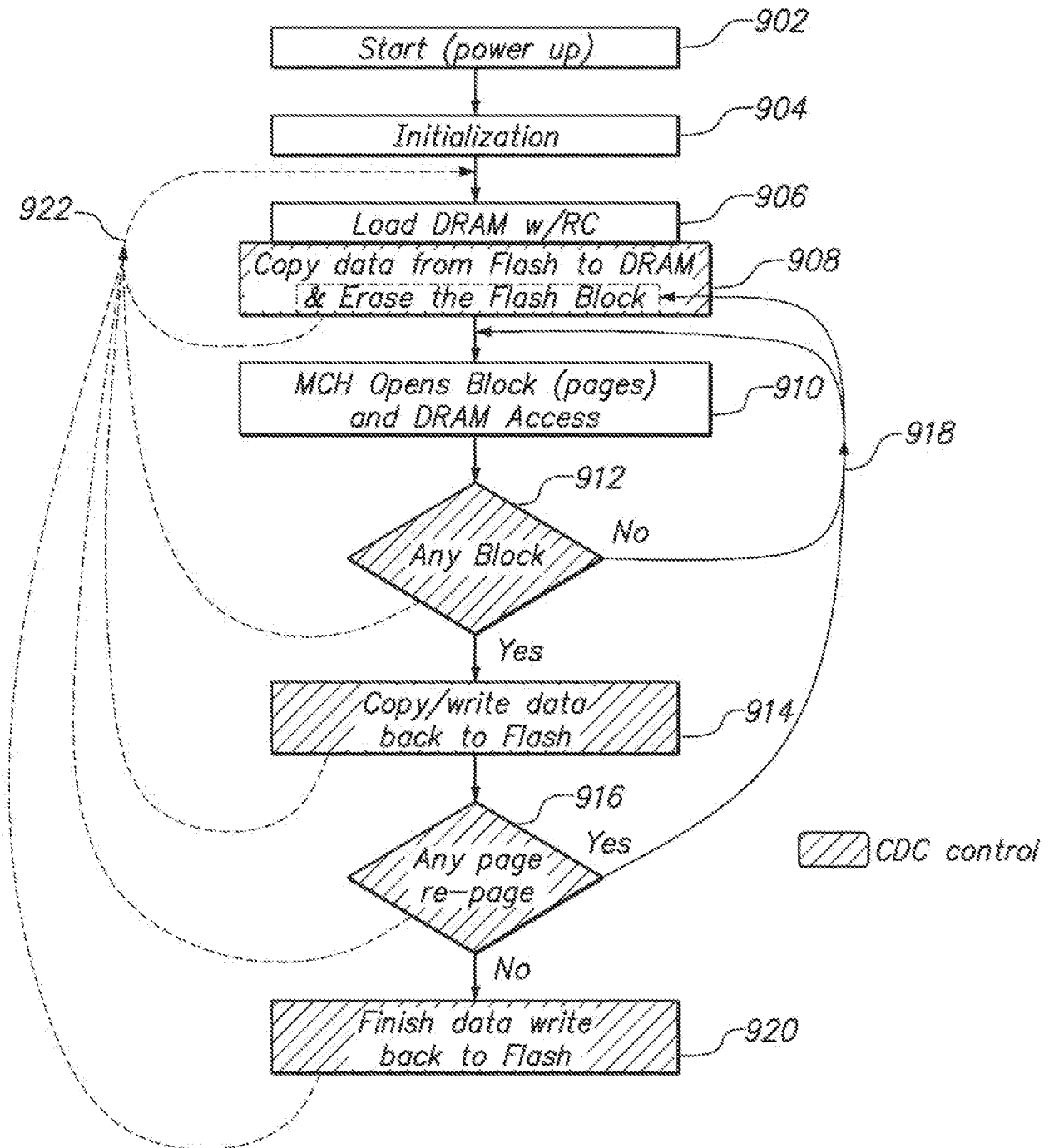
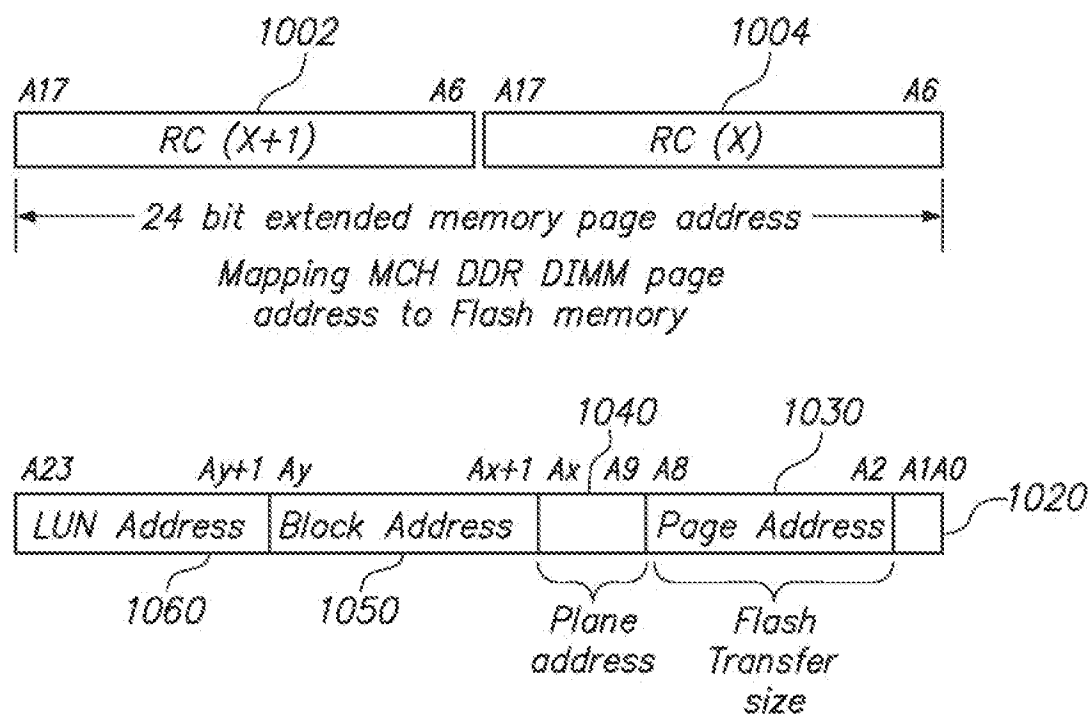


FIG. 8A

**FIG. 8B**

**FIG. 9**

**FIG. 10**

10/10

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTOR

Inventor: Jeffrey C. Solomon

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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LEGAL NAME OF INVENTOR

Inventor: Scott H. Milton

Date (Optional): 10/17/2014

Signature: Scott H. Milton

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Hyun Lee

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Jayesh Bhakta

Date (Optional): 9-17-2014

Signature: Jayesh Bhakta

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Chi-She Chen

Date (Optional): Oct-17-2014

Signature: Chi-She Chen

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/840,865

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/840,865	08/31/2015	2139	0.00	062453-036	1	1

CONFIRMATION NO. 2445

FILING RECEIPT



22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Date Mailed: 09/16/2015

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/489,269 09/17/2014
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access - A proper **Authorization to Permit Access to Application by Participating Offices** (PTO/SB/39 or its equivalent) has been received by the USPTO.

If Required, Foreign Filing License Granted: 09/14/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/840,865**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

**** SMALL ENTITY ****

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/840,865	08/31/2015	Hyun Lee	062453-036

CONFIRMATION NO. 2445

FORMALITIES LETTER



22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Date Mailed: 09/16/2015

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ 70 basic filing fee.
- \$ 70 surcharge.
- \$ 300 search fee.
- \$ 360 examination fee.
- \$(0) previous unapplied payment amount.
- \$ 800 TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/kxaysana/

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun LEE CONFIRMATION NO.: 2445
APPLICATION NO.: 14/840,865
FILING DATE: August 31, 2015
TITLE: FLASH-DRAM HYBRID MEMORY MODULE
EXAMINER: NOT ASSIGNED
ART UNIT: 2139

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

RESPONSE TO NOTICE TO FILE MISSING PARTS

Sir:

In response to the Notice to File Missing Parts of Nonprovisional Application, mailed September 16, 2015, Applicant submits the following fees:

Basic Filing Fee	\$ 280.00
Surcharge	\$ 140.00
Search Fee	\$ 600.00
Examination Fee	\$ 720.00
Four Months Extension	<u>\$2,200.00</u>
TOTAL	\$3,940.00

Please charge any additional required fees or credit any overpayment not otherwise credited, to our Deposit Account No. 19-2380.

Respectfully submitted,

NIXON PEABODY LLP

Dated: March 16, 2016

/Khaled Shami/

Khaled Shami

Reg. No. 38,745

NIXON PEABODY LLP
799 9TH STREET, N.W., SUITE 500
WASHINGTON, D.C. 20001-4501
TEL: 202-585-8000
FAX: 202-585-8080

Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Sheila M. Mattingly			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	280	280
Utility Search Fee	1111	1	600	600
Utility Examination Fee	1311	1	720	720
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	1051	1	140	140
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 4 months with \$0 paid	1254	1	2200	2200
Miscellaneous:				
Total in USD (\$)				3940

Electronic Acknowledgement Receipt

EFS ID:	25216000
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Sheila M. Mattingly
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	16-MAR-2016
Filing Date:	31-AUG-2015
Time Stamp:	16:08:52
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$3940
RAM confirmation Number	2879
Deposit Account	502686
Authorized User	COSTELLIA, JEFF

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	062453_036_Response_to_Missing_Parts.pdf	54944	no	2
			fdfead67040d793636ab69eb4e03e64f07130e06		

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	38624	no	2
			63291843d0dde33baa084ea20807c9f9f62a8889		

Warnings:

Information:

Total Files Size (in bytes):			93568
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/840,865

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	280
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	600
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	720
TOTAL CLAIMS (37 CFR 1.16(j))	1	minus 20 = *			OR	x 80 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1	minus 3 = *				x 420 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1600

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/840,865	08/31/2015	2139	1740	062453-036	1	1

CONFIRMATION NO. 2445

UPDATED FILING RECEIPT

22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001



Date Mailed: 03/28/2016

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/489,269 09/17/2014 PAT 9158684
which is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 09/14/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/840,865**

Projected Publication Date: 07/07/2016

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

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SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop

technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

22204 7590 04/21/2016
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT	PAPER NUMBER
----------	--------------

2138

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

04/21/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

nppatent@nixonpeabody.com
ipairlink@nixonpeabody.com

Office Action Summary	Application No. 14/840,865	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2138	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 8/31/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 8/31/2015 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☐ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>8/31/2015</u> | 4) ☐ Other: _____ |

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

1. This Office action responds to the Continuation application filed 8/31/2015.
2. Claim 1 is presented for examination.

Priority

3. Acknowledgment is made of applicant's claim for Domestic Benefit/National Stage priority based on applications filed 6/1/2007, 6/2/2008, 9/29/2008, 7/28/2011, 7/26/2012, and 9/17/2014. The priority requests are granted for the presently claimed invention of claim 1.

Information Disclosure Statement

4. The information disclosure statement filed 8/31/2015 contains listed items which fail to comply with 37 CFR § 1.98(a)(2), which requires a legible copy of all foreign patents, publications, or other information submitted for consideration by the Office. The listed items which fail to comply with 37 CFR § 1.98(a)(2) have been "lined-through" or "crossed-out".

Double Patenting

5. A rejection based on double patenting of the "same invention" type finds its support in the language of 35 U.S.C. § 101 which states that "whoever invents or discovers any new and useful process ... may obtain a patent therefor ..." (Emphasis added). Thus, the term "same invention," in this context, means an invention drawn to identical subject matter. See *Miller v. Eagle Mfg. Co.*, 151 U.S. 186 (1894); *In re Ockert*, 245 F.2d 467, 114 USPQ 330 (CCPA 1957); and *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970).

A statutory type (35 U.S.C. § 101) double patenting rejection can be overcome by canceling or amending the conflicting claims so they are no longer coextensive in scope. The

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filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. § 101.

Claim 1 is rejected under 35 U.S.C. 101 as claiming the same invention as that of claim 1 of prior U.S. Patent No. 9,158,684. This is a statutory double patenting rejection.

Conclusion

6. The prior art made of record and not relied upon is cited to establish the level of skill in applicant's art and those arts considered reasonably pertinent to applicant's disclosure. See MPEP 707.05(c).

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mano Padmanabhan can be reached on (571) 272-4210. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2138

April 17, 2016

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		2015-08-31
	First Named Inventor	Hyun Lee	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		062453-036

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	2043099		1936-02-02	Hanna	
	2	3562555		1971-02-09	Ahrns	
	3	3916390		1975-10-28	Chang et al.	
	4	4234920		1980-11-18	Van Ness et al.	
	5	4420821		1983-12-13	Hoffman	
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	7	4965828		1990-10-23	Ergott, Jr. et al.	
	8	5430742		1995-07-04	Jeddeloh et al.	

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Attorney Docket Number	062453-036	

	9	5519663		1996-05-21	Harper, Jr. et al.	
	10	5519831	A	1996-05-21	Holzhammer	
	11	5563839		1996-10-08	Herd et al.	
	12	5577213		1996-11-19	Avery et al.	
	13	5619644		1997-04-08	Crockett et al.	
	14	5675725		1997-10-07	Malcolm	
	15	5870350		1999-02-09	Bertin et al.	
	16	5874995		1999-02-23	Naimpally et al.	
	17	5890192		1999-03-30	Lee et al.	
	18	5953215		1999-09-14	Karabatsos	
	19	6023421		2000-02-08	Clinton et al.	

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	20	6112310		2000-08-29	Jun et al.	
	21	6145068		2000-11-07	Lewis	
	22	6158015		2000-12-05	Klein	
	23	6199142	B1	2001-03-06	Saulsbury et al.	
	24	6216247	B1	2001-04-10	Creta et al.	
	25	6269382		2001-07-31	Cabrera et al.	
	26	6336174		2002-01-01	Li et al.	
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	32	6691209		2004-02-10	O'Connell	
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	34	6769081	B1	2004-07-27	Parulkar	
	35	6799241	B2	2004-09-28	Kahn et al.	
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	37	6816982		2004-11-09	Ravid	
	38	6487102	B1	2002-11-26	Halbert et al.	
	39	6944042	B2	2005-09-13	Komatsuzaki	
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53	7411859	B2	2008-08-12	Sohn et al.	
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	64	8412879		2013-04-02	Chang et al.	
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	71	8904099	B2	2014-12-02	Chen et al.	

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	1	20020083368		2002-06-27	Abe et al.	

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25	20130086309		2013-04-04	Lee et al.	
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	1	2013016723	WO	A3	2013-01-31	NETLIST INC.		☐
	2	2737383	EP	A2	2014-06-04	NETLIST INC.		☐

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	1	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011.	☐
	2	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012.	☐
	3	Advisory Action in U.S. Patent Application No. 12/240,916, mailed March 13, 2012.	☐
	4	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012.	☐
	5	Office Action in U.S. Patent Application No. 13/536,173, mailed April 15, 2013.	☐
	6	Office Action in U.S. Patent Application No. 13/905,048, mailed August 1, 2013.	☐
	7	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed May 6, 2014.	☐
	8	International Search Report and Written Opinion in PCT/US12/48750, dated October 10, 2012.	☐
	9	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014.	☐

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Examiner Name	
Attorney Docket Number	062453-036

10	Office Action in U.S. Patent Application No. 13/625,563, mailed August 5, 2013.	☐
11	Office Action in U.S. Patent Application No. 13/625,563, mailed May 9, 2014.	☐
12	Office Action in U.S. Patent Application No. 13/905,053, mailed August 1, 2013.	☐
13	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014.	☐
14	Office Action in U.S. Patent Application No. 14/173,212, mailed March 14, 2014.	☐
15	JEDEC Standard 21-C, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.	☐
16	Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889-YGR, dated June 6, 2014.	☐
17	Smart Storage Systems, Inc.'s Invalidity Contentions, Case No. 4:13-cv-05889-YGR, dated June 6, 2014.	☐
18	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, August 1999, 114 pages.	☐
19	WONG, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.	☐
20	American National Standard Dictionary of Electrical and Electrical Terms, IEEE, Fourth Edition, Revised, ANSI/IEEE Std 100-1988, Institute of Electrical Engineers, November 3, 1988, pp. 215, 722, 964 and 1103.	☐

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21	Webster's New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1415.	☐
22	Bruce, J., "Synchronous DRAM Architectures, Organizations, and Alternate Technologies", Electrical and Computer Engineering Dept., Univ. of Maryland, December 10, 2002, 22 pages.	☐
23	David, H. et al., "Fully Buffered DIMM (FB-DIMM) Design Considerations", Intel Developer Forum, Intel Corp., February 18, 2004, 36 pages.	☐
24	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.	☐
25	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.	☐
26	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322.	☐
27	Jandhyala, S. et al., "Design-For-Test Analysis of a Buffered SDRAM DIMM", Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, August 13014, 1996, 15 pages.	☐
28	JEDEC Standard, Double Data Rate (DDR): SDRAM Specification: JESD79C (Revision JESD79B), March 2003, pp. 1-75.	☐
29	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC SOLID STATE TECH. ASSOC., March 2007, pp. 1-129.	☐
30	Mutnary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.	☐
31	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed June 19, 2014.	☐

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2015-08-31

First Named Inventor

Hyun Lee

Art Unit

Examiner Name

Attorney Docket Number

062453-036

32

Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.

☐

33

"Using Two Chip Selects to Enable Quad Rank", an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: December 17, 2005, 2 pages.

☐

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Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.

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Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.

☐

36

"Out of Stealth Mode, Start-Up MetaRAM Unveils New Technology That Quadruples DRAM Capacity", Press Release edited by Storage Newsletter on February 28, 2008 at StorageNewsLetter.com, 8 pages.

☐

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"MetaRAM Develops New Technology That Quadruples Memory Capacity of Servers and Workstations; Reduces Price by Up to 90 Percent", Press Release provided by MetaRAM and published on MarketWired.com, February 25, 2008, 3 pages.

☐

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"240pin DDR2 MetaSDRAM Registered DIMM based on 1 GB version C", Hynix Semiconductor, Product Description Rev. 0.2, September 2008, 32 pages.

☐

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Notice of Allowance in U.S. Patent Application No. 14/489,269, mailed October 8, 2015.

☐

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EXAMINER SIGNATURE

Examiner Signature

/Stephen Elmore/

Date Considered

04/17/2016

*EXAMINER: Initial reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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Art Unit			
Examiner Name			
Attorney Docket Number	062453-036		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-08-31
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

<i>Index of Claims</i> 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47									
CLAIM		DATE							
Final	Original	04/17/2016							
	1	✓							

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	11370	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L2	2789	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L3	21178	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L4	33146	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L5	6379	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L6	666	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L7	16	L5 and L6	US-PGPUB; USPAT;	OR	ON	2016/04/17 18:28

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L9	2	(US-20070136523-\$).did. or (US-8412879-\$).did.	US- PGPUB; USPAT	OR	ON	2016/04/17 18:28
L10	185600	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L11	100	L6 and L10	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L12	4	L5 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L13	6	L4 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L14	373	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L15	4	L14 with (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/04/17 18:28

			EPO; JPO; IBM_TDB;			
L16	24	(data near3 (port or input-output or I/O or IO)) and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
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L18	15	L16 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L19	0	L4 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
L20	15	L6 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB;	OR	ON	2016/04/17 18:28
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		metadata) near3 controller	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
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L25	3	L6 and L23 and L10 and L4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L26	4	L6 and L23 and L10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L27	12	L6 and L23	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L28	7045	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/04/17 18:28
L29	24	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/04/17 18:28
L30	107	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/04/17 18:28
L31	176	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/04/17 18:28
L32	97	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/04/17 18:28
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		or L14)	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
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L36	0	L35 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L37	10	(data adj manager and memory adj controller) and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L38	0	L37 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L39	2	(data adj manager) and L34 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
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L44	0	L16 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L45	0	L6 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L46	1	"8874831".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L47	8	"8301833".pn.	US-PGPUB; USPAT;	OR	ON	2016/04/17 18:28

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L54	1	(US-8102614-\$).did.	USPAT	OR	ON	2016/04/17 18:28
L55	188	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L56	3	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L55	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
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L62	3078	G06F13/4243.CPC.	US-	OR	ON	2016/04/17

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L66	4	L55 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
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L68	11	L5 and L57	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L69	2	L6 and L68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/04/17 18:28

			IBM_TDB			
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L72	104	(data adj manager) and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
L73	2	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L72	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/04/17 18:28
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EAST Search History (Interference)

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L75	107	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L76	132	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L77	97	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L78	7045	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L79	7289	L74 or L75 or L76 or L77 or L78	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L80	10	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L81	4	L79 and L80	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L82	13234	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L83	49	L79 and L82	US-PGPUB; USPAT;	OR	ON	2016/04/17 18:28

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L85	11327	711/103.ccls.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L86	1583	(data adj manager).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L87	603	G06F12/0638.CPC.	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L88	20	L79 and L85	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28
L89	3	L86 and L88	US-PGPUB; USPAT; UPAD	OR	ON	2016/04/17 18:28

4/ 17/ 2016 6:37:55 PM
C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 14840865.wsp

Search Notes 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243; G11C 7/1072; G11C 14/0018;	4/17/2016	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/17/2016	SE
365	185.33	4/17/2016	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/17/2016	SE
Inventor Name Search for DP	4/17/2016	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/840,865	08/31/2015	Hyun Lee	062453-036

CONFIRMATION NO. 2445

PUBLICATION NOTICE



OC000000084194570

22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2016-0196223-A1

Publication Date:07/07/2016

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun LEE CONFIRMATION NO.: 2445
APPLICATION NO.: 14/840,865
FILING DATE: August 31, 2015
TITLE: FLASH-DRAM HYBRID MEMORY MODULE
EXAMINER: Stephen C. ELMORE
ART UNIT: 2138

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In response to the Office Action mailed April 21, 2016, please amend the subject application as indicated.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 7.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1. (Canceled)
2. (New) A memory module comprising:
 - a data manager configured to be coupled to a memory controller of a host system using a data bus, the data manager is operable to communicate data signals with the memory controller of the host system by way of the data bus in accordance with a first protocol;
 - a first volatile memory subsystem coupled to the data manager using a first data bus, the first volatile memory subsystem is operable to communicate data signals with the data manager by way of the first data bus in accordance with the first protocol;
 - a second volatile memory subsystem coupled to the data manager using a second data bus, the second volatile memory subsystem is operable to communicate data signals with the data manager by way of the second data bus in accordance with the first protocol;
 - a non-volatile memory subsystem coupled to the data manager using a third data bus, the non-volatile memory subsystem is operable to communicate data signals with the data manager by way of the third data bus using a second protocol; and
 - a controller operable to receive one or more commands from the memory controller of the host system in accordance with the first protocol, and in response to the one or more commands received from the memory controller of the host system, the controller generates and transmits one or more control signals to the data manager, wherein, in response to the one or more control signals, the data manager transfers data between the first volatile memory subsystem and the memory controller of the host system and transfers data signals between the non-volatile memory subsystem and the second volatile memory subsystem.
3. (New) The memory module of claim 2, wherein the second protocol is different from the first protocol.

4. (New) The memory module of claim 3, wherein the first protocol is selected from the group consisting of DDR, DDR2, DDR3, and DDR4 protocols.
5. (New) The memory module of claim 2, wherein the data manager controls data traffic between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-volatile memory subsystem and the memory controller of the host system based on one or more commands received from the controller.
6. (New) The memory module of claim 2, wherein the data manager controls a starting time of a data transfer between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-volatile memory subsystem and the memory controller of the host system based on one or more commands or control signals received from the controller.
7. (New) The memory module of claim 2, wherein the data manager is configured as a bi-directional data transfer fabric.
8. (New) The memory module of claim 2, wherein the data manager is operable to concurrently transfer data (i) between the first volatile memory subsystem and the memory controller of the host system, and (ii) between the non-volatile memory subsystem and the second volatile memory subsystem.
9. (New) The memory module of claim 2, wherein the data manager further comprises a data formatting subsystem operable to format data to be transferred via the data bus, the first data bus, the second data bus and the third data bus.
10. (New) A memory module couplable to a memory controller of a host system, the memory module comprising:
 - a non-volatile memory (NVM) subsystem coupled to a NVM controller;
 - a data manager coupled to a data manager controller and to the NVM subsystem;
 - a volatile memory (VM) subsystem coupled to a VM controller and to the data manager;and
 - a controller operable to receive a command from the memory controller of the host system, the controller include the NVM controller, the VM controller, and the data manager

controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command.

11. (New) The memory module of claim 10, wherein the controller further includes an interface scheduler, the interface scheduler schedules the VM subsystem operation and the NVM subsystem operation such that no resource conflict occurs in the data manager.

12. (New) The memory module of claim 10, wherein the controller further includes an interface scheduler, the interface scheduler assigns time slots for the VM controller and the NVM controller based on current command status and a pending command received or to be received from the memory controller of the host system.

13. (New) The memory module of claim 10, wherein the VM subsystem is operable to exchange data with the NVM subsystem by way of the data manager in response to the second command received from the VM controller.

14. (New) The memory module of claim 10, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller (i) to abort the current operation of VM subsystem, and (ii) to execute an operation in response to the first command.

15. (New) The memory module of claim 10, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the

host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller to complete the current operation of VM subsystem before executing an operation in response to the first command.

16. (New) The memory module of claim 10, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller (i) to halt the current operation of VM subsystem, (ii) to execute an operation in response to the first command, and (iii) to resume the operation in response to the second command.

17. (New) A memory module couplable to a memory controller of a host system, the memory module comprising:

- a non-volatile memory (NVM) subsystem;
- a data manager coupled to a data manager controller and to the NVM subsystem;
- a volatile memory (VM) subsystem coupled to the data manager; and
- a controller operable to receive a read command from the memory controller of the host system, the controller is configured to determine, in response to the read command, a target address for a requested data, and if requested data is not stored in the VM subsystem, then the controller (i) initiates data transfer operations from the NVM subsystem to the VM subsystem by way of the data manager, and (ii) initiates data transfer operations from the VM subsystem to the memory controller of the host system by way of the data manager.

18. (New) The memory module of claim 17, wherein the controller is operable to direct (i) operation of the NVM subsystem by way of a NVM controller, (ii) operation of the VM subsystem by way of a VM controller, and (iii) operation of the data manager by way of a data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command.

19. (New) The memory module of claim 17, wherein the data manager controls a starting time of a data transfer between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command.
20. (New) The memory module of claim 17, wherein the VM subsystem includes first and second VM subsystems.
21. (New) The memory module of claim 20, wherein the data manager controls a starting time of a data transfer between any one of the first VM subsystem and the second VM subsystem and any one of the NVM subsystem and the memory controller of the host system based on the read command.
22. (New) The memory module of claim 20, wherein the data manager is operable to concurrently transfer data (i) between the first VM subsystem and the memory controller of the host system, and (ii) between the NVM subsystem and the second VM subsystem.

REMARKS

The Office Action mailed April 21, 2016 has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Canceled Claims

Claim 1 has been canceled without prejudice or disclaimer of the subject matter contained therein.

Rejection(s) Pursuant to Judicially-Created Double Patenting

Claim 1 stands rejected pursuant to the judicially-created doctrine of obviousness-type double patenting as allegedly being unpatentable over claim 1 of prior U.S. Patent No. 9,158,684.

Claim 1 has been canceled and the rejection thereof is moot.

Newly-Added Claims

Claims 2-22 have been added to further particularly point out and distinctly claim the subject matter regarded as the invention.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our Deposit Account No. 19-2380.

Respectfully submitted,

NIXON PEABODY LLP

Dated: October 20, 2016

/Khaled Shami/

Khaled Shami

Reg. No. 38,745

NIXON PEABODY LLP
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TEL: 202-585-8000
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Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Sheila M. Mattingly			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	1	80	80
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Extension - 3 months with \$0 paid	1253	1	1400	1400
Miscellaneous:				
Total in USD (\$)				1480

Electronic Acknowledgement Receipt

EFS ID:	27277502
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Sheila M. Mattingly
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	20-OCT-2016
Filing Date:	31-AUG-2015
Time Stamp:	16:45:33
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1480
RAM confirmation Number	102116INTEFSW00003532192380
Deposit Account	
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1		062453_036_Amendment.pdf	78779	yes	8
			52e162faad2f98a699c0fe96dad45c508a5b8cb7		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	8	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14840865
	Filing Date	2015-08-31
	First Named Inventor	Hyun LEE
	Art Unit	2138
	Examiner Name	Stephen C. ELMORE
	Attorney Docket Number	062453-036

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	2	20110161569	A1	2011-06-30	Shan et al.	

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Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	Hyun LEE
Art Unit	2138
Examiner Name	Stephen C. ELMORE
Attorney Docket Number	062453-036

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	Hyun LEE
Art Unit	2138
Examiner Name	Stephen C. ELMORE
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami, Reg. No. 38,745/	Date (YYYY-MM-DD)	2016-10-20
Name/Print	Khaled Shami	Registration Number	38,745

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The information provided by you in this form will be subject to the following routine uses:

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Sheila M. Mattingly			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	1806	1	180	180
Total in USD (\$)				180

Electronic Acknowledgement Receipt

EFS ID:	27278548
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Sheila M. Mattingly
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	20-OCT-2016
Filing Date:	31-AUG-2015
Time Stamp:	17:35:32
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 180
RAM confirmation Number	102116INTEFSW00004472192380
Deposit Account	
Authorized User	

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1	Information Disclosure Statement (IDS) Form (SB08)	062453_036_IDS.pdf	1120800	no	4
			887a5ff2e5b948cf553cff6e433a682a445176d4		
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30214	no	2
			a5fab6095c978497439e4c075befb8e43b94a273		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/840,865	Filing Date 08/31/2015	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA		RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A		N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*		X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*		X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).				
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))					
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	10/20/2016	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 21	Minus	** 20	= 1		X \$80 =	80
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0		X \$420 =	0
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
	TOTAL ADD'L FEE							80

(Column 1)

(Column 2)

(Column 3)

		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=		X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=		X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
	TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
PARTHENIA D. MERRILL

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

22204 7590 11/18/2016
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT

PAPER NUMBER

2133

DATE MAILED: 11/18/2016

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/840,865

08/31/2015

Hyun Lee

062453-036

2445

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	02/21/2017

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

22204 7590 11/18/2016
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	02/21/2017

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2133	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
Typed or printed name _____ Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

22204 7590 11/18/2016
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT	PAPER NUMBER
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2133

DATE MAILED: 11/18/2016

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.** Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/840,865	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the Amendment filed 10/20/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2-22, renumbered 1-21. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>10/20/2016</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2133

The present application is being examined under the pre-AIA first to invent provisions.

EXAMINER'S COMMENT, AMENDMENT, and REASONS FOR ALLOWANCE

Priority

1. Applicant claims Domestic Benefit to the following parent patent applications based upon the following continuity claim dependencies:

- a. Continuation of prior application 14/489,269 filed 9/17/2014;
- b. Continuation of prior application 13/559,476 filed 7/26/2012;
- c. Non-provisional 61/512,871 filed 7/28/2011;
- d. Continuation-in-part of 12/240,916 filed 6/2/2008;
- e. Continuation of 12/131,873 filed June 2, 2008;
- f. Non-provisional 60/941,586 filed 6/1/2007.

Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, 365(c), or 386(c) is acknowledged. However, Applicant has not complied with one or more conditions for receiving the benefit of an earlier filing date under 35 U.S.C. 120 as follows:

The later-filed application must be an application for a patent for an invention which is also disclosed in the prior application (the parent or original non-provisional application or provisional application). The disclosure of the invention in the parent application and in the later-filed application must be sufficient to comply with the requirements of 35 U.S.C. 112(a) or the first paragraph of pre-AIA 35 U.S.C. 112, except for the best mode requirement. See *Transco Products, Inc. v. Performance Contracting, Inc.*, 38 F.3d 551, 32 USPQ2d 1077 (Fed. Cir. 1994).

The disclosures of the prior-filed applications, 1.(d)-(f), i.e., Continuation-in-part of 12/240,916, Continuation of 12/131,873, and Non-provisional of 60/941,586, fail to provide adequate support or enablement in the manner provided by 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph for one or more claims of this application.

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Specifically, Claims 2-22 of the present application recite the claimed feature "data manager" which lacks proper antecedent basis or clear support under 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph in the above identified parent applications, specifically, parent items 1.(d)-(f).

The effective priority date that claims 2-22 are entitled to therefore becomes 7/28/2011 based upon the earliest proper priority claim being that to parent application Non-provisional 61/512,871.

EXAMINER'S AMENDMENT

2. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it **MUST** be submitted no later than the payment of the issue fee.

The application has been amended as follows to correct an obvious typographical error:
IN THE CLAIMS

In claim 10, line 8, replace "include" with -- includes --.

REASONS FOR ALLOWANCE

3. The following is an examiner's statement of reasons for allowance:

In independent claims 2, 10, and 17 the following features taken **in combination with the** remaining limitations of the independent claim are not found in and/or are not obvious in view of the closest prior art of record, Shan et al., U.S. 2011/0161569 A1, giving the feature "data

Art Unit: 2133

manager" the scope of meaning disclosed in the specification, page 26, paragraphs [0087-0088], which identifies the feature "data manager" additionally as element **DMgr 504**,

Claim 2,

“a first volatile memory subsystem coupled to the data manager using a first data bus, the first volatile memory subsystem is operable to communicate data signals with the data manager by way of the first data bus in accordance with the first protocol; a second volatile memory subsystem coupled to the data manager using a second data bus, the second volatile memory subsystem is operable to communicate data signals with the data manager by way of the second data bus in accordance with the first protocol; a non-volatile memory subsystem coupled to the data manager using a third data bus, the non-volatile memory subsystem is operable to communicate data signals with the data manager by way of the third data bus using a second protocol” and “a controller operable to receive one or more commands from the memory controller of the host system in accordance with the first protocol, and in response to the one or more commands received from the memory controller of the host system, the controller generates and transmits one or more control signals to the data manager, wherein, in response to the one or more control signals, the data manager transfers data between the first volatile memory subsystem and the memory controller of the host system and transfers data signals between the non-volatile memory subsystem and the second volatile memory subsystem”;

Claim 10,

“a data manager coupled to a data manager controller and to the NVM subsystem; a volatile memory (VM) subsystem coupled to a VM controller and to the data manager” and “a controller operable to receive a command from the memory controller of the host system, the

Art Unit: 2133

controller includes the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command”;

Claim 17,

“a data manager coupled to a data manager controller and to the NVM subsystem; a volatile memory (VM) subsystem coupled to the data manager” and “a controller operable to receive a read command from the memory controller of the host system, the controller is configured to determine, in response to the read command, a target address for a requested data, and if requested data is not stored in the VM subsystem, then the controller (i) initiates data transfer operations from the NVM subsystem to the VM subsystem by way of the data manager, and (ii) initiates data transfer operations from the VM subsystem to the memory controller of the host system by way of the data manager.”

Art Unit: 2133

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jared Rutz can be reached on (571) 272-5535. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2133

November 13, 2016

Notice of References Cited	Application/Control No. 14/840,865	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-7,873,750 B2	01-2011	Yabuta; Keizo	G06F13/4027	709/250
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
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Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	Hyun LEE
Art Unit	2138
Examiner Name	Stephen C. ELMORE
Attorney Docket Number	062453-036

U.S.PATENTS

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	9158684	B2	2015-10-13	Lee et al.	

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	1	20100110748	A1	2010-05-06	Best	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	Hyun LEE
Art Unit	2138
Examiner Name	Stephen C. ELMORE
Attorney Docket Number	062453-036

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
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EXAMINER SIGNATURE

Examiner Signature	/STEPHEN C ELMORE/	Date Considered	11/13/2016
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	Hyun LEE
Art Unit	2138
Examiner Name	Stephen C. ELMORE
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

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See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami, Reg. No. 38,745/	Date (YYYY-MM-DD)	2016-10-20
Name/Print	Khaled Shami	Registration Number	38,745

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BIB DATA SHEET

CONFIRMATION NO. 2445

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
14/840,865	08/31/2015	711	2133	062453-036		
RULE						
APPLICANTS Netlist, Inc., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;						
** CONTINUING DATA ***** This application is a CON of 14/489,269 09/17/2014 PAT 9158684 which is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007						
** FOREIGN APPLICATIONS *****						
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 09/14/2015						
Foreign Priority claimed	☐ Yes ☒ No	☐ Met after Allowance	STATE OR COUNTRY	SHEETS DRAWINGS	TOTAL CLAIMS	INDEPENDENT CLAIMS
35 USC 119(a-d) conditions met	☐ Yes ☒ No		CA	10	21	3
Verified and	/STEPHEN C ELMORE/	Initials				
Acknowledged	Examiner's Signature					
ADDRESS NIXON PEABODY, LLP 799 Ninth Street, NW SUITE 500 WASHINGTON, DC 20001 UNITED STATES /SE/						
TITLE FLASH-DRAM HYBRID MEMORY MODULE						
FILING FEE RECEIVED 1820	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
				☐ Credit		

Search Notes 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243; G11C 7/1072; G11C 14/0018;	4/17/2016	SE
Search updated;	11/13/2016	SE
G06F 13/28; G06F 1/185; G06F 3/0613; G06F 3/0659;	11/13/2016	SE
G06F 3/0685; G06F 13/1694; G06F 13/4027;	11/13/2016	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/17/2016	SE
365	185.33	4/17/2016	SE
Search updated		11/13/2016	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/17/2016	SE
Inventor Name Search for DP	4/17/2016	SE
Search Updated	11/13/2016	SE
Assignee searched	11/13/2016	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	11/13/2016	SE
G06F	13/28	11/13/2016	SE

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INTERFERENCE SEARCH

US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
PGPUB Searched		11/13/2016	SE
Assignee Searched		11/13/2016	SE

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	11580	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L2	2789	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L3	21373	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L4	33536	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L5	7171	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L6	705	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L7	18	L5 and L6	US-PGPUB; USPAT;	OR	ON	2016/11/13 11:21

			USOCR; FPRS; EPO; JPO; IBM_TDB			
L8	3	L7 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L9	2	(US-20070136523-\$).did. or (US-8412879-\$).did.	US- PGPUB; USPAT	OR	ON	2016/11/13 11:21
L10	194576	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L11	107	L6 and L10	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L12	5	L5 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L13	6	L4 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L14	388	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L15	5	L14 with (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/11/13 11:21

			EPO; JPO; IBM_TDB			
L16	26	(data near3 (port or input-output or I/O or IO)) and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L17	5	L5 and L16	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L18	15	L16 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L19	0	L4 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L20	15	L6 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L21	0	L14 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L22	0	L5 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L23	36794	(control adj information or control adj data or control adj meta-data or control	US- PGPUB;	OR	ON	2016/11/13 11:21

		adj metadata) near3 controller	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L24	5	((control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller) and L16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L25	3	L6 and L23 and L10 and L4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L26	5	L6 and L23 and L10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L27	13	L6 and L23	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L28	7556	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:21
L29	25	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:21
L30	111	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:21
L31	182	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:21
L32	100	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:21
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		or L14)	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L34	794	(hybrid near3 memory).ti.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L35	11	L6 and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L36	0	L35 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L37	11	(data adj manager and memory adj controller) and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L38	0	L37 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L39	2	(data adj manager) and L34 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
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L42	114	("2043099" "20030158995" "20040163027" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20090031099" "3562555" "3916390" "4234920" "4965828" "5430742" "5519831" "5563839" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6199142" "6216247" "6421279" "6459647" "6487102" "6769081" "6799241" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7155627" "7200021" "7234099" "7409491" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8233303").PN.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
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L46	1	"8874831".pn.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L47	8	"8301833".pn.	US- PGPUB;	OR	ON	2016/11/13 11:21

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			IBM_TDB			
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L54	1	(US-8102614-\$).did.	USPAT	OR	ON	2016/11/13 11:21
L55	203	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L56	4	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L55	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
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L58	8	L57 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L59	0	L37 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L60	1426	G06F12/0638.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L61	17059	G06F12/0246.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21

L62	3169	G06F13/4243.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L63	7236	G11C7/1072.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L64	500	G11C14/0018.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L65	28322	L60 or L61 or L62 or L63 or L64	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L66	5	L55 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L67	0	L66 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L68	12	L5 and L57	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L69	3	L6 and L68	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/11/13 11:21

			EPO; JPO; IBM_TDB			
L70	3	(data adj manager) and L68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L71	3	L57 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L72	113	(data adj manager) and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L73	3	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L72	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L74	10	("9158684" or "8874831" or "8301833").PN.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L75	211	("20020083368" "20020199061" "20030158995" "20040088508" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20120271990" "20130019076" "20130086309" "20130254456" "20130254497" "20140059170" "20140156919" "20140156920" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213"	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21

		"5619644" "5675725" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6721860" "6769081" "6799241" "6799244" "6816982" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7353325" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8407395" "8412879" "8516187" "8671243" "8677060" "8874831" "8880791" "8904098" "8904099").PN.				
L76	3	L72 and L75	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:21
L93	9333	G06F13/28.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:40
L94	2896	G06F1/185.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:40
L95	6474	G06F3/0613.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:40
L96	8802	G06F3/0659.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:40
L97	4068	G06F3/0685.CPC.	US- PGPUB;	OR	ON	2016/11/13 11:41

			USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L98	2174	G06F13/1694.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:41
L99	3977	G06F13/4027.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:41
L101	34040	93 or 94 or 95 or 96 or 97 or 98 or 99	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:42
L102	227	(data adj manager) and 101	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:42
L103	27	6 and 102	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:42
L104	7	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and 103	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:44
L105	2	104 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:45

L106	0	((read\$3 near3 command) with (memory adj controller)) and 105	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:48
L107	5	((read\$3 near3 command) with (memory adj controller)) and 104	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:48
L108	0	107 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:48
L109	0	((first adj command) with (second adj command) with (memory adj controller)) and 105	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:49
L110	4	("20100110748" "20110161569" "9158684").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:50
L111	3	(US-20110161569-\$ or US-20100110748-\$).did. or (US-9158684-\$).did.	US-PGPUB; USPAT	OR	ON	2016/11/13 11:51
L112	2	((read\$3 near3 command) with (memory adj controller)) and 111	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:52
L113	1	112 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:52
L114	0	113 and (data adj manager)	US-PGPUB;	OR	ON	2016/11/13 11:54

			USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L115	1	111 and (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:54
L116	0	115 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 11:55
L125	0	("Netlist, Inc.").AANM.	US- PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:59
L126	39	("Netlist").AANM.	US- PGPUB; USPAT; USOCR	OR	ON	2016/11/13 11:59
L134	614	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and 93	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 12:02
L135	5	(data adj manager) and 134	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 12:02
L136	0	135 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 12:02
L137	31	((read\$3 near3 command) with (memory adj controller)) and 134	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/11/13 12:03

			IBM_TDB			
L138	5	(data adj manager) and 137	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 12:03
L139	0	138 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2016/11/13 12:03

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L77	25	((Chi-She) near2 (Chen)).INV.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L78	111	((Jeffrey) near2 (Solomon)).INV.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L79	138	((Scott) near2 (Milton)).INV.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L80	100	((Jayesh) near2 (Bhakta)).INV.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L81	7556	((Hyun) near2 (Lee)).INV.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L82	7809	L77 or L78 or L79 or L80 or L81	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L83	12	(data adj manager with controller with memory adj controller).clm.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L84	5	L82 and L83	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L85	13683	(data adj path or memory adj segment).clm.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L86	53	L82 and L85	US-	OR	ON	2016/11/13

			PGPUB; USPAT; UPAD			11:21
L87	4	L83 and L86	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L88	11537	711/103.ccls.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L89	1642	(data adj manager).clm.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L90	698	G06F12/0638.CPC.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L91	21	L82 and L88	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L92	3	L89 and L91	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:21
L100	4183	G06F13/28.CPC.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:41
L117	11	82 and 100	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:56
L118	5	83 and 117	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:56
L119	1159	((read\$3 near3 command) with (memory adj controller)).clm.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:56
L120	0	118 and 119	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:57
L121	33867	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller).clm.	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:57
L122	4	117 and 121	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:58
L123	4	83 and 122	US- PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:58

L124	0	123 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:58
L127	39	("Netlist").AANM.	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:59
L128	39	82 and 127	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:59
L129	3	89 and 128	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 11:59
L130	0	119 and 129	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 12:00
L131	2	121 and 129	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 12:00
L132	4	83 and 121	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 12:00
L133	2	83 and 131	US-PGPUB; USPAT; UPAD	OR	ON	2016/11/13 12:00

11/ 13/ 2016 12:03:50 PM
C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 14840865.wsp

Issue Classification 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	

CPC						
Symbol					Type	Version
G06F		13	/	28	F	2013-01-01
G06F		12	/	0246	I	2013-01-01
G06F		1	/	185	I	2013-01-01
G06F		2212	/	205	A	2013-01-01
G06F		13	/	1694	I	2013-01-01
G06F		12	/	0638	I	2013-01-01
G06F		13	/	4243	I	2013-01-01
G06F		2212	/	7208	A	2013-01-01
G11C		7	/	1072	I	2013-01-01
G11C		14	/	0018	I	2013-01-01
G06F		3	/	0613	I	2013-01-01
G06F		3	/	0659	I	2013-01-01
G06F		3	/	0685	I	2013-01-01
G06F		13	/	4027	I	2013-01-01
			/			

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE		Total Claims Allowed:	
		21	
(Assistant Examiner)	(Date)	O.G. Print Claim(s)	O.G. Print Figure
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2133	11/13/2016	1	6
(Primary Examiner)	(Date)		

Issue Classification 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	2		18												
	3		19												
	4		20												
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	12														
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	16														
	17														

NONE		Total Claims Allowed: 21	
(Assistant Examiner)	(Date)		
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2133	11/13/2016	O.G. Print Claim(s) 1	O.G. Print Figure 6
(Primary Examiner)	(Date)		

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL **(Submitted Only via EFS-Web)**

Application Number	14/840,865	Filing Date	2015-08-31	Docket Number (if applicable)	062453-000036	Art Unit	2133
First Named Inventor	Hyun LEE			Examiner Name	ELMORE, STEPHEN C		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☐ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
 (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 192380

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	Khaled Shami/	Date (YYYY-MM-DD)	2017-02-21
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14840865
	Filing Date	2015-08-31
	First Named Inventor	LEE, HYUN
	Art Unit	2133
	Examiner Name	ELMORE, STEPHEN C.
	Attorney Docket Number	062453-036

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4882709	A	1989-11-21	WYLAND	
	2	5490155	A	1996-02-06	ABDOO et al.	
	3	5799200	A	1998-08-25	BRANT et al.	
	4	6026465	A	2000-02-15	MILLS et al.	
	5	6065092	A	2000-05-16	ROY	
	6	6571244	B1	2003-05-27	LARSON	
	7	6614685	B2	2003-09-02	WONG	
	8	6693840	B2	2004-02-17	SHIMADA et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

9	6810513	B1	2006-10-26	VEST	
10	7136978	B2	2006-11-14	MIURA et al.	
11	7519754	B2	2009-04-14	WANG et al.	Related to CN 101017460 A (SILICON STORAGE TECH INC) 2007-08-15
12	9043677	B2	2015-05-26	KONG et al.	
13	9361250	B2	2016-06-07	SHAN et al.	Related to CN 102110057 A (LANQI SEMICONDUCTOR SHANGHAI CO.,LTD) 2011-06-29

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20020053944	A1	2002-05-09	BRASS et al.	
	2	20030028733	A1	2003-02-06	TSUNODA et al.	
	3	20050249011	A1	2005-11-10	MAEDA	
	4	20050273548	A1	2005-12-08	ROOHPARVAR	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		14840865
Filing Date		2015-08-31
First Named Inventor	LEE, HYUN	
Art Unit		2133
Examiner Name	ELMORE, STEPHEN C.	
Attorney Docket Number		062453-036

5	20060212651	A1	2006-09-21	ASHMORE	
6	20070070669	A1	2007-03-29	TSEHN	
7	20070147115	A1	2007-06-28	LIN et al.	
8	20070255898	A1	2007-11-01	NISHIDE et al.	
9	20070288683	A1	2007-12-13	PANABAKER et al.	
10	20080147968	A1	2008-06-19	LEE et al.	
11	20080235443	A1	2008-09-25	CHOW et al.	
12	20080291727	A1	2008-11-27	SEO et al.	
13	20100322020	A1	2010-12-23	KIM	
14	20110078496	A1	2011-03-31	JEDDELOH	
15	20120317433	A1	2012-12-13	ELLIS et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

16	20140032820	A1	2014-01-30	HARASAWA et al.
17	20150058701	A1	2015-02-06	XING et al.

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FOREIGN PATENT DOCUMENTS

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	CATSOULIS, "Designing Embedded Hardware: Create New Computers and Devices", O'Reilly Media, Inc. (2005). (67 pages)	
	2	ELMHURST et al., "A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write", IEEE Journal of Solid-State Circuits 38(11):1929-1933 (2003).	
	3	Extended European Search Report for European Application No. 12 817 751.6, mailed June 9, 2015.	
	4	Final Office Action, dated June 15, 2016, issued in U.S. Patent Application No. 14/489,281, 10 pages.	

**INFORMATION DISCLOSURE
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Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

5	Notice of Allowance in U.S. Patent Application No. 13/536,173, mailed July 2, 2013.
6	Notice of Allowance in U.S. Patent Application No. 13/905,053, mailed December 11, 2013.
7	Notice of Allowance in U.S. Patent Application No. 13/905,048, mailed December 19, 2013, 8 pages.
8	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed September 29, 2014.
9	Notice of Allowance in U.S. Patent Application No. 14/173,219 dated July 7, 2014.
10	Office Action in U.S. Application No. 13/536,176, mailed on April 15, 2013.
11	Office Action in U.S. Patent Application No. 14/302,292, mailed December 21, 2015.
12	Petition for Inter Partes Review of U.S. Patent No. 8,301,833, filed September 29, 2008.
13	PATTERSON et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998).
14	Petition for Inter Partes Review of U.S. Patent No. 8,874,831, filed July 26, 2012.
15	Petition for Inter Partes Review of U.S. Patent No. 8,671,243, filed May 29, 2013.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

16	Restriction Requirement in U.S. Patent Application No. 12/240,916, mailed March 31, 2011.
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Examiner Signature		Date Considered	
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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OR

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-02-21
Name/Print	Khaled Shami	Registration Number	38,745

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Jacqueline Allen			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 1st Request	1801	1	1200	1200
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	28416999
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Jacqueline Allen
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	21-FEB-2017
Filing Date:	31-AUG-2015
Time Stamp:	16:17:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	022217INTEFSW00003805192380
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Authorized User	

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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	062453-036_RCE.pdf	1364852	no	3
			145008cdbac8ce8873a855ab435110e1e21d1d68		
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	062453-036_IDS.pdf	1061597	no	8
			8b611629a7b7e9b7cbb0d7b41cd073f5dc8bcb85		
Warnings:					
Information:					
3	Non Patent Literature	NPL_Catsoulis_2005.pdf	26009454	no	67
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Information:					
4	Non Patent Literature	NPL_Elmhurst_2003_1929.pdf	644893	no	5
			b18b5188ef5ac1d4d7e9347bfafbbb01136eb03		
Warnings:					
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5	Non Patent Literature	NPL_Petition_8301833_1_and_2new.pdf	941788	no	73
			280866d0c7ff97a2485a1c83dd9419c94964f4c0		
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6	Non Patent Literature	NPL_Petition_8671243_1_and_2new.pdf	2385310	no	85
			8a0a67b6ca0b5c3e3d6448d599ca53c09e84d97d		
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7	Non Patent Literature	NPL_Petitioner_8874831_1_and_2new.pdf	1894705	no	81
			f4a0a0e60a6a074187be2614267a2d9ff8d981f		

Warnings:

Information:

8	Non Patent Literature	NPL_Patterson_1998.pdf	14503482	no	71
			af5a0494e64b877707d19ca7984f6b2773bdf1c0		

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9	Fee Worksheet (SB06)	fee-info.pdf	30078	no	2
			d81c65ff993ee019a0fa01621784b762e09020e0		

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14840865
	Filing Date	2015-08-31
	First Named Inventor	LEE, HYUN
	Art Unit	2133
	Examiner Name	ELMORE, STEPHEN C.
	Attorney Docket Number	062453-036

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Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

1	File History 12/240,916 filed September 29, 2008
2	Bonella, Provisional Application for "ADVANCED DYNAMIC DISK MEMORY MODULE", 53 pages.
3	File History US Patent No. 8,671,243
4	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18).
5	Provisional Application No. 60/941,586 filed June 1, 2007
6	JEDEC STANDARD, Double Data Rate (DDR) SDRAM Specification, JESD79, June 2000, 77 pages
7	JEDEC STANDARD, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A) January 2005, 113 pages.
8	Microsoft Computer Dictionary Fifth Edition, 10 pages.
9	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
10	Data Sheet, 74F257A Quad 2-line to 1-line selector/multiplexer, non-inverting (3-State), Product specification, IC15 Data Handbook 1995 Mar 31, 10 pages.
11	Ex. 1004 - IPR2017-00587 Ron Maltiel CV, 7 pages.

**INFORMATION DISCLOSURE
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Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

12	Provisional Application No. 60/912,321 filed April 17, 2007
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Attorney Docket Number	062453-036

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-04-20
Name/Print	Khaled Shami	Registration Number	38,745

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	28976069
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami
Filer Authorized By:	
Attorney Docket Number:	062453-036
Receipt Date:	20-APR-2017
Filing Date:	31-AUG-2015
Time Stamp:	14:21:37
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	062453-036_SB08.pdf	1099799	no	5
			7989dfc192c8b53133609ce96644a1372a573d88		

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2	Non Patent Literature	12240916_File_History.pdf	6653677	no	320
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Warnings:					
Information:					
6	Non Patent Literature	IPR2017_00692_Provisional_60941586.pdf	462055	no	23
			c4bd0ef93ee01c683c38ddb6b3f0a767ec455055		
Warnings:					
Information:					
7	Non Patent Literature	JEDECDDR1SPEC.pdf	4541877	no	77
			673edc787869e7628f50c3bdc000c6bd22630625		
Warnings:					
Information:					

8	Non Patent Literature	JESD79_2B.pdf	5888025	no	113
			5c42cbc59b0ae7c1584dfda9da74afde2d827417		
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9	Non Patent Literature	MS_Computer_Dictionary_2002.pdf	208166	no	10
			35a6b9480e36f5d4b6cb5c027e6b93ec9c7207e8		
Warnings:					
Information:					
10	Non Patent Literature	MS_Windows_2000_Profl_Res_Kit.pdf	2536464	no	76
			285f1ae6d52ec5979b431fa3c5d6f4a4dc46a09d		
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11	Non Patent Literature	MUX_from_Philips.pdf	1015908	no	10
			3a240f97340b98b521f89e2cd8e2c6dc461171d8		
Warnings:					
Information:					
12	Non Patent Literature	NPL_Ron_Maltiel.pdf	109242	no	7
			e1f4559cfc279059acd31c229e48aa8b6469f772		
Warnings:					
Information:					
13	Non Patent Literature	Provisional_60912321.pdf	807076	no	42
			4c02829492a416a437bafc44e025007f422328f0		
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Information:					
Total Files Size (in bytes):			31813008		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

22204 7590 05/05/2017
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT	PAPER NUMBER
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2133

NOTIFICATION DATE	DELIVERY MODE
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05/05/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

nppatent@nixonpeabody.com
ipairlink@nixonpeabody.com

Office Action Summary	Application No. 14/840,865	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 2/21/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 2-22 is/are pending in the application.
5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 2-13 and 17-22 is/are rejected.
- 8) ☒ Claim(s) 10 and 14-16 is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 8/31/2015 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>2/21/2017 and 4/20/2017</u> . | 4) ☐ Other: ____. |

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

1. This Office action responds to the RCE application filed 2/21/2017.
2. Claim 1 was canceled and claims 2-22 were added by the amendment filed 10/20/2016.
3. Claims 2-22 remain for examination.

Priority

4. Applicant claims Domestic Benefit to the following parent patent applications based upon the following continuity claim dependencies:

- a. Continuation of prior application 14/489,269 filed 9/17/2014;
- b. Continuation of prior application 13/559,476 filed 7/26/2012;
- c. Non-provisional 61/512,871 filed 7/28/2011;
- d. Continuation-in-part of 12/240,916 filed 6/2/2008;
- e. Continuation of 12/131,873 filed June 2, 2008;
- f. Non-provisional 60/941,586 filed 6/1/2007.

Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, 365(c), or 386(c) is acknowledged. However, Applicant has not complied with one or more conditions for receiving the benefit of an earlier filing date under 35 U.S.C. 120 as follows:

The later-filed application must be an application for a patent for an invention which is also disclosed in the prior application (the parent or original non-provisional application or provisional application). The disclosure of the invention in the parent application and in the later-filed application must be sufficient to comply with the requirements of 35 U.S.C. 112(a) or the first paragraph of pre-AIA 35 U.S.C. 112, except for the best mode requirement. See *Transco Products, Inc. v. Performance Contracting, Inc.*, 38 F.3d 551, 32 USPQ2d 1077 (Fed. Cir. 1994).

The disclosures of prior-filed applications, 1(d)-1(f), i.e., Continuation-in-part of 12/240,916, Continuation of 12/131,873, and non-provisional 60/941,586, fail to provide adequate support or enablement in the manner provided by 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph for one or more claims of this application. Specifically, present claims 2-22 recite the feature and corresponding recited functionality for the element "data manager" which lacks proper antecedent basis or clear support, and further lacks adequate written description, under 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph in the above noted parent applications.

Therefore, claims 2-22 are only entitled to the effective priority date of 7/28/2011 based upon the earliest proper priority claim being that to parent application 61/512,871 filed 7/28/2011.

Information Disclosure Statement

5. The Information Disclosure Statement (IDS) filed 4/2017 contains two "lined-through" entries in the category Non-Patent Literature Documents, which have not been considered. The IDS entries fail to comply with 37 CFR § 1.98(a)(1) because they are and have been identified as "File History..." in "Non-Patent Literature Documents" of the IDS and therefore do not comply with 37 CFR § 1.98(a)(1). Each publication requested to be considered under 37 CFR § 1.98(a)(1) needs to be an individual publication listed individually on form PTO-1449. Aggregated listings (i.e., an electronic "file history") do not comply because the aggregate "history" is a compilation of multiple documents with each document having distinct and/or different publication dates, and so is not an individual document publication according to 37 CFR § 1.98(a)(1).

Specification

6. The disclosure is objected to because in paragraph [0001] the status of application 14/489,269 should be updated to reflect that it is now a U.S. patent.

Claim Objections

7. Claim 10 is objected to because of the informality: at line 8, typographic error “include” should be “includes”.

Claim Rejections - 35 USC § 102

8. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of pre-AIA 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(e) the invention was described in (1) an application for patent, published under section 122(b), by another filed in the United States before the invention by the applicant for patent or (2) a patent granted on an application for patent by another filed in the United States before the invention by the applicant for patent, except that an international application filed under the treaty defined in section 351(a) shall have the effects for purposes of this subsection of an application filed in the United States only if the international application designated the United States and was published under Article 21(2) of such treaty in the English language.

Claim(s) 2-13 and 17-22 are rejected under pre-AIA 35 U.S.C. 102(e) as being anticipated by Best, U.S. 2010/0110748 A1, (which claims priority to provisional application 60/912,321 filed April 17, 2007).

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Best discloses a composite, hybrid memory device including a first volatile storage die and a second non-volatile storage die disposed within an integrated circuit package, see Abstract, and discloses a shared interface circuit to receive memory access commands directed to the first storage die and to convey read and write data between an external data path and the first and second storage die, see Figure 1A as an embodiment of the hybrid memory device, described in paragraphs [0013-0015].

Best then teaches the claimed memory module (claims 2, 10, and 17) as claimed comprising:

Best teaches, as per claims 2-9, *a memory module comprising: a data manager* (the data manager is interpreted under Broadest Reasonable Interpretation (BRI) as a data transfer interconnect operable in two directions, see Figure 2, the combination of elements 131, 133, and 144 functioning through the connected data busses 142, 144, 140, and 128 to read, write, and thereby transfer data bi-directionally in response to multiple memory commands from the host over the external DQ I/F interface as described through paragraphs [0017-0020], where first protocol and second protocols (being different from each other, for example, as per claim 4, first protocol is of a DDR protocol for DRAM 103, and second protocol is a different) are disclosed in paragraphs [0015, 0016, and 0030] (as teaching claim 3) including the disclosure of “performing any necessary protocol conversion”) *configured to be coupled to a memory controller of a host system* (host instructions for operating the data transfer functions of the hybrid memory are disclosed in paragraph [0033], teaching claim 5) *using a data bus, the data manager is operable to communicate data signals with the memory controller of the host system* (the host memory controller is inherent to the providing of host memory instructions, and is also

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taught by element command decoder 122 (read and write commands), and element data control 151, Figures 2 and 3) *by way of the data bus in accordance with a first protocol; a first volatile memory subsystem coupled to the data manager using a first data bus, the first volatile memory subsystem is operable to communicate data signals with the data manager by way of the first data bus in accordance with the first protocol; a second volatile memory subsystem coupled to the data manager using a second data bus, the second volatile memory subsystem is operable to communicate data signals with the data manager by way of the second data bus in accordance with the first protocol; a non-volatile memory subsystem coupled to the data manager using a third data bus, the non-volatile memory subsystem is operable to communicate data signals with the data manager by way of the third data bus using a second protocol; and a controller operable to receive one or more commands from the memory controller of the host system in accordance with the first protocol, and in response to the one or more commands received from the memory controller of the host system, the controller generates and transmits one or more control signals to the data manager, wherein, in response to the one or more control signals, the data manager transfers data between the first volatile memory subsystem and the memory controller of the host system and transfers data signals between the non-volatile memory subsystem and the second volatile memory subsystem;*

Best teaches, as per claim 3, *wherein the second protocol is different from the first protocol*, as noted above, and teaches as per claim 4, *wherein the first protocol is selected from the group consisting of DDR, DDR2, DDR3, and DDR4 protocols*, as noted above, and teaches as per claim 5, *wherein the data manager controls data traffic between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-*

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volatile memory subsystem and the memory controller of the host system based on one or more commands received from the controller, as noted above;

Best discloses as per claim 6, *wherein the data manager controls a starting time of a data transfer between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-volatile memory subsystem and the memory controller of the host system based on one or more commands or control signals received from the controller, as noted above and paragraph [0028];*

Best discloses as per claim 7, *wherein the data manager is configured as a bi- directional data transfer fabric, as noted above, and as disclosed by the data control/steering circuit 131, Figure 2;*

Best discloses as per claim 8, *wherein the data manager is operable to concurrently transfer data (i) between the first volatile memory subsystem and the memory controller of the host system, and (ii) between the non-volatile memory subsystem and the second volatile memory subsystem, See paragraph [0018];*

Best discloses as per claim 9, *wherein the data manager further comprises a data formatting subsystem operable to format data to be transferred via the data bus, the first data bus, the second data bus and the third data bus, as the subsystem consisting of the underlying logic for a serializing/deserializing function contained within the data steering/control circuit 131 and the external data interface 133;*

Best teaches, as noted above, as per claims 10-13, *a memory module couplable to a memory controller of a host system, the memory module comprising: a non-volatile memory (NVM) subsystem coupled to a NVM controller (element 137); a data manager coupled to a data*

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manager controller and to the NVM subsystem; a volatile memory (VM) subsystem coupled to a VM controller (element 129) and to the data manager; and a controller operable to receive a command from the memory controller of the host system, the controller include(s) the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command;

Best discloses as per claims 11 and 12, wherein the controller further includes an interface scheduler, the interface scheduler schedules the VM subsystem operation and the NVM subsystem operation such that no resource conflict occurs in the data manager, and wherein the controller further includes an interface scheduler, the interface scheduler assigns time slots for the VM controller and the NVM controller based on current command status and a pending command received or to be received from the memory controller of the host system, see paragraph [0018], and Best teaches as per claim 13, wherein the VM subsystem is operable to exchange data with the NVM subsystem by way of the data manager in response to the second command received from the VM controller, as noted above;

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Best teaches as per claims 17-22, *a memory module couplable to a memory controller of a host system, the memory module comprising: a non-volatile memory (NVM) subsystem; a data manager coupled to a data manager controller and to the NVM subsystem; a volatile memory (VM) subsystem coupled to the data manager; and a controller operable to receive a read command from the memory controller of the host system, the controller is configured to determine, in response to the read command, a target address for a requested data, and if requested data is not stored in the VM subsystem, then the controller (i) initiates data transfer operations from the NVM subsystem to the VM subsystem by way of the data manager, and (ii) initiates data transfer operations from the VM subsystem to the memory controller of the host system by way of the data manager, as noted above; and discloses as per claims 18-22, wherein the controller is operable to direct (i) operation of the NVM subsystem by way of a NVM controller, (ii) operation of the VM subsystem by way of a VM controller, and (iii) operation of the data manager by way of a data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command, and wherein the data manager controls a starting time of a data transfer between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command, and wherein the VM subsystem includes first and second VM subsystems (see paragraph [0025], SRAM); and discloses wherein the data manager controls a starting time of a data transfer between any one of the first VM subsystem and the second VM subsystem and any one of the NVM subsystem and the memory controller of the host system based on the read command, paragraph [0028], and wherein the data manager is operable to concurrently transfer data (i) between the first VM subsystem and the memory*

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controller of the host system, and (ii) between the NVM subsystem and the second VM subsystem, paragraph [0018], as noted above.

Allowable Subject Matter

9. Claims 14-16 are objected to as being dependent upon a rejected base claim, but would be allowable over the prior art of record if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

10. The prior art made of record and not relied upon is cited to establish the level of skill in applicant's art and those arts considered reasonably pertinent to applicant's disclosure. See MPEP 707.05(c).

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jared Rutz can be reached on (571) 272-5535. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2133

May 1, 2017

Notice of References Cited	Application/Control No. 14/840,865	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2010/0110748 A1	05-2010	Best; Scott C.	G06F12/0638	365/51
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIM		DATE							
Final	Original	04/17/2016	05/01/2017						
	1	✓							
	2		✓						
	3		✓						
	4		✓						
	5		✓						
	6		✓						
	7		✓						
	8		✓						
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	12		✓						
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	17		✓						
	18		✓						
	19		✓						
	20		✓						
	21		✓						
	22		✓						

Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14840865
	Filing Date		2015-08-31
	First Named Inventor		LEE, HYUN
	Art Unit		2133
	Examiner Name		ELMORE, STEPHEN C.
	Attorney Docket Number		062453-036

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4882709	A	1989-11-21	WYLAND	
	2	5490155	A	1996-02-06	ABDOO et al.	
	3	5799200	A	1998-08-25	BRANT et al.	
	4	6026465	A	2000-02-15	MILLS et al.	
	5	6065092	A	2000-05-16	ROY	
	6	6571244	B1	2003-05-27	LARSON	
	7	6614685	B2	2003-09-02	WONG	
	8	6693840	B2	2004-02-17	SHIMADA et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
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First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

9	6810513	B1	2006-10-26	VEST	
10	7136978	B2	2006-11-14	MIURA et al.	
11	7519754	B2	2009-04-14	WANG et al.	Related to CN 101017460 A (SILICON STORAGE TECH INC) 2007-08-15
12	9043677	B2	2015-05-26	KONG et al.	
13	9361250	B2	2016-06-07	SHAN et al.	Related to CN 102110057 A (LANQI SEMICONDUCTOR SHANGHAI CO.,LTD) 2011-06-29

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	1	20020053944	A1	2002-05-09	BRASS et al.	
	2	20030028733	A1	2003-02-06	TSUNODA et al.	
	3	20050249011	A1	2005-11-10	MAEDA	
	4	20050273548	A1	2005-12-08	ROOHPARVAR	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		14840865
Filing Date		2015-08-31
First Named Inventor	LEE, HYUN	
Art Unit		2133
Examiner Name	ELMORE, STEPHEN C.	
Attorney Docket Number		062453-036

5	20060212651	A1	2006-09-21	ASHMORE	
6	20070070669	A1	2007-03-29	TSEERN	
7	20070147115	A1	2007-06-28	LIN et al.	
8	20070255898	A1	2007-11-01	NISHIDE et al.	
9	20070288683	A1	2007-12-13	PANABAKER et al.	
10	20080147968	A1	2008-06-19	LEE et al.	
11	20080235443	A1	2008-09-25	CHOW et al.	
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13	20100322020	A1	2010-12-23	KIM	
14	20110078496	A1	2011-03-31	JEDDELOH	
15	20120317433	A1	2012-12-13	ELLIS et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

16	20140032820	A1	2014-01-30	HARASAWA et al.
17	20150058701	A1	2015-02-06	XING et al.

If you wish to add additional U.S. Published Application citation information please click the Add button.

FOREIGN PATENT DOCUMENTS

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	CATSOU LIS, "Designing Embedded Hardware: Create New Computers and Devices", O'Reilly Media, Inc. (2005). (67 pages)	
	2	ELMHURST et al., "A 1.8-V 128-Mb 125-MHz Multilevel Cell Flash Memory With Flexible Read While Write", IEEE Journal of Solid-State Circuits 38(11):1929-1933 (2003).	
	3	Extended European Search Report for European Application No. 12 817 751.6, mailed June 9, 2015.	
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Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

5	Notice of Allowance in U.S. Patent Application No. 13/536,173, mailed July 2, 2013.
6	Notice of Allowance in U.S. Patent Application No. 13/905,053, mailed December 11, 2013.
7	Notice of Allowance in U.S. Patent Application No. 13/905,048, mailed December 19, 2013, 8 pages.
8	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed September 29, 2014.
9	Notice of Allowance in U.S. Patent Application No. 14/173,219 dated July 7, 2014.
10	Office Action in U.S. Application No. 13/536,176, mailed on April 15, 2013.
11	Office Action in U.S. Patent Application No. 14/302,292, mailed December 21, 2015.
12	Petition for Inter Partes Review of U.S. Patent No. 8,301,833, filed September 29, 2008.
13	PATTERSON et al., "Computer Organization & Design: The Hardware/Software Interface" Morgan Kaufmann Publishers, Inc. (1998).
14	Petition for Inter Partes Review of U.S. Patent No. 8,874,831, filed July 26, 2012.
15	Petition for Inter Partes Review of U.S. Patent No. 8,671,243, filed May 29, 2013.

**INFORMATION DISCLOSURE
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Application Number	14840865
Filing Date	2015-08-31
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Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

16	Restriction Requirement in U.S. Patent Application No. 12/240,916, mailed March 31, 2011.
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Examiner Signature	/STEPHEN C ELMORE/	Date Considered	05/01/2017
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-02-21
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	11599	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L2	2789	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L3	21406	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L4	33587	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L5	7870	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L6	743	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L7	20	L5 and L6	US-PGPUB; USPAT;	OR	ON	2017/05/01 09:21

			USOCR; FPRS; EPO; JPO; IBM_TDB			
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L9	2	(US-20070136523-\$).did. or (US-8412879-\$).did.	US- PGPUB; USPAT	OR	ON	2017/05/01 09:21
L10	201568	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L11	112	L6 and L10	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L12	5	L5 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L13	6	L4 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L14	396	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L15	5	L14 with (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2017/05/01 09:21

			EPO; JPO; IBM_TDB			
L16	29	(data near3 (port or input-output or I/O or IO)) and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L17	5	L5 and L16	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L18	15	L16 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
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L20	15	L6 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L21	0	L14 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L22	0	L5 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L23	38362	(control adj information or control adj data or control adj meta-data or control	US- PGPUB;	OR	ON	2017/05/01 09:21

		adj metadata) near3 controller	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L24	5	((control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller) and L16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L25	3	L6 and L23 and L10 and L4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
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L28	7927	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L29	26	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L30	113	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L31	183	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L32	103	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
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		or L14)	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
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L35	11	L6 and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L36	0	L35 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L37	11	(data adj manager and memory adj controller) and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L38	0	L37 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L39	2	(data adj manager) and L34 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
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L42	115	("2043099" "20030158995" "20040163027" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20090031099" "3562555" "3916390" "4234920" "4965828" "5430742" "5519831" "5563839" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6199142" "6216247" "6421279" "6459647" "6487102" "6769081" "6799241" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7155627" "7200021" "7234099" "7409491" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8233303").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L43	0	L14 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L44	0	L16 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L45	0	L6 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L46	1	"8874831".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L47	8	"8301833".pn.	US-PGPUB;	OR	ON	2017/05/01 09:21

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L51	82	("20020083368" "20030158995" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20140059170" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213" "5619644" "5675725" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6769081" "6799241" "6799244" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8412879").PN.	US- PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L52	8	"8301833".pn.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2017/05/01 09:21

			IBM_TDB			
L53	1	(US-8301833-\$).did.	USPAT	OR	ON	2017/05/01 09:21
L54	1	(US-8102614-\$).did.	USPAT	OR	ON	2017/05/01 09:21
L55	212	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
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L61	18686	G06F12/0246.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21

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L65	31181	L60 or L61 or L62 or L63 or L64	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L66	5	L55 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L67	0	L66 and ((@pd or @ad)< "20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L68	12	L5 and L57	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L69	3	L6 and L68	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2017/05/01 09:21

			EPO; JPO; IBM_TDB			
L70	3	(data adj manager) and L68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L71	3	L57 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L72	122	(data adj manager) and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L73	3	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L72	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L74	10	("9158684" or "8874831" or "8301833").PN.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L75	212	("20020083368" "20020199061" "20030158995" "20040088508" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20120271990" "20130019076" "20130086309" "20130254456" "20130254497" "20140059170" "20140156919" "20140156920" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213"	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21

		"5619644" "5675725" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6721860" "6769081" "6799241" "6799244" "6816982" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7353325" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8407395" "8412879" "8516187" "8671243" "8677060" "8874831" "8880791" "8904098" "8904099").PN.				
L76	3	L72 and L75	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L77	10992	G06F13/28.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L78	3097	G06F1/185.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L79	7290	G06F3/0613.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L80	11035	G06F3/0659.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L81	4886	G06F3/0685.CPC.	US- PGPUB;	OR	ON	2017/05/01 09:21

			USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L82	2394	G06F13/1694.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L83	4501	G06F13/4027.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L84	39528	L77 or L78 or L79 or L80 or L81 or L82 or L83	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L85	245	(data adj manager) and L84	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L86	34	L6 and L85	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L87	7	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and L86	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L88	2	L87 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21

L89	0	((read\$3 near3 command) with (memory adj controller)) and L88	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L90	5	((read\$3 near3 command) with (memory adj controller)) and L87	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L91	0	L90 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L92	0	((first adj command) with (second adj command) with (memory adj controller)) and L88	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L93	4	("20100110748" "20110161569" "9158684").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L94	3	(US-20110161569-\$ or US-20100110748-\$).did. or (US-9158684-\$).did.	US-PGPUB; USPAT	OR	ON	2017/05/01 09:21
L95	2	((read\$3 near3 command) with (memory adj controller)) and L94	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L96	1	L95 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L97	0	L96 and (data adj manager)	US-PGPUB;	OR	ON	2017/05/01 09:21

			USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L98	1	L94 and (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L99	0	L98 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L100	0	("Netlist, Inc.").AANM.	US- PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L101	43	("Netlist").AANM.	US- PGPUB; USPAT; USOCR	OR	ON	2017/05/01 09:21
L102	658	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and L77	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L103	5	(data adj manager) and L102	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L104	0	L103 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L105	33	((read\$3 near3 command) with (memory adj controller)) and L102	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2017/05/01 09:21

			IBM_TDB			
L106	5	(data adj manager) and L105	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L107	0	L106 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:21
L141	30	("20020053944" "20030028733" "20050249011" "20050273548" "20060212651" "20070070669" "20070147115" "20070255898" "20070288683" "20080147968" "20080235443" "20080291727" "20100322020" "20110078496" "20120317433" "20140032820" "20150058701" "4882709" "5490155" "5799200" "6026465" "6065092" "6571244" "6614685" "6693840" "6810513" "7136978" "7519754" "9043677" "9361250").PN.	US- PGPUB; USPAT	OR	ON	2017/05/01 09:30
L142	1	("20100110748").PN.	US- PGPUB; USPAT	OR	ON	2017/05/01 09:37
L143	1	142 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:37
L144	1	142 and protocol and bus	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:38
L145	13	141 and protocol and bus	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:39
L146	0	(data adj manager) and 145	US- PGPUB; USPAT; USOCR;	OR	ON	2017/05/01 09:40

			FPRS; EPO; JPO; IBM_TDB			
L147	11	(data near5 transfer\$3) and 145	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:44
L148	1	(data near5 transfer\$3) and 142	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:44
L149	1	(data near5 width) and 142	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/05/01 09:45

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L108	26	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L109	113	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L110	139	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L111	103	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L112	7927	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L113	8182	L108 or L109 or L110 or L111 or L112	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L114	12	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L115	5	L113 and L114	US-PGPUB; USPAT; *No UPAD	OR	ON	2017/05/01 09:21
L116	14031	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT;	OR	ON	2017/05/01 09:21

			* No UPAD			
L117	56	L113 and L116	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L118	4	L114 and L117	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L119	11556	711/103.ccls.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L120	1694	(data adj manager).clm.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L121	753	G06F12/0638.CPC.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L122	21	L113 and L119	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L123	3	L120 and L122	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L124	4433	G06F13/28.CPC.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L125	12	L113 and L124	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L126	5	L114 and L125	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L127	1226	((read\$3 near3 command) with (memory adj controller)).clm.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L128	0	L126 and L127	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L129	35069	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller).clm.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L130	4	L125 and L129	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L131	4	L114 and L130	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L132	0	L131 and (((@pd or @ad)< "20110728")	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L133	43	("Netlist").AANM.	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L134	43	L113 and L133	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L135	3	L120 and L134	US-PGPUB; USPAT;	OR	ON	2017/05/01 09:21

			* No UPAD			
L136	0	L127 and L135	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L137	2	L129 and L135	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L138	4	L114 and L129	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21
L139	2	L114 and L137	US-PGPUB; USPAT; * No UPAD	OR	ON	2017/05/01 09:21

5/ 1/ 2017 9:45:38 AM
C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 14840865.w sp

Search Notes 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243; G11C 7/1072; G11C 14/0018;	4/17/2016	SE
Search updated;	11/13/2016	SE
G06F 13/28; G06F 1/185; G06F 3/0613; G06F 3/0659;	11/13/2016	SE
G06F 3/0685; G06F 13/1694; G06F 13/4027;	11/13/2016	SE
Search updated	5/1/2017	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/17/2016	SE
365	185.33	4/17/2016	SE
Search updated		11/13/2016	SE
Search updated		5/1/2017	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/17/2016	SE
Inventor Name Search for DP	4/17/2016	SE
Search Updated	11/13/2016	SE
Assignee searched	11/13/2016	SE
Search updated	5/1/2017	SE

INTERFERENCE SEARCH	
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US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	11/13/2016	SE
G06F	13/28	11/13/2016	SE
PGPUB Searched		11/13/2016	SE
Assignee Searched		11/13/2016	SE
Search updated		5/1/2017	SE

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Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14840865
	Filing Date	2015-08-31
	First Named Inventor	LEE, HYUN
	Art Unit	2133
	Examiner Name	ELMORE, STEPHEN C.
	Attorney Docket Number	062453-036

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

1	File History 12/240,916 filed September 29, 2008
2	Bonella, Provisional Application for "ADVANCED DYNAMIC DISK MEMORY MODULE", 53 pages.
3	File History US Patent No. 8,671,243
4	Intel 1.8 Volt Intel StrataFlash Wireless Memory (L18).
5	Provisional Application No. 60/941,586 filed June 1, 2007
6	JEDEC STANDARD, Double Data Rate (DDR) SDRAM Specification, JESD79, June 2000, 77 pages
7	JEDEC STANDARD, DDR2 SDRAM Specification, JESD79-2B (Revision of JESD79-2A) January 2005, 113 pages.
8	Microsoft Computer Dictionary Fifth Edition, 10 pages.
9	Microsoft Windows 2000 Professional Resource Kit, 76 pages.
10	Data Sheet, 74F257A Quad 2-line to 1-line selector/multiplexer, non-inverting (3-State), Product specification, IC15 Data Handbook 1995 Mar 31, 10 pages.
11	Ex. 1004 - IPR2017-00587 Ron Maltiel CV, 7 pages.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

12	Provisional Application No. 60/912,321 filed April 17, 2007
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EXAMINER SIGNATURE

Examiner Signature	/STEPHEN C ELMORE/	Date Considered	05/01/2017
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-04-20
Name/Print	Khaled Shami	Registration Number	38,745

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14840865
	Filing Date	2015-08-31
	First Named Inventor	LEE, HYUN
	Art Unit	2133
	Examiner Name	ELMORE, STEPHEN C.
	Attorney Docket Number	062453-036

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	5813029	A	1998-09-22	KLEIN	
	2	5991885	A	1999-11-23	CHANG et al.	
	3	9436600	B2	2016-09-06	LEE	

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	1	20120265952	A1	2012-10-18	KURITA	
	2	20120117402	A1	2012-05-10	MACHNICKI et al.	

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

1							
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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	File History for Application No. 13/905,048 filed May 29, 2013, 181 pages.	
	2	File History for Provisional Application 60/941,586, filed June 1, 2007, 23 pages.	
	3	Office Action dated August 19, 2016 of the Chinese Patent Application No. 201280047758.X	

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-09-11
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Jacqueline Allen			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	1806	1	180	180
Total in USD (\$)				180

Electronic Acknowledgement Receipt

EFS ID:	30324418
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Jacqueline Allen
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	11-SEP-2017
Filing Date:	31-AUG-2015
Time Stamp:	15:33:57
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 180
RAM confirmation Number	091217INTEFSW00002343192380
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	062453-036_SB08_2.pdf	1149912	no	4
			cb0acb1c08991dda1a0e3294607c818ee9190655		
Warnings:					
Information:					
2	Non Patent Literature	062453-036_CNOA.pdf	742500	no	9
			a47f7557fdb02ed5ade389a879e6e2f31c98b44d		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30340	no	2
			980a6757ab3a927521b6ff22e9610b1e8a221566		
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Information:					
Total Files Size (in bytes):			1922752		
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun LEE CONFIRMATION NO.: 2445
APPLICATION NO.: 14/840,865
FILING DATE: August 31, 2015
TITLE: FLASH-DRAM HYBRID MEMORY MODULE
EXAMINER: Stephen C. ELMORE
ART UNIT: 2133

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

AMENDMENT AND/OR REPLY TO OFFICE ACTION

Sir:

In response to the Office Action mailed May 5, 2017, please amend the subject application as indicated.

Amendments to the Specification begin on page **2**.

Amendments to the Claims begin on page **3**.

Remarks begin on page **10**.

It is not believed that extensions of time are required beyond those that may otherwise be provided for in documents accompanying this paper. However, if additional extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. § 1.136(a), and any additional fees required to continue prosecution or appeal of this application (including issue fee, fees for net addition of claims or forwarding to appeal) are hereby authorized to be charged to our Deposit Account No. 60-2034.

Electronic Patent Application Fee Transmittal

Application Number:	14840865			
Filing Date:	31-Aug-2015			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Casey Berger			
Attorney Docket Number:	062453-036			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	3	420	1260
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Extension - 3 months with \$0 paid	1253	1	1400	1400
Miscellaneous:				
Total in USD (\$)				2660

Electronic Acknowledgement Receipt

EFS ID:	30861819
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	06-NOV-2017
Filing Date:	31-AUG-2015
Time Stamp:	13:15:38
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 2660
RAM confirmation Number	110617INTEFSW13165400
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		0016001000D_OAResponse_11062017.pdf	135900	yes	14
			a849b433ea2184fd7dc6d9d05a0c459f02de5781		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Arguments/Remarks Made in an Amendment		10	14	
	Claims		3	9	
	Specification		2	2	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	32346	no	2
			6a36885af0920d5b78808cd163785f683b718206		
Warnings:					
Information:					
Total Files Size (in bytes):			168246		

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If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

REMARKS

The Office Action mailed May 5, 2017 has been carefully considered. Reconsideration in view of the following remarks is respectfully requested.

Specification

The specification has been amended in order to address the objections raised in the Office Action.

Subject Matter Indicated Allowed or Allowable

Claims 14-16 are objected to as being dependent upon a rejected base claim, but would be allowable over the prior art of record if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Claims 14-16 have been rewritten in independent form to include the limitations of base claim 10 and are now in condition for allowance.

Claim Objection

Claim 10 has been amended to change “include” to “including” for grammatical accuracy.

Rejection(s) Under 35 U.S.C. § 102

Claims 2-13 and 17-22 stand rejected under pre-AIA 35 U.S.C. § 102(e) as allegedly being anticipated by U.S. Pat. Publ. No. 2010/0110748 to **Best** (hereinafter, "**Best**").

Applicants respectfully traverse.

Claim 2 has been amended to recite, *inter alia*,

a controller operable to receive one or more commands from the memory controller of the host system in accordance with the first protocol, and in response to the one or more commands received from the memory controller of the host system, the controller generates and transmits one or more control signals to the data manager, wherein, in response to the one or more control signals, the data manager transfers data between the first volatile memory subsystem and the memory controller of the host system, and transfers data signals ~~between from~~ the non-volatile memory subsystem ~~and to~~ the second volatile memory subsystem by way of the third data bus and the second data bus.

Best does not disclose transferring data between a **first** volatile memory subsystem and a memory controller of a host system, and from a non-volatile memory subsystem to a **second** volatile memory subsystem. Rather, **Best** discloses transfer of data between the DRAM and an external controller device (by way of dedicated data buses 142 and 155 and shared data path 140); and transfer of data from the NV Memory **to the same DRAM** (by way of the dedicated data buses 144, 171, and 142). Thus in both these cases in **Best**, the same DRAM is involved, whereas in the claimed arrangement different, first and second, volatile memory subsystems are involved.

In addition, in the claimed arrangement, data is transferred from the non-volatile memory subsystem to the volatile memory subsystem **via a third data bus in a second protocol and via second data bus in first protocol**. In this transfer, two different protocols are utilized. In **Best**,

there is no discussion of different protocols in the description of the transfer of data from the NV Memory to the DRAM. While *Best* mentions protocol conversion at [0015], this is in the context of data transfer between the entire hybrid memory system 100 on the one hand, and the external controller device (not shown) on the other. It does not relate to conversion of protocols internally, within the hybrid memory system, during transfer of data from the NV Memory to the DRAM.

It will be appreciated that, a claim is anticipated under 35 U.S.C. § 102 only if each and every claim element is found, either expressly or inherently described, in a single prior art reference.¹ The aforementioned reasons clearly indicate the contrary, and withdrawal of the 35 U.S.C. § 102 rejection of claims 2-9 based on *Best* is respectfully urged.

With respect to claim 10, it recites a VM controller that is operable to generate a first command to the VM subsystem in response to the command received from the memory controller of the host system; and a second command to the VM subsystem that is independent from the command received from the memory controller of the host system. *Best* does not disclose two such commands: one in response to, and one independent from, commands received from the external controller device. Moreover, in the claimed arrangement, both of these commands can be used by the controller, which includes the VM controller, to direct the transfer of data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM. *Best* does not disclose this capability—that is, the use of an independent command and a dependent command to direct such data transfer.

In view of the above, Applicants respectfully submit that claim 10, along with claims 11-13 dependent therefrom, are not anticipated by *Best* and the rejection under 35 U.S.C. § 102 should be withdrawn.

Turning to claim 17, it has been amended to recite, *inter alia*,

¹ Manual of Patent Examining Procedure (MPEP) § 2131. See also *Verdegaal Bros. v. Union Oil Co. of California*, 814 F.2d 628, 631, 2 USPQ2d 1051, 1053 (Fed. Cir. 1987).

wherein transfer of data between the memory controller of the host system and the VM subsystem is by way of a first data bus and a first protocol, and transfer of data between the NVM and the VM subsystem is by way of the first and a second protocol respectively on a second data bus and a third data bus

As explained above with respect to claim 2, in this arrangement, data is transferred from the non-volatile memory subsystem to the volatile memory subsystem **via a third data bus in a second protocol and via second data bus in first protocol**. In this transfer, two different protocols are utilized. In *Best*, there is no discussion of different protocols in the description of the transfer of data from the NV Memory to the DRAM. While *Best* mentions protocol conversion at [0015], this is in the context of data transfer between the entire hybrid memory system 100 on the one hand, and the external controller device (not shown) on the other. It does not relate to conversion of protocols internally, within the hybrid memory system, during transfer of data from the NV Memory to the DRAM. Accordingly, Applicants respectfully submit that claim 17, along with claims 18-22 dependent therefrom, are not anticipated by *Best* and the rejection under 35 U.S.C. § 102 should be withdrawn.

Conclusion

In view of the preceding discussion, Applicants respectfully urge that the claims of the present application define patentable subject matter and should be passed to allowance.

If the Examiner believes that a telephone call would help advance prosecution, the Examiner is kindly invited to call the undersigned attorney at the number below.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 60-2034.

Respectfully submitted,

SHAMI MESSINGER PLLC

Dated: November 6, 2017

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

SHAMI MESSINGER PLLC
1000 Potomac Street, NW
Fifth Floor
Washington, D.C. 20007
Tel: 202-791-9025

Amendments to the Claims

The following Listing of Claims replaces all prior versions in the application:

1. (Canceled)
2. (Currently amended) A memory module comprising:
 - a data manager configured to be coupled to a memory controller of a host system using a data bus, the data manager is operable to communicate data signals with the memory controller of the host system by way of the data bus in accordance with a first protocol;
 - a first volatile memory subsystem coupled to the data manager using a first data bus, the first volatile memory subsystem is operable to communicate data signals with the data manager by way of the first data bus in accordance with the first protocol;
 - a second volatile memory subsystem coupled to the data manager using a second data bus, the second volatile memory subsystem is operable to communicate data signals with the data manager by way of the second data bus in accordance with the first protocol;
 - a non-volatile memory subsystem coupled to the data manager using a third data bus, the non-volatile memory subsystem is operable to communicate data signals with the data manager by way of the third data bus using a second protocol; and
 - a controller operable to receive one or more commands from the memory controller of the host system in accordance with the first protocol, and in response to the one or more commands received from the memory controller of the host system, the controller generates and transmits one or more control signals to the data manager, wherein, in response to the one or more control signals, the data manager transfers data between the first volatile memory subsystem and the memory controller of the host system, and transfers data signals ~~between from~~ the non-volatile memory subsystem ~~and to~~ the second volatile memory subsystem by way of the third data bus and the second data bus.
3. (Previously Presented) The memory module of claim 2, wherein the second protocol is different from the first protocol.
4. (Previously Presented) The memory module of claim 3, wherein the first protocol is selected from the group consisting of DDR, DDR2, DDR3, and DDR4 protocols.

5. (Previously Presented) The memory module of claim 2, wherein the data manager controls data traffic between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-volatile memory subsystem and the memory controller of the host system based on one or more commands received from the controller.
6. (Previously Presented) The memory module of claim 2, wherein the data manager controls a starting time of a data transfer between any one of the first volatile memory subsystem and the second volatile memory subsystem and any one of the non-volatile memory subsystem and the memory controller of the host system based on one or more commands or control signals received from the controller.
7. (Previously Presented) The memory module of claim 2, wherein the data manager is configured as a bi-directional data transfer fabric.
8. (Previously Presented) The memory module of claim 2, wherein the data manager is operable to concurrently transfer data (i) between the first volatile memory subsystem and the memory controller of the host system, and (ii) between the non-volatile memory subsystem and the second volatile memory subsystem.
9. (Previously Presented) The memory module of claim 2, wherein the data manager further comprises a data formatting subsystem operable to format data to be transferred via the data bus, the first data bus, the second data bus and the third data bus.
10. (Currently amended) A memory module couplable to a memory controller of a host system, the memory module comprising:
 - a non-volatile memory (NVM) subsystem coupled to a NVM controller;
 - a data manager coupled to a data manager controller and to the NVM subsystem;
 - a volatile memory (VM) subsystem coupled to a VM controller and to the data manager;and
 - a controller operable to receive a command from the memory controller of the host system, the controller ~~include~~ including the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host

system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command.

11. (Previously Presented) The memory module of claim 10, wherein the controller further includes an interface scheduler, the interface scheduler schedules the VM subsystem operation and the NVM subsystem operation such that no resource conflict occurs in the data manager.

12. (Previously Presented) The memory module of claim 10, wherein the controller further includes an interface scheduler, the interface scheduler assigns time slots for the VM controller and the NVM controller based on current command status and a pending command received or to be received from the memory controller of the host system.

13. (Previously Presented) The memory module of claim 10, wherein the VM subsystem is operable to exchange data with the NVM subsystem by way of the data manager in response to the second command received from the VM controller.

14. (Currently amended) A memory module couplable to a memory controller of a host system, the memory module comprising:

a non-volatile memory (NVM) subsystem coupled to a NVM controller;

a data manager coupled to a data manager controller and to the NVM subsystem;

a volatile memory (VM) subsystem coupled to a VM controller and to the data manager;

and

a controller operable to receive a command from the memory controller of the host system, the controller ~~include~~ including the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system,

and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command~~The memory module of claim 10~~, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller (i) to abort the current operation of VM subsystem, and (ii) to execute an operation in response to the first command.

15. (Currently amended) A memory module couplable to a memory controller of a host system, the memory module comprising:

a non-volatile memory (NVM) subsystem coupled to a NVM controller;

a data manager coupled to a data manager controller and to the NVM subsystem;

a volatile memory (VM) subsystem coupled to a VM controller and to the data manager;

and

a controller operable to receive a command from the memory controller of the host system, the controller ~~include~~ including the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command

~~and the second command~~~~The memory module of claim 10~~, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller to complete the current operation of VM subsystem before executing an operation in response to the first command.

16. (Currently amended) A memory module couplable to a memory controller of a host system, the memory module comprising:

a non-volatile memory (NVM) subsystem coupled to a NVM controller;

a data manager coupled to a data manager controller and to the NVM subsystem;

a volatile memory (VM) subsystem coupled to a VM controller and to the data manager;

and

a controller operable to receive a command from the memory controller of the host system, the controller include-including the NVM controller, the VM controller, and the data manager controller, the VM controller is operable (i) to generate and transmit a first command to the VM subsystem in response to the command received from the memory controller of the host system, and (ii) to generate and transmit a second command to the VM subsystem, wherein the second command is generated by the VM controller independently from the command received from the memory controller of the host system, and wherein the controller is operable to direct (i) operation of the NVM subsystem by way of the NVM controller, (ii) operation of the VM subsystem by way of the VM controller, and (iii) operation of the data manager by way of the data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on any one of the first command and the second command~~The memory module of claim 10~~, wherein the controller further includes an interface scheduler, if the command received by the controller from the memory controller of the host system is a read or write operation from/to the VM subsystem that conflicts with an operation being executed by VM subsystem in response to the second command, then the interface scheduler instructs the VM controller (i) to halt the current operation of VM subsystem, (ii) to execute an operation in response to the first command, and (iii) to resume the operation in response to the second command.

17. (Currently amended) A memory module couplable to a memory controller of a host system, the memory module comprising:
- a non-volatile memory (NVM) subsystem;
 - a data manager coupled to a data manager controller and to the NVM subsystem;
 - a volatile memory (VM) subsystem coupled to the data manager; and
 - a controller operable to receive a read command from the memory controller of the host system, the controller is configured to determine, in response to the read command, a target address for a requested data, and if requested data is not stored in the VM subsystem, then the controller (i) initiates data transfer operations from the NVM subsystem to the VM subsystem by way of the data manager, and (ii) initiates data transfer operations from the VM subsystem to the memory controller of the host system by way of the data manager, wherein transfer of data between the memory controller of the host system and the VM subsystem is by way of a first data bus and a first protocol, and transfer of data between the NVM and the VM subsystem is by way of the first and a second protocol respectively on a second data bus and a third data bus.
18. (Previously Presented) The memory module of claim 17, wherein the controller is operable to direct (i) operation of the NVM subsystem by way of a NVM controller, (ii) operation of the VM subsystem by way of a VM controller, and (iii) operation of the data manager by way of a data manager controller to transfer data between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command.
19. (Previously Presented) The memory module of claim 17, wherein the data manager controls a starting time of a data transfer between any two or more of the memory controller of the host system, the VM subsystem, and the NVM subsystem based on the read command.
20. (Previously Presented) The memory module of claim 17, wherein the VM subsystem includes first and second VM subsystems.
21. (Previously Presented) The memory module of claim 20, wherein the data manager controls a starting time of a data transfer between any one of the first VM subsystem and the

second VM subsystem and any one of the NVM subsystem and the memory controller of the host system based on the read command.

22. (Previously Presented) The memory module of claim 20, wherein the data manager is operable to concurrently transfer data (i) between the first VM subsystem and the memory controller of the host system, and (ii) between the NVM subsystem and the second VM subsystem.

Amendments to the Specification

Please amend paragraph [0001] as follows:

[0001] This application is a continuation of U.S. Patent Application No. 14/489,269, filed September 17, 2014, titled, “FLASH-DRAM HYBRID MEMORY MODULE”, now U.S. Patent No. 9,158,684, issued October 13, 2015, which is a continuation of U.S. Patent No. 8,874,831, issued, October 28, 2014, titled , “FLASH-DRAM HYBRID MEMORY MODULE“, which claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and is a continuation-in-part of US Patent No. 8,301,833, issued October 30, 2012, which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/840,865	Filing Date 08/31/2015	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	11/06/2017	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 21	Minus	** 21	= 0	X \$80 =	0
	Independent (37 CFR 1.16(h))	* 6	Minus	***3	= 3	X \$420 =	1260
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	1260

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
AMANDA FORD

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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NOTICE OF ALLOWANCE AND FEE(S) DUE

22204 7590 11/17/2017
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT

PAPER NUMBER

2133

DATE MAILED: 11/17/2017

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/840,865

08/31/2015

Hyun Lee

062453-036

2445

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	02/20/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

22204 7590 11/17/2017
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	02/20/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2133	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☐ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).
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5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

22204 7590 11/17/2017
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

EXAMINER

ELMORE, STEPHEN C

ART UNIT	PAPER NUMBER
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2133

DATE MAILED: 11/17/2017

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/840,865	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the Amendment filed 11/6/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2-22. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date 9/11/2017
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material
4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____.
5. ☒ Examiner's Amendment/Comment
6. ☐ Examiner's Statement of Reasons for Allowance
7. ☐ Other _____.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2133

The present application is being examined under the pre-AIA first to invent provisions.

EXAMINER'S COMMENT

Information Disclosure Statement

1. The Information Disclosure Statement (IDS) filed 9/11/2017 contains two "lined-through" entries in the category Non-Patent Literature Documents, Cites No. 1 and 2, which have not been considered.

The IDS entries fail to comply with 37 CFR § 1.98(a)(1) because they are and have been identified as "File History..." in "Non-Patent Literature Documents" of the IDS and therefore do not comply with 37 CFR § 1.98(a)(1).

Each publication requested to be considered under 37 CFR § 1.98(a)(1) is required to be an individual publication listed individually on form PTO-1449. Aggregated listings (i.e., such as, an electronic "file history") do not comply because the aggregate "history" is a compilation of multiple documents with each document having distinct and/or different publication dates, and so is not an individual document publication according to 37 CFR § 1.98(a)(1).

2. The Information Disclosure Statement (IDS) filed 9/11/2017 also contains a "lined-through" entry in the category Non-Patent Literature Documents, Cite No. 3, which has not been considered because the listed document is not in the English language, and the listing fails to comply with 37 CFR 1.98 (a)(3).

Doc code: IDS

PTO/SB/08a (03-15)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2016. OMB 0651-0031

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14840865
	Filing Date		2015-08-31
	First Named Inventor	LEE, HYUN	
	Art Unit	2133	
	Examiner Name	ELMORE, STEPHEN C.	
	Attorney Docket Number	062453-036	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	5813029	A	1998-09-22	KLEIN	
	2	5991885	A	1999-11-23	CHANG et al.	
	3	9436600	B2	2016-09-06	LEE	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20120265952	A1	2012-10-18	KURITA	
	2	20120117402	A1	2012-05-10	MACHNICKI et al.	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	File History for Application No. 13/905,048 filed May 29, 2013, 181 pages.	
	2	File History for Provisional Application 60/941,586, filed June 1, 2007, 23 pages.	
	3	Office Action dated August 19, 2016 of the Chinese Patent Application No. 201280047758 X	

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/STEPHEN C ELMORE/	Date Considered	11/13/2017
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14840865
Filing Date	2015-08-31
First Named Inventor	LEE, HYUN
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C.
Attorney Docket Number	062453-036

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2017-09-11
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
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Search Notes 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243; G11C 7/1072; G11C 14/0018;	4/17/2016	SE
Search updated;	11/13/2016	SE
G06F 13/28; G06F 1/185; G06F 3/0613; G06F 3/0659;	11/13/2016	SE
G06F 3/0685; G06F 13/1694; G06F 13/4027;	11/13/2016	SE
Search updated	5/1/2017	SE
Search Updated	11/13/2017	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/17/2016	SE
365	185.33	4/17/2016	SE
Search updated		11/13/2016	SE
Search updated		5/1/2017	SE
Search Updated		11/13/2017	SE

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/17/2016	SE
Inventor Name Search for DP	4/17/2016	SE
Search Updated	11/13/2016	SE

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SEARCH NOTES

Search Notes	Date	Examiner
Assignee searched	11/13/2016	SE
Search updated	5/1/2017	SE
Search Updated	11/13/2017	SE

INTERFERENCE SEARCH

US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	11/13/2016	SE
G06F	13/28	11/13/2016	SE
PGPUB Searched		11/13/2016	SE
Assignee Searched		11/13/2016	SE
Search updated		5/1/2017	SE
Search Updated		11/13/2017	SE

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BIB DATA SHEET

CONFIRMATION NO. 2445

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
14/840,865	08/31/2015	711	2133	062453-036	
RULE					
APPLICANTS Netlist, Inc., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;					
** CONTINUING DATA ***** This application is a CON of 14/489,269 09/17/2014 PAT 9158684 which is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 09/14/2015					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /STEPHEN C ELMORE/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 10	TOTAL CLAIMS 21	INDEPENDENT CLAIMS 6
ADDRESS NIXON PEABODY, LLP 799 Ninth Street, NW SUITE 500 WASHINGTON, DC 20001 UNITED STATES			/SCE/		/SCE/
TITLE FLASH-DRAM HYBRID MEMORY MODULE					
FILING FEE RECEIVED 3080	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Issue Classification 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2133	

CPC						
Symbol					Type	Version
G06F		13		28	F	2013-01-01
G06F		12		0246	I	2013-01-01
G06F		1		185	I	2013-01-01
G06F		2212		205	A	2013-01-01
G06F		13		1694	I	2013-01-01
G06F		12		0638	I	2013-01-01
G06F		13		4243	I	2013-01-01
G06F		2212		7208	A	2013-01-01
G11C		7		1072	I	2013-01-01
G11C		14		0018	I	2013-01-01
G06F		3		0613	I	2013-01-01
G06F		3		0659	I	2013-01-01
G06F		3		0685	I	2013-01-01
G06F		13		4027	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE		Total Claims Allowed:	
		21	
(Assistant Examiner)	(Date)	O.G. Print Claim(s)	O.G. Print Figure
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2133	11/13/2017	1	6
(Primary Examiner)	(Date)		

Issue Classification 	Application/Control No. 14840865	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2133

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	2		18												
	3		19												
	4		20												
	5		21												
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	14														
	15														
	16														
	17														

NONE		Total Claims Allowed: 21	
(Assistant Examiner)	(Date)		
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2133	11/13/2017	O.G. Print Claim(s) 1	O.G. Print Figure 6
(Primary Examiner)	(Date)		

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	11612	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L2	2789	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L3	21421	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L4	33615	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L5	8719	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L6	797	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L7	24	L5 and L6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L8	3	L7 and (((@pd or @ad)<"20120726")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L9	2	(US-20070136523-\$).did. or (US- 8412879-\$).did.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41

L10	209916	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L11	121	L6 and L10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L12	8	L5 and L11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L13	6	L4 and L11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L14	404	(bi-direction or bi-directional) near3 fabric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L15	5	L14 with (data adj manager)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L16	30	(data near3 (port or input-output or I/O or IO)) and L11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L17	5	L5 and L16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
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L19	0	L4 and L18	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L20	15	L6 and L18	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L21	0	L14 and L20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L22	0	L5 and L20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L23	40297	(control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L24	5	((control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller) and L16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L25	3	L6 and L23 and L10 and L4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L26	5	L6 and L23 and L10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
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L28	8387	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L29	26	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L30	114	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L31	183	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41

L32	104	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L33	5	(data adj manager same controller same memory adj controller) and L5 and (L10 or L14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L34	906	(hybrid near3 memory).ti.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
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L37	11	(data adj manager and memory adj controller) and L34	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L38	0	L37 and ((@pd or @ad)< "20120726")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L39	2	(data adj manager) and L34 and ((@pd or @ad)< "20120726")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L40	38	("20080195806" "6658507" "5675725" "20040190210" "6336176" "7409590" "20100274953" "6336174" "5519663" "6487623" "20080104344" "4420821" "6799244" "20020083368" "4449205" "8301833" "7111142" "20070192627" "6158015" "20120204079").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L41	1	L6 and L40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L42	117	("2043099" "20030158995" "20040163027" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20090031099" "3562555" "3916390" "4234920" "4965828" "5430742" "5519831" "5563839" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6199142" "6216247" "6421279" "6459647" "6487102" "6769081" "6799241" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7155627" "7200021" "7234099" "7409491" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8233303").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L43	0	L14 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L44	0	L16 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L45	0	L6 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L46	1	"8874831".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L47	8	"8301833".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L48	1	(US-8301833-\$).did.	USPAT	OR	ON	2017/11/13 07:41
L49	3	L7 and ((@pd or @ad)< "20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L50	1	"8874831".pn.	US-PGPUB; USPAT; USOCR;	OR	ON	2017/11/13 07:41

			FPRS; EPO; JPO; IBM_TDB			
L51	82	("20020083368" "20030158995" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20140059170" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213" "5619644" "5675725" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6769081" "6799241" "6799244" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8412879").PN.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L52	8	"8301833".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L53	1	(US-8301833-\$).did.	USPAT	OR	ON	2017/11/13 07:41
L54	1	(US-8102614-\$).did.	USPAT	OR	ON	2017/11/13 07:41
L55	221	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L56	4	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L55	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L57	12	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L34	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L58	8	L57 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L59	0	L37 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L60	1672	G06F12/0638.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L61	19610	G06F12/0246.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L62	3460	G06F13/4243.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L63	8401	G11C7/1072.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L64	704	G11C14/0018.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L65	32575	L60 or L61 or L62 or L63 or L64	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L66	5	L55 and L65	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L67	0	L66 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L68	12	L5 and L57	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L69	3	L6 and L68	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L70	3	(data adj manager) and L68	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L71	3	L57 and L65	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L72	127	(data adj manager) and L65	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L73	3	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L72	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L74	10	("9158684" or "8874831" or "8301833").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L75	215	("20020083368" "20020199061" "20030158995" "20040088508" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20120271990" "20130019076" "20130086309" "20130254456" "20130254497" "20140059170" "20140156919" "20140156920" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213" "5619644" "5675725" "5870350"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

		"5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6721860" "6769081" "6799241" "6799244" "6816982" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7353325" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8407395" "8412879" "8516187" "8671243" "8677060" "8874831" "8880791" "8904098" "8904099").PN.				
L76	3	L72 and L75	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L77	11382	G06F13/28.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L78	3187	G06F1/185.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L79	7554	G06F3/0613.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L80	12366	G06F3/0659.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L81	5394	G06F3/0685.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L82	2476	G06F13/1694.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L83	4661	G06F13/4027.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L84	42031	L77 or L78 or L79 or L80 or L81 or L82 or L83	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L85	251	(data adj manager) and L84	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L86	36	L6 and L85	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L87	7	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and L86	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L88	2	L87 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L89	0	((read\$3 near3 command) with (memory adj controller)) and L88	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L90	5	((read\$3 near3 command) with (memory adj controller)) and L87	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L91	0	L90 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L92	0	((first adj command) with (second adj command) with (memory adj controller)) and L88	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L93	4	("20100110748" "20110161569" "9158684").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L94	3	(US-20110161569-\$ or US-20100110748-\$).did. or (US-9158684-\$).did.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L95	2	((read\$3 near3 command) with (memory adj controller)) and L94	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L96	1	L95 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L97	0	L96 and (data adj manager)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L98	1	L94 and (data adj manager)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L99	0	L98 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L100	0	("Netlist, Inc.").AANM.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L101	47	("Netlist").AANM.	US-PGPUB; USPAT; USOCR	OR	ON	2017/11/13 07:41
L102	671	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller) and L77	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L103	5	(data adj manager) and L102	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L104	0	L103 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR;	OR	ON	2017/11/13 07:41

			FPRS; EPO; JPO; IBM_TDB			
L105	33	((read\$3 near3 command) with (memory adj controller)) and L102	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L106	5	(data adj manager) and L105	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L107	0	L106 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L108	30	("20020053944" "20030028733" "20050249011" "20050273548" "20060212651" "20070070669" "20070147115" "20070255898" "20070288683" "20080147968" "20080235443" "20080291727" "20100322020" "20110078496" "20120317433" "20140032820" "20150058701" "4882709" "5490155" "5799200" "6026465" "6065092" "6571244" "6614685" "6693840" "6810513" "7136978" "7519754" "9043677" "9361250").PN.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L109	1	("20100110748").PN.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L110	1	L109 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L111	1	L109 and protocol and bus	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L112	13	L108 and protocol and bus	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L113	0	(data adj manager) and L112	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41

L114	11	(data near5 transfer\$3) and L112	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L115	1	(data near5 transfer\$3) and L109	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L116	1	(data near5 width) and L109	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 07:41
L149	5	("20120117402" "20120265952" "5813029" "5991885" "9436600").PN.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:47
L150	7260	(first adj2 bus) and (second adj2 bus) and (third adj2 bus)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:02
L151	29195	(first adj2 protocol) and (second adj2 protocol)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:03
L152	126	150 and 151	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:03
L153	23	((first adj2 bus) with (first adj2 protocol)) and ((third adj2 bus) with (second adj2 protocol))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:05
L154	23	152 and 153	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:05
L155	0	149 and 154	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2017/11/13 08:05

			DERWENT; IBM_TDB			
L156	8	154 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 08:05
L157	5	(US-7895388-\$ or US-7801162-\$ or US-7073008-\$ or US-6574330-\$ or US-6085269-\$).did.	USPAT	OR	ON	2017/11/13 08:07
L158	0	6 and 157	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:08
L159	2	host and (memory adj controller) and 157	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:09
L160	16831	(non-volatile adj memory or NVM) with ((non-volatile adj memory or NVM) near3 controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:12
L161	17824	(volatile adj memory or VM) with ((volatile adj memory or VM) near3 controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:12
L162	13317	160 and 161	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:13
L163	36	host and (memory adj controller) and (data adj manager) and 162	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:13
L164	1940	(volatile adj memory or VM) with ((volatile adj memory or VM) near3 controller) with (command or instruction)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:14

L165	8	163 and 164	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/13 08:14
L166	3	165 and ((@pd or @ad)<"20110728")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2017/11/13 08:15

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L117	26	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L118	114	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L119	139	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L120	104	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L121	8387	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L122	8643	L117 or L118 or L119 or L120 or L121	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L123	12	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L124	5	L122 and L123	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L125	14433	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L126	60	L122 and L125	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L127	4	L123 and L126	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L128	11569	711/103.ccls.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L129	1752	(data adj manager).clm.	US-PGPUB; USPAT	OR	ON	2017/11/13 07:41
L130	802	G06F12/0638.CPC.	US-	OR	ON	2017/11/13

			PGPUB; USPAT			07:41
L131	21	L122 and L128	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L132	3	L129 and L131	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L133	4694	G06F13/28.CPC.	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L134	12	L122 and L133	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L135	5	L123 and L134	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L136	1307	((read\$3 near3 command) with (memory adj controller)).clm.	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L137	0	L135 and L136	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L138	36516	((control adj information or control adj data or control adj signal or control adj metadata) near3 controller).clm.	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L139	4	L134 and L138	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L140	4	L123 and L139	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L141	0	L140 and (((@pd or @ad)<"20110728")	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L142	47	("Netlist").AANM.	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L143	47	L122 and L142	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L144	3	L129 and L143	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L145	0	L136 and L144	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L146	2	L138 and L144	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L147	4	L123 and L138	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41
L148	2	L123 and L146	US- PGPUB; USPAT	OR	ON	2017/11/13 07:41

11/ 13/ 2017 8:16:41 AM

C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 14840865.wsp

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

151145

OR

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

☒ The address associated with the above-mentioned Customer Number

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Netlist, Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

12/5/17

Name

Noel B. Whitley

Title

VP, IP & Licensing

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	14/840,865
Filing Date	08-31-2015
First Named Inventor	Hyun Lee
Title	FLASH-DRAM HYBRID MEMORY MODULE
Art Unit	2133
Examiner Name	ELMORE, STEPHEN C
Attorney Docket Number	0016.001000D

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	
Name	Khaled Shami	Registration Number	38,745
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	31167900
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-036
Receipt Date:	08-DEC-2017
Filing Date:	31-AUG-2015
Time Stamp:	10:15:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	0016001000D_POA.pdf	831888	no	2
			1845b55fc638335ec9f43b4e2b3005c5cd49d774		

Warnings:

Information:

Total Files Size (in bytes):

831888

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/840,865	08/31/2015	Hyun Lee	0016.001000D

151145
Shami Messinger PLLC
1000 Potomac Street NW
Fifth Floor
Washington, DC 20007

CONFIRMATION NO. 2445
POA ACCEPTANCE LETTER



OC000000096025171

Date Mailed: 12/13/2017

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/08/2017.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/nhassani/

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

22204 7590 11/17/2017
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	08/31/2015	Hyun Lee	062453-036	2445

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	02/20/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2133	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Shami Messinger PLLC

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☒ Payment by credit card. Form PTO-2038 is attached.
- ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature Khaled Shami

Date February 16, 2018

Typed or printed name Khaled Shami

Registration No. 38,745

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

Your fee transmittal sheet cannot be generated at this time. Please complete your submission.
After completing your submission, please contact the Electronic Business Center (EBC) to request the fee transmittal sheet for this submission.

We apologize for the inconvenience.

Electronic Business Center (EBC)
1-866-217-9197 (toll-free)
or 571-272-4100
6 a.m. to 12 Midnight Eastern Time
Monday -Friday

First Named Inventor/Applicant Name:

Filer:

Attorney Docket Number:

Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				

Electronic Acknowledgement Receipt

EFS ID:	31813339
Application Number:	14840865
International Application Number:	
Confirmation Number:	2445
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	151145
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	0016.001000D
Receipt Date:	16-FEB-2018
Filing Date:	31-AUG-2015
Time Stamp:	11:54:59
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$960
RAM confirmation Number	021618INTEFSW11553300
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	0016001000D_IssueFeePayment.pdf	121798	no	1
			b07e45595df873356edd087cd512116240e4d8bb		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30061	no	2
			769380a49c92bc8e06d1c1dfb5959d2273810ce5		

Warnings:

Information:

Total Files Size (in bytes):			151859
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

	42	7053470	B1	2006-05-30	Sellers et al.	
	43	7062618	B2	2006-06-13	Tsunoda et al.	
	44	7089412	B2	2006-08-08	Chen	
	45	7102391	B1	2006-09-05	Sun et al.	
	46	7111142		2006-09-19	Spencer et al.	
	47	7155627	B2	2006-12-26	Matsui	
	48	7200021	B2	2007-04-03	Raghuram	
	49	7234099	B2	2007-06-19	Gower et al.	
	50	7353325		2008-04-01	Lofgren et al.	
Change(s) applied to document, /LH/ 12/1/2016	51	7409491	B2	08/2008 2008-12-05	Doblar et al.	
	52	7409590		2008-08-05	Moshayedi et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2015-08-31
First Named Inventor	Hyun Lee	
Art Unit		
Examiner Name		
Attorney Docket Number	062453-036	

	53	7411859	B2	2008-08-12	Sohn et al.	
	54	7421552	B2	2008-09-02	Long	
	55	7467251	B2	2008-12-16	Park et al.	
Change(s) applied to document, /R.L./ 12/2/2016	56	7600142		1999-10-17 10/2009	Cross Ichikawa	
	57	7716411	B2	2010-05-11	Panabaker et al.	
	58	7818488	B2	2010-10-19	Park et al.	
	59	8086955		2011-12-27	Zhou et al.	
	60	8102614		2012-01-24	Song et al.	
	61	8233303	B2	2012-07-31	Best et al.	
	62	8301833		2012-10-30	Chen et al.	
	63	8407395		2013-03-26	Kim et al.	



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/840,865	03/27/2018	9928186	0016.001000D	2445

151145 7590 03/07/2018
Shami Messinger PLLC
1000 Wisconsin Ave. NW
Suite 200
Washington, DC 20007

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

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FLASH-DRAM HYBRID MEMORY MODULE

PRIORITY CLAIM

[0001] This application is a continuation of U.S. Patent Application No. 13/559,476, filed July 26, 2012, titled , “FLASH-DRAM HYBRID MEMORY MODULE“ which claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and is a continuation-in-part of US patent application serial no. 12/240,916, filed September 29, 2008 which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

[0002] This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

TECHNICAL FIELD

[0003] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0004] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0005] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0006] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or

solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0007] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem

using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MBs for the EcoRAM vs. 104MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™), which occupies one of the server's CPU sockets is used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0008] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0009] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0010] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read

operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

OVERVIEW

[0011] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0012] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory

subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0013] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0014] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0015] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0016] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0017] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0018] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0019] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0020] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0021] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0022] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0023] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0024] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0025] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0026] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0027] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0028] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0029] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0030] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0031] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0032] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0033] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0034] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0035] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0036] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0037] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0038] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0039] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0040] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0041] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0042] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0043] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0044] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0045] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0046] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0047] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0048] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0049] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0050] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0051] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0052] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0053] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0054] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0055] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0056] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0057] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0058] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0059] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0060] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating

commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0061] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0062] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0063] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0064] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0065] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0066] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0067] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0068] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be

a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0069] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0070] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0071] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one

arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0072] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction, scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0073] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0074] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or

is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0075] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly,

when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0076] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0077] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0078] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0079] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term

“read/write access” means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0080] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0081] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion, Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0082] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly

(while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0083] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH 510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0084] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0085] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0086] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr

504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0087] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0088] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating

at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0089] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0090] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0091] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[0092] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502 may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[0093] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash–DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[0094] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM) 801. As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 803 and the other for the DRAM memory DRAM 805. The MCH 810 treats the DRAM 805 and Flash 803 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However, the on-DIMM data path 807 between Flash 803 and DRAM 805 allows for direct data transfer to occur between the Flash 803 and the DRAM 805 in response to control information from Ctrl 830. In this embodiment, this data transfer mechanism provides direct support for executing commands from the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 803 to the DRAM 805. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[0095] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate)

protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[0096] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[0097] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in

accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[0098] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one

embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[0099] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which

was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[00100] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906 and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502 monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00101] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately

4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig. 10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1060 field, a Block address 1050 field, a Plane address 1040, a Page address 1030, and a group of least significant address bits A_0A_1 1020. The Plane address 1040 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00102] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00103] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508

to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00104] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'})) * (\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00105] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00106] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00107] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module couplable to a memory controller of a host system, comprising:
a non-volatile memory subsystem;
a data manager coupled to the non-volatile memory subsystem;
a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager; and
a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller, wherein:
at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments, each memory segment comprising at least one memory circuit, memory device, or memory die, and
the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems, the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile

memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

ABSTRACT

A memory module that is couplable to a memory controller hub (MCH) of a host system includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive read/write commands from the MCH and to direct transfer of data between any two or more of the MCH, the volatile memory subsystem, and the non-volatile memory subsystem based on the commands.

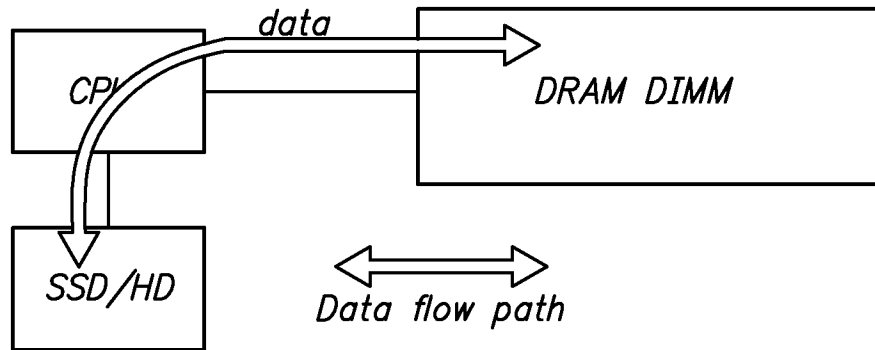


FIG. 1
(PRIOR ART)

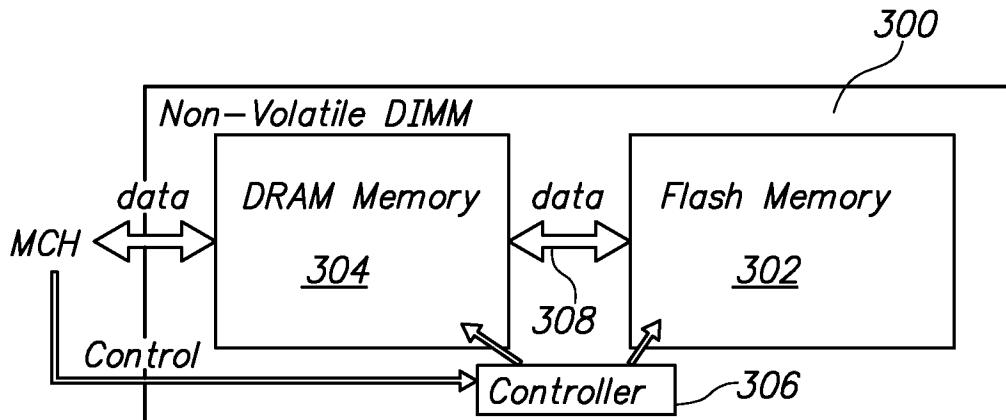
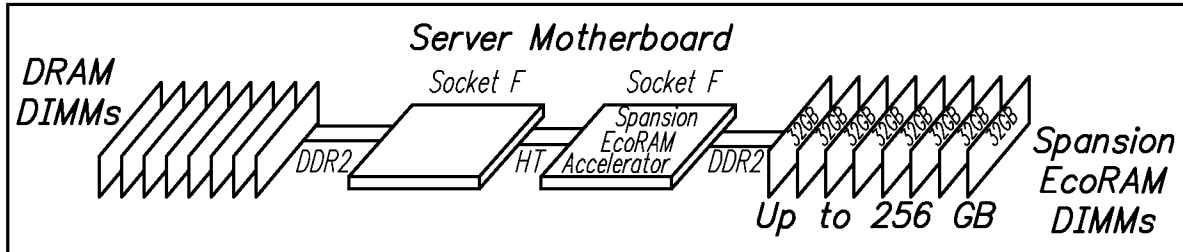


FIG. 3A

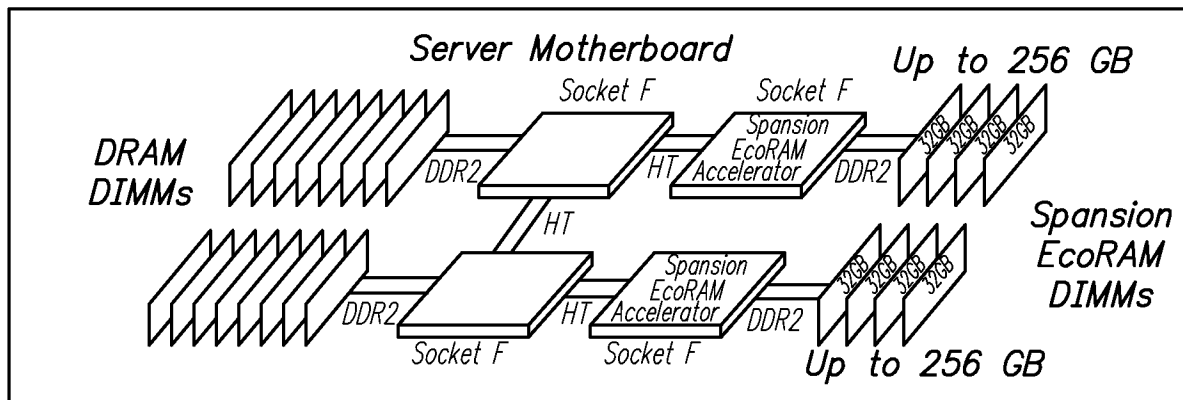
Spancion EcoRAM Configurations

256GB Spancion EcoRAM Solution – Single Accelerator

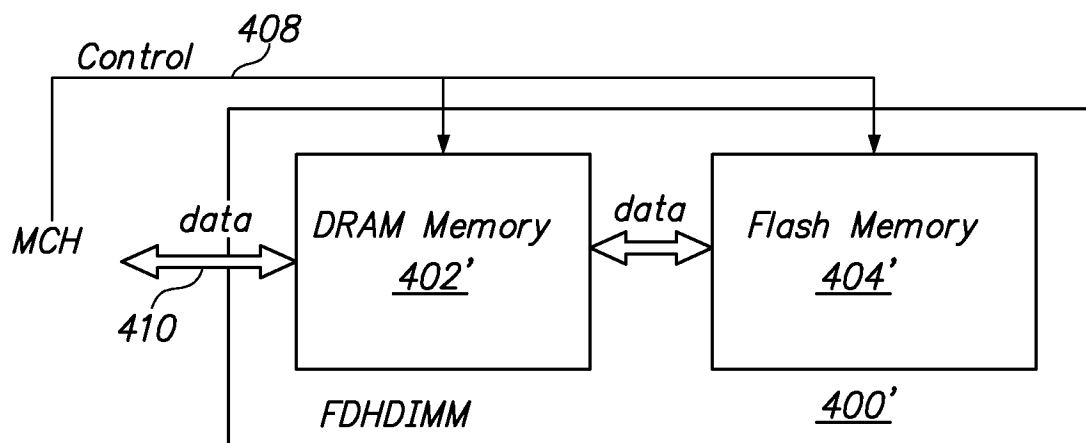


256GB Single Accelerator Spancion EcoRAM Solution

256GB Spancion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

**FIG. 2
(PRIOR ART)****FIG. 4B**

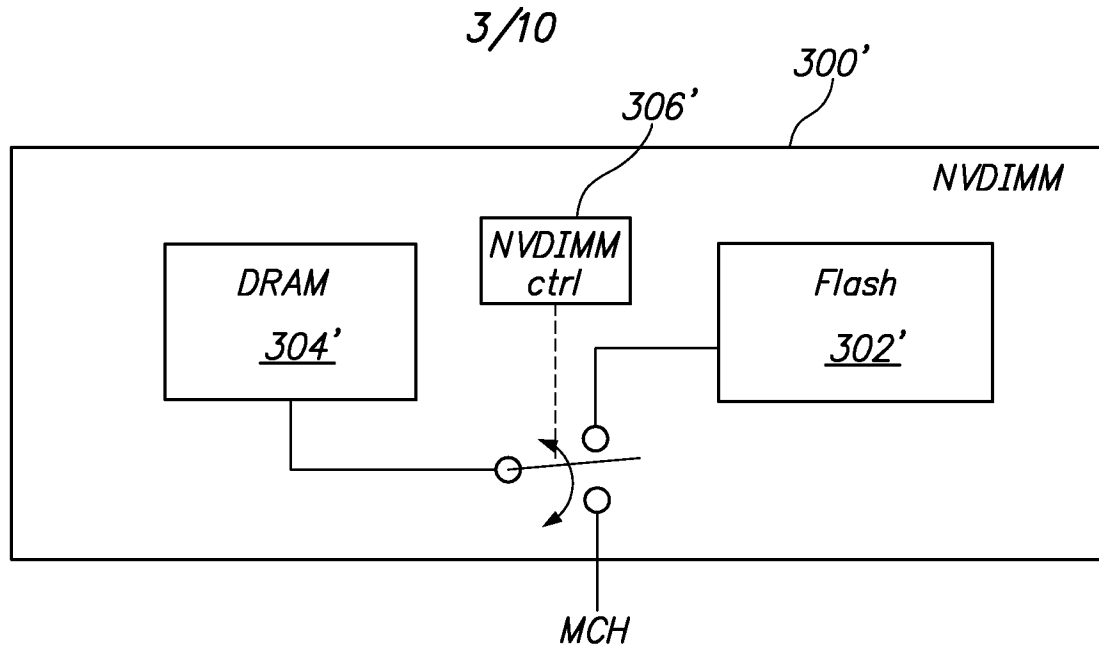


FIG. 3B

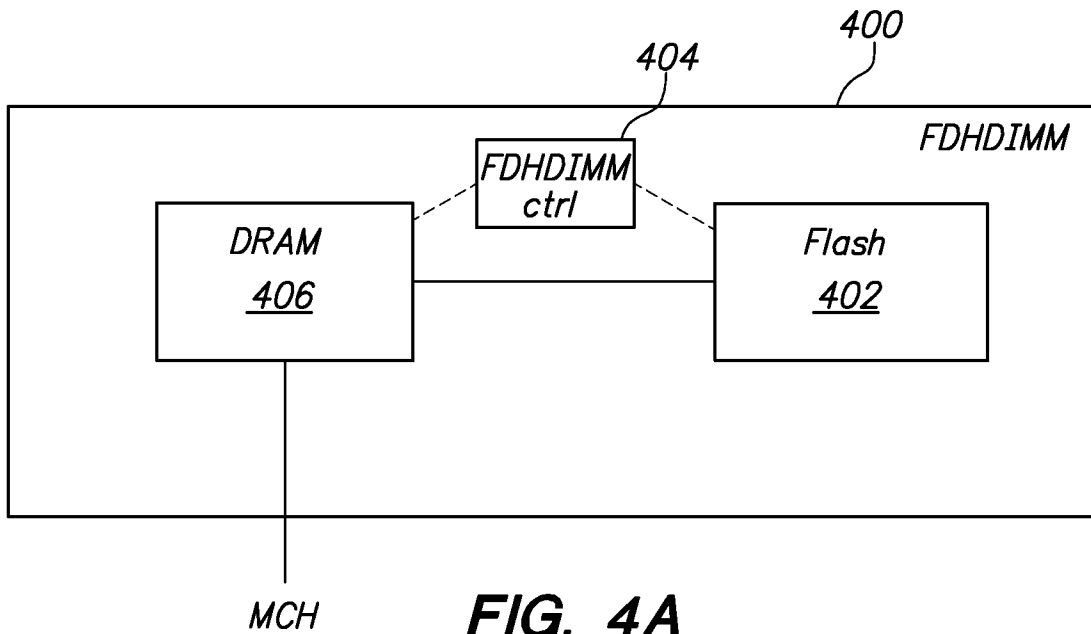


FIG. 4A

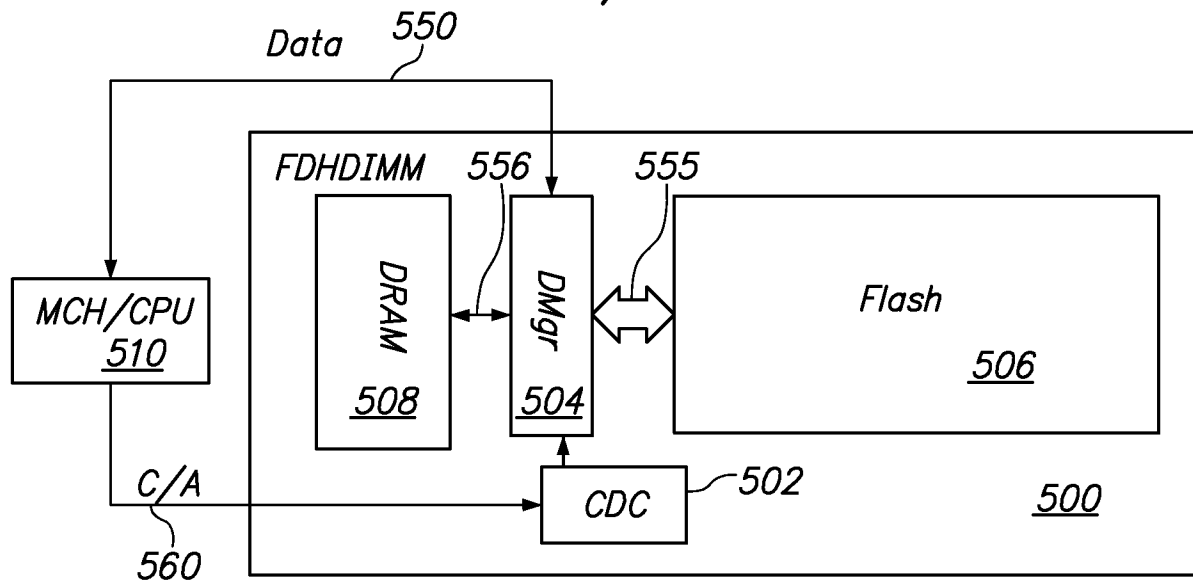


FIG. 5A

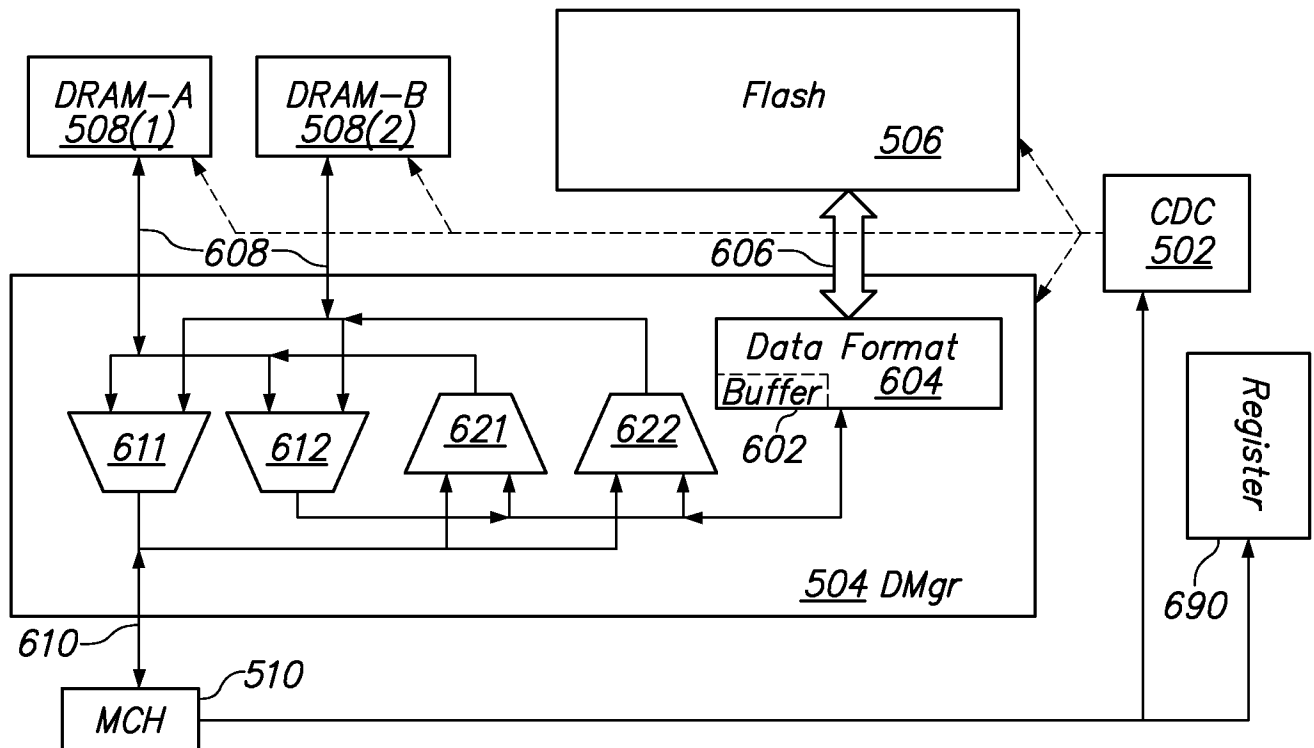


FIG. 6

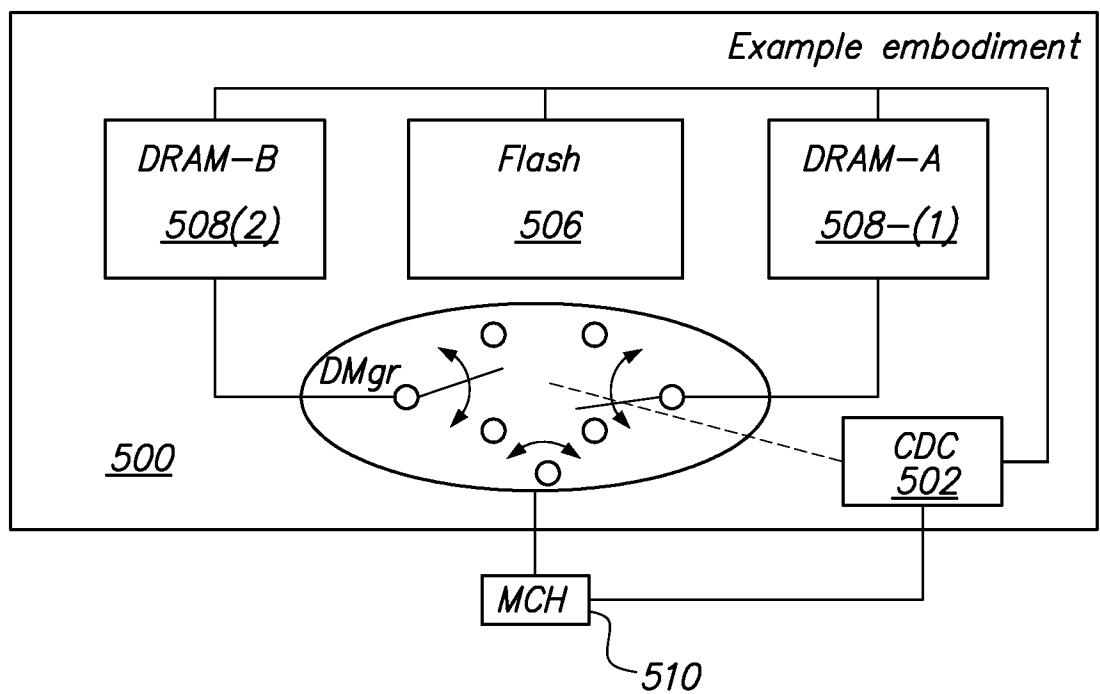
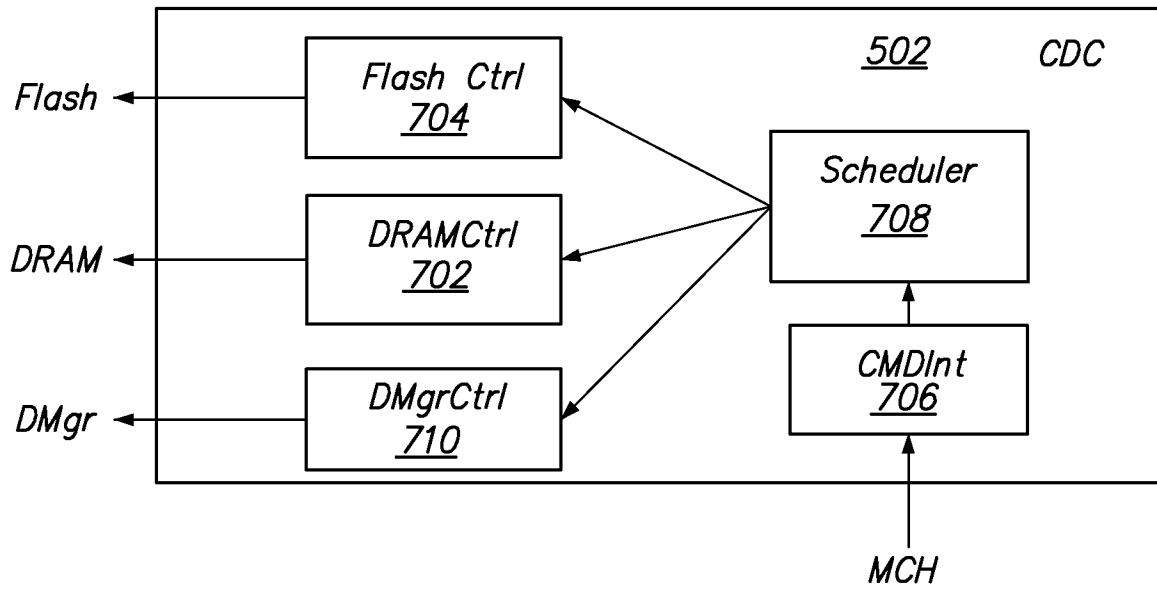
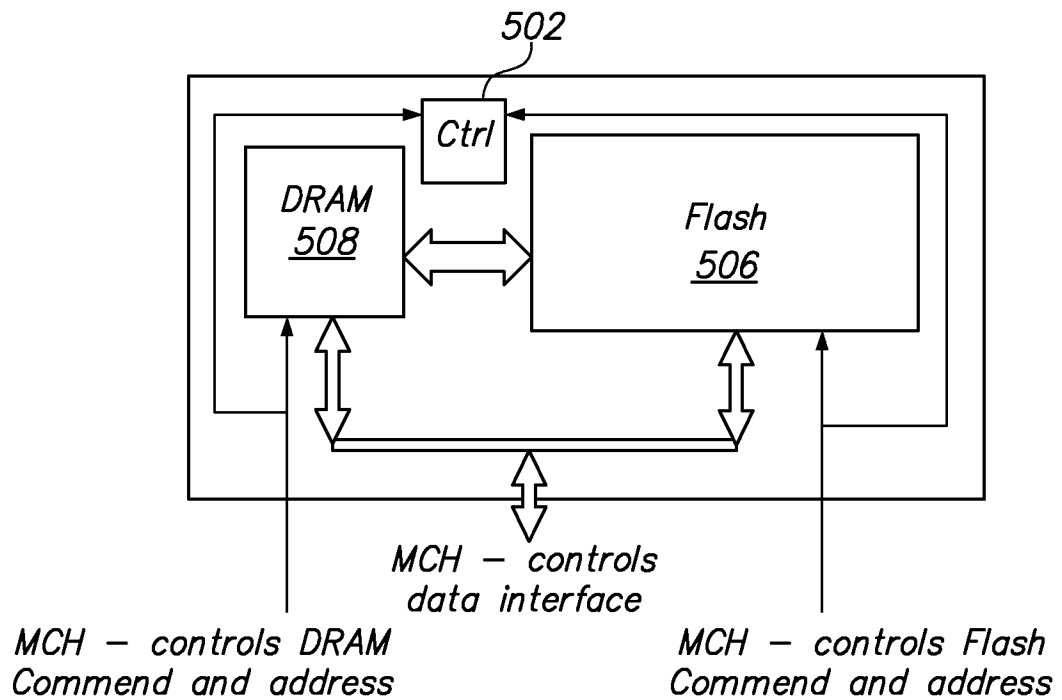


FIG. 5B

**FIG. 7****FIG. 8A**

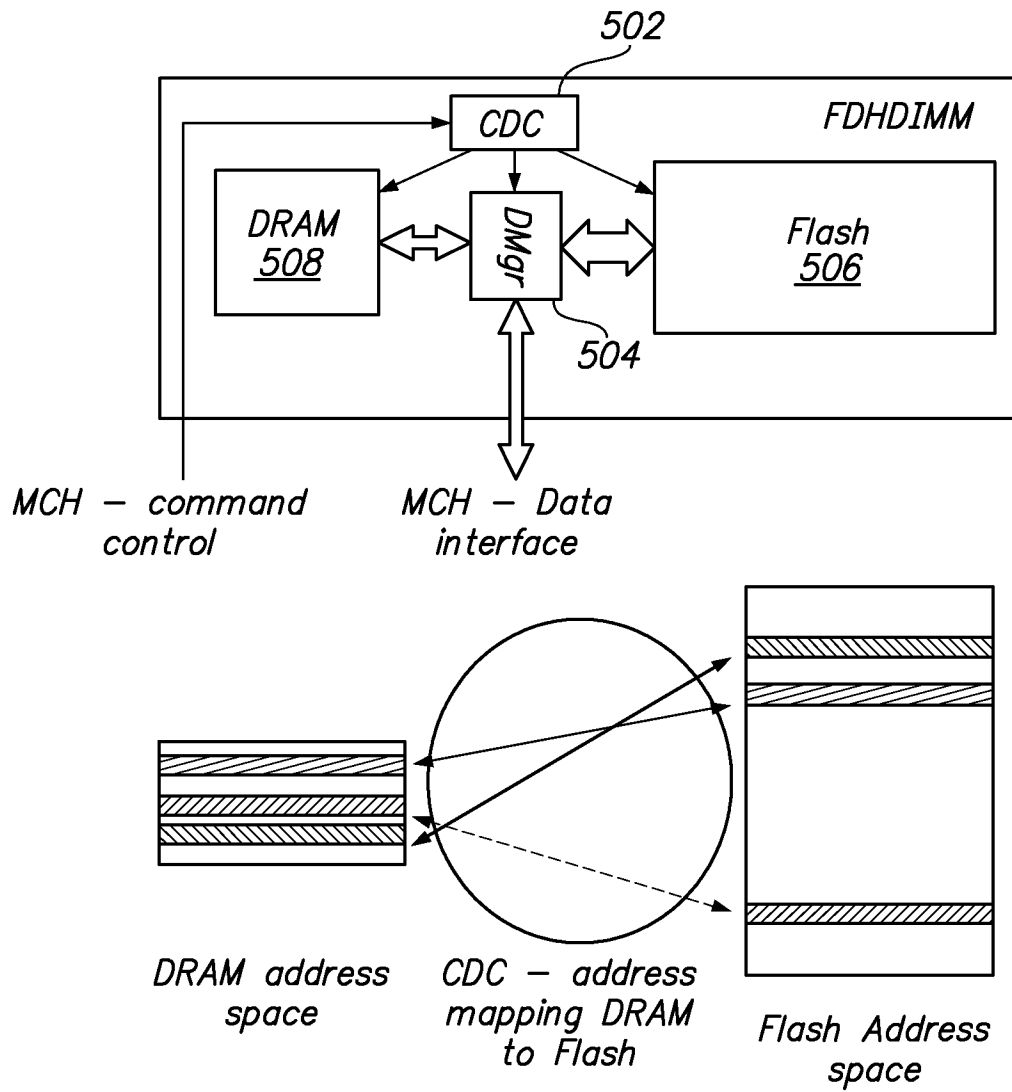
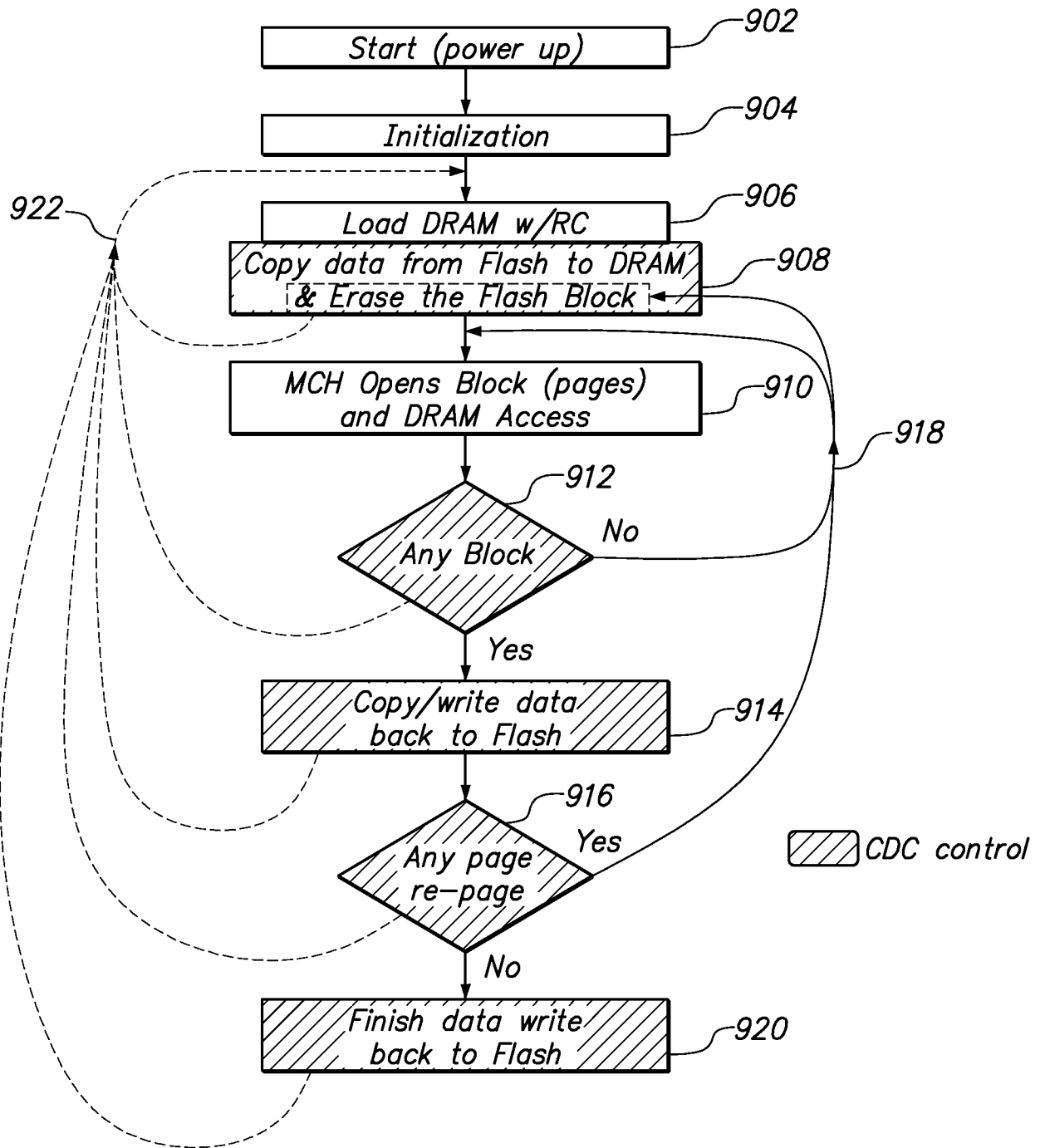
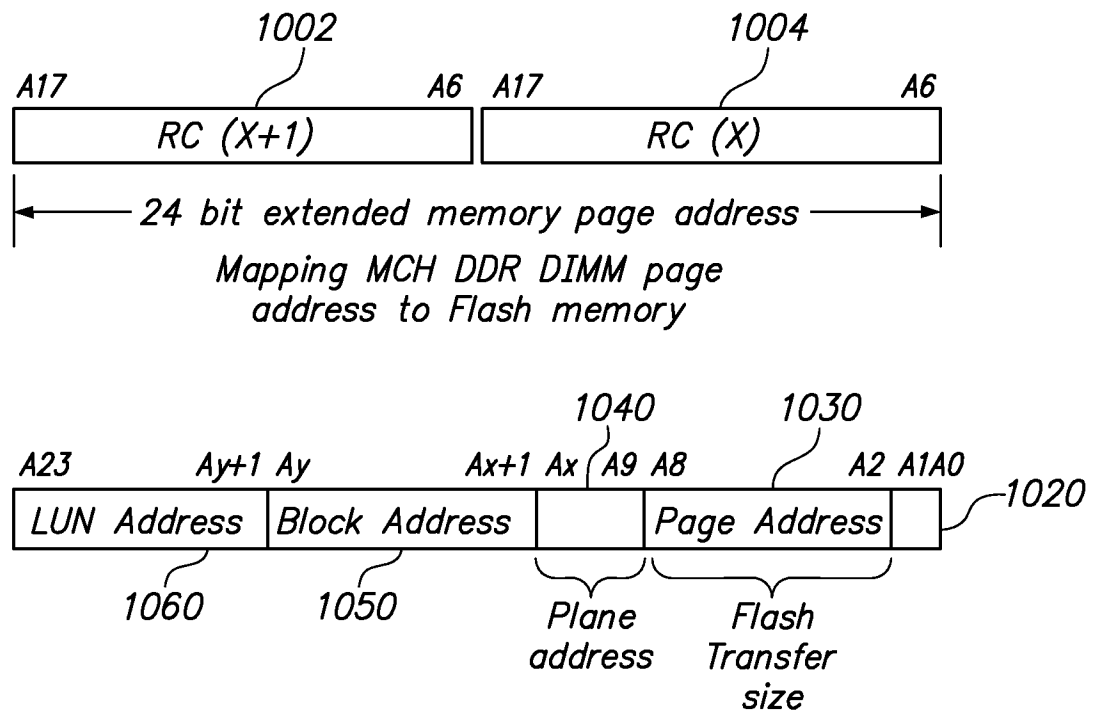


FIG. 8B

**FIG. 9**

**FIG. 10**

10/10

<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

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☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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	Jeffrey	C.	Solomon			
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

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City	Irvine	State/Province	CA	Country of Residence i	US
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For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Customer Number	46188		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	062453-032	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	10	Suggested Figure for Publication (if any)	

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	13559476	2012-07-26
Prior Application Status	Expired	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032		
		Application Number			
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE				

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
13559476	non provisional of	61512871	2011-07-28		
Prior Application Status	Patented	Remove			

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13559476	Continuation in part of	12240916	2008-09-29	8301833	2012-10-30
Prior Application Status	Abandoned	Remove			

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12240916	Continuation of	12131873	2008-06-02		
Prior Application Status	Expired	Remove			

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12131873	non provisional of	60941586	2007-06-01		

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
			Remove
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117		
☐ Person to whom the inventor is obligated to assign.	☐ Person who shows sufficient proprietary interest		
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor :			
If the Assignee is an Organization check here. ☒			
Organization Name	Netlist, Inc.		
Mailing Address Information:			
Address 1	175 Technology Drive		
Address 2	Suite 150		
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-032
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Email Address	
---------------	--

Additional Applicant Data may be generated within this form by selecting the Add button.

Add

Signature:

Remove

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications

Signature	/Khaled Shami/			Date (YYYY-MM-DD)	2014-09-17
First Name	Khaled	Last Name	Shami	Registration Number	38745

Additional Signature may be generated within this form by selecting the Add button.

Add

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	20170114
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	Flash-Dram Hybrid Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	17-SEP-2014
Filing Date:	
Time Stamp:	18:37:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453-032_Specification.pdf	179926 55ae34cd798357d30619643fdba04d1a75789cd	yes	39

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	36	
	Claims		37	38	
	Abstract		39	39	
Warnings:					
Information:					
2	Drawings-only black and white line drawings	032_Drawings.pdf	193289	no	10
			69585bc7e182e94597beb0985bb7cea3871856cf		
Warnings:					
Information:					
3	Application Data Sheet	062453-032_ADS.pdf	999310	no	7
			ce9bb22282542386ec7b650185b12422cabf6492		
Warnings:					
Information:					
Total Files Size (in bytes):			1372525		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/489,269

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/489,269	09/17/2014	Hyun Lee	062453-032

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 1077
FORMALITIES LETTER



Date Mailed: 09/24/2014

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **70** basic filing fee.
- \$ **70** surcharge.
- \$ **300** search fee.
- \$ **360** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **800** TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's

oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Chi-She Chen
Jeffrey C. Solomon
Scott H. Milton
Jayesh Bhakta

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/tmekuria/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/489,269	09/17/2014	2189	0.00	062453-032	1	1

CONFIRMATION NO. 1077

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 09/24/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/559,476 07/26/2012
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access - A proper **Authorization to Permit Access to Application by Participating Offices** (PTO/SB/39 or its equivalent) has been received by the USPTO.

If Required, Foreign Filing License Granted: 09/23/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/489,269**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

**** SMALL ENTITY ****

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

POWER OF ATTORNEY TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I hereby revoke all previous powers of attorney given in the application identified in the attached statement under 37 CFR 3.73(b).

I hereby appoint:

☒ Practitioners associated with the Customer Number:

46188

OR

☐ Practitioner(s) named below (if more than ten patent practitioners are to be named, then a customer number must be used):

Name	Registration Number	Name	Registration Number

as attorney(s) or agent(s) to represent the undersigned before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned only to the undersigned according to the USPTO assignment records or assignment documents attached to this form in accordance with 37 CFR 3.73(b).

Please change the correspondence address for the application identified in the attached statement under 37 CFR 3.73(b) to:

☒ The address associated with Customer Number:

46188

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

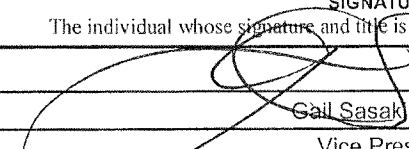
Assignee Name and Address:

Netlist, Inc.
51 Discovery, Suite 150
Irvine, CA 92618

A copy of this form, together with a statement under 37 CFR 3.73(b) (Form PTO/SB/96 or equivalent) is required to be filed in each application in which this form is used. The statement under 37 CFR 3.73(b) may be completed by one of the practitioners appointed in this form if the appointed practitioner is authorized to act on behalf of the assignee, and must identify the application in which this Power of Attorney is to be filed.

SIGNATURE of Assignee of Record

The individual whose signature and title is supplied below is authorized to act on behalf of the assignee

Signature		Date	7/10/12
Name	Gail Sasak	Telephone	949-435-0025
Title	Vice President and Chief Financial Officer		

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

ASSIGNMENT

WHEREAS, we, Hyun Lee, a citizen of the United States, residing at Ladera Ranch, California, Chi-She Chen, a citizen of Taiwan, residing at Walnut, California, Jeffrey C Solomon, a citizen of the United States, residing at Irvine, California, Scott H. Milton, a citizen of the United States, residing at Irvine, California, and Jayesh Bhakta, a citizen of the United States, residing at Cerritos, California, have invented a **"FLASH-DRAM HYBRID MEMORY MODULE"** for which we have executed application papers for a U.S. patent thereon, which was filed on July 26, 2012, Application No. 13/559,476; and

WHEREAS, ASSIGNEE., Netlist, Inc., a Delaware corporation, having a place of business located at 51 Discovery, Suite 150, Irvine, California 92618, is desirous of acquiring the exclusive right, title and interest in and to said invention and in and to the Letters Patent to be granted and issued therefor in the United States of America and its territories and possessions, and all countries foreign thereto;

NOW, THEREFORE, for a valuable consideration, the receipt of which is hereby acknowledged, we, Hyun Lee, Chi-She Chen, Jeffrey C Solomon, Scott H. Milton, and Jayesh Bhakta, do sell, assign, transfer and set over unto the said Assignee, its successors and assigns, the full and exclusive right, title and interest in and to said invention, and in and to any and all Letters Patent to be granted and issued therefor, not only for, to and in the United States of America, its territories and possessions, but also for, to and in all other countries including all priority rights under the International Convention, in any or all continuation, continuation-in-part, divisional and/or converted utility and design patent applications and

[ASSIGNMENT CONTINUATION]

Re: U.S. Patent Application entitled: **FLASH-DRAM HYBRID MEMORY MODULE**

resulting Letters Patents; and we hereby authorize and request the Commissioner of Patents and Trademarks to issue said Letters Patent to said ASSIGNEE Netlist, Inc., its successors and assigns, in accordance with this Assignment.

The undersigned hereby grants the firm of Nixon Peabody LLP, 2 Palo Alto Square, 3000 El Camino Real, Suite 500, Palo Alto, CA 94306 the power to insert on this assignment any further identification, including the application number and filing date, which may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.


Hyun Lee

2/19/2013
DATE

State of California, County of Orange
On February 19, 2013 before me, Judi Lowenthal, Notary Public,
Personally appeared HYUN LEE
who proved to me on the basis of satisfactory evidence to be the person(s)
whose name(s) is/are subscribed to the within instrument and acknowledged
to me that he/she/they executed the same in his/her/their authorized capacity(ies),
and that by his/her/their signature(s) on the instrument the person(s), or the entity
upon behalf of which the person(s) acted, executed the instrument.
I certify under PENALTY OF PERJURY under the laws of the State of California
that the foregoing paragraph is true and correct.
WITNESS my hand and official seal.

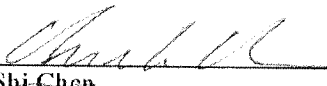


[ASSIGNMENT CONTINUATION]

Re: U.S. Patent Application entitled: **FLASH-DRAM HYBRID MEMORY MODULE**

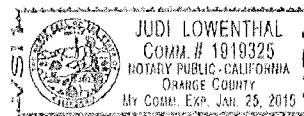
resulting Letters Patents; and we hereby authorize and request the Commissioner of Patents and Trademarks to issue said Letters Patent to said ASSIGNEE Netlist, Inc., its successors and assigns, in accordance with this Assignment.

The undersigned hereby grants the firm of Nixon Peabody LLP, 2 Palo Alto Square, 3000 El Camino Real, Suite 500, Palo Alto, CA 94306 the power to insert on this assignment any further identification, including the application number and filing date, which may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.


Chi-Shi Chen
Chi-She Chen

02-19-2013
DATE

State of California, County of Orange
On February 19, 2013 before me, Judi Lowenthal, Notary Public,
Personally appeared CHI-SHE CHEN
who proved to me on the basis of satisfactory evidence to be the person(s)
whose name(s) is/are subscribed to the within instrument and acknowledged
to me that he/she/they executed the same in his/her/their authorized capacity(ies),
and that by his/her/their signature(s) on the instrument the person(s), or the entity
upon behalf of which the person(s) acted, executed the instrument.
I certify under PENALTY OF PERJURY under the laws of the State of California
that the foregoing paragraph is true and correct.
WITNESS my hand and official seal.

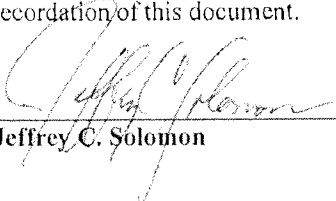


[ASSIGNMENT CONTINUATION]

Re: U.S. Patent Application entitled: **FLASH-DRAM HYBRID MEMORY MODULE**

resulting Letters Patents; and we hereby authorize and request the Commissioner of Patents and Trademarks to issue said Letters Patent to said ASSIGNEE Netlist, Inc., its successors and assigns, in accordance with this Assignment.

The undersigned hereby grants the firm of Nixon Peabody LLP, 2 Palo Alto Square, 3000 El Camino Real, Suite 500, Palo Alto, CA 94306 the power to insert on this assignment any further identification, including the application number and filing date, which may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.


Jeffrey C. Solomon

02-19-2013
DATE

State of California, County of Orange
On February 19, 2013 before me, Judi Lowenthal, Notary Public,
Personally appeared Jeffrey C. Solomon
who proved to me on the basis of satisfactory evidence to be the person(s)
whose name(s) is/are subscribed to the within instrument and acknowledged
to me that he/she/they executed the same in his/her/their authorized capacity(ies),
and that by his/her/their signature(s) on the instrument the person(s), or the entity
upon behalf of which the person(s) acted, executed the instrument.
I certify under PENALTY OF PERJURY under the laws of the State of California
that the foregoing paragraph is true and correct.
WITNESS my hand and official seal.



[ASSIGNMENT CONTINUATION]

Re: U.S. Patent Application entitled: **FLASH-DRAM HYBRID MEMORY MODULE**

resulting Letters Patents; and we hereby authorize and request the Commissioner of Patents and Trademarks to issue said Letters Patent to said ASSIGNEE Netlist, Inc., its successors and assigns, in accordance with this Assignment.

The undersigned hereby grants the firm of Nixon Peabody LLP, 2 Palo Alto Square, 3000 El Camino Real, Suite 500, Palo Alto, CA 94306 the power to insert on this assignment any further identification, including the application number and filing date, which may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.

Scott H. Milton
Scott Milton

02/19/2013
DATE

State of California, County of Orange
On February 17, 2013 before me, Judi Lowenthal, Notary Public,
Personally appeared SCOTT MILTON
who proved to me on the basis of satisfactory evidence to be the person(s)
whose name(s) is/are subscribed to this within instrument and acknowledged
to me that he/she/they executed the same in his/her/their authorized capacity(ies),
and that by his/her/their signature(s) on the instrument the person(s), or the entity
upon behalf of which the person(s) acted, executed the instrument.
I certify under PENALTY OF PERJURY under the laws of the State of California
that the foregoing paragraph is true and correct.
WITNESS my hand and official seal.

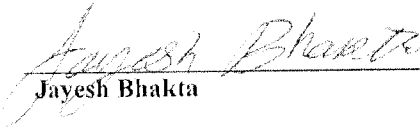


[ASSIGNMENT CONTINUATION]

Re: U.S. Patent Application entitled: **FLASH-DRAM HYBRID MEMORY MODULE**

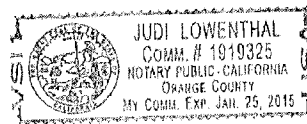
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The undersigned hereby grants the firm of Nixon Peabody LLP, 2 Palo Alto Square, 3000 El Camino Real, Suite 500, Palo Alto, CA 94306 the power to insert on this assignment any further identification, including the application number and filing date, which may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.


Jayesh Bhakta

2-19-2013
DATE

State of California, County of Orange
On FEBRUARY 19, 2013 before me, Judi Lowenthal, Notary Public,
Personally appeared JAYESH BHAKTA
who proved to me on the basis of satisfactory evidence to be the person(s)
whose name(s) is/are subscribed to the within instrument and acknowledged
to me that he/she/they executed the same in his/her/their authorized capacity(ies),
and that by his/her/their signature(s) on the instrument the person(s), or the entity
upon behalf of which the person(s) acted, executed the instrument.
I certify under PENALTY OF PERJURY under the laws of the State of California
that the foregoing paragraph is true and correct.
WITNESS my hand and official seal. Judi Lowenthal



STATEMENT UNDER 37 CFR 3.73(c)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 14/489,269Filed/Issue Date: 09/17/2014Titled: Flash-Dram Hybrid Memory ModuleNetlist, Inc., a Delaware Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that, for the patent application/patent identified above, it is (choose one of options 1, 2, 3 or 4 below):

1. ☒ The assignee of the entire right, title, and interest.
2. ☐ An assignee of less than the entire right, title, and interest (check applicable box):
- ☐ The extent (by percentage) of its ownership interest is _____%. Additional Statement(s) by the owners holding the balance of the interest must be submitted to account for 100% of the ownership interest.
- ☐ There are unspecified percentages of ownership. The other parties, including inventors, who together own the entire right, title and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

3. ☐ The assignee of an undivided interest in the entirety (a complete assignment from one of the joint inventors was made). The other parties, including inventors, who together own the entire right, title, and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

4. ☐ The recipient, via a court proceeding or the like (e.g., bankruptcy, probate), of an undivided interest in the entirety (a complete transfer of ownership interest was made). The certified document(s) showing the transfer is attached.

The interest identified in option 1, 2 or 3 above (not option 4) is evidenced by either (choose one of options A or B below):

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 034531, Frame 0385, or for which a copy thereof is attached.
- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:
1. From: _____ To: _____
- The document was recorded in the United States Patent and Trademark Office at
- Reel _____, Frame _____, or for which a copy thereof is attached.
2. From: _____ To: _____
- The document was recorded in the United States Patent and Trademark Office at
- Reel _____, Frame _____, or for which a copy thereof is attached.

[Page 1 of 2]

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(c)

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

4. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

5. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

6. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(c)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Khaled Shami/

Signature

Khaled Shami

Printed or Typed Name

April 24, 2015

Date

38,745

Title or Registration Number

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Susan Tamada			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	2051	1	70	70
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 5 months with \$0 paid	2255	1	1500	1500
Miscellaneous:				
Total in USD (\$)				2300

Electronic Acknowledgement Receipt

EFS ID:	22163642
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Susan Tamada
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	24-APR-2015
Filing Date:	17-SEP-2014
Time Stamp:	15:34:24
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 2300
RAM confirmation Number	2034
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Authorization for Extension of Time all replies	062453_032_FeeTrans_sb0017.pdf	256219	no	2
			ea1e6bc0b9505b0ec5640ba8603efd80ba4079f7		
Warnings:					
Information:					
2	Oath or Declaration filed	062453_032_Declarations.pdf	1575040	no	5
			03c2465164e5b57a9eecd8f058665658ae119fc3		
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
3	Transmittal Letter	Transmittal_for_POA_aia0082.pdf	156997	no	1
			bf1624c19a2eb5b0c3dcb6d9a9a0ea15a99b0d33		
Warnings:					
Information:					
4	Power of Attorney	063452_032_POA.pdf	285565	no	7
			5bd8dabff8c877a854d584c49e092b84cca5c374		
Warnings:					
Information:					
5	Assignee showing of ownership per 37 CFR 3.73	062453_032_373c_Statement.pdf	97017	no	2
			c79841400ac5b8d61672ee629e25d4175cd35fc3		
Warnings:					
Information:					
6	Fee Worksheet (SB06)	fee-info.pdf	38713	no	2
			a0288e0bda05fd583f3e7209d230ec5019bf45f0		
Warnings:					
Information:					
Total Files Size (in bytes):			2409551		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995 no persons are required to respond to a collection of information unless it displays a valid OMB control number

FEE TRANSMITTAL		Complete if known	
		Application Number	14/489,269
		Filing Date	September 17, 2014
☒ Applicant asserts small entity status. See 37 CFR 1.27.		First Named Inventor	Hyun LEE
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously.		Examiner Name	Not yet assigned
		Art Unit	2189
TOTAL AMOUNT OF PAYMENT	(\$ 800.00)	Practitioner Docket No.	062453-032

METHOD OF PAYMENT (check all that apply)
☐ Check ☐ Credit Card ☐ Money Order ☐ None ☐ Other (please identify): _____

☒ Deposit Account Deposit Account Number: 50-3557 Deposit Account Name: Nixon Peabody LLP

For the above-identified deposit account, the Director is hereby authorized to (check all that apply):

☒ Charge fee(s) indicated below ☐ Charge fee(s) indicated below, **except for the filing fee**
☒ Charge any additional fee(s) or underpayment of fee(s) ☒ Credit any overpayment of fee(s)
under 37 CFR 1.16 and 1.17

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

FEE CALCULATION**1. BASIC FILING, SEARCH, AND EXAMINATION FEES (U = undiscounted fee; S = small entity fee; M = micro entity fee)**

Application Type	FILING FEES			SEARCH FEES			EXAMINATION FEES			Fees Paid (\$)
	U (\$)	S (\$)	M (\$)	U (\$)	S (\$)	M (\$)	U (\$)	S (\$)	M (\$)	
Utility	280	140*	70	600	300	150	720	360	180	730.00
Design	180	90	45	120	60	30	460	230	115	
Plant	180	90	45	380	190	95	580	290	145	
Reissue	280	140	70	600	300	150	2,160	1,080	540	
Provisional	260	130	65	0	0	0	0	0	0	

* The \$140 small entity status filing fee for a utility application is further reduced to \$70 for a small entity status applicant who files the application via EFS-Web.

2. EXCESS CLAIM FEES

Fee Description	Undiscounted Fee (\$)	Small Entity Fee (\$)	Micro Entity Fee (\$)
Each claim over 20 (including Reissues)	80	40	20
Each independent claim over 3 (including Reissues)	420	210	105
Multiple dependent claims	780	390	195
Total Claims			
_____ - 20 or HP = _____ x _____ = _____			
HP = highest number of total claims paid for, if greater than 20.			
Indep. Claims			
_____ - 3 or HP = _____ x _____ = _____			
HP = highest number of independent claims paid for, if greater than 3.			

3. APPLICATION SIZE FEE

If the specification and drawings exceed 100 sheets of paper (excluding electronically filed sequence or computer listings under 37 CFR 1.52(e)), the application size fee due is \$400 (\$200 for small entity) (\$100 for micro entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).

Total Sheets	Extra Sheets	Number of each additional 50 or fraction thereof	Fee (\$)	Fee Paid (\$)
_____ - 100 = _____ / 50 = _____ (round up to a whole number) x _____ = _____				

4. OTHER FEE(S)

Non-English specification, \$130 fee (no small or micro entity discount)

Non-electronic filing fee under 37 CFR 1.16(t) for a utility application, \$400 fee (\$200 small or micro entity)

Other (e.g., late filing surcharge): Surcharge for late filing 70.00

SUBMITTED BY			
Signature	/Khaled Shami/	Registration No. (Attorney/Agent) 38,745	Telephone 650-320-7749
Name (Print/Type)	Khaled Shami	Date April 24, 2015	

This collection of information is required by 37 CFR 1.136. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

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LEGAL NAME OF INVENTOR

Inventor: Jeffrey C. Solomon

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

FLASH-DRAM HYBRID MEMORY MODULE

As the below named inventor, I hereby declare that:

This declaration
is directed to:



The attached application, or



United States application or PCT international application number 14/489269

filed on September 17, 2014

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: Scott H. Milton

Date (Optional): 10/17/2014

Signature: Scott H. Milton

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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LEGAL NAME OF INVENTOR

Inventor: Hyun Lee

Date (Optional) : _____

Signature: _____

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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LEGAL NAME OF INVENTOR

Inventor: Jayesh BhaktaDate (Optional): 9-17-2014Signature: Jayesh Bhakta

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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United States application or PCT international application number 14/489269

filed on September 17, 2014

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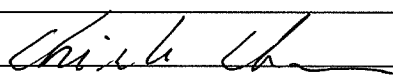
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LEGAL NAME OF INVENTOR

Inventor: Chi-She Chen

Date (Optional): Oct-17-2014

Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

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NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	14/489,269
Filing Date	September 17, 2015
First Named Inventor	Hyun LEE
Title	FLASH-DRAM HYBRID MEMORY MODULE
Art Unit	2189
Examiner Name	Not yet assigned
Attorney Docket Number	062453-032

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	April 24, 2015
Name	Khaled Shami	Registration Number	38,745
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee
SERIAL NO.: 14/489,269
FILING DATE: September 17, 2014 CONFIRMATION NO.: 1077
TITLE: Flash-Dram Hybrid Memory Module
EXAMINER: not assigned
ART UNIT: 2189

**Mail Stop AMENDMENT
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

PRELIMINARY AMENDMENT

Dear Sir:

Prior to the examination of the Subject Application, please amend the application as follows.

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 7.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1. (Canceled)
2. (New) A memory module comprising:
 - a first plurality of data signal lines forming a first data bus;
 - a second plurality of data signal lines forming a second data bus;
 - a third plurality of data signal lines forming a data bus;
 - a data manager coupled to the data bus, the first data bus, and the second data bus,wherein the memory module is couplable to a memory controller of a host system using the data bus, a control bus, and an address bus;
 - a non-volatile memory subsystem coupled to the data manager using the first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus;
 - a volatile memory subsystem coupled to the data manager using the second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus; and
 - a controller operable to receive one or more memory access commands from the memory controller of the host system by way of the control bus and the address bus, the controller operable to generate at least one of a first, second and third plurality of signals in response to the one or more memory access commands received from the memory controller of the host system, the controller operable to direct (i) operation of the non-volatile memory subsystem using the first plurality of signals, (ii) operation of the volatile memory subsystem using the second plurality of signals, and (iii) operation of the data manager using the third plurality of signals.
3. (New) The memory module of claim 2, wherein, in response to the first plurality of signals, data signals are communicated between the non-volatile memory subsystem and the data manager by way of the first data bus.

4. (New) The memory module of claim 2, wherein, in response to the second plurality of signals, data signals are communicated between the volatile memory subsystem and the data manager by way of the second data bus.

5. (New) The memory module of claim 2, wherein, in response to the third plurality of signals, data signals are communicated between the data manager and at least one of (i) the memory controller of the host system by way of the data bus, (ii) the non-volatile memory subsystem by way of the first data bus, and (iii) the volatile memory subsystem by way of the second data bus.

6. (New) The memory module of claim 2, wherein each of the data bus, the first data bus, and the second data bus includes bi-directional data signal lines.

7. (New) The memory module of claim 2, wherein the data manager further comprises a data formatting subsystem operable, based on control information received from the controller, to format data to be transferred between any two or more of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem.

8. (New) The memory module of claim 7, wherein the data manager further comprises a data formatting subsystem operable, in response to the third plurality of signals received from the controller, to format data to be transferred between any two or more of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem.

9. (New) The memory module of claim 2, wherein the controller is operable to receive the one or more memory access commands from the memory controller of the host system using a protocol of the volatile memory subsystem.

10. (New) The memory module of claim 6, wherein the protocol of the volatile memory subsystem is a JEDEC standard double data rate (DDR) protocol.

11. (New) The memory module of claim 10, wherein the volatile memory subsystem includes at least one memory circuit, memory device, or memory die, and the controller communicates with the volatile memory subsystem using the protocol of the volatile memory subsystem.

12. (New) The memory module of claim 10, wherein the controller communicates with the non-volatile memory subsystem using a protocol of the non-volatile memory subsystem.

13. (New) The memory module of claim 12, wherein the protocol of the non-volatile memory subsystem is different from the protocol of the volatile memory subsystem.

14. (New) The memory module of claim 2, wherein the memory module is in communication with the memory controller of the host system by way of a JEDEC standard double data rate (DDR) protocol.

15. (New) The memory module of claim 2, wherein the controller is operable to configure a unified memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, the controller operable to present the unified memory space as a volatile memory space to the memory controller of the host system.

16. (New) The memory module of claim 15, wherein the memory controller of the host system accesses the unified memory space using a standard protocol of the volatile memory subsystem.

17. (New) The memory module of claim 16, wherein the standard protocol of the volatile memory subsystem is a JEDEC standard double data rate (DDR) protocol used for accessing standard DDR DRAM.

18. (New) A memory module comprising:

a data manager couplable to a memory controller of a host system using a data bus, the data manager operable to communicate data signals with the memory controller of the host system by way of the data bus using a first protocol;

a non-volatile memory subsystem coupled to the data manager using a first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus using a second protocol;

a volatile memory subsystem coupled to the data manager using a second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus using the first protocol; and

a controller operable to receive one or more commands from the memory controller of the host system via a control bus and an address bus using the first protocol, and in response to the one or more commands from the memory controller of the host system, the controller is operable to direct (i) operation of the non-volatile memory subsystem using a first plurality of signals, (ii) operation of the volatile memory subsystem using a second plurality of signals, and (iii) operation of the data manager using a third plurality of signals,

wherein in response to any one or more of the first, second, and third plurality of signals the data manager communicates data signals with at least one of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem.

19. (New) The memory module of claim 18, wherein the first, second and third plurality of signals are generated by the controller in response to the one or more commands from the memory controller of the host system.

20. (New) The memory module of claim 18, wherein the first protocol is a JEDEC standard double data rate (DDR) protocol, and wherein the second protocol is different from the first protocol.

21. (New) The memory module of claim 18, wherein, based on instructions received from the controller using the third plurality of signals, the data manager is operable to control data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem.

22. (New) The memory module of claim 18, wherein, in response to the third plurality of signals received from the controller, the data manager is operable to control data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem.

23. (New) The memory module of claim 21, wherein the data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and a starting time of a data transfer.

24. (New) The memory module of claim 18, wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one or more of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem.

25. (New) The memory module of claim 24, wherein at least one set of data ports is operated by the data manager to independently or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

REMARKS

In this Amendment, claim 1 has been cancelled and claims 2-25 have been added. Entry and consideration of this Amendment are respectfully requested.

The USPTO is directed and authorized to charge all required fees, except for the Issue Fee and the Publication Fee, to Deposit Account No. 50-3557. Please also credit any overpayments to said Deposit Account.

Respectfully submitted,

Dated: May 15, 2015

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306
Tel. (650) 320-7700
Fax. (650) 320-7701

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Sherri Hale			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Claims in excess of 20	2202	4	40	160
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	22360318
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Sherri Hale
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	15-MAY-2015
Filing Date:	17-SEP-2014
Time Stamp:	14:39:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 160
RAM confirmation Number	867
Deposit Account	503557
Authorized User	SHAMI, KHALED

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.19 (Document supply fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453-032-Preliminary-Amendment-as-filed-May-15-2015.pdf	82723 f869f66e7f07efa5dfabf8bf86a1b5251aa4e38d	yes	7
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	7	

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30331 23a5f2e91de629916ed01bebad19930a58e096ef	no	2
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Warnings:

Information:

Total Files Size (in bytes):			113054
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/489,269

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	24 minus 20 =	* 4
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	160
x 210 =	0.00
	0.00
TOTAL	890

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/489,269	09/17/2014	2189	960	062453-032	24	2

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 1077
UPDATED FILING RECEIPT



Date Mailed: 05/20/2015

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 46188

Domestic Priority data as claimed by applicant

This application is a CON of 13/559,476 07/26/2012 PAT 8874831
which claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access - A proper **Authorization to Permit Access to Application by Participating Offices** (PTO/SB/39 or its equivalent) has been received by the USPTO.

If Required, Foreign Filing License Granted: 09/23/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/489,269**

Projected Publication Date: 08/27/2015

Non-Publication Request: No

Early Publication Request: No

**** SMALL ENTITY ****

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/489,269	09/17/2014	Hyun Lee	062453-032

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

CONFIRMATION NO. 1077
POA ACCEPTANCE LETTER



Date Mailed: 05/20/2015

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 04/24/2015.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ylueng/



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NOTICE OF ALLOWANCE AND FEE(S) DUE

46188 7590 07/08/2015
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

EXAMINER

ELMORE, STEPHEN C

ART UNIT

PAPER NUMBER

2138

DATE MAILED: 07/08/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/489,269

09/17/2014

Hyun Lee

062453-032

1077

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	10/08/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46188 7590 07/08/2015
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/489,269	09/17/2014	Hyun Lee	062453-032	1077

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	10/08/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2138	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/489,269	09/17/2014	Hyun Lee	062453-032	1077
46188	7590	07/08/2015	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ELMORE, STEPHEN C	
			ART UNIT	PAPER NUMBER
			2138	
DATE MAILED: 07/08/2015				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/489,269	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2138	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the Continuation Application filed 9/17/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2-25. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|---|
| <ol style="list-style-type: none"> 1. ☒ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | <ol style="list-style-type: none"> 5. ☒ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____. |
|---|---|

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2138

The present application is being examined under the pre-AIA first to invent provisions.

EXAMINER'S COMMENT and AMENDMENT, and REASONS FOR ALLOWANCE

Priority

Applicant claims Domestic Benefit to the following parent patent applications based upon the following continuity claim dependencies:

- a. Continuation of prior application 13/559,476 filed 7/26/2012;
- b. Non-provisional 61/512,871 filed 7/28/2011;
- c. Continuation-in-part of 12/240,916 filed 6/2/2008;
- d. Continuation-in-part of 12/131,873 filed June 2, 2008;
- e. Non-provisional 60/941,586 filed 6/1/2007.

Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, 365(c), or 386(c) is acknowledged. However, Applicant has not complied with one or more conditions for receiving the benefit of an earlier filing date under 35 U.S.C. 120 as follows:

The later-filed application must be an application for a patent for an invention which is also disclosed in the prior application (the parent or original non-provisional application or provisional application). The disclosure of the invention in the parent application and in the later-filed application must be sufficient to comply with the requirements of 35 U.S.C. 112(a) or the first paragraph of pre-AIA 35 U.S.C. 112, except for the best mode requirement. See *Transco Products, Inc. v. Performance Contracting, Inc.*, 38 F.3d 551, 32 USPQ2d 1077 (Fed. Cir. 1994).

The disclosures of the prior-filed applications, (c)-(e), i.e., Continuation-in-part of 12/240,916, Continuation-in-part of 12/131,873, and Non-provisional 60/941,586, fail to provide adequate support or enablement in the manner provided by 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph for one or more claims of this application.

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Specifically, Claims 1-25 of the present application recite the claimed feature "data manager" which lacks proper antecedent basis or clear support under 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph in the above identified parent applications (c)-(e).

The effective priority date that the present claims are entitled to therefore becomes 7/28/2011 based upon the earliest priority claim to Non-provisional 61/512,871.

EXAMINER'S AMENDMENT

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

The application has been amended as follows to update the current status of the parent applications identified in the Specification:

IN THE SPECIFICATION

At page 1, paragraph [0001], line 2, after "July 26, 2012," insert

-- now U.S. Patent 8,874,831, --;

At page 1, paragraph [0001], line 2, replace "titled ," with -- titled, --;

At page 1, paragraph [0001], line 4, after "September 29, 2008" insert

-- , now U.S. Patent 8,301,833, --;

At page 1, paragraph [0001], line 5, after "June 2, 2008," insert

-- now abandoned, --.

REASONS FOR ALLOWANCE

The following is an examiner's statement of reasons for allowance:

In independent claims 2 and 18 the following features taken **in combination with** the remaining limitations of the independent claim are not found in and/or are not obvious in view of the closest prior art of record, Song et al., U.S. Patent 8,102,614, giving the feature "data manager" the scope of meaning disclosed in the specification, page 26, paragraph [0088], which identifies the feature "data manager" additionally as element **DMgr 504**,

Claim 2,

"a data manager coupled to the data bus, the first data bus, and the second data bus, wherein the memory module is couplable to a memory controller of a host system using the data bus, a control bus, and an address bus; a non-volatile memory subsystem coupled to the data manager using the first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus; a volatile memory subsystem coupled to the data manager using the second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus" and "a controller operable to receive one or more memory access commands from the memory controller of the host system by way of the control bus and the address bus, the controller operable to generate at least one of a first, second and third plurality of signals in response to the one or more memory access commands received from the memory controller of the host system, the controller operable to direct (i) operation of the non-volatile memory subsystem using the first plurality of signals, (ii) operation of the volatile memory subsystem using the

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second plurality of signals, and (iii) operation of the data manager using the third plurality of signals";

Claim 18,

"a data manager couplable to a memory controller of a host system using a data bus"
and *"a non-volatile memory subsystem coupled to the data manager using a first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus using a second protocol; a volatile memory subsystem coupled to the data manager using a second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus using the first protocol; and a controller operable to receive one or more commands from the memory controller of the host system via a control bus and an address bus using the first protocol, and in response to the one or more commands from the memory controller of the host system, the controller is operable to direct (i) operation of the non-volatile memory subsystem using a first plurality of signals, (ii) operation of the volatile memory subsystem using a second plurality of signals, and (iii) operation of the data manager using a third plurality of signals, wherein in response to any one or more of the first, second, and third plurality of signals the data manager communicates data signals with at least one of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem".*

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

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Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mano Padmanabhan can be reached on (571) 272-4210. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2138

June 28, 2015

Notice of References Cited	Application/Control No. 14/489,269		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE		Art Unit 2138	Page 1 of 7

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2,043,099 A	06-1936	HANNA WILLIAM M	361/62
*	B	US-3,562,555 A	02-1971	Ahrons, Richard W.	365/229
*	C	US-3,916,390 A	10-1975	Chang et al.	365/183
*	D	US-4,234,920 A	11-1980	Van Ness et al.	714/22
*	E	US-4,420,821 A	12-1983	Hoffman, Charles R.	365/185.08
*	F	US-4,449,205 A	05-1984	Hoffman, Charles R.	365/185.08
*	G	US-4,965,828 A	10-1990	Ergott et al.	713/193
*	H	US-5,430,742 A	07-1995	Jeddeloh et al.	714/764
*	I	US-5,519,663 A	05-1996	Harper et al.	365/229
*	J	US-5,519,831 A	05-1996	Holzhammer, Gerald S.	714/22
*	K	US-5,563,839 A	10-1996	Herd et al.	365/227
*	L	US-5,577,213 A	11-1996	Avery et al.	710/100
*	M	US-5,619,644 A	04-1997	Crockett et al.	714/45

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Notice of References Cited	Application/Control No. 14/489,269		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE		Art Unit 2138	Page 2 of 7

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-5,675,725 A	10-1997	Malcolm, Peter B.	714/6.12
*	B	US-5,870,350 A	02-1999	Bertin et al.	365/233.11
*	C	US-5,874,995 A	02-1999	Naimpally et al.	375/240.25
*	D	US-5,890,192 A	03-1999	Lee et al.	711/103
*	E	US-5,953,215 A	09-1999	Karabatsos, Chris	361/767
*	F	US-6,023,421 A	02-2000	Clinton et al.	365/63
*	G	US-6,112,310 A	08-2000	Jun et al.	713/501
*	H	US-6,145,068 A	11-2000	Lewis, Timothy	711/170
*	I	US-6,158,015 A	12-2000	Klein, Dean A.	714/2
*	J	US-6,199,142 B1	03-2001	Saulsbury et al.	711/118
*	K	US-6,216,247 B1	04-2001	Creta et al.	714/763
*	L	US-6,269,382 B1	07-2001	Cabrera et al.	1/1
*	M	US-6,336,174 B1	01-2002	Li et al.	711/162

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U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-6,336,176 B1	01-2002	Leyda et al.	711/170
*	B	US-2002/0083368 A1	06-2002	Abe et al.	714/24
*	C	US-6,421,279 B1	07-2002	Tobita et al.	365/189.011
*	D	US-6,459,647 B1	10-2002	Kengeri, Subramani	365/230.03
*	E	US-6,487,102 B1	11-2002	Halbert et al.	365/51
*	F	US-6,487,623 B1	11-2002	Emerson et al.	710/302
*	G	US-2003/0158995 A1	08-2003	Lee et al.	711/105
*	H	US-6,658,507 B1	12-2003	Chan, Wan-Kan	710/100
*	I	US-6,691,209 B1	02-2004	O'Connell, Mark Andrew	711/114
*	J	US-6,769,081 B1	07-2004	Parulkar, Ishwardutt	714/733
*	K	US-2004/0163027 A1	08-2004	MacLaren et al.	714/764
*	L	US-2004/0190210 A1	09-2004	Leete, Brian A.	361/090
*	M	US-6,799,241 B2	09-2004	Kahn et al.	711/105

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-6,799,244 B2	09-2004	Tanaka et al.	711/113
*	B	US-2005/0044302 A1	02-2005	Pauley et al.	711/001
*	C	US-2005/0060488 A1	03-2005	Poechmueller, Peter	711/106
*	D	US-2005/0132250 A1	06-2005	Hansen et al.	714/005
*	E	US-2005/0141273 A1	06-2005	Park et al.	365/185.12
*	F	US-6,944,042 B2	09-2005	Komatsuzaki, Katsuo	365/145
*	G	US-6,948,029 B2	09-2005	Yano, Masatoshi	711/106
*	H	US-6,952,368 B2	10-2005	Miura et al.	365/185.11
*	I	US-2006/0039197 A1	02-2006	Khoury et al.	365/185.12
*	J	US-2006/0069896 A1	03-2006	Sanders, Richard	711/168
*	K	US-2006/0080515 A1	04-2006	Spiers et al.	711/162
*	L	US-7,053,470 B1	05-2006	Sellers et al.	257/678
*	M	US-7,062,618 B2	06-2006	Tsunoda et al.	711/154

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-7,089,412 B2	08-2006	Chen, Kong-Chen	713/2
*	B	US-7,102,391 B1	09-2006	Sun et al.	327/10
*	C	US-7,111,142 B2	09-2006	Spencer et al.	711/170
*	D	US-7,155,627 B2	12-2006	Matsui, Yoshinori	713/401
*	E	US-2006/0294295 A1	12-2006	Fukuzo, Yukio	711/105
*	F	US-7,200,021 B2	04-2007	Raghuram, Siva	365/51
*	G	US-7,234,099 B2	06-2007	Gower et al.	714/767
*	H	US-2007/0136523 A1	06-2007	Bonella et al.	711/113
*	I	US-2007/0192627 A1	08-2007	Oshikiri, Takashi	713/191
*	J	US-2008/0104344 A1	05-2008	Shimozono et al.	711/162
*	K	US-7,409,491 B2	08-2008	Doblar et al.	711/103
*	L	US-7,409,590 B2	08-2008	Moshayedi et al.	714/20
*	M	US-7,411,859 B2	08-2008	Sohn et al.	365/230.05

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*	A	US-2008/0195806 A1	08-2008	Cope, Bryan	711/111
*	B	US-7,421,552 B2	09-2008	Long, Matthew	711/162
*	C	US-7,467,251 B2	12-2008	Park et al.	710/307
*	D	US-2009/0031099 A1	01-2009	Sartore, Ronald H.	711/162
*	E	US-7,600,142 B2	10-2009	Ichikawa, Takeshi	713/322
*	F	US-7,716,411 B2	05-2010	Panabaker et al.	711/101
*	G	US-7,818,488 B2	10-2010	Park et al.	711/5
*	H	US-2010/0274953 A1	10-2010	LEE et al.	711/103
*	I	US-8,086,955 B2	12-2011	Zhou et al.	715/234
*	J	US-2011/0320804 A1	12-2011	Chan et al.	713/150
*	K	US-8,102,614 B2	01-2012	Song et al.	360/31
*	L	US-8,233,303 B2	07-2012	Best et al.	365/51
*	M	US-2012/0204079 A1	08-2012	Takefman et al.	714/758

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Notice of References Cited	Application/Control No. 14/489,269		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE		Art Unit 2138	Page 7 of 7

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-8,301,833 B1	10-2012	Chen et al.	711/104
*	B	US-8,412,879 B2	04-2013	Chang et al.	711/103
*	C	US-2014/0059170 A1	02-2014	Gasparakis et al.	709/217
*	D	US-8,874,831 B2	10-2014	Lee et al.	711/103
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Search Notes 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243;	6/28/2015	SE
G11C 7/1072; G11C 14/0018;	6/28/2015	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	6/28/2015	SE
365	185.33	6/28/2015	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	6/28/2015	SE
Inventor Name Search for DP	6/28/2015	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	6/28/2015	SE
G06F	12/0638	6/28/2015	SE
PGPUB Searched		6/28/2015	SE

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BIB DATA SHEET

CONFIRMATION NO. 1077

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
14/489,269	09/17/2014	711	2138	062453-032		
RULE						
APPLICANTS Netlist, Inc., Irvine, CA;						
INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;						
** CONTINUING DATA ***** This application is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007						
** FOREIGN APPLICATIONS *****						
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 09/23/2014						
Foreign Priority claimed	☐ Yes ☒ No	☐ Met after Allowance	STATE OR COUNTRY	SHEETS DRAWINGS	TOTAL CLAIMS	INDEPENDENT CLAIMS
35 USC 119(a-d) conditions met	☐ Yes ☒ No		CA	10	24	2
Verified and	/STEPHEN C ELMORE/	Initials				
Acknowledged	Examiner's Signature					
ADDRESS Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306 UNITED STATES						
TITLE FLASH-DRAM HYBRID MEMORY MODULE						
FILING FEE RECEIVED 960	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
			☐ Credit			

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	10570	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L2	2787	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L3	20672	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L4	31875	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L5	5435	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L6	604	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L7	12	L5 and L6	US-PGPUB; USPAT;	OR	ON	2015/06/28 18:17

			USOCR; FPRS; EPO; JPO; IBM_TDB			
L8	3	L7 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L9	2	(US-20070136523-\$).did. or (US-8412879-\$).did.	US- PGPUB; USPAT	OR	ON	2015/06/28 18:17
L10	173369	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L11	90	L6 and L10	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L12	2	L5 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L13	4	L4 and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L14	356	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L15	2	L14 with (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS;	OR	ON	2015/06/28 18:17

			EPO; JPO; IBM_TDB			
L16	20	(data near3 (port or input-output or I/O or IO)) and L11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L17	2	L5 and L16	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L18	15	L16 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L19	0	L4 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L20	15	L6 and L18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L21	0	L14 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L22	0	L5 and L20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L23	31939	(control adj information or control adj data or control adj meta-data or control	US- PGPUB;	OR	ON	2015/06/28 18:17

		adj metadata) near3 controller	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L24	2	((control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller) and L16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L25	2	L6 and L23 and L10 and L4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L26	2	L6 and L23 and L10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L27	8	L6 and L23	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L28	6304	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:17
L29	19	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:17
L30	101	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:17
L31	165	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:17
L32	90	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:17
L33	2	(data adj manager same controller same memory adj controller) and L5 and (L10	US-PGPUB;	OR	ON	2015/06/28 18:17

		or L14)	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
L34	624	(hybrid near3 memory).ti.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L35	7	L6 and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L36	0	L35 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L37	7	(data adj manager and memory adj controller) and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L38	0	L37 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L39	2	(data adj manager) and L34 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L40	37	("20080195806" "6658507" "5675725" "20040190210" "6336176" "7409590" "20100274953" "6336174" "5519663" "6487623" "20080104344" "4420821" "6799244" "20020083368" "4449205" "8301833" "7111142" "20070192627" "6158015" "20120204079").PN.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17

L41	1	L6 and L40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L42	110	("2043099" "20030158995" "20040163027" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20090031099" "3562555" "3916390" "4234920" "4965828" "5430742" "5519831" "5563839" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6199142" "6216247" "6421279" "6459647" "6487102" "6769081" "6799241" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7155627" "7200021" "7234099" "7409491" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8233303").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L43	0	L14 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L44	0	L16 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L45	0	L6 and L42	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L46	1	"8874831".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:17
L47	8	"8301833".pn.	US-PGPUB; USPAT;	OR	ON	2015/06/28 18:17

			USOCR; FPRS; EPO; JPO; IBM_TDB			
L48	1	(US-8301833-\$).did.	USPAT	OR	ON	2015/06/28 18:17
L60	3	7 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:20
L61	1	"8874831".pn.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:22
L62	82	("20020083368" "20030158995" "20040163027" "20040190210" "20050044302" "20050060488" "20050132250" "20050141273" "20060039197" "20060069896" "20060080515" "20060294295" "20070136523" "20070192627" "20080104344" "20080195806" "20090031099" "20100274953" "20110320804" "20120204079" "20140059170" "2043099" "3562555" "3916390" "4234920" "4420821" "4449205" "4965828" "5430742" "5519663" "5519831" "5563839" "5577213" "5619644" "5675725" "5870350" "5874995" "5890192" "5953215" "6023421" "6112310" "6145068" "6158015" "6199142" "6216247" "6269382" "6336174" "6336176" "6421279" "6459647" "6487102" "6487623" "6658507" "6691209" "6769081" "6799241" "6799244" "6944042" "6948029" "6952368" "7053470" "7062618" "7089412" "7102391" "7111142" "7155627" "7200021" "7234099" "7409491" "7409590" "7411859" "7421552" "7467251" "7600142" "7716411" "7818488" "8086955" "8102614" "8233303" "8301833" "8412879").PN.	US- PGPUB; USPAT; USOCR	OR	ON	2015/06/28 18:22
L63	8	"8301833".pn.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:24

L64	1	(US-8301833-\$).did.	USPAT	OR	ON	2015/06/28 18:24
L65	1	(US-8102614-\$).did.	USPAT	OR	ON	2015/06/28 18:26
L66	181	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:29
L67	1	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and 66	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:31
L68	9	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and 34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:32
L69	8	68 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:32
L70	0	37 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:33
L72	1053	G06F12/0638.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:34
L73	13038	G06F12/0246.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:34
L74	2893	G06F13/4243.CPC.	US-	OR	ON	2015/06/28

			PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			18:34
L75	6209	G11C7/1072.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:34
L76	308	G11C14/0018.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:35
L77	23101	72 or 73 or 74 or 75 or 76	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:35
L78	2	66 and 77	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:35
L79	0	78 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:35
L80	9	5 and 68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:37
L81	0	6 and 80	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2015/06/28 18:38

			IBM_TDB			
L82	0	(data adj manager) and 80	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:38
L88	0	68 and 77	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:41
L89	91	(data adj manager) and 77	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:41
L90	0	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and 89	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/06/28 18:41

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L49	20	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L50	101	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L51	121	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L52	91	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L53	6309	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L54	6543	L49 or L50 or L51 or L52 or L53	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L55	7	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L56	2	L54 and L55	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L57	12646	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L58	43	L54 and L57	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17

L59	2	L55 and L58	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:17
L71	10543	711/103.ccls.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:33
L83	1520	(data adj manager).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:38
L84	468	G06F12/0638.CPC.	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:38
L85	14	54 and 71	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:39
L87	2	83 and 85	US-PGPUB; USPAT; UPAD	OR	ON	2015/06/28 18:39

6/ 28/ 2015 6:42:09 PM
C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 14489269.wsp

Issue Classification 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2138	

CPC						
Symbol					Type	Version
G06F		12	/	0638	F	2013-01-01
G06F		12	/	0246	I	2013-01-01
G11C		14	/	0018	I	2013-01-01
G11C		7	/	1072	I	2013-01-01
G06F		13	/	4243	I	2013-01-01
G06F		2212	/	7208	A	2013-01-01
G06F		2212	/	205	A	2013-01-01
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CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE		Total Claims Allowed: 24	
(Assistant Examiner) /STEPHEN ELMORE/ Primary Examiner.Art Unit 2138 (Primary Examiner)	(Date) 6/28/2015 (Date)		

Issue Classification 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	2		18												
	3		19												
	4		20												
	5		21												
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	10														
	11														
	12														
	13														
	14														
	15														
	16														
	17														

NONE	Total Claims Allowed: 24	
(Assistant Examiner)	(Date)	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2138	6/28/2015	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 6

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Application Number	14/489,269
Filing Date	2014-09-17
First Named Inventor	Hyun Lee
Art Unit	2138
Examiner Name	Stephen C. Elmore
Attorney Docket Number	062453-032

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Applicant

Attorney or agent of record. Registration Number 38,745.

Registered practitioner named in the application transmittal papers who acts in a representative capacity under 37 CFR 1.34. See 37 CFR 1.33(a)(1). Registration Number _____.

Signature /Khaled Shami/

Typed or Printed
Name Khaled Shami

Date July 16, 2015

Telephone (202) 585-8000

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.
Submit multiple forms if more than one signature is required, see below*.

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: **Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	22933547
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/George Hinton
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	16-JUL-2015
Filing Date:	17-SEP-2014
Time Stamp:	10:31:25
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Change of Address	062453-032_Change_of_Correspondence_Address_14489269.pdf	203324 d9a26bd56724723bc9cdffa718/fa86e3a89d583	no	2

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** Mail Stop ISSUE FEE
 Commissioner for Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46186 25091 07/08/2015
 Nixon Peabody LLP
 P.O. Box 60610
 Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/489,269	09/17/2014	Hyun Lee	062451-032	1077

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPL. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	10/08/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2138	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (for Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
 (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

Nixon Peabody LLP

Khaled Shami

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE (CITY and STATE OR COUNTRY)

NETLIST, INC.

Irvine, California

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☒ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number 19-2380 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.37
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Khaled Shami/
 Typed or printed name Khaled Shami

Date August 3, 2015
 Registration No. 38,745

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/George Hinton			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	2501	1	480	480

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	23077446
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/George Hinton
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	03-AUG-2015
Filing Date:	17-SEP-2014
Time Stamp:	17:58:17
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$480
RAM confirmation Number	5061
Deposit Account	192380
Authorized User	NIXON PEABODY LLP

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	20150803_Issue_Fee_062453-00032.pdf	339997	no	1
			381a77526629c8c6defc905aa896f1db7a96c529		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30350	no	2
			af90268355be3b77ccfce0b240b0e176aaea041a		

Warnings:

Information:

Total Files Size (in bytes):			370347
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/489,269	09/08/2015	9128830	062453-032	1077

22204 7590 08/19/2015
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/489,269	09/17/2014	Hyun Lee	062453-032

CONFIRMATION NO. 1077

PUBLICATION NOTICE



OC000000077119706

22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2015-0242313-A1

Publication Date:08/27/2015

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	14/489,269	Filing Date	2014-09-17	Docket Number (if applicable)	062453-032	Art Unit	2138
First Named Inventor	Hyun Lee			Examiner Name	Stephen C. Elmore		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☐ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____.
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No. 192380

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Doc code: RCEX

Doc description: Request for Continued Examination (RCE)

PTO/SB/30EFS (07-14)

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-08-31
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Doc Code: PET.AUTO Document Description: Petition automatically granted by EFS-Web		PTO/SB/140 U.S. Patent and Trademark Office Department of Commerce	
Electronic Petition Request	PETITION TO WITHDRAW AN APPLICATION FROM ISSUE AFTER PAYMENT OF THE ISSUE FEE UNDER 37 CFR 1.313(c)		
Application Number	14489269		
Filing Date	17-Sep-2014		
First Named Inventor	Hyun Lee		
Art Unit	2138		
Examiner Name	STEPHEN ELMORE		
Attorney Docket Number	062453-032		
Title	FLASH-DRAM HYBRID MEMORY MODULE		
An application may be withdrawn from issue for further action upon petition by the applicant. To request that the Office withdraw an application from issue, applicant must file a petition under this section including the fee set forth in § 1.17(h) and a showing of good and sufficient reasons why withdrawal of the application from issue is necessary. APPLICANT HEREBY PETITIONS TO WITHDRAW THIS APPLICATION FROM ISSUE UNDER 37 CFR 1.313(c). A grantable petition requires the following items: (1) Petition fee; and (2) One of the following reasons: (a) Unpatentability of one or more claims, which must be accompanied by an unequivocal statement that one or more claims are unpatentable, an amendment to such claim or claims, and an explanation as to how the amendment causes such claim or claims to be patentable; (b) Consideration of a request for continued examination in compliance with § 1.114 (for a utility or plant application only); or (c) Express abandonment of the application. Such express abandonment may be in favor of a continuing application, but not a CPA under 37 CFR 1.53(d).			
Petition Fee			
☒ Small Entity			
☐ Micro Entity			
☐ Regular Undiscounted			
Reason for withdrawal from issue			

- ☐ One or more claims are unpatentable
- ☒ Consideration of a request for continued examination (RCE) (List of Required Documents and Fees)
- ☐ Applicant hereby expressly abandons the instant application (any attorney/agent signing for this reason must have power of attorney pursuant to 37 CFR 1.32(b)).

RCE request, submission, and fee.

- ☐ I certify, in accordance with 37 CFR 1.4(d)(4) that:
- ☐ The RCE request, submission, and fee have already been filed in the above-identified application on
- ☒ Are attached.

THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

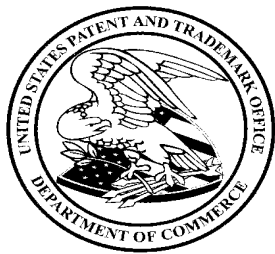
- ☒ An attorney or agent registered to practice before the Patent and Trademark Office who has been given power of attorney in this application.
- ☐ An attorney or agent registered to practice before the Patent and Trademark Office, acting in a representative capacity.
- ☐ A sole inventor
- ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application
- ☐ A joint inventor; all of whom are signing this e-petition

Signature	/Khaled Shami/
Name	Khaled Shami
Registration Number	38745

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/George Hinton			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Petition Fee-37CFR 1.17(h) (Group II)	2464	1	70	70
Request for Continued Examination	2801	1	600	600
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				670



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Decision Date : August 31, 2015

In re Application of :

Hyun Lee

DECISION ON PETITION

UNDER CFR 1.313(c)(2)

Application No : 14489269

Filed : 17-Sep-2014

Attorney Docket No : 062453-032

This is an electronic decision on the petition under 37 CFR 1.313(c)(2), filed August 31, 2015 , to withdraw the above-identified application from issue after payment of the issue fee.

The petition is **GRANTED**.

The above-identified application is withdrawn from issue for consideration of a submission under 37 CFR 1.114 (request for continued examination). See 37 CFR 1.313(c)(2).

Petitioner is advised that the issue fee paid in this application cannot be refunded. If, however, this application is again allowed, petitioner may request that it be applied towards the issue fee required by the new Notice of Allowance.

Telephone inquiries concerning this decision should be directed to the Patent Electronic Business Center (EBC) at 866-217-9197.

This application file is being referred to Technology Center AU 2138 for processing of the request for continuing examination under 37 CFR 1.114 .

Office of Petitions

Electronic Acknowledgement Receipt

EFS ID:	23348298
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/George Hinton
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	31-AUG-2015
Filing Date:	17-SEP-2014
Time Stamp:	16:50:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$670
RAM confirmation Number	3731
Deposit Account	192380
Authorized User	NIXON PEABODY LLP

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	20150831_RCE_062453-032.pdf	1363955	no	3
			229f403e0cba510d08b0143d5500557fa8d0e6b8		
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	20150831_IDS_062453-032.pdf	625879	no	16
			5f7306eedc4ceda528c759f5c25fd059994bba86		
Warnings:					
Information:					
3	Quick Path Information Disclosure Statement	20150831_QPIDS_Statement_062453-032.pdf	166106	no	2
			6c8974189d3e6e709c26aa1ae45a47f4dfab961b		
Warnings:					
Information:					
4	Petition automatically granted by EFS	petition-request.pdf	31498	no	2
			6ba94e9dab0198fe4e072b9045d023ccfefa18d3		
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	32325	no	2
			021c4fd92dee92967ea038cfc791205225fe181		
Warnings:					
Information:					
Total Files Size (in bytes):			2219763		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14489269
	Filing Date		2014-09-17
	First Named Inventor	Hyun Lee	
	Art Unit	2138	
	Examiner Name	Stephen C. Elmore	
	Attorney Docket Number	062453-032	

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	1	20020083368		2002-06-27	Abe et al.	

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Attorney Docket Number	062453-032

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- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☒ A certification statement is not submitted herewith.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-08-31
Name/Print	Khaled Shami	Registration Number	38745

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30323

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14489269

PATENT NUMBER

9128830

TITLE

FLASH-DRAM HYBRID MEMORY MODULE

NAME AND ADDRESS

HYUN LEE
Ladera Ranch, CA

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Auto-petition to withdraw - Granted

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14/489,269	09/17/2014	Hyun Lee	062453-032	1077

22204 7590 09/15/2015
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799 Ninth Street, NW
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Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Corrected Notice of Allowability	Application No. 14/489,269	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2138	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the RCE and QPIDS Form PTO/SB/09 filed 8/31/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2-25. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>8/31/2015</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other ____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. | |

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2138

The present application is being examined under the pre-AIA first to invent provisions.

EXAMINER'S COMMENT and REASONS FOR ALLOWANCE

Priority

1. Applicant claims Domestic Benefit to the following parent patent applications based upon the following continuity claim dependencies:

- a. Continuation of prior application 13/559,476 filed 7/26/2012;
- b. Non-provisional 61/512,871 filed 7/28/2011;
- c. Continuation-in-part of 12/240,916 filed 6/2/2008;
- d. Continuation-in-part of 12/131,873 filed June 2, 2008;
- e. Non-provisional 60/941,586 filed 6/1/2007.

Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, 365(c), or 386(c) is acknowledged. However, Applicant has not complied with one or more conditions for receiving the benefit of an earlier filing date under 35 U.S.C. 120 as follows:

The later-filed application must be an application for a patent for an invention which is also disclosed in the prior application (the parent or original non-provisional application or provisional application). The disclosure of the invention in the parent application and in the later-filed application must be sufficient to comply with the requirements of 35 U.S.C. 112(a) or the first paragraph of pre-AIA 35 U.S.C. 112, except for the best mode requirement. See *Transco Products, Inc. v. Performance Contracting, Inc.*, 38 F.3d 551, 32 USPQ2d 1077 (Fed. Cir. 1994).

The disclosures of the prior-filed applications, 1.(c)-(e), i.e., Continuation-in-part of 12/240,916, Continuation-in-part of 12/131,873, and Non-provisional 60/941,586, fail to provide adequate support or enablement in the manner provided by 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph for one or more claims of this application.

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Specifically, Claims 2-25 of the present application recite the claimed feature "data manager" which lacks proper antecedent basis or clear support under 35 U.S.C. 112(a) or pre-AIA 35 U.S.C. 112, first paragraph in the above identified parent applications, specifically, items 1.(c)-(e).

The effective priority date that claims 2-25 are entitled to therefore becomes 7/28/2011 based upon the earliest proper priority claim being that to Non-provisional 61/512,871.

***Request for Continuation Examination (RCE) under 37 CFR 1.114
and Certification and Request For Consideration of an Information Disclosure Statement
Filed After Payment of the Issue Fee Under the QPIDS Pilot Program***

2. Applicant filed a Request for Continued Examination (RCE) of the present application under 37 CFR 1.114 on 8/31/2015 which included Certification Form PTO/SB/09 and an Information Disclosure Statement (IDS), also filed 8/31/2015.

3. Applicant's Certification Form PTO/SB/09 is improper because Applicant failed to check the appropriate selection in item 2 of the form by which the Applicant certifies the status of the information contained in the IDS as complying with one of either 37 CFR 1.97(e)(1) or 37 CFR 1.97(e)(2). Applicant must check one of the two boxes in item 2 otherwise Form PTO/SB/09, lacking such certification, means the form is incomplete and improper.

Information Disclosure Statement

4. The information disclosure statement filed 8/31/2015 fails to comply with 37 CFR 1.97(d) because it lacks a statement as specified in 37 CFR 1.97(e), see paragraph 3 above.

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It has been placed in the application file, but the information referred to therein has not been considered.

REASONS FOR ALLOWANCE

The following is an examiner's statement of reasons for allowance:

In independent claims 2 and 18 the following features taken **in combination with** the remaining limitations of the independent claim are not found in and/or are not obvious in view of the closest prior art of record, Song et al., U.S. Patent 8,102,614, giving the feature "data manager" the scope of meaning disclosed in the specification, page 26, paragraph [0088], which identifies the feature "data manager" additionally as element **DMgr 504**,

Claim 2,

"a data manager coupled to the data bus, the first data bus, and the second data bus, wherein the memory module is couplable to a memory controller of a host system using the data bus, a control bus, and an address bus; a non-volatile memory subsystem coupled to the data manager using the first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus; a volatile memory subsystem coupled to the data manager using the second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus" and "a controller operable to receive one or more memory access commands from the memory controller of the host system by way of the control bus and the address bus, the controller operable to generate at least one of a first, second and third plurality of signals in response to the one or more memory access commands received from the memory controller of the host system, the controller operable to direct (i) operation of the non-volatile memory subsystem

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using the first plurality of signals, (ii) operation of the volatile memory subsystem using the second plurality of signals, and (iii) operation of the data manager using the third plurality of signals";

Claim 18,

"a data manager couplable to a memory controller of a host system using a data bus"
and *"a non-volatile memory subsystem coupled to the data manager using a first data bus, the non-volatile memory subsystem operable to communicate data signals with the data manager by way of the first data bus using a second protocol; a volatile memory subsystem coupled to the data manager using a second data bus, the volatile memory subsystem operable to communicate data signals with the data manager by way of the second data bus using the first protocol; and a controller operable to receive one or more commands from the memory controller of the host system via a control bus and an address bus using the first protocol, and in response to the one or more commands from the memory controller of the host system, the controller is operable to direct (i) operation of the non-volatile memory subsystem using a first plurality of signals, (ii) operation of the volatile memory subsystem using a second plurality of signals, and (iii) operation of the data manager using a third plurality of signals, wherein in response to any one or more of the first, second, and third plurality of signals the data manager communicates data signals with at least one of the memory controller of the host system, the volatile memory subsystem, and the non-volatile memory subsystem".*

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue

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fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mano Padmanabhan can be reached on (571) 272-4210. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2138

September 6, 2015

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number 14489269

Filing Date 2014-09-17

First Named Inventor Hyun Lee

Art Unit 2138

Examiner Name Stephen C. Elmore

Attorney Docket Number 062453-032

U.S. PATENTS

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	2043099		1986-02-02	Hanna	
	2	3562555		1971-02-09	Ahrong	
	3	3916390		1975-10-28	Chang et al.	
	4	4234920		1980-11-18	Van Ness et al.	
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	7	4965828		1990-10-23	Ergott, Jr. et al.	
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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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Filing Date	2014-09-17
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Art Unit	2138
Examiner Name	Stephen C. Elmore
Attorney Docket Number	062453-032

9	5519663		1996-05-21	Harper, Jr. et al.	
10	5519831	A	1996-05-21	Holzhammer	
11	5563839		1996-10-08	Herd et al.	
12	5577213		1996-11-19	Avery et al.	
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14	5675725		1997-10-07	Malcolm	
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16	5874995		1999-02-23	Naimpally et al.	
17	5890192		1999-03-30	Lee et al.	
18	5953215		1999-09-14	Karabatsos	
19	6023421		2000-02-08	Clinton et al.	

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21	6145068		2000-11-07	Lewis	
22	6158015		2000-12-05	Klein	
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24	6216247	B1	2001-04-10	Oreta et al.	
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57	7716411	B2	2010-05-11	Panabaker et al.	
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62	8301833		2012-10-30	Chen et al.	
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64	8412879		2013-04-02	Chang et al.	
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15	20070136523		2007-06-14	Bonella et al.	
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17	20080104344		2008-05-01	Shimozono et al.	
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20	20100274953		2010-10-28	Lee et al.	
21	20110320804		2011-12-29	Chan et al.	
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	1	2013016723	WO	A3	2013-01-31	NETLIST INC.		☐
	2	2737383	EP	A2	2014-06-04	NETLIST INC.		☐

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	1	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011.	☐
	2	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012.	☐
	3	Advisory Action in U.S. Patent Application No. 12/240,916, mailed March 13, 2012.	☐
	4	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012.	☐
	5	Office Action in U.S. Patent Application No. 13/536,173, mailed April 15, 2013.	☐
	6	Office Action in U.S. Patent Application No. 13/905,048, mailed August 1, 2013.	☐
	7	Notice of Allowance in U.S. Patent Application No. 13/559,476, mailed May 6, 2014.	☐
	8	International Search Report and Written Opinion in PCT/US12/48750, dated October 10, 2012.	☐
	9	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014.	☐

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10	Office Action in U.S. Patent Application No. 13/625,563, mailed August 5, 2013.	☐
11	Office Action in U.S. Patent Application No. 13/625,563, mailed May 9, 2014.	☐
12	Office Action in U.S. Patent Application No. 13/905,053, mailed August 1, 2013.	☐
13	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014.	☐
14	Office Action in U.S. Patent Application No. 14/173,212, mailed March 14, 2014.	☐
15	JEDEC Standard 21-C, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.	☐
16	Diablo Technologies, Inc.'s Invalidity Contentions, Case No. 13-CV-05889-YGR, dated June 6, 2014.	☐
17	Smart Storage Systems, Inc.'s Invalidity Contentions, Case No. 4:13-cv-05889-YGR, dated June 6, 2014.	☐
18	JEDEC Standard, "Configurations for Solid State Memories", JEDEC Standard 21-C, Release 9, August 1999, 114 pages.	☐
19	WONG, A. "The BIOS Optimization Guide", Adrian's Rojak Pot, Rev. 6.2, 1998-2001, 67 pages.	☐
20	American National Standard Dictionary of Electrical and Electrical Terms, IEEE, Fourth Edition, Revised, ANSI/IEEE Std 100-1988, Institute of Electrical Engineers, November 3, 1988, pp. 215, 722, 964 and 1103.	☐

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Attorney Docket Number	062453-032

21	Webster's New College Dictionary, Houghton Mifflin Company, Boston, MA, 2001, pp. 259, 1415.	☐
22	Bruce, J., "Synchronous DRAM Architectures, Organizations, and Alternate Technologies", Electrical and Computer Engineering Dept., Univ. of Maryland, December 10, 2002, 22 pages.	☐
23	David, H. et al., "Fully Buffered DIMM (FB-DIMM) Design Considerations", Intel Developer Forum, Intel Corp., February 18, 2004, 36 pages.	☐
24	Horowitz, P. et al., "The Art of Electronics", Cambridge University Press 2nd Ed. 1989, pp. 471, 495-496.	☐
25	Innis, J., "MPC8560 PowerQUICC III Compact Flash Interface Design", Freescale Semiconductor, Inc., 2004-2006, pp. 1-23.	☐
26	Jacob, B., "Memory Systems Cache, DRAM, Disk", Morgan Kaufman Publishers, Burlington, MA, 2008, Preface and Ch. 7 pp. 315-322.	☐
27	Jandhyala, S. et al., "Design-For-Test Analysis of a Buffered SDRAM DIMM", Semiconductor Group, Texas Instruments, Proceedings of International Workshop in Memory Technology, Design and Testing, Singapore, August 13014, 1996, 15 pages.	☐
28	JEDEC Standard, Double Data Rate (DDR): SDRAM Specification: JESD79C (Revision JESD79B), March 2003, pp. 1-75.	☐
29	JEDEC Standard, FBDIMM Specification: DDR2 SDRAM Fully Buffered DIMM (FBDIMM) Design Specification: JESD205, JEDEC SOLID STATE TECH. ASSOC., March 2007, pp. 1-129.	☐
30	Mutnary, B. et al., "Analysis of Fully Buffered DIMM Interface in High-speed Server Applications", IBM Corp, xSeries eServer Development, 2006 Electronic Components and Technology Conference, pp. 203-208.	☐
31	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SanDisk, Corp.), filed June 19, 2014.	☐

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	14489269
Filing Date	2014-09-17
First Named Inventor	Hyun Lee
Art Unit	2138
Examiner Name	Stephen C. Elmore
Attorney Docket Number	062453-032

32	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.	☐
33	"Using Two Chip Selects to Enable Quad Rank", an IP.com Prior Art Database Technical Disclosure, IP.com Electronic Publication: December 17, 2005, 2 pages.	☐
34	Petition for Inter Partes Review of U.S. Patent No. 8,516,187 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	☐
35	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SMART Modular Technologies, Inc.), filed August 22, 2014.	☐
36	"Out of Stealth Mode, Start-Up MetaRAM Unveils New Technology That Quadruples DRAM Capacity", Press Release edited by Storage Newsletter on February 28, 2008 at StorageNewsLetter.com, 8 pages.	☐
37	"MetaRAM Develops New Technology That Quadruples Memory Capacity of Servers and Workstations; Reduces Price by Up to 90 Percent", Press Release provided by MetaRAM and published on MarketWired.com, February 25, 2008, 3 pages.	☐
38	"240pin DDR2 MetaSDRAM Registered DIMM based on 1 GB version C", Hynix Semiconductor, Product Description Rev. 0.2, September 2008, 32 pages.	☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	09/06/2015
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	14489269
Filing Date	2014-09-17
First Named Inventor	Hyun Lee
Art Unit	2138
Examiner Name	Stephen C. Elmore
Attorney Docket Number	062453-032

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-08-31
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

Search Notes 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

CPC- SEARCHED		
Symbol	Date	Examiner
G06F 12/0638; G06F 12/0246; G06F 13/4243;	6/28/2015	SE
G11C 7/1072; G11C 14/0018;	6/28/2015	SE
Search Updated	9/6/2015	SE

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	6/28/2015	SE
365	185.33	6/28/2015	SE
Search Updated		9/6/2015	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	6/28/2015	SE
Inventor Name Search for DP	6/28/2015	SE
Search Updated	9/6/2015	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	6/28/2015	SE
G06F	12/0638	6/28/2015	SE
PGPUB Searched		6/28/2015	SE

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INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
Search Updated		9/6/2015	SE

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BIB DATA SHEET

CONFIRMATION NO. 1077

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
14/489,269	09/17/2014	711	2138	062453-032	
RULE					
APPLICANTS Netlist, Inc., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott H. Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;					
** CONTINUING DATA ***** This application is a CON of 13/559,476 07/26/2012 PAT 8874831 which claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 09/23/2014					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /STEPHEN C ELMORE/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 10	TOTAL CLAIMS 24	INDEPENDENT CLAIMS 2
ADDRESS NIXON PEABODY, LLP 799 Ninth Street, NW SUITE 500 WASHINGTON, DC 20001 UNITED STATES					
TITLE FLASH-DRAM HYBRID MEMORY MODULE					
FILING FEE RECEIVED 960	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Issue Classification 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2138	

CPC						
Symbol					Type	Version
G06F		12	/	0638	F	2013-01-01
G06F		12	/	0246	I	2013-01-01
G06F		13	/	4243	I	2013-01-01
G11C		7	/	1072	I	2013-01-01
G11C		14	/	0018	I	2013-01-01
G06F		2212	/	205	A	2013-01-01
G06F		2212	/	7208	A	2013-01-01
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CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 24	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2138 (Primary Examiner) _____ (Date) _____		O.G. Print Claim(s) 1	O.G. Print Figure 6

Issue Classification 	Application/Control No. 14489269	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2138

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	2		18												
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NONE	Total Claims Allowed: 24	
(Assistant Examiner)	(Date)	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2138	9/6/2015	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 6

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	10754	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
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L3	20768	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L4	32142	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L5	5651	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L6	614	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L7	14	L5 and L6	US-PGPUB; USPAT;	OR	ON	2015/09/06 16:11

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L29	21	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/09/06 16:11
L30	103	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/09/06 16:11
L31	171	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2015/09/06 16:11
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		or L14)	USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			
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L47	8	"8301833".pn.	US-PGPUB; USPAT;	OR	ON	2015/09/06 16:11

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L53	1	(US-8301833-\$).did.	USPAT	OR	ON	2015/09/06 16:11
L54	1	(US-8102614-\$).did.	USPAT	OR	ON	2015/09/06 16:11
L55	184	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing) same (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L56	2	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L55	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L57	10	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L34	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L58	8	L57 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L59	0	L37 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L60	1095	G06F12/0638.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L61	13731	G06F12/0246.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L62	2919	G06F13/4243.CPC.	US-	OR	ON	2015/09/06

			PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			16:11
L63	6394	G11C7/1072.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L64	332	G11C14/0018.CPC.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L65	24008	L60 or L61 or L62 or L63 or L64	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L66	3	L55 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L67	0	L66 and ((@pd or @ad)<"20110728")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L68	10	L5 and L57	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L69	1	L6 and L68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2015/09/06 16:11

			IBM_TDB			
L70	1	(data adj manager) and L68	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L71	1	L57 and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L72	94	(data adj manager) and L65	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11
L73	1	((first adj3 signals) and (second adj3 signals) and (third adj3 signals)) and L72	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2015/09/06 16:11

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L74	22	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L75	103	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L76	127	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L77	93	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L78	6505	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L79	6744	L74 or L75 or L76 or L77 or L78	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L80	8	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11

L81	3	L79 and L80	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L82	12801	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L83	45	L79 and L82	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L84	3	L80 and L83	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L85	10727	711/103.ccls.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L86	1534	(data adj manager).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L87	489	G06F12/0638.CPC.	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L88	17	L79 and L85	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11
L89	3	L86 and L88	US-PGPUB; USPAT; UPAD	OR	ON	2015/09/06 16:11

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	Filing Date		2014-09-17
	First Named Inventor	Hyun Lee	
	Art Unit	2138	
	Examiner Name	Stephen C. Elmore	
	Attorney Docket Number	062453-032	

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32	Petition for Inter Partes Review of U.S. Patent No. 8,301,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.	☐
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OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-09-17
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/George Hinton			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 2nd and Subsequent Request	2820	1	850	850
Total in USD (\$)				850

Electronic Acknowledgement Receipt

EFS ID:	23518598
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/George Hinton
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	17-SEP-2015
Filing Date:	17-SEP-2014
Time Stamp:	19:14:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$850
RAM confirmation Number	5672
Deposit Account	192380
Authorized User	NIXON PEABODY LLP

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	20150917_RCE_062453-032.pdf	1350033	no	3
			6ed38219256008cd18d0814208001ecf32eef364		
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	20150917_IDS_062453-032.pdf	624900	no	16
			4583433c3a86925c1329507bf91919b7e7fba18b		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30557	no	2
			f799c885979cc5f6fb5a531bc04427f6206fd421		
Warnings:					
Information:					
Total Files Size (in bytes):			2005490		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	14/489,269	Filing Date	2014-09-17	Docket Number (if applicable)	062453-032	Art Unit	2138
First Named Inventor	Hyun Lee			Examiner Name	Stephen C. Elmore		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☐ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____.
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No. 192380

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Doc code: RCEX

Doc description: Request for Continued Examination (RCE)

PTO/SB/30EFS (07-14)

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2015-09-17
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	23348298
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/George Hinton
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	31-AUG-2015
Filing Date:	17-SEP-2014
Time Stamp:	16:50:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$670
RAM confirmation Number	3731
Deposit Account	192380
Authorized User	NIXON PEABODY LLP

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Electronic Patent Application Fee Transmittal

Application Number:	14489269			
Filing Date:	17-Sep-2014			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/George Hinton			
Attorney Docket Number:	062453-032			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Petition Fee-37CFR 1.17(h) (Group II)	2464	1	70	70
Request for Continued Examination	2801	1	600	600
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Adjustment date: 09/21/2015 SDIRETA1
09/01/2015 INTEFSW 00003731 192300 14489269
02 FC:2801 600.00 CR



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/489,269	10/13/2015	9158684	062453-032	1077

22204 7590 09/23/2015
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott H. Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

151145

OR

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

☒ The address associated with the above-mentioned Customer Number

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

Netlist, Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

12/5/17

Name

Noel B. Whitley

Title

VP, IP & Licensing

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	14/489,269
Filing Date	09-17-2014
First Named Inventor	Hyun Lee
Title	FLASH-DRAM HYBRID MEMORY MODULE
Art Unit	2138
Examiner Name	ELMORE, STEPHEN C
Attorney Docket Number	0016.001000A

SIGNATURE of Applicant or Patent Practitioner

Signature	/Khaled Shami/	Date (Optional)	
Name	Khaled Shami	Registration Number	38,745
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	31175751
Application Number:	14489269
International Application Number:	
Confirmation Number:	1077
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	22204
Filer:	Khaled Shami/Casey Berger
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-032
Receipt Date:	08-DEC-2017
Filing Date:	17-SEP-2014
Time Stamp:	16:12:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	0016001000A_POA.pdf	880630	no	2
			04d2e1354fab45a74a63e2367df0f60dcdc792a6		

Warnings:

Information:

Total Files Size (in bytes):

880630

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/489,269	09/17/2014	Hyun Lee	062453-032

22204
NIXON PEABODY, LLP
799 Ninth Street, NW
SUITE 500
WASHINGTON, DC 20001

CONFIRMATION NO. 1077
POWER OF ATTORNEY NOTICE



Date Mailed: 12/14/2017

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/08/2017.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/megga/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/489,269	09/17/2014	Hyun Lee	0016.001000A

151145
Shami Messinger PLLC
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NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 12/08/2017.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/megga/



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Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 07/09/2018 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

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UNITED STATES PATENT APPLICATION

FOR

TITLE

FLASH-DRAM HYBRID MEMORY MODULE

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Flash-DRAM Hybrid Memory Module

PRIORITY CLAIM

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, titled “HIGH-DENSITY DIMMS”, and is a continuation-in-part (CIP) of US patent application serial no. 12/240,916, filed September 29, 2008, titled “NON-VOLATILE MEMORY MODULE”, the contents of both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to computer memory devices, and more particularly, to devices that employ different types of memory devices such as combinations of Flash and random access memories.

BACKGROUND

[0003] As technology advances and the usage of portable computing devices, such as tablet notebook computers, increases, more data needs to be transferred among data centers and to/from end users. In many cases, data centers are built by clustering multiple servers that are networked to increase performance.

[0004] Although there are many types of networked servers that are specific to the types applications envisioned, the basic concept is generally to increase server performance by dynamically allocating computing and storage resources. In recent years, server technology has evolved to be specific to particular applications such as ‘finance transactions’ (for example, point-of-service, inter-bank transaction, stock market transaction), ‘scientific computation’ (for example, fluid dynamic for automobile and ship design, weather prediction, oil and gas expeditions), ‘medical diagnostics’ (for example, diagnostics based on the fuzzy logic, medical data processing), ‘simple information sharing and searching’ (for example, web search, retail store website, company home page), ‘email’ (information distribution and archive), ‘security service’, ‘entertainment’ (for example, video-on-demand), and so on. However, all of these applications suffer from the same information transfer bottleneck due to the inability of a high speed CPU (central processing unit) to efficiently transfer data in and out of relatively slower speed storage or memory subsystems, particularly since data transfers typically pass through the CPU input/output (I/O) channels.

[0005] The data transfer limitations by the CPU are exemplified by the arrangement shown in FIG. 1, and apply to data transfers between main storage (for example the hard disk (HD) or solid state drive (SSD) and the memory subsystems (for example DRAM DIMM (Dynamic Random Access Memory Dual In-line Memory Module) connected to the front side bus (FSB)). In arrangements such as that of FIG. 1, the SSD/HD and DRAM DIMM of a conventional memory arrangement are connected to the CPU via separate memory control ports (not shown). FIG. 1 specifically shows, through the double-headed arrow, the data flow path between the computer or server main storage (SSD/HD) to the DRAM DIMMs. Since the SSD/HD data I/O and the DRAM DIMM data I/O are controlled by the CPU, the CPU needs to allocate its process

cycles to control these I/Os, which may include the IRQ (Interrupt Request) service which the CPU performs periodically. As will be appreciated, the more time a CPU allocates to controlling the data transfer traffic, the less time the CPU has to perform other tasks. Therefore, the overall performance of a server will deteriorate with the increased amount of time the CPU has to expend in performing data transfer.

[0006] There have been various approaches to increase the data transfer throughput rates from/to the main storage, such as SSD/HD, to local storage, such as DRAM DIMM. In one example as illustrated in FIG. 2, EcoRAM™ developed by Spansion provides a storage SSD based system that assumes a physical form factor of a DIMM. The EcoRAM™ is populated with Flash memories and a relatively small memory capacity using DRAMs which serve as a data buffer. This arrangement is capable of delivering higher throughput rate than a standard SSD based system since the EcoRAM™ is connected to the CPU (central processing unit) via a high speed interface, such as the HT (Hyper Transport) interface, while an SSD/HD is typically connected via SATA (serial AT attachment), USB (universal serial bus), or PCI Express (peripheral component interface express). For example, the read random access throughput rate of EcoRAM™ is near 3GB/s compared with 400MB/s for a NAND SSD memory subsystem using the standard PCI Express-based. This is a 7.5X performance improvement. However, the performance improvement for write random access throughput rate is less than 2X (197MBs for the EcoRAM vs. 104MBs for NAND SSD). This is mainly due to the fact that the write speed is cannot be faster than the NAND Flash write access time. Figure 2 is an example of EcoRAM™ using SSD with the form factor of a standard DIMM such that it can be connected to the FSB (front side bus). However, due to the interface protocol difference between DRAM and Flash, an interface device, EcoRAM Accelerator™), which occupies one of the server's CPU sockets is

used, and hence further reducing server's performance by reducing the number of available CPU sockets available, and in turn reducing the overall computation efficiency. The server's performance will further suffer due to the limited utilization of the CPU bus due to the large difference in the data transfer throughput rate between read and write operations.

[0007] The EcoRAM™ architecture enables the CPU to view the Flash DIMM controller chip as another processor with a large size of memory available for CPU access.

[0008] In general, the access speed of a Flash based system is limited by four items: the read/write speed of the Flash memory, the CPU's FSB bus speed and efficiency, the Flash DIMM controller's inherent latency, and the HT interconnect speed and efficiency which is dependent on the HT interface controller in the CPU and Flash DIMM controller chip.

[0009] The published results indicate that these shortcomings are evident in that the maximum throughput rate is 1.56 GBs for the read operation and 104 MBs for the write operation. These access rates are 25% of the DRAM read access speed, and 1.7% of the DRAM access speed at 400MHz operation. The disparity in the access speed (15 to 1) between the read operation and write operation highlight a major disadvantage of this architecture. The discrepancy of the access speed between this type of architecture and JEDEC standard DRAM DIMM is expected to grow wider as the DRAM memory technology advances much faster than the Flash memory.

OVERVIEW

[0010] Described herein is a memory module couplable to a memory controller of a host system. The memory module includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.

[0011] Also described herein is a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems. The method includes receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem. The method further includes identifying a data path to be used for transferring data to or from the memory module using the received control information, and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

[0012] Also described herein is a memory module wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring,

and data error correction in response to receiving at least one of a control signal and control information from the controller.

[0013] Also described herein is a memory module wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

[0014] Also described herein is a memory module wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

[0015] Also described herein is a memory module wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

[0016] Also described herein is a memory module wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0017] Also described herein is a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0018] Also described herein is a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0019] Also described herein is a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0020] Also described herein is a memory module wherein the non-volatile memory subsystem comprises flash memory.

[0021] Also described herein is a memory module wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0022] Also described herein is a memory module wherein the data manager and controller are configured to effect data transfer between the memory controller and the non-volatile memory subsystem in response to memory access commands received by the controller from the memory controller.

[0023] Also described herein is a memory module wherein the volatile memory subsystem is operable as a buffer for the data transfer between the memory controller and non-volatile memory.

[0024] Also described herein is a memory module wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the

memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

[0025] Also described herein is a memory module wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.

[0026] Also described herein is a memory module wherein the controller is operable to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0027] Also described herein is a memory module wherein the controller is configured to effect operation with the host system in accordance with a prescribed protocol.

[0028] Also described herein is a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0029] Also described herein is a memory module wherein the controller is operable to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0030] Also described herein is a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0031] Also described herein is a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0032] Also described herein is a memory module wherein the controller is operable to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0033] Also described herein is a memory module wherein the controller includes a volatile memory control module, a non-volatile memory control module, data manager control module, a command interpreter module, and a scheduler module.

[0034] Also described herein is a memory module wherein commands from the volatile memory control module to the volatile memory subsystem are subordinated to commands from the memory controller to the controller.

[0035] Also described herein is a memory module wherein the controller effects pre-fetching of data from the non-volatile to the volatile memory.

[0036] Also described herein is a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0037] Also described herein is a memory module wherein the controller is operable to initiate a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0038] Also described herein is a memory module wherein, if the closed block is re-opened, the controller is operable to abort the copy operation and to erase the target block from the non-volatile memory subsystem.

[0039] Also described herein is a method for managing a memory module wherein the transfer of data includes a bidirectional transfer of data between the non-volatile and the volatile memory subsystems.

[0040] Also described herein is a method for managing a memory module further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

[0041] Also described herein is a method for managing a memory module further comprising operating the data manager to control data traffic between the memory controller and at least one of the volatile and non-volatile memory subsystems.

[0042] Also described herein is a method for managing a memory module wherein data traffic control relates to any one or more of data transfer size, formatting information, direction of data flow, and the starting time of the transfer of data.

[0043] Also described herein is a method for managing a memory module wherein data traffic control by the data manager is based on instructions received from the controller.

[0044] Also described herein is a method for managing a memory module further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.

[0045] Also described herein is a method for managing a memory module wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

[0046] Also described herein is a method for managing a memory module wherein each memory segment comprises at least one memory circuit, memory device, or memory die.

[0047] Also described herein is a method for managing a memory module wherein the volatile memory subsystem comprises DRAM memory.

[0048] Also described herein is a method for managing a memory module wherein the non-volatile memory subsystem comprises Flash memory.

[0049] Also described herein is a method for managing a memory module further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

[0050] Also described herein is a method for managing a memory module further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

[0051] Also described herein is a method for managing a memory module further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

[0052] Also described herein is a method for managing a memory module further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

[0053] Also described herein is a method for managing a memory module further comprising using the controller to effect communication with a host system by the volatile memory subsystem in accordance with a prescribed protocol.

[0054] Also described herein is a method for managing a memory module wherein the prescribed protocol is selected from one or more of DDR, DDR2, DDR3, and DDR4 protocols.

[0055] Also described herein is a method for managing a memory module further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a

value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

[0056] Also described herein is a method for managing a memory module wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

[0057] Also described herein is a method for managing a memory module wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

[0058] Also described herein is a method for managing a memory module further comprising using the controller to copy booting information from the non-volatile to the volatile memory subsystem during power up.

[0059] Also described herein is a method for managing a memory module wherein the controller includes a volatile memory control module, the method further comprising generating commands by the volatile memory control module in response to commands from the memory controller, and transmitting the generated commands to the volatile memory subsystem.

[0060] Also described herein is a method for managing a memory module further comprising pre-fetching of data from the non-volatile memory subsystem to the volatile memory subsystem.

[0061] Also described herein is a method for managing a memory module wherein the pre-fetching is initiated by the memory controller writing an address of requested data into a register of the controller.

[0062] Also described herein is a method for managing a memory module further comprising initiating a copy operation of data of a closed block in the volatile memory subsystem to a target block in the non-volatile memory subsystem.

[0063] Also described herein is a method for managing a memory module further comprising aborting the copy operation when the closed block of the volatile memory subsystem is re-opened, and erasing the target block in the non-volatile memory subsystem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0064] The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more examples of embodiments and, together with the description of example embodiments, serve to explain the principles and implementations of the embodiments.

[0065] In the drawings:

FIG. 1 is a block diagram illustrating the path of data transfer, via a CPU, of a conventional memory arrangement;

FIG. 2 is a block diagram of a known EcoRAM™ architecture;

FIGS. 3A and 3B are block diagrams of a non-volatile memory DIMM or NVDIMM;

FIGS. 4A and 4B are block diagrams of a Flash-DRAM hybrid DIMM or FDHDIMM;

FIG. 5A is a block diagram of a memory module 500 in accordance with certain embodiments described herein;

FIG. 5B is a block diagram showing some functionality of a memory module such as that shown in FIG. 5A;

FIG. 6 is a block diagram showing some details of the data manager (DMgr);

FIG. 7 is a functional block diagram of the on-module controller (CDC);

FIG. 8A is a block diagram showing more details of the prior art Flash-DRAM hybrid DIMM (FDHDIMM) of FIGS. 4A and 4B;

FIG. 8B is a block diagram of a Flash-DRAM hybrid DIMM (FDHDIMM) in accordance with certain embodiments disclosed herein;

FIG. 9 is a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM;

FIG. 10 is a block diagram showing an example of mapping of DRAM address space to Flash memory address space; and

FIG. 11 is a table showing estimates of the maximum allowed closed blocks in a queue to be written back to Flash memory for different DRAM densities using various average block use time.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0066] Example embodiments are described herein in the context of a system of computers, servers, controllers, memory modules, hard disk drives and software. Those of ordinary skill in the art will realize that the following description is illustrative only and is not intended to be in any way limiting. Other embodiments will readily suggest themselves to such skilled persons having the benefit of this disclosure. Reference will now be made in detail to implementations of the example embodiments as illustrated in the accompanying drawings. The same reference indicators will be used to the extent possible throughout the drawings and the following description to refer to the same or like items.

[0067] In the interest of clarity, not all of the routine features of the implementations described herein are shown and described. It will, of course, be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions must be made in order to achieve the developer's specific goals, such as compliance with application- and business-related constraints, and that these specific goals will vary from one implementation to another and from one developer to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of engineering for those of ordinary skill in the art having the benefit of this disclosure.

[0068] In accordance with this disclosure, the components, process steps, and/or data structures described herein may be implemented using various types of operating systems, computing platforms, computer programs, and/or general purpose machines. In addition, those

of ordinary skill in the art will recognize that devices of a less general purpose nature, such as hardwired devices, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), or the like, may also be used without departing from the scope and spirit of the inventive concepts disclosed herein. Where a method comprising a series of process steps is implemented by a computer or a machine and those process steps can be stored as a series of instructions readable by the machine, they may be stored on a tangible medium such as a computer memory device (e.g., ROM (Read Only Memory), PROM (Programmable Read Only Memory), EEPROM (Electrically Erasable Programmable Read Only Memory), Flash memory, Jump Drive, and the like), magnetic storage medium (e.g., tape, magnetic disk drive, and the like), optical storage medium (e.g., CD-ROM, DVD-ROM, paper card, paper tape and the like) and other types of program memory.

[0069] The term “exemplary” where used herein is intended to mean “serving as an example, instance or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0070] Disclosed herein are arrangements for improving memory access rates and addressing the high disparity (15 to 1 ratio) between the read and write data throughput rates. In one arrangement, a Flash-DRAM-hybrid DIMM (FDHDIMM) with integrated Flash and DRAM is used. Methods for controlling such an arrangement are described.

[0071] In certain embodiments, the actual memory density (size or capacity) of the DIMM and/or the ratio of DRAM memory to Flash memory are configurable for optimal use with a particular application (for example, POS, inter-bank transaction, stock market transaction,

scientific computation such as fluid dynamics for automobile and ship design, weather prediction, oil and gas expeditions, medical diagnostics such as diagnostics based on the fuzzy logic, medical data processing, simple information sharing and searching such as web search, retail store website, company home page, email or information distribution and archive, security service, and entertainment such as video-on-demand).

[0072] In certain embodiments, the device contains a high density Flash memory with a low density DRAM, wherein the DRAM is used as a data buffer for read/write operation. The Flash serves as the main memory. Certain embodiments described herein overcome the needs of having a long separation period between an Activate command (may be referred to as RAS) and a corresponding read or write command (may be referred to as first CAS command).

[0073] In accordance with one embodiment, described with reference to FIGS. 3A and 3B, a memory system 300 includes a non-volatile (for example Flash) memory subsystem 302 and a volatile (for example DRAM) memory subsystem 304. The examples of FIGS. 3A and 3B are directed to architectures of a non-volatile DIMM (NVDIMM) NVDIMM system that may use a power subsystem (not shown) that can include a battery or a capacitor as a means for energy storage to copy DRAM memory data into Flash memory when power loss occurs, is detected, or is anticipated to occur during operation. When normal power is restored, a restore NVDIMM operation is initiated and the data stored in the Flash memory is properly restored to the DRAM memory. In this architecture, the density of the Flash is about the same as the DRAM memory size or within a few multiples, although in some applications it may be higher. This type of architecture may also be used to provide non-volatile storage that is connected to the FSB (front side bus) to support RAID (Redundant Array of Independent Disks) based systems or other type

of operations. An NVDIMM controller 306 receives and interprets commands from the system memory controller hub (MCH). The NVDIMM controller 306 control the NVDIMM DRAM and Flash memory operations. In FIG. 3A, the DRAM 304 communicates data with the MCH, while an internal bus 308 is used for data transfer between the DRAM and Flash memory subsystems. In FIG. 3B, the NVDIMM controller 306' of NVDIMM 300' monitors events or commands and enables data transfer to occur in a first mode between the DRAM 304' and Flash 302' or in a second mode between the DRAM and the MCH.

[0074] In accordance with one embodiment, a general architecture for a Flash and DRAM hybrid DIMM (FDHDIMM) system 400 is shown in FIG. 4A. The FDHDIMM interfaces with an MCH (memory controller hub) to operate and behave as a high density DIMM, wherein the MCH interfaces with the non-volatile memory subsystem (for example Flash) 402 is controlled by an FDHDIMM controller 404. Although the MCH interfaces with the Flash via the FDHDIMM controller, the FDHDIMM overall performance is governed by the Flash access time. The volatile memory subsystem (for example DRAM) 406 is primarily used as a data buffer or a temporary storage location such that data from the Flash memory 402 is transferred to the DRAM 406 at the Flash access speed, and buffered or collected into the DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. Similarly, when the MCH transfers data to the DRAM 406, the FDHDIMM controller 404 manages the data transfer from the DRAM 406 to the Flash 402. Since the Flash memory access speed (both read and write) is relatively slower than DRAM, (e.g. for example a few hundred microseconds for read access), the average data throughput rate of FDHDIMM 400 is limited by the Flash access speed. The DRAM 406 serves as a data buffer stage that buffers the MCH read or write data. Thus, the DRAM 406 serves as a temporary storage for the data to be transferred from/to

the Flash 402. Furthermore, in accordance with one embodiment, the MCH recognizes the physical density of an FDHDIMM operating as a high density DIMM as the density of Flash alone.

[0075] In accordance with one embodiment, a read operation can be performed by the MCH by sending an activate command (may be simply referred to as RAS, or row address strobe) to the FDHDIMM 400 to conduct a pre-fetch read data operation from the Flash 402 to the DRAM 406, with the pre-fetch data size being for example a page (1KB or 2KB, or may be programmable to any size). The MCH then sends a read command (may be simply referred to as CAS, or column address strobe) to read the data out input of the DRAM. In this embodiment, the data transfer from Flash to DRAM occurs at Flash access speed rates, while data transfer from DRAM to MCH occurs at DRAM access speed rates. In this example, data latency and throughput rates are the same as any DRAM operation as long as the read operations are executed onto the pages that were opened with the activate command previously sent to pre-fetch data from the Flash to DRAM. Thus, a longer separation time period between the RAS (e.g. Activate command) and the first CAS (column address strobe e.g. read or write command) is required to account for the time it takes to pre-fetch data from the Flash to DRAM.

[0076] An example of FDHDIMM operating as a DDR DIMM with SSD is shown in FIG. 4B, wherein the FDHDIMM 400' supports two different interface interpretations to the MCH. In the first interface interpretation, the MCH views the FDHDIMM 400' as a combination of DRAM DIMM and SSD (not illustrated). In this mode the MCH needs to manage two address spaces, one for the DRAMs 402' and one for the Flash 404'. The MCH is coupled to, and controls, both of the DRAM and Flash memory subsystems. One advantage of this mode is that

the CPU does not need to be in the data path when data is moved from DRAM to Flash or from Flash to DRAM. In the second interface interpretation, the MCH views the FDHDIMM 400' as an on-DIMM Flash with the SSD in an extended memory space that is behind the DRAM space. Thus, in this mode, the MCH physically fetches data from the SSD to the DDR DRAM and then the DRAM sends the data to the MCH. Since all data movement occurs on the FDHDIMM, this mode will provide better performance than if the data were to be moved through or via the CPU.

[0077] In accordance with one embodiment and as shown in FIG. 4B, the FDHDIMM 400' receives control signals 408 from the MCH, where the control signals may include one or more control signals specifically for the DRAM 402' operation and one or more control signals specifically for the Flash 404' operation. In this embodiment, the MCH or CPU is coupled to the FDHDIMM via a single data bus interface 410 which couples the MCH to the DRAM.

[0078] FIGS. 5A and 5B are block diagrams of a memory module 500 that is couplable to a host system (not shown). The host system may be a server or any other system comprising a memory system controller or an MCH for providing and controlling the read/write access to one or more memory systems, wherein each memory system may include a plurality of memory subsystems, a plurality of memory devices, or at least one memory module. The term "read/write access" means the ability of the MCH to interface with a memory system or subsystem in order to write data into it or read data from it, depending on the particular requirement at a particular time.

[0079] In certain embodiments, memory module 500 is a Flash-DRAM hybrid memory subsystem which may be integrated with other components of a host system. In certain

embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor, and may be referred to as a FDHDIMM, although it is to be understood that in both structure and operation it may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B. Memory module 500 includes two on-module intermediary components: a controller and a data manager. These on-module intermediary components may be physically separate components, circuits, or modules, or they may be integrated onto a single integrated circuit or device, or integrated with other memory devices, for example in a three dimensional stack, or in any one of several other possible expedients for integration known to those skilled in the art to achieve a specific design, application, or economic goal. In the case of a DIMM, these on-module intermediary components are an on-DIMM Controller (CDC) 502 and an on-DIMM data manager (DMgr) 504. While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well. CDC 502 and data manager DMgr 504 are operative to manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.

[0080] In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM memory from/to MCH 510. Read/write access between DRAM 508, Flash 506 and MCH 510 may be referred to herein generally as communication, wherein control and address information C/A 560 is sent from MCH 510 to CDC 502, and possible data transfers follow as indicated by Data 550, Data 555, and/or Data 556. In certain embodiments, the CDC 502 performs specific functions for memory address transformation, such as address translation, mapping, or address domain conversion,

Flash access control, data error correction, manipulation of data width or data formatting or data modulation between the Flash memory and DRAM, and so on. In certain embodiments, the CDC 502 ensures that memory module 500 provides transparent operation to the MCH in accordance with certain industry standards, such as DDR, DDR2, DDR3, DDR4 protocols. In the arrangement shown in FIGS. 5A and 5B, there is no direct access from the MCH 510 to the Flash 506 memory subsystem. Thus in accordance with certain embodiments, the Flash access speed has minimal impact on the overall FDHDIMM access speed. In the schematic illustration of FIG. 5B and in accordance with one embodiment, the CDC controller 502 receives standard DDR commands from the MCH, interprets, and produces commands and/or control signals to control the operation of the Data manager (DMgr), the Flash memory and the DRAM memory. The DMgr controls the data path routing amongst DRAMs, Flash and MCH, as detailed below. The data path routing control signals are independently operated without any exclusivity.

[0081] An exemplary role of DMgr 504 is described with reference to FIG. 6. In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction. For example, these functions or operations can be performed on-the-fly (while data is being transferred via the DMgr 504) or performed on buffered or stored data in DRAM or a buffer. In addition, one role of DMgr 504 is to provide interoperability among various memory subsystems or components and/or MCH 510.

[0082] In one embodiment, an exemplary host system operation begins with initialization. The CDC 502 receives a first command from the MCH 510 to initialize FDHDIMM 500 using a certain memory space. The memory space as would be controlled by MCH 510 can be

configured or programmed during initialization or after initialization has completed. The MCH 510 can partition or parse the memory space in various ways that are optimized for a particular application that the host system needs to run or execute. In one embodiment, the CDC 502 maps the actual physical Flash 506 and DRAM 508 memory space using the information sent by MCH 510 via the first command. In one embodiment, the CDC 502 maps the memory address space of any one of the Flash 506 and DRAM 508 memory subsystems using memory address space information that is received from the host system, stored in a register within FDHDIMM 500, or stored in a memory location of a non-volatile memory subsystem, for example a portion of Flash 506 or a separate non-volatile memory subsystem. In one embodiment, the memory address space information corresponds to a portion of initialization information of the FDHDIMM 500.

[0083] In one embodiment, MCH 510 may send a command to restore a certain amount of data information from Flash 506 to DRAM 508. The CDC 502 provides control information to DMgr 504 to appropriately copy the necessary information from Flash 506 to the DRAM 508. This operation can provide support for various host system booting operations and/or a special host system power up operation.

[0084] In one embodiment, MCH 510 sends a command which may include various fields comprising control information regarding data transfer size, data format options, and/or startup time. CDC 502 receives and interprets the command and provides control signals to DMgr 504 to control the data traffic between the Flash 506, the DRAM 508, and the MCH 510. For example, DMgr 504 receives the data transfer size, formatting information, direction of data flow (via one or more multiplexers such as 611, 612, 621, 622 as detailed below), and the starting time of the actual data transfer from CDC 502. DMgr 504 may also receive additional control information

from the CDC 502 to establish a data flow path and/or to correctly establish the data transfer fabric. In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric. For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508. Multiplexers 611 and 612 provide controllable data paths from any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506. Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B). In one embodiment, DRAM 508(1) is a segment of DRAM 508, while in other embodiments, DRAM 508(1) is a separate DRAM memory subsystem. It will be understood that each memory segment can comprise one or more memory circuits, a memory devices, and/or memory integrated circuits. Of course other configurations for DRAM 508 are possible, and other data transfer fabrics using complex data paths and suitable types of multiplexing logic are contemplated.

[0085] In accordance with one embodiment, the two sets of multiplexors 611, 612 and 621, 622 allow independent data transfer to Flash 506 from DRAM-A 508(1) and DRAM-B 508(2). For example, in response to one or more control signals or a command from CDC 502, DMgr 504 can transfer data from DRAM-A 508(1) to MCH 510, via multiplexer 611, at the same time as from DRAM-B 508(2) to the Flash 506, via multiplexer 612; or data is transferred from DRAM-B 508(2) to MCH 510, via multiplexer 611, and simultaneously data is transferred from the Flash 506 to DRAM-A 508(1), via multiplexer 621. Further, in the same way that data can be transferred to or from the DRAM in both device-wide or segment-by-segment fashion, data can be transferred to or from the flash memory in device-wide or segment-by-segment fashion, and the flash memory can be addressed and accessed accordingly.

[0086] In accordance with one embodiment the illustrated arrangement of data transfer fabric of DMgr 504 also allows the CDC 502 to control data transfer from the Flash memory to the MCH by buffering the data from the Flash 506 using a buffer 602, and matching the data rate and/or data format of MCH 510. The buffer 602 is shown in FIG. 6 as a portion of a data format module 604; however, buffer 602 may also be a distributed buffer such that one buffer is used for each one of the set of multiplexer logic elements shown as multiplexers 611, 612, 621, and 622. Various buffer arrangements may be used, such as a programmable size buffer to meet the requirement of a given system design requirement, for example the disparity between read/write access time; or overall system performance, for example latency. In certain embodiments, the buffer 604 may introduce one or more clock cycle delays into a data communication path between MCH 510, DRAM 508, and Flash 506.

[0087] In certain embodiments, data format module 604 contains a data formatting subsystem (not shown) to enable DMgr 504 to format and perform data transfer in accordance with control information received from CDC502. Data buffer 604 of data format module 602, discussed above, also supports a wide data bus 606 coupled to the Flash memory 506 operating at a first frequency, while receiving data from DRAM 508 using a relatively smaller width data bus 608 operating at a second frequency, the second frequency being larger than the first frequency in certain embodiments. The buffer 602 is designed to match the data flow rate between the DRAM 508 and the Flash 506.

[0088] A register 690 provides the ability to register commands received from MCH 510 via C/A 560 (FIG. 5A). The register 690 may communicate these commands to CDC 502 and/or to the DRAM 508 and/or Flash 506. The register 690 communicates these registered commands to

CDC 502 for processing. The register 690 may also include multiple registers (not shown), such that it can provide the ability to register multiple commands, a sequence of commands, or provide a pipeline delay stage for buffering and providing a controlled execution of certain commands received from MCH 510.

[0089] In certain embodiments, the register 690 may register commands from MCH 510 and transmit the registered commands to DRAM 508 and/or Flash 506 memory subsystems. In certain embodiments, the CDC 502 monitors commands received from MCH 510, via control and address bus C/A 560, and provides appropriate control information to DMgr 504, DRAM 508, or Flash 506 to execute these commands and perform data transfer operations between MCH 510 and FDHDIMM 500 via MCH data bus 610.

[0090] FIG. 7 illustrates a functional block diagram of the CDC 502. In certain embodiments, the major functional blocks of the CDC 502 are a DRAM control block DRAMCtrl 702, Flash control block FlashCtrl 704, MCH command interpreter CmdInt 706, DRAM-Flash interface scheduler Scheduler 708, and DMgr control block (DMgrCtrl) 710.

[0091] In accordance with one embodiment, DRAMCtrl 702 generates DRAM commands that are independent from the commands issued by the MCH 510. In accordance with one embodiment, when the MCH 510 initiates a read/write operation from/to the same DRAM 508 that is currently executing a command from the DRAMCtrl 702, then the CDC 502 may choose to instruct DRAMCtrl 702 to abort its operation in order to execute the operation initiated by the MCH. However, the CDC 502 may also pipeline the operation so that it causes DRAMCtrl 702 to either halt or complete its current operation prior to executing that of the MCH. The CDC 502

may also instruct DRAMCtrl 702 to resume its operation once the command from MCH 510 is completed.

[0092] In accordance with one embodiment, the FlashCtrl 704 generates appropriate Flash commands for the proper read/write operations. The CmdInt 706 intercepts commands received from MCH 510 and generates the appropriate control information and control signals and transmit them to the appropriate FDHDIMM functional block. For example, CmdInt 706 issues an interrupt signal to the DRAMCtrl 702 when the MCH issues a command that collides (conflicts) with the currently executing or pending commands that DRAMCtrl 702 has initiated independently from MCH 510, thus subordinating these commands to those from the MCH. The Scheduler 708 schedules the Flash–DRAM interface operation such that there is no resource conflict in the DMgr 504. In accordance with one embodiment, the Scheduler 708 assigns time slots for the DRAMCtrl 702 and FlashCtrl 704 operation based on the current status and the pending command received or to be received from the MCH. The DMgrCtrl 710 generates and sends appropriate control information and control signals for the proper operation and control of the data transfer fabric to enable or disable data paths between Flash 506, DRAM 508, and the MCH 510.

[0093] FIG. 8A is a block diagram showing a Flash-DRAM hybrid DIMM (FDHDIMM) 801. As seen from FIG. 8A, this Flash-DRAM hybrid DIMM requires two separate and independent address buses to separately control the address spaces: one for the Flash memory Flash 803 and the other for the DRAM memory DRAM 805. The MCH 810 treats the DRAM 805 and Flash 803 as separate memory subsystems, for example DRAM and SSD/HD memory subsystems. The memory in each address space is controlled directly by the MCH. However,

the on-DIMM data path 807 between Flash 803 and DRAM 805 allows for direct data transfer to occur between the Flash 803 and the DRAM 805 in response to control information from Ctrl 830. In this embodiment, this data transfer mechanism provides direct support for executing commands from the MCH without having the MCH directly controlling the data transfer, and thus improving data transfer performance from Flash 803 to the DRAM 805. However, the MCH needs to manage two address spaces and two different memory protocols simultaneously. Moreover, the MCH needs to map the DRAM memory space into the Flash memory space, and the data interface time suffers due to the difference in the data access time between the Flash memory and the DRAM memory.

[0094] In accordance with one embodiment, a memory space mapping of a Flash-DRAM hybrid DIMM is shown in FIG. 8B. A memory controller of a host system (not shown) controls both of the DRAM 508 address space and the Flash 506 address space using a single unified address space. The CDC 502 receives memory access commands from the MCH and generates control information for appropriate mapping and data transfer between Flash and DRAM memory subsystem to properly carry out the memory access commands. In one embodiment, the memory controller of the host system views the large Flash memory space as a DRAM memory space, and accesses this unified memory space with a standard DDR (double data rate) protocol used for accessing DRAM. The unified memory space in this case can exhibit overlapping memory address space between the Flash 506 and the DRAM 508. The overlapping memory address space may be used as a temporary storage or buffer for data transfer between the Flash 506 and the DRAM 508. For example, the DRAM memory space may hold a copy of data from the selected Flash memory space such that the MCH can access this data normally via DDR memory access commands. The CDC 502 controls the operation of the Flash 506 and

DRAM 508 memory subsystems in response to commands received from a memory controller of a host system.

[0095] In one embodiment, the unified memory space corresponds to a contiguous address space comprising a first portion of the address space of the Flash 506 and a first portion of the address space of the DRAM 508. The first portion of the address space of the Flash 506 can be determined via a first programmable register holding a first value corresponding to the desired Flash memory size to be used. Similarly, the first portion of the address space of the DRAM 508 can be determined via a second programmable register holding a second value corresponding to the desired DRAM memory size to be used. In one embodiment, any one of the first portion of the address space of the Flash 506 and the first portion of the address space of the DRAM 508 is determined via a first value corresponding to a desired performance or memory size, the first value being received by the CDC 502 via a command sent by memory controller of the host system.

[0096] In accordance with one embodiment, a flow diagram directed to the transfer of data from Flash memory to DRAM memory and vice versa in an exemplary FDHDIMM is shown in Fig. 9. In certain embodiments, data transfer from the Flash 506 to the DRAM 508 occurs in accordance with memory access commands which the CDC 502 receives from the memory controller of the host system. In certain embodiments, the CDC 502 controls the data transfer from the DRAM 508 to the Flash 506 so as to avoid conflict with any memory operation that is currently being executed. For example, when all the pages in a particular DRAM memory block are closed. The CDC 502 partitions the DRAM memory space into a number of blocks for the purpose of optimally supporting the desired application. The controller can configure memory

space in the memory module based on at least one of one or more commands received from the MCH, instructions received from the MCH, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value. Furthermore, the block size can be configurable by the memory controller of the host system, such that the number pages in a block can be optimized to support a particular application or a task. Furthermore, the block size may be configured on-the-fly, e.g. CDC 502 can receive instruction regarding a desired block size from the memory controller via a memory command, or via a programmable value.

[0097] In certain embodiments, a memory controller can access the memory module using a standard access protocol, such as JEDEC's DDR DRAM, by sending a memory access command to the CDC 502 which in turn determines what type of a data transfer operation it is and the corresponding target address where the data information is stored, e.g. data information is stored in the DRAM 508 or Flash 506 memory subsystems. In response to a read operation, if the CDC 502 determines that data information, e.g. a page (or block), does not reside in the DRAM 508 but resides in Flash 506, then the CDC 502 initiates and controls all necessary data transfer operations from Flash 506 to DRAM 508 and subsequently to the memory controller. In one embodiment, once the CDC 502 completes the data transfer operation of the requested data information from the Flash 506 to the DRAM 508, the CDC 502 alerts the memory controller to retrieve the data information from the DRAM 508. In on embodiment, the memory controller initiates the copying of data information from Flash 506 to DRAM 508 by writing, into a register in the CDC 502, the target Flash address along with a valid block size. The CDC 502 in turn, executes appropriate operations and generates control information to copy the data information

to the DRAM 508. Consequently, the memory controller can access or retrieve the data information using standard memory access commands or protocol.

[0098] An exemplary flow chart is shown in FIG. 9, a starting step or power up 902, is followed by an initialization step 904, the memory controller initiates, at step 906, a data move from the Flash 506 to the DRAM 508 by writing target address and size, to a control register in the CDC 502, which then copies, at 908, data information from the Flash 506 to the DRAM 508 and erases the block in the Flash. Erasing the data information from Flash may be accomplished independently from (or concurrently with) other steps that CDC 502 performs in this flow chart, i.e. other steps can be executed concurrently with the Erase the Flash block step. Once the data information or a block of data information is thus moved to the DRAM 508, the memory controller can operate on this data block using standard memory access protocol or commands at 910. The CDC 502 checks, at 912, if any of the DRAM 508 blocks, or copied blocks, are closed. If the memory controller closed any open blocks in DRAM 508, then the CDC 502 initiate a Flash write to write the closed block from the DRAM 508 to the Flash 506, at 914. In addition, the memory controller, at 916, reopens the closed block that is currently being written into the Flash 506, then the CDC 502 stops the Flash write operation and erases the Flash block which was being written to, as shown at 918. Otherwise, the CDC 502 continues and completes the writing operation to the Flash at 920.

[0099] The dashed lines in FIG. 9 indicate independent or parallel activities that can be performed by the CDC 502. At any time the CDC 502 receives a DRAM load command from a memory controller which writes a Flash target address and/or block size information into the RC register(s) at 922, as described above, then the CDC 502 executes a load DRAM w/RC step 906

and initiates another branch (or a thread) of activities that includes steps 908 – 922. In one embodiment, the CDC 502 controls the data transfer operations between DRAM 508 and Flash 506 such that the Flash 506 is completely hidden from the memory controller. The CDC 502 monitors all memory access commands sent by the memory controller using standard DRAM protocol and appropriately configures and manipulate both Flash 506 and DRAM 508 memory subsystems to perform the requested memory access operation and thus achieve the desired results. The memory controller does not interface directly with the Flash memory subsystem. Instead, the memory controller interfaces with the CDC 502 and/or DMgr 504 as shown in Fig. 5 and Fig. 6. Moreover, the memory controller may use one or more protocol, such as DDR, DDR2, DDR3, DDR4 protocols or the like.

[00100] In accordance with one embodiment, an example of mapping a DRAM address space to Flash memory address space is shown in FIG. 10. Two sets (1002, 1004) of address bits AD6 to AD17, forming a 24 bit extended memory page address, are allocated for the block address. For example, assuming a Block size of 256K Bytes, then a 24-bit block address space (using the two sets of AD6 to AD17 1002 and 1004) would enable access to 4TB of Flash memory storage space. If a memory module has 1GB of DRAM storage capacity, then it can hold approximately 4K Blocks of data in the DRAM memory, each Block comprise 256 K Bytes of data. The DRAM address space, corresponding to the 4K blocks, can be assigned to different virtual ranks and banks, where the number of virtual ranks and banks is configurable and can be manipulated to meet a specific design or performance needs. For example, if a 1G Bytes memory module is configured to comprise two ranks with eight banks per rank, then each bank would hold two hundred fifty (250) blocks or the equivalent of 62 M Bytes or 62K pages, where each page correspond to a 1K Bytes. Other configurations using different page, block, banks, or ranks

numbers may also be used. Furthermore, an exemplary mapping of 24-bit DDR DIMM block address to Flash memory address, using Block addressing as described above, is shown in Fig.

10. The 24-bit can be decomposed into fields, such as a logical unit number LUN address 1060 field, a Block address 1050 field, a Plane address 1040, a Page address 1030, and a group of least significant address bits A_0A_1 1020. The Plane address 1040 is a sub address of the block address, and it may be used to support multiple page IO so as to improve Flash memory subsystem operation. In this example, it is understood that different number of bits may be allocated to each field of the 24-bit

[00101] The CDC 502 manages the block write-back operation by queuing the blocks that are ready to be written back to the Flash memory. As described above, if any page in a queued block for a write operation is reopened, then the CDC 502 will stop the queued block write operation, and remove the block from the queue. Once all the pages in a block are closed, then the CDC 502 restarts the write-back operation and queue the block for a write operation.

[00102] In accordance with one embodiment, an exemplary read operation from Flash 506 to DRAM 508 can be performed in approximately 400 μ s, while a write operation from DRAM 508 to Flash 506 can be performed in approximately 22ms resulting in a read to write ratio of 55 to 1. Therefore, if the average time a host system's memory controller spends accessing data information in a Block of DRAM is about 22ms (that is the duration that a Block comprises one or more pages that are open), then the block write-back operation from DRAM to Flash would not impact performance and hence the disparity between read and write access may be completely hidden from the memory controller. If the block usage time is 11ms instead of 22ms, then the CDC 502 control the data transfer operation between DRAM 508 and Flash 506 such

that there are no more than 9 closed blocks in the queue to be written-back to the Flash memory, hence approximately an average of 100ms can be maintained for a standard DDR DRAM operation. Moreover, the number of closed Blocks in the queue to be written-back to the Flash memory subsystem varies with the average block usage time and the desired performance for a specific host system or for a specific application running using the host system resources.

[00103] Consequently, the maximum number of closed Blocks to be written-back to Flash can be approximated to be

$$((\text{\#of blocks per bank})/(\text{ratio of 'Flash_block_write_time' to 'Flash_read_time'}))*(\text{Block usage time})/(\text{'Flash_block_write_time'})$$

[00104] In order to maintain less than 100ms time period for queued write-back Blocks, then using a Flash memory subsystem having 22ms write access time per Block would results in a maximum number of four Blocks to be queued for write operation to Flash 506. Therefore, on average approximately 88ms (= 22ms * 4) for blocks means that each bank should not have more than four Blocks that need to be written back to the Flash 506.

[00105] The above equation also indicates that bigger DRAM memory space can support shorter block usage times. For example, 2GB of DRAM memory allows the 8 closed blocks to be written-back to Flash. The table in FIG. 11 provides an estimation of the maximum allowed closed blocks in the queue to be written back to the Flash memory for different DRAM density using various average block use time.

[00106] While embodiments and applications have been shown and described, it would be apparent to those skilled in the art having the benefit of this disclosure that many more modifications than mentioned above are possible without departing from the inventive concepts disclosed herein. The invention, therefore, is not to be restricted except in the spirit of the appended claims.

CLAIMS

What is claimed is:

1. A memory module couplable to a memory controller of a host system, comprising:
a non-volatile memory subsystem;
a data manager coupled to the non-volatile memory subsystem;
a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager; and
a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller.
2. The memory module of claim 1, wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.
3. The memory module of claim 1, wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

4. The memory module of claim 3, wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.
5. The memory module of claim 1, wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.
6. The memory module of claim 1, wherein the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.
7. The memory module of claim 6, wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.
8. The memory module of claim 7, wherein each memory segment comprises at least one memory circuit, memory device, or memory die.
9. The memory module of claim 1, wherein the volatile memory subsystem comprises DRAM memory.

10. The memory module of claim 7, wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.
11. The memory module of claim 6, wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.
12. The memory module of 11, wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem.
13. A method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems, the method comprising:
 - receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem;
 - identifying a data path to be used for transferring data to or from the memory module using the received control information; and
 - using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path.

14. The method of claim 13, further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.
15. The method of claim 13, further comprising operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems.
16. The memory module of claim 13, wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.
17. The method of claim 15, further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.
18. The method of claim 13, further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.
19. The method of claim 18, further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

20. The method of claim 13, further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and non-volatile memory subsystems.

21. The method of claim 13, further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

22. The method of claim 21, wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

23. The method of claim 21, wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

24. The method of claim 13, further comprising:

operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system;

operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency.

ABSTRACT

A memory module that is couplable to a memory controller hub (MCH) of a host system includes a non-volatile memory subsystem, a data manager coupled to the non-volatile memory subsystem, a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager, and a controller operable to receive read/write commands from the MCH and to direct transfer of data between any two or more of the MCH, the volatile memory subsystem, and the non-volatile memory subsystem based on the commands.

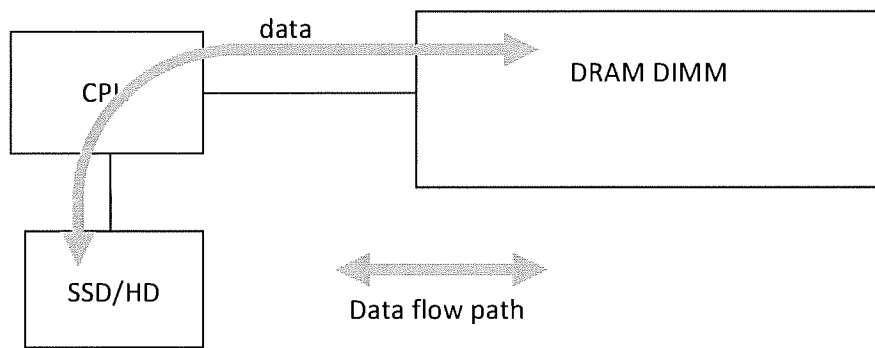


FIG. 1
(PRIOR ART)

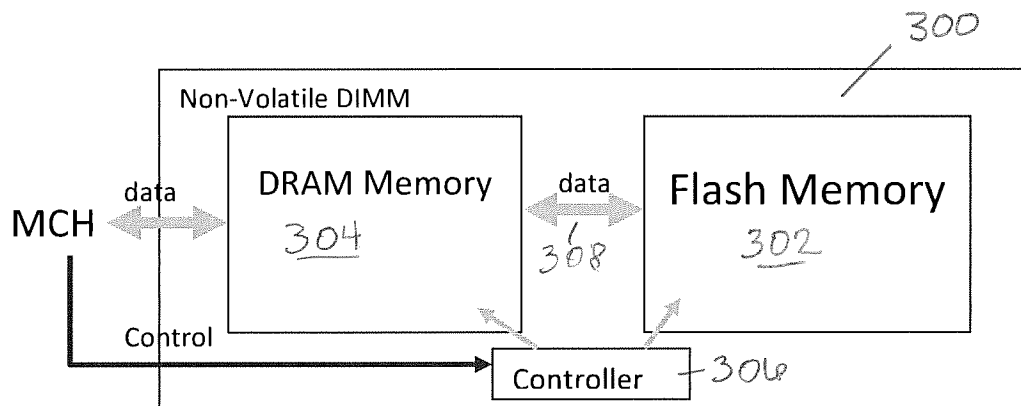
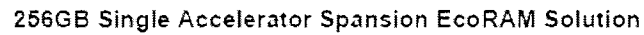


FIG. 3A

256GB Spansion EcoRAM Solution – Single Accelerator



The diagram illustrates a server motherboard architecture. At the top, it is labeled "Server Motherboard". Two Xeon processors are shown, each connected to a "Spanion EcoRAM Accelerator" via a HyperTransport (HT) link. Each accelerator is then connected to a set of DDR2 DIMMs, labeled "Up to 256 GB". The accelerators are also connected to the processors via HT links. The DIMMs are labeled "Spanion EcoRAM DIMMs".

256GB Dual Accelerator Spansion EcoRAM Solution

FIG. 2
(PRIOR ART)

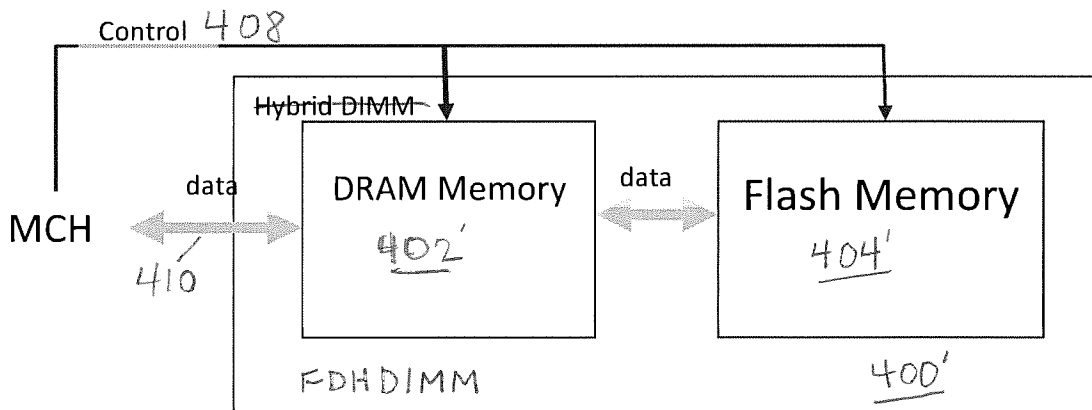


FIG. 4B

FIG. 3B

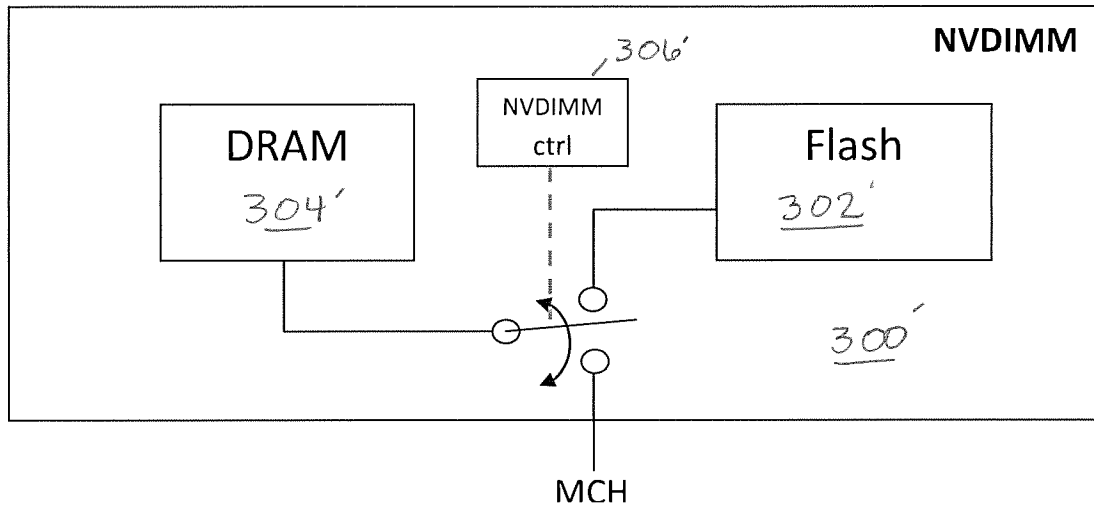
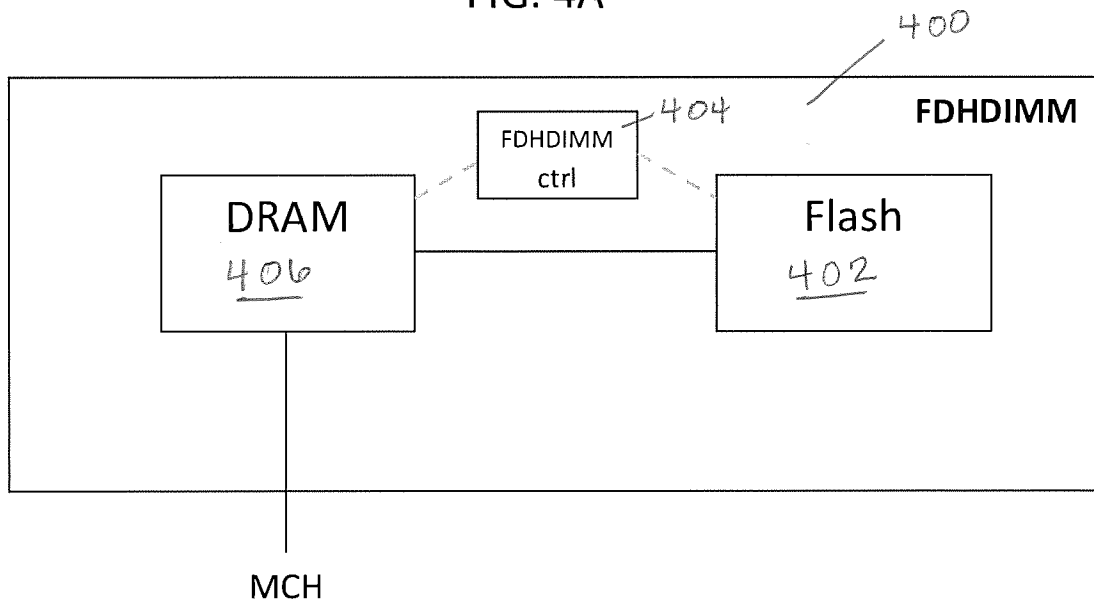


FIG. 4A



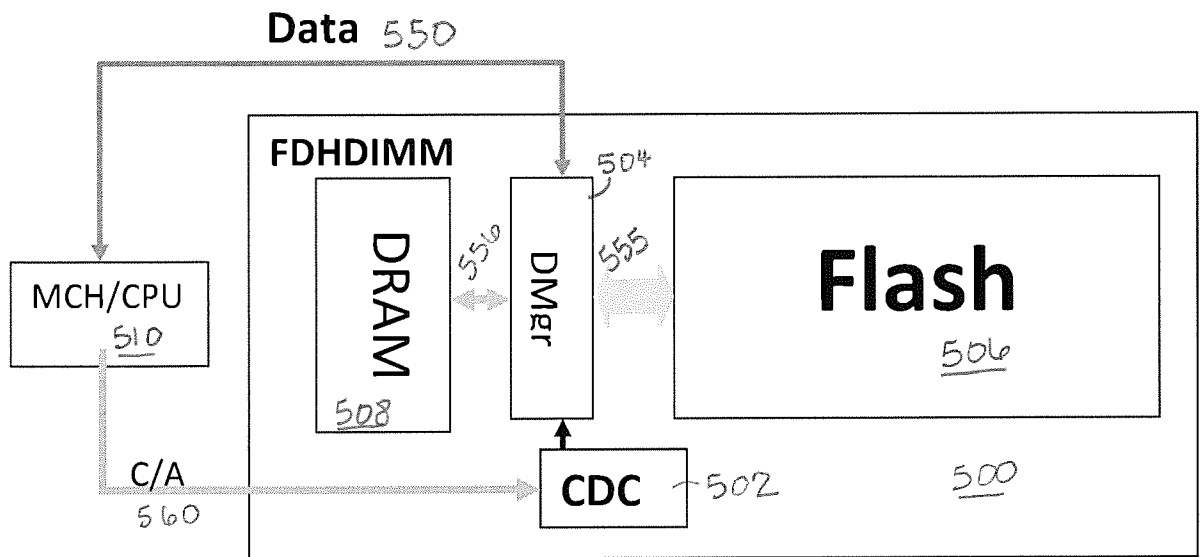


FIG. 5A

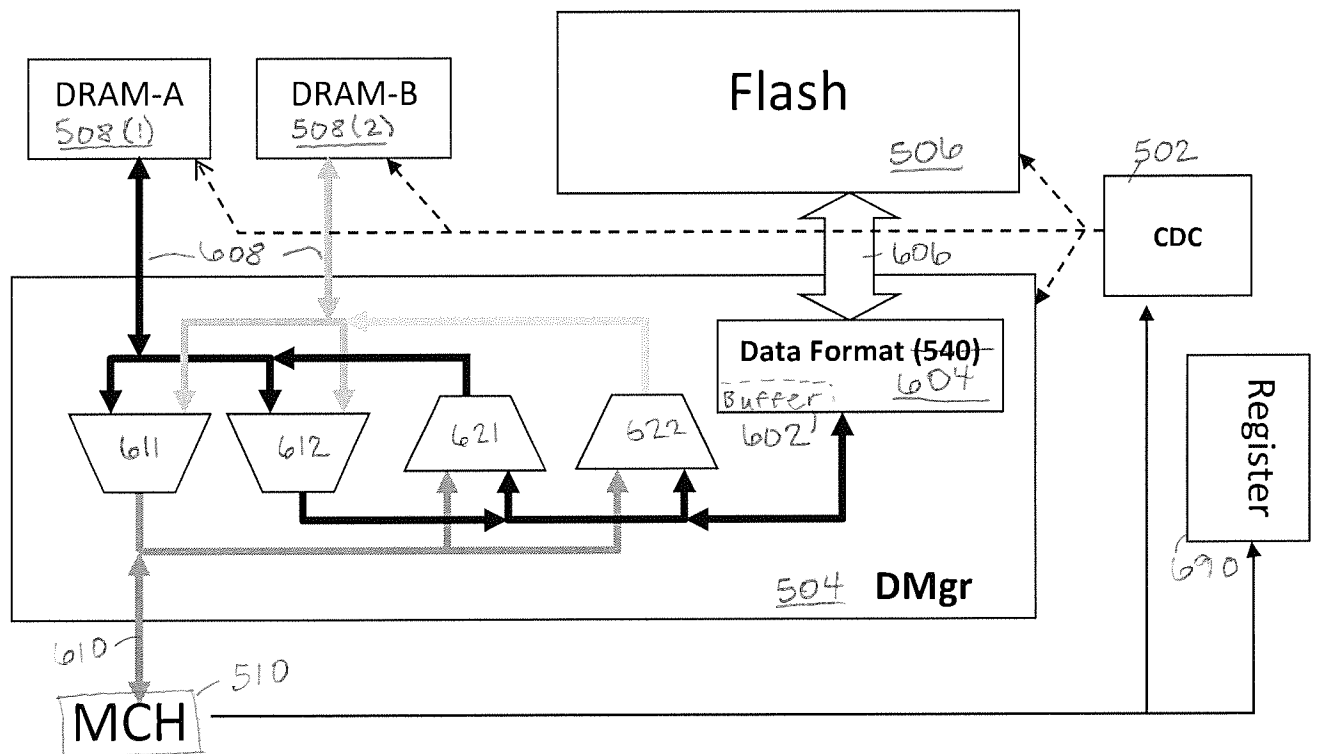
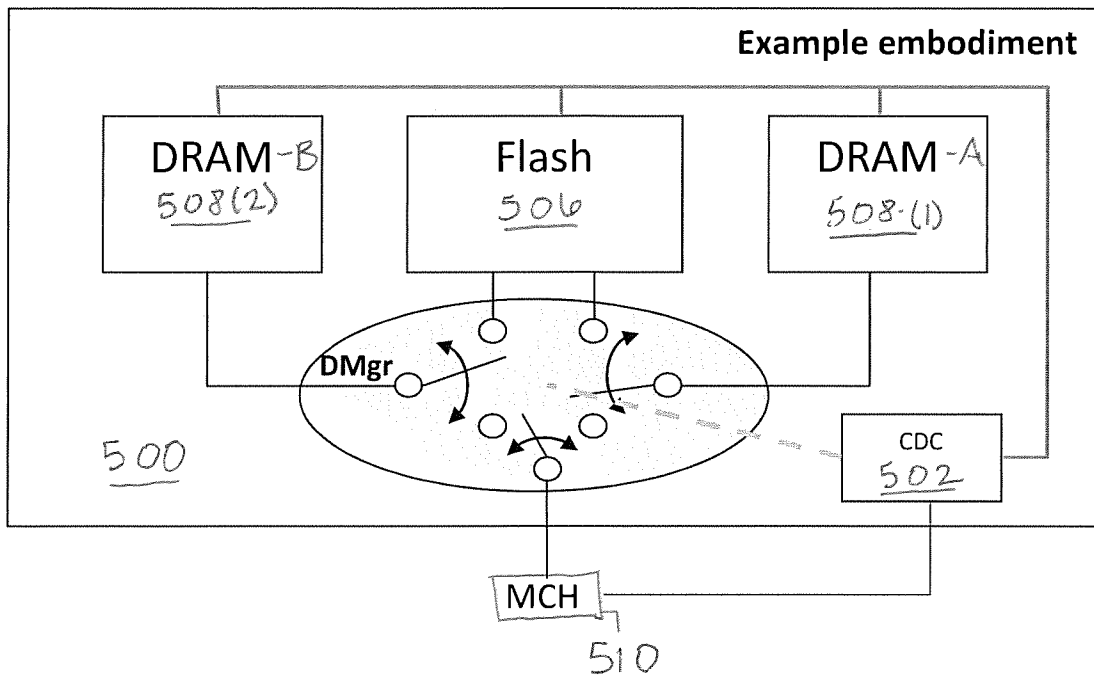


FIG. 6

FIG. 5B



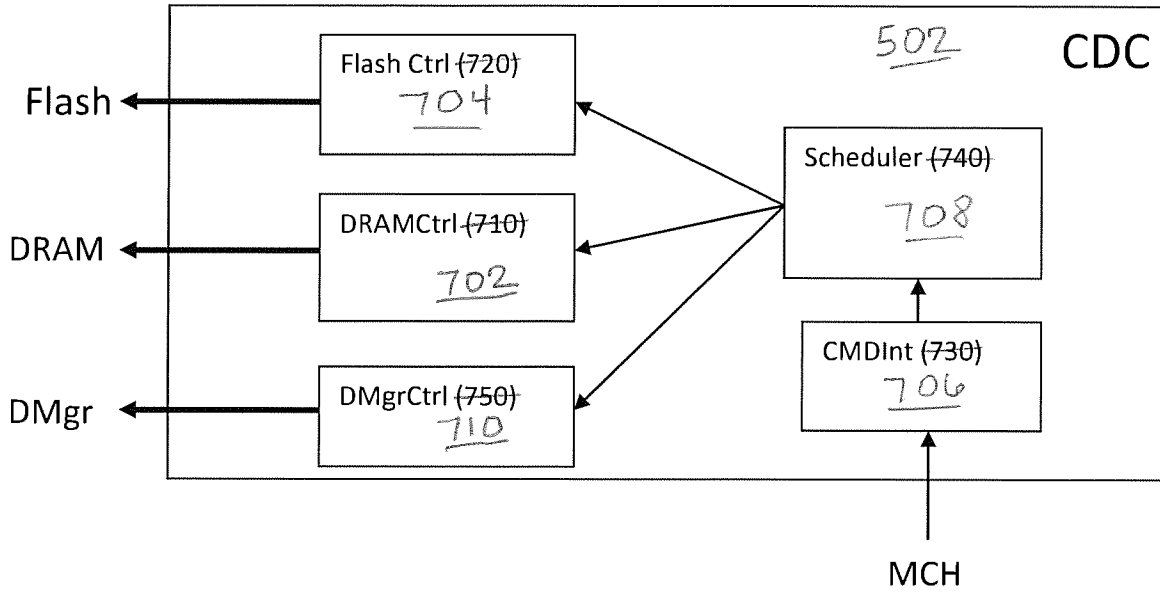


FIG. 7

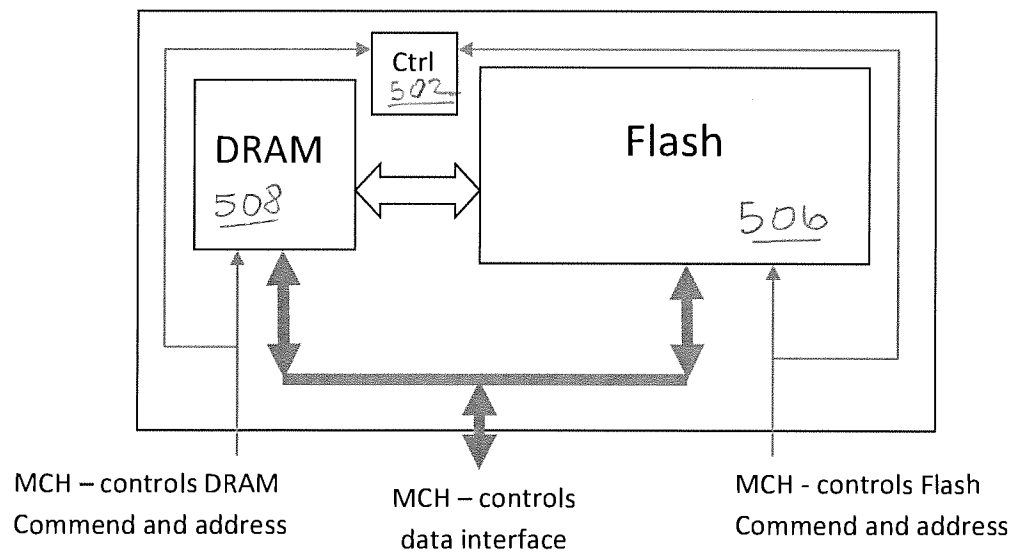


Figure 8A

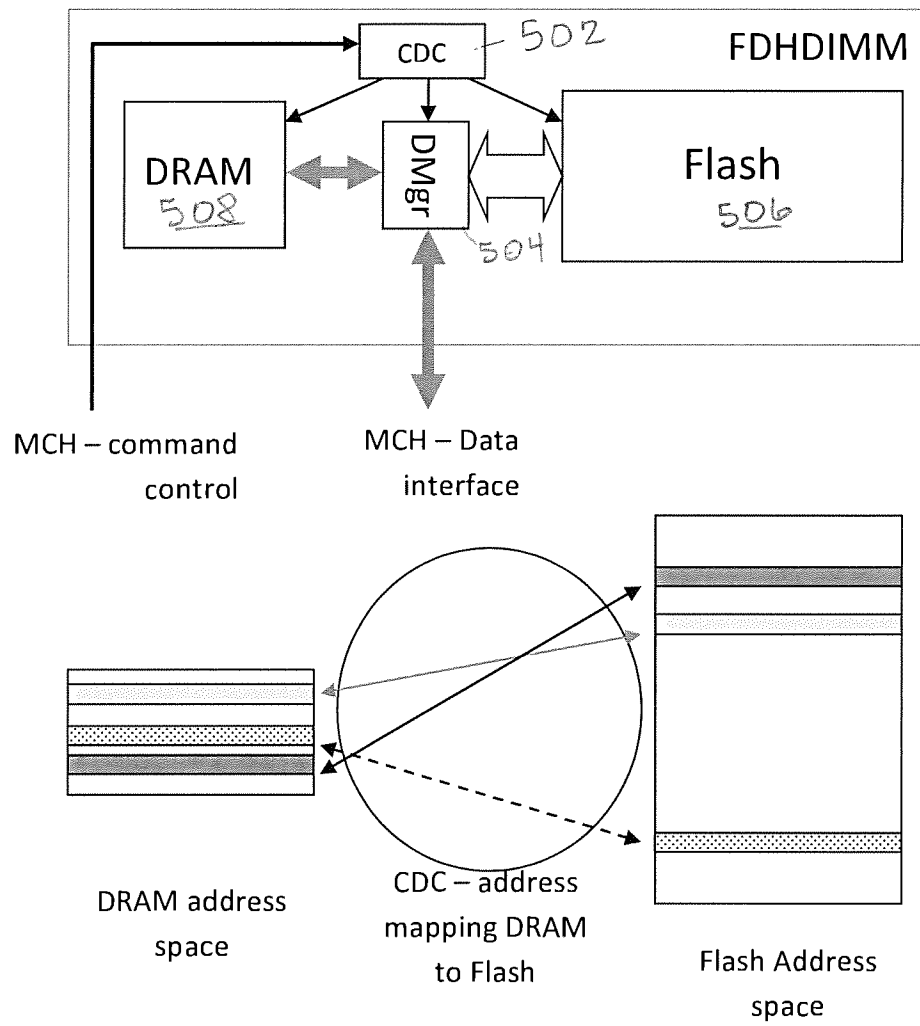


FIG. 8B

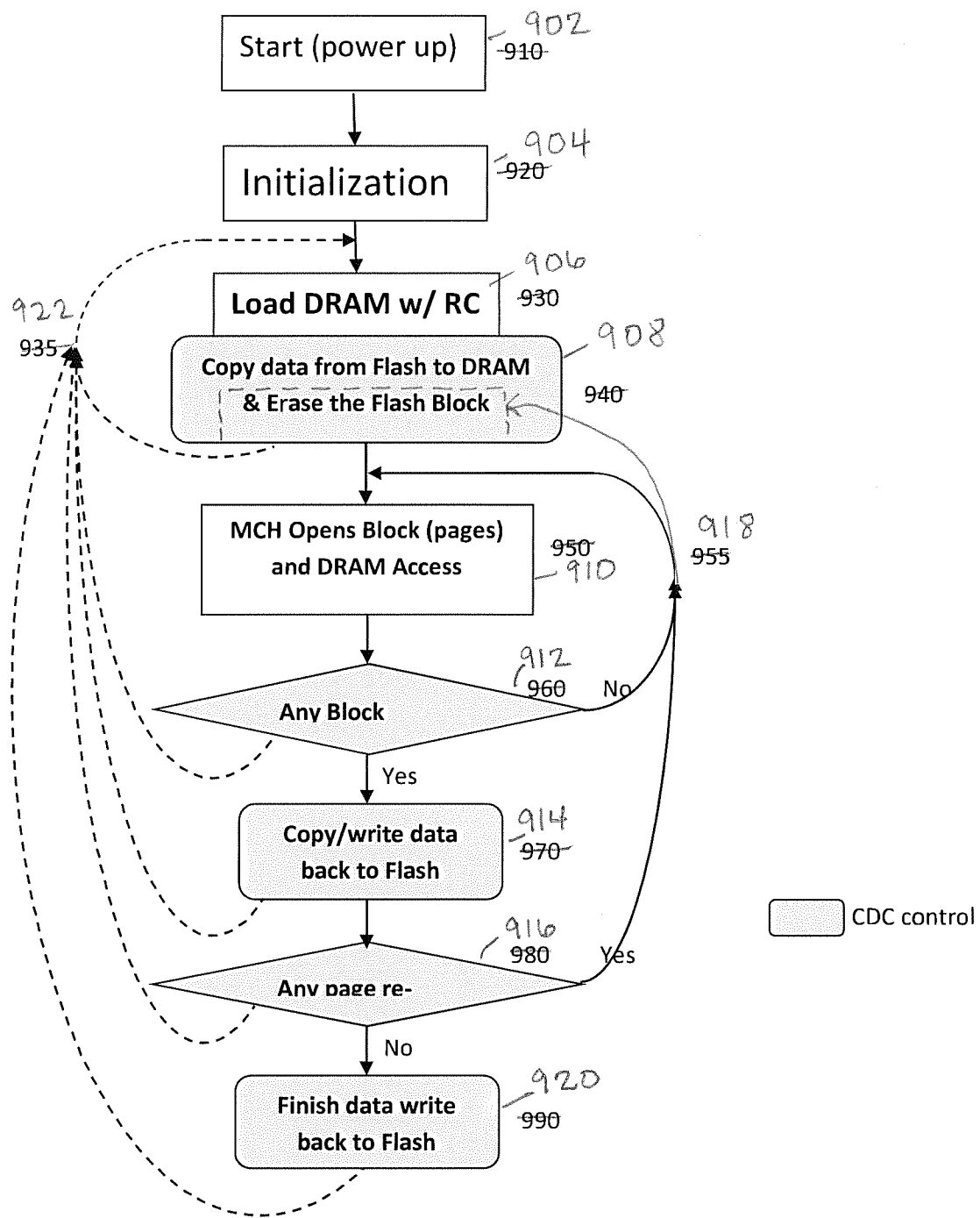


FIG. 9

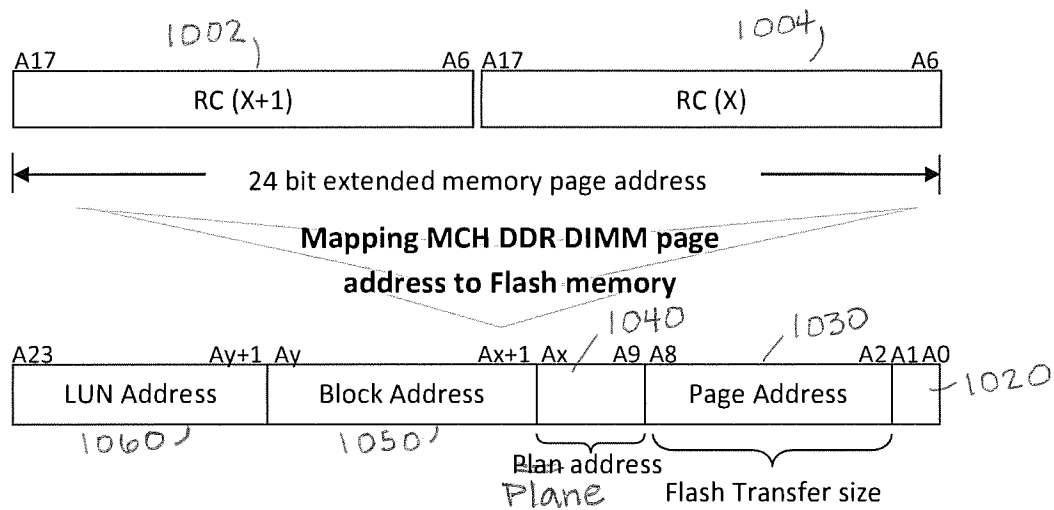


FIG. 10

DRAM density (GB)	# of blocks per bank	Flash wr-time to rd-time ratio	Avg block use time (sec)	Flash write time (sec)	Max allowed Closed Blk in queue to be written back to Flash
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-010
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

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☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

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Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence i	US
Citizenship under 37 CFR 1.41(b) i		US			
Mailing Address of Applicant:					
Address 1		21 THALIA ST			
Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country i	US		
Applicant 2 Remove					
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Chi-She		Chen		
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City	Walnut	State/Province	CA	Country of Residence i	US
Citizenship under 37 CFR 1.41(b) i		TW			
Mailing Address of Applicant:					
Address 1		944 Crystal Water Lane			
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country i	US		
Applicant 3 Remove					
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence i	US

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-010
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Citizenship under 37 CFR 1.41(b) i		US	
Mailing Address of Applicant:			
Address 1	6 Silver Fir		
Address 2			
City	Irvine	State/Province	CA
Postal Code	92604	Country ⁱ	US

Applicant 4					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Scott		Milton		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence ⁱ	US
Citizenship under 37 CFR 1.41(b) i		US			

Mailing Address of Applicant:					
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Address 2					
City	Irvine	State/Province	CA		
Postal Code	92602	Country ⁱ	US		

Applicant 5					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
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	Jayesh		Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US
Citizenship under 37 CFR 1.41(b) i		US			

Mailing Address of Applicant:					
Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country ⁱ	US		
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☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	46188

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-010
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	FLASH-DRAM HYBRID MEMORY MODULE		
Attorney Docket Number	062453-010	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	10	Suggested Figure for Publication (if any)	

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S. C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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Customer Number	46188		

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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61512871	2011-07-28
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation in part of	12240916	2008-09-29

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-010
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

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This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

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Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☐ No

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Assignee Information:

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Assignee 1

Remove

If the Assignee is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information:

Address 1	51 Discovery, Suite 150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			

Additional Assignee Data may be generated within this form by selecting the **Add** button.

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Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/		Date (YYYY-MM-DD)	2012-07-26	
First Name	Khaled	Last Name	Shami	Registration Number	38745

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	062453-010
		Application Number	
Title of Invention	FLASH-DRAM HYBRID MEMORY MODULE		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

FLASH-DRAM HYBRID MEMORY MODULE

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Khaled Shami/Pamela Wilson

Attorney Docket Number:

062453-010

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	380	380
Utility Search Fee	1111	1	620	620
Utility Examination Fee	1311	1	250	250

Pages:

Claims:

Claims in excess of 20	1202	4	60	240
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Miscellaneous-Filing:

Petition:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1490

Electronic Acknowledgement Receipt

EFS ID:	13353335
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	26-JUL-2012
Filing Date:	
Time Stamp:	19:31:36
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 1490
RAM confirmation Number	6870
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_010_Spec.pdf	202079	yes	44
			c60dd948101904d70a8443a4a64c47ae69da3e71		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	37	
	Claims		38	43	
	Abstract		44	44	
Warnings:					
Information:					
2	Drawings-only black and white line drawings	062453_010_Drawings.pdf	190034	no	10
			073f5c7db3f6af50483c4fbc3e361c5806bb3acf		
Warnings:					
Information:					
3	Application Data Sheet	data_sheet.pdf	1537065	no	6
			c32664fd53e98c643bf5c173e5101bfce57e9255		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	36510	no	2
			9f8fce9ec382ceb6b6db27f9144ce79078ff5de5		
Warnings:					
Information:					
Total Files Size (in bytes):			1965688		

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/559,476	07/26/2012	2189	1490	062453-010	24	2

CONFIRMATION NO. 1046

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 08/13/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/559,476**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/559,476

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	380
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	620
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	250
TOTAL CLAIMS (37 CFR 1.16(j))	24 minus 20 =	*			OR	x 60 =	240
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*				x 250 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1490

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/559,476	07/26/2012	Hyun Lee	062453-010

CONFIRMATION NO. 1046

FORMALITIES LETTER



46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Date Mailed: 08/13/2012

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The oath or declaration is missing.

A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.

Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of \$130 for a non-small entity, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$130 for a non-small entity

- \$130 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/eggolla/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT:	Hyun Lee.	CONFIRMATION NO.: 1046
APPLICATION NO.:	13/559,476	
FILING DATE:	July 26, 2012	
TITLE:	FLASH-DRAM HYBRID MEMORY MODULE	
EXAMINER:	unassigned	
ART UNIT:	unassigned	

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

PRELIMINARY AMENDMENT

Sir:

Please amend the above-identified application as follows:

Amendments to the Specification begin on page **2** of this paper.

Remarks begin on page **3** of this paper.

In the Specification

Please amend paragraph [0001] as follows:

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, ~~itled “HIGH DENSITY DIMMS”~~, and of U.S. patent application serial 13/559,476, filed July 26, 2012, and which is a continuation-in-part (CIP) of US patent application serial no. 12/240,916, filed September 29, 2008, ~~itled “NON-VOLATILE MEMORY MODULE”~~which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of both of which are incorporated herein by reference in their entirety.

This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

REMARKS

The specification has been amended to more accurately reflect priority and related application information.

Early consideration and allowance of this application is earnestly solicited.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: August 20, 2012

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX. (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	13539422
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	20-AUG-2012
Filing Date:	26-JUL-2012
Time Stamp:	18:21:55
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_010_Prelim_amendme nt.pdf	116869 078f71fa178386ecc77a09770b11ebcf028b c89	yes	3

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Specification	2	2
	Applicant Arguments/Remarks Made in an Amendment	3	3
Warnings:			
Information:			
Total Files Size (in bytes):		116869	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name		
	Attorney Docket Number	062453-010	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4420821		1983-12-13	Hoffman	
	2	4449205		1984-05-15	Hoffman	
	3	5519663		1996-05-21	Harper, Jr. et al.	
	4	6158015		2000-12-05	Klein	
	5	6336174	B1	2002-01-01	Li et al.	
	6	6336176	B1	2002-01-01	Leyda et al.	
	7	6487623	B1	2002-11-26	Emerson et al.	
	8	6658507	B1	2003-12-02	Chan	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit		2189	
	Examiner Name			
	Attorney Docket Number		062453-010	

	9	6799244	B2	2004-09-28	Tanaka et al.	
	10	7409590	B2	2008-08-05	Moshayedi et al.	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20020083368	A1	2002-06-27	Abe et al.	
	2	20040190210	A1	2004-09-30	Leete	
	3	20070192627	A1	2007-08-16	Oshikiri	
	4	20080195806	A1	2008-08-14	Cope	

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FOREIGN PATENT DOCUMENTS								Remove
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² ;	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name		
	Attorney Docket Number	062453-010	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012.	☐
	2	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012.	☐
	3	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011.	☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	
Attorney Docket Number	062453-010

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2012-09-27
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	13853673
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Tadas Narauskas
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	27-SEP-2012
Filing Date:	26-JUL-2012
Time Stamp:	14:19:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	062453-010_IDS_FORM.pdf	613227 6d6b8421757a2e96e22bbcd22ffe7e3b557d812	no	5

Warnings:

Information:

2	Non Patent Literature	OA_12240916_02-01-2012.pdf	478286 4c543efdb9b4cbd435ce0fe9eff687d64b283f53	no	14
Warnings:					
Information:					
3	Non Patent Literature	OA_12240916_04-03-2012.pdf	414082 45d9b1f93055200016b162ea2b824edecfefaf2a	no	12
Warnings:					
Information:					
4	Non Patent Literature	OA_12240916_07-29-2011.pdf	264654 1c9f68a01a64c8fec8e2c0d219a52aaa9c284d60	no	8
Warnings:					
Information:					
Total Files Size (in bytes):			1770249		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

DECLARATION & POWER OF ATTORNEY

As a below-named inventor, I hereby declare that:

My correct city and state of residence, my post office address and my citizenship are stated below next to my name.

I believe myself to be an original and joint inventor of the subject matter which is disclosed and claimed and for which a patent is sought on the invention entitled:

"FLASH-DRAM HYBRID MEMORY MODULE"

The specification of this subject matter:

- ☐ is attached hereto.
- ☒ was filed on July 26, 2012 as United States Application Number 13/559,476
- ☒ and was amended on August 20, 2012 (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified patent application, including the claims, as amended by any amendment(s) referred to above. I do not know and do not believe that the claimed invention was ever known or used in the United States of America before my invention thereof, or patented or described in any printed publication in any country before my invention thereof or more than one year prior to this application, that the same was not in public use or on sale in the United States of America more than one year prior to this application, and that the invention has not been patented or made the subject of an inventor's certificate issued before the date of this application in any country foreign to the United States of America on an application filed by me or my legal representatives or assigns more than twelve months (for a utility patent application) or six months (for a design patent application) prior to this application.

I acknowledge the duty to disclose information which is material to the patentability of this application in accordance with Title 37, Code of Federal Regulations § 1.56.

I hereby claim foreign priority benefits under 35 U.S.C. §119 of any foreign application(s) for patent or inventor's certificate listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed.

PRIOR FOREIGN APPLICATION(S)

Priority Claimed

Certified Copy Attached?

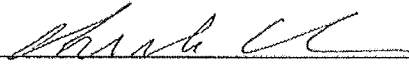
Number	Country	Month/Day/Year Filed	Yes	No	Yes	No

I hereby appoint practitioners associated with **Customer Number 46188** as attorneys of record with full power of substitution and revocation, to prosecute this application and transact all business in the United States Patent and Trademark Office connected therewith. If this application is assigned by me I agree and understand that the above-named attorneys will represent the assignee and not me.

Please send all correspondence and direct all telephone calls to the address associated with **Customer Number 46188**.

I, the undersigned, declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code, and that such willful false statements may jeopardize the validity of the application or any patent issuing therefrom.

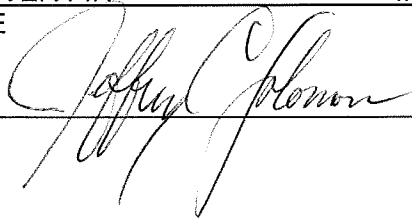
FULL NAME OF INVENTOR 1	FIRST Name	MIDDLE Initial(s)	LAST Name	
HUYN			LEE	
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship	
LADERA RANCH		CALIFORNIA	US	
POST OFFICE ADDRESS	Number and Street	City	State or Country	Zip Code
21 THALIA ST.	LADERA RANCH	CA	92694	
SIGNATURE	Date			
	11/12/2012.			

FULL NAME OF INVENTOR 2	FIRST Name	MIDDLE Initial(s)	LAST Name
CHI-SHE			CHEN
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
WALNUT		CALIFORNIA	TW
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
944 CRYSTAL WATER LANE	WALNUT	CA	91789
SIGNATURE	Date		
	Sep. 09, 2012		

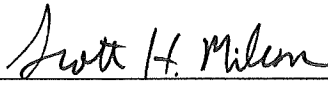
FULL NAME OF INVENTOR 3	FIRST Name	MIDDLE Initial(s)	LAST Name
JEFFREY		C.	SOLOMON
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
IRVINE		CALIFORNIA	US
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
6 SILVER FIR	IRVINE	CA	92604
SIGNATURE	Date		

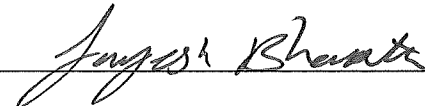
FULL NAME OF INVENTOR 2	FIRST Name	MIDDLE Initial(s)	LAST Name
CHI-SHE			CHEN
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
WALNUT		CALIFORNIA	TW
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
944 CRYSTAL WATER LANE	WALNUT	CA	91789
SIGNATURE		Date	

FULL NAME OF INVENTOR 3	FIRST Name	MIDDLE Initial(s)	LAST Name
JEFFREY		C.	SOLOMON
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
IRVINE		CALIFORNIA	US
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
16 SILVER FIR	IRVINE	CA	92604
SIGNATURE		Date	



11-12-2012

FULL NAME OF Name	FIRST Name	MIDDLE Initial(s)	LAST
INVENTOR 4			
SCOTT			MILTON
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
IRVINE		CALIFORNIA	US
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
49 STATEHOUSE PLACE	IRVINE	CA	92602
SIGNATURE	Date		
	09/06/12		

FULL NAME OF Name	FIRST Name	MIDDLE Initial(s)	LAST
INVENTOR 5			
JAYESH			BHAKTA
RESIDENCE AND CITIZENSHIP	City	State or Foreign Country	Country of Citizenship
CERRITOS		CALIFORNIA	US
POST OFFICE ADDRESS	Number and Street	City	State or Country Zip Code
12220 ROSE STREET	CERRITOS	CA	90703
SIGNATURE	Date		
	9/6/12		

37 C.F.R. §1.56**Duty to disclose information material to patentability**

(a) A patent by its very nature is affected with a public interest. The public interest is best served, and the most effective patent examination occurs when, at the time an application is being examined, the Office is aware of and evaluates the teachings of all information material to patentability. Each individual associated with the filing and prosecution of a patent application has a duty of candor and good faith in dealing with the Office which includes a duty to disclose to the Office all information known to that individual to be material to patentability as defined in this section. The duty to disclose information exists with respect to each pending claim until the claim is cancelled or withdrawn from consideration, or the application becomes abandoned. Information material to the patentability of a claim that is cancelled or withdrawn from consideration need not be submitted if the information is not material to the patentability of any claim remaining under consideration in the application. There is no duty to submit information which is not material to the patentability of any existing claim. The duty to disclose all information known to be material to patentability is deemed to be satisfied if all information known to be material to patentability of any claim issued in a patent was cited by the Office or submitted to the Office in the manner prescribed by §§1.97(b)-(d) and 1.98. However, no patent will be granted on an application in connection with which fraud on the Office was practiced or attempted or the duty of disclosure was violated through bad faith or intentional misconduct. The Office encourages applicants to carefully examine:

(1) Prior art cited in search reports of a foreign patent office in a counterpart application, and

(2) The closest information over which individuals associated with the filing or prosecution of a patent application believe any pending claim patentably defines, to make sure that any material information contained therein is disclosed to the Office.

(b) Under this section, information is material to patentability when it is not cumulative to information already of record or being made of record in the application, and

(1) It establishes, by itself or in combination with other information, a prima facie case of unpatentability of a claim; or

(2) It refutes, or is inconsistent with, a position the applicant takes in:

- (i) Opposing an argument of unpatentability relied on by the Office, or
- (ii) Asserting an argument of patentability.

A prima facie case of unpatentability is established when the information compels a conclusion that a claim is unpatentable under the preponderance of evidence, burden-of-proof standard, giving each term in the claim its broadest reasonable construction consistent with the specification, and before any consideration is given to evidence which may be submitted in an attempt to establish a contrary conclusion of patentability.

(c) Individuals associated with the filing or prosecution of a patent application within the meaning of this section are:

(1) Each inventor named in the application;

(2) Each attorney or agent who prepares or prosecutes the application; and

(3) Every other person who is substantively involved in the preparation or prosecution of the application and who is associated with the inventor, with the assignee or with anyone to whom there is an obligation to assign the application.

(d) Individuals other than the attorney, agent or inventor may comply with this section by disclosing information to the attorney, agent, or inventor.

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-010			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	150	150
Miscellaneous:				
Total in USD (\$)				280

Electronic Acknowledgement Receipt

EFS ID:	14212528
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	13-NOV-2012
Filing Date:	26-JUL-2012
Time Stamp:	13:27:04
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 280
RAM confirmation Number	20035
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	062453_010_Dec_and_POA.pdf	539426 6546714f83a331355ba19d601961fe0506dc5c84	no	6

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	32225 63dc82500c4030088f04c95b9b587daca31c087e	no	2
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Warnings:**Information:**

Total Files Size (in bytes):			571651
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	8301833		2012-10-30	Chen et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

Add

U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20080104344		2008-05-01	Shimozono et al.	
	2	20100274953		2010-10-28	Lee et al.	

If you wish to add additional U.S. Published Application citation information please click the Add button.

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FOREIGN PATENT DOCUMENTS								Remove
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

Add

NON-PATENT LITERATURE DOCUMENTS								Remove
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name		
	Attorney Docket Number	062453-010	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion in PCT/US12/48750, dated October 10, 2012	☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	
Attorney Docket Number	062453-010

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2012-12-11
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PATENT COOPERATION TREATY

RECEIVED

OCT 15 2012

From the INTERNATIONAL SEARCHING AUTHORITY

PCT NIXON PEABODY LLP

To: Khaled Shami
P.O. Box 60610
Palo Alto, CA 94306
United States of America

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

Date of mailing
(day/month/year)

10 OCT 2012

Applicant's or agent's file reference
062453-0011

FOR FURTHER ACTION See paragraphs 1 and 4 below

International application No.
PCT/US12/48750

International filing date
(day/month/year) 28 July 2012 (28.07.2012)

Applicant
Netlist, Inc.

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.

Where? Directly to the International Bureau of WIPO, 34 chemin des Colombettes
1211 Geneva 20, Switzerland, Facsimile No.: +41 22 338 82 70

For more detailed instructions, see *PCT Applicant's Guide*, International Phase, paragraphs 9.004 – 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:

☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.

☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. Reminders

The applicant may submit comments on an informal basis on the written opinion of the International Searching Authority to the International Bureau. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established. Following the expiration of 30 months from the priority date, these comments will also be made available to the public.

Shortly after the expiration of **18 months** from the priority date, the international application will be published by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90bis.1 and 90bis.3).

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for entry into the national phase before those designated Offices.

In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months.

For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.

Name and mailing address of the ISA/
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Authorized officer

Shane Thomas

PCT Helpdesk: 571-272-4300

Telephone No. PCT OSP: 571-272-7774

PATENT COOPERATION TREATY

RECEIVED

From the INTERNATIONAL SEARCHING AUTHORITY

OCT 15 2012

PCT

NIXON PEABODY LLP

NOTIFICATION OF TRANSMITTAL OF
THE INTERNATIONAL SEARCH REPORT AND
THE WRITTEN OPINION OF THE INTERNATIONAL
SEARCHING AUTHORITY, OR THE DECLARATION

(PCT Rule 44.1)

To: Khaled Shami P.O. Box 60610 Palo Alto, CA 94306 United States of America	
Date of mailing (day/month/year)	
Applicant's or agent's file reference 062453-0011	FOR FURTHER ACTION See paragraphs 1 and 4 below
International application No. PCT/US12/48750	International filing date (day/month/year) 28 July 2012 (28.07.2012)
Applicant Netlist, Inc.	

1. ☒ The applicant is hereby notified that the international search report and the written opinion of the International Searching Authority have been established and are transmitted herewith.

Filing of amendments and statement under Article 19:

The applicant is entitled, if he so wishes, to amend the claims of the international application (see Rule 46):

When? The time limit for filing such amendments is normally two months from the date of transmittal of the international search report.

Where? Directly to the International Bureau of WIPO, 34 chemin des Colombettes
1211 Geneva 20, Switzerland, Facsimile No.: +41 22 338 82 70

For more detailed instructions, see *PCT Applicant's Guide*, International Phase, paragraphs 9.004 – 9.011.

2. ☐ The applicant is hereby notified that no international search report will be established and that the declaration under Article 17(2)(a) to that effect and the written opinion of the International Searching Authority are transmitted herewith.

3. ☐ **With regard to any protest** against payment of (an) additional fee(s) under Rule 40.2, the applicant is notified that:
- ☐ the protest together with the decision thereon has been transmitted to the International Bureau together with any request to forward the texts of both the protest and the decision thereon to the designated Offices.
- ☐ no decision has been made yet on the protest; the applicant will be notified as soon as a decision is made.

4. Reminders

The applicant may submit comments on an informal basis on the written opinion of the International Searching Authority to the International Bureau. The International Bureau will send a copy of such comments to all designated Offices unless an international preliminary examination report has been or is to be established. Following the expiration of 30 months from the priority date, these comments will also be made available to the public.

Shortly after the expiration of **18 months** from the priority date, the international application will be published by the International Bureau. If the applicant wishes to avoid or postpone publication, a notice of withdrawal of the international application, or of the priority claim, must reach the International Bureau before the completion of the technical preparations for international publication (Rules 90bis.1 and 90bis.3).

Within **19 months** from the priority date, but only in respect of some designated Offices, a demand for international preliminary examination must be filed if the applicant wishes to postpone the entry into the national phase **until 30 months** from the priority date (in some Offices even later); otherwise, the applicant must, **within 20 months** from the priority date, perform the prescribed acts for entry into the national phase before those designated Offices.

In respect of other designated Offices, the time limit of **30 months** (or later) will apply even if no demand is filed within 19 months.

For details about the applicable time limits, Office by Office, see www.wipo.int/pct/en/texts/time_limits.html and the *PCT Applicant's Guide*, National Chapters.

Name and mailing address of the ISA/ Mail Stop PCT, Attn: ISA/US Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-3201	Authorized officer Shane Thomas PCT Helpdesk: 571-272-4300 Telephone No. PCT OSP: 571-272-7774
---	---

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference 062453-0011	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/US12/48750	International filing date (<i>day/month/year</i>) 28 July 2012 (28.07.2012)	(Earliest) Priority Date (<i>day/month/year</i>) 28 July 2011 (28.07.2011)
Applicant Netlist, Inc.		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 2 sheets.

☐ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed.
☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (see Box No. II).

3. ☐ **Unity of invention is lacking** (see Box No. III).

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant.
☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☒ the text is approved as submitted by the applicant.
☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority.

6. With regard to the **drawings**,

a. the figure of the **drawings** to be published with the abstract is Figure No. 5

- ☒ as suggested by the applicant.
☐ as selected by this Authority, because the applicant failed to suggest a figure.
☐ as selected by this Authority, because this figure better characterizes the invention.

b. ☐ none of the figures is to be published with the abstract.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US12/48750

A. CLASSIFICATION OF SUBJECT MATTER

IPC(8) - G06F 12/00 (2012.01)

USPC - 711/118, 103

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC(8): G06F 12/00, 12/08, 13/16 (2012.01)

USPC: 711/170, 104, 118, 103

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

MicroPatent (US-G, US-A, EP-A, EP-B, WO, JP-bib, DE-C,B, DE-A, DE-T, DE-U, GB-A, FR-A); DialogPRO; IEEE/IEEExplore; Google/Google Scholar; IP.com; volatile, non-volatile, memory, controller, manager, performance, frequency, clock, plural, multiple, port, segment, subset

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X -- Y	US 2008/0104344 A1, (SHIMOZONO, N., et al.), May 1, 2008, figures 19, paragraphs [0017], [0053], [0055], [0057], [0061], [0065], [0067], [0080]	1-6, 9, 11-15, 18-23 ----- 7, 8, 10, 16, 17, 24
Y	US 2010/0274953 A1, (LEE, T., et al.), October 28, 2010, figures 1, 2, 4B, 10, paragraphs [0005], [0007], [0046], [0053], [0058], [0062], [0065], [0066], [0071], [0080], [0115]	7, 8, 10, 16, 17, 24

☐ Further documents are listed in the continuation of Box C.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

21 September 2012 (21.09.2012)

Date of mailing of the international search report

10 OCT 2012

Name and mailing address of the ISA/US

Mail Stop PCT, Attn: ISA/US, Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450

Facsimile No. 571-273-3201

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300

PCT OSP: 571-272-7774

KS

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43bis.1)

To: Khaled Shami
P.O. Box 60610
Palo Alto, CA 94306
United States of America

Date of mailing
(day/month/year)

1 0 OCT 2012

Applicant's or agent's file reference
062453-0011

FOR FURTHER ACTION

See paragraph 2 below

International application No.

PCT/US12/48750

International filing date (day/month/year)

28 July 2012 (28.07.2012)

Priority date (day/month/year)

28 July 2011 (28.07.2011)

International Patent Classification (IPC) or both national classification and IPC

IPC(8) - G06F 12/00 (2012.01)

USPC - 711/118, 103

Applicant
Netlist, Inc.

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☐ Box No. VIII Certain observations on the international application

2. **FURTHER ACTION**

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1 bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/US
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Date of completion of this opinion

21 September 2012 (21.09.2012)

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US12/48750

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43 bis.1(a))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:
 - a. (means)
 - ☐ on paper
 - ☐ in electronic form
 - b. (time)
 - ☐ in the international application as filed
 - ☐ together with the international application in electronic form
 - ☐ subsequently to this Authority for the purposes of search
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US12/48750

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

I. Statement

Novelty (N)	Claims	7, 8, 10, 16, 17, 24	YES
	Claims	1-6, 9, 11-15, 18-23	NO
Inventive step (IS)	Claims	NONE	YES
	Claims	1-24	NO
Industrial applicability (IA)	Claims	1-24	YES
	Claims	NONE	NO

2. Citations and explanations:

Claims 1-6, 9, 11-15 and 18-23 lack novelty under PCT Article 33(2) as being anticipated by US 2008/0104344 A1, SHIMOZONO, N., et al (hereinafter 'Shimozono').

As per Claim 1, Shimozono discloses a memory module (storage system 101 (a memory module), Figure 19, paragraph [0053]) couplable to a memory controller of a host system (is connected to storage system input/output (memory controller) of host 200 through front end I/F 307, Figure 19, paragraphs [0017] and [0080]), comprising: a non-volatile memory subsystem (non-volatile memory 313 (non-volatile memory subsystem), NM, Figure 19, paragraph [0055]); a data manager coupled to the non-volatile memory subsystem (microprocessor 303, MP, comprising the functionality of a data manager, is connected to non-volatile memory 313, through switch 309, SW, Figure 19, paragraph [0055]); a volatile memory subsystem coupled to the data manager (cache memory 311, CM, is volatile memory (a volatile memory subsystem) connected to microprocessor 303 through switch 309, SW, Figure 19, paragraphs [0055] and [0057]) and operable to exchange data with the non-volatile memory subsystem by way of the data manager (microprocessor 303 can copy data on the volatile memory, CM311, to the non-volatile memory 313, paragraphs [0055], [0057], [0065]); and a controller operable to receive commands from the memory controller (controller 301 communicates with the storage system input/output (the memory controller) of host 200 and uses microprocessor 303 to process read and write commands, paragraphs [0017], [0055], [0056], [0061]) and to direct (i) operation of the non-volatile memory subsystem (processes read and write commands with non-volatile memory 313, paragraphs [0055], [0065], [0067]), (ii) operation of the volatile memory subsystem (processes read and write commands with volatile memory 311, paragraphs [0055], [0057], [0065], [0067]), and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller (based on a read command from the storage system input/output (memory controller) of host 200, transmits data between volatile memory 311 and non-volatile memory 313 with the storage system input/output of host 200, paragraphs [0017], [0055], [0057], [0065], [0067]).

As per Claim 2, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller (microprocessor 303 (the data manager) processes commands received from the storage system input/output (the memory controller) of host 200 and allocates the volatile memory into a fixed length segment size (controls data buffer size), paragraphs [0017], [0057], [0061], [0114]).

As per Claim 3, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller (microprocessor 303 (the data manager) specifies the access address information from the read and write commands of the storage system input/output (the memory controller) of host 200 and controls the transmission of data (controls data traffic) between volatile memory 311, non-volatile memory 313, and the storage system input/output of host 200, paragraphs [0017], [0055], [0057], [0065], [0067]).

As per Claim 4, Shimozono discloses the memory module of claim 3; additionally, Shimozono discloses wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer (microprocessor 303 processes host 200 commands indicating the type of command, transfer address, and transfer length (data transfer size), paragraphs [0055], [0061], [0083]).

As per Claim 5, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module (the host 200 through the storage system input/output (the memory controller) can control the memory regions of volatile sub-regions (a first memory address space of the volatile memory subsystem) and non-volatile sub-regions (a second memory address space of the non-volatile memory subsystem), in units called host volumes when the microprocessor executes commands from the controller 301 regarding the configuration of the memory regions, paragraph [0017], [0055]-[0059], [0086], [0102]).

As per Claim 6, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager is configured as a bidirectional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems (microprocessor 303 (the data manager) provides read and write command processing with the storage system input/output of host 200 by utilizing one or more ports (two or more sets of data ports) to the volatile cache memory of the multiplexed storage devices (a bidirectional data transfer fabric), paragraphs [0017], [0061], [0077], claim 15).

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**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US12/48750

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

---Continued from Box No. V: Citations and Explanations---

As per Claim 9, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the volatile memory subsystem comprises DRAM memory (CM 311 is volatile memory may be DRAM, Dynamic Random Access Memory, paragraph [0057]).

As per Claim 11, Shimozono discloses the memory module of claim 6; additionally, Shimozono discloses wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller (microprocessor 303 (the data manager) processes volatile memory 311 and non-volatile memory 313 commands between the storage system input/output (the memory controller) of host 200, using Fiber Channel Protocol, Ethernet and iSCSI, Internet Small Computer System Interface, protocols (a data format module to format data), paragraphs [0017], [0057], [0061], [0077], [0078]).

As per Claim 12, Shimozono discloses the memory module of 11; additionally, Shimozono discloses wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem (microprocessor 303 (the data manager) copies volatile memory 311 to the non-volatile cache region 391 (data buffer for buffering data) of the non-volatile memory 313, Figure 19, paragraphs [0055], [0057], [0068]).

As per Claim 13, Shimozono discloses a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems (a process of receiving commands on storage system 101 (a memory module) from storage system input/output (a memory controller) of host 200 and reading or writing data to volatile memory 311 and non-volatile memory 313, Figure 19, paragraph [0017], [0055], [0057], [0061]), the method comprising: receiving control information from the memory controller (commands are transceived between the storage system input/output (the memory controller) of host 200 and storage system 101 using a selected protocol, paragraphs [0017], [0077]-[0079]), wherein the control information is received using a protocol of the volatile memory subsystem (using the protocol, the storage system 101 provides volatile memory, CM 311 sub-regions, for processing read and write commands and data, paragraphs [0057], [0065], [0067] [0077]); identifying a data path to be used for transferring data to or from the memory module using the received control information (microprocessor 303 specifies access address information (identifying a data path) from the read and write commands sent from storage system input/output (the memory controller) of host 200, paragraph [0017], [0055], [0065], [0067]); and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path (storage system 101 (the memory module) comprises controller 301 and using microprocessor 303, recognized as comprising the functionality of a data manager, specifies access address information (the identified data path) and transmits data between volatile memory 311 and non-volatile memory 313 with the storage system input/output (the memory controller) of host 200, Figure 19, paragraphs [0017], [0055], [0057], [0065], [0067]).

As per Claim 14, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data (microprocessor 303 (the data manager) processes commands received from the storage system input/output (the memory controller) of host 200 and allocates the volatile memory into a fixed length segment size (controls data buffer size), paragraphs [0017], [0057], [0061], [0114]).

As per Claim 15, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising operating the data manager as a bidirectional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems (microprocessor 303 (the data manager) provides read and write command processing with the storage system input/output of host 200 by utilizing one or more ports (two or more sets of data ports) to the volatile cache memory of the multiplexed storage devices (a bidirectional data transfer fabric), paragraphs [0017], [0061], [0077], claim 15).

As per Claim 18, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller (microprocessor 303 (the data manager) copies data from volatile memory 311 to non-volatile memory 313 and to volatile memory from non-volatile memory 313 (directing transfer of data bi-directionally) using the access address information from the storage system input/output (the memory controller) of host 200, [0017], [0055]-[0058]).

As per Claim 19, Shimozono discloses the method of claim 18; additionally, Shimozono discloses further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem (microprocessor 303 processes the commands received from the storage system input/output (the memory controller) of host 200 in volatile memory 311 and copy the data to non-volatile memory 313 by executing the specific computer program, paragraphs [0017], [0055], [0057], [0061], [0065]).

As per Claim 20, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and nonvolatile memory subsystems (controller 301 comprises an access control unit in storage system 101 and when executed by microprocessor 303, allows memory to be used (memory access control), Figure 19, paragraphs [0014] and [0055]).

As per Claim 21, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value (the host 200 through the storage system input/output (the memory controller) can control the memory regions of volatile sub-regions and non-volatile sub-regions (configure memory space) in units called host volumes through the microprocessor executing commands from the controller 301 regarding the configuration of the memory regions, paragraph [0017], [0055]-[0059], [0086], [0102]).

---Continued Within the Next Supplemental Box---

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US12/48750

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

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As per Claim 22, Shimozone discloses the method of claim 21; additionally, Shimozone discloses wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller (controller 301 executes a program with microprocessor 303 to control volatile memory 311 or non-volatile memory 313 sub-region address information (configures the memory space using a first portion of the memory subsystem) in storage system 101 (the memory module) by converting the access address information designated in an access address command (the controller presents a unified memory space) from the storage system input/output (the memory controller) of host 200, Figure 19, paragraphs [0017], [0055], [0057], [0058]).

As per Claim 23, Shimozone discloses the method of claim 21; additionally, Shimozone discloses wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific (microprocessor executes the controller 301 cache control program 454 to control the forming of volatile memory 311 sub-region in storage system 101 (configures the memory space in the memory module), based on the command processing program 451 and the RAID control program 453 (instructions that are application-specific), Figures 3 and 19, paragraphs [0057], [0102]-[0106]).

Claims 7, 8, 10, 16, 17 and 24 lack an inventive step under PCT Article 33(3) as being obvious over Shimozone in view of US 2010/0274953 A1, LEE, T., et al (hereinafter 'Lee').

As per Claim 7, Shimozone discloses the memory module of claim 6. Shimozone does not disclose wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments. Lee discloses wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments (the memory system of solid state disk 120 comprises a non-volatile, flash memory interface 240 connected to a plurality of non-volatile flash memory devices groups 123 through 125 (non-volatile memory subsystem), each containing memory devices (one or more memory segments), Figures 1 and 2, paragraphs [0007] and [0046]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

As per Claim 8, Shimozone, in combination with Lee, discloses the memory module of claim 7; additionally, Lee discloses wherein each memory segment comprises at least one memory circuit, memory device, or memory die (each of the memory device groups 123 through 125 contains memory devices MEM_11 through MEM_1m, which may be NAND flash memory devices, Figure 3, paragraphs [0058] and [0071]).

As per Claim 10, Shimozone, in combination with Lee, discloses the memory module of claim 7; additionally, Lee discloses wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems (solid state disk controller 121 (the data manager) can activate some or all (independently and/or concurrently) of the channels (one set of data ports) to the non-volatile memory devices (non-volatile memory subsystem) for read and write transfer control, paragraphs [0053] and [0062]).

As per Claim 16, Shimozone discloses the memory module of claim 13. Shimozone does not disclose wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments. Lee discloses wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments (the memory system of solid state disk 120 comprises a non-volatile, flash memory interface 240 connected to a plurality of non-volatile flash memory devices groups 123 through 125 (non-volatile memory subsystem), each containing memory devices (one or more memory segments), Figures 1 and 2, paragraphs [0007] and [0046]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

As per Claim 17, Shimozone discloses the method of claim 15. Shimozone does not disclose further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or nonvolatile memory subsystems. Lee discloses further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or nonvolatile memory subsystems (solid state disk controller 121 (the data manager) can activate some or all (independently and/or concurrently) of the channels (one set of data ports) to the non-volatile memory devices (non-volatile memory subsystem) for read and write transfer control, paragraphs [0053] and [0062]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

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WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US12/48750

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

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As per Claim 24, Shimozono discloses the method of claim 13. Shimozono does not disclose further comprising: operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency. Lee discloses further comprising: operating the volatile memory subsystem at a first clock frequency, when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system (SDRAM, volatile buffer memory 522 (the volatile memory subsystem) clock frequencies, fb, (a first clock frequency) are controlled by different internal clock signals in the solid state disk 520 (the memory system) when temporarily storing data involving read and write operations with host 510 (the host system), Figures 4B and 10, paragraphs [0065], [0080], [0115]); operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem (the flash clock frequency, fi, (a second clock frequency) controls the rate of the non-volatile flash interface 240 for exchanging data between the SDRAM volatile buffer memory 122 (the volatile memory subsystem) and the non-volatile flash memory device groups (non-volatile memory subsystem), Figure 4B, paragraphs [0005], [0066], [0080]); and operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency (solid state disk (the memory system) controller 121 can adjust the clock frequency driving SDRAM volatile memory 122 (the volatile memory subsystem) by lowering its clock frequency (the third clock frequency being less than the first clock frequency) to reduce its power consumption, when activating some or all of the non-volatile flash memory device groups (the second mode of operation), paragraphs [0005], [0053], [0065]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozono and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozono to include the data storage performance system and method of Lee because of selective component clock frequencies. The suggestion/motivation for doing so would have been to provide a method to lower clock frequencies of selective components to enable better power consumption and redirect the limited available power to other storage components to ensure proper data storage and retention.

Claims 1-24 have industrial applicability as defined by PCT Article 33(4) because the subject matter can be made or used in industry.

Electronic Acknowledgement Receipt

EFS ID:	14430454
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	10-DEC-2012
Filing Date:	26-JUL-2012
Time Stamp:	19:25:20
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453_010_preliminary_2.pdf	115826 3af9e0a4c1857da157bf9ae528c0ba4df3e9d7f9	yes	3

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Specification		2	2	
	Applicant Arguments/Remarks Made in an Amendment		3	3	
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	062453-010_IDS_FORM.pdf	612328 fc61e91c21cd5d37de2144fd901556a94dcecd7	no	4
Warnings:					
Information:					
3	Non Patent Literature	062453_0011_ISR.pdf	1134384 2ec404a332aec427be1b2f18b7d1df7a3581fd41	no	10
Warnings:					
Information:					
Total Files Size (in bytes):			1862538		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT:	Hyun Lee	CONFIRMATION NO.: 1046
APPLICATION NO.:	13/559,476	
FILING DATE:	July 26, 2012	
TITLE:	FLASH-DRAM HYBRID MEMORY MODULE	
EXAMINER:	unassigned	
ART UNIT:	2189	

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

PRELIMINARY AMENDMENT

Sir:

Please amend the above-identified application as follows:

Amendments to the Specification begin on page **2** of this paper.

Remarks begin on page **3** of this paper.

In the Specification

Please amend paragraph [0001] as follows:

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and ~~of U.S. patent application serial 13/559,476, filed July 26, 2012, which is a continuation-in-part (CIP)~~ of US patent application serial no. 12/240,916, filed September 29, 2008 which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

REMARKS

The specification has been amended to more accurately reflect priority and related application information.

Early consideration and allowance of this application is earnestly solicited.

Please charge any additional required fees, including those necessary to obtain extensions of time to render timely the filing of the instant Amendment and/or Reply to Office Action, or credit any overpayment not otherwise credited, to our deposit account no. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: December 10, 2012

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/559,476		Filing Date 07/26/2012		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL				
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	12/10/2012	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 24	Minus	** 20	= 4	X \$	=		X \$62=	248
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$	=		X \$250=	0
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE			TOTAL ADD'L FEE	248
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=		X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE			TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/ANDREW JAMES JR/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/559,476

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	390
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	620
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	250
TOTAL CLAIMS (37 CFR 1.16(j))	24 minus 20 =	*			OR	x 62 =	248
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*				x 250 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1508

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/559,476	07/26/2012	2189	1620	062453-010	24	2

CONFIRMATION NO. 1046

UPDATED FILING RECEIPT

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306



Date Mailed: 12/20/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 46188

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 08/07/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/559,476**

Projected Publication Date: Perfected

Non-Publication Request: No

Early Publication Request: No
Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/559,476	07/26/2012	2188	1620	062453-010	24	2

CONFIRMATION NO. 1046

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

UPDATED FILING RECEIPT



Date Mailed: 12/28/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

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Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 46188

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 08/07/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/559,476**

Projected Publication Date: 04/04/2013

Non-Publication Request: No

Early Publication Request: No
Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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NOT GRANTED

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SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage, facilitate, and accelerate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/559,476

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	390
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	620
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MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1508

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/559,476	07/26/2012	Hyun Lee	062453-010

CONFIRMATION NO. 1046

PUBLICATION NOTICE



OC00000060250294

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Title:FLASH-DRAM HYBRID MEMORY MODULE

Publication No.US-2013-0086309-A1

Publication Date:04/04/2013

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee. CONFIRMATION NO.: 1046
APPLICATION NO.: 13/559,476
FILING DATE: July 26, 2012
TITLE: FLASH-DRAM HYBRID MEMORY MODULE
EXAMINER: Reginald Glenwood Bragdon
ART UNIT: 2189

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REQUEST TO CORRECT FILING RECEIPT

Sir:

Our office received the attached filing receipt for the above-identified case. Pursuant to the Supplemental Amendments filed August 20, 2012 and December 10, 2012, please amend the Filing Receipt to reflect the correct domestic priority application numbers as follows:

This application claims the benefit of 61/512,871, 07/28/11,
and of 12/240,916, 9/29/2008
which is a continuation of 12/131,873, 6/2/08,
which claims the benefit of 60/941,586, 6/1/07

(see attached copy of the filing receipt with the changes shown on a second page).

The Commissioner is hereby authorized to charge any additional fees or credit any overpayment to Deposit Account No. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: April 19, 2013

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/559,476	07/26/2012	2188	1620	062453-010	24	2

CONFIRMATION NO. 1046

UPDATED FILING RECEIPT



46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Date Mailed: 12/28/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 46188

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833

*See attached
comments*

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

In the Specification

(August 20, 2012)

Please amend paragraph [0001] as follows:

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and of U.S. patent application serial 13/559,476, filed July 26, 2012, which is a continuation-in-part (CIP) of US patent application serial no. 12/240,916, filed September 29, 2008 which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

(December 10, 2012)

Please amend paragraph [0001] as follows:

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and of US patent application serial no. 12/240,916, filed September 29, 2008 which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

Electronic Acknowledgement Receipt

EFS ID:	15570410
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	19-APR-2013
Filing Date:	26-JUL-2012
Time Stamp:	19:29:43
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Corrected Filing Receipt	062453_010_req_corr_fr.pdf	216375 b8198dc59f6e2cdd908413f392fe6fdbe78524d	no	3

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/559,476	07/26/2012	2189	1620	062453-010	24	2

CONFIRMATION NO. 1046

CORRECTED FILING RECEIPT

46188
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306



Date Mailed: 04/29/2013

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

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Chi-She Chen, Walnut, CA;
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Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 46188

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/512,871 07/28/2011
and is a CIP of 12/240,916 09/29/2008 PAT 8301833
which is a CON of 12/131,873 06/02/2008 ABN
which claims benefit of 60/941,586 06/01/2007

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 08/07/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/559,476**

Projected Publication Date: Not Applicable

Non-Publication Request: No

Early Publication Request: No

Title

FLASH-DRAM HYBRID MEMORY MODULE

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications:

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

U.S.PATENTS						
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

If you wish to add additional U.S. Patent citation information please click the Add button.

U.S.PATENT APPLICATION PUBLICATIONS						
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20120204079	A1	2012-08-09	Takefman et al.	

If you wish to add additional U.S. Published Application citation information please click the Add button.

FOREIGN PATENT DOCUMENTS								
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

NON-PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit		2189
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number		062453-010

	1		☐
If you wish to add additional non-patent literature document citation information please click the Add button			
EXAMINER SIGNATURE			
Examiner Signature		Date Considered	
*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	Bragdon, Reginald Glenwood
Attorney Docket Number	062453-010

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2013-05-21
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Electronic Acknowledgement Receipt

EFS ID:	15836017
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Yang (Aaron) Zhang
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	21-MAY-2013
Filing Date:	26-JUL-2012
Time Stamp:	18:13:18
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	062453_010_IDS__ss.pdf	92624 31353328b4773c808095aa170dfd6004929350c	no	3

Warnings:

Information:

Total Files Size (in bytes):

92624

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

U.S.PATENTS						
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	5675725		1997-10-07	Malcolm	
	2	7111142		2006-09-19	Spencer et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

U.S.PATENT APPLICATION PUBLICATIONS						
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

If you wish to add additional U.S. Published Application citation information please click the Add button.

FOREIGN PATENT DOCUMENTS								
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

NON-PATENT LITERATURE DOCUMENTS								
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	U.S. Office Action in U.S. Application No. 13/536,173, mailed on April 15, 2013.	☐
	2	Notice of Allowance in U.S. Application No. 12/240,916 dated September 17, 2012.	☐

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ²Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	Bragdon, Reginald Glenwood
Attorney Docket Number	062453-010

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2013-07-12
Name/Print	Khaled Shami	Registration Number	38745

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Electronic Acknowledgement Receipt

EFS ID:	16307297
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Yang (Aaron) Zhang
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	12-JUL-2013
Filing Date:	26-JUL-2012
Time Stamp:	18:20:33
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	062453_010_IDS__JUL_12_2013__ss.pdf	93341 400c5940d490964376e3aa3f0c62fb832c2c62d3	no	3

Warnings:

Information:

This is not an USPTO supplied IDS fillable form					
2	Non Patent Literature	OA__003__APR_15_2013.pdf	361822	no	10
			95898d369cdbcda76d8d225741008677ab4d4f82		
Warnings:					
Information:					
3	Non Patent Literature	NoA__002__SEP_17_2012.pdf	387574	no	7
			f51e469063c3f9737f844e1fc95d514c7846b2da		
Warnings:					
Information:					
Total Files Size (in bytes):			842737		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee et al.
SERIAL NO.: 13/559,476 CONFIRMATION NO: 1046
FILING DATE: July 26, 2012
TITLE: Flash-DRAM Hybrid Memory Module
EXAMINER: Elmore, Stephen C.
ART UNIT: 2188

**Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

PRELIMINARY AMENDMENT

Dear Sir:

Please amend the above-identified application as follows:

Amendments to the Claims, if any, are reflected in the Listing of Claims beginning on page 2.

Remarks begin on page 7 of this paper.

In the Claims

The following Listing of Claims replaces all prior versions in the application:

LISTING OF CLAIMS

1. (Currently Amended) A memory module couplable to a memory controller of a host system, comprising:
 - a non-volatile memory subsystem;
 - a data manager coupled to the non-volatile memory subsystem;
 - a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager; and
 - a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller, wherein:
 - at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments, each memory segment comprising at least one memory circuit, memory device, or memory die, and
 - the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems, the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

2. (Previously Presented) The memory module of claim 1, wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller.

3. (Previously Presented) The memory module of claim 1, wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller.

4. (Previously Presented) The memory module of claim 3, wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer.

5. (Previously Presented) The memory module of claim 1, wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module.

6-8. (Cancelled)

9. (Previously Presented) The memory module of claim 1, wherein the volatile memory subsystem comprises DRAM memory.

10-12. (Cancelled)

13. (Currently Amended) A method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems, the method comprising:

receiving control information from the memory controller, wherein the control information is received using a protocol of the volatile memory subsystem;

identifying a data path to be used for transferring data to or from the memory module using the received control information; ~~and~~

using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path;

operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports, wherein a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystems, and a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem;

operating the two or more sets of data ports to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems based on control information received from the controller of the memory module; and

using the controller of the memory module to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem.

14. (Previously Presented) The method of claim 13, further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data.

15. (Cancelled)

16. (Previously Presented) The memory module of claim 13, wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments.

17. (Cancelled)

18. (Previously Presented) The method of claim 13, further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller.

19. (Previously Presented) The method of claim 18, further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem.

20. (Cancelled)

21. (Previously Presented) The method of claim 13, further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value.

22. (Previously Presented) The method of claim 21, wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller.

23. (Previously Presented) The method of claim 21, wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific.

24. (Currently Amended) The method of claim 13, further comprising:
operating the volatile memory subsystem at a first clock frequency when the memory ~~system-module~~ is in a first mode of operation in which data is communicated between the volatile memory subsystem and the ~~host-system~~ memory controller;

operating the non-volatile memory subsystem at a second clock frequency when the memory ~~system~~module is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and

operating the volatile memory subsystem at a third clock frequency when the memory ~~system~~module is in the second mode of operation, the third clock frequency being less than the first clock frequency.

REMARKS

Applicants have amended Claims 1, 13 and 24 and have cancelled Claims 6-8, 10-12, 15, 17 and 20. Applicants reserve the right to further pursue the cancelled claims in a continuation and/or divisional application as well as for appeal purposes. Early consideration and allowance of this application are earnestly solicited.

The Commissioner is hereby authorized to charge any fees which may be required, or credit any overpayment to deposit account No. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: March 28, 2014

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

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P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX (650) 320-7701

Electronic Acknowledgement Receipt

EFS ID:	18619309
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Monica Pizarro
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	28-MAR-2014
Filing Date:	26-JUL-2012
Time Stamp:	20:13:57
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		062453-010_Preliminary_Amen dment.pdf	99801 931544b89091d35af0ad5f98a298b9cffe157300	yes	7

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Claims	2	6
	Applicant Arguments/Remarks Made in an Amendment	7	7
Warnings:			
Information:			
		Total Files Size (in bytes):	99801
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 13/559,476	Filing Date 07/26/2012	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	03/28/2014	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 15	Minus	** 24	= 0	X \$80 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$420 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						0	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
/DOROTHY BELL/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

46188 7590 05/06/2014
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

EXAMINER

ELMORE, STEPHEN C

ART UNIT

PAPER NUMBER

2188

DATE MAILED: 05/06/2014

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/559,476

07/26/2012

Hyun Lee

062453-010

1046

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	08/06/2014

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax **(571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46188 7590 05/06/2014
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	08/06/2014

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2188	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☐ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).
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5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046
46188	7590	05/06/2014	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ELMORE, STEPHEN C	
			ART UNIT	PAPER NUMBER
			2188	
DATE MAILED: 05/06/2014				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 13/559,476	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2188	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the Application filed 7/26/2012.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-5,9,13,14,16,18,19 and 21-24. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☒ **CORRECTED DRAWINGS** (as "replacement sheets") must be submitted.
☒ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ **DEPOSIT OF and/or INFORMATION** about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>See Continuation Sheet</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

Continuation of Attachment(s) 2. Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date: 9/27/12, 12/10/12, 5/21/13, 7/12/13.

EXAMINER'S COMMENT and REASONS FOR ALLOWANCE

The present application is being examined under the pre-AIA first to invent provisions.

Priority

Applicant's claim for the benefit of one or more prior-filed applications under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, or 365(c) is acknowledged. However, Applicant has not complied with one or more conditions for receiving the benefit of an earlier filing date under 35 U.S.C. 119(e) or 120 as follows: the inventive entity of each of the prior-filed applications is/are not the same inventive entity of the present application. Prior filed provisional application 61/512871 has single inventor, Hyun Lee (a distinct inventive entity), and prior filed CIP application 12/240916 has four inventors: Chi-She Chen, Jeffrey Solomon, Scott Milton, and Jayesh Bhakta (a different distinct inventive entity), while the present application has five inventors: Hyun Lee, Chi-She Chen, Jeffrey Solomon, Scott Milton, and Jayesh Bhakta (another different distinct inventive entity). The priority date granted to the examination of the present application is 7/26/2012.

Drawings

The drawings Figures 2, 8B, and 9 filed on 7/26/2012 are objected-to subject to correction of the informalities indicated below. In order to avoid abandonment of this application, correction is required in reply to the Office action. The correction will not be held in abeyance.

Figure 2 is objected to because graphic shading elements in the figure obscure text in the drawing.

Art Unit: 2188

Figure 8B is objected-to because the multiple drawing elements composed of graphical shading are unidentified.

Figure 9 is objected-to because flowchart element 916 contains an incomplete descriptive labeling/statement.

REASONS FOR ALLOWANCE

The following is an examiner's statement of reasons for allowance:

In independent claims 1 and 13 the following features taken **in combination with** the remaining limitations of the independent claim are not found in and/or are not obvious in view of the closest prior art of record, Song et al., US 8,102,614,

Claim 1,

"a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller, wherein: at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments, each memory segment comprising at least one memory circuit, memory device, or memory die, and the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems, the data

Art Unit: 2188

manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem" and "a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller";

Claim 13,

"identifying a data path to be used for transferring data to or from the memory module using the received control information; and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path; operating the data manager as a bi-directional data transfer fabric with two or more sets of data ports, wherein a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystems, and a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem; operating the two or more sets of data ports to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems based on control information received from the controller of the memory module" and "using the controller of the memory module to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem".

Art Unit: 2188

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mano Padmanabhan can be reached on (571) 272-4210. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

April 20, 2014

Notice of References Cited	Application/Control No. 13/559,476		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE		Art Unit 2188	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-5,577,213 A	11-1996	Avery et al.	710/100
*	B	US-5,619,644 A	04-1997	Crockett et al.	714/45
*	C	US-6,269,382 B1	07-2001	Cabrera et al.	1/1
*	D	US-6,691,209 B1	02-2004	O'Connell, Mark Andrew	711/114
*	E	US-2007/0136523 A1	06-2007	Bonella et al.	711/113
*	F	US-8,086,955 B2	12-2011	Zhou et al.	715/234
*	G	US-2011/0320804 A1	12-2011	Chan et al.	713/150
*	H	US-8,102,614 B2	01-2012	Song et al.	360/31
*	I	US-8,412,879 B2	04-2013	Chang et al.	711/103
*	J	US-2014/0059170 A1	02-2014	Gasparakis et al.	709/217
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

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	N					
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	P					
	Q					
	R					
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NON-PATENT DOCUMENTS

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	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 12/10/2012

13559476 - GAU: 2188

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	8301833		2012-10-30	Chen et al.	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20080104344		2008-05-01	Shimozono et al.	
	2	20100274953		2010-10-28	Lee et al.	

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Receipt date: 12/10/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion in PCT/US12/48750, dated October 10, 2012 pp 1-10.	☐

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EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	04/20/2014
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Receipt date: 12/10/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2012-12-11
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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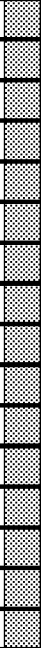
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

Issue Classification 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

CPC		
Symbol	Type	Version
		
		

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
				
				

NONE	Total Claims Allowed: 15	
(Assistant Examiner)	(Date)	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2188	4/20/2014	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 6

Issue Classification 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
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NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	15	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2188	4/20/2014	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	6

Search Notes 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/20/2014	SE
365	185.33	4/20/2014	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/20/2014	SE
Inventor Name Search for DP	4/20/2014	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	4/20/2014	SE
PGPUB Searched		4/20/2014	SE

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Receipt date: 09/27/2012

13559476 - GAU: 2188

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

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	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name		
	Attorney Docket Number	062453-010	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	4420821		1983-12-13	Hoffman	
	2	4449205		1984-05-15	Hoffman	
	3	5519663		1996-05-21	Harper, Jr. et al.	
	4	6158015		2000-12-05	Klein	
	5	6336174	B1	2002-01-01	Li et al.	
	6	6336176	B1	2002-01-01	Leyda et al.	
	7	6487623	B1	2002-11-26	Emerson et al.	
	8	6658507	B1	2003-12-02	Chan	

Receipt date: 09/27/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
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	Art Unit		2189	
	Examiner Name			
	Attorney Docket Number		062453-010	

	9	6799244	B2	2004-09-28	Tanaka et al.	
	10	7409590	B2	2008-08-05	Moshayedi et al.	

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	1	20020083368	A1	2002-06-27	Abe et al.	
	2	20040190210	A1	2004-09-30	Leete	
	3	20070192627	A1	2007-08-16	Oshikiri	
	4	20080195806	A1	2008-08-14	Cope	

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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Receipt date: 09/27/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Office Action in U.S. Patent Application No. 12/240,916, mailed April 3, 2012. pp 1-12.	☐
	2	Office Action in U.S. Patent Application No. 12/240,916, mailed February 1, 2012. pp 1-14.	☐
	3	Office Action in U.S. Patent Application No. 12/240,916, mailed July 29, 2011. pp 1-8.	☐

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Examiner Signature	/Stephen Elmore/	Date Considered	04/20/2014
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Receipt date: 09/27/2012 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name			
	Attorney Docket Number	062453-010		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2012-09-27
Name/Print	Khaled Shami	Registration Number	38,745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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BIB DATA SHEET

CONFIRMATION NO. 1046

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
13/559,476	07/26/2012	711	2188	062453-010	
APPLICANTS INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;					
** CONTINUING DATA ***** This appln claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 08/07/2012					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /STEPHEN C ELMORE/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 10	TOTAL CLAIMS 24	INDEPENDENT CLAIMS 2
ADDRESS Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306 UNITED STATES					
TITLE FLASH-DRAM HYBRID MEMORY MODULE					
FILING FEE RECEIVED 1620	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	8720	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:03
L2	2611	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:04
L3	18749	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:04
L4	28259	1 or 2 or 3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:04
L5	4365	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:14
L6	522	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:15
L7	6	5 and 6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:15
L8	3	7 and ((@pd or @ad)<"20120726")	US-	OR	ON	2014/04/20

			PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			19:16
L9	2	(US-20070136523-\$).did. or (US-8412879-\$).did.	US- PGPUB; USPAT	OR	ON	2014/04/20 19:18
L10	154543	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:22
L11	79	6 and 10	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:23
L12	1	5 and 11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:23
L13	2	4 and 11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:23
L14	344	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:25
L15	1	14 with (data adj manager)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:25
L16	17	(data near3 (port or input-output or I/O or IO)) and 11	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:27
L17	1	5 and 16	US- PGPUB; USPAT;	OR	ON	2014/04/20 19:27

			USOCR; FPRS; EPO; JPO; IBM_TDB			
L18	14	16 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:27
L19	0	4 and 18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:29
L20	14	6 and 18	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:30
L21	0	14 and 20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:30
L22	0	5 and 20	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:30
L23	28107	(control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:32
L24	1	((control adj information or control adj data or control adj meta-data or control adj metadata) near3 controller) and 16	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:33
L25	1	6 and 23 and 10 and 4	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:33
L26	1	6 and 23 and 10	US-	OR	ON	2014/04/20

			PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB			19:33
L27	6	6 and 23	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:33
L28	5305	((Hyun) near2 (Lee)).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2014/04/20 19:35
L29	11	((Chi-She) near2 (Chen)).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2014/04/20 19:35
L30	89	((Jeffrey) near2 (Solomon)).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2014/04/20 19:35
L31	154	((Scott) near2 (Milton)).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2014/04/20 19:35
L32	76	((Jayesh) near2 (Bhakta)).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2014/04/20 19:35
L42	1	(data adj manager same controller same memory adj controller) and 5 and (10 or 14)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:39
L47	503	(hybrid near3 memory).ti.	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:43
L48	2	6 and 47	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:43
L49	0	48 and ((@pd or @ad)<"20120726")	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2014/04/20 19:43

			IBM_TDB			
L50	2	(data adj manager and memory adj controller) and 47	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:44
L51	0	50 and ((@pd or @ad)<"20120726")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:44
L52	1	(data adj manager) and 47 and ((@pd or @ad)<"20120726")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:44
L53	36	("20080195806" "6658507" "5675725" "20040190210" "6336176" "7409590" "20100274953" "6336174" "5519663" "6487623" "20080104344" "4420821" "6799244" "20020083368" "4449205" "8301833" "7111142" "20070192627" "6158015" "20120204079").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:46
L54	1	6 and 53	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/04/20 19:46

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L33	12	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:36
L34	89	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:36
L35	110	((Scott) near2 (Milton)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:36
L36	77	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:36
L37	5310	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:36
L38	5528	33 or 34 or 35 or 36 or 37	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:37
L40	6	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:38
L41	1	38 and 40	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:38

L44	11629	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:40
L45	36	38 and 44	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:41
L46	1	40 and 45	US-PGPUB; USPAT; UPAD	OR	ON	2014/04/20 19:41

4/ 20/ 2014 7:47:30 PM
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Receipt date: 05/21/2013

13559476 - GAU: 2188

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

U.S.PATENTS						
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20120204079	A1	2012-08-09	Takefman et al.	

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	Filing Date		2012-07-26	
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	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

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Examiner Signature	/Stephen Elmore/		Date Considered 04/20/2014
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	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

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OR

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☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2013-05-21
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Receipt date: 07/12/2013

13559476 - GAU: 2188

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

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	Attorney Docket Number	062453-010		

U.S.PATENTS						
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	5675725		1997-10-07	Malcolm	
	2	7111142		2006-09-19	Spencer et al.	

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U.S.PATENT APPLICATION PUBLICATIONS						
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	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	U.S. Office Action in U.S. Application No. 13/536,173, mailed on April 15, 2013. pp 1-10.	☐
	2	Notice of Allowance in U.S. Application No. 12/240,916 dated September 17, 2012. pp 1-7.	☐

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EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	04/29/2014
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ²Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵Applicant is to place a check mark here if English language translation is attached.

Receipt date: 07/12/2013 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476 13559476 - GAU: 2188
	Filing Date		2012-07-26
	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2013-07-12
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

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Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

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NON-PATENT LITERATURE DOCUMENTS			Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

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STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	Bragdon, Reginald Glenwood
Attorney Docket Number	062453-010

1	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014.	☐
2	Office Action in U.S. Patent Application No. 14/173,242, mailed March 14, 2014.	☐

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Examiner Signature		Date Considered	
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STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	13559476
Filing Date	2012-07-26
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	Bragdon, Reginald Glenwood
Attorney Docket Number	062453-010

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☒ See attached certification statement.

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☐ A certification statement is not submitted herewith.

SIGNATURE

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-05-20
Name/Print	Khaled Shami	Registration Number	38745

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Tiffany Weeks			
Attorney Docket Number:	062453-010			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	1806	1	180	180
Total in USD (\$)				180

Electronic Acknowledgement Receipt

EFS ID:	19070990
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Tiffany Weeks
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	20-MAY-2014
Filing Date:	26-JUL-2012
Time Stamp:	13:09:32
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 180
RAM confirmation Number	10597
Deposit Account	503557
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Information Disclosure Statement (IDS) Form (SB08)	062453-010_IDS_dated_05-20- 2014_US_Certify.pdf	584336 287a75239810988fd443f4152cdc9939cf16 4c12	no	4
Warnings:					
Information:					
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2	Non Patent Literature	OA_14173219_03_13_2014.pdf	234253 c0a3c9a61fe5d7ac4bbaa13398360b4025a 240a2	no	7
Warnings:					
Information:					
3	Non Patent Literature	OA_14173242_03_14_2014.pdf	238405 9a9bf0b9be1d253060968f09d66d0206c89 e7eca	no	7
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	30324 648fc6349613b6c0a17d2417b1b5fb0a78d c319b	no	2
Warnings:					
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Total Files Size (in bytes):			1087318		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

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	First Named Inventor	Hyun Lee	
	Art Unit	2189	
	Examiner Name	Bragdon, Reginald Glenwood	
	Attorney Docket Number	062453-010	

	1	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014.	☐
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Examiner Signature		Date Considered	
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☐ A certification statement is not submitted herewith.

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Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-05-20
Name/Print	Khaled Shami	Registration Number	38745

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TRANSMITTAL OF COPY OF INTERNATIONAL
PRELIMINARY REPORT ON PATENTABILITY
(CHAPTER I OF THE PATENT COOPERATION
TREATY)

(PCT Rule 44bis.1(c))

To:

SHAMI, Khaled
P.O. Box 60610
Palo Alto, CA 94306
ETATS-UNIS D'AMERIQUE

Date of mailing (day/month/year)

03 April 2014 (03.04.2014)

Applicant's or agent's file reference

062453-0011

IMPORTANT NOTICE

International application No.

PCT/US2012/048750

International filing date (day/month/year)

28 July 2012 (28.07.2012)

Priority date (day/month/year)

28 July 2011 (28.07.2011)

Applicant

NETLIST, INC. et al

The International Bureau transmits herewith a copy of the international preliminary report on patentability (Chapter I of the Patent Cooperation Treaty)

The International Bureau of WIPO
34, chemin des Colombettes
1211 Geneva 20, Switzerland

Facsimile No. +41 22 338 82 70

Authorized officer

Yukari Nakamura

e-mail: pt07.pct@wipo.int

PATENT COOPERATION TREATY

PCT

INTERNATIONAL PRELIMINARY REPORT ON PATENTABILITY

(Chapter I of the Patent Cooperation Treaty)

(PCT Rule 44bis)

Applicant's or agent's file reference 062453-0011	FOR FURTHER ACTION	See item 4 below
International application No. PCT/US2012/048750	International filing date (<i>day/month/year</i>) 28 July 2012 (28.07.2012)	Priority date (<i>day/month/year</i>) 28 July 2011 (28.07.2011)
International Patent Classification (8th edition unless older edition indicated) See relevant information in Form PCT/ISA/237		
Applicant NETLIST, INC.		

1. This international preliminary report on patentability (Chapter I) is issued by the International Bureau on behalf of the International Searching Authority under Rule 44 bis.1(a).

2. This REPORT consists of a total of 7 sheets, including this cover sheet.

In the attached sheets, any reference to the written opinion of the International Searching Authority should be read as a reference to the international preliminary report on patentability (Chapter I) instead.

3. This report contains indications relating to the following items:

- | | | |
|-------------------------------------|--------------|---|
| ☒ | Box No. I | Basis of the report |
| ☐ | Box No. II | Priority |
| ☐ | Box No. III | Non-establishment of opinion with regard to novelty, inventive step and industrial applicability |
| ☐ | Box No. IV | Lack of unity of invention |
| ☒ | Box No. V | Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement |
| ☐ | Box No. VI | Certain documents cited |
| ☐ | Box No. VII | Certain defects in the international application |
| ☐ | Box No. VIII | Certain observations on the international application |

4. The International Bureau will communicate this report to designated Offices in accordance with Rules 44bis.3(c) and 93bis.1 but not, except where the applicant makes an express request under Article 23(2), before the expiration of 30 months from the priority date (Rule 44bis .2).

The International Bureau of WIPO 34, chemin des Colombettes 1211 Geneva 20, Switzerland Facsimile No. +41 22 338 82 70	Date of issuance of this report 25 March 2014 (25.03.2014) Authorized officer Yukari Nakamura e-mail: pt07.pct@wipo.int
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PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To: Khaled Shami
P.O. Box 60610
Palo Alto, CA 94306
United States of America

PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43bis.1)

Date of mailing
(day/month/year)

10 OCT 2012

Applicant's or agent's file reference
062453-0011

FOR FURTHER ACTION

See paragraph 2 below

International application No.

PCT/US12/48750

International filing date (day/month/year)

28 July 2012 (28.07.2012)

Priority date (day/month/year)

28 July 2011 (28.07.2011)

International Patent Classification (IPC) or both national classification and IPC

IPC(8) - G06F 12/00 (2012.01)

USPC - 711/118, 103

Applicant
Netlist, Inc.

I. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☐ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1 bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/US
Mail Stop PCT, Attn: ISA/US
Commissioner for Patents
P.O. Box 1450, Alexandria, Virginia 22313-1450
Facsimile No. 571-273-3201

Date of completion of this opinion

21 September 2012 (21.09.2012)

Authorized officer:

Shane Thomas

PCT Helpdesk: 571-272-4300
PCT OSP: 571-272-7774

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

International application No.

PCT/US12/48750

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:
- ☒ the international application in the language in which it was filed.
- ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).
2. ☐ This opinion has been established taking into account ~~there~~**rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(a))
3. With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing filed or furnished:
- a. (means)
- ☐ on paper
- ☐ in electronic form
- b. (time)
- ☐ in the international application as filed
- ☐ together with the international application in electronic form
- ☐ subsequently to this Authority for the purposes of search
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that in the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

PCT/US12/48750

Box No. V	Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement			
1. Statement				
Novelty (N)	Claims	<u>7, 8, 10, 16, 17, 24</u>	YES	
	Claims	<u>1-6, 9, 11-15, 18-23</u>	NO	
Inventive step (IS)	Claims	<u>NONE</u>	YES	
	Claims	<u>1-24</u>	NO	
Industrial applicability (IA)	Claims	<u>1-24</u>	YES	
	Claims	<u>NONE</u>	NO	
2. Citations and explanations:				
Claims 1-6, 9, 11-15 and 18-23 lack novelty under PCT Article 33(2) as being anticipated by US 2008/0104344 A1, SHIMOZONO, N., et al (hereinafter 'Shimozono'). As per Claim 1, Shimozono discloses a memory module (storage system 101 (a memory module), Figure 19, paragraph [0053]) couplable to a memory controller of a host system (is connected to storage system input/output (memory controller) of host 200 through front end I/F 307, Figure 19, paragraphs [0017] and [0080]), comprising: a non-volatile memory subsystem (non-volatile memory 313 (non-volatile memory subsystem), NM, Figure 19, paragraph [0055]); a data manager coupled to the non-volatile memory subsystem (microprocessor 303, MP, comprising the functionality of a data manager, is connected to non-volatile memory 313, through switch 309, SW, Figure 19, paragraph [0055]); a volatile memory subsystem coupled to the data manager (cache memory 311, CM, is volatile memory (a volatile memory subsystem) connected to microprocessor 303 through switch 309, SW, Figure 19, paragraphs [0055] and [0057]) and operable to exchange data with the non-volatile memory subsystem by way of the data manager (microprocessor 303 can copy data on the volatile memory, CM311, to the non-volatile memory 313, paragraphs [0055], [0057], [0065]); and a controller operable to receive commands from the memory controller (controller 301 communicates with the storage system input/output (the memory controller) of host 200 and uses microprocessor 303 to process read and write commands, paragraphs [0017], [0055], [0056], [0061]) and to direct (i) operation of the non-volatile memory subsystem (processes read and write commands with non-volatile memory 313, paragraphs [0055], [0065], [0067]), (ii) operation of the volatile memory subsystem (processes read and write commands with volatile memory 311, paragraphs [0055], [0057], [0065], [0067]), and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller (based on a read command from the storage system input/output (memory controller) of host 200, transmits data between volatile memory 311 and non-volatile memory 313 with the storage system input/output of host 200, paragraphs [0017], [0055], [0057], [0065], [0067]). As per Claim 2, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager is operable to control one or more of data flow rate, data transfer size, data buffer size, data error monitoring, and data error correction in response to receiving at least one of a control signal and control information from the controller (microprocessor 303 (the data manager) processes commands received from the storage system input/output (the memory controller) of host 200 and allocates the volatile memory into a fixed length segment size (controls data buffer size), paragraphs [0017], [0057], [0061], [0114]). As per Claim 3, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager controls data traffic between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on instructions received from the controller (microprocessor 303 (the data manager) specifies the access address information from the read and write commands of the storage system input/output (the memory controller) of host 200 and controls the transmission of data (controls data traffic) between volatile memory 311, non-volatile memory 313, and the storage system input/output of host 200, paragraphs [0017], [0055], [0057], [0065], [0067]). As per Claim 4, Shimozono discloses the memory module of claim 3; additionally, Shimozono discloses wherein data traffic control relates to any one or more of data flow rate, data transfer size, data buffer size, data transfer bit width, formatting information, direction of data flow, and the starting time of data transfer (microprocessor 303 processes host 200 commands indicating the type of command, transfer address, and transfer length (data transfer size), paragraphs [0055], [0061], [0083]). As per Claim 5, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the controller configures at least one of a first memory address space of the volatile memory subsystem and a second memory address space of the non-volatile memory subsystem in response to at least one of a received command from the memory controller and memory address space initialization information of the memory module (the host 200 through the storage system input/output (the memory controller) can control the memory regions of volatile sub-regions (a first memory address space of the volatile memory subsystem) and non-volatile sub-regions (a second memory address space of the non-volatile memory subsystem), in units called host volumes when the microprocessor executes commands from the controller 301 regarding the configuration of the memory regions, paragraph [0017], [0055]-[0059], [0086], [0102]). As per Claim 6, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the data manager is configured as a bidirectional data transfer fabric having two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems (microprocessor 303 (the data manager) provides read and write command processing with the storage system input/output of host 200 by utilizing one or more ports (two or more sets of data ports) to the volatile cache memory of the multiplexed storage devices (a bidirectional data transfer fabric), paragraphs [0017], [0061], [0077], claim 15). ***-Continued Within the Next Supplemental Box-***				

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.

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Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

Continuation of:

-----Continued from Box No. V: Citations and Explanations-----

As per Claim 9, Shimozono discloses the memory module of claim 1; additionally, Shimozono discloses wherein the volatile memory subsystem comprises DRAM memory (CM 311 is volatile memory may be DRAM, Dynamic Random Access Memory, paragraph [0057]).

As per Claim 11, Shimozono discloses the memory module of claim 6; additionally, Shimozono discloses wherein the data manager further includes a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller (microprocessor 303 (the data manager) processes volatile memory 311 and non-volatile memory 313 commands between the storage system input/output (the memory controller) of host 200, using Fiber Channel Protocol, Ethernet and iSCSI, Internet Small Computer System Interface, protocols (a data format module to format data), paragraphs [0017], [0057], [0061], [0077], [0078]).

As per Claim 12, Shimozono discloses the memory module of 11; additionally, Shimozono discloses wherein the data manager further includes a data buffer for buffering data delivered to or from the non-volatile memory subsystem (microprocessor 303 (the data manager) copies volatile memory 311 to the non-volatile cache region 391 (data buffer for buffering data) of the non-volatile memory 313, Figure 19, paragraphs [0055], [0057], [0068]).

As per Claim 13, Shimozono discloses a method for managing a memory module by a memory controller, the memory module including volatile and non-volatile memory subsystems (a process of receiving commands on storage system 101 (a memory module) from storage system input/output (a memory controller) of host 200 and reading or writing data to volatile memory 311 and non-volatile memory 313, Figure 19, paragraph [0017], [0055], [0057], [0061]), the method comprising: receiving control information from the memory controller (commands are transceived between the storage system input/output (the memory controller) of host 200 and storage system 101 using a selected protocol, paragraphs [0017], [0077]-[0079]), wherein the control information is received using a protocol of the volatile memory subsystem (using the protocol, the storage system 101 provides volatile memory, CM 311 sub-regions, for processing read and write commands and data, paragraphs [0057], [0065], [0067] [0077]); identifying a data path to be used for transferring data to or from the memory module using the received control information (microprocessor 303 specifies access address information (identifying a data path) from the read and write commands sent from storage system input/output (the memory controller) of host 200, paragraph [0017], [0055], [0065], [0067]); and using a data manager and a controller of the memory module to transfer data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one of the received control information and the identified data path (storage system 101 (the memory module) comprises controller 301 and using microprocessor 303, recognized as comprising the functionality of a data manager, specifies access address information (the identified data path) and transmits data between volatile memory 311 and non-volatile memory 313 with the storage system input/output (the memory controller) of host 200, Figure 19, paragraphs [0017], [0055], [0057], [0065], [0067]).

As per Claim 14, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising operating the data manager to control one or more of data flow rate, data transfer size, data width size, data buffer size, data error monitoring, data error correction, and the starting time of the transfer of data (microprocessor 303 (the data manager) processes commands received from the storage system input/output (the memory controller) of host 200 and allocates the volatile memory into a fixed length segment size (controls data buffer size), paragraphs [0017], [0057], [0061], [0114]).

As per Claim 15, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising operating the data manager as a bidirectional data transfer fabric with two or more sets of data ports coupled to any one of the volatile and non-volatile memory subsystems (microprocessor 303 (the data manager) provides read and write command processing with the storage system input/output of host 200 by utilizing one or more ports (two or more sets of data ports) to the volatile cache memory of the multiplexed storage devices (a bidirectional data transfer fabric), paragraphs [0017], [0061], [0077], claim 15).

As per Claim 18, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising directing transfer of data bi-directionally between the volatile and non-volatile memory subsystems using the data manager and in response to memory access commands received by the controller from the memory controller (microprocessor 303 (the data manager) copies data from volatile memory 311 to non-volatile memory 313 and to volatile memory from non-volatile memory 313 (directing transfer of data bi-directionally) using the access address information from the storage system input/output (the memory controller) of host 200, [0017], [0055]-[0058]).

As per Claim 19, Shimozono discloses the method of claim 18; additionally, Shimozono discloses further comprising buffering the data transferred between the memory controller and non-volatile memory subsystem using the volatile memory subsystem (microprocessor 303 processes the commands received from the storage system input/output (the memory controller) of host 200 in volatile memory 311 and copy the data to non-volatile memory 313 by executing the specific computer program, paragraphs [0017], [0055], [0057], [0061], [0065]).

As per Claim 20, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising using the controller to perform one or more of memory address translation, memory address mapping, address domain conversion, memory access control, data error correction, and data width modulation between the volatile and nonvolatile memory subsystems (controller 301 comprises an access control unit in storage system 101 and when executed by microprocessor 303, allows memory to be used (memory access control), Figure 19, paragraphs [0014] and [0055]).

As per Claim 21, Shimozono discloses the method of claim 13; additionally, Shimozono discloses further comprising using the controller to configure memory space in the memory module based on at least one of a command received from the memory controller, a programmable value written into a register, a value corresponding to a first portion of the volatile memory subsystem, a value corresponding to a first portion of the non-volatile memory subsystem, and a timing value (the host 200 through the storage system input/output (the memory controller) can control the memory regions of volatile sub-regions and non-volatile sub-regions (configure memory space) in units called host volumes through the microprocessor executing commands from the controller 301 regarding the configuration of the memory regions, paragraph [0017], [0055]-[0059], [0086], [0102]).

-----Continued Within the Next Supplemental Box-----

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

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Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

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-***-Continued from Previous Supplemental Box-***-

As per Claim 22, Shimozone discloses the method of claim 21; additionally, Shimozone discloses wherein the controller configures the memory space of the memory module using at least a first portion of the volatile memory subsystem and a first portion of the non-volatile memory subsystem, and the controller presents a unified memory space to the memory controller (controller 301 executes a program with microprocessor 303 to control volatile memory 311 or non-volatile memory 313 sub-region address information (configures the memory space using a first portion of the memory subsystem) in storage system 101 (the memory module) by converting the access address information designated in an access address command (the controller presents a unified memory space) from the storage system input/output (the memory controller) of host 200, Figure 19, paragraphs [0017], [0055], [0057], [0058]).

As per Claim 23, Shimozone discloses the method of claim 21; additionally, Shimozone discloses wherein the controller configures the memory space in the memory module using partitioning instructions that are application-specific (microprocessor executes the controller 301 cache control program 454 to control the forming of volatile memory 311 sub-region in storage system 101 (configures the memory space in the memory module), based on the command processing program 451 and the RAID control program 453 (instructions that are application-specific), Figures 3 and 19, paragraphs [0057], [0102]-[0106]).

Claims 7, 8, 10, 16, 17 and 24 lack an inventive step under PCT Article 33(3) as being obvious over Shimozone in view of US 2010/0274953 A1, LEE, T., et al (hereinafter 'Lee').

As per Claim 7, Shimozone discloses the memory module of claim 6. Shimozone does not disclose wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments. Lee discloses wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments (the memory system of solid state disk 120 comprises a non-volatile, flash memory interface 240 connected to a plurality of non-volatile flash memory devices groups 123 through 125 (non-volatile memory subsystem), each containing memory devices (one or more memory segments), Figures 1 and 2, paragraphs [0007] and [0046]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

As per Claim 8, Shimozone, in combination with Lee, discloses the memory module of claim 7; additionally, Lee discloses wherein each memory segment comprises at least one memory circuit, memory device, or memory die (each of the memory device groups 123 through 125 contains memory devices MEM_11 through MEM_1m, which may be NAND flash memory devices, Figure 3, paragraphs [0058] and [0071]).

As per Claim 10, Shimozone, in combination with Lee, discloses the memory module of claim 7; additionally, Lee discloses wherein at least one set of data ports is operated by the data manager to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems (solid state disk controller 121 (the data manager) can activate some or all (independently and/or concurrently) of the channels (one set of data ports) to the non-volatile memory devices (non-volatile memory subsystem) for read and write transfer control, paragraphs [0053] and [0062]).

As per Claim 16, Shimozone discloses the memory module of claim 13. Shimozone does not disclose wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments. Lee discloses wherein at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments (the memory system of solid state disk 120 comprises a non-volatile, flash memory interface 240 connected to a plurality of non-volatile flash memory devices groups 123 through 125 (non-volatile memory subsystem), each containing memory devices (one or more memory segments), Figures 1 and 2, paragraphs [0007] and [0046]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

As per Claim 17, Shimozone discloses the method of claim 15. Shimozone does not disclose further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or nonvolatile memory subsystems. Lee discloses further comprising operating the data ports to independently and/or concurrently transfer data to or from one or more memory segments of the volatile or nonvolatile memory subsystems (solid state disk controller 121 (the data manager) can activate some or all (independently and/or concurrently) of the channels (one set of data ports) to the non-volatile memory devices (non-volatile memory subsystem) for read and write transfer control, paragraphs [0053] and [0062]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozone and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozone to include the data storage performance system and method of Lee because of more granular control of memory storage functions. The suggestion/motivation for doing so would have been to provide a process to selectively activate memory channels and adjust the degree of interleaving of the activated channels to provide a more efficient and reliable memory access.

-***-Continued Within the Next Supplemental Box-***-

**WRITTEN OPINION OF THE
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PCT/US12/48750

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.

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As per Claim 24, Shimozono discloses the method of claim 13. Shimozono does not disclose further comprising: operating the volatile memory subsystem at a first clock frequency when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system; operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem; and operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency. Lee discloses further comprising: operating the volatile memory subsystem at a first clock frequency, when the memory system is in a first mode of operation in which data is communicated between the volatile memory subsystem and the host system (SDRAM, volatile buffer memory 522 (the volatile memory subsystem) clock frequencies, fb, (a first clock frequency) are controlled by different internal clock signals in the solid state disk 520 (the memory system) when temporarily storing data involving read and write operations with host 510 (the host system), Figures 4B and 10, paragraphs [0065], [0080], [0115]); operating the non-volatile memory subsystem at a second clock frequency when the memory system is in a second mode of operation in which data is communicated between the volatile memory subsystem and the non-volatile memory subsystem (the flash clock frequency, fi, (a second clock frequency) controls the rate of the non-volatile flash interface 240 for exchanging data between the SDRAM volatile buffer memory 122 (the volatile memory subsystem) and the non-volatile flash memory device groups (non-volatile memory subsystem), Figure 4B, paragraphs [0005], [0066], [0080]); and operating the volatile memory subsystem at a third clock frequency when the memory system is in the second mode of operation, the third clock frequency being less than the first clock frequency (solid state disk (the memory system) controller 121 can adjust the clock frequency driving SDRAM volatile memory 122 (the volatile memory subsystem) by lowering its clock frequency (the third clock frequency being less than the first clock frequency) to reduce its power consumption, when activating some or all of the non-volatile flash memory device groups (the second mode of operation), paragraphs [0005], [0053], [0065]). At the time of the invention, it would have been obvious to one of ordinary skill in the art, having the teachings of Shimozono and Lee before him or her, to modify the volatile cache and non-volatile storage system and method of Shimozono to include the data storage performance system and method of Lee because of selective component clock frequencies. The suggestion/motivation for doing so would have been to provide a method to lower clock frequencies of selective components to enable better power consumption and redirect the limited available power to other storage components to ensure proper data storage and retention.

Claims 1-24 have industrial applicability as defined by PCT Article 33(4) because the subject matter can be made or used in industry.

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Tiffany Weeks			
Attorney Docket Number:	062453-010			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	1806	1	180	180
Total in USD (\$)				180

Electronic Acknowledgement Receipt

EFS ID:	19071082
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Tiffany Weeks
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	20-MAY-2014
Filing Date:	26-JUL-2012
Time Stamp:	13:26:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 180
RAM confirmation Number	10803
Deposit Account	503557
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Information Disclosure Statement (IDS) Form (SB08)	062453-010_IDS_dated_05-20- 2014_Foreign_Certify.pdf	590758 9c766ce4d9580852d24f74af71bfd361a1845618	no	4
Warnings:					
Information:					
A U.S. Patent Number Citation or a U.S. Publication Number Citation is required in the Information Disclosure Statement (IDS) form for autoloading of data into USPTO systems. You may remove the form to add the required data in order to correct the Informational Message if you are citing U.S. References. If you chose not to include U.S. References, the image of the form will be processed and be made available within the Image File Wrapper (IFW) system. However, no data will be extracted from this form. Any additional data such as Foreign Patent Documents or Non Patent Literature will be manually reviewed and keyed into USPTO systems.					
2	Non Patent Literature	IPRP_PCTUS1248750_mailed_0 4_03_2014.pdf	494897 b8a072d0cdcedbc842ace7cc23fed63e2dc5e795	no	8
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30325 45e625d91703c76c66e8af7b92a2deb40ee1fae6	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			1115980		
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046
46188	7590	05/23/2014	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ELMORE, STEPHEN C	
			ART UNIT	PAPER NUMBER
			2188	
			NOTIFICATION DATE	DELIVERY MODE
			05/23/2014	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 13/559,476	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2188	

All participants (applicant, applicant's representative, PTO personnel):

(1) STEPHEN ELMORE. (3) ____.

(2) Mr Khaled Shami, Reg. No. 38,745. (4) ____.

Date of Interview: 16 May 2014.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: ____.

Issues Discussed ☐101 ☐112 ☐102 ☐103 ☒Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: None.

Identification of prior art discussed: None.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

This interview summary pertains to Applicant's claim for priority based on prior provisional application 61/512,871, filed 7/28/2011, and on prior application 12/240,916, filed 9/29/2008. The Examiner and Applicant's representative reached agreement that the present application is entitled to the benefit of the priority dates claimed above.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/STEPHEN ELMORE/ Primary Examiner, Art Unit 2188	
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Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

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Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

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The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

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- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
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- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
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**UNITED STATES DEPARTMENT OF COMMERCE****U.S. Patent and Trademark Office**

Address : COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR / PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
13/559,476	26 July, 2012	LEE ET AL.	062453-010

Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306		EXAMINER	
		STEPHEN ELMORE	
		ART UNIT	PAPER
		2188	20140516

DATE MAILED:

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

The Applicant-Initiated Interview held on 5/16/2014 (see attached PTOL-413) confirms the determination that the present application is entitled to Applicant's claim for priority based on prior provisional application 61/512,871, filed 7/28/2011, and on prior application 12/240,916, filed 9/29/2008.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

<i>Applicant-Initiated Interview Summary</i>	Application No. 13/559,476	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2188	

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/STEPHEN ELMORE/ Primary Examiner, Art Unit 2188	
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UNITED STATES PATENT AND TRADEMARK OFFICE

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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046
46188	7590	06/24/2014	EXAMINER	
Nixon Peabody LLP			ELMORE, STEPHEN C	
P.O. Box 60610			ART UNIT	
Palo Alto, CA 94306			PAPER NUMBER	
			2188	
			NOTIFICATION DATE	DELIVERY MODE
			06/24/2014	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Application No. : 13559476
Applicant : Lee
Filing Date : 07/26/2012
Date Mailed : 06/24/2014

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The application is not in compliance with 37 CFR 1.78, as indicated in the attachment. The consequences of failure to respond within the above-identified time period are set forth in the attachment.

Even if the Office has recognized a benefit claim and has entered it into the Office's database and included it on applicant's filing receipt, the benefit claim is not a proper benefit claim unless the reference in compliance with 37 CFR 1.78 is included, depending upon the application's filing date and as indicated in the attachment, in an application data sheet or in the first sentence(s) of the specification and all other requirements are met.

See attachment.

*A copy of this notice **MUST** be returned with the reply. Please address response to
"Mail Stop Issue Fee, Commissioner for Patents,
P.O. Box 1450, Alexandria, VA 22313-1450".*

/Kam Sin/
Publication Branch
Office of Data Management
(571) 272-4200

**APPLICATION FILED PRIOR TO SEPTEMBER 16, 2012,
NOT IN COMPLIANCE WITH 37 CFR 1.78**

- ☒ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification does not indicate the relationship (continuation, division, continuation-in-part) to the prior U.S. nonprovisional application or international application designating the U.S. See document coded SPEC dated 12/10/2012, listing application number(s) 12/240916.
- ☐ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification following the title does not provide the U.S. nonprovisional application number (series code and serial number) or, with respect to an international PCT application designating the U.S., it provides the international application number or international filing date but not both. See document coded dated , in which the following is missing: .
- ☐ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification following the title shows an incorrect, incomplete, or illegible U.S. nonprovisional application number, international PCT application number, or international PCT filing date. See document coded dated , in which the following error was made: .
- ☐ The 37 CFR 1.78(a)(2) reference to the prior U.S. nonprovisional application or international application designating the U.S. is not present on an application data sheet or in the first sentence(s) of the specification following the title, thus removing the validating link under 35 U.S.C. 119(a)-(d) to a prior foreign application or under 35 U.S.C. 119(e) to a prior U.S. provisional application.
- ☐ The 37 CFR 1.78(a)(2) reference to the prior U.S. nonprovisional application or international application designating the U.S. is not present on an application data sheet or in the first sentence(s) of the specification following the title.
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application is not present on an application data sheet or in first sentence(s) of the specification following the title.
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application on an application data sheet or in first sentence(s) of the specification following the title does not provide the provisional application number (series code and serial number). See document coded dated , in which the following is missing: .
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application on an application data sheet or in first sentence(s) of the specification following the title shows an incorrect, incomplete, or illegible U.S. provisional application number. See document coded dated , in which the following error was made: .
- ☐ Other: .

HOW TO RESPOND

A proper response to this notice would include any one of: (1) a supplemental Application Data Sheet (ADS) pursuant to 37 CFR 1.76(c) which provides benefit information that complies with 37 CFR 1.78(a)(2) or 37 CFR 1.78(a)(5); (2) an amendment to the first sentence(s) of the specification which provides benefit information that complies with 37 CFR 1.78(a)(2) or 37 CFR 1.78(a)(5); or (3) a petition filed pursuant to the provisions of 37 CFR 1.78(a)(3) or 37 CFR 1.78(a)(6) if the benefit information from the document identified above by code and date does not accurately reflect the benefits under 35 U.S.C. 119(e), 120, 121 or 365(c) as claimed by applicant (a grantable petition would include either a supplemental ADS or an amendment to the first sentence(s) of the specification as required by 37 CFR 1.78(a)(3)(i) or 37 CFR 1.78(a)(6)(i)). Such amendments to the specification or supplemental ADS submission may be filed after payment of the issue fee if limited to informalities noted herein. See Waiver of 37 CFR 1.312 for Document Required by Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004).

WARNING: If Applicant fails to timely submit a proper response, the benefit information will be deleted and the patent will be printed without the benefit information present.



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46188	7590	07/02/2014	EXAMINER	
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			ELMORE, STEPHEN C	
			ART UNIT	PAPER NUMBER
			2188	
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patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

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APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR / PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
13/559,476	26 July, 2012	LEE ET AL.	062453-010

Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306		EXAMINER	
		STEPHEN ELMORE	
		ART UNIT	PAPER
		2188	20140627

DATE MAILED:

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Commissioner for Patents

The two IDS documents filed 5/20/2014 have been entered and considered. A copy of each signed IDS is enclosed.

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

Receipt date: 05/20/2014

13559476 - GAU: 2188

Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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	1							☐

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NON-PATENT LITERATURE DOCUMENTS		Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.

T⁵

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	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

	1	International Preliminary Report on Patentability in PCT/US12/48750, mailed April 3, 2014. pp. 1-8.	☐
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If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	06/27/2014
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

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	Examiner Name	Bragdon, Reginald Glenwood		
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CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☒ See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☐ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-05-20
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

Receipt date: 05/20/2014

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

13559476 - GAU-2188

PTO/SB/08a (01-10)

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

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Add

U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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If you wish to add additional U.S. Published Application citation information please click the Add button.

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FOREIGN PATENT DOCUMENTS							Remove	
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

Add

NON-PATENT LITERATURE DOCUMENTS		Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.
		T ⁵

Receipt date: 05/20/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

1	Office Action in U.S. Patent Application No. 14/173,219, mailed March 13, 2014. pp. 1-7.	☐
2	Office Action in U.S. Patent Application No. 14/173,242, mailed March 14, 2014. pp. 1-7.	☐

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	06/27/2014
--------------------	------------------	-----------------	------------

***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Receipt date: 05/20/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	Hyun Lee		
	Art Unit	2189		
	Examiner Name	Bragdon, Reginald Glenwood		
	Attorney Docket Number	062453-010		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☒ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☒ See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☐ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-05-20
Name/Print	Khaled Shami	Registration Number	38745

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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

46188 7590 05/06/2014
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046

TITLE OF INVENTION: FLASH-DRAM HYBRID MEMORY MODULE

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960 \$480	\$0	\$0	\$960 \$480	08/06/2014

EXAMINER	ART UNIT	CLASS-SUBCLASS
ELMORE, STEPHEN C	2188	711-103000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 **Nixon Peabody LLP**
 2 **Khaled Shami**
 3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

NETLIST, INC.

IRVINE, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number **50-3557** (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☒ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature **/Khaled Shami/**

Date **August 6, 2014**

Typed or printed name **Khaled Shami**

Registration No. **38,745**

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-010			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	2501	1	480	480
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	19789878
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	06-AUG-2014
Filing Date:	26-JUL-2012
Time Stamp:	14:04:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 480
RAM confirmation Number	253
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Post Allowance Communication - Incoming	062453-010_resp_to_correct_a ppln_papers.pdf	202616 634082c521e1918897b8c655035f807a4d9d40de	no	5
Warnings:					
Information:					
2	Drawings-only black and white line drawings	062453-010_Drawings1.pdf	192592 57b191387c053dfc8143ba94f099f34dff324b85	no	10
Warnings:					
Information:					
3	Post Allowance Communication - Incoming	062453-010_comments_on_all owance.pdf	84400 4c26bbc2a6f090ad584030db8ba9553090401f5	no	2
Warnings:					
Information:					
4	Issue Fee Payment (PTO-85B)	062453_010_IF_Pment.pdf	114304 4500842d284d9f045028b33ab2db645e47cfd640	no	1
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	30036 1733fb7e5f200424bbebd1e7a82b731313bb049a	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			623948		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee et al.
SERIAL NO.: 13/559,476 CONFIRMATION NO: 1046
FILING DATE: July 26, 2012
TITLE: Flash-DRAM Hybrid Memory Module
EXAMINER: Elmore, Stephen C.
ART UNIT: 2188

**Mail Stop Issue Fee
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

RESPONSE TO NOTICE TO FILE CORRECTED APPLICATION PAPERS

Dear Sir:

This paper is responsive to the notice mailed June 24, 2014. Please amend the above-identified application as follows:

Amendments to the Specification begin on page **2** of this paper.

Remarks begin on page **3** of this paper.

In the Specification

Please amend paragraph [0001] as follows:

[0001] This application claims the benefit of provisional patent application serial no. 61/512,871, filed July 28, 2011, and is a continuation-in-part of US patent application serial no. 12/240,916, filed September 29, 2008 which is a continuation of U.S. patent application serial no. 12/131,873, filed June 2, 2008, which claims the benefit of U.S. provisional patent application serial no. 60/941,586, filed June 1, 2007, the contents of all of which are incorporated herein by reference in their entirety.

This application may also be considered to be related to co-pending U.S. patent application serial no. 13/536,173, filed on June 28, 2012, and commonly owned herewith.

REMARKS

In response to the Notice to File Corrected Application Papers mailed June 24, 2014, Applicants amend the specification as set out in the Amendments of the Specification section herein to include an amended paragraph [0001] to amend the priority listing for the purposes of improving the readability of the application. Applicants added no new matter through the amendment.

Please charge any additional required fee or credit any overpayment not otherwise paid or credited to our deposit account No. 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: August 6, 2014

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX (650) 320-7701



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Application No. : 13559476
Applicant : Lee
Filing Date : 07/26/2012
Date Mailed : 06/24/2014

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The application is not in compliance with 37 CFR 1.78, as indicated in the attachment. The consequences of failure to respond within the above-identified time period are set forth in the attachment.

Even if the Office has recognized a benefit claim and has entered it into the Office's database and included it on applicant's filing receipt, the benefit claim is not a proper benefit claim unless the reference in compliance with 37 CFR 1.78 is included, depending upon the application's filing date and as indicated in the attachment, in an application data sheet or in the first sentence(s) of the specification and all other requirements are met.

See attachment.

*A copy of this notice **MUST** be returned with the reply. Please address response to
"Mail Stop Issue Fee, Commissioner for Patents,
P.O. Box 1450, Alexandria, VA 22313-1450".*

/Kam Sin/
Publication Branch
Office of Data Management
(571) 272-4200

**APPLICATION FILED PRIOR TO SEPTEMBER 16, 2012,
NOT IN COMPLIANCE WITH 37 CFR 1.78**

- ☒ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification does not indicate the relationship (continuation, division, continuation-in-part) to the prior U.S. nonprovisional application or international application designating the U.S. See document coded SPEC dated 12/10/2012, listing application number(s) 12/240916.
- ☐ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification following the title does not provide the U.S. nonprovisional application number (series code and serial number) or, with respect to an international PCT application designating the U.S., it provides the international application number or international filing date but not both. See document coded dated , in which the following is missing: .
- ☐ The 37 CFR 1.78(a)(2) reference on the application data sheet or in the first sentence(s) of the specification following the title shows an incorrect, incomplete, or illegible U.S. nonprovisional application number, international PCT application number, or international PCT filing date. See document coded dated , in which the following error was made: .
- ☐ The 37 CFR 1.78(a)(2) reference to the prior U.S. nonprovisional application or international application designating the U.S. is not present on an application data sheet or in the first sentence(s) of the specification following the title, thus removing the validating link under 35 U.S.C. 119(a)-(d) to a prior foreign application or under 35 U.S.C. 119(e) to a prior U.S. provisional application.
- ☐ The 37 CFR 1.78(a)(2) reference to the prior U.S. nonprovisional application or international application designating the U.S. is not present on an application data sheet or in the first sentence(s) of the specification following the title.
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application is not present on an application data sheet or in first sentence(s) of the specification following the title.
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application on an application data sheet or in first sentence(s) of the specification following the title does not provide the provisional application number (series code and serial number). See document coded dated , in which the following is missing: .
- ☐ The 37 CFR 1.78(a)(5) reference to the prior U.S. provisional application on an application data sheet or in first sentence(s) of the specification following the title shows an incorrect, incomplete, or illegible U.S. provisional application number. See document coded dated , in which the following error was made: .
- ☐ Other: .

HOW TO RESPOND

A proper response to this notice would include any one of: (1) a supplemental Application Data Sheet (ADS) pursuant to 37 CFR 1.76(c) which provides benefit information that complies with 37 CFR 1.78(a)(2) or 37 CFR 1.78(a)(5); (2) an amendment to the first sentence(s) of the specification which provides benefit information that complies with 37 CFR 1.78(a)(2) or 37 CFR 1.78(a)(5); or (3) a petition filed pursuant to the provisions of 37 CFR 1.78(a)(3) or 37 CFR 1.78(a)(6) if the benefit information from the document identified above by code and date does not accurately reflect the benefits under 35 U.S.C. 119(e), 120, 121 or 365(c) as claimed by applicant (a grantable petition would include either a supplemental ADS or an amendment to the first sentence(s) of the specification as required by 37 CFR 1.78(a)(3)(i) or 37 CFR 1.78(a)(6)(i)). Such amendments to the specification or supplemental ADS submission may be filed after payment of the issue fee if limited to informalities noted herein. See Waiver of 37 CFR 1.312 for Document Required by Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004).

WARNING: If Applicant fails to timely submit a proper response, the benefit information will be deleted and the patent will be printed without the benefit information present.

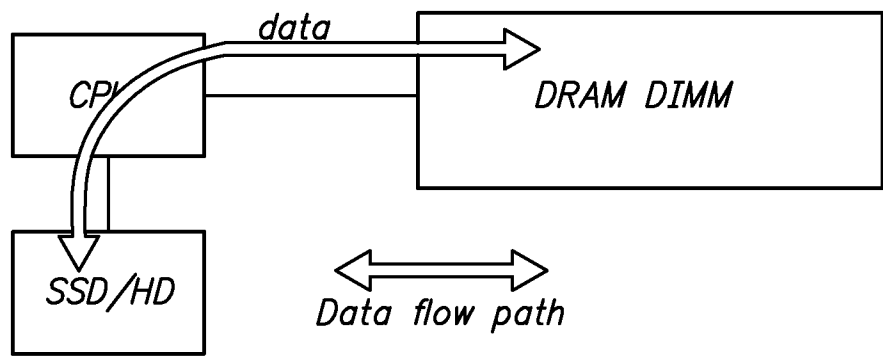


FIG. 1
(PRIOR ART)

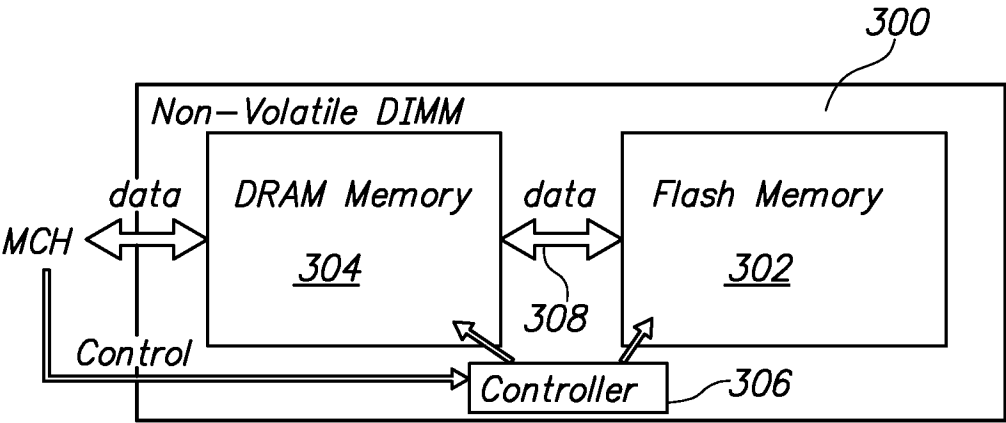
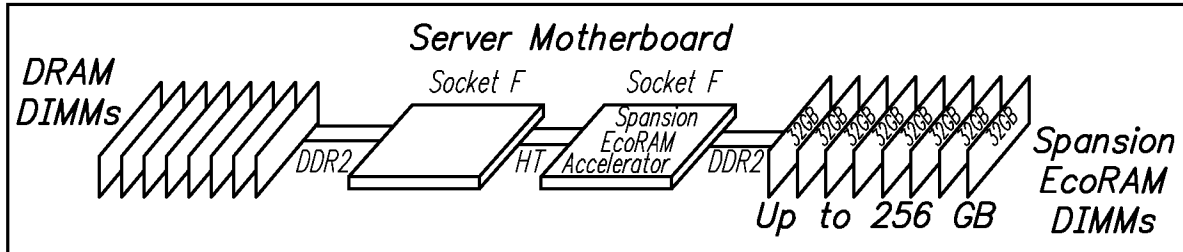


FIG. 3A

2/10

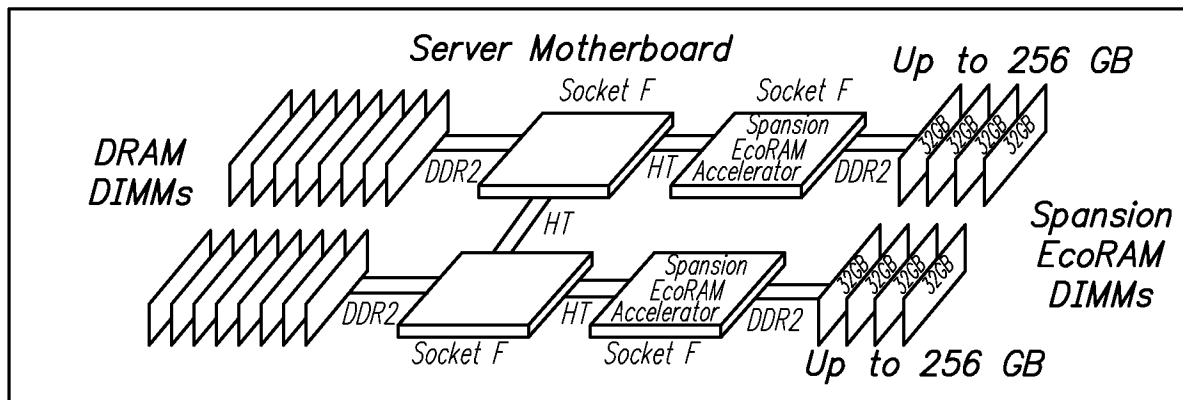
Spancion EcoRAM Configurations

256GB Spancion EcoRAM Solution – Single Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

256GB Spancion EcoRAM Solution – Dual Accelerator



256GB Single Accelerator Spancion EcoRAM Solution

FIG. 2
(PRIOR ART)

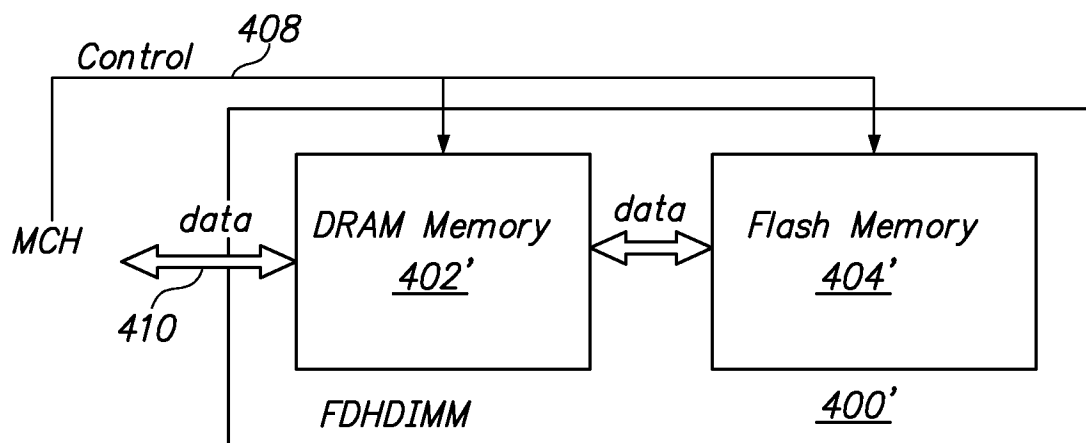


FIG. 4B

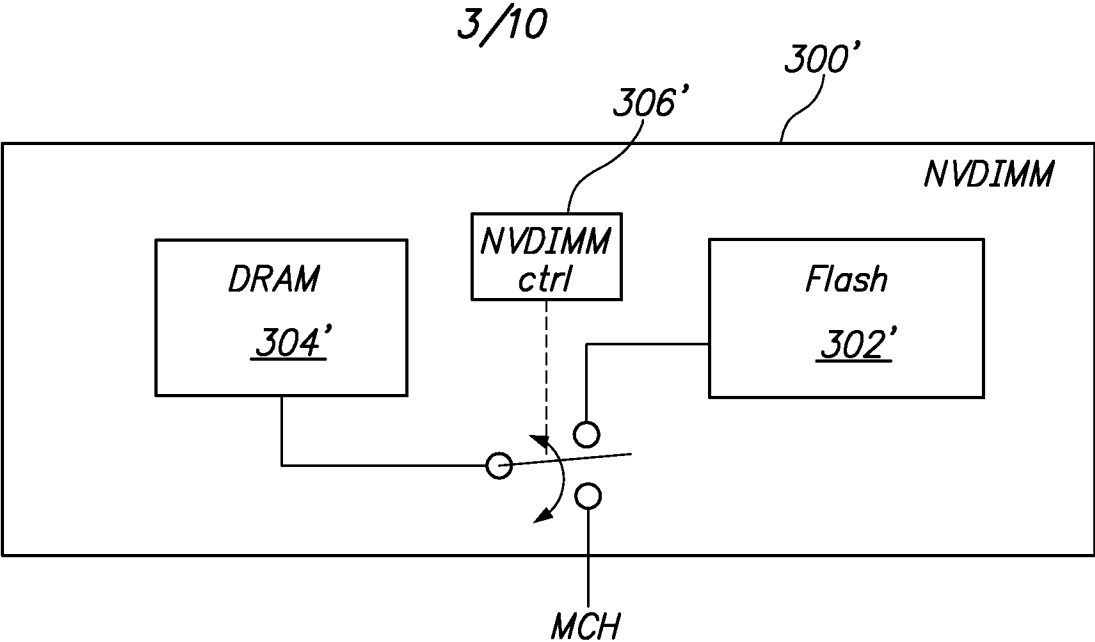


FIG. 3B

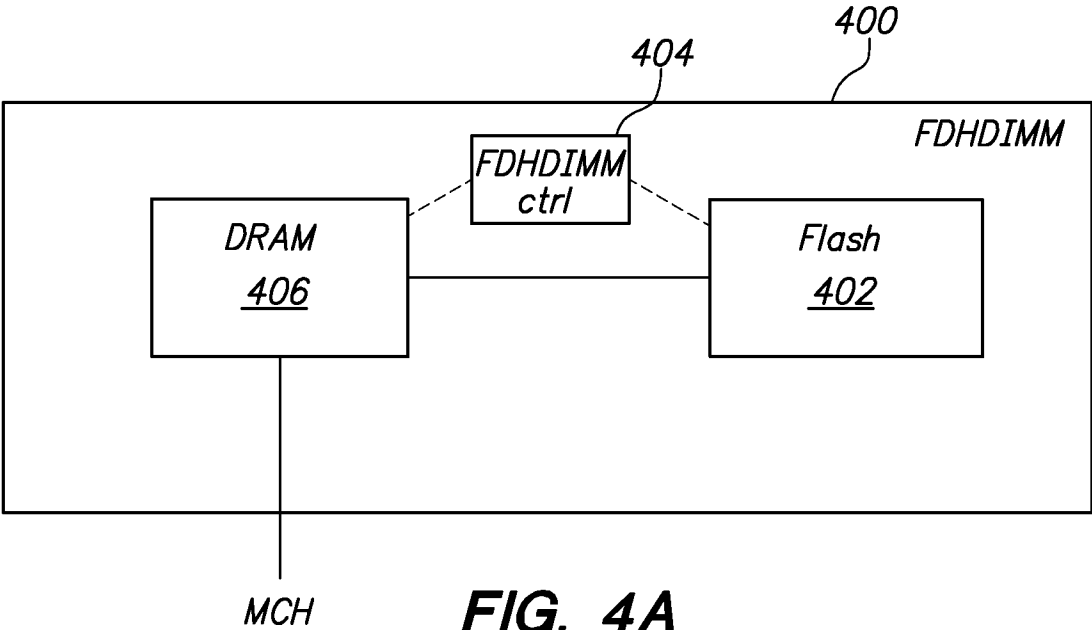
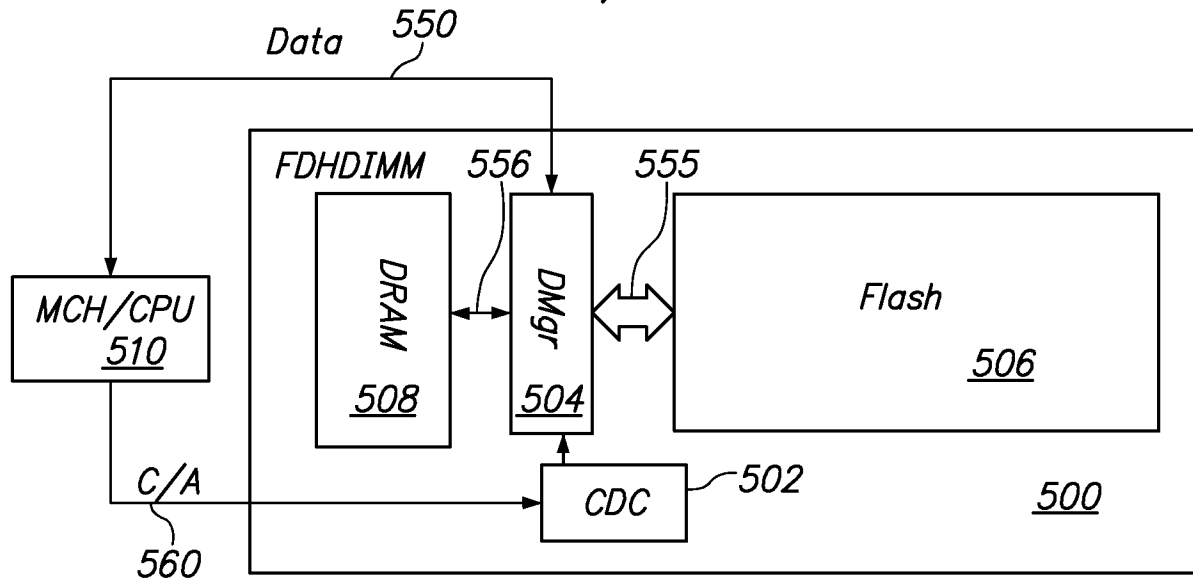
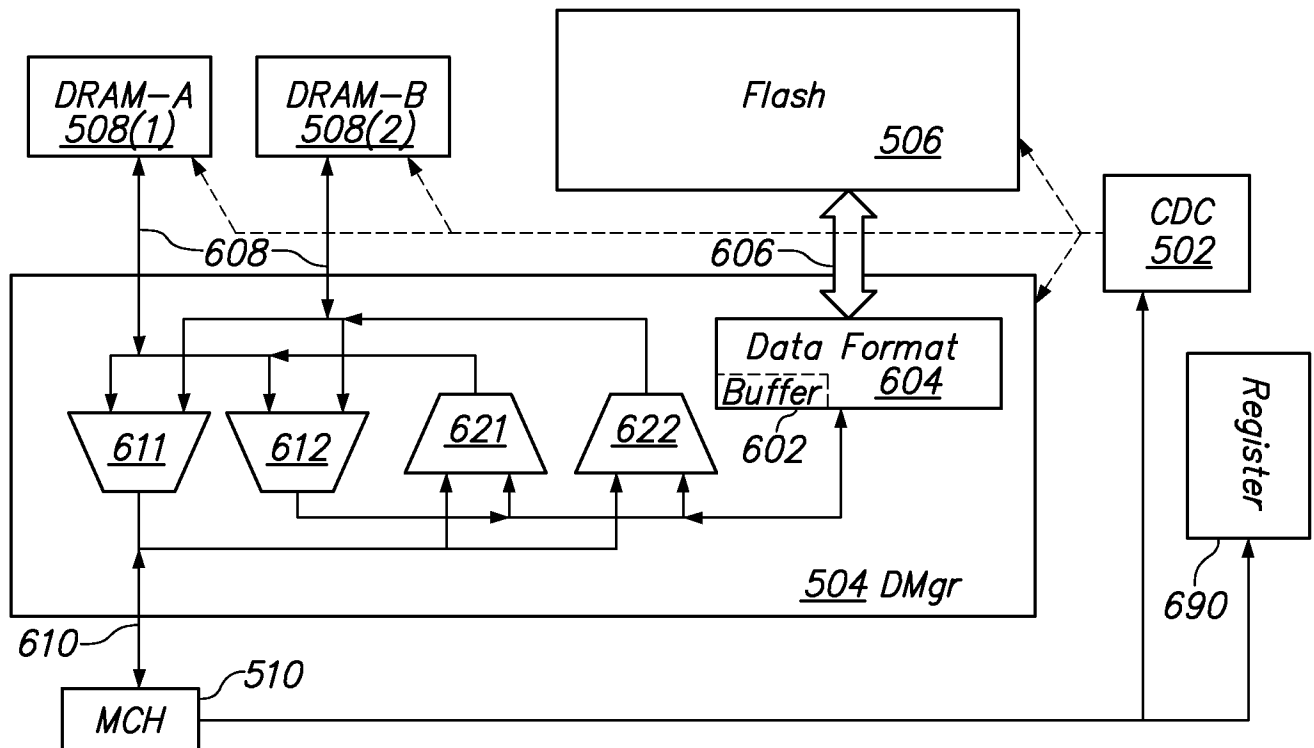
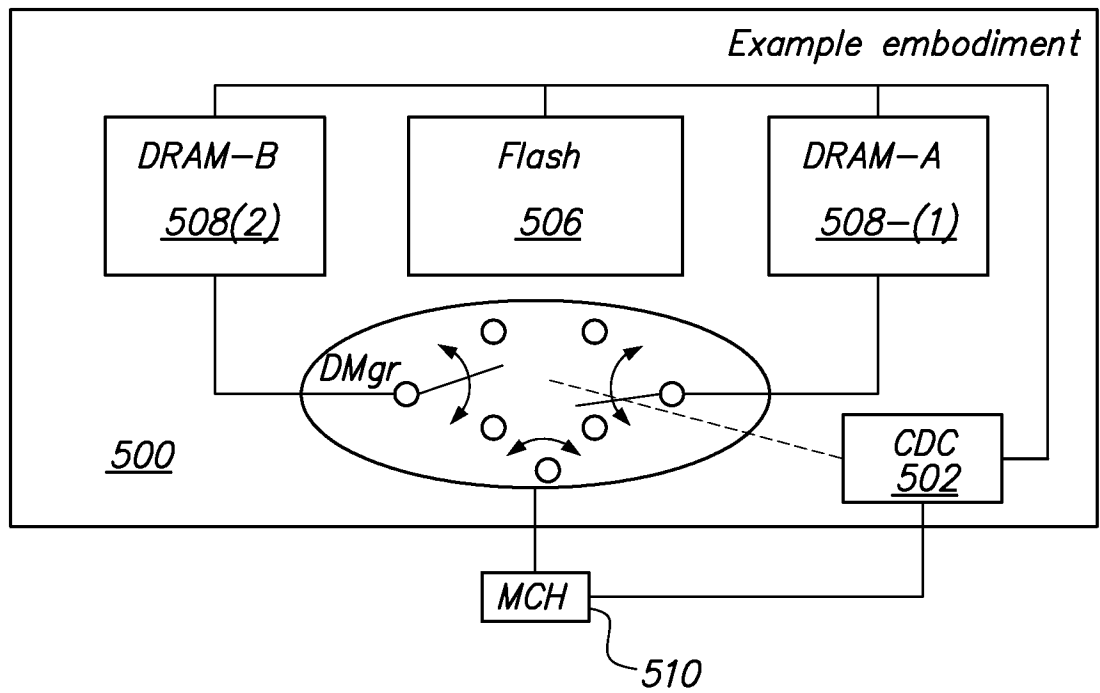


FIG. 4A

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**FIG. 5A****FIG. 6**

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**FIG. 5B**

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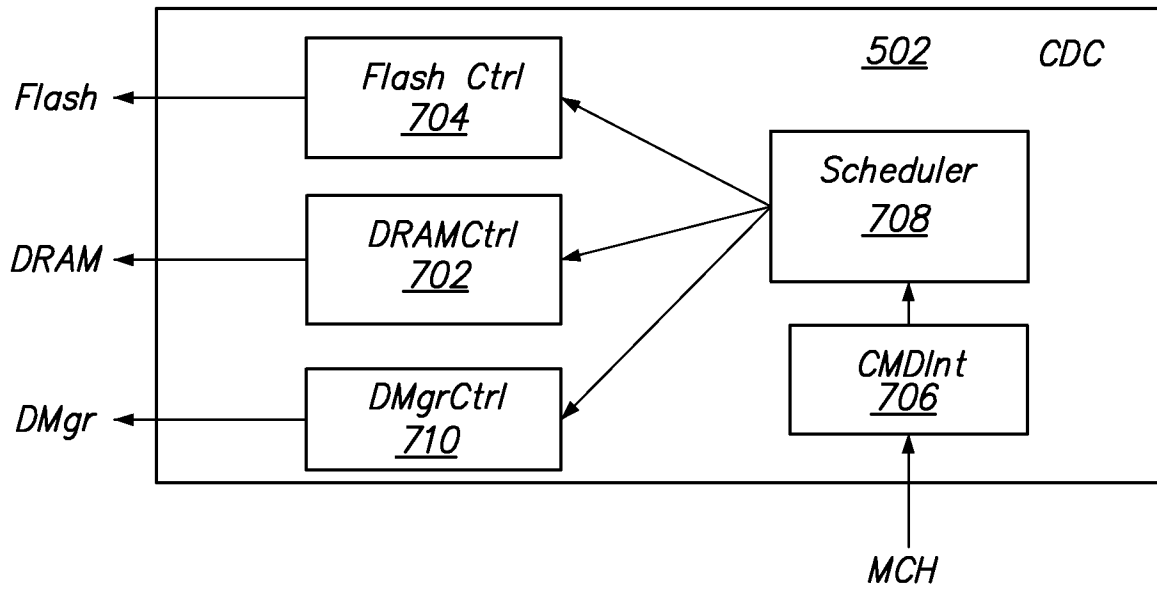


FIG. 7

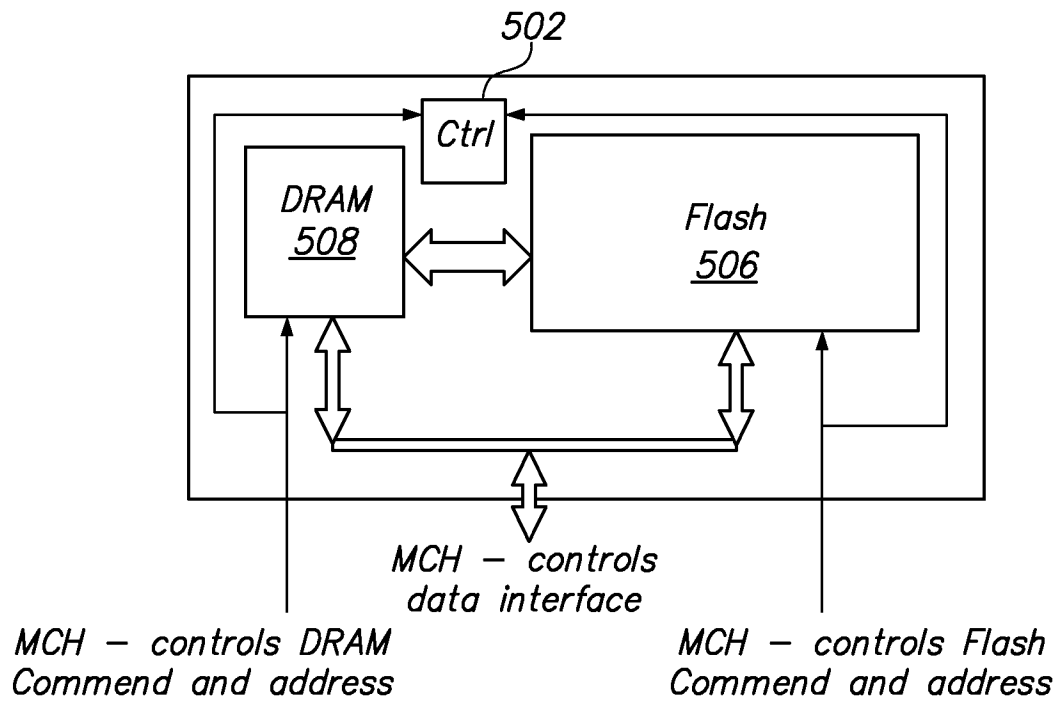


FIG. 8A

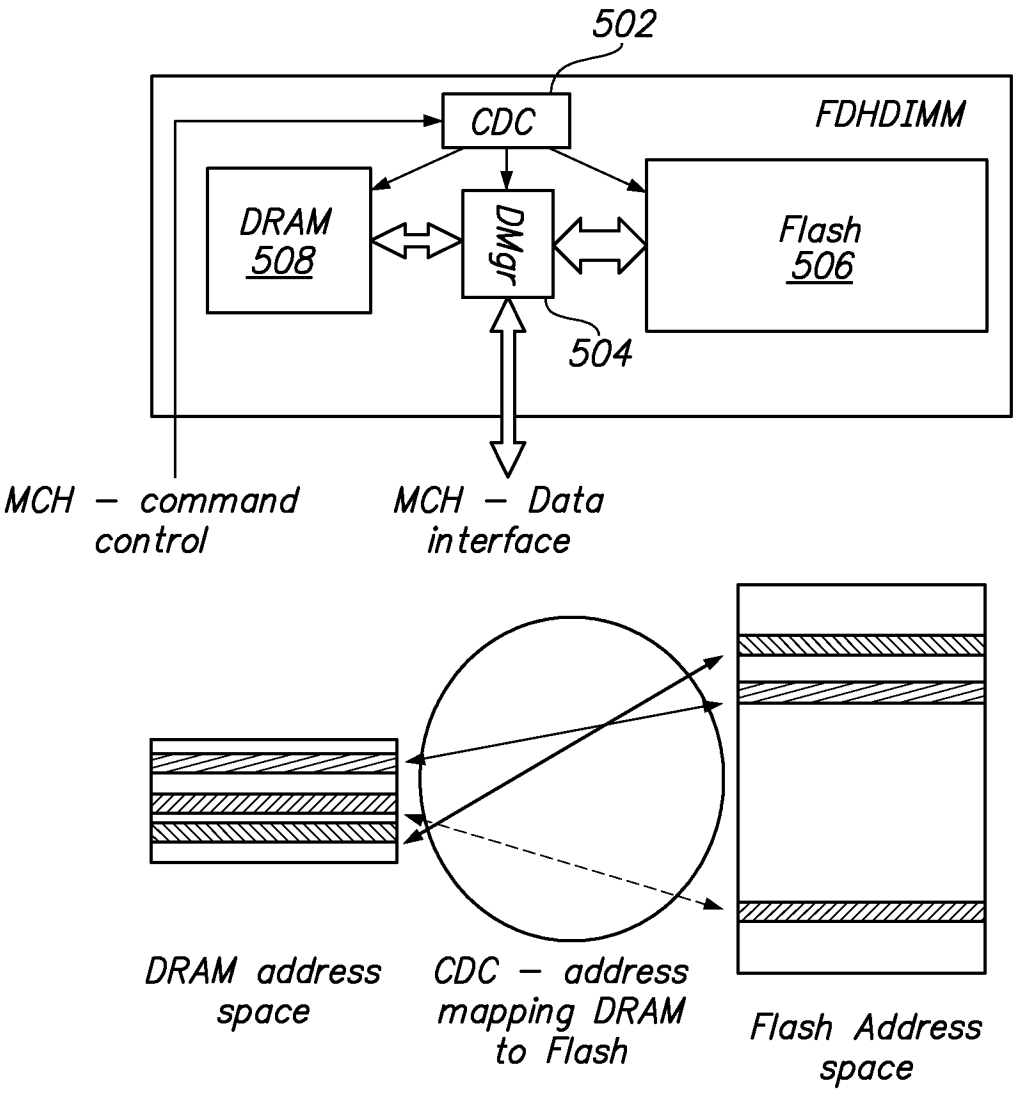
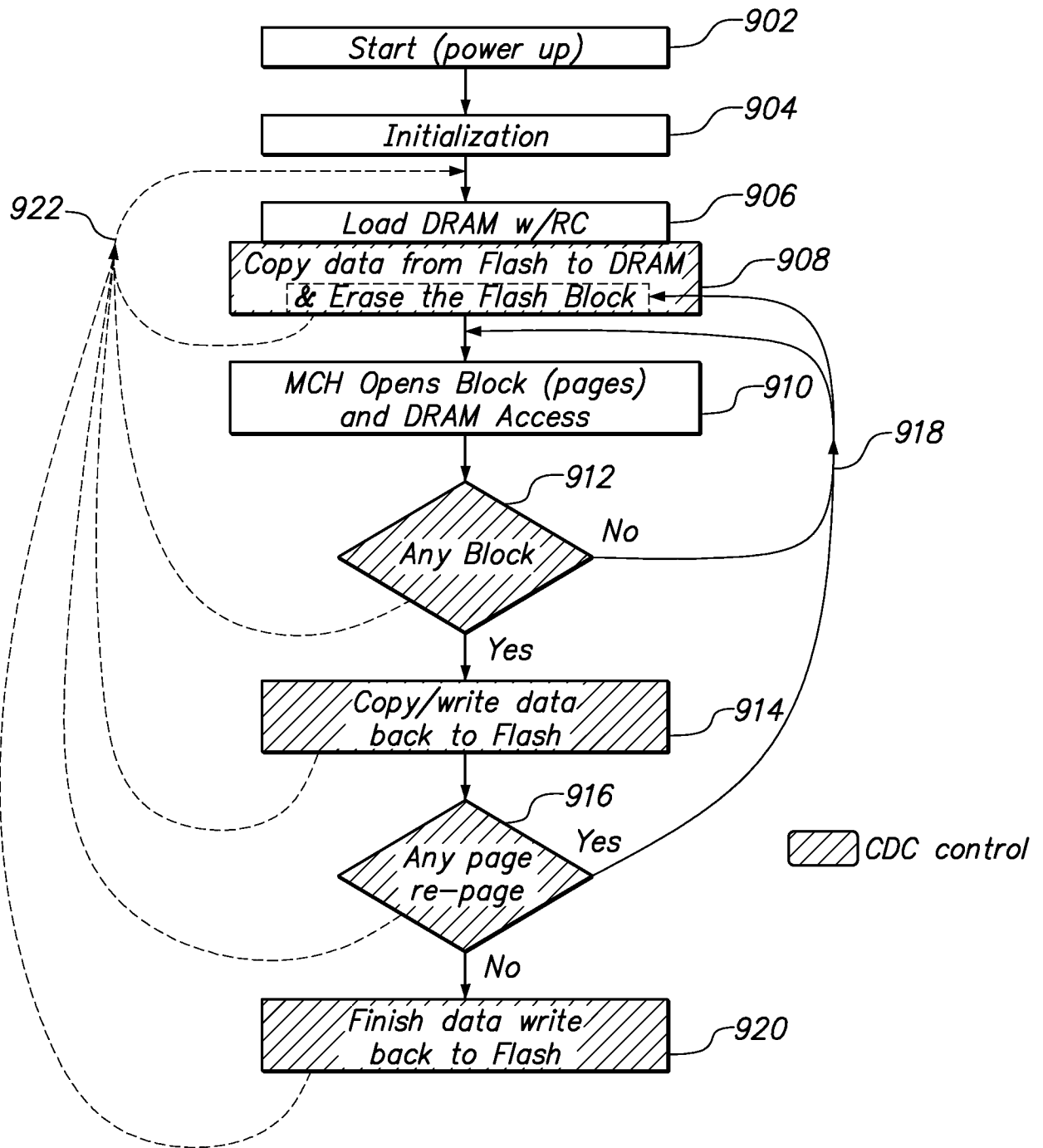
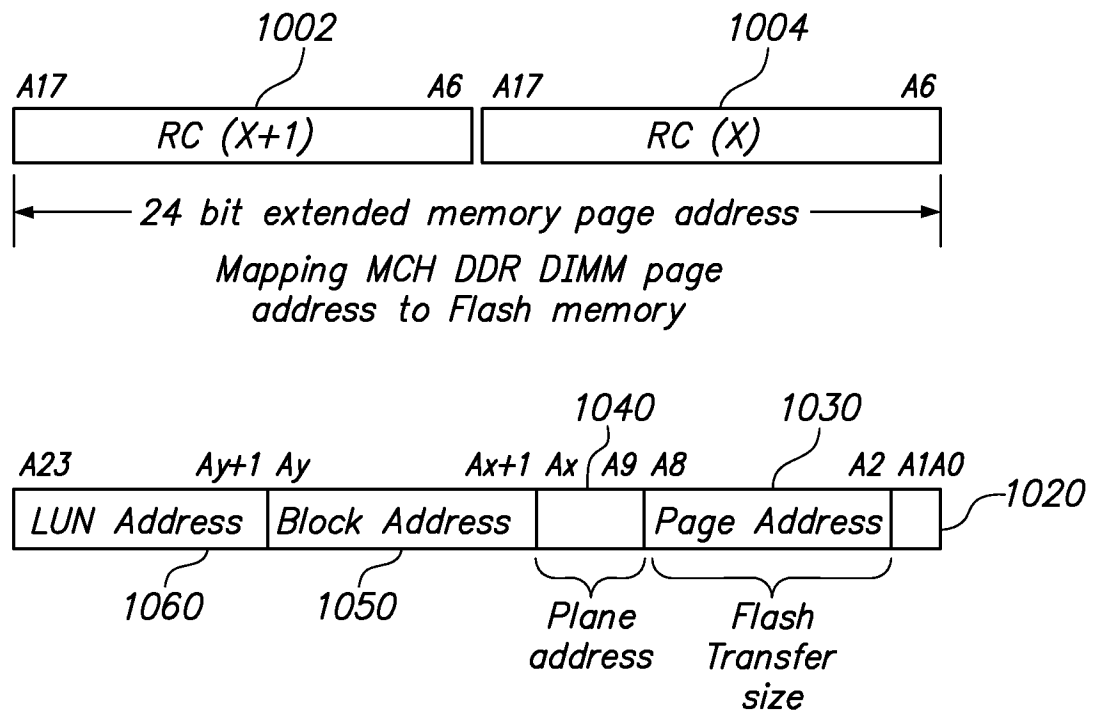


FIG. 8B

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**FIG. 9**

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**FIG. 10**

REPLACEMENT SHEET

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<i>DRAM density (GB)</i>	<i># of blocks per bank</i>	<i>Flash wr-time to rd-time ratio</i>	<i>Avg block use time (sec)</i>	<i>Flash write time (sec)</i>	<i>Max allowed Closed Blk in queue to be written back to Flash</i>
1	250	55	1.00E-03	2.00E-02	0
1	250	55	1.00E-02	2.00E-02	2
1	250	55	2.00E-02	2.00E-02	5
1	250	55	5.00E-02	2.00E-02	11
2	500	55	1.00E-03	2.00E-02	0
2	500	55	1.00E-02	2.00E-02	5
2	500	55	2.00E-02	2.00E-02	9
2	500	55	5.00E-02	2.00E-02	23
4	1000	55	1.00E-03	2.00E-02	1
4	1000	55	1.00E-02	2.00E-02	9
4	1000	55	2.00E-02	2.00E-02	18
4	1000	55	5.00E-02	2.00E-02	45

FIG. 11

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee et al.
SERIAL NO.: 13/559,476 CONFIRMATION NO: 1046
FILING DATE: July 26, 2012
TITLE: Flash-DRAM Hybrid Memory Module
EXAMINER: Elmore, Stephen C.
ART UNIT: 2188

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

APPLICANT'S COMMENTS ON REASONS FOR ALLOWANCE

These comments are responsive to the Notice of Allowance, mailed on May 6, 2014.

Applicant gratefully acknowledges the indication of allowance of Claims 1-5, 9, 13, 14, 16, 18, 19 and 21-24. Applicant respectfully urges that additional and/or alternative reasons for allowance may exist apart from those advanced by the Examiner and the Applicant, and these reasons may each be independently sufficient to establish the patentability of the allowed claims.

Applicant respectfully reserves the right to introduce, articulate, or otherwise comment on any such additional reasons for allowance as may be appropriate in any future proceedings concerning the one or more claimed embodiments.

Please charge any additional required fee or credit any overpayment to our Deposit

Account number 50-3557.

Respectfully submitted,
NIXON PEABODY LLP

Dated: August 6, 2014

/Khaled Shami/
Khaled Shami
Reg. No. 38,745

NIXON PEABODY LLP
P.O. BOX 60610
PALO ALTO, CA 94306
TEL. (650) 320-7700
FAX (650) 320-7701



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	09/16/2014	8838886	062453-010	1046

46188 7590 08/27/2014
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 176 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL **(Submitted Only via EFS-Web)**

Application Number	13559476	Filing Date	2012-08-26	Docket Number (if applicable)	062453-010	Art Unit	2188
First Named Inventor	Hyun LEE			Examiner Name	Stephen C. ELMORE		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☐ Amendment/Reply

☒ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☒ Other Certification and Request for Consideration of the Information Disclosure Statement Filed After Payment of Issue Fee Under the QPIDS Pilot Program

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
 (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 503557

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

☐ Applicant Signature

Doc code: RCEX

Doc description: Request for Continued Examination (RCE)

PTO/SB/30EFS (07-09)

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Signature of Registered U.S. Patent Practitioner			
Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-09-05
Name	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

**CERTIFICATION AND REQUEST FOR CONSIDERATION OF AN INFORMATION DISCLOSURE
STATEMENT FILED AFTER PAYMENT OF THE ISSUE FEE UNDER THE QPIDS PILOT PROGRAM**

Non-Provisional Application Number: 13/559,476	Filing Date: 2012-08-26
First Named Inventor: Hyun LEE	Title of Invention: FLASH-DRAM HYBRID MEMORY MODULE

**THE UNDERSIGNED HEREBY CERTIFIES AND REQUESTS THE FOLLOWING FOR THE ABOVE-
IDENTIFIED APPLICATION.**

1. Consideration is requested of the information disclosure statement (IDS) submitted herewith, which is being filed after payment of the issue fee.
2. Check the box next to the appropriate selection:
☐ Each item of information contained in the IDS was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the IDS. See 37 CFR 1.97(e)(1).
OR
☒ No item of information contained in the IDS was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the IDS was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the IDS. See 37 CFR 1.97(e)(2).
3. Please charge the IDS fee set forth in 37 CFR 1.17(p) to Deposit Account No. 50-3557.
4. A Petition to Withdraw from Issue After Payment of the Issue Fee (37 CFR 1.313(c)(2)), including the petition fee set forth in 37 CFR 1.17(h), is submitted herewith as a **Web-based ePetition**.
WARNING: Do not submit the petition as a follow-on paper via EFS-Web. Submit the petition as a Web-based ePetition by signing on to EFS-Web as a registered user, selecting the radio button next to "Existing application/patent," and then selecting the radio button next to "ePetition (for automatic processing and immediate grant, if all petitions requirements are met)." Failure to use the Web-based ePetition interface will result in automatic entry of the RCE.
5. A request for continued examination (RCE) under 37 CFR 1.114 and the RCE fee under 37 CFR 1.17(e) are submitted herewith.
6. The RCE will be treated as a "conditional" RCE. In the event the examiner determines that any item of information contained in the IDS necessitates the reopening of prosecution in the application, the undersigned understands that (i) the RCE will be processed and treated as an RCE under 37 CFR 1.114 and therefore (ii) the IDS fee under 37 CFR 1.17(p) will be returned in accordance with 37 CFR 1.97(b)(4). In the event that no item of information in the IDS necessitates reopening prosecution, the undersigned understands that the RCE will not be processed and the RCE fee under 37 CFR 1.17(e) will be returned.
7. This certification and request is being filed as a **Web-based ePetition** and is not accompanied by an amendment to the application. Inclusion of an amendment will result in automatic entry of the RCE.

Signature <u>/Khaled Shami/</u>	Date 2014-09-05
Name (Print/Typed) Khaled Shami	Practitioner Registration Number (If applicable) 38,745
Note: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature. If necessary, submit multiple forms for more than one signature, see below.*	
☐ *Total of _____ forms are submitted.	

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Doc Code: PET.AUTO Document Description: Petition automatically granted by EFS-Web		PTO/SB/140 U.S. Patent and Trademark Office Department of Commerce	
Electronic Petition Request	PETITION TO WITHDRAW AN APPLICATION FROM ISSUE AFTER PAYMENT OF THE ISSUE FEE UNDER 37 CFR 1.313(c)		
Application Number	13559476		
Filing Date	26-Jul-2012		
First Named Inventor	Hyun Lee		
Art Unit	2188		
Examiner Name	STEPHEN ELMORE		
Attorney Docket Number	062453-010		
Title	FLASH-DRAM HYBRID MEMORY MODULE		
An application may be withdrawn from issue for further action upon petition by the applicant. To request that the Office withdraw an application from issue, applicant must file a petition under this section including the fee set forth in § 1.17(h) and a showing of good and sufficient reasons why withdrawal of the application from issue is necessary. APPLICANT HEREBY PETITIONS TO WITHDRAW THIS APPLICATION FROM ISSUE UNDER 37 CFR 1.313(c). A grantable petition requires the following items: (1) Petition fee; and (2) One of the following reasons: (a) Unpatentability of one or more claims, which must be accompanied by an unequivocal statement that one or more claims are unpatentable, an amendment to such claim or claims, and an explanation as to how the amendment causes such claim or claims to be patentable; (b) Consideration of a request for continued examination in compliance with § 1.114 (for a utility or plant application only); or (c) Express abandonment of the application. Such express abandonment may be in favor of a continuing application, but not a CPA under 37 CFR 1.53(d).			
Petition Fee			
☒ Small Entity			
☐ Micro Entity			
☐ Regular Undiscounted			
Reason for withdrawal from issue			

- ☐ One or more claims are unpatentable
- ☒ Consideration of a request for continued examination (RCE) (List of Required Documents and Fees)
- ☐ Applicant hereby expressly abandons the instant application (any attorney/agent signing for this reason must have power of attorney pursuant to 37 CFR 1.32(b)).

RCE request, submission, and fee.

- ☐ I certify, in accordance with 37 CFR 1.4(d)(4) that :
- ☐ The RCE request ,submission, and fee have already been filed in the above-identified application on
- ☒ Are attached.

THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

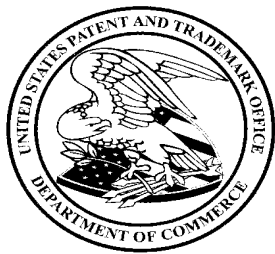
- ☒ An attorney or agent registered to practice before the Patent and Trademark Office who has been given power of attorney in this application.
- ☐ An attorney or agent registered to practice before the Patent and Trademark Office, acting in a representative capacity.
- ☐ A sole inventor
- ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application
- ☐ A joint inventor; all of whom are signing this e-petition

Signature	/Khaled Shami/
Name	Khaled Shami
Registration Number	38745

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-010			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Petition Fee-37CFR 1.17(h) (Group II)	2464	1	70	70
Request for Continued Examination	2801	1	600	600
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				670



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Decision Date : September 5, 2014

In re Application of :

Hyun Lee

DECISION ON PETITION

UNDER CFR 1.313(c)(2)

Application No : 13559476

Filed : 26-Jul-2012

Attorney Docket No : 062453-010

This is an electronic decision on the petition under 37 CFR 1.313(c)(2), filed September 5, 2014 to withdraw the above-identified application from issue after payment of the issue fee.

The petition is **GRANTED**.

The above-identified application is withdrawn from issue for consideration of a submission under 37 CFR 1.114 (request for continued examination). See 37 CFR 1.313(c)(2).

Petitioner is advised that the issue fee paid in this application cannot be refunded. If, however, this application is again allowed, petitioner may request that it be applied towards the issue fee required by the new Notice of Allowance.

Telephone inquiries concerning this decision should be directed to the Patent Electronic Business Center (EBC) at 866-217-9197.

This application file is being referred to Technology Center AU 2188 for processing of the request for continuing examination under 37 CFR 1.114 .

Office of Petitions

Electronic Acknowledgement Receipt

EFS ID:	20055150
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	05-SEP-2014
Filing Date:	26-JUL-2012
Time Stamp:	11:59:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$670
RAM confirmation Number	9407
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	RCE.pdf	722759	no	3
			ae884bd7800065e952966a7ff6f3915abf883705		
Warnings:					
Information:					
2	Transmittal Letter	QPIDS_petition.pdf	153108	no	2
			28f1d8909a9c172ac30a3e9c43b5399353b64e38		
Warnings:					
Information:					
3	Quick Path Information Disclosure Statement	062453-010_IDS_1_of_2_dated_09-05-2014_f.pdf	78148	no	5
			903ebadb3aed372a1303dcd39ef1adc8727fb96a		
Warnings:					
Information:					
4	Quick Path Information Disclosure Statement	062453-010_IDS_2_of_2_dated_09-05-2014_f.pdf	93863	no	9
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Warnings:					
Information:					
5	Petition automatically granted by EFS	petition-request.pdf	31539	no	2
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6	Fee Worksheet (SB06)	fee-info.pdf	32092	no	2
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Warnings:					
Information:					
Total Files Size (in bytes):			1111509		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Electronic Acknowledgement Receipt

EFS ID:	20052493
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	05-SEP-2014
Filing Date:	26-JUL-2012
Time Stamp:	12:01:24
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	JEDEC21-C.pdf	427496 724cded39aa33581516ae1037c08fd3bf329c93f	no	18

Warnings:

Information:

2	Non Patent Literature	Diablo_INVALIDITY_CONTENTIONS_Redacted_Part_1_of_3.pdf	9884128	no	128
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13	Non Patent Literature	Horowitz_The_Art_of_Electronics.pdf	1277181 5888c452b6dfcec9affe0fe1af52fc3b1217dea4	no	15
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14	Non Patent Literature	Innis_MPC8560_PowerQUICCIII_Compact_Flash_Interface_Design.pdf	2723361 9393028aebc43a647a8984d424aab67abed24706	no	24
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15	Non Patent Literature	Jacob_Memory_Systems_Cache_DRAM_Disk.pdf	1576215 b8c7166a007a5ad79a9ec15fbb674e383bbbb24c	no	15
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16	Non Patent Literature	Jandhyala_Design_For_Test_Analysis_Buffered_DRAM_DIMM_IEEE_Semiconductor_Group.pdf	1434512 682e1320112a3bb7b6fc2df93eb39d12c6e2c0fe	no	15
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36	Non Patent Literature	SMART_Inter_PARTES_Petition_For_Review_8301833_Part_3_of_6.pdf	2861988	no	128
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37	Non Patent Literature	SMART_Inter_PARTES_Petition_For_Review_8301833_Part_4_of_6.pdf	10558618	no	179
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40	Non Patent Literature	Using_Two_Chip_Selects_to_Enable_Quad_Rank.pdf	884607	no	2
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41	Non Patent Literature	MetaRAM_Develops_New_Technology_SNDK-NET-0000386.pdf	483215	no	3
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42	Non Patent Literature	HYNIX_240pin_DDR2_MetaSDRAM_SNDK-NET-0000389.pdf	2020886	no	32
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43	Non Patent Literature	NPL_WONG_1998-2001_BIOS_Optimization_Guide_67_pages.pdf	1301358	no	67
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44	Non Patent Literature	ANS_1988_Dictionary_Electrical_Electrical_Terms_pp_215_722_964_1103.pdf	1504338	no	6
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The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
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45	Non Patent Literature	Websters_New_College_Dictionary_2001_pp_259_1115.pdf	969448	no	4
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46	Non Patent Literature	Out_Of_Stealth_Mode_Start_Up_Storage_Newsletter_SNDK-NET-0000361.pdf	2727068 94d7509d790659409ee276bf946c6f90e5abe8	no	8
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47	Non Patent Literature	JEDEC_Configurations_For_Solid_State_Std_21-C_Release_9_August_1999.pdf	21706341 e00b93ebef389deae120f48b7d629935d40c36ee	no	114
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 09/11/2014

TROBERTS	SALE	#00000003	Mailroom Dt:	09/05/2014	503557	13559476
		01	FC : 1806	180.00	DA	

U.S. DEPARTMENT OF COMMERCE
PATENT AND TRADEMARK OFFICE

PATENT WITHDRAWAL NOTICE

DATE WITHDRAWN

9/9/2014

WITHDRAWAL NUMBER

26649

The following application has been **WITHDRAWN** from the

9/16/2014 issue.

SERIAL NO.

13559476

PATENT NUMBER

8838886

TITLE

FLASH-DRAM HYBRID MEMORY MODULE

NAME AND ADDRESS

HYUN LEE
Ladera Ranch, CA

REASON FOR WITHDRAWAL

Auto-petition to withdraw - Granted

APPROVED

/Kimberly Terrell/, Manager

Patent Publication Branch
Office of Data Management

U.S. DEPARTMENT OF COMMERCE
PATENT AND TRADEMARK OFFICE

PATENT WITHDRAWAL NOTICE

DATE WITHDRAWN

9/9/2014

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The following application has been **WITHDRAWN** from the

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13559476

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FLASH-DRAM HYBRID MEMORY MODULE

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Auto-petition to withdraw - Granted

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/Kimberly Terrell/, Manager

Patent Publication Branch
Office of Data Management



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	07/26/2012	Hyun Lee	062453-010	1046
46188	7590	09/29/2014		
Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306			EXAMINER ELMORE, STEPHEN C	
			ART UNIT 2188	PAPER NUMBER
			NOTIFICATION DATE 09/29/2014	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentsv@nixonpeabody.com
ocastanon@nixonpeabody.com

Corrected Notice of Allowability	Application No. 13/559,476	Applicant(s) LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2188	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to See Continuation Sheet.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-5,9,13,14,16,18,19 and 21-24. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| <ol style="list-style-type: none"> 1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>9/5/2014 and 9/5/2014</u> 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | <ol style="list-style-type: none"> 5. ☒ Examiner's Amendment/Comment 6. ☐ Examiner's Statement of Reasons for Allowance 7. ☒ Other <u>Response to Rule 312 Amendment (PTOL-271)</u>. |
|--|--|

/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

Continuation of Item 1. This communication is responsive to : the Petition to Withdraw an Application From Issue Under 37 CFR 1.313(c) and QPIDS Pilot Program submission both filed 9/5/2014, and Rule 312 Amendment and Replacement Drawings both filed 8/6/2014.

The present application is being examined under the pre-AIA first to invent provisions.

Drawings

The replacement drawings were received on 8/6/2014. These drawings are approved.

Information Disclosure Statement

The two (2) IDS [PTOL-1449's] submitted 9/5/2014 includes listed publications that have been "lined-through" because the listings fail to comply with 37 CFR § 1.98 (b)(5), which requires the following information: publisher, author (if any), title, relevant pages of the publication, date, and place of publication.

Applicant is advised that the date of any re-submission of any item of information contained in this information disclosure statement or the submission of any missing element(s) will be the date of submission for purposes of determining compliance with the requirements based on the time of filing the statement, including all certification requirements for statements under 37 CFR 1.97(e). See MPEP § 609.05(a).

A copy of the submitted PTOL-1449(s) otherwise initialed and dated by the examiner is attached to the instant office action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEPHEN ELMORE whose telephone number is (571)272-4436. The examiner can normally be reached on Mon-Fri from 9:30-5:00.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Mano Padmanabhan can be reached on (571) 272-4210. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2188

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/STEPHEN ELMORE/
Primary Examiner, Art Unit 2188

September 19, 2014

Receipt date: 09/05/2014

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

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13559476 - GAU: 2188

PTO/SB/08a (01-10)

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476
	Filing Date		2012-07-26
	First Named Inventor	LEE, Hyun	
	Art Unit	2188	
	Examiner Name	ELMORE, STEPHEN C	
	Attorney Docket Number	062453-010 2 of 2	

U.S.PATENTS						
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	2043099		1936-02-02	Hanna	
	2	3562555		1971-02-09	Ahrns	
	3	3916390		1975-10-28	Chang et al.	
	4	4234920		1980-11-18	Van Ness et al.	
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	Art Unit		2188	
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	9	5890192		1999-03-30	Lee et al.	
	10	5953215		1999-09-14	Karabatsos	
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	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number		062453-010 2 of 2	

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Receipt date: 09/05/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	LEE, Hyun		
	Art Unit		2188	
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number		062453-010 2 of 2	

Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	20040163027	A1	2004-08-19	MacLaren et al.	
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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NON-PATENT LITERATURE DOCUMENTS

Receipt date: 09/05/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	LEE, Hyun		
	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 2 of 2		

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T5
	1	Bruce, J., "Synchronous DRAM Architectures, Organizations, and Alternate Technologies", Electrical and Computer Engineering Dept., Univ. of Maryland, December 10, 2002, 22 pages.	☐
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	7	JEDEC Standard, Double Data Rate (DDR): SDRAM Specification: JESD79C (Revision JESD79B), March 2003, pp. 1-75.	☐
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	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 2 of 2		

11	Petition for Inter Partes Review of U.S. Patent No. 8,001,833 (on behalf of SanDisk, Corp.), filed June 20, 2014.	☐
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	First Named Inventor	LEE, Hyun		
	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 2 of 2		

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

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- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
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SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-09-05
Name/Print	Khaled Shami	Registration Number	38745

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Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	9438	711/103.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
L2	2687	365/185.33.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
L3	19492	711/111,112,114,154,156.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
L4	29661	L1 or L2 or L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
L5	4794	hybrid near3 memory	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
L6	566	data adj manager and controller and memory adj controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
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L10	162084	(format\$3 near3 data) with (transfer or transferred or transferring or move or moved or moving or transmitted or transmission or write or written or writing or store or stored or storing)	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
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L14	349	(bi-direction or bi-directional) near3 fabric	US- PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:48
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L54	0	14 and 53	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:52
L55	0	16 and 53	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:52
L56	0	6 and 53	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; IBM_TDB	OR	ON	2014/09/19 17:53

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L42	15	((Chi-She) near2 (Chen)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L43	94	((Jeffrey) near2 (Solomon)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L44	114	((Scott) near2 (Milton)).INV.	US-PGPUB;	OR	ON	2014/09/19

			USPAT; UPAD			17:48
L45	81	((Jayesh) near2 (Bhakta)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L46	5660	((Hyun) near2 (Lee)).INV.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L47	5883	L42 or L43 or L44 or L45 or L46	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L48	6	(data adj manager with controller with memory adj controller).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L49	1	L47 and L48	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L50	12077	(data adj path or memory adj segment).clm.	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L51	38	L47 and L50	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48
L52	1	L48 and L51	US-PGPUB; USPAT; UPAD	OR	ON	2014/09/19 17:48

9/ 19/ 2014 5:53:35 PM
C:\ Users\ selmore\ Documents\ EAST\ Workspaces\ 13559476.wsp

<i>Response to Rule 312 Communication</i>	Application No. 13/559,476	Applicant(s) LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

1. ☒ The amendment filed on 06 August 2014 under 37 CFR 1.312 has been considered, and has been:

a) ☐ entered.

b) ☒ entered as directed to matters of form not affecting the scope of the invention.

c) ☐ disapproved because the amendment was filed after the payment of the issue fee.

Any amendment filed after the date the issue fee is paid must be accompanied by a petition under 37 CFR 1.313(c)(1) and the required fee to withdraw the application from issue.

d) ☐ disapproved. See explanation below.

e) ☐ entered in part. See explanation below.

	/STEPHEN ELMORE/ Primary Examiner, Art Unit 2188
--	---

Receipt date: 09/05/2014

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

13559476 - GAU: 2188

PTO/SB/08a (01-10)

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		13559476
Filing Date		2012-07-26
First Named Inventor	LEE, Hyun	
Art Unit	2188	
Examiner Name	ELMORE, STEPHEN C	
Attorney Docket Number	062453-010 1 of 2	

U.S.PATENTS

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	6145068		2000-11-07	Lewis	
	2	6199142		2001-03-06	Saulsbury et al.	
	3	6421279		2002-07-16	Tobita et al.	
	4	6944042		2005-09-13	Komatsuzaki	
	5	6952368		2005-10-04	Miura et al.	
	6	7062618		2006-06-13	Tsunoda et al.	
	7	7411859		2008-08-12	Sohn et al.	
	8	7421552		2008-09-02	Long	

Receipt date: 09/05/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	LEE, Hyun		
	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 1 of 2		

	9	7467251		2008-12-16	Park et al.	
	10	7600142		1990-10-17	Groos	
	11	7716411		2010-05-11	Panabaker et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

U.S.PATENT APPLICATION PUBLICATIONS

Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20030158995		2003-08-21	Lee et al.	
	2	20050141273		2005-06-30	Park et al.	
	3	20060039197		2006-02-23	Khoury et al.	
	4	20060069896		2006-03-30	Sanders	
	5	20060294295		2006-12-28	Fukuzo	

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FOREIGN PATENT DOCUMENTS

Receipt date: 09/05/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	LEE, Hyun		
	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 1 of 2		

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	JEDEC Standard 21-C, "Configurations for Solid State Memories," pp. 4.5.5-1 to 4.5.5-18.	☐
	2	Diablo Technologies, Inc.'s Invalidation Contentions, Case No. 13-CV-05889 YGR, dated June 6, 2014.	☐
	3	Smart Storage Systems, Inc.'s Invalidation Contentions, Case No. 4:13-cv-05889-YGR, dated June 6, 2014.	☐

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/Stephen Elmore/	Date Considered	09/22/2014
--------------------	------------------	-----------------	------------

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ²Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵Applicant is to place a check mark here if English language translation is attached.

Receipt date: 09/05/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13559476	13559476 - GAU: 2188
	Filing Date		2012-07-26	
	First Named Inventor	LEE, Hyun		
	Art Unit	2188		
	Examiner Name	ELMORE, STEPHEN C		
	Attorney Docket Number	062453-010 1 of 2		

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☒ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached certification statement.
- ☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.
- ☐ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Khaled Shami/	Date (YYYY-MM-DD)	2014-09-05
Name/Print	Khaled Shami	Registration Number	38745

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /S.E./



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 www.uspto.gov

BIB DATA SHEET

CONFIRMATION NO. 1046

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
13/559,476	07/26/2012	711	2188	062453-010	
APPLICANTS INVENTORS Hyun Lee, Ladera Ranch, CA; Chi-She Chen, Walnut, CA; Jeffrey C. Solomon, Irvine, CA; Scott Milton, Irvine, CA; Jayesh Bhakta, Cerritos, CA;					
** CONTINUING DATA ***** This appln claims benefit of 61/512,871 07/28/2011 and is a CIP of 12/240,916 09/29/2008 PAT 8301833 which is a CON of 12/131,873 06/02/2008 ABN which claims benefit of 60/941,586 06/01/2007					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 08/07/2012					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /STEPHEN C ELMORE/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 10	TOTAL CLAIMS 24	INDEPENDENT CLAIMS 2
ADDRESS Nixon Peabody LLP P.O. Box 60610 Palo Alto, CA 94306 UNITED STATES					
TITLE FLASH-DRAM HYBRID MEMORY MODULE					
FILING FEE RECEIVED 1620	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Issue Classification 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner STEPHEN ELMORE	Art Unit 2188	

CPC					
Symbol				Type	Version
G06F	12	0246		F	2013-01-01
G06F	1	185		I	2013-01-01
G06F	2212	205		A	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 15	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2188 (Primary Examiner) _____ (Date) _____		O.G. Print Claim(s) 1	O.G. Print Figure 6

Issue Classification 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1														
	2														
	3														
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	9														
	13														
	14														
	16														
	18														
	19														
	21														
	22														
	23														
	24														

NONE	Total Claims Allowed: 15	
(Assistant Examiner)	(Date)	
/STEPHEN ELMORE/ Primary Examiner.Art Unit 2188	9/19/2014	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 6

Search Notes 	Application/Control No. 13559476	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner STEPHEN ELMORE	Art Unit 2188

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	103, 111, 112, 114, 154, 156	4/20/2014	SE
365	185.33	4/20/2014	SE
Search Updated		9/19/2014	SE

SEARCH NOTES		
Search Notes	Date	Examiner
EAST	4/20/2014	SE
Inventor Name Search for DP	4/20/2014	SE
Search Updated	9/19/2014	SE

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	103	4/20/2014	SE
PGPUB Searched		4/20/2014	SE
Search Updated		9/19/2014	SE

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/S.E./ 09/19/2014

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT: Hyun Lee et al.

SERIAL NO.: 13/559,476

CONFIRMATION NO: 1046

FILING DATE: July 26, 2012

TITLE: Flash-DRAM Hybrid Memory Module

EXAMINER: Elmore, Stephen C.

ART UNIT: 2188

**Mail Stop Issue Fee
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

RESPONSE TO NOTICE TO FILE CORRECTED APPLICATION PAPERS

Dear Sir:

This paper is responsive to the notice mailed June 24, 2014. Please amend the above-identified application as follows:

Amendments to the Specification begin on page **2** of this paper.

Remarks begin on page **3** of this paper.

Electronic Acknowledgement Receipt

EFS ID:	20055150
Application Number:	13559476
International Application Number:	
Confirmation Number:	1046
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	46188
Filer:	Khaled Shami/Pamela Wilson
Filer Authorized By:	Khaled Shami
Attorney Docket Number:	062453-010
Receipt Date:	05-SEP-2014
Filing Date:	26-JUL-2012
Time Stamp:	11:59:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$670
RAM confirmation Number	9407
Deposit Account	503557
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

Electronic Patent Application Fee Transmittal

Application Number:	13559476			
Filing Date:	26-Jul-2012			
Title of Invention:	FLASH-DRAM HYBRID MEMORY MODULE			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Khaled Shami/Pamela Wilson			
Attorney Docket Number:	062453-010			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Petition Fee-37CFR 1.17(h) (Group II)	2464	1	70	70
Request for Continued Examination	2801	1	600	600
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Adjustment date: 09/29/2014 SDIRETAL
09/05/2014 INTEFSW 00009407 503557 13559476
02 FC:2801 600.00 CR



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/559,476	10/28/2014	8874831	062453-010	1046

46188 7590 10/08/2014
Nixon Peabody LLP
P.O. Box 60610
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 176 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Chi-She Chen, Walnut, CA;
Jeffrey C. Solomon, Irvine, CA;
Scott Milton, Irvine, CA;
Jayesh Bhakta, Cerritos, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00692
Patent 8,874,831 B2

Before STEPHEN C. SIU, MATTHEW R. CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

CLEMENTS, *Administrative Patent Judge*.

DECISION
Instituting *Inter Partes* Review
35 U.S.C. § 314 and 37 C.F.R. § 42.108

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc. and SK hynix memory solutions Inc. (“Petitioner”) filed a Petition requesting *inter partes* review of claims 1–15 (“the challenged claims”) of U.S. Patent No. 8,874,831 B2 (Ex. 1001, “the ’831 patent”). Paper 1 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response. Paper 6 (“Prelim. Resp.”). We review the Petition pursuant to 35 U.S.C. § 314, which provides that an *inter partes* review may be authorized only if “the information presented in the petition . . . and any [preliminary] response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a); 37 C.F.R. § 42.4(a).

Upon consideration of the Petition and the Preliminary Response, we determine that the information presented by Petitioner establishes that there is a reasonable likelihood that Petitioner would prevail in showing the unpatentability of at least one of the challenged claims of the ’831 patent. Accordingly, pursuant to 35 U.S.C. § 314, we institute an *inter partes* review of claims 1–15 of the ’831 patent.

A. Related Proceedings

The ’831 patent is involved in *Netlist, Inc. v. Smart Modular Technologies, Inc. et al*, Case No. 2:13-cv-02613 (E.D. Cal.). Paper 4, 3. Related patents have been asserted in *Netlist, Inc. v. SMART Modular Technologies, Inc.*, Case No. 8-13-cv-00996 (C.D. Cal.), *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. 4-13-cv-03916 (N.D. Cal.), *Diablo Technologies, Inc. v. Netlist, Inc.*, Case No. 4-13-cv-03901 (N.D. Cal.), and *Netlist, Inc. v. Smart Modular Technologies, Inc.*, 4-13-cv-05889

(N.D. Cal.). Pet. 2. Related patents are also the subject of *SanDisk Corp. v. Netlist, Inc.*, Case No. IPR2014-00982 (PTAB) (institution denied), *SanDisk Corp. v. Netlist, Inc.*, Case No. IPR2014-00994 (PTAB) (institution denied), *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01370 (PTAB) (institution denied); *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01371 (PTAB) (institution denied), *SK hynix Inc., et al. v. Netlist, Inc.*, Case No. IPR2017-00587 (PTAB) (instituted June 22, 2017), and *SK hynix Inc., et al. v. Netlist, Inc.*, Case No. IPR2017-00649 (PTAB). Pet. 2; Paper 4, 3.

B. The '831 patent

The '831 patent, titled “Flash-Dram Hybrid Memory Module,” issued October 28, 2014, from U.S. Patent Application No. 13/559,476. Ex. 1001 at [54], [45], [21]. The '831 patent generally relates to a memory module with a non-volatile memory, a volatile memory, and a data manager through which the volatile memory and non-volatile memory may exchange data, and a controller to receive read/write commands from a memory controller hub (“MCH”) and transfer data between any two or more of the MCH, volatile memory, and non-volatile memory. *Id.* at Abstract. Figure 4A is reproduced below.

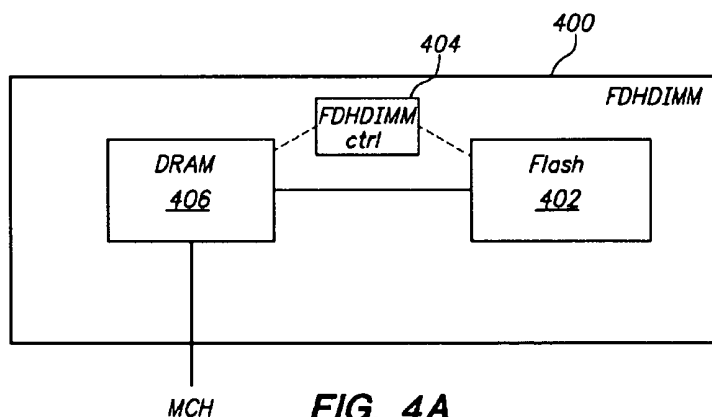


Figure 4A is a block diagram of a Flash-DRAM hybrid dynamic random access memory dual in-line memory module (DIMM). In this embodiment, volatile memory subsystem 406 (e.g. DRAM) is used as a data buffer such that data from Flash memory 402 is transferred to DRAM 406 at the Flash access speed, and buffered or collected into DRAM 406, which then transfers the buffered data to the MCH based on the access time of DRAM. *Id.* at 9:15–21. Similarly, when the MCH transfers data to DRAM 406, controller 404 manages the data transfer from DRAM 406 to Flash 402. *Id.* at 9:21–23.

Figure 5 is reproduced below.

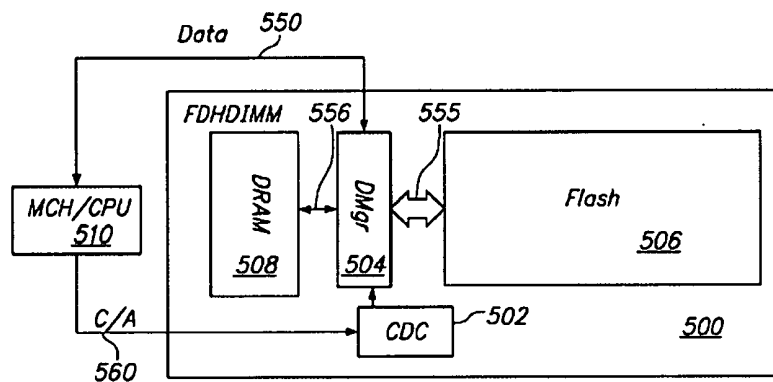


FIG. 5A

Figure 5A is a block diagram of memory module 500 in accordance with certain embodiments. Ex. 1001, 7:7–8. As shown in Figure 5, memory module 500 includes two on-module intermediary components: controller (CDC) 502 and data manager (DMgr) 504. *Id.* at 10:35–46. These components “manage the interface between a non-volatile memory subsystem such as a Flash 506, a volatile memory subsystem such as a DRAM 508, and a host system represented by MCH 510.” *Id.* at 10:49–53. “In certain embodiments, CDC 502 controls the read/write access to/from Flash memory 506 from/to DRAM memory 508, and to/from DRAM

memory from/to MCH 510.” *Id.* at 10:54–56. “In certain embodiments and in response to communication from CDC 502, DMgr 504 provides a variety of functions to control data flow rate, data transfer size, data buffer size, data error monitoring or data error correction.” *Id.* at 11:18–21.

Figure 6 is reproduced below.

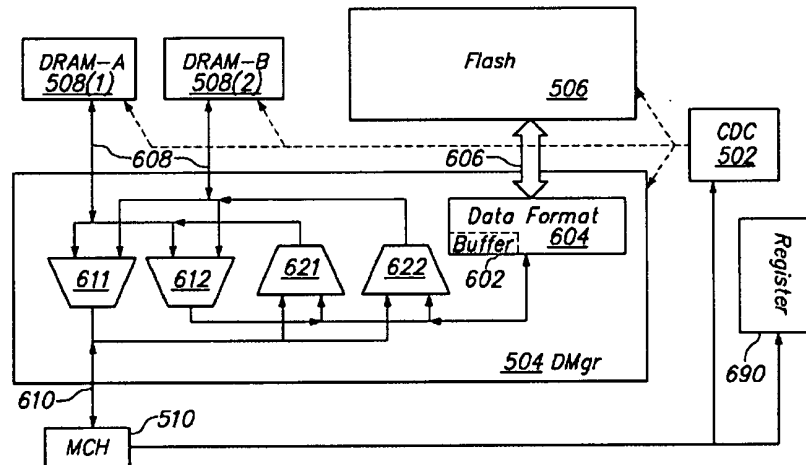


FIG. 6

Figure 6 is a block diagram showing some details of data manager 504. Ex. 1001, 7:11–12. “In certain embodiments, DMgr 504 also functions as a bi-directional data transfer fabric.” *Id.* at 12:1–3. “For example, DMgr 504 may have more than 2 sets of data ports facing the Flash 506 and the DRAM 508.” *Id.* at 12:3–5. “Multiplexers 611 and 612 provide controllable data paths from any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B) to any one of the MCH 510 and the Flash 506.” *Id.* at 12:5–8. “Similarly multiplexers 621 and 622 provide controllable data paths from any one of the MCH and the Flash memory to any one of the DRAMs 508(1) and 508(2) (DRAM-A and DRAM-B).” *Id.* at 12:8–11.

C. Illustrative Claim

Of the challenged claims, claims 1 and 7 are independent, claims 2–6 depend, directly or indirectly, from claim 1, and claims 8–15 depend, directly or indirectly, from claim 7. Independent claim 1 is illustrative of the challenged claims and is reproduced below:

1. A memory module couplable to a memory controller of a host system, comprising:

a non-volatile memory subsystem;

a data manager coupled to the non-volatile memory subsystem;

a volatile memory subsystem coupled to the data manager and operable to exchange data with the non-volatile memory subsystem by way of the data manager; and

a controller operable to receive commands from the memory controller and to direct (i) operation of the non-volatile memory subsystem, (ii) operation of the volatile memory subsystem, and (iii) transfer of data between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on at least one received command from the memory controller, wherein:

at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments, each memory segment comprising at least one memory circuit, memory device, or memory die, and

the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems, the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a

data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

Ex. 1001, 17:44–18:13.

D. Evidence Relied Upon

Petitioner relies upon the following prior art references:

Best	US 2010/0110748 A1	May 6, 2010	Ex. 1006
Roy	US 6,065,092	May 16, 2000	Ex. 1008
Tsunoda	US 2003/0028733 A1	Feb. 6, 2003	Ex. 1009
Ashmore	US 2006/0212651 A1	Sept. 21, 2006	Ex. 1011
Bonella	US 2007/0136523 A1	June 14, 2007	Ex. 1013
Roohparvar	US 2005/0273548 A1	Dec. 8, 2005	Ex. 1019

Pet. 3. Petitioner also relies upon the Declaration of Ron Maltiel (“Maltiel Decl.”) (Ex. 1003).

E. Asserted Grounds of Unpatentability

Petitioner asserts that the challenged claims are unpatentable based on the following grounds (Pet. 3):

Reference(s)	Basis	Claim(s) challenged
Best	§ 102	1–14
Best and Roy	§ 103	1–14
Best and Tsunoda, with or without Roy	§ 103	2 and 8
Best and Roohparvar, with or without Roy	§ 103	5 and 12–14
Best and Bonella, with or without Roy	§ 103	15
Best, Bonella, and Ashmore, with or without Roy	§ 103	15

Petitioner also relies upon the Declaration of Ron Maltiel. Ex. 1003 (“Maltiel Decl.”).

II. ANALYSIS

A. Claim Construction

In an *inter partes* review, a claim in an unexpired patent shall be given its broadest reasonable construction in light of the specification of the patent in which it appears. 37 C.F.R. § 42.100(b). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term must be set forth in the specification with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994). We must be careful not to read a particular embodiment appearing in the written description into the claim if the claim language is broader than the embodiment. *See In re Van Geuns*, 988 F.2d 1181, 1184 (Fed. Cir. 1993). Only terms that are in controversy need to be construed, and then only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999).

Petitioner proposes constructions for “bi-directional data transfer fabric,” “set of data ports,” “format data,” and “operable at a . . . clock frequency.” Pet. 10–14. Patent Owner argues that no construction is necessary for “bi-directional data transfer fabric,” “set of data ports,” and “operable at a . . . clock frequency.” Prelim. Resp. 21–24. Patent Owner proposes a different construction of “format data,” and offers constructions

for the terms “memory module,” “memory address mapping,” “address domain conversion,” and “data width modulation.” *Id.* at 21, 23–28. On this record, and for purposes of this Decision, we determine that only the term “memory module” requires express construction.

1. “memory module” (claim 1)

Petitioner does not propose a construction for “memory module.” Patent Owner contends that “memory module” should be construed to mean “a carrier that contains one or more memory chips.” Prelim. Resp. 21. Specifically, Patent Owner contends that a “memory module” is a removable circuit board and does not refer to an integrated circuit by itself. *Id.* (citing Ex. 2001 (Microsoft Computer Dictionary, 5th ed.), 334 (“**memory module** *n.* A removable circuit board, cartridge, or other carrier that contains one or more RAM memory chips.”); *see also* Prelim. Resp. 32–33 (arguing that Best’s integrated circuit package is not a “memory module” because it is not a circuit board). Patent Owner also relies upon our construction of “memory module” in a prior proceeding involving a different patent to mean “a carrier that contains one or more memory chips.” *Id.* (citing *Diablo Technologies, Inc. v. Netlist, Inc.*, Case IPR2014-00882 (PTAB Dec. 14, 2015) (Paper 33 (“882 FWD”), 8–11 (construing “memory module” as used in U.S. Patent No. 7,881,150 B2) (“the ’150 patent”))).

The ’831 patent does not define a “memory module.” The ’831 patent depicts a memory module 500 in Figures 5A and 5B, and describes how “[i]n certain embodiments, memory module 500 is a Flash-DRAM hybrid memory module that has the DIMM (dual-inline memory module) form factor” (Ex. 1001, 10:28–31), but the ’831 patent also expressly contemplates other form factors (*see, e.g., id.* at 10:32–34 (“it is to be

understood that in both structure and operation [memory module 500] may be different from the FDHDIMM discussed above and described with reference to FIGS. 4A and 4B”), 10:46–49 (“While the DIMM form factor will predominate the discussion herein, it should be understood that this is for illustrative purposes only and memory systems using other form factors are contemplated as well.”).

Moreover, our construction of “memory module” in IPR2014-00882 is informative, but not dispositive, because the ’150 patent has a different Specification than the ’831 patent, and our construction in that case was based, in part, on disclosures in the ’150 patent that are not found in the ’831 patent. *See, e.g.*, 882 FWD, 10 (citing teachings of a printed circuit board on which memory devices are mounted). The ’831 patent, in contrast, does not use the terms “printed circuit board” or “circuit board” even once, much less limit explicitly a “memory module” to a removable printed circuit board, as Patent Owner suggests. Because the ’831 patent is open-ended regarding the form factor that a “memory module” may take, and because it makes no mention of a printed circuit board, much less a removable printed circuit board, we also are not persuaded by Patent Owner’s extrinsic evidence (Ex. 2001) such that we should import such a limitation into our construction of this term.

On this record, and for purposes of this decision, we construe “memory module” to mean “a carrier that contains one or more memory chips,” where “carrier” encompasses an integrated circuit package.

B. Claims 1–14: Anticipation by Best

Petitioner argues that the claims 1–14 are unpatentable under 35 U.S.C. § 102(e) as anticipated by Best. Pet. 11–35. In light of the

arguments and evidence of record, we are persuaded that Petitioner has established a reasonable likelihood that the claims 1–14 are unpatentable as anticipated by Best.

1. *Best (Ex. 1006)*

Best is directed to a hybrid volatile and non-volatile memory device. Ex. 1006, Abstract. Specifically, Best discloses “[a] composite, hybrid memory device including a first storage die having an array of volatile storage cells and a second storage die having an array of non-volatile storage cells disposed within an integrated circuit package.” *Id.* “The hybrid memory device includes a shared interface circuit to receive memory access commands directed to the first storage die and the second storage die and to convey read and write data between an external data path and the first and second storage dice.” *Id.*

Figure 2 of Best is reproduced below.

FIG. 2

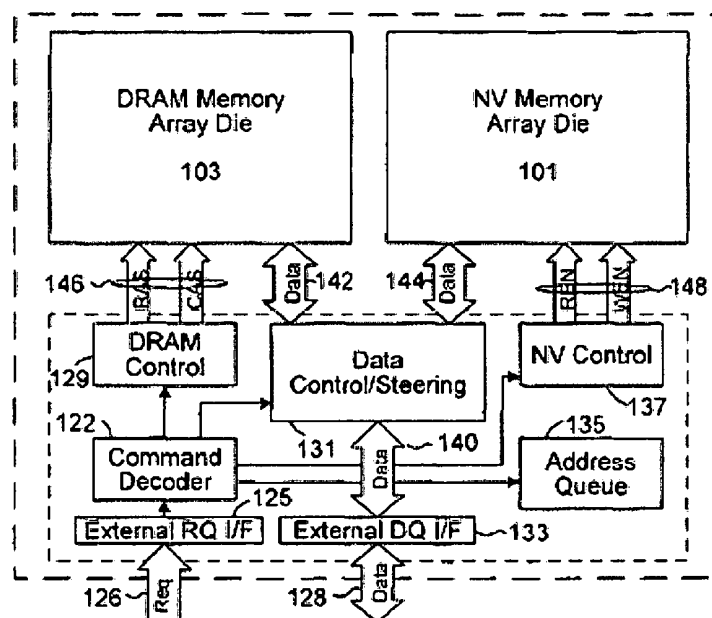


Figure 2 illustrates an embodiment of a hybrid, composite memory device with shared interface circuitry, volatile memory, and non-volatile memory. *Id.* ¶ 17. “[T]he shared interface circuitry includes an external request interface 125, external data interface 133, command decoder 122, address queue 135, DRAM control circuit 129, Flash control circuit 137, and data control/steering circuit 131.” *Id.* “[I]ncoming control signals and addresses . . . are received in the external request interface 125 via control/address (CA) path 126, reformatted as necessary (e.g., deserialized to form a parallel command word and one or more address values) and then forwarded to the command decoder 122.” *Id.* “The command decoder 122 in turn forwards address to the address queue 135 and stores memory access commands.” *Id.* “[M]emory access operations may be automatically directed to either the volatile storage die or non-volatile storage die according to the memory address to be accessed.” *Id.* “[C]ommand decoder 122 outputs, from the head of the command queue, an enable signal and corresponding memory access control signals to the DRAM control circuit 129 and NV control circuit 137.” *Id.* ¶ 18. “[D]ata control/steering circuit 131 is used to control the transfer of data between a shared internal data bus and dedicated internal data buses associated with the volatile and non-volatile storage dice, respectively.” *Id.* ¶ 20.

Figure 3 of Best is reproduced below.

FIG. 3

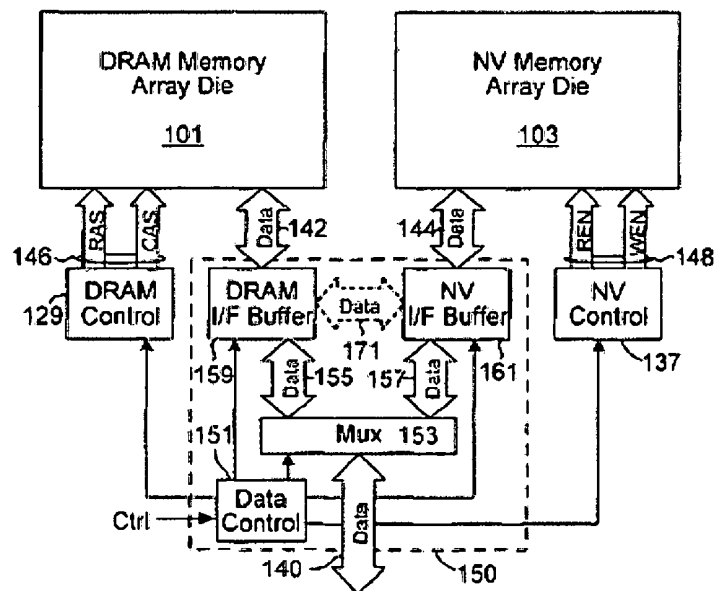


Figure 3 illustrates an embodiment of a data control/steering circuit 150 that may be used to implement the data control/steering circuit 131 of Figure 2.

Id. ¶ 21. “[D]ata control circuit 151 receives control signals from the command decoder that indicate the direction of data flow during a memory access operation (read or write) and whether the volatile or non-volatile storage die is the target of the memory access.” *Id.*

2. Independent claims 1 and 7

Claim 1 recites “[a] memory module couplable to a memory controller of a host system.” Petitioner relies upon Best’s disclosure of “a ‘hybrid composite memory device having non-volatile and volatile memories implemented in distinct integrated circuit (IC) dice that are packaged together and accessed through a shared interface.’” Pet. 20 (quoting Ex. 1006 ¶ 12). Claim 1 further recites “a non-volatile memory subsystem.”

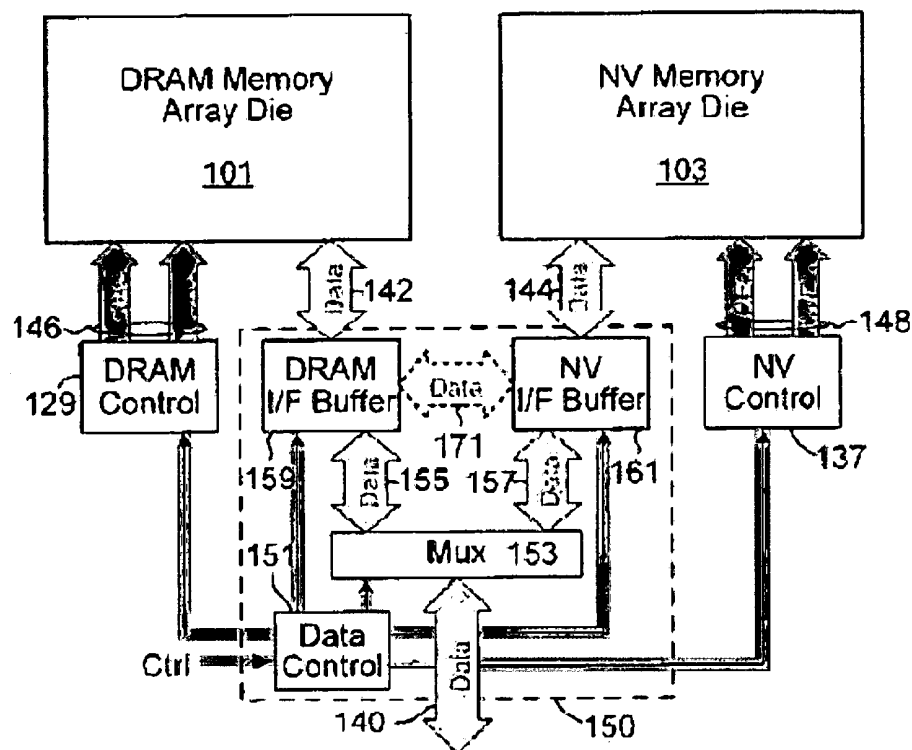
Petitioner relies upon Best's disclosure of a Flash memory. Pet. 21. Claim 1 further recites "a data manager coupled to the non-volatile memory subsystem." Petitioner relies upon Best's data control/steering circuit in combination with the external interface. Pet. 21–22 (citing Ex. 1006, Figure 3). Claim 1 further recites "a volatile memory subsystem." Petitioner relies upon Best's disclosure of a DRAM. Pet. 22–23. Claim 1 further recites "a controller." Petitioner relies upon Best's command decoder 122, which receives "incoming control signals and addresses" and directs operation of the volatile and non-volatile memories by "output[ting] . . . an enable signal and corresponding memory access control signals to the DRAM control circuit . . . and NV control circuit," and transfers data between the memories and memory controller. Pet. 23–24 (citing Ex. 1006 ¶¶ 17–19, 21, 29). Claim 1 further recites "at least one of the volatile and non-volatile memory subsystems comprises one or more memory segments." Petitioner relies upon Best's disclosure that "the volatile and non-volatile memories [are] implemented by a DRAM die 103 and Flash memory die 101, respectively," each of which, according to Petitioner, is a memory segment. Pet. 24–25. Claim 1 further recites

the data manager is configured as a bi-directional data transfer fabric having two or more sets of data ports, a first set of data ports of the two or more sets of data ports is coupled to the volatile memory subsystem, a second set of data ports of the two or more sets of data ports is coupled to the non-volatile memory subsystem, the two or more sets of data ports being operable by the data manager to transfer data to or from one or more memory segments of the volatile or non-volatile memory subsystems.

Petitioner relies upon Best's disclosure of an interface to the primary volatile data path 142 between data control/steering circuit 150 and DRAM 101 ("a first set of data ports . . . coupled to the volatile memory subsystem"), and of an

interface to primary non-volatile data path 144 between data control steering circuit 150 and NV memory 103 (“a second set of data ports . . . coupled to the non-volatile memory subsystem”). Pet. 25–26. Petitioner contends that each set of interconnections is “bi-directional” because data can flow in either direction, and illustrates the point with an annotated version of Figure 3, reproduced below:

FIG. 3



Finally, claim 1 recites

the data manager further including a data buffer for buffering data delivered to or from the non-volatile memory subsystem, and a data format module configured to format data to be transferred between any two or more of the memory controller, the volatile memory subsystem, and the non-volatile memory subsystem based on control information received from the controller.

Petitioner relies upon Best’s disclosure of “non-volatile-storage-die interface buffer 161” as the recited “data buffer” (Pet. 27–28), and relies

upon Best's disclosure of logic for serializing/deserializing within data steering/control circuit and the external data interface as the recited "data format module" (*id.* at 28–30). We are persuaded that Petitioner's citations support its contentions.

We also are persuaded that Petitioner's citations support its contentions with respect to claim 7, for which Petitioner relies upon largely the same disclosures in Best (Pet. 38–43).

Patent Owner argues that Best does not disclose a "memory module" because it is implemented on a single integrated circuit package—i.e., a single chip. Prelim. Resp. 32–33. This argument is not persuasive, however, because it is based upon Patent Owner's proposed construction of "memory module," which we do not adopt. Because our broadest reasonable construction of "memory module," as discussed above, encompasses an integrated circuit package, we are persuaded that Best discloses this limitation.

Patent Owner also argues that "neither [Best's interface to the single volatile data path 142 nor its interface to the single nonvolatile data path 144 is] a 'set of data ports' because each interface is to a single data bus." Prelim. Resp. 33–34. This argument also is not persuasive. Patent Owner argues that no construction is necessary for the phrase "set of data ports." Prelim. Resp. 23. The claims require that the "first set of data ports" is "coupled to the volatile memory subsystem," and that the "second set of data ports" is "coupled to the non-volatile memory subsystem." The '831 patent explicitly describes non-volatile memory subsystem 506 as coupled to a single data bus, i.e., data bus 606. Ex. 1001, Fig. 6, 12:58 ("wide data bus

606 coupled to the Flash memory 506”). As a result, we are not persuaded that “set of data ports” excludes a single data bus, as disclosed by Best.

3. Dependent claims 2–6 and 8–14

We have reviewed Petitioner’s explanations and supporting evidence regarding dependent claims 2–6 and 8–14, and find them persuasive. *See* Pet. 30–49. Patent Owner does not argue separately these claims. Based on the record before us, Petitioner has demonstrated a reasonable likelihood that it would prevail on its assertion that claims 2–6 and 8–14 are anticipated by Best.

4. Conclusion

On this record, we are persuaded that Petitioner has established a reasonable likelihood that it would prevail in showing that claims 1–14 are unpatentable as anticipated by Best.

C. Claim 1–14: Obviousness over Best¹

As discussed above, we are persuaded that Petitioner has established a reasonable likelihood that it would prevail in showing that claims 1–14 are unpatentable as anticipated by Best. Inasmuch as “anticipation is the epitome of obviousness” (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)) and because there is no evidence of objective indicia of non-obviousness on the record, we determine that Petitioner has also established a reasonable likelihood of prevailing in showing that claims 1–14 would have been obvious over Best.

¹ Although Petitioner challenges claims 1–14 as obvious over “Best in view of Roy” (Pet. 3), its other obviousness challenges are “with or without Roy” (*id.*). We, therefore, interpret the Petition to allege that claims 1–14 also are obvious over Best alone.

*D. Claims 1–14: Obviousness over
Best and Roy*

Petitioner argues that claims 1–14 are unpatentable under 35 U.S.C. § 103(a) as obvious over the combination of Best and Roy. Pet. 49–53. In light of the arguments and evidence of record, we are persuaded that Petitioner has established a reasonable likelihood that the claims 1–14 are unpatentable as obvious over the combination of Best and Roy.

1. Roy (Ex. 1008)

Roy is directed generally to an “independent and cooperative multichannel memory architecture” that includes a plurality of independent channels between a master device and one or more memory clusters. Ex. 1008, Abstract. Figure 1 of Roy is reproduced below.

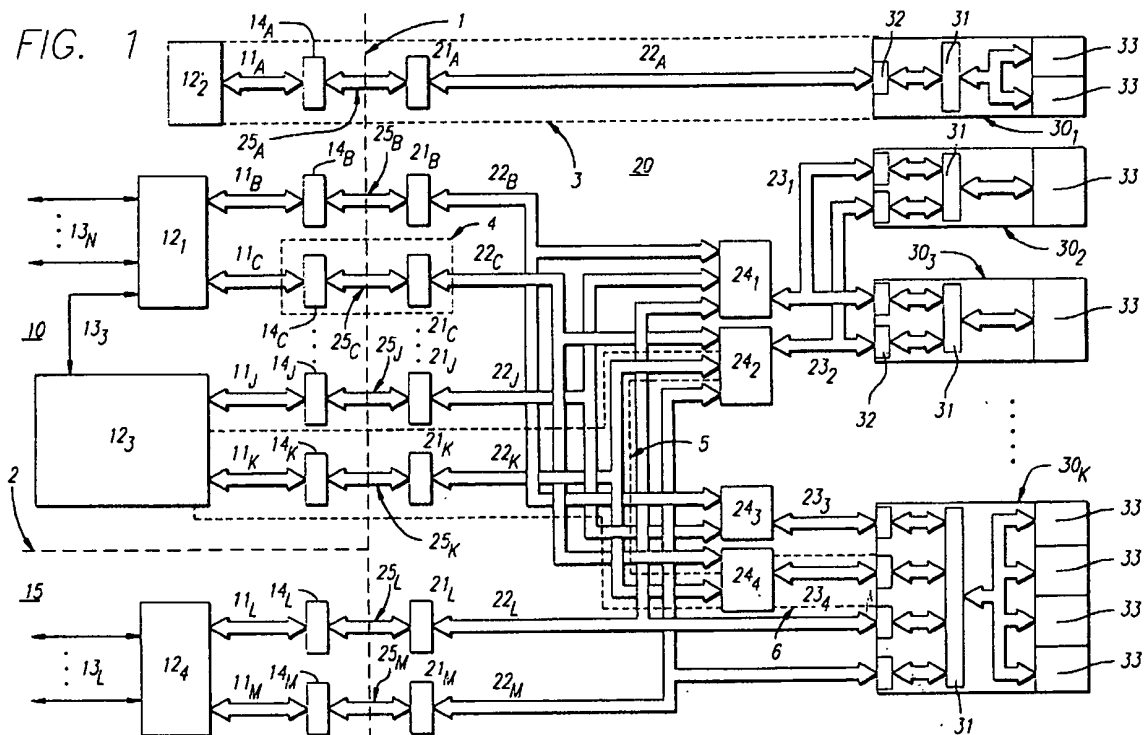


Figure 1 of Roy is a system level block diagram of a multichannel memory architecture disclosed in Roy. *Id.* at 7:59–60. As shown, “memory device 20 further includes multiplexer units 24_{1–4} which couple individual ones of a